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Franc Forstnerič

Stein Manifolds and Holomorphic Mappings

The Homotopy Principle in Complex Analysis

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The Homotopy Principle in Complex Analysis

Springer

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Preface

This book is an attempt to present a coherent account of Oka theory, from the classical Oka-Grauert theory originating in the works of Kiyoshi Oka and Hans Grauert to the contemporary developments initiated by Mikhael Gromov.

At the core of Oka theory lies the heuristic *Oka principle*, a term coined by Jean-Pierre Serre in 1951: *Analytic problems on Stein manifolds admit analytic solutions if there are no topological obstructions.* The Cartan-Serre Theorems A and B are primary examples. The main exponent of the classical Oka-Grauert theory is the equivalence between topological and holomorphic classification of principal fiber bundles over Stein spaces. On the interface with affine algebraic geometry the Oka principle holds only rarely, while in projective geometry we have Serre's GAGA principle, the equivalence of analytic and algebraic coherent sheaves on compact projective algebraic varieties. In smooth geometry there is the analogous *homotopy principle* originating in the Smale-Hirsch homotopy classification of smooth immersions.

Modern Oka theory focuses on those properties of a complex manifold Y which insure that any continuous map $X \to Y$ from a Stein source space X can be deformed to a holomorphic map; the same property is considered for sections of a holomorphic submersion $Y \to X$. By including the Runge approximation and the Cartan extension condition one obtains several ostensibly different Oka properties. Gromov's main result is that a geometric condition called ellipticity – the existence of a dominating holomorphic spray on Y – implies all forms of the Oka principle for maps or sections $X \to Y$. Subsequent research culminated in the result that all Oka properties of a complex manifold Y are equivalent to the following Runge approximation property:

A complex manifold Y is said to be an Oka manifold *if every holomorphic map $f\colon K \to Y$ from a neighborhood of a compact convex set $K \subset \mathbb{C}^n$ to Y can be approximated uniformly on K by entire maps $\mathbb{C}^n \to Y$.*

The related concept of an *Oka map* pertains to the Oka principle for lifting holomorphic maps from Stein sources. The class of Oka manifolds is dual to

the class of Stein manifolds in a sense that can be made precise by means of abstract homotopy theory. Finnur Lárusson constructed a model category containing all complex manifolds in which Stein manifolds are cofibrant, Oka manifolds are fibrant, and Oka maps are fibrations. This means that

Stein manifolds are the natural sources of holomorphic maps, while Oka manifolds are the natural targets.

Oka manifolds seem to be few and special; in particular, no compact complex manifold of Kodaira general type is Oka. However, special and highly symmetric objects are often more interesting than average generic ones.

A few words about the content. Chapter 1 contains some preparatory material, and Chapter 2 is a brief survey of Stein space theory. In Chapter 3 we construct open Stein neighborhoods of certain types of sets in complex spaces that are used in Oka theory. Chapter 4 contains an exposition of the theory of holomorphic automorphisms of Euclidean spaces and of the density property, a subject closely intertwined with our main theme. In Chapter 5 we develop Oka theory for stratified fiber bundles with Oka fibers (this includes the classical Oka-Grauert theory), and in Chapter 6 we treat Oka-Gromov theory for stratified subelliptic submersions over Stein spaces. Chapters 7 and 8 contain applications ranging from classical to the recent ones. In Chapter 8 we present results on regular holomorphic maps of Stein manifolds; highlights include the optimal embedding theorems for Stein manifolds and Stein spaces, proper holomorphic embeddings of some bordered Riemann surfaces into $\mathbb{C}^2$, and the construction of noncritical holomorphic functions, submersions and foliations on Stein manifolds. In Chapter 9 we explore implications of Seiberg-Witten theory to the geometry of Stein surfaces, and we present the Eliashberg-Gompf construction of Stein structures on manifolds with suitable handlebody decomposition. A part of this story is the *Soft Oka principle*.

This book would not have existed without my collaboration with Jasna Prezelj who explained parts of Gromov's work on the Oka principle in her dissertation (University of Ljubljana, 2000). Josip Globevnik suggested that we look into this subject, while many years earlier Edgar Lee Stout proposed that I study the Oka-Grauert principle. My very special thanks go to the colleagues who read parts of the text and offered suggestions for improvements: Barbara Drinovec-Drnovšek, Frank Kutzschebauch, Finnur Lárusson, Takeo Ohsawa, Marko Slapar, and Erlend Fornæss Wold. I am grateful to Reinhold Remmert for his invitation to write a volume for the Ergebnisse series, and to the staff of Springer-Verlag for their professional help.

Finally, I thank Angela Gheorghiu for all those incomparably beautiful arias, and my family for their patience.

Ljubljana, *Franc Forstnerič*
May 1, 2011

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'Say at last – who art thou?'
'That Power I serve which wills forever evil yet does forever good.'

J. W. Goethe, *Faust*

'Forgive me, but I don't believe you,' said Woland. 'That cannot be. Manuscripts don't burn.'

M. A. Bulgakov, *The Master and Margarita*

1

Preliminaries

This preliminary chapter is a brief review of the basic notions and constructions that are indispensable for reading the book. A comprehensive account is available in a number of excellent sources; for smooth manifolds see [5] and [503]; for complex and algebraic manifolds see [103, 233, 241, 370, 229, 508], among others; and for the theory of Stein manifolds and Stein spaces see the monographs [228, 241], and [267].

1.1 Complex Manifolds and Holomorphic Mappings

We denote by $\mathbb{R}$ the field of real numbers and by $\mathbb{C}$ the field of complex numbers. Let $n \in \mathbb{N} = \{1, 2, 3, \ldots\}$ be a positive integer. The model n-dimensional complex manifold is the complex Euclidean space $\mathbb{C}^n$, the Cartesian product of n copies of $\mathbb{C}$. Let $z = (z_1, \ldots, z_n)$ denote the complex coordinates on $\mathbb{C}^n$. Write $z_j = x_j + \mathrm{i}\, y_j$, where $x_j, y_j \in \mathbb{R}$ and $\mathrm{i} = \sqrt{-1}$. Given a differentiable complex valued function $f\colon D \to \mathbb{C}$ on a domain $D \subset \mathbb{C}^n$, the differential $\mathrm{d}f$ splits as the sum of the $\mathbb{C}$-linear part ∂f and the $\mathbb{C}$-antilinear part $\overline{\partial} f$:

$$\mathrm{d}f = \partial f + \overline{\partial} f = \sum_{j=1}^{n} \frac{\partial f}{\partial z_j}\, \mathrm{d}z_j + \sum_{j=1}^{n} \frac{\partial f}{\partial \bar{z}_j}\, \mathrm{d}\bar{z}_j. \tag{1.1}$$

Here $\mathrm{d}z_j = \mathrm{d}x_j + \mathrm{i}\,\mathrm{d}y_j$, $\mathrm{d}\bar{z}_j = \mathrm{d}x_j - \mathrm{i}\mathrm{d}y_j$, and

$$\frac{\partial f}{\partial z_j} = \frac{1}{2}\left(\frac{\partial f}{\partial x_j} - \mathrm{i}\frac{\partial f}{\partial y_j}\right), \qquad \frac{\partial f}{\partial \bar{z}_j} = \frac{1}{2}\left(\frac{\partial f}{\partial x_j} + \mathrm{i}\frac{\partial f}{\partial y_j}\right). \tag{1.2}$$

The function f is *holomorphic* if $\mathrm{d}f = \partial f$ on D; that is, if the differential $\mathrm{d}f_z$ is $\mathbb{C}$-linear at every point $z \in D$. Equivalently, f is holomorphic if and only if $\overline{\partial} f = 0$, and this is equivalent to the n equations

$$\frac{\partial f}{\partial \bar{z}_j} = 0, \quad j = 1, \ldots, n.$$

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Writing $f = u + \mathrm{i}v$ with u and v real, the equation $\partial f/\partial \bar{z}_j = 0$ is equivalent to the system of Cauchy-Riemann equations

$$\frac{\partial u}{\partial x_j} = \frac{\partial v}{\partial y_j}, \qquad \frac{\partial u}{\partial y_j} = -\frac{\partial v}{\partial x_j}. \tag{1.3}$$

A mapping $f = (f_1, f_2, \ldots, f_m)\colon D \to \mathbb{C}^m$ is holomorphic if each component function f_j is such. When $m = n$, f is *biholomorphic* onto its image $D' = f(D) \subset \mathbb{C}^n$ if it is bijective and its inverse $f^{-1}\colon D' \to D$ is also holomorphic. An injective holomorphic map of a domain $D \subset \mathbb{C}^n$ to $\mathbb{C}^n$ is always biholomorphic onto its image [233, p. 19].

A *topological manifold of dimension* n is a second countable Hausdorff topological space which is locally Euclidean, in the sense that each point has an open neighborhood homeomorphic to an open set in $\mathbb{R}^n$. Such a space is metrizable, countably compact, and paracompact.

Assume now that X is a topological manifold of even dimension $2n$. A *complex atlas* on X is a collection $\mathcal{U} = \{(U_\alpha, \phi_\alpha)\}_{\alpha \in A}$, where $\{U_\alpha\}_{\alpha \in A}$ is an open cover of X and ϕ_α is a homeomorphism of U_α onto an open subset U'_α in $\mathbb{R}^{2n} = \mathbb{C}^n$ such that for every pair of indexes $\alpha, \beta \in A$ the *transition map*

$$\phi_{\alpha,\beta} = \phi_\alpha \circ \phi_\beta^{-1}\colon \phi_\beta(U_{\alpha,\beta}) \to \phi_\alpha(U_{\alpha,\beta}) \tag{1.4}$$

is biholomorphic. Here $U_{\alpha,\beta} = U_\alpha \cap U_\beta$. An element (U_α, ϕ_α) of a complex atlas is called a *complex chart*, or a *local holomorphic coordinate system* on X. We also say that charts in a complex atlas are *holomorphically compatible.* For any three indexes $\alpha, \beta, \gamma \in A$ we have

$$\phi_{\alpha,\alpha} = \mathrm{Id}, \quad \phi_{\alpha,\beta} = \phi_{\beta,\alpha}^{-1}, \qquad \phi_{\alpha,\beta} \circ \phi_{\beta,\gamma} = \phi_{\alpha,\gamma} \tag{1.5}$$

on the respective domains of these maps. Two complex atlases $\mathcal{U}$, $\mathcal{V}$ on a topological manifold X are said to be *holomorphically compatible* if their union $\mathcal{U} \cup \mathcal{V}$ is also a complex atlas. This is an equivalence relation on the set of all complex atlases on X. Each equivalence class contains a unique *maximal complex atlas* – the union of all complex atlases in the given class.

A *complex manifold of complex dimension* n is a topological manifold X of real dimension $2n$ equipped with a complex atlas. Two complex atlases determine the same complex structure on X if and only if they are holomorphically compatible. We write $n = \dim_{\mathbb{C}} X$. A complex manifold of dimension one is called a *Riemann surface*, or a *complex curve* when it is seen as a complex submanifold in another complex manifold. A *complex surface* is a complex manifold of dimension $n = 2$.

A function $f\colon X \to \mathbb{C}$ on a complex manifold is said to be holomorphic if for any chart (U, ϕ) from the maximal atlas on X the function $f \circ \phi^{-1}\colon \phi(U) \to \mathbb{C}$ is holomorphic on the open set $\phi(U) \subset \mathbb{C}^n$. We denote by $\mathcal{O}(X)$ the Fréchet algebra of all holomorphic functions on X with the compact-open topology.

If D is a relatively compact domain with $\mathcal{C}^r$ smooth boundary in a complex manifold X for some $r \in \mathbb{N}$ then $\mathcal{A}^r(D)$ denotes the Banach algebra of all functions $\bar{D} \to \mathbb{C}$ of class $\mathcal{C}^r$ that are holomorphic on D.

Let X and Y be complex manifolds of dimensions n and m, respectively. A continuous map $f\colon X \to Y$ is said to be *holomorphic* if for any point $p \in X$ there are complex charts (U, ϕ) on X and (V, ψ) on Y such that $p \in U$, $f(U) \subset V$, and the map $\widetilde{f} = \psi \circ f \circ \phi^{-1}\colon \phi(U) \longrightarrow \psi(V) \subset \mathbb{C}^m$ is holomorphic on the open set $\phi(U) \subset \mathbb{C}^n$. Since the charts in a complex atlas are holomorphically compatible, the choice of charts is not important.

We adopt the convention that a map $X \to Y$ is *holomorphic on a compact set* K in X if it is holomorphic in an open neighborhood of K in X; two such maps are identified if they agree in some neighborhood of K. A family of maps is holomorphic on K if every map in the family is holomorphic in an open neighborhood of K that is independent of the map.

A map $f\colon X \to Y$ is *biholomorphic* if it is bijective and if both f and its inverse $f^{-1}\colon Y \to X$ are holomorphic. (This requires that $\dim X = \dim Y$.) As before, the latter condition is superfluous — a bijective holomorphic map between complex manifolds is actually biholomorphic. Note that every local chart $\phi\colon U \to \mathbb{C}^n$ on X is a biholomorphic map of U onto $\phi(U) \subset \mathbb{C}^n$.

A biholomorphic self-map $f\colon X \to X$ is called a *holomorphic automorphism* of X; the collection of all automorphisms is the *holomorphic automorphism group* $\operatorname{Aut} X = \operatorname{Aut}_{\mathrm{hol}} X$. We denote by $\operatorname{Aut}_{\mathrm{alg}} X$ the group of all algebraic automorphisms of an algebraic manifold. In many cases $\operatorname{Aut} X$ has the structure of a real or complex *Lie group* (see Example 1.2.4 below). For instance, $\operatorname{Aut} \mathbb{C}$ consists of all affine linear maps $z \mapsto \alpha z + \beta$ ($\alpha \in \mathbb{C}^* = \mathbb{C} \backslash \{0\}$, $\beta \in \mathbb{C}$) and is a complex two dimensional Lie group. The automorphism group of any bounded domain $D \subset \mathbb{C}^n$ is a finite dimensional real Lie group; the maximal dimension is obtained when D is the ball $\mathbb{B}^n = \{z \in \mathbb{C}^n\colon |z|^2 = \sum_{j=1}^n |z_j|^2 < 1\}$. The group $\operatorname{Aut} \mathbb{B}^n$ acts transitively on $\mathbb{B}^n$, and the isotropy group of the origin $0 \in \mathbb{B}^n$ is the unitary group $U(n)$ (see [423]). Most bounded domains $D \subset \mathbb{C}^n$ have no automorphisms other than the identity. On the other hand, for $n > 1$ the group $\operatorname{Aut} \mathbb{C}^n$ is infinite dimensional (see Chapter 4).

Given a holomorphic map $f = (f_1, \ldots, f_m)\colon D \to \mathbb{C}^m$ on a domain $D \subset \mathbb{C}^n$, we denote by $\operatorname{rank}_p f$ the complex rank of f at a point $p \in D$; that is, the rank of the complex $m \times n$ Jacobian matrix

$$f'(p) = \left(\frac{\partial f_j}{\partial z_k}(p) \right). \tag{1.6}$$

This matrix represents the differential $\mathrm{d}f_p = \partial f_p\colon \mathrm{T}_p\mathbb{C}^n \to \mathrm{T}_{f(p)}\mathbb{C}^m$ in standard bases on the tangent spaces $\mathrm{T}_p\mathbb{C}^n$, $\mathrm{T}_{f(p)}\mathbb{C}^m$, respectively. (See §1.6.) Clearly $\operatorname{rank}_p f \le \min\{m, n\}$. The map f is an *immersion* at p if $\operatorname{rank}_p f = n$, and is a *submersion* at p if $\operatorname{rank}_p f = m$. These notions coincide when $n = m$, and in this case f is said to be *locally biholomorphic* at p. These notions, being local, extend to holomorphic maps between complex manifolds.

Let X be a complex manifold of dimension n. A subset M of X is a *complex submanifold* of dimension $m \in \{0, 1, \ldots, n\}$ (and codimension $d = n - m$) if every point $p \in M$ admits an open neighborhood $U \subset X$ and a holomorphic chart $\phi: U \to U' \subset \mathbb{C}^n$ such that $\phi(U \cap M) = U' \cap (\mathbb{C}^m \times \{0\}^{n-m})$. Any such chart (U, ϕ) on X is said to be *adapted* to M. Let $\pi: \mathbb{C}^n \to \mathbb{C}^m$ denote the coordinate projection $\pi(z_1, \ldots, z_m, \ldots, z_n) = (z_1, \ldots, z_m)$. For each adapted complex chart (U, ϕ) on X we get a complex chart $(U \cap M, \pi \circ \phi|_{U \cap M})$ on M with values in $\mathbb{C}^m$. The collection of all such charts is a complex atlas on M, and the corresponding complex structure on M is the *complex submanifold structure* induced by the inclusion map $\iota: M \hookrightarrow X$. Considering M with this submanifold structure as a complex manifold in its own right, the inclusion ι is a *holomorphic embedding* of M in X, that is, an injective holomorphic immersion of M onto the complex submanifold $\iota(M)$ of X.

The image of an injective holomorphic immersion $f: M \to X$ need not be a submanifold of X, not even a topological one, due to possible accumulation of the image on itself. The following important property prevents this behavior.

Definition 1.1.1. *A continuous map $f: X \to Y$ of topological spaces is said to be* proper *if the preimage $f^{-1}(K)$ of any compact set $K \subset Y$ is compact.*

A map $f: X \to Y$ between manifolds is proper if and only if it maps any discrete sequence in X to a discrete sequence in Y. If X and Y are complex manifolds and $f: X \to Y$ is a proper injective holomorphic immersion, then $f(X)$ is a closed complex submanifold of Y; such f is called a *proper holomorphic embedding*. More generally, if X and Y are complex spaces (see §1.3 below) and $f: X \to Y$ is a proper holomorphic map then $f(X)$ is a closed complex subvariety of Y according to a theorem of Remmert [412].

1.2 Examples of Complex Manifolds

Example 1.2.1. (*Riemann surfaces.*) These are one dimensional complex manifolds. By the Riemann-Koebe uniformization theorem [303, 250, 251] the only connected and simply connected Riemann surfaces up to a biholomorphism are the complex plane $\mathbb{C}$, the *Riemann sphere* $\mathbb{C} \cup \{\infty\} = \mathbb{P}^1$, and the disc $\mathbb{D} = \{z \in \mathbb{C}: |z| < 1\}$. If R is a connected Riemann surface then its universal covering space X is one of the surfaces $\mathbb{P}^1, \mathbb{C}, \mathbb{D}$, and R is biholomorphic to the quotient X/Γ for some group $\Gamma \subset \operatorname{Aut} X$ acting without fixed points and properly discontinuously on X. The automorphism group $\operatorname{Aut} \mathbb{P}^1 = \{z \mapsto \frac{az+b}{cz+d}: ad - bc = 1\}$ does not contain any nontrivial subgroups with these properties; hence $\mathbb{P}^1$ has no nontrivial holomorphic quotients. The only subgroups $\Gamma \subset \operatorname{Aut} \mathbb{C}$ with the required properties are *lattices*, i.e., discrete $\mathbb{Z}$-submodules of $\mathbb{C}$ acting on $\mathbb{C}$ by translations. Such Γ has either one or two generators: $\Gamma = \mathbb{Z}a$ $(a \neq 0)$, or $\Gamma = \mathbb{Z}a + \mathbb{Z}b$ where $a, b \in \mathbb{C}$ are nonzero

numbers with $ab^{-1} \notin \mathbb{R}$. The quotient $\mathbb{C}/\Gamma$ is $\mathbb{C}^* = \mathbb{C}\backslash\{0\}$ in the case of a single generator, and is a complex one dimensional torus in the case of two generators. All other Riemann surfaces are quotients of the disc $\mathbb{D}$. $\square$

Example 1.2.2. (*Affine algebraic manifolds.*) An *affine algebraic variety* in $\mathbb{C}^n$ is the common zero set of finitely many holomorphic polynomials in n complex variables. An affine algebraic variety without singular points is called an *affine algebraic manifold.* $\square$

Example 1.2.3. (*Stein manifolds.*) The class of Stein manifolds was introduced by Karl Stein in 1951 [460] (under the name of *holomorphically complete manifolds*) by a system of three axioms postulating the existence of many global holomorphic functions, in analogy to the properties of domains of holomorphy (see Def. 2.2.1 on p. 47). The simplest characterization of this class is given by the Remmert embedding theorem [411]: A complex manifold is Stein if and only if it is biholomorphic to a closed complex submanifold of a Euclidean space $\mathbb{C}^N$. (For a more precise result see Theorem 2.2.8.) Hence Stein manifolds are holomorphic analogues of affine algebraic manifolds, a fact that is made precise by the algebraic approximations theorems (see p. 50). Analytic properties of Stein manifolds are in many aspects close to those of smooth manifolds, and are very different from those of compact complex manifolds. The main topic of this book is the theory of holomorphic mappings from Stein manifolds and Stein spaces to other complex manifolds. $\square$

Example 1.2.4. (*Lie groups and homogeneous manifolds.*) A complex manifold G that is also a group with holomorphic group operations is called a *complex Lie group.* The main examples include the *general linear group* $GL_n(\mathbb{C})$ (the group of invertible complex $n \times n$ matrices) and its subgroups such as $SL_n(\mathbb{C}) = \{A \in GL_n(\mathbb{C}) \colon \det A = 1\}$; the symplectic subgroup $Sp_n(\mathbb{C}) \subset GL_{2n}(\mathbb{C})$; certain quotients such as the *projective linear group* $PGL_n(\mathbb{C}) = GL_{n+1}(\mathbb{C})/\mathbb{C}^* = \operatorname{Aut}\mathbb{P}^n$ (the holomorphic automorphism group of $\mathbb{P}^n$); universal coverings of Lie groups, etc. A complex manifold X is said to be *G-homogeneous* if there exists a transitive holomorphic action $G \times X \to X$ of G on X by holomorphic automorphisms. Fixing a point $p \in X$, we see that X is biholomorphic to the quotient G/H where $H = \{g \in G \colon g(p) = p\}$ is the isotropy subgroup of the point p. For results on this subject see e.g. [7, 56]. $\square$

Example 1.2.5. (*Complex projective spaces.*) The complex projective spaces $\mathbb{P}^n = \mathbb{CP}^n = \mathbb{P}(\mathbb{C}^{n+1})$ play the analogous role in algebraic geometry as the Euclidean spaces play in affine and Stein geometry. As a set, $\mathbb{P}^n$ consists of all complex lines through the origin in $\mathbb{C}^{n+1}$. A complex line $\lambda \subset \mathbb{C}^{n+1}$ is determined by any point $0 \neq z = (z_0, \dots, z_n) \in \lambda$; we denote this line by $[z] = [z_0 \colon z_1 \colon \cdots \colon z_n]$ and call these the *homogeneous coordinates* on $\mathbb{P}^n$. Clearly $[z] = [w]$ if and only if $w = tz$ for some $t \in \mathbb{C}^*$. There is a unique complex manifold structure on $\mathbb{P}^n$ in which the projection $\pi \colon \mathbb{C}^{n+1}_* = \mathbb{C}^{n+1}\backslash\{0\} \to \mathbb{P}^n$,

$\pi(z) = [z] \in \mathbb{P}^n$ is holomorphic. A complex atlas is given by the collection (U_j, ϕ_j) $(j = 0, 1, \ldots, n)$ where $U_j = \{[z_0 \colon z_1 \colon \cdots \colon z_n] \in \mathbb{P}^n \colon z_j \neq 0\}$ and

$$\phi_j\left([z_0 \colon z_1 \colon \cdots \colon z_n]\right) = \left(\frac{z_0}{z_j}, \ldots, \frac{z_{j-1}}{z_j}, \frac{z_{j+1}}{z_j}, \ldots, \frac{z_n}{z_j}\right) \in \mathbb{C}^n.$$

It is immediate that ϕ_j maps U_j bijectively onto $\mathbb{C}^n$ and that the transition maps $\phi_i \circ \phi_j^{-1}$ are linear fractional. $\square$

Example 1.2.6. (*Projective manifolds and varieties.*) A nonzero holomorphic polynomial $P(z_0, \ldots, z_n)$ is *homogeneous* of degree $d \in \mathbb{N}$ if $P(tz_0, \ldots, tz_n) = t^d P(z_0, \ldots, z_n)$ for all $t \in \mathbb{C}$. Such P determines a complex hypersurface

$$V = V(P) = \{[z_0 \colon z_1 \colon \cdots \colon z_n] \in \mathbb{P}^n \colon P(z_0, \ldots, z_n) = 0\}.$$

More generally, homogeneous polynomials $P_1, \ldots, P_m$ on $\mathbb{C}^{n+1}$ determine a complex subvariety $V(P_1, \ldots, P_m) = V(P_1) \cap \cdots \cap V(P_m) \subset \mathbb{P}^n$. Subvarieties of this type in $\mathbb{P}^n$ are called *projective varieties*, or *projective manifolds* when they are nonsingular. A *quasi-projective variety* is a variety of the form $V = X \backslash Y$, where X and Y are closed complex subvarieties of $\mathbb{P}^n$. By Chow's theorem [85, 233, 241] every closed complex subvariety of $\mathbb{P}^n$ equals $V(P_1, \ldots, P_m)$ for some homogeneous polynomials in $n+1$ variables. A compact complex manifold (resp. a complex space) is said to be *projective algebraic* if it is biholomorphic to a projective manifold (resp. to projective subvariety) in some $\mathbb{P}^n$. A considerable extension of Chow's theorem is the *GAGA principle* of J.-P. Serre [441] concerning the equivalence between analytic and algebraic coherent sheaves over projective algebraic varieties. $\square$

Example 1.2.7. (*Stiefel manifolds.*) Pick integers $1 \leq k \leq n$. The complex *Stiefel manifold* $V_{k,n}$ consists of all complex $k \times n$ matrices $A \in M_{k,n}(\mathbb{C}) \cong \mathbb{C}^{kn}$ with $\operatorname{rank} A = k$. Clearly $V_{k,n}$ is an open subset of $M_{k,n} = M_{k,n}(\mathbb{C})$. The group $GL_n(\mathbb{C})$ acts transitively on $V_{k,n}$ by right multiplication, so $V_{k,n}$ is a complex homogeneous manifold. We have $V_{k,n} = M_{k,n} \backslash \Sigma_{k,n}$ where $\Sigma_{k,n}$ consists of all complex $k \times n$ matrices of less than maximal rank. Note that $\Sigma_{k,n}$ is an algebraic subvariety of $M_{k,n}(\mathbb{C}) \cong \mathbb{C}^{kn}$ defined by the vanishing of all maximal $k \times k$ minors; these are homogeneous polynomial equations of order k (so $\Sigma_{k,n}$ is a complex cone in $\mathbb{C}^{kn}$), and at every point of $\Sigma_{k,n}$ at least $n - k + 1$ of these equations are independent. In fact we have a stratification $\Sigma_{k,n} = \Sigma_{k,n}^1 \supset \Sigma_{k,n}^2 \supset \cdots$ where for every $i = 1, \ldots, k$ the set

$$\Sigma_{k,n}^i = \{A \in M_{k,n} \colon \operatorname{rank} A = k - i\}$$

is an algebraic subvariety of complex codimension $\operatorname{codim} \Sigma_{k,n}^i = i(n - k + i)$. (See e.g. [214, Proposition 5.3, p. 60].) In particular, $\operatorname{codim} \Sigma_{k,n} = n - k + 1$. It follows from the transversality theorem that the homotopy groups of $V_{k,n}$ vanish in the range up to $2(n - k)$:

$$\pi_q(V_{k,n}) = 0, \quad q = 1, 2, \ldots, 2n - 2k. \tag{1.7}$$

Example 1.2.8. (*Grassmann manifolds.*) The complex *Grassmann manifold* $G_{k,n} = G_k(\mathbb{C}^n)$ is the set of all k-dimensional complex linear subspaces of $\mathbb{C}^n$. (Thus $G_{1,n} = \mathbb{P}^{n-1}$.) Let $V_{k,n}$ be the Stiefel manifold (Example 1.2.7 above). We have a surjective map $\pi\colon V_{k,n} \to G_{k,n}$ which sends $A \in V_{k,n}$ to the $\mathbb{C}$-linear span of the row vectors in A. There is a unique complex structure on $G_{k,n}$ which makes this projection holomorphic. The group $GL_k(\mathbb{C})$ acts on $V_{k,n}$ by left multiplication, and we have $\pi(A) = \pi(B)$ for $A, B \in V_{k,n}$ if and only if $A = GB$ for some $G \in GL_k(\mathbb{C})$, so the Grassman manifold $G_{k,n}$ is the leaf space of this action. Grassmann manifolds are projective algebraic; the *Plücker embedding* $G_{k,n}(\mathbb{C}) \to \mathbb{P}(\wedge^k \mathbb{C}^n)$ is induced by the map $V_{k,n} \to \wedge^k \mathbb{C}^n$ sending a matrix $A \in V_{k,n}$ with rows $a_1, \ldots, a_k$ to $a_1 \wedge \cdots \wedge a_k \in \wedge^k \mathbb{C}^n$ [508, p. 11]. An important property for us is that every point in $G_{k,n}$ contains a Zariski open neighborhood isomorphic to $\mathbb{C}^{k(n-k)}$. $\square$

Example 1.2.9. (Complexifications.) For every real analytic manifold M there exists a complex manifold X obtained by complexifying the transition maps defining M [58]. Such X contains M as a maximal totally real submanifold, and it can be chosen Stein according to Grauert [224, §3]. (See §3.5.) $\square$

Example 1.2.10. (Hyperbolic manifolds.) The *Kobayashi-Royden pseudometric* on a complex manifold X is the largest pseudometric which equals the Poincaré metric on the unit disc and such that holomorphic maps are distance decreasing. A complex manifold X is said to be *Kobayashi hyperbolic* if the Kobayashi-Royden pseudometric on X is a metric, and is *complete hyperbolic* if this metric is complete (see [299, 300] for precise definitions). A complex manifold X is *Brody k-hyperbolic* for some $k \in \{1, \ldots, \dim X\}$ if every holomorphic map $\mathbb{C}^k \to X$ has rank $< k$; for $k = 1$ this means that every map $\mathbb{C} \to X$ is constant. For $k = \dim X$ this property is called (Brody) *volume hyperbolicity.* A compact complex manifold is Brody 1-hyperbolic if and only if it is Kobayashi hyperbolic [55]. For the notion of the Kobayashi-Eisenman form and hyperbolicity see [126, 287]. $\square$

1.3 Subvarieties and Complex Spaces

Let X be a complex manifold. We denote by $\mathcal{O}_x = \mathcal{O}_{X,x}$ the ring of germs of holomorphic functions at a point $x \in X$. A germ $[f]_x \in \mathcal{O}_x$ is represented by a holomorphic function in an open neighborhood of x; two such functions determine the same germ at x if and only if they agree in some neighborhood of x. The ring $\mathcal{O}_{X,x}$ is isomorphic to the ring $\mathcal{O}_{\mathbb{C}^n,0}$ via any holomorphic coordinate map sending x to 0. We can identify $\mathcal{O}_{\mathbb{C}^n,0}$ with the ring of convergent power series in n complex variables $(z_1, \ldots, z_n)$. This ring is Noetherian and a unique factorization domain. Its units are precisely the germs that do not vanish at 0. The set of germs vanishing at 0 is the unique maximal ideal $\mathfrak{m}_0 \subset \mathcal{O}_{\mathbb{C}^n,0}$ and $\mathcal{O}_{\mathbb{C}^n,0}/\mathfrak{m}_0 = \mathbb{C}$. For further properties of the local ring see

[103, 229, 233, 241]. The disjoint union $\mathcal{O}_X = \cup_{x\in X}\mathcal{O}_{X,x}$ is equipped with the topology whose basis is given by sets $\{[f]_x : x \in U\}$, where $f: U \to \mathbb{C}$ is a holomorphic function on an open set $U \subset X$. This makes $\mathcal{O}_X$ into a sheaf of commutative rings, called the *sheaf of germs of holomorphic functions* or *the structure sheaf* of X. The identity principle shows that the sheaf $\mathcal{O}_X$ is Hausdorff. We denote by $\mathcal{C}_X$ the sheaf of germs of continuous functions on X.

Since the ring $\mathcal{O}_x$ has no zero divisors, we can form its quotient field $\mathcal{M}_x$, called the field of germs of *meromorphic functions* on X at the point x. Thus a meromorphic function on X is locally at every point $x \in X$ given as the quotient f/g of two holomorphic functions whose germs at x are coprime. Such function is holomorphic off the zero locus of g, also called the *polar set* of f/g, and its *indeterminacy set* is $\{f = 0,\ g = 0\}$.

A subset A of a complex manifold X is a *complex (analytic) subvariety* of X if for every point $p \in A$ there exist a neighborhood $U \subset X$ of p and functions $f_1, \ldots, f_d \in \mathcal{O}(U)$ such that

$$A \cap U = \{x \in U : f_1(x) = 0, \ldots, f_d(x) = 0\}.$$

If such A is topologically closed in X then A is a *closed complex subvariety* of X. Since the local ring $\mathcal{O}_x$ is Noetherian, a subset of X that is locally defined by infinitely many holomorphic equations is still a subvariety and can be locally defined by finitely many equations.

A point p in a subvariety A is a *regular* (or *smooth*) point if A is a complex submanifold near p; the set of all regular points is denoted A_{reg}. A point $p \in A \backslash A_{\mathrm{reg}} = A_{\mathrm{sing}}$ is a *singular point* of A.

Let A be a closed complex subvariety X. For every point $x \in X$ we denote by $\mathcal{J}_{A,x}$ the ideal in $\mathcal{O}_x$ consisting of all holomorphic function germs at x whose restriction to A vanishes. In particular, $\mathcal{J}_{A,x} = \mathcal{O}_x$ for every $x \in X \backslash A$. The corresponding sheaf $\mathcal{J}_A = \cup_{x\in X}\mathcal{J}_{A,x}$ is the *sheaf of ideals* (or the *ideal sheaf*, or simply the *ideal*) of A in X. The restriction of the quotient sheaf $\mathcal{O}_X/\mathcal{J}_A = \cup_{x\in X}\mathcal{O}_{X,x}/\mathcal{J}_{A,x}$ to A is the sheaf of germs of holomorphic functions on A, denoted $\mathcal{O}_A$ and called the structure sheaf of A.

The notion of a *complex space* was first introduced in 1951 by H. Behnke and K. Stein [42] and H. Cartan [75]; their definitions correspond to what is now called a *normal complex space* (see [227]). The definition which is accepted as the standard one, and which is also used in this book, was given by J.-P. Serre in his GAGA paper [441]:

A *reduced complex space* is a pair $(X, \mathcal{O}_X)$, where X is a paracompact Hausdorff topological space and $\mathcal{O}_X$ is a sheaf of rings of continuous functions on X (a subsheaf of the sheaf $\mathcal{C}_X$ of germs of continuous functions) such that for every point $x \in X$ there is a neighborhood $U \subset X$ and a homeomorphism $\phi: U \to A \subset \mathbb{C}^n$ onto a locally closed complex subvariety of $\mathbb{C}^n$ so that the homomorphism $\phi^*: \mathcal{C}_A \to \mathcal{C}_X$, $f \mapsto f \circ \phi$, induces an isomorphism of $\mathcal{O}_A$ onto $\mathcal{O}_U = \mathcal{O}_X|_U$. Intuitively speaking, X is obtained by gluing pieces of subvarieties in Euclidean spaces using biholomorphic transition maps. Similarly one

defines an *algebraic spaces* [441]. We get a *nonreduced complex space* by allowing local models $(A, \mathcal{F})$, where A is a closed complex subvariety in an open set $\Omega \subset \mathbb{C}^n$ and $\mathcal{F} = (\mathcal{O}_\Omega/\mathcal{I})|_A$ for some sheaf of ideals $\mathcal{I} \subset \mathcal{J}_A$ supported on A (i.e., $\mathcal{I}_x = \mathcal{O}_x$ for $x \notin A$). The ring $\mathcal{F}_x$ may have nilpotent elements. By the Nullstellensatz the radical $\sqrt{\mathcal{I}_x}$ of any such ideal equals $\mathcal{J}_{A,x}$.

Let $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ be complex spaces. A continuous map $f\colon X \to Y$ is said to be holomorphic if for every $x \in X$ the composition $\mathcal{C}_{Y,f(x)} \ni g \mapsto g \circ f \in \mathcal{C}_{X,x}$ defines a homomorphism $f_x^*\colon \mathcal{O}_{Y,f(x)} \to \mathcal{O}_{X,x}$. For such map we can define the differential $\mathrm{d}f_x\colon \mathrm{T}_x X \to \mathrm{T}_{f(x)} Y$ as a $\mathbb{C}$-linear map on the *Zariski tangent space* (see (1.29) on p. 21). This is the usual differential at smooth points, while at singular points we locally embed the two spaces as complex subvarieties of Euclidean space of minimal dimension $n_x = \mathrm{embdim}_x X$, $m_y = \mathrm{embdim}_y Y$, respectively, and take the differential $\mathrm{d}F_x$ of the local holomorphic extension F of f (a $\mathbb{C}$-linear map $\mathbb{C}^{n_x} \to \mathbb{C}^{m_y}$).

Among the most fundamental results in the theory of complex spaces is Remmert's theorem [412] saying that the image $f(X)$ of a proper holomorphic map $f\colon X \to Y$ is a closed analytic subvariety of Y. A more general result of Grauert (see [229]) gives the coherence of the direct image $f_*\mathcal{F}$ of any coherent analytic sheaf $\mathcal{F}$ under a proper holomorphic map. (See also p. 53 below.)

Definition 1.3.1. *Let Z and X be reduced complex spaces. A holomorphic map $\pi\colon Z \to X$ is a* holomorphic submersion *if for every point $z_0 \in Z$ there exist an open neighborhood $V \subset Z$ of z_0, an open neighborhood $U \subset X$ of $x_0 = \pi(z_0)$, an open set W in $\mathbb{C}^p$, and a biholomorphic map $\phi\colon V \to U \times W$ such that $pr_1 \circ \phi = \pi$. (Here $pr_1\colon U \times W \to U$ is the projection on the first factor.) Each such local chart ϕ will be called* adapted to π.

Note that each fiber $Z_x = \pi^{-1}(x)$ $(x \in X)$ of a holomorphic submersion is a closed complex submanifold of Z, and the dimension $\dim Z_x$ is constant on every connected component of Z.

Definition 1.3.2. *Assume that $h\colon Z \to X$ is a holomorphic submersion onto a complex space X, X' is a closed complex subvariety of X, and $\mathcal{S} \subset \mathcal{O}_X$ is a sheaf of ideals with support X', i.e., $\mathcal{S}_x = \mathcal{O}_{X,x}$ when $x \in X \backslash X'$. Local holomorphic sections f_0, f_1 of $h\colon Z \to X$ in a neighborhood of a point $x \in X'$ are* $\mathcal{S}$-tangent at x *if there is a neighborhood $V \subset Z$ of the point $z = f_0(x) = f_1(x) \in Z$ and a holomorphic embedding $\phi\colon V \hookrightarrow \mathbb{C}^N$ such that the germ at x of any component of the map $\phi f_0 - \phi f_1\colon U \to \mathbb{C}^N$ belongs to $\mathcal{S}_x$. If f_0 and f_1 are holomorphic in a neighborhood of X' and $\mathcal{S}$-tangent at each point $x \in X'$, then we say that f_0 and f_1 are $\mathcal{S}$-tangent and write $\delta(f_0, f_1) \in \mathcal{S}$. If this holds for the r-th power of the ideal sheaf $\mathcal{J}_{X'}$ of the subvariety X' then f_0 and f_1 are said to be* tangent to order r along X'.

Definition 1.3.3. *A* stratification *of a finite dimensional complex space X is a finite descending sequence $X = X_0 \supset X_1 \supset \cdots \supset X_m = \emptyset$ of closed complex*

subvarieties such that each connected component S (stratum) of a difference $X_k \backslash X_{k+1}$ is a complex manifold and $\overline{S} \backslash S \subset X_{k+1}$.

Every finite dimensional complex space admits a stratification [514, p. 227]: Take X_1 to be the union of the singular locus of $X = X_0$ and of all irreducible components of X_0 of less than maximal dimension; define X_2 in the same way with respect to X_1, etc. By considering substratifications we can ask for many additional properties. For example, a finite dimensional Stein space admits a stratification whose strata are Stein manifolds. Whitney's condition (a) is used in transversality theorems proved in §7.8.

1.4 Holomorphic Fiber Bundles

Fiber bundles represent one of the most important constructions of new manifolds from the existing ones.

Definition 1.4.1. *A* holomorphic fiber bundle *is a triple (Z, π, X), where X and Z are complex spaces and $\pi\colon Z \to X$ is a holomorphic map of Z onto X such that there exist a complex manifold Y, an open cover $\mathcal{U} = \{U_\alpha\}$ of X, and for every α a biholomorphic map*

$$\theta_\alpha\colon Z|_{U_\alpha} = \pi^{-1}(U_\alpha) \to U_\alpha \times Y, \quad \theta_\alpha(z) = (\pi(z), \vartheta_\alpha(z)). \tag{1.8}$$

The manifold Z is the total space*, X is the* base space*, and Y is the* fiber*.*

A holomorphic submersion $\pi\colon Z \to X$ is a stratified holomorphic fiber bundle *if there is a stratification $X = X_0 \supset X_1 \supset \cdots \supset X_m = \emptyset$ of X such that the restriction of Z to every stratum $S \subset X_k \backslash X_{k+1}$ is a holomorphic fiber bundle over S. (Fibers over different strata may be different.)*

The simplest example is a *product bundle* $\pi\colon Z = X \times Y \to X$, $(x, y) \mapsto x$. By definition, every fiber bundle is isomorphic to the product bundle over small open sets in X. The fiber $Z_x = \pi^{-1}(x)$ over any point $x \in X$ is biholomorphic to Y. A map θ_α (1.8) is a *fiber bundle chart* on Z, and the collection $\{(U_\alpha, \theta_\alpha)\}$ is a *holomorphic fiber bundle atlas* on Z. The *transition maps* $\theta_{\alpha,\beta} = \theta_\alpha \circ \theta_\beta^{-1}\colon U_{\alpha,\beta} \times Y \to U_{\alpha,\beta} \times Y$ are of the form

$$\theta_{\alpha,\beta}(x, y) = \big(x, \vartheta_{\alpha,\beta}(x, y)\big), \quad x \in U_{\alpha,\beta},\ y \in Y, \tag{1.9}$$

and they satisfy the *cocycle condition*

$$\theta_{\alpha,\alpha} = \mathrm{Id}, \quad \theta_{\alpha,\beta} \circ \theta_{\beta,\gamma} \circ \theta_{\gamma,\alpha} = \mathrm{Id} \quad \text{on } U_{\alpha,\beta,\gamma} \times Y. \tag{1.10}$$

For every fixed $x \in U_{\alpha,\beta}$ we have $\vartheta_{\alpha,\beta}(x, \cdot\,) \in \operatorname{Aut} Y$. Conversely, given an open cover $\mathcal{U} = \{U_\alpha\}$ of X and a collection of biholomorphic self-maps (1.9)

satisfying the cocycle condition (1.10), we get a holomorphic fiber bundle $Z \to X$ with these transition maps by taking Z to be the disjoint union of all $U_\alpha \times Y$, modulo the identifications provided by the transition maps.

A *section* of $\pi\colon Z \to X$ is a map $f\colon X \to Z$ such that $\pi \circ f$ is the identity on X; that is, $f(x) \in Z_x$ for every $x \in X$. Any section of the product bundle $X \times Y \to X$ is of the form $f(x) = (x, g(x))$, where $g\colon X \to Y$ is a map to the fiber. If $\{(U_\alpha, \theta_\alpha)\}$ is a holomorphic fiber bundle atlas on $Z \to X$ with the transition maps $\theta_{\alpha,\beta}$ (1.9), then a holomorphic section $f\colon X \to Z$ is given by a collection of holomorphic maps $f_\alpha\colon U_\alpha \to Y$ satisfying the compatibility conditions

$$f_\alpha(x) = \vartheta_{\alpha,\beta}\big(x, f_\beta(x)\big), \quad x \in U_{\alpha,\beta}. \tag{1.11}$$

Definition 1.4.2. *A* holomorphic isomorphism *of holomorphic fiber bundles $\pi\colon Z \to X$, $\pi'\colon Z' \to X$ is a biholomorphic map $\Phi\colon Z \to Z'$ such that $\pi' \circ \Phi = \pi$; if such Φ exists then the bundles are* holomorphically isomorphic. *A fiber bundle is* trivial *if it is isomorphic to the product bundle.*

Isomorphisms of a fiber bundle onto itself are *fiber bundle automorphisms.* Holomorphic automorphisms of a product bundle $X \times Y \to X$ are biholomorphic self-maps of $X \times Y$ of the form $(x, y) \mapsto (x, \varphi(x, y))$, with $\varphi(x, \cdot\,) \in \operatorname{Aut} Y$ for every $x \in X$. In general we choose an open cover $\mathcal{U} = \{U_\alpha\}$ of X and fiber bundle atlases $\{(U_\alpha, \theta_\alpha)\}$ for (Z, π, X), and $\{(U_\alpha, \theta'_\alpha)\}$ for (Z', π', X). A fiber bundle isomorphism Φ of (Z, π, X) to (Z', π', X) is then given by a collection of fiber preserving biholomorphic self-maps $\phi_\alpha\colon U_\alpha \times Y \to U_\alpha \times Y$ of the form $\phi_\alpha(x, y) = (x, \varphi_\alpha(x, y))$ so that the following diagrams commute:

$$\begin{tikzcd}
U_{\alpha,\beta} \times Y \arrow[d, "\phi_\alpha"'] & U_{\alpha,\beta} \times Y \arrow[l, "\theta_{\alpha,\beta}"'] \arrow[d, "\phi_\beta"] \\
U_{\alpha,\beta} \times Y & U_{\alpha,\beta} \times Y \arrow[l, "\theta'_{\alpha,\beta}"']
\end{tikzcd}$$

If there exists a fiber bundle atlas on $Z \to X$ (in the given isomorphism class) such that all transition maps belong to a certain subgroup G of $\operatorname{Aut} Y$, then we say G is the *structure group* of the bundle, or that the structure group of the bundle has been reduced to G.

If $\pi\colon Z \to X$ is a holomorphic fiber bundle and X' is a complex subvariety of X then the restriction $\pi\colon Z' = Z|_{X'} \to X'$ is a holomorphic fiber bundle over X', called the *restricted bundle.*

Given a holomorphic fiber bundle $\pi\colon Z \to X$ with fiber Y and a holomorphic map $f\colon W \to X$, the *pull-back bundle* $\pi'\colon f^*Z \to W$ and the map $F\colon f^*Z \to Z$ are defined as follows:

$$f^*Z = \{(w, z) \in W \times Z\colon f(w) = \pi(z)\}, \quad \pi'(w, z) = w, \quad F(w, z) = z. \tag{1.12}$$

Let $\{(U_\alpha, \theta_\alpha)\}$ be a fiber bundle atlas for $\pi\colon Z \to X$ with

$$\theta_\alpha(z) = (\pi(z), \vartheta_\alpha(z)) \in U_\alpha \times Y.$$

Set $V_\alpha = f^{-1}(U_\alpha) \subset W$ and define a map $\theta'_\alpha\colon f^*Z|_{V_\alpha} \to V_\alpha \times Y$ by $\theta'_\alpha(w, z) = (w, \vartheta_\alpha(z))$. (The map f appears implicitly in the above definition by the condition $\pi(z) = f(w)$.) This is a holomorphic fiber bundle atlas on $\pi'\colon f^*Z \to W$ with transition maps $\vartheta'_{\alpha,\beta}(w, y) = \vartheta_{\alpha,\beta}(f(w), y)$; hence $f^*Z \to W$ is indeed a holomorphic fiber bundle with fiber Y.

Example 1.4.3. (Vector bundles.) These are fiber bundles with fiber $\mathbb{C}^n$ and structure group $GL_n(\mathbb{C})$. They are considered in the following section. □

Example 1.4.4. (*Affine bundles.*) These are fiber bundles with fiber $\mathbb{C}^n$ and structure group consisting of affine linear maps:

$$v \mapsto a + Bv, \quad a \in \mathbb{C}^n,\ B \in GL_n(\mathbb{C}).$$

By [280] every projective analytic variety X carries an affine bundle $E \to X$ whose total space E is a Stein space. (This will be of interest in §5.16.) We recall the construction in the basic case $X = \mathbb{P}^n$. Consider the Segre embedding $\rho\colon \mathbb{P}^n \times \mathbb{P}^n \hookrightarrow \mathbb{P}^{n^2+2n}$,

$$\big([z_0\colon\cdots\colon z_n], [w_0\colon\cdots\colon w_n]\big) \longmapsto [z_0w_0\colon\cdots\colon z_iw_j\colon\cdots\colon z_nw_n].$$

Let Σ denote the quadratic hypersurface in $\mathbb{P}^n \times \mathbb{P}^n$ defined by the equation $z_0w_0 + \cdots + z_nw_n = 0$ and set $E = \mathbb{P}^n \times \mathbb{P}^n \backslash \Sigma$. Choose homogeneous coordinates $[\zeta_0\colon \zeta_1\colon\cdots\colon \zeta_{n^2+2n}]$ on $\mathbb{P}^{n^2+2n}$ such that $\zeta_j = z_jw_j$ for $j = 0, \ldots, n$ under the embedding ρ. Let H denote the hyperplane in $\mathbb{P}^{n^2+2n}$ given by the equation $\sum_{j=0}^n \zeta_j = 0$. Then ρ embeds E properly into $\mathbb{P}^{n^2+2n}\backslash H = \mathbb{C}^{n^2+2n}$, so E is Stein. The projection $\pi\colon E \to \mathbb{P}^n$ onto the first component is an affine bundle with fiber $\mathbb{C}^n$. The restriction $E|_X$ to any closed complex subvariety $X \subset \mathbb{P}^n$ is still an affine bundle whose total space (being a closed analytic subvariety of E) is Stein. For quasi-projective varieties there is some work to see that one can get an affine bundle with Stein total space. □

Example 1.4.5. (*Principal bundles.*) Let G be a finite dimensional complex Lie group. For every $g \in G$ let $\sigma_g \in \operatorname{Aut} G$ be the left multiplication on G by g: $\sigma_g(g') = gg'$ $(g' \in G)$. Set $\Gamma = \{\sigma_g\colon g \in G\} \subset \operatorname{Aut} G$. A *holomorphic principal G-bundle* over X is a holomorphic fiber bundle $\pi\colon Z \to X$ with fiber G and structure group Γ. Such bundle is determined by a 1-cocycle $g_{\alpha,\beta}\colon U_{\alpha,\beta} \to G$ over an open cover $\mathcal{U} = \{U_\alpha\}$ of X; the corresponding transition maps are $\phi_{\alpha,\beta}(x, g) = \big(x, g_{\alpha,\beta}(x)g\big)$. The group G acts holomorphically on the total space Z by the right multiplication on the fibers Z_x, and these fibers are precisely the orbits of the action. See §7.1 and §7.2 for further results. □

Example 1.4.6. (*Fiber bundles associated to principal bundles.*) Assume that a complex Lie group G acts holomorphically on a complex manifold Y. Every holomorphic 1-cocycle $g_{\alpha,\beta}\colon U_{\alpha,\beta} \to G$ on an open cover $\mathcal{U} = \{U_\alpha\}$ of X determines a holomorphic fiber bundle with fiber Y, structure group G and transition maps $\theta_{\alpha,\beta}(x,v) = \big(x, g_{\alpha,\beta}(x)v)\big)$. In this way we associate to a principal G-bundle $Z \to X$ a fiber bundle $E \to X$ with fiber Y, structure group G and the same structure cocycle that determines the bundle Z. In particular, to a principal $GL_n(\mathbb{C})$-bundle we associate a holomorphic vector bundle of rank n. Conversely, to a holomorphic vector bundle $E \to X$ of rank n we associate the principal $GL_n(\mathbb{C})$-bundle $Z = F(E) \to X$, called *the frame bundle* of the vector bundle $E \to X$. The elements of $Z_x = F(E_x)$ are *frames* (complex bases) on the vector space $E_x \cong \mathbb{C}^n$. $\square$

Example 1.4.7. (*Bundles with Euclidean fibers.*) These are fiber bundles with fiber $\mathbb{C}^n$ and structure group $\operatorname{Aut}\mathbb{C}^n$, the holomorphic automorphism group of $\mathbb{C}^n$. For $n = 1$ these are just affine bundles, but for $n > 1$ the group $\operatorname{Aut}\mathbb{C}^n$ is very large and we get many non-affine fiber bundles. In §4.21 we mention examples, due to Skoda, Demailly and Rosay, of such bundles over the disc or $\mathbb{C}$ whose total space is non-Stein. $\square$

Example 1.4.8. (*Flat bundles.*) A holomorphic fiber bundle $Z \to X$ with fiber Y is said to be *flat* if it admits a holomorphic fiber bundle atlas whose transition maps $\vartheta_{\alpha,\beta}\colon U_{\alpha,\beta} \to \operatorname{Aut} Y$ are locally constant, and hence constant on every connected component of $U_{\alpha,\beta}$. The trivial horizontal foliations of $U_\alpha \times Y$ with leaves $U_\alpha \times \{y\}$ $(y \in Y)$ patch together to a horizontal holomorphic foliation of the total space Z. Flat bundles arise naturally when considering the Chern connection with vanishing curvature tensor on a Hermitian holomorphic vector bundle. A flat holomorphic fiber bundle is trivial over every simply connected open set in the base X, and every holomorphic isomorphism class of flat bundles is determined by a representation of the fundamental group $\pi_1(X,p)$ in the automorphism group $\operatorname{Aut} Y$. $\square$

1.5 Holomorphic Vector Bundles

Vector bundles are a principal tool used to linearize problems in analysis and geometry. They are also a subject of intrinsic investigation with a profound impact on modern mathematics. We focus on holomorphic vector bundles, recalling those constructions that will be important to us. Similar constructions apply to other classes of vector bundles (topological, smooth, and with $\mathbb{C}$ replaced by another field such as $\mathbb{R}$).

Definition 1.5.1. *A* holomorphic vector bundle of rank n *over a complex space X is a holomorphic fiber bundle $E \to X$ (Def. 1.4.1) with fiber $Y = \mathbb{C}^n$ and structure group $GL_n(\mathbb{C})$. A vector bundle of rank $n = 1$ is also called a (holomorphic)* line bundle.

This means that we have an open cover $\mathcal{U} = \{U_\alpha\}$ of X and vector bundle charts $\theta_\alpha\colon E|_{U_\alpha} \xrightarrow{\cong} U_\alpha \times \mathbb{C}^n$ with transition maps of the form

$$\theta_\alpha \circ \theta_\beta^{-1}(x,v) = \theta_{\alpha,\beta}(x,v) = \big(x, g_{\alpha,\beta}(x)\, v\big), \quad x \in U_{\alpha,\beta},\ v \in \mathbb{C}^n, \tag{1.13}$$

where $g_{\alpha,\beta}\colon U_{\alpha,\beta} \to GL_n(\mathbb{C})$ is a holomorphic multiplicative 1-cocycle:

$$g_{\alpha,\alpha} = 1, \quad g_{\alpha,\beta}\, g_{\beta,\alpha} = 1, \quad g_{\alpha,\beta}\, g_{\beta,\gamma}\, g_{\gamma,\alpha} = 1. \tag{1.14}$$

Every fiber E_x is a complex vector space such that the fiber bundle charts $\theta_\alpha\colon Z|_{U_\alpha} \to U_\alpha \times \mathbb{C}^n$ are $\mathbb{C}$-vector space isomorphisms on each fiber.

If X is a real manifold of class $\mathcal{C}^r$ and $E \to X$ is a complex vector bundle whose transition maps $g_{\alpha,\beta}\colon U_{\alpha,\beta} \to GL_n(\mathbb{C})$ are of class $\mathcal{C}^r$, then we have a *complex vector bundle of class* $\mathcal{C}^r$ over X. For $r = 0$ we have a *topological complex vector bundle* over X. Replacing $\mathbb{C}$ by any field F we get topological F-vector bundles over topological spaces, or smooth F-vector bundles over smooth manifolds. For $F = \mathbb{R}$ we speak of *real vector bundles*.

Every vector bundle has the *zero section* sending each point $x \in X$ to the origin $0_x \in E_x$. Given a holomorphic vector bundle atlas $\{(U_\alpha, \theta_\alpha)\}$ on E with transition maps $g_{\alpha,\beta}$ (1.13), a section $f\colon X \to E$ is determined by a collection of maps $f_\alpha\colon U_\alpha \to \mathbb{C}^n$ satisfying the compatibility conditions

$$f_\alpha = g_{\alpha,\beta} f_\beta \quad \text{on } U_{\alpha,\beta}. \tag{1.15}$$

Example 1.5.2. (*The tangent bundle.*) Let X be a $\mathcal{C}^r$ manifold of dimension n. Given an atlas $\{(U_\alpha, \phi_\alpha)\}$ on X with transition maps $\phi_{\alpha,\beta}$, the tangent bundle $\mathrm{T}X \to X$ is a real vector bundle of rank n and of class $\mathcal{C}^{r-1}$ with vector bundle charts $\theta_\alpha\colon \mathrm{T}X|_{U_\alpha} \to U_\alpha \times \mathbb{R}^n$ and the transition cocycle $g_{\alpha,\beta} = (\phi_{\alpha,\beta})' \circ \phi_\beta$, where $(\phi_{\alpha,\beta})'$ is the Jacobian matrix of $\phi_{\alpha,\beta}$. The cotangent bundle T^*X is the dual bundle of $\mathrm{T}X$. For complex manifolds see §1.6 – §1.7. □

Example 1.5.3. (*The universal bundle.*) Let $G_{k,n}$ be the Grassmann manifold whose points are k-dimensional subspace of $\mathbb{C}^n$ (Example 1.2.8), and set

$$U_{k,n} = \{(\lambda, z) \in G_{k,n} \times \mathbb{C}^n \colon z \in \lambda\}.$$

The projection $\pi\colon U_{k,n} \to G_{k,n}$, $\pi(\lambda, z) = \lambda$, admits the structure of a holomorphic vector bundle (a holomorphic vector subbundle of the trivial bundle $G_{k,n} \times \mathbb{C}^n$), called the universal bundle over $G_{k,n}$.

In particular, $U_{1,n+1} \to G_{1,n+1} = \mathbb{P}^n$ is a holomorphic line bundle over the projective space $\mathbb{P}^n$. This bundle is trivial over every coordinate neighborhood $V_j = \{[z_0\colon\cdots\colon z_n]\colon z_j \neq 0\} \cong \mathbb{C}^n$ in $\mathbb{P}^n$; a local vector bundle chart is given by

$$\theta_j([z_0\colon\cdots\colon z_n], (v_0, \dots, v_n)) = ([z_0\colon\cdots\colon z_n], v_j) \in V_j \times \mathbb{C}.$$

The colinearity condition $v \in [z_0\colon\cdots\colon z_n]$ defining $U_{1,n+1}$ implies $v_i z_j = v_j z_i$, which shows that the transition maps equal $g_{i,j}\big([z_0\colon\cdots\colon z_n]\big) = \frac{z_i}{z_j}$. □

Definition 1.5.4. *Let $\pi\colon E \to X$ and $\pi'\colon E' \to X$ be holomorphic vector bundles. A* holomorphic morphism *of (E, π, X) to (E', π', X) is a holomorphic map $\Phi\colon E \to E'$ such that $\pi' \circ \Phi = \pi$, $\Phi_x\colon E_x \to E'_x$ is $\mathbb{C}$-linear for every $x \in X$. If in addition $\dim\ker\Phi_x$ is independent of the point $x \in X$ then Φ is called a* holomorphic vector bundle map. *Such Φ is a* isomorphism *if $\Phi_x\colon E_x \to E'_x$ is an isomorphism of $\mathbb{C}$-vector spaces for every $x \in X$. A $\mathcal{C}^r$* morphism *is a $\mathcal{C}^r$ map $\Phi\colon E \to E'$ that is $\mathbb{C}$-linear on every fiber.*

The *kernel* and *cokernel* of a morphism $\Phi\colon E \to E'$ are defined by

$$\ker\Phi = \bigcup_{x\in X} \ker\Phi_x \subset E, \quad \operatorname{im}\Phi = \bigcup_{x\in X} \operatorname{im}\Phi_x \subset E'.$$

Definition 1.5.5. *Let $\pi\colon E \to X$ be a holomorphic vector bundle of rank n. A* holomorphic vector subbundle *of rank $m \in \{0, 1, \ldots, n\}$ of (E, π, X) is a complex submanifold $E' \subset E$, with the restricted projection $\pi' = \pi|_{E'}\colon E' \to X$ onto X, such that every point $x_0 \in X$ admits an open neighborhood $U \subset X$ and a holomorphic vector bundle chart $\theta\colon E|_U \xrightarrow{\cong} U \times \mathbb{C}^n$ satisfying the condition*

$$\theta(E'|_U) = U \times (\mathbb{C}^m \times \{0\}^{n-m}). \tag{1.16}$$

Any such chart θ is said to be *adapted* to E'. Denote by

$$pr_1\colon \mathbb{C}^n = \mathbb{C}^m \times \mathbb{C}^{n-m} \to \mathbb{C}^m, \quad pr_2\colon \mathbb{C}^n = \mathbb{C}^m \times \mathbb{C}^{n-m} \to \mathbb{C}^{n-m}$$

the projections onto the first and the second factor, respectively. For every θ as above the map $pr_1 \circ \theta\colon E'|_U \to U \times \mathbb{C}^m$ is a vector bundle chart on E', and the collection of all such charts is a holomorphic vector bundle atlas on E'. In this structure the inclusion map $E' \hookrightarrow E$ is a holomorphic vector bundle morphism. We have the following elementary result.

Proposition 1.5.6. *Let $\Phi\colon E \to E'$ be a holomorphic morphism of holomorphic vector bundles $E \to X$, $E' \to X$. If $\dim\ker\Phi_x$ is independent of the point $x \in X$, then the kernel $\ker\Phi$ is a holomorphic vector subbundle of E and the image $\operatorname{im}\Phi$ is a holomorphic vector subbundle of E'.*

We give a description of morphisms in local charts, beginning with the simplest case of product bundles. Let $\operatorname{Hom}(\mathbb{C}^n, \mathbb{C}^m)$ denote the set of all $\mathbb{C}$-linear maps $\mathbb{C}^n \to \mathbb{C}^m$. With respect to any pair of complex bases on the two space this equals $M_{m,n}(\mathbb{C}) \cong \mathbb{C}^{mn}$, the set of all complex $m \times n$ matrices. A morphism $\Phi\colon X \times \mathbb{C}^n \to X \times \mathbb{C}^m$ of product bundles is of the form $(x, v) \to (x, \varphi(x)v)$ for a holomorphic map $\varphi\colon X \to M_{m,n}(\mathbb{C})$. In particular, an automorphism of $X \times \mathbb{C}^n$ is given by a map $X \to GL_n(\mathbb{C}) \subset M_{n,n}(\mathbb{C})$.

Assume now that $E \to X$ and $E' \to X$ are holomorphic vector bundles of rank n, m, respectively. Choose holomorphic vector bundle atlases $\{(U_\alpha, \theta_\alpha)\}$, $\{(U_\alpha, \theta'_\alpha)\}$ for E, E', with the transition maps $g_{\alpha,\beta}$, $g'_{\alpha,\beta}$, respectively. A mor-

phism $\Phi: E \to E'$ is given by a collection of maps $\varphi_\alpha: U_\alpha \to M_{m,n}(\mathbb{C})$ satisfying the compatibility conditions

$$\varphi_\alpha \, g_{\alpha,\beta} = g'_{\alpha,\beta} \, \varphi_\beta \quad \text{on } U_{\alpha,\beta}.$$

If Φ is an isomorphism, then $\varphi_\alpha: U_\alpha \to GL_n(\mathbb{C})$ and we can write

$$g'_{\alpha,\beta} = \varphi_\alpha \, g_{\alpha,\beta} \, \varphi_\beta^{-1}. \tag{1.17}$$

We say that the 1-cocycle $g' = (g'_{\alpha,\beta})$ is obtained by *twisting the 1-cocycle* $g = (g_{\alpha,\beta})$ *by the* 0*-cochain* $\varphi = (\varphi_\alpha)$, and we write $g' = \varphi \,\square\, g$. This leads to the observation that the isomorphism classes of holomorphic vector bundles of rank n over X are given by elements of the cohomology group $H^1(X; \mathcal{O}^{GL_n(\mathbb{C})})$ with coefficients in the multiplicative sheaf of germs of holomorphic maps $X \to GL_n(\mathbb{C})$. (See §7.1 for a further discussion of this topic.)

The group $H^1(X; \mathcal{O}^{GL_n(\mathbb{C})})$ is Abelian only for $n = 1$ when it equals $H^1(X; \mathcal{O}^*)$. The multiplicative group $H^1(X; \mathcal{O}^*) = \mathrm{Pic}(X)$ of equivalence classes of holomorphic line bundles on X is called the *Picard group* of X. The product on $\mathrm{Pic}(X)$ corresponds to the tensor product of line bundles.

Example 1.5.7. (*Line bundles and divisors.*) A *divisor* D on a complex manifold X is determined by an open cover $\mathcal{U} = \{U_\alpha\}_{\alpha \in A}$ of X and a collection of meromorphic functions $f_\alpha \in \mathcal{M}(U_\alpha)$ that are not identically zero on any connected component of U_α such that for any pair of indexes $\alpha, \beta \in A$ there exists a nowhere vanishing holomorphic function $f_{\alpha,\beta} \in \mathcal{O}^*(U_{\alpha,\beta})$ satisfying

$$f_\alpha = f_{\alpha,\beta} f_\beta \quad \text{on } U_{\alpha,\beta}. \tag{1.18}$$

The 1-cocycle $(f_{\alpha,\beta})$ determines a holomorphic line bundle $E = [D]$ over X, and the collection (f_α) is a meromorphic section of $[D]$ in view of (1.18). In particular, a meromorphic function $f \in \mathcal{M}(X)$ that is not identically zero on any connected component determines a trivial line bundle on X. Conversely, if a line bundle E is presented over an open cover $\mathcal{U} = \{U_\alpha\}_{\alpha \in A}$ of X by a 1-cocycle $(f_{\alpha,\beta})$ with coefficients in the sheaf $\mathcal{O}_X^*$, then each meromorphic section f of E is given in the respective holomorphic trivializations of $E|_{U_\alpha} \cong U_\alpha \times \mathbb{C}$ by a collection of meromorphic functions $f_\alpha \in \mathcal{M}(U_\alpha)$ satisfying (1.18). If D is the divisor determined by (f_α) then clearly $E \cong [D]$.

A complex hypersurface $V \subset X$ determines a divisor D given by a collection of local defining functions for V. Conversely, every divisor D on X can be represented by a locally finite formal combination $D = \sum_i a_i V_i$ of irreducible complex hypersurfaces $V_i \subset X$ with integer coefficients $a_i \in \mathbb{Z}$ [233, p. 130]. The divisors on X form an Abelian group, $\mathrm{Div}(X)$, and by the above discussion we have a natural homomorphism $\mathrm{Div}(X) \to \mathrm{Pic}(X)$. This homomorphism is surjective on any quasi-projective manifold. The line bundle $[D]$ determined by a divisor D is trivial if and only if D is given by a global meromorphic function on X. (In this connection see the discussion in

§5.2 concerning the *second Cousin problem.*) Two divisors D, D' are *linearly equivalent* if $D = D' + (f)$ for some $f \in \mathcal{M}(X)$; thus linear equivalence of a pair of divisors corresponds to holomorphic equivalence of the corresponding line bundles $[D]$ and $[D']$. For a sheaf theoretic interpretation of divisors and linear equivalence see [233]. □

Definition 1.5.8. *Given holomorphic vector bundles $\pi: E \to X$, $\pi': E' \to X'$, a morphism of the first to the second bundle is a pair of holomorphic maps $f: X \to X'$, $F: E \to E'$ such that $\pi' \circ F = f \circ \pi$ and F is $\mathbb{C}$-linear on fibers.*

An example is the tangent map $F = \mathrm{T}f: \mathrm{T}X \to \mathrm{T}X'$ of a holomorphic map $f: X \to X'$; in this case $F_x: \mathrm{T}_x X \to \mathrm{T}_{f(x)} X'$ is the differential $\mathrm{d}f_x$ of f at x. The analogous definition applies to $\mathcal{C}^r$ vector bundles.

Given a holomorphic vector bundle $\pi: E \to X$ and a holomorphic map $f: W \to X$, the *pull-back bundle* $f^*E \to W$ (see (1.12) on p. 11) is a holomorphic vector bundle over W. We have a natural morphism $f^*E \to E$ over f which maps each fiber $(f^*E)_x$ isomorphically onto the fiber $E_{f(x)}$.

Assume that $E \to X$ and $F \to X$ are complex (or holomorphic) vector bundles. Using standard functors on complex vector spaces we obtain the following derived complex (resp. holomorphic) vector bundles over X:

(a) $E \oplus F = \cup_{x \in X} E_x \oplus F_x$, the *direct sum* or the *Whitney sum*,
(b) $E \otimes F = \cup_{x \in X} E_x \otimes F_x$, the *tensor product*,
(c) $E^* = \cup_{x \in X} E_x^*$, the *dual bundle* of E,
(d) $\mathrm{Hom}(E, F) = \cup_{x \in X} \mathrm{Hom}(E_x, F_x) = E^* \otimes F$,
(e) $\wedge^k E = \cup_{x \in X} \wedge^k E_x$, the *k-th exterior power* of E,
(f) $S^k(E) = \cup_{x \in X} S^k(E_x)$, the *k-th symmetric power* of E.

The transition maps in these bundles are obtained by applying the respective functor fiberwise to the transition maps of the original bundles. For example, if E and E' are given by cocycles $g_{\alpha,\beta}$, $g'_{\alpha,\beta}$ over the same open cover $\mathcal{U} = \{U_\alpha\}$ of X then the direct sum $E \oplus E'$ is given by the cocycle

$$\begin{pmatrix} g_{\alpha,\beta} & 0 \\ 0 & g'_{\alpha,\beta} \end{pmatrix}.$$

Given a subbundle E' of E, the *quotient bundle* $E/E' \to X$ is defined by $E/E' = \cup_{x \in X} E_x/E'_x$. For any vector bundle chart θ on E satisfying (1.16) the map $pr_2 \circ \theta: E|_U \to U \times \mathbb{C}^{n-m}$ factors through $(E/E')|_U$ and induces a bijective map $\widetilde{\theta}: (E/E')|_U \to U \times \mathbb{C}^{n-m}$. The collection of all such maps is a complex (resp. holomorphic) vector bundle atlas on E/E'. If $E = E' \oplus E''$ is a direct sum of its subbundles $E', E'' \subset E$ then the projection $\tau: E \to E''$ with the kernel $\ker \tau = E'$ induces an isomorphism of E/E' onto E''.

A sequence of vector bundle maps over X,

$$\cdots \longrightarrow E_{k-1} \xrightarrow{\sigma_{k-1}} E_k \xrightarrow{\sigma_k} E_{k+1} \longrightarrow \cdots$$

is a *complex* if $\sigma_k \circ \sigma_{k-1} = 0$ (equivalently, $\operatorname{im} \sigma_{k-1} \subset \ker \sigma_k$) for every k. The sequence is *exact* at E_k if $\operatorname{im} \sigma_{k-1} = \ker \sigma_k$. A *short exact sequence* is an exact sequence of the form

$$0 \longrightarrow E' \xrightarrow{\sigma} E \xrightarrow{\tau} E'' \longrightarrow 0. \tag{1.19}$$

This means that σ is injective, τ is surjective, and $\operatorname{im} \sigma = \ker \tau$. Hence τ induces an isomorphism of the quotient bundle $E/\sigma(E')$ onto E''.

A short exact sequence (1.19) *splits* if there exists a vector bundle homomorphism $\rho\colon E'' \to E$ such that $\tau \circ \rho$ is the identity on E''. Such ρ is called a *splitting map* for the sequence. In this case E is isomorphic to the Whitney sum $E = \sigma(E') \oplus \rho(E'')$ of its subbundles $\sigma(E')$ and $\rho(E'')$. Note that every short exact sequence splits locally over small open subsets of the base, and any convex linear combination of splittings is again a splitting. By patching local splittings with a partition of unity one gets the following.

Proposition 1.5.9. *Every short exact sequence (1.19) of complex vector bundle maps of class $\mathcal{C}^r$ ($r \in \{0, 1, \ldots, \infty\}$) admits a $\mathcal{C}^r$ splitting. In particular, we have $E \cong E' \oplus E''$ as complex vector bundles of class $\mathcal{C}^r$.*

The analogous result for holomorphic vector bundles over Stein spaces follows from Cartan's Theorem B; see Corollary 2.4.5 on p. 54.

1.6 The Tangent Bundle

We assume that the reader is familiar with the construction of the real tangent bundle $\mathrm{T}X$ of a smooth manifold X (Example 1.5.2). A tangent vector $V_x \in \mathrm{T}_x X$ is viewed as a derivation $\mathcal{C}_x^\infty \ni f \mapsto V_x(f) \in \mathbb{R}$ on the algebra of germs of smooth functions at x. Sections $X \to \mathrm{T}X$ are called *vector fields* on X. The complexification $\mathbb{C}\mathrm{T}X = \mathrm{T}X \otimes_{\mathbb{R}} \mathbb{C}$ of $\mathrm{T}X$ is the *complexified tangent bundle* of X; its sections are called *complex vector fields* on X.

Assume now that X is a complex manifold. There is a unique real linear endomorphism $J \in \operatorname{End}_{\mathbb{R}} \mathrm{T}X$, called the *almost complex structure operator*, which is given in any local holomorphic coordinate system $z = (z_1, \ldots, z_n)$ ($z_j = x_j + \mathrm{i}y_j$) on X by

$$J\frac{\partial}{\partial x_j} = \frac{\partial}{\partial y_j}, \qquad J\frac{\partial}{\partial y_j} = -\frac{\partial}{\partial x_j}. \tag{1.20}$$

The operator J extends to $\mathbb{C}\mathrm{T}X$ by $J(v \otimes \alpha) = J(v) \otimes \alpha$ for $v \in \mathrm{T}X$ and $\alpha \in \mathbb{C}$. From $J^2 = -\mathrm{Id}$ we infer that the eigenvalues of J are $+\mathrm{i}$ and $-\mathrm{i}$. Hence we have a decomposition

$$\mathbb{C}\mathrm{T}X = \mathrm{T}^{1,0}X \oplus \mathrm{T}^{0,1}X \tag{1.21}$$

into the $+\mathrm{i}$ eigenspace $\mathrm{T}^{1,0}X$ and the $-\mathrm{i}$ eigenspace $\mathrm{T}^{0,1}X$ of J. In holomorphic coordinates $z = (z_1, \ldots, z_n)$ on an open subset $U \subset X$ we have

$$\mathrm{T}^{1,0}X|_U = \mathrm{Span}_{\mathbb{C}}\left\{\frac{\partial}{\partial z_1}, \ldots, \frac{\partial}{\partial z_n}\right\}, \quad \mathrm{T}^{0,1}X|_U = \mathrm{Span}_{\mathbb{C}}\left\{\frac{\partial}{\partial \bar{z}_1}, \ldots, \frac{\partial}{\partial \bar{z}_n}\right\},$$

where

$$\frac{\partial}{\partial z_j} = \frac{1}{2}\left(\frac{\partial}{\partial x_j} - \mathrm{i}\frac{\partial}{\partial y_j}\right), \quad \frac{\partial}{\partial \bar{z}_j} = \frac{1}{2}\left(\frac{\partial}{\partial x_j} + \mathrm{i}\frac{\partial}{\partial y_j}\right).$$

Note that $\mathrm{T}^{1,0}X$ is a holomorphic vector bundle whose transition functions are the complex Jacobians (1.6) of the holomorphic transition maps (1.4) between complex charts on X. The conjugation $v \otimes \alpha \to v \otimes \bar{\alpha}$ induces a $\mathbb{C}$-antilinear isomorphism of $\mathrm{T}^{1,0}X$ onto $\mathrm{T}^{0,1}X$, and hence $\mathrm{T}^{0,1}X = \overline{\mathrm{T}^{1,0}X}$ is an antiholomorphic complex vector bundle.

We have an $\mathbb{R}$-linear isomorphism $\Phi\colon \mathrm{T}X \xrightarrow{\cong} \mathrm{T}^{1,0}X$ given by

$$\mathrm{T}X \ni V \longmapsto \Phi(V) = \frac{1}{2}(V - \mathrm{i}JV) \in \mathrm{T}^{1,0}X \tag{1.22}$$

with the inverse $\Phi^{-1}(W) = 2\Re W$ $(W \in \mathrm{T}^{1,0}X)$. We have $\Phi(JV) = \mathrm{i}\,\Phi(V)$ for $V \in \mathrm{T}X$; i.e., the following diagram commutes:

$$\begin{tikzcd} \mathrm{T}X \arrow[r, "J"] \arrow[d, "\Phi"'] & \mathrm{T}X \arrow[d, "\Phi"] \\ \mathrm{T}^{1,0}X \arrow[r, "\mathrm{i}"] & \mathrm{T}^{1,0}X \end{tikzcd}$$

In local holomorphic coordinates the isomorphism Φ is given by

$$\sum_{j=1}^{n}\left(a_j\frac{\partial}{\partial x_j} + b_j\frac{\partial}{\partial y_j}\right) \overset{\Phi}{\longmapsto} \sum_{j=1}^{n}(a_j + \mathrm{i}b_j)\frac{\partial}{\partial z_j}. \tag{1.23}$$

Definition 1.6.1. *A real vector field V on X is said to be* holomorphic *if $\Phi(V) = \frac{1}{2}(V - \mathrm{i}JV)$ is a holomorphic section of $\mathrm{T}^{1,0}X$.*

We see from (1.23) that the vector field $V = \sum_{j=1}^{n} a_j\frac{\partial}{\partial x_j} + b_j\frac{\partial}{\partial y_j}$ is holomorphic if and only if the functions $a_j + \mathrm{i}b_j$ $(j = 1, \ldots, n)$ are holomorphic.

Denote by $\aleph_r(X)$ the space of all real vector fields of class $\mathcal{C}^r$ on a (real or complex) manifold X $(r = 0, 1, \ldots, \infty, \omega)$, by $\aleph_{\mathcal{O}}(X)$ the space of all holomorphic vector fields on a complex manifold X, and by $\aleph_{\mathcal{A}}(X)$ the space of all algebraic vector fields on a complex algebraic manifold X (algebraic sections of the tangent bundle $\mathrm{T}X$).

The *commutator* or *Lie bracket* of vector fields V, W is defined by

$$[V,W](f) = V(W(f)) - W(V(f)), \quad f \in \mathcal{C}^\infty(X). \tag{1.24}$$

If V, W are of class $\mathcal{C}^r$ for some $r \geq 1$ then $[V,W]$ is a vector field of class $\mathbb{C}^{r-1}$. If

$$V = \sum_{j=1}^n a_j(x)\frac{\partial}{\partial x_j}, \quad W = \sum_{j=1}^n b_j(x)\frac{\partial}{\partial x_j} \tag{1.25}$$

then

$$[V,W] = \sum_{j=1}^n \big(V(b_j) - W(a_j)\big)\frac{\partial}{\partial x_j}. \tag{1.26}$$

This operation is linear in both factors, anticommutative ($[V,W] = -[W,V]$, hence $[V,V] = 0$), and it satisfies the *Jacobi identity*:

$$\big[U,[V,W]\big] + \big[V,[W,U]\big] + \big[W,[U,V]\big] = 0.$$

The $\mathbb{R}$-vector spaces $\aleph_\infty(X)$ and $\aleph_\omega(X)$, endowed with the commutator bracket, are real Lie algebras. The commutator operation extends to sections of $\mathbb{C}\mathrm{T}X$, $\mathrm{T}^{1,0}X$ and $\mathrm{T}^{0,1}X$, with the same properties.

Proposition 1.6.2. *The space $\aleph_{\mathcal{O}}(X)$ of all holomorphic vector fields is a Lie subalgebra of $\aleph_\infty(X)$. The map Φ (1.22) induces a complex Lie algebra isomorphism of $\aleph_{\mathcal{O}}(X)$ onto the space of all holomorphic sections of $\mathrm{T}^{1,0}X$.*

Proof. The Cauchy-Riemann equations (1.3) (p. 2) show that for a real vector field V and a holomorphic function f we have $\Phi(V)(f) = V(f)$. Hence V is holomorphic if and only if, as a derivation $f \to V_x f$ on the algebra $\mathcal{C}_x^\infty$ of germs of smooth functions at any point x, it maps the subalgebra $\mathcal{O}_x \subset \mathcal{C}_x^\infty$ consisting of holomorphic germs to itself. The proposition now follows by applying (1.24) to holomorphic test functions f on small open sets in X. □

In the sequel we tacitly use the identification $\mathrm{T}X \cong \mathrm{T}^{1,0}X$ and the corresponding identification of $\aleph_{\mathcal{O}}(X)$ with the holomorphic sections of $\mathrm{T}^{1,0}X$.

To a smooth map $f\colon X \to Y$ we associate its *tangent map* $\mathrm{T}f\colon \mathrm{T}X \to \mathrm{T}Y$; this is a vector bundle morphism over f that is given on any fiber T_xX by the differential $\mathrm{d}f_x\colon \mathrm{T}_xX \to \mathrm{T}_{f(x)}Y$. If (X, J_X) and (Y, J_Y) are complex manifolds then a map $f\colon X \to Y$ is holomorphic precisely when $\mathrm{d}f$ commutes with the almost complex structure operators on X and Y:

$$\mathrm{d}f \circ J_X = J_Y \circ \mathrm{d}f.$$

In this case $\mathrm{d}f$ respects the decomposition $\mathbb{C}\mathrm{T}X = \mathrm{T}^{1,0}X \oplus \mathrm{T}^{0,1}X$ (1.21) and $\mathrm{d}f_x\colon \mathrm{T}^{1,0}_xX \to \mathrm{T}^{1,0}_{f(x)}Y$ is $\mathbb{C}$-linear for every $x \in X$.

If $f\colon X \to Y$ is biholomorphic then every holomorphic vector field V on X is pushed forward by f to a holomorphic vector field $(f_*V)_{f(x)} = \mathrm{d}f_x \cdot V_x$ on Y. Conversely, a holomorphic vector field W on Y pulls back to a holomorphic vector field $f^*W = (f^{-1})_*W$.

If M is a complex submanifold of a complex manifold X then its tangent bundle $\mathrm{T}M$ is a holomorphic vector subbundle of the restricted bundle $\mathrm{T}X|_M$. The *normal bundle* of M in X is the quotient bundle $\mathrm{N}_{M,X} = \mathrm{T}X|_M/\mathrm{T}M$. We thus have a short exact sequence

$$0 \longrightarrow \mathrm{T}M \longrightarrow \mathrm{T}X|_M \xrightarrow{\beta} \mathrm{N}_{M/X} \longrightarrow 0. \tag{1.27}$$

By Proposition 1.5.9 we can realize $\mathrm{N}_{M/X}$ as a smooth complex vector subbundle of $\mathrm{T}X|_M$ such that

$$\mathrm{T}X|_M = \mathrm{T}M \oplus \mathrm{N}_{M/X}. \tag{1.28}$$

If the submanifold M is Stein, it admits an open Stein neighborhood $\Omega \subset X$ (Theorem 3.1.1 on p. 57); hence Cartan's Theorem B (Theorem 2.4.1 on p. 52) implies that the sequence (1.27) also splits holomorphically and we get a holomorphic direct sum (1.28).

The notion of tangent space is also defined for a complex space X with singularities. We get a *linear space* $\pi\colon \mathrm{T}X \to X$, that is, a (not necessarily reduced) complex space $L = \mathrm{T}X$ and a holomorphic projection π onto X such that the fibers $L_x = \mathrm{T}_x X$ are vector spaces whose dimension depends on x. Here is a local description; for further details see [140]. Fix a point $x_0 \in X$ and represent an open neighborhood $W \subset X$ of x_0 as a closed complex subvariety in an open set $U \subset \mathbb{C}^n$. After shrinking U we find holomorphic functions $f_1, \dots, f_k \in \mathcal{O}(U)$ which generate the ideal sheaf $\mathcal{J}_W$ at each point. Let $(z_1, \dots, z_n, w_1, \dots, w_n)$ be complex coordinates on $\mathbb{C}^n \times \mathbb{C}^n$. Then $\mathrm{T}W = \mathrm{T}X|_W$ is the complex subspace of $U \times \mathbb{C}^n$ determined by the functions

$$f_1, \dots, f_k \quad \text{and} \quad \frac{\partial f_i}{\partial z_1} w_1 + \cdots + \frac{\partial f_i}{\partial z_n} w_n \quad \text{for} \quad i = 1, \dots, k. \tag{1.29}$$

The projection $\mathrm{T}W \to W$ is the coordinate projection $W \times \mathbb{C}^m \to W$ restricted to $\mathrm{T}W$. Different local representations of X as subvarieties of $\mathbb{C}^n$ give isomorphic representations of the tangent space. The restriction of $\mathrm{T}X$ to the regular locus X_{reg} is the usual tangent bundle of X_{reg}. The fiber $\mathrm{T}_x X$ is called the *Zariski tangent space* of X at the point $x \in X$. A holomorphic map $f\colon X \to Y$ of complex spaces induces the Zariski differential $\mathrm{d}f_x\colon \mathrm{T}_x X \to \mathrm{T}_{f(x)} Y$.

If $h\colon Z \to X$ is a holomorphic submersion of a complex space Z onto a complex space X (Def. 1.3.1) then the fibers $Z_x = h^{-1}(x)$ are smooth, and the tangent spaces $\mathrm{T}_z Z_{h(z)} = \ker \mathrm{d}h_z$ define a holomorphic vector bundle $\mathrm{VT}Z \to Z$ called the *vertical tangent bundle* of Z, or the *relative tangent bundle*, sometimes denoted $\mathrm{T}Z/X$.

Assume now that M is a real submanifold of a complex manifold (X, J). Let $m = \dim_{\mathbb{R}} M$, $n = \dim_{\mathbb{C}} X$ and $d = 2n - m$ (the real codimension of M in X). For every point $x \in M$ we define subspaces $\mathrm{T}_x^{\mathbb{C}} M, L_x \subset \mathrm{T}_x M$ by

$$\mathrm{T}_x^{\mathbb{C}} M = \mathrm{T}_x M \cap J(\mathrm{T}_x M), \quad \mathrm{T}_x M = \mathrm{T}_x^{\mathbb{C}} M \oplus L_x.$$

Clearly $\mathrm{T}_x^{\mathbb{C}}M$ is the unique maximal complex subspace of T_xM. The choice of the complementary real subspace L_x can be specified by requiring that the direct sum is orthogonal with respect to a chosen Hermitian metric on $\mathrm{T}X$. If $k = \dim_{\mathbb{C}} \mathrm{T}_x^{\mathbb{C}}M$ is independent of the point $x \in M$ then M is called a *CR submanifold* of X of *CR dimension* k. If $k + d = n$ (hence $\dim_{\mathbb{R}} M = 2k + d$) then M is said to be *generic*. If $\dim M = 2k + 1$ then M is *of hypersurface type*. (For analysis on CR manifolds see [30].)

1.7 The Cotangent Bundle and Differential Forms

Let X be a complex manifold of dimension n. We denote by T^*X its real cotangent bundle, the real dual of the tangent bundle $\mathrm{T}X$. The $\mathbb{R}$-linear endomorphism J of $\mathbb{C}\mathrm{T}X$ induces the dual endomorphism J^* of the complexified cotangent bundle $\mathbb{C}\mathrm{T}^*X = \mathrm{T}^*X \otimes_{\mathbb{R}} \mathbb{C}$ with $(J^*)^2 = -\mathrm{Id}$. We have a splitting

$$\mathbb{C}\mathrm{T}^*X = \mathrm{T}^{*1,0}X \oplus \mathrm{T}^{*0,1}X$$

into the $+\mathrm{i}$ and $-\mathrm{i}$ eigenspaces of J^*. In local coordinates $z = (z_1, \ldots, z_n)$ on an open subset U of X we have

$$\mathrm{T}^{*1,0}X|_U = \mathrm{Span}\,\{\mathrm{d}z_1, \ldots, \mathrm{d}z_n\}\,, \quad \mathrm{T}^{*0,1}X|_U = \mathrm{Span}\,\{\mathrm{d}\bar{z}_1, \ldots, \mathrm{d}\bar{z}_n\}\,.$$

Note that $\mathrm{T}^{*1,0}X$ and $\mathrm{T}^{*0,1}X$ are the complex dual bundles of $\mathrm{T}^{1,0}X$ and $\mathrm{T}^{0,1}X$, respectively.

The complex exterior algebra bundles $\wedge\mathrm{T}^{*1,0}X$, $\wedge\mathrm{T}^{*0,1}X$ admit natural bundle injections into the exterior algebra bundle $\wedge\mathbb{C}\mathrm{T}^*X$. We have $\wedge\mathbb{C}\mathrm{T}^*X = \oplus_{k=0}^{2n} \wedge^k \mathbb{C}\mathrm{T}^*X$, $\wedge\mathrm{T}^{*1,0}X = \oplus_{k=0}^{n} \wedge^k \mathrm{T}^{*1,0}X$, and $\wedge\mathrm{T}^{*0,1}X = \oplus_{k=0}^{n} \wedge^k \mathrm{T}^{*0,1}X$. For every $k = 0, 1 \ldots, 2n$ we have a decomposition

$$\wedge^k\mathbb{C}\mathrm{T}^*X = \bigoplus_{p+q=k} \left(\wedge^p\mathrm{T}^{*1,0}X\right) \wedge \left(\wedge^q\mathrm{T}^{*0,1}X\right).$$

Let $\mathcal{E}_k(X)$ denote the space of smooth sections of the bundle $\wedge^k\mathbb{C}\mathrm{T}^*X$; these are smooth complex valued k-forms on X. In particular, $\mathcal{E}_0(X) = \mathcal{C}^\infty(X)$ is the space of smooth functions on X. We have $\mathcal{E}_k(X) = \bigoplus_{p+q=k} \mathcal{E}_{p,q}(X)$ where $\mathcal{E}_{p,q}(X)$ is the space of smooth sections of $(\wedge^p\mathrm{T}^{*1,0}X) \wedge (\wedge^q\mathrm{T}^{*0,1}X)$; these are called differential (p,q)-forms. In complex coordinates $z = (z_1, \ldots, z_n)$ every (p,q)-form has the expression

$$\alpha = \sum_{|I|=p,\ |J|=q} a_{I,J}(z)\, \mathrm{d}z_I \wedge \mathrm{d}\bar{z}_J \tag{1.30}$$

where $I = (i_1, \ldots, i_p)$, $J = (j_1, \ldots, j_q)$ $(1 \le i_1 < \cdots < i_p \le n,\ 1 \le j_1 < \cdots < j_q \le n)$, $\mathrm{d}z_I = \mathrm{d}z_{i_1} \wedge \cdots \wedge \mathrm{d}z_{i_p}$, and $\mathrm{d}\bar{z}_J = \mathrm{d}\bar{z}_{j_1} \wedge \cdots \wedge \mathrm{d}\bar{z}_{j_q}$.

A *holomorphic p-form* is a holomorphic section of the holomorphic vector bundle $\wedge^p \mathrm{T}^{*1,0}X$. In local holomorphic coordinates,

$$\alpha = \sum_{|I|=p} a_I(z)\,\mathrm{d}z_I = \sum_{|I|=p} a_I(z)\,\mathrm{d}z_{i_1} \wedge \cdots \wedge \mathrm{d}z_{i_p},$$

where the coefficients a_I are holomorphic functions. The space of all holomorphic p-forms on X is denoted $\Omega_p(X)$; hence $\Omega_0(X) = \mathcal{O}(X)$.

The *pull-back* of a form (1.30) by a smooth map $f = (f_1, \ldots, f_n)$ equals

$$f^*\alpha = \sum_{|I|=p,\ |J|=q} (a_{I,J} \circ f)\,\mathrm{d}f_I \wedge \mathrm{d}\bar{f}_J.$$

We have $\mathrm{d}(f^*\alpha) = f^*(\mathrm{d}\alpha)$, where $\mathrm{d}\colon \mathcal{E}_k(X) \to \mathcal{E}_{k+1}(X)$ is the exterior derivative. If $z = f(w)$ is holomorphic in w then

$$\mathrm{d}f_j = \sum_k \frac{\partial f_j}{\partial w_k}\,\mathrm{d}w_k, \qquad \mathrm{d}\bar{f}_j = \sum_k \frac{\partial \bar{f}_j}{\partial \bar{w}_k}\,\mathrm{d}\bar{w}_k$$

for every j, and we see that $f^*\alpha$ is again a (p,q)-form. Thus the pull-back by a holomorphic map respects the decomposition (1.30) and takes holomorphic forms to holomorphic forms. The exterior derivative splits as $\mathrm{d} = \partial + \overline{\partial}$, where

$$\partial\colon \mathcal{E}_{p,q}(X) \to \mathcal{E}_{p+1,q}(X), \quad \overline{\partial}\colon \mathcal{E}_{p,q}(X) \to \mathcal{E}_{p,q+1}(X)$$

are defined in local holomorphic coordinates by applying the corresponding operators (1.1) to the coefficient functions of α (1.30):

$$\partial\alpha = \sum_{|I|=p,\ |J|=q} \partial a_{I,J} \wedge \mathrm{d}z_I \wedge \mathrm{d}\bar{z}_J, \quad \overline{\partial}\alpha = \sum_{|I|=p,\ |J|=q} \overline{\partial} a_{I,J} \wedge \mathrm{d}z_I \wedge \mathrm{d}\bar{z}_J.$$

The definition is independent of the choice of holomorphic coordinates since the pull-back by a holomorphic map f commutes with ∂ and $\overline{\partial}$:

$$f^*(\partial\alpha) = \partial(f^*\alpha), \quad f^*(\overline{\partial}\alpha) = \overline{\partial}(f^*\alpha).$$

The operators d, ∂ and $\overline{\partial}$ are local and satisfy the properties

$$\mathrm{d}^2 = \mathrm{d} \circ \mathrm{d} = 0, \quad \partial^2 = 0, \quad \overline{\partial}^2 = 0, \quad \partial\overline{\partial} = -\overline{\partial}\partial.$$

The *conjugate differential* is defined by

$$\mathrm{d}^c = -J^*\mathrm{d}. \tag{1.31}$$

Explicitly, on functions we have $(\mathrm{d}^c f)_x \cdot v = -\mathrm{d}f_x \cdot Jv$ for $v \in \mathrm{T}_x X$. In local holomorphic coordinates $z = (z_1, \ldots, z_n)$, $z_j = x_j + \mathrm{i}y_j$, we have

$$\mathrm{d}^c = \sum_{j=1}^n -\frac{\partial}{\partial y_j}\,\mathrm{d}x_j + \frac{\partial}{\partial x_j}\,\mathrm{d}y_j. \tag{1.32}$$

Clearly d^c is a real operator, and we have the following relations:

$$\mathrm{d} = \partial + \overline{\partial}, \quad \mathrm{d}^c = \mathrm{i}(\overline{\partial} - \partial), \quad \mathrm{d}\mathrm{d}^c = 2\mathrm{i}\,\partial\overline{\partial}.$$

A differential form α is *closed* if $\mathrm{d}\alpha = 0$, and is $\overline{\partial}$*-closed* if $\overline{\partial}\alpha = 0$. We denote by $\mathcal{Z}_k(X)$ the space of all closed k-forms, and by $\mathcal{Z}_{p,q}(X)$ the space of all $\overline{\partial}$-closed (p,q)-forms. Note that $\mathcal{Z}^{\overline{\partial}}_{0,0}(X) = \mathcal{O}(X)$ is the space of holomorphic functions, and $\mathcal{Z}^{\overline{\partial}}_{p,0}(X) = \Omega_p(X)$ is the space of holomorphic p-forms.

A form α of degree $k \geq 1$ is *exact* if $\alpha = \mathrm{d}\beta$ for some form β (this implies $\mathrm{d}\alpha = 0$); α is $\overline{\partial}$*-exact* if $\alpha = \overline{\partial}\beta$ for some form β (this implies $\overline{\partial}\alpha = 0$).

Each of the operators $\mathrm{d}, \partial, \overline{\partial}$ is *locally exact*. For d this is the *Poincaré lemma* which states that for every closed k-form α the equation $\mathrm{d}\beta = \alpha$ has a solution on any contractible open set. For $\overline{\partial}$ we have the following lemma of Grothendieck, also called the $\overline{\partial}$*-Poincaré lemma* [233, p. 25].

Lemma 1.7.1. *For every smooth (p,q)-form α on a polydisc $\mathbb{D}^n \subset \mathbb{C}^n$ $(q \geq 1)$ such that $\overline{\partial}\alpha = 0$ there exists a smooth $(p, q-1)$-form β on $\mathbb{D}^n$) satisfying $\overline{\partial}\beta = \alpha$. The analogous result holds for the operator ∂.*

The complex vector spaces

$$H^p_{\mathrm{d}}(X) = \frac{\mathcal{Z}_p(X)}{\mathrm{d}(\mathcal{E}_{p-1}(X))}, \qquad p = 0, 1, \ldots$$

are called the *de Rham cohomology groups* of X. (For $p < 0$ set $\mathcal{E}_p(X) = \{0\}$.) In view of Stokes' theorem, the integration of closed p-forms over singular p-cycles in X defines a map $H^p_{\mathrm{d}}(X) \to H^p_{\mathrm{sing}}(X;\mathbb{C})$ to the singular cohomology groups of X with complex coefficients; this map is an isomorphism by de Rham's theorem [233, 503, p. 43]. Hence the de Rham cohomology ring $H^*_{\mathrm{d}}(X) = \oplus_{p\geq 0} H^p_{\mathrm{d}}(X)$ is a topological invariant of X. For $p > \dim_{\mathbb{R}} X$ the group $H^p_{\mathrm{d}}(X)$ vanishes by dimension reasons. The vector spaces

$$H^{p,q}_{\overline{\partial}}(X) = \frac{\mathcal{Z}_{p,q}(X)}{\overline{\partial}(\mathcal{E}_{p,q-1}(X))}, \qquad p, q \geq 0 \tag{1.33}$$

are called the *Dolbeault cohomology groups* of X. They depend on the holomorphic structure of X. Clearly $H^{p,0}_{\overline{\partial}}(X) = \Omega_p(X)$ is the space of holomorphic p-forms on X, and $H^{p,q}_{\overline{\partial}}(X) = 0$ if $\max\{p,q\} > \dim_{\mathbb{C}} X$. The Grothendieck Lemma 1.7.1 says that $H^{p,q}_{\overline{\partial}}(P) = 0$ for all $p \geq 0,\ q \geq 1$ when P is a polydisc in $\mathbb{C}^n$. It is a much deeper result that

$$X \text{ Stein} \implies H^{p,q}_{\overline{\partial}}(X) = 0, \quad p \geq 0,\ q \geq 1$$

(Theorem 2.4.6). If X is a compact complex manifold then the cohomology groups $H^{p,q}_{\overline{\partial}}(X)$ are finite dimensional vector spaces.

1.8 Plurisubharmonic Functions and the Levi Form

Plurisubharmonic functions were introduced in 1942 by K. Oka [385] and P. Lelong [333]. This is undoubtedly one of the most important classes of functions in complex analysis and geometry. A complex manifold with a strongly plurisubharmonic exhaustion function is a Stein manifold (see Theorem 2.3.2 on p. 51). Plurisubharmonic functions play a major role in the L^2-theory for solving the $\overline{\partial}$-problem (see §2.5). Pluripotential theory provides an important link with Kähler geometry via the complex Monge-Ampère equation. For more on this subject see [268, 297, 334].

An upper semicontinuous function u on a complex manifold X which is not identically $-\infty$ on any connected component is said to be *plurisubharmonic* if for every continuous map $f\colon \overline{\mathbb{D}} \to X$ which is holomorphic in the disc $\mathbb{D} = \{z \in \mathbb{C}\colon |z| < 1\}$ (such f is called an *analytic disc* in X) we have the submeanvalue property

$$u(f(0)) \le \int_0^{2\pi} u(f(\mathrm{e}^{\mathrm{i}\theta}))\,\frac{\mathrm{d}\theta}{2\pi}.$$

This is equivalent to asking that $u \circ f$ is subharmonic on $\mathbb{D}$ for every such f. For a domain $X \subset \mathbb{C}^n$ it suffices to assume that this holds for small linear complex discs in X. The definition extends to functions on a complex space X by local plurisubharmonic extendibility from local embeddings of X in $\mathbb{C}^N$. We denote by $\mathrm{Psh}(X)$ the set of all plurisubharmonic functions on X.

Here are some basic examples of plurisubharmonic functions:

- $\log|f|$ where f is holomorphic and not identically zero.
- $|f|^p$ for any $p > 0$, where f is holomorphic.
- $\log\left(\sum_{j=1}^m |f_j|^2\right)$ where the f_j are holomorphic and not all zero.

We collect the main properties of the class $\mathrm{Psh}(X)$:

- Plurisubharmonicity is a local property: A function is plurisubharmonic if and only if it is such in an open neighborhood of any point.
- *The maximum principle:* If $u \in \mathrm{Psh}(X)$ has a local maximum at some point, then u is constant on the respective connected component of X.
- If $u_j \in \mathrm{Psh}(X)$ and $c_j \ge 0$ for $j = 1, \ldots, k$ then $\sum_{j=1}^k c_j u_j \in \mathrm{Psh}(X)$.
- If $u_j \in \mathrm{Psh}(X)$ for $j = 1, \ldots, k$ then $\max\{u_1, \ldots, u_k\} \in \mathrm{Psh}(X)$.
- If $u_1 \le u_2 \le \cdots \le u = \lim_{j\to\infty} u_j < +\infty$ and $u_j \in \mathrm{Psh}(X)$ for every $j = 1, 2, \ldots$, then u is plurisubharmonic provided that it is upper semicontinuous. If u is locally bounded from above then its upper regularization $u^*(z) = \limsup_{\zeta\to z} u(\zeta)$ is plurisubharmonic.
- If $u_1 \ge u_2 \ge \cdots \ge u = \lim_{j\to\infty} u_j$ and $u_j \in \mathrm{Psh}(X)$ for every $j = 1, 2, \ldots$, then u is either plurisubharmonic or identically $-\infty$ on any connected component of X.

- Every $u \in \mathrm{Psh}(X)$ is the pointwise limit of a decreasing sequence of smooth plurisubharmonic functions on any relatively compact subset $\Omega \Subset X$.
- If $u \in \mathrm{Psh}(X)$ and $h\colon [-\infty, \infty) \to [-\infty, \infty)$ is an increasing convex function then $h \circ \rho \in \mathrm{Psh}(X)$.
- If u is plurisubharmonic and f is a holomorphic map then $u \circ f$ is plurisubharmonic or $\equiv -\infty$ on each connected component of its domain.

In order to place plurisubharmonicity in its proper geometric context we recall the notion of the *Levi form*. Let $z = (z_1, \dots, z_n)$ be complex coordinates on $\mathbb{C}^n$. Using the isomorphism $\mathrm{T}_z\mathbb{C}^n \cong \mathrm{T}_z^{1,0}\mathbb{C}^n \cong \mathbb{C}^n$ (1.22) we identify $v = 2\Re \sum_{j=1}^n v_j \frac{\partial}{\partial z_j} \in \mathrm{T}_z\mathbb{C}^n$ with $(v_1, \dots, v_n) \in \mathbb{C}^n$. Given a $\mathcal{C}^2$ function ρ in a neighborhood of a point $z \in \mathbb{C}^n$, we have

$$\rho(z+v) = \rho(z) + 2\Re \sum_{j=1}^n \frac{\partial \rho}{\partial z_j}(z)\, v_j + \Re \sum_{j,k=1}^n \frac{\partial^2 \rho}{\partial z_j \partial z_k}(z)\, v_j v_k$$
$$+ \frac{1}{2} \sum_{j,k=1}^n \frac{\partial^2 \rho}{\partial z_j \partial \bar{z}_k}(z)\, v_j \overline{v}_k + o(|v|^2).$$

The *complex Hessian* of ρ at z is the Hermitian bilinear form

$$\mathrm{T}_z\mathbb{C}^n \times \mathrm{T}_z\mathbb{C}^n \ni (v, w) \mapsto H_{\rho,z}(v, w) = \sum_{j,k=1}^n \frac{\partial^2 \rho}{\partial z_j \partial \bar{z}_k}(z)\, v_j \overline{w}_k. \tag{1.34}$$

The associated Hermitian quadratic form is called the *Levi form* of ρ at z:

$$\mathcal{L}_{\rho,z}(v) = H_{\rho,z}(v, v) = \sum_{j,k=1}^n \frac{\partial^2 \rho}{\partial z_j \partial \bar{z}_k}(z)\, v_j \bar{v}_k = \frac{\partial^2}{\partial \zeta\, \partial \bar{\zeta}}\bigg|_{\zeta=0} \rho(z + \zeta v). \tag{1.35}$$

Clearly this equals one quarter of the Laplacian $\triangle = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$ of the function $\zeta = x + \mathrm{i}y \mapsto \rho_{z,v}(\zeta) = \rho(z + \zeta v)$ at $\zeta = 0$:

$$\mathcal{L}_{\rho,z}(v) = \frac{1}{4} \triangle \rho_{z,v}(0). \tag{1.36}$$

Hence *a $\mathcal{C}^2$ function $\rho\colon \Omega \to \mathbb{R}$ on a domain $\Omega \subset \mathbb{C}^n$ is plurisubharmonic if and only if $\mathcal{L}_{\rho,z} \geq 0$ is nonnegative definite on $\mathrm{T}_z\mathbb{C}^n$ for every $z \in \Omega$.*

Definition 1.8.1. *A $\mathcal{C}^2$ function $\rho\colon \Omega \to \mathbb{R}$ is* strongly plurisubharmonic *if its Levi form $\mathcal{L}_{\rho,z}$ is positive definite for every point $z \in \Omega$.*

Clearly this condition is stable under small $\mathcal{C}^2$ deformations.

Example 1.8.2. If $f_1, \dots, f_k$ are holomorphic functions then $\rho = \sum_{j=1}^k |f_j|^2$ is plurisubharmonic, and is strongly plurisubharmonic at each point z where the complex differentials $(\mathrm{d}f_j)_z$ span the cotangent space. $\square$

Consider now the effect of coordinate changes. Assume that $f\colon \Omega \to \Omega'$ is a holomorphic map between domains $\Omega \subset \mathbb{C}^n$, $\Omega' \subset \mathbb{C}^m$. Let $\rho'\colon \Omega' \to \mathbb{R}$ be a $\mathcal{C}^2$ function and set $\rho = \rho' \circ f\colon \Omega \to \mathbb{R}$. A computation gives

$$H_{\rho,z}(v,w) = H_{\rho',f(z)}\big(\mathrm{d}f_z\cdot v, \mathrm{d}f_z\cdot w\big), \quad v,w \in \mathrm{T}_z\mathbb{C}^n.$$

The Levi form transforms accordingly:

$$\mathcal{L}_{\rho,z}(v) = \mathcal{L}_{\rho',f(z)}(\mathrm{d}f_z\cdot v), \quad v \in \mathrm{T}_z\mathbb{C}^n.$$

In particular, if ρ' is plurisubharmonic on Ω' then $\rho = \rho' \circ f$ is plurisubharmonic on Ω. If f is a holomorphic immersion and ρ' is strongly plurisubharmonic on Ω', then $\rho' \circ f$ is strongly plurisubharmonic on Ω.

We now give a coordinate free expression for the Levi form. It is natural to look at the more general case of the *canonical form* associated to a Hermitian metric. Let h be a contravariant Hermitian tensor field of type $(1,1)$ on a complex manifold X. In local holomorphic coordinates, h is of the form

$$h = \sum_{j,k=1}^{n} h_{jk}(z)\,\mathrm{d}z_j \otimes \mathrm{d}\bar{z}_k, \qquad \overline{h}_{jk} = h_{kj}. \tag{1.37}$$

It defines a Hermitian bilinear form $\langle h_z, v\otimes w\rangle = \sum_{j,k=1}^{n} h_{jk}(z)v_j\overline{w}_k$ on each tangent space T_zX. Write $h = S - \mathrm{i}\omega$ with S,ω real. Then

$$2\langle S, v\otimes w\rangle = \sum_{j,k=1}^{n} h_{jk}v_j\overline{w}_k + \sum_{j,k=1}^{n} h_{kj}w_k\overline{v}_j = \sum_{j,k=1}^{n} h_{jk}(v_j\overline{w}_k + \overline{v}_k w_j).$$

Thus S is a symmetric tensor field given by

$$S = \Re h = \frac{1}{2}\sum_{j,k=1}^{n} h_{jk}\big(\mathrm{d}z_j \otimes \mathrm{d}\bar{z}_k + \mathrm{d}\bar{z}_k \otimes \mathrm{d}z_j\big).$$

Similarly, $2\mathrm{i}\,\omega = \bar{h} - h$, and hence

$$-2\mathrm{i}\langle \omega, v\otimes w\rangle = \sum_{j,k=1}^{n} (h_{jk}v_j\overline{w}_k - h_{kj}w_k\overline{v}_j) = \sum_{j,k=1}^{n} h_{jk}\big(v_j\overline{w}_k - w_j\overline{v}_k\big).$$

Thus ω is antisymmetric, whence a 2-form. Since

$$\langle \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k, v\wedge w\rangle = \big\langle \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k, (V+\overline{V})\wedge(W+\overline{W})\big\rangle = v_j\overline{w}_k - w_j\overline{v}_k,$$

we have

$$\omega = -\Im h = \frac{\mathrm{i}}{2}\sum_{j,k=1}^{n} h_{jk}\mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k. \tag{1.38}$$

We can express h and S in terms of ω as follows. Note that

$$\langle h, v \otimes w\rangle = \langle S, v \otimes w\rangle - \mathrm{i}\langle \omega, v \wedge w\rangle.$$

Replacing the vector $w = W + \overline{W} \in \mathrm{T}X$ by $Jw = \mathrm{i}W - \mathrm{i}\overline{W}$ gives

$$\langle h, v \otimes Jw\rangle = \langle S, v \otimes Jw\rangle - \mathrm{i}\langle \omega, v \wedge Jw\rangle.$$

The Hermitian property of h, together with the previous identity, implies

$$\langle h, v \otimes Jw\rangle = -\mathrm{i}\langle h, v \otimes w\rangle = \langle \omega, v \wedge w\rangle - \mathrm{i}\langle S, v \otimes w\rangle.$$

Comparing the imaginary parts in the last two equations gives $\langle S, v \otimes w\rangle = \langle \omega, v \wedge Jw\rangle$. If $v = w$ then $\langle h, v \otimes v\rangle$ is real, and hence

$$\langle h, v \otimes v\rangle = \langle S, v \otimes v\rangle = \langle \omega, v \wedge Jv\rangle.$$

When $h = H_\rho$ is the complex Hessian (1.34) of a $\mathcal{C}^2$ function ρ in some local holomorphic coordinates $z = (z_1, \ldots, z_n)$, we have

$$\omega = \frac{\mathrm{i}}{2} \sum_{j,k=1}^{n} \frac{\partial^2 \rho}{\partial z_j \partial \bar{z}_k} \, \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_k = \frac{\mathrm{i}}{2} \partial\overline{\partial}\rho = \frac{1}{4} \, \mathrm{dd}^c \rho. \tag{1.39}$$

Comparing (1.35) and (1.39) gives the following coordinate free definition of the Levi form on any complex manifold (X, J):

$$\mathcal{L}_{\rho,x}(v) = \frac{1}{4} \left\langle \mathrm{dd}^c \rho|_x, v \wedge Jv \right\rangle, \qquad v \in \mathrm{T}_x X. \tag{1.40}$$

This is the restriction to the diagonal $v = w$ of the symmetric bilinear form

$$\mathcal{L}_{\rho,x}(v, w) = \frac{1}{4} \left\langle \mathrm{dd}^c \rho|_x, v \wedge Jw \right\rangle, \qquad v, w \in \mathrm{T}_x X,$$

which is the coordinate free expression for the complex Hessian $H_{\rho,x}(v, w)$. Clearly $\mathcal{L}_{\rho,x}(Jv) = \mathcal{L}_{\rho,x}(v)$ and $\mathcal{L}_{\rho,x}(Jv, Jw) = \mathcal{L}_{\rho,x}(v, w)$.

It follows that a $\mathcal{C}^2$ function $\rho\colon X \to \mathbb{R}$ on a complex manifold is plurisubharmonic if the Levi form $\mathcal{L}_{\rho,x} \geq 0$ is nonnegative for every $x \in X$, and is strongly plurisubharmonic if $\mathcal{L}_{\rho,x}$ is positive definite for every $x \in X$. These conditions are commonly written as $\mathrm{dd}^c\rho \geq 0$, resp. $\mathrm{dd}^c\rho > 0$.

Assume now that h is a Hermitian metric on $\mathrm{T}X$. Then any matrix (h_{jk}), representing h in a local holomorphic coordinate system on X, is positive definite. The associated form ω (1.38) is called the *canonical form* of h. Since $\langle \omega, v \wedge Jv\rangle > 0$ for every $0 \neq v \in \mathrm{T}_x X$, ω is said to be a *positive* $(1,1)$*-form*. We say that h is a *Kähler metric*, and that $\omega = -\Im h$ is a *Kähler form*, if $\mathrm{d}\omega = 0$; in such case (X, h) is a *Kähler manifold*. Note that $\omega = \mathrm{dd}^c\rho$ is a Kähler form if and only if the function ρ is strongly plurisubharmonic. If $\dim X = n$ and ω is a Kähler form on X then its n-th power ω^n is a *volume form* on X, that is, a nonvanishing (n, n)-form.

Example 1.8.3. The standard Kähler form on $\mathbb{C}^n$, associated to the Euclidean metric $h = \sum_{j=1}^n \mathrm{d}z_j \otimes \mathrm{d}\bar{z}_j$, is $\omega = \frac{1}{4}\,\mathrm{dd}^c|z|^2 = \frac{\mathrm{i}}{2}\sum_{j=1}^n \mathrm{d}z_j \wedge \mathrm{d}\bar{z}_j$. The *Fubini-Study metric* on $\mathbb{P}^n$ is given in any affine coordinates $z \in \mathbb{C}^n \subset \mathbb{P}^n$ by the Kähler form $\omega = \frac{1}{4}\mathrm{dd}^c \log(1+|z|^2)$. $\square$

For a general plurisubharmonic function ρ we consider $\mathrm{dd}^c\rho$ as a positive current of bidimension $(1,1)$. This generalizes the fact that the Laplacian of a subharmonic function of one complex variable is a positive measure.

Definition 1.8.4. *A set E in a complex manifold X is* pluripolar *if for each point $p \in E$ there exist a neighborhood $U \subset X$ of p and a plurisubharmonic function $u\colon U \to [-\infty, +\infty)$ such that $E \cap U \subset \{x \in U\colon\ u(x) = -\infty\}$. The set E is* complete pluripolar *if it is locally near each point $p \in E$ the exact $-\infty$ level set of a plurisubharmonic function.*

By a theorem of B. Josefson [279], if $E \subset \mathbb{C}^n$ is pluripolar then there is a plurisubharmonic function u on $\mathbb{C}^n$ such that $E \subset \{u = -\infty\}$. That is, a set $E \subset \mathbb{C}^n$ is locally pluripolar if and only if it is globally pluripolar.

Example 1.8.5. Every closed complex analytic set A in a complex manifold X is complete pluripolar. If A is defined locally in an open set $U \subset X$ by holomorphic functions $f_1, \dots, f_k \in \mathcal{O}(U)$ then $u = \log(|f_1|^2 + \cdots + |f_k|^2)$ is plurisubharmonic in U and $\{u = -\infty\} = A \cap U$. $\square$

Pluripolar sets are removable singularities for plurisubharmonic functions:

Theorem 1.8.6. *If E is a closed pluripolar set in a complex manifold X then every plurisubharmonic function on $X \backslash E$ which is locally bounded at every point of E extends to a plurisubharmonic function on X.*

Functions satisfying the equation $\mathrm{dd}^c u = 0$ are also of great interest.

Lemma 1.8.7. *A real $\mathcal{C}^2$ function u satisfies $\mathrm{dd}^c u = 0$ if and only if u is locally the real part of a holomorphic function. Such u is called* pluriharmonic.

Proof. If $f = u + \mathrm{i}v$ is holomorphic then $\mathrm{dd}^c f = 2\mathrm{i}\,\partial\overline{\partial} f = 0$. Since dd^c is a real operator, we get $\mathrm{dd}^c u = \mathrm{dd}^c v = 0$. Conversely, if $\mathrm{dd}^c u = 0$ then $\mathrm{d}^c u$ is a closed 1-form, and hence on any simply connected domain we have $\mathrm{d}^c u = \mathrm{d}v$ for some $\mathcal{C}^1$ function v. Hence $\langle \mathrm{d}u, -J\xi\rangle = \langle \mathrm{d}^c u, \xi\rangle = \langle \mathrm{d}v, \xi\rangle$ for every tangent vector ξ. Replacing ξ by $-J\xi$ in this identity gives $\mathrm{d}u = -\mathrm{d}^c v$. It follows that $2\overline{\partial}(u + \mathrm{i}v) = (\mathrm{d} - \mathrm{i}\mathrm{d}^c)(u + \mathrm{i}v) = 0$, so the function $u + \mathrm{i}v$ is holomorphic. $\square$

Note that a real $\mathcal{C}^2$ function u is pluriharmonic if and only if both u and $-u$ are plurisubharmonic. This definition also applies of u is merely continuous, but the conclusion of the lemma remains the same.

The following lemma gives an expression for the Levi form in terms of the commutator of complex tangential vector fields.

Lemma 1.8.8. *Let $M = \{\rho = 0\}$ be a real hypersurface in a complex manifold X with a $\mathcal{C}^2$ defining function ρ satisfying $\mathrm{d}\rho \neq 0$ on M. For every pair of vector field V, W on X that are tangential to $\mathrm{T}^{\mathbb{C}}M = \mathrm{T}M \cap J\mathrm{T}M$ we have*

$$4\mathcal{L}_\rho(V, W) = \langle \mathrm{dd}^c\rho, V \wedge JW\rangle = -\langle \mathrm{d}^c\rho, [V, JW]\rangle. \tag{1.41}$$

Proof. Since V and JW are tangential to M, so is their commutator $[V, JW]$. For every 1-form η and vector fields V, V' we have [5, p. 432]

$$\langle \mathrm{d}\eta, V \wedge V'\rangle = V\langle \eta, V'\rangle - V'\langle \eta, V\rangle - \langle \eta, [V, V']\rangle.$$

Taking $\eta = \mathrm{d}^c\rho$, $V' = JW$, and observing that $\ker \mathrm{d}^c\rho|_{\mathrm{T}M} = \mathrm{T}^{\mathbb{C}}M$ (hence $\langle \eta, V\rangle = 0 = \langle \eta, JW\rangle$) gives (1.41). □

We also consider functions whose Levi form has no more that a prescribed number of negative or zero eigenvalues. The following notions were introduced by Andreotti and Grauert [20]. (For complex spaces see Def. 3.1.2.)

Definition 1.8.9. *Let X be a complex manifold of dimension n.*

(i) *A $\mathcal{C}^2$ function $\rho\colon X \to \mathbb{R}$ is (strongly)* q-convex *if its Levi form $\mathcal{L}_\rho$ has at most $q-1$ negative or zero eigenvalues (so it has at least $n-q+1$ positive eigenvalues) at each point of X.*

(ii) *The manifold X is* q-convex *if there exists a smooth exhaustion function $\rho\colon X \to \mathbb{R}$ that is q-convex on $\{\rho > c\}$ for some $c \in \mathbb{R}$.*

(iii) *The manifold X is* q-complete *if there exists a smooth exhaustion function $\rho\colon X \to \mathbb{R}$ that is q-convex on X.*

A 1-complete manifold is the same thing as a Stein manifold (see Theorem 2.3.2). Every open connected complex manifold of dimension n is n-convex [379]. For surveys on q-convexity see Colţoiu [89] and Grauert [225].

1.9 Vector Fields, Flows and Foliations

We recall the basic notions on vector fields that are used in this book. Let X be a smooth manifold. Given a differentiable path $\phi\colon I \to X$ from an interval I in the t-axis $\mathbb{R}$, we denote its velocity vector at time t by $\dot\phi(t) = (\mathrm{d}\phi)_t \cdot \frac{\partial}{\partial t} \in \mathrm{T}_{\phi(t)}X$. An *integral curve* of a vector field V on X is a path $\phi\colon I \to X$ such that $\dot\phi(t) = V_{\phi(t)}$ for all $t \in I$. (We also write $V_x = V(x)$ when there is no danger of confusing the latter notation with the value of the vector field V on a function x.) The initial value problem

$$\dot x = V_x, \qquad x(0) = x^0 \tag{1.42}$$

asks for an integral curve which passes through the point x^0 at time $t = 0$. In local coordinates $x = (x_1, \dots, x_n)$ on X, with $V_x = \sum_{j=1}^n a_j(x)\frac{\partial}{\partial x_j}$ and $x^0 = (x_1^0, \dots, x_n^0)$, the flow equation (1.42) is equivalent to the system of autonomous ordinary differential equations

$$\dot{x}_j = a_j(x_1, \dots, x_n), \quad x_j(0) = x_j^0, \qquad j = 1, \dots, n.$$

If V is Lipschitz continuous then for every point $p \in X$ there exist a neighborhood $U \subset X$ of p and a number $t_0 > 0$ such that (1.42) has a unique solution $x(t, x^0) = \phi_t(x^0)$ for every $x^0 \in U$ and every $|t| < t_0$. This solution and its t-derivative are continuous in (t, x^0). The map $t \to \phi_t(x)$ is called the (local) *flow* of V. If V is of class $\mathcal{C}^k$ for some $k \geq 1$ then $\phi_t(x)$ and $\dot{\phi}_t(x)$ are also of class $\mathcal{C}^k$ in (t, x). (See e.g. [5, §4.1].) For every fixed $t \in \mathbb{R}$ the map ϕ_t is a diffeomorphism of its domain $\Omega_t \subset X$ onto $\phi_t(\Omega_t) \subset X$, called the *time-t map*. These maps satisfy the group law

$$\phi_t \circ \phi_s = \phi_{t+s}, \quad t, s \in \mathbb{R}$$

on the set in X where both sides are defined.

By uniqueness of trajectories the flow $t \mapsto \phi_t(x)$ extend to the maximal interval $I_x = (\alpha(x), \omega(x)) \subset \mathbb{R}$ around the origin. The functions $-\alpha, \omega \colon X \to (0, +\infty]$ are lower semicontinuous. If $\omega(x) < +\infty$ for some $x \in X$ then the point $\phi_t(x)$ leaves every compact set in X as $t \nearrow \omega(x)$. Similarly, if $\alpha(x) > -\infty$ then the point $\phi_t(x)$ leaves every compact set in X as $t \searrow \alpha(x)$. The open set

$$\Omega = \{(t, x) \in \mathbb{R} \times X \colon \alpha(x) < t < \omega(x)\} \subset \mathbb{R} \times X \tag{1.43}$$

is the *fundamental domain* of the vector field V.

Definition 1.9.1. *A vector field V is* $\mathbb{R}$-complete *(or* complete in real time*) if $\Omega = \mathbb{R} \times X$; that is, if the flow $\phi_t(x)$ exists for all $t \in \mathbb{R}$ and $x \in X$.*

Every vector field on a compact manifold without boundary is complete. The flow $\{\phi_t\}_{t \in \mathbb{R}}$ of an $\mathbb{R}$-complete vector field is a real one-parameter subgroup of the diffeomorphism group $\mathrm{Diff}\, X$ of X, representing an action of the group $(\mathbb{R}, +)$ on X by diffeomorphisms. Conversely, every action $\phi \colon \mathbb{R} \times X \to X$ of $(\mathbb{R}, +)$ by diffeomorphisms of X is the flow of the vector field $V_x = \frac{\partial}{\partial t}\phi(t, x)|_{t=0}$, called the *infinitesimal generator* of ϕ.

A diffeomorphism $f \colon X \to Y$ pushes a vector field V on X to the vector field $W = f_* V$ on Y (explicitly, $W_{f(x)} = \mathrm{d}f_x V_x$); a vector field W on Y pulls back to the vector field $V = f^* W = (f^{-1})_* W$ on X. The chain rule shows that if $\phi_t(x)$ is the flow of V then $\psi_t(y) = f \circ \phi_t(f^{-1}(y))$ is the flow of W.

More generally, if $f \colon X \to Y$ is a smooth map, V is a vector field on X, and W is a vector fields on Y such that $\mathrm{d}f_x V_x = W_{f(x)}$ holds for all $x \in X$ (in such case we write $V = f^* W$), then f conjugates the flow ϕ_t of V to the flow ψ_t of W: $f(\phi_t(x)) = \psi_t(f(x))$. In particular, if V is tangential to a submanifold $M \subset X$ then for each $x \in M$ the flow $\phi_t(x)$ remains in M.

Lemma 1.9.2. *Let V be a $\mathcal{C}^1$ vector field on an n-dimensional manifold X. If $V_p \neq 0$ for some $p \in X$ then there exist local coordinates $x = (x_1, \dots, x_n)$ in a neighborhood of p such that $V = \frac{\partial}{\partial x_1}$.*

Proof. A suitable choice of local coordinates ensures that $p = 0 \in \mathbb{R}^n$ and that $V_0 = \frac{\partial}{\partial x_1}\big|_0$. Let ϕ_t denote the flow of V. Consider the map

$$g(x_1, x_2, \dots, x_n) = \phi_{x_1}(0, x_2, \dots, x_n).$$

It is immediate that $g_* \frac{\partial}{\partial x_1} = V$ and that $\mathrm{d}g_0$ is the identity map. Hence g is a diffeomorphism near the origin. □

Let $\mathbb{R} \times X$ be the extended phase space with the projection $\pi\colon \mathbb{R} \times X \to X$. The tangent bundle $\mathrm{T}X$ pulls back to a bundle on $\mathbb{R} \times X$, still denoted $\mathrm{T}X$. A *time dependent* (non-autonomous) vector field on a domain $\Omega \subset \mathbb{R} \times X$ is a section V of $\mathrm{T}X|_\Omega$. For a fixed t, V_t is a vector field on the domain

$$\Omega_t = \{x \in X\colon (t, x) \in \Omega\}. \tag{1.44}$$

In local coordinates $x = (x_1, \dots, x_n)$ on an open set $U \subset X$ we have $V_t(x) = \sum_{j=1}^n v_j(t, x) \frac{\partial}{\partial x_j}$, where the coefficient functions $v_j(t, x)$ are defined for $(t, x) \in \Omega \cap \pi^{-1}(U)$. Fix a point $x \in \Omega_s$ for some $s \in \mathbb{R}$. If V is continuous in (t, x) and Lipschitz continuous in x then the equation

$$\frac{\partial}{\partial t} \phi_{t,s}(x) = V_t(\phi_{t,s}(x)), \quad \phi_{s,s}(x) = x \tag{1.45}$$

has a unique local solution. The flow $\phi_{t,s}(x)$ is defined as long as the trajectory remains in the domain of the vector field. We have the semigroup property

$$\phi_{t,u} \circ \phi_{u,s} = \phi_{t,s}, \quad \phi_{s,t} = \phi_{t,s}^{-1}.$$

The *Grönwall inequality* says that if nonnegative continuous functions $f, g\colon [a, b) \to \mathbb{R}_+$ satisfy the inequality $f(t) \leq A + \int_a^t f(\tau) g(\tau) \mathrm{d}\tau$ for some $A \geq 0$, then $f(t) \leq A \exp(\int_a^t g(\tau) \mathrm{d}\tau))$ for all $t \in [a, b)$ [5, Lemma 4.1.8]. This inequality gives the following estimate on the distance between trajectories.

Lemma 1.9.3. *Let V be a time dependent continuous vector field on a domain $\Omega \subset \mathbb{R}^{1+n}$ satisfying a uniform Lipschitz estimate*

$$\big|V_t(x) - V_t(y)\big| \leq B|x - y|. \tag{1.46}$$

Then for any $s \in \mathbb{R}$ and any pair of points $x, y \in \Omega_s$ (1.44) we have

$$|\phi_{t,s}(x) - \phi_{t,s}(y)| \leq \mathrm{e}^{B|t-s|}|x - y| \tag{1.47}$$

for all t such that the trajectories exist and remain in the domain Ω_t (1.44).

Proof. Let $f(t) = |\phi_{t,s}(x) - \phi_{t,s}(y)|$. Assume first that $t \geq s$. By integrating the equation (1.45) we get

$$\begin{aligned} f(t) &= \left| x + \int_s^t V_\tau(\phi_{\tau,s}(x))\,\mathrm{d}\tau - y - \int_s^t V_\tau(\phi_{\tau,s}(y))\,\mathrm{d}\tau \right| \\ &\leq |x-y| + \int_s^t \left| V_\tau(\phi_{\tau,s}(x)) - V_\tau(\phi_{t,\tau(y)}) \right| \mathrm{d}\tau \\ &\leq |x-y| + B\int_s^t |\phi_{\tau,s}(x) - \phi_{\tau,s}(y)|\,\mathrm{d}\tau \\ &= |x-y| + B\int_s^t f(\tau)\,\mathrm{d}\tau. \end{aligned}$$

Since $f(s) = |x-y|$, Grönwall's inequality implies $f(t) \leq \mathrm{e}^{B|t-s|}|x-y|$ which gives (1.47). The estimate for $t < s$ is obtained by replacing V by $-V$. $\square$

Assume now that $\Omega_0 \neq \emptyset$. Fix a compact subset $K \subset \Omega_0$ and let $t_0 > 0$ be such that the flow $\phi_t(x) = \phi_{t,0}(x)$ exists and remains in Ω_t when $x \in K$ and $t \in [0, t_0]$. Set $K_t = \phi_t(K) \subset \Omega_t$. For any $\epsilon > 0$ we let

$$K(\epsilon) = \{x \in \mathbb{R}^n\colon \operatorname{dist}(x, K) = \inf_{y \in K} |x-y| < \epsilon\},$$

$$S(\epsilon) = \{(t,x) \in \mathbb{R} \times \mathbb{R}^n \colon 0 \leq t \leq t_0,\ \operatorname{dist}(x, K_t) < \epsilon\}.$$

Set $\eta_0 = (1+t_0)\mathrm{e}^{Bt_0} > 1$, where B is the Lipschitz constant (1.46). Choose $\epsilon_0 > 0$ sufficiently small such that $S(\epsilon_0\eta_0) \Subset \Omega$.

Lemma 1.9.4. (Notation as above.) *Assume that for some $\epsilon \in (0, \epsilon_0)$ we have a continuous map $V^\epsilon\colon \Omega \to \mathbb{R}^n$ (a time-dependent vector field) satisfying*

$$||V - V^\epsilon||_{L^\infty(S(\epsilon\eta_0))} \leq \epsilon.$$

Then the flow $\phi_t^\epsilon(x)$ of V_t^ϵ with $\phi_0^\epsilon(x) = x$ exists for all $x \in K(\epsilon)$ and for all $t \in [0, t_0]$, and it satisfies the estimate

$$||\phi_t - \phi_t^\epsilon||_{L^\infty(K(\epsilon))} \leq t_0 \mathrm{e}^{Bt} ||V - V^\epsilon||_{L^\infty(S(\epsilon\eta_0))}, \quad 0 \leq t \leq t_0. \tag{1.48}$$

Proof. Let $A(\epsilon) = ||V - V^\epsilon||_{L^\infty(S(\epsilon\eta_0))}$. Fix a point $x \in K(\epsilon)$ and set $f(t) = \left|\phi_t(x) - \phi_t^\epsilon(x)\right|$ for $t \geq 0$. We have $f(0) = 0$ and

$$\begin{aligned} f(t) &= \left| \int_0^t (V_s(\phi_s(x)) - V_s^\epsilon(\phi_s^\epsilon(x))\,\mathrm{d}s \right| \leq \int_0^t \left| V_s(\phi_s(x)) - V_s^\epsilon(\phi_s^\epsilon(x)) \right| \mathrm{d}s \\ &\leq \int_0^t \left| V_s(\phi_s(x)) - V_s(\phi_s^\epsilon(x)) \right| \mathrm{d}s + \int_0^t \left| V_s(\phi_s^\epsilon(x)) - V_s^\epsilon(\phi_s^\epsilon(x)) \right| \mathrm{d}s \\ &\leq B\int_0^t f(s)\,\mathrm{d}s + t_0 A(\epsilon). \end{aligned}$$

The last inequality holds for all $t \in [0, t_0]$ such that $\phi_t^\epsilon(x) \in K_t(\epsilon\eta_0)$. For such values of t, Grönwall's inequality implies $f(t) \le A(\epsilon)t_0 \mathrm{e}^{Bt}$ which gives (1.48). For $x \in K(\epsilon)$, Lemma 1.9.3 also gives $\phi_t(x) \in K_t(\epsilon \mathrm{e}^{Bt})$. Hence

$$\begin{aligned}\operatorname{dist}\big(\phi_t^\epsilon(x), K_t\big) &\le |\phi_t^\epsilon(x) - \phi_t(x)| + \operatorname{dist}(\phi_t(x), K_t)\\ &\le A(\epsilon)t_0\mathrm{e}^{Bt} + \epsilon \mathrm{e}^{Bt} \le \epsilon(t_0+1)\mathrm{e}^{Bt_0} = \epsilon\eta_0.\end{aligned}$$

We used the assumption $A(\epsilon) \le \epsilon$ and the definition of η_0. This shows that $\phi_t^\epsilon(x) \in K_t(\epsilon\eta_0)$ for all $t \in [0, t_0]$, so the estimate (1.48) holds. □

The k-th power of a smooth autonomous vector field V is a differential operator of order k defined inductively by $V^k(f) = V(V^{k-1}(f))$, where f is a smooth function. Let $\phi_t(x)$ be the flow of V. By differentiating the function $t \mapsto f(\phi_t(x))$ for a fixed $x \in X$ we get $\frac{\mathrm{d}^k}{\mathrm{d}t^k} f(\phi_t(x)) = V^k(f)(\phi_t(x))$. This gives the formal Taylor expansion of $t \mapsto f(\phi_t(x))$, also called the *Lie series*:

$$f(\phi_t(x)) = \sum_{k\ge 0} \frac{1}{k!} t^k V^k(f)(x) = f(x) + tV(f)(x) + \frac{1}{2}t^2 V^2(f)(x) + \cdots. \tag{1.49}$$

Let $x = (x_1, \ldots, x_n)$ be a coordinate map and $V_x = \sum_{j=1}^n a_j(x)\frac{\partial}{\partial x_j}$. By applying V to the j-th component x_j of x we get $V(x_j) = a_j$ (the j-th coefficient function of V). Thus $V(x_1, \ldots, x_n) := (V(x_1), \ldots, V(x_n)) = (a_1, \ldots, a_n)$. Using (1.49) with the identity map $f(x) = x$ we obtain the Lie series

$$\phi_t(x) = x + tV(x) + \frac{1}{2}t^2V^2(x) + \cdots + \frac{1}{k!}t^kV^k(x) + \cdots. \tag{1.50}$$

Note that the j-th component function of $V^k(x)$ equals $V^{k-1}(a_j)$.

Definition 1.9.5. *Let ϕ_t be the flow of a vector field V. The* Lie derivative *$L_V W$ of a tensor field W is defined by*

$$L_V W = \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} \phi_t^* W. \tag{1.51}$$

The Lie derivative of a function f equals $L_V f = V(f) = \mathrm{d}f \cdot V$. Suppose now that W is a vector field. Writing $s = t + u$ we have $\phi_s^* W = \phi_t^*(\phi_u^* W)$ by the group property of the flow. Differentiating on s at $s = t$ gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\phi_t^* W = \frac{\mathrm{d}}{\mathrm{d}u}\bigg|_{u=0} \phi_t^*(\phi_u^* W) = \phi_t^*\left(\frac{\mathrm{d}}{\mathrm{d}u}\bigg|_{u=0} \phi_u^* W\right) = \phi_t^*(L_V W). \tag{1.52}$$

Hence $L_V W = 0$ implies $\phi_t^* W = W$ for all t. If $f\colon X \to Y$ is a smooth map and $\widetilde{V}$, $\widetilde{W}$ are vector fields on Y such that $V = f^*\widetilde{V}$ and $W = f^*\widetilde{W}$, then

$$f^*\left(L_{\widetilde{V}}\widetilde{W}\right) = L_V W. \tag{1.53}$$

To see this, let $\widetilde{\phi}_t$ denote the flow of $\widetilde{V}$. Then $f \circ \phi_t = \widetilde{\phi}_t \circ f$ and hence

$$\phi_t^* W = \phi_t^* f^* \widetilde{W} = (f \circ \phi_t)^* \widetilde{W} = (\widetilde{\phi}_t \circ f)^* \widetilde{W} = f^* \left(\widetilde{\phi}_t^* \widetilde{W} \right).$$

Differentiation at $t = 0$ gives (1.53). If f is a diffeomorphism then (1.53) implies $f_* (L_V W) = L_{f_* V} f_* W$, so the Lie derivative can be calculated in any coordinate system. It is now easy to show that

$$L_V W = [V, W] = VW - WV,$$

where the commutator $[V, W]$ is defined by (1.24) on p. 20. Indeed, at a point $p \in X$ where $V_p \neq 0$ Lemma 1.9.2 gives local coordinates $x = (x_1, \ldots, x_n)$ such that $V = \frac{\partial}{\partial x_1}$. Its flow is $\phi_t(x_1, \ldots, x_n) = (x_1 + t, x_2, \ldots, x_n)$. If $W = \sum_{j=1}^n b_j(x) \frac{\partial}{\partial x_j}$, a trivial calculation gives

$$L_V W = \sum_{j=1}^n \frac{\partial b_j}{\partial x_1}(x) \frac{\partial}{\partial x_j} = [V, W].$$

If V equals zero in some neighborhood of p then $L_V W = 0 = [V, W]$ near p. The equality at all points then follows by the continuity of both fields.

Lemma 1.9.6. *If V, W are vector fields with flows ϕ_t, ψ_t, then $[V, W] = 0$ if and only if $\phi_t \circ \psi_s = \psi_s \circ \phi_t$ holds on the domain of the compositions.*

Proof. Suppose that $\phi_t \circ \psi_s = \psi_s \circ \phi_t$. Differentiating on s at $s = 0$ gives $(\phi_t)_* W = W \circ \phi_t$ which is equivalent to $\phi_t^* W = W$. Differentiation at $t = 0$ gives $L_V W = 0$. Conversely, if $L_V W = 0$ then $(\phi_t)_* W = W \circ \phi_t$ by (1.52). Consider the path $\mathbb{R} \ni s \mapsto \gamma(s) = \phi_t(\psi_s(x))$. Then

$$\frac{\mathrm{d}\gamma}{\mathrm{d}s}(s) = (\phi_t)_* \frac{\mathrm{d}}{\mathrm{d}s} \psi_s(x) = (\phi_t)_* W(\psi_s(x)) = W(\phi_t \circ \psi_s(x)) = W(\theta(s)),$$

so γ is an integral curve of W. Also, $\sigma(s) = \psi_s \circ \phi_t(x)$ is obviously an integral curve of W. Since $\gamma(0) = \phi_t(x) = \sigma(0)$, the uniqueness of integral curves implies $\gamma(s) = \sigma(s)$. This means that $\phi_t \circ \psi_s = \psi_s \circ \phi_t$. $\square$

Lemma 1.9.6 can also be seen as follows. Denote by $a(x), b(x) \in \mathbb{R}^n$ the coefficient vectors of V, W, respectively. Using the Lie series (1.50) we get

$$\begin{aligned} \psi_s(\phi_t(x)) &= x + ta(x) + sb(x) + stV(b)(x) + O(s^2, t^2), \\ \phi_t(\psi_s(x)) &= x + ta(x) + sb(x) + stW(a)(x) + O(s^2, t^2), \end{aligned}$$

$$\frac{\partial^2}{\partial s \, \partial t}\bigg|_{s=t=0} \left[\psi_s(\phi_t(x)) - \phi_t(\psi_s(x)) \right] = V(b)(x) - W(a)(x) = [V, W](x).$$

If the flows ϕ_t and ψ_s commute then clearly $[V, W] = 0$. The converse is shown in the same way as above by using uniqueness of integral curves.

A similar calculation, using the Lie series (1.50) of the flow, implies

Lemma 1.9.7. *If V and W are vector fields with flows ψ_t, ψ_t, resp., then*

$$[V,W]_x = \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} \psi_{-\sqrt{t}} \circ \phi_{-\sqrt{t}} \circ \psi_{\sqrt{t}} \circ \phi_{\sqrt{t}}(x). \tag{1.54}$$

Lemma 1.9.6 gives a simple proof of the classical Frobenius theorem. Let X be a smooth manifold of dimension n, and let E be a smooth vector subbundle of $\mathrm{T}X$ of rank m and corank $d = n-m$. A vector field V on an open set $U \subset X$ is *tangential to* E if $V_x \in E_x$ for all $x \in U$. The subbundle E of $\mathrm{T}X$ is said to be *involutive* if for any pair of local vector fields V, W that are tangential to E, their commutator $[V,W]$ is also tangential to E.

A *nonsingular smooth foliation* $\mathcal{F}$ of dimension $m \in \{1,\dots,n\}$ on X is a subdivision of X into a disjoint union $X = \cup_{\alpha\in A} F_\alpha$ of m dimensional, locally closed, connected smooth submanifolds $F_\alpha \subset X$, called the *leaves* of $\mathcal{F}$, such that every point $x^0 \in X$ has an open neighborhood $U \subset X$ and a diffeomorphism $f = (f', f'')\colon U \to D^m \times D^{n-m} \subset \mathbb{R}^n$ such that for every $\alpha \in A$, the intersection $F_\alpha \cap U$ is a union of plaques $f'' = const$. It is evident that the tangent bundle to a foliation is an involutive subbundle of $\mathrm{T}X$.

Theorem 1.9.8. (Frobenius) *Every involutive rank m smooth subbundle E of $\mathrm{T}X$ is the tangent bundle of a smooth foliation of X.*

Proof. We may work in local coordinates around $0 \in \mathbb{R}^n$. Assume that $E_0 = \mathbb{R}^m \times \{0\}^d$. Choose vector fields $V_j = \sum_{k=1}^n a_{j,k}(x)\frac{\partial}{\partial x_k}$ $(j = 1,\dots,m)$ spanning E near 0. We may assume that the $m \times m$ matrix $A(x) = (a_{j,k}(x))_{j,k=1}^m$ is invertible for x near 0. Replacing the V_j's by their linear combinations obtained by applying the inverse matrix $A(x)^{-1}$ we get a new system of spanning vector fields of the simpler form

$$V_j = \frac{\partial}{\partial x_j} + \sum_{k=m+1}^n a_{j,k}(x)\frac{\partial}{\partial x_k}, \quad j = 1,\dots,m.$$

Assuming that E is involutive, each commutator $[V_j, V_l]$ is a linear combination of the vector fields $V_1,\dots,V_m$. On the other hand, a direct computation shows that $[V_j, V_l]$ is a linear combination of $\frac{\partial}{\partial x_{m+1}},\dots,\frac{\partial}{\partial x_n}$. It follows that $[V_j, V_l] = 0$ for all $j, l = 1,\dots,m$. Hence the flows $\phi_{j,t}$ of V_j commute by Lemma 1.9.6. The map

$$g(x_1,\dots,x_n) = \phi_{1,x_1} \circ \cdots \circ \phi_{m,x_m}(\underbrace{0,\dots,0}_{m}, x_{m+1},\dots,x_n)$$

is then a diffeomorphism near $0 \in \mathbb{R}^n$ that satisfies $g_*\frac{\partial}{\partial x_j} = V_j$ for $j = 1,\dots,m$. For $j = 1$ this is immediate from the definition of g and of the flow. For other values of j we use commutativity of the flows to bring ϕ_{j,x_j} to the left-most place in the definition of g, and hence the same argument applies. This shows that the foliation with leaves $x_{m+1} = c_{m+1},\dots,x_n = c_n$ is mapped by g to a foliation whose leaves are tangential to E. $\square$

Given a vector field V and a differential r-form α on a manifold X, we denote by $V\rfloor\alpha$ the *contraction* (or the *interior product*) of α by V. This is an $(r-1)$-form defined at a point $x \in X$ by

$$\langle V\rfloor\alpha, V_1 \wedge \cdots V_{r-1}\rangle = \langle \alpha, V_x \wedge V_1 \wedge \cdots V_{r-1}\rangle, \quad V_1, \ldots, V_{r-1} \in \mathrm{T}_x X. \tag{1.55}$$

Some texts use the notation $V\rfloor\alpha = i_V\alpha$. It is easily seen that $i_V = V\rfloor\cdot$ is an antiderivation with respect to $\wedge$. The properties relating the interior product i_V, the Lie derivative L_V, and the exterior derivative d, are summarized by [5, Theorem 6.4.8.]. In particular, we have the Cartan formulas:

$$V\rfloor \mathrm{d}f = V(f), \quad L_V\alpha = V\rfloor\mathrm{d}\alpha + \mathrm{d}(V\rfloor\alpha), \quad \mathrm{d}(L_V\alpha) = L_V\mathrm{d}\alpha. \tag{1.56}$$

Assume now that V is a holomorphic vector field on a complex manifold X (Def. 1.6.1 on p. 19). Let $W = \frac{1}{2}(V - \mathrm{i}JV)$ be the associated holomorphic section of $\mathrm{T}^{1,0}X$. The flow equation (1.42) (p. 30) can now be considered for a complex time variable $\zeta = t + \mathrm{i}s$. In local holomorphic coordinates $z = (z_1, \ldots, z_n)$ with $z_j = x_j + \mathrm{i}y_j$ we have $V = \sum_{j=1}^n a_j \frac{\partial}{\partial x_j} + b_j \frac{\partial}{\partial y_j}$ and $W = \sum_{j=1}^n \alpha_j \frac{\partial}{\partial z_j}$ with $\alpha_j = a_j + \mathrm{i}b_j$. We are looking for holomorphic functions $z_j = z_j(\zeta)$ of a complex variable $\zeta = t + \mathrm{i}s \in \mathbb{C}$ satisfying

$$\frac{\mathrm{d}z_j}{\mathrm{d}\zeta}(\zeta) = \alpha_j\big(z_1(\zeta), \ldots, z_n(\zeta)\big), \quad z_j(0) = z_j^0, \qquad j = 1, \ldots, n. \tag{1.57}$$

This system has a unique local solution $\zeta \mapsto \phi_\zeta(z^0)$ that is holomorphic in all variables (ζ, z^0). Write $\zeta = t + \mathrm{i}s$ with $t, s \in \mathbb{R}$. By the Cauchy-Riemann equations, (1.57) is equivalent to the system of $2n$ real equations

$$\frac{\partial x}{\partial t} = V(x), \quad \frac{\partial x}{\partial s} = JV(x), \qquad x(0) = x^0.$$

In fact, $[V, JV] = [W + \overline{W}, \mathrm{i}W - \mathrm{i}\overline{W}] = -2\mathrm{i}[W, \overline{W}] = 0$ by (1.26) (p. 20); hence the flows ϕ_t, ψ_t commute, and the flow in complex time is given by

$$\phi_{t+\mathrm{i}s}(x) = \phi_t \circ \psi_s(x) = \psi_s \circ \phi_t(x). \tag{1.58}$$

Definition 1.9.9. *A holomorphic vector field V on a complex manifold X is* $\mathbb{C}$-complete *(or* completely integrable*) if its flow $\phi_{t+\mathrm{i}s}(x)$ exists for all $x \in X$ and $t + \mathrm{i}s \in \mathbb{C}$.*

A holomorphic vector field V is $\mathbb{C}$-complete if and only if V and JV are both $\mathbb{R}$-complete. The flow of an $\mathbb{R}$-complete (resp. $\mathbb{C}$-complete) holomorphic vector field is an action of the additive group $(\mathbb{R}, +)$ (resp. of $(\mathbb{C}, +)$) on X by holomorphic automorphisms, and every action is of this form. Two such actions $\phi, \psi\colon \mathbb{R} \times X \to X$ are *holomorphically equivalent* if there exists a holomorphic automorphism $f \in \operatorname{Aut} X$ such that $f(\phi(t, x)) = \psi(t, f(x))$ for all $t \in \mathbb{R}$ (resp. $t \in \mathbb{C}$). Letting V, W denote the infinitesimal generator of ϕ, ψ, respectively, this holds if and only if $\mathrm{d}f_x \cdot V_x = W_{f(x)}$.

Remark 1.9.10. If X is a connected Stein manifold without nonconstant bounded plurisubharmonic functions (a complex manifold with the latter property is called *Liouville*) then every $\mathbb{R}$-complete holomorphic vector field on X is also $\mathbb{C}$-complete [167, Corollary 2.2]. This holds in particular when $X = \mathbb{C}^n$, or more generally when X is a Stein Oka manifold (Def. 5.4.1 on p. 192). Furthermore, if X is as above, then a holomorphic vector field on X that is complete in positive real time is also $\mathbb{C}$-complete [8]. $\square$

A *holomorphic volume form* on an n-dimensional complex manifold X is a nowhere vanishing holomorphic n-form, i.e., a nonzero section of the *canonical bundle* $K_X = \wedge^n \mathrm{T}^{*1,0}X$. In local holomorphic coordinates such a form equals $\omega = a(z)\mathrm{d}z_1 \wedge \cdots \wedge \mathrm{d}z_n$ where a is a nonvanishing holomorphic function. A holomorphic map $f\colon D \to X$ on a domain $D \subset X$ is *volume preserving* if $f^*\omega = \omega$. For the standard volume form $\omega = \mathrm{d}z_1 \wedge \cdots \wedge \mathrm{d}z_n$ on $\mathbb{C}^n$ we get $f^*\omega = \mathrm{J}f \cdot \omega$, where $\mathrm{J}f = \det f'$ is the complex Jacobian of f.

The *divergence* of a holomorphic vector field V with respect to ω is the holomorphic function $\mathrm{div}_\omega V$ satisfying the equation

$$L_V\omega = \mathrm{d}(V\rfloor\omega) = \mathrm{div}_\omega V \cdot \omega, \tag{1.59}$$

where $L_V\omega$ is the Lie derivative of ω (p. 34) and $V\rfloor\omega$ is the contraction of ω by V (p. 37). The first equality follows from Cartan's formula (1.56). For the standard volume form $\omega = \mathrm{d}z_1 \wedge \cdots \wedge \mathrm{d}z_n$ on $\mathbb{C}^n$ we get

$$\mathrm{div}_\omega\left(\sum_{j=1}^n a_j(z)\frac{\partial}{\partial z_j}\right) = \sum_{j=1}^n \frac{\partial a_j}{\partial z_j}(z).$$

Assume now that ϕ_t is the local flow of a holomorphic vector field V and ω is a holomorphic volume form. Cartan's formula (1.56) gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\,(\phi_t^*\omega)(z) = \mathrm{d}(V\rfloor\omega)(\phi_t(z)) = (\mathrm{div}_\omega V)(\phi_t(z))\,\omega.$$

Since $\phi_t^*\omega = (\mathrm{J}\phi_t)\,\omega$ and $\phi_0 = \mathrm{Id}$, we conclude that

$$\mathrm{J}\phi_t \equiv 1 \iff \mathrm{div}_\omega V = 0; \tag{1.60}$$

that is, divergence zero vector fields correspond to volume preserving flows. The analogous result holds for time dependent holomorphic vector fields.

Let X be a complex manifold of dimension n. A *nonsingular holomorphic foliation* $\mathcal{F}$ of dimension $m \in \{1, \ldots, n\}$ on X is a subdivision of X into a disjoint union $X = \cup_{\alpha\in A} F_\alpha$ of m dimensional, locally closed, connected complex submanifolds $F_\alpha \subset X$, called the *leaves* of $\mathcal{F}$, such that every point $x^0 \in X$ has an open neighborhood $U \subset X$ and a biholomorphic map $f = (f', f'')\colon U \to \mathbb{D}^m \times \mathbb{D}^{n-m} \subset \mathbb{C}^n$ such that for every $\alpha \in A$, the intersection $F_\alpha \cap U$ is a union of plaques $f'' = const$. Frobenius' theorem (Theorem 1.9.8 on p. 36) is valid also in the holomorphic case.

1.10 Jet Bundles, Holonomic Sections and the Homotopy Principle

We recall the notion of jet spaces of smooth and holomorphic mappings and discuss a simple case of Gromov's h-principle that will be used in the book. For a more complete introduction to h-principle we refer to Gromov's paper [235] and the monographs [236, 135].

The discussion can be done either in the setting of smooth real manifolds and smooth mappings, or in the setting of complex manifolds and holomorphic mappings. We begin with the former. Let $f = (f_1, \dots, f_m): U \to \mathbb{R}^m$ be a smooth map in an open neighborhood $U \subset \mathbb{R}^n$ of a point $x \in \mathbb{R}^n$. The k-jet $\mathrm{j}^k_x f$ of f at x is determined by the Taylor polynomial of f of order k at x; that is, by the partial derivatives

$$c_{j,\alpha} = \frac{\partial^{|\alpha|} f_j}{\partial x_1^{\alpha_1} \cdots \partial x_n^{\alpha_n}}(x), \quad 0 \le |\alpha| \le k,\ j = 1, \dots, m.$$

The manifold of all k-jets $\mathrm{J}^k(\mathbb{R}^n, \mathbb{R}^m)$ is a Euclidean space with the coordinates $x = (x_1, \dots, x_n)$ and $c_{j,\alpha} \in \mathbb{R}$ for all $|\alpha| \le k,\ j = 1, \dots, m$. In particular, the 0-jet is $\mathrm{j}^0_x f = (x, f(x))$, and $\mathrm{J}^0(\mathbb{R}^n, \mathbb{R}^m) = \mathbb{R}^n \times \mathbb{R}^m$. The 1-jet of f at x is $\mathrm{j}^k_x f = (x, f(x), \mathrm{d}f_x)$, and the space of 1-jets

$$\mathrm{J}^1(\mathbb{R}^n, \mathbb{R}^m) = \big\{(x, y, \lambda): x \in \mathbb{R}^n,\ y \in \mathbb{R}^m,\ \lambda \in \mathrm{Hom}(\mathrm{T}_x\mathbb{R}^n, \mathrm{T}_y\mathbb{R}^m)\big\}$$

is isomorphic to $\mathbb{R}^{n+m+nm}$.

Assume now that X and Y are smooth manifolds. For a fixed pair of points $x \in X$, $y \in Y$ we consider smooth maps $f: U \to Y$ in neighborhoods of x such that $f(x) = y$. Choosing local coordinates around x and y brings us back to the Euclidean case. Clearly the maps f and g have the same k-jet at $x \in X$ if and only if their difference $f - g$ vanishes to order k at x in some (and hence in any) pair of local coordinates. We denote the space of k-jets of maps $X \to Y$ by $\mathrm{J}^k(X, Y)$; thus $\mathrm{J}^0(X, Y) = X \times Y$ and

$$\mathrm{J}^1(X, Y) = \big\{(x, y, \lambda): x \in X,\ y \in Y,\ \lambda \in \mathrm{Hom}(\mathrm{T}_x X, \mathrm{T}_y Y)\big\}.$$

If X and Y are complex manifolds of dimension n, m, respectively, then we consider the complex (holomorphic) derivatives with respect to the local holomorphic coordinates $z = (z_1, \dots, z_n)$ on X:

$$c_{j,\alpha} = \frac{\partial^{|\alpha|} f_j}{\partial z_1^{\alpha_1} \cdots \partial z_n^{\alpha_n}}(z), \quad |\alpha| \le k,\ j = 1, \dots, m.$$

In this case we denote by $\mathrm{J}^k(X, Y) = \mathrm{J}^k_{\mathbb{C}}(X, Y)$ the manifold of complex k-jets and by $\mathrm{J}^k_{\mathbb{R}}(X, Y)$ the manifold of real k-jets.

The chain rule shows that $\mathrm{J}^k(X, Y)$ is a smooth manifold, resp. a complex manifold in the holomorphic case. Every smooth (resp. holomorphic) map

$f\colon X \to Y$ determines the k-jet map $\mathrm{j}^k f\colon X \to \mathrm{J}^k(X,Y)$, $x \mapsto \mathrm{j}^k_x f$. We have natural source point and image point projections

$$p^k\colon \mathrm{J}^k(X,Y) \to X, \quad q^k\colon \mathrm{J}^k(X,Y) \to Y.$$

We denote by $\mathrm{J}^k_x(X,Y) = (p^k)^{-1}(x)$ the set of all k-jets with the source point x, and by $\mathrm{J}^k_{x,y}(X,Y)$ the set of all k-jets with the source x and the target y.

For every $k \in \mathbb{N}$ we have the projection

$$\tau_k\colon \mathrm{J}^k(X,Y) \to \mathrm{J}^{k-1}(X,Y) \tag{1.61}$$

which forgets the partial derivatives of pure order k. The chain rule shows that (1.61) carries a natural structure of an affine bundle; this no longer holds for the composition of two or more of these projections.

Similarly, if $\pi\colon Z \to X$ is a smooth or a holomorphic submersion, we let $Z^{(k)}$ denote the manifold of all k-jets of smooth (resp. holomorphic) sections $f\colon X \to Z$, $\pi \circ f = \mathrm{Id}_X$. (It will always be clear from the context whether we are in the real or in the complex case.) These have similar properties as the jet manifolds $\mathrm{J}^k(X,Y)$. In particular, the natural 'forgetful' projection $\tau_k\colon Z^{(k)} \to Z^{(k-1)}$ is an affine smooth (resp. holomorphic) bundle.

Definition 1.10.1. *A map $g\colon X \to \mathrm{J}^k(X,Y)$ such that $g(x) \in \mathrm{J}^k_x(X,Y)$ (i.e., a section of the projection $p^k\colon \mathrm{J}^k_x(X,Y) \to X$) is said to be* holonomic *if there exists a smooth (resp. holomorphic) map $f\colon X \to Y$ such that $g(x) = \mathrm{j}^k_x f$ for all $x \in X$. The analogous definition applies to sections $X \to Z^{(k)}$ when $\pi\colon Z \to X$ is a submersion.*

Clearly every section $g\colon X \to \mathrm{J}^0(X,Y)$ is of the form $g(x) = (x, f(x))$ for some map $f\colon X \to Y$, and hence g is holonomic. However, for every $k \geq 1$, being holonomic is a nontrivial condition. A section $g\colon X \to \mathrm{J}^1(X,Y)$ is of the form $g(x) = (x, f(x), \lambda(x))$ with $\lambda(x)\colon \mathrm{T}_x X \to \mathrm{T}_{f(x)} Y$ an arbitrary linear map; g is holonomic if $\lambda(x) = \mathrm{d}f_x$.

The following notions are due to Gromov [236]. They apply both in the smooth and in the holomorphic category.

Definition 1.10.2. *A* differential relation of order $k \in \mathbb{N}$ *for maps $X \to Y$ (resp. for section $X \to Z$ of a submersion $\pi\colon Z \to X$) is a subset $\Omega \subset \mathrm{J}^k(X,Y)$ (resp. $\Omega \subset Z^{(k)}$). A map $f\colon X \to Y$ is* solution *of Ω over a subset $V \subset X$ if the k-jet extension $\mathrm{j}^k f\colon X \to \mathrm{J}^k(X,Y)$ maps V to Ω.*

Recall that a continuous map $\phi\colon X \to Y$ is said to be a *weak homotopy equivalence* if the induced map of homotopy groups

$$\pi_k(\phi)\colon \pi_k(X) \to \pi_k(Y)$$

is an isomorphism for all $k = 0, 1, 2, \ldots$.

Definition 1.10.3. *Let $\Omega \subset \mathrm{J}^k(X,Y)$ be a differential relation of order k.*

(i) *Ω satisfies the* basic h-principle *if every section $X \to \Omega$ of the projection $p^k\colon \mathrm{J}^k(X,Y) \to X$ is homotopic to a holonomic section by a continuous homotopy of sections $X \to \Omega$.*

(ii)) *Ω satisfies the* one-parametric h-principle *if, in addition to (i), any homotopy $g_t\colon X \to \Omega$ ($t \in [0,1]$) between a pair of holonomic sections $g_0 = \mathrm{j}^k f_0$, $g_1 = \mathrm{j}^k f_1$ can be deformed, with fixed ends at $t = 0,1$, to a homotopy $\mathrm{j}^k f_t$ consisting of holonomic sections of Ω.*

(iii) *Ω satisfies the* weak homotopy equivalence principle *if the inclusion*

$$\{\mathrm{j}^k f\colon X \to \Omega \mid f\colon X \to Y\} \hookrightarrow \{g\colon X \to \Omega\}$$

of the space of holonomic sections of Ω into the space of all sections of Ω is a weak homotopy equivalence. (We use the $\mathcal{C}^k$ topology on the space of maps $X \to Y$ and $\mathcal{C}^0$ topology on the space of maps $X \to \Omega$.)

The analogous definitions apply to differential relations $\Omega \subset Z^{(k)}$ for sections of a submersion $Z \to X$.

The validity of the h-principle for a certain differential relation means that the corresponding analytic or geometric problem has a solution provided that there are no topological obstructions (these would amount to absence of a nonholonomic section). There are many natural examples when the h-principle applies. For example, the Smale-Hirsch principle [455, 264] says that regular homotopy classes of smooth immersions $X \looparrowright Y$ of a manifold X to a higher dimensional manifold Y are in one-to-one correspondence with the homotopy classes of fiberwise injective vector bundle maps $\mathrm{T}X \hookrightarrow \mathrm{T}Y$ of their tangent bundles. In particular, immersions $X \looparrowright \mathbb{R}^m$ for $m > n = \dim X$ are classified by the homotopy classes of vector bundle injections $\mathrm{T}X \hookrightarrow X \times \mathbb{R}^m$. The same holds when $m = n = \dim X$ provided that X is an open manifold.

Similarly, if X is a smooth open manifold then smooth submersions $X \to Y$ satisfy the h-principle which is expressed in terms of the existence of fiberwise surjective vector bundle maps $\mathrm{T}X \to \mathrm{T}Y$ [394, 234]. The same principle applies to k-mersions, that is, maps whose rank is bounded from below by a certain integer k.

The differential relations describing immersion and submersions, as well as many other natural relations arising from geometric problems, are particular examples in a certain class of first order open differential relations that we now describe. The condition is local and can be explained for sections of a trivial bundle $Z = \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^n$. Every local section is of the form $x \mapsto (x, f(x))$ for a map $f\colon U \to \mathbb{R}^m$ on an open set $U \subset \mathbb{R}^n$. A 1-jet $z^{(1)} \in Z^{(1)}$ is of the form $z^{(1)} = (z, v_1, \ldots, v_n)$ where $z = (x,y)$, $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$, $y = (y_1, \ldots, y_m) \in \mathbb{R}^m$, and $v_j \in \mathbb{R}^m$ for $j = 1, \ldots, n$. In a holonomic section we have $v_j = \frac{\partial f}{\partial x_j}(x)$. The restriction of a jet to a coordinate hyperplane $\{x_k = const\} \subset \mathbb{R}^n$ is determined by omitting the k-th vector v_k of the jet:

$$z_k^{(1)} = (z, v_1, \dots, v_{k-1}, v_{k+1}, \dots, v_m).$$

Definition 1.10.4. [235] *An open differential relation $\Omega \subset Z^{(1)}$ is* ample in the coordinate directions *if for every restricted 1-jet $z_k^{(1)}$ as above, the subset $\{v \in \mathbb{R}^m : (z, v_1, \dots, v_{k-1}, v, v_{k+1}, \dots) \in \Omega\} \subset \mathbb{R}^m$ is either empty or else the convex hull of each of its connected components equals $\mathbb{R}^m$.*

Theorem 1.10.5. (H-principle for ample differential relations) *Assume that $\pi : Z \to X$ is a smooth submersion and $\Omega \subset Z^{(1)}$ is an open subset of the manifold of 1-jets of sections $X \to Z$. Assume that for any point $z^0 \in Z$ we can find local coordinates (x, y) near z^0 such that, in these coordinates, $\pi(x, y) = x$ and Ω is ample in the coordinate directions. Then solutions of Ω satisfy the weak homotopy equivalence principle.*

For proof see [235], [236, Sec. 2.4], or [135, §18.2] (in particular Corollary 18.2.2). The construction of a holonomic section proceeds by triangulating the manifold X and extending a solution inductively from a neighborhood of a skeleton of certain dimension $q < \dim X$ to the skeleton of dimension $q + 1$. The main point is that a solution of a differential relation that is ample in the coordinate directions can be extended from the boundary of a cube in $\mathbb{R}^n$ to the interior of the cube if there are no topological obstructions. Here is the precise statement.

Lemma 1.10.6. [235, p. 339, Lemma 3.1.3] *Let $P = [0, 1]^n \subset \mathbb{R}^n$, let $Z = P \times \mathbb{R}^m \to P$ be the trivial bundle, and let the relation $\Omega \subset Z^{(1)}$ be ample in the coordinate directions. Suppose that $f_0 : P \to Z$ and $\phi_0 : P \to \Omega$ are smooth sections such that $\tau_1(\phi_0) = f_0$ on P (see (1.61)) and $\mathrm{j}^1_x f_0 = \phi_0(x)$ holds for every $x \in bP$. Then for any $\epsilon > 0$ there exists a smooth section $f : P \to Z$ satisfying the following properties:*

(i) $\mathrm{j}^1_x f = \phi_0(x)$ *for every* $x \in bP$,
(ii) $\mathrm{j}^1_x f \in \Omega$ *for every* $x \in P$,
(iii) $|f(x) - f_0(x)| < \epsilon$ *for every* $x \in P$, *and*
(iv) *there is a homotopy of sections* $\phi_t : P \to \Omega$ $(t \in [0, 1])$ *that is fixed on* bP *such that* ϕ_0 *is the given section and* $\phi_1 = \mathrm{j}^1 f$.

This lemma is proved by Gromov's *convex integration method*. Theorem 1.10.5 and Lemma 1.10.6 are used in several places in the text.

2

Stein Manifolds

This chapter is a brief survey of the theory of Stein manifolds and Stein spaces, with emphasis on the results that are frequently used in this book. After the initial developments by K. Weierstrass, B. Riemann, F. Hartogs, E. E. Levi, K. Reinhardt, H. Kneser, H. Cartan, P. Thullen and others, the main contributions were made in the period 1942–1965 by Kiyoshi Oka, by the French school around Henri Cartan including Pierre Dolbeault, Alexander Grothendieck and Jean-Pierre Serre, and by the Münster school founded by Heinrich Behnke and including Karl Stein, Hans Grauert, Reinhold Remmert and Friedrich Hirzebruch. In 1942 Oka [385, 388] published the first solution to the Levi problem on two dimensional domains, while the year 1965 marks the publication of Hörmander's fundamental paper [266] in which the $\overline{\partial}$-equation was solved by L^2-methods. (Another contemporary work in this direction is due to Andreotti and Vesentini [22].) Together with the works of Kohn [304, 305] these provide the basis for quantitative methods in complex analysis. Comprehensive accounts of Stein theory are available in [228, 241, 267]. The article of Schumacher [434] provides a solid historical survey. A quick introduction to interesting topics in L^2-theory can be found in [380].

2.1 Domains of Holomorphy

A basic notion in complex analysis is that of analytic continuation. Karl Weierstrass knew already in 1841 that a holomorphic function in an annulus in the complex plane $\mathbb{C}$ admits a development into what is now called a Laurent series. By estimating the coefficients in this series, Bernhard Riemann showed in his dissertation in 1851 that a function which is analytic in a punctured neighborhood of a point $p \in \mathbb{C}$ and is bounded near p extends to a holomorphic function in a full neighborhood of p. It was known early on that on any open set $D \subset \mathbb{C}$ there exist holomorphic functions that do not extend holomorphically across any boundary point of D. An explicit example on the disc

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$\mathbb{D} = \{|z| < 1\}$ is Kronecker's function $f(z) = \sum_{n=1}^{\infty} z^{n^2}$; further examples were given by Weierstrass.

A fundamental discovery was the phenomenon of *simultaneous analytic continuation*. In 1897 Adolph Hurwitz showed in his lecture at the first International Congress of Mathematicians that a holomorphic function of two or more variables does not have any isolated singularities. Much more interesting examples of analytic continuation were found by Friedrich Hartogs in 1906 [247]. The simplest *Hartogs figure* is the domain H in the bidisc $\mathbb{D}^2 \subset \mathbb{C}^2$ defined by

$$H = \left\{(z,w) \in \mathbb{D}^2 \colon |z| < \frac{1}{2} \ \text{ or } \ |w| > \frac{1}{2}\right\}. \tag{2.1}$$

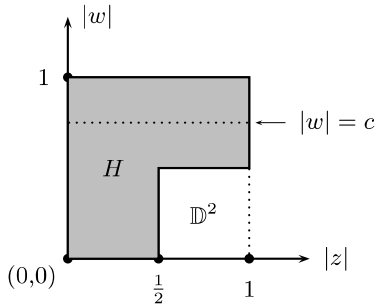

Fig. 2.1. A Hartogs figure in the bidisc

Every function $f \in \mathcal{O}(H)$ extends to a holomorphic function on the bidisc $\mathbb{D}^2$. Indeed, pick a number $\frac{1}{2} < c < 1$ and consider the Cauchy integral

$$F(z,w) = \frac{1}{2\pi \mathrm{i}} \int_{|\zeta|=c} \frac{f(z,\zeta)}{\zeta - w}\, \mathrm{d}\zeta. \quad |z| < 1,\ |w| < c,$$

Then F is a holomorphic function on $D = \mathbb{D} \times c\mathbb{D}$ which agrees with f on $H \cap D$. (Since the disc $\{z\} \times c\mathbb{D}$ is contained in H when $|z| < \frac{1}{2}$, we have $f = F$ there by the Cauchy integral formula; the equality elsewhere follows by the identity principle.) This extends f to a holomorphic function on $H \cup D = \mathbb{D}^2$.

Fifteen years later, Karl Reinhardt [410] studied domains of convergence of power series $\sum_{\alpha \in \mathbb{Z}_+^n} c_\alpha z^\alpha$ in several variables $z = (z_1, \ldots, z_n)$. It is immediate that the domain of convergence is a union of open polydiscs centered at the origin. By introducing the map $\phi \colon \mathbb{C}^n \to (\{-\infty\} \cup \mathbb{R})^n$,

$$\phi(z_1, \ldots, z_n) = (\log |z_1|, \ldots, \log |z_n|),$$

we see that each union of polydiscs is of the form $\Omega = \phi^{-1}(D)$ where D is a domain in $(\{-\infty\} \cup \mathbb{R})^n$ such that $(x_1, \ldots, x_n) \in D$ and $y_j \le x_j$ for $j = 1, \ldots, n$ implies that $(y_1, \ldots, y_n) \in D$. Reinhardt showed that Ω is the domain of convergence of a power series if and only of the corresponding

domain $D \subset (\{-\infty\} \cup \mathbb{R})^n$ is *convex*. This gives analytic continuation of holomorphic functions from a complete Reinhardt domain $\Omega \subset \mathbb{C}^n$ to the smallest logarithmically convex complete Reinhardt domain $\widetilde{\Omega} \subset \mathbb{C}^n$ containing Ω.

In 1932 Hellmuth Kneser reformulated Hartogs's result into a more useful form known as the *Kontinuitätssatz*: Given an embedded family of closed analytic discs $D_t \subset \mathbb{C}^n$ ($t \in [0,1]$) such that D_0 and all the boundaries bD_t belong to a domain $\Omega \subset \mathbb{C}^n$, every holomorphic function on Ω admits an analytic continuation along this family to a neighborhood of the disc D_1.

Hartogs' discovery initiated research on 'natural domains' of holomorphic functions. Analytic continuation in general yields a multivalued function. Following an idea of Riemann, multivalued functions are considered as singlevalued functions on *Riemann domains* over $\mathbb{C}^n$: A complex manifold X together with a locally biholomorphic map $\pi: X \to \mathbb{C}^n$. The central concept became that of a *domain of holomorphy* – a domain in $\mathbb{C}^n$, or over $\mathbb{C}^n$, with a holomorphic function that does not extend holomorphically to any larger domain. Much of the classical theory developed around the problem of characterizing domains of holomorphy, and of constructing the *envelope of holomorphy* $\widetilde{\Omega}$ of a given domain $\Omega \subset \mathbb{C}^n$ – the largest domain such that every holomorphic function on Ω extends to a holomorphic function on $\widetilde{\Omega}$.

Another important discovery was made by Eugenio E. Levi in 1911 [337]. He investigated domains $D \Subset \mathbb{C}^n$ with $\mathcal{C}^2$ boundaries. Let $\rho: \mathbb{C}^n \to \mathbb{R}$ be a $\mathcal{C}^2$ defining function for D, i.e., $D = \{z \in \mathbb{C}^n : \rho(z) < 0\}$ and $\mathrm{d}\rho_z \neq 0$ for every $z \in bD = \{\rho = 0\}$. Levi noticed that, if for some boundary point $p \in bD$ and some vector $v \in \mathrm{T}_p^{\mathbb{C}} bD$ that is complex tangential to the boundary (i.e., such that $\sum_{j=1}^n \frac{\partial \rho}{\partial z_j}(p) v_j = 0$) the Levi form $\mathcal{L}_{\rho,p}(v) < 0$ is negative, then holomorphic functions on D continue to a neighborhood of p in $\mathbb{C}^n$. The condition $\mathcal{L}_{\rho,p}(v) < 0$ implies that we can embed a Hartogs pair $(H, \mathbb{D}^n)$ in $\mathbb{C}^n$ such that H is mapped into D but the image of $\mathbb{D}^n$ contains a neighborhood of p. Levi conjectured that any domain $D \subset \mathbb{C}^n$ as in the following definition is a domain of holomorphy; this became known as the *Levi problem*.

Definition 2.1.1. *A domain $D = \{\rho < 0\}$ with a $\mathcal{C}^2$ defining function ρ such that $\mathrm{d}\rho \neq 0$ on $bD = \{\rho = 0\}$ is said to be* (weakly) Levi pseudoconvex *if $\mathcal{L}_{\rho,p}(v) \geq 0$ for every $p \in bD$ and $v \in \mathrm{T}_z^{\mathbb{C}} bD$. The domain D is* strongly pseudoconvex *if $\mathcal{L}_{\rho,p}(v) > 0$ for every $p \in bD$ and $0 \neq v \in \mathrm{T}_p^{\mathbb{C}} bD$.*

It is easily seen that the definition is independent of the choice of a defining function. A strongly pseudoconvex domain is locally at each boundary point biholomorphic to a piece of a strongly convex domain, and is osculated by a ball in suitable coordinates. This is commonly known as *Narasimhan's lemma*, although it was already known to Kneser in 1936 [298].

An important characterization of domains of holomorphy was obtained by Henri Cartan and Peter Thullen in 1932. To a compact set K in a complex space X we associate its $\mathcal{O}(X)$*-hull*

$$\widehat{K}_{\mathcal{O}(X)} = \{p \in X : |f(p)| \le \max_{x \in K} |f(x)|,\ \forall f \in \mathcal{O}(X)\}. \tag{2.2}$$

If K is a compact set in $\mathbb{C}^n$ then $\widehat{K} = \widehat{K}_{\mathcal{O}(\mathbb{C}^n)}$ is the *polynomial hull* of K.

Definition 2.1.2. *A compact set K in a complex space X is $\mathcal{O}(X)$-convex if $K = \widehat{K}_{\mathcal{O}(X)}$; if $X = \mathbb{C}^n$ then such K is said to be* polynomially convex. *A complex space X is* holomorphically convex *if for every compact set $K \subset X$ its $\mathcal{O}(X)$-hull $\widehat{K}_{\mathcal{O}(X)}$ is also compact.*

Theorem 2.1.3. (Cartan and Thullen [80]) *A Riemann domain over $\mathbb{C}^n$ is a domain of holomorphy if and only if it is holomorphically convex.*

The hull of any compact set in a domain $\Omega \subset \mathbb{C}^n$ is a bounded closed subset of Ω, but it may fail to be compact as is seen in the Hartogs figure (2.1): Since every $f \in \mathcal{O}(H)$ extends to a function in $\mathcal{O}(\mathbb{D}^2)$, the maximum principle shows that the $\mathcal{O}(H)$-hull of the circle $\{(z_0, w) : |w| = \frac{3}{4}\}$ is the intersection of the disc $\{(z_0, w) : |w| \le \frac{3}{4}\}$ with Ω; clearly this set is not compact if $\frac{1}{2} < |z_0| < 1$.

Theorem 2.1.3 is not difficult to prove. On the one hand, the derivatives of a holomorphic function $f \in \mathcal{O}(\Omega)$ satisfy the same bounds on $\widehat{K}_{\mathcal{O}(\Omega)}$ as on K, and hence the Taylor series of f centered around a point $p \in \widehat{K}_{\mathcal{O}(\Omega)}$ has the same domain of convergence as for points in K. If Ω is a domain of holomorphy, it follows that for any compact set $K \subset \Omega$ we have

$$\operatorname{dist}\big(\widehat{K}_{\mathcal{O}(\Omega)}, b\Omega\big) = \operatorname{dist}(K, b\Omega), \tag{2.3}$$

so $\widehat{K}_{\mathcal{O}(\Omega)}$ is compact. Conversely, using holomorphic convexity one can easily construct holomorphic functions tending to infinity along a given discrete sequence, so Ω is a domain of holomorphy.

A more challenging problem was to find a geometric characterization of domains of holomorphy. It follows from (2.3) that any closed holomorphic disc D in a domain of holomorphy Ω satisfies $\operatorname{dist}(D, b\Omega) = \operatorname{dist}(bD, b\Omega)$. This condition, which can be formulated in terms of *Hartogs pairs* (biholomorphic images of a standard pair $H \subset \mathbb{D}^n$, where H is a Hartogs figure in the polydisc $\mathbb{D}^n$), is known as *Hartogs pseudoconvexity* of Ω. Essentially it means that an analytic disc in $\overline{\Omega}$ with boundary in Ω must be contained in Ω. Oka showed that in such case the function $\Omega \ni z \mapsto -\log \operatorname{dist}(z, b\Omega)$ is plurisubharmonic on Ω. Clearly this function blows up at $b\Omega$, so by adding the term $|z|^2$ we get a strongly plurisubharmonic exhaustion function on Ω. Similarly, Levi pseudoconvexity of a domain $\Omega \Subset \mathbb{C}^n$ easily implies that the function $-\log \operatorname{dist}(\cdot, b\Omega)$ is plurisubharmonic on Ω.

Could this be a characterization of domains of holomorphy?

This Levi problem was solved in the affirmative by Oka in 1942 for domains in $\mathbb{C}^2$ [385]; the higher dimensional case followed ten years later. In summary we have the following result [267, Theorem 2.6.7].

Theorem 2.1.4. [385, 387, 388, 54, 378] *The following conditions are equivalent for a domain Ω in $\mathbb{C}^n$ (or a domain over $\mathbb{C}^n$):*

(a) *Ω is a domain of holomorphy.*

(b) *Ω is Hartogs pseudoconvex.*

(c) *The function $-\log \operatorname{dist}(\cdot\,, b\Omega)$ is plurisubharmonic.*

(d) *There exists a (strongly) plurisubharmonic exhaustion function on Ω.*

A domain $\Omega \subset \mathbb{C}^n$ with $\mathcal{C}^2$ boundary is a domain of holomorphy if and only if it is Levi pseudoconvex.

Every domain in (or over) $\mathbb{C}^n$ admits an envelope of holomorphy which can be constructed by 'pushing analytic discs' countably many times. A construction of the envelope in one step for domains in $\mathbb{C}^2$, and also in any two dimensional Stein manifold, was given by B. Jöricke in 2009 [283]. For results in this direction see also Merker and Porten [354].

2.2 Stein Manifolds and Stein Spaces

In 1951 Karl Stein introduced the following class of complex manifolds.

Definition 2.2.1. (K. Stein [460]) *A complex manifold X is said to be a* Stein manifold *(or a* holomorphically complete manifold*) if the following hold:*

(a) *For every pair of distinct points $x \neq y$ in X there is a holomorphic function $f \in \mathcal{O}(X)$ such that $f(x) \neq f(y)$.*

(b) *For every point $p \in X$ there exist functions $f_1, \ldots, f_n \in \mathcal{O}(X)$, $n = \dim X$, whose differentials $\mathrm{d}f_j$ are $\mathbb{C}$-linearly independent at p.*

(c) *X is holomorphically convex (see Def. 2.1.2).*

Property (b) means that global holomorphic functions provide local charts at each point. Here are some observation and examples:

- An open set in $\mathbb{C}^n$ is Stein if and only if it is a domain of holomorphy. (This follows from the Cartan-Thullen Theorem 2.1.3.)
- A Stein manifold does not contain any compact complex subvariety of positive dimension. (Apply axiom (a) and the maximum principle.)
- The Cartesian product $X \times Y$ of a pair of Stein manifolds is Stein.
- A closed complex submanifold X of $\mathbb{C}^N$ is Stein. (Use coordinate functions restricted to X. For the converse see Theorem 2.2.8 below.) More generally, every closed complex submanifold of a Stein manifold is Stein.
- An open Riemann surface is a Stein manifold. (This nontrivial result is due to Behnke and Stein [40, 41].)

- If $X \to Y$ is a holomorphic covering space and Y is Stein then X is Stein. (This is due to K. Stein [461].)
- If $E \to X$ is a holomorphic vector bundle over a Stein base X then the total space E is also Stein. However, fiber bundles with fiber $\mathbb{C}^n$ $(n > 1)$ and with a nonlinear transition group may fail to be Stein; see §4.21.

The notion of a Stein space was first introduced by Hans Grauert in 1955 [220]. The standard definition is the following one.

Definition 2.2.2. *A second countable complex space X is a* Stein space *if it satisfies properties (a), (c) in Def. 2.2.1 and also*

(b') *Every local ring $\mathcal{O}_{X,x}$ is generated by functions in $\mathcal{O}(X)$.*

Condition (b') means that there is a holomorphic map $X \to \mathbb{C}^N$ which embeds a neighborhood of x as a local complex subvariety of $\mathbb{C}^N$. Grauert showed in [220] that one gets an equivalent definition by keeping (c) and replacing (a) and (b) (resp. (b')) by the following property.

Definition 2.2.3. *A complex space X is called* K-complete *if for every point $x \in X$ there is a holomorphic map $f\colon X \to \mathbb{C}^N$ (with $N = N_x$) such that x is an isolated point of the fiber $f^{-1}(f(x))$.*

It is immediate that axiom (a) implies K-completeness. In summary:

Theorem 2.2.4. [220] *A complex space X is a Stein space if and only if it is holomorphically convex and it satisfies one of the following two properties:*

(i) *Holomorphic functions separate points on X (axiom (a) in Def. 2.2.1).*
(ii) *X is K-complete in the sense of Def. 2.2.3.*

For further characterizations of Stein spaces see [228, p. 152].

The following *Oka-Weil theorem* generalizes the classical Runge theorem. The analogous result holds for sections of holomorphic vector bundles and, more generally, for sections of coherent analytic sheaves (see §2.4).

Theorem 2.2.5. *If X is a Stein space and K is a compact $\mathcal{O}(X)$-convex subset of X then every holomorphic function in an open neighborhood of K can be approximated uniformly on K by functions in $\mathcal{O}(X)$.*

Theorem 2.2.5 was first proved for domains of holomorphy by Oka [382] using his *Oka lemma* [267, Lemma 2.7.5]. It is immediate that an $\mathcal{O}(X)$-convex set K can be approximated from the outside by *analytic polyhedra*, i.e., by Stein open sets of the form

$$U = \{x \in X\colon |h_j(x)| < 1,\ j = 1, \dots, m\}, \quad h_1, \dots, h_m \in \mathcal{O}(X).$$

By adding more functions we can insure that $h = (h_1, \dots, h_m): X \to \mathbb{C}^m$ embeds U properly in the polydisc $\mathbb{D}^m \subset \mathbb{C}^m$. The key point proved by Oka is that for any function $f \in \mathcal{O}(U)$ there is a $g \in \mathcal{O}(\mathbb{D}^m)$ such that $g \circ h = f$. (This is a special case of Cartan's extension theorem, Corollary 2.4.3.) By expanding g in power series and approximating it by Taylor polynomials $P \in \mathbb{C}[z_1, \dots, z_m]$ we get functions $P \circ h \in \mathcal{O}(X)$ approximating f on K. Another proof can be given by L^2-methods (see [267]).

Definition 2.2.6. *A domain Ω in a complex space X is* Runge in X *if every holomorphic function $f \in \mathcal{O}(\Omega)$ can be approximated uniformly on compacts in Ω by functions in $\mathcal{O}(X)$; equivalently, if the subalgebra $\{f|_\Omega : f \in \mathcal{O}(X)\}$ of $\mathcal{O}(\Omega)$ is dense in $\mathcal{O}(\Omega)$.*

Theorem 2.2.7. [267, p. 91] *A Stein domain Ω in a Stein space X is Runge in X if and only if for every compact set $K \subset \Omega$ we have $\widehat{K}_{\mathcal{O}(\Omega)} = \widehat{K}_{\mathcal{O}(X)}$.*

An important characterization of Stein manifolds is that they are embeddable in Euclidean spaces. It is an immediate consequence of Definition 2.2.1 that for every relatively compact domain Ω in a Stein manifold X there is a holomorphic map $f: X \to \mathbb{C}^N$ for a big enough N such that $f|_\Omega : \Omega \to \mathbb{C}^N$ is an injective holomorphic immersion. In 1956 Reinhold Remmert proved a substantially stronger result that every Stein manifold admits a *proper* holomorphic embedding in some Euclidean space $\mathbb{C}^N$ [411]. In 1960–61, Errett Bishop and Raghavan Narasimhan independently showed that if $\dim_{\mathbb{C}} X = n$ then the number N in Remmert's theorem can be taken to be $2n + 1$.

Theorem 2.2.8. ([48], [366, Theorem 5])

(a) *If X is a Stein manifold of dimension n then the set of proper holomorphic embeddings of X into $\mathbb{C}^{2n+1}$ is dense in $\mathcal{O}(X)^{2n+1}$.*

(b) *If X is a Stein space of dimension n then the set of holomorphic maps $X \to \mathbb{C}^{2n+1}$ which are proper, injective and regular on the regular part X_{reg} is dense in $\mathcal{O}(X)^{2n+1}$.*

(c) *If X is a Stein space of dimension n and of finite embedding dimension m then for $N = \max\{n+m, 2n+1\}$ the set of proper holomorphic embeddings $X \hookrightarrow \mathbb{C}^N$ is dense in $\mathcal{O}(X)^N$.*

More precise embedding theorems for Stein manifolds and Stein spaces are proved in §8.2 – §8.4, and for Riemann surfaces in §8.9 – §8.10. Since every real analytic manifold admits a Stein complexification [224], we get the following consequence:

Corollary 2.2.9. [224, Theorem 3] *Every real analytic manifold admits a proper real analytic embedding into a Euclidean space $\mathbb{R}^N$.*

Since Stein manifolds are complex submanifolds of Euclidean spaces, it is not surprising that they can be approximated by affine algebraic manifolds. It was proved by E. L. Stout [472] that any relatively compact domain in a Stein manifold is biholomorphically equivalent to a domain in an affine algebraic manifold. (For the real algebraic case see Nash [371].) More precise algebraic approximation results were obtained by Demailly, Lempert and Schiffman [105, 336] and by Lisca and Matič [343] (see Theorem 9.8.1 on p. 434).

2.3 Characterization by Plurisubharmonic Functions

It is a fundamental fact that Stein manifolds and Stein spaces are characterized by plurisubharmonicity (Theorem 2.3.2 below). Often the most efficient way to show that a complex space is Stein is to produce a strongly plurisubharmonic exhaustion function on it. For example, this is how Y.-T. Siu proved in 1976 [446] that a Stein subvariety of any complex space has a basis of open Stein neighborhoods (Theorem 3.1.1 on p. 57). Stein neighborhood constructions often allow us to transfer a problem on a complex space to Euclidean space where it becomes tractable; Chapter 3 focuses on such methods.

It follows from holomorphic convexity that every Stein space X is exhausted by an increasing sequence of compacts $K_1 \subset K_2 \subset \cdots \subset \cup_j K_j = X$ such that $K_j = \widehat{K}_j$ for every j. Using such exhaustions and axioms (a), (b') one can easily find strongly plurisubharmonic exhaustion functions of the form

$$\rho = \sum_{j=1}^{\infty} |f_j|^2 \colon X \to \mathbb{R}_+, \quad f_j \in \mathcal{O}(X),\ j = 1, 2, \ldots.$$

By a more precise argument one obtains the following result approximating $\mathcal{O}(X)$-convex sets by sublevel sets of strongly plurisubharmonic functions (see [267, Theorem 5.1.5, p. 117]).

Proposition 2.3.1. *If K is a compact $\mathcal{O}(X)$-convex set in a Stein space X then for every open set $U \subset X$ containing K there exists a smooth strongly plurisubharmonic function $\rho \colon X \to \mathbb{R}$ such that $\rho < 0$ on K and $\rho > 1$ on $X \backslash U$. Furthermore, there exists a plurisubharmonic exhaustion function $\rho \colon X \to \mathbb{R}_+$ such that $\rho^{-1}(0) = K$ and ρ is strongly plurisubharmonic on $X \backslash K = \{\rho > 0\}$.*

Note that the function $\rho_a \colon \mathbb{C}^N \to \mathbb{R}_+$ given by $\rho_a = |z - a|^2$ is strongly plurisubharmonic on any complex subvariety $X \subset \mathbb{C}^N$; if X is closed then this is an exhaustion function on X. Furthermore, if X is smooth then $\rho_a|_X$ is a Morse function on X for most choices of the point $a \in \mathbb{C}^N$.

These observations show that a Stein space admits plenty of smooth strongly plurisubharmonic exhaustion functions. The following converse is the most useful characterization of Stein manifolds and Stein spaces.

Theorem 2.3.2. (a) [224, 113] *A complex manifold which admits a strongly plurisubharmonic exhaustion function is a Stein manifold.*

(b) [368, 145] *A complex space which admits a strongly plurisubharmonic exhaustion function is a Stein space.*

Furthermore, if $\rho\colon X \to \mathbb{R}$ is a strongly plurisubharmonic exhaustion function then each sublevel set $\{x \in X\colon \rho(x) \le c\}$ is $\mathcal{O}(X)$-convex.

Corollary 2.3.3. *For every compact set K in a Stein space X the $\mathcal{O}(X)$-hull of K coincides with its plurisubharmonic hull:*

$$\widehat{K}_{\mathcal{O}(X)} = \widehat{K}_{\mathrm{Psh}(X)}.$$

Hence every holomorphic function in a neighborhood of a compact $\mathrm{Psh}(X)$*-convex set $K = \widehat{K}_{\mathrm{Psh}(X)}$ is a uniform limit on K of functions in $\mathcal{O}(X)$.*

An efficient proof of these results is given by the L^2-method for solving nonhomogeneous $\overline{\partial}$-equations with weights of the form $\mathrm{e}^{-\rho}$ with $\rho \in \mathrm{Psh}(X)$ [266, 267]. Theorem 2.3.2 implies the following solution of the Levi problem.

Corollary 2.3.4. *A domain Ω in a Stein space X which admits a plurisubharmonic exhaustion function $\rho\colon \Omega \to \mathbb{R}$ is Stein. In particular, every Levi (or Hartogs) pseudoconvex domain in a Stein manifold is Stein.*

The existence of a Morse strongly plurisubharmonic function implies that every Stein manifold X is homotopy equivalent to a CW complex of real dimension at most $n = \dim_{\mathbb{C}} X$ (the Lefschetz theorem; see §3.11). The same holds for any finite dimensional Stein space.

There exist several notions of *ambient holomorphic convexity* of a compact set (see [474]). We shall use the following properties.

Definition 2.3.5. *Assume that K is a compact set in a complex space X.*

(i) *K is a* Stein compactum *if it admits a basis of Stein neighborhoods in X.*

(ii) *K is* holomorphically convex *if it admits an open Stein neighborhood Ω in X such that K is $\mathcal{O}(\Omega)$-convex.*

Proposition 2.3.1 and Theorem 2.3.2 imply the following.

Proposition 2.3.6. *A compact set K in a Stein space X is holomorphically convex if and only if there exists a plurisubharmonic function $\rho\colon U \to \mathbb{R}_+$ in an open neighborhood U of K such that $\rho^{-1}(0) = K$ and ρ is strongly plurisubharmonic on $U \backslash K = \{\rho > 0\}$.*

The sets $\Omega_c = \{x \in U\colon \rho(x) < c\}$ for small $c > 0$ then form a basis of Stein neighborhoods of K such that K is $\mathcal{O}(\Omega_c)$-convex.

Proposition 2.3.7. *Let K be a compact set in a complex space X. Assume that there exist a neighborhood $U \subset X$ of K, a strongly plurisubharmonic function $\rho: U \to \mathbb{R}$, and a weakly plurisubharmonic function $\tau: U \to \mathbb{R}_+$ such that $K = \{\tau = 0\}$. Then K is a Stein compactum.*

Proof. Pick an open neighborhood $V \Subset U$ of K and choose a fast growing convex increasing function $\chi: \mathbb{R} \to \mathbb{R}$ such that the strongly plurisubharmonic function $\phi = \rho + \chi \circ \tau: U \to \mathbb{R}$ satisfies $\phi|_K < 0$ and $K \subset V_c = \{\phi < c\} \Subset V$ for some $c > 0$. By Theorem 2.3.2 the domain V_c is Stein. □

The closure of a smooth weakly pseudoconvex domain $D \Subset \mathbb{C}^n$ need not be a Stein compactum; an example is the *worm domain* [110]. For the existence of bounded strongly plurisubharmonic exhaustion functions on certain weakly Levi pseudoconvex domains see [111].

2.4 Cartan-Serre Theorems A & B

The famous Theorems A and B of Henri Cartan were proved in his seminar in 1951–52. It would be impossible to overstate the importance of these results to the development of analytic and algebraic geometry.

Theorem 2.4.1. (Theorems A and B; [75, 77, 228].) *For every coherent analytic sheaf $\mathcal{F}$ on a Stein space $(X, \mathcal{O}_X)$ the following hold:*

(A) *The stalks $\mathcal{F}_x$ are generated as $\mathcal{O}_{X,x}$-modules by global sections of $\mathcal{F}$.*
(B) $H^p(X; \mathcal{F}) = 0$ *for all* $p = 1, 2, \ldots$.

The corresponding results hold for every coherent algebraic sheaf over an affine algebraic variety $X \subset \mathbb{C}^N$ [440, p. 237, Théorème 2]. *For coherent sheaves with continuous boundary values on strongly pseudoconvex domains see* [262, 330].

We recall the relevant notions; a comprehensive account is available in [229]. An *analytic sheaf* (or $\mathcal{O}_X$-sheaf) on a complex space X is a sheaf $\mathcal{F}$ of $\mathcal{O}_X$-modules; that is, a sheaf whose stalk $\mathcal{F}_x$ over any point $x \in X$ is a module over the ring $\mathcal{O}_{X,x}$. The sheaf $\mathcal{F}$ is *locally finitely generated* if for every point $x_0 \in X$ there exist an open neighborhood $U \subset X$ and finitely many sections $f_1, \ldots, f_k \in \mathcal{F}(U) = \Gamma(U, \mathcal{F})$ whose germs at any point $x \in U$ generate $\mathcal{F}_x$ as an $\mathcal{O}_{X,x}$-module. The simplest example is $\mathcal{O}_X^k$, the direct sum of k copies of the structure sheaf $\mathcal{O}_X$; this is the sheaf of holomorphic sections of the trivial bundle $X \times \mathbb{C}^k \to X$. An analytic sheaf is *coherent* if it is locally finitely generated and if for any set of local sections $f_1, \ldots, f_k \in \mathcal{F}(U)$ the corresponding *sheaf of relations* $\mathcal{R} = \mathcal{R}(f_1, \ldots, f_k)$ is also locally finitely generated. The latter sheaf has stalks

$$\mathcal{R}_x = \left\{ g_{1,x}, \ldots, g_{k,x} \in \mathcal{O}_{X,x} \colon \sum_{j=1}^{k} g_{j,x} f_{j,x} = 0 \right\}, \quad x \in U. \tag{2.4}$$

From the above description we see that on small open sets $U \subset X$ we have a short exact sequence of analytic sheaf homomorphisms

$$\mathcal{O}_U^m \xrightarrow{\alpha} \mathcal{O}_U^k \xrightarrow{\beta} \mathcal{F}|_U \longrightarrow 0, \tag{2.5}$$

where $\beta(g_{1,x}, \ldots, g_{k,x}) = \sum_{j=1}^{k} g_{j,x} f_{j,x}$. Hence β maps the standard basis sections $e_j = (0, \ldots, 1, \ldots, 0)$ of $\mathcal{O}_U^k$ onto the generators f_j of $\mathcal{F}|_U$ and $\mathcal{R} = \ker \beta = \operatorname{im} \alpha$ is the sheaf of relations (2.4). If X is a Stein space then such resolution exists over any relatively compact open subset $U \Subset X$.

Here are the main examples of coherent sheaves on a complex space X:

- The structure sheaf $\mathcal{O}_X$ ([386], [229, p. 59]).
- The sheaf of ideals $\mathcal{O}_A$ of a complex subvariety A in X (Cartan's coherence theorem, [74], [79, p. 631], [229, p. 84]).
- A locally free analytic sheaf. (Locally free analytic sheaves are sheaves of holomorphic sections of holomorphic vector bundles.)
- The Whitney sum $\mathcal{E} \oplus \mathcal{F}$ and the tensor product $\mathcal{E} \otimes \mathcal{F}$ of coherent sheaves.
- The sheaf $\mathcal{H}om(\mathcal{E}, \mathcal{F})$ of $\mathcal{O}_X$-homomorphisms $\mathcal{E} \to \mathcal{F}$ between a pair of coherent analytic sheaves. In particular, the dual $\mathcal{E}^*$ of a coherent sheaf.
- The kernel $\ker \beta$ and the image $\operatorname{im}\beta$ of an $\mathcal{O}_X$-homomorphism $\beta \colon \mathcal{F} \to \mathcal{G}$ of coherent analytic sheaves. In summary, given a short exact sequence of homomorphisms of $\mathcal{O}_X$-analytic sheaves

$$0 \longrightarrow \mathcal{E} \xrightarrow{\alpha} \mathcal{F} \xrightarrow{\beta} \mathcal{G} \longrightarrow 0, \tag{2.6}$$

  if two sheaves are coherent then so is the third [229, p. 236].
- The direct image of an $\mathcal{O}_X$-coherent sheaf by a proper holomorphic map $X \to Y$ of complex spaces is a coherent $\mathcal{O}_Y$-sheaf [229, p. 207].

Each coherent analytic sheaf $\mathcal{F}$ can be represented as the sheaf of germs of fiberwise linear holomorphic functions on a linear space $\pi \colon L \to X$ [140]. More precisely, there is a contravariant equivalence between the category of coherent analytic sheaves and the category of linear spaces such that locally free sheaves correspond to vector bundles. The sheaf of germs of holomorphic sections $X \to L$ of any linear space is also coherent [140, p. 53, Corollary].

We now mention some applications of Theorems A and B; see [228, Chapter V] for more on this subject.

Corollary 2.4.2. *Let $\mathcal{F} \xrightarrow{\beta} \mathcal{G}$ be an epimorphism of analytic sheaves over a Stein space X. If the kernel $\mathcal{E} = \ker \beta$ is coherent then the induced map on sections $\mathcal{F}(X) \to \mathcal{G}(X)$, $f \mapsto \beta(f)$, is surjective.*

Proof. Since $H^1(X;\mathcal{E}) = 0$ by Theorem B, the conclusion follows from the exact cohomology sequence $\mathcal{F}(X) \to \mathcal{G}(X) \to H^1(X;\mathcal{E}) = 0$. $\square$

Applying this to the exact sequence $0 \to \mathcal{J}_A \to \mathcal{O}_X \to \mathcal{O}_X/\mathcal{J}_A \to 0$ where A is a closed complex subvariety of X one obtains

Corollary 2.4.3. (Cartan's extension theorem.) *Every holomorphic function on a closed complex subvariety in a Stein space X extends to a holomorphic function on X.*

Corollary 2.4.4. (Cartan's division theorem.) *If $\mathcal{F}$ is a coherent analytic sheaf on a Stein space X and if $f_1, \ldots, f_k \in \mathcal{F}(X)$ generate each stalk $\mathcal{F}_x$, then every section $f \in \mathcal{F}(X)$ is of the form $f = \sum_{j=1}^k g_j f_j$ for some $g_j \in \mathcal{O}(X)$.*

Proof. Consider the exact sequence $0 \to \mathcal{R} \to \mathcal{O}^k \xrightarrow{\beta} \mathcal{F} \to 0$ as in (2.5). Since $\mathcal{R} = \ker \beta$ is coherent, the conclusion follows from Corollary 2.4.2. $\square$

Corollary 2.4.5. *Given a short exact sequence (2.6) of analytic sheaves on a Stein space, if $\mathcal{G}$ is locally free then there exists a sheaf homomorphism $\phi\colon \mathcal{G} \to \mathcal{F}$ such that $\beta \circ \phi = \mathrm{Id}_{\mathcal{G}}$. In particular, a short exact sequence of homomorphisms of holomorphic vector bundles over a Stein space splits.*

Proof. Consider the induced exact sequence

$$0 \longrightarrow \mathcal{H}om(\mathcal{G},\mathcal{E}) \longrightarrow \mathcal{H}om(\mathcal{G},\mathcal{F}) \xrightarrow{\beta} \mathcal{H}om(\mathcal{G},\mathcal{G}) \longrightarrow 0.$$

Surjectivity of β is due to $\mathcal{G}$ being locally free. By Theorem B we have $H^1(X;\mathcal{H}om(\mathcal{G},\mathcal{E})) = 0$ and hence β is surjective also on the level of sections. Hence $\mathrm{Id}_{\mathcal{G}}$ lifts to a homomorphism $\phi\colon \mathcal{G} \to \mathcal{F}$ with $\beta \circ \phi = \mathrm{Id}_{\mathcal{G}}$. $\square$

Theorem 2.4.6. *On any Stein manifold X the Dolbeault cohomology groups vanish: $H^{p,q}_{\overline{\partial}}(X) = 0$ for all $p \geq 0,\ q \geq 1$.*

Proof. The sheaf Ω_p of holomorphic p-forms on X admits a resolution

$$0 \to \Omega_p \hookrightarrow \mathcal{E}_{p,0} \xrightarrow{\overline{\partial}} \mathcal{E}_{p,1} \xrightarrow{\overline{\partial}} \mathcal{E}_{p,2} \cdots \xrightarrow{\overline{\partial}} \mathcal{E}_{p,n} \longrightarrow 0.$$

Since the sheaves $\mathcal{E}_{p,q}$ of smooth (p,q)-forms on X are fine, their cohomology vanishes. Leray's theorem implies that $H^{p,q}_{\overline{\partial}}(X) \cong H^q(X;\Omega_p)$. Since the sheaf Ω_p is coherent analytic, these groups are zero. $\square$

J.-P. Serre proved that each element of a de Rham cohomology group $H^p(X;\mathbb{C})$ of a Stein manifold is represented by a closed holomorphic p-form (see [439, Theorem 1], [228, p. 155]). The de Rham cohomology of an affine algebraic manifold is represented by algebraic forms.

The space of sections of a coherent analytic sheaf with the topology of uniform convergence on compacts is a Fréchet space [228, p. 167]. We have the Runge approximation theorem [228, p. 170]:

Theorem 2.4.7. *Let $\mathcal{F}$ be a coherent analytic sheaf over a Stein space X. If K is a compact $\mathcal{O}(X)$-convex set in X then any section over an open neighborhood of K can be approximated uniformly on K by sections in $\mathcal{F}(X)$.*

2.5 The $\overline{\partial}$-Problem

The $\overline{\partial}$-problem asks for a solution of the equation $\overline{\partial}u = f$ for a given $\overline{\partial}$-closed form f. By Theorem 2.4.6 this problem is always solvable on a Stein manifold. A more direct proof which also gives L^2 estimates is provided by the L^2 theory of Hörmander [266, 267], Andreotti and Vesentini [22] and Kohn [304, 305]. (A comprehensive account of this subject can be found in [83].) We quote the following result for $(0,1)$-forms that is used in the present text. Let $\mathrm{d}\lambda$ denote the Lebesgue measure on $\mathbb{C}^n = \mathbb{R}^{2n}$.

Theorem 2.5.1. [267, Theorem 4.4.2, p. 94] *Let Ω be a pseudoconvex (Stein) domain in $\mathbb{C}^n$ and ϕ a plurisubharmonic function in Ω. For every $(0,1)$-form $f = \sum f_j \,\mathrm{d}\bar{z}_j$ such that $f_j \in L^2_{loc}(\Omega)$ and $\overline{\partial}f = 0$ (in the weak sense) there exists $u \in L^2_{loc}(\Omega)$ such that*

$$\overline{\partial}u = f \quad \text{and} \quad \int_\Omega \frac{|u|^2}{(1+|z|^2)^2}\mathrm{e}^{-\phi}\mathrm{d}\lambda \le \int_\Omega \sum_{j=1}^n |f_j|^2 \mathrm{e}^{-\phi}\mathrm{d}\lambda.$$

By taking Ω bounded and $\phi = 0$ we get the estimate

$$\overline{\partial}u = f \quad \text{and} \quad \int_\Omega |u|^2 \mathrm{d}\lambda \le C \int_\Omega \sum_j |f_j|^2 \,\mathrm{d}\lambda \tag{2.7}$$

where the constant C depends only on the radius of Ω and on the dimension n. The analogous results hold on relatively compact domains in Stein manifolds.

To pass from L^2 to $\mathcal{C}^k$ estimates one needs the following well known lemma which follows from the Bochner-Martinelli formula [189, Lemma 3.2].

Lemma 2.5.2. (Interior elliptic regularity estimates.) *Let $\mathbb{B}^n$ denote the open unit ball in $\mathbb{C}^n$. For each integer $s \in \mathbb{Z}_+$ there is a constant $c_s > 0$ such that if $f \in \mathcal{C}^{s+1}(\epsilon\mathbb{B})$ for some $\epsilon > 0$ and $\alpha \in \mathbb{Z}_+^{2n}$ is a multiindex with $|\alpha| = s$ then*

$$c_s\,|\partial^\alpha f(0)| \le \epsilon^{-n-s}||f||_{L^2(\epsilon\mathbb{B})} + \sum_{|\beta|\le s} \epsilon^{|\beta|+1-s}||\partial^\beta(\overline{\partial}f)||_{L^\infty(\epsilon\mathbb{B})}.$$

In particular we have the sup-norm estimate

$$c_0\,|f(0)| \le \epsilon^{-n}||f||_{L^2(\epsilon\mathbb{B})} + \epsilon||\overline{\partial}f||_{L^\infty(\epsilon\mathbb{B})}.$$

On bounded strongly pseudoconvex domains in Stein manifolds the $\overline{\partial}$-equation can also be solved by means of integral formulas with holomorphic kernels. This kernel method gives optimal Hölder estimates. The first results of this type were obtained by G. Henkin [253] and R. de Arellano (see [256]). We shall use the following result due to Range and Siu [408] and Lieb and Range [339, Theorem 1]; see also [356, Theorem 1'] and [338]. We denote by $\mathcal{C}^l_{p,q-1}$ the space of (p,q)-forms with coefficients of class $\mathcal{C}^l$.

Theorem 2.5.3. *For every bounded strongly pseudoconvex domain D with $\mathcal{C}^2$ boundary in a Stein manifold there exists a linear operator $T\colon \mathcal{C}^0_{0,1}(D) \to \mathcal{C}^{1/2}(D)$ such that, if $f \in \mathcal{C}^0_{0,1}(\bar{D}) \cap \mathcal{C}^1_{0,1}(D)$ and $\overline{\partial} f = 0$ in D then*

$$\overline{\partial}(Tf) = f, \quad ||Tf||_{\mathcal{C}^{1/2}(\bar{D})} \le c_D \, ||f||_{\mathcal{C}^0_{0,1}(\bar{D})}.$$

The constant c_D can be chosen uniform for all domains sufficiently $\mathcal{C}^2$-close to D. If D has boundary of class $\mathcal{C}^\ell$ for some $\ell \in \{2,3,\ldots\}$ then there exists a linear operator $T\colon \mathcal{C}^0_{0,1}(D) \to \mathcal{C}^0(D)$ satisfying the following properties:

(i) *If $f \in \mathcal{C}^0_{0,1}(\bar{D}) \cap \mathcal{C}^1_{0,1}(D)$ and $\overline{\partial} f = 0$ then $\overline{\partial}(Tf) = f$.*

(ii) *If $f \in \mathcal{C}^0_{0,1}(\bar{D}) \cap \mathcal{C}^r_{0,1}(D)$ for some $r \in \{1,\ldots,\ell)$ then*

$$||Tf||_{\mathcal{C}^{l,1/2}(\bar{D})} \le C_{l,D} ||f||_{\mathcal{C}^l_{0,1}(\bar{D})}, \quad l = 0,1,\ldots,r. \tag{2.8}$$

Although these results are stated in the original papers for domains with $\mathcal{C}^\infty$ boundaries, one only needs $\mathcal{C}^\ell$ boundary to get estimates up to order ℓ; this is implicitly contained in the paper by Michel and Perotti [356].

In [189, Theorem 3.1] the authors constructed special integral kernels for solving the $\overline{\partial}$-equation on thin tubular neighborhoods of totally real submanifolds. The precise result is the following.

Theorem 2.5.4. *Let $M \subset \mathbb{C}^n$ be a totally real submanifold of class $\mathcal{C}^1$ and $c \in (0,1)$. Denote by $M(\delta)$ the tube of radius $\delta > 0$ around M. Then there is a number $\delta_0 > 0$ and for each integer $l \ge 1$ a constant $C_l > 0$ such that the following hold for $0 < \delta \le \delta_0$, $p \ge 0$, $q \ge 1$: For each $u \in \mathcal{C}^l_{p,q}(M(\delta))$ with $\overline{\partial} u = 0$ there is a $v \in \mathcal{C}^l_{p,q-1}(M(\delta))$ satisfying $\overline{\partial} v = u$ in $M(c\delta)$ and the following estimate for all $\alpha \in \mathbb{Z}^{2n}_+$ with $|\alpha| \le l$:*

$$||\partial^\alpha v||_{L^\infty(M(c\delta))} \le C_l \left(\delta ||\partial^\alpha u||_{L^\infty(M(\delta))} + \delta^{1-|\alpha|} ||u||_{L^\infty(M(\delta))} \right). \tag{2.9}$$

In particular, we have the sup-norm estimate

$$\|v\|_{L^\infty(M(c\delta))} \le C\delta \|u\|_{L^\infty(M(\delta))}.$$

The integral kernels used to prove this theorem are of Henkin-de Arellano type, but are made especially for tubes around totally real submanifolds. This theorem was used in [189] to find optimal results on approximating diffeomorphisms between totally real submanifolds by biholomorphisms of their tubular neighborhoods (see §4.13).

3

Stein Neighborhoods and Holomorphic Approximation

In this chapter we find Stein neighborhoods of certain types of sets in arbitrary complex manifolds and complex spaces, focusing on results that are essential for our constructions in Oka theory.

We begin with Siu's theorem that every Stein subvariety in a complex space admits an open Stein neighborhood. The analogous result for q-convex spaces (Theorem 3.1.4), due to Colţoiu and Demailly, is obtained without much additional effort. In §3.2 we give some extensions of Siu's theorem. An important application in §3.3 is the existence of local holomorphic retractions onto Stein submanifolds, or a family of such retraction in the foliated case. In §3.4 we obtain semiglobal extension of holomorphic sections, an essential result in Oka theory for singular Stein spaces. In §3.5 we consider Stein neighborhoods of totally real submanifolds. In §3.6 we find Stein neighborhoods of certain compact sets in laminated manifolds. In §3.7 we construct Stein neighborhoods of holomorphically convex compact sets with attached totally real manifolds, and we prove a holomorphic approximation theorem on such objects. In §3.8 we construct thin strongly pseudoconvex handlebodies around the union of a strongly pseudoconvex domain with a totally real submanifold; this is used in the Eliashberg-Gompf construction of Stein manifold structures on smooth manifolds with the correct handlebody decomposition (see Chapter 9). In §3.9 – §3.10 we analyze the geometry of a q-convex function at a Morse critical point. In §3.11 we consider the topological structure of Stein spaces and of q-complete complex spaces.

3.1 Q-Complete Neighborhoods

This and the following section are devoted to the proof and some generalizations of the following result of Yum-Tong Siu.

Theorem 3.1.1. [446] *Every locally closed Stein subvariety Y of any complex space X admits a basis of open Stein neighborhoods in X.*

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The complex space X in Siu's theorem is not necessarily reduced. After a simple reduction to the case in which X is reduced and finite dimensional, the first step consists in showing that every compact subset K of Y admits an open Stein neighborhood in X; this was proved independently in [430].

Here we follow a slightly different proof given by Demailly [101, Theorem 1] and Colţoiu [88]. Both obtained the analogous result in the more general case of a q-complete subvariety in a complex space; Siu's theorem corresponds to the case $q = 1$. We begin by recalling the following notions due to Andreotti and Grauert [20].

Definition 3.1.2. *Let $(X, \mathcal{O}_X)$ be a complex space, possibly nonreduced.*

(a) *A function $\phi\colon X \to \mathbb{R}$ is said to be* q-convex *if there exists a cover of X by open patches $A_\lambda \subset X$, isomorphic to closed analytic sets in open sets $\Omega_\lambda \subset \mathbb{C}^{N_\lambda}$, such that each restriction $\phi|_{A_\lambda}$ admits an extension $\widetilde{\phi}_\lambda$ to Ω_λ that is q-convex (Def. 1.8.9), i.e., its Levi form $\mathrm{dd^c}\widetilde{\phi}_\lambda = \mathrm{i}\partial\overline{\partial}\widetilde{\phi}_\lambda$ has at most $q-1$ negative or zero eigenvalues at each point of Ω_λ.*

(b) *X is* q-convex *if it admits a smooth exhaustion function $\rho\colon X \to \mathbb{R}_+$ that is q-convex on $\{\rho > c\}$ for some $c > 0$; X is* q-complete *if ρ can be chosen q-convex on all of X.*

Remark 3.1.3. By a theorem of Narasimhan (Theorem 2.3.2 on p. 51) every 1-complete complex space is a Stein space. A 1-convex complex space X contains a maximal compact subvariety of positive dimension, S, called the *exceptional variety* of X, and there exists a proper holomorphic map $\pi\colon X \to \widehat{X}$ onto a Stein space $\widehat{X}$ that maps each connected component of S to a point in $\widehat{X}$ and is biholomorphic outside of $\pi^{-1}(\pi(S))$; furthermore, the composition operator $\mathcal{O}(\widehat{X}) \ni f \mapsto f \circ \pi \in \mathcal{O}(X)$ is an isomorphism of $\mathcal{O}(\widehat{X})$ onto $\mathcal{O}(X)$. The Stein space $\widehat{X}$ is called the *Remmert reduction* of X. $\square$

Theorem 3.1.4. [88, 101] *A q-complete complex subvariety Y of a complex space X has a basis of open q-complete neighborhoods in X.*

By essentially the same proof Colţoiu obtained Theorem 3.1.4 when Y is a complete locally pluripolar set in X which is *ambiently q-complete*, in the sense that there exist an open neighborhood U of Y in X and a q-convex function ϕ on U such that $\phi|_Y$ is an exhaustion function on Y [88, Theorem 2]. We begin by showing that every q-complete complex subvariety is also ambiently q-complete.

Lemma 3.1.5. [101, Theorem 4] *Let Y be an analytic subvariety of a complex space X and let ψ be a smooth q-convex function on Y. For every continuous function $\delta > 0$ on Y there exists a smooth q-convex function ϕ on a neighborhood V of Y in X such that $\psi \le \phi|_Y < \psi + \delta$.*

Proof. Replacing X by a suitably chosen open neighborhood of Y we may assume that Y is closed in X. Choose a stratification $Y = Y_n \supset Y_{n-1} \supset \cdots \supset Y_0 \supset Y_{-1} = \emptyset$ by closed complex subvarieties such that $S_k = Y_k \backslash Y_{k-1}$ is a complex k-dimensional submanifold of Y for every $k = 0, \ldots, n = \dim Y$. We prove the following statement by induction on k (for $k = n$ we obtain the statement of Lemma 3.1.5):

(*) *There exists a smooth function ϕ_k on X which is q-convex along Y and on a neighborhood $\overline{V}_k$ of Y_k in X, such that $\psi \le \phi_k|_Y < \psi + \delta$.*

Any smooth extension ϕ_{-1} of ψ to X satisfies the stated requirements with $Y_{-1} = V_{-1} = \emptyset$. Assume inductively that V_{k-1} and ϕ_{k-1} have been chosen. The closed set $Y_k \backslash V_{k-1} \subset S_k = Y_k \backslash Y_{k-1}$ admits a locally finite cover by open coordinate patches $A_\lambda \subset \Omega_\lambda \subset \mathbb{C}^{N_\lambda}$ as in Def. 3.1.2 (p. 58) such that $Y_k \cap A_\lambda$ is given by equations $z'_\lambda = (z_{\lambda,k+1}, \ldots, z_{\lambda,N_\lambda}) = 0$. (Here, $z_\lambda = (z_{\lambda,1}, \ldots, z_{\lambda,N_\lambda})$ denote the complex coordinates on $\mathbb{C}^{N_\lambda}$, as well as their restrictions to $A_\lambda \subset X$.) Let $\theta_\lambda \colon A_\lambda \to [0,1]$ be smooth functions with compact support in A_λ such that $\sum_\lambda \theta_\lambda = 1$ on $Y_k \backslash V_{k-1}$. For $\epsilon_\lambda > 0$ chosen small enough the function

$$\phi_k(x) = \phi_{k-1}(x) + \sum_\lambda \theta_\lambda(x)\, \epsilon_\lambda^3 \log\left(1 + \epsilon_\lambda^{-4} |z'_\lambda|^2\right), \quad x \in X$$

satisfies $\psi \le \phi_k|_Y \le \phi_k|_Y < \psi + \delta$. We now check that ϕ_k remains q-convex along Y and on $\overline{V}_{k-1}$ (where ϕ_{k-1} was such), and that ϕ_k becomes q-convex near every point $x_0 \in Y_k \backslash V_{k-1}$. If x_0 does not belong to the support of θ_λ for any λ then $\phi_k = \phi_{k-1}$ in a neighborhood of x_0 and the above holds. Assume now that $x_0 \in \operatorname{Supp} \theta_\mu$. Choose a small neighborhood $W \Subset \Omega_\mu$ of x_0 such that the following properties hold:

(a) if $x_0 \subset Y_k \backslash V_{k-1} \subset S_k$ then $\theta_\mu(x_0) > 0$ and $A_\mu \cap W \Subset \{\theta_\mu > 0\}$,
(b) if $x_0 \in A_\lambda$ for some λ (there is a finite such of such λ's) then $A_\mu \cap W \Subset A_\lambda$ and $z_\lambda|_{A_\mu \cap W}$ admits a holomorphic extension $\widetilde{z}_\lambda$ to $\overline{W}$,
(c) if $x_0 \in \overline{V}_{k-1}$ then $\phi_{k-1}|_{A_\mu \cap W}$ admits a q-convex extension $\widetilde{\phi}_{k-1}$ to $\overline{W}$, and
(d) if $x_0 \in Y \backslash \overline{V}_{k-1}$ then $\phi_{k-1}|_{Y \cap W}$ admits a q-convex extension $\widetilde{\phi}_{k-1}$ to $\overline{W}$.

If $x \notin Y \cup \overline{V}_{k-1}$, we take an arbitrary smooth extension $\widetilde{\phi}_{k-1}$ of $\phi_{k-1}|_{A_\mu \cap W}$ to $\overline{W}$. Let $\widetilde{\theta}_\lambda$ denote a smooth extension of $\theta_\lambda|_{A_\mu \cap W}$ to $\overline{W}$. Then the function

$$\widetilde{\phi}_k = \widetilde{\phi}_{k-1} + \sum_\lambda \widetilde{\theta}_\lambda\, \epsilon_\lambda^3 \log\left(1 + \epsilon_\lambda^{-4} |\widetilde{z}'_\lambda|^2\right)$$

is a smooth extension of $\phi_k|_{A_\mu \cap W}$ to $\overline{W}$, resp. of $\phi_k|_{Y \cap W}$ to $\overline{W}$ in case (d). Since the function $\log\left(1 + \epsilon_\lambda^{-4} |\widetilde{z}'_\lambda|^2\right)$ is plurisubharmonic and

$$\partial \log\left(1 + \epsilon_\lambda^{-4} |\widetilde{z}'_\lambda|^2\right) = \left(\epsilon_\lambda^4 + |\widetilde{z}'_\lambda|^2\right)^{-1} \sum_{j=k+1}^{N_\lambda} \overline{\widetilde{z}}_{\lambda,j}\, \partial \widetilde{z}_{\lambda,j}$$

is bounded by $O(\epsilon_\lambda^{-2})$, we obtain $\mathrm{i}\partial\overline{\partial}\widetilde{\phi}_k \geq \mathrm{i}\partial\overline{\partial}\widetilde{\phi}_{k-1} - O(\sum \epsilon_\lambda)$. Therefore for sufficiently small numbers $\epsilon_\lambda > 0$ the function $\widetilde{\phi}_k$ remains q-convex on $\overline{W}$ in cases (c) and (d). Since all functions $\widetilde{z}'_\lambda$ vanish along $Y_k \cap W$, the following holds at every point of $Y_k \cap W$:

$$\mathrm{i}\partial\overline{\partial}\widetilde{\phi}_k \geq \mathrm{i}\partial\overline{\partial}\widetilde{\phi}_{k-1} + \sum_\lambda \theta_\lambda \epsilon_\lambda^{-1}\, \mathrm{i}\partial\overline{\partial}|\widetilde{z}'_\lambda|^2 \geq \mathrm{i}\partial\overline{\partial}\widetilde{\phi}_{k-1} + \theta_\mu \epsilon_\mu^{-1}\, \mathrm{i}\partial\overline{\partial}|\widetilde{z}'_\mu|^2.$$

Since $S_k \subset Y$, $\mathrm{i}\partial\overline{\partial}\widetilde{\phi}_{k-1}$ has at most $q-1$ nonpositive eigenvalues on $\mathrm{T}S_k$, whereas $\mathrm{i}\partial\overline{\partial}|\widetilde{z}'_\lambda|^2$ is positive definite in the normal directions to S_k; hence in case (a) the function $\widetilde{\phi}_k$ is q-convex on $\overline{W}$ for small enough $\epsilon_\mu > 0$. Observe that only finitely many conditions are required on each number ϵ_λ if we choose a locally finite cover of $\cup_\lambda \mathrm{Supp}\,\theta_\lambda$ by neighborhoods W chosen as above. Hence, for small enough $\epsilon_\lambda > 0$, ϕ_k is q-convex on a neighborhood $\overline{V}'_k$ of $Y_k \backslash V_{k-1}$. The function ϕ_k and the set $V_k = V_{k-1} \cup V'_k$ then satisfy (*) and the induction step is complete. □

The second main ingredient are *almost plurisubharmonic functions* with a logarithmic pole along a closed complex subvariety.

Definition 3.1.6. *A function $v\colon X \to \mathbb{R} \cup \{-\infty\}$ on a complex manifold X is* almost plurisubharmonic *if every point $x_0 \in X$ admits an open neighborhood $U \subset X$ such that $v|_U = w_U + h_U$, where w_U is plurisubharmonic on U and h_U is smooth on U. A function on a complex space is almost plurisubharmonic if it can be locally extended to an almost plurisubharmonic function on the ambient manifold of a local holomorphic embedding.*

An equivalent definition is that v is almost plurisubharmonic if it is upper semicontinuous and its Levi form has a bounded negative part on each compact subset of X (i.e., its eigenvalues are bounded away from $-\infty$).

Lemma 3.1.7. ([88, Theorem 1], [101, Lemma 5]) *For every closed analytic subvariety Y in a complex space X there exists an almost plurisubharmonic function v on X which is smooth on $X \backslash Y$ and has a logarithmic pole on $Y = \{v = -\infty\}$.*

Proof. Let $\mathcal{J}_Y$ denote the sheaf of ideals of Y in X. There is a locally finite cover of X by patches A_λ, isomorphic to closed analytic subsets $A_\lambda \subset \mathbb{B}_\lambda$ in unit balls $\mathbb{B}_\lambda \subset \mathbb{C}^{N_\lambda}$, such that $\mathcal{J}_Y$ admits a finite system of generators $g_\lambda = (g_{\lambda,j})$ on an open neighborhood of each set $\bar{A}_\lambda$ in X. Let z denote the complex coordinate on $\mathbb{B}_\lambda$ and $z(x)$ its value at $x \in A_\lambda$. The function

$$v_\lambda(x) = \log|g_\lambda(x)|^2 - \frac{1}{1-|z(x)|^2}, \qquad x \in A_\lambda$$

is almost plurisubharmonic on A_λ since the first summand is plurisubharmonic and the second one is smooth. Note that $v_\lambda = -\infty$ on $(A_\lambda \cap Y) \cup (\bar{A}_\lambda \cap bB_\lambda)$.

We patch the functions v_λ into an almost plurisubharmonic function v on X by using a *regularized maximum function* defined as follows. Select a nonnegative smooth function ρ on $\mathbb{R}$, with support in $[-\frac{1}{2}, \frac{1}{2}]$, such that $\int \rho(u)\mathrm{d}u = 1$ and $\int_{\mathbb{R}} u\rho(u)\mathrm{d}u = 0$, and set

$$\operatorname{rmax}\{t_1, \dots, t_p\} = \int_{\mathbb{R}^p} \max\{t_1 + u_1, \dots, t_p + u_p\} \prod_{j=1,\dots,p} \rho(u_j)\mathrm{d}u_j. \tag{3.1}$$

Since rmax is clearly increasing and convex in all variables, the composition with rmax preserves plurisubharmonicity. Moreover, we have

$$\operatorname{rmax}\{t_1, \dots, t_j, \dots, t_p\} = \operatorname{rmax}\{t_1, \dots, t_{j-1}, t_{j+1}, \dots, t_p\}$$

as soon as $t_j < \max\{t_1, \dots, t_{j-1}, t_{j+1}, \dots, t_p\} - 1$. Set

$$v(x) = \operatorname{rmax}\{v_\lambda(x) \colon x \in A_\lambda\}, \qquad x \in X.$$

Near every point x there are only finitely many terms under rmax as the cover A_λ is locally finite. Since the generators g_λ and g_μ for $\mathcal{J}_Y$ can be expressed in terms of each other in a neighborhood of $\bar{A}_\lambda \cap \bar{A}_\mu$, the quotient $|g_\lambda|/|g_\mu|$ is bounded on this set, and hence none of the values $v_\lambda(x)$ for $x \in A_\lambda$ sufficiently near bA_λ (where v_λ is large negative) contributes to the value of $v(x)$. It follows that v is smooth on $X \backslash Y$, it has a logarithmic pole on $Y = \{v = -\infty\}$, and it is almost plurisubharmonic on X since it is locally the regularized maximum of finitely many almost plurisubharmonic functions v_λ. $\square$

Proof (of Theorem 3.1.4). By Lemma 3.1.5, applied to a q-convex exhaustion of Y and $\delta = 1$, there exists a q-convex function ρ on a neighborhood W_0 of Y in X such that $\rho|_Y$ is an exhaustion of Y. Choose a neighborhood W of Y such that $\overline{W} \subset W_0$ and $\rho|_{\overline{W}}$ is an exhaustion (every sufficiently thin neighborhood of Y satisfies these requirements). Let v be an almost plurisubharmonic function on X with a logarithmic pole along $Y = \{v = -\infty\}$, furnished by Lemma 3.1.7. Choose a smooth convex increasing function $g \colon \mathbb{R} \to \mathbb{R}$ and set

$$\widetilde{\rho} = v + g \circ \rho, \quad V = \{x \in W \colon \widetilde{\rho}(x) < 0\}.$$

Observe that $Y \subset V$ since $v|_V = -\infty$. Choosing g to grow fast enough we get $\widetilde{\rho} > 0$ on bW and hence $\overline{V} \subset W$. Fast growth of g also insures that the positive Levi form of $g \circ \rho$ on a suitable family of $(q-1)$-codimensional subspaces of the tangent bundle (on which $\mathrm{dd}^c\rho > 0$) compensates the locally bounded negative part of the Levi form of v. For such g the function $\widetilde{\rho}$ is q-convex on W. Let $\tau \colon (-\infty, 0) \to \mathbb{R}_+$ be a smooth convex increasing function such that $\tau(t) = 0$ for $t \le -1$ and $\lim_{t \to 0} \tau(t) = +\infty$. Then

$$\psi = \rho + \tau \circ \widetilde{\rho} \colon V \to \mathbb{R}$$

is q-convex exhaustion of V, and hence V is q-complete. $\square$

3.2 Stein Neighborhoods of Stein Subvarieties

In this section we generalize Siu's theorem (Theorem 3.1.1 on p. 57) to the form that will be used in Oka theory. Recall that a compact set K in a complex space X is holomorphically convex if K is $\mathcal{O}(\Omega)$-convex in an open Stein neighborhood $\Omega \subset X$ (Def. 2.3.5 on p. 51).

Theorem 3.2.1. [175, Theorem 1.2] *Let Y be a closed Stein subvariety of a complex space X. Assume that K is a compact holomorphically convex set in X such that $K \cap Y$ is $\mathcal{O}(Y)$-convex. Then $K \cup Y$ has a basis of open Stein neighborhoods $V \subset X$ such that K is $\mathcal{O}(V)$-convex.*

Remark 3.2.2. If V is a Stein neighborhood of $K \cup Y$ such that K is $\mathcal{O}(V)$-convex, then $K \cup Y$ has a basis of open Stein neighborhoods $W \subset V$ that are Runge in V [91, Proposition 2.1]. The necessity of $\mathcal{O}(Y)$-convexity of $K \cap Y$ in Theorem 3.2.1 is seen by taking $X = \mathbb{C}^2$, $Y = \mathbb{C} \times \{0\}$, and $K = \{(z,w) \in \mathbb{C}^2 \colon 1 \le |z| \le 2,\ |w| \le 1\}$: Every Stein neighborhood of $K \cup Y$ also contains the bidisc $\{(z,w) \in \mathbb{C}^2 \colon |z| \le 2,\ |w| \le 1\}$. $\square$

Proof. Let U be an open set in X containing $K \cup Y$. We shall find an open Stein set V in X, with $K \cup Y \subset V \subset U$, such that K is $\mathcal{O}(V)$-convex.

By the assumption K has an open Stein neighborhood $\Omega \subset X$ such that K is $\mathcal{O}(\Omega)$-convex. Hence K has a basis of open Stein neighborhoods, and replacing Ω by one of them we may assume that $\Omega \subset U$.

Since the set $K_0 = K \cap X_0$ is $\mathcal{O}(Y)$-convex, it has a compact $\mathcal{O}(Y)$-convex neighborhood K_0' in Y that is contained in Ω. Choose a compact neighborhood K' of K such that $K' \subset \Omega$ and $K' \cap Y = K_0'$. As K is $\mathcal{O}(\Omega)$-convex, Proposition 2.3.1 furnishes a smooth strongly plurisubharmonic function ρ_0 on Ω such that $\rho_0 < 0$ on K and $\rho_0 > 1$ on $\Omega \backslash K'$. Set $U_c = \{x \in \Omega \colon \rho_0(x) < c\}$. Fix $c \in (0, \frac{1}{2})$; then $K \subset U_c \subset U_{2c} \subset K'$ (Fig. 3.1).

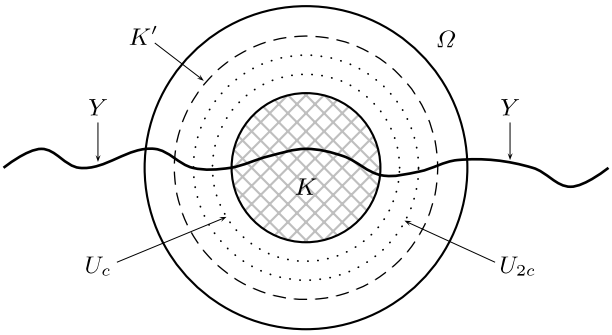

Fig. 3.1. Sets in the proof of Theorem 3.2.1

The restriction $\rho_0|_{Y\cap\Omega}$ is smooth and strongly plurisubharmonic. Since $K_0' = K' \cap Y$ is $\mathcal{O}(Y)$-convex, there is a smooth strongly plurisubharmonic exhaustion function $\rho_0' \colon Y \to \mathbb{R}$ that agrees with ρ_0 on K_0', and that satisfies $\rho_0' > c$ on $Y \backslash \overline{U}_c$. (To obtain such ρ_0', take a smooth strongly plurisubharmonic exhaustion function $\tau \colon Y \to \mathbb{R}$ such that $\tau < 0$ on K_0' and $\tau > 1$ on $Y \backslash \Omega$; also choose a smooth convex increasing function $\xi \colon \mathbb{R} \to \mathbb{R}_+$ with $\xi(t) = 0$ for $t \le 0$, and a smooth function $\chi \colon X \to [0,1]$ such that $\chi = 1$ on $\{x \in Y \colon \tau(x) \le 1/2\}$ and $\chi = 0$ on $\{x \in Y \colon \tau(x) \ge 1\}$. The function $\rho_0' = \chi\rho_0 + \xi \circ \tau$ then satisfies the stated properties provided that $\xi(t)$ is chosen to grow sufficiently fast for $t > 0$.) Let $\widetilde{\rho}_0 \colon K' \cup Y \to \mathbb{R}$ be defined by $\widetilde{\rho}_0|_{K'} = \rho_0|_{K'}$ and $\widetilde{\rho}_0|_Y = \rho_0'$.

Choose a smooth convex increasing function $h \colon \mathbb{R} \to \mathbb{R}$ satisfying $h(t) \ge t$ for all $t \in \mathbb{R}$, $h(t) = t$ for $t \le c$, and $h(t) > t + 1$ for $t \ge 2c$. The function $\rho_1 := h \circ \widetilde{\rho}_0$ is smooth strongly plurisubharmonic on $K' \cup Y$; on the set $\overline{U}_c = \{\rho_0 \le c\}$ we have $\rho_1 = \widetilde{\rho}_0 = \rho_0$, while outside of U_{2c} we have $\rho_1 > \widetilde{\rho}_0 + 1$.

By Lemma 3.1.5 there exists a smooth strongly plurisubharmonic function ρ_2 in an open neighborhood of Y, satisfying

$$\rho_1(x) - 1 < \rho_2(x) < \rho_1(x), \qquad x \in Y.$$

On a neighborhood of $\overline{U}_c \cap Y = \{x \in Y \colon \widetilde{\rho}_0(x) \le c\}$ we have $\rho_2 < \rho_1 = \widetilde{\rho}_0$, while on $Y \backslash U_{2c}$ we have $\rho_2 > \rho_1 - 1 > \widetilde{\rho}_0$. It follows that the function

$$\rho = \max\{\widetilde{\rho}_0, \rho_2\}$$

is well defined and strongly plurisubharmonic in an open set $W \subset X$ satisfying $\overline{U}_c \cup Y \subset W \subset U$. (To see this, observe that the union of the domains of $\widetilde{\rho}_0$ and ρ_2 contains a neighborhood of $\overline{U}_c \cup Y$, and before running out of the domain of one of these two functions, the second function is the larger one and hence takes over.) After shrinking W around $\overline{U}_c \cup Y$, the function $\rho \colon W \to \mathbb{R}$ satisfies the following properties:

(i) $\rho = \widetilde{\rho}_0 = \rho_0$ on $\overline{U}_c$ (hence $\rho < 0$ on K),
(ii) $\rho > c$ on $W \backslash \overline{U}_c$, and
(iii) $\rho = \rho_2$ on $W \backslash U_{2c}$.

Using rmax instead of max (see the proof of Theorem 3.1.4) we can also insure that ρ is smooth. After shrinking W we may assume that ρ is a strongly plurisubharmonic exhaustion of $\overline{W} \supset \overline{U}_c \cup Y$ satisfying $\rho > c$ on bW. The compact set $L = \{x \in W \colon \rho(x) \le 0\}$ contains K in its interior.

Let v be an almost plurisubharmonic function on X with a logarithmic pole on $Y = \{v = -\infty\}$, furnished by Lemma 3.1.7. By subtracting a constant from v we may assume that $v|_K < 0$. Let $g \colon \mathbb{R} \to \mathbb{R}$ be a convex increasing function with $g(t) = t$ for $t \le 0$. For a small $\epsilon > 0$ we set

$$\widetilde{\rho} = \epsilon\, v + g \circ \rho, \qquad V = \{x \in W \colon \widetilde{\rho}(x) < 0\}.$$

Clearly $\widetilde{\rho}|_Y = -\infty$ and hence $Y \subset V$. As $v|_K < 0$ and $g \circ \rho|_K = \rho|_K < 0$, we have $K \subset V$. To complete the proof we show that for suitable choices of ϵ and g the set V is Stein and K is $\mathcal{O}(V)$-convex.

On the set $L = \{\rho \leq 0\}$ we have $g \circ \rho = \rho$ which is strongly plurisubharmonic. Since the Levi form of v is bounded on L, $\widetilde{\rho}$ is strongly plurisubharmonic on L for a sufficiently small $\epsilon > 0$; fix such ϵ. Choosing g to grow sufficiently fast on $(0, +\infty)$ we can insure that the Levi form of $g \circ \rho$ compensates the bounded negative part of the Levi form of $\epsilon\, v$ on each compact in $\overline{W}$, and hence $\widetilde{\rho}$ is strongly plurisubharmonic on $\overline{W}$. Furthermore, rapid growth of g insures that $\widetilde{\rho}|_{bW} > 0$, and hence $\overline{V} \subset W$.

Let $\tau\colon (-\infty, 0) \to \mathbb{R}_+$ be a smooth convex increasing function such that $\tau(t) = 0$ for $t \leq -1$ and $\lim_{t\to 0} \tau(t) = +\infty$. Then $\psi = \rho + \tau \circ \widetilde{\rho}$ is a strongly plurisubharmonic exhaustion function on V, and hence V is Stein according to Theorem 2.3.2 on p. 51.

It remains to see that V can be chosen such that K is $\mathcal{O}(V)$-convex. Since K is $\mathcal{O}(\Omega)$-convex, there is a plurisubharmonic function $\theta\colon \Omega \to \mathbb{R}_+$ that vanishes precisely on K and is strongly plurisubharmonic on $\Omega \backslash K$ (Proposition 2.3.1 on p. 50). As before we find an open set $W \subset X$ containing $K \cup Y$ and a smooth plurisubharmonic function $\widetilde{\theta}\colon W \to \mathbb{R}_+$ such that, for some $c > 0$, we have $\theta = \widetilde{\theta}$ in $O_c = \{x \in \Omega\colon \theta(x) < c\} \supset K$ and $\widetilde{\theta} > c$ on $W \backslash \overline{O}_c$. (See (i) and (ii) above.) The first part of the proof gives an open Stein domain V with $K \cup Y \subset V \subset W$. The restriction $\widetilde{\theta}|_V \geq 0$ is then a plurisubharmonic function on V that vanishes precisely on the set K, so K is $\mathcal{O}(V)$-convex. $\square$

A related result was obtained by J. Prezelj [402]; it concerns the following situation. Let X be a 1-convex complex space with the exceptional subvariety S (the maximal compact complex subvariety of positive dimension in X). Assume that $h\colon Z \to X$ is a holomorphic submersion of a complex space Z onto X and $f\colon X \to Z$ is a holomorphic section. The subvariety $f(X) \subset Z$ does not have a Stein neighborhood in Z since it contains the compact subvariety $f(S)$. However, if $g \in \mathcal{O}(X)$ is a holomorphic functions that vanishes on S then $X_g = X \backslash \{g = 0\}$ is an open Stein subset of X, and hence $f(X_g)$ admits an open Stein neighborhood $\Omega_g \subset Z$ by Siu's theorem (Theorem 3.1.1 on p. 57). Call such a neighborhood Ω_g *conic along* $g^{-1}(0)$ if its diameter in the fiber direction decays at most polynomially in $|g(x)|$ as the point $x \in X$ approaches the zero locus $g^{-1}(0)$. Over any compact subset of X this notion is independent of the choice of a distance function on Z.

Theorem 3.2.3. [402, Theorem 3.2] (Hypotheses as above.) *For every compact $\mathcal{O}(X)$-convex subset K of X there exists an open Stein neighborhood Ω_g of $f(K \backslash g^{-1}(0))$ in Z which is conic along $g^{-1}(0)$.*

Note that finitely many functions $g_1, \ldots, g_k \in \mathcal{O}(X)$ define the exceptional set S; the corresponding conic Stein neighborhoods Ω_{g_j} then cover $f(X \backslash S)$.

Proof (sketch). There exists a plurisubharmonic exhaustion $\phi\colon X \to \mathbb{R}_+$ such that $\phi^{-1}(0) = S$, ϕ is strictly plurisubharmonic on $X\backslash S$, and the Levi form of ϕ decreases to zero polynomially with the distance from the exceptional set S. Such a function is obtained by taking $\phi = \varphi \circ \pi$, where $\pi\colon X \to \widehat{X}$ is the Remmert reduction of X and φ is a suitably chosen strongly plurisubharmonic exhaustion on the Stein space $\widehat{X}$ such that $\varphi^{-1}(0) = \pi(S)$. Now ϕ and g can be regarded as functions on the subvariety $f(X) \subset Z$. We extend them as constants along the fibers of the submersion $h\colon Z \to X$ to functions on Z and denote the extensions by the same letters. The function g is holomorphic and the function ϕ is plurisubharmonic on Z.

Pick a number $c > 0$ such that the domain $U = \{x \in X\colon \phi(x) < c\}$ contains $K \cup S$. By adding to ϕ a function of type $\xi = \sum \xi_i \overline{\xi_i}$, where the functions ξ_i on Z are asymptotically holomorphic to a high order in the fiber directions along $f(X)$ and vanish on $f(X) \cup h^{-1}(S)$, we obtain a strictly plurisubharmonic function $\phi + \xi$ on a conic neighborhood of $f(\overline{U}\backslash g^{-1}(0))$ in Z such that $\phi + \xi$ is an exhaustion in the horizontal direction.

The function $\log \xi$ has logarithmic poles (only) along $f(X) \cup h^{-1}(S)$. In a suitable neighborhood of $f(X\backslash S)$ in Z that is conic along $f(S)$, the negative part of the Levi form of $\log \xi$ decreases at most polynomially with the distance from $f(S)$. If the functions ξ_i have been chosen asymptotically holomorphic to a sufficiently high order along $f(X)$ then the function $\phi + \xi + \log \xi$ is plurisubharmonic. The problem is that this function equals $-\infty$ on $h^{-1}(S)$. By adding a pluriharmonic term $-\log |g|^{2M}$ we get the function

$$G = \phi + \xi + \log \xi - \log |g|^{2M}.$$

which is plurisubharmonic away from $g^{-1}(0)$, and for $M > 0$ chosen large enough we have $\liminf_{z \to z_0} G(z) > 0$ for each $z_0 \in h^{-1}(S)\backslash f(S)$. For $t \ll 0$ the sublevel set $G^{-1}([-\infty, t))$ is conic along $g^{-1}(0)$ and it lies in the conic neighborhood where G is plurisubharmonic. Choose a convex increasing function $\chi\colon (-\infty, t) \to \mathbb{R}_+$ with a pole at t such that χ is zero on $(-\infty, 2t]$. The function $\chi \circ G$ is then plurisubharmonic and exhausts a conic neighborhood in the vertical direction. Hence the function

$$\widetilde{G} = \phi + \xi - \log |g|^{2M} + \chi \circ G$$

is a strictly plurisubharmonic exhaustion function on a conic neighborhood of the set $f(U\backslash g^{-1}(0))$ in Z. $\square$

Remark 3.2.4. It is a natural question when does a compact subvariety $A \subset X$ with smooth boundary bA and Stein interior $A\backslash bA$ in a complex space X admit an open Stein neighborhood in X, or a basis of Stein neighborhoods. This problem is highly nontrivial even for domains in $\mathbb{C}^n$; the answer is negative in general as is shown by the *worm domain* of Diederich and Fornæss [111]. A positive answer was given for Stein subvarieties with strongly pseudoconvex boundaries by Starčič [457, Theorem 1.1]; for the special case of curves with

boundaries see [121, Theorem 2.1]. (It is also possible to include a compact holomorphically convex set K as in Theorem 3.2.1 above.) Furthermore, if D is a relatively compact strongly pseudoconvex domain in a Stein manifold S and $f\colon \bar{D} \to X$ is a continuous map that is holomorphic in D, then the graph $\Gamma_f = \{(z, f(z)) \colon z \in \bar{D}\}$ admits a basis of Stein neighborhoods in $S \times X$ [178, Theorem 1.2]. These results, which we shall not need in this book, are proved by the technique of gluing holomorphic sprays developed in §5.8 – §5.9. □

3.3 Holomorphic Retractions onto Stein Submanifolds

In this section we prove the Docquier-Grauert tubular neighborhood theorem for Stein submanifolds in arbitrary complex manifolds (Theorem 3.3.3).

Proposition 3.3.1. *Let X be a Stein manifold. There exist an open Stein neighborhood Ω of the zero section in the tangent bundle $\mathrm{T}X$ and a holomorphic map $s\colon \Omega \to X$ that maps the fiber Ω_x over any point $x \in X$ biholomorphically onto a neighborhood of x in X, with $s(0_x) = x$. It is possible to choose Ω to be Runge in $\mathrm{T}X$ and to have convex fibers.*

Proof. By Cartan's Theorem B there are finitely many holomorphic vector fields $V_1, \ldots, V_N$ on X spanning the tangent space at every point. Let ϕ_t^j denote the local flow of V_j for complex values of t (1.57). The map

$$F(x, t_1, \ldots, t_N) = \phi_{t_1}^1 \circ \cdots \circ \phi_{t_N}^N(x), \qquad x \in X,\ t_j \in \mathbb{C} \tag{3.2}$$

is defined and holomorphic in an open neighborhood of $X \times \{0\}^N$ in $X \times \mathbb{C}^N$, and it satisfies

$$F(x, 0) = x, \qquad \frac{\partial}{\partial t_j} F(x, t)|_{t=0} = V_j(x)$$

for every $x \in X$ and $j = 1, \ldots, N$. As the V_j's span the tangent space $\mathrm{T}_x X$ at every point, the partial differential

$$\Theta_x = \partial_t|_{t=0} F(x, t) \colon \mathbb{C}^N \to \mathrm{T}_x X \tag{3.3}$$

is surjective for every $x \in X$. Hence the subset $E' \subset X \times \mathbb{C}^N$ with fibers $E'_x = \ker \Theta_x$ is a holomorphic vector subbundle of $X \times \mathbb{C}^N$. Since X is Stein, we have $X \times \mathbb{C}^N = E \oplus E'$ for a holomorphic vector subbundle E of $X \times \mathbb{C}^N$. Then $\Theta_x \colon E_x \to \mathrm{T}_x X$ is bijective for every $x \in X$, and hence $E \cong \mathrm{T}X$; we identify E with $\mathrm{T}X$. The implicit function theorem gives an open neighborhood $\Omega \subset \mathrm{T}X$ of the zero section such that the restricted map $s = F|_\Omega$ satisfies the conclusion of the proposition. It is possible to choose Ω to be Stein, Runge in $\mathrm{T}X$, and to have convex fibers: Simply take $\Omega_x = \{v \in E_x \colon \mathrm{e}^{\phi(x)}|v|^2 < 1\}$, where $\phi \colon X \to \mathbb{R}_+$ is a fast growing strongly plurisubharmonic function. □

Essentially the same proof gives the following result.

Proposition 3.3.2. *Assume that $\pi\colon Z \to X$ is a holomorphic submersion of a complex space Z onto a Stein space X. There is a Stein neighborhood Ω of the zero section in the vertical tangent bundle $\mathrm{VT}Z = \ker \mathrm{d}\pi$ (see p. 21) and a holomorphic map $s\colon \Omega \to Z$ such that for every $z \in Z$,*

(i) *s maps the fiber Ω_z of Ω over z biholomorphically onto a neighborhood of z in the fiber $Z_{\pi(z)}$, and*
(ii) *$s(0_z) = z$. (Here 0_z is the origin in $\mathrm{VT}_z Z$.)*

We may choose Ω to be Runge in $\mathrm{VT}Z$ and to have convex fibers. (A map s with these properties is a local fiber dominating spray*; see Def. 5.9.1.)*

As a consequence of Proposition 3.3.1 and of Siu's theorem (Corollary 3.1.1) we obtain the following *tubular neighborhood theorem* of Docquier and Grauert ([113], [241, p. 257, Theorem 8]).

Theorem 3.3.3. (Tubular Neighborhood Theorem.) *Let S be a Stein submanifold of a complex manifold X. Denote by $\mathrm{N}_{S/X}$ the normal bundle of S in X. There exist an open Stein neighborhood U of S in X, biholomorphic to an open neighborhood Ω of the zero section in $\mathrm{N}_{S/X}$, and a homotopy of holomorphic maps $\iota_t\colon U \to U$ $(t \in [0,1])$ such that ι_0 is the identity map on U, $\iota_t|_S$ is the identity map on S for all $t \in [0,1]$, and $\iota_1(U) = S$.*

The family $\{\iota_t\}_{t\in[0,1]}$ is a strong deformation retraction of U onto S consisting of holomorphic mappings. There is no analogue of Theorem 3.3.3 when S is a Stein space with singularities.

Proof. By Corollary 3.1.1 we may assume that X is Stein. Choose vector fields $V_1, \ldots, V_N$ as in the proof of Proposition 3.3.1 and let F and Θ_x be defined by (3.2), (3.3), respectively. The subset $E' \subset S \times \mathbb{C}^N$ with the fibers

$$E'_x = \Theta_x^{-1}(\mathrm{T}_x S), \qquad x \in S$$

is a holomorphic vector subbundle of $S \times \mathbb{C}^N$. Since S is Stein, there is holomorphic vector subbundle $E \subset S \times \mathbb{C}^N$ such that $S \times \mathbb{C}^N = E \oplus E'$. Then $\Theta\colon E \to \mathrm{T}X|_S$ is an injective holomorphic vector bundle map such that $\mathrm{T}X|_S = \mathrm{T}S \oplus \Theta(E)$; thus $\Theta\colon E \xrightarrow{\cong} \mathrm{N}_{S/X}$ is an isomorphism of E onto the normal bundle of S in X. By the inverse function theorem F maps an open neighborhood Ω of the zero section in E biholomorphically onto an open neighborhood $U = F(\Omega)$ of S in X. Then F conjugates the family of radial dilations

$$\iota'_t(v) = (1-t)v, \quad v \in \Omega,\ t \in [0,1]$$

to a family of holomorphic maps $\iota_t\colon U \to U$ satisfying the stated properties. As in the proof of Proposition 3.3.1 we can choose the domain Ω to be Stein, Runge in $\mathrm{N}_{S/X}$, and with convex fibers. $\square$

Similarly one proves the following foliated version of Theorem 3.3.3.

Theorem 3.3.4. [174, Lemma 3.4] *Assume that $\pi\colon Z \to X$ is a holomorphic submersion of a complex space Z onto a complex space X. Let S be a locally closed Stein subvariety of Z whose fibers $S_x = S \cap Z_x$ are complex manifolds of dimension independent of $x \in X$. Then there exist an open Stein neighborhood U of S in Z and a homotopy of holomorphic maps $\iota_t\colon U \to U$ satisfying properties (a)–(c) in Theorem 3.3.3 and $\pi \circ \iota_t = \pi$ $(t \in [0,1])$. In particular, $\iota_1\colon U \to S$ is a fiber preserving holomorphic retraction of U onto S.*

3.4 A Semiglobal Holomorphic Extension Theorem

In this section we prove the following semiglobal extension and approximation theorem that is used in Oka theory.

Theorem 3.4.1. ([175, Theorem 3.1], [180]) *Assume that X is a complex space, X' is a closed Stein subvariety of X, and K is a compact holomorphically convex set in X (Def. 2.3.5) such that $K \cap X'$ is $\mathcal{O}(X')$-convex. Assume that $\pi\colon Z \to X$ is a holomorphic submersion of a complex space Z onto X. Given an open set $U \subset X$ containing K and a section $f\colon U \cup X' \to Z|_{U \cup X'}$ whose restrictions to U and to X' are holomorphic, there exist for every $\epsilon > 0$ an open Stein neighborhood $V = V^\epsilon$ of $K \cup X'$ in X and a holomorphic section $f^\epsilon\colon V \to Z|_V$ such that*

$$f_\epsilon|_{X'} = f|_{X'}, \qquad \sup_{x \in K} \operatorname{dist}(f^\epsilon(x), f(x)) < \epsilon.$$

Theorem 3.4.1 fails in general if $\pi\colon Z \to X$ is a holomorphic map that is not a submersion. For example, the map $\mathbb{C}^2 \ni (z,w) \mapsto z^p w^q \in \mathbb{C}$ for coprime integers $p, q \in \mathbb{Z}$ admits a continuous section at $(0,0)$, but there are no local holomorphic sections.

The following corollary follows immediately by replacing the submersion $\pi\colon Z \to X$ by $\widetilde{\pi}\colon Z \times S \to X \times S$, where $\widetilde{\pi}(z,s) = (\pi(z), s)$.

Corollary 3.4.2. *Assume that $\pi\colon Z \to X$ is a holomorphic submersion of a complex space Z onto a complex space X. Fix a point $x_0 \in X$. Let S be a Stein space and $f\colon S \to Z_{x_0} = \pi^{-1}(x_0)$ a holomorphic map into the fiber over x_0. For every relatively compact domain $U \Subset S$ there exist an open neighborhood $B \subset X$ of x_0 and a holomorphic map $F\colon B \times U \to Z$ such that $F(x,s) \in Z_x$ for all $s \in U$ and $F(x_0, \cdot\,) = f|_U$.*

Proof (of Theorem 3.4.1). By Theorem 3.2.1 (p. 62) there is an open Stein neighborhood $V \subset X$ of $K \cup X'$ such that K is $\mathcal{O}(V)$-convex. We replace X by V and assume in the sequel that K is $\mathcal{O}(X)$-convex. By shrinking $U \supset K$ we may also assume that U is Stein and $\overline{U}$ is compact.

Consider first the case when $Z = X \times \mathbb{C}^p$ and $\pi\colon Z \to X$ is the projection $\pi(x,\zeta) = x$. We identify sections $X \to Z$ with maps $f\colon X \to \mathbb{C}^p$ to the fiber. By the Cartan extension theorem (Corollary 2.4.3) there is a holomorphic map $\phi\colon X \to \mathbb{C}^p$ such that $\phi|_{X'} = f|_{X'}$. By Theorem A (p. 52) there exist finitely many functions $h_1, \ldots, h_m \in \mathcal{O}(X)$ that vanish on X' and generate the ideal sheaf $\mathcal{J}_{X'}$ of X' at every point of U. Since $f - \phi$ vanishes on $X' \cap U$, Theorem B gives holomorphic maps $g_j\colon U \to \mathbb{C}^p$ $(j = 1, \ldots, m)$ such that

$$f = \phi + \sum_{j=1}^{m} g_j h_j \quad \text{on } U.$$

By the Oka-Weil theorem we can approximate each g_j, uniformly on K, by a holomorphic map $\widetilde{g}_j\colon X \to \mathbb{C}^p$. The map

$$\widetilde{f} = \phi + \sum_{j=1}^{m} \widetilde{g}_j h_j \colon X \to \mathbb{C}^p \tag{3.4}$$

then approximates f uniformly on K and it agrees with f on X'.

With the aid of Theorems 3.2.1 (p. 62) and 3.3.4 (p. 68) we now reduce the general case to this special case. Consider the following subsets of Z:

$$\widetilde{K} = f(K), \quad \widetilde{X}' = f(X'), \quad \widetilde{U} = f(U).$$

Since U is Stein and $f\colon U \to \widetilde{U}$ is biholomorphic, $\widetilde{U}$ is a closed Stein subvariety of $Z|_U$. Theorem 3.2.1 furnishes an open Stein set $\widetilde{\Omega} \subset Z|_U$ containing $\widetilde{U}$ as a closed subvariety. As K is $\mathcal{O}(X)$-convex and hence $\mathcal{O}(U)$-convex, we infer that $\widetilde{K}$ is $\mathcal{O}(\widetilde{U})$-convex, and hence also $\mathcal{O}(\widetilde{\Omega})$-convex. Since X' is Stein and $f\colon X' \to \widetilde{X}'$ is biholomorphic, $\widetilde{X}'$ is also Stein. Finally, since $K \cap X'$ is $\mathcal{O}(X')$-convex, the set $\widetilde{K} \cap \widetilde{X}' = f(K \cap X')$ is $\mathcal{O}(\widetilde{X}')$-convex. By Theorem 3.2.1 the set $\widetilde{K} \cup \widetilde{X}'$ admits a basis of open Stein neighborhoods in Z.

Lemma 3.4.3. *There exist an open Stein neighborhood W of $\widetilde{K} \cup \widetilde{X}'$ in Z and a holomorphic embedding $G\colon W \hookrightarrow X \times \mathbb{C}^N$ for some $N \in \mathbb{N}$ such that for every $x \in X$, G embeds the fiber $W_x = W \cap \pi^{-1}(x)$ onto a locally closed submanifold $G(W_x)$ of $\{x\} \times \mathbb{C}^N$. ($W_x$ may be empty for some $x \in X$.)*

Proof. Let $\mathrm{VT}Z = \mathrm{T}Z/X$ denote the vertical tangent bundle, a holomorphic vector bundle of rank p over Z (p. 21). For each holomorphic function g on an open subset of Z we denote by $\mathrm{V}\mathrm{d}g$ the differential of g restricted to $\mathrm{VT}Z$; thus $\mathrm{V}\mathrm{d}g$ is a holomorphic section of the vertical cotangent bundle VT^*Z. Choose an open Stein neighborhood $W_0 \subset Z$ of $\widetilde{K} \cup \widetilde{X}'$. By Cartan's Theorem A there exist functions $g_1, \ldots, g_N \in \mathcal{O}(W_0)$ whose vertical differentials $\mathrm{V}\mathrm{d}g_j$ span VT_z^*Z at each point $z \in W_0$. Consider the holomorphic map

$$G\colon W_0 \to X \times \mathbb{C}^N, \quad G(z) = \big(\pi(z), g_1(z), \ldots, g_N(z)\big). \tag{3.5}$$

Our choice of the g_j's implies that G embeds an open Stein neighborhood $W \subset W_0$ of $\widetilde{K} \cup \widetilde{X}'$ biholomorphically onto a locally closed subvariety $G(W)$ of $X \times \mathbb{C}^N$. Clearly G respects the fibers, i.e., $pr_1 \circ G = \pi$ where $pr_1 \colon X \times \mathbb{C}^N \to X$ is the projection $pr_1(x, \zeta) = x$, and $G(W_x)$ is a locally closed Stein submanifold of $\{x\} \times \mathbb{C}^N$ for every $x \in \pi(W)$. $\square$

Let $G \colon W \hookrightarrow X \times \mathbb{C}^N$ be a holomorphic embedding furnished by Lemma 3.4.3 (p. 69). Theorem 3.3.4 (p. 68), applied to the Stein subvariety $S = G(W)$ of $X \times \mathbb{C}^N$, gives an open Stein neighborhood $\Omega \subset X \times \mathbb{C}^N$ of $G(W)$ and a fiber preserving holomorphic retraction $\iota \colon \Omega \to S$ which retracts the fiber $\Omega_x = \Omega \cap (\{x\} \times \mathbb{C}^N)$ onto $G(W_x)$ for each $x \in \pi(W)$. After shrinking $U \supset K$ we may assume that $f(U) \subset W$. Consider the composed section

$$G \circ f \colon U \cup X' \to (U \cup X') \times \mathbb{C}^N \subset X \times \mathbb{C}^N.$$

By the special case proved above we can approximate $G \circ f$ uniformly on K by a section $F \colon V \to V \times \mathbb{C}^N$ which is holomorphic in an open neighborhood V of $K \cup X'$ such that $F = G \circ f$ on X'. If the approximation is sufficiently close and if V is chosen small enough then $F(V) \subset \Omega$, and the holomorphic section $f^\epsilon = G^{-1} \circ \iota \circ F \colon V \to Z|_V$ fulfills Theorem 3.4.1. $\square$

Remark 3.4.4. Our proof gives the following addition. Assume that f satisfies the hypotheses of Theorem 3.4.1, and that $\phi \colon W \to Z|_W$ is a holomorphic section in an open neighborhood W of X' such that $f = \phi$ on X' and f and ϕ are tangent to order $\ell \in \mathbb{N}$ along $K \cap X'$. Then the maps f^ϵ in Theorem 3.4.1 can be chosen tangent to ϕ to a given order ℓ along X'. This follows from the setup (3.4) by choosing $h_1, \ldots, h_m \in \mathcal{O}(X)$ that vanish to order ℓ on the subvariety X' and that generate the sheaf $\mathcal{J}_{X'}^\ell$ (the ℓ-th power of the ideal sheaf $\mathcal{J}_{X'}$) at every point of the compact set K. $\square$

We have the analogous result for a family of sections depending continuously on a parameter in a compact Hausdorff space.

Theorem 3.4.5. *Assume the setup of Theorem 3.4.1. Let L be a compact set in X with $K \subset L$. Given an open neighborhood U of K and a family of holomorphic section $f_p \colon U \cup X' \to Z|_{U \cup X'}$, depending continuously on a parameter p in a compact Hausdorff space P, there exist for every $\epsilon > 0$ an open Stein neighborhood $V \subset X$ of $K' = K \cup (L \cap X')$ and holomorphic sections $f_p^\epsilon \colon V \to Z|_V$, depending continuously on $p \in P$, such that*

$$f_p^\epsilon|_{X'} = f_p|_{X'} \ (p \in P), \qquad \sup_{x \in K,\ p \in P} \operatorname{dist}(f_p^\epsilon(x), f_p(x)) < \epsilon.$$

Proof. For every fixed value of the parameter $p \in P$ we set

$$\widetilde{K}_p = f_p(K), \quad \widetilde{X}'_p = f_p(X'), \quad \widetilde{U}_p = f_p(U).$$

Lemma 3.4.3 (p. 69) gives a holomorphic embedding $G_p\colon W_p \hookrightarrow X \times \mathbb{C}^N$ of an open Stein neighborhood $W_p \subset Z$ of $\widetilde{K}_p \cup \widetilde{X}'_p$ (with $N = N_p \in \mathbb{N}$), and a fiber preserving holomorphic retraction $\iota_p\colon \Omega_p \to G_p(W_p)$. Set $K' = K \cup (L \cap X')$. Every point $p \in P$ admits an open neighborhood $P'_p \subset P$ such that for every $q \in P'_p$ we have $f_q(K') \subset W_p$. By the compactness of P we obtain a finite open cover $\{P'_j\}$ of P and points $p_j \in P'_j$ such that $f_q(K') \subset W_{p_j}$ for all $q \in \bar{P}'_j$. Write $N_j = N_{p_j}$, $G_j = G_{p_j}$, $W_j = W_{p_j}$, $\Omega_j = \Omega_{p_j}$ and $\iota_j = \iota_{p_j}$. As in the proof of Theorem 3.4.1 we obtain for every j an open neighborhood $V_j \supset K'$ and a continuous family of holomorphic sections $F_{j,p}\colon V_j \to V_j \times \mathbb{C}^{N_j}$ $(p \in \bar{P}'_j)$ such that $F_{j,p}$ approximates f_p uniformly on K and it agrees with f_p on $V_j \cap X'$. If the approximation is sufficiently close on K and the neighborhood $V_j \supset K'$ is chosen sufficiently small, we also have $F_{j,p}(V_j) \subset \Omega_j$. Set

$$g_{j,p} = \iota_j \circ F_{j,p}, \quad f_{j,p} = G_j^{-1} \circ g_{j,p}\,, \qquad p \in \bar{P}'_j.$$

Each of the families $\{f_{j,p}\colon p \in \bar{P}'_j\}$ satisfies the conclusion of Theorem 3.4.5 for the respective values of the parameter p. To complete the proof we patch these families of sections into a global family depending continuously on $p \in P$. The following method will be used in many subsequent places (e.g. in §6.5).

The method of successive patching. Choose compact sets $P_j \subset P'_j$ such that $\cup_j P_j = P$. Also choose a continuous function $\chi\colon P \to [0,1]$ with support in P'_1 which equals 1 in a neighborhood $\widetilde{P}_1 \subset P'_1$ of P_1. For $p \in \widetilde{P}_1 \cup P'_2$ we set

$$f_p = G_2^{-1} \circ \iota_2 \circ \big(\chi(p)\, G_2 \circ f_{1,p} + (1-\chi(p)) g_{2,p}\big). \tag{3.6}$$

The expression is well defined on a neighborhood of K' in X provided that all approximations were sufficiently close on K. For $p \in \widetilde{P}_1$ we have $\chi(p) = 1$ and hence $f_p = f_{1,p}$; for $p \in P'_2 \backslash P'_1$ we have $\chi(p) = 0$ and hence $f_p = f_{2,p}$. A nontrivial convex linear combination of the sections $G_2 \circ f_{1,p}$ and $g_{2,p}$ (with values in $\mathbb{C}^{N_2}$) only occurs for $p \in P'_2 \backslash \widetilde{P}_1 \subset P'_2$, and our choices insure that these sections lie in the domain of the retraction ι_2. This gives a continuous family of holomorphic sections for p in a neighborhood of $P_1 \cup P_2$ in P. We continue in the same way to patch the family (3.6) with $\{f_{3,p}\colon p \in P'_3\}$ into a new family defined for p in a neighborhood of $P_1 \cup P_2 \cup P_3$. After finitely many steps we obtain a desired solution. $\square$

3.5 Totally Real Submanifolds

Assume that M is a real submanifold of a complex manifold (X, J). The question whether a compact set $K \subset M$ admits an open Stein neighborhood in X, or a basis of such neighborhoods, is rather intricate. The simplest case are totally real submanifolds. Recall that a submanifold $M \subset X$ is *totally real* (or J-real) if for every $p \in M$ the tangent space T_pM contains no complex line: $\mathrm{T}_pM \cap J(\mathrm{T}_pM) = \{0\}$. The following result and its corollary are well known ([409], [474, p. 281, Theorem 6.1.6]).

Proposition 3.5.1. *Let M be a $\mathcal{C}^1$ totally real submanifold of a complex manifold X. Assume that $\rho \geq 0$ is a $\mathcal{C}^2$ function in a neighborhood of M such that $\rho^{-1}(0) = M$. If for every point $p \in M$ there are a neighborhood U_p and a number $c = c_p > 0$ such that $\rho(x) \geq c\,\mathrm{dist}(x, M)^2$ for $x \in U_p$ then ρ is strongly plurisubharmonic near M.*

Proof. Let $m = \dim_{\mathbb{R}} M \leq n = \dim_{\mathbb{C}} X$. Assume first that $m = n$. Fix a point $p \in M$ and choose local holomorphic coordinates $z = (z_1, \ldots, z_n) = x + \mathrm{i}y$ near p such that $z(p) = 0$ and $\mathrm{T}_0 M = \mathbb{R}^n \oplus \{\mathrm{i}0\}^n = \{y = 0\}$. We calculate the Levi form $\mathcal{L}_{\rho,0}$ of ρ at 0. We have

$$4\frac{\partial^2 \rho}{\partial z_j \partial \bar{z}_k} = \frac{\partial^2 \rho}{\partial x_j \partial x_k} + \frac{\partial^2 \rho}{\partial y_j \partial y_k} + \mathrm{i}\left(\frac{\partial^2 \rho}{\partial x_j \partial y_k} - \frac{\partial^2 \rho}{\partial x_k \partial y_j}\right).$$

The conditions imply that all first order partial derivatives of ρ vanish identically on M; hence each second order derivative of ρ that includes a differentiation with respect to one of the variables $x_1, \ldots, x_n$ vanishes at 0 (since these directions are tangential to M at 0). It follows that $4\dfrac{\partial^2 \rho}{\partial z_j \partial \bar{z}_k}(0) = \dfrac{\partial^2 \rho}{\partial y_j \partial y_k}(0)$. Thus $\mathcal{L}_{\rho,0}$ equals a quarter of the real Hessian of $y \to \rho(\mathrm{i}y)$ at $y = 0$. Since $\rho(\mathrm{i}y) \geq c|y|^2$, its Hessian is positive definite, and hence $\mathcal{L}_{\rho,0} > 0$.

If $m < n$, set $d = n - m$ and choose local coordinates at $p = 0$ with $\mathrm{T}_0 M = \mathbb{R}^m \oplus \{\mathrm{i}0\}^m \oplus \{0\}^d$. The previous argument shows that $\mathcal{L}_{\rho,0}$ is positive on $\mathbb{C}^m \oplus \{0\}^d$. Consider now a vector $v \in \mathbb{C}^n \backslash (\mathbb{C}^m \oplus \{0\}^d)$. The complex line $\mathbb{C}v$ intersects M transversely at 0, and hence the function $\zeta \to \lambda_v(\zeta) = \rho(\zeta v) \geq 0$ vanishes precisely to order two at $\zeta = 0$; thus its Hessian at 0 is positive definite. The Laplacian of λ_v at 0 equals the trace of the Hessian and hence is positive. But the Laplacian of λ_v at 0 equals $4\mathcal{L}_{\rho,0}(v)$. $\square$

Corollary 3.5.2. *For every $\mathcal{C}^1$ totally real submanifold M of a complex manifold X there exists a strongly plurisubharmonic function $\rho \geq 0$ in a neighborhood V of M satisfying*

$$\frac{1}{2}\,\mathrm{dist}(x, M)^2 \leq \rho(x) \leq \mathrm{dist}(x, M)^2, \quad x \in V.$$

Further, every compact set $K \subset M$ admits a neighborhood $U \subset X$ and a strongly plurisubharmonic function $\tau\colon U \to \mathbb{R}_+$ such that $\tau^{-1}(0) = K$. The sets $\{\rho < c\}$ for small $c > 0$ form a basis of open Stein neighborhoods of K.

Proof. If M is of class $\mathcal{C}^2$ then the squared distance function $\rho = \mathrm{dist}(\cdot\,, M)^2$ satisfies the hypothesis in Lemma 3.5.1, and hence is strongly plurisubharmonic near M. If M is merely $\mathcal{C}^1$, we can still find a $\mathcal{C}^2$ function $\rho \geq 0$ that is asymptotic to $\mathrm{dist}(\cdot\,, M)^2$ near M by using Whitney's extension theorem [474, Theorem 6.1.6.]. For the second statement, choose a smooth function $\chi \geq 0$ on X such that $\chi^{-1}(0) = K$ and χ vanishes to at least third order at every point of K. Then $\tau = \rho + \chi$ satisfies the stated properties near K. $\square$

By a refinement of this proof it can be shown that even a noncompact $\mathcal{C}^1$ totally real submanifold M in a complex manifold X admits a basis of open tubular Stein neighborhoods (see [224, §3]). Since a real analytic manifold M embeds as a totally real submanifold of maximal dimension in its complexification, we have the following corollary.

Corollary 3.5.3. ([224, §3], [228, p. 157]) *Every real analytic manifold is real analytically isomorphic to a real analytic totally real submanifold M of maximal dimension in a Stein manifold X. Furthermore, X can be chosen diffeomorphic to the normal bundle $\mathrm{N}_{M/X}$.*

Continuous or smooth functions on totally real submanifolds can be approximated by functions that are holomorphic in open neighborhoods. A classical case is the Weierstrass theorem on approximation of continuous functions on the real axis $\mathbb{R} \subset \mathbb{C}$ by holomorphic polynomials. H. Alexander proved that every continuous function on a rectifiable arc in $\mathbb{C}^n$ can be approximated uniformly by functions that are holomorphic in open neighborhoods of the arc [13, 474]. The first general results on totally real submanifolds go back to the papers [269] and [248]. The following optimal result is due to Range and Siu; see also [474, §6.3].

Theorem 3.5.4. [409, Theorem 1] *If M is a totally real submanifold of class $\mathcal{C}^k$ ($k \in \mathbb{N} \cup \{\infty\}$) in a complex manifold X, then there exists a Stein open neighborhood U of M in X such that the set of restrictions to M of all holomorphic functions on U is dense in the Fréchet space $\mathcal{C}^k(M)$.*

Theorem 3.5.4 also follows from [189, Theorem 3.1] and the following lemma on the existence of $\overline{\partial}$-flat extensions. We omit the proof.

Lemma 3.5.5. [269, Lemma 4.3] *Let M be a totally real submanifold of class $\mathcal{C}^k$ ($k \in \{1, 2, \ldots, \infty\}$) in a complex manifold X. Every function $f \in \mathcal{C}^k(M)$ extends to a function $F \in \mathcal{C}^k(X)$ that is smooth on $X \backslash M$ and is $\overline{\partial}$-flat to order $k-1$ along M, in the sense that all partial derivatives of order $\leq k-1$ of the coefficients of $\overline{\partial} F$ vanish on M.*

Theorem 3.5.4 extends to maps with values in a complex manifold, except that the neighborhood may depend on the map. Here is the precise result. For a more general approximation result see Theorem 3.7.2 on p. 81.

Corollary 3.5.6. *If M is a totally real submanifold of class $\mathcal{C}^r$ ($r \geq 1$) in a complex manifold X and if Y is a complex manifold, then every map $f\colon M \to Y$ of class $\mathcal{C}^s$ for some $s \leq r$ can be approximated in the $\mathcal{C}^s$ topology by maps $F\colon U \to Y$ in open Stein neighborhoods $U = U_F$ of M.*

Proof. By an initial approximation we may assume that f is of class $\mathcal{C}^r$. The graph $\widetilde{M} = \{(x, f(x)) \colon x \in M\}$ is a totally real submanifold of $X \times Y$,

and hence it has an open Stein neighborhood $\Omega \subset X \times Y$. The domain Ω embeds as a closed Stein submanifold of some $\mathbb{C}^N$, and hence we can view $M \ni x \mapsto (x, f(x))$ as a $\mathbb{C}^N$-valued map. Approximate it by a holomorphic map, compose with a holomorphic retraction onto the submanifold Ω (here comes a restriction of the domain of the map), and project it back to X. $\square$

In the same direction we mention the following result of E. L. Stout.

Theorem 3.5.7. [476] *Assume that K is a compact set in a complex space X such that every continuous function $f\colon K \to \mathbb{C}$ can be approximated uniformly on K by holomorphic functions on open neighborhoods of K. Then every continuous map $f\colon K \to Y$ to an arbitrary complex manifold Y can be uniformly approximated by maps $F\colon U_F \to Y$ in open neighborhoods $U_F \supset K$.*

Proof. Choose a smooth embedding $\phi\colon Y \hookrightarrow \mathbb{R}^m$ for some $m \in \mathbb{N}$. Considering $\mathbb{R}^m$ as the real subspace of $\mathbb{C}^m$, the graph $Z = \{(y, \phi(y))\colon y \in Y\} \subset Y \times \mathbb{C}^m$ is a totally real submanifold of $Y \times \mathbb{C}^m$, and hence it has an open Stein neighborhood Ω in $Y \times \mathbb{C}^m$ by Corollary 3.5.2. Embed Ω as a complex submanifold of some $\mathbb{C}^N$. Given a continuous map $f\colon K \to Y$, the continuous map $K \ni x \mapsto (f(x), \phi(f(x)) \in \Omega \subset \mathbb{C}^N$ can be approximated by holomorphic maps $U \to \mathbb{C}^N$ in open neighborhoods $U \subset X$ of K. The proof is now concluded exactly as in Corollary 3.5.6. $\square$

Theorem 3.5.7 gives the following corollary to a theorem of H. Alexander [13] concerning the approximation of continuous function on rectifiable arcs in $\mathbb{C}^n$ by holomorphic functions. For results on approximation of maps into almost complex manifolds see Chakrabarti [82].

Corollary 3.5.8. *If C is a rectifiable arc in $\mathbb{C}^n$, and if f is a continuous map from C to a complex manifold Y, then f can be approximated uniformly on C by maps to Y that are holomorphic on neighborhoods of C.*

The next simplest case where things are fairly well understood are real two dimensional surfaces in a complex surface; we consider these in §9.2–9.8.

Generic CR submanifolds of positive CR dimension (see p. 22) are only rarely locally holomorphically convex. If M is *minimal* at a point $p \in M$, in the sense that there is no local smooth CR submanifold $N \subset M$ passing through p such that $\dim_p N < \dim_p M$ but $\mathrm{CRdim}_p N = \mathrm{CRdim}_p M$, then there exist analytic discs in X with boundaries in M that sweep out a wedge-like domain W with edge M, and any CR function on M near p extends to a holomorphic function on W (see [30] for a survey of results on this subject). Clearly M is not locally holomorphically convex at such a point. In spite of this, CR functions on any CR submanifold of $\mathbb{C}^n$ are locally approximable by holomorphic polynomials [31].

3.6 Stein Neighborhoods of Certain Laminated Sets

The main result of this section is used in the proof of Theorem 5.13.3 (p. 230) on the equivalence of the basic and the parametric Oka property.

Let Z be a complex space. Consider the product $\mathbb{C}^n \times Z$ with the projection $\pi\colon \mathbb{C}^n \times Z \to \mathbb{C}^n$. Let S be a compact subset of $\mathbb{C}^n \times Z$. Under what conditions on the projection $\pi(S) \subset \mathbb{C}^n$ and on the fibers $S_\zeta = \{z \in Z\colon (\zeta, z) \in S\}$ is S a Stein compactum in $\mathbb{C}^n \times Z$? We have the following precise answer when the projection $\pi(S)$ is contained in $\mathbb{R}^n$, the real subspace of $\mathbb{C}^n$.

Theorem 3.6.1. [198] *Let S be a compact set in $\mathbb{C}^n \times Z$ such that $P = \pi(S) \subset \mathbb{R}^n$. Then S is a Stein compactum in $\mathbb{C}^n \times Z$ if and only if for any open neighborhood $U \subset \mathbb{R}^n \times Z$ of S there exist open sets $V, \Omega \subset \mathbb{R}^n \times Z$, with $S \subset V \Subset \Omega \subset U$, such that for any $u \in P$ the fiber Ω_u is Stein and $\widehat{(S_u)}_{\mathcal{O}(\Omega_u)} \subset V$. The same holds if $\pi(S)$ belongs to a totally real subset of $\mathbb{C}^n$.*

The following simple example illustrates why it is not enough to assume that each fiber S_ζ is a Stein compactum.

Example 3.6.2. (Thin Hartogs' figure) Let $Z = \mathbb{C}$ and $P = \{\zeta \in \mathbb{C}\colon 0 \le \zeta \le 1\}$. For $0 \le \zeta \le \frac{1}{2}$ let $S_\zeta = \{z \in \mathbb{C}\colon |z| \le 1\}$, and for $\frac{1}{2} \le \zeta \le 1$ let $S_\zeta = \{z \in \mathbb{C}\colon \frac{1}{2} \le |z| \le 1\}$. Clearly each fiber is a Stein compactum, but due to the continuity principle S does not have a Stein neighborhood basis in $\mathbb{C}^2$. $\square$

Proof. Assume that S is a Stein compact. Then for every open neighborhood U of S in $\mathbb{R}^n \times Z$ there exists a Stein neighborhood $\widetilde{\Omega}$ of S in $\mathbb{C}^n \times Z$ such that $\Omega := \widetilde{\Omega} \cap (\mathbb{R}^n \times Z) \subset U$. The hull $\widehat{S}_{\mathcal{O}(\widetilde{\Omega})} \subset \widetilde{\Omega}$ is compact, and $\pi(\widehat{S}_{\mathcal{O}(\widetilde{\Omega})}) = \pi(S) \subset \mathbb{R}^n$ (since every compact subset of $\mathbb{R}^n$ is polynomially convex in $\mathbb{C}^n$). Taking $V \Subset \Omega$ to be an open neighborhood of $\widehat{S}_{\mathcal{O}(\widetilde{\Omega})}$ in $\mathbb{R}^n \times Z$ we see that the conditions are satisfied.

Conversely, assume that the conditions hold. Fix an open set $\widetilde{U} \subset \mathbb{C}^n \times Z$ containing S and let $U = \widetilde{U} \cap (\mathbb{R}^n \times Z)$. Choose open sets $V \Subset \Omega \subset U$ satisfying the hypotheses. Choose an open set V', $V \Subset V' \Subset \Omega$. For $\epsilon > 0$ set

$$V(\epsilon) = \big\{(u + \mathrm{i}v, z) \in \mathbb{C}^n \times Z\colon (u, z) \in V,\ |v| < \epsilon\big\}.$$

Choose $\epsilon_0 > 0$ sufficiently small such that $V'(\epsilon_0) \Subset \widetilde{U}$. We construct a (Stein) plurisubharmonic polyhedral neighborhood of S contained in $V(\epsilon_0)$. For $\delta > 0$ and $u \in \mathbb{R}^n$ let $\mathbb{B}(u, \delta)$ denote the open ball in $\mathbb{R}^n$ of radius δ centered at u.

Lemma 3.6.3. *There exists a positive strongly plurisubharmonic function ρ_0 in an open neighborhood of S in $\mathbb{C}^n \times Z$.*

Proof. Fix a point $s_0 = (u_0, z_0) \in S$. By compactness it suffices to show that there exists a plurisubharmonic function in a neighborhood of S that is strongly plurisubharmonic near s_0. Since $\overline{V}' \subset \Omega$ and Ω is open in $\mathbb{R}^n \times Z$, there exists a $\delta > 0$ such that $V'_u \Subset \Omega_{u_0}$ for all u in the ball $\mathbb{B}(u_0, 2\delta) = \{|u - u_0| < 2\delta\}$. As $W = (\mathbb{B}(u_0, 2\delta) + \mathrm{i}\mathbb{R}^n) \times \Omega_{u_0}$ is Stein, there exists a positive strongly plurisubharmonic function ρ_1 on W. We may assume that $\rho_1(s_0) = 1$. Let

$$M = \sup\{\rho_1(u, z) \colon |u - u_0| = \delta,\ z \in V'_u\}.$$

Let ρ_2 be a plurisubharmonic function on $\mathbb{C}^n$ such that $\rho_2(u_0) = 0$ and $\rho_2(u) > M$ for all $u \in b\mathbb{B}(u_0, \delta)$. If $\epsilon > 0$ is chosen small enough then the function $\max\{\rho_1, \rho_2\}$ is well defined and plurisubharmonic on $V'(\epsilon)$, and is strongly plurisubharmonic near the point s_0. $\square$

By shrinking the initial neighborhood $\widetilde{U} \subset Z$ of S we may assume by Lemma 3.6.3 that there exists a positive strongly plurisubharmonic function ρ_0 on $\widetilde{U}$. We adjust the sets V, V' and Ω to the new $U = \widetilde{U} \cap (\mathbb{R}^n \times Z)$.

Lemma 3.6.4. *For any point q belonging to the boundary of V in $\mathbb{R}^n \times Z$ there exist an $\epsilon > 0$ and a continuous plurisubharmonic function ρ on $V'(\epsilon)$ such that $\rho(q) = 2$ and $\rho < \frac{1}{2}$ on S.*

Proof. If $\pi(q) \notin \pi(S) = P$, we may find a holomorphic function $g \in \mathcal{O}(\mathbb{C}^n)$ such that $\rho = |g \circ \pi|$ will work.

Assume now that $q = (u, z)$ with $u \in P$. Since $\widehat{(S_u)}_{\mathcal{O}(\Omega_u)} \subset V_u$ by the assumption, there is a holomorphic function $f \in \mathcal{O}(\Omega_u)$ such that $f(z) = 2$ and $|f| < \frac{1}{2}$ on S_u. By continuity of f and since $\overline{V'} \subset \Omega$ there is a $\delta > 0$ such that for all $u' \in \overline{\mathbb{B}(u, \delta)}$ we have $V'_{u'} \Subset \Omega_u$ and $|f| < \frac{1}{2}$ on $S_{u'}$. Let

$$M = \sup\big\{|f(z)| \colon z \in \overline{V}'_{u'},\ |u' - u| \le \delta\big\}.$$

We consider f as a holomorphic function on $\mathbb{C}^n \times \Omega_u$ which is independent of the first variable. Let $\chi \in \mathcal{C}_0^\infty(\mathbb{B}(u, \delta))$ be such that $0 \le \chi \le 1$ and $\chi(u) = 1$. Let $g \in \mathcal{O}(\mathbb{C}^n)$ be a holomorphic function approximating χ close enough on $\overline{\mathbb{B}(u, \delta)}$ such that $|g|_{b\mathbb{B}(u,\delta)} < 1/3M$ and $g(u) = 1$. Then $\rho_1 = |fg|$ is plurisubharmonic on $[\overline{\mathbb{B}(u, \delta)} \oplus \mathrm{i}\mathbb{R}^n] \times \Omega_u$, $\rho_1(q) = |f(z)g(u)| = 2$, and $\rho_1(w) < \frac{1}{3}$ for all $w = (u', z)$ such that $u' \in b\mathbb{B}(u, \delta)$ and $z \in V'_{u'}$.

Let $h \in \mathcal{O}(\mathbb{C}^n)$ be a holomorphic function such that $h(u) = 0$, $0 \le |h| \le \frac{1}{2}$ on $\mathbb{B}(u, \delta)$, and $\frac{1}{3} < |h| < \frac{1}{2}$ on $\mathbb{R}^n \backslash \mathbb{B}(u, \delta)$. The function $\rho_2 = |h \circ \pi|$ is plurisubharmonic on $\mathbb{C}^n \times Z$. If $\epsilon > 0$ is small enough then $\rho = \max\{\rho_1, \rho_2\}$ is well defined on $V'(\epsilon)$ and satisfies the conclusion of the lemma. $\square$

We now complete the proof of Theorem 3.6.1. By compactness and Lemma 3.6.4 (p. 76) there exist an $\epsilon > 0$ with $2\epsilon < \epsilon_0$ and plurisubharmonic functions

$\rho_1, \dots, \rho_m$ on $V'(2\epsilon)$ such that $\rho_j < \frac{1}{2}$ on S for $j = 1, \dots, m$, and such that for every $q = (u, z) \in bV \subset \mathbb{R}^n \times Z$ we have $\rho_j(q) > \frac{3}{2}$ for at least one $j \in \{1, \dots, m\}$. Denote the variables on $\mathbb{C}^n$ by $\zeta = u + \mathrm{i}v$. If we replace ρ_j by $\rho_j + C|v|^2$ for a sufficiently large $C > 0$, then we have

$$b[V(\epsilon)] \subset \bigcup_{j=1}^{m} \left\{ w \in V'(2\epsilon) \colon \rho_j(w) > \frac{3}{2} \right\}.$$

Let

$$W_0 = \{w \in V'(2\epsilon) \colon \rho_j(w) < 1,\ j = 1, \dots, m\},$$

and let W be the union of all connected components of W_0 which intersect S; then $S \subset W \Subset V(\epsilon)$. Choose a convex increasing function $\varphi \in \mathcal{C}^\infty((-\infty, 1))$ with $\lim_{t \to 1} \varphi(t) = +\infty$. Recall that ρ_0 is a strongly plurisubharmonic function on $\widetilde{U} \supset S$. The function ρ defined by

$$\rho = \rho_0 + \max_{1 \le j \le m} \varphi \circ \rho_j$$

is a strongly plurisubharmonic exhaustion of W, and hence W is Stein by Theorem 2.3.2 (p. 51). Since $\overline{W} \subset V(\epsilon_0)$, this concludes the proof. □

Corollary 3.6.5. *Assume that S is a compact set in $\mathbb{R}^n \times Z$ with the following property. Given an open neighborhood U of S, there exist for each point $u \in \pi(S) \subset \mathbb{R}^n$ a Stein neighborhood $\Omega_u \Subset U_u \subset Z$ of the fiber S_u and a number $\delta = \delta_u > 0$ such that for every point $u' \in \mathbb{B}(u, \delta)$ the fiber $S_{u'}$ is holomorphically convex in Ω_u. Then S is a Stein compactum in $\mathbb{C}^n \times Z$.*

Proof. We verify the assumptions in Theorem 3.6.1. By compactness there is a finite number of u_j's, Ω_{u_j}'s and δ_j's such that $\{\mathbb{B}(u_j, \delta_j)\}$ is an open cover of $\pi(S)$ in $\mathbb{C}^n$. If all δ_j's are small enough then $\bigcup_j \mathbb{B}(u_j, \delta_j) \times \Omega_{u_j}$ is contained in U. Let Ω be a neighborhood of S in $\mathbb{C}^n \times Z$ with the fibers

$$\Omega_u = \bigcap \left\{ \Omega_{u_k} \colon u \in \overline{\mathbb{B}(u_k, \delta_k)} \right\}.$$

Then Ω is contained in U and S_u is $\mathcal{O}(\Omega_u)$-convex for every u. So let $V \Subset \Omega$ be any open neighborhood that contains S. □

Corollary 3.6.6. *Assume that $h \colon Z \to X$ is a holomorphic submersion of a complex space Z onto a Stein space X, K is a compact $\mathcal{O}(X)$-convex subset of X, $U \subset X$ is an open set containing K, and P is a compact set in $\mathbb{R}^n$ (the real subspace of $\mathbb{C}^n$). Let $f \colon P \times U \to Z$ be a continuous map such that $h \circ f(p, x) = x$ for all $(p, x) \in P \times U$, and $f_p = f(p, \cdot) \colon U \to Z$ is holomorphic for every fixed $p \in P$. Then the set*

$$\Sigma = \left\{ (p, f(p, x)) \colon p \in P,\ x \in K \right\} \subset \mathbb{C}^n \times Z$$

admits an open Stein neighborhood Θ in $\mathbb{C}^n \times Z$ such that Σ is $\mathcal{O}(\Theta)$-convex.

Proof. By shrinking U around K we may assume that U is Stein. For every $p \in P$ the set $V_p = f_p(U)$ is then a closed Stein subvariety of the complex space $Z|_U = h^{-1}(U)$, and the set $K_p = f_p(K)$ is $\mathcal{O}(V_p)$-convex.

Fix a point $p \in P$. By Siu's theorem (Corollary 3.1.1, p. 57) there exists an open Stein neighborhood $\Omega_p \subset Z$ of K_p such that $\overline{V_p \cap \Omega_p} \subset V_p$. If $q \in P$ is sufficiently near p then $\Sigma_q \subset V_q \cap \Omega_p$ due to the continuity of f. Since Σ_q is $\mathcal{O}(V_q)$-convex, it is also $\mathcal{O}(V_q \cap \Omega_p)$-convex. As $V_q \cap \Omega_p$ is a closed subvariety of the Stein domain Ω_p, it follows that Σ_q is also $\mathcal{O}(\Omega_p)$-convex. (Indeed, for any point $z \in \Omega_p \backslash V_q$ there exists by Cartan's theorem a holomorphic function on Ω_p that equals one at z and that vanishes on $V_q \cap \Omega_p$; hence no such point can belong to the $\mathcal{O}(\Omega_p)$-hull of Σ_q.)

This shows that Σ satisfies the assumptions of Corollary 3.6.5, and hence it admits a basis of open Stein neighborhoods in $\mathbb{C}^n \times Z$.

Since P is contained in $\mathbb{R}^n$, $\mathcal{O}(\mathbb{C}^n)|_P$ is dense in the space of complex valued continuous functions on P by the Weierstrass approximation theorem. By using cut-off functions in the Euclidean variable and approximating them by entire functions we see that Σ is $\mathcal{O}(\Theta)$-convex in every open Stein set $\Theta \subset \mathbb{C}^n \times Z$ containing Σ. $\square$

3.7 Stein Compacts with Totally Real Handles

In this section we construct Stein neighborhoods of certain Stein compacts with attached totally real submanifolds. Configurations of this type play an important role in various geometric constructions. We also prove a Mergelyan type approximation theorem for mappings defined on such sets.

The following result was proved by Hörmander and Wermer when $X = \mathbb{C}^n$ [269]; the general case is [174, Theorem 3.1].

Theorem 3.7.1. *Let $S = K_0 \cup E$ be a closed subset of a complex space X, where K_0 is compact holomorphically convex (Def. 2.3.5) and $E = S \backslash K_0$ is a $\mathcal{C}^1$ totally real submanifold contained in X_{reg}. Assume that there exists a compact holomorphically convex set $K_1 \subset S$ that is a relative neighborhood of K_0 in S. Then every compact set K with $K_1 \subset K \subset S$ is holomorphically convex. For any such K and any neighborhood $N \supset K_0$ there exists for every sufficiently small $\epsilon > 0$ an open Stein domain $\Omega_\epsilon \subset X$ such that*

(i) *Ω_ϵ contains all points with distance $< \epsilon$ from K, and*
(ii) *all points $x \in \Omega_\epsilon \backslash N$ have distance $< 2\epsilon$ from S.*

The holomorphically convex set K_1 in the theorem contains a collar in E around K_0; when such K_1 exists, we say that the totally real submanifold E is *well attached to* K_0. (Of course $\overline{E}$ might be disjoint from K_0 in which case the condition is trivially satisfied.) In a typical application the set K_0 will

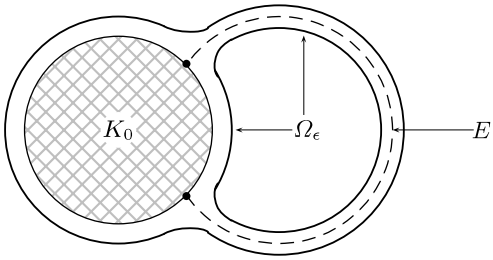

Fig. 3.2. A Stein neighborhood Ω_ϵ of $S = K_0 \cup E$. (Modified from [194, p. 632, Fig. 7] and also [195, p. 348, Fig. 3])

be the closure of a strongly pseudoconvex domain $D \Subset X$, and $\overline{E}$ will be a compact totally real submanifold in $X \backslash D$ attached to bD along a part (or all) of its boundary bE. For more precise results in this direction see §3.8 below.

Proof. By the hypothesis there exist for $j = 0, 1$ an open set $U_j \supset K_j$ and a plurisubharmonic exhaustion function $\rho_j \colon U_j \to \mathbb{R}_+$ such that $K_j = \{\rho_j = 0\}$. We may choose $U_0 \subset U_1$. Choose $c_2 > 0$ sufficiently small such that

$$\{x \in U_0 \colon \rho_0(x) \le c_2\} \subset N, \quad K_1' := S \cap \{\rho_0 \le c_2\} \subset K_1.$$

Choose a convex increasing function $h \colon \mathbb{R}_+ \to \mathbb{R}_+$ which vanishes on $[0, c_2]$, is positive on $(c_2, +\infty)$ and $\lim_{t\to\infty} h(t) = +\infty$. The function

$$\rho_1' = \rho_1 + h \circ \rho_0 \colon U_0 \to \mathbb{R}_+$$

is a plurisubharmonic exhaustion with $\{\rho_1' = 0\} = K_1'$, and hence K_1' is $\mathcal{O}(U_0)$-convex. We now replace K_1 by K_1' and ρ_1 by ρ_1'. Choose constants $0 < c_0 < c_0' < c_1 < c_1' < c_2$. Let $\operatorname{dist}(x, E) = \inf\{\operatorname{dist}(x, y) \colon y \in E\}$. The function $\rho_1 \ge 0$ has a local minimum at each point $x \in K_1$. After multiplying ρ_1 by a suitably chosen positive constant we have

$$\rho_1(x) < \frac{1}{2} \operatorname{dist}(x, K_1)^2, \quad x \in U_0.$$

Fix a compact subset K of S with $K_1 \subset K$. Since E is a totally real submanifold, Corollary 3.5.2 (p. 72) gives a $\mathcal{C}^2$ strongly plurisubharmonic function $\tau \ge 0$ in an open set $V_0 \supset E \cap K$ that vanishes precisely on $E \cap K$ and satisfies

$$\frac{1}{2} \operatorname{dist}(x, E)^2 \le \tau(x) \le \operatorname{dist}(x, E \cap K)^2, \quad x \in V_0.$$

If E is of class $\mathcal{C}^r$ for some $r > 1$, we can take $\tau(x) = \frac{1}{2}\operatorname{dist}(x, E)^2 + \tau'(x)$ where $\tau' \ge 0$ is a smooth function that vanishes to third order on $E \cap K$ and is small positive on $E \backslash K$. Observe that $K \subset U_0 \cup V_0$.

Choose a smooth function $\chi \ge 0$ on X such that $\chi = 1$ on $\{\rho_0 \le c_0\}$ and $\operatorname{supp} \chi \Subset \{\rho_0 < c_0'\}$. For small $\delta > 0$ the function $\tau_\delta = \tau - \delta\chi$ is strongly

plurisubharmonic on V_0; fix such δ. Choose an open set V satisfying $K \subset V \subset U_0 \cup V_0$ and define a function $\rho\colon V \to \mathbb{R}_+$ by

$$\rho = \begin{cases} \rho_1 & \text{on } V \cap \{\rho_0 < c_0\}, \\ \max\{\rho_1, \tau_\delta\} & \text{on } V \cap \{c_0 \le \rho_0 \le c_0'\}, \\ \tau & \text{on } V \cap \{\rho_0 > c_0'\}. \end{cases}$$

These choices are compatible if the neighborhood $V \supset K$ is sufficiently small:

- near $E \cap \{\rho_0 = c_0\}$ we have $\rho_1 \ge 0$ and $\tau_\delta < 0$, hence $\rho = \rho_1$ there;
- for x near $E \cap \{\rho_0 = c_0'\}$ we have $\rho_1(x) < \frac{1}{2}\mathrm{dist}(x, E)^2 \le \tau(x) = \tau_\delta(x)$, hence $\rho = \tau$ there.

By the construction, $\rho \ge 0$ is a continuous plurisubharmonic function on V that vanishes precisely on K and satisfies the following properties:

- $\rho(x) \le \mathrm{dist}(x, K)^2$ for all $x \in V$, and
- $\rho(x) = \tau(x) \ge \frac{1}{2}\mathrm{dist}(x, E \cap K)^2$ for $x \in V \cap \{\rho_0 \ge c_0'\} \supset V \backslash N$.

Hence for every sufficiently small $\epsilon > 0$ the set $\Omega_\epsilon = \{x \in V\colon \rho(x) < \epsilon^2\}$ is a pseudoconvex open neighborhood of K satisfying

$$\{x \in X\colon \mathrm{dist}(x, K) < \epsilon\} \subset \Omega_\epsilon, \quad \Omega_\epsilon \backslash N \subset \{x \in X\colon \mathrm{dist}(x, E \cap K) < \sqrt{2}\,\epsilon\}.$$

It remains to show that the sets Ω_ϵ are Stein. Fix an ϵ and choose an increasing convex function $h_\epsilon\colon (-\infty, \epsilon^2) \to \mathbb{R}$ with $\lim_{t \to \epsilon^2} h(t) = +\infty$; then $h_\epsilon \circ \rho\colon \Omega_\epsilon \to \mathbb{R}$ is a plurisubharmonic exhaustion. Choose a smooth strongly plurisubharmonic function ξ on the Stein domain U_0. Let $\chi \ge 0$ be a smooth cut-off function on X such that $\chi = 1$ on $\{\rho_0 \le c_1\}$ and $\mathrm{supp}\,\chi \subset \{\rho_0 < c_1'\}$. We claim that, for sufficiently small $\delta > 0$, the function $\widetilde{\rho} = \rho + \delta\chi\xi$ is strongly plurisubharmonic on V. Indeed,

- on $V \cap \{\rho_0 \le c_1\}$ we have $\widetilde{\rho} = \rho + \delta\xi$ that is strongly plurisubharmonic,
- on $V \cap \{\rho_0 > c_1\}$ the function ρ is strongly plurisubharmonic, and hence so is $\widetilde{\rho}$ provided that $\delta > 0$ is chosen sufficiently small.

For such δ, $h_\epsilon \circ \rho + \widetilde{\rho}\colon \Omega_\epsilon \to \mathbb{R}$ is a strongly plurisubharmonic exhaustion and hence Ω_ϵ is Stein. Since ρ is plurisubharmonic on $\Omega_{\epsilon_0} = \{\rho < \epsilon_0\}$ for a small $\epsilon_0 > 0$, the domain Ω_ϵ is Runge in $\Omega_{\epsilon'}$ for all $0 < \epsilon < \epsilon' \le \epsilon_0$. $\square$

We now prove the following Mergelyan type approximation theorem in the situation of Theorem 3.7.1. Although the stated result is not optimal, it will suffice for our applications. For a precise result without loss of derivatives one must use the optimal holomorphic approximation theorem in tubular neighborhoods of totally real submanifolds [189, Theorem 1.1] (see Theorem 2.5.4 on p. 56) and then solve a Cousin problem with bounds.

Theorem 3.7.2. ([269, Th. 4.1], [174, Th. 3.2]) *Assume that X and Z are complex spaces, $\pi\colon Z \to X$ is a holomorphic submersion, and X' is a closed complex subvariety of X containing $X_{\rm sing}$. Let K_0 and $S = K_0 \cup E$ be compact holomorphically convex subsets of X, where $E \subset X \backslash X'$ is a compact totally real m-dimensional submanifold of class $\mathcal{C}^r$. Assume that $r \geq \frac{m}{2} + 1$ and that k is an integer satisfying $0 \leq k \leq r - \frac{m}{2} - 1$. Given an open set $U \supset K_0$ and a section $f\colon U \cup E \to Z|_{U\cup E}$ such that $f|_U$ is holomorphic and $f|_E \in \mathcal{C}^r(E)$, there exist for every $s \in \mathbb{N}$ open sets $V_j \supset S$ and a sequence of holomorphic sections $f_j\colon V_j \to Z|_{V_j}$ $(j \in \mathbb{N})$ such that f_j agrees with f to order s along $X' \cap V_j$ for each $j \in \mathbb{N}$, and $\lim_{j\to\infty} f_j = f$ in the $\mathcal{C}^k$-topology on $K_0 \cup E$.*

Proof. We begin with the special case when X is a manifold and $Z = X \times \mathbb{C}$; we identify its sections with functions $X \to \mathbb{C}$. Lemma 3.5.5 (p. 73) shows that f extends to a $\mathcal{C}^r$ function in an open neighborhood of S, still denoted f, that agrees with the original function near K_0 and that satisfies

$$|\overline{\partial} f(x)| = o\big(\mathrm{dist}(x, E)^{r-1}\big), \quad x \text{ near } E.$$

Let $\Omega_\epsilon \supset S$ be a Stein neighborhood basis of S furnished by Theorem 3.7.1. Every point of Ω_ϵ has distance at most 2ϵ from S, and the volume of the ϵ-tube $U_\epsilon \supset E$ around E is proportional to ϵ^{2n-m} (the area of the cross section). Thus $||\overline{\partial} f||_{L^\infty(\Omega_\epsilon)} = o(\epsilon^{r-1})$ and

$$||\overline{\partial} f||_{L^2(\Omega_\epsilon)} \leq ||\overline{\partial} f||_{L^\infty(\Omega_\epsilon)} \sqrt{\mathrm{Vol}(U_{2\epsilon})} = o(\epsilon^{r-1+n-m/2}).$$

By [266] (see (2.7) on p. 55) there exists a solution u_ϵ of

$$\begin{cases} \overline{\partial} u_\epsilon = \overline{\partial} f \quad \text{on} \quad \Omega_\epsilon, \\ ||u_\epsilon||_{L^2(\Omega_\epsilon)} \leq \text{const}\cdot\, ||\overline{\partial} f||_{L^2(\Omega_\epsilon)} = o(\epsilon^{r-1+n-m/2}). \end{cases}$$

Recall that every $\mathcal{C}^1$ function on the ball $\epsilon\mathbb{B} = \{\zeta \in \mathbb{C}^n\colon |\zeta| < \epsilon\}$ satisfies the following estimate (see Lemma 2.5.2 on p. 55):

$$|u(0)| \leq \text{const}\cdot \left(\epsilon^{-n}||u||_{L^2(\epsilon\mathbb{B})} + \epsilon||\overline{\partial} u||_{L^\infty(\epsilon\mathbb{B})}\right).$$

Applying this estimate to u_ϵ at points $z \in S$ gives

$$||u_\epsilon||_{L^\infty(S)} \leq o(\epsilon^{r-1-m/2}) + \epsilon\, o(\epsilon^{r-1}) = o(\epsilon^{r-1-m/2}).$$

The function $f_\epsilon = f - u_\epsilon$ is holomorphic on Ω_ϵ, and assuming that $r \geq 1 + \frac{m}{2}$ we get $\lim_{\epsilon\to 0} ||f - f_\epsilon||_{L^\infty(S)} = 0$. By repeated differentiation we also get $\lim_{\epsilon\to 0} ||\partial^\alpha(f - f_\epsilon)||_{L^\infty(S)} = 0$ for $|\alpha| \leq r - 1 - \frac{m}{2}$.

Interpolation along a subvariety X' is easily built into the construction as follows. By the proof of Theorem 3.4.1 (p. 68) we have $f = \phi + \sum_\nu g_\nu h_\nu$ on U, where ϕ is a holomorphic function in a Stein neighborhood $V \subset X$ of S, h_ν is a finite collection of holomorphic functions in V that vanish to order s along X' and whose common zero set is $X' \cap V$, and the coefficients g_ν are

holomorphic on U and of class $\mathcal{C}^r$ on E. By the special case proved above, g_ν is the limit of a a sequence of holomorphic functions $g_{\nu,j}$ in open neighborhoods V_j of S. The sequence $f_j = \phi + \sum_\nu g_{\nu,j} h_\nu$ $(j \in \mathbb{N})$ then fulfills Theorem 3.7.2.

For $Z = X \times \mathbb{C}^N$ the result follows immediately. Suppose now that X is a complex space and $\pi\colon Z \to X$ is a holomorphic submersion.

Lemma 3.7.3. *Let X and Z be complex spaces and $\pi\colon Z \to X$ a holomorphic submersion. Assume that $S = K_0 \cup E$ and $f\colon U \cup E \to Z$ satisfy the hypotheses of Theorem 3.7.2. Then $f(S)$ has a basis of Stein neighborhoods in Z.*

Proof. We may assume that U is a Stein domain in X and K_0 is $\mathcal{O}(U)$-convex. Since the real submanifold $f(E)$ of Z is projected by π onto the totally real submanifold E of X, $f(E)$ is totally real in Z. Since $f|_U$ is holomorphic, $f(U)$ is a closed Stein subvariety of $Z|_U = \pi^{-1}(U)$ and hence it has an open Stein neighborhood $\widetilde{U} \subset Z|_U$ by Corollary 3.1.1 (p. 57). For any compact $\mathcal{O}(U)$-convex subset K in U the set $f(K)$ is holomorphically convex in $f(U)$, and hence also in $\widetilde{U}$ (since $f(U)$ is a closed complex submanifold of $\widetilde{U}$). Applying this to $K = K_0$, and also to $K = S \cap N$ for some compact holomorphically convex neighborhood $N \subset U$ of K_0, we see that $f(S)$ satisfies the hypothesis of Theorem 3.7.1; hence it has a basis of Stein neighborhoods in Z. $\square$

We now complete the proof of Theorem 3.7.2. By Lemma 3.4.3 (p. 69) there exist a Stein neighborhood W_0 of $f(S)$ in Z and a fiber preserving holomorphic embedding $G\colon W_0 \hookrightarrow X \times \mathbb{C}^N$ (3.5). By the special case proved above we can approximate the section $G \circ f$ (with values in $X \times \mathbb{C}^N$) by a sequence of holomorphic sections F_j in open neighborhoods of S. Let $\iota\colon \Omega \to G(W_0)$ be a fiber preserving holomorphic retraction from an open neighborhood $\Omega \subset X \times \mathbb{C}^N$ onto $G(W_0)$, furnished by Theorem 3.3.4 on p. 68. The sequence of holomorphic sections $f_j = G^{-1} \circ \iota \circ F_j$ in open neighborhoods V_j of S then fulfills the conclusion of Theorem 3.7.2. $\square$

3.8 Thin Strongly Pseudoconvex Handlebodies

Let W be a relatively compact, smoothly bounded domain in a complex manifold X. Assume that M is a smooth submanifold of X that is contained in $X \backslash W$ and intersects the boundary $\Sigma = bW$ transversely along its boundary $bM \subset \Sigma$. By smoothly connecting the boundary of a tube around M with Σ we obtain a smoothly bounded domain $\Omega \subset X$ which contains W and agrees with W outside a small neighborhood of M, and such that $\Omega \backslash W$ is diffeomorphic to the normal bundle of M in X. Such Ω (or its closure) is called a *handlebody with core* $W \cup M$. Replacing W in this construction with a slightly larger set $W' \supset W$ diffeomorphic to W, we still call the resulting set, obtained by thickening $W' \cup M$, a handlebody with core $\overline{W} \cup M$. We address the following problem which is important in applications.

Problem 3.8.1. Let $W \cup M \subset X$ be as above, with M totally real and W strongly pseudoconvex. Given a neighborhood $V \subset X$ of M, does there exist a handlebody $\Omega \subset W \cup V$ with core $W \cup M$ and with strongly pseudoconvex boundary $b\Omega$?

Theorem 3.7.1 (p. 78) shows that the existence of a Stein neighborhood basis of $\overline{W} \cup M$ depends only on the attachment of M to $\overline{W}$ along the boundary $bM \subset bW$. At least for handles of maximal dimension $\dim_{\mathbb{R}} M = \dim_{\mathbb{C}} X$, a necessary condition is that bM is a complex tangential (Legendrian) submanifold of $\Sigma = bW$, that is,

$$\mathrm{T}_p bM \subset \mathrm{T}_p^{\mathbb{C}} \Sigma = \mathrm{T}_p(\Sigma) \cap J(\mathrm{T}_p \Sigma), \quad p \in bM. \tag{3.7}$$

There are reasons to believe that this might suffice for a positive answer. In this section we prove a positive results under the additional assumption that M is an embedded real analytic disc which is *J-orthogonal to Σ* along bM:

$$J(\mathrm{T}_p M) \subset \mathrm{T}_p \Sigma, \quad p \in bM. \tag{3.8}$$

In view of (3.7) this orthogonality condition is equivalent to $Jw \in \mathrm{T}\Sigma$, where w is a tangential vector field to M along bM that is normal to bM.

The following result is used in the Eliashberg-Gompf construction of Stein structures on an almost complex manifold (X, J) with suitable handlebody decomposition (see §9.9 – §9.10). Results in this direction were obtained in [129], [418, Lemma 2], [186].

Theorem 3.8.2. *Let W be a relatively compact strongly pseudoconvex domain with smooth boundary $\Sigma = bW$ in a complex manifold X, and let $M \subset X \backslash W$ be an embedded, real analytic, totally real disc (a diffeomorphic image of the standard disc $D^k \subset \mathbb{R}^k$ for some $1 \le k \le n = \dim_{\mathbb{C}} X$). If the conditions (3.7) and (3.8) hold then for every open set $U \supset \overline{W} \cup M$ there is a smooth strongly pseudoconvex handlebody $\widetilde{W}$ with core $\overline{W} \cup M$ such that $\overline{W} \cup M \subset \widetilde{W} \Subset U$. If W is Stein then $\widetilde{W}$ can also be chosen Stein and such that the sets $\overline{W}$ and $\overline{W} \cup M$ are $\mathcal{O}(\widetilde{W})$-convex.*

A construction of such 'well attached' totally real discs is given in §9.9. These two results together give a strong tool for constructing Stein manifold structures (see §9.10). The conditions that M be real analytic and diffeomorphic to the disc are mainly out of convenience; however, it is unclear whether the orthogonality condition (3.8) can be entirely dispensed with.

We begin by considering a certain model situation in $\mathbb{C}^n$. Let J_{st} denote the standard complex structure on $\mathbb{C}^n$. For a fixed $k \in \{1, \ldots, n\}$ let $z = (z_1, \ldots, z_n) = x + \mathrm{i}y = (x' + \mathrm{i}y', x'' + \mathrm{i}y'')$, with $z_j = x_j + \mathrm{i}y_j$, denote the coordinates on $\mathbb{C}^n$ corresponding to the decomposition

$$\mathbb{C}^n = \mathbb{C}^k \oplus \mathbb{C}^{n-k} = \mathbb{R}^k \oplus \mathrm{i}\mathbb{R}^k \oplus \mathbb{R}^{n-k} \oplus \mathrm{i}\mathbb{R}^{n-k}.$$

Let $D = D^k \subset \mathbb{R}^k$ be the closed unit ball in $\mathbb{R}^k$ and $S = S^{k-1} = bD$ its boundary $(k-1)$-sphere. Identifying D^k with its image in $\mathbb{R}^k \oplus \{0\}^{2n-k} \subset \mathbb{C}^n$ we obtain the core of the standard index k handle

$$H_\delta = (1+\delta)D^k \times \delta D^{2n-k} \subset \mathbb{C}^n, \quad \delta > 0. \tag{3.9}$$

A *standard handlebody of index* k in $\mathbb{C}^n$ is a set $K_{\lambda,\delta} = Q_\lambda \cup H_\delta$ for some $0 < \lambda < 1$ and $0 < \delta < \frac{2\lambda}{1-\lambda}$ (Fig. 3.3), where

$$Q_\lambda = \big\{z = (x' + \mathrm{i}y', z'') \in \mathbb{C}^k \oplus \mathbb{C}^{n-k} \colon |y'|^2 + |z''|^2 < \lambda(|x'|^2 - 1)\big\}. \tag{3.10}$$

The condition $\lambda < 1$ insures that Q_λ is strongly pseudoconvex, and the bound on δ implies $(1+\delta)bD^k \times \delta D^{2n-k} \subset Q_\lambda$.

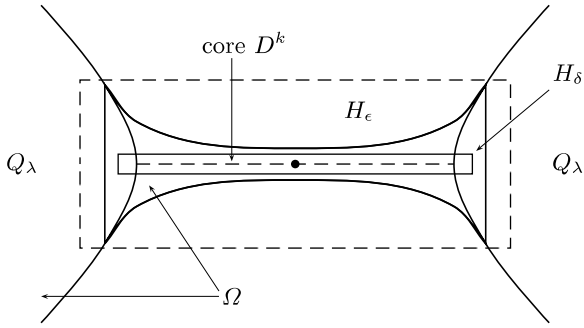

Fig. 3.3. A strongly pseudoconvex handlebody Ω. (Modified from [194, p. 620, Fig. 1])

Lemma 3.8.3. [129] *Let Q_λ be the domain (3.10), $0 < \lambda < 1$. For every $\epsilon > 0$ there exist a number $\delta \in (0, \epsilon)$ and a smooth strongly pseudoconvex handlebody $\Omega \subset \mathbb{C}^n$ with core $Q_\lambda \cup D^k$ such that $K_{\lambda,\delta} \subset \Omega \subset K_{\lambda,\epsilon}$ (Fig. 3.3).*

Proof. The main case to consider is $k = n$. In this case we have

$$Q_\lambda = \{x + \mathrm{i}y \in \mathbb{C}^n \colon |y|^2 \le \lambda(|x|^2 - 1)\}, \quad D^n = \{x + \mathrm{i}0 \colon x \in \mathbb{R}^n,\ |x| \le 1\}.$$

Given $\epsilon > 0$ we shall find a number $\sigma = \sigma(\epsilon) \in (0, \epsilon)$ and a smooth increasing function $f \colon \mathbb{R}_+ \to (0, \infty)$ such that the domain

$$\Omega = \{x + \mathrm{i}y \in \mathbb{C}^n \colon |y|^2 \le f(|x|^2)\} \tag{3.11}$$

is strongly pseudoconvex and we have

$$f(t) = \sigma \text{ for } t \le 1 - \epsilon, \qquad f(t) = \lambda(t-1) \text{ for } t \ge 1 + \epsilon. \tag{3.12}$$

Hence for every neighborhood $V \subset \mathbb{C}^n$ of the disc D^n we can choose a handlebody Ω as above such that $Q_\lambda \cup D^n \subset \Omega \subset Q_\lambda \cup V$ and $\Omega \backslash V = Q_\lambda \backslash V$.

The construction proceeds in two steps. First we identify conditions on a function $f\colon \mathbb{R}_+ \to (0,\infty)$ that characterize strong pseudoconvexity of the tubular domain (3.11). Let U be a nonempty open set in $\mathbb{R}^n$ $(n > 1)$ that is invariant under the action of the orthogonal group $O(n)$. Set $I = \{|x|^2 \colon x \in U\} \subset \mathbb{R}_+$. Assume that $f\colon I \to (0,+\infty)$ is a positive function of class $\mathcal{C}^2$.

Proposition 3.8.4. [186, Proposition 2.1] *The domain*

$$\Omega = \{x + \mathrm{i}y \in \mathbb{C}^n \colon x \in U,\ |y|^2 < f(|x|^2)\} \tag{3.13}$$

is strongly pseudoconvex along the hypersurface $\Sigma = \{|y|^2 = f(|x|^2)\} \subset b\Omega$ *if and only if* f *satisfies the following differential inequalities for* $t \in I$*:*

$$f'(t) < 1, \qquad 2tf(t)f''(t) < \big[1 - f'(t)\big] \cdot \big[tf'(t)^2 + f(t)\big]. \tag{3.14}$$

Proof. Set $\rho(x + \mathrm{i}y) = |y|^2 - f(|x|^2)$. A calculation gives for $1 \le j \ne k \le n$

$$-\rho_{z_k} = x_k f' + \mathrm{i}y_k,\ -2\rho_{z_k \bar z_k} = 2x_k^2 f'' + f' - 1,\ -2\rho_{z_j \bar z_k} = 2x_j x_k f''.$$

Here f and its derivatives are evaluated at $t = |x|^2$. The calculation of the Levi form of $\Sigma = \{\rho = 0\}$ can be simplified by observing that ρ is invariant under the action of $O(n)$ on $\mathbb{C}^n$ given by $A(x + \mathrm{i}y) = Ax + \mathrm{i}Ay$ $(A \in O(n))$. Fix a point $p \in \Sigma$. After an orthogonal rotation we have $p = (x_1 + \mathrm{i}y_1, \mathrm{i}\widetilde{y}_2, \dots, \mathrm{i}\widetilde{y}_n) \in \mathbb{C} \times \mathrm{i}\mathbb{R}^{n-1}$. Applying another orthogonal map fixing $\mathbb{C} \times \{0\}^{n-1}$ we may further assume that $p = (x_1 + \mathrm{i}y_1, \mathrm{i}y_2, 0, \dots, 0)$. At this point we have

$$\rho_{z_1}(p) = -x_1 f' - \mathrm{i}y_1, \quad \rho_{z_2}(p) = -\mathrm{i}y_2, \quad \rho_{z_k}(p) = 0 \text{ for } k = 3, \dots, n.$$

Hence the complex tangent space $\mathrm{T}_p^{\mathbb{C}}\Sigma = \{v \in \mathbb{C}^n \colon \sum_{k=1}^n \frac{\partial \rho}{\partial z_k}(p) v_k = 0\}$ consists of all $v \in \mathbb{C}^n$ satisfying $v_1 = -\lambda \mathrm{i}y_2$, $v_2 = \lambda(x_1 f' + \mathrm{i}y_1)$ for arbitrary choices of $\lambda \in \mathbb{C}$ and $v'' = (v_3, \dots, v_n) \in \mathbb{C}^{n-2}$. We also have

$$\begin{aligned} &2\rho_{z_1 \bar z_1}(p) = 1 - f' - 2x_1^2 f'', \\ &2\rho_{z_k \bar z_k}(p) = 1 - f', && k = 2, \dots, n, \\ &2\rho_{z_j \bar z_k}(p) = 0, && 1 \le j \ne k \le n. \end{aligned}$$

For $v \in \mathrm{T}_p^{\mathbb{C}}\Sigma$ we thus get (noting that $y_1^2 + y_2^2 = f(x_1^2)$) the following:

$$\begin{aligned} 2\mathcal{L}_{\rho,p}(v) &= \big(1 - f' - 2x_1^2 f''\big)|\lambda|^2 y_2^2 + (1 - f')|\lambda|^2 (x_1^2 f'^2 + y_1^2) + (1 - f')|v''|^2 \\ &= |\lambda|^2 \Big(-2x_1^2 y_2^2 f'' + (1 - f')(x_1^2 f'^2 + f)\Big) + (1 - f')|v''|^2. \end{aligned} \tag{3.15}$$

The function f and its derivatives are evaluated at x_1^2. Thus $\mathcal{L}_{\rho,p}(v) > 0$ for all choices of $\lambda \in \mathbb{C}$ and $v'' \in \mathbb{C}^{n-2}$ with $|\lambda|^2 + |v''|^2 > 0$ if and only if

$$f' < 1, \qquad 2x_1^2 y_2^2 f'' < (1 - f')(x_1^2 f'^2 + f).$$

Observe that $0 \le y_2^2 \le |s|^2 = f(x_1^2)$, and y_2^2 assumes both extreme values 0 and $f(x_1^2)$ when (y_1, y_2) traces the circle $y_1^2 + y_2^2 = f(x_1^2)$. Thus the second inequality above holds at all points of this circle precisely when it holds at the point $y_1 = 0$, $y_2 = \sqrt{f(x_1^2)}$. This gives the conditions

$$f' < 1, \qquad 2x_1^2 f f'' < (1 - f')(x_1^2 f'^2 + f)$$

characterizing strong pseudoconvexity of Ω along the mentioned circle in Σ. Since $x_1^2 = |x(p)|^2$, this is equivalent to the pair of inequalities (3.14). Similarly we see that $\Omega' = \{|y|^2 > f(|x|^2)\}$ is strongly pseudoconvex along the hypersurface Σ if and only if the reverse inequalities hold in (3.14). □

The second step in the proof of Lemma 3.8.3 amounts to finding functions $f\colon \mathbb{R} \to (0, \infty)$ satisfying Proposition 3.8.4 and the boundary conditions (3.12). A detailed construction of such functions is given in [186, §3]; we do not reproduced it here since it only involves elementary calculus. □

For handles of lower dimension we obtain strongly pseudoconvex handlebodies satisfying Lemma 3.8.3 by taking

$$\Omega = \{(x' + \mathrm{i}y', z'')\colon |y'|^2 + |z''|^2 \le f(|x'|^2)\},$$

where f has the same properties as above.

Proof (of Theorem 3.8.2). We reduce to the model situation in Lemma 3.8.3. Consider the case when $\dim_{\mathbb{R}} M = \dim_{\mathbb{C}} X = n$. Let ρ be a smooth defining function for $W = \{\rho < 0\}$ which is strongly plurisubharmonic near $\Sigma = bW = \{\rho = 0\}$ and $\mathrm{d}\rho \ne 0$ on Σ. (If W is Stein then we can choose ρ to be strongly plurisubharmonic in a neighborhood of $\overline{W}$.) After a small perturbation of W that enlarges the domain and keeps $\Sigma = bW$ fixed along the real analytic submanifold $bM \subset \Sigma$ we may assume that ρ and Σ are real analytic. The rest of the construction will be performed in a neighborhood of M.

Since M is real analytic and totally real, a biholomorphic change of coordinates along M brings us to the situation when M is the disc $D^n \subset \mathbb{R}^n \subset \mathbb{C}^n$. We use the coordinates $z = x + \mathrm{i}y$ on $\mathbb{C}^n$ in which $\mathbb{R}^n = \{y = 0\}$ and $D^n = \{|x| \le 1\}$. In a neighborhood of the sphere $S = bD^n = \{|x|^2 = 1\}$ the defining function of $W = \{\rho < 0\}$ is of the form

$$\rho(z) = r(x)\big(1 - |x|^2\big) + \sum_{j=1}^n y_j g_j(z) + \sum_{j,k=1}^n y_j y_k g_{jk}(z) + o\left(|y|^2, (|x|^2 - 1)^2\right)$$

for some real analytic function $r > 0$. Note that a real analytic diffeomorphism of the disc D^n fixing the boundary sphere S complexifies to a biholomorphic

change of coordinates in neighborhood of D^n in $\mathbb{C}^n$. By using a coordinate change of this type, induced by the map

$$x \mapsto \big(1 + (|x|^2 - 1)\tau(x)\big)\, x, \qquad x \in D^n$$

for a suitably chosen real analytic function τ on D^n, we can normalize to $r(x) = 1$ and $\mathrm{d}r(x) = 0$ for every $x \in S$. (The first condition is satisfied by taking $\tau(x) = \frac{1}{2}(r(x) - 1)$ on $|x| = 1$, and the second condition relates the differentials $\mathrm{d}\tau_x$ and $\mathrm{d}r_x$ at points $|x| = 1$.) Thus the first term on the right hand side can be replaced by $1 - |x|^2$. Note that ρ remains strongly plurisubharmonic since we only made a biholomorphic coordinate change.

The Legendre condition (3.7) and the orthogonality condition (3.8) together imply that the tangent space $\mathrm{T}_x\Sigma$ at any point $x \in S$ contains $\{x\} \times \mathrm{i}\mathbb{R}^n$. Hence the functions $g_j(z)$ in the expansion of ρ vanish on S, so $\sum_{j=1}^n y_j g_j(z)$ can be put in the remainder term. This simplifies ρ to

$$\rho(z) = 1 - |x|^2 + \sum_{j,k=1}^n y_j y_k g_{jk}(z) + o(|y|^2, (|x|^2 - 1)^2).$$

At points $z \in S$ the Levi form of ρ in the direction $w \in \mathbb{C}^n$ equals

$$\mathcal{L}_{\rho,z}(w) = -\frac{1}{2}|w|^2 + \frac{1}{2}\sum_{j,k=1}^n g_{jk}(z) w_j \overline{w}_k.$$

Since this is positive and S is compact, there is a number $\lambda < 1$ such that

$$\sum_{j,k=1}^n g_{jk}(z) w_j \overline{w}_k > \lambda^{-1}|w|^2, \quad z \in S.$$

Setting $w = y$ we get the following estimate in a neighborhood of S:

$$\rho(z) \geq \rho_\lambda(z) = 1 - |x|^2 + \lambda^{-1}|y|^2.$$

Note that $\{\rho_\lambda < 0\} = \{|y|^2 < \lambda(|x|^2 - 1)\} = Q_\lambda$ is the quadric model domain (3.10) on p. 84. Hence Q_λ osculates W from the outside along the sphere S.

Pick $\epsilon > 0$. The domain $W_\epsilon = \{\rho < \epsilon\}$ contains S, and a generic choice of ϵ insures that the hypersurfaces $bW_\epsilon = \{\rho = \epsilon\}$ and $\{\rho_\lambda = 0\} = bQ_\lambda$ intersect transversely. By smoothing the corners we find a smooth strongly pseudoconvex domain W' which coincides with Q_λ in a neighborhood U of S and satisfies $W \subset W' \subset W_\epsilon \cap Q_\lambda$. By gluing onto Q_λ a thin strongly pseudoconvex tube Ω (3.11) as in the model case (Lemma 3.8.3 on p. 84), insuring that the connection of Ω onto Q_λ is realized within U, we get a smooth strongly pseudoconvex handlebody $W' \cup \Omega$ with core $W \cup M$. This proves Theorem 3.8.2 for handles of maximal dimension. Essentially the same proof applies to handles of lower dimension. $\square$

3.9 Morse Critical Points of q-Convex Functions

We describe the quadratic normal form of a q-convex function ρ at a quadratically nondegenerate (Morse) critical point. (Recall that a smooth function is said to be a *Morse function* if all its critical points are Morse points.)

We begin with the case when ρ is strongly plurisubharmonic ($q = 1$). The following lemma appeared in a 1924 paper of T. Takagi [487, Theorem II]; it was rediscovered in [435] and [248, p. 166].

Lemma 3.9.1. *Let ρ be a strongly plurisubharmonic function of class $\mathcal{C}^2$ in a neighborhood of the origin in $\mathbb{C}^n$ with a Morse critical point of index k at $0 \in \mathbb{C}^n$. Then $k \in \{0, 1, \ldots, n\}$. Write $z = (z', z'') = (x' + \mathrm{i}y', x'' + \mathrm{i}y'') \in \mathbb{C}^k \oplus \mathbb{C}^{n-k}$. After a $\mathbb{C}$-linear change of coordinates on $\mathbb{C}^n$ we have*

$$\rho(z) = \rho(0) - |x'|^2 + |x''|^2 + \sum_{j=1}^{n} \lambda_j y_j^2 + o(|z|^2) \tag{3.16}$$

where $\lambda_j > 1$ for $j \in \{1, \ldots, k\}$ and $\lambda_j \geq 1$ for $j \in \{k+1, \ldots, n\}$.

Proof. We follow the proof in [248]. We may assume that $\rho(0) = 0$. Since the Levi form of ρ at 0 is positive definite, a $\mathbb{C}$-linear change of coordinates normalizes it to $|z|^2$. In these coordinates we have

$$\rho(z) = |z|^2 + \Re\left(z^t S z\right) + o(|z|^2)$$

where $S = A + \mathrm{i}B$ is a symmetric $n \times n$ matrix with entries $S_{j,l} = \frac{\partial^2 \rho(0)}{\partial z_j \partial z_l}$. The expression for ρ in terms of the real coordinates $s = (x, y)^t \in \mathbb{R}^{2n}$ is

$$\rho(s) = |s|^2 + s^t T s + o(|s|^2), \quad T = \begin{pmatrix} A & -B \\ -B & -A \end{pmatrix}.$$

Let J denote the standard complex structure on $\mathbb{C}^n$. For every tangent vector $v = (a, b)^t \in \mathrm{T}_0\mathbb{R}^{2n} \cong \mathbb{R}^{2n}$ we have $Jv = J(a, b)^t = (-b, a)^t$. If $Tv = \lambda v$ then a calculation gives $T(Jv) = -\lambda Jv$. Thus for each eigenvector v of T, with the eigenvalue λ, Jv is also an eigenvector of T with the eigenvalue $-\lambda$. Since T is symmetric, it follows that there exists an orthonormal basis $v_1, \ldots, v_n, Jv_1, \ldots, Jv_n$ of $\mathbb{R}^{2n}$ consisting of eigenvectors of T. Interchanging v_j and Jv_j if necessary we may assume that the eigenvalue of v_j is $-\mu_j \leq 0$.

Let $e_1, \ldots, e_{2n}$ denote the standard basis of $\mathbb{R}^{2n}$, chosen so that $Je_j = e_{n+j}$ for $j = 1, \ldots, n$. Let C denote the orthogonal linear transformation on $\mathbb{R}^{2n}$ determined by $Ce_j = v_j$ and $Ce_{n+j} = Jv_j$ for $j = 1, \ldots, n$. Then $CJ = JC$, and hence C determines a unitary transformation $\widetilde{C}$ of $\mathbb{C}^n$. We introduce new coordinates $\sigma = (x, y)^t \in \mathbb{R}^{2n}$ by $s = C\sigma$. In these coordinates

$$\begin{aligned}\rho(C\sigma) &= (C\sigma)^t(I^{2n}+T)C\sigma + o(|\sigma|^2)\\ &= \sigma^t(I + C^tTC)\sigma + o(|\sigma|^2)\\ &= |\sigma|^2 + \sigma^t D\sigma + o(|\sigma|^2),\end{aligned}$$

where D is a diagonal matrix with entries $-\mu_1, \dots, -\mu_n, \mu_1, \dots, \mu_n$ on its main diagonal. Denoting again $z = x + \mathrm{i}y \in \mathbb{C}^n$, the above reads

$$\rho(\widetilde{C}z) = \sum_{j=1}^{n} \big((1-\mu_j)x_j^2 + (1+\mu_j)y_j^2\big) + o(|z|^2).$$

Since 0 is a Morse critical point of ρ, we have $\mu_j \neq 1$ or all j. Let $k \in \{0, 1, \dots, n\}$ denote the number of indexes $j \in \{1, \dots, n\}$ for which $\mu_j > 1$. Thus the Hessian of ρ at 0 has precisely k negative eigenvalues, so k is the *Morse index* of ρ at 0. After a permutation of the variables we may assume that $\mu_j > 1$ precisely for $j = 1, \dots, k$. Setting

$$\lambda_j = \frac{1+\mu_j}{|1-\mu_j|} \geq 1, \quad j = 1, \dots, n$$

and replacing $\sqrt{|1-\mu_j|}\, z_j$ by z_j we get the normal form (3.16). □

Definition 3.9.2. *A critical point p of a strongly plurisubharmonic function ρ is* nice *if, in some local holomorphic coordinates in a neighborhood of p, ρ is of the form (3.16) without a remainder term.*

Lemma 3.9.3. *Every strongly plurisubharmonic function $\rho\colon X \to \mathbb{R}$ can be approximated in the fine $\mathcal{C}^2$ Whitney topology by a Morse strongly plurisubharmonic function all of whose critical points are nice.*

Proof. By the Morse lemma we can approximate ρ in the fine $\mathcal{C}^2$ topology on X by a strongly plurisubharmonic Morse function. Let $p \in X$ be a (Morse) critical point of ρ. In local holomorphic coordinates z in an open set $p \in U \subset X$, with $z(p) = 0$, ρ is of the form (3.16). Choose a smooth function $\chi\colon \mathbb{C}^n \to [0,1]$ which equals zero on the unit ball $\mathbb{B} \subset \mathbb{C}^n$ and equals one on $\mathbb{C}^n \backslash 2\mathbb{B}$. For sufficiently small $\epsilon > 0$ the function

$$\widetilde{\rho}(z) = \rho(0) - |x'|^2 + |x''|^2 + \sum_{j=1}^{n} \lambda_j y_j^2 + \chi(\epsilon^{-1}z)\, o(|z|^2)$$

is strongly plurisubharmonic, it has a nice critical point at 0 and no other critical points nearby, and $\widetilde{\rho} = \rho$ outside of a small neighborhood of p. Performing such local change at every critical point we obtain a new strongly plurisubharmonic function with nice critical points. □

We now consider the general case when ρ is q-convex near $0 \in \mathbb{C}^n$ for some $q \in \{1, 2, \dots, n+1\}$, i.e., its Levi form $\mathcal{L}_{\rho,0}$ has at least $r = n - q + 1$

positive eigenvalues; the remaining $s = q - 1$ eigenvalues can be of any sign. By a complex linear change of coordinates on $\mathbb{C}^n$ we can achieve that the subspace $\mathbb{C}^r \times \{0\}^s$ is spanned by some of the eigenvectors corresponding to the positive eigenvalues of $\mathcal{L}_{\rho,0}$ and 0 is a Morse critical point of $\rho(\cdot\,,0)$. We denote the coordinates on $\mathbb{C}^n = \mathbb{C}^r \times \mathbb{C}^s = \mathbb{C}^r \times \mathbb{R}^{2s}$ by $z = (\zeta, u)$ where $\zeta = x + \mathrm{i}y \in \mathbb{C}^r$ $(x, y \in \mathbb{R}^r)$ and $u \in \mathbb{R}^{2s}$. By shrinking the domain of ρ to a sufficiently small polydisc $P = P^r \times P^s \subset \mathbb{C}^n$ around 0 we can assume that the function $\zeta \to \rho(\zeta, u)$ is strongly plurisubharmonic on P^r for each $u \in P^s$.

Lemma 3.9.1 applied to the strongly plurisubharmonic function $\zeta \mapsto \rho(\zeta, 0)$ gives a complex linear change of coordinates on $\mathbb{C}^r$ and an integer $k \in \{0, 1, \ldots, r\}$ such that, in the new coordinates,

$$\rho(\zeta, 0) = \sum_{j=1}^{r} \big(\delta_j x_j^2 + \lambda_j y_j^2\big) + o(|\zeta|^2)$$

where $\lambda_j > 1$, $\delta_j = -1$ for $j = 1, \ldots, k$, and $\lambda_j \geq 1$, $\delta_j = +1$ for $j = k+1, \ldots, r$. Note that k is the Morse index of $\rho(\zeta, 0)$ at $\zeta = 0$. Writing $x' = (x_1, \ldots, x_k)$ and $x'' = (x_{k+1}, \ldots, x_r)$ we obtain

$$\rho(\zeta, 0) = -|x'|^2 + |x''|^2 + \sum_{j=1}^{r} \lambda_j y_j^2 + o(|\zeta|^2).$$

We now consider the full second order Taylor expansion of ρ at $0 \in \mathbb{C}^n$:

$$\rho(z) = \rho(\zeta, u) = \rho(\zeta, 0) + \sum_{j=1}^{2s} u_j a_j(x, y) + \sum_{i,j=1}^{2s} c_{ij}\, u_i u_j + o(|z|^2).$$

Here $a_j(x, y) = \sum_{l=1}^{r} (\alpha_{jl} x_l + \beta_{jl} y_l)$ are real-valued linear functions on $\mathbb{C}^r = \mathbb{R}^{2r}$ and $c_{ij} = c_{ji}$ are real constants. Our next aim is to remove the mixed terms $u_j a_j(x, y)$ by a shear of the form $(\zeta, u) \mapsto (\zeta + h(u), u)$ for a suitable $\mathbb{R}$-linear map $h\colon \mathbb{R}^{2s} \to \mathbb{C}^r$; such transformation clearly preserves plurisubharmonicity in the ζ variable. Consider the critical point equation $\partial_\zeta \rho^{(2)}(\zeta, u) = 0$, where $\rho^{(2)}$ is the 2nd order homogeneous polynomial of ρ:

$$\frac{\partial \rho^{(2)}}{\partial x_i}(\zeta, u) = 2\delta_i x_i + \sum_{j=1}^{2s} u_j \alpha_{ji} = 0; \quad \frac{\partial \rho^{(2)}}{\partial y_i}(\zeta, u) = 2\lambda_i y_i + \sum_{j=1}^{2s} u_j \beta_{ji} = 0.$$

This system has a unique (linear) solution $\zeta = x + \mathrm{i}y = h(u)$, and the quadratic map $\zeta \to \rho^{(2)}(\zeta + h(u), u)$ has a unique critical point at $\zeta = 0$ for every u. Writing $\rho(\zeta, u) = \widetilde{\rho}(\zeta + h(u), u)$, the function $\widetilde{\rho}$ is of the same form as ρ but with $a_j(x, y) = 0$ for all $j = 1, \ldots, 2s$. We drop the tilde and denote the new function again by ρ. Sylvester's theorem furnishes an $\mathbb{R}$-linear transformation of the u-coordinates which puts $\sum_{i,j=1}^{2s} c_{ij}\, u_i u_j$ into a normal form $-|u'|^2 + |u''|^2$ where $u' = (u_1, \ldots, u_m)$ and $u'' = (u_{m+1}, \ldots, u_{2s})$ for some $m \in \{0, 1, \ldots, 2s\}$. This gives $\rho(\zeta, u) = \widetilde{\rho}(\zeta, u) + o(|\zeta|^2 + |u|^2)$ where

$$\widetilde{\rho}(\zeta, u) = -|x'|^2 - |u'|^2 + |x''|^2 + |u''|^2 + \sum_{j=1}^{r} \lambda_j y_j^2 \tag{3.17}$$

and

$$\lambda_j > 1 \text{ for } j = 1, \ldots, k, \qquad \lambda \geq 1 \text{ for } j = k+1, \ldots, r. \tag{3.18}$$

We say that a function (3.17) satisfying conditions (3.18) is a *q-convex quadratic normal form* at a critical point. Note that $k+m$ is the Morse index at 0. We summarize the above discussion in the following lemma.

Lemma 3.9.4. *Let X be an n-dimensional complex manifold, and let $\rho\colon X \to \mathbb{R}$ be a $\mathcal{C}^2$ function with a Morse critical point at $p_0 \in X$. If ρ is q-convex at p_0 for some $q \in \{1, \ldots, n+1\}$ then there exist*

- *a local holomorphic coordinate map $z = (\zeta, w)\colon U \to \mathbb{C}^r \times \mathbb{C}^s$ on an open neighborhood $U \subset X$ of p_0, with $z(p_0) = 0$, $r = n - q + 1$ and $s = q - 1$,*
- *a change of coordinates $\psi(z) = \psi(\zeta, w) = (\zeta + h(w), g(w))$ on $\mathbb{C}^n$ which is $\mathbb{R}$-linear in $w \in \mathbb{C}^s = \mathbb{R}^{2s}$, and*
- *a quadratic normal form $\widetilde{\rho}(\zeta, u)$ of type (3.17), (3.18)*

such that, setting $\phi(p) = \psi(z(p)) \in \mathbb{C}^n$ for $p \in U$, we have

$$\rho(p) = \rho(p_0) + \widetilde{\rho}(\phi(p)) + o(|\phi(p)|^2), \quad p \in U.$$

Furthermore, we can approximate ρ as close as desired in the $\mathcal{C}^2$ topology by a q-convex function ρ' that agrees with ρ outside of U and has a nice critical point *at p_0, in the sense that $\rho'(p) = \rho(p_0) + \widetilde{\rho}(\phi(p))$ near p_0.*

Proof. Everything except the claim in the last sentence has already been proved. The latter is seen by taking

$$\rho'(p) = \rho(p_0) + \widetilde{\rho}(\phi(p)) + \chi(\epsilon^{-1}\phi(p))o(|\phi(p)|^2)$$

where $\chi\colon \mathbb{C}^n \to [0, 1]$ is a smooth function which equals zero in the unit ball $\mathbb{B} \subset \mathbb{C}^n$ and equals one outside $2\mathbb{B}$. When $\epsilon > 0$ decreases to zero, the $\mathcal{C}^2$-norm of the last summand tends to zero uniformly on U. $\square$

3.10 Crossing a Critical Level of a q-Convex Function

The results of this section serve to pass a critical level of a strongly plurisubharmonic function or, more generally, of a q-convex function, in many subsequent analytic constructions. We begin with the case $q = 1$. Let $\rho\colon \mathbb{C}^n \to \mathbb{R}$ be a strongly plurisubharmonic quadratic normal form

$$\rho(z) = -|x'|^2 + |x''|^2 + \sum_{j=1}^{n} \lambda_j y_j^2 \tag{3.19}$$

where $z = (z', z'') = (x' + \mathrm{i}y', x'' + \mathrm{i}y'') \in \mathbb{C}^k \oplus \mathbb{C}^{n-k}$, $k \in \{1, \dots, n\}$, $\lambda_j > 1$ for $j \in \{1, \dots, k\}$, and $\lambda_j \geq 1$ for $j \in \{k+1, \dots, n\}$ (Lemma 3.9.1). This function has a unique (Morse) critical point with Morse index k at the origin. Set $\lambda = \min\{\lambda_1, \dots, \lambda_k\} > 1$. Choose a number $c_0 > 0$ and let

$$E = \{(x' + \mathrm{i}y', z'') \in \mathbb{C}^n \colon y' = 0,\ z'' = 0, |x'|^2 \leq c_0\}. \tag{3.20}$$

Choose a number $\mu \in (1, \lambda)$ and set

$$t_0 = t_0(c_0, \mu) = c_0\big(1 - \frac{1}{\mu}\big)^2 \in (0, c_0). \tag{3.21}$$

Lemma 3.10.1. *There exists a smooth strongly plurisubharmonic function τ on $\mathbb{C}^n$ satisfying the following properties (see Fig. 3.5):*

(a) $\{\rho \leq -c_0\} \cup E \subset \{\tau \leq 0\} \subset \{\rho \leq -t_0\} \cup E$,
(b) $\{\rho \leq c_0\} \subset \{\tau \leq 2c_0\} \subset \{\rho < 3c_0\}$,
(c) $\tau = \rho + t_1$ *on* $\{|x'|^2 \geq c_0\}$ *for some* $t_1 \in (t_0, c_0)$, *and*
(d) τ *has no critical values in* $(0, +\infty)$.

Proof. We shall find τ of the form

$$\tau(z) = -h(|x'|^2) + |x''|^2 + \sum_{j=1}^{n} \lambda_j y_j^2 \tag{3.22}$$

where $h \colon \mathbb{R} \to [0, +\infty)$ is a smooth convex increasing function satisfying

(i) $h(t) = 0$ for $t \leq t_0$,
(ii) for $t \geq c_0$ we have $h(t) = t - t_1$ with $t_1 = c_0 - h(c_0) \in (t_0, c_0)$,
(iii) for $t_0 \leq t \leq c_0$ we have $t - t_1 \leq h(t) \leq t - t_0$, and
(iv) for all $t \in \mathbb{R}$ we have $0 \leq \dot{h}(t) \leq 1$ and $2t\ddot{h}(t) + \dot{h}(t) < \lambda$.

To construct h, we first consider the function

$$\xi(t) = \begin{cases} 0, & \text{if } t \leq t_0; \\ \mu\big(\sqrt{t} - \sqrt{t_0}\big)^2, & \text{if } t_0 \leq t \leq c_0, \\ t - c_0(1 - \frac{1}{\mu}), & \text{if } c_0 \leq t. \end{cases}$$

On $[t_0, c_0]$, ξ solves the initial value problem $2t\,\ddot{\xi} + \dot{\xi} = \mu$, $\xi(t_0) = \dot{\xi}(t_0) = 0$. It is $\mathcal{C}^1$ and piecewise $\mathcal{C}^2$, with discontinuities of the second derivative $\ddot{\xi}$ at t_0 and c_0. The value of t_0 is chosen such that $\dot{\xi}(c_0) = 1$. We have $\ddot{\xi}(t) = \mu\sqrt{t_0}/2\sqrt{t}^3 > 0$ for $t \in [t_0, c_0]$, $\ddot{\xi}(t) = 0$ for t outside this interval, and $\int_{t_0}^{c_0} \ddot{\xi}(t)\mathrm{d}t = 1$.

Choose a smooth function $\chi \geq 0$ which vanishes outside $[t_0, c_0]$, equals $\ddot{\xi} + \epsilon$ on $[t_0 + \delta, c_0 - \delta]$ for small $\epsilon, \delta > 0$, and interpolates between 0 and $\ddot{\xi}$ on the intervals $[t_0, t_0 + \delta]$ and $[c_0 - \delta, c_0]$. We can find $\delta, \epsilon > 0$ arbitrarily small

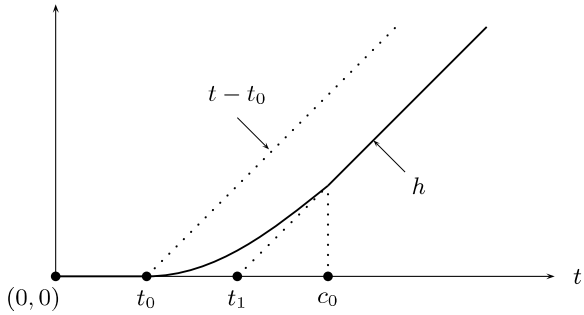

Fig. 3.4. The function h

such that $\int_{t_0}^{c_0} \chi(t)dt = \int_{t_0}^{c_0} \ddot{\xi}(t)\mathrm{d}t = 1$. The function $h\colon \mathbb{R}_+ \to \mathbb{R}_+$ obtained by integrating χ twice with the initial conditions $h(t_0) = \dot{h}(t_0) = 0$ will satisfy the properties (i)–(iii) provided that ϵ and δ were chosen sufficiently small (since h is then $\mathcal{C}^1$-close to ξ and $\ddot{h} \le \ddot{\xi} + \epsilon$). In particular, $t_1 = c_0 - h(c_0) \approx c_0 - \xi(c_0) = (1 - \frac{1}{\mu})c_0$ and hence $t_0 < t_1 < c_0$.

Lemma 3.10.2. *Let $A = (a_{jl}) > 0$ be a symmetric real $n \times n$ matrix with the smallest eigenvalue $\lambda > 0$. If a $\mathcal{C}^2$ function $h\colon I \subset \mathbb{R}_+ \to \mathbb{R}$ satisfies*

$$\dot{h} < \lambda \quad \text{and} \quad 2t\ddot{h} + \dot{h} < \lambda, \quad t \in I, \tag{3.23}$$

then the function $\tau(z) = \tau(x + \mathrm{i}y) = \langle Ay, y\rangle - h(|x|^2)$ is strongly plurisubharmonic on the set $\{z = x + \mathrm{i}y \in \mathbb{C}^n\colon |x|^2 \in I\}$. (Here $\langle \cdot, \cdot \rangle$ denotes the Euclidean inner product on $\mathbb{R}^n$.)

Proof. Let $A = (\, a_{jl}\,)$. A calculation gives $-\tau_{z_j} = x_j\dot{h} + \mathrm{i}\sum_{s=1}^n a_{js}y_s$ and

$$-2\tau_{z_j\bar{z}_l} = \begin{cases} 2x_j^2\ddot{h} + \dot{h} - a_{jj} & \text{if } j = l; \\ 2x_jx_l\ddot{h} - a_{jl} & \text{if } j \ne l. \end{cases}$$

Thus the complex Hessian $H_\tau = (\frac{\partial^2\tau}{\partial z_j \partial \bar{z}_l})$ of τ satisfies

$$-2H_\tau = 2\ddot{h}\cdot xx^t + \dot{h}I - A,$$

where xx^t is the matrix product of the column $x \in \mathbb{R}^n$ with the row x^t and I denotes the identity matrix. For any $v \in \mathbb{R}^n$ we have $\langle (xx^t)v, v\rangle = v^txx^tv = |\langle x, v\rangle|^2$ which lies between 0 and $|x|^2|v|^2$. Hence $0 \le xx^t \le |x|^2 I$. (Here we write $A \le B$ if $B - A$ is nonnegative definite.) At points $|x|^2 = t$ where $\ddot{h}(t) \ge 0$ we thus get

$$-2H_\tau \le (2t\ddot{h} + \dot{h})I - A < \lambda I - A \le 0$$

and hence $H_\tau > 0$ (we used the second inequality in (3.23)). At points where $\ddot{h} < 0$ we can omit $2\ddot{h}xx^t \le 0$ to get $-2H_\tau \le \dot{h}I - A \le (\dot{h} - \lambda)I < 0$, so $H_\tau > 0$ as before. Thus τ is strongly plurisubharmonic. □

We continue with the proof of Lemma 3.10.1. Let h be the function constructed above. By Lemma 3.10.2 the associated function τ (3.22) is strongly plurisubharmonic on $\mathbb{C}^n$. The properties (i)–(iii) of h imply

(α) $\rho \le \tau \le \rho + t_1$ (since $t - t_1 \le h(t) \le t$ for all $t \ge 0$),
(β) $\rho + t_0 \le \tau$ on the set $\{|x'|^2 \ge t_0\}$ (from (ii) and (iii)), and
(γ) $\tau = \rho + t_1$ on $\{|x'|^2 \ge c_0\}$ (from (ii)).

The properties (a)–(c) in Lemma 3.10.1 now follow immediately. To see that τ has no critical values in $(0, +\infty)$ (property (d)), note that $\tau(z) > 0$ implies (in view of $h \ge 0$) that $|x''|^2 + \sum_{j=1}^n \lambda_j y_j^2 > 0$, and this function has nonvanishing differential except where it equals zero. □

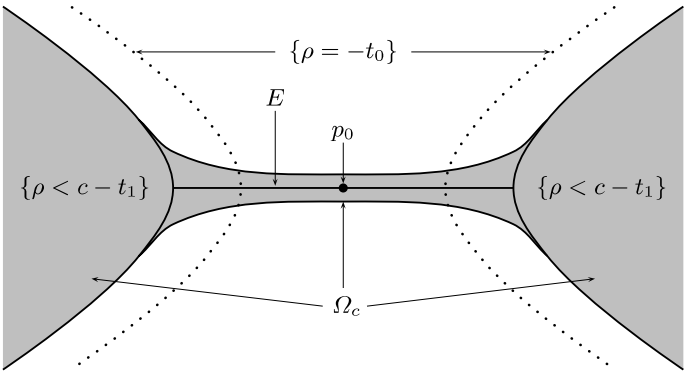

Fig. 3.5. The set $\Omega_c = \{\tau < c\}$. (Modified from [121, p. 246, Fig. 5] and also in [187, p. 12, Fig. 1])

We obtain the analogous result for the q-convex normal form (3.17) on $\mathbb{C}^n = \mathbb{C}^r \oplus \mathbb{R}^{2s}$ ($r = n - q + 1$, $s = n - r = q - 1$). Using the coordinates $\zeta = (\zeta', \zeta'') = (x' + \mathrm{i}y', x'' + \mathrm{i}y'') \in \mathbb{C}^k \oplus \mathbb{C}^{r-k}$, $u = (u', u'') \in \mathbb{R}^{2s}$, we have

$$\rho(\zeta, u) = -|x'|^2 - |u'|^2 + Q(y, x'', u'') \tag{3.24}$$

where

$$Q(y, x'', u'') = |x''|^2 + |u''|^2 + \sum_{j=1}^{r} \lambda_j y_j^2,$$

with $\lambda_j > 1$ for $j = 1, \ldots, k$, and $\lambda \ge 1$ for $j = k+1, \ldots, r$. Set $\lambda = \min\{\lambda_1, \ldots, \lambda_k\} > 1$; if $k = 0$ we take $\lambda = 2$. Choose $c_0 > 0$ and set

$$E = \{(\zeta, u)\colon y' = 0,\ \zeta'' = 0,\ u'' = 0,\ |x'|^2 + |u'|^2 \le c_0\}. \tag{3.25}$$

Choose a number $\mu \in (1, \lambda)$ and let $t_0 \in (0, c_0)$ be given by (3.21). With $h\colon \mathbb{R} \to \mathbb{R}_+$ chosen as in the proof of Lemma 3.10.1 we set

$$\tau(\zeta, u) = -h(|x'|^2 + |u'|^2) + |x''|^2 + |u''|^2 + \sum_{j=1}^{r} \lambda_j y_j^2. \tag{3.26}$$

Lemma 3.10.3. *For every fixed $u \in \mathbb{R}^{2s}$ the function $\tau(\cdot\,, u)$ (3.26) is strongly plurisubharmonic on $\mathbb{C}^r$ (hence τ is q-convex on $\mathbb{C}^n$), and τ satisfies the properties (a)–(d) in Lemma 3.10.1 with respect to the set E given by (3.25), except that (c) must be replaced by*

(c') $\tau = \rho + t_1$ *on* $\{|x'|^2 + |u'|^2 \ge c_0\}$ *for some number* $t_1 \in (t_0, c_0)$.

Proof. For a fixed $u \in \mathbb{R}^{2s}$ we set $c = |u'|^2 \ge 0$ and $h_c(t) = h(t + c)$ $(t \in \mathbb{R})$. We have $\dot{h}_c(t) = \dot{h}(t + c) \le 1 < \lambda$ and

$$2t\ddot{h}_c(t) + \dot{h}_c(t) \le 2(t + c)\ddot{h}(t + c) + \dot{h}(t + c) < \lambda$$

(we have used $\ddot{h} \ge 0$ and the property (iv) of h). Lemma 3.10.2 now shows that the function

$$\mathbb{C}^r \ni \zeta \mapsto -h_c(|x'|^2) + Q(y, x'', u'') = \tau(\zeta, u)$$

is strongly plurisubharmonic. The other properties are verified as before. □

Although Lemmas 3.10.1 and 3.10.3 only apply locally near a nice q-convex critical point, the property (c) (resp. (c')) of the resulting function τ gives the same construction in the following global setting. Assume that ρ is a q-convex function in an open subset Ω of an n-dimensional complex manifold X, with a nice critical point at the point $p_0 \in \Omega$. Without loss of generality we may assume $\rho(p_0) = 0$. Assume in addition that for some $c_0 > 0$ the set

$$\Omega_{c_0} = \{p \in \Omega\colon -c_0 < \rho(p) < 3c_0\}$$

does not contain any critical point of ρ other than p_0. Choose an open neighborhood $U \subset \Omega$ of p_0 and a coordinate map $\phi\colon U \to P$ onto a polydisc P in $\mathbb{C}^n$ such that the function $\widetilde{\rho} = \rho \circ \phi^{-1}\colon P \to \mathbb{R}$ is a q-convex normal form (3.24). By decreasing $c_0 > 0$ if necessary we may assume that

$$\big\{(x + \mathrm{i}y, u) \in \mathbb{C}^r \times \mathbb{R}^{2s}\colon |x'|^2 + |u'|^2 \le c_0,\ Q(y, x'', u'') \le 4c_0\big\} \subset P.$$

Let $\widetilde{E} \subset \mathbb{C}^n$ be given by (3.25). Its preimage $E = \phi^{-1}(\widetilde{E}) \subset U$ is an embedded disc of dimension $k + m$ (the Morse index of ρ at p_0) that is attached from the outside to the sublevel set $\{\rho \le -c_0\}$ along the sphere $bE \subset \{\rho = -c_0\}$. (In the metric on U inherited by ϕ from the standard metric in $\mathbb{C}^n$, E is the local stable manifold of p_0 for the gradient flow of ρ.) Let $t_0 \in (0, c_0)$ be given by (3.21). Also set

$$U' = \big\{p \in U\colon |x'(p)|^2 + |u'(p)|^2 < c_0\big\}.$$

Proposition 3.10.4. *There is a q-convex function $\tau\colon \{\rho < 3c_0\} \to \mathbb{R}$ satisfying the following properties:*

(a) $\{\rho \le -c_0\} \cup E \subset \{\tau \le 0\} \subset \{\rho \le -t_0\} \cup E$,
(b) $\{\rho \le c_0\} \subset \{\tau \le 2c_0\} \subset \{\rho < 3c_0\}$,
(c) $\tau = \rho + t_1$ *holds outside of* U' *for some* $t_1 \in (t_0, c_0)$, *and*
(d) τ *has no critical values in the open interval* $(0, 3c_0) \subset \mathbb{R}$.

Proof. Set $V = \{\rho < 3c_0\}$. Let $\widetilde{\tau}$ be a q-convex function on $\mathbb{C}^n$ given by Lemma 3.10.3. We define a function $\tau\colon V \to \mathbb{R}$ by

$$\tau = \begin{cases} \widetilde{\tau} \circ \phi & \text{on } U \cap V, \\ \rho + t_1 & \text{on } V \backslash U. \end{cases}$$

By Lemma 3.10.3 (c) these two definitions agree on $(U \backslash U') \cap V$; hence τ is well defined, and the stated properties follow immediately. □

3.11 The Topological Structure of a Stein Space

A complex manifold X of dimension n, being triangulable, has the homotopy type of a CW complex of real dimension $\le 2n$. Since a Stein manifold admits a strongly plurisubharmonic Morse exhaustion function and each critical point of such function has Morse index at most $\dim_{\mathbb{C}} X$ by Proposition 3.9.1, Morse theory implies that an n-dimensional Stein manifold is homotopy equivalent to a CW complex of real dimension $\le n$. More precisely, assume that $\rho\colon X \to \mathbb{R}$ is a smooth Morse exhaustion function. Given real numbers $r' < r$, set

$$X_r = \{x \in X\colon \rho(x) \le r\}, \quad X_{r',r} = \{x \in X\colon r' \le \rho(x) \le r\} = \overline{X_r \backslash X_{r'}}.$$

If ρ is strongly plurisubharmonic on $X_{r',r}$ then the pair $(X_r, X_{r'})$ is a relative CW complex of dimension $\le n$, which means that X_r is obtained from $X_{r'}$ by attaching cells of index $\le n$. (See [456, p. 401] for the precise definition.) This was first observed by S. Lefschetz in the proof of his theorem on hyperplane sections of projective algebraic varieties ([19], [357, §7]). By the same token, Lemma 3.9.4 shows that a q-complete complex manifold of dimension n has the homotopy type of a CW complex of dimension $n + q - 1$. These results were generalized by H. Hamm to complex spaces, both in the absolute and in the relative form as in the following theorem.

Theorem 3.11.1. [244, 245] *Let X be a complex space of dimension n, $\rho\colon X \to \mathbb{R}$ a real analytic q-convex exhaustion function on X, and $A \subset X$ a closed complex subvariety. Then for every $r \in \mathbb{R}$ the pair $(X, A \cup X_r)$ is a relative CW complex of dimension (at most) $n + q - 1$. In particular, if ρ is strongly plurisubharmonic (and hence X is Stein) then $(X, A \cup X_r)$ is a relative CW complex of dimension n.*

Corollary 3.11.2. *A q-complete complex space X of complex dimension n has the homotopy type of a CW complex of dimension $n+q-1$. In particular, an n-dimensional Stein space is homotopic to an n-dimensional CW complex.*

Proof. We describe the main idea and refer to the paper [245] for the details.

Let us consider the case when X is Stein. By a finite induction on strata it suffices to prove the result when the difference $X\backslash A$ is smooth (without singularities) and of pure dimension n. Pick functions $f_1, \ldots, f_k \in \mathcal{O}(X)$ such that $A = \{x \in X: f_1(x) = 0, \ldots, f_k(x) = 0\}$ and set $\varphi = \sum_{j=1}^k |f_j|^2$. (By a substratification one can even assume that $A = \{f = 0\}$ is a hypersurface and that $X\backslash A$ is a Stein manifold.) It suffices to show that for any given number $r' > r$ such that r' is a regular values of $\rho|_{X\backslash A}$ the set $X_{r'} = \{\rho \le r'\}$ is obtained from $K = (A \cap X_{r'}) \cup X_r$ by attaching cells of index $\le n$. Choose $M > 0$ large enough such that $X_{r'} \subset U = \{x \in X: \varphi(x) < M\}$. Consider the function $\chi: U \to \mathbb{R}_+$ given by $\chi = 0$ on $A \cup X_r$ any by

$$\chi(x) = \frac{\rho(x) - r}{\rho(x) - r + \log \frac{M}{\varphi(x)}}, \qquad x \in U' = U\backslash(A \cup X_r).$$

Clearly χ is continuous on U and smooth on U'. A calculation shows that the Levi form $\mathcal{L}_\chi$ on U' is positive definite in the directions tangential to the level set of χ [245, Lemma 1]. Furthermore, there is a number $\alpha > 0$ such that χ and its restriction to the hypersurface $\{\rho = r'\}\backslash A$ have no critical points x with $0 < \chi(x) \le \alpha$ [245, Lemma 2]. It follows that the set $\{x \in X_{r'}: \chi(x) \le \alpha\}$ admits a strong deformation retraction onto K. (That K is a strong neighborhood deformation retract also follows from the fact that it is a semianalytic set.) The proof is completed in [245] by applying Morse theory on manifolds with smooth boundaries; in our case $X_{r'}\backslash A$.

Here is an alternative argument for the last step. (Compare with the proof of Proposition 5.12.1 on p. 224.) After decreasing $\alpha > 0$ we may assume that α is a regular value of χ and the hypersurfaces $\{\rho = r'\}$ and $\{\chi = \alpha\}$ intersect transversely along the codimension two submanifold $\Sigma = \{\rho = r'\} \cap \{\chi = \alpha\}$ contained in $X\backslash A$. Hence $D_0 = \{\chi \le \alpha\} \cap X_{r'}$ is a piecewise smooth compact strongly pseudoconvex domain. Choosing $\lambda >> 0$ big enough the function $\tau = \mathrm{e}^{\lambda\chi}$ is strongly plurisubharmonic on the set where $\tau \ge \alpha' = \mathrm{e}^{\lambda\alpha}$. For each number $s \in [0, 1]$ let

$$\rho_s = (1 - s)(\tau - \alpha') + s(\rho - r'), \quad D_s = \{\rho_s \le 0\} \cap X_{r'}.$$

(The sets D_s are similar to those in Fig. 5.4 on p. 226.) As the parameter s increases from 0 to 1, the compact sets D_s increase from D_0 to $D_1 = X_{r'}$. For any $0 \le s < s' \le 1$ we have $D_{s'}\backslash D_s \subset D_1\backslash D_0 \Subset X\backslash A$. There are at most finitely many values of the parameter s at which the topology of D_s changes, and this change is described by attaching a handle of index $\le n = \dim X$. (See the proof of Proposition 5.12.1.) The conclusion follows. A similar proof applies in the case of a q-complete complex space. $\square$

Theorem 3.11.1 together with standard topological arguments implies the following corollary generalizing results of Narasimhan [369, Theorem 3] and Kaup [295, Satz 1].

Corollary 3.11.3. [244, Korollar] *If X is a q-complete complex space and $A \subset X$ is a closed complex subvariety of X then for any abelian group G we have*

$$H^k(X, A; G) = 0 = H_k(X, A; G), \quad k \geq n + q,$$

and $H_{n+q-1}(X; G)$ is free. The same holds if A is replaced by $A \cup X_r$, where $X_r = \{\rho \leq r\}$ for some real analytic q-convex exhaustion function $\rho\colon X \to \mathbb{R}$.

In the case $q = 1$ (when X is Stein) a simpler proof of these cohomology vanishing theorems was given by Colţoiu and Michalache [91].

Morse theoretic methods also give information on the topology of complements of q-convex sets. The following case is of particular interest.

Theorem 3.11.4. ([166], [474, p. 93]) *If K is a compact polynomially convex set in $\mathbb{C}^n$ then its complement $\mathbb{C}^n \backslash K$ is $(n-1)$-connected:*

$$\pi_k(\mathbb{C}^n \backslash K) = 0, \qquad k = 0, 1, \ldots, n-1.$$

Here $\pi_0(U) = 0$ means that U is path connected.

Proof. Pick an open set $U \subset \mathbb{C}^n$ with $K \subset U$. Using standard techniques we find a strongly plurisubharmonic Morse exhaustion function $\rho\colon \mathbb{C}^n \to \mathbb{R}$ that equal $|z|^2$ near infinity and such that $K \subset \{\rho \leq 0\} \subset U$. We may assume that 0 is a regular value of ρ. For $r > 0$ large enough we have $\{\rho \geq r\} = \{|z| \geq \sqrt{r}\}$, and this set is clearly homotopic to the sphere S^{2n-1}. Morse theory applied with the function $-\rho$ (which has Morse indexes $\geq n$) shows that the set $\{\rho \geq 0\}$ is obtained from $\{\rho \geq r\} \simeq S^{2n-1}$ by attaching cells of indexes $\geq n$. Such cells do not add any nontrivial element to the homotopy groups in the range up to $n-1$, and hence the set $\{\rho \geq 0\}$ is $(n-1)$-connected. By exhaustion we get the same conclusion for $\mathbb{C}^n \backslash K$. $\square$

Theorem 3.11.4, coupled with the Alexander duality theorem, gives the cohomology vanishing theorem of A. Browder [57]: For any polynomially convex set $K \subset \mathbb{C}^n$ and an arbitrary abelian group G we have

$$H^k(K; G) = 0, \qquad k = n, n+1, \ldots.$$

(See also [474, p. 96].) Furthermore, E. Zeron [530] proved that for any compact set $K \subset \mathbb{C}^n$ with a basis of open Stein neighborhoods, in particular for any compact rationally convex set, we have $\pi_k(\mathbb{C}^n \backslash K) = 0$ for $k = 0, \ldots, n-2$. It follows that $H^k(K; G) = 0$ for $k \geq n+1$. Many further results on this topic can be found in Stout's monograph [474].

4

Automorphisms of Complex Euclidean Spaces

In this chapter we develop the theory of holomorphic automorphisms of complex Euclidean spaces and of other complex manifolds with large automorphism groups. These results, which appear for the first time in a book form, play an important role in Oka theory. On the one hand, any Stein manifold with the *holomorphic density property* (a condition implying that it has a large group of holomorphic automorphisms) is also an Oka manifold. On the other hand, constructions using automorphisms give a rich family of examples and counterexamples which delineate the borderline between holomorphic flexibility (when the Oka principle holds) and holomorphic rigidity.

The holomorphic automorphism group $\mathrm{Aut}\,\mathbb{C}^n$ is large and complicated when $n > 1$. Following pioneering works of Dixon and Esterle and Rosay and Rudin, Andérsen and Lempert showed that the group generated by shears is dense in $\mathrm{Aut}\,\mathbb{C}^n$ (Theorem 4.9.1). The key observation is that every polynomial holomorphic vector field on $\mathbb{C}^n$ is a finite sum of complete polynomial fields whose flows consist of shears (Proposition 4.9.5 on p. 127). It follows that every isotopy of biholomorphic maps $F_t\colon \Omega \to \Omega_t = F_t(\Omega)$ between Runge domains in $\mathbb{C}^n$, beginning at $t = 0$ with the identity map, can be approximated by holomorphic automorphisms of $\mathbb{C}^n$ (Theorem 4.9.2).

These methods are used in many interesting constructions: of nonstraightenable embedded complex lines in $\mathbb{C}^n$ (Corollary 4.18.8), contrasting the Abhyankar-Moh-Suzuki theorem on straightening of algebraic embeddings $\mathbb{C} \hookrightarrow \mathbb{C}^2$; of proper holomorphic embeddings with interpolation along discrete sets (Theorem 4.17.1); of proper holomorphic embeddings $f\colon \mathbb{C}^k \hookrightarrow \mathbb{C}^n$ whose complement $\mathbb{C}^n \setminus f(\mathbb{C}^k)$ is $(n-k)$-hyperbolic (Theorem 4.18.10); of nonlinearizable periodic holomorphic automorphisms (Theorem 4.19.1); of non-Runge Fatou-Bieberbach domains (Theorem 4.20.2), to mention just a few. The Andersén-Lempert theory applies to any complex manifold with sufficiently many complete holomorphic vector fields. The relevant *density property* was first introduced and studied by Varolin and was further developed by many authors, in particular by Kaliman and Kutzschebauch.

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4.1 Shears

We denote by $\operatorname{Aut}\mathbb{C}^n$ the group of all holomorphic automorphisms of $\mathbb{C}^n$, endowed with the topology of uniform convergence on compact sets.

Every automorphism of $\mathbb{C}$ is affine linear: $z \to az + b$ $(a, b, \in \mathbb{C},\ a \neq 0)$. From now on we assume that $n > 1$. Let $z = (z_1, \ldots, z_n) = (z', z_n)$ denote the coordinates on $\mathbb{C}^n$. Among the simplest automorphisms of $\mathbb{C}^n$ are maps

$$z \mapsto \big(z', z_n + f(z')\big), \tag{4.1}$$

$$z \mapsto \big(z', \mathrm{e}^{f(z')} z_n\big) \tag{4.2}$$

and their conjugates by the elements of $GL_n(\mathbb{C})$. Their inverses are of the same form, with f replaced by $-f$. Automorphisms of type (4.1) and their $SL_n(\mathbb{C})$-conjugates are called *additive shears*, or *volume preserving shears* since they preserve the *standard complex volume form* on $\mathbb{C}^n$,

$$\omega = \mathrm{d}z_1 \wedge \mathrm{d}z_2 \wedge \cdots \wedge \mathrm{d}z_n. \tag{4.3}$$

Automorphisms of type (4.2), and their $GL_n(\mathbb{C})$-conjugates, are called *multiplicative shears* (also called *overshears* in [18]). Here the term *shear* will refer to maps of both types. By composing one shear of each type we get a *generalized shear*:

$$z \mapsto \big(z', \mathrm{e}^{f(z')} z_n + g(z')\big). \tag{4.4}$$

Every additive shear (4.1) is a limit of polynomial additive shears obtained by approximating the entire function f by polynomials. A nontrivial multiplicative shear (4.2) is of course never polynomial.

Each shear is the time-one map of a one-parameter group of automorphisms, with the complex parameter $t \in \mathbb{C}$, consisting entirely of shears:

$$\Phi_t(z) = \big(z', z_n + tf(z')\big), \tag{4.5}$$

$$\Psi_t(z) = \big(z', \mathrm{e}^{tf(z')} z_n\big). \tag{4.6}$$

This means that $\Phi_0 = \mathrm{Id}$, $\Phi_t \circ \Phi_s = \Phi_{t+s}$ for $t, s \in \mathbb{C}$, and similarly for Ψ_t.

Let us write down a coordinate free expression for shears. Recall that $\langle z, w\rangle = \sum_{i=1}^n z_i \overline{w}_i$, the standard Hermitian form on $\mathbb{C}^n$. Given a $\mathbb{C}$-linear map $\lambda\colon \mathbb{C}^n \to \mathbb{C}^k$ for some $k < n$, a vector $v = (v_1, \ldots, v_n) \in \ker\lambda$ and an entire function $f \in \mathcal{O}(\mathbb{C}^k)$, we set for $t \in \mathbb{C}$

$$\Phi_t(z) = z + tf(\lambda z)\, v, \tag{4.7}$$

$$\Psi_t(z) = z + \big(\mathrm{e}^{tf(\lambda z)} - 1\big)\langle z, v\rangle\, v. \tag{4.8}$$

These are one-parameter groups of holomorphic automorphisms of $\mathbb{C}^n$ whose infinitesimal generators are the holomorphic vector fields

$$V_z = \frac{\mathrm{d}}{\mathrm{d}t}\Phi_t(z)|_{t=0} = f(\lambda z)\, v, \tag{4.9}$$

$$W_z = \frac{\mathrm{d}}{\mathrm{d}t}\Psi_t(z)|_{t=0} = f(\lambda z)\langle z, v\rangle\, v. \tag{4.10}$$

We have identified the vector $v = (v_1, \dots, v_n)$ with the constant vector field $\sum_{j=1}^n v_j \frac{\partial}{\partial z_j}$; occasionally we shall tacitly do the same in the sequel. When $\pi(z_1, \dots, z_n) = (z_1, \dots, \hat{z}_j, \dots, z_n) \in \mathbb{C}^{n-1}$ and $v = e_j$, we have

$$V_z = f(\pi(z))\frac{\partial}{\partial z_j}, \qquad W_z = z_j f(\pi(z))\frac{\partial}{\partial z_j}. \tag{4.11}$$

The following lemma is seen by a simple calculation.

Lemma 4.1.1. *Let Φ_t (resp. Ψ_t) be given by (4.7) (resp. (4.8)). Let $A \in SL_n(\mathbb{C})$ be such that $Av = e_n$. For every $t \in \mathbb{C}$ we then have*

$$A \circ \Phi_t(z) = \widetilde{\Phi}_t(Az), \quad A \circ \Psi_t(z) = \widetilde{\Psi}_t(Az),$$

where $\widetilde{\Phi}_t$ (resp. $\widetilde{\Psi}_t$) is given by (4.5) (resp. (4.6)) with f replaced by $f \circ \lambda \circ A^{-1}$.

Given a holomorphic map $F = (F_1, \dots, F_n)$ from a domain in $\mathbb{C}^n$ to $\mathbb{C}^n$, we denote by $F' = (\,\partial F_j/\partial z_k\,)$ its complex Jacobian matrix and by $\mathrm{J}F = \det F'$ its Jacobian determinant. Then $F^*\omega = (\mathrm{J}F)\,\omega$. We say that F is *volume preserving* if $F^*\omega = \omega$; equivalently, $\mathrm{J}F = 1$. Let

$$\mathrm{Aut}_1\mathbb{C}^n = \{F \in \mathrm{Aut}\,\mathbb{C}^n\colon\ \mathrm{J}F = 1\}.$$

All additive shears (4.1) and their $SL_n(\mathbb{C})$-conjugates are volume preserving; in particular, the family Φ_t (4.7) is a one-parameter subgroup of $\mathrm{Aut}_1\mathbb{C}^n$.

Recall from §1.9 (see in particular (1.55) and (1.59)) that the divergence of a holomorphic vector field V with respect to a holomorphic volume form ω is the holomorphic function $\mathrm{div}_\omega V$ satisfying the equation

$$L_V\omega = \mathrm{d}(V\rfloor\omega) = \mathrm{div}_\omega V \cdot \omega.$$

If $\omega = \mathrm{d}z_1 \wedge \cdots \wedge \mathrm{d}z_n$ then

$$\mathrm{div}_\omega\left(\sum_{j=1}^n a_j(z)\frac{\partial}{\partial z_j}\right) = \sum_{j=1}^n \frac{\partial a_j}{\partial z_j}(z).$$

For the shear vector fields we have

$$\mathrm{div}_\omega\,(f(\lambda z)\, v) = 0, \quad \mathrm{div}_\omega\,(f(\lambda z)\langle z, v\rangle\, v) = f(\lambda z). \tag{4.12}$$

This is obvious when $v = e_n$, and the general case follows by observing that conjugation by maps in $SL_n(\mathbb{C})$ preserves the divergence.

We mention a few other types of automorphisms. Choose nonnegative integers $k_1, \dots, k_n$ and complex numbers $c_1, \dots, c_n$ such that $\sum_{j=1}^n c_j k_j = 0$, and let $f\colon \mathbb{C} \to \mathbb{C}$ be an entire function. The map $w = F(z)$ given by

$$w_j = z_j \, \mathrm{e}^{c_j f(z_1^{k_1} \cdots z_n^{k_n})}, \quad j = 1, \dots, n$$

is an automorphism satisfying $w_1^{k_1} \cdots w_n^{k_n} = z_1^{k_1} \cdots z_n^{k_n}$, and its Jacobian is $w_1 \cdots w_n / z_1 \cdots z_n$. In the special case $n = 2$, $k_1 = k_2 = 1$, $c_1 = 1$ and $c_2 = -1$ we get a one-parameter group of automorphisms $\{\Theta_t\}_{t \in \mathbb{C}} \in \mathrm{Aut}_1 \mathbb{C}^2$ given by

$$\Theta_t(z_1, z_2) = \big(z_1 \mathrm{e}^{t f(z_1 z_2)}, z_2 \mathrm{e}^{-t f(z_1 z_2)}\big).$$

According to Nishimura [377] every injective holomorphic map $\mathbb{C}^2 \to \mathbb{C}^2$ with Jacobian one and sending the coordinate axes to themselves is of this form.

Let $n = 2$. By composing a shear (4.1) with an affine linear map $(z_1, z_2) \to (z_2, \alpha z_1 + \beta)$ $(\alpha, \beta \in \mathbb{C},\ \alpha \neq 0)$ we get a *Hénon map*

$$(z_1, z_2) \mapsto \big(z_2 + f(z_1), \alpha z_1 + \beta\big). \tag{4.13}$$

4.2 Automorphisms of $\mathbb{C}^2$

In this section we review some classical results on the polynomial automorphism group $\mathrm{Aut}_{\mathrm{alg}} \mathbb{C}^2$ of the plane, and we mention generalizations to the subgroups of $\mathrm{Aut}\, \mathbb{C}^2$ generated by entire shears.

Let $z = (x, y)$ be complex coordinates on $\mathbb{C}^2$. We denote by $\mathcal{A}$ the group of all affine automorphisms of $\mathbb{C}^2$:

$$a(x, y) = \big(\alpha x + \beta y + \xi, \gamma x + \delta y + \eta\big), \quad \alpha\delta - \beta\gamma \neq 0. \tag{4.14}$$

Let $\mathcal{E} \subset \mathrm{Aut}_{\mathrm{alg}} \mathbb{C}^2$ consist of all *elementary automorphisms*

$$e(x, y) = \big(\alpha x + p(y), \beta y + \gamma\big), \tag{4.15}$$

where $\alpha, \beta \in \mathbb{C}^*$, $\gamma \in \mathbb{C}$, and $p \in \mathbb{C}[x, y]$ is a polynomial. Observe that every affine linear map is a composition of elementary linear maps (in fact, of shear and dilations in the coordinate directions).

More generally, automorphisms of $\mathbb{C}^n$ whose k-th component only depends on the first k variables are called *elementary*, or *de Jonquières*; they form a group under composition, called *de Jonquières group*. These notions extend to fields other than $\mathbb{C}$, and (in the case of $\mathbb{C}$) to entire automorphisms.

According to a theorem of Jung [284], for any field k, the group $\mathrm{Aut}\,(k^2)$ of polynomial automorphisms of the plane k^2 is a free product of the affine group $\mathcal{A}$ and the elementary group $\mathcal{E}$, amalgamated over $\mathcal{A} \cap \mathcal{E}$. (See also [498], [414], and [204].) In particular:

Theorem 4.2.1. *The polynomial automorphism group* $\mathrm{Aut}_{\mathrm{alg}}\,\mathbb{C}^2$ *is an amalgamated free product* $\mathcal{A} * \mathcal{E}$ *of its subgroups* $\mathcal{A}$ *(affine) and* $\mathcal{E}$ *(elementary).*

The amalgamated free product structure $\mathrm{Aut}_{\mathrm{alg}}\,\mathbb{C}^2 = \mathcal{A} * \mathcal{E}$ means that every automorphism $g \in \mathrm{Aut}_{\mathrm{alg}}\,\mathbb{C}^2 \backslash (\mathcal{A} \cap \mathcal{E})$ is a represented by a reduced word $g = g_m \circ g_{m-1} \circ \cdots \circ g_1$, where $g_j \in \mathcal{A} \cup \mathcal{E} \backslash (\mathcal{A} \cap \mathcal{E})$ for every j and every two consecutive elements g_j, g_{j+1} belong to different groups $\mathcal{A}$, $\mathcal{E}$. Such representation is unique up to elements in $\mathcal{A} \cap \mathcal{E}$; in particular, no reduced word gives the identity map.

We do not give a detailed proof of Theorem 4.2.1 here, but only outline the main steps. The first main point is that the subgroups $\mathcal{A}$ and $\mathcal{E}$ generate $\mathrm{Aut}_{\mathrm{alg}}\,\mathbb{C}^2$. This is accomplished by showing that, if P and Q are holomorphic polynomials in two complex variables (x, y) such that the map $(x, y) \mapsto F(x, y) = (P(x, y), Q(x, y))$ is an automorphism of $\mathbb{C}^2$ then the degree of one of the polynomials divides the degree of the other one. One can proceed via the Abhyankar-Moh-Suzuki theorem [3, 482] as follows. (See also §4.18 below.) Consider the polynomial embedding $\mathbb{C} \ni x \mapsto (P(x, 0), Q(x, 0)) \in \mathbb{C}^2$ of $\mathbb{C}$ into $\mathbb{C}^2$. Let p, q be the degrees of the two components, considered as polynomials in a single complex variable x. We may assume that $p \leq q$. The key step is to show that p divides q, $q = dp$. There exist several algebraic proofs of this result, but perhaps the simplest proof, due to L. Rudolph [426], uses elementary knot theory. Once this is known, we can decrease the degree by composing the given embedding with a polynomial shear $(P, Q) \mapsto (P, Q - \alpha P^d)$ for a suitable choice of $\alpha \in \mathbb{C}^*$. Continuing inductively we show that by composing F with a finite composition of polynomial shears we obtain an automorphism of $\mathbb{C}^2$ which preserves the line $\mathbb{C} \times \{0\}$. Clearly the other lines $\mathbb{C} \times \{y\}$ for $y \neq 0$ get mapped onto lines $\mathbb{C} \times \{h(y)\}$ with $h(y) \neq 0$ (since they get mapped to $\mathbb{C} \times \mathbb{C}^*$ and the projection to the second coordinate is a nowhere vanishing polynomial on $\mathbb{C}$, hence a constant). So h is a linear function of y, and the new map $(x, y) \mapsto \big(\alpha x + g(y), by\big)$ is elementary.

To prove uniqueness of the decomposition into reduced words, one shows that the degree of a reduced word equals the product of the degrees of its characters g_j. (See [204] for this. The degree of $g \in \mathrm{Aut}_{\mathrm{alg}}\,\mathbb{C}^2$ is the larger of the degrees of its polynomial components.) It follows that every $g \in \mathrm{Aut}_{\mathrm{alg}}\,\mathbb{C}^2$ is either conjugate to an element of $\mathcal{E}$, or else is conjugate to a *cyclically reduced word* $a_r \circ e_r \circ \cdots \circ a_1 \circ e_1$ of even length, with $a_j \in \mathcal{A}$ and $e_j \in \mathcal{E}$ for every j. In the latter case we can arrange that every a_j equals the map $\sigma(x, y) = (y, x)$. The length of a cyclically reduced word is invariant under conjugation, and the degrees of its characters are invariant up to cyclic permutations. The simplest examples are $g = a \circ e$ with $a \in \mathcal{A} \backslash \mathcal{E}$ and $e \in \mathcal{E} \backslash \mathcal{A}$. These include all Hénon mappings

$$h(x, y) = \big(y, p(y) - \delta x\big),$$

where $\delta \in \mathbb{C}^*$ and p is a polynomial (take $a = \sigma$ and $e(x, y) = (-\delta x + p(y), y)$).

While maps in $\mathcal{E} \cup \mathcal{A}$ have simple dynamical properties, automorphisms represented by cyclically reduced words of even length exhibit nontrivial dynamical properties; see e.g. [260, 204, 37, 38, 147, 148].

If a group $\mathcal{G}$ is an amalgamated free product $\mathcal{A} * \mathcal{E}$ of its subgroups $\mathcal{A}$ and $\mathcal{E}$, then certain types of subgroups $G \subset \mathcal{G}$ are conjugate to a subgroup of $\mathcal{A}$ or to a subgroup of $\mathcal{E}$. This holds if G has bounded length with respect to the amalgamated product decomposition of $\mathcal{G}$ [442, Chap. I, No. 4.3, Théorèm 8]. If $\mathcal{G}$ is a topological group and $\mathcal{A}$, $\mathcal{E}$ are closed subgroups then the same holds for every connected Lie group G that is a subgroup of $\mathcal{G}$ (see [342]). In particular, we have the following result [360, 529].

Theorem 4.2.2. *Every subgroup of* $\mathrm{Aut}_{\mathrm{alg}}\,\mathbb{C}^2$ *isomorphic to* $\mathbb{R}$ *or to* $\mathbb{C}$ *is conjugate to a subgroup in the affine group* $\mathcal{A}$ *or in the elementary group* $\mathcal{E}$.

There is a simple reason why Hénon maps (4.13) and their composition do not belong to any flow. Denote by h^k the k-th iterate of an automorphism (or a diffeomorphism) h, defined inductively by

$$h^1 = h, \qquad h^k = h \circ h^{k-1} \text{ for } k \in \mathbb{N}, \qquad h^{-k} = (h^{-1})^k. \tag{4.16}$$

A point p is a *periodic point* of h is $h^k(p) = p$ for some $k \in \mathbb{N}$; the smallest such k is the *period* of p. If this holds for $k = 1$ then p is a *fixed point* of h.

Lemma 4.2.3. *Assume that X is a smooth manifold and that h is a diffeomorphism of X with the following two properties:*

(i) *the set of periodic points of h does not contain any nontrivial curves, and*
(ii) *there exists a periodic point that is not a fixed point.*

Then h is not the time-one map of any complete flow on X.

Proof. Assume that $h = \phi_1$ where $\{\phi_t\}_{t\in\mathbb{R}}$ is a complete flow. Let p be a periodic point of h of period $k > 1$. Using the group property of ϕ_t and the assumption $h = \phi_1$ (hence $h^k = \phi_k$), we have

$$h^k\big(\phi_t(p)\big) = \phi_k\big(\phi_t(p)\big) = \phi_t\big(\phi_k(p)\big) = \phi_t\big(h^k(p)\big) = \phi_t(p).$$

Thus the orbit $\{\phi_t(p) \colon t \in \mathbb{R}\}$ (which is a nontrivial curve since $\phi_1(p) \neq p$) consists of periodic points of period k, a contradiction (i). □

Corollary 4.2.4. [64] *If $\{\phi_t\}_{t\in\mathbb{C}} \subset \mathrm{Aut}\,\mathbb{C}^2$ is a flow such that $\phi_1 \in \mathrm{Aut}_{\mathrm{alg}}\,\mathbb{C}^2$, then ϕ_1 is conjugate in $\mathrm{Aut}_{\mathrm{alg}}\,\mathbb{C}^2$ to an elementary map in $\mathcal{E}$. In particular, a nontrivial Hénon map (4.13) does not belong to any flow.*

Proof (sketch). Using the explicit form of polynomial Hénon maps, one can see by Bezout's theorem that every composition of Hénon maps has finitely many

periodic points of any fixed period k, and this number grows as $k \to +\infty$. Hence we find a periodic point that is not a fixed point; Lemma 4.2.3 implies that such map does not belong to any flow. Hence, if $h = \phi_1 \in \operatorname{Aut}_{\mathrm{alg}} \mathbb{C}^2$ then h is conjugate in $\operatorname{Aut}_{\mathrm{alg}} \mathbb{C}^2$ to an element of $\mathcal{A}$ or to an element of $\mathcal{E}$. Further, every affine map $z \mapsto Az + b$ is linearly conjugate to a map in $\mathcal{E}$ (by conjugating the matrix A to its lower triangular Jordan form). $\square$

Complete flows on $\mathbb{C}^2$ consisting of polynomial maps were classified by M. Suzuki [484]; another proof was given in [34]. (Suzuki also classified complete flows on $\mathbb{C}^2$ such that the induced foliation admits a meromorphic first integral.) The classification was extended by Ahern, Forstnerič and Varolin [9, 10] to holomorphic flows on $\mathbb{C}^2$ with polynomial time-one map. It follows from their classification that any holomorphic flow $\{\phi_t\}_{t\in\mathbb{C}} \subset \operatorname{Aut} \mathbb{C}^2$ such that $\phi_1 \in \operatorname{Aut}_{\mathrm{alg}} \mathbb{C}^2$ is conjugate to a flow in the polynomial group $\operatorname{Aut}_{\mathrm{alg}} \mathbb{C}^2$. More recently, M. Brunella [59] obtained a full classification of complete polynomial vector fields on $\mathbb{C}^2$ with entire flows.

A result analogous to Theorem 4.2.1 hold for subgroups of $\operatorname{Aut} \mathbb{C}^2$ generated by entire shears. Denote by $\mathcal{E} \subset \operatorname{Aut} \mathbb{C}^2$ the set of all maps

$$e(x, y) = \big(\mathrm{e}^{g(y)}x + h(y), \beta y + \gamma\big), \tag{4.17}$$

where g and h are entire functions on $\mathbb{C}$, $\beta \in \mathbb{C}^*$, and $\gamma \in \mathbb{C}$. In analogy to the polynomial case we call such maps elementary. Let $\mathcal{E}_1$ denote the set of maps (4.17) with Jacobian one; this requires that g is constant (so $\alpha = \mathrm{e}^g \in \mathbb{C}^*$) and that $\alpha\beta = 1$. Let $\mathcal{E}_c$ denote the set of maps (4.17) with constant nonvanishing Jacobian; this requires that g is constant. Denote by $\mathcal{S}(2)$ the subgroup of $\operatorname{Aut} \mathbb{C}^2$ generated by $\mathcal{E}$ and by the affine group $\mathcal{A}$. Similarly, let $\mathcal{S}_1(2)$ be the subgroup of $\operatorname{Aut} \mathbb{C}^2$ generated by $\mathcal{E}_1$ and $\mathcal{A}_1$, where $\mathcal{A}_1$ contains all affine maps with Jacobian one. Finally, let $\mathcal{S}_c(2) = \mathcal{S}_1(2) \times \mathbb{C}^*$ be the group generated by $\mathcal{E}_c$ and $\mathcal{A}$; $\mathcal{S}_c(2)$ consists of maps in $\mathcal{S}(2)$ with constant Jacobian. It was proved by Andérsen [16] that for every $n \geq 2$ the shear group $\mathcal{S}_1(n)$ is a proper subgroup of $\operatorname{Aut}_1 \mathbb{C}^n$; similarly, $\mathcal{S}(n)$ is a proper subgroup of $\operatorname{Aut} \mathbb{C}^n$ for every $n > 1$ [18]. Ahern and Rudin proved [12] that $\mathcal{S}(2)$ is a free product of its subgroups $\mathcal{E}$ and $\mathcal{A}$, amalgamated over their intersection $\mathcal{A} \cap \mathcal{E}$. Earlier C. de Fabritiis showed [97] that the groups $\mathcal{S}_1(2)$, resp. $\mathcal{S}_c(2)$, are amalgamated free products of their subgroups $\mathcal{E}_1$ and $\mathcal{A}_1$, resp. of $\mathcal{E}_c$ and $\mathcal{A}$. So in these cases we have an analogue of Theorem 4.2.1 of Jung and Van der Kulk.

The polynomial automorphism group $\operatorname{Aut}_{\mathrm{alg}} \mathbb{C}^n$ for $n \geq 3$ is much more complicated and contains automorphisms that are not a composition of elementary ones; such are called wild. In particular, Shestakov and Umirbaev proved in [444] that the following *Nagata automorphism* of $\mathbb{C}^3$ is wild:

$$\begin{aligned} x' &= x + (x^2 - yz)z, \\ y' &= y + 2(x^2 - yz)x + (x^2 - yz)^2 z, \\ z' &= z. \end{aligned}$$

However, this automorphism becomes tame after adding sufficiently many additional variables.

The main open problem in this area is the *Jacobian conjecture* which claims that every polynomial map $F\colon \mathbb{C}^n \to \mathbb{C}^n$ with nonvanishing (hence constant) Jacobian is an automorphism of $\mathbb{C}^n$. This holds if F is injective [425].

4.3 Attracting Basins and Fatou-Bieberbach Domains

By the Riemann mapping theorem and Liouville's theorem, the complex number field $\mathbb{C}$ does not contain any proper subsets biholomorphic to itself. The situation is entirely different in higher dimensional Euclidean spaces. In a couple of classical papers, Fatou [137] and Bieberbach [47] investigated basins of attraction of holomorphic automorphisms of $\mathbb{C}^2$ which were proper subdomains but nevertheless biholomorphic to the whole space.

Definition 4.3.1. *A proper subdomain Ω of $\mathbb{C}^n$ that is biholomorphic to $\mathbb{C}^n$ is called a* Fatou-Bieberbach domain. *A biholomorphic map $F\colon \mathbb{C}^n \to \Omega$ onto such Ω (and its inverse map) is called a* Fatou-Bieberbach map.

According to Rudin [425] all injective polynomial maps $\mathbb{C}^n \to \mathbb{C}^n$ are automorphisms, so each Fatou-Bieberbach map is non-algebraic.

Let $F \in \operatorname{Aut}\mathbb{C}^n$. We denote by F^k its k-th iterate (4.16). A point $p \in \mathbb{C}^n$ for which $F(p) = p$ is called a *fixed point* of F; a fixed point of an iterate F^k is a *periodic point*. A fixed point p is said to be *attracting* if all eigenvalues λ_j of the derivative $F'(p)$ satisfy $|\lambda_j| < 1$, and is said to be *repellent* if all eigenvalues satisfy $|\lambda_j| > 1$. In the attracting case the set

$$\Omega_{F,p} = \{z \in \mathbb{C}^n \colon \lim_{k\to\infty} F^k(z) = p\} \tag{4.18}$$

is called the *basin* of p. In this section we prove the following result due to Sternberg [464] and Rosay and Rudin [420].

Theorem 4.3.2. *If $n > 1$ and $p \in \mathbb{C}^n$ is an attracting fixed point of a holomorphic automorphism $F \in \operatorname{Aut}\mathbb{C}^n$ then there exists a biholomorphic map ψ from the basin Ω (4.18) onto $\mathbb{C}^n$. If moreover the Jacobian $\mathrm{J}F$ is constant, then ψ can be chosen such that $\mathrm{J}\psi \equiv 1$.*

This theorem is related to the issue of finding normal forms for germs of biholomorphic maps, a subject with a long history. The proof of Theorem 4.3.2 that we give here is due to D. E. Buchmann, T. A. Nærland and E. F. Wold [60]. A similar result holds for an attracting periodic point as is easily seen by replacing F with the suitable iterate F^k.

Remark 4.3.3. (A) A Fatou-Bieberbach domain $\Omega \subset \mathbb{C}^n$ is Runge in $\mathbb{C}^n$ if and only if the associated Fatou-Bieberbach map $F\colon \mathbb{C}^n \to \Omega$ is a locally uniform limit of holomorphic automorphisms of $\mathbb{C}^n$. (The direct implication follows from the Andérsen-Lempert theorem; see Corollary 4.9.7 on p. 129.) Hence the proof of Theorem 4.3.2 shows that an attracting basin of a holomorphic automorphism of $\mathbb{C}^n$ is Runge in $\mathbb{C}^n$. (For examples of non-Runge Fatou-Bieberbach domains see §4.20 below.)

(B) Our proof of Theorem 4.3.2 only uses the local behavior of F at a fixed point; hence it applies to biholomorphic automorphisms $F\colon X \to X$ of any complex manifold and shows that the basin of any attracting fixed or periodic point of F is biholomorphic to $\mathbb{C}^n$ with $n = \dim X$. $\square$

We begin with some preparations. We may assume that the fixed point p is the origin $0 \in \mathbb{C}^n$. By writing $F\colon \mathbb{C}^n_0 \to \mathbb{C}^n_0$ we mean that F is an injective holomorphic map near the origin in $\mathbb{C}^n$ satisfying $F(0) = 0$. Such F is *attracting* at 0 if the eigenvalues $\lambda_1, \dots, \lambda_n$ of $F'(0)$ are all less than one in absolute value. (Injectivity implies that no eigenvalue equals zero.) Two such maps $F, G\colon \mathbb{C}^n_0 \to \mathbb{C}^n_0$ are said to be *conjugate* if there exists a holomorphic map $\psi\colon \mathbb{C}^n_0 \to \mathbb{C}^n_0$ such that $F = \psi^{-1} \circ G \circ \psi$ holds near the origin.

Lemma 4.3.4. *Let $F, G \in \operatorname{Aut}\mathbb{C}^n$ satisfy $F(0) = G(0) = 0$, assume that F and G are attracting at the origin, and assume that F and G are locally conjugate at the origin. Then F and G are globally conjugate on their basins of attraction. In particular, $\Omega_{F,0}$ is biholomorphic to $\Omega_{G,0}$.*

Proof. Let ψ denote the local conjugating map, i.e., $\psi = G^{-1} \circ \psi \circ F$ on $r\mathbb{B}$ for some $r > 0$. Clearly ψ extends to the basin of F by observing that $\psi = G^{-j} \circ \psi \circ F^j$ and letting $j \to \infty$. $\square$

In view of this lemma, it suffices to prove that any F in Theorem 4.3.2 is locally conjugate to an automorphism whose basin of attraction is all of $\mathbb{C}^n$. We will actually show that any such F is locally conjugate to a *lower triangular polynomial automorphism* of the form

$$G(z) = Az + \big(0, g_2(z), \dots, g_n(z)\big), \quad z \in \mathbb{C}^n,$$

where A is a lower triangular matrix with the same eigenvalues as $F'(0)$ and each g_j is a polynomial of degree at least two in the variables $z_1, \dots, z_{j-1}$. It is evident that a lower triangular polynomial map $G\colon \mathbb{C}^n_0 \to \mathbb{C}^n_0$ is an automorphism of $\mathbb{C}^n$, and if G is attracting at the origin then $\Omega_{G,0} = \mathbb{C}^n$. In view of Lemma 4.3.4 this will prove Theorem 4.3.2.

Let $F = (f_1, \dots, f_n)\colon \mathbb{C}^n_0 \to \mathbb{C}^n_0$ be attracting at the origin. After a linear change of coordinates we may assume that

$(*)$ the matrix $A = F'(0) = (\,a_{ij}\,)$ is lower triangular, and the eigenvalues $\lambda_j = a_{jj}$ of A are ordered so that $1 > |\lambda_1| \geq |\lambda_2| \geq \cdots \geq |\lambda_n| > 0$.

Theorem 4.3.2 is a simple consequence of the following two results.

Theorem 4.3.5. *Let $F\colon \mathbb{C}^n_0 \to \mathbb{C}^n_0$ be attracting at the origin. If $k \in \mathbb{N}$ is chosen such that the eigenvalues λ_j of $F'(0)$ satisfy the condition*

$$|\lambda_l|^{k+1} < |\lambda_m|, \qquad l, m = 1, \dots, n, \tag{4.19}$$

then F is locally conjugate at $0 \in \mathbb{C}^n$ to its k-jet at the origin.

A monomial $a_\mu z^\mu$ in a component f_j of F is said to be *resonant* if $\lambda_1^{\mu_1} \cdots \lambda_n^{\mu_n} = \lambda_j$.

Theorem 4.3.6. *Let $F\colon \mathbb{C}^n_0 \to \mathbb{C}^n_0$ be attracting at the origin. For any $k \in \mathbb{N}$, F is locally conjugate at the origin to a map $G\colon \mathbb{C}^n_0 \to \mathbb{C}^n_0$ satisfying $(*)$ and whose k-jet contains only resonant monomials.*

Let us first see how these results imply Theorem 4.3.2.

Proof (of Theorem 4.3.2). We may assume that $(*)$ holds. Choose a sufficiently large integer $k \in \mathbb{N}$ such that (4.19) is satisfied. By Theorem 4.3.6, the map F is conjugate at the origin to a map $G\colon \mathbb{C}^n_0 \to \mathbb{C}^n_0$ whose k-jet G_k contains only resonant terms. Theorem 4.3.5 implies that G is locally conjugate to its k-jet, and hence we may assume that G is a polynomial map.

Since the eigenvalues λ_j decrease in size according to $(*)$, any resonant monomial in a component g_j of G only depends on the variables $z_1, \dots, z_{j-1}$. Hence G is a lower triangular polynomial automorphism whose basin equals $\mathbb{C}^n$. Since F is locally conjugate to G at 0, they are globally conjugate on their basins by Lemma 4.3.4. In particular, $\Omega_{F,0}$ is biholomorphic to $\mathbb{C}^n$.

Next assume that the Jacobian of F is constant, $\mathrm{J}F \equiv c$. Since G is polynomial, $\mathrm{J}G$ is constant, and clearly $\mathrm{J}G(0) = c$. On the basin $\Omega_{F,0}$ we have $G^{-j}\psi F^j = \psi$ for all $j \in \mathbb{N}$, so for $z \in \Omega_{F,0}$ we have

$$\mathrm{J}\psi(z) = c^{-j}\mathrm{J}\psi(F^j(z))\, c^j = \mathrm{J}\psi(F^j(z)), \quad j = 1, 2, \dots.$$

Since $F^j(z) \to 0$ as $j \to \infty$, we obtain $\mathrm{J}\psi(z) = \mathrm{J}\psi(0)$, and by rescaling ψ we may arrange that this constant equals one. $\square$

Proof (of Theorem 4.3.5). Let F satisfy $(*)$. Then (4.19) is equivalent to

$$|\lambda_1|^k < |\lambda_n|.$$

The matrix $A = F'(0)$ is linearly conjugate to a lower triangular matrix a_{ij} with $\tilde{a}_{ii} = \lambda_i$. Moreover, by conjugating with a diagonal matrix (ϵ_{ii}) where all ratios $\frac{\epsilon_{ii}}{\epsilon_{jj}}$ are small for $j > i$, we can arrange that all off-diagonal entries in A are arbitrary small. Therefore, by choosing numbers $\alpha, \beta \in \mathbb{R}$ such that

$$0 < \alpha < |\lambda_n|, \quad |\lambda_1| < \beta < 1, \quad \beta^{k+1} < \alpha$$

we may assume that for a sufficiently small $r > 0$ we have

(a) $A = F'(0) = (a_{ij})$ is lower triangular, $|a_{ii}| = |\lambda_i| \geq |\lambda_j| = |a_{jj}|$ for all $j \geq i$, and
(b) $\alpha|z| \leq |F(z)| \leq \beta|z|$ for all $z \in r\mathbb{B}$.

Since the preliminary conjugacies performed so far were all linear, it is enough to prove the theorem under these assumptions. Let G denote the k-jet of F at the origin. Choosing $r > 0$ small enough and $C > 0$ large enough, the following hold for $z, z' \in r\mathbb{B}$:

(1) $|G^{-1} \circ F(z) - z| \leq C|z|^{k+1}$,
(2) $|F(z)| \leq \beta|z|$, and
(3) $|G^{-1}(z) - G^{-1}(z')| \leq \alpha^{-1}|z - z'|$.

To see (3), we write

$$G^{-1}(z') = G^{-1}(z) + (G^{-1})'(z)(z - z') + O(|z - z'|^2).$$

By continuity of partial derivatives and the fact that $(G^{-1})'(0) = F'(0)^{-1}$ we get (3) by decreasing α from (b) slightly and choosing $r > 0$ small enough.

We claim that the sequence $\psi_j = G^{-j} \circ F^j$ converges uniformly on some ball $\rho\mathbb{B}$ centered at the origin. We will show that if ρ is chosen such that $\rho + \rho^k C \sum_{j=1}^{\infty} (\beta^{k+1}\alpha^{-1})^j < r$ then

(4) $|\psi_j(z) - \psi_{j-1}(z)| \leq C(\beta^{k+1}\alpha^{-1})^j \, |z|^{k+1}$ for all $z \in \rho\mathbb{B}$.

We prove this by induction. Observe that (4) holds for $j = 1$ in view of (1). Suppose that (4) holds for all $j = 1, \ldots, m$. Write $w_m = \psi_m(F(z))$, $w_{m-1} = \psi_{m-1}(F(z))$. By (4) and the choice of ρ we see that both w_m and w_{m-1} are contained in $r\mathbb{B}$, and we get

$$\begin{aligned} |G^{-j-1} \circ F^{j+1}(z) - G^{-j} \circ F^j(z)| &= |G^{-1}(w_j) - G^{-1}(w_{j-1})| \\ &\leq \alpha^{-1}|\psi_m(F(z)) - \psi_{m-1}(F(z))| \\ &\leq \alpha^{-1} C \left(\beta^{k+1}\alpha^{-1}\right)^m (\beta|z|)^{k+1}. \end{aligned}$$

This is the desired estimate for $m + 1$. The induction may proceed.

The estimate (4) shows that ψ_j converges uniformly on $\rho\mathbb{B}$ to an injective holomorphic map $\psi\colon \rho\mathbb{B} \to \mathbb{C}^n_0$. Clearly ψ satisfies $G^{-1} \circ \psi \circ F = \psi$. $\square$

Proof (of Theorem 4.3.6). Write

$$F(z) = Q_m(z) + (P_1^m(z), \ldots, P_n^m(z)) + O(|z|^{m+1}),$$

where Q_m is a polynomial map of degree less than m and P_j^m is homogeneous of degree m. Let l_j denote the j-th component of $A = F'(0)$. For each component f_j of F, let $f_{j,k}$ denote its k-jet at the origin. For some j, let $c \in \mathbb{C}$ be a constant, let $|\mu| = m$, and let φ_c denote the map

$$\varphi_c(z) = \big(z_1, \dots, z_j + cz^\mu, \dots, z_n\big).$$

The key to the proof is to describe the m-jet of $\varphi_c^{-1} \circ F \circ \varphi_c$.

Write $F \circ \varphi_c = (\tilde{f}_1, \dots, \tilde{f}_n)$. Since $A = F'(0)$ is lower triangular, we have

$$\tilde{f}_{i,m} = f_{i,m} \text{ for } i < j, \quad \tilde{f}_{j,m} = f_{j,m} + \lambda_j c z^\mu. \tag{4.20}$$

Write $\varphi_c^{-1} \circ F \circ \varphi_c = (\widehat{f}_1, \dots, \widehat{f}_n)$. Then

$$\widehat{f}_{i,m} = \tilde{f}_{i,m} = f_{i,m} \quad \text{for } i < j$$

and

$$\widehat{f}_{j,m}(z) = f_{j,m}(z) + c(\lambda_j - \lambda^\mu) z^\mu + \sum_{|\beta|=m} a_\beta z^\beta, \tag{4.21}$$

where we have

$$a_\beta = 0 \text{ unless } \beta_s < \mu_s, \text{ where } s = \max\{j : \mu_j \neq 0\}. \tag{4.22}$$

If z^μ is not a resonant monomial, the constant c may be chosen such that $\widehat{f}_{j,m}$ does not contain any monomial of type z^μ.

The proof will now consist of performing a suitable sequence of conjugacies of this type. To this end we define an ordering of monomials in the variables $z_1, \dots, z_n$ of degree m. We do this by induction on n. For $n = 2$ we say that $z_1^{\mu_1} z_2^{\mu_2} > z_1^{\tilde{\mu}_1} z_2^{\tilde{\mu}_2}$ if $\mu_2 > \tilde{\mu}_2$. For a general n we say that $z^\mu > z^{\tilde{\mu}}$ if either $\mu_n > \tilde{\mu}_n$, or $\mu_n = \tilde{\mu}_n$ and $z_1^{\mu_1} \cdots z_{n-1}^{\mu_{n-1}} > z_1^{\tilde{\mu}_1} \cdots z_{n-1}^{\tilde{\mu}_{n-1}}$. Then (4.22) amounts to saying

$$a_\beta = 0 \text{ unless } z^\beta < z^\mu. \tag{4.23}$$

For fixed j and m we define the following procedure $R_{j,m}$ for creating a conjugation. Consider the largest monomial az^μ in $P_j^m(z)$. If z^μ is not resonant, apply a φ_{c_1} as above to cancel the z^μ-term. If z^μ is resonant then simply ignore it and proceed to the largest monomial smaller than z^μ. Repeat and define conjugating maps φ_{c_l} for $l = 1, \dots, N$ until there are no more monomials of degree m left. Set $\phi_{j,m} = \varphi_{c_1} \circ \cdots \circ \varphi_{c_N}$ and write $\phi_{j,m}^{-1} F \phi_{j,m} = (f_1', \dots, f_n')$. It follows from (4.20) that $f_{i,m}' = f_{i,m}$ for all $i < j$, and from (4.21) that $f_{j,m-1}' = f_{j,m-1}$. Now (4.23) implies that any monomial of degree m in f_j' is resonant.

For a fixed m we define a procedure S_m for creating a conjugation. Let $\Phi_m = \phi_{1,m} \circ \cdots \circ \phi_{n,m}$, where $\phi_{j,m}$ is obtained by following procedure $R_{j,m}$ with $(\phi_{1,m} \circ \cdots \circ \phi_{j-1,m})^{-1} \circ F \circ (\phi_{1,m} \circ \cdots \circ \phi_{j-1,m})$ as input. Then the $(m-1)$-jet of $\Phi_m^{-1} \circ F \circ \Phi_m$ equals the $(m-1)$-jet of F, and its homogeneous part of degree m contains only resonant monomials.

We begin by following procedure S_2 to create a map Φ_2. We inductively follow procedure S_{m+1} with $(\Phi_2 \circ \cdots \circ \Phi_m)^{-1} \circ F \circ (\Phi_2 \circ \cdots \circ \Phi_m)$ as data to create a map Φ_{m+1}. Then $\psi = \Phi_2 \circ \cdots \circ \Phi_k$ accomplishes the task. $\square$

Example 4.3.7. [420, Example 9.4] (*A Fatou-Bieberbach domain $\Omega \subset \mathbb{C}^2$ whose intersection with every affine complex line is bounded.*) Choose a number $\alpha \in \mathbb{C}$, $0 < |\alpha| < 1$. The polynomial automorphism

$$F(z,w) = (\alpha w, \alpha z + w^2)$$

fixes $(0,0)$, and the eigenvalues of $F'(0,0)$ are $\pm\alpha$. Let Ω denote its basin at $(0,0)$. Set $E = \{(z,w)\colon |w| > 1 + 2|\alpha| + |z|\}$. If $(z,w) \in E$ then

$$\begin{aligned} |w'| \geq |w^2| - |\alpha z| > |w|^2 - |\alpha w| &= |w|(|w| - |\alpha|) \\ &> |w|(1 + |\alpha|) > 1 + 2|\alpha| + |z'|, \end{aligned}$$

which means that $(z',w') = F(z,w) \in E$. Hence $E \cap \Omega = \emptyset$. It is a simple exercise to see that for every affine complex line L in $\mathbb{C}^2$ the set $F(L)\backslash E$ is bounded; hence $F(L) \cap \Omega \subset F(L)\backslash E$ is also bounded. Since $F(\Omega) \subset \Omega$, we also have $F(L \cap \Omega) \subset F(L) \cap \Omega$, and hence $L \cap \Omega$ is bounded. □

Bedford and Smillie showed that this is actually a general phenomenon.

Theorem 4.3.8. [37, Theorem 1] *Let Ω be a basin of a polynomial automorphism of $\mathbb{C}^2$. If $\Omega \neq \mathbb{C}^2$ then Ω intersects every algebraic curve $A \subset \mathbb{C}^2$, and the set $\overline{\Omega \cap A}$ is compact.*

We shall not prove this here, but we mention the following consequence. Let Ω be as above, with a Fatou-Bieberbach map $\Phi\colon \Omega \to \mathbb{C}^2$. If L is an affine complex line in $\mathbb{C}^2$ then each connected component D of $L \cap \Omega$ is a bounded domain which is Runge in L, and hence is biholomorphic to the disc. The image $\Phi(D)$ of D is a closed embedded holomorphic disc in $\mathbb{C}^2$.

Example 4.3.9. [420, Example 9.5] (*A Fatou-Bieberbach domain in $\mathbb{C}^2$ which does not intersect a complex line.*) Let $F(z,w) = (z',w')$ be given by

$$z' = z + w, \qquad w' = \frac{1}{2}(1 - w - \mathrm{e}^{z+w}). \tag{4.24}$$

Its fixed points are $p_m = (2m\pi\mathrm{i}, 0)$ for $m \in \mathbb{Z}$, and the eigenvalues of $F'(p_m)$ are $\pm\frac{1}{\sqrt{2}}$. Let Ω_m be the basin at p_m. Clearly these domains are pairwise disjoint. Furthermore, $F((z,w) + p_m) = F(z,w) + p_m$ and hence

$$\Omega_m = \Omega_0 + p_m, \quad m = 0, \pm 1, \pm 2, \ldots. \tag{4.25}$$

Thus there exist countably many pairwise disjoint Fatou-Bieberbach domain $\Omega_m \subset \mathbb{C}^2$ ($m \in \mathbb{Z}$) which are translates of each other.

Consider now the map $E\colon \mathbb{C}^2 \to \mathbb{C}^2$ given by $E(z,w) = \left(\mathrm{e}^z, w\mathrm{e}^{-z}\right)$. Note that $\mathrm{J}E \equiv 1$ and that $E(z,w) = E(z',w')$ if and only if $w = w'$ and $z' = z + 2m\pi\mathrm{i}$ for some $m \in \mathbb{Z}$. Since the domains Ω_m are pairwise disjoint, it follows from (4.25) that E is injective on each Ω_m and the image $\Omega^* = E(\Omega_m)$ is independent of m. Note that Ω^* is Fatou-Bieberbach domain in $\mathbb{C}^2$ which does not intersect the complex line $\{z = 0\}$. □

It is easy to generalize the construction in Example 4.3.11 to find Fatou-Bieberbach domains in the complement of a hyperplane in $\mathbb{C}^n$. This is no longer the case if we replace a hyperplane by a more general analytic subvariety, even by a zero dimensional one (see Theorem 4.7.2), but it is true for tame subvarieties that do not contain any hypersurfaces (Proposition 4.11.8 on p. 141). Further results in this direction are given in §4.6 and §4.11.

Example 4.3.10. [420, Examples 9.5 and 9.6] (*A Fatou-Bieberbach domain omitting countably many real 2-planes.*) Let F be the map (4.24). Set

$$\Pi_k = \big\{(x + (2k+1)\pi\mathrm{i}, y) \colon x, y \in \mathbb{R}\big\}, \quad k = 0, \pm 1, \pm 2, \dots .$$

Note that $\Pi_k = \Pi_0 + (2k+1)\pi\mathrm{i}, 0)$, and Π_0 is the standard $\mathbb{R}^2$ in $\mathbb{C}^2$. Then $F(\Pi_k) = \Pi_k$, and no p_m lies in any Π_k; therefore no point of any Π_k is attracted to any p_m by F. Hence *no Π_k intersects any Ω_m.*

The same can be done with finitely many rotated copies of $\mathbb{R}^2$ in $\mathbb{C}^2$. Choose $N \in \mathbb{N}$, put $\alpha = \mathrm{e}^{\pi\mathrm{i}/2N}$ and $E_k = \alpha^k \mathbb{R}^2$ for $k = 0, 1, \dots, 2N-1$. (Here we identified $\mathbb{R}^2$ with the subset of $\mathbb{C}^2$ consisting of points with real coordinates.) Define $F(z, w) = (z', w')$ by

$$z' = z + w, \quad w' = \frac{1}{2N+1}\big(z + (z+w)^{2N+1}\big).$$

Then $F \in \operatorname{Aut}\mathbb{C}^2$, $F(E_k) = E_k$ for all k, and the fixed points of F are $(0, 0)$ and $p_m = (\alpha^m, 0)$ for odd m. The eigenvalues of $F'(p_m)$ are $\pm\frac{1}{\sqrt{2N+1}}$. By Theorem 4.3.2 on p. 106, the basin Ω_m at p_m is a Fatou-Bieberbach domain. These N domains (one for each odd m between 1 and $2N-1$) are pairwise disjoint and are contained in $\mathbb{C}^2 \backslash (E_0 \cup E_2 \cup \cdots E_{2N-2})$. Since $F(\alpha^2 z, \alpha^2 w) = \alpha^2 F(z, w)$, the rotation $(z, w) \to (\alpha^2 z, \alpha^2 w)$ permutes the domains Ω_m. □

Example 4.3.11. [420, Example 9.7] (*A Fatou-Bieberbach domain in $\mathbb{C}^2$ whose closure misses a complex line.*) Choose a number $\alpha \in \mathbb{C}$, $0 < |\alpha| < 1$. Let f be an entire function on $\mathbb{C}$ such that

$$\mathrm{e}^{f(0)} = \frac{1}{\alpha}, \quad f'(0) = 0, \quad f(1) = 1, \quad f'(1) = \frac{1+\alpha^2}{1-\alpha^2}.$$

Define a map $F(z, w) = (z', w')$ by

$$z' = 1 - \alpha^2 + \alpha^2 z \mathrm{e}^{f(zw)}, \quad w' = w\mathrm{e}^{-f(zw)}.$$

Then $F \in \operatorname{Aut}\mathbb{C}^2$, $\mathrm{J}F \equiv \alpha^2$, $F(1, 1) = (1, 1)$, and the eigenvalues of $F'(1, 1)$ are $\pm\alpha\mathrm{i}$. Let Ω_0 be the basin of attraction of $(1, 1)$. F also fixes the point $(1+\alpha, 0)$ and $F'(1+\alpha, 0) = \alpha I$. Let Ω_1 denote the basin of $(1+\alpha, 0)$. Since $F(z, 0) = (1 - \alpha^2 + \alpha z, 0)$ for all $z \in \mathbb{C}$, we see that Ω_1 contains the complex line $\{w = 0\}$, and hence $\overline{\Omega}_0$ does not intersect this line. □

The image of a nondegenerate holomorphic self-map of $\mathbb{C}^2$ can not omit a pair of parallel complex lines (by Picard), and it can not omit three complex lines [231]. It is a long standing open problem whether an *injective* self-map of $\mathbb{C}^2$ can omit a pair of intersecting complex lines:

Problem 4.3.12. Is there a Fatou-Bieberbach domain contained in $(\mathbb{C}^*)^2$? Is there a holomorphic automorphism of $\mathbb{C}^2$ fixing both coordinate axes and with an attracting fixed point at $(1,1) \in (\mathbb{C}^*)^2$? In this connection see the paper of Nishimura [377].

We now mention without proof some generalizations of Theorem 4.3.2. Suppose that X is a complex manifold and $F\colon X \to X$ is a holomorphic automorphism which is hyperbolic on some invariant compact set $K \subset X$. (For the definition of hyperbolicity see e.g. [433]. Note that F is hyperbolic at a fixed point $p \in X$ if the derivative $F'(p)$ does not have any eigenvalue of modulus one; for a periodic point we consider the eigenvalues of a suitable iterate.) The *stable manifold* W_p^s of a point $p \in K$ is defined by

$$W_p^s = \big\{q \in X\colon \lim_{n\to\infty} \operatorname{dist}(F^n(q), F^n(p)) = 0\big\};$$

this is an immersed complex submanifold of X without self-intersections. It was asked by E. Bedford whether all stable manifolds of points $p \in K$ are biholomorphic to $\mathbb{C}^k$, where k is the stable dimension of F. At fixed and periodic points this holds by applying Theorem 4.3.2 and Remark 4.3.3 to the restricted map $F\colon W_p^s \to W_p^s$. Jonsson and Varolin proved in [278] that the stable manifold is biholomorphic to $\mathbb{C}^k$ for almost all points $p \in K$ with respect to any invariant measure. Although it is still an open question whether every stable manifold is a $\mathbb{C}^k$, the following result due to Fornæss and Stensønes gives a more precise information.

Theorem 4.3.13. [149, Theorem 3.5] *Let $F\colon X \to X$ be a holomorphic automorphism of a complex manifold X and $K \subset X$ an invariant compact hyperbolic set of stable dimension k. Then the stable manifold of F at any point $p \in K$ is biholomorphic to a domain in $\mathbb{C}^k$ which is a Stein increasing union of balls and has vanishing Kobayashi-Royden pseudometric.*

Whether the basin is always a $\mathbb{C}^k$ or not remains unknown. To prove this result the authors analyze the (non-autonomous) basin of attraction of a sequence of compositions of local holomorphic self-maps $F_j\colon \mathbb{B}^k \to \mathbb{B}^k$ of the unit ball $\mathbb{B}^k \subset \mathbb{C}^k$ under the assumption that there exist numbers $0 < s \le r < 1$ so that for $z, w \in \mathbb{B}^k$ and $j = 1, 2, \ldots$ we have the estimates

$$s|z-w| \le |F_j(z) - F_j(w)| \le r|z-w| \quad \text{and} \quad |F_j(z)| \le r.$$

Under these hypotheses they prove that the basin is one of the manifolds in Theorem 4.3.13. If all maps F_j in the sequence are close to a given map F

then every stable manifold is biholomorphic to $\mathbb{C}^k$ [393]. The same is true if $r^2 < s$ [522]; the proof is similar to the special case of Theorem 4.3.2.

A domain Ω in $\mathbb{C}^k$ which is an increasing union of (biholomorphic images of) balls is called a *short* $\mathbb{C}^k$. If $k = 2$ and the Kobayashi-Royden pseudometric of Ω is not identically zero then Ω is biholomorphic to the ball or to $\mathbb{D} \times \mathbb{C}$, where $\mathbb{D}$ is the unit disc [146]. Fornæss gave an example of a short $\mathbb{C}^k$ different from $\mathbb{C}^k$, with vanishing Kobayashi pseudometric and supporting a nonconstant bounded plurisubharmonic function [144].

4.4 Random Iterations and the Push-Out Method

A sequences of compositions $\Psi_m \circ \cdots \circ \Psi_1$ $(m = 1, 2, \ldots)$ of holomorphic automorphisms $\Psi_k \in \operatorname{Aut} \mathbb{C}^n$ is called a *random sequence* of automorphisms. We prove that, under rather mild assumptions on the maps Ψ_k, the domain of convergence of such a random sequence is a Fatou-Bieberbach domain (Corollary 4.4.2). We begin with the following more general result; the analogous result holds in any complex manifold.

Proposition 4.4.1. *Let D be a connected open set in $\mathbb{C}^n$ that is exhausted by a sequence of compact sets $K_0 \subset K_1 \subset \cdots \subset \cup_{j=0}^{\infty} K_j = D$ such that $K_{j-1} \subset \operatorname{Int} K_j$ for each $j \in \mathbb{N}$. Choose numbers ϵ_j $(j \in \mathbb{N})$ such that*

$$0 < \epsilon_j < \operatorname{dist}(K_{j-1}, \mathbb{C}^n \backslash K_j) \quad (\forall j \in \mathbb{N}), \qquad \sum_{j=1}^{\infty} \epsilon_j < \infty. \tag{4.26}$$

Suppose that Ψ_j $(j \in \mathbb{N})$ is a holomorphic automorphism of $\mathbb{C}^n$ satisfying

$$|\Psi_j(z) - z| < \epsilon_j, \qquad z \in K_j, \ j \in \mathbb{N}. \tag{4.27}$$

Set $\Phi_m = \Psi_m \circ \Psi_{m-1} \circ \cdots \circ \Psi_1$. Then there is an open set $\Omega \subset \mathbb{C}^n$ such that $\lim_{m\to\infty} \Phi_m = \Phi$ exists uniformly on compacts in Ω, and Φ is a biholomorphic map of Ω onto D. In fact, $\Omega = \cup_{m=1}^{\infty} \Phi_m^{-1}(K_m)$.

Proof. Set $\Phi_0(z) = z$ for $z \in \mathbb{C}^n$. Let

$$L_m = \Phi_m^{-1}(K_m) \quad (\forall m \in \mathbb{N}), \quad \Omega = \bigcup_{m=1}^{\infty} L_m \subset \mathbb{C}^n. \tag{4.28}$$

From (4.26) and (4.27) we get $\Psi_{m+1}(K_m) \subset \operatorname{Int} K_{m+1}$ for all $m \geq 0$, and hence

$$\Phi_{m+1}(L_m) = \Psi_{m+1}(\Phi_m(L_m)) = \Psi_{m+1}(K_m) \subset \operatorname{Int} K_{m+1}.$$

Thus $L_m \subset \operatorname{Int} L_{m+1}$ for all $m \in \mathbb{N}$, and hence Ω is an open set in $\mathbb{C}^n$. By induction on j we also get that $\Phi_j(L_m) \subset K_j$ for all $j \geq m$. Hence (4.27) gives the following for all $l > m \geq 1$ and $z \in L_m$:

$$|\Phi_l(z) - \Phi_m(z)| \le \sum_{j=m+1}^{l} |\Phi_j(z) - \Phi_{j-1}(z)|$$
$$= \sum_{j=m+1}^{l} |\Psi_j(\Phi_{j-1}(z)) - \Phi_{j-1}(z)| < \sum_{j=m+1}^{l} \epsilon_j. \quad (4.29)$$

This shows that $\lim_{l\to\infty} \Phi_l = \Phi$ exists on the set L_m, and it satisfies

$$|\Phi(z) - \Phi_m(z)| \le \sum_{j=m+1}^{\infty} \epsilon_j < \operatorname{dist}(K_m, \mathbb{C}^n \backslash D), \quad z \in L_m. \quad (4.30)$$

The last inequality above follows from (4.26). Thus $\Phi(L_m) \subset D$ for each $m \in \mathbb{N}$, and hence $\Phi(\Omega) \subset D$.

For $z \in L_m$ we write $\Phi_m(z) = w \in K_m$. Then (4.30) implies

$$|\Phi \circ \Phi_m^{-1}(w) - w| \le \sum_{j=m+1}^{\infty} \epsilon_j, \quad w \in K_m. \quad (4.31)$$

We claim that Φ maps Ω onto D. Fix an integer $m \in \mathbb{N}$ and then choose $l > m$ large enough such that $\sum_{j=l+1}^{\infty} \epsilon_j < \operatorname{dist}(K_m, \mathbb{C}^n \backslash K_{m+1})$. The map $\widetilde{\Phi}_l = \Phi \circ \Phi_l^{-1}$ is holomorphic in a neighborhood of K_l. From (4.31) we get that

$$|\widetilde{\Phi}_l(w) - w| < \sum_{j=l+1}^{\infty} \epsilon_j < \operatorname{dist}(K_m, \mathbb{C}^n \backslash K_{m+1}), \quad w \in K_l.$$

Hence the map $\widetilde{\Phi}_l|_{bK_l} : bK_l \to \mathbb{C}^n \backslash K_m$ has the same degree as the identity map with respect to any point of K_m, and hence it's degree equals one. Rouché's theorem [85, p. 110] implies that $\widetilde{\Phi}_l(K_l) \supset K_m$. From $\widetilde{\Phi}_l(K_l) = \Phi(\Phi_l^{-1}(K_l)) = \Phi(L_l)$ we infer that $K_m \subset \Phi(L_l) \subset \Phi(\Omega)$. Since $\cup_{m\in\mathbb{N}} K_m = D$, it follows that $\Phi(\Omega) = D$. Since Φ is a locally uniform limit of injective holomorphic maps and Φ is nondegenerate, it follows that Φ is also injective. □

Corollary 4.4.2. *Let $K_0 \subset K_1 \subset \cdots \subset \cup_{j=0}^{\infty} K_j = \mathbb{C}^n$ be compacts sets such that $K_{j-1} \subset \operatorname{Int} K_j$ for each $j \in \mathbb{N}$. Let $\epsilon_j > 0$, $\Psi_j \in \operatorname{Aut} \mathbb{C}^n$ and Φ_j be as in Proposition 4.4.1, satisfying (4.26) and (4.27). Let $\Omega \subset \mathbb{C}^n$ consist of all points $z \in \mathbb{C}^n$ such that the sequence $\{\Phi_m(z) : m \in \mathbb{N}\} \subset \mathbb{C}^n$ is bounded. Then the limit $\lim_{m\to\infty} \Phi_m = \Phi$ exists uniformly on compacts in Ω, and Φ is a biholomorphic map of Ω onto $\mathbb{C}^n$.*

Although Corollary 4.4.2 is analogous to the result of Fornæss and Stensønes [149] mentioned at the end of §4.3 (replace the maps by their inverses and consider the attracting basin of a random sequence), it will be more convenient for our purposes to use the result in this particular form.

Corollary 4.4.2 lies at the heart the *push-out method* first used in [112] to construct Fatou-Bieberbach domains with certain properties. By taking Ψ_k to be suitably chosen shears in coordinate directions, Globevnik [209] and Stensønes [463] constructed Fatou-Bieberbach domains with smooth boundaries by a method originating in their joint paper [212] on embedding certain plane domains into $\mathbb{C}^2$ (see §8.9).

Theorem 4.4.3. [463] *For every $n > 1$ there exists a Fatou-Bieberbach domain with $\mathcal{C}^\infty$-smooth boundary in $\mathbb{C}^n$.*

It is unknown whether this phenomenon can occur also for basins of an attracting or semi-attracting fixed point. The existing information suggests to the contrary. Bedford and Smillie showed in [38] that for any basins of attraction $\Omega_1, \Omega_2, \ldots$ of a polynomial automorphism of $\mathbb{C}^2$, their boundaries coincide; and if such a mapping has at least two basins then the boundaries can not be 3-manifolds. The interested reader should consult Bedford's very informative review MR1441879 of the paper [463].

Problem 4.4.4. Is there a Fatou-Bieberbach domain in $\mathbb{C}^n$ with real analytic boundary?

The boundary of a smooth Fatou-Bieberbach domain in $\mathbb{C}^2$ is easily seen to be pseudoconvex from both sides, and hence Levi-flat. Globevnik [209] constructed Fatou-Bieberbach domains with less regular boundary, but with a more precise behavior in a given compact set. For instance, he found Fatou-Bieberbach domains $\Omega \subset \mathbb{C}^2$ such that for a given $r > 1$ the intersection $\Omega \cap r\mathbb{D}^2$ is a $\mathcal{C}^2$-small perturbation of the tube $\mathbb{D} \times r\mathbb{D}$. Recently E. F. Wold constructed a Fatou-Bieberbach domain in $\mathbb{C}^2$ such that a connected component of its intersection with the z-axis is the unit disc $\{|z| < 1\}$ (to appear in Proc. Amer. Math. Soc.).

4.5 Mittag-Leffler Theorem for Entire Maps

In this and the following two sections we present some of the results of Rosay and Rudin [420] concerning the action of holomorphic self-mappings of $\mathbb{C}^n$ on different types of countable subsets. The first result is motivated by the classical Mittag-Leffler interpolation theorem: *If $\{a_j\}$ is a discrete sequence in $\mathbb{C}$ without repetition then for every sequence $\{b_j\} \subset \mathbb{C}$ there exists an entire function $f\colon \mathbb{C} \to \mathbb{C}$ such that $f(a_j) = b_j$ for $j = 1, 2, \ldots$.*

Theorem 4.5.1. [420, Theorem 1.1] *Let $n > 1$ and assume that $\{a_j\}_{j\in\mathbb{N}}$ is a discrete sequence without repetition in $\mathbb{C}^n$. For every sequence $\{b_j\}_{j\in\mathbb{N}}$ in $\mathbb{C}^n$ there exists an entire map $\Phi\colon \mathbb{C}^n \to \mathbb{C}^n$ satisfying*

(i) *$\Phi(a_j) = b_j$ for $j = 1, 2, \ldots$, and*
(ii) *$(\mathrm{J}\Phi)(z) = 1$ for every $z \in \mathbb{C}^n$.*

The proof of Theorem 4.5.1 relies on the following simple lemma.

Lemma 4.5.2. *Assume that $n > 1$, $a_1, \dots, a_m$ are points in a compact convex set $K \subset \mathbb{C}^n$, and p, q are points in a complex affine hyperplane $\Lambda \subset \mathbb{C}^n \backslash K$. For every $\epsilon > 0$ there exists a volume preserving shear τ on $\mathbb{C}^n$ that moves p to q, fixes every point a_j, and satisfies $|\tau(z) - z| < \epsilon$ for all $z \in K$.*

Proof. Choose a $\mathbb{C}$-linear functional $\lambda\colon \mathbb{C}^n \to \mathbb{C}$ that is constant on Λ. The point $\lambda(p) = \lambda(q)$ then lies outside the compact convex set $\lambda(K) \subset \mathbb{C}$. Choose a unit vector $v \in \ker \lambda$; then $q = p + cv$ for some $c \in \mathbb{C}$. Choose a holomorphic polynomial g on $\mathbb{C}$ such that $g(\lambda(p)) = c$, $g(\lambda(a_j)) = 0$ for $j = 1, \dots, m$, and $|g| < \epsilon$ on $\lambda(K)$. The shear $\tau(z) = z + g(\lambda(z))v$ does the job. $\square$

Corollary 4.5.3. *If $a_1, \dots, a_m$, K and ϵ are as in Lemma 4.5.2, and if p, q are points in $\mathbb{C}^n \backslash K$, then some composition of two volume preserving shears moves p to q, fixes every a_j, and moves no point of K by as much as ϵ.*

Proof. There are affine complex hyperplanes $p \in \Lambda$, $q \in \Lambda'$ that do not intersect K and are not parallel. Pick $w \in \Lambda \cap \Lambda'$ and apply Lemma 4.5.2 twice to move p to w and then w to q, fixing the points a_j and moving no point of K by as much as $\frac{\epsilon}{2}$ at every step. $\square$

Proof (of Theorem 4.5.1). By an affine change of coordinates of $\mathbb{C}^n$ we may assume that $0 < |a_1| < |a_2| < \cdots$ and that the hyperplane $\{z_1 = 0\}$ does not contain any points of the b_j sequence. Let $\Phi = E \circ F$ where

$$E(z_1, z_2, \dots, z_n) = \left(\mathrm{e}^{z_1}, z_2 \mathrm{e}^{-z_2}, z_3, \dots, z_n\right)$$

and F will be found as a limit of a certain sequence of compositions of additive shears (hence $\mathrm{J}F = 1$). Observe that every point $w = (w_1, \dots, w_n)$ with $\pi_1(w) = w_1 \neq 0$ belongs to the range of E, and we can choose $v \in \mathbb{C}^n$ such that $E(v) = w$ and $|\pi_1(v)|$ is arbitrary large.

Set $F_0(z) = z$. Assume inductively that $F_{k-1} \in \operatorname{Aut} \mathbb{C}^n$ has been chosen for some $k \in \mathbb{N}$. Choose $v_k \in \mathbb{C}^n$ so that $E(v_k) = b_k$ and $|\pi_1(v_k)|$ is so large that v_k lies outside of the compact set $F_{k-1}(|a_k|\overline{\mathbb{B}})$. Hence there exists $q_k \in \mathbb{C}^n$ such that $F_{k-1}(q_k) = v_k$ and $|q_k| > |a_k|$. Choose a number δ_k, $0 < \delta_k < |a_k| - |a_{k-1}|$, such that for all $z, w \in |a_k|\overline{\mathbb{B}}$ with $|z - w| < \delta_k$ we have $|F_{k-1}(z) - F_{k-1}(w)| < 2^{-k}$. Corollary 4.5.3 furnishes $G_k \in \operatorname{Aut} \mathbb{C}^n$ (a composition of two additive shears) such that $G_k(a_j) = a_j$ for $j = 1, \dots, k-1$, $G_k(a_k) = b_k$, and $|G_k(z) - z)| < \delta_k$ for $|z| \leq |a_{k-1}|$. The map $F_k = F_{k-1} \circ G_k$ then satisfies

$$F_k(a_k) = v_k, \qquad F_k(a_j) = F_{k-1}(a_j) \text{ for } j = 1, \dots, k-1,$$

and

$$|F_k(z) - F_{k-1}(z)| < 2^{-k}, \qquad |z| \leq |a_{k-1}|.$$

The limit $F = \lim_{k \to \infty} F_k$ then exists uniformly on compacts in $\mathbb{C}^n$, F is injective holomorphic, $\mathrm{J}F \equiv 1$, and $F(a_k) = v_k$ for all $k = 1, 2, \dots$. Since $E(v_k) = b_k$, we see that $\Phi(a_k) = b_k$ and the proof is complete. $\square$

4.6 Tame Discrete Sets in $\mathbb{C}^n$

A discrete set is always assumed to be closed, and hence without limit points. Rosay and Rudin [420] showed that, in general, one can not map an infinite discrete set in $\mathbb{C}^n$ onto another discrete set by an automorphism of $\mathbb{C}^n$. Indeed, the infinite discrete sets in $\mathbb{C}^n$ form uncountably many different equivalence classes under $\operatorname{Aut}\mathbb{C}^n$-equivalence (see Corollary 4.7.7 on p. 122 below). They introduced the following notion.

Definition 4.6.1. *A discrete set $E \subset \mathbb{C}^n$ is* tame *if there exists a holomorphic automorphism $F \in \operatorname{Aut}\mathbb{C}^n$ such that*

$$F(E) = N = \{je_1 : j \in \mathbb{N}\} \subset \mathbb{C}^n. \tag{4.32}$$

The set E is very tame *if the above holds for some $F \in \operatorname{Aut}_1\mathbb{C}^n$.*

It is easily seen that any permutation of the standard tame set $N \subset \mathbb{C}^n$ (4.32) is achieved by a composition of three additive shears (4.1) [420, Proposition 3.1]. Hence we can permute points in any tame set by holomorphic automorphisms, and for every tame sequence $\{a_j\}_{j\in\mathbb{N}} \subset \mathbb{C}^n$ there is an automorphism $F \in \operatorname{Aut}\mathbb{C}^n$ such that

$$F(a_j) = je_1 = (j, 0, \ldots, 0), \qquad j = 1, 2, \ldots.$$

Let $\mathbb{C}^n = \mathbb{C}^k \oplus \mathbb{C}^m$ where $k, m \in \mathbb{N}$. Write the coordinates on $\mathbb{C}^n$ accordingly as $z = (z', z'') \in \mathbb{C}^k \oplus \mathbb{C}^m$ and set $\pi'(z', z'') = z'$, $\pi''(z', z'') = z''$. The following result combines Theorems 3.3 and 3.9 in [420].

Theorem 4.6.2. [420] *Suppose that E is an infinite discrete set in $\mathbb{C}^n$.*

(a) *If $\pi'(E)$ is discrete in $\mathbb{C}^k$, then E is tame.*
(b) *If in addition to (a) the set $(\{p\} \times \mathbb{C}^m) \cap E$ is finite for every $p \in \mathbb{C}^k$, then E is very tame.*

Proof. Let $\pi'(E) = \{p_1, p_2, \ldots\} \subset \mathbb{C}^k$. We first prove part (b). By induction we find $w_1, w_2, \ldots \in \mathbb{C}^m$ such that for all points $(p_j, q) \in E$ and for all $(p_i, z'') \in E$ with $i < j$ we have $|q + w_j| > j + |z'' + w_i|$. By the Mittag-Leffler interpolation theorem there is a holomorphic map $f : \mathbb{C}^k \to \mathbb{C}^m$ such that $f(p_i) = w_i$ for $i = 1, 2, \ldots$. Define

$$\sigma_1(z', z'') = \big(z', z'' + f(z')\big), \quad E_1 = \sigma_1(E).$$

Our choice of $\{w_j\}$ implies that π'' is one-to-one on E_1 and that $\pi''(E_1)$ is discrete in $\mathbb{C}^m$. Let $z = (z_1, \ldots, z_n)$. The above implies that there is a function $\varphi : \pi''(E_1) \to \mathbb{C}$ such that $\{z_1 + \varphi(z'') : z \in E_1\} = \mathbb{N}$. Choose a holomorphic function g on $\mathbb{C}^m$ such that $g = \varphi$ on $\pi''(E_1)$ and set

$$\sigma_2(z) = \big(z_1 + g(z''), z_2, \ldots, z_n\big), \quad E_2 = \sigma_2(E_1).$$

The first coordinate projection $\pi_1(z) = z_1$ then maps E_2 bijectively onto $\mathbb{N}$. Choose holomorphic functions $g_2, \ldots, g_n$ on $\mathbb{C}$ such that

$$E_2 = \big\{(k, g_2(k), \ldots, g_n(k)) \colon\ k \in \mathbb{N}\big\}.$$

The shear

$$\sigma_3(z) = \big(z_1, z_2 - g_2(z_1), \ldots, z_n - g_n(z_1)\big)$$

then takes E_2 onto the standard tame set N (4.32). Setting $\sigma = \sigma_3 \circ \sigma_2 \circ \sigma_1$ we have $\sigma(E) = N$ and $\mathrm{J}\sigma = 1$. This proves (b).

To prove (a) we may suppose by a translation that $z' \neq 0$ and $z'' \neq 0$ for every point $z = (z', z'') \in E$. As before let $\pi'(E) = \{p_1, p_2, \ldots\}$. Put

$$\delta_j = \min\{|z''| \colon (p_j, z'') \in E\} > 0, \qquad j \in \mathbb{N}.$$

Choose a holomorphic function h on $\mathbb{C}^k$ such that $\Re h(p_j) > \log \frac{|p_j|}{\delta_j}$ for $j \in \mathbb{N}$. Define $F \in \operatorname{Aut} \mathbb{C}^n$ by $F(z', z'') = (z', \mathrm{e}^{h(z')} z'')$. If $(p_j, z'') \in E$ then

$$|\mathrm{e}^{h(p_j)} z''| \geq \mathrm{e}^{\Re h(p_j)}\, \delta_j > |p_j|.$$

This shows that each point of $F(E)$ has the form (p_j, w'') with $|w''| > |p_j|$. Hence the projection π'' is finite-to-one on $F(E)$ and the set $\pi''(F(E))$ is discrete in $\mathbb{C}^m$. By part (b), $F(E)$ is tame in $\mathbb{C}^n$, and hence E is tame. □

Corollary 4.6.3. *Let $n > 1$ be an integer.*

(a) *Every infinite discrete set in $\mathbb{C}^{n-1}$ is very tame in $\mathbb{C}^n$.*

(b) *The union of a finite set and a (very) tame set is (very) tame.*

(c) *Every infinite discrete set in $\mathbb{C}^n$ is the union of two very tame sets.*

Proof. Part (a) follows from part (b) in Theorem 4.6.2 with $k = 1$. To prove (b), it suffices to consider the union with the set N (4.32) and to apply Theorem 4.6.2. To prove (c), write $n = k + m$ with $k, m \in \mathbb{N}$ and put

$$E_1 = \big\{(z', z'') \in E \colon |z''| \leq |z'|\big\}, \quad E_2 = \big\{(z', z'') \in E \colon |z''| > |z'|\big\}.$$

Theorem 4.6.2 directly applies to E_1, and it also applies to E_2 with the roles of π' and π'' reversed. Thus both E_1 and E_2 are very tame in $\mathbb{C}^n$. If one of these two sets is finite, it suffices to apply part (b) above.

Corollary 4.6.4. *If A is a proper algebraic subvariety of $\mathbb{C}^n$ for some $n > 1$ then every discrete set contained in A is very tame.*

Proof. After a linear change of coordinates on $\mathbb{C}^n$ we achieve that the coordinate projection $\pi' \colon \mathbb{C}^n \to \mathbb{C}^{n-1}$, $\pi(z', z_n) = z'$, is proper when restricted to A. Hence every discrete set $E \subset A$ satisfies condition (b) in Theorem 4.6.2. □

Corollary 4.6.5. *For every infinite discrete set in $\mathbb{C}^n$ $(n > 1)$ there is an injective holomorphic map $H\colon \mathbb{C}^n \to \mathbb{C}^n$ such that $\mathrm{J}H \equiv 1$ and $H(E) = N$.*

Proof. In the proof of Theorem 4.5.1 on p. 116 we found an injective holomorphic map $F\colon \mathbb{C}^n \to \mathbb{C}^n$ with $\mathrm{J}F \equiv 1$ such that the restriction of the first coordinate projection $\pi_1\colon \mathbb{C}^n \to \mathbb{C}$ to the set $F(E)$ was one-to-one and $\pi_1(F(E))$ was discrete in $\mathbb{C}$. Theorem 4.6.2 (b) (with $k = 1$) shows that $F(E)$ is very tame in $\mathbb{C}^n$. Let $G \in \mathrm{Aut}_1\mathbb{C}^n$ be such that $G(F(E)) = N$. Then $H = G \circ F$ satisfies the conclusion of Corollary 4.6.5. □

J. Winkelmann [520] found the following sufficient growth condition for tameness of a discrete set. The second part of Theorem 4.6.6 is due to D. Kolarič [307, Theorem 2.3] who obtained several further results on tameness of discrete sets in the presence of an algebraic subvariety.

Theorem 4.6.6. *A discrete set $\{a_k\}_{k\in\mathbb{N}} \subset \mathbb{C}^n$ $(n > 1)$ satisfying the growth condition $\sum_{k=1}^{\infty} 1/|a_k|^{2n-2} < +\infty$ is tame. Furthermore, given an algebraic subvariety $A \subset \mathbb{C}^n$ of codimension ≥ 2 and discrete sets $\{a_j\}$, $\{b_j\}$ in $\mathbb{C}^n$ contained in $\mathbb{C}^n \backslash A$ and satisfying the above growth condition, there exists an $F \in \mathrm{Aut}\,\mathbb{C}^n$ such that $F(z) = z$ for all $z \in A$ and $F(a_j) = b_j$ for $j = 1, 2, \ldots$.*

4.7 Unavoidable and Rigid Discrete Sets

In this section we consider certain classes of nontame discrete sets that were introduced by Rosay and Rudin [420].

Definition 4.7.1. *A subset E of $\mathbb{C}^n$ is* unavoidable *for a certain class $\mathcal{F}$ of holomorphic maps $\mathbb{C}^n \to \mathbb{C}^n$ if $F(\mathbb{C}^n) \cap E \neq \emptyset$ for every $F \in \mathcal{F}$.*

By Picard's theorem, every set in $\mathbb{C}$ containing at least two points is unavoidable by nonconstant entire functions. On the other hand, complex hyperplanes in $\mathbb{C}^n$ for $n > 1$ are avoidable by injective holomorphic maps $\mathbb{C}^n \to \mathbb{C}^n$ (Example 4.3.7 on p. 111), and hence every tame discrete set is avoidable by injective holomorphic maps $\mathbb{C}^n \to \mathbb{C}^n$. However, we now show that this fails for some discrete sets in $\mathbb{C}^n$.

Theorem 4.7.2. [420, Theorem 4.5] *Let $n > 1$. There exists a discrete set in $\mathbb{C}^n$ which is unavoidable by nondegenerate holomorphic self-maps of $\mathbb{C}^n$.*

We need a couple lemmas. Denote by $\mathbb{B}_n$ the open unit ball in $\mathbb{C}^n$.

Lemma 4.7.3. *Let $1 \leq k \leq n$ be integers. Given numbers $0 < a_1 < a_2$, $0 < r_1 < r_2$, $c > 0$, let Γ be the class of all holomorphic maps $F = (f_1, \ldots, f_k)\colon a_2\mathbb{B}_n \to r_2\mathbb{B}_k$ such that*

(i) $|F(0)| \leq \frac{r_1}{2}$, *and*

(ii) $\left|\frac{\partial(f_1,\ldots,f_k)}{\partial(z_1,\ldots,z_k)}\right| \geq c$ *at some point of* $a_1\bar{\mathbb{B}}_n$.

Then there is a finite set $E = E(a_1, a_2, r_1, r_2, c) \subset b(r_1\mathbb{B}_k)$ *such that if* $F \in \Gamma$ *and* $F(a_1\mathbb{B}_n)$ *intersects* $b(r_1\mathbb{B}_k)$, *then* $F(a_2\mathbb{B}_n)$ *intersects* E.

Proof. Let $E_1 \subset E_1 \subset \cdots$ be finite subsets of the sphere $b(r_1\mathbb{B}_k)$ whose union $\cup_{j=1}^{\infty} E_j$ is dense in in $b(r_1\mathbb{B}_k)$. To reach a contradiction, assume that no E_j satisfies the conclusion of the lemma. This means that there exist $F_j \in \Gamma$ and $z_j \in a_1\mathbb{B}_n$ $(j = 1, 2, \ldots)$ such that

$$F_j(z_j) \in b(r_1\mathbb{B}_k), \quad F_j(a_2\mathbb{B}_n) \cap E_j = \emptyset, \quad j = 1, 2, \ldots.$$

Note that Γ is a normal family. Passing to a subsequence we may therefore assume that, as $j \to \infty$, $z_j \to w \in a_1\overline{\mathbb{B}}_n$ and $F_j \to F \in \Gamma$, uniformly on compacts in $a_2\mathbb{B}_n$. Hence $F(w) = \lim_{j\to\infty} F_j(z_j) \in b(r_1\mathbb{B}_k)$. Condition (ii) in the lemma shows that the open set

$$\Omega = \{z \in a_2\mathbb{B}_n \colon \mathrm{rank} F'(z) = k\}$$

is not empty. The complement of Ω is clearly a subvariety of $a_2\mathbb{B}_n$, and hence $F(\Omega)$ is a connected open dense set in $F(a_2\mathbb{B}_n)$.

Since $w \in a_2\mathbb{B}_n$ and $|F(w)| = r_1$, the maximum principle shows that $F(a_2\mathbb{B}_n)$ contains points outside $r_1\bar{\mathbb{B}}_k$, and hence so does $F(\Omega)$. Further, condition (i) in the lemma shows that $F(\Omega)$ intersects $r_1\mathbb{B}_k$. Being connected, $F(\Omega)$ must therefore intersect $b(r_1\mathbb{B}_k)$. Choose a point $p \in \Omega$ such that $F(p) = q \in b(r_1\mathbb{B}_k)$. Since $\mathrm{rank} F'(p) = k$, there is a compact set K in $a_2\mathbb{B}_n$, contained in a k-dimensional affine complex subspace, such that F maps K bijectively onto a closed ball $\overline{\mathbb{B}_k(q,\delta)}$ for some $\delta > 0$. Since $F_j \to F$ uniformly on K, Rouché's theorem implies that $\mathbb{B}_k(q, \frac{\delta}{2}) \subset F_j(K)$ for all large $j \in \mathbb{N}$. But $\mathbb{B}_k(q, \frac{\delta}{2})$ intersects E_j for large j (since $\cup_{j=1}^{\infty} E_j$ is dense in in $b(r_1\mathbb{B}_k)$), a contradiction to our assumption. $\square$

We now use the previous lemma with $k = n$. Set $\mathbb{B} = \mathbb{B}_n$ and $S = b\mathbb{B}$.

Lemma 4.7.4. *For each* $m \in \mathbb{N}$ *there is a discrete set* $E_m \subset \mathbb{C}^n \backslash m\mathbb{B}$ *such that for every holomorphic map* $F \colon m\mathbb{B} \to \mathbb{C}^n$ *satisfying*

(i) $|F(0)| \leq \frac{m}{2}$,

(ii) $|\mathrm{J}F(z)| \geq \frac{1}{m}$ *at some point* $z \in \frac{m}{2}\overline{\mathbb{B}}$, *and*

(iii) $F(m\mathbb{B}) \cap E_m = \emptyset$

we have $F(\frac{m}{2}\mathbb{B}) \subset m\mathbb{B}$.

Proof. Choose real sequences a_j and r_j such that

$$\frac{m}{2} = a_1 < a_2 < a_3 < \cdots < \frac{3m}{4}, \quad m = r_1 < r_2 < \cdots, \quad \lim_{j\to\infty} r_j = +\infty.$$

Using the notation of Lemma 4.7.3, we define

$$E_m = \bigcup_{j=1}^{\infty} E\left(a_j, a_{j+1}, r_j, r_{j+1}, \frac{1}{m}\right). \tag{4.33}$$

Since E_m is a union of finite sets lying in the spheres r_jS and $r_j \to \infty$, we conclude that E_m is discrete. If F satisfies the conditions of the lemma, then F is bounded on $\frac{3m}{4}\mathbb{B}$, and hence $F(a_{j+1}\mathbb{B}) \subset r_{j+1}\mathbb{B}$ for some $j \in \mathbb{N}$. Condition (iii) implies that $F(a_{j+1}\mathbb{B})$ does not intersect $E(a_j, a_{j+1}, r_j, r_{j+1}, 1/m)$, and hence Lemma 4.7.3 shows that $F(a_j\mathbb{B}) \cap r_jS = \emptyset$. Thus $F(a_j\mathbb{B}) \subset r_j\mathbb{B}$. By a downward induction on j we conclude that $F(a_1\mathbb{B}) \subset r_1\mathbb{B}$. $\square$

Proof (of Theorem 4.7.2). Let E_m be given by (4.33) and set $E = \cup_{m=1}^{\infty} E_m$. Since $E_m \subset \mathbb{C}^n \backslash m\mathbb{B}$, we see that $E \cap m\mathbb{B}$ is finite for each m, and hence E is discrete. We claim that the range of every nondegenerate entire map $F\colon \mathbb{C}^n \to \mathbb{C}^n$ intersects E. Assume to the contrary. Then F satisfies the hypotheses of Lemma 4.7.4 for all large $m \in \mathbb{N}$, and hence $F(\frac{m}{2}\mathbb{B}) \subset m\mathbb{B}$ for such m. Liouville's theorem implies that F is affine linear. Since F is nondegenerate, we have $F(\mathbb{C}^n) = \mathbb{C}^n$, a contradiction to the assumption. $\square$

By a more elaborate version of the above construction, Rosay and Rudin also proved the following result.

Theorem 4.7.5. [420, Theorem 5.1] *For every $n \in \mathbb{N}$ there is a discrete set $D \subset \mathbb{C}^n$ such that the only nondegenerate holomorphic map $F\colon \mathbb{C}^n \to \mathbb{C}^n$ that maps $\mathbb{C}^n \backslash D$ to itself is the identity map. In particular, no automorphism of $\mathbb{C}^n$ other than identity can map D onto itself.*

A set D as in Theorem 4.7.5 is called *rigid*. The proof in [420, §5] actually produces a rigid set D with the following additional property.

Proposition 4.7.6. *For every $n >$ there exists a rigid set D in $\mathbb{C}^n$ such that, if $p' \neq p''$ are points in $\mathbb{C}^n \backslash D$, then there is no automorphism of $\mathbb{C}^n$ mapping $D' = D \cup \{p'\}$ onto $D'' = D \cup \{p''\}$.*

As there exist continuum many choices of p'' for a given p', we get

Corollary 4.7.7. [420, Corollary 5.3] *There exist continuum many discrete sets in $\mathbb{C}^n$ no two of which are* $\operatorname{Aut}\mathbb{C}^n$*-equivalent.*

By replacing a discrete set of points in Theorem 4.7.2 by a discrete set of closed balls in affine subspaces of $\mathbb{C}^n$, we have the following theorem which summarizes results from [65], [168, §6], and [52].

Theorem 4.7.8. *Let $1 < k < n$ and $m \geq n - k$ be integers. There exists a closed set $B = \cup_{j=1}^{\infty} B_j$ in $\mathbb{C}^n$, where B_j are pairwise disjoint balls contained in k-dimensional affine subspaces Λ_j of $\mathbb{C}^n$, such that for every entire map $f\colon \mathbb{C}^m \to \mathbb{C}^n$ with $\operatorname{rank}_p f \geq n-k$ for some $p \in \mathbb{C}^m$, the set $\{z \in \mathbb{C}^m \colon f(z) \in B\}$ is infinite. It is possible to choose B such that $\mathbb{C}^n \backslash B$ is Kobayashi-Eisenman $(n-k)$-hyperbolic.*

Furthermore, the balls B_j in Theorem 4.7.8 can be chosen such that any sufficiently small smooth perturbation B'_j of B_j (depending on j) still gives a set $B' = \cup_{j=1}^{\infty} B'_j \subset \mathbb{C}^n$ whose complement $\mathbb{C}^n \backslash B'$ is Eisenman $(n-k)$-hyperbolic. This will be used in the proof of Theorem 4.18.8 on p. 170 below.

4.8 Algorithms for Vector Fields

We show that the flow of a vector field can be approximated by iterates of a fixed map. This is essential in the Andersén-Lempert theory.

Definition 4.8.1. [5, p. 254] *Let V be a continuous vector field on a manifold X, and let $A_t(x)$ be a continuous map from an open set in $\mathbb{R}_+ \times X$ containing $\{0\} \times X$ to X such that its t-derivative exists and is continuous. We say that A is an* algorithm for V *if for all $x \in X$ we have*

$$A_0(x) = x, \qquad \left.\frac{\partial}{\partial t}\right|_{t=0} A(t,x) = V_x. \tag{4.34}$$

In the following result we focus on positive time $t \geq 0$; the same result for $t \leq 0$ is obtained by replacing V by $-V$.

Theorem 4.8.2. [5, Theorem 4.1.26, p. 254] *Let V be a Lipschitz continuous vector field with flow ϕ_t on a manifold X. Let Ω be the fundamental domain of V (1.43) and $\Omega_+ = \Omega \cap (\mathbb{R}_+ \times X)$. If A is an algorithm for V, then for all $(t,x) \in \Omega_+$ the n-th iterate $A^n_{t/n}(x)$ of the map $A_{t/n}$ is defined for sufficiently large $n \in \mathbb{N}$ (depending on x and t), and we have*

$$\lim_{n\to\infty} A^n_{t/n}(x) = \phi_t(x). \tag{4.35}$$

The convergence is uniform on compacts in Ω_+. Conversely, if $t_0 > 0$ is such that $A^n_{t/n}(x)$ is defined for all $t \in [0,t_0]$ and all sufficiently large $n \in \mathbb{N}$, and if $A^n_{t/n}(x)$ converges as $n \to \infty$, then $(t_0, x) \in \Omega_+$ and (4.35) holds.

Proof. We give details for $X = \mathbb{R}^m$ as we only use this case; a similar argument applies in general. Fix a point $p \in \mathbb{R}^m$ and suppose that the flow $\phi_t(p)$ exists for $t \in [0,t_0]$. Let $C = \{\phi_t(p) \colon t \in [0,t_0]\}$. Choose compact sets $L_1 \subset L_2 \subset \mathbb{R}^m$ such that $C \subset \operatorname{Int} L_1$ and $L_1 \subset \operatorname{Int} L_2$. There is a compact neighborhood

$K \subset \mathrm{Int} L_1$ of p such that for every $x \in K$ the flow $\phi_t(x)$ exists for $t \in [0, t_0]$ and remains in L_1. From (4.34) we see that $|\phi_t(x) - A_t(x)| = o(t)$ as $t \to 0$, uniformly on L_2. Fix $n \in \mathbb{N}$. Given a point $x \in K$, assume for the moment that the orbit

$$y_0 = x,\ y_1 = A_{t/n}(y_0),\ y_2 = A_{t/n}(y_1), \ldots, y_n = A_{t/n}(y_{n-1}) \tag{4.36}$$

exists and lies in L_2. If β is a Lipschitz constant for V on L_2 then Lemma 1.9.3 (p. 32) gives the estimate $|\phi_t(x) - \phi_t(y)| \le \mathrm{e}^{\beta t}|x - y|$. Since $\phi_t(x) = \phi^n_{t/n}(x)$, we have

$$\phi_t(x) - A^n_{t/n}(x) = \sum_{j=1}^{n} \phi^{n-j}_{t/n}\left(\phi_{t/n}(y_{j-1})\right) - \phi^{n-j}_{t/n}\left(A_{t/n}(y_{j-1})\right).$$

Applying the inequality (1.47) (p. 32) to each term we get

$$|\phi_t(x) - A^n_{t/n}(x)| \le \sum_{j=1}^{n} \mathrm{e}^{\beta t(n-j)/n}|\phi_{t/n}(y_{j-1}) - A_{t/n}(y_{j-1})| \le n\mathrm{e}^{\beta t} o(t/n).$$

This converges to 0 as $n \to \infty$, uniformly for $x \in K$. A similar estimate gives

$$|\phi_{kt/n}(x) - A^k_{t/n}(x)| \le k\mathrm{e}^{\beta t} o(t/n), \quad k = 1, \ldots, n.$$

Using this estimate inductively we see that for every sufficiently large $n \in \mathbb{N}$ the orbit (4.36) exists and remains in L_2 for each $x \in K$ and $t \in [0, t_0]$. This proves the main implication. The converse is left as an exercise. □

Proposition 4.8.3. *Let V and W be vector fields with flows ϕ_t, ψ_t. Then*

(i) *$\phi_t \circ \psi_t$ is an algorithm for $V + W$.*
(ii) *$\psi_{-\sqrt{t}} \circ \phi_{-\sqrt{t}} \circ \psi_{\sqrt{t}} \circ \phi_{\sqrt{t}}$ is an algorithm for $[V, W]$.*

The same holds if ϕ_t, ψ_t are algorithms for V, W, respectively.

Proof. This follows from the Taylor expansion of flows (see 1.50 on p. 34). The second statement is Lemma 1.9.7 on p. 36. □

By a repeated application of Proposition 4.8.3, using sums and Lie brackets, Theorem 4.8.2 gives the following result.

Corollary 4.8.4. *Let $V_1, \ldots, V_m$ be $\mathbb{R}$-complete holomorphic vector fields on a complex manifold X. Denote by $\mathfrak{g}$ the Lie subalgebra of $\aleph_{\mathcal{O}}(X)$ genenerated by the V_j's and let $V \in \mathfrak{g}$. Assume that K is a compact set in X and $t_0 > 0$ is such that the flow $\phi_t(x)$ of V exists for every $x \in K$ and for all $t \in [0, t_0]$. Then ϕ_{t_0} is a uniform limit on K of a sequence of compositions of time-forward maps of the vector fields $V_1, \ldots, V_m$. In particular, ϕ_{t_0} can be approximated uniformly on K by holomorphic automorphisms of X.*

Example 4.8.5. If $V = V_1 + \cdots + V_m$ then its flow ϕ_t is given by

$$\phi_t(x) = \lim_{n\to\infty} \left(\phi^1_{t/n} \circ \cdots \circ \phi^m_{t/n}\right)^n (x)$$

for all (t, x) in the fundamental domain of the vector field V. $\square$

4.9 The Andersén-Lempert Theorem

A major breakthrough in understanding the automorphism groups $\operatorname{Aut}\mathbb{C}^n$ was made by Andersén [16] and Andersén and Lempert [18] in early 1990's. Their main result is the following.

Theorem 4.9.1. *Let $n > 1$.*

(i) [18] *Every holomorphic automorphism of $\mathbb{C}^n$ can be approximated uniformly on compacts by compositions of shears.*

(ii) [16] *Every automorphism of $\mathbb{C}^n$ with Jacobian one can be approximated uniformly on compacts by compositions of additive polynomial shears.*

Andersén and Lempert also proved that for every $n > 1$ there exists a holomorphic automorphism of $\mathbb{C}^n$ that is not a composition of finitely many shears. For example, they showed that the automorphism

$$(z_1, z_2) \mapsto \left(z_1 \mathrm{e}^{z_1 z_2}, z_2 \mathrm{e}^{-z_1 z_2}\right)$$

of $\mathbb{C}^2$ is not a composition of shears.

Theorem 4.9.1 is a special case of the following result that is much better adapted to applications.

Theorem 4.9.2. [193, Theorem 1.1 & Erratum] *Let Ω be a Runge domain (not necessarily pseudoconvex) in $\mathbb{C}^n$ for some $n > 1$, and let*

$$\Phi_t \colon \Omega \to \Omega_t = \Phi_t(\Omega) \subset \mathbb{C}^n, \qquad t \in [0, 1]$$

be a $\mathcal{C}^1$ isotopy consisting of injective holomorphic maps, with Φ_0 the identity map on $\Omega = \Omega_0$. If $\Phi_t(\Omega)$ is Runge in $\mathbb{C}^n$ for each $t \in [0, 1]$, then Φ_1 can be approximated uniformly on compacts in Ω by compositions of shears. If in addition Ω is a domain of holomorphy satisfying $H^{n-1}(\Omega; \mathbb{C}) = 0$ and if $\mathrm{J}\Phi_t = 1$ for every $t \in [0, 1]$, then Φ_1 can be approximated by compositions of additive shears (4.1) and their $SL(n)$-conjugates.

Proof. Let $\widetilde{\Omega}$ denote the trace of the isotopy $\{\Phi_t\}_{t\in[0,1]}$ in $\mathbb{R} \times \mathbb{C}^n$:

$$\widetilde{\Omega} = \left\{(t, z) \colon t \in [0, 1],\ z \in \Omega_t\right\} \subset \mathbb{R} \times \mathbb{C}^n.$$

We consider Φ_t as the flow of the continuous time dependent vector field

$$V(t,z) = \dot{\Phi}_t\big(\Phi_t^{-1}(z)\big), \quad (t,z) \in \widetilde{\Omega}.$$

(The dot denotes the derivative on t.) Note that V is continuous on $\widetilde{\Omega}$ and is holomorphic on Ω_t for every fixed $t \in [0,1]$. Choose an integer $N \in \mathbb{N}$ and subdivide $[0,1]$ into subintervals $I_k = [t_k, t_{k+1}]$ of length $1/N$, where $t_k = k/N$ for $k = 0, 1, \ldots, N-1$. Consider the locally constant holomorphic vector field $\widetilde{V}(t,z)$ that equals $V(t_k, z)$ for $t \in I_k$ and $z \in \Omega_{t_k}$; let ϕ_t denote its flow. Since the flows of $V_{t_k} = V(t_k, \cdot\,)$ and of V are tangential at $t = t_k$, these two flows differ on the interval $t \in I_k$ only by a term of order $o(N^{-1})$. We see as in the proof Theorem 4.8.2 (p. 123) that the total error is estimated by

$$|\phi_t(z) - \Phi_t(z)| = N o(N^{-1}) = o(1), \qquad t \in [0,1],\ z \in \Omega.$$

As $N \to +\infty$, ϕ_t converges to Φ_t uniformly on compacts in Ω for all $t \in [0,1]$. Further, given a compact set $K \subset \Omega$, for every N large enough we have

$$\{\phi_t(z) \colon z \in K,\ t \in I_k\} \Subset \Omega_{t_k}, \qquad k = 0, \ldots, N-1.$$

Hence it suffices to show that the flow ϕ_t^k of the autonomous holomorphic vector field V_{t_k} (now chosen so that $\phi_0^k(z) = z$ for $z \in \Omega_{t_k}$) can be approximated, uniformly on any compact subset $L \subset \Omega_{t_k}$ whose trace $\widetilde{L} = \{\phi_t^k(z) \colon z \in L,\ t \in [0, 1/N]\}$ remains in Ω_{t_k}, by holomorphic automorphisms of $\mathbb{C}^n$. This is done in two steps: First we approximate by the flow of a suitably chosen polynomial vector field (the easy part); we then approximate the latter by compositions of automorphisms of $\mathbb{C}^n$. The second step contains the essential observation of the Andersén-Lempert theory.

Step 1: Approximation by the flow of a polynomial vector field. Since the domain Ω_{t_k} is Runge in $\mathbb{C}^n$, we can approximate the holomorphic vector field V_{t_k}, uniformly on a compact set in Ω_{t_k} that contains $\widetilde{L}$ in its interior, by a polynomial holomorphic vector field. Lemma 1.9.4 on p. 33 shows that ϕ_t can be approximated uniformly on L by flows of polynomial vector fields.

A more careful argument is needed when $\mathrm{J}\Phi_t = 1$ for all $t \in [0,1]$. Let $V_t = \sum_{i=1}^n V_{t,i} \frac{\partial}{\partial z_i}$ and $\omega = \mathrm{d}z_1 \wedge \cdots \wedge \mathrm{d}z_n$. We see from (1.60) (p. 38) that

$$\mathrm{J}\Phi_t = 1 \ (\forall t \in [0,1]) \iff \mathrm{div}_\omega V_t = 0 \ (\forall t \in [0,1]);$$

that is, divergence zero vector fields correspond to volume preserving flows. From the Cartan formula $\mathrm{d}(V_t \rfloor \omega) = \mathrm{div}_\omega V_t \cdot \omega$ ((1.59) on p. 38) we see that $\mathrm{div} V_t = 0$ if and only if the holomorphic $(n-1)$-form $\alpha_t = V_t \rfloor \omega$ on Ω_t is closed; assume this is so. If Ω is a domain of holomorphy and $H^{n-1}(\Omega; \mathbb{C}) = 0$, then the same holds for every $\Omega_t = F_t(\Omega)$. Since the de Rham cohomology classes of a Stein manifold are represented by holomorphic forms [439, Theorem 1], we get $\alpha_t = \mathrm{d}\beta_t$ for some holomorphic $(n-2)$-form β_t on Ω_t. Approximating the coefficients of β_t by holomorphic polynomials we get a $(n-2)$-form $\widetilde{\beta}_t$ on

$\mathbb{C}^n$. The polynomial vector field W_t defined by $W_t \rfloor \omega = \mathrm{d}\widetilde{\beta}_t$ has divergence zero and it approximates the vector field V_t.

Step 2: Approximating the flow of a polynomial vector field by automorphisms. The following key lemma is due to Andersén [16] (for divergence zero vector fields) and Andersén and Lempert [18].

Proposition 4.9.3. *Let $n > 1$. Every polynomial holomorphic vector field V on $\mathbb{C}^n$ is a finite sum of complete polynomial shear fields (4.9) and (4.10) (p. 101). If $\mathrm{div}_\omega V = 0$ then V is a finite sum of shear fields (4.9).*

In view of Corollary 4.8.4 and Example 4.8.5 we get

Corollary 4.9.4. *Let $V = \sum_{i=1}^n V_i \frac{\partial}{\partial z_i}$ be a polynomial (or entire) vector field on $\mathbb{C}^n$ for some $n > 1$. Assume that $t_0 > 0$ and the flow $\phi_t(x)$ of V exists for all $t \in [0, t_0]$ and all points z in a compact set $K \subset \mathbb{C}^n$. Then $\phi_{t_0}|_K$ is a uniform limit of compositions of shear (resp. of additive shears if $\mathrm{div}_\omega V = 0$).*

Proof (of Proposition 4.9.3). Write $V = V_0 + V_1 + \cdots + V_m$ where each V_k is a vector field on $\mathbb{C}^n$ whose coefficients are homogeneous polynomials of degree k. Then $\mathrm{div}_\omega V = 0$ if and only if $\mathrm{div}_\omega V_k = 0$ for all k, and hence it suffices to prove the result for homogeneous vector fields. This is accomplished by the following lemma in which we identify a vector $v = (v_1, \ldots, v_n) \in \mathbb{C}^n$ with the constant vector field $\sum_{j=1}^n v_j \frac{\partial}{\partial z_j}$. Note that the summands in (4.37) are homogeneous shear vector fields of the form (4.9), (4.10).

Lemma 4.9.5. *For each integer $k \in \mathbb{N}$ there exist finitely many linear forms $\lambda_1, \ldots, \lambda_r$ on $\mathbb{C}^n$ and vectors $v_1, \ldots, v_r \in \mathbb{C}^n$, with $\lambda_i(v_i) = 0$ and $|v_i| = 1$ for all $i \in \{1, \ldots, r\}$, such that every holomorphic polynomial map $V \colon \mathbb{C}^n \to \mathbb{C}^n$ that is homogeneous of degree k is of the form*

$$V(z) = \sum_{i=1}^r c_i\, \lambda_i(z)^k\, v_i + d_i\, \lambda_i(z)^{k-1} \langle z, v_i \rangle\, v_i \tag{4.37}$$

for some numbers $c_i, d_i \in \mathbb{C}$. If $\mathrm{div}_\omega V = 0$, we may take $d_i = 0$ for all i. The λ_i's can be chosen from any nonempty open subset of $(\mathbb{C}^n)^*$ (the dual of $\mathbb{C}^n$).

Proof. For each $k \in \mathbb{N}$ there exist linear forms $\lambda_i(z) = \sum_{j=1}^n a_{i,j} z_j$ on $\mathbb{C}^n$ $(i = 1, \ldots, m = m(k,n))$ such that every homogeneous polynomial $P \in \mathbb{C}[z_1, \ldots, z_n]$ of degree k can be written in the form

$$P(z) = \sum_{i=1}^m c_i\, \lambda_i(z)^k, \quad c_i \in \mathbb{C}. \tag{4.38}$$

The forms λ_i may be chosen from any nonempty open set $U \subset (\mathbb{C}^*)^n$. (See e.g. [16, Lemma 5.6].) For convenience we replace U by a cone with vertex

at the origin, so that any nonzero multiple of an element in U remains in U (this requires a rescaling of coefficients in (4.37) and (4.37)). By a linear change of coordinates we may assume that $e_1^*(z) = z_1$ belongs to U. Choose $\lambda_1, \ldots, \lambda_m \in U$ satisfying (4.38) for homogeneous polynomials of degrees k and $k-1$ and such that $a_{i,n} = \lambda_i(e_n) = 1$ for $i = 1, \ldots, m$. (It suffices to insure that $\lambda_i(e_n) \neq 0$ and then rescale them.) Applying (4.38) to the homogeneous polynomial $\mathrm{div}_\omega V$ of degree $k-1$ we get $(\mathrm{div}_\omega V)(z) = \sum_{i=1}^m d_i\, \lambda_i(z)^{k-1}$ for some $d_i \in \mathbb{C}$. Choose vectors $v_i \in \ker \lambda_i$ with $|v_i| = 1$ and set

$$V_i(z) = d_i\, \lambda_i(z)^{k-1} \langle z, v_i \rangle\, v_i, \quad i = 1, \ldots, m.$$

These vector fields are taken as the second part of the expression (4.37). Since $\mathrm{div}_\omega V_i = d_i\, \lambda_i(z)^{k-1}$, the vector field $X = V - \sum_{i=1}^m V_i$ (that is homogeneous of degree k) has divergence zero.

It remains to show that every divergence zero homogeneous polynomial vector field X on $\mathbb{C}^n$ of degree k is of the form

$$X(z) = \sum_{i=1}^r c_i\, \lambda_i(z)^k\, v_i, \quad c_i \in \mathbb{C}, \tag{4.39}$$

where $\lambda_i \in U$ and $v_i \in \ker \lambda_i$. Applying (4.38) to the component X_j of X for $j \in \{1, \ldots, n-1\}$ we get $X_j(z) = \sum_{i=1}^m c_{i,j} \lambda_i(z)^k$. Each vector field

$$V_{i,j}(z) = c_{i,j}\, \lambda_i(z)^k\, (e_j - a_{i,j}\, e_n)$$

is a shear (4.9), with zero divergence, that may be used as one of the summands in (4.39). Set $W = \sum V_{i,j} = (W_1, \ldots, W_n)$ (summation over $i = 1, \ldots, m$ and $j = 1, \ldots, n-1$). Then $W_l = X_l$ for $l = 1, \ldots, n-1$, and hence $X = W + (X_n - W_n)\, e_n$. Since $\mathrm{div}_\omega X = 0 = \mathrm{div}_\omega W$, we get $0 = \mathrm{div}_\omega (X - W) = \partial(X_n - W_n)/\partial z_n$. Thus $X_n - V_n$ is independent of z_n, and hence $(X_n - W_n)\, e_n$ is a shear field (4.9). By choosing additional linear forms $\lambda_j' \in U$ that only depend on the variables $z' = (z_1, \ldots, z_{n-1})$ (such exist due to our choice of coordinates) we write $X_n - W_n = \sum c_j' \lambda_j'(z')$. Adding the corresponding vector fields $c_j' \lambda_j'(z')\, e_n$ to W we obtain the desired expression (4.39) for X.

This concludes the proof of Lemma 4.9.5, and hence of Proposition 4.9.3 and Theorem 4.9.2. □

Theorem 4.9.1 follows immediately from Theorem 4.9.2 and the following

Lemma 4.9.6. *For every $\Phi \in \mathrm{Aut}\,\mathbb{C}^n$ there is a smooth isotopy of automorphisms $\Phi_t \in \mathrm{Aut}\,\mathbb{C}^n$ ($t \in [0,1]$) such that $\Phi_0 = \mathrm{Id}$ and $\Phi_1 = \Phi$. When $\Phi_1 \in \mathrm{Aut}_1 \mathbb{C}^n$, the isotopy can be chosen in $\mathrm{Aut}_1 \mathbb{C}^n$.*

Proof. By a family of translations of $\mathbb{C}^n$ we connect Φ to $\Psi = \Phi - \Phi(0)$ satisfying $\Psi(0) = 0$. The isotopy $\Psi_t(z) = \frac{1}{t}\Psi(tz)$ ($t \in (0,1]$) connects $\Psi = \Psi_1$

to $\Psi_0 = \Phi'(0) \in GL_n(\mathbb{C})$. Finally, there is an isotopy in $GL_n(\mathbb{C})$ from Ψ_0 to Id. Combining these isotopies and reparametrizing the interval $[0,1]$ we obtain a smooth isotopy of automorphisms of $\mathbb{C}^n$ connecting Φ to the identity. The analogous arguments apply when $\Phi \in \mathrm{Aut}_1(\mathbb{C}^n)$. $\square$

The same proof gives the following corollary [18, Theorem 2.1].

Corollary 4.9.7. *If D is a starshaped domain in $\mathbb{C}^n$ $(n > 1)$ and $\Phi\colon D \xrightarrow{\cong} \Omega \subset \mathbb{C}^n$ is a biholomorphic map onto a Runge domain $\Omega \subset \mathbb{C}^n$, then Φ can be approximated uniformly on compacts by holomorphic automorphisms of $\mathbb{C}^n$. In particular, every biholomorphic map $\Phi\colon \mathbb{C}^n \to \Omega$ onto a Runge domain $\Omega \subset \mathbb{C}^n$ is a locally uniform limit of a sequence of holomorphic automorphisms of $\mathbb{C}^n$.*

Example 4.9.8. Theorem 4.9.2 fails in general for volume preserving maps if $H^{n-1}(\Omega;\mathbb{C}) \neq 0$; here is an example. The family $\Phi_t(z,w) = (z, w + tz^{-1})$ is a one-parameter group of automorphism of $\mathbb{C}^* \times \mathbb{C}$ preserving the volume form $\mathrm{d}z \wedge \mathrm{d}w$. The circle $T = \{(z,\bar{z}) \in \mathbb{C}^2 \colon |z| = 1\}$ is polynomially convex and hence admits a pseudoconvex tubular neighborhood $\Omega \subset \mathbb{C}^* \times \mathbb{C}$ that is Runge in $\mathbb{C}^2$. Clearly $H^1(\Omega;\mathbb{C}) = H^1(T;\mathbb{C}) = \mathbb{C}$. Thus all conditions except the cohomological one are satisfied. However, Φ_t for $t \neq 0$ is not approximable by volume preserving automorphisms of $\mathbb{C}^2$ in any neighborhood of T. Indeed, $\int_{\Phi_t(T)} w\,\mathrm{d}z = 2\pi\mathrm{i}(1+t)$ depends on t, while Stokes' theorem implies (in view of $\mathrm{d}(w\mathrm{d}z) = -\mathrm{d}z \wedge \mathrm{d}w$) that this integral is independent of t for any volume preserving automorphism Φ_t. $\square$

Theorem 4.9.2 has the following parametric version.

Theorem 4.9.9. [315, Theorem 2.3] *Let Ω be an open set in $\mathbb{C}^n = \mathbb{C}^k \times \mathbb{C}^m$ with $m > 1$. For every $t \in [0,1]$ let Φ_t be a biholomorphic map from Ω into $\mathbb{C}^n$, of the form*

$$\Phi_t(z,w) = \big(z, \varphi_t(z,w)\big), \quad z \in \mathbb{C}^k,\ w \in \mathbb{C}^m \tag{4.40}$$

and of class $\mathcal{C}^1$ in $(t,z,w) \in [0,1] \times \Omega$. Assume that each domain $\Phi_t(\Omega)$ is Runge in $\mathbb{C}^n$. If Φ_0 can be approximated uniformly on compacts in Ω by automorphisms of the form (4.40), then the same holds for every Φ_t $(t \in [0,1])$.

Proof. Consider a holomorphic vector field on $\mathbb{C}^n = \mathbb{C}^k \times \mathbb{C}^m$ of the form

$$V(z,w) = \sum_{j=1}^{m} a_j(z,w) \frac{\partial}{\partial w_j}, \qquad z \in \mathbb{C}^k,\ w \in \mathbb{C}^m. \tag{4.41}$$

Its flow remains in the level sets $w = \mathrm{const}$, and V is complete on $\mathbb{C}^n$ if and only if $V(z,\cdot)$ is complete on $\mathbb{C}^m$ for each fixed $z \in \mathbb{C}^k$. We conclude the proof in the same way as in Theorem 4.9.2, but replacing Proposition 4.9.3 with the following lemma.

Lemma 4.9.10. *If $m \geq 2$ then every polynomial vector field V (4.41) on $\mathbb{C}^n = \mathbb{C}^k \times \mathbb{C}^m$ is a finite sum of complete polynomial fields of the same type.*

Proof. We have $V(z,w) = \sum_\alpha z^\alpha V_\alpha(w)$ (a finite sum), where $z^\alpha = z_1^{\alpha_1} \cdots z_k^{\alpha_k}$ and $V_\alpha = \sum_{j=1}^m a_{\alpha,j}(w) \frac{\partial}{\partial w_j}$ is a polynomial vector field on $\mathbb{C}^m$. By Corollary 4.9.3 each V_α is a finite sum of complete polynomial fields on $\mathbb{C}^m$. The products of such fields with z^α are complete on $\mathbb{C}^n$. $\square$

4.10 The Density Property

Many results of the Andersén-Lempert theory extend to Lie algebras of holomorphic vector fields which are densely generated by complete fields. The relevant *density property* (Def. 4.10.1) was introduced and first systematically investigated by Varolin [500, 501]; it has since become one of the standard notions of elliptic geometry with close ties to Oka theory.

Recall (p. 19) that $\aleph_{\mathcal{O}}(X)$ denotes the Lie algebra of all holomorphic vector fields on a complex manifold X. Let $\mathfrak{g} \subset \aleph_{\mathcal{O}}(X)$ be a Lie subalgebra of $\aleph_{\mathcal{O}}(X)$. Given a subset $\mathfrak{g}_0$ of $\mathfrak{g}$, we denote by $\mathrm{Span}(\mathfrak{g}_0)$ the linear span of $\mathfrak{g}_0$ (a vector subspace of $\mathfrak{g}$), and by $\mathrm{Lie}(\mathfrak{g}_0)$ the Lie subalgebra of $\mathfrak{g}$ generated by $\mathfrak{g}_0$. Obviously $\mathrm{Span}(\mathfrak{g}_0) \subset \mathrm{Lie}(\mathfrak{g}_0)$. We also let

$$\mathfrak{g}_{\mathrm{int}} = \{V \in \mathfrak{g} \colon V \text{ is completely integrable}\}. \tag{4.42}$$

In the sequel a holomorphic vector field is called complete, or completely integrable, if it is $\mathbb{C}$-complete (Def. 1.9.9 on 37).

Definition 4.10.1. *Let X be a a complex manifold.*

(a) *A Lie algebra of holomorphic vector fields $\mathfrak{g} \subset \aleph_{\mathcal{O}}(X)$ has the* density property *if the Lie subalgebra* $\mathrm{Lie}(\mathfrak{g}_{\mathrm{int}})$ *of $\mathfrak{g}$, generated by the complete vector fields in $\mathfrak{g}$, is dense in $\mathfrak{g}$. The manifold X has the (holomorphic)* density property *if this holds for $\mathfrak{g} = \aleph_{\mathcal{O}}(X)$.*

(b) *If X is an algebraic manifold then a Lie algebra $\mathfrak{g} \subset \aleph_{\mathcal{A}}(X)$ has the* algebraic density property *if $\mathfrak{g} = \mathrm{Lie}(\mathfrak{g}_{\mathrm{int}})$. If this holds for $\mathfrak{g} = \aleph_{\mathcal{A}}(X)$ then X has the* algebraic density property.

(c) *A complex manifold X with a holomorphic volume form ω has the* volume density property *if the Lie algebra*

$$\aleph_{\mathcal{O}}^{\omega}(X) = \left\{V \in \aleph_{\mathcal{O}}(X) \colon \mathrm{div}_\omega V = 0\right\}$$

of divergence zero vector fields has the density property.

(d) *An algebraic manifold X with an algebraic volume form ω has the* algebraic volume density property *if the Lie algebra*

$$\aleph_{\mathcal{A}}^{\omega}(X) = \left\{V \in \aleph_{\mathcal{A}}(X) \colon \mathrm{div}_\omega V = 0\right\}$$

has the algebraic density property.

Recall that ω-divergence $\mathrm{div}_\omega V$ of a vector field V is the function satisfying the equation $L_V\omega = \mathrm{d}(V\rfloor\omega) = \mathrm{div}_\omega V\cdot\omega$. (See (1.59) on p. 38.) Every compact complex manifold enjoys the density property since each vector field is complete; however, $\aleph_{\mathcal{O}}(X)$ is finite dimensional. By the same token, a compact algebraic manifold has the algebraic density property. On the other hand, the density property is a very restrictive condition on noncompact manifolds. A Stein manifold with the density property is elliptic (Def. 5.5.11, p. 203), and an affine algebraic manifold with the algebraic density property is algebraically elliptic (Theorem 5.5.18 on p. 206). In particular, every such manifold is an Oka manifold (Def. 5.4.1 on p. 192).

Remark 4.10.2. The algebraic density property implies the holomorphic density property on any algebraic manifold on which holomorphic vector fields can be approximated by algebraic vector fields; this includes all compact algebraic manifolds and all affine algebraic manifolds.

The *algebraic volume density property* (Def. 4.10.1 (d)) was introduced by Kaliman and Kutzschebauch [290] who proved that, on affine algebraic manifolds, it also implies the holomorphic volume density property. The proof is similar to the approximation argument in the proof of Theorem 4.9.2 and goes as follows:

Assume that V is a holomorphic vector field on X such that $\mathrm{div}_\omega V = 0$. We need to approximate V by algebraic divergence zero vector fields. Note that $\mathrm{d}(V\rfloor\omega) = \mathrm{div}_\omega V\cdot\omega = 0$, so $\alpha = V\rfloor\omega$ is a closed holomorphic $(n-1)$-form. By Grothendieck there exists a closed algebraic $(n-1)$-form $\widetilde{\alpha}$ in the cohomology class of α, which means that $\alpha = \widetilde{\alpha} + \mathrm{d}\beta$ for some holomorphic $(n-2)$-form β. Approximating β by an algebraic form $\widetilde{\beta}$ we get a closed algebraic $(n-1)$-form $\theta = \widetilde{\alpha} + \mathrm{d}\widetilde{\beta}$ that approximates α. The algebraic vector field determined by $W\rfloor\omega = \theta$ then approximates V, and $\mathrm{div}_\omega W = 0$. $\square$

In Definition 4.10.1 (i) we can replace $\mathrm{Lie}(\mathfrak{g}_{\mathrm{int}})$ by $\mathrm{Span}(\mathfrak{g}_{\mathrm{int}})$ due to the following elementary observation.

Lemma 4.10.3. *For every Lie algebra $\mathfrak{g} \subset \aleph(X)$ we have*

$$\overline{\mathrm{Span}(\mathfrak{g}_{\mathrm{int}})} = \overline{\mathrm{Lie}(\mathfrak{g}_{\mathrm{int}})},$$

where the closures are in the topology of uniform convergence on compacts.

Proof. Since $\mathrm{Lie}(\mathfrak{g}_{\mathrm{int}})$ is the smallest Lie subalgebra of $\aleph_{\mathcal{O}}(X)$ that contains $\mathrm{Span}(\mathfrak{g}_{\mathrm{int}})$, it suffices to prove that $\overline{\mathrm{Span}(\mathfrak{g}_{\mathrm{int}})}$ is a Lie algebra. This is so if the commutator of any two complete vector fields $V, W \in \mathfrak{g}_{\mathrm{int}}$ belongs to $\overline{\mathrm{Span}(\mathfrak{g}_{\mathrm{int}})}$. Let $\{\phi_t\}_{t\in\mathbb{R}}$ denote the flow of V. The Lie bracket

$$[V, W] = L_V W = \lim_{t\to 0} \frac{1}{t}\left(\phi_t^* W - W\right)$$

can be approximated by $t^{-1}(\phi_t^* W - W)$ for small $t \neq 0$, uniformly on compacts in X. If V and W are complete, the desired conclusion follows from the fact that the fields $t^{-1}\phi_t^* W$ and $t^{-1}W$ are both complete since global change of variables and multiplication by a constant preserve integrability. □

The significance of the density property is captured by the following result that follows from Corollary 4.8.4 (p. 124) and (the proof of) Theorem 4.9.2.

Proposition 4.10.4. *Let $\mathfrak{g} \subset \aleph_{\mathcal{O}}(X)$ be a Lie algebra of holomorphic vector fields with the density property, and let $V \in \mathfrak{g}$. Assume that for some $t_0 > 0$ the flow $\phi_t(x)$ of V exists for all $t \in [0, t_0]$ and all x in an open set $\Omega \subset X$. Then $\phi_{t_0}\colon \Omega \to X$ can be approximated uniformly on compacts in Ω by compositions of flows of complete vector fields contained in $\mathfrak{g}$. The same conclusion holds for the flow of a continuous time dependent vector field $V_t \in \mathfrak{g}$ $(t \in [0, t_0])$.*

Remark 4.10.5. In Corollary 4.8.4 and Proposition 4.10.4 we only consider flows along a real time interval. The analogous results hold along any path in the complex time domain since the flow in complex time is a composition of real time flows of V and JV (see (1.58) on p. 37). □

We also have the following analogue of Theorem 4.9.2 for Stein manifolds with the holomorphic density property.

Theorem 4.10.6. *Let $\Phi_t\colon \Omega_0 \to \Omega_t = \Phi_t(\Omega_0) \subset X$ $(t \in [0,1])$ be a $\mathcal{C}^1$ isotopy consisting of injective holomorphic maps between pseudoconvex Runge domains in a Stein manifold X, with $\Phi_0 = \mathrm{Id}_{\Omega_0}$. If X has the holomorphic density property, then Φ_1 can be approximated uniformly on compacts in Ω by holomorphic automorphisms of X.*

To see this, note that every vector field on Ω_t can be approximated by global holomorphic vector fields on X by using the Oka-Weil theorem; the rest of the proof is exactly as before.

The approximation question becomes rather delicate for proper Lie subalgebras of $\aleph_{\mathcal{O}}$ as is seen for divergence zero and Hamiltonian vector fields on $\mathbb{C}^n$; see Example 4.9.8 on p. 129.

Proposition 4.9.3 (the main point of the Andersén-Lempert theory) says that $\mathbb{C}^n$ for $n > 1$ has the algebraic density property, and also the algebraic volume density property with respect to $\omega = \mathrm{d}z_1 \wedge \cdots \wedge \mathrm{d}z_n$. We give another short proof, due to Varolin [500] and Kaliman and Kutzschebauch [289, Corollary 2.2], that effectively uses commutators.

Proposition 4.10.7. *If $n \geq 2$ then $\mathbb{C}^n$ has the algebraic density property.*

Proof. Let $z = (z_1, \ldots, z_n)$ be complex coordinates on $\mathbb{C}^n$. Let $V_i = \frac{\partial}{\partial z_i}$ be viewed as a derivation on the polynomial ring $\mathbb{C}[z]$; hence $\ker V_i$ is the ring of

polynomials independent of z_i. For any $f_i \in \ker V_i$ the vector fields $f_i V_i$ and $z_j f_i V_i$ (any j) are complete (these are shears (4.11)). Hence the vector field

$$[f_1 V_1, z_1 f_2 V_2] - [z_1 f_1 V_1, f_2 V_2] = f_1 f_2 V_2$$

belongs to $\mathrm{Lie}_{\mathrm{alg}}(\mathbb{C}^n)$, the Lie subalgebra of $\aleph_{\mathcal{A}}(\mathbb{C}^n)$ generated by the complete algebraic vector fields. Since $\mathbb{C}[z]$ is spanned as a vector space by elements $f_1 f_2 \in \ker V_1 \cdot \ker V_2$, we conclude that $\mathrm{Lie}_{\mathrm{alg}}(\mathbb{C}^n)$ contains all polynomial fields proportional to V_2. Now repeat the same argument for $V_1, V_3, \ldots, V_n$. □

Note that $\mathbb{C}$ does not have the density property (since the only complete holomorphic vector fields are the linear ones), but it has the volume density property (since every divergence zero vector field is constant).

The Andersén-Lempert theorem also holds for symplectic holomorphic automorphisms of $\mathbb{C}^{2n}$; see [167, Proposition 5.2].

The volume density property of a complex Lie group G is always meant with respect to the unique left-invariant holomorphic volume form ω that equals the standard volume form on $\mathrm{T}_1 G$; this ω is called the *Haar form* on G. The Haar form on $(\mathbb{C}^*)^n$ is

$$(z_1 \cdots z_n)^{-1} \mathrm{d}z_1 \wedge \cdots \wedge \mathrm{d}z_n. \tag{4.43}$$

The following theorem summarizes the main results of Varolin's first paper on the density property. The proof involves elementary but clever calculations using sums and Lie brackets, similar to those in the proof of Proposition 4.10.7.

Theorem 4.10.8. (Varolin [500])

(a) *If X and Y are Stein manifolds with the density property then so are the manifolds $X \times Y$, $X \times \mathbb{C}$, and $X \times \mathbb{C}^*$.*

(b) *For every complex Lie group G, $G \times \mathbb{C}$ has the density property. In particular, $\mathbb{C}^k \times (\mathbb{C}^*)^l$ has the density property when $k \geq 1$ and $k + l \geq 2$.*

(c) *If a complex Lie group G has the volume density property then so does $G \times \mathbb{C}^*$. In particular, $(\mathbb{C}^*)^n$ has the volume density property with respect to the Haar volume form (4.43).*

(d) *If $0 < k < n$ then the Lie algebra $\mathfrak{g}_0^{n,k}$ of holomorphic vector fields on $\mathbb{C}^n = \mathbb{C}^k \times \mathbb{C}^{n-k}$ that vanish on $\mathbb{C}^k \times \{0\}^{n-k}$ has the density property.*

(e) *If $n > k \geq 2$ then the Lie algebra $\mathfrak{g}_T^{n,k}$ of holomorphic vector fields on $\mathbb{C}^n$ that are tangent to $\mathbb{C}^k \times \{0\}^{n-k}$ has the density property.*

The algebraic density property fails on $(\mathbb{C}^*)^n$ since every globally integrable algebraic vector field has zero divergence with respect to the form (4.43) [17]. However, the following has been a long standing open problem (see e.g. [420]).

Problem 4.10.9. Does $(\mathbb{C}^*)^n$ enjoy the density property when $n > 1$? Does every $F \in \operatorname{Aut}(\mathbb{C}^*)^n$ preserve the volume form (4.43) up to a sign?

The next result, due to Tóth and Varolin, deserves to be stated separately.

Theorem 4.10.10. [493, 494] *Every complex semisimple Lie group G has the density property. If the center of G is trivial and if K is a reductive subgroup of G then the homogeneous space G/K has the density property.*

Since every complex semisimple Lie group is Stein, Theorem 4.10.10 together with the main results of [501] gives the following conclusions (Corollaries 1–3 in [493]).

Corollary 4.10.11. *A complex semisimple Lie group G of dimension n admits an open subset biholomorphic to $\mathbb{C}^n$ (a Fatou-Bieberbach domain of the first kind), and a proper open subset biholomorphic to G (a Fatou-Bieberbach domain of the second kind).*

We now present some of the results of Kaliman and Kutzschebauch who introduced new methods for establishing the algebraic density property.

Theorem 4.10.12. [289, Theorem 3] *A linear algebraic group whose connected component of the identity is different from $\mathbb{C}_+$ and from $(\mathbb{C}^*)^n$ has the algebraic density property.*

This result subsumes Theorem 4.10.10 of Tóth and Varolin since every semisimple complex Lie group is reductive, and every reductive group admits a unique structure of a linear algebraic group (i.e., a subgroup of $GL_n(\mathbb{C})$ that is also an algebraic subvariety). The details in the proof are considerable; we present here some of the main ingredients of independent interest.

Let X denote an algebraic manifold, $\mathbb{C}[X]$ the algebra of regular functions on X, $\operatorname{Aut}_{\mathrm{alg}} X$ the group of all algebraic automorphisms of X, and

$$\operatorname{Lie}_{\mathrm{alg}}(X) = \operatorname{Lie}(\aleph_{\mathcal{A}}(X)_{\mathrm{int}}) \subset \aleph_{\mathcal{A}}(X)$$

the Lie algebra generated by $\mathbb{C}$-complete algebraic vector fields on X.

Definition 4.10.13. *Let X be an algebraic manifold and $x_0 \in X$. A finite subset M of the tangent space $\mathrm{T}_{x_0}X$ is called* a generating set *if the image of M under the action of the isotropy subgroup of x_0 (in $\operatorname{Aut}_{\mathrm{alg}} X$) spans $\mathrm{T}_{x_0}X$ as a complex vector space. X is called* tangentially semihomogeneous *if it is $\operatorname{Aut}_{\mathrm{alg}} X$-homogeneous and it admits a generating set consisting of one vector.*

Theorem 4.10.14. [289, Theorem 1] *Let X be an affine algebraic manifold that is $\operatorname{Aut}_{\mathrm{alg}} X$-homogeneous. Let L be a $\mathbb{C}[X]$-submodule of $\aleph_{\mathcal{A}}(X)$ such that $L \subset \operatorname{Lie}_{\mathrm{alg}}(X)$. If the fiber $L_{x_0} = \{V_{x_0} : V \in L\} \subset \mathrm{T}_{x_0}X$ over some point $x_0 \in X$ contains a generating set then X has the algebraic density property.*

Proof. The $\mathbb{C}[X]$-modules $\aleph_{\mathcal{A}}(X)$ and L generate coherent sheaves $\mathcal{T}$ (the tangent sheaf of X) and $\mathcal{L}$, respectively, with $\mathcal{L}$ a subsheaf of $\mathcal{T}$. The action of $\alpha \in \operatorname{Aut}_{\mathrm{alg}} X$ on $\aleph_{\mathcal{A}}(X)$, taking V to $\alpha_* V$, maps $\mathcal{L}$ onto another coherent subsheaf $\mathcal{L}_\alpha$ of $\mathcal{T}$. The sum of any finite collection of $\mathcal{L}_\alpha$'s is a coherent subsheaf $\mathcal{E}$ of $\mathcal{T}$. Let $\mathfrak{m}_{x_0} \subset \mathbb{C}[X]$ denote the maximal ideal of a point $x_0 \in X$. Definition 4.10.13 implies that $\mathcal{E}$ can be chosen such that $\mathcal{E}/\mathfrak{m}_{x_0}\mathcal{E} = \mathrm{T}_{x_0}X$; i.e., sections of $\mathcal{E}$ span the tangent space $\mathrm{T}_{x_0}X$. Furthermore, since X is homogeneous, we can suppose (by induction on dimension of the exceptional variety) that this holds for every point in X; hence global sections of $\mathcal{E}$ span the tangent space to X at every point. A standard application of Serre's Theorem B then shows that every section of $\mathcal{T}$ (algebraic vector field on X) is a linear combination with coefficients in $\mathbb{C}[X]$ of vector fields that are sections of $\mathcal{E}$; in other words, $\mathcal{E} = \mathcal{T}$. Since the push-forward of a vector field by an automorphism preserves complete integrability, all global sections of $\mathcal{E}$ are in $\mathrm{Lie}_{\mathrm{alg}}(X)$; thus $\mathrm{Lie}_{\mathrm{alg}}(X) = \aleph_{\mathcal{A}}(X)$. $\square$

Definition 4.10.15. *Let X be a smooth affine algebraic variety and V an algebraic vector field on X.*

(a) V *is* semisimple *if its flow generates an algebraic $\mathbb{C}^*$ action on X.*

(b) V *is* locally nilpotent *if for every $f \in \mathbb{C}[X]$ there is an integer $k = k(f) \in \mathbb{N}$ such that $V^k(f) = 0$.*

Note that an algebraic vector field V is locally nilpotent if and only if its flow $\{\phi_t\}_{t\in\mathbb{C}}$ is an algebraic $\mathbb{C}_+$ action on X; this can be seen from the Lie series expansion (1.49) (p. 34) of a function along the flow.

Definition 4.10.16. *Let V_1 and V_2 be nontrivial algebraic vector fields on an affine algebraic manifold X such that V_1 is a locally nilpotent, and V_2 is either locally nilpotent or semisimple. We say that V_1 and V_2 are* compatible *if*

(i) *the vector space* $\operatorname{Span}(\ker V_1 \cdot \ker V_2)$ *generated by elements from the set* $\ker V_1 \cdot \ker V_2$ *contains a nonzero ideal in $\mathbb{C}[X]$, and*

(ii) *there exists an element $a \in \ker V_2$ that is of degree one with respect to V_1, i.e., $V_1(a) \neq 0$ but $V_1^2(a) = 0$.*

Theorem 4.10.17. [289, Theorem 2] *Let X be a homogeneous affine algebraic manifold with finitely many pairs $\{(V_{k,1}, V_{k,2})\}_{k=1}^m$ of compatible vector fields such that for some $x_0 \in X$, $\{(V_{k,2})_{x_0}\}_{k=1}^m \subset \mathrm{T}_{x_0}X$ is a generating set (Def. 4.10.13). Then X has the algebraic density property.*

Proof. Let V_1, V_2 be one of our compatible pairs. Choose $a \in \ker V_2$ of degree one with respect to V_1 and set $b = V_1(a) \in \ker V_1 \backslash \{0\}$. If $f_i \in \ker V_i$ $(i = 1, 2)$ then the vector field

$$[af_1V_1, f_2V_2] - [f_1V_1, af_2V_2] = -bf_1f_2V_2$$

belongs to $\mathrm{Lie}_{\mathrm{alg}}(X)$. Since V_1 and V_2 are compatible, Definition 4.10.16 (i) implies that sums of such vector fields include every vector field in IV_2 for some nonzero ideal I in $\mathbb{C}[X]$. Applying this argument to all compatible pairs $V_{k,1}, V_{k,2}$ we see that there is a nonzero ideal $J \subset \mathbb{C}[X]$ such that $\mathrm{Lie}_{\mathrm{alg}}(X)$ contains the $\mathbb{C}[X]$-submodule $L = \{\sum_{k=1}^m \alpha_k V_{k,2} \colon \alpha_1, \ldots, \alpha_m \in J\}$. Since $\{(V_{k,2})_{x_0}\}_{k=1}^m$ remains a generating set under small perturbations of the base point x_0, we can suppose that x_0 does not belong to the zero locus of J. For such x_0 the set $\{V_{x_0} \colon V \in L\} \subset \mathrm{T}_{x_0}X$ contains a generating set, and by Theorem 4.10.14 it follows that X has the algebraic density property. □

If X is tangentially semihomogenous and if every nonzero vector $0 \neq v \in \mathrm{T}X$ is a generating set, then by Theorem 4.10.17 the existence of a single pair of compatible vector fields implies the algebraic density property of X.

Corollary 4.10.18. [289, Corollary 2.8] *Let X_1 and X_2 be homogeneous affine algebraic manifolds. Assume that each X_i $(i = 1, 2)$ admits $\mathbb{C}$-complete algebraic vector fields $\{V_{i,k}\}_{k=1}^{m_i}$ whose values at some point $x_i \in X_i$ form a generating set in $\mathrm{T}_{x_i}X_i$, and the fields $\{V_{1,k}\}_{k=1}^{m_1}$ are locally nilpotent. Then $X_1 \times X_2$ has the algebraic density property.*

Proof. Applying the isotropy groups one can suppose that $\{V_{i,k}(x_i)\}_{k=1}^{m_i}$ is a basis of $\mathrm{T}_{x_i}X_i$ for $i = 1, 2$. Note that $V_{1,k}$ and $V_{2,j}$ define compatible integrable vector fields on $X = X_1 \times X_2$ which we denote by the same symbols. In order to show that $M = \{0 \oplus V_{2,k}(x_2)\}_k$ is a generating set in $\mathrm{T}_{(x_1,x_2)}X$ we need the following fact that is obvious in a local coordinate system.

Lemma 4.10.19. *Let V be a holomorphic vector field on a complex manifold X. Suppose that f is a holomorphic function such that $V(f) = 0$ and $f(x_0) = 0$ for some $x_0 \in X$. Then the flow of fV generates a linear action on the tangent space $\mathrm{T}_{x_0}X$ given by the formula*

$$\mathrm{T}_{x_0}X \ni w \to w + \mathrm{d}f_{x_0}(w)V_{x_0}.$$

The span of the orbit of w under this flow contains the vector $\mathrm{d}f_{x_0}(w)V_{x_0}$.

Applying this to $V = V_{1,j}$ we see that the orbit of M under the isotropy group of (x_1, x_2) contains all vectors $(V_{1,j})_{x_1} \oplus (V_{2,k})_{x_2}$. Thus M is a generating set, and hence Corollary 4.10.18 follows from Theorem 4.10.17. □

Example 4.10.20. The manifold $X = \mathbb{C}^k \times (\mathbb{C}^*)^l$ with $k \geq 1$ and $k + l \geq 2$ has the algebraic density property by Corollary 4.10.18. □

Example 4.10.21. As an application of Theorem 4.10.17 we give a simple proof, due to Kaliman and Kutzschebauch [289], that

$SL_n(\mathbb{C})$ for $n \geq 2$ has the algebraic density property.

Every simple complex Lie group G is tangentially semihomogeneous since the adjoint action of G generates an irreducible representation on the Lie algebra $\mathfrak{g} = \mathrm{T}_1 G$ (i.e., any nonzero vector in $\mathfrak{g}$ is a generating set). In particular, $SL_n(\mathbb{C})$ for $n \geq 2$ is tangentially semihomogeneous. Every element of $SL_n(\mathbb{C})$ is a matrix $x = (c_{k,j})$ with determinant 1. We use the coefficients c_{jk} as coordinates. Consider the vector fields

$$V_1 = \sum_{j=1}^{n} c_{2,j} \frac{\partial}{\partial c_{1,j}}, \quad V_2 = \sum_{j=1}^{n} c_{1,j} \frac{\partial}{\partial c_{2,j}}.$$

Note that constants and functions depending on $c_{k,j}$ for $k \neq 1$ are in $\ker V_1$, while constants and functions depending on $c_{k,j}$ for $k \neq 2$ are in $\ker V_2$; therefore condition (i) of Definition 4.10.16 holds. Taking c_{11} as a in condition (ii) we see that V_1 and V_2 are compatible; hence Theorem 4.10.17 applies. □

We describe another type of affine hypersurfaces with the density property.

Theorem 4.10.22. [288] *If f is a holomorphic function on $\mathbb{C}^n$ such that $\mathrm{d}f \neq 0$ on the zero locus $\{f = 0\}$ then the hypersurface*

$$X_f = \left\{(z_1, \ldots, z_n, u, v) \in \mathbb{C}^{n+2} \colon uv = f(z_1, \ldots, z_n)\right\} \tag{4.44}$$

has the holomorphic density property. If $f \in \mathbb{C}[z_1, \ldots, z_n]$ is a polynomial and $\mathrm{d}f \neq 0$ on $\{f = 0\}$ then X_f enjoys the algebraic density property.

Proof (Sketch). For any complete holomorphic (resp. polynomial) vector field θ on $\mathbb{C}^n_z$ and for any $h \in \mathcal{O}(\mathbb{C}^n_z)$ (resp. $h \in \mathbb{C}[z_1, \ldots, z_n]$) we get the following complete vector fields on $\mathbb{C}^{n+2} = \mathbb{C}^n_z \times \mathbb{C}_u \times \mathbb{C}_v$ that are clearly tangential to the hypersurface $X_f = \{uv - f(z) = 0\}$:

$$u\theta + \theta(f)\frac{\partial}{\partial v}, \quad v\theta + \theta(f)\frac{\partial}{\partial u}, \quad h(z)\left(u\frac{\partial}{\partial u} - v\frac{\partial}{\partial v}\right).$$

It is immediate that finitely many such vector fields span the tangent space to X_f at every point, so X_f is elliptic (resp. algebraically elliptic) in the sense of Def. 5.5.11 on p. 203. In particular, X_f is an Oka manifold by Theorem 5.5.18 (p. 206). With some more work it can be seen that vector fields of these types densely generate the Lie algebra of holomorphic vector fields $\aleph_{\mathcal{O}}(X_f)$; in the algebraic case they generate the Lie algebra $\aleph_{\mathcal{A}}(X_f)$. □

Remark 4.10.23. If the zero fiber $\{f = 0\}$ is contractible then the hypersurface X_f (4.44) is diffeomorphic to $\mathbb{C}^{n+1}$, but it is not known for which f is X_f biholomorphic (or algebraically isomorphic if f is a polynomial) to $\mathbb{C}^{n+1}$. These are examples of *Danielewski surfaces*. In [316] the authors define the notion of an overshear on such a surface and prove that the group generated by overshears is dense in the path connected component of the identity of the holomorphic automorphism group. □

4.11 Automorphisms Fixing a Subvariety

Let $n > 1$. We are interested in the question for which closed complex subvarieties A of $\mathbb{C}^n$ are the automorphism group $\mathrm{Aut}\,(\mathbb{C}^n \backslash A)$ and its subgroup

$$\mathrm{Aut}\,(\mathbb{C}^n, A) = \big\{F \in \mathrm{Aut}\,\mathbb{C}^n \colon F(z) = z,\ \forall z \in A\big\} \tag{4.45}$$

'large' in the sense of the Andersén-Lempert theory. To this end we study the Lie algebra $\aleph_{\mathcal{O}}(\mathbb{C}^n, A)$ of all holomorphic vector fields on $\mathbb{C}^n$ that vanish on A. If A is algebraic, we consider the corresponding algebraic automorphism group $\mathrm{Aut}_{\,\mathrm{alg}}(\mathbb{C}^n, A)$ and the Lie algebra $\aleph_{\mathcal{A}}(\mathbb{C}^n, A)$ of all algebraic vector fields on $\mathbb{C}^n$ that vanish on A. The main questions are the following:

Problem 4.11.1. When does the Lie algebra $\aleph_{\mathcal{O}}(\mathbb{C}^n, A)$ satisfy the density property (Def. 4.10.1 on p. 130)? If A is algebraic, when does $\aleph_{\mathcal{A}}(\mathbb{C}^n, A)$ satisfy the algebraic density property?

It is easily seen that $\aleph_{\mathcal{A}}(\mathbb{C}^n, A)$ has the algebraic density property if A is a complex affine subspace of $\mathbb{C}^n$ ([500]; see Theorem 4.10.8 (iv)). On the other hand, there exist algebraic hypersurfaces A in $\mathbb{C}^n$ whose complement $\mathbb{C}^n \backslash A$ is hyperbolic, and consequently $\mathrm{Aut}\,(\mathbb{C}^n \backslash A)$ is small. For example, the complement of a generic set of four lines in $\mathbb{C}^2$ is hyperbolic by Green's theorem [231]. By removing additional hypersurfaces one can find examples when the group $\mathrm{Aut}\,(\mathbb{C}^n \backslash A)$ is trivial. Looking at lower dimensional non-algebraic subvarieties, we have seen in §4.6 that there even exist discrete sets $A \subset \mathbb{C}^n$ for which $\mathrm{Aut}\,(\mathbb{C}^n \backslash A)$ is trivial.

Here is the main observation. Assume that $\pi \colon \mathbb{C}^n \to \mathbb{C}^{n-1}$ is a linear projection such that the restriction $\pi|_A \colon A \to \mathbb{C}^{n-1}$ is proper; then $\pi(A)$ is a closed analytic subvariety of $\mathbb{C}^{n-1}$. If $\dim A < n-1$ then $\pi(A)$ is a proper subvariety of $\mathbb{C}^{n-1}$, and hence there exist nonconstant holomorphic functions on $\mathbb{C}^{n-1}$ (polynomials if A is algebraic) that vanish on A. Let f be such a function. Choose a vector $0 \neq v \in \ker \pi$ and consider it as a constant holomorphic vector field on $\mathbb{C}^n$. The shear vector field $(f \circ \pi)\, v$ on $\mathbb{C}^n$ then vanishes on A and generates the $\mathbb{C}_+$ action

$$F_t(z) = z + t f(\pi(z))\, v, \qquad t \in \mathbb{C}. \tag{4.46}$$

This action fixes the subvariety $\pi^{-1}(\pi(A)) = A + \mathbb{C}v$ of $\mathbb{C}^n$ and hence it belongs to the group $\mathrm{Aut}\,(\mathbb{C}^n, A)$ (resp. to $\mathrm{Aut}_{\mathrm{alg}}\,(\mathbb{C}^n, A)$ if A is algebraic).

This suggests to consider subvarieties with $\dim A \leq n-2$ such that, in suitable coordinates on $\mathbb{C}^n$, there exist sufficiently many linear projections $\pi \colon \mathbb{C}^n \to \mathbb{C}^{n-1}$ that are proper when restricted to A. We analyze this condition more carefully. Given a point $v \in \mathbb{C}^n \backslash \{0\}$, we denote by $[v]$ the complex line $\mathbb{C}v$ considered as a point in $\mathbb{P}^{n-1}$ (the hyperplane at infinity in $\mathbb{P}^n$). For a closed analytic subvariety $A \subset \mathbb{C}^n$ we consider the limit set of A at infinity:

$$\omega(A) = \big\{ \lim_{j\to\infty} [a^j] \colon a^j \in A,\ |a^j| \to \infty \big\} \subset \mathbb{P}^{n-1}. \tag{4.47}$$

Clearly $\omega(A)$ is a closed subset of $\mathbb{P}^{n-1}$, and the closure of A in $\mathbb{P}^n$ equals $\overline{A} = A \cup \omega(A)$. A subvariety $A \subset \mathbb{C}^n$ of pure dimension p is algebraic if and only if $\omega(A)$ is an algebraic subset of $\mathbb{P}^{n-1}$ of dimension $p-1$. A complex hypersurface $A \subset \mathbb{C}^n$ is algebraic if and only if $\omega(A) \neq \mathbb{P}^{n-1}$ (see [85, §I.7]).

Lemma 4.11.2. *If $0 \neq v \in \mathbb{C}^n$ is such that $[v] \notin \omega(A)$, then for any surjective linear map $\pi \colon \mathbb{C}^n \to \mathbb{C}^{n-1}$ with $\pi(v) = 0$ the map $\pi|_A \colon A \to \mathbb{C}^{n-1}$ is proper.*

Proof. By a $\mathbb{C}$-linear change of coordinates we may assume that $v = e_n = (0, \ldots, 0, 1)$ and $\pi(z) = (z_1, \ldots, z_{n-1})$. If $\pi|_A$ is not proper, there exists a sequence $\{a^j\} \subset A$ such that $|a^j| \to \infty$ but $\pi(a^j)$ is bounded. Thus $|a^j_n| \to \infty$ and therefore $[a^j] \to [e_n]$ as $j \to \infty$. Hence $[v] = [e_n] \in \omega(A)$. $\square$

We now introduce a class of varieties for which the above groups are large; this class contains all algebraic subvarieties of codimension at least two.

Definition 4.11.3. *A closed analytic subset $A \subset \mathbb{C}^n$ is* tame *if there exists a holomorphic automorphism $\Phi \in \operatorname{Aut} \mathbb{C}^n$ such that $\omega(\Phi(A)) \neq \mathbb{P}^{n-1}$.*

Every closed complex subvariety A of $\mathbb{C}^n$ contained in a proper algebraic subvariety is tame. (Such A is said to be *algebraically degenerate*.) A hypersurface A is tame if and only if there exists a $\Phi \in \operatorname{Aut} \mathbb{C}^n$ such that $\Phi(A)$ is algebraic. For discrete sets, this notion of tameness coincides with the one of Rosay-Rudin (Def. 4.6.1 on p. 118) in view of Theorem 4.6.2 (i) (p. 118) and the following elementary lemma that is left as an exercise.

Lemma 4.11.4. *An analytic subvariety A of $\mathbb{C}^n$ is tame if and only if there exist holomorphic coordinates $z = (z', z_n)$ on $\mathbb{C}^n$ and a $C > 0$ such that*

$$A \subset \big\{ (z', z_n) \in \mathbb{C}^n \colon |z_n| \leq C(1 + |z'|) \big\}. \tag{4.48}$$

Lemma 4.11.5. *Assume that $A \subset \mathbb{C}^n$ is a closed complex subvariety with $\dim A \leq n-2$ and $\omega(A) \neq \mathbb{P}^{n-1}$. For every point $p \in \mathbb{C}^n \backslash A$ and every vector $0 \neq v \in \mathbb{C}^n$ such that $[v] \neq \omega(A)$ there exists a vector $w \in \mathbb{C}^n$ arbitrary close to v such that the affine complex line $\Lambda = p + \mathbb{C}w$ does not intersect A, and we can move p to any point $q \in \Lambda$ by a single shear of the form (4.46).*

Proof. We may assume that $p = 0$. Let $\mathbb{C}^n_* = \mathbb{C}^n \backslash \{0\}$, and let $\rho \colon \mathbb{C}^n_* \to \mathbb{P}^{n-1}$ denote the projection $\rho(z) = [z] \in \mathbb{P}^{n-1}$. For every neighborhood $U \subset \mathbb{C}^n_*$ of v we let $\Gamma_U = \{tw \colon w \in U,\ t \in \mathbb{C}^*\} \subset \mathbb{C}^n_*$, the open cone spanned by U. Since $[z] \notin \omega(A)$, we can choose U small enough such that the set $A \cap \Gamma_U$ is bounded, and hence the restricted projection $\rho \colon A \cap \Gamma_U \to \rho(U)$ is proper. Since $\dim A \leq n-2$, it follows that $\rho(A \cap \Gamma_U)$ is a proper subvariety of the open set $\rho(U) \subset \mathbb{P}^{n-1}$. For every point $w \in U$ such that $[w] \neq \rho(A \cap \Gamma_U)$ the complex line $\Lambda = \mathbb{C}w$ misses A. This proves the first statement; the second one is an immediate consequence. $\square$

We are now ready for the main results of this section. The following result is essentially due to Gromov [237] and Winkelmann [518, Proposition 1]; for the last statement see [288, Lemma 4.1].

Proposition 4.11.6. *If $A \subset \mathbb{C}^n$ is a tame analytic subvariety of codimension at least two then* $\mathrm{Aut}\,(\mathbb{C}^n, A)$ *acts transitively on $\mathbb{C}^n \backslash A$, and for any point $p \in \mathbb{C}^n \backslash A$ the image of any vector $v \in \mathrm{T}_p\mathbb{C}^n$ under the isotropy group* $\mathrm{Aut}\,(\mathbb{C}^n, A)_p$ *generates $\mathrm{T}_p\mathbb{C}^n$ (compare with Definition 4.10.13). If A is algebraic then the same conclusions hold for the group* $\mathrm{Aut}_{\mathrm{alg}}(\mathbb{C}^n, A)$.

Proof. Replacing A by $\Phi(A)$ for a suitable $\Phi \in \mathrm{Aut}\,(\mathbb{C}^n)$ we can assume that $\mathbb{P}^{n-1} \backslash \omega(A) \neq \emptyset$. Since this set is open, we can choose a basis $v_1, \ldots, v_n$ of $\mathbb{C}^n$ such that $[v_j] \notin \omega(A)$ for $j = 1, \ldots, n$. Let $\pi_j \colon \mathbb{C}^n \to \mathbb{C}^{n-1}$ be a surjective linear map such that $\pi_j(v_j) = 0$. Lemma 4.11.5 shows that for every point $p \in \mathbb{C}^n \backslash A$ the vectors v_j can be chosen such that for all points q sufficiently near p, none of the affine lines $q + \mathbb{C}v_j$ intersect the set A. Choose a function $f_j \in \mathcal{O}(\mathbb{C}^{n-1})$ that vanishes on $\pi_j(A)$ but is nonzero at $\pi_j(p)$. The shears (4.46), generated by the vector fields $V_j = (f_j \circ \pi_j)\, v_j$ $(j = 1, \ldots, n)$, then act transitively in a neighborhood of p, i.e., we can map p to any nearby point by their compositions.

Consider now the problem of moving p to q for an arbitrary pair of points $p, q \in \mathbb{C}^n \backslash A$. Choose $v_1, \ldots, v_n$ as above, write $q - p = \sum_{j=1}^n \lambda_j v_j$ $(\lambda_j \in \mathbb{C})$, and define

$$p_0 = p,\ p_1 = p_0 + \lambda_1 v_1,\ p_2 = p_1 + \lambda_2 v_2, \ldots,\ p_n = p_{n-1} + \lambda_n v_n = q.$$

We wish to move p_{j-1} to p_j by a shear in the direction v_j. The trouble is that the affine line through these two points may intersect A. To avoid this problem, choose small pairwise disjoint balls $U_j \ni p_j$ for $j = 0, 1, \ldots, n$ such that for every pair of points $a \in U_{j-1},\ b \in U_j$ we have $[a - b] \neq \omega(A)$ (this is possible since $\omega(A)$ is a closed set and $[p_{j-1} - p_j] = [v_j] \neq \omega(A)$.) By Lemma 4.11.5 we can move $p = p_0$ to some point $p_1' \in U_1$ by a shear (4.46) that is fixed on A. By the same argument we can move p_1' to some point $p_2' \in U_2$ by a shear that is fixed on A. Continuing in this way, we move p to some point $p_n' \in U_n$ by a composition of n shears. Assuming that U_n is a sufficiently small neighborhood of $p_n = q$, we can move p_n' to q by the first part of the proof. This proves that $\mathbb{C}^n \backslash A$ is $\mathrm{Aut}\,(\mathbb{C}^n, A)$-homogeneous. If A is algebraic then all shears in the proof can be chosen polynomial.

Fix a point $p \in \mathbb{C}^n \backslash A$. Choose a vector $v \neq 0$ with $[v] \notin \omega(A)$, and then choose a projection $\pi \colon \mathbb{C}^n \to \mathbb{C}^{n-1}$ with $\ker \pi = \mathbb{C}v$ such that $\pi(p) \notin \pi(A)$. Choose $f \in \mathcal{O}(\mathbb{C}^{n-1})$ that vanishes on $\pi(A)$ and has a simple zero at $\pi(p)$. Then p is a fixed point of the $\mathbb{C}_+$ action $(t, z) \mapsto z + tf(\pi(z))v$. By Lemma 4.10.19 (p. 136) the induced $\mathbb{C}_+$ action on $\mathrm{T}_p\mathbb{C}^n$ is given by $w \mapsto w + \mathrm{d}f_p(w)\, v$. This shows that the isotropy group $\mathrm{Aut}\,(\mathbb{C}^n, A)_p$ induces an irreducible representation on $\mathrm{T}_p\mathbb{C}^n$ which gives the second statement. $\square$

The following result will be used in Chapters 5 and 6; it says that, under the stated conditions, $\mathbb{C}^n \backslash A$ is an (algebraically) elliptic manifold.

Proposition 4.11.7. *If $A \subset \mathbb{C}^n$ is a tame analytic subvariety of codimension at least two, then there exist finitely many holomorphic vector fields on $\mathbb{C}^n$ that vanish on A and that span the tangent space $\mathrm{T}_z\mathbb{C}^n$ at every point $z \in \mathbb{C}^n \backslash A$. If A is algebraic, then this holds with algebraic vector fields.*

Proof. Choose a basis $v_1, \ldots, v_n$ of $\mathbb{C}^n$, linear projections π_j, and functions f_j as in the proof of Proposition 4.11.6. The corresponding complete (shear) vector fields $V_j = (f_j \circ \pi_j)v_j$ then span $\mathrm{T}_z\mathbb{C}^n$ at every point outside the proper complex subvariety $A' = \cup_{j=1}^n \pi_j^{-1}(\pi_j(A)) \supset A$ of $\mathbb{C}^n$. Choosing a point $p \in A' \backslash A$ and adding to the previous collection finitely many shears of the same form that span $\mathrm{T}_p\mathbb{C}^n$, we reduce A' to a smaller subvariety containing A. In the algebraic case we can complete the proof in finitely many steps by reducing the dimension of $A' \backslash A$ at every step. In the analytic case the variety $A' \backslash A$ may have countably many connected components, and a more precise argument is needed to show that the dimension can be reduced at each step. The main point is that, due to its special form, the subvariety A' still has sufficiently many proper projections π to $\mathbb{C}^{n-1}$, and any discrete set of points in A' projects to a discrete set in $\mathbb{C}^{n-1}$ under such a projection. It is then possible to choose the coefficient function $f \in \mathcal{O}(\mathbb{C}^{n-1})$ in the corresponding shear $(f \circ \pi)v$ such that f vanishes on $\pi(A)$ but does not vanish at those projected points that belong to $\mathbb{C}^{n-1} \backslash \pi(A)$. $\square$

Another evidence that a tame subvariety of codimension ≥ 2 in $\mathbb{C}^n$ has a holomorphically 'large' complement is provided by the following results. (See also Corollaries 4.12.2 and 4.12.7 below.)

Proposition 4.11.8. *Let A be a tame subvariety of $\mathbb{C}^n$ of codimension at least two. For every point $p \in \mathbb{C}^n \backslash A$ there exists an injective holomorphic map $F\colon \mathbb{C}^n \to \mathbb{C}^n \backslash A$ (a Fatou-Bieberbach map) such that $F(0) = p$.*

Proof. We may assume that $p = 0$. By Lemma 4.11.5 there is a basis $v_1, \ldots, v_n$ of $\mathbb{C}^n$ such that $[v_j] \notin \omega(A)$, and such that the complex line $\mathbb{C}v_j$ does not intersect A for $j = 1, \ldots, n$. After a linear change of coordinates we may assume that these properties hold for the standard basis $v_j = e_j$. Let $\pi_j\colon \mathbb{C}^n \to \mathbb{C}_j^{n-1}$ denote the projection onto the hyperplane $\mathbb{C}_j^{n-1} = \{z_j = 0\}$, so that $\pi_j(e_j) = 0$. By Lemma 4.11.2 the restricted projection $\pi_j|_A\colon A \to \mathbb{C}_j^{n-1}$ is proper, and hence $A_j = \pi_j(A)$ is a proper closed analytic subset of $\mathbb{C}_j^{n-1}$. By the construction we have $0 \notin A_j$ for $j = 1, \ldots, n$. Choose a holomorphic function g_j on $\mathbb{C}_j^{n-1}$ such that $g_j(0) = -\log 2$ and $g_j = 0$ on A_j, and set

$$\Phi_j(z) = \left(z_1, \ldots, z_{j-1}, \mathrm{e}^{g_j(\pi_j(z))} z_j, z_{j+1}, \ldots, z_n\right).$$

Then $\Phi = \Phi_1 \circ \Phi_2 \circ \cdots \circ \Phi_n \in \operatorname{Aut}\mathbb{C}^n$, $\Phi|_A$ is the identity, $\Phi(0) = 0$, and $\Phi'(0) = \frac{1}{2}I$. Thus $0 \in \mathbb{C}^n$ is an attracting fixed point of Φ whose basin $\Omega \subset \mathbb{C}^n \backslash A$ is a Fatou-Bieberbach domain in view of Theorem 4.3.2 (p. 106). The corresponding Fatou-Bieberbach map $F\colon \mathbb{C}^n \to \Omega$ satisfies the conclusion. □

Remark 4.11.9. In the situation of Proposition 4.11.8 with A algebraic, Buzzard and Hubbard [68] constructed a Fatou-Bieberbach domain $\Omega \subset \mathbb{C}^n$ such that even the closure $\overline{\Omega}$ of Ω does not intersect A. □

The above results come close to, but still do not answer Problem 4.11.1 on p. 138. We now describe a partial answer in the algebraic case, due to Kaliman and Kutzschebauch [288], that enables us to approximate isotopies of biholomorphic maps in the complement of an algebraic subvariety A by holomorphic automorphisms of $\mathbb{C}^n$ that are fixed on A.

Given an ideal $I \subset \mathbb{C}^{[n]} = \mathbb{C}[z_1, \ldots, z_n]$, we denote by $\aleph_I(\mathbb{C}^n)$ the Lie algebra of vector fields $V = \sum_{j=1}^n a_j \frac{\partial}{\partial z_j}$ whose coefficient functions a_j belong to I. If I is the ideal of an algebraic subvariety $A \subset \mathbb{C}^n$, then $\aleph_I(\mathbb{C}^n) = \aleph_{\mathcal{A}}(\mathbb{C}^n, A)$. Let $\mathrm{Lie}_{\mathrm{alg}}(\mathbb{C}^n, A)$ denote the Lie subalgebra of $\aleph_{\mathcal{A}}(\mathbb{C}^n, A)$ generated by the complete algebraic vector fields on $\mathbb{C}^n$ that vanish on A.

Theorem 4.11.10. [288, Theorem 4] *Let A be a closed algebraic subvariety of codimension at least two in $\mathbb{C}^n$ and $I \subset \mathbb{C}^{[n]}$ the ideal of functions vanishing on A. There is an ideal $L \subset \mathbb{C}^{[n]}$ with radical $\sqrt{L} = I$ such that $\aleph_L(\mathbb{C}^n) \subset \mathrm{Lie}_{\mathrm{alg}}(\mathbb{C}^n, A)$. That is, every vector field with coefficients in L is generated by complete algebraic vector fields that vanish on A.*

Proof. Let $z = (z_1, \ldots, z_n)$ be coordinates on $\mathbb{C}^n$ and $\pi_i\colon \mathbb{C}^n \to \mathbb{C}^{n-1}$ the projection onto the hyperplane $\mathbb{C}_i^{n-1} = \{z_i = 0\}$. As in the proof of Proposition 4.11.8 we may assume that $\pi_i|_A$ is proper for every $i = 1, \ldots, n$.

Let $V_i = \frac{\partial}{\partial z_i}$. Given functions $f_i, h_i \in \ker V_i$ (i.e., independent of z_i) and such that h_i vanishes on $\pi_i(A)$, the algebraic vector field $f_i h_i V_i$ (a shear) is $\mathbb{C}$-complete and it vanishes on A; hence it generates a $\mathbb{C}_+$ action on $\mathbb{C}^n \backslash A$ by elements of the group $\operatorname{Aut}(\mathbb{C}^n, A)$). The vector field

$$\big[f_1 h_1 V_1, z_1 f_2 h_2 V_2\big] - \big[z_1 f_1 h_1 V_1, f_2 h_2 V_2\big] = f_1 f_2 h_1 h_2 V_2$$

then belongs to $\mathrm{Lie}_{\mathrm{alg}}(\mathbb{C}^n, A)$. Since $\ker V_1 \cdot \ker V_2$ spans $\mathbb{C}^{[n]}$ as a vector space, we see that $\mathrm{Lie}_{\mathrm{alg}}(\mathbb{C}^n, A)$ contains all algebraic fields proportional to $V_2 = \frac{\partial}{\partial z_2}$ whose coefficients belong to the principal ideal $(h_1 h_2)$ generated by $h_1 h_2$. (Note that A is contained in the zero locus of $(h_1 h_2)$.) Since one can perturb z_2 (as a linear function), we infer that $\mathrm{Lie}_{\mathrm{alg}}(\mathbb{C}^n, A)$ contains all algebraic vector fields whose coordinates are in some nonzero ideal L of $\mathbb{C}^{[n]}$. Since $\mathbb{C}^n \backslash A$ is $\operatorname{Aut}(\mathbb{C}^n, A)$-homogenous, arguing as in the proof of Theorem 4.10.14 (moving the point and the projection direction) one can arrange that the zero locus of L equals A. By the Nullstellensatz this means that $\sqrt{L} = I$. □

Since the polynomial ring $\mathbb{C}^{[n]}$ is Noetherian, a simple argument shows that for any ideal $L \subset \mathbb{C}^{[n]}$ with radical $\sqrt{L} = I$ there is an integer $k \in \mathbb{N}$ such that $I^k \subset L$. Hence Theorem 4.11.10 implies the following.

Corollary 4.11.11. *If A is an algebraic subvariety of $\mathbb{C}^n$ with $\dim A \le n-2$ then every polynomial vector field V vanishing on A to a sufficiently high order is a Lie combination of complete polynomial vector fields vanishing on A.*

Although this suffices for the approximation purposes (see Theorem 4.12.1 on p. 143 below), Kaliman and Kutzschebauch also gave the following more precise result that establishes the algebraic density property of the Lie algebra $\aleph_{\mathcal{A}}(\mathbb{C}^n, A)$ for sufficiently nice algebraic subvarieties A of $\mathbb{C}^n$.

Theorem 4.11.12. [288, Theorem 6] *Let A be an algebraic subvariety of $\mathbb{C}^n$ of codimension at least two such that the Zariski tangent space $\mathrm{T}_z A$ (the intersection of kernels of differentials $\mathrm{d}f_z$ of all polynomials $f \in \mathbb{C}[z]$ that vanish on A) has dimension at most $n-1$ for any point $z \in A$. Then $\mathrm{Lie}_{\mathrm{alg}}(\mathbb{C}^n, A) = \aleph_{\mathcal{A}}(\mathbb{C}^n, A)$; that is, the Lie algebra $\aleph_{\mathcal{A}}(\mathbb{C}^n, A)$ of polynomial vector fields on $\mathbb{C}^n$ that vanish on A has the algebraic density property.*

4.12 Moving Polynomially Convex Sets

We consider the problem of moving polynomially convex sets in $\mathbb{C}^n$ by holomorphic automorphisms. The following result is an analogue of Theorem 4.9.2 for compact sets.

Theorem 4.12.1. [193, Theorem 2.1] *Let Ω be an open subset of $\mathbb{C}^n$ $(n > 1)$ and $\Phi_t \colon \Omega \to \mathbb{C}^n$ $(t \in [0,1])$ a $\mathcal{C}^1$-isotopy of injective holomorphic maps such that $\Phi_0 = \mathrm{Id}$. Assume that K is a compact set in Ω such that $K_t = \Phi_t(K)$ is polynomially convex for every $t \in [0,1]$. Then the map Φ_1 can be approximated uniformly on K by holomorphic automorphisms of $\mathbb{C}^n$. Furthermore, if A is an algebraic subvariety of $\mathbb{C}^n$ with $\dim A \le n-2$ and if $\Phi_t(K) \cap A = \emptyset$ for all $t \in [0,1]$, then Φ_1 can be approximated uniformly on K by holomorphic automorphisms belonging to the group $\mathrm{Aut}\,(\mathbb{C}^n, A)$ (4.45) (i.e., fixing A pointwise). The approximating automorphism can be chosen to agree with Φ_1 at finitely many points of K.*

Proof. Since a compact polynomially convex set admits a basis of Stein neighborhoods that are Runge in $\mathbb{C}^n$, the proof of Theorem 4.9.2 (p. 125) easily adapts to give the first statement. (This is [193, Theorem 2.1].)

To get the second part, observe that a holomorphic function on a neighborhood of a compact polynomially convex set K in the complement of an affine algebraic subvariety $A \subset \mathbb{C}^n$ can be approximated, uniformly on K,

by polynomials that vanish to any chosen order on A. (Choose polynomials $h_1, \dots, h_m$ whose common zero set equals A and let $\ell \in \mathbb{N}$. Since $A \cap K = \emptyset$ and K is polynomially convex, we can write the given function near K as $f = \sum_{j=1}^m f_j h_j^\ell$ by applying Cartan's division theorem, where f_j are holomorphic functions near K. It remains to approximate f_j by holomorphic polynomials.) The same holds for vector fields by applying this to each coefficient. By Corollary 4.11.11 (p. 143) every polynomial vector field on $\mathbb{C}^n$ that vanishes to a sufficiently high order on A is a Lie combination of complete algebraic vector fields vanishing A. Hence the proof of Theorem 4.9.2 (p. 125) gives approximation by automorphisms that restrict to the identity on A. $\square$

Using the second part of Theorem 4.12.1 (instead of Theorem 4.9.2) we can strengthen many results in this and subsequent sections; we leave it to the reader to formulate these stronger results and to fully explore their consequences. Here is an example of this type.

Corollary 4.12.2. *Let $A \subset \mathbb{C}^n$ be an algebraic subvariety with $\dim A \le n-2$. Given a compact convex set $K \subset \mathbb{C}^n \backslash A$ and $\epsilon > 0$, there is an injective holomorphic map $F\colon \mathbb{C}^n \to \mathbb{C}^n$ with $F(\mathbb{C}^n) \cap A = \emptyset$ and $\sup_{z\in K} |F(z) - z| < \epsilon$.*

Proof. We may assume that $0 \in K$. Consider the isotopy $\Phi_t(z) = tz$ for z near K and $t \in (0,1]$. By Theorem 4.12.1 there exists an automorphism $\Phi \in \operatorname{Aut}(\mathbb{C}^n, A)$ that approximates the dilation $z \mapsto \Phi_{\frac{1}{2}}(z) = \frac{1}{2}z$ uniformly on K. We can also insure that $\Phi(0) = 0$. Then 0 is an attracting fixed point of Φ whose basin Ω satisfies $K \subset \Omega \subset \mathbb{C}^n \backslash A$. Let $G\colon \mathbb{C}^n \to \Omega$ be a Fatou-Bieberbach map onto Ω furnished by Theorem 4.3.2 (p. 106). Since G is a limit of holomorphic automorphisms of $\mathbb{C}^n$, the set $L = G^{-1}(K)$ is polynomially convex. Since G^{-1} maps the convex set K biholomorphically onto a polynomially convex set L, we can approximate G uniformly on a neighborhood of K by an automorphism $H \in \operatorname{Aut}\mathbb{C}^n$. Then $F = G \circ H$ maps $\mathbb{C}^n$ biholomorphically onto Ω and it approximates the identity map on K. $\square$

Applying Theorem 4.9.9 (p. 129) instead of Theorem 4.9.2 gives the following parametric version of Theorem 4.12.1.

Theorem 4.12.3. *Let Ω be an open set in $\mathbb{C}^n = \mathbb{C}^k \times \mathbb{C}^m$ $(m > 1)$, and let $\Phi_t\colon \Omega \to \mathbb{C}^n$ be a $\mathcal{C}^1$-isotopy of biholomorphic maps as in Theorem 4.9.9 such that Φ_0 is the identity map. Let $K \subset \Omega$ be a compact polynomially convex subset of $\mathbb{C}^n$. If $\Phi_t(K)$ is polynomially convex for every $t \in [0,1]$, then for every $t \in [0,1]$ the map Φ_t can be approximated uniformly on K by holomorphic automorphisms of $\mathbb{C}^n$ of the form (4.40) (p. 129).*

The next corollary shows that, under suitable conditions, we can simultaneously approximate a collection of independent holomorphic motions of compact sets by holomorphic automorphism of $\mathbb{C}^n$.

Corollary 4.12.4. *Let $K_1, K_2, \ldots, K_m$ be pairwise disjoint, compact, polynomially convex sets in $\mathbb{C}^n$ such that all except one of them are starshaped. Let $\Phi^j \in \operatorname{Aut}\mathbb{C}^n$ $(j = 1, \ldots, m)$ be such that the images $K'_j = \Phi^j(K_j)$ are also pairwise disjoint. If $K = \cup_{j=1}^m K_j$ and $K' = \cup_{j=1}^m K'_j$ are polynomially convex, then for every $\epsilon > 0$ there exists an automorphism $\Phi \in \operatorname{Aut}\mathbb{C}^n$ such that*

$$|\Phi(z) - \Phi^j(z)| < \epsilon, \qquad z \in K_j,\ j = 1, \ldots, m.$$

Proof. Assume that each of the sets $K_2, \ldots, K_m$ is starshaped with respect to a point $a_j \in K_j$. In view of Theorem 4.12.1 it suffices to find for each $j = 1, 2, \ldots, m$ an isotopy of biholomorphic maps Φ_t^j $(t \in [0,1])$, defined on a neighborhood of K_j in $\mathbb{C}^n$, such that Φ_0^j is the identity map, $\Phi_1^j = \Phi^j$, and for each $t \in [0,1]$ the sets $K_{j,t} = \Phi_t^j(K_j)$ $(j = 1, \ldots, m)$ are pairwise disjoint and their union $\cup_{j=1}^m K_{j,t}$ is polynomially convex.

Lemma 4.9.6 (p. 128) gives for each j a smooth isotopy of automorphisms $\Psi_t^j \in \operatorname{Aut}\mathbb{C}^n$ $(t \in [0,1])$ such that $\Psi_0^j = \mathrm{Id}$ and $\Psi_1^j = \Phi^j$. We take $\Phi_t^1 = \Psi_t^1$, but the other isotopies need to be modified.

We first replace Ψ_t^j by a smooth isotopy $(z,t) \mapsto \widetilde{\Psi}_t^j(z) = \Psi_t^j(z) + \gamma_j(t)$, where the maps $\gamma_j \colon [0,1] \to \mathbb{C}^n$ are chosen such that $\gamma_j(0) = 0 = \gamma_j(1)$ and the points $b_{j,t} = \widetilde{\Psi}_t^j(a_j)$ for $j = 2, \ldots, m$ are pairwise distinct and contained in $\mathbb{C}^n \backslash \Phi_t^1(K_1)$. In the sequel we drop the tilde's.

Note that the union of a compact polynomially convex set and a finite set of points is still polynomially convex. The same holds if we replace points by small closed balls around them. Hence we can find a small enough $\delta > 0$ such that the closed balls $\overline{\mathbb{B}}(b_{j,t}, \delta)$ $(j = 2, \ldots, m)$ and the set $\Psi_t^1(K_1)$ are pairwise disjoint, and their union is polynomially convex for every $t \in [0,1]$.

Choose a number $\eta > 0$ such that Ψ_t^j maps the ball $\overline{\mathbb{B}}(a_j, \eta)$ into the ball $\mathbb{B}(b_{j,t}, \delta)$ for every $t \in [0,1]$ and $j = 2, \ldots, m$. Let $R > 0$ be sufficiently large such that for every $j = 2, \ldots, m$ the set K_j is contained in the ball $\mathbb{B}(a_j, R)$. Set $\Theta_t^j(z) = a_j + t(z - a_j)$. An isotopy Φ_t^j with the required properties is now obtained by first squeezing K_j into $\mathbb{B}(a_j, \eta)$ by using the contractions Θ_t^j, then following the isotopy Ψ_t^j restricted to the ball $\mathbb{B}(a_j, \eta)$, and finally expanding the image of K_j onto K'_j. To give explicit formulas, we determine $c > 0$ by $1 - \frac{c}{3} = \frac{\eta}{R}$ (hence $\Theta^j(K_j) \subset \mathbb{B}(a_j, \eta)$), and we define the isotopy Φ_t^j on a small open neighborhood of K_j as follows:

$$\Phi_t^j = \begin{cases} \Theta_{1-ct}^j, & \text{if } 0 \le t \le \frac{1}{3}; \\ \Psi_{3t-1}^j \circ \Theta_{\eta/R}^j, & \text{if } \frac{1}{3} < t \le \frac{2}{3}; \\ \Phi^j \circ \Theta_{1+c(t-1)}^j, & \text{if } \frac{2}{3} < t \le 1. \end{cases}$$

For each $t \in [0,1]$ the set $\cup_{j=1}^m \Phi_t^j(K_j)$ is polynomially convex: For $0 \le t \le \frac{1}{3}$ this holds since $\Phi_t^j(K_j) \subset K_j$ $(2 \le j \le m)$, and similarly for $\frac{2}{3} \le t \le 1$ we have $\Phi_t^j(K_j) \subset K'_j$. (Here we need that the sets $K_2, \ldots, K_m$ are starshaped.)

On $\frac{1}{3} \leq t \leq \frac{2}{3}$ we have $\Phi_t^j(K_j) \subset \overline{\mathbb{B}}(b_{j,t}, \delta)$, and the union of these balls together with $\Phi_t^1(K_1)$ is polynomially convex. By a reparametrization of the interval $[0,1]$ we obtain a smooth isotopy with the same properties. □

Remark 4.12.5. The union of two disjoint compact convex sets in $\mathbb{C}^n$ is polynomially convex and hence Corollary 4.12.4 applies. The union of three disjoint compact convex sets in general fails to be polynomially convex [294, 417], unless we are talking of three disjoint balls [294, 470] or three disjoint polydiscs in $\mathbb{C}^2$ with sides parallel to the coordinate axes [417]. □

Corollary 4.12.6. *Let K be a compact polynomially convex set in $\mathbb{C}^n$, and let $p_1, \ldots, p_m$ and $q_1, \ldots, q_m$ be m-tuples of pairwise distinct points in $\mathbb{C}^n \backslash K$. Given $\epsilon > 0$, there exists a holomorphic automorphism $\Phi \in \operatorname{Aut} \mathbb{C}^n$ such that*

$$|\Phi(z) - z| < \epsilon \ (\forall z \in K), \qquad \Phi(p_j) = q_j \ (j = 1, \ldots, m).$$

Using Corollary 4.12.6 inductively, the push-out method (Corollary 4.4.2 on p. 115) gives the following consequence.

Corollary 4.12.7. *Let K be a compact polynomially convex set in $\mathbb{C}^n$ $(n > 1)$. Given a finite set $A \subset \mathbb{C}^n \backslash K$ and a number $\epsilon > 0$, there is a biholomorphic map F of $\mathbb{C}^n$ onto a subset $\Omega \subset \mathbb{C}^n \backslash A$ such that $\sup_{z \in K} |F(z) - z| < \epsilon$.*

Note that the image $\Omega = F(\mathbb{C}^n)$ of a map F in Corollary 4.12.7 is a Fatou-Bieberbach domain in $\mathbb{C}^n$ such that $K \subset \Omega \subset \mathbb{C}^n \backslash A$.

4.13 Moving Totally Real Submanifolds

The Andersén-Lempert theory, combined with the $\overline{\partial}$-methods, gives results on approximation of isotopies of polynomially convex totally real submanifolds of $\mathbb{C}^n$ by holomorphic automorphisms of $\mathbb{C}^n$ [188, 189, 193]. The most precise result without any loss of derivatives is the following one.

Theorem 4.13.1. [189, Theorem 1.3] *Let $n, r \geq 2$ be integers, and let M be a compact smooth manifold. Given a $\mathcal{C}^r$-isotopy of embeddings $f_t \colon M \hookrightarrow \mathbb{C}^n$ $(t \in [0,1])$ such that the submanifold $f_t(M) \subset \mathbb{C}^n$ is totally real and polynomially convex for each $t \in [0,1]$, there exists a sequence of automorphisms $\Phi_j \in \operatorname{Aut} \mathbb{C}^n$ $(j = 1, 2, \ldots)$ such that*

$$\lim_{j \to \infty} ||\Phi_j \circ f_0 - f_1||_{\mathcal{C}^r(M)} = 0, \qquad \lim_{j \to \infty} ||\Phi_j^{-1} \circ f_1 - f_0||_{\mathcal{C}^r(M)} = 0.$$

The proof is essentially the same as that of Theorem 4.13.2 below, the only difference being that one solves the $\overline{\partial}$-problem with the integral kernel method from [189] (see Theorem 2.5.4 on p. 56). The following result which relies on the simpler L^2 method suffices for the applications in this book.

Theorem 4.13.2. [188, Theorem 2.1] *Let $n \geq 2$ and $r \geq 3$. Let $K \subset \mathbb{C}^n$ be a compact polynomially convex set and let $U \subset \mathbb{C}^n$ be an open set containing K. For each $t \in [0,1]$ let $M_t \subset \mathbb{C}^n$ be a compact totally real submanifold of class $\mathcal{C}^r$, and let $f_t \colon M_0 \to M_t$ for $t \in [0,1]$ be a $\mathcal{C}^r$ diffeomorphism satisfying the following conditions:*

(i) *$K \cup M_t$ is polynomially convex for all $0 \leq t \leq 1$,*
(ii) *$f_0(z) = z$ for all $z \in M_0$,*
(iii) *$f_t(z) = z$ for all $z \in M_0 \cap U$ and $t \in [0,1]$, and*
(iv) *f_t and $\partial f_t/\partial t$ are of class $\mathcal{C}^r$ in $(t,z) \in [0,1] \times M_0$.*

Let $m = \dim M$ and choose an integer k with $1 \leq k \leq r - \frac{m+3}{2}$. Given $\epsilon > 0$ there is an automorphism $\Phi \in \operatorname{Aut} \mathbb{C}^n$ satisfying

(a) $||\Phi|_{M_0} - f_1||_{\mathcal{C}^k(M_0)} < \epsilon$,
(b) $||\Phi^{-1}|_{M_1} - f_1^{-1}||_{\mathcal{C}^k(M_1)} < \epsilon$,
(c) $||\Phi - \mathrm{Id}||_{\mathcal{C}^k(K)} < \epsilon$.

If E is a finite subset of $K \cup M_0$, then we can choose Φ such that, in addition to the above, it agrees with the identity to order k at each point of $E \cap K$, and $\Phi|_{M_0}$ agrees with f_1 to order k at each point of $E \cap M_0$.

In part (c), $\mathcal{C}^k(K)$ denotes the k-jet norm on a compact set $K \subset \mathbb{C}^n$:

$$||f||_{\mathcal{C}^k(K)} = \max\big\{|\partial^\alpha f(z)| \colon z \in K,\ |\alpha| \leq k\big\},$$

where $\alpha \in \mathbb{Z}_+^{2n}$ and ∂^α denotes the corresponding partial derivative.

Proof. We think of the isotopy $\{f_t\}$ as the flow of a time dependent vector field. We extend t to a complex variable. Thus, let $\mathbb{C}^{n+1} = \mathbb{C} \times \mathbb{C}^n$ be the extended phase space with complex coordinates $w = (t,z)$, with $t \in \mathbb{C}$ and $z \in \mathbb{C}^n$. We define the following sets in $\mathbb{C}^{n+1}$:

$$L_0 = [0,1] \times K, \quad S = \bigcup_{0 \leq t \leq 1} \{t\} \times M_t, \quad L = L_0 \cup S.$$

Since $K \cup M_t$ is polynomially convex for each $t \in [0,1]$, L is also polynomially convex. Clearly $S = L \backslash L_0$ is a totally real submanifold with boundary of class $\mathcal{C}^r$. We extend each f_t to $M_0 \cup U$ as the identity map on U. For each $t \in [0,1]$ let $V_t \colon M_t \cup U \to \mathbb{C}^n$ be the velocity vector field of f_t, defined by the equation

$$\dot{f}_t(z) = V_t\big(f_t(z)\big), \quad z \in M_0 \cup U,\ t \in [0,1].$$

The map $V(t,z) = V_t(z)$ with values in $\mathbb{C}^n$ is defined on $S \cup ([0,1] \times U) \subset \mathbb{C}^{n+1}$. Property (iii) of $\{f_t\}$ implies that V vanishes on $[0,1] \times U$, and the restriction $V|_S$ is of class $\mathcal{C}^r(S)$. After shrinking U slightly around K we can extend V to

a $\mathcal{C}^r$ map $V\colon \mathbb{C}^{n+1} \to \mathbb{C}^n$ with compact support that vanishes on $\mathbb{C}\times U \subset \mathbb{C}^{n+1}$ and satisfies the estimates

$$|\partial^\alpha \overline{\partial} V(w)| = o\left(\operatorname{dist}(w,S)^{r-1-|\alpha|}\right), \quad 0 \le |\alpha| \le r-1.$$

The derivative ∂^α (with respect to the real coordinates) is applied to each component of the $(0,1)$-form $\overline{\partial} V$, and $|\partial^\alpha \overline{\partial} V|$ denotes the Euclidean norm.

The following lemma follows immediately from Hörmander's L^2-method (see Theorem 2.5.1 on p. 55) and the interior regularity estimates (Lemma 2.5.2 on p. 55). Let $L(\epsilon)$ denote the ϵ-neighborhood of L.

Lemma 4.13.3. [188, Proposition 2.3] *Let $r \in \mathbb{Z}_+$, $r \ge \frac{m+5}{2}$, and $k = r - \frac{m+3}{2} \ge 1$. For each sufficiently small $\epsilon > 0$ there exists an entire map $V^\epsilon\colon \mathbb{C}^{n+1} \to \mathbb{C}^n$ such that for all multiindexes $\alpha \in \mathbb{Z}_+^{2n+2}$ with $|\alpha| \le k$ we have*

$$||\partial^\alpha (V - V^\epsilon)||_{L^\infty(L(\epsilon))} = o(\epsilon^{k-|\alpha|}), \quad \epsilon \to 0.$$

Let ψ_t^ϵ be the flow of the time dependent holomorphic vector field V^ϵ:

$$\dot{\psi}_t^\epsilon(z) = V^\epsilon\big(t, \psi_t^\epsilon(z)\big), \quad \psi_0^\epsilon(z) = z.$$

From Lemma 4.13.3 and standard results on perturbations of flows (see Lemma 1.9.4 on p. 33) we obtain the following.

Proposition 4.13.4. *There are a neighborhood U_0 of $K \cup M_0$ and an $\epsilon_0 > 0$ such that $\psi_t^\epsilon(z)$ exists for all $z \in U_0$, $t \in [0,1]$ and $0 < \epsilon < \epsilon_0$, and it satisfies*

$$\lim_{\epsilon\to 0} ||f_t - \psi_t^\epsilon||_{\mathcal{C}^k(K\cup M_0)} = 0, \qquad \lim_{\epsilon\to 0} ||(f_t)^{-1} - (\psi_t^\epsilon)^{-1}||_{\mathcal{C}^k(K\cup M_t)} = 0.$$

Theorem 4.13.2 follows by combining Proposition 4.13.4 (for $t = 1$) and Proposition 4.9.4 (p. 127); the latter one says that for each fixed $\epsilon > 0$, ψ_1^ϵ is a uniform limit of automorphisms of $\mathbb{C}^n$ in a neighborhood of $K \cup M_0$. The requirement on matching f at finitely many points is a trivial addition. □

Corollary 4.13.5. *Let K be a compact polynomially convex set in $\mathbb{C}^n$ for $n > 1$, and let $C \subset \mathbb{C}^n$ be a smooth embedded arc attached to K in a single point of K. Given numbers $r \in \mathbb{Z}_+$, $\epsilon > 0$ and a smooth diffeomorphism $F\colon C \to C' \subset \mathbb{C}^n$ which is the identity on $C \cap U$ for some open neighborhood U of K (Fig. 4.1), there exist a neighborhood W of K and an automorphism $\Phi \in \operatorname{Aut}\mathbb{C}^n$ satisfying*

$$||\Phi - \mathrm{Id}||_{\mathcal{C}^r(W)} < \epsilon, \quad ||\Phi - F||_{\mathcal{C}^r(C)} < \epsilon.$$

Note that the diffeomorphism F in Corollary 4.13.5 can be connected to the identity map on C by an isotopy of diffeomorphisms $F_t\colon C \to C_t$ ($t \in [0,1]$) satisfying the hypotheses of Theorem 4.13.2. The union $K \cup C$ of a compact polynomially convex set K and a smooth embedded arc C, attached to K at a single point or disjoint from K, is polynomially convex [467].

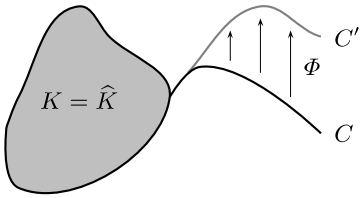

Fig. 4.1. Moving an arc

4.14 Controlling Unbounded Curves

In this section we show how to control holomorphic automorphisms of $\mathbb{C}^n$ on certain unbounded arcs. For simplicity we work with $\mathcal{C}^\infty$ arcs (this restriction is inessential for our applications), although much less smoothness is actually needed. The main idea first appeared in the proof of following result.

Theorem 4.14.1. [66, Theorem 1.1] *Let $n > 1$ and $r \geq 0$ be integers. Given a proper smooth embedding $\lambda\colon \mathbb{R} \hookrightarrow \mathbb{C}^n$ and a continuous function $\eta\colon \mathbb{R} \to (0,\infty)$, there is a proper holomorphic embedding $f\colon \mathbb{C} \hookrightarrow \mathbb{C}^n$ such that*

$$\left|f^{(s)}(t) - \lambda^{(s)}(t)\right| < \eta(t), \quad t \in \mathbb{R}, \quad s = 0, 1, \ldots, r.$$

The proof uses the push-out method from §4.4 with a key addition that insures the control of unbounded arcs at infinity. This method was further developed and fully exploited by Wold [523, 524], and it was also used in [197]. The following notion, introduced in a similar form in [523], is the key geometric assumption in subsequent constructions.

Definition 4.14.2. *A family $\ell = \cup_{i=1}^m \ell_i \subset \mathbb{C}^n$ of pairwise disjoint embedded smooth curves $\ell_i = \{\gamma_i(t)\colon t \in [0,\infty)$ or $t \in (-\infty,\infty)\}$ has the* nice projection property *if there is a linear projection $\pi\colon \mathbb{C}^n \to \mathbb{C}$ such that*

(i) $\lim_{|t|\to\infty} |\pi(\gamma_i(t))| = \infty$ *for $i = 1, 2, \ldots, m$, and*

(ii) *there is a number $r_0 > 0$ such π is injective on $\ell \backslash \pi^{-1}(\overline{\mathbb{D}}_{r_0})$.*

This essentially corresponds to what is called the *immediate nice projection property* in the papers [318, 351]; the nice projection property for them means that there is an automorphism $\alpha \in \operatorname{Aut}\mathbb{C}^n$ such that $\alpha(\ell)$ has the immediate nice projection property. In most sources it is additionally required that for all large $r > 0$ the set $\mathbb{C}\backslash(\pi(\ell) \cup \overline{\mathbb{D}}_r)$ has no bounded connected component. We replace the latter condition by the following lemma.

Lemma 4.14.3. *Assume that $\ell = \cup_{i=1}^m \ell_i \subset \mathbb{C}^n$ has the nice projection property with respect to a projection $\pi\colon \mathbb{C}^n \to \mathbb{C}$. Then for every $r > 0$ there exists $r' \geq r$ such that the polynomial hull of $\overline{\mathbb{D}}_r \cup \pi(\ell)$ is contained in $\overline{\mathbb{D}}_{r'} \cup \pi(\ell)$.*

Proof. We may assume that $r \geq r_0$, where r_0 satisfies property (ii) in Def. 4.14.2. Property (i) implies that each of the projected curves $\pi(\ell_i) \subset \mathbb{C}$ can leave and return back to the disc $\overline{\mathbb{D}}_r$ at most finitely many times, and by (ii) these curves do not intersect outside of $\overline{\mathbb{D}}_r$. It follows that the polynomial hull of $\overline{\mathbb{D}}_r \cup \pi(\ell)$ is the union of this set with at most finitely many (bounded) connected components of $\mathbb{C}\backslash(\overline{\mathbb{D}}_r \cup \pi(\ell))$. $\square$

The next result is [351, Theorem 9]; similar results were proved beforehand in [66, §3] and [523, Lemma 1].

Lemma 4.14.4. *Let K and L be compact polynomially convex sets in $\mathbb{C}^n$ $(n \geq 2)$. Assume that $\ell = \{\gamma(t) : t \in [0,\infty)\}$ is a smooth embedded curve in $\mathbb{C}^n$ such that $K \cap \ell = \{\gamma(0)\}$, and such that $\alpha(\ell)$ has the nice projection property for some $\alpha \in \operatorname{Aut}\mathbb{C}^n$. Set $C = \gamma([0,1])$, and let $F : C \to C'$ be a smooth diffeomorphism onto an arc $C' \subset \mathbb{C}^n$ such that $F|_{C\cap U} = \mathrm{Id}$ for some neighborhood U of K and $F(\gamma(1)) \notin L$. Given numbers $\epsilon > 0$ and $s \in \mathbb{Z}_+$ there exists an automorphism $\Phi \in \operatorname{Aut}(\mathbb{C}^n)$ satisfying*

$$||\Phi - \mathrm{Id}||_{\mathcal{C}^0(K)} < \epsilon, \quad ||\Phi - F||_{\mathcal{C}^s(C)} < \epsilon, \quad \Phi(\ell\backslash C) \cap L = \emptyset. \tag{4.49}$$

The analogous result holds if $\ell = \cup_{i=1}^m \ell_i$ is the union of finitely many curves.

Proof. We consider the case when ℓ is a single curve; the same proof applies to finitely many curves. It suffices to prove the result for $\alpha = \mathrm{Id}$; in the general case we obtain Φ satisfying (4.49) by taking $\Phi = \alpha^{-1} \circ \Phi' \circ \alpha$, where $\Phi' \in \operatorname{Aut}\mathbb{C}^n$ is chosen such that

- Φ' approximates the identity map on $\alpha(K)$,
- Φ' approximates the diffeomorphisms $F' = \alpha F \alpha^{-1}$ on the arc $\alpha(C)$, and
- $\Phi'\big(\alpha(\ell\backslash C)\big) \cap \alpha(L) = \emptyset$.

Thus we assume that ℓ has the nice projection property with respect to the first coordinate projection $\pi(z_1,\ldots,z_n) = z_1$. Let $r_0 > 0$ be as in Def. 4.14.2 (ii). Choose a number $r \geq r_0$ such that $\pi_1(K \cup C) \subset \mathbb{D}_r$, and let $r' \geq r$ be chosen such that the conclusion of Lemma 4.14.3 holds. Write $\gamma(t) = (\gamma_1(t), \gamma_2(t), \ldots)$. Choose a number $t_1 \geq 1$ such that $|\gamma_1(t)| > r'$ for all $t \geq t_1$ and $\gamma_1'(t_1) \neq 0$ (this holds for almost every large t_1 since $\lim_{t\to\infty} |\gamma_1(t)| = \infty$). Set $C_1 = \gamma([0,t_1])$. We extend the given diffeomorphism $F : C \to C'$ to a smooth diffeomorphism $F : C_1 \to C_1'$, where the arc C_1' is an extension of C' across the endpoint $F(\gamma(1)) \in \mathbb{C}^n\backslash L$ such that $C_1'\backslash C_1 \subset \mathbb{C}^n\backslash L$.

We seek $\Phi \in \operatorname{Aut}\mathbb{C}^n$ satisfying (4.49) as a composition $\Phi = \phi \circ \psi$ of two automorphisms. Corollary 4.13.5 furnishes $\phi \in \operatorname{Aut}\mathbb{C}^n$ such that

- ϕ is uniformly close to the identity on a neighborhood of K, and
- $\phi|_{C_1}$ approximates the diffeomorphism $F : C_1 \to C_1'$ in the $\mathcal{C}^s$ topology.

If the approximation is close enough then $\phi(C_1 \backslash C) \cap L = \emptyset$. However, the problem is that $\phi(\ell \backslash C_1) \cap L$ need not be empty, so the last property in (4.49) need not hold. We remove this intersection by precomposing ϕ by a suitably chosen shear of the form

$$\psi(z) = z + h(z_1)e_2 = (z_1, z_2 + h(z_1), \ldots), \quad h \in \mathcal{O}(\mathbb{C}). \tag{4.50}$$

Let $A = \phi^{-1}(L)$. It suffices to choose ψ such that

$$\psi \approx \mathrm{Id} \text{ on } \overline{\mathbb{D}}_r \cup \gamma_1([0, t_1]), \quad \psi(\gamma(t)) \notin A \text{ for all } t \geq t_1. \tag{4.51}$$

The first property holds if h is small on $\overline{\mathbb{D}}_r \cup \gamma_1([0, t_1])$. Observe that the polynomial hull of this set is contained in $\overline{\mathbb{D}}_{r'} \cup \gamma_1([0, t_1])$. We now explain how to achieve the second property. Choose a number $M > r'$ such that A is contained in the polydisc $\overline{\mathbb{D}}^n_M$ of radius M centered at the origin, and let $t' > t_1$ be such that $|\gamma_1(t)| > M$ for $t \geq t'$. Let $c_2 = \sup\{|\gamma_2(t)| : t \in [0, t']\}$. Choose $b \in \mathbb{C}$ such that $M + c_2 < |b|$. Let $q = \gamma(t_1) = (q_1, \ldots, q_n)$ and $q' = (q_1, b, q_3, \ldots)$; then $q, q' \notin A$. Consider the affine complex line $\Lambda = q + \mathbb{C}e_2 \cong \mathbb{C}$. The intersection $A \cap \Lambda$ is polynomially convex in Λ, and hence $\Lambda \backslash A$ is connected. Therefore the points q and q' can be connected in $\Lambda \backslash A$ by an arc $\{q + \lambda(s)e_2 : s \in [0, 1]\}$, with $\lambda(0) = 0$ and $\lambda(1) = b - q_2$. Choose a number $t_2 > t_1$ close enough to t_1 such that the arc $\gamma_1([t_1, t_2]) \subset \mathbb{C}$ is embedded (this is possible since $\gamma_1'(t_1) \neq 0$) and such that

$$\big\{\gamma(t) + \lambda(s)e_2 : t \in [t_1, t_2],\ s \in [0, 1]\big\} \subset \mathbb{C}^n \backslash A.$$

Choose a smooth increasing function $\chi : \mathbb{R}_+ \to [0, 1]$ such that $\chi(t) = 0$ for $t \leq t_1$ and $\chi(t) = 1$ for $t \geq t_2$. Each point $\zeta \in \gamma_1([t_1, t_2])$ equals $\zeta = \zeta(t)$ for a unique $t \in [t_1, t_2]$. Define a function $\widetilde{h} : \overline{\mathbb{D}}_{r'} \cup \pi_1(\ell) \to \mathbb{C}$ by setting

$$\widetilde{h}(\zeta) = \begin{cases} 0, & \text{if } \zeta \in \overline{\mathbb{D}}_{r'} \cup \gamma_1([0, t_1]), \\ \lambda(\chi(t)), & \text{if } \zeta = \zeta(t),\ t \in [t_1, t_2], \\ \lambda(1), & \text{if } \zeta \in \gamma_1([t_2, \infty)). \end{cases}$$

Then $\gamma(t) + \widetilde{h}(\gamma_1(t))e_2 \notin A$ for all $t \in [0, \infty)$. By Mergelyan's theorem we can approximate $\widetilde{h}$ uniformly on the polynomial hull of the set $\overline{\mathbb{D}}_r \cup \gamma([0, t'])$ (which is contained in $\overline{\mathbb{D}}_{r'} \cup \gamma([0, t'])$) by entire functions $h \in \mathcal{O}(\mathbb{C})$. If the approximation is close enough then the shear (4.50) satisfies the desired properties (4.51). The values of h on the unbounded arc $\gamma_1([t', \infty)) \subset \mathbb{C} \backslash \overline{\mathbb{D}}_M$ are unimportant since the set A projects to the disc $\mathbb{D}_M$. $\square$

Corollary 4.14.5. *Let K and L be compact polynomially convex sets in $\mathbb{C}^n$ ($n \geq 2$), and let $\ell = \{\gamma(t) : t \in [0, \infty)$ or $t \in (-\infty, \infty)\}$ be a smooth embedded curve in $\mathbb{C}^n$ such that $K \cap \ell = \emptyset$. Assume that there exist an automorphism $\alpha \in \mathrm{Aut}\,\mathbb{C}^n$ such that $\alpha(\ell)$ has the nice projection property (Def. 4.14.2). Given $\epsilon > 0$ there exists an automorphism $\Phi \in \mathrm{Aut}\,(\mathbb{C}^n)$ satisfying*

$$||\Phi - \mathrm{Id}||_{\mathcal{C}^0(K)} < \epsilon, \quad \Phi(\ell) \cap L = \emptyset.$$

The same result holds if ℓ is a finite union of smooth pairwise disjoint curves.

Proof. Assume that $\ell = \{\gamma(t): t \in \mathbb{R}\}$. Choose real numbers $t_1 < t_2$ such that $\gamma([t_1, t_2]) \cap L = \emptyset$. By a theorem of Stolzenberg [467] the set $K' = K \cup \gamma([t_1, t_2])$ is polynomially convex. It now suffices to apply Lemma 4.14.4 with K' in place of K and with the two arcs $\ell_1 = \gamma((-\infty, t_1])$ and $\ell_2 = \gamma([t_2, +\infty))$ attached to K'. Similarly we deal with the other cases. $\square$

Using Corollary 4.14.5 and the push-out method from §4.4 we now prove the following result of E. Wold [524]. This is used in §8.9 to construct proper holomorphic embeddings of certain bordered Riemann surfaces in $\mathbb{C}^2$.

Theorem 4.14.6. [524] *Assume that X is a complex curve in $\mathbb{C}^n$ $(n > 1)$ (possibly with singularities) with smooth boundary $bX = \cup_{j=1}^m \ell_j$ consisting of finitely many unbounded curves $\ell_j \cong \mathbb{R}$. If $\alpha(bX)$ has the nice projection property for some $\alpha \in \operatorname{Aut} \mathbb{C}^n$ then there exists a Fatou-Bieberbach domain $\Omega \subset \mathbb{C}^n$ such that $X \subset \Omega$ and $bX \subset b\Omega$. Furthermore, a Fatou-Bieberbach map $\Phi: \Omega \to \mathbb{C}^n$ can be chosen as closed as desired to the identity map on a given compact set in X. In particular, $\Phi|_X: X \hookrightarrow \mathbb{C}^n$ is a proper holomorphic embedding of X into $\mathbb{C}^n$.*

Proof. We claim that X enjoys the following properties (1)–(3) which are invariant under the composition by holomorphic automorphisms.

(1) *X admits an exhaustion $K_1 \subset K_2 \subset \cdots \subset \cup_{j=1}^\infty K_j = X$ by compact polynomially convex sets K_j.*

To see this, it suffices to show that any smoothly bounded compact set $K \subset X$ that is holomorphically convex in X is also polynomially convex in $\mathbb{C}^n$. Since $\widehat{K} = \widehat{bK}$ and bK is a union of smooth curves, the set $A = \widehat{K} \backslash bK$ is an analytic subvariety of $\mathbb{C}^n \backslash bK$ containing $K \backslash bK$ [467]. If $A \neq K \backslash bK$, then A contains a local extension of K in X near a boundary component of K; hence $\widehat{K}$ contains at least one connected component of $X \backslash K$, a contradiction since each of these components is unbounded in $\mathbb{C}^n$.

A similar argument shows that

(2) *For any compact polynomially convex set $K \subset \mathbb{C}^n \backslash bX$, the set $K \cup K_j$ is polynomially convex for all large $j \in \mathbb{N}$.* (In fact, this holds when $K \cap X \subset K_j$.)

(3) *For every compact polynomially convex set L contained in $\mathbb{C}^n \backslash bX$ and for every pair of numbers $\epsilon > 0$ (small) and $R > 0$ (large) there exists a holomorphic automorphism ϕ of $\mathbb{C}^n$ such that*

$$\sup_{x \in L} |\phi(x) - x| < \epsilon \quad \text{and} \quad \phi(bX) \subset \mathbb{C}^2 \backslash R\mathbb{B}.$$

(Here $\mathbb{B}$ is the unit ball in $\mathbb{C}^n$.) Such ϕ is furnished by Corollary 4.14.5.

Using these properties we find a sequence of holomorphic automorphisms

$$\Phi_j = \phi_j \circ \phi_{j-1} \circ \cdots \circ \phi_1 \in \operatorname{Aut} \mathbb{C}^n, \qquad j = 1, 2, \ldots \tag{4.52}$$

whose domain of convergence is a Fatou-Bieberbach domain Ω with the stated properties. The inductive step is the following. Fix $j \in \mathbb{N}$ and assume that $\Phi_j(bX) \cap j\overline{\mathbb{B}} = \emptyset$. (This trivially holds for $j = 0$ with $\Phi_0 = \mathrm{Id}$.) Choose $m_j \in \mathbb{N}$ large enough such that $L_j = j\overline{\mathbb{B}} \cup \Phi_j(K_{m_j})$ is polynomially convex (such m_j exists by property (2)). Given a number $\epsilon_j > 0$, property (3) furnishes an automorphism $\phi_{j+1} \in \mathrm{Aut}\,\mathbb{C}^n$ such that

- $|\phi_{j+1}(x) - x| < \epsilon_j$ for all $x \in L_j$, and
- $|\phi_{j+1}(x)| > j + 1$ for all $x \in \Phi_j(bX)$.

Setting $\Phi_{j+1} = \phi_{j+1} \circ \Phi_j$ completes the induction step. Suitable choices of sequences $\epsilon_j \to 0$ and $m_j \to +\infty$ insure that the sequence $\Phi_j \in \mathrm{Aut}\,\mathbb{C}^n$ converges locally uniformly on the domain $\Omega = \cup_{j=1}^{\infty} \Phi_j^{-1}(j\overline{\mathbb{B}}) \subset \mathbb{C}^n$ to a biholomorphic map $\Phi\colon \Omega \to \mathbb{C}^n$ onto $\mathbb{C}^n$ (see Corollary 4.4.2 on p. 115). By the construction we have $X \subset \Omega$ and $bX \subset b\Omega$. $\square$

4.15 Automorphisms with Given Jets

In this section we prove a jet interpolation theorem for automorphisms of $\mathbb{C}^n$ at finitely many points. This is a preparation for the Mittag-Leffler type interpolation theorems in the following section. For interpolation theorems on more general affine algebraic varieties we refer to the papers [25, 26].

Definition 4.15.1. *Let $m, n \in \mathbb{N}$.*

(i) *An A-jet of order m at $0 \in \mathbb{C}^n$ is a holomorphic polynomial $P\colon \mathbb{C}^n \to \mathbb{C}^n$ of degree $\leq m$ such that $P(0) = 0$ and $\mathrm{J}P(0) = \det P'(0) \neq 0$.*

(ii) *P as in (i) is an A_1-jet of order m if $\mathrm{J}P(z) = 1 + O(|z|^m)$ as $z \to 0$.*

Given an open set $U \subset \mathbb{C}^n$, a holomorphic map $F\colon U \to \mathbb{C}^n$, a point $a \in U$, and an integer $m \in \mathbb{N}$, we denote by $F_{m,a}$ the Taylor polynomial of F of order m at a without the constant term:

$$F(z) = F(a) + F_{m,a}(z - a) + O(|z - a|^{m+1}). \tag{4.53}$$

If F is nondegenerate at a (i.e., $\mathrm{J}F(a) \neq 0$), then $F_{m,a}$ is an A-jet, called *the A-jet of F of order m at a.* If $\mathrm{J}F \equiv 1$, then $F_{m,a}$ is an A_1-jet for each $a \in U$. The following lemma is evident by composing the power series of the respective mappings.

Lemma 4.15.2. *If F and G are holomorphic maps defined on open subsets of $\mathbb{C}^n$, with values in $\mathbb{C}^n$, then for each integer $m \in \mathbb{N}$ we have*

$$(G \circ F)_{m,a}(z) = G_{m,F(a)} \circ F_{m,a}(z) + O(|z|^{m+1}), \quad z \to 0$$

at each point a in the domain of $G \circ F$. If F is locally invertible at a, then

$$(F^{-1})_{m,F(a)} \circ F_{m,a}(z) = z + O(|z|^{m+1}), \quad z \to 0.$$

The following is the main result of this section (see [18, Proposition 6.2] and [168, Proposition 2.1]).

Proposition 4.15.3. *Let $n, N \in \mathbb{N}$, $n > 1$. Assume that*

(a) *K is a compact polynomially convex set in $\mathbb{C}^n$,*
(b) *$\{a_j\}_{j=1}^s$ is a finite set of points in K,*
(c) *p and q are points in $\mathbb{C}^n \backslash K$, and*
(d) *$P\colon \mathbb{C}^n \to \mathbb{C}^n$ is an A-jet of order m.*

Then for each $\epsilon > 0$ there exists an automorphism $\Phi \in \operatorname{Aut} \mathbb{C}^n$ satisfying the following properties:

(i) *$\Phi(p) = q$ and $\Phi(z) = q + P(z-p) + O(|z-p|^{m+1})$ as $z \to p$,*
(ii) *$\Phi(z) = z + O(|z-a_j|^N)$ as $z \to a_j$ for each $j = 1, 2, \ldots, s$, and*
(iii) *$|\Phi(z) - z| + |\Phi^{-1}(z) - z| < \epsilon$ for each $z \in K$.*

If in addition P is an A_1-jet of order m, then there exists a polynomial automorphism Φ with $J\Phi \equiv 1$ satisfying the properties (i)–(iii).

Proof. Set $K_0 = K$. Choose compact polynomially convex sets $K_1 \subset K_2 \subset K_3 \subset \mathbb{C}^n \backslash \{p, q\}$ and a number $\epsilon_0 > 0$ such that

$$K_0 \subset K_1, \quad \operatorname{dist}(K_j, \mathbb{C}^n \backslash K_{j+1}) > \epsilon_0, \quad j = 0, 1, 2.$$

Condition (iii) in Proposition 4.15.3 will be satisfied if $0 < \epsilon < \epsilon_0$ and

$$|\Phi(z) - z| < \frac{\epsilon}{2}, \qquad z \in K_1. \tag{4.54}$$

Indeed, if (4.54) holds then Rouché's theorem [85, p. 110] implies that $\Phi(K_1) \supset K$, and hence $\Phi^{-1}(z) \in K_1$ for each $z \in K_0$. Setting $w = \Phi^{-1}(z) \in K_1$, we have $|\Phi^{-1}(z) - z| = |w - \Phi(w)| < \frac{\epsilon}{2}$ by (4.54), and hence (iii) holds.

We seek $\Phi \in \operatorname{Aut} \mathbb{C}^n$ of the form

$$\Phi = H^{-1} \circ S \circ G \circ H \in \operatorname{Aut} \mathbb{C}^n. \tag{4.55}$$

Each of the automorphisms G, H, H^{-1}, and S will move points of K_2 less than $\frac{\epsilon}{8}$; clearly this will imply (4.54).

Choose a ball B containing the set K_3, and then choose points $p', q' \in \mathbb{C}^n$ belonging to an affine complex hyperplane Σ that does not intersect $\bar{B}$. (If $p = q$, we choose $p' = q'$.) By Corollary 4.12.6 (p. 146) there exists a holomorphic automorphism H satisfying $H(p) = p'$, $H(q) = q'$, and

$$|H(z) - z| + |H^{-1}(z) - z| < \frac{\epsilon}{8}, \quad z \in K_2.$$

Set $b_j = H(a_j)$ for $j = 1, \ldots, s$. Next we construct a polynomial additive shear G satisfying $|G(z) - z| < \frac{\epsilon}{8}$ for all $z \in K_2$, $G(p') = q'$, and

$$G(z) = z + O(|z - b_j|^{N+1}) \text{ as } z \to b_j, \quad j = 1, \ldots, s.$$

To find G, choose a linear form $\lambda\colon \mathbb{C}^n \to \mathbb{C}$ that is constant on Σ; hence $\lambda(p') = \lambda(q') \notin \lambda(\bar{B})$. Choose a holomorphic polynomial $f\colon \mathbb{C} \to \mathbb{C}$ such that $f(\lambda(p')) = 1$, f vanishes to order N at each of the points $\lambda(b_j)$ $(j = 1, \ldots, s)$, and $|f(\zeta)(q' - p')| < \frac{\epsilon}{8}$ for each $\zeta \in \lambda(K_2)$. The shear

$$G(z) = z + f(\lambda(z))(q' - p')$$

then satisfies the required properties.

It remains to find $S \in \operatorname{Aut} \mathbb{C}^n$ satisfying the following properties:

(i) $S(q') = q'$,
(ii) $S(z) = z + O(|z - b_j|^{N+1})$ as $z \to b_j$ for $j = 1, \ldots, s$,
(iii) $|S(z) - z| < \frac{\epsilon}{8}$ for all $z \in K_2$, and
(iv) the A-jet $S_{m,q'}$ of S at the point q' is determined by

$$S_{m,q'}(z) = H_{m,q} \circ P \circ (H^{-1})_{m,p'} \circ (G^{-1})_{m,q'}(z) + O(|z|^{m+1}).$$

The automorphism Φ (4.55) then clearly satisfies (ii) and (iii) in Proposition 4.15.3; condition (i) follows by Lemma 4.15.2 and the choice of $S_{m,q'}$.

We may assume that $q' = 0$ and that Σ is a complex hyperplane through the origin. Choose a number $\eta > 0$ (to be fixed later). We seek S as a composition $S = S_m \circ S_{m-1} \circ \cdots \circ S_1$, where each $S_k \in \operatorname{Aut} \mathbb{C}^n$ satisfies the following properties:

$$\begin{aligned} &S_k(0) = 0, \\ &S_k(z) = z + O(|z - b_j|^{N+1}), \quad z \to b_j, \quad j = 1, \ldots, s, \\ &|S_k(z) - z| < \eta, \quad z \in K_3. \end{aligned} \tag{4.56}$$

Let Q be the A-jet $S_{m,q'}$ in condition (iv) above. Write $Q(z) = Q_1(z) + O(|z|^2)$ as $z \to 0$. For $k = 1$ we shall find an automorphism $S_1 \in \operatorname{Aut} \mathbb{C}^n$ such that $S_1(z) = Q_1(z) + O(|z|^2)$ as $z \to 0$. Then

$$Q \circ S_1^{-1}(z) = z + Q_2(z) + O(|z|^3), \quad z \to 0,$$

where $Q_2\colon \mathbb{C}^n \to \mathbb{C}^n$ is homogeneous of degree 2. Next we shall find $S_2 \in \operatorname{Aut} \mathbb{C}^n$ satisfying (4.56) and $S_2(z) = z + Q_2(z) + O(|z|^3)$ as $z \to 0$. Then

$$Q \circ S_1^{-1} \circ S_2^{-1}(z) = z + Q_3(z) + O(|z|^4), \quad z \to 0,$$

where Q_3 is homogeneous of degree 3. Continuing in this way we obtain in m steps an automorphism $S = S_m \circ S_{m-1} \circ \cdots \circ S_1$ satisfying conditions (i)–(iv), provided that $\eta > 0$ was chosen small enough.

Construction of S_1: Choose $\mathbb{C}$-linearly independent forms $\lambda_1, \ldots, \lambda_n \in (\mathbb{C}^n)^*$ satisfying $0 \notin \lambda_j(\bar{B})$ for $j = 1, 2, \ldots, n$ (this holds if $\ker \lambda_j$ is close to Σ for every j), and then choose vectors $e_1, \ldots, e_n$ of $\mathbb{C}^n$ such that $\lambda_j(e_k) = \delta_{j,k}$. The group $SL_n(\mathbb{C})$ is generated by linear shears of the form

$$z \mapsto z + \alpha \lambda_j(z)\, e_k, \qquad 1 \le j \ne k \le n, \quad \alpha \in \mathbb{C}.$$

(See [276, p. 357].) In order to satisfy (4.56) we interpolate each of these shears at the origin by a polynomial shear of the form $z \mapsto z + f(\lambda_j(z))\, e_k$, where f is a polynomial on $\mathbb{C}$ such that $f(\zeta) = \alpha\zeta + O(|\zeta|^2)$ as $\zeta \to 0$, f vanishes to order N at all points $\lambda_j(b_l)$ for $l = 1, 2, \ldots, s$, and $|f|$ is small on the set $\lambda_j(\bar{B}) \subset \mathbb{C}\backslash\{0\}$. A suitable composition of such shears will give a desired S_1 in the special case when $S_1'(0) = Q_1 \in SL_n(\mathbb{C})$.

In the general case when $Q_1 \in GL_n(\mathbb{C})$ we let S_0 be a multiplicative shear

$$S_0(z) = z + \big(\mathrm{e}^{f(\lambda_1(z))} - 1\big)\langle z, e_2\rangle\, e_2,$$

where $f\colon \mathbb{C} \to \mathbb{C}$ is a polynomial that vanishes to order N at all points $\lambda_1(b_l)$ $(l = 1, 2, \ldots, s)$, $|f|$ is small on $\lambda_1(\bar{B})$, and $\mathrm{e}^{f(0)} = \mathrm{J}Q(0)$. Since $\mathrm{J}S_0(0) = \mathrm{e}^{f(0)}$, we have $\mathrm{J}S_0(0) = \mathrm{J}Q(0)$. The map $Q \circ S_0^{-1}$ has Jacobian one at the origin and we are back in the previous case. Choosing $\widetilde{S}_1$ as in the special case above such that its derivative at $0 \in \mathbb{C}^n$ matches that of $Q \circ S_0^{-1}$, the composition $S_1 = \widetilde{S}_1 \circ S_0$ satisfies all required conditions.

The inductive step. Suppose that $k \ge 2$ and that we have already found automorphisms $S_1, \ldots, S_{k-1}$ satisfying (4.56) and such that

$$Q \circ S_1^{-1} \circ \cdots \circ S_{k-1}^{-1}(z) = z + V_z + O(|z|^{k+1}), \quad z \to 0 \tag{4.57}$$

for a homogeneous polynomial map $V\colon \mathbb{C}^n \to \mathbb{C}^n$ of degree k. We now construct the next automorphism S_k of $\mathbb{C}^n$ satisfying (4.56) and such that

$$S_k(z) = z + V_z + O(|z|^{k+1}), \quad z \to 0. \tag{4.58}$$

From (4.57) and (4.58) it will follow that

$$Q \circ S_1^{-1} \circ \cdots \circ S_k^{-1}(z) = z + O(|z|^{k+1}), \quad z \to 0.$$

This will conclude the inductive step in the construction of S.

To construct S_k, we write V in the form (4.37):

$$V_z = \sum_{j=1}^{r} c_j \lambda_j(z)^k\, v_j + d_j \lambda_j(z)^{k-1} \langle z, v_j\rangle\, v_j$$

for some constants $c_j, d_j \in \mathbb{C}$, where the linear forms $\lambda_j \in (\mathbb{C}^n)^*$ are chosen such that $0 \notin \lambda_j(\bar{B})$. If $\operatorname{div} V = 0$ then the above holds with $d_j = 0$ for all j.

Choose polynomials $f_j, g_j\colon \mathbb{C} \to \mathbb{C}$ $(j = 1, \ldots, r)$ that are close to 0 on $\lambda_j(\bar{B})$, that vanish to order N at points $\lambda_j(b_l)$ $(l = 1, \ldots, s)$, and that satisfy

$$f_j(\zeta) = c_j \zeta^k + O(|\zeta|^{k+1}), \quad g_j(\zeta) = d_j \zeta^{k-1} + O(|\zeta|^k), \quad \zeta \to 0.$$

We then set

$$\begin{aligned} \Phi_j(z) &= z + f_j(\lambda_j(z))\, v_j = z + c_j \lambda_j(z)^k\, v_j + O(|z|^{k+1}), \\ \Psi_j(z) &= z + \big(e^{g_j(\lambda_j(z))} - 1\big)\langle z, v_j\rangle\, v_j \\ &= z + d_j(\lambda_j(z))^{k-1}\langle z, v_j\rangle\, v_j + O(|z|^{k+1}). \end{aligned}$$

These are the time-one maps of $f_j(\lambda_j z)v_j$, resp. of $g_j(\lambda_j(z))\langle z, v_j\rangle\, v_j$. Take S_k to be the composition of all maps Φ_j and Ψ_j (in any order). The Taylor expansion of each of these maps begins with z+homogeneous terms of order k. When composing such maps, their homogeneous parts of degree k add up and we get (4.58). This finishes the inductive step.

Since each of the m automorphism S_k moves points of $K_3 \subset \bar{B}$ by less than η, we can achieve (by choosing $\eta > 0$ sufficiently small) that their composition S moves points of K_2 by less than $\frac{\epsilon}{8}$ as required by (iii). This completes the proof of Proposition 4.15.3 when P is an A-jet.

Suppose now that P is an A_1-jet, $\mathrm{J}P(z) = 1 + O(|z|^m)$. Recall that H and G are finite compositions of polynomial additive shears. The A-jet $Q = S_{m,q'}$, being conjugate to the A_1-jet P by a volume preserving automorphism, is itself an A_1-jet of order m. Thus $\mathrm{J}Q(0) = 1$, and hence S_1 constructed above is volume preserving. Assuming inductively that map S_l is volume preserving for $l = 1, \ldots, k-1$, it follows from chain rule that the Jacobian of the map in (4.57) agrees with 1 to order m at the origin. This implies

$$1 + O(|z|^m) = \det\big(I + V_z' + O(|z|^k)\big) = 1 + (\mathrm{div} V)(z) + O(|z|^k).$$

Since $\mathrm{div} V$ is homogeneous of degree $k-1$, we get $\mathrm{div} V = 0$. Consequently V is a sum of vector fields whose time-one maps Φ_j are additive shears (4.1). This shows that each S_k for $k = 1, 2, \ldots, m$ is a finite composition of additive shears, and hence the same is true for the maps S and Φ. $\square$

Remark 4.15.4. It is easy to obtain the following improvement of Proposition 4.15.3. Let $\Psi \in \mathrm{Aut}\,\mathbb{C}^n$ be such that the set $K \cup \Psi(K)$ does not contain the points p and q. Then there exists $\Phi \in \mathrm{Aut}\,\mathbb{C}^n$ satisfying

(i') $\Phi(z) = q + P(z-p) + O(|z-p|^{m+1})$ as $z \to p$,

(ii') $\Phi(z) = \Psi(z) + O(|z - a_j|^N)$ as $z \to a_j$ for each $j = 1, 2, \ldots, s$, and

(iii') $|\Phi(z) - \Psi(z)| < \epsilon$ for each $z \in K$, and $|\Phi^{-1}(z) - \Psi^{-1}(z)| < \epsilon$ for each $z \in \Psi(K)$.

We find such Φ as follows. Choose a compact set K_1 containing K in its interior and such that $p, q \notin K_1 \cup \Psi(K_1)$. Let $p' = \Psi(p)$, $q' = \psi(q)$. Set $\Phi = G \circ H \circ \Psi$, where $G, H \in \mathrm{Aut}\,\mathbb{C}^n$ are chosen such that

- H is close to the identity on $\Psi(K_1)$, it interpolates the identity map to order N at each of the points $a'_j = \Psi(a_j)$, and $H(p') = p$, $H(q') = q$;
- G is close to the identity on $\Psi(K_1)$, it interpolates the identity map to order N at each of the points a'_j, $G(p) = q$, and its A-jet $G_{m,p}$ satisfies $P = G_{m,p} \circ H_{m,p'} \circ \Psi_{m,p}$.

Automorphisms G and H with these properties exist by Proposition 4.15.3.

Furthermore, instead of a single pair p, q, we can choose finite collections $\{p_1, \dots, p_r\}$ and $\{q_1, \dots, q_r\}$ of pairwise disjoint points in the complement of $K \cup \Psi(K)$ and then find an automorphism $\Phi \in \operatorname{Aut} \mathbb{C}^n$ such that $\Phi(p_j) = q_j$ and the interpolation property (i) above holds for each $j = 1, \dots, r$ (with the polynomial $P = P_j$ depending on j). $\square$

A special case of the following corollary is due to Buzzard [62]. In §4.16 below we present analogous results for certain countable discrete sets.

Corollary 4.15.5. *Let $n > 1$. Given finite subsets $\{a_j\}_{j=1}^r$ and $\{b_j\}_{j=1}^r$ of $\mathbb{C}^n$ (without repetition), and for each $j = 1, \dots, r$ a holomorphic polynomial map $P_j \colon \mathbb{C}^n \to \mathbb{C}^n$ of degree $m_j \in \mathbb{N}$ satisfying $P_j(0) = 0$ and $\mathrm{J}P_j(0) \neq 0$, there exists an automorphism $\Phi \in \operatorname{Aut} \mathbb{C}^n$ such that for each $j = 1, 2, \dots, r$ we have*

$$\Phi(z) = b_j + P_j(z - a_j) + O(|z - a_j|^{m_j+1}), \quad z \to a_j.$$

If $\mathrm{J}P_j(z) = 1 + O(|z|^{m_j})$ as $z \to 0$ for each j, then we may choose Φ to be a polynomial automorphism with Jacobian one. In addition, we may choose Φ as close as desired to the identity map on any compact polynomially convex set $K \subset \mathbb{C}^n$ that does not contain any of the points a_j and b_j.

Proof. By Proposition 4.15.3 there is for each $j = 1, 2, \dots, r$ an automorphism $\Phi_j \in \operatorname{Aut} \mathbb{C}^n$ satisfying the following properties:

$$\begin{aligned}
\Phi_j(z) &= b_j + P_j(z - a_j) + O(|z - a_j|^{m_j+1}), \quad z \to a_j, \\
\Phi_j(z) &= z + O(|z - b_k|^{m_k+1}), \quad z \to b_k,\ 1 \le k \le j-1, \\
\Phi_j(z) &= z + O(|z - a_k|^{m_k+1}), \quad z \to a_k,\ j+1 \le k \le r.
\end{aligned}$$

The composition $\Phi = \Phi_r \circ \Phi_{r-1} \circ \cdots \circ \Phi_1$ then satisfies Corollary 4.15.5. $\square$

4.16 A Mittag-Leffler Theorem for Automorphisms of $\mathbb{C}^n$

In this section we construct automorphisms mapping tame discrete sequences of $\mathbb{C}^n$ one onto another with a prescribed finite jet at each point, thereby solving a general *Mittag-Leffler interpolation problem* for automorphisms.

Theorem 4.16.1. [67, Theorem 1.1] *Assume that $n > 1$, that $\{a_j\}$ and $\{b_j\}$ ($j \in \mathbb{N}$) are tame sequences in $\mathbb{C}^n$ without repetitions, and that $P_j\colon \mathbb{C}^n \to \mathbb{C}^n$ is an A-jet of order $m_j \in \mathbb{N}$ (Definition 4.15.1). Then there exists $F \in \operatorname{Aut}\mathbb{C}^n$ such that for every $j = 1, 2, \dots$ we have $F(a_j) = b_j$ and*

$$F(z) = b_j + P_j(z - a_j) + O(|z - a_j|^{m_j+1}), \quad z \to a_j. \tag{4.59}$$

If in addition every P_j is an A_1-jet of order m_j and if the sequences $\{a_j\}$ and $\{b_j\}$ are very tame, then there exists an $F \in \operatorname{Aut}_1\mathbb{C}^n$ with these properties.

If $a_j = b_j = je_1$ for all $j \in \mathbb{N}$ and if the degrees of the polynomials P_j are uniformly bounded, one obtains a desired F as a finite composition of shears (see [168, Theorem 3.1]). The restriction to tame sequences is justified in view of the results in §4.5.

Under certain circumstances the above interpolation result holds while at the same time approximating a given automorphism on a polynomially convex set (compare with Remark 4.15.4 on p. 157).

Theorem 4.16.2. [66, Theorem 1.2] *Assume, in addition to the hypotheses of Theorem 4.16.1, that $\Phi \in \operatorname{Aut}\mathbb{C}^n$ and $K \subset \mathbb{C}^n \backslash \{a_j\}_{j=1}^\infty$ is a compact polynomially convex set such that $\Phi(K) \subset \mathbb{C}^n \backslash \{b_j\}_{j=1}^\infty$. Then for each $\epsilon > 0$ there exists an $F \in \operatorname{Aut}\mathbb{C}^n$ satisfying Theorem 4.16.1 and also*

$$|F(z) - \Phi(z)| < \epsilon \ (\forall z \in K), \quad |F^{-1}(w) - \Phi^{-1}(w)| < \epsilon \ (\forall w \in \Phi(K)). \tag{4.60}$$

If the volume preserving assumptions in Theorem 4.16.1 hold and if $\Phi \in \operatorname{Aut}_1\mathbb{C}^n$, then we may choose $F \in \operatorname{Aut}_1\mathbb{C}^n$.

Corollary 4.16.3. *Let $n > 1$. For every tame discrete set $\{a_j\} \subset \mathbb{C}^n$ there exists $\Phi \in \operatorname{Aut}\mathbb{C}^n$ with an attracting fixed point at each a_j.*

Here is the key ingredient in the proof of Theorems 4.16.1 and 4.16.2.

Lemma 4.16.4. *Given P_j and m_j as in Theorem 4.16.1, there exists a discrete sequence $\{c_j\}$ contained in the z_1-axis and and an automorphism $F \in \operatorname{Aut}\mathbb{C}^n$ such that for all $j \in \mathbb{N}$ we have*

$$F(z) = c_j + P_j(z - c_j) + O(|z - c_j|^{m_j+1}), \quad z \to c_j. \tag{4.61}$$

If in addition each polynomial P_j is an A_1-jet, then there exists an $F \in \operatorname{Aut}_1\mathbb{C}^n$ with the above property.

Proof. For $j \geq 1$, let $\epsilon_j = 2^{-j}$. We will construct the sequence $\{c_j\}$ inductively and will find F as the limit of a sequence of compositions of automorphisms.

Let $K_0 = \emptyset$ and $K_1 = 2\overline{\mathbb{B}}$. Let $c_1 = 3e_1$ where $e_1 = (1, 0, \dots, 0)$. By Proposition 4.15.3, applied to $K = K_1$, there exists $\Psi_1 \in \operatorname{Aut}\mathbb{C}^n$ such that

$$\begin{aligned} \Psi_1(z) &= c_1 + P_1(z-c_1) + O(|z-c_1|^{m_1+1}), \quad && z \to c_1, \\ |\Psi_1(z) - z| + |\Psi_1^{-1}(z) - z| &< \epsilon_1, && z \in K_1. \end{aligned}$$

If P_1 is an A_1-jet, there exists a $\Psi_1 \in \mathrm{Aut}_1\mathbb{C}^n$ with these properties.

Let $F_0(z) = z$ and $F_1 = \Psi_1$. We shall inductively choose automorphisms $F_k = \Psi_k \circ \cdots \circ \Psi_1$. For the induction, suppose that we have the following:

1. Compact convex sets $K_0 \subset K_1 \subset \cdots \subset K_k$ with $j\mathbb{B} \cup F_{j-1}(j\mathbb{B}) \subset K_j$ and
$$\mathrm{dist}(K_{j-1} \cup F_{j-1}(j\mathbb{B}), \mathbb{C}^n \backslash K_j) > \epsilon_j, \quad j = 1, \ldots, k.$$
2. Points $c_j \in K_{j+1} \backslash K_j$ for $j = 1, \ldots, k-1$, and $c_k \in \mathbb{C}^n \backslash K_k$, such that each c_j is contained in the z_1-axis.
3. Automorphisms $\Psi_j \in \mathrm{Aut}\,\mathbb{C}^n$ for $j = 1, \ldots, k$, satisfying
$$|\Psi_j(z) - z| + |\Psi_j^{-1}(z) - z| < \epsilon_j, \quad z \in K_j,$$
with $\Psi_j \in \mathrm{Aut}_1\mathbb{C}^n$ if P_j is an A_1-jet.
4. Automorphisms $F_k = \Psi_k \circ \Psi_{k-1} \circ \cdots \circ \Psi_1$ satisfying (4.61) for $j = 1, \ldots, k$.

Given these, let K_{k+1} be a compact convex set in $\mathbb{C}^n$ such that

$$\begin{aligned} (k+1)\mathbb{B} \cup K_k \cup F_k((k+1)\mathbb{B}) \cup \{c_k\} &\subset K_{k+1}, \\ \mathrm{dist}\big(K_k \cup F_k((k+1)\mathbb{B}), \mathbb{C}^n \backslash K_{k+1}\big) &> \epsilon_{k+1}. \end{aligned}$$

Since K_{k+1} is compact, we can choose a point c_{k+1} in the z_1-axis so that $c_{k+1}, F_k(c_{k+1}) \in \mathbb{C}^n \backslash K_{k+1}$. Let $N = \max\{m_1, \ldots, m_k\} + 1$. Let $d_{k+1} = F_k(c_{k+1})$ and let $Q_{k+1} = P_{k+1} \circ (F_k^{-1})_{m_{k+1}, d_k}$. Proposition 4.15.3 furnishes an automorphism $\Psi_{k+1} \in \mathrm{Aut}\,\mathbb{C}^n$ (resp. $\Psi_{k+1} \in \mathrm{Aut}_1\mathbb{C}^n$ if P_{k+1} is an A_1-jet and $F_k \in \mathrm{Aut}_1\mathbb{C}^n$)) such that the following hold:

(i) $\Psi_{k+1}(z) = c_{k+1} + Q_{k+1}(z - d_{k+1}) + O(|z - d_{k+1}|^{m_{k+1}+1})$ as $z \to d_{k+1}$,
(ii) $\Psi_{k+1}(z) = z + O(|z - c_j|^N)$ as $z \to c_j$ for $j = 1, \ldots, k$, and
(iii) $|\Psi_{k+1}(z) - z| + |\Psi_{k+1}^{-1}(z) - z| < \epsilon_{k+1}$ for each $z \in K_{k+1}$.

Taking $F_{k+1} = \Psi_{k+1} \circ F_k$, we obtain the induction hypotheses at stage $k+1$, and hence the induction proceeds.

By Proposition 4.4.1 (p. 114) the sequence $\{F_j\}$ converges uniformly on compact subsets of $\Omega = \cup_{j=1}^{\infty} F_j^{-1}(K_j)$ to a biholomorphic map F from Ω onto the domain $D = \cup_{j=1}^{\infty} K_j$. Since $j\mathbb{B} \subset K_j$, we see that $D = \mathbb{C}^n$. Moreover, we have $F_j^{-1}(K_j) = F_{j-1}^{-1}\Psi_j^{-1}(K_j)$. By Rouché's theorem [85, p. 110] and the induction hypotheses (1) and (3) we see that $F_{j-1}(j\mathbb{B}) \subset \Psi_j^{-1}(K_j)$. Hence $j\mathbb{B} \subset F_j^{-1}(K_j)$, so we also have $\Omega = \mathbb{C}^n$.

Hence $F \in \mathrm{Aut}\,\mathbb{C}^n$, and F satisfies (4.61) for all $j \in \mathbb{N}$ since each F_k satisfies this condition for $j = 1, \ldots, k$. Finally, if all P_j are A_1-jets then $\Psi_j \in \mathrm{Aut}_1\mathbb{C}^n$ for each j, and hence $F \in \mathrm{Aut}_1\mathbb{C}^n$. $\square$

Remark 4.16.5. Given numbers $R > 0$ and $\epsilon \in (0,1)$, we can replace the set K_1 by $(R+1)\overline{\mathbb{B}}$, the point c_1 by $(R+2)e_1$, and ϵ_j by $\epsilon\epsilon_j/2$, to construct an F satisfying Lemma 4.16.4 and also $|F(z) - z| + |F^{-1}(z) - z| < \epsilon$ on $R\mathbb{B}$. □

We next use a classical one-variable interpolation result to find an automorphism fixing each point je_1 and having a prescribed jet at this point. The technique of using shears to map a given discrete set in the z_1-axis to another is similar to the one used by Rosay and Rudin (see §4.6 above). For the current application, we need the map to be tangent to a translation to high order at each point in the discrete set.

Lemma 4.16.6. *Let P_j and m_j be as in Theorem 4.16.1. For each $R > 0$ and $\epsilon > 0$ there exists an automorphism $F \in \operatorname{Aut}\mathbb{C}^n$ such that*

$$\begin{aligned} &F(z) = je_1 + P_j(z - je_1) + O(|z - je_1|^{m_j+1}), \quad z \to je_1,\ j > R, \\ &|F(z) - z| + |F^{-1}(z) - z| < \epsilon, \quad |z| \le R. \end{aligned}$$

If all P_j's are A_1-jets then we can choose $F \in \operatorname{Aut}_1\mathbb{C}^n$.

Proof. We shall first prove the lemma without the approximation condition. By Lemma 4.16.4 there exist an automorphism $G \in \operatorname{Aut}\mathbb{C}^n$ and a sequence of points $\{c_j\} \subset \mathbb{C}^n$ in the z_1-axis such that

$$G(z) = c_j + P_j(z - c_j) + O(|z - c_j|^{m_j+1}), \quad z \to c_j.$$

(We can take $G \in \operatorname{Aut}_1\mathbb{C}^n$ if each P_j is an A_1-jet.) We need to find an automorphism $\Psi \in \operatorname{Aut}_1\mathbb{C}^n$ mapping the point je_1 to the point c_j for each $j \in \mathbb{N}$, with $\Psi_{m_j,je_1}(z) = z - je_1$; Lemma 4.15.2 (p. 153) then shows that the automorphism $\Psi^{-1} \circ G \circ \Psi$ has the desired properties. To find such Ψ, let $\xi_j \in \mathbb{C}$ be such that $c_j = \xi_j e_1$. By a standard one-variable interpolation theorem (see [267, Corollary 1.5.4]) there exists an entire function f_1 on $\mathbb{C}$ with $f_1(\zeta) = j + O(|\zeta - j|^{m_j+1})$ as $\zeta \to j$ for all $j \in \mathbb{N}$. Let $\Psi_1(z) = z + f_1(z_1)e_2$, where $e_2 = (0, 1, 0, \ldots, 0)$. Then $\Psi_1 \in \operatorname{Aut}_1\mathbb{C}^n$ and

$$\Psi_1(z) = j(e_1 + e_2) + (z - je_1) + O(|z - je_1|^{m_j+1}), \quad z \to je_1.$$

Thus Ψ_1 maps the point je_1 to the point $j(e_1 + e_2)$, and it agrees with a translation to order $m_j + 1$ at je_1.

Likewise, choosing f_2 entire with $f_2(\zeta) = \xi_j - j + O(|\zeta - j|^{m_j+1})$ as $\zeta \to j$, and taking $\Psi_2(z) = z + f_2(z_2)e_1$, we see that Ψ_2 maps the point $j(e_1 + e_2)$ to the point $\xi_j e_1 + je_2$, and it agrees with a translation to order $m_j + 1$ at $j(e_1 + e_2)$. Similarly, we can find $\Psi_3(z) = z_1 + f_3(z_1)e_2$ that maps $\xi_j e_1 + je_2$ to $\xi_j e_1 = c_j$ and that agrees with a translation to order $m_j + 1$ at $\xi_j e_1 + je_2$.

Let $\Psi = \Psi_3 \circ \Psi_2 \circ \Psi_1$. Then

$$\Psi(z) = c_j + (z - je_1) + O(|z - je_1|^{m_j+1}), \quad z \to je_1,$$

and by Lemma 4.15.2 we see that

$$\Psi^{-1}(z) = je_1 + (z - c_j) + O(|z - c_j|^{m_j+1}), \quad z \to c_j.$$

Also, $\Psi \in \mathrm{Aut}_1\mathbb{C}^n$ since each Ψ_l has Jacobian one. By Lemma 4.15.2, the automorphism $F = \Psi^{-1} \circ G \circ \Psi$ satisfies Lemma 4.16.6.

The proof also shows that the sequence $\{je_1\}$ could be replaced by any discrete sequence $\{d_j\}$ without repetition contained in the z_1-axis. Moreover, to get the approximation condition on F, suppose that $\{d_j\}$ and $\{c_j\}$ lie outside the ball $R\overline{\mathbb{B}}$ (in the z_1-axis). A simple argument (using Runge's theorem and the Weierstrass theorem) shows that we can choose each f_l as above and such that $|f_l|$ is small on a neighborhood of the closed disc of radius R in $\mathbb{C}$. Hence, by the remark after the proof of Lemma 4.16.4, we see that we may choose F satisfying the conclusions of Lemma 4.16.6 with d_j in place of je_1 and such that $|F(z) - z| + |F^{-1}(z) - z| < \epsilon$ for $z \in R\mathbb{B}$. □

Proof (of Theorem 4.16.1). Since $\{a_j\}$ and $\{b_j\}$ are tame, there exist $H_1, H_2 \in \mathrm{Aut}\,\mathbb{C}^n$ such that $H_1(a_j) = H_2(b_j) = je_1$, and if the sequences are very tame, this holds with $H_1, H_2 \in \mathrm{Aut}_1\mathbb{C}^n$. Hence it suffices to construct $G \in \mathrm{Aut}\,\mathbb{C}^n$ (resp. $G \in \mathrm{Aut}_1\mathbb{C}^n$) such that $G(je_1) = je_1$ and $(H_2^{-1}GH_1)_{m_j,a_j} = P_j$. By Lemma 4.15.2 this latter condition can be satisfied by making

$$G_{m_j,je_1}(z) = (H_2)_{m_j,b_j} \circ P_j \circ (H_1^{-1})_{m_j,je_1}(z) + O(|z|^{m_j+1}),\ z \to 0$$

for each $j \in \mathbb{N}$. By Lemma 4.16.6 we can find $G \in \mathrm{Aut}\,\mathbb{C}^n$ satisfying this condition and also $G(je_1) = je_1$. Moreover, if each P_j is an A_1-jet and if $H_1, H_2 \in \mathrm{Aut}_1\mathbb{C}^n$ then we can choose $G \in \mathrm{Aut}_1\mathbb{C}^n$. Finally, taking $F = H_2^{-1} \circ G \circ H_1$, we obtain the desired automorphism. □

In the proof of Theorem 4.16.2 we need the following lemma.

Lemma 4.16.7. *Let $H \in \mathrm{Aut}\,\mathbb{C}^n$, let K be a compact polynomially convex set in $\mathbb{C}^n$, and let $\{a_j\}_{j=1}^{\infty}$ be a discrete sequence disjoint from K and contained in the z_1-axis. Given numbers $R > 0$ with $H(K) \subset R\mathbb{B}$ and $\epsilon > 0$, there exists $\Psi \in \mathrm{Aut}\,\mathbb{C}^n$ such that $|\Psi(z) - H(z)| < \epsilon$ on K, $|\Psi^{-1}(z) - H^{-1}(z)| < \epsilon$ on $H(K)$, and $\Psi(a_j) = (R + j)e_1$ for all $j \in \mathbb{N}$. If $H \in \mathrm{Aut}_1\mathbb{C}^n$ then we can choose $\Psi \in \mathrm{Aut}_1\mathbb{C}^n$.*

Proof. Choose $r > 0$ such that $K \subset r\mathbb{B}$. Let $K_0 \subset r\mathbb{B}$ be a compact polynomially convex set such that $K \subset \mathrm{Int}K_0$, $K_0 \cap \{a_j\} = \emptyset$, and $H(K_0) \subset R\mathbb{B}$. Let $a_{j_1}, \ldots, a_{j_m}$ be the points in $\{a_j\} \cap r\overline{\mathbb{B}}$, and let $\delta > 0$. Since the union of K_0 with finitely many points is again polynomially convex, we can apply Proposition 4.15.3 m times to find $\Psi_1 \in \mathrm{Aut}_1\mathbb{C}^n$ such that $|\Psi_1(z) - z| + |\Psi_1^{-1}(z) - z| < \delta$ for $z \in K_0$ and $\Psi_1(a_{j_k}) = H^{-1}((R + j_k)e_1)$ for $k = 1, \ldots, m$.

Let $\pi_2(z) = z_2$. For a fixed vector $v \in \mathbb{C}^n \backslash \{0\}$ and $j \in \mathbb{N}$ we consider the 1-variable function $g_j(\zeta) = \pi_2 H \Psi_1(a_j + \zeta v)$. Since the kernel of the derivative

$\mathrm{d}(\pi_2 H\Psi_1)_{a_j}$ is an $(n-1)$-dimensional subspace for each j, we may choose v arbitrarily near e_2 such that g_j is nonconstant for each $j \in \mathbb{N}$ (and hence the image of g_j omits at most one point in $\mathbb{C}$), and such that there exists a $\mathbb{C}$-linear form λ satisfying $\lambda(v) = 0$, $\lambda(e_1) = 1$, and $\lambda(a_j) \notin \lambda(r\overline{\mathbb{B}})$ if $j \notin \{j_1, \dots, j_m\}$.

Choose f entire on $\mathbb{C}$ such that $|f| < \frac{\delta}{2}$ on $\lambda(r\mathbb{B})$, $f(\lambda(a_{j_k})) = 0$ for each $k = 1, \dots, m$, and $|\pi_2 H\Psi_1(a_j + f(\lambda(a_j))v)| = R + j$ for each $j \notin \{j_1, \dots, j_m\}$. Let $\Psi_2(z) = z + f(\lambda(z))v$. Then $\Psi_2 \in \mathrm{Aut}_1\mathbb{C}^n$, $|\Psi_2(z) - z| + |\Psi_2^{-1}(z) - z| < \delta$ on $r\mathbb{B}$, $H\Psi_1\Psi_2(a_{j_k}) = (R + j_k)e_1$ for $k = 1, \dots, m$, and $|\pi_2 H\Psi_1\Psi_2(a_j)| = R + j$ for $j \notin \{j_1, \dots, j_m\}$.

Using a composition of two shears as in the proof of Lemma 4.16.6, we can find $\Psi_3 \in \mathrm{Aut}_1\mathbb{C}^n$ such that $|\Psi_3(z) - z| + |\Psi_3^{-1}(z) - z| < \delta$ on $R\mathbb{B}$, $\Psi_3((R + j_k)e_1) = (R + j_k)e_1$ for $k = 1, \dots, m$, and $\Psi_3 H\Psi_1\Psi_2(a_j) = (R + j)e_1$ for each $j \notin \{j_1, \dots, j_m\}$.

Let $\Psi = \Psi_3 \circ H \circ \Psi_1 \circ \Psi_2$. Then $\Psi(a_j) = (R + j)e_1$ for all $j \in \mathbb{N}$, and for δ sufficiently small, we have $|\Psi(z) - H(z)| < \epsilon$ on K and $|\Psi^{-1}(z) - H^{-1}(z)| < \epsilon$ on $H(K)$, so the lemma follows. Since each $\Psi_l \in \mathrm{Aut}_1\mathbb{C}^n$, we see that $\Psi \in \mathrm{Aut}_1\mathbb{C}^n$ if $H \in \mathrm{Aut}_1\mathbb{C}^n$. $\square$

Proof (of Theorem 4.16.2). Since the sequences $\{a_j\}$ and $\{b_j\}$ are tame, there exist automorphisms H_1, H_2 of $\mathbb{C}^n$ such that $H_1(a_j) = je_1$ and $H_2(b_j) = je_1$ for each $j \in \mathbb{N}$. Replacing Φ by $H_2 \circ \Phi \circ H_1^{-1}$, K by $H_1(K)$, and adjusting the jets P_j as in the proof of Theorem 4.16.1 we reduce the problem to the case when $a_j = b_j = je_1$ for all $j \in \mathbb{N}$.

Choose a larger polynomially convex set $L \subset \mathbb{C}^n \backslash \{je_1 : j \in \mathbb{N}\}$ such that $K \subset \mathrm{Int}\, L$. Fix $\eta > 0$, and choose an integer $r > 0$ such that $L \cup \Phi(L) \subset r\mathbb{B}$. By Lemma 4.16.7 there exist automorphisms $\Psi, \Theta \in \mathrm{Aut}\,\mathbb{C}^n$ such that

$$\begin{aligned}
&\Psi(je_1) = \Theta(je_1) = (r + j)e_1, \quad j = 1, 2, \dots, \\
&|\Psi(z) - \Phi(z)| < \eta, \quad z \in L, \\
&|\Theta(z) - z| + |\Theta^{-1}(z) - z| < \eta, \quad z \in \Phi(L).
\end{aligned}$$

By Lemma 4.16.6 there exists $G \in \mathrm{Aut}\,\mathbb{C}^n$ such that $G((r + j)e_1) = (r + j)e_1$ for $j \in \mathbb{N}$, $|G(z) - z| < \eta$ for $|z| \le r$, and such that for each $j \in \mathbb{N}$ the jet $Q_j = G_{m_j,(r+j)e_1}$ satisfies

$$P_j(z) = (\Theta^{-1})_{m_j,(r+j)e_1} \circ Q_j \circ \Psi_{m_j,je_1}(z) + O(|z|^{m_j+1}), \quad z \to 0.$$

Let

$$F = \Theta^{-1} \circ G \circ \Psi.$$

Then $F(je_1) = je_1$ for each $j \in \mathbb{N}$, F satisfies (4.60) provided that $\eta > 0$ is chosen sufficiently small (depending on ϵ and $\mathrm{dist}(K, \mathbb{C}^n \backslash L)$), and Lemma 4.15.2 shows that F satisfies (4.59), with $a_j = b_j = je_1$.

Finally, if the sequences a_j and b_j are very tame, the polynomials P_j are A_1-jets, and $\Phi \in \mathrm{Aut}_1\mathbb{C}^n$, then the automorphisms Ψ, Θ, and G can be chosen to have Jacobian one. $\square$

4.17 Interpolation by Fatou-Bieberbach Maps

In this section we prove the following interpolation theorem that summarizes several results in the literature; see the papers [184] and [168, Theorem 6.1] for the first part, and [185, Theorem 1.3] for the second part.

Theorem 4.17.1. *Let X be a closed complex subvariety of $\mathbb{C}^n$ $(n > 1)$.*

(i) *For every discrete sequence $\{b_j\}_{j\in\mathbb{N}} \subset \mathbb{C}^n$ without repetition there exist a domain $\Omega \subset \mathbb{C}^n$ containing X and a biholomorphic map $\Phi\colon \Omega \to \mathbb{C}^n$ onto $\mathbb{C}^n$ such that $b_j \in \Phi(X)$ for $j = 1, 2, \ldots$.*

(ii) *If $\{a_j\}_{j\in\mathbb{N}} \subset X$ is a tame discrete sequence without repetition then the map Φ in (i) can be chosen such that $\Phi(a_j) = b_j$ for $j = 1, 2, \ldots$.*

In case (i) we can insure that $b_j \in \Phi(X_{\mathrm{reg}})$ for all j. In both cases one can prescribe finite order jets of $\Phi(X)$ at all points $b_j \in \Phi(X_{\mathrm{reg}})$.

The domain Ω in Theorem 4.17.1 is either all of $\mathbb{C}^n$, or a Fatou-Bieberbach domain. The image $\Phi(X)$ is a closed complex subvariety of $\mathbb{C}^n$ that is biholomorphic to X and contains the sequence $\{b_j\}_{j\in\mathbb{N}}$.

Example 4.17.2. Part (ii) of Theorem 4.17.1 fails if $\mathbb{C}^n\backslash\{a_j\}_{j\in\mathbb{N}}$ is volume hyperbolic, while the sequence $\{b_j\}_{j\in\mathbb{N}}$ is tame in $\mathbb{C}^n$. Indeed, assume that there is a biholomorphic map $\Phi\colon \Omega \to \mathbb{C}^n$ from a domain $\Omega \subset \mathbb{C}^n$ containing $\{a_j\}$ onto $\mathbb{C}^n$ satisfying $\Phi(a_j) = b_j$ for all $j = 1, 2, \ldots$. The set $\Omega\backslash\{a_j\}_{j\in\mathbb{N}} \subset \mathbb{C}^n\backslash\{a_j\}_{j\in\mathbb{N}}$ is volume hyperbolic, and hence its Φ-image $\mathbb{C}^n\backslash\{b_j\}_{j\in\mathbb{N}}$ is volume hyperbolic as well, a contradiction since $\{b_j\}$ is tame. $\square$

Proof (of Theorem 4.17.1, part (i)). Choose compact $\mathcal{O}(X)$-convex subsets $K_1 \subset K_2 \subset \cdots \subset \cup_{k=1}^{\infty} K_k = X$. Let $B_k = k\overline{\mathbb{B}}$ denote the closed ball of radius k in $\mathbb{C}^n$. Choose a number $\epsilon \in (0, 1)$ and let $\epsilon_k = 2^{-k}\epsilon$ for $k \in \mathbb{N}$. A Fatou-Bieberbach map $\Phi\colon \Omega \to \mathbb{C}^n$ satisfying Theorem 4.17.1 will be found as a limit $\Phi = \lim_{k\to\infty} \Phi_k$ of a sequence $\Phi_k = \Psi_k \circ \Psi_{k-1} \circ \cdots \circ \Psi_1 \in \operatorname{Aut}\mathbb{C}^n$.

We begin by choosing $\Psi_1 \in \operatorname{Aut}\mathbb{C}^n$ such that the subvariety $X_1 = \Psi_1(X)$ contains in its regular locus all points of the b_j sequence that belong to the first ball B_1, and X_1 contains none of the points b_j in the shell $B_2\backslash B_1$; in addition, we can prescribe finite order jets of X_1 at all points $b_j \in B_1$. Such Ψ_1 exists by Proposition 4.15.3 (p. 154): Choose a suitable number of points $a_j \in X_{\mathrm{reg}}$ and bring them to the points $b_j \in B_1$ by an automorphism with a prescribed jet at every point; if the image variety contains any of the points $b_j \in B_2\backslash B_1$, it is pushed away by another automorphism, close to the identity, that is fixed to a sufficiently high order at the points $b_j \in B_1$.

Choose $k_1 \geq 1$ such that the $B_1 \cap X_1 \subset \Psi_1(K_{k_1})$; the set $B_1 \cup \Psi_1(K_{k_1})$ is then polynomially convex. In the second step we apply Proposition 4.15.3 (p. 154): to get an automorphism $\Psi_2 \in \operatorname{Aut}\mathbb{C}^n$ such that

(i) $|\Psi_2(z) - z| < \epsilon_2$ for all $z \in B_1 \cup \Psi_1(K_{k_1})$,

(ii) Ψ_2 is tangent to the identity at every point $b_j \in B_1$,

(iii) the subvariety $X_2 = \Psi_2(X_1)$ contains all points $b_j \in B_2 \backslash B_1$ in its regular locus, with prescribed jets at these points, and

(iv) X_2 does not contain any of the points $b_j \in B_3 \backslash B_2$.

Set $\Phi_2 = \Psi_2 \circ \Psi_1$. The subvariety $X_2 = \Phi_2(X) = \Psi_2(X_1)$ then contains all points $b_j \in B_2$ (with the correct jets at these points) and none of the points $b_j \in B_3 \backslash B_2$. We now choose $k_2 > k_1$ such that $B_2 \cap X_2 \subset \Phi_2(K_{k_2})$, and then choose $\Psi_3 \in \operatorname{Aut} \mathbb{C}^n$ satisfying the above properties with all indexes increased by one. Proceed inductively. Proposition 4.4.1 (p. 114) shows that the sequence $\Phi_k = \Psi_k \circ \Psi_{k-1} \circ \cdots \circ \Psi_1 \in \operatorname{Aut} \mathbb{C}^n$, obtained in this way, converges to a limit map with the stated properties. This proves part (i) of Theorem 4.17.1. □

The following lemma will provide the key step in the proof of part (ii).

Lemma 4.17.3. *Let $\{a_j\} \subset X \subset \mathbb{C}^n$ and $\{b_j\} \subset \mathbb{C}^n$ satisfy the hypotheses of Theorem 4.17.1 (ii). Let $B \subset B'$ be closed balls in $\mathbb{C}^n$ and let $L = X \cap B'$. Assume that all points of the $\{b_j\}$ sequence that belong to $B \cup L$ coincide with the corresponding points of the $\{a_j\}$ sequence, and all remaining points of the $\{a_j\}$ sequence are contained in $X \backslash L$. Given $\epsilon > 0$ and a compact set $K \subset X$, there exist a ball $B'' \subset \mathbb{C}^n$ containing B' (B'' may be chosen as large as desired), a compact polynomially convex set $M \subset X$ with $K \cup L \subset M$, and a holomorphic automorphism θ of $\mathbb{C}^n$ satisfying the following properties:*

(i) *$|\theta(z) - z| < \epsilon$ for all $z \in B \cup L$,*

(ii) *if $a_j \in M$ for some index j then $\theta(a_j) = b_j \in B''$,*

(iii) *if $b_j \in B' \backslash (B \cup L)$ for some j then $a_j \in M$ and $\theta(a_j) = b_j$,*

(iv) *$\theta(M) \subset \operatorname{Int} B''$, and*

(v) *if $a_j \in X \backslash M$ for some j then $\theta(a_j) \in \mathbb{C}^n \backslash B''$.*

Remark 4.17.4. If θ satisfies the conclusion of Lemma 4.17.3 then the set $L' = \{z \in X : \theta(z) \in B''\}$ contains M (and hence $K \cup L$), and $L' \backslash M$ does not contain any points of the $\{a_j\}$ sequence (since the θ-image of any point $a_j \in X \backslash M$ lies outside of B'' according to (v)). □

Proof. An automorphism θ of $\mathbb{C}^n$ with the required properties will be constructed in two steps, $\theta = \psi \circ \phi$.

Since $X \cap B \subset L$ and the sets B and L are polynomially convex, $B \cup L$ is also polynomially convex. By applying a preliminary automorphism of $\mathbb{C}^n$ that is very close to the identity map on $B \cup L$ we may assume that X does not contain any points of the $\{b_j\}$ sequence, except those that coincide with the corresponding points $a_j \in X$. The same procedure will be repeated whenever necessary during later stages of the construction without mentioning it again.

Choose a pair of compact polynomially convex neighborhoods $D_0 \subset D \subset \mathbb{C}^n$ of $B \cup L$, with $D_0 \subset \mathrm{Int}D$, such that D does not contain any additional points of the $\{a_j\}$ or the $\{b_j\}$ sequence. Choose $\epsilon_0 > 0$ so small that

$$\mathrm{dist}(B \cup L, \mathbb{C}^n \backslash D_0) > \epsilon_0, \quad \mathrm{dist}(D_0, \mathbb{C}^n \backslash D) > \epsilon_0.$$

We may assume that $0 < \epsilon < \epsilon_0$. Choose a compact polynomially convex set $M \subset X$ containing $K \cup (X \cap D)$ (and hence the set L), and also containing all those points of the $\{a_j\}$ sequence for which the corresponding point b_j is contained in the ball B'. (Of course M may also contain some other points of the $\{a_j\}$ sequence for which $b_j \in \mathbb{C}^n \backslash B'$.) Theorem 4.16.2 (applied with $\Phi = \mathrm{Id}$) furnishes an automorphism ϕ of $\mathbb{C}^n$ satisfying the following:

(a) $\sup_{z \in D} |\phi(z) - z| < \frac{\epsilon}{2}$ and $\sup_{z \in D} |\phi^{-1}(z) - z| < \frac{\epsilon}{2}$,
(b) $\phi(a_j) = b_j$ for all $a_j \in M$, and
(c) $\phi(a_j) = a_j$ for all $a_j \in X \backslash M$.

Condition (a) and the choice of ϵ imply $\phi(D_0) \subset D$ and $\phi(\mathbb{C}^n \backslash D) \cap D_0 = \emptyset$; the latter condition also implies that $\phi(X) \cap D_0 \subset \phi(M)$. Since the sets $\phi(M)$ and D_0 are polynomially convex, their union $\phi(M) \cup D_0$ is also polynomially convex. Choose a large ball $B'' \subset \mathbb{C}^n$ containing $\phi(M) \cup B'$. Theorem 4.16.2 furnishes an automorphism ψ of $\mathbb{C}^n$ satisfying the following:

(a') $|\psi(z) - z| < \frac{\epsilon}{2}$ when $z \in \phi(M) \cup D_0$,
(b') $\psi(\phi(a_j)) = \phi(a_j) = b_j$ for all $a_j \in M$, and
(c') $\psi(a_j) \in \mathbb{C}^n \backslash B''$ for all $a_j \in X \backslash M$.

We can also insure that ψ fixes all points $\phi(a_j) \in \phi(X) \backslash B''$. It is immediate that $\theta = \psi \circ \phi$ satisfies the conclusion of Lemma 4.17.3. $\square$

The above proof of Lemma 4.17.3 is illustrated in Fig. 4.2. The first drawing shows the initial situation – the thick dots on X indicate the points $b_j \in B \cup L$ that agree with the corresponding points a_j, while the crosses indicate the remaining points $b_j \in B'$ that are matched with the images of a_j by applying the automorphism ϕ. The second drawing shows the situation after the application of ϕ: The black dots in $\phi(X) \cap B'$ indicate the points $b_j = \phi(a_j) \in B'$, while the crossed dots on the subvariety $\phi(X)$ inside the set $B'' \backslash B'$ will be expelled from the ball B'' by the next automorphism ψ.

Proof (of Theorem 4.17.1 (ii)). Choose an exhaustion $K_1 \subset K_2 \subset \cdots \subset \cup_{j=1}^{\infty} K_j = X$ by compact sets. Fix a number ϵ with $0 < \epsilon < 1$. We inductively construct the following:

(a) a sequence of holomorphic automorphisms Φ_k of $\mathbb{C}^n$ ($k \in \mathbb{N}$),
(b) an exhaustion $L_1 \subset L_2 \subset \cdots \subset \cup_{k=1}^{\infty} L_k = X$ by compact polynomially convex sets, and

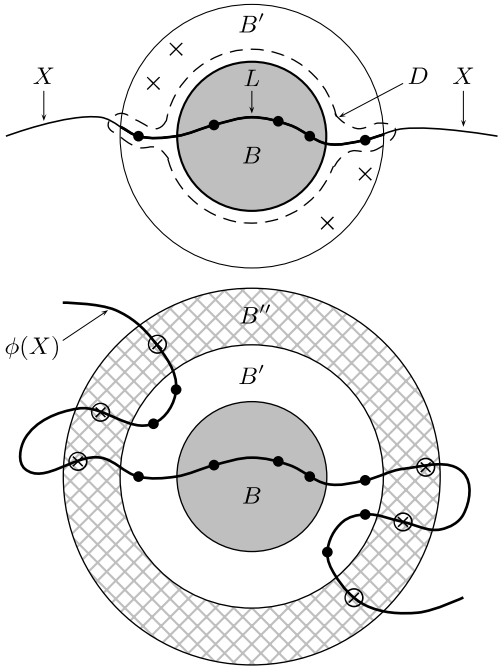

Fig. 4.2. The proof of Lemma 4.17.3. ([185, p. 550, Fig. 1])

(c) a sequence of balls $B_1 \subset B_2 \subset \cdots \subset \cup_{k=1}^{\infty} B_k = \mathbb{C}^n$ centered at $0 \in \mathbb{C}^n$ whose radii satisfy $r_{k+1} > r_k + 1$ for $k = 1, 2, \ldots$,

such that the following hold for all $k = 1, 2, \ldots$ (conditions (iv) and (v) are vacuous for $k = 1$):

(i) $\Phi_k(L_k) = \Phi_k(X) \cap B_{k+1}$,
(ii) if $a_j \in L_k$ for some j then $\Phi_k(a_j) = b_j$,
(iii) if $b_j \in \Phi_k(L_k) \cup B_k$ for some j then $a_j \in L_k$ and $\Phi_k(a_j) = b_j$,
(iv) $L_{k-1} \cup K_{k-1} \subset \mathrm{Int} L_k$,
(v) $|\Phi_k(z) - \Phi_{k-1}(z)| < 2^{-k}\epsilon$ for all $z \in B_{k-1} \cup L_{k-1}$.

To begin we set $B_0 = \emptyset$ and choose a pair of balls $B_1 \subset B_2 \subset \mathbb{C}^n$ whose radii satisfy $r_2 \geq r_1 + 1$. Theorem 4.16.2 (p. 159) furnishes an automorphism Φ_1 of $\mathbb{C}^n$ such that $\Phi_1(a_j) = b_j$ for all those (finitely many) indexes j for which $b_j \in B_2$, and $\Phi_1(a_j) \in \mathbb{C}^n \backslash B_2$ for the remaining indexes j. Setting $L_1 = \{z \in X : \Phi_1(z) \in B_2\}$, the properties (i), (ii) and (iii) are satisfied for $k = 1$, and the remaining two properties (iv), (v) are void.

Assume inductively that we have already found sets $L_1, \ldots, L_k \subset X$, balls $B_1, \ldots, B_{k+1} \subset \mathbb{C}^n$, and automorphisms $\Phi_1, \ldots, \Phi_k$ of $\mathbb{C}^n$ such that properties (i)–(v) hold up to index k. We now apply Lemma 4.17.3 with $B = B_k$, $B' = B_{k+1}$, X replaced by $X_k = \Phi_k(X)$, and $L = \Phi_k(L_k) \subset X_k$. This gives us a compact polynomially convex set $M = M_k \subset X_k$ containing $\Phi_k(K_k \cup L_k)$, an automorphism $\theta = \theta_k$ of $\mathbb{C}^n$, and a ball $B'' = B_{k+2} \subset \mathbb{C}^n$ of radius $r_{k+2} \geq r_{k+1}+1$ such that the conclusion of Lemma 4.17.3 holds. In particular, $\theta_k(M_k) \subset B_{k+2}$, the interpolation condition is satisfied for all points $b_j \in \theta_k(M_k) \cup B_{k+1}$, and the remaining points in the sequence $\{\Phi_k(a_j)\}_{j\in\mathbb{N}}$ are sent by θ_k out of the ball B_{k+2}. Setting

$$\Phi_{k+1} = \theta_k \circ \Phi_k, \quad L_{k+1} = \{z \in X : \Phi_{k+1}(z) \in B_{k+2}\}$$

one easily checks that the properties (i)–(v) hold for the index $k+1$ as well. (Note that L_{k+1} corresponds to the set L' in Remark 4.17.4.) The induction may now continue.

Let $\Omega = \cup_{k=1}^{\infty} \Phi_k^{-1}(B_k)$ consist of all points $z \in \mathbb{C}^n$ for which the sequence $\{\Phi_k(z)\}_{k\in\mathbb{N}}$ remains bounded. Proposition 4.4.1 (p. 114) implies that the limit $\lim_{k\to\infty} \Phi_k = \Phi$ exists on Ω, and $\Phi : \Omega \to \mathbb{C}^n$ is a biholomorphic map of Ω onto $\mathbb{C}^n$. From (v) we see that $X \subset \Omega$, and properties (ii), (iii) imply that $\Phi(a_j) = b_j$ for $j = 1, 2, \ldots$. This completes the proof of Theorem 4.17.1. □

4.18 Twisted Holomorphic Embeddings $\mathbb{C}^k \hookrightarrow \mathbb{C}^n$

In this section we survey results on the existence of 'twisted' holomorphic embeddings of Euclidean spaces, and of their complex subvarieties, to Euclidean spaces. A main tool in these constructions is Theorem 4.17.1 (p. 164).

Definition 4.18.1. *Let $0 < k < n$ be integers.*

(A) *A proper holomorphic embedding $f : \mathbb{C}^k \hookrightarrow \mathbb{C}^n$ is (holomorphically)* straightenable *if there is a holomorphic automorphism $\Phi \in \operatorname{Aut} \mathbb{C}^n$ such that*

$$\Phi(f(\mathbb{C}^k)) = \mathbb{C}^k \times \{0\}^{n-k}.$$

A proper algebraic (polynomial) embedding $f : \mathbb{C}^k \hookrightarrow \mathbb{C}^n$ is algebraically straightenable *if the above holds for some polynomial automorphism $\Phi \in \operatorname{Aut}_{\mathrm{alg}} \mathbb{C}^n$.*

(B) *Holomorphic embeddings $f, g : \mathbb{C}^k \hookrightarrow \mathbb{C}^n$ are* $\operatorname{Aut} \mathbb{C}^n$-equivalent *if $\Phi \circ f = g$ for some $\Phi \in \operatorname{Aut} \mathbb{C}^n$, and are* $(\operatorname{Aut} \mathbb{C}^k, \operatorname{Aut} \mathbb{C}^n)$*-equivalent if $\Phi \circ f = g \circ \Psi$ for some $\Phi \in \operatorname{Aut} \mathbb{C}^n$ and $\Psi \in \operatorname{Aut} \mathbb{C}^k$.*

Thus an embedding $\mathbb{C}^k \hookrightarrow \mathbb{C}^n$ is straightenable if and only if it is $(\operatorname{Aut} \mathbb{C}^k, \operatorname{Aut} \mathbb{C}^n)$-equivalent to the inclusion $z \mapsto (z, 0)$. The analogous notions make sense in the algebraic case.

Let us first briefly survey the situation in the algebraic category.

Question 4.18.2. Let $0 < k < n$. Is every algebraic (polynomial) embedding $\mathbb{C}^k \hookrightarrow \mathbb{C}^n$ algebraically straightenable?

Theorem 4.18.3. [3, 482] *Every polynomial embedding $\mathbb{C} \hookrightarrow \mathbb{C}^2$ is algebraically straightenable.*

There is a number of different proofs in the literature, including a relatively simple proof by L. Rudolph [426] using elementary knot theory.

A positive answer to Question 4.18.2 is known if $n > 2k + 1$ [286, 277]; in particular, every polynomial embedding $\mathbb{C} \hookrightarrow \mathbb{C}^n$ for $n \geq 4$ is algebraically straightenable. Furthermore:

Theorem 4.18.4. [286] *Every polynomial embedding $\mathbb{C} \hookrightarrow \mathbb{C}^3$ is holomorphically straightenable.*

It is still an open question whether every polynomial embedding $\mathbb{C} \hookrightarrow \mathbb{C}^3$ is algebraically straightenable. Kaliman also proved the following results for general affine algebraic varieties.

Theorem 4.18.5. [285, 286] *Assume that X is an affine algebraic variety and $f, g\colon X \hookrightarrow \mathbb{C}^n$ are two proper algebraic embeddings.*

(a) *If $n \geq \max\{1 + 2\dim X, \dim \mathrm{T}X\}$ then f and g are isotopic via proper algebraic embeddings.*
(b) *If $n > \max\{1 + 2\dim X, \dim \mathrm{T}X\}$ then there exists an algebraic automorphism $\alpha \in \mathrm{Aut}_{\mathrm{alg}}\, \mathbb{C}^n$ such that $f = \alpha \circ g$.*

Here $\mathrm{T}X$ is the Zariski tangent bundle of X (an affine variety). An automorphism $\alpha \in \mathrm{Aut}_{\mathrm{alg}}\, \mathbb{C}^n$ is tame if it is a composition of polynomial shears.

In codimension one, Question 4.18.2 is also known as the

Conjecture 4.18.6. (Abhyankar-Sathaye Conjecture) Every polynomial embedding $\mathbb{C}^{n-1} \hookrightarrow \mathbb{C}^n$ is algebraically straightenable [2, 427].

A more general version of Question 4.18.2 considered in the literature is to understand which algebraic subvarieties A of $\mathbb{C}^n$ have the *Abhyankar-Moh property* (AMP): For any polynomial embedding $f\colon A \hookrightarrow \mathbb{C}^n$ there exists a polynomial automorphism $\Phi \in \mathrm{Aut}_{\mathrm{alg}}\, \mathbb{C}^n$ such that $f = \Phi|_A$. In this formulation, Conjecture (4.18.6) says that a hyperplane in $\mathbb{C}^n$ has the AMP.

We now show that Question 4.18.2 in the holomorphic category has a negative answer for every pair of integers $0 < k < n$.

Let us begin with embeddings of the complex line $\mathbb{C}$ into $\mathbb{C}^n$. Parts (a) and (b) of the following theorem were proved in [420] for $n > 2$ (see also [286, Theorem 7]), and in [184] for $n = 2$. Part (c) is due to Buzzard and Fornæss [65]. All known proofs of these results for $n = 2$ use the Andérsen-Lempert theorem (Theorem 4.9.1 on p. 125).

Theorem 4.18.7. *Let $n > 1$.*

(a) *There exists a nonstraightenable proper holomorphic embedding $\mathbb{C} \hookrightarrow \mathbb{C}^n$.*

(b) *The set of $\operatorname{Aut}\mathbb{C}^n$-equivalence classes of proper holomorphic embeddings $\mathbb{C} \hookrightarrow \mathbb{C}^n$ is uncountable.*

(c) *There exists a proper holomorphic embedding $f\colon \mathbb{C} \hookrightarrow \mathbb{C}^2$ such that $\mathbb{C}^2 \backslash f(\mathbb{C})$ is Kobayashi hyperbolic.*

Proof. (a) Choose a rigid discrete set $D = \{a_1, a_2, \ldots\}$ in $\mathbb{C}^n$ (see Theorem 4.7.2 on p. 120). Theorem 4.17.1 furnishes a proper holomorphic embedding $f\colon \mathbb{C} \hookrightarrow \mathbb{C}^n$ such that $f(j) = a_j$ for $j = 1, 2, \ldots$. If $\Phi(f(\mathbb{C})) = \mathbb{C} \times \{0\}^{n-1}$ for some $\Phi \in \operatorname{Aut}\mathbb{C}^n$, then Φ maps D to the affine line $\mathbb{C} \times \{0\}^{n-1}$. But implies that D is tame in contradiction to our assumption.

(b) Let $D = \{a_1, a_2, \ldots\}$ be a rigid set as in Proposition 4.7.6 (p. 122). Choose points $p \neq q$ in $\mathbb{C}^n \backslash D$. By Theorem 4.17.1 there exist proper holomorphic embeddings $f, g\colon \mathbb{C} \hookrightarrow \mathbb{C}^n$ such that

(i) $f(0) = p$ and $f(j) = a_j$ for $j = 1, 2, \ldots$;

(ii) $g(0) = q$ and $g(j) = a_j$ for $j = 1, 2, \ldots$.

If f and g are $\operatorname{Aut}\mathbb{C}^n$-equivalent then there is an automorphism of $\mathbb{C}^n$ which fixes D and which maps p to q; a contradiction to the choice of D. This gives uncountably many pairwise inequivalent embeddings $\mathbb{C} \hookrightarrow \mathbb{C}^n$, one for each choice of $q \in \mathbb{C}^n \backslash D \cup \{p\}$ (for a fixed rigid set D and point $p \in \mathbb{C}^n \backslash D$).

(c) We sketch the main idea and refer to [65] for the details. One begins with a suitably chosen discrete sequence of pairwise disjoint closed complex discs $\Delta_1, \Delta_2, \ldots$, contained in affine lines in $\mathbb{C}^2$ parallel to the coordinate axes, such that for each sufficiently small holomorphic perturbation Δ'_j of Δ_j in $\mathbb{C}^2$ the set $\mathbb{C}^2 \backslash \cup_{j=1}^{\infty} \Delta'_j$ is Kobayashi hyperbolic. By the proof of Theorem 4.17.1 there is a proper holomorphic embedding $f\colon \mathbb{C} \hookrightarrow \mathbb{C}^2$ such that $f(\mathbb{C})$ contains a sequence of disjoint complex discs Δ'_j (images of suitably chosen discs in $\mathbb{C}$) that are as close as desired to the linear discs Δ_j for all $j = 1, 2, \ldots$. If the approximations are close enough then $\mathbb{C}^2 \backslash f(\mathbb{C})$ is Kobayashi hyperbolic since it is contained in the hyperbolic domain $\mathbb{C}^2 \backslash \cup_{j=1}^{\infty} \Delta'_j$. $\square$

The idea from part (a) above also gives the following result.

Theorem 4.18.8. ([168, Corollary 5.3], [52]) *For every pair of integers $1 \leq k < n$ there exists a proper holomorphic embedding $f\colon \mathbb{C}^k \hookrightarrow \mathbb{C}^n$ such that $\mathbb{C}^n \backslash f(\mathbb{C}^k)$ is $(n-k)$-hyperbolic in the sense of Eisenman* [126].

In particular, if f is as in Theorem 4.18.8 then every entire map $g\colon \mathbb{C}^m \to \mathbb{C}^n$ such that $g(\mathbb{C}^m) \cap f(\mathbb{C}^k) = \emptyset$ is degenerate, in the sense that its rank is less than $n - k$ (the codimension of $f(\mathbb{C}^k)$ in $\mathbb{C}^n$) at each point.

Using these results one can see that the set of $\operatorname{Aut}\mathbb{C}^n$-equivalence classes of proper holomorphic embeddings $\mathbb{C}^k \hookrightarrow \mathbb{C}^n$ is uncountable. The following result, applied with $X = \mathbb{C}$, shows that the smaller set of $(\operatorname{Aut}\mathbb{C}^k, \operatorname{Aut}\mathbb{C}^n)$-equivalence classes of such embeddings is also uncountable.

Theorem 4.18.9. [52, Theorem 1.2] *Assume that X is a positive dimensional Stein space whose group of holomorphic automorphisms is a Lie group, possibly with countably many components. If there is a proper holomorphic embedding of X into $\mathbb{C}^n$ for some $n > \dim X$, then for any $k \geq 0$ there exist uncountably many proper holomorphic embeddings $X \times \mathbb{C}^k \hookrightarrow \mathbb{C}^n \times \mathbb{C}^k$ such that any two are inequivalent with respect to* $(\operatorname{Aut} X \times \mathbb{C}^k, \operatorname{Aut}\mathbb{C}^n \times \mathbb{C}^k)$.

Kutzschebauch and Lodin [317] constructed holomorphic families of proper holomorphic embeddings $\mathbb{C}^k \hookrightarrow \mathbb{C}^n$ $(0 < k < n-1)$ so that no two embeddings in the family are $(\operatorname{Aut}\mathbb{C}^k, \operatorname{Aut}\mathbb{C}^n)$-equivalent. As an application they derived the existence of families of holomorphic $\mathbb{C}^*$ actions on $\mathbb{C}^n$ for $n \geq 5$ so that different actions in the family are not conjugate.

The idea in part (c) of Theorem 4.18.7 applies to any complex subvariety of $\mathbb{C}^n$. The following result of Borell and Kutzschebauch uses the construction developed in the proof of [168, Theorem 5.1].

Theorem 4.18.10. [52, Proposition 2.5] *Let X be a closed k-dimensional complex subvariety of $\mathbb{C}^n$ with $0 < k = \dim X < n$. Given a discrete set $A = \{a_1, a_2, \ldots\}$ in $\mathbb{C}^n$, there is an open set $\Omega \subset \mathbb{C}^n$ containing X and a biholomorphic map $\Phi\colon \Omega \to \mathbb{C}^n$ onto $\mathbb{C}^n$ such that $A \subset \Phi(X)$ and $\mathbb{C}^n \backslash \Phi(X)$ is Eisenman $(n-k)$-hyperbolic.*

These results do not answer the question whether there exist topologically twisted (knotted if $n = 2$) proper holomorphic embeddings of $\mathbb{C}$ in $\mathbb{C}^n$.

Problem 4.18.11. Is there a proper holomorphic embedding $f\colon \mathbb{C} \hookrightarrow \mathbb{C}^n$ such that $\mathbb{C}^n \backslash f(\mathbb{C})$ is not homeomorphic to $\mathbb{C}^n \backslash (\mathbb{C} \times \{0\}^{n-1})$?

A positive answer is known for discs in $\mathbb{C}^2$:

Theorem 4.18.12. [28] *There exists a topologically knotted proper holomorphic embedding of the unit disc $\mathbb{D}$ into $\mathbb{C}^2$.*

The construction in [28] is based on the existence of locally well behaved Fatou-Bieberbach domains and the existence of knotted holomorphic discs in the 4-ball. We already mentioned in §4.4 that Globevnik [209] constructed Fatou-Bieberbach domains $\Omega \subset \mathbb{C}^2$ whose intersections with $\mathbb{C} \times \mathbb{D}$ are small $\mathcal{C}^1$ perturbations of the bidisc $\mathbb{D}^2 = \mathbb{D} \times \mathbb{D}$. In [28] the authors use the theory of complex algebraic curves in $\mathbb{C}^2$ to find a knotted proper holomorphic embedding $\varphi\colon \overline{\mathbb{D}} \to \overline{\mathbb{D}}^2$ that maps the circle $b\mathbb{D}$ to $b\mathbb{D} \times \mathbb{D}$. After a small correction of ϕ they get a knotted holomorphic embedding $\phi\colon \mathbb{D} \hookrightarrow \Omega$ such that $\phi(b\mathbb{D}) \subset b\Omega$. By composing ϕ with a biholomorphism $h\colon \Omega \to \mathbb{C}^2$ they obtain a knotted proper embedding $\mathbb{D} \hookrightarrow \mathbb{C}^2$.

Problem 4.18.13. Does there exist an *unknotted* proper holomorphic embedding $\mathbb{D} \hookrightarrow \mathbb{C}^2$?

4.19 Nonlinearizable Periodic Automorphisms of $\mathbb{C}^n$

A holomorphic automorphism $\Phi \in \operatorname{Aut}\mathbb{C}^n$ is *linearizable* if it is conjugate to a linear automorphism $A \in GL(n,\mathbb{C})$; that is, $\Psi \circ \Phi = A \circ \Psi$ for some $\Psi \in \operatorname{Aut}\mathbb{C}^n$. If this is not the case then Φ is *nonlinearizable*. Similarly one defines linearizability of a group of holomorphic automorphisms of $\mathbb{C}^n$ – the entire group should be conjugate to a subgroup of $GL_n(\mathbb{C})$ by a $\Psi \in \operatorname{Aut}\mathbb{C}^n$. In this section we prove the following result of Derksen and Kutzschebauch which settled a long standing question.

Theorem 4.19.1. [107] *For every integer $n \geq 2$ there exists a nonlinearizable holomorphic automorphism of period n on $\mathbb{C}^{2+n}$. In particular, there is a nonlinearizable holomorphic involution on $\mathbb{C}^4$.*

Derksen and Kutzschebauch actually proved that every nontrivial complex reductive Lie group G admits an effective nonlinearizable holomorphic action of G on $\mathbb{C}^N$ for all sufficiently large N (depending on G).

An important ingredient in their proof are proper holomorphic embeddings $\phi\colon \mathbb{C} \hookrightarrow \mathbb{C}^2$ such that every holomorphic map $\mathbb{C}^2 \to \mathbb{C}^2 \backslash \phi(\mathbb{C})$ is degenerate (Theorem 4.18.7 on p. 170). Choose such ϕ. For every $k \in \mathbb{Z}_+$ define the embedding $\phi_k\colon \mathbb{C} \times \mathbb{C}^k \to \mathbb{C}^2 \times \mathbb{C}^k$ by $\phi_k(z,y) = (\phi(z), y)$.

Corollary 4.19.2. *The embedding ϕ_k is not straightenable for every $k \in \mathbb{Z}_+$.*

Proof. If ϕ_k is straightenable then $\mathbb{C}^{2+k} \backslash \phi_k(\mathbb{C}^{1+k}) \cong (\mathbb{C}^2 \backslash \phi(\mathbb{C})) \times \mathbb{C}^k$ is biholomorphic to $\mathbb{C}^* \times \mathbb{C}^{k+1}$. Hence there exists a nondegenerate holomorphic map $f\colon \mathbb{C}^{2+k} \to \mathbb{C}^{2+k} \backslash \phi_k(\mathbb{C}^{1+k})$, for instance the universal covering map. By composing f with the projection $(\mathbb{C}^2 \backslash \phi(\mathbb{C})) \times \mathbb{C}^k \to \mathbb{C}^2 \backslash \phi(\mathbb{C})$ we get an entire map $\mathbb{C}^{k+2} \to \mathbb{C}^2 \backslash \phi(\mathbb{C})$ with rank two, contradicting the choice of ϕ. $\square$

We now describe certain pseudo affine modifications that are used in the proof. Let $X \subset Y$ and Z be Stein spaces, and let f be a holomorphic function on Z. Suppose that the ideal sheaf of X in Y is generated by finitely many functions $h_1, \ldots, h_r \in \mathcal{O}(Y)$ (this always holds if X and Y are nonsingular [150, Satz 5.5]). Consider the map

$$\psi\colon Y \times (Z \backslash \{f = 0\}) \to Y \times Z \times \mathbb{C}^r, \quad \psi(y,z) = \left(y, z, \frac{h_1(y)}{f(z)}, \ldots, \frac{h_r(y)}{f(z)}\right),$$

and let

$$\mathcal{R}(X,Y,Z,f) = \overline{\mathrm{Im}(\psi)} \subset Y \times Z \times \mathbb{C}^r$$

denote the closure of the image of ψ in $Y \times Z \times \mathbb{C}^r$. (Note that the topological closure equals the holomorphic Zariski closure. In [107, p. 44] these are called *Rees spaces*.) It is easily seen that, up to a biholomorphism, $\mathcal{R}(X,Y,Z,f)$ does not depend on the choice of the generators h_j for the ideal $\mathcal{J}_X(Y)$.

Lemma 4.19.3. *The spaces $\mathcal{R}(X,Y,Z,f)$ have the following properties:*

1. $\mathcal{R}(X,Y,Z,f) \cong \mathcal{R}(\phi(X),Y,Z,f)$ *for* $\phi \in \mathrm{Aut}\, Y$.
2. $\mathcal{R}(X \times W, Y \times W, Z, f) \cong \mathcal{R}(X,Y,Z,f)$ *for any Stein space* W.
3. $\mathcal{R}(X \times \{0\}, Y \times \mathbb{C}^m, Z, f) \cong \mathcal{R}(X,Y,Z,f) \times \mathbb{C}^m$.

Proof. The first two properties are immediate. To show the third property, suppose that the ideal $\mathcal{J}_X(Y)$ is generated by the functions $h = (h_1, \dots, h_r) \in \mathcal{O}(Y)^r$. Then $\mathcal{J}_{X\times\{0\}}(Y \times \mathbb{C}^m)$ is generated by (h,u) where $u = (u_1, \dots, u_m)$ are the coordinates on $\mathbb{C}^m$. Consider the map

$$\psi\colon Y \times \mathbb{C}^m \times (Z \backslash \{f = 0\}) \to Y \times \mathbb{C}^m \times Z \times \mathbb{C}^r \times \mathbb{C}^m,$$

$$\psi(y,u,z) = \big(y, u, z, f(z)^{-1}h, f(z)^{-1}u\big)\,.$$

Define $\phi \in \mathrm{Aut}\,(Y \times \mathbb{C}^m \times Z \times \mathbb{C}^r \times \mathbb{C}^m)$ by

$$\phi(y,u,z,v,w) = \big(y, u - f(z)w, z, v, w\big).$$

Then we have

$$\phi \circ \psi(y,u,z) = \big(y, 0, z, f(z)^{-1}h, f(z)^{-1}\big).$$

It follows that

$$\mathcal{R}(X \times \{0\}, Y \times \mathbb{C}^m, Z, f) = \overline{\mathrm{Im}(\psi)} \cong \overline{\mathrm{Im}(\phi \circ \psi)} \cong \mathcal{R}(X,Y,Z,f) \times \mathbb{C}^m.$$

This proves Lemma 4.19.3. □

The following example will be used in the proof of Theorem 4.19.1. Suppose that $X \subset Y$ is given by $X = \{h = 0\}$ for some $h \in \mathcal{O}(Y)$ which generates the ideal $\mathcal{J}_X(Y)$. Let $\psi\colon Y \times Z \backslash \{f = 0\} \to Y \times Z \times \mathbb{C}$ be given by $\psi(y,z) = (y, z, h(y)/f(z))$. Then

$$\mathcal{R}(X,Y,Z,f) = \overline{\mathrm{Im}(\psi)} = \{(y,z,w) \in Y \times Z \times \mathbb{C}\colon h(y) = f(z)w\}.$$

Lemma 4.19.4. *If $\phi_1\colon X \hookrightarrow \mathbb{C}^n$ and $\phi_2\colon X \hookrightarrow \mathbb{C}^m$ are proper holomorphic embeddings, then*

$$\mathcal{R}(\phi_1(X), \mathbb{C}^n, Z, f) \times \mathbb{C}^m \cong \mathcal{R}(\phi_2(X), \mathbb{C}^m, Z, f) \times \mathbb{C}^n.$$

Proof. It is easily see that there is an automorphism $\alpha \in \mathrm{Aut}\,(\mathbb{C}^n \times \mathbb{C}^m)$ such that $\alpha \circ (\phi_1 \times \{0\}^m) = \{0\}^n \times \phi_2$. From Lemma 4.19.3 it follows that

$$\mathcal{R}(\phi_1(X), \mathbb{C}^n, Z, f) \cong \mathcal{R}(\phi_1(X) \times \{0\}, \mathbb{C}^n \times \mathbb{C}^m, Z, f)$$
$$\cong \mathcal{R}(\{0\} \times \phi_2(X), \mathbb{C}^n \times \mathbb{C}^m, Z, f) \cong \mathcal{R}(\phi_2(X), \mathbb{C}^m, Z, f).$$

This proves Lemma 4.19.4. □

Applying Lemma 4.19.4 with $X = \mathbb{C}^k$ for some $1 \le k < n$ and taking $\phi_2 \colon \mathbb{C}^k \to \mathbb{C}^k$ to be the identity map we obtain

Corollary 4.19.5. *For an embedding $\phi \colon \mathbb{C}^k \hookrightarrow \mathbb{C}^n$ and $f \in \mathcal{O}(\mathbb{C}^l)$ we have*

$$\mathcal{R}(\phi(X), \mathbb{C}^n, \mathbb{C}^l, f) \times \mathbb{C}^k \cong \mathcal{R}(\mathbb{C}^k, \mathbb{C}^k, \mathbb{C}^l, f) \times \mathbb{C}^n \cong \mathbb{C}^{k+l+n}.$$

Proof (of Theorem 4.19.1). Let $G = \langle g \rangle$ be cyclic group of order $n > 1$. Choose a nonstraightenable proper holomorphic embedding $\phi \colon \mathbb{C} \hookrightarrow \mathbb{C}^2$ as in Corollary 4.19.2 above. Let $h \in \mathcal{O}(\mathbb{C}^2)$ generate the ideal sheaf of the smooth curve $\phi(\mathbb{C})$ in $\mathbb{C}^2$. For every $n \in \mathbb{N}$ the set

$$Y_n = \{(x, y, z) \in \mathbb{C}^3 \colon h(x, y) = z^n\}$$

is a then smooth complex hypersurface in $\mathbb{C}^3$.

Let ζ be a primitive n-th root of 1. Define the action $\tau \colon G \times Y_n \to Y_n$ by

$$g \cdot (x, y, z) = (x, y, \zeta z).$$

For each $k = 1, 2, \ldots$ we extend τ to an action $\tau_k \colon G \times (Y_n \times \mathbb{C}^k) \to Y_n \times \mathbb{C}^k$ by acting trivially on $\mathbb{C}^k$. Note that

$$\mathrm{Fix}\,\tau_k = (\mathrm{Fix}\,\tau) \times \mathbb{C}^k = \phi(\mathbb{C}) \times \{0\} \times \mathbb{C}^k$$

is a smooth complex hypersurface in $Y_n \times \mathbb{C}^k$.

Lemma 4.19.6. *If $Y_n \times \mathbb{C}^k \cong \mathbb{C}^{2+k}$ then the action τ_k is nonlinearizable.*

Proof. Assume that $Y_n \times \mathbb{C}^k \cong \mathbb{C}^{2+k}$ and that τ_k is linearizable (when considered as an action on $\mathbb{C}^{2+k}$). The linearization map takes the fixed point set of τ_k onto a linear hyperplane in $\mathbb{C}^{2+k}$, and hence the set

$$\widetilde{Y} = (Y_n \times \mathbb{C}^k) \backslash \mathrm{Fix}\,\tau_k = (Y_n \backslash \mathrm{Fix}\,\tau) \times \mathbb{C}^k$$

is biholomorphic to $\mathbb{C}^* \times \mathbb{C}^{1+k}$. The projection $\pi \colon \mathbb{C}^{3+k} \to \mathbb{C}^{2+k}$, $\pi(x, y, z, w) = (x, y, w)$, gives a proper ramified (n-to-one) covering $\pi \colon Y_n \times \mathbb{C}^k \to \mathbb{C}^{2+k}$ whose branching locus equals $\phi(\mathbb{C}) \times \{0\} \times \mathbb{C}^k = \mathrm{Fix}\,\tau_k$. The restriction

$$\pi \colon \widetilde{Y} = (Y_n \times \mathbb{C}^k) \backslash \mathrm{Fix}\,\tau_k \to (\mathbb{C}^2 \backslash \phi(\mathbb{C})) \times \mathbb{C}^k$$

is an unramified covering, and hence both sets have the same universal covering space X. Let $\eta \colon X \to (\mathbb{C}^2 \backslash \phi(\mathbb{C})) \times \mathbb{C}^k$ be the universal covering projection. Since $\widetilde{Y} \cong \mathbb{C}^* \times \mathbb{C}^{1+k}$, we have $X \cong \mathbb{C}^{2+k}$. Hence the restriction of η to a suitable two-plane in X, followed by the projection $\mathbb{C}^2 \times \mathbb{C}^k \to \mathbb{C}^2$, is an entire map $\mathbb{C}^2 \to \mathbb{C}^2 \backslash \phi(\mathbb{C})$ of rank 2. This contradicts the choice of ϕ. □

Lemma 4.19.7. *For each pair of integers $n \geq 2$ and $k \geq 1$ there exists an action $\sigma_k\colon G \times \mathbb{C}^{1+n+k} \to \mathbb{C}^{1+n+k}$ by holomorphic automorphisms with the fixed point set* $\operatorname{Fix}\sigma_k \cong Y_n \times \mathbb{C}^k$.

Proof. Let ϕ and h be as above. The set

$$X_n = \big\{(x, y, w_1, w_2, \ldots, w_n) \in \mathbb{C}^{2+n}\colon h(x, y) = w_1 w_2 \cdots w_n\big\} \tag{4.62}$$

is a smooth hypersurface in $\mathbb{C}^{2+n}$. Observe that $X_n = \mathcal{R}\big(\phi(\mathbb{C}), \mathbb{C}^2, \mathbb{C}^{n-1}, f\big)$, where $f \in \mathcal{O}(\mathbb{C}^{n-1})$ equals $f(w_1, \ldots, w_{n-1}) = w_1 w_2 \cdots w_{n-1}$.

Let $\sigma\colon G \times \mathbb{C}^{2+n} \to \mathbb{C}^{2+n}$ be the action

$$g\cdot(x, y, w_1, w_2, \ldots, w_n) = (x, y, w_2, \ldots, w_{n-1}, w_1).$$

Since X_n is clearly invariant under σ, we have an induced action on X_n which we still denote by σ. For each integer $k \geq 1$ we extend σ to an action

$$\sigma_k\colon G \times (X_n \times \mathbb{C}^k) \to X_n \times \mathbb{C}^k$$

by acting trivially on $\mathbb{C}^k$. We have that

$$\operatorname{Fix}\sigma = \{(x, y, w) \in \mathbb{C}^{2+n}\colon w_1 = w_2 = \cdots = w_n,\ h(x, y) = (w_1)^n\} \cong Y_n,$$

$$\operatorname{Fix}\sigma_k = \operatorname{Fix}\sigma \times \mathbb{C}^k \cong Y_n \times \mathbb{C}^k.$$

To complete the proof, it suffices to observe that $X_n \times \mathbb{C}^k$ is biholomorphic to $\mathbb{C}^{1+n+k}$ for each $k \geq 1$. Indeed, by Corollary 4.19.5 (p. 174) we have

$$X_n \times \mathbb{C}^k \cong \mathcal{R}(\phi(\mathbb{C}), \mathbb{C}^2, \mathbb{C}^{n-1}, f) \times \mathbb{C}^k \cong \mathbb{C}^{1+n+k}.$$

This proves Lemma 4.19.7. □

Comparing Lemmas 4.19.6 and 4.19.7 we draw the following conclusion.

Corollary 4.19.8. *For each $n \geq 2$ and $k \geq 1$, at least one of the actions σ_k, resp. τ_k, is a nonlinearizable action of the cyclic group $\mathbb{Z}/n\mathbb{Z}$ by holomorphic automorphism on $\mathbb{C}^{1+n+k}$, resp. on $\mathbb{C}^{2+k}$.*

Theorem 4.19.1 is an immediate consequence of Corollary 4.19.8. □

Problem 4.19.9. Is the hypersurface X_n (4.62) biholomorphic to $\mathbb{C}^{n+1}$?

This is a special case of the *holomorphic Zariski cancellation problem*:

Problem 4.19.10. Let X be a complex manifold such that $X \times \mathbb{C}$ is biholomorphic to $\mathbb{C}^{n+1}$ ($n \geq 2$); does it follow that $X \cong \mathbb{C}^n$?

In conclusion we describe a construction of *non-linearizable* $\mathbb{C}^*$ *actions* on $\mathbb{C}^l$ for every $l \geq 4$ [107, §4]. Let $\phi\colon \mathbb{C} \hookrightarrow \mathbb{C}^2$ be a nonstraightenable proper holomorphic embedding furnished by Theorem 4.18.7 (p. 170). Choose an entire function $h \in \mathcal{O}(\mathbb{C}^2)$ with $\{h = 0\} = \phi(\mathbb{C})$ and $\mathrm{d}h \neq 0$ on $\{h = 0\}$. Let X be the hypersurface in $\mathbb{C}^3$ defined by

$$X = \big\{(x, y, z, w) \in \mathbb{C}^4 \colon h(x, y) = zw\big\}. \tag{4.63}$$

Note that $X = \mathcal{R}(\phi(\mathbb{C}), \mathbb{C}^2, \mathbb{C}, z)$. By Corollary 4.19.5 we have $X \times \mathbb{C} \cong \mathbb{C}^4$. Now $X \times \mathbb{C}^k \subset \mathbb{C}^4 \times \mathbb{C}^k$ is stable under the linear $\mathbb{C}^*$ action on $\mathbb{C}^4 \times \mathbb{C}^k$,

$$\lambda \cdot (x, y, z, w, t_1, \ldots, t_k) = (x, y, \lambda z, \lambda^{-1} w, t_1, \ldots, t_k).$$

The restriction of this action to $X \times \mathbb{C}^k$ induces via α_k a holomorphic $\mathbb{C}^*$ action $\sigma_k\colon \mathbb{C}^* \times \mathbb{C}^{k+3} \to \mathbb{C}^{k+3}$.

Proposition 4.19.11. *The action σ_k is not linearizable for every $k \geq 1$. So for every $l \geq 4$ there exists a non-linearizable $\mathbb{C}^*$ action on $\mathbb{C}^l$.*

Proof. Suppose that $\alpha_k\colon X \times \mathbb{C}^k \xrightarrow{\cong} \mathbb{C}^{3+k}$ is a biholomorphic $\mathbb{C}^*$-equivariant map, where $\mathbb{C}^*$ acts linearly on $\mathbb{C}^{3+k}$. This representation of $\mathbb{C}^*$ on $\mathbb{C}^{k+3}$ must be isomorphic to the representation of $\mathbb{C}^*$ on the tangent space of some fixed point of $X \times \mathbb{C}^k$. With respect to some coordinates, this action is given by

$$\lambda \cdot (z, w, u_1, \ldots, u_{k+1}) = (\lambda z, \lambda^{-1} w, u_1, \ldots, u_{k+1}).$$

The categorical quotient of $X \times \mathbb{C}^k$ is given by $\pi_1\colon X \times \mathbb{C}^k \to \mathbb{C}^{2+k}$,

$$\pi_1(x, y, z, w, t_1, \ldots, t_k) = (x, y, t_1, \ldots, t_k),$$

and the categorical quotient of $\mathbb{C}^{3+k}$ is given by $\pi_2\colon \mathbb{C}^{3+k} \to \mathbb{C}^{2+k}$,

$$\pi_2(z, w, u_1, \ldots, u_{k+1}) = (zw, u_1, \ldots, u_{k+1}).$$

The fixed point set $(X \times \mathbb{C}^k)^{\mathbb{C}^*}$ equals

$$\big\{(x, y, z, w, t_1, \ldots, t_k) \in X \times \mathbb{C}^k \colon f(x, y) = z = w = 0\big\}.$$

Its image under π_1 is $\phi(\mathbb{C}) \times \mathbb{C}^k \subset \mathbb{C}^2 \times \mathbb{C}^k$. On the other hand $(\mathbb{C}^{3+k})^{\mathbb{C}^*}$ is

$$\big\{(z, w, u_1, \ldots, u_{k+1}) \in \mathbb{C}^{3+k} \mid z = w = 0\big\}$$

and its image under π_2 is $\{0\} \times \mathbb{C}^{1+k} \subset \mathbb{C}^{2+k}$. Now $\alpha_k\colon X \times \mathbb{C}^k \xrightarrow{\cong} \mathbb{C}^{3+k}$ induces a biholomorphism $\gamma\colon \mathbb{C}^{2+k} \xrightarrow{\cong} \mathbb{C}^{2+k}$ of the categorical quotients such that $\gamma(\phi(\mathbb{C}) \times \mathbb{C}^k) = \{0\} \times \mathbb{C}^{k+1}$. In view of Corollary 4.19.2 this contradicts our choice of ϕ. $\square$

4.20 Non-Runge Fatou-Bieberbach Domains and Long $\mathbb{C}^n$'s

Recall from §4.3 that a Fatou-Bieberbach domain $\Omega \subset \mathbb{C}^n$ is Runge in $\mathbb{C}^n$ if and only if any Fatou-Bieberbach map $F\colon \mathbb{C}^n \to \Omega$ onto Ω is a limit of a sequence of holomorphic automorphisms of $\mathbb{C}^n$. It was a long standing problem whether every Fatou-Bieberbach domain in $\mathbb{C}^n$ is Runge. We describe a counterexample by E. F. Wold [526]. The following is the key lemma.

Lemma 4.20.1. [526] *There exists a compact set Y in $\mathbb{C}^* \times \mathbb{C}$ such that*

(i) *Y is holomorphically convex in $\mathbb{C}^* \times \mathbb{C}$,*

(ii) *the polynomial hull $\widehat{Y}$ contains the origin $(0,0) \in \mathbb{C}^2$, and*

(iii) *for every nonempty open set $U \subset \mathbb{C}^* \times \mathbb{C}$ there exists an automorphism $G \in \operatorname{Aut}(\mathbb{C}^* \times \mathbb{C})$ such that $Y \subset G(U)$.*

Proof. A set Y with properties (i) and (ii) is found as a union of two disjoint totally real discs, following [466] or [470, pp. 392–396]. Choose smoothly bounded simply connected domains $\Omega_1, \Omega_2 \subset \mathbb{C}$ as in Fig. 4.3. (The set Ω_2 is obtained by reflecting Ω_1 by the map $x + \mathrm{i}y \mapsto -x + \mathrm{i}y$.) Their boundaries intersect only at the points $\pm\mathrm{i}$, the segment $I_+ = [1, \sqrt{3}]$ is contained in $\Omega_1 \cap b\Omega_2$, and the segment $I_- = [-\sqrt{3}, -1]$ is contained in $\Omega_2 \cap b\Omega_1$.

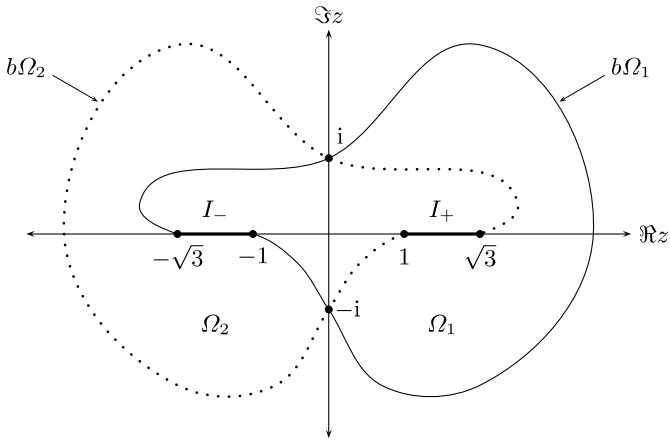

Fig. 4.3. Simply connected domains Ω_1 and Ω_2. (Modified from [526, p. 777, Fig. 1])

Consider the following sets in $\mathbb{C}^* \times \mathbb{C}$:

$$X_1 = \big\{(z,w)\colon z \in b\Omega_1,\ w = z^2 - c,\ c \in [0,1]\big\},$$
$$X_2 = \big\{(z,w)\colon z \in b\Omega_2,\ w \in [1,2]\big\}.$$

Observe that X_1 and X_2 are disjoint totally real annuli, X_1 is foliated by the boundaries of analytic discs $\Gamma_c = \{(z, z^2 - c) : z \in \overline{\Omega}_1\}$ $(c \in [0,1])$, and X_2 is foliated by the boundaries of analytic discs $\Lambda_c = \{(z, c) : z \in \overline{\Omega}_2\}$ $(c \in [1,2])$. Hence the curves Γ_c and Λ_c lie in the polynomial hulls $\widehat{X}_1$, $\widehat{X}_2$, respectively. In particular, the origin $(0,0) \in \Gamma_0$ belongs to the polynomial hull $\widehat{X}_1$. Note that all these curves intersect the complex line $\{0\} \times \mathbb{C}$, and it can be seen that the union $X_1 \cup X_2$ is $\mathcal{O}(\mathbb{C}^* \times \mathbb{C})$-convex. We now remove from X_1 and X_2 suitably chosen discs so that the remaining sets become discs whose union still contains $(0,0)$ in its polynomial hull. Take

$$S_1 = \big\{(z,w) : z \in I_-,\ w \in [1,2]\big\},\ \ S_2 = \big\{(z,w) : z \in I_+,\ w = z^2 - c,\ c \in [0,1]\big\}$$

and set

$$Y_1 = \overline{X_1 \backslash S_1}, \quad Y_2 = \overline{X_2 \backslash S_2}, \quad Y = Y_1 \cup Y_2.$$

It is easily verified that Y_1 and Y_2 are discs and that $S_1 \subset \widehat{X}_2$, $S_2 \subset \widehat{X}_1$. Hence neither S_1 nor S_2 can contain any peak points for the polynomial algebra $\mathcal{P}(X_1 \cup X_2)$. It follows that $\widehat{X_1 \cup X_2} = \widehat{Y_1 \cup Y_2} = \widehat{Y}$; in particular, $(0,0) \in \widehat{Y}$. (Note however that each of the two discs Y_1 and Y_2 is polynomially convex!) This establishes properties (i) and (ii) in the lemma.

The manifold $\mathbb{C}^* \times \mathbb{C}$ enjoys the holomorphic density property (Example 4.10.20 on p. 136). Let U be any nonempty open set in $\mathbb{C}^* \times \mathbb{C}$. Since Y is the disjoint union of two totally real discs, there exists a $\mathcal{O}(\mathbb{C}^* \times \mathbb{C})$-convex neighborhood $V \subset \mathbb{C}^* \times \mathbb{C}$ of Y and a holomorphic isotopy $\theta_t : V \to V_t \subset \mathbb{C}^* \times \mathbb{C}$ $(t \in [0,1])$ such that $\theta_0 = \mathrm{Id}_V$, $\theta_1(Y) \subset U$, and V_t is $\mathcal{O}(\mathbb{C}^* \times \mathbb{C})$-convex for every $t \in [0,1]$. (Essentially we contract each of the two discs Y_1, Y_2 to a small neighborhood of a point and then push them into U.) By Theorem 4.10.6 we can approximate θ_1 by an automorphism $g \in \mathrm{Aut}\,(\mathbb{C}^* \times \mathbb{C})$. If the approximation is close enough then $g(Y) \subset U$, and the inverse map $G = g^{-1}$ then satisfies property (iii). $\square$

Corollary 4.20.2. *There exists a Fatou-Bieberbach domain in $\mathbb{C}^* \times \mathbb{C}$ that is not Runge in $\mathbb{C}^2$.*

Proof. Pick a Fatou-Bieberbach domain D contained in $\mathbb{C}^* \times \mathbb{C}$ (see Example 4.3.9). Let Y be as in Lemma 4.20.1, and let $G \in \mathrm{Aut}\,(\mathbb{C}^* \times \mathbb{C})$ be such that the Fatou-Bieberbach domain $\Omega = G(D)$ satisfies $Y \subset \Omega \subset \mathbb{C}^* \times \mathbb{C}$. Since $\widehat{Y}_{\mathcal{O}(\mathbb{C}^2)}$ contains the origin $0 \in \mathbb{C}^2$, Ω is not Runge in $\mathbb{C}^2$. $\square$

By using non-Runge Fatou-Bieberbach domains, Wold also constructed non-Stein long $\mathbb{C}^2$'s, thereby resolving a question posed by J.-E. Fornæss [143, 144]. A complex manifold X of dimension n is said to be a *long* $\mathbb{C}^n$ if X is exhausted by an increasing sequence $X_0 \subset X_1 \subset X_2 \subset \cdots X = \cup_{j=0}^{\infty} X_j$ of domains X_j biholomorphic to $\mathbb{C}^n$.

Theorem 4.20.3. [527] *For every $n > 1$ there exists a non-Stein long $\mathbb{C}^n$.*

The question whether an increasing union of Stein manifolds is Stein has been called the *union problem*. Fornæss [143] gave a negative answer in dimension ≥ 3 by using Wermer's non-Runge polydiscs in $\mathbb{C}^3$ [510].

Proof. It suffices to give the proof for $n = 2$. Let $Y \subset \mathbb{C}^* \times \mathbb{C}$ be the compact set in Lemma 4.20.1. Choose a Fatou-Bieberbach domain $D \subset \mathbb{C}^* \times \mathbb{C}$ and a biholomorphic map $F\colon \mathbb{C}^2 \to D$ (see Example 4.3.9). Pick a compact set $K \subset \mathbb{C}^2$ with nonempty interior. By Lemma 4.20.1 (c) there exists an automorphism $G_0 \in \mathrm{Aut}\,(\mathbb{C}^* \times \mathbb{C})$ such that $Y \subset G_0(F(K))$. Then $\varphi_0 = G_0 \circ F\colon \mathbb{C}^2 \to \mathbb{C}^* \times \mathbb{C}$ is a Fatou-Bieberbach map with a non-Runge image. Continuing inductively we construct a sequence of injective holomorphic maps $\varphi_k = G_k \circ F\colon \mathbb{C}^2 \to \mathbb{C}^2$, where $G_k \in \mathrm{Aut}\,(\mathbb{C}^* \times \mathbb{C})$, such that

$$Y \subset K_{k+1} := \varphi_k \circ \cdots \circ \varphi_0(K), \quad k = 0, 1, \ldots. \tag{4.64}$$

Assume that the maps φ_k have been defined for $k = 0, 1, \ldots, n-1$. Since the set $F(K_n) \subset \mathbb{C}^* \times \mathbb{C}$ has nonempty interior, Lemma 4.20.1 (iii) furnishes an automorphism $G_n \in \mathrm{Aut}\,(\mathbb{C}^* \times \mathbb{C})$ such that $Y \subset G_n(F(K_n))$. Setting $\varphi_n = G_n \circ F$ we get (4.64) also for $k = n$, and the induction may proceed.

Let X be the direct limit of the system $\{\varphi_j\colon \mathbb{C}^2 \to \mathbb{C}^2 \colon j = 0, 1, \ldots\}$. Precisely, X is the disjoint union of countably many copies of $\mathbb{C}^2$, one for each $j = 0, 1, \ldots$, modulo the equivalence relation that identifies a point z in the j-th copy $X_j \cong \mathbb{C}^2$ with the point $\varphi_j(z) \in X_{j+1} \cong \mathbb{C}^2$, and hence also with $\varphi_{k-1} \circ \cdots \varphi_j(z) \in X_k$ whenever $k > j$. More precisely, for every $j = 0, 1, \ldots$ we have a biholomorphic map $\psi_j\colon X_j \to \mathbb{C}^2$ (a local chart on X) and an injective holomorphic map $\phi_j\colon X_j \to X_{j+1}$ (considered as an inclusion of X_j into X_{j+1}) such that the following diagram commutes:

$$\begin{array}{ccc} X_j & \xrightarrow{\phi_j} & X_{j+1} \\ {\scriptstyle \psi_j}\downarrow & & \downarrow{\scriptstyle \psi_{j+1}} \\ \mathbb{C}^2 & \xrightarrow{\varphi_j} & \mathbb{C}^2 \end{array}$$

Thus X contains each X_j as an open subset, and these sets exhaust X.

Let L denote the compact set $L = \psi_0^{-1}(K) \subset X_0 \subset X$. Using the identifications in X we have $L = \phi_{n-1} \circ \cdots \circ \phi_0(L) \subset X_n$ and

$$\psi_n(L) = \varphi_{n-1} \circ \cdots \circ \varphi_0(K) = K_n \subset \mathbb{C}^* \times \mathbb{C}, \quad n = 1, 2, \ldots.$$

Since $Y \subset \psi_n(L)$ by (4.64), it follows that the polynomial hull $\widehat{\psi_n(L)}$ contains the origin, and hence it is not contained in the domain $\psi_n(X_{n-1}) \subset \mathbb{C}^* \times \mathbb{C}$. Passing back to X_n via ψ_n we conclude that

$$\widehat{L}_{\mathcal{O}(X_n)} \not\subset X_{n-1}, \qquad n = 1, 2, \ldots.$$

Since we clearly have $\widehat{L}_{\mathcal{O}(X_n)} \subset \widehat{L}_{\mathcal{O}(X)}$ for every $n \in \mathbb{N}$, we see that $\widehat{L}_{\mathcal{O}(X)}$ is not contained in any of the domains X_n. As these domains exhaust X, the hull $\widehat{L}_{\mathcal{O}(X)}$ is noncompact, and hence X is not Stein. $\square$

On the other hand, assuming a Runge pair hypothesis we get the following.

Proposition 4.20.4. [522, Proposition 3] *If $X = \cup_{j=0}^{\infty} X_j$ is a long $\mathbb{C}^2$ (with $X_j \cong \mathbb{C}^2$ for every j) and if (X_j, X_{j+1}) is a Runge pair for every $j = 0, 1, \ldots$, then X is biholomorphic to $\mathbb{C}^2$.*

Proof (sketch). By using the Andersen-Lempert theorem (Theorem 4.9.1) one can inductively build a biholomorphism between X and $\mathbb{C}^2$. If $\psi_j \colon X_j \xrightarrow{\cong} \mathbb{C}^2$ and $\psi_{j+1} \colon X_{j+1} \xrightarrow{\cong} \mathbb{C}^2$ are biholomorphisms, then the Fatou-Bieberbach map $\varphi_{j+1} \circ \varphi_j^{-1}$ has Runge image and hence it can be approximated by an automorphism of $\mathbb{C}^2$; so φ_{j+1} can be corrected to approximate φ_j. For the details see [522, Proposition 3]. $\square$

Using the above construction and the fact that the automorphism group of any Stein manifold with the density property (such as $\mathbb{C}^* \times \mathbb{C}$) contains plenty of entire curves of automorphisms one can easily show the following result concerning holomorphic families of long $\mathbb{C}^n$'s.

Theorem 4.20.5. [183] *Fix an integer $n > 1$. Assume that S is a Stein manifold, $A = \cup_j A_j$ is a finite or countable union of closed complex subvarieties of S, and $B = \{b_j\}$ is a finite or countable set in $S \backslash A$. Then there exists a complex manifold X and a holomorphic submersion $\pi \colon X \to S$ onto S with the following properties:*

(i) *the fiber $X_s = \pi^{-1}(s)$ is a long $\mathbb{C}^n$ for every $s \in S$,*
(ii) *X_s is biholomorphic to $\mathbb{C}^n$ for every $s \in A$, and*
(iii) *X_s is non-Stein for every $s \in B$.*

In particular, for any two disjoint countable sets $A, B \subset \mathbb{C}$ there is a holomorphic family $\{X_s\}_{s \in \mathbb{C}}$ of long $\mathbb{C}^2$'s such that X_s is biholomorphic to $\mathbb{C}^2$ for all $s \in A$ and is non-Stein for all $s \in B$. This is particularly striking if the sets A and B are chosen to be everywhere dense in $\mathbb{C}$.

Problem 4.20.6. Is there a holomorphic family X_s of long $\mathbb{C}^2$'s, parametrized by the disc $\mathbb{D} = \{s \in \mathbb{C} \colon |s| < 1\}$ or the plane $\mathbb{C}$, such that X_s is not biholomorphic to $X_{s'}$ whenever $s \neq s'$?

4.21 Serre's Problem on Stein Bundles

The Oka-Grauert principle fails rather spectacularly for fiber bundles with Euclidean fibers whose structure group is not a finite dimensional Lie group. The following important problem in the development of Stein manifold theory was posed by J.-P. Serre:

If $E \to X$ is a holomorphic fiber bundle whose base X and fiber Y are Stein manifolds, is the total space E also Stein?

This holds e.g. for holomorphic vector bundles. The first counterexamples with the fiber $\mathbb{C}^n$ $(n > 1)$ and nonlinear transition automorphisms were given by H. Skoda [450, 452]. In the first example [450, Theorem] the base is the unit disc with eight discs removed. Around each deleted disc the bundle is defined by a holonomy automorphism of the fiber which, up to a possible permutation of the coordinates, is a multiplicative shear of the form (4.2). In Skoda's second example [452, Theorem 1] the base is the disc with two removed discs; around one of the deleted discs the holonomy automorphism of the fiber $\mathbb{C}^2$ is the entire shear $(z_1, z_2) \mapsto (z_1, z_2 e^{z_1})$, and for the other deleted disc it is the affine map $(z_1, z_2) \mapsto (\mathrm{i}z_2, z_1)$.

Next, Demailly [99, 100] gave the following example.

Theorem 4.21.1. *Let $\Omega = \{\zeta \in \mathbb{C}\colon 1 < |\zeta| < R\}$, and let $E \to \Omega$ be a holomorphic fiber bundle with fiber $\mathbb{C}^2$ whose holonomy automorphism is the polynomial Hénon map $h(z_1, z_2) = (z_2, -z_1 + z_2^k)$. If $k \in \mathbb{N}$ is large enough (for a fixed r), or if $r > 1$ is large enough (for a fixed $k \geq 2$) then E is non-Stein.*

Demailly showed that, under stated conditions, E does not admit any plurisubharmonic functions other than those coming from the base Ω. He also gave counterexamples where the base is the disc or the plane $\mathbb{C}$, the fiber is $\mathbb{C}^2$, and the gluing automorphisms are entire. (See also [86].)

The question that remained open for another twenty years was whether there exists a non-Stein fiber bundle with fiber $\mathbb{C}^n$ and a contractible Stein base whose transition maps are polynomial automorphisms of the fiber. Such examples were constructed in 2007 by J.-P. Rosay.

Theorem 4.21.2. [419] *There exists a holomorphic fiber bundle with fiber $\mathbb{C}^2$ over the disc $\mathbb{D}$ (or the plane $\mathbb{C}$) whose transition maps are polynomial automorphisms of $\mathbb{C}^2$ and whose total space is not Stein.*

Hence Rosay's bundle is nontrivial even though the base is contractible.

Here is the main idea of the construction. Consider the Hénon map $h(z_1, z_2) = (z_2, -z_1 + z_2^k)$ used in Demailly's example of a non-Stein fiber bundle with fiber $\mathbb{C}^2$ over an annulus. Note that h is the composition of the linear map $\alpha(z_1, z_2) = (z_2, -z_1)$ and of the shear $\beta(z_1, z_2) = (z_1, z_2 + z_1^k)$. Each of these maps is an element of a one-parameter group of polynomial automorphisms of $\mathbb{C}^2$. If we remove a disc from Ω, we can define the same bundle (now over a disc with two holes) by using each of the maps α, β as the holonomy map around one of the two holes. The main observation is that any hole whose holonomy automorphism belongs to a one-parameter group of automorphisms can be 'filled in', in the sense that the bundle extends trivially over the hole by using the transition maps from the given one-parameter group of automorphisms. Here is the precise result.

Proposition 4.21.3. [419, Proposition] *Let Ω be a circle domain in $\mathbb{C}$ (a disc with finitely many closed pairwise disjoint discs removed). Assume that $E \to \Omega$ is a holomorphic fiber bundle with fiber $\mathbb{C}^n$ and with transition maps that are locally independent of the base point and belong to a group $G \subset \operatorname{Aut}\mathbb{C}^n$ that is generated by complex one-parameter subgroups. Then there exists a holomorphic fiber bundle $\widetilde{E} \to \mathbb{P}^1$ with fiber $\mathbb{C}^n$ and with transition maps in G (allowed to locally depend on the base point) such that $\widetilde{E}|_\Omega \cong E$.*

Remark 4.21.4. In Rosay's paper, Ω is a domain bounded by finitely many Jordan curves; however, every such is conformally equivalent to a circle domain by a theorem of Koebe [302], so we stick to the simpler formulation. A bundle satisfying the stated conditions is *flat* (Example 1.4.8 on p. 13); such bundle is holomorphically trivial over every simply connected domain. $\square$

Proof. Let $\Omega = \mathbb{P}^1 \setminus \cup_{j=0}^N \overline{D}_j$, where $\overline{D}_j$ are pairwise disjoint discs. Fix $j \in \{0, 1, \ldots, N\}$ and let $D = D_j$; we explain how to extend E over D. We change the coordinates in $\mathbb{P}^1$ so that the hole D becomes the unit disc $\mathbb{D}$. Let $V = \{\zeta \in \mathbb{C} : 1 < |\zeta| < 1 + \epsilon\}$, where $\epsilon > 0$ is chosen small enough such that $V \subset \Omega$ and $\overline{V}$ does not intersect any of the other closed discs $\overline{D}_k$ with $k \neq j$. Write $V = V^+ \cup V^-$ where

$$V^+ = \left\{\zeta \in V : \Im\zeta > -\frac{1}{2}\right\}, \quad V^- = \left\{\zeta \in V : \Im\zeta < +\frac{1}{2}\right\}.$$

Then $V^+ \cap V^- = \omega_0 \cup \omega_1$ where ω_0, ω_1 are disjoint contractible open sets.

Since E is defined by locally constant transition maps and the sets $V^\pm$ are simply connected, the restricted bundles $E|_{V^+}$ and $E|_{V^-}$ are trivial. Furthermore, the conditions imply that there are trivializations

$$\Theta^+ : E|_{V^+} \xrightarrow{\cong} V^+ \times \mathbb{C}^n, \quad \Theta^- : E|_{V^-} \xrightarrow{\cong} V^- \times \mathbb{C}^n$$

such that the transition map equals

$$\big(\Theta^- \circ (\Theta^+)^{-1}\big)(\zeta, z) = \widetilde{T}_j(\zeta, z) = \big(\zeta, T_j(z)\big), \quad \zeta \in \omega_j \ (j = 0, 1),$$

where $T_0, T_1 \in G$. Replacing the trivialization Θ^+ by $\widetilde{T}_0 \circ \Theta^+$, T_1 gets replaced by $T = T_1 \circ T_0^{-1} \in G$ and T_0 becomes the identity. Letting $\widetilde{T}(\zeta, z) = (\zeta, T(z))$, the above compatibility conditions read:

$$\Theta^+ = \Theta^- \text{ on } \omega_0 \times \mathbb{C}^n, \quad \widetilde{T} \circ \Theta^+ = \Theta^- \text{ on } \omega_1 \times \mathbb{C}^n.$$

Note that T is the holonomy automorphism of the fiber associated to the loop $\gamma(t) = re^{2\pi i t}$ for any $r \in (1, 1 + \epsilon)$ (see Example 1.4.8).

At this point we split the proof in two cases.

Case 1: There is a one-parameter group $(S^t)_{t \in \mathbb{C}} \subset G$ such that $T = S^1$. In this case the bundle extends trivially over $\overline{D}$. For gluing the trivial bundle

$(\overline{D}\cup V)\times\mathbb{C}^n$ with $E|_V$ (using the local trivializations $\Theta^{\pm}$ on the latter bundle) we need to find holomorphic automorphisms

$$\Phi^+\colon V^+\times\mathbb{C}^n \xrightarrow{\cong} V^+\times\mathbb{C}^n, \quad \Phi^-\colon V^-\times\mathbb{C}^n \xrightarrow{\cong} V^-\times\mathbb{C}^n$$

whose fiber maps belong to G and we have the compatibility conditions

$$\Phi^+ = \Phi^- \quad \text{on } \omega_0\times\mathbb{C}^n, \quad \widetilde{T}\circ\Phi^+ = \Phi^- \quad \text{on } \omega_1\times\mathbb{C}^n. \tag{4.65}$$

Let λ^+ and λ^- be holomorphic functions defined respectively on V^+ and V^-, such that $\lambda^+ + \lambda^- = 0$ on ω_0 and $\lambda^+ + \lambda^- = -1$ on ω_1. (This is a classical Cousin problem that here is elementary to solve by using logarithms.) Set

$$\Phi^+(\zeta,z) = \big(\zeta, S^{\lambda^+(\zeta)}(z)\big), \quad \Phi^-(\zeta,z) = \big(\zeta, S^{-\lambda^-(\zeta)}(z)\big).$$

(Here comes the dependence on ζ.) For $\zeta\in\omega_0$ we have $S^{\lambda^+(\zeta)} = S^{-\lambda^-(\zeta)}$ which gives the first equality in (4.65). For $\zeta\in\omega_1$ we have

$$T\circ S^{\lambda^+(\zeta)} = S^1\circ S^{\lambda^+(\zeta)} = S^{1+\lambda^+(\zeta)} = S^{-\lambda^-(\zeta)}.$$

This gives the second equality in (4.65) and settles Case 1.

Case 2: The general case. Using the notation introduced above, we have

$$T_1\circ T_0^{-1} = T = A_k\circ\cdots\circ A_1,$$

where each A_p $(p = 1,\dots,k)$ is an element of a one-parameter group of automorphisms in G. Choose the annulus V as before. Fix points $b_0 = -1-\epsilon$, $a_{k+1} = 1+\epsilon$, and

$$a_1 = -1 < b_1 < a_2 < b_2 < \cdots < a_p < b_p < a_{p+1} < \cdots < b_k = 1.$$

Let W be the complement in $\mathbb{D}_{1+\epsilon}$ of the closed discs Δ_p of diameter $[a_p,b_p]$ $(p = 1,\dots,k)$; so W is a k-connected region that contains V and that is contained in $V\cup\overline{D}$. Set

$$W^+ = W\cap\{\Im\zeta\ge 0\}, \quad W^- = W\cap\{\Im\zeta\le 0\}.$$

Observe that $E|_V$ extends to a fiber bundle E' over W that is obtained by gluing $W^+\times\mathbb{C}^n$ with $W^-\times\mathbb{C}^n$ over each component $[b_p,a_{p+1}]$ of the intersection $W^+\cap W^-$ by the automorphism $A_p\circ\cdots\circ A_1$. (This equals the identity if $p=0$, and equals T if $p=k$.) For the extended bundle E', over each hole Δ_p of W we are in Case 1 since the corresponding holonomy automorphism

$$(A_p\circ\cdots A_1)\circ(A_{p-1}\circ\cdots A_1)^{-1} = A_p$$

belongs to a one-parameter subgroup of G. By Case 1, E' extends to a fiber bundle over $\Omega\cup\overline{D}$ with transition automorphisms belonging to G. $\square$

'But the novel, the novel,' she shouted to the Master, 'take the novel with you wherever you are flying.'

'I don't have to,' replied the Master, 'I remember it by heart.'

M. A. Bulgakov, *The Master and Margarita*

5

Oka Manifolds

We begin by outlining the main ideas and results of the Oka-Grauert theory and then explain how these lead to the theory of Oka manifolds. The main result of the chapter, Theorem 5.4.4, subsumes the classical Oka-Grauert principle concerning principal bundles over Stein spaces. Gromov's theory for non-locally trivial holomorphic submersions is treated in the following chapter.

5.1 A Historical Introduction to the Oka Principle

The Oka principle has its origin in seminal works of Kiyoshi Oka (1901-78), one of the pioneers of modern complex analysis. In his impressive series of papers during 1936–1953 Oka invented new methods of constructing global analytic objects from local ones. In the sequel we mention several of his key papers; they are also available in Springer's 1984 edition [388].

The notion of cohomology with coefficients in a sheaf has its origin in the paper of P. Cousin [94] after whom the *Cousin problems* are named. Following the motivating works of Henri Cartan [71, 72], Oka solved the first Cousin problem on a reasonably general class of domains [382]. The sheaf theoretic formulation of this result is a special case of Cartan's Theorem B: On a domain of holomorphy (or, more generally, on a Stein space) we have $H^1(X;\mathcal{O}_X) = 0$.

The beginning of Oka theory is Oka's theorem from 1939 [384] that a second (multiplicative) Cousin problem on a domain of holomorphy is solvable with holomorphic functions if it is solvable with continuous functions; hence a holomorphic line bundle over such a domain is holomorphically trivial if it is topologically trivial. In modern language, the inclusion $\mathcal{O}_X^* \hookrightarrow \mathcal{C}_X^*$ of the sheaf of nonvanishing holomorphic functions into the sheaf of nonvanishing continuous functions induces an isomorphism $H^1(X;\mathcal{O}_X^*) \cong H^1(X;\mathcal{C}_X^*)$ on any complex space satisfying $H^1(X;\mathcal{O}_X) = H^2(X;\mathcal{O}_X) = 0$; this holds in particular on any Stein space (see §5.2).

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In the 1930's major new ideas were brought to analysis and topology by the introduction of Morse theory and fiber bundles. In 1940 Cartan started studying local bases of ideals of holomorphic functions and matrices transforming one set of generators to another set of generators [73]. Based on these developments and the cohomological formulation of Oka's result [384] by Cartan and Serre, certain non-abelian generalizations were obtained by J. Frenkel [201, 202, 203].

In his seminal papers of 1957-58, Grauert extended Oka's and Frenkel's theorem to principal holomorphic fiber bundles over Stein spaces with an arbitrary Lie group as the fiber, showing that the holomorphic classification of such bundles coincides with the topological classification [221, 222, 223]. Henri Cartan called this the *Oka-Grauert principle*, and Rene Thom extolled it as *the most beautiful principle in analysis.* (An exposition of Grauert's work is found in Cartan's paper [78] which is also available in [79].) In the monograph [228] one finds on p. 145 the following formulation of the Oka-Grauert principle:

Analytic problems on Stein manifolds which can be cohomologically formulated have only topological obstructions.

During 1960's Forster and Ramspott and others found many extensions and applications of the Oka-Grauert principle (see §7.4 – §7.5).

Another major event of mid-1960's was the development of L^2-methods for solving the nonhomogeneous $\overline{\partial}$-equation by Kohn [304, 305], Hörmander [266] and Andreotti and Vesentini [22]. This is a powerful tool for solving analytic problems by correcting a smooth solution to a holomorphic one via an inhomogeneous Cauchy-Riemann equation. The L^2-method has the potential of playing an important role in quantitative Oka theory (see §7.11).

In spite of the progress in 1960's, many natural problems in Stein geometry remained elusive since they could not be formulated as cohomological problems involving fiber bundles with homogeneous fibers. In his paper [152] on embedding Stein manifolds in Euclidean spaces, Forster made the correct conjecture on the minimal embedding dimension (which was confirmed much later by Eliashberg and Gromov [133] and Schürmann [436]), but he realized that the solution would entail constructing holomorphic sections of certain holomorphic submersions with nonhomogeneous fibers.

An important contribution from the conceptual point of view was made in 1986 by Henkin and Leiterer [258, 259] who proved Grauert's theorem by using the 'bumping method'.

The subject was revitalized in 1989 by Mikhael Gromov whose paper [237] marks the beginning of modern Oka theory. The emphasis shifted from the cohomological to the homotopy theoretic aspect, focusing on those analytic properties of a complex manifold Y which insure that every continuous map from a Stein space X to Y is homotopic to a holomorphic map, with natural additions concerning approximation and interpolation of maps that are motivated by the extension and approximation theorems for holomorphic functions

on Stein spaces. Specifically, for which complex manifolds Y is the natural inclusion $\mathcal{O}(X,Y) \hookrightarrow \mathcal{C}(X,Y)$ of the space of holomorphic maps into the space of continuous maps a weak homotopy equivalence? The analogous questions are considered for sections of holomorphic submersions $Z \to X$ over Stein spaces. Gromov's main result is that the existence of a holomorphic fiber dominating spray on $Z|_U$ over small open subsets U of the Stein base manifold X implies all forms of the Oka principle for sections $X \to Z$ (Theorem 6.2.2 on p. 243). Dominating holomorphic sprays are used as a replacement of the exponential map in the linearization and patching problems in Oka-Grauert theory. Gromov's work also illuminates connections between Oka theory and the *homotopy principle*, a major subject in differential topology in geometry originating in the works of Smale and Hirsch on classification of smooth immersions; see [236].

Although ellipticity is a useful geometric sufficient condition for validity of the Oka principle, it is not known whether it is also necessary and whether it satisfies any nontrivial functorial properties. Gromov asked whether the Oka principle for maps from Stein spaces to a certain complex manifold Y is characterized by a Runge approximation property for entire maps $\mathbb{C}^n \to Y$ on a certain family of simple compact subsets of $\mathbb{C}^n$ [237, 3.4.(D), p. 881]. This was confirmed in [176], and in a more decisive way in [179]. The main result of this chapter is that a complex manifold Y satisfies all natural Oka properties if (and only if) any holomorphic map $K \to Y$ from a compact convex set K in $\mathbb{C}^n$ is a uniform limit of entire maps $\mathbb{C}^n \to Y$ (Theorem 5.4.4). A complex manifold enjoying any of these equivalent properties is said to be an *Oka manifold*. Since every complex homogeneous manifold is an Oka manifold, Theorem 5.4.4 includes the classical Oka-Grauert theory and also its extensions by Forster and Ramspott.

5.2 Cousin Problems and Oka's Theorem

To motivate the discussion leading to Oka's theorem, we begin by considering the *first Cousin problem* (see [94]) whose origin is the following *Mittag-Leffler problem*: Given a collection $\{(U_j, m_j)\}$, where $\mathcal{U} = \{U_j\}$ is an open cover of a complex space X and $m_j \in \mathcal{M}(U_j)$ is a meromorphic function on U_j such that every difference $f_{ij} = m_i|_{U_{ij}} - m_j|_{U_{ij}}$ is holomorphic on $U_{ij} = U_i \cap U_j$, find a global meromorphic function $m \in \mathcal{M}(X)$ on X such that $m|_{U_j} - m_j$ is holomorphic on U_j for every j. The standard solution proceeds via the associated *Cousin I problem*. The collection $f_{ij} \in \mathcal{O}(U_{ij})$ is a 1-cocycle on $\mathcal{U}$ with values in the sheaf $\mathcal{O}$, meaning that for each triple of indexes i, j, k we have

$$f_{ij} + f_{jk} + f_{ki} = 0 \quad \text{on } U_{ijk} = U_i \cap U_j \cap U_k.$$

Given such a 1-cocycle $\{f_{ij}\}$, the Cousin I problem asks for a collection of holomorphic functions $f_j \in \mathcal{O}(U_j)$ (a 0-cochain) such that

$$f_i - f_j = f_{ij} \quad \text{on } U_{ij}.$$

If a solution exists, then $\{f_{ij}\}$ is said to be a *1-coboundary*, $\{f_{ij}\} = \delta(\{f_j\})$. From $f_i - f_j = f_{ij} = m_i - m_j$ on U_{ij} we see that the collection of meromorphic functions $m'_j = m_j - f_j \in \mathcal{M}(U_j)$ satisfies $m'_i = m'_j$ on U_{ij}, and hence it defines a global meromorphic function on X solving the Mittag-Leffler problem.

Let us denote by $Z^1(\mathcal{U};\mathcal{O})$ the Abelian group of all 1-cocycles on $\mathcal{U}$ with coefficients in $\mathcal{O}$. By $C^0(\mathcal{U};\mathcal{O})$ we denote the Abelian group of all collections $\{f_j\}$ of holomorphic functions $f_j \in \mathcal{O}(U_j)$ (these are called 0-cochains on $\mathcal{U}$ with coefficients in $\mathcal{O}$). The coboundary homomorphism $\delta\colon C^0(\mathcal{U};\mathcal{O}) \to Z^1(\mathcal{U};\mathcal{O})$ is $\delta(\{f_j\}) = \{f_{ij}\}$ where $f_{ij} = f_i - f_j$. The quotient group

$$H^1(\mathcal{U};\mathcal{O}) = Z^1(\mathcal{U};\mathcal{O})/\delta(C^0(\mathcal{U};\mathcal{O}))$$

is the *1-st Čech cohomology group* of the cover $\mathcal{U}$ with coefficients in $\mathcal{O}$. Given a refinement $\mathcal{V} = \{V_k\}$ of the cover $\mathcal{U}$ (every set $V_k \in \mathcal{V}$ is contained in some $U_{i(k)} \in \mathcal{U}$), there is a natural homomorphism $H^1(\mathcal{U};\mathcal{O}) \to H^1(\mathcal{V};\mathcal{O})$ induced by restricting a cochain (or a cocycle) over $\mathcal{U}$ to one over $\mathcal{V}$. The *1-st cohomology group of X with coefficients in $\mathcal{O}$* is the direct limit

$$H^1(X;\mathcal{O}) = \varinjlim_{\mathcal{U}} H^1(\mathcal{U};\mathcal{O}).$$

Thus $H^1(X;\mathcal{O}) = 0$ means that every 1-cocycle becomes a 1-coboundary on a refinement, so every Mittag-Leffler problem is solvable. If X is a complex manifold then the group $H^1(X;\mathcal{O})$ is isomorphic to the Dolbeault group $H^{0,1}_{\bar\partial}(X)$, and these groups vanish on any Stein manifold (Theorem 2.4.6). If $\mathcal{U}$ is a *Leray cover* (meaning that the cohomology groups of all sets $U_j \in \mathcal{U}$ and of their finite intersections vanish; this holds if every U_j is Stein) then the natural homomorphism $H^1(\mathcal{U};\mathcal{O}) \to H^1(X;\mathcal{O})$ is an isomorphism.

Let us now look at the *second (multiplicative) Cousin problem* which arises from the problem of finding holomorphic functions with prescribed zeros, or meromorphic functions with prescribed zeros and poles. Consider an open cover $\mathcal{U} = \{U_i\}_{i\in I}$ of a complex space X and a collection of holomorphic functions $g_i \in \mathcal{O}(U_i)$ that do not vanish identically on any connected component of U_i. The collection (g_i) defines a divisor D on X if for any pair of indexes $i, j \in I$ there exists a nowhere vanishing holomorphic function $f_{ij} \in \mathcal{O}(U_{ij})$ such that $g_i = f_{ij} g_j$ holds on U_{ij}. (See Example 1.5.7 on p. 16.) The problem is to find a holomorphic function $f \in \mathcal{O}(X)$ that defines the divisor D; that is, such that f/g_i is a nonvanishing holomorphic function on U_i for every $i \in I$. A solution proceeds via the second Cousin problem: Given a collection $\{f_{ij}\}$ of nonvanishing holomorphic functions $f_{ij}\colon U_{ij} \to \mathbb{C}^*$ satisfying the 1-cocycle condition

$$f_{ii} = 1, \quad f_{ij} f_{ji} = 1, \quad f_{ij} f_{jk} f_{ki} = 1$$

on the respective sets where all functions are defined, find nonvanishing holomorphic functions $f_j\colon U_j \to \mathbb{C}^*$ such that

$$f_i = f_{ij} f_j \quad \text{on } U_{ij}.$$

If such f_i exist then $g_i/f_i = g_j/f_j$ on U_{ij} for every $i, j \in I$ which defines a solution $f \in \mathcal{O}(X)$. As before we define the cohomology groups

$$H^1(\mathcal{U}; \mathcal{O}^*) = Z^1(\mathcal{U}; \mathcal{O}^*)/\delta(C^0(\mathcal{U}; \mathcal{O}^*)), \quad H^1(X; \mathcal{O}^*) = \varinjlim H^1(\mathcal{U}; \mathcal{O}^*).$$

We can now state (an extension of) Oka's theorem.

Theorem 5.2.1. [384] *Assume that X is a complex space satisfying the condition $H^1(X; \mathcal{O}) = 0$. Let $\mathcal{U} = \{U_j\}$ be a cover of X by simply connected Stein open sets, and let $\{f_{ij}\} \in Z^1(\mathcal{U}; \mathcal{O}^*)$ be a multiplicative 1-cocycle. If there exist nonvanishing continuous functions $c_j \colon U_j \to \mathbb{C}^*$ satisfying $c_i = f_{ij} c_j$ on U_{ij} for every $i, j \in I$, then there also exist nonvanishing holomorphic functions $f_j \colon U_j \to \mathbb{C}^*$ such that $f_i = f_{ij} f_j$ (i.e., $\{f_{ij}\}$ is a coboundary).*

Proof. Since the sets U_j are simply connected, there exist continuous functions $g_j \colon U \to \mathbb{C}$ such that $c_j = \mathrm{e}^{g_j}$. Setting $g_{ij} = g_i - g_j$ on U_{ij}, we have $\mathrm{e}^{g_{ij}} = \mathrm{e}^{g_i - g_j} = c_i/c_j = f_{ij}$ and hence g_{ij} is holomorphic. The collection g_{ij} clearly satisfies the additive 1-cocycle condition, so $\{g_{ij}\} \in Z^1(\mathcal{U}; \mathcal{O})$. From $H^1(X; \mathcal{O}) = 0$ and the assumption that U_j are Stein we infer that $H^1(\mathcal{U}; \mathcal{O}) = 0$. Hence $g_{ij} = h_i - h_j$ for some holomorphic functions $h_j \colon U_j \to \mathbb{C}$. The collection $f_j = \mathrm{e}^{h_j} \colon U_j \to \mathbb{C}^*$ of nonvanishing holomorphic functions then satisfies $f_i/f_j = \mathrm{e}^{h_i - h_j} = \mathrm{e}^{g_{ij}} = f_{ij}$. $\square$

We now give a cohomological formulation and proof of Oka's theorem. Let $\mathcal{C} = \mathcal{C}_X$ denote the sheaf of continuous functions and $\mathcal{C}^* = \mathcal{C}_X^*$ the sheaf of nonvanishing continous functions on a complex space X.

Theorem 5.2.2. *If $(X, \mathcal{O})$ is a complex space satisfying $H^1(X; \mathcal{O}) = 0$ then the homomorphism $H^1(X; \mathcal{O}^*) \to H^1(X; \mathcal{C}^*)$, induced by the sheaf inclusion $\mathcal{O}^* \hookrightarrow \mathcal{C}^*$, is injective. In particular, if a Cousin II problem is solvable by continuous functions then it is solvable by holomorphic functions. If in addition we have $H^2(X; \mathcal{O}) = 0$ then the above map is an isomorphism.*

Proof. Let $\sigma(f) = \mathrm{e}^{2\pi \mathrm{i} f}$. Consider the *exponential sheaf sequence*:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \mathbb{Z} & \longrightarrow & \mathcal{O} & \xrightarrow{\sigma} & \mathcal{O}^* & \longrightarrow & 1 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \mathbb{Z} & \longrightarrow & \mathcal{C} & \xrightarrow{\sigma} & \mathcal{C}^* & \longrightarrow & 1 \end{array} \tag{5.1}$$

Since $H^1(X; \mathcal{C}) = 0 = H^2(X; \mathcal{C})$, the relevant portion of the long exact sequence on cohomology reads:

$$\begin{array}{ccccccccc} H^1(X; \mathbb{Z}) & \longrightarrow & H^1(X; \mathcal{O}) & \longrightarrow & H^1(X; \mathcal{O}^*) & \xrightarrow{c_1} & H^2(X; \mathbb{Z}) & \longrightarrow & H^2(X; \mathcal{O}) \\ & & \downarrow & & \downarrow & & \| & & \downarrow \\ & & 0 & \longrightarrow & H^1(X; \mathcal{C}^*) & \xrightarrow{c_1} & H^2(X; \mathbb{Z}) & \longrightarrow & 0 \end{array} \tag{5.2}$$

Hence the map in the bottom row is an isomorphism $H^1(X;\mathcal{C}^*) \cong H^2(X;\mathbb{Z})$. If $H^1(X;\mathcal{O}) = 0$ then the *1-st Chern class map* c_1 in the first row is injective:

$$0 \longrightarrow H^1(X;\mathcal{O}^*) \xrightarrow{c_1} H^2(X;\mathbb{Z}) \cong H^1(X;\mathcal{C}^*).$$

If also $H^2(X;\mathcal{O})$ then this map is an isomorphism. If $H^q(X;\mathcal{O}) = 0$ for all $q > 0$ then we have $H^p(X;\mathcal{O}^*) \cong H^p(X;\mathcal{C}^*) \cong H^{p+1}(X;\mathbb{Z})$ for all $p > 0$. $\square$

Recall from §1.5 that $\mathrm{Vect}^1_{\mathrm{holo}}(X) = H^1(X;\mathcal{O}^*) = \mathrm{Pic}(X)$ is the group of all equivalence classes of holomorphic line bundles over X, also called the *Picard group* of X. Likewise, $\mathrm{Vect}^1_{\mathrm{top}}(X) = H^1(X;\mathcal{C}^*)$ is the group of all equivalence classes of topological complex line bundles over X. Hence Theorem 5.2.2 has the following interpretation.

Corollary 5.2.3. *If $H^1(X;\mathcal{O}) = H^2(X;\mathcal{O}) = 0$ then the natural map*

$$\mathrm{Vect}^1_{\mathrm{holo}}(X) \longrightarrow \mathrm{Vect}^1_{\mathrm{top}}(X)$$

is an isomorphism. In particular, a holomorphic line bundle over X is topologically trivial if and only if it is holomorphically trivial.

5.3 The Oka-Grauert Principle

Oka's theorem was generalized to vector bundles of arbitrary rank over Stein spaces by Grauert [222, 223] (see also Cartan [78]). The following is one of the most frequently used results in complex geometry.

Theorem 5.3.1. (i) *Every topological complex vector bundle over a Stein space admits an equivalent holomorphic vector bundle structure.*

(ii) *Two holomorphic vector bundles over a Stein space that are equivalent as topological complex vector bundles are also holomorphically equivalent.*

In short, the natural map $\mathrm{Vect}^r_{\mathrm{holo}}(X) \to \mathrm{Vect}^r_{\mathrm{top}}(X)$ of equivalence classes of rank r complex vector bundles over X is a bijection for every $r \in \mathbb{N}$.

Every holomorphic vector bundle on an open Riemann surface is holomorphically trivial.

For vector bundles of rank $r > 1$ the cohomological proof in §5.2 fails due to noncommutativity of the relevant cohomological groups. We describe an alternative approach. Assume that $E \to X$ is a topological complex vector bundle with fiber $\mathbb{C}^r$. Using a partition of unity one can embed E as a topological complex vector subbundle in a trivial bundle $X \times \mathbb{C}^N$. Let $G_{r,N}$ denote the complex Grassmann manifold whose elements are r-dimensional complex linear subspaces of $\mathbb{C}^N$ (Example 1.2.8 on p. 7). Over $G_{r,N}$ we have the universal bundle $U_{r,N} \to G_{r,N}$ whose fiber over $\lambda \in G_{r,N}$ consists of all vectors

$v \in \mathbb{C}^N$ belonging to the subspace $\lambda \subset \mathbb{C}^N$ (Example 1.5.3). Let $f\colon X \to G_{r,N}$ denote the continuous map sending the point $x \in X$ to the subspace $E_x \subset \mathbb{C}^N$, considered as an element of $G_{r,N}$. Then E is isomorphic to $f^*U_{r,N}$, the pull-back by f of the universal bundle $U_{r,N} \to G_{r,N}$. If $f_t\colon X \to G_{r,N}$ ($t \in [0,1]$) is a homotopy of continuous maps then the bundles $f_t^*U_{r,N}$ are topologically isomorphic to each other (see e.g. [270, 458]). Furthermore, the pull-back by a holomorphic map is a holomorphic vector bundle over X. Therefore Theorem 5.3.1 (i) holds if we can prove that

Every continuous map $X \to G_{r,N}$ is homotopic to a holomorphic map.

Since $G_{r,N}$ is $GL_N(\mathbb{C})$-homogeneous, the above statement is a special case of Grauert's results; see Theorem 5.3.2 below.

Consider now the equivalence problem for holomorphic vector bundles E, E' of rank r over X. There is an open cover $\mathcal{U} = \{U_j\}$ of X and holomorphic vector bundle isomorphisms $\theta_j\colon E|_{U_j} \to U_j \times \mathbb{C}^r$, $\theta'_j\colon E'|_{U_j} \to U_j \times \mathbb{C}^r$. Let

$$g_{ij}\colon U_{ij} = U_i \cap U_j \to GL_r(\mathbb{C}), \quad g'_{ij}\colon U_{ij} \to GL_r(\mathbb{C})$$

denote the holomorphic transition maps for the two bundles. Precisely,

$$\theta_i \circ \theta_j^{-1}(x,v) = \big(x, g_{ij}(x)v\big), \quad x \in U_{ij},\ v \in \mathbb{C}^r,$$

and likewise for E'. A complex vector bundle isomorphism $\Phi\colon E \to E'$ is given by a collection of isomorphisms $\Phi_j\colon U_j \times \mathbb{C}^r \to U_j \times \mathbb{C}^r$,

$$\Phi_j(x,v) = \big(x, \phi_j(x)v\big), \quad x \in U_j,\ v \in \mathbb{C}^r,$$

with $\phi_j(x) \in GL_r(\mathbb{C})$, satisfying the compatibility conditions

$$\phi_i = g'_{ij}\phi_j g_{ij}^{-1} = g'_{ij}\phi_j g_{ji} \quad \text{on } U_{ij}. \tag{5.3}$$

Let $Z \to X$ denote the holomorphic fiber bundle with fiber $G = GL_r(\mathbb{C})$ which is defined by the transition maps (5.3). Precisely, $Z|_{U_j} \cong U_j \times G$ for each j, and an element $(x,v) \in U_j \times G$ with $x \in U_{ij}$ is identified with the element $(x,v') \in U_i \times G$, where $v' = g'_{ij}(x)\, v\, g_{ji}(x)$. A collection of maps $\phi_j\colon U_j \to G$ satisfying (5.3) is then a section $X \to Z$; hence complex vector bundle isomorphisms $E \to E'$ correspond to section $X \to Z$. Part (ii) of Theorem 5.3.1 now follows from the following result of Grauert [222, 223] (for Lie group fibers) and Grauert and Kerner [226] and Ramspott [406] (for bundles with homogeneous fibers).

Theorem 5.3.2. *If X is a Stein space and if $\pi\colon Z \to X$ is a holomorphic fiber bundle with a complex homogeneous fiber whose structure group is (reducible to) a complex Lie group acting transitively on the fiber, then the inclusion $\Gamma_{\mathcal{O}}(X;Z) \hookrightarrow \Gamma(X;Z)$ of the space of holomorphic sections into the space of continuous sections is a weak homotopy equivalence. In particular, every continuous section $X \to Z$ is homotopic to a holomorphic section.*

Theorem 5.3.2 is a special case of Theorem 5.4.4 in the following section; see also Corollary 5.4.6. The most direct proof of Grauert's theorem was given by Leiterer [329]. Heunemann [261] and Sebbar [437] obtained the Oka principle for holomorphic vector bundles on strongly pseudoconvex Stein domains which are continuous (resp. smooth) up to the boundary (see also Theorem 5.4.12 below). An extension of Theorem 5.3.1 to 1-convex manifolds is due to Henkin and Leiterer [259]; for complex Banach-valued vector bundles over Stein spaces see Bungart [61] and Leiterer and Vâjâitu [332]. An equivariant version of the Oka-Grauert principle was proved by Heinzner and Kutzschebauch [252]. For further results on Oka-Grauert theory see §7.2–§7.6.

5.4 What is an Oka Manifold?

The simplest of several equivalent characterizations of an Oka manifold, a class of complex manifolds that was introduced in [179], is the following Runge approximation property for maps from Euclidean spaces.

Definition 5.4.1. *A complex manifold Y is an* Oka manifold *if every holomorphic map $f\colon K \to Y$ from (a neighborhood of) a compact convex set $K \subset \mathbb{C}^n$ can be approximated uniformly on K by entire maps $\mathbb{C}^n \to Y$.*

It is convenient to reduce the testing family of compact convex sets in Def. 5.4.1. Let $z = (z_1, \ldots, z_n)$ be complex coordinates on $\mathbb{C}^n$, with $z_j = x_j + \mathrm{i}y_j$.

Definition 5.4.2. *A* special convex set *in $\mathbb{C}^n$ is a compact convex set of the form*

$$K = \{z \in Q\colon y_n \le h(z_1, \ldots, z_{n-1}, x_n)\}, \tag{5.4}$$

where Q is a cube of the form

$$Q = \{z \in \mathbb{C}^n\colon |x_j| \le a_j,\ |y_j| \le b_j,\ j = 1, \ldots, n\} \tag{5.5}$$

and h is a continuous concave function with values in $(-b_n, b_n)$. Such pair (K, Q) is called a special convex pair *in $\mathbb{C}^n$.*

Definition 5.4.3. [176] *A complex manifold Y enjoys the n-*dimensional Convex Approximation Property (CAP_n) *if for every special convex pair (K, Q) in $\mathbb{C}^n$, any holomorphic map $f\colon K \to Y$ can be uniformly approximated by holomorphic maps $Q \to Y$. A manifold Y satisfies* CAP *if it satisfies* CAP_n *for all $n \in \mathbb{N}$.*

By definition an Oka manifold satisfies CAP. The converse implication follows from Theorem 5.4.4 applied with $X = \mathbb{C}^n$, K a compact convex set in $\mathbb{C}^n$ and $P = \{p\}$ is a singleton. Hence a complex manifold is an Oka manifold if and only if it enjoys CAP.

The following main result of this chapter says that maps $X \to Y$ from any reduced Stein space X to an Oka manifold Y satisfy all forms of the Oka principle that have been considered in the literature; the same holds for sections of stratified holomorphic fiber bundles with Oka fibers over reduced Stein spaces. Proposition 5.15.1 on p. 236 gives the equivalence of several ostensibly different Oka type properties that characterize the class of Oka manifolds. Examples of Oka manifolds are given in §5.5.

Theorem 5.4.4. *Let $\pi\colon Z \to X$ be a holomorphic submersion of a complex space Z onto a reduced Stein space X. Assume that X is exhausted by a sequence of open subsets $U_1 \subset U_2 \subset \cdots \subset \cup_j U_j = X$ such that each restriction $Z|_{U_j} \to U_j$ is a stratified holomorphic fiber bundle whose fibers enjoy* CAP. *Then sections $X \to Z$ satisfy the following*

Parametric Oka Property (POP): *Given a compact $\mathcal{O}(X)$-convex subset K of X, a closed complex subvariety X' of X, compact sets $P_0 \subset P$ in a Euclidean space $\mathbb{R}^m$, and a continuous map $f\colon P \times X \to Z$ such that*

(a) *for every $p \in P$, $f(p,\cdot)\colon X \to Z$ is a section of $Z \to X$ that is holomorphic on a neighborhood of K (independent of p) and such that $f(p,\cdot)|_{X'}$ is holomorphic on X', and*

(b) *$f(p,\cdot)$ is holomorphic on X for every $p \in P_0$,*

there is a homotopy $f_t\colon P \times X \to Z$ ($t \in [0,1]$), with $f_0 = f$, such that f_t enjoys properties (a) and (b) for all $t \in [0,1]$, and also the following hold:

(i) *$f_1(p,\cdot)$ is holomorphic on X for all $p \in P$,*

(ii) *f_t is uniformly close to f on $P \times K$ for all $t \in [0,1]$, and*

(iii) *$f_t = f$ on $(P_0 \times X) \cup (P \times X')$ for all $t \in [0,1]$.*

If, in addition to (a) and (b), every section $f(p,\cdot)\colon X \to Z$ is holomorphic on a neighborhood of X' (independent of $p \in P$), then for every coherent analytic sheaf of ideals $\mathcal{S} \subset \mathcal{O}_X$ supported on X' the homotopy f_t can be chosen such that $\delta(f_t(p,\cdot), f(p,\cdot)) \in \mathcal{S}$ for all $t \in [0,1]$ and $p \in P$ (see Def. 1.3.2). In particular, f_t may be chosen tangent to f to a given finite order along X'.

If X has finite dimension n then the above conclusions hold if each fiber Y enjoys CAP_N *with $N = n + \dim Y$.*

Theorem 5.4.4 answers Gromov's question [237, p. 881, 3.4.(D)] whether Runge approximation on a certain class of compact sets in Euclidean spaces suffices to infer the Oka property. While it may conceivably be possible to reduce the testing family to balls by more careful geometric considerations, this would not simplify the verification of CAP in concrete examples.

Theorem 5.4.4 summarizes the main results from the papers [176, 179, 180, 181, 182]. The proof builds upon the works of Grauert [222, 223], Henkin and Leiterer [258, 259], Gromov [237], and my papers with J. Prezelj [190, 191,

192]. The proof is outlined at the end of this section; the details are given in §5.8 – §5.13, proceeding in steps from the simplest to the most general case.

The statement of Theorem 5.4.4 defines various Oka type properties of a holomorphic map $\pi\colon Z \to X$. The full conclusion is expressed by saying that sections $X \to Z$ satisfy the *parametric Oka principle with approximation and jet interpolation*. By omitting approximation and interpolation, it means that a continuous map $f\colon P \to \Gamma_{\mathcal{C}}(X,Z)$ with $f(P_0) \subset \Gamma_{\mathcal{O}}(X,Z)$ can be deformed to a continuous map $f_1\colon P \to \Gamma_{\mathcal{O}}(X,Z)$ by a homotopy that is fixed on P_0:

$$\begin{tikzcd} P_0 \arrow[r] \arrow[d, "incl"'] & \Gamma_{\mathcal{O}}(X,Z) \arrow[d, "incl"] \\ P \arrow[r, "f"] \arrow[ur, "f_1"] & \Gamma_{\mathcal{C}}(X,Z) \end{tikzcd}$$

In the *nonparametric case* (with P a singleton and $P_0 = \emptyset$) the conclusion is called the *basic Oka principle with approximation and (jet) interpolation*.

Theorem 5.4.4 also pertains to maps $X \to Y$, considered as sections of the trivial bundle $\pi\colon X \times Y \to X$, $\pi(x,y) = x$. The corresponding forms of the Oka principle are called *Oka properties* of the manifold Y. We will show that all Oka properties considered above are pairwise equivalent (Proposition 5.15.1 on p. 236), and hence any of them characterizes the class of Oka manifolds. This class is dual to the class of Stein manifolds in a sense that can be made precise by means of abstract homotopy theory [321, 322]. Intuitively speaking, Lárusson's results show that *Stein manifolds and Stein spaces are the natural sources of holomorphic maps, while Oka manifolds are the natural targets.*

The definition of an Oka manifold implies that the Kobayashi-Eisenman volume form vanishes identically; hence no compact Oka manifold is of Kodaira general type (see Corollary 5.15.3).

Remark 5.4.5. We say that mappings $X \to Y$ between a pair of complex manifolds satisfy the *basic Oka property* if every continuous map is homotopic to a holomorphic map, but without insisting on the approximation or the interpolation condition. Jörg Winkelmann [519] gave a complete answer to the question which pairs of Riemann surfaces satisfy the basic Oka principle. His classification shows that this property is strictly weaker than any of the properties in Theorem 5.4.4. □

We mention some consequences of Theorem 5.4.4. The following result of Ramspott [405, 406] follows from Theorem 5.4.4 and Proposition 5.5.1 to the effect that every complex homogeneous manifold is Oka. In his theorem the structure group of the bundle is assumed to be a complex Lie group; this condition is now omitted, it is only the fiber that matters.

Corollary 5.4.6. *If X is a Stein space and $\pi\colon Z \to X$ is a stratified holomorphic fiber bundle with complex homogeneous fibers, then its sections satisfy the parametric Oka principle with approximation and jet interpolation.*

Corollary 5.4.7. *Assume that X is a Stein space, Y is an Oka manifold, K is a compact $\mathcal{O}(X)$-convex subset of X, U is an open neighborhood of K in X, and $f\colon U \to Y$ is a null-homotopic holomorphic map. Then f can be approximated uniformly on K by holomorphic maps $X \to Y$.*

Proof. Since $f\colon U \to Y$ is null-homotopic, there exists a continuous map $F\colon X \to Y$ that equals f in some neighborhood of K. The conclusion now follows from Theorem 5.4.4. □

Corollary 5.4.8. (The weak homotopy equivalence principle.) *If $\pi\colon Z \to X$ is a stratified holomorphic fiber bundle over a reduced Stein space X whose fibers are Oka manifolds, then the inclusion*

$$\iota\colon \Gamma_{\mathcal{O}}(X,Z) \hookrightarrow \Gamma_{\mathcal{C}}(X,Z) \tag{5.6}$$

of the space of holomorphic sections of π into the space of continuous sections is a weak homotopy equivalence*, i.e., the induced map of homotopy groups*

$$\pi_k(\iota)\colon \pi_k(\Gamma_{\mathcal{O}}(X,Z)) \to \pi_k(\Gamma_{\mathcal{C}}(X,Z))$$

is an isomorphism for every $k = 0, 1, 2, \ldots$.

Proof. Denote by D^{k+1} the closed unit ball in $\mathbb{R}^{k+1}$ and by $S^k = bD^{k+1}$ the k-sphere. Applying Theorem 5.4.4 with $P = S^k$, $P_0 = \emptyset$, $K = \emptyset$ and $X' = \emptyset$ we see that every continuous map $S^k \to \Gamma_{\mathcal{C}}(X,Z)$ can be deformed to a continuous map $S^k \to \Gamma_{\mathcal{O}}(X,Z)$, so $\pi_k(\iota)$ is surjective. Applying Theorem 5.4.4 with $P = D^{k+1}$, $P_0 = S^k$, $K = \emptyset$ and $X' = \emptyset$ we conclude that every map $f\colon D^{k+1} \to \Gamma_{\mathcal{C}}(X,Z)$ such that $f(S^k) \subset \Gamma_{\mathcal{O}}(X,Z)$ can be deformed to a map $D^{k+1} \to \Gamma_{\mathcal{O}}(X,Z)$ by a homotopy that is fixed on S^k. This means that $\pi_k(\iota)$ is injective. □

The analogous result holds for the spaces of sections that agree with a given holomorphic section $f\colon X' \to Z|_{X'}$ on a closed complex subvariety X' of X. Let us denote by

$$\iota\colon \Gamma_{\mathcal{O}}(X,Z;X',f) \hookrightarrow \Gamma_{\mathcal{C}}(X,Z;X',f) \tag{5.7}$$

the inclusion of the space of holomorphic sections $F\colon X \to Z$ with $F|_{X'} = f$ into the space of continuous sections satifying the same condition. The following result is proved in the same way as Corollary 5.4.8.

Corollary 5.4.9. *If $\pi\colon Z \to X$ satisfies the hypotheses of Theorem 5.4.4 and $f\colon X' \to Z|_{X'}$ is a holomorphic section over a closed complex subvariety X' of X, then the inclusion (5.7) is a weak homotopy equivalence.*

Applying Theorem 5.4.4 with $K = \emptyset$, $P = \{p\}$ and $P_0 = \emptyset$ gives

Corollary 5.4.10. (The Oka principle for extensions.) *Suppose that X is a reduced Stein space, X' is a closed complex subvariety of X, and Y is an Oka manifold. Given a continuous map $F\colon X \to Y$ such that $F|_{X'} = f\colon X' \to Y$ is holomorphic, there is a homotopy $F_t\colon X \to Y$ $(t \in [0,1])$ such that $F_0 = F$, F_1 is holomorphic, and the homotopy is fixed on X'.*

Given a map $\pi\colon E \to B$, we say that a map $F\colon X \to E$ is a *lifting* of a map $f\colon X \to B$ if $\pi \circ F = f$. Similarly one defines homotopies of liftings.

Corollary 5.4.11. (The Oka principle for liftings.) *Assume that $\pi\colon E \to B$ is a stratified holomorphic fiber bundle all of whose fibers are Oka manifolds. If X is a reduced Stein space and $f\colon X \to B$ is a holomorphic map, then any continuous lifting $F_0\colon X \to Z$ of f admits a homotopy of liftings $F_t\colon X \to E$ $(t \in [0,1])$ such that F_1 is holomorphic. Furthermore, if the lifting F_0 is holomorphic on a subset $K \cup X' \subset X$ as in Theorem 5.4.4, then the homotopy $\{F_t\}$ can be chosen to satisfy properties (i) and (ii) in Theorem 5.4.4.*

Proof. Assume first that $\pi\colon E \to B$ is a holomorphic fiber bundle with Oka fiber Y. Let $h\colon f^*E \to X$ denote the pull-back bundle whose fiber over a point $x \in X$ is $E_{f(x)} \cong Y$.

$$\begin{tikzcd} f^*E \arrow[r] \arrow[d, "h"] & E \arrow[d, "\pi"] & Y \arrow[l] \\ X \arrow[r, "f"'] \arrow[ur, "F_t"] & B & \end{tikzcd}$$

Sections $X \to f^*E$ are in one-to-one correspondence with liftings $F\colon X \to E$ of the map $f\colon X \to B$. Since the fiber Y of $h\colon f^*E \to X$ is Oka, the conclusion follows from Theorem 5.4.4. In the general case we stratify X so that the strata are mapped by f to the strata of B; then f^*E is also a stratified fiber bundle over X and we conclude the proof as before. $\square$

Outline of proof of Theorem 5.4.4. Assume first that the base X is a Stein manifold. We exhaust X by an increasing sequence of compact strongly pseudoconvex domains A_k such that $A_{k+1} = A_k \cup B_k$ is obtained by attaching to A_k either a small convex bump B_k, or a special handle whose core is a totally real disc attached to bA_k along a complex tangential sphere, so that the fiber bundle is trivial over a neighborhood of B_k. A holomorphic section $f\colon X \to Z$ in a given homotopy class is found as a locally uniform limit $f = \lim_{k\to\infty} f_k$ of a sequence of continuous sections $f_k\colon X \to Z$ such that f_k is holomorphic over a neighborhood of A_k. In the induction step we find f_{k+1} that approximates f_k uniformly on a neighborhood of A_k and is homotopic to it. We treat separately the extension across a convex bump (the *noncritical case*) and the crossing of a critical level of an exhaustion function on X (the *critical case*).

In the *noncritical case* (which is the more difficult of the two, and is the only one which requires a special condition on the fiber) the problem

is subdivided in two substeps. First we approximate f_k on a neighborhood of $C_k = A_k \cap B_k$ by a holomorphic section g_k defined on a neighborhood of B_k. Since the bundle is trivial there, this is a Runge approximation problem for maps to the fiber Y, and here CAP of the fiber is invoked. We then glue f_k and g_k into a section f_{k+1} which is holomorphic over a neighborhood of $A_{k+1} = A_k \cup B_k$. In the classical case when f_k and g_k are maps to a complex Lie group G, one has $f_k = g_k \cdot \gamma$ on C_k, where $\gamma = g_k^{-1} \cdot f_k \colon C_k \to G$ is a holomorphic map close to the identity. By Cartan's lemma we split γ as a product $\gamma = \beta \cdot \alpha^{-1}$ of holomorphic maps $\alpha \colon A_k \to G$, $\beta \colon B_k \to G$, and we take $f_k \cdot \alpha = g_k \cdot \beta$ as the next map f_{k+1}.

In the general case we work with *thick sections*, also called *local holomorphic sprays*. These are families of holomorphic sections, depending holomorphically on a parameter in a neighborhood of the origin in a Euclidean space, which are submersive with respect to the parameter. Given a thick section f_k over A_k, we approximate f_k over C_k by a thick section g_k over B_k as before. We then find a fiber preserving biholomorphic transition map γ, close to the identity map and satisfying $f_k = g_k \circ \gamma$. Next we split $\gamma = \beta \circ \alpha^{-1}$ as a composition of two holomorphic maps, α over A_k, resp. β over B_k (Proposition 5.8.1, p. 211). Then $f_k \circ \alpha = g_k \circ \beta$ over C_k, and hence these two thick sections amalgamate into a thick section f_{k+1} that is holomorphic over a neighborhood of A_{k+1}. The induction may proceed. For the details see §5.10.

The *critical case* is accomplished by using Theorem 3.7.2 (p. 81) to approximately extend the holomorphic section across the stable manifold at the critical point, thereby reducing the problem to the noncritical case (§5.11).

In the case of a stratified fiber bundle the proof proceeds by induction on the strata; the main step is furnished by Proposition 5.12.1 on p. 224. Every step of the proof can also be done in the parametric case, and this shows that a parametric version of CAP (called PCAP) of the fibers of $Z \to X$ implies the parametric Oka principle for sections $X \to Z$. The proof is completed by showing that CAP implies PCAP (see §5.13).

For a nonstratified holomorphic fiber bundle with Oka fiber over a Stein space one can also proceed as in the classical Oka-Grauert theory by induction over cubes, also called *stones* in the works of Grauert and Remmert.

The same techniques give various 'up to the boundary' versions of Theorem 5.4.4. We quote a simplified version of the main result from [122].

Theorem 5.4.12. (The Oka principle for sections of $\mathcal{A}^r(D)$-bundles.) *Assume that $D \Subset X$ is a strongly pseudoconvex domain with $\mathcal{C}^\ell$ boundary ($\ell \geq 2$) in a Stein manifold X and $\pi \colon Z \to \bar{D}$ is a fiber bundle of class $\mathcal{A}^r(D)$ for some $r \in \{0, 1, \ldots, \ell\}$ (holomorphic on D and of class $\mathcal{C}^r$ over $\bar{D}$). If the fiber $Y = \pi^{-1}(z)$ ($z \in \bar{D}$) enjoys* CAP *then every continuous section $f_0 \colon \bar{D} \to Z$ is homotopic to a section $f_1 \colon \bar{D} \to Z$ of class $\mathcal{A}^r(D)$, and every homotopy $\{f_t\}_{t \in [0,1]}$ of continuous sections such that f_0, f_1 are of class $\mathcal{A}^r(D)$ can be deformed with fixed ends to a homotopy of sections of class $\mathcal{A}^r(D)$.*

A more precise result with approximation on certain compact sets in $\bar{D}$ is [122, Theorem 6.1]. This result also applies if $\bar{D}$ is a compact complex manifold with Stein interior and smooth strongly pseudoconvex boundary; such manifold is diffeomorphic to a strongly pseudoconvex domain in a Stein manifold by a diffeomorphism which is holomorphic in the interior (see Catlin's boundary version [81] of the Newlander-Nirenberg integrability theorem).

5.5 Examples of Oka Manifolds

Complex Homogeneous Manifolds

Proposition 5.5.1. [221] *Every complex Lie group and, more generally, every complex homogeneous manifold is an Oka manifold.*

Proof. Let G be a complex Lie group. Denote by $1 \in G$ the identity element and by $\exp\colon \mathfrak{g} = \mathrm{T}_1 G \cong \mathbb{C}^n \to G$ the exponential map. Assume that K is a compact convex set in $\mathbb{C}^n$, $U \subset \mathbb{C}^n$ is an open convex neighborhood of K, and $f\colon U \to G$ is a holomorphic map.

If $f(K)$ lies sufficiently close to $1 \in G$ then there is a unique holomorphic map $h\colon U' \to \mathfrak{g}$ on a smaller neighborhood of K such that $f = \exp h$. Approximating h uniformly on K by an entire map $\widetilde{h}\colon \mathbb{C}^n \to \mathfrak{g}$ and taking $\widetilde{f} = \exp \widetilde{h}\colon \mathbb{C}^n \to G$ gives a desired entire approximation of f.

In general we proceed as follows. By translation of coordinates we may assume that the origin $0 \in \mathbb{C}^n$ is contained in $\mathrm{Int}K$. Set $f_t(z) = f(tz)$ for $t \in [0,1]$ and $z \in U$; then $f_1 = f$ and f_0 is the constant map $\mathbb{C}^n \ni z \to f(0) \in G$. Choose $N \in \mathbb{N}$. Using the group structure on G we can write

$$f = f_1 = f_1\big(f_{\frac{N-1}{N}}\big)^{-1} \cdot f_{\frac{N-1}{N}}\big(f_{\frac{N-2}{N}}\big)^{-1} \cdots f_{\frac{1}{N}}(f_0)^{-1} \cdot f_0.$$

If N is sufficiently large then each of the quotients $f_{\frac{k}{N}}(f_{\frac{k-1}{N}})^{-1}$ $(k = 1, \ldots, N)$ is sufficiently close to 1 so that it admits a holomorphic logarithm $h_k\colon K \to \mathfrak{g}$. Approximating h_k by an entire map $\widetilde{h}_k\colon \mathbb{C}^n \to \mathfrak{g}$ and taking $\widetilde{g}_k = \exp \widetilde{h}_k$ and $\widetilde{f} = \widetilde{g}_N \widetilde{g}_{N-1} \cdots \widetilde{g}_1 f_0\colon \mathbb{C}^n \to G$ gives a desired approximation of f.

The proof for a complex homogeneous manifold Y is quite similar. Let G be a complex Lie group acting holomorphically and transitively on Y. Let $K \subset \mathbb{C}^n$ be a compact convex set and $f\colon K \to Y$ a holomorphic map. Let f_t be a homotopy defined as above. If we choose a sufficiently large integer $N \in \mathbb{N}$, then there is for every $k = 1, 2, \ldots, N$ a holomorphic map $h_k\colon K \to \mathfrak{g}$ such that $f_{\frac{k}{N}} = \exp h_k \cdot f_{\frac{k-1}{N}}$. (We used that the holomorphic map $s\colon Y \times \mathfrak{g} \to Y$, $s(y,v) = \exp v \cdot y$, is submersive in an neighborhood of the zero section in $Y \times \mathfrak{g}$, so the existence of h_k follows from the implicit function theorem provided that the maps $f_{\frac{k}{N}}$ and $f_{\frac{k-1}{N}}$ are sufficiently close. Further details can be found in §6.6 in the wider context of elliptic manifolds.) Approximating each h_k by a holomorphic map $\widetilde{h}_k\colon \mathbb{C}^n \to \mathfrak{g}$ completes the proof as before. $\square$

Ascent and Descent in Holomorphic Fiber Bundles

We now prove one of the main known functorial properties of the class of Oka manifolds that concerns holomorphic fiber bundles with Oka fibers. The result is especially simple for covering projections.

Proposition 5.5.2. *If $\pi\colon E \to B$ is a holomorphic covering map of complex manifolds then B is an Oka manifold if and only if E is an Oka manifold.*

Proof. Assume that E is an Oka manifold. Let K be compact convex set in $\mathbb{C}^n$ and $f\colon K \to B$ a holomorphic map. By the lifting property for coverings there exists a holomorphic map $h\colon K \to E$ such that $\pi \circ h = f$. Since E is an Oka manifold, we can approximate h, uniformly on K, by a holomorphic map $\widetilde{h}\colon \mathbb{C}^n \to E$. The map $\widetilde{f} = \pi \circ \widetilde{h}\colon \mathbb{C}^n \to B$ then approximates f on K, so B is an Oka manifold. A similar argument proves the converse implication. $\square$

Corollary 5.5.3. *The following are equivalent for a Riemann surface Y:*

(a) *Y is an Oka manifold.*

(b) *Y is non-hyperbolic.*

(c) *Y is one of the surfaces $\mathbb{P}^1$, $\mathbb{C}$, $\mathbb{C}^*$, or a torus $\mathbb{C}/\Gamma$.*

Proof. The universal covering of any Riemann surface is one of the surfaces $\mathbb{P}^1$, $\mathbb{C}$, or the disc $\mathbb{D} = \{z \in \mathbb{C}\colon |z| < 1\}$. $\mathbb{P}^1$ is homogeneous and hence Oka; it has no nontrivial complex quotients. The complex plane $\mathbb{C}$ covers $\mathbb{C}^*$ and the tori $\mathbb{C}/\Gamma$; hence these are Oka. The disc and its quotients are hyperbolic. $\square$

Theorem 5.5.4. *If E and X are complex manifolds and $\pi\colon E \to X$ is a holomorphic fiber bundle whose fiber is an Oka manifold, then X is an Oka manifold if and only if E is an Oka manifold.*

Proof. Assume first that E is an Oka manifold. Let $K \subset Q$ be a pair of compact convex sets in $\mathbb{C}^n$ and $f\colon U \to X$ a holomorphic map from an open convex neighborhood $U \subset \mathbb{C}^n$ of K. By the homotopy lifting theorem there exists a continuous lifting $h\colon U \to E$ of f. By Corollary 5.4.11 we can replace h by a holomorphic lifting. Since E is Oka, we can approximate h uniformly on K by a holomorphic map $\widetilde{h}\colon Q \to E$. The map $\widetilde{f} = \pi \circ \widetilde{h}\colon Q \to X$ then approximates f on K. This shows that X is also an Oka manifold.

Conversely, assume that X is Oka. Choose $K \subset Q \subset \mathbb{C}^n$ and a holomorphic map $h\colon U \to E$ as above. Let $f = \pi \circ h\colon U \to X$. Since X is Oka, we can approximate f uniformly on K by a holomorphic map $f_1\colon V \to X$ in an open set $V \supset Q$. If the approximation is close enough, we can also approximate h, uniformly on K, by a holomorphic map $h_1\colon K \to E$ that is a lifting of f_1. To do this, recall that Lemma 3.3.4 (p. 68) provides for every $x \in V$ a holomorphic retraction ρ_x of an open neighborhood $\Omega_x \subset E$ of the fiber

$R_x = \pi^{-1}(f_1(x)) \subset E$ onto R_x, with ρ_x depending holomorphically on $x \in V$. If f_1 is sufficiently uniformly close to f on K, then for every x in an open neighborhood $U' \subset V$ of K we have $h(x) \in \Omega_x$, and for such x we can define $h_1(x) = \rho_x(h(x)) \in R_x$. The map $h_1 \colon U' \to E$ is holomorphic, it is uniformly close to h on K, and $\pi \circ h_1(x) = f_1(x)$ for $x \in U'$ as required.

Since $\pi \colon E \to X$ is a fiber bundle and the set $K \subset \mathbb{C}^n$ is convex, the map h_1 extends from a smaller neighborhood of K in $\mathbb{C}^n$ to a continuous map $h_1 \colon Q \to E$ that satisfies $\pi \circ h_1(x) = f_1(x)$ for all $x \in Q$. Since the fiber of π is an Oka manifold, Corollary 5.4.11 (p. 196) shows that we can deform h_1 to a holomorphic lifting $\widetilde{h} \colon Q \to E$ of $f_1 \colon Q \to X$ by a homotopy of liftings which remains uniformly close to h_1 on K. In particular, $\widetilde{h}$ approximates h uniformly on K. This shows that E is an Oka manifold. □

Corollary 5.5.5. *If $\pi \colon E \to X$ is a holomorphic fiber bundle whose fiber Y and base X are one of the Riemann surfaces $\mathbb{P}^1$, $\mathbb{C}$, $\mathbb{C}^*$, or a torus $\mathbb{T} = \mathbb{C}/\Gamma$, then the total space E is an Oka manifold. In particular, all* Hirzebruch surfaces *H_l $(l = 0, 1, 2, \ldots)$ are Oka manifolds.*

Proof. The list $\{\mathbb{P}^1, \mathbb{C}, \mathbb{C}^*, \mathbb{T} = \mathbb{C}/\Gamma\}$ contains all Riemann surfaces which are Oka; the others are uniformized by the disc $\mathbb{D}$ and therefore hyperbolic. Hence the first statement follows from Proposition 5.5.4 (p. 199). For the second statement note that Hirzebruch surfaces H_l are $\mathbb{P}^1$-bundles over $\mathbb{P}^1$ (see [33, p. 191]). Each H_l is birationally equivalent to $\mathbb{P}^2$. □

Increasing Unions

Proposition 5.5.6. *If a complex manifold Y is exhausted by open domains $D_1 \subset D_2 \subset \cdots \subset \cup_{j=1}^{\infty} D_j = Y$ such that every D_j is an Oka manifold, then Y is an Oka manifold. In particular, every long $\mathbb{C}^n$ is an Oka manifold.*

Recall that a *long* $\mathbb{C}^n$ is a complex manifold exhausted by an increasing sequence of domains biholomorphic to $\mathbb{C}^n$. By Theorem 4.20.3 there exists for every $n > 1$ a long $\mathbb{C}^n$ which is not Stein.

Proof. Let $f \colon K \to Y$ be a holomorphic map from a compact convex set $K \subset \mathbb{C}^n$. Then $f(K)$ is compact and hence is contained in D_j for some j. As D_j is Oka, f can be approximated uniformly on K by entire maps $\mathbb{C}^n \to D_j$. Thus Y is an Oka manifold. □

Complements of Algebraic Subvarieties

Question: If $\widehat{Y}$ is an algebraic Oka manifold and A is a closed algebraic subvariety of $\widehat{Y}$, when is the complement $Y = \widehat{Y} \backslash A$ an Oka manifold?

The answer is negative in general if A is a hypersurface. In fact, there exist examples of algebraic hypersurfaces in $\mathbb{C}^n$, $\mathbb{P}^n$, abelian varieties, etc. whose complement is hyperbolic, and hence is not Oka. It is conjectured that this holds for the complement of a generic algebraic hypersurface in $\mathbb{C}^n$ or $\mathbb{P}^n$ of sufficiently large degree; see the papers [104, 109, 447, 448, 449].

Definition 5.5.7. *A closed complex subvariety A in a complex manifold X is* thin *if at every point $p \in A$ we have $\dim_p A \le \dim_p X - 2$.*

Proposition 5.5.8. *Let $\widehat{Y}$ denote $\mathbb{C}^k$, $\mathbb{P}^k$, or a complex Grassmannian. If A is a thin algebraic subvariety of $\widehat{Y}$ then $Y = \widehat{Y}\backslash A$ is an Oka manifold.*

Proof. Each of the manifolds $\mathbb{C}^k$, $\mathbb{P}^k$ or a complex Grassmannian is complex homogeneous, and hence an Oka manifold by Proposition 5.5.1 (p. 198). We shall use the equivalence CAP$\Leftrightarrow$Oka.

Consider first the case $\widehat{Y} = \mathbb{C}^k$. Given a special convex pair (K, Q) in $\mathbb{C}^n$ (Def. 5.4.2 on p. 192) and a holomorphic map $f_0 \colon U \to Y$ from an open neighborhood U of K to $Y = \mathbb{C}^k\backslash A$, we need to find a holomorphic map $\widetilde{f} \colon Q \to Y$ that approximates f_0 as close as desired on K; this will verify that Y enjoys CAP, and hence it is an Oka manifold. Choose a compact convex set $L \subset U$ with $K \subset \mathrm{Int} L$. By the Oka-Weil theorem we can approximate f_0 uniformly on L by a holomorphic polynomial map $f \colon \mathbb{C}^n \to \mathbb{C}^k$. By the jet transversality theorem (Corollary 7.8.7, p. 317), f can be chosen such that the algebraic subvariety $\Sigma = f^{-1}(A) \subset \mathbb{C}^n$ is thin (of codimension at least two) in $\mathbb{C}^n$. Assuming that the approximations were sufficiently close we have $L \cap \Sigma = \emptyset$. By Corollary 4.12.2 (p. 144) there exists for any $\delta > 0$ a holomorphic automorphism ψ of $\mathbb{C}^n$ satisfying $\sup_{z \in K} |\psi(z) - z| < \delta$ and $\psi(Q) \cap \Sigma = \emptyset$. If $\delta > 0$ is small enough then $\widetilde{f} = f \circ \psi \colon \mathbb{C}^n \to \mathbb{C}^k$ maps Q into Y, and it approximates the initial map f_0 uniformly on K.

Next we consider that case $\widehat{Y} = \mathbb{P}^k$ for $k \ge 2$. (When $k = 1$, A must be empty.) The quotient projection $\pi \colon \mathbb{C}^{k+1}_* = \mathbb{C}^{k+1}\backslash\{0\} \to \mathbb{P}^k$ is a holomorphic fiber bundle with fiber $\mathbb{C}^* = \mathbb{C}\backslash\{0\}$, and by adding the zero section we obtain the universal line bundle $L \to \mathbb{P}^k$. Assume that (K, Q) is a special convex pair in $\mathbb{C}^n$, $U \subset \mathbb{C}^n$ is an open convex neighborhood of K, and $f \colon U \to Y = \mathbb{P}^k\backslash A$ is a holomorphic map. Since U is contractible, the bundle $f^*L \to U$ is topologically trivial, and hence it admits a nowhere vanishing section. Pushing this section forward gives a continuous lifting $h \colon U \to \mathbb{C}^{k+1}_*$ of f, i.e., $\pi \circ h = f$. Since the fiber $\mathbb{C}^*$ of π is a Lie group and hence enjoys CAP, Corollary 5.4.11 (p. 196) allows us to replace h by a holomorphic lifting of f (still denoted h). The set $A' = \pi^{-1}(A) \cup \{0\}$ is a thin algebraic subvariety of $\mathbb{C}^{k+1}$ satisfying $h(U) \cap A' = \emptyset$. By the first part of the proposition we can approximate h arbitrarily well on K by a holomorphic map $\widetilde{h} \colon Q \to \mathbb{C}^{k+1}\backslash A'$. The map $\widetilde{f} = \pi \circ \widetilde{h} \colon Q \to Y$ is then holomorphic and close to f on K, so Y enjoys CAP.

The case when $\widehat{Y} = G_{k,m}$ for some $1 \le k < m$ is proved in essentially the same way. Consider the fiber bundle $\pi \colon V_{k,m} \to G_{k,m}$, where $V_{k,m}$ is the Stiefel

manifold of k-frames in $\mathbb{C}^m$ (Example 1.2.7). We have $V_{k,m} = \mathbb{C}^{km} \backslash \Sigma$ where $\Sigma \subset \mathbb{C}^{km}$ is an algebraic subvariety of codimension $n - k + 1 \geq 2$, hence is thin in $\mathbb{C}^{km}$. Therefore the algebraic subset $A' = \pi^{-1}(A) \cup \Sigma \subset \mathbb{C}^{km}$ is also thin and we can complete the proof as in the projective case. □

Corollary 5.5.9. *Every Hopf manifold is an Oka manifold.*

Proof. A Hopf manifold is a nonramified holomorphic quotient of $\mathbb{C}^n_*$ for some $n > 1$. Since $\mathbb{C}^n_*$ is an Oka manifold by Proposition 5.5.8, the conclusion follows from Proposition 5.5.2 (p. 199). □

Proposition 5.5.8 fails in general for non-algebraic subvarieties of $\mathbb{C}^n$ irrespectively of their dimension (however, see Proposition 5.5.14 below). Indeed, Theorem 4.7.2 furnishes a discrete infinite set in $\mathbb{C}^n$ whose complement is volume hyperbolic, and hence it fails to be Oka. Here is another example.

Example 5.5.10. For each pair of integers $0 < k < n$ there exists a proper holomorphic embedding $\phi\colon \mathbb{C}^k \hookrightarrow \mathbb{C}^n$ such that $Y = \mathbb{C}^n \backslash \phi(\mathbb{C}^k)$ is not an Oka manifold. By Theorem 4.18.8 (p. 170) we can choose ϕ such that every entire map $\mathbb{C}^N \to Y$ has rank $< n - k$ at each point. The same holds for maps $X = (\mathbb{C}^*)^N \to Y$ as is seen by precomposing with the universal covering map $\mathbb{C}^N \to (\mathbb{C}^*)^N$. A general position argument then shows that every holomorphic map $X \to Y$ is null homotopic. However, if we take $N = 2(n-k) - 1$ then there exists a homotopically nontrivial smooth map $X \to Y$: Choose a smooth embedding $g\colon T \to Y$ of the real torus T of dimension $2(n-k)-1$ (the product of unit circles in each copy of $\mathbb{C}^*$ in $X = (\mathbb{C}^*)^{2(n-k)-1}$) into Y such that $g(T)$ links the complex submanifold $\phi(\mathbb{C}^k)$ of $\mathbb{C}^n$. Let $\iota\colon X \to T$ denote a retraction of X onto T. The map $f = g \circ \iota\colon X \to Y$ is then homotopically nontrivial. Hence the basic Oka principle fails for maps $X \to Y$. □

Elliptic and Subelliptic Manifolds

We motivate the definition of a *dominating spray* on a complex manifold Y by the following consideration. Choose an integer $n \geq \dim Y$ and let $\mathbb{B} = \mathbb{B}^n$ denote the open unit ball in $\mathbb{C}^n$. For any point $y \in Y$ there is a holomorphic submersion $g_y\colon \mathbb{B} \to Y$ with $g_y(0) = y$ (simply map $\mathbb{B}$ to a coordinate neighborhood of y). If Y is an Oka manifold then g_y is approximable, uniformly on any smaller ball $r\mathbb{B}$ $(0 < r < 1)$, by entire maps $s_y\colon \mathbb{C}^n \to Y$. If the approximation is sufficiently close then s_y is also a submersion on $r\mathbb{B}$, and we can arrange $s_y(0) = y$. In summary, $s_y\colon \mathbb{C}^n \to Y$ is an entire map satisfying

(i) $s_y(0) = y$, and
(ii) the differential $\mathrm{d}s_y(0)\colon \mathrm{T}_0\mathbb{C}^n \cong \mathbb{C}^n \to \mathrm{T}_y Y$ is surjective.

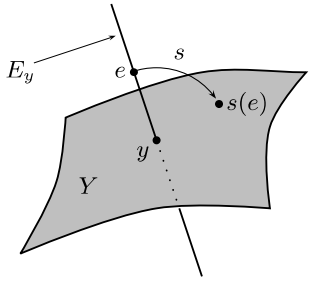

Fig. 5.1. A spray on Y. (Modified from [187, p. 13, Fig. 2])

More generally, given an open Stein subset U of Y, there are an integer $n \in \mathbb{N}$, an open Stein set $\Omega \subset U \times \mathbb{C}^n$ containing $U \times \{0\}$, and a holomorphic map $g\colon \Omega \to Y$ such that $g_y = g(y, \cdot\,)$ satisfies properties (i) and (ii) for each $y \in U$. Now extend g to a continuous map $U \times \mathbb{C}^n \to Y$ without changing its values near $U \times \{0\}$. If Y is an Oka manifold then there is a holomorphic map $s\colon U \times \mathbb{C}^n \to Y$ which agrees with g to second order at $U \times \{0\}$; hence the entire map $s_y = s(y, \cdot\,)\colon \mathbb{C}^n \to Y$ satisfies properties (i) and (ii) above. Such an s is called a *dominating holomorphic spray* on Y over the subset $U \subset Y$.

Definition 5.5.11. *Let Y be a complex manifold.*

(a) *A* (holomorphic) spray *on Y is a triple (E, π, s) consisting of a holomorphic vector bundle $\pi\colon E \to Y$* (a spray bundle) *and a holomorphic map $s\colon E \to Y$* (a spray map) *such that for each $y \in Y$ we have $s(0_y) = y$. A spray (E, π, s) is algebraic if $\pi\colon E \to Y$ is an algebraic vector bundle over an algebraic manifold Y and $s\colon E \to Y$ is an algebraic map.*

(b) *A spray (E, π, s) on Y is* dominating *on a subset $U \subset Y$ if the differential $\mathrm{d}s_{0_y}\colon \mathrm{T}_{0_y}E \to \mathrm{T}_yY$ maps the vertical subspace E_y of $\mathrm{T}_{0_y}E$ surjectively onto T_yY for every $y \in U$; s is* dominating *if this holds for all $y \in Y$.*

(c) *A family of sprays (E_j, π_j, s_j) on Y $(j = 1, \ldots, m)$ is* dominating *on a subset $U \subset Y$ if for all $y \in U$ we have*

$$(\mathrm{d}s_1)_{0_y}(E_{1,y}) + (\mathrm{d}s_2)_{0_y}(E_{2,y}) + \cdots + (\mathrm{d}s_m)_{0_y}(E_{m,y}) = \mathrm{T}_yY. \tag{5.8}$$

The family is dominating *if this holds for every $y \in Y$.*

(d) *A complex manifold Y is* elliptic *if it admits a dominating holomorphic spray, and is* subelliptic *if it admits a finite dominating family of holomorphic sprays.*

(e) *An algebraic manifold Y is* algebraically elliptic (A-elliptic) *if it admits a dominating algebraic spray, and is* algebraically subelliptic (A-subelliptic) *if it admits a finite dominating family of algebraic sprays.*

(f) *A complex manifold Y is* weakly elliptic *(resp.* weakly subelliptic*) if for every compact set $K \subset Y$ there exists a holomorphic spray on Y (resp. a family of holomorphic sprays) that is dominating at every point of K.*

The class of elliptic manifolds was introduced in [237], while subelliptic manifolds were introduced in [170]. Clearly we have the following implications, where the first row pertains only to algebraic manifolds:

$$\begin{array}{ccccc} A\text{-elliptic} & \Longrightarrow & A\text{-subelliptic} & & \\ \Downarrow & & \Downarrow & & \\ \text{elliptic} & \Longrightarrow & \text{subelliptic} & \Longrightarrow & \text{weakly subelliptic} \end{array}$$

It is not known whether these implications can be reversed in general. An example of a subelliptic manifold which is not known to be elliptic is the complement of a thin algebraic subvariety in a projective space (Proposition 6.4.1 on p. 249). An example of a weakly subelliptic manifold is $\mathbb{C}^n$ blown up at all points of a tame discrete sequence (Proposition 6.4.11 on p. 254). A Stein Oka manifold is elliptic by Proposition 5.15.2 on p. 237 below.

The relevance of these notions is shown by the following result which is essentially a corollary to Theorem 6.6.1 (p. 263).

Corollary 5.5.12. *Every weakly subelliptic manifold is an Oka manifold. In particular, every elliptic manifold is an Oka manifold.*

Proof. Let K be a compact convex set in $\mathbb{C}^n$ and let $f\colon U \to Y$ be a holomorphic map from an open convex neighborhood $U \subset \mathbb{C}^n$ of K. We may assume that $0 \in K$. Let $f_t(z) = f(tz)$ for $z \in U$ and $t \in [0,1]$; this is a homotopy from the constant map $f_0(z) = f(0) \in Y$ ($z \in \mathbb{C}^n$) to the map $f = f_1$. If Y is subelliptic then Theorem 6.6.1 (p. 263), applied to the projection $Z = \mathbb{C}^n \times Y \to \mathbb{C}^n$, shows that f can be approximated uniformly on K by entire maps $\mathbb{C}^n \to Y$. By Remark 6.6.4 (p. 265) the same holds if Y is weakly subelliptic. In each case we conclude that Y is an Oka manifold. $\square$

We give some elementary examples of elliptic and subelliptic manifolds; for further examples see §6.4.

Example 5.5.13. (A) *Every complex homogeneous manifold is elliptic.* Indeed, assume that a complex Lie group G acts on a complex manifold Y transitively by holomorphic automorphisms of Y. Let $\mathfrak{g} \cong \mathbb{C}^p$ denote the Lie algebra of G and $\exp\colon \mathfrak{g} \to G$ the exponential map. The holomorphic map $s\colon Y \times \mathfrak{g} \cong Y \times \mathbb{C}^p \to Y$, given by

$$s(y,v) = \exp v \cdot y \in Y, \quad y \in Y,\ v \in \mathfrak{g}, \tag{5.9}$$

is a dominating holomorphic spray on Y.

(B) *If the tangent bundle of a complex manifold Y is spanned by finitely many $\mathbb{C}$-complete holomorphic vector fields, then Y is elliptic.*

Let $V_1, \ldots, V_m$ be $\mathbb{C}$-complete holomorphic vector fields on Y. Denote by $\mathbb{C} \ni t \mapsto \phi_j^t(y)$ the flow of V_j. The map $s\colon Y \times \mathbb{C}^m \to Y$, given by

$$s(y,t) = s(y, t_1, \ldots, t_m) = \phi_1^{t_1} \circ \phi_2^{t_2} \circ \cdots \circ \phi_m^{t_m}(y), \tag{5.10}$$

satisfies $s(y,0) = y$ (hence it is a spray on Y), and

$$\frac{\partial}{\partial t_j} s(y,0) = V_j(y), \quad y \in Y,\ j = 1, \ldots, m.$$

Thus s is dominating at the point y precisely when the vectors $V_1(y), \ldots, V_m(y)$ span $\mathrm{T}_y Y$. In particular, if these fields span the tangent space at every point then s is a dominating spray on Y.

(C) The following example is related to Proposition 5.5.8 (p. 201).

Proposition 5.5.14. *If $A \subset \mathbb{C}^n$ is a tame (Def. 4.48, p. 139) analytic subvariety with $\dim A \le n-2$ then the complement $\mathbb{C}^n \backslash A$ is elliptic. If in addition A is algebraic then $\mathbb{C}^n \backslash A$ is algebraically elliptic.*

Proof. By the proof of Proposition 4.11.7 (p. 141) there exist finitely many $\mathbb{C}$-complete vector fields $V_1, \ldots, V_m$ on $\mathbb{C}^n$ that vanish on A and span the tangent space $\mathrm{T}_z\mathbb{C}^n$ at every point $z \in \mathbb{C}^n \backslash A$. Let ϕ_j^t denote the flow of V_j. The associated spray $s\colon \mathbb{C}^n \times \mathbb{C}^m \to \mathbb{C}^n$, defined by (5.10), satisfies $s(z,t) \equiv z$ for $z \in A$, and $s(z,t) \in \mathbb{C}^n \backslash A$ for $z \in \mathbb{C}^n \backslash A$ and $t \in \mathbb{C}^m$. Since the vectors $V_j(z)$ span $\mathrm{T}_z\mathbb{C}^n$ for every $z \in \mathbb{C}^n \backslash A$, the restricted map $s\colon (\mathbb{C}^n \backslash A) \times \mathbb{C}^m \to \mathbb{C}^n \backslash A$ is a dominating spray over $\mathbb{C}^n \backslash A$. If A is an algebraic subvariety then this construction can be done with algebraic vector fields. □

Corollary 5.5.15. *Let $X = \mathbb{C}^n / \Gamma$ where Γ is a lattice in $\mathbb{C}^n$ $(n \ge 2)$. Then the complement $X \backslash \{x_1, \ldots, x_m\}$ of any finite set in X is an Oka manifold.*

Proof. Let $\pi\colon \mathbb{C}^n \to X$ denote the quotient projection. Choose points $q_j \in \mathbb{C}^n$ with $\pi(q_j) = x_j$ $(j = 1, \ldots, m)$. The discrete set $\Gamma_0 = \cup_{j=1}^m (\Gamma + q_j) \subset \mathbb{C}^n$ is tame (see [69, Proposition 4.1] and [63]). Therefore $\widetilde{Y} = \mathbb{C}^p \backslash \Gamma_0$ is elliptic and hence Oka. Since $\pi\colon \widetilde{Y} \to Y = X \backslash \{x_1, \ldots, x_m\}$ is an nonramified covering, Y is an Oka manifold by Proposition 5.5.2 (p. 199). □

(D) *If Y is a complex Grassmann manifold and $A \subset Y$ is a closed algebraic subvariety of codimension at least two, then $Y \backslash A$ is algebraically subelliptic* (Proposition 6.4.1). It is not known whether these manifolds are elliptic. □

Further examples can be found in §6.4. We prove there that *every compact rational surface is an Oka manifold* (Corollary 6.4.8 on p. 253).

Holomorphically Flexible Manifolds

Definition 5.5.16. [26] *A complex manifold Y is* (holomorphically) flexible at a point $y \in Y$ *if the values at y of $\mathbb{C}$-complete holomorphic vector fields on Y span the tangent space $\mathrm{T}_y Y$. The manifold Y is* flexible *if it is flexible at every point $y \in Y$.*

Clearly a connected manifold Y is flexible if it is flexible at one point $y_0 \in Y$ and the holomorphic automorphism group $\mathrm{Aut}\, Y$ acts transitively on Y. In [25, 26] the authors also study the analogous notion of *algebraic flexibility* of affine algebraic varieties.

Proposition 5.5.17. ([291, Lemma 4.1], [26]) *A flexible Stein manifold is elliptic, and hence an Oka manifold.*

Proof. Assume first that Y is affine algebraic manifold that is algebraically flexible, in the sense that finitely many complete algebraic vector fields span the tangent space at every point of Y. Choose finitely many such vector fields spanning $\mathrm{T}_{y_0} Y$ at some point; the same fields then span $\mathrm{T}_y Y$ at all points outside a proper algebraic subset A of Y. By adding new vector fields we see by induction over the dimension of the exceptional variety A that finitely many complete algebraic fields span the tangent space at each point of Y. The spray (5.10) obtained by composing their flows is dominating on Y. A similar but more precise argument is needed to lower the dimension of the exceptional variety if Y is a flexible Stein manifold; see [289, Theorem 4] for the details. Alternatively, the above argument gives finitely many complete holomorphic vector fields that span the tangent space at every point of a given compact set K in Y; the composition of their flows (5.10) is a spray on Y that is dominating over K. Hence Y is weakly elliptic and therefore an Oka manifold by Corollary 5.5.12. □

Manifolds with the Density Property

Recall (Def. 4.10.1, p. 130) that a complex manifold Y enjoys the *density property* if the Lie algebra $\mathrm{Lie}(Y)$ generated by all $\mathbb{C}$-complete holomorphic vector fields on Y is dense in the Lie algebra $\aleph_{\mathcal{O}}(Y)$ of all holomorphic vector fields. An algebraic manifold Y has the *algebraic density property* if the Lie algebra $\mathrm{Lie}_{\mathrm{alg}}(Y)$ generated by all complete algebraic vector fields coincides with the Lie algebra $\aleph_{\mathcal{A}}(Y)$ of all algebraic vector fields.

Theorem 5.5.18. (Kaliman and Kutzschebauch [289, Theorem 4].)

(a) *An affine algebraic manifold with the algebraic density property admits a dominating algebraic spray (5.10), and hence is algebraically elliptic.*

(b) *A Stein manifold with the density property is elliptic.*

Proof. Assume first that Y is an affine algebraic manifold enjoying the algebraic density property. Lemma 4.10.3 on p. 131 says that for any Lie algebra $\mathfrak{g}$ of vector fields on Y we have $\overline{\mathrm{Span}(\mathfrak{g}_{\mathrm{int}})} = \overline{\mathrm{Lie}(\mathfrak{g}_{\mathrm{int}})}$, where $\mathfrak{g}_{\mathrm{int}}$ is the set of all $\mathbb{R}$-complete vector fields in $\mathfrak{g}$. Taking $\mathfrak{g} = \aleph_{\mathcal{A}}(Y)$ this shows that every algebraic vector field on Y can be approximated by $\mathbb{C}$-linear combinations of complete algebraic vector fields. By Serre's Theorem A the algebraic vector fields on Y span the tangent space at every point; hence the same is true for the complete algebraic vector fields. Thus Y is algebraically flexible in the sense of [26], and hence elliptic by Proposition 5.5.17. Essentially the same argument applies on a flexible Stein manifold. □

5.6 An Application of Michael's Selection Theorem

In this section we use Michael's *Convex Selection Theorem* [355] to show that one can replace the parameter subset P_0 of P in Theorem 5.4.4 (p. 193) by a neighborhood of P_0 in P. This will enable us to use cut-off functions on P in order to patch together families of sections.

Given topological spaces P and B and a set-valued map $\phi\colon P \to 2^B$ whose values are subsets of B, we say that a map $f\colon P \to B$ is a *selection* of ϕ if $f(p) \in \phi(p)$ for every $p \in P$. Such ϕ is said to be lower semicontinuous if for every open set $V \subset B$ the set $\{p \in P\colon \phi(p) \cap V \neq \emptyset\}$ is open in P. The following is a special case of Michael's theorem; a similar result was proved by H. Cartan [78, Appendix].

Theorem 5.6.1. [355] *Assume that B is a Banach space, P is a paracompact Hausdorff space, and $\phi\colon P \to 2^B$ is a lower semicontinuous set-valued map such that $\phi(p)$ is a nonempty closed convex subset of B for every $p \in P$. For every closed subset P_0 of P and every continuous selection $f\colon P_0 \to B$ of $\phi|_{P_0}$ there exists a continuous selection $F\colon P \to B$ of ϕ extending f.*

We shall need the following lemma from [192]. Denote by $H^\infty(D)$ the Banach space of all bounded holomorphic functions on D.

Lemma 5.6.2. *Assume that X is a reduced Stein space, X' is a closed complex subvariety of X and $\Omega \Subset \Omega'$ are relatively compact Stein domains in X. There exists a bounded linear extension operator $S\colon H^\infty(X' \cap \Omega') \to H^\infty(\Omega)$ such that $(Sf)(x) = f(x)$ for each $f \in H^\infty(X' \cap \Omega')$ and $x \in X' \cap \Omega$.*

Proof. We replace X by a relatively compact Stein subdomain containing $\overline{\Omega}'$ and embed it as a closed complex subvariety in a Euclidean space $\mathbb{C}^n$. Since every Stein domain $D \Subset X$ is the intersection $D = X \cap \widetilde{D}$ of X with an open Stein domain $\widetilde{D} \Subset \mathbb{C}^n$, it suffices to prove the lemma for the case $X = \mathbb{C}^n$.

Since Ω' is Stein, the restriction operator $R\colon \mathcal{O}(\Omega') \to \mathcal{O}(X' \cap \Omega')$ is surjective by Cartan's extension theorem. Since these are Fréchet spaces,

the open mapping theorem applies. Choose a domain $\Omega_1 \subset \mathbb{C}^n$ such that $\Omega \Subset \Omega_1 \Subset \Omega'$. By the open mapping theorem the image by R of the set $\{f \in \mathcal{O}(\Omega'): ||f||_{L^\infty(\Omega_1)} < 1\}$ contains a neighborhood of the origin in $\mathcal{O}(X' \cap \Omega')$. This means that there are a relatively compact subset $Y \Subset X' \cap \Omega'$ and a constant $M < +\infty$ such that every $h \in \mathcal{O}(X' \cap \Omega')$ extends to a function $h' \in \mathcal{O}(\Omega')$ satisfying the estimate

$$||h'||_{L^\infty(\Omega_1)} \leq M||h||_{L^\infty(Y)}.$$

We may assume that $\Omega_1 \cap X' \subset Y$. The restriction $h'|_{\Omega_1}$, being bounded, belongs to the Bergman space $H = L^2(\Omega_1) \cap \mathcal{O}(\Omega_1)$. Note that H is a Hilbert space containing the closed subspace $H_0 = \{f \in H: f|_{X'} = 0\}$.

Let H_1 be the orthogonal complement of H_0 in H. Projecting h' orthogonally to H_1 we get a function $\widetilde{h} \in H_1$ such that $\widetilde{h}|_{X \cap \Omega_1} = h|_{X \cap \Omega_1}$ and $\widetilde{h}$ has the minimal $L^2(\Omega_1)$-norm among all L^2-holomorphic extensions of h to Ω_1. Clearly such $\widetilde{h}$ is unique, and $S: h \to \widetilde{h}$ gives a bounded linear operator $S: H^\infty(X' \cap \Omega') \to L^2(\Omega_1)$. By restricting $\widetilde{h}$ to $\Omega \Subset \Omega_1$ we get a bounded linear extension operator $S: H^\infty(X' \cap \Omega') \to H^\infty(\Omega)$. $\square$

Proposition 5.6.3. *Let $\pi: Z \to X$ and (K, X', P_0, P, f) be as in Theorem 5.4.4 (p. 193). Given a domain $D \Subset X$, there exist a neighborhood P_0' of P_0 in P and a homotopy $f_t: P \times X \to Z$ $(t \in [0,1])$ satisfying the conclusion of Theorem 5.4.4, except that Condition (i) is replaced by*

(i') *$f_1(p, \cdot)$ is holomorphic on D for all $p \in P_0'$.*

Proof. We begin by considering the special case when $\pi: Z = X \times \mathbb{C} \to X$ is the trivial line bundle. We have $f(p,x) = (x, g_p(x))$ where g_p is holomorphic on X for $p \in P_0$, and is holomorphic on $K \cup X'$ for all $p \in P$.

We replace X by a relatively compact subset containing $\bar{D}$ and consider it as a closed complex subvariety of a Euclidean space $\mathbb{C}^N$. Choose bounded pseudoconvex domains $\Omega \Subset \Omega' \Subset \mathbb{C}^N$ such that $\bar{D} \subset \Omega \cap X$. Lemma 5.6.2 furnishes bounded linear extension operators

$$S: H^\infty(X \cap \Omega') \longrightarrow H^\infty(\Omega), \quad S': H^\infty(X' \cap \Omega') \longrightarrow H^\infty(\Omega)$$

such that $S(g)|_{X \cap \Omega} = g|_{X \cap \Omega}$ and $S'(g)|_{X' \cap \Omega} = g|_{X' \cap \Omega}$. Set

$$h_p = S(g_p|_{X \cap \Omega'}) - S'(g_p|_{X' \cap \Omega'}) \in H^\infty(\Omega), \qquad p \in P_0.$$

Then h_p belongs to the closed subspace $H^\infty_{X'}(\Omega)$ consisting of all functions in $H^\infty(\Omega)$ that vanish on $X' \cap \Omega$. Since these are Banach spaces, Theorem 5.6.1 furnishes a continuous extension of the map $P_0 \to H^\infty_{X'}(\Omega)$, $p \to h_p$, to a map $P \ni p \to \widetilde{h}_p \in H^\infty_{X'}(\Omega)$. Set

$$G_p = \widetilde{h}_p + S'(g_p|_{X' \cap \Omega'}) \in H^\infty(\Omega), \qquad p \in P.$$

Then we clearly have

$$G_p|_{X'\cap\Omega} = g_p|_{X'\cap\Omega}\ (\forall p \in P), \quad G_p|_{X\cap\Omega} = g_p|_{X\cap\Omega}\ (\forall p \in P_0).$$

By continuity G_p approximates g_p uniformly on K for the values of $p \in P$ close enough to P_0, so this solves the problem in the special case. (By an addition argument we could easily insure that G_p approximates g_p uniformly on K for all $p \in P$, but this will not be needed.)

The general case reduces to the special case by using that for every $p_0 \in P_0$ the Stein subvariety $\{f(p_0, x): x \in X\}$ of Z admits an open Stein neighborhood in Z by Corollary 3.1.1 (p. 57). Embedding these neighborhoods in Euclidean spaces and using holomorphic retractions onto fibers of π furnished by Theorem 3.3.4 (p. 68), the special case gives neighborhoods $U_{p_0} \subset U'_{p_0}$ of p_0 in P and a continuous family of sections $f_1: P \times \bar{D} \to Z$, homotopic to f, such that the following conditions hold:

(i) $f_1(p, \cdot\,)$ is holomorphic on $\bar{D}$ for $p \in U_{p_0}$,
(ii) $f_1(p, \cdot\,) = f(p, \cdot\,)$ for $p \in P_0 \cup (P\backslash U'_{p_0})$,
(iii) $f_1(p, x) = f(p, x)$ for all $x \in X' \cap D$ and $p \in P$, and
(iv) f_1 approximates f uniformly on $K \times P$.

The special case is first used for the values p in a neighborhood U'_{p_0} of p_0; the resulting family of holomorphic maps $\bar{D} \times U'_{p_0} \to \mathbb{C}^N$ is then patched with f by using a cut-off function $\chi(p)$ with support in U'_{p_0} that equals one on a smaller neighborhood U_{p_0} of p_0, and applying holomorphic retractions onto the fibers of π. In finitely many steps of this kind we complete the proof. □

5.7 Cartan Pairs

Definition 5.7.1. (I) *A pair (A, B) of compact subsets in a complex space X is a* Cartan pair *if it satisfies the following two conditions:*

(i) *A, B, $D = A \cup B$ and $C = A \cap B$ are Stein compacta, and*
(ii) *A, B are* separated *in the sense that $\overline{A\backslash B} \cap \overline{B\backslash A} = \emptyset$.*

(II) *A pair (D_0, D_1) of open relatively compact subsets in a complex manifold X is a* strongly pseudoconvex Cartan pair of class $\mathcal{C}^\ell$ $(\ell \geq 2)$ *if $(\bar{D}_0, \bar{D}_1)$ is a Cartan pair and D_0, D_1, $D = D_0 \cup D_1$, $D_{0,1} = D_0 \cap D_1$ are strongly pseudoconvex Stein domains with $\mathcal{C}^\ell$ boundaries.*

The following lemma is an immediate consequence of Siu's theorem (Corollary 3.1.1 on p. 57). It shows that the notion of a Cartan pair is invariant with respect to the ambient space.

Lemma 5.7.2. *If X is a complex subvariety of a complex space $\widetilde{X}$, then a pair of compact subsets $A, B \subset X$ is a Cartan pair in X if and only if it is a Cartan pair in $\widetilde{X}$.*

We shall need the following result on approximating Cartan pairs in complex manifolds by smooth strongly pseudoconvex Cartan pairs.

Proposition 5.7.3. *Given a Cartan pair (A, B) in a complex manifold X and open sets $U \supset A$, $V \supset B$ in X, there exists a smooth strongly pseudoconvex Cartan pair (D_0, D_1) satisfying $A \subset D_0 \Subset U$ and $B \subset D_1 \Subset V$.*

Proof. Fix a distance function dist on X induced by a smooth Riemannian metric. Given a subset $A \subset X$ and a number $r > 0$, we set

$$A(r) = \{x \in X \colon \mathrm{dist}(x, y) < r \text{ for some } y \in A\}.$$

Lemma 5.7.4. *Given $A, B \subset X$ and $r > 0$ we have*

$$(A \cup B)(r) = A(r) \cup B(r), \quad (A \cap B)(r) \subset A(r) \cap B(r).$$

If A and B are (relatively) compact and separated in X then for all sufficiently small $r > 0$ we also have that $(A \cap B)(r) = A(r) \cap B(r)$, and the sets $\overline{A(r)}$, $\overline{B(r)}$ are separated.

Proof. The first two properties are immediate. Now write $A = (A \backslash B) \cup (A \cap B)$, $B = (B \backslash A) \cup (A \cap B)$ and apply the first property to get

$$A(r) = (A \backslash B)(r) \cup (A \cap B)(r), \quad B(r) = (B \backslash A)(r) \cup (A \cap B)(r).$$

If $\overline{A \backslash B} \cap \overline{B \backslash A} = \emptyset$ then for all sufficiently small $r > 0$ we have $(A \backslash B)(r) \cap (B \backslash A)(r) = \emptyset$ (in fact, even the closures of $(A \backslash B)(r)$ and $(B \backslash A)(r)$ are disjoint). Hence the previous display gives $A(r) \cap B(r) = (A \cap B)(r)$, and also the separation property for the pair $\overline{A(r)}$, $\overline{B(r)}$. $\square$

Since A and B are Stein compacts, there exist smooth strongly pseudoconvex domains D_0', D_1' in X such that $A \subset D_0' \Subset U$ and $B \subset D_1' \Subset V$. Choose $r > 0$ such that $A(r) \Subset D_1'$, $B(r) \Subset D_1'$ and the conclusions of Lemma 5.7.4 hold. Since $A \cup B$ is a Stein compact, there is a closed strongly pseudoconvex Stein domain $\Omega \subset X$ such that $A \cup B \subset \Omega \Subset A(r) \cup B(r)$. Set

$$A' = D_0' \cap \Omega, \quad B' = D_1' \cap \Omega.$$

Then $A' \cup B' = \Omega$ and the sets A' and B' are separated. By a small deformation of Ω we may assume that the intersections of $b\Omega$ with bD_0' and bD_1' are transverse, and hence A' and B' are piecewise smooth strongly pseudoconvex domains. By smoothing the corners of A' and B' we get a pair of smooth strongly pseudoconvex domains D_0, D_1 satisfy the stated properties. $\square$

5.8 A Splitting Lemma

One of the main analytic ingredients in the proof of Theorem 5.4.4 is a method of splitting and gluing holomorphic sprays which are local in the parameter. (Such sprays can be thought of as *thick maps* or *thick holomorphic sections*; they should not be confused with the global sprays in the sense of Def. 5.5.11.) Proposition 5.8.1, which generalizes Cartan's lemma on factoring invertible holomorphic matrices [228, p. 88], applies to any fiber preserving holomorphic map that is close enough to the identity, with control up to the boundary. This technique has already found a number of applications which unfortunately can not be presented here due to a self-imposed bound on the total length; the reader is referred to the papers [121, 122, 123, 124, 178]. For similar gluing techniques in almost complex geometry see e.g. Donaldson [115] and McDuff and Salamon [353].

Given a compact subset K in a complex space X and an open set $W \subset \mathbb{C}^N$ we shall consider maps $\gamma\colon K \times W \to K \times \mathbb{C}^N$ of the form

$$\gamma(x,w) = \big(x, \psi(x,w)\big), \quad x \in K,\ w \in W. \tag{5.11}$$

We say that $\gamma \in \mathcal{A}(K \times W)$ if γ is continuous on $K \times W$ and holomorphic in $\mathrm{Int}K \times W$. Let $\mathrm{Id}(x,w) = (x,w)$, the identity map on $X \times \mathbb{C}^N$. Set

$$\mathrm{dist}_{K\times W}(\gamma, \mathrm{Id}) = \sup\{|\psi(x,w) - w| \colon x \in K,\ w \in W\}.$$

Proposition 5.8.1. *Let (D_0, D_1) be a strongly pseudoconvex Cartan pair of class $\mathcal{C}^2$ in a complex manifold X (Def. 5.7.1 on p. 209). Set $D_{0,1} = D_0 \cap D_1$ and $D = D_0 \cup D_1$. Given a bounded open convex set $0 \in W \subset \mathbb{C}^N$ and a number $r \in (0,1)$, there is a $\delta > 0$ satisfying the following. For every map $\gamma\colon \bar{D}_{0,1} \times W \to \bar{D}_{0,1} \times \mathbb{C}^N$ of the form (5.11) and of class $\mathcal{A}(D_{0,1} \times W)$, satisfying $\mathrm{dist}_{D_{0,1}\times W}(\gamma, \mathrm{Id}) < \delta$, there exist maps*

$$\alpha_\gamma\colon \bar{D}_0 \times rW \to \bar{D}_0 \times \mathbb{C}^N, \quad \beta_\gamma\colon \bar{D}_1 \times rW \to \bar{D}_1 \times \mathbb{C}^N$$

of the form (5.11) and of class $\mathcal{A}(D_0 \times rW)$ and $\mathcal{A}(D_1 \times rW)$, respectively, depending smoothly on γ, such that $\alpha_{\mathrm{Id}} = \mathrm{Id}$, $\beta_{\mathrm{Id}} = \mathrm{Id}$ and

$$\gamma \circ \alpha_\gamma = \beta_\gamma \quad \text{on } \bar{D}_{0,1} \times rW.$$

If γ agrees with Id to order $m \in \mathbb{N}$ along $w = 0$ then so do α_γ and β_γ. Furthermore, if X' is a closed complex subvariety of X such that $X' \cap \bar{D}_{0,1} = \emptyset$ then we can choose α_γ and β_γ to be tangent to Id to any given finite order along $(X' \cap \bar{D}_0) \times rW$ and $(X' \cap \bar{D}_1) \times rW$, respectively.

Proof. Denote by C_r and Γ_r the Banach space consisting of all continuous maps $K \times rW \to \mathbb{C}^N$ which are holomorphic in $D_{0,1} \times rW$ and satisfy

$$||\phi||_{C_r} = \sup\{|\phi(x,w)|: x \in \bar{D}_{0,1},\ w \in rW\} < +\infty,$$
$$||\phi||_{\Gamma_r} = \sup\{\big(|\phi(x,w)| + |\partial_w \phi(x,w)|\big): x \in \bar{D}_{0,1},\ w \in rW\} < +\infty.$$

Here, ∂_w denotes the partial differential with respect to the variable $w \in \mathbb{C}^N$, and $|\partial_w \phi(x,w)|$ is the Euclidean operator norm. Similarly we denote by A_r, B_r the Banach space of all continuous maps $\bar{D}_0 \times rW \to \mathbb{C}^N$, $\bar{D}_1 \times rW \to \mathbb{C}^N$, respectively, that are holomorphic in the interior, endowed with the sup-norm.

Choose a number $r_1 \in (r,1)$. By the hypotheses $\gamma(x,w) = (x, \psi(x,w))$ with $\psi \in C_1$. Set $\psi_0(x,w) = w$. By the Cauchy estimates the restriction map $C_1 \to \Gamma_{r_1}$ is continuous, hence $\psi|_{K \times r_1 W} \in \Gamma_{r_1}$ and $||\psi - \psi_0||_{\Gamma_{r_1}} \le \mathrm{const}||\psi - \psi_0||_{C_1}$.

Lemma 5.8.2. *There exist bounded linear operators $\mathcal{A}: C_r \to A_r$, $\mathcal{B}: C_r \to B_r$ satisfying*

$$c = \mathcal{A}c - \mathcal{B}c, \qquad c \in C_r. \tag{5.12}$$

If $c(x,w)$ vanishes to order $m \in \mathbb{N}$ at $w = 0$ then so do $\mathcal{A}c$ and $\mathcal{B}c$. If X' is a closed complex subvariety of X such that $X' \cap \bar{D}_{0,1} = \emptyset$ then we can also insure that $\mathcal{A}c$ and $\mathcal{B}c$ vanish to a given finite order along $(X' \cap \bar{D}_0) \times rW$ and $(X' \cap \bar{D}_1) \times rW$, respectively.

Proof. By the separation condition (ii) in Def. 5.7.1 there is a smooth function $\chi: X \to [0,1]$ such that $\chi = 0$ in a neighborhood of $\overline{D_0 \backslash D_1}$, and $\chi = 1$ in a neighborhood of $\overline{D_1 \backslash D_0}$. Thus for any $c \in C_r$ the product $\chi(x)c(x,w)$ extends to a continuous function on $\bar{D}_0 \times W$ which vanishes on $\overline{D_0 \backslash D_1} \times W$, and $(\chi(x) - 1)c(x,w)$ extends to a continuous function on $\bar{D}_1 \times W$ which vanishes on $\overline{D_1 \backslash D_0} \times W$. Furthermore,

$$\overline{\partial}(\chi c) = \overline{\partial}((\chi - 1)c) = c\overline{\partial}\chi$$

is a $(0,1)$-form on $\bar{D}$ with continuous coefficients and with support in $\bar{D}_{0,1} \times W$, depending holomorphically on $w \in W$.

Choose functions $f_1, \ldots, f_m \in \mathcal{O}(X)$ that vanish to order k along X' and have no common zero on $\bar{D}_{0,1}$. Since $D_{0,1}$ is strongly pseudoconvex, Cartan's division theorem (Corollary 2.4.4) gives holomorphic functions $g_1, \ldots, g_m$ in a neighborhood of $\bar{D}_{0,1}$ such that $\sum_{j=1}^m f_j g_j = 1$ there.

Since the domain $D = D_0 \cup D_1$ is strongly pseudoconvex, there exists a sup-norm bounded linear solution operator $T: \mathcal{C}^0_{0,1}(D) \to \mathcal{C}^0(D)$ to the $\overline{\partial}$-equation on D at the level of $(0,1)$-forms (Theorem 2.5.3, p. 56). For any $c \in C_r$ and $w \in rW$ we set

$$(\mathcal{A}c)(x,w) = \chi(x)\, c(x,w) - \sum_{j=1}^m f_j(x)\, T\big(g_j\, c(\cdot\,,w)\overline{\partial}\chi\big)(x), \qquad x \in \bar{D}_0;$$

$$(\mathcal{B}c)(x,w) = \big(\chi(x) - 1\big)c(x,w) - \sum_{j=1}^m f_j(x)\, T\big(g_j\, c(\cdot\,,w)\overline{\partial}\chi\big)(x), \quad x \in \bar{D}_1.$$

Then $\mathcal{A}c - \mathcal{B}c = c$ on $\bar{D}_{0,1} \times rW$ and $\overline{\partial}_x(\mathcal{A}c) = 0$, $\overline{\partial}_x(\mathcal{B}c) = 0$ in the interior of their respective domains. Since $\overline{\partial}_w \big(c(x,w)\overline{\partial}\chi(x)\big) = 0$ and the operator T commutes with $\overline{\partial}_w$, we also have $\overline{\partial}_w(\mathcal{A}c) = 0$ and $\overline{\partial}_w(\mathcal{B}c) = 0$. The estimates follow from boundedness of T. $\square$

Given $\psi \in \Gamma_{r_1}$ sufficiently near ψ_0 and $c \in C_r$ sufficiently near 0, we define for $x \in \bar{D}_{0,1}$ and $w \in rW$:

$$\Phi(\psi, c)(x,w) = \psi\big(x, w + (\mathcal{A}c)(x,w)\big) - \big(w + (\mathcal{B}c)(x,w)\big).$$

Then $(\psi, c) \to \Phi(\psi, c)$ is a smooth map from an open neighborhood of $(\psi_0, 0)$ in the Banach space $\Gamma_{r_1} \times C_r$ to the Banach space C_r. Indeed, Φ is linear in ψ, and its partial differential with respect to the second variable equals

$$\partial_c \Phi(\psi, c_0)c(x,w) = \partial_c \psi\big(x, w + (\mathcal{A}c_0)(x,w)\big) \cdot (\mathcal{A}c)(x,w) - (\mathcal{B}c)(x,w).$$

This is again linear in ψ and continuous in all variables. A similar argument applies to higher order differentials of Φ. Since

$$\Phi(\psi_0, c) = \mathcal{A}(c) - \mathcal{B}(c) = c, \quad c \in C_r$$

by (5.12), we see that $\partial_c \Phi(\psi_0, 0)$ is the identity map on C_r. By the implicit function theorem there exists a smooth map $\psi \to \mathcal{C}(\psi) \in C_r$, defined in an open neighborhood of ψ_0 in Γ_{r_1} and satisfying $\Phi(\psi, \mathcal{C}(\psi)) = 0$ and $\mathcal{C}(\psi_0) = 0$. The maps

$$a_\psi(x,w) = w + \mathcal{A} \circ \mathcal{C}(\psi)(x,w), \quad b_\psi(x,w) = w + \mathcal{B} \circ \mathcal{C}(\psi)(x,w)$$

then satisfy $a_\psi \in A_r$, $a_{\psi_0} = \psi_0$, $b_\psi \in B_r$, $b_{\psi_0} = \psi_0$ and

$$\psi\big(x, a_\psi(x,w)\big) = b_\psi(x,w), \quad (x,w) \in \bar{D}_{0,1} \times rW.$$

The associated maps

$$\alpha_\gamma(x,w) = \big(x, a_\psi(x,w)\big), \quad \beta_\gamma(x,w) = \big(x, b_\psi(x,w)\big)$$

depend smoothly on γ and satisfy the conclusion of Proposition 5.8.1. $\square$

Remark 5.8.3. (A) The solutions $\alpha_\gamma = \alpha(\gamma)$ and $\beta_\gamma = \beta(\gamma)$ in in Proposition 5.8.1 are given by smooth nonlinear operators defined in a neighborhood of the identity map. If γ depends continuously or smoothly on some parameters, we get the same type of dependence in solutions.

(B) The analogous splitting lemma holds in any Banach spaces of holomorphic functions for which there exists a bounded linear solution operator to the $\overline{\partial}$-equation on the level of $(0,1)$-forms on strongly pseudoconvex domains; an example is H^∞ or Sobolev spaces. Further, if (D_0, D_1) is a strongly pseudoconvex Cartan pair of class $\mathcal{C}^\ell$ for some $\ell \geq 2$ then for every $l \in \{0, 1, \ldots, \ell\}$ the

analogous result holds for maps α, β, γ of class $\mathcal{A}^l$ on their respective domains (holomorphic in the interior and of class $\mathcal{C}^l$ up to the boundary), with $\mathcal{C}^l$ estimates [121, Theorem 3.2]. One uses appropriate function spaces and (in the proof of Lemma 5.8.2) a bounded linear solution operator $T\colon \mathcal{C}^l_{0,1}(D) \to \mathcal{C}^l(D)$ to the $\overline{\partial}$-equation, furnished by Theorem 2.5.3 on p. 56.

(C) Proposition 5.8.1 easily extends to the case when $\bar{D} \times \mathbb{C}^N$ is replaced by the total space of a complex vector bundle $\pi\colon E \to \bar{D}$ which is holomorphic over D and continuous (or smooth) over $\bar{D}$. Such E can be embedded as a complex vector subbundle of class $\mathcal{A}^l(D) = \mathcal{C}^l(\bar{D}) \cap \mathcal{O}(D)$ in a trivial bundle $\bar{D} \times \mathbb{C}^N$, and we have $\bar{D} \times \mathbb{C}^N = E \oplus E'$ for another subbundle E' of the same class. (These facts follows from Cartan's Theorem B for $\mathcal{A}^l(D)$-bundles; see the paper [329, 330, 261, 262].) $\square$

We now prove a version of Proposition 5.8.1 in which (A, B) is a Cartain pair in an arbitrary complex space.

Proposition 5.8.4. *Assume that (A, B) is a Cartan pair in a complex space X. Set $C = A \cap B$. Given an open set $U \supset C$ in X, a bounded open convex set $0 \in W \subset \mathbb{C}^N$ and a number $r \in (0, 1)$, there exist arbitrarily small open neighborhoods $U_A \supset A$, $U_B \supset B$ with $U_{A,B} = U_A \cap U_B \subset U$, and a number $\delta > 0$ satisfying the following. Given a holomorphic map $\gamma\colon U \times W \to U \times \mathbb{C}^N$ (5.11) with $\mathrm{dist}_{U\times W}(\gamma, \mathrm{Id}) < \delta$, there exist holomorphic maps*

$$\alpha_\gamma\colon U_A \times rW \to U_A \times \mathbb{C}^N, \quad \beta_\gamma\colon U_B \times rW \to U_B \times \mathbb{C}^N$$

of the form (5.11), depending continuously on γ, with $\alpha_{\mathrm{Id}} = \mathrm{Id}$, $\beta_{\mathrm{Id}} = \mathrm{Id}$ and

$$\gamma \circ \alpha_\gamma = \beta_\gamma \quad \text{on } U_{A,B} \times rW.$$

If γ agrees with Id to order $m \in \mathbb{N}$ along $w = 0$ then so do α_γ and β_γ. Further, if X' is a closed complex subvariety of X such that $X' \cap C = \emptyset$ then we can choose α_γ to be tangent to Id to any given finite order along $(X' \cap A) \times rW$.

Proof. Since $D = A \cup B$ is a Stein compact in X, an open Stein neighborhood $U_0 \subset X$ of D embeds holomorphically as a closed complex analytic subvariety in an open set $\widetilde{D} \subset \mathbb{C}^n$. We replace X by U_0 and identify it with the image subvariety in the domain $\widetilde{D} \subset \mathbb{C}^n$; then (A, B) is a Stein compact in $\mathbb{C}^n$. We may assume that the open set $U \supset C$ in the proposition is of the form $U = X \cap \widetilde{U}$ for some pseudoconvex domain $\widetilde{U} \subset \mathbb{C}^n$.

Choose a number $r_0 \in (r, 1)$ and a pseudoconvex domain $\widetilde{C} \Subset \widetilde{U}$ such that $C \subset \widetilde{C} \cap X \Subset U$. Lemma 5.6.2 furnishes a bounded extension operator $S\colon H^\infty(U \times W) \to H^\infty(\widetilde{C} \times r_0 W)$. Given a map $\gamma(x, w) = \big(x, w + c(x, w)\big)$ $(x \in U,\ w \in W)$ as in Proposition 5.8.4, we set

$$\widetilde{\gamma}(x, w) = \big(x, w + S(c)(x, w)\big), \quad x \in \widetilde{C},\ w \in r_0 W.$$

Then $\widetilde{\gamma}$ is a holomorphic map of the same type (5.11), it agrees with γ on the set $\widetilde{C} \cap X$, and we have an estimate

$$\mathrm{dist}_{\widetilde{C}\times(r_0W)}(\widetilde{\gamma}, \mathrm{Id}) \le M \mathrm{dist}_{(\widetilde{C}\cap X)\times W}(\gamma, \mathrm{Id}) \le M\delta,$$

where the constant $M < +\infty$ comes from the bounded extension operator.

Choose open sets $\widetilde{A}, \widetilde{B} \subset \mathbb{C}^n$ with $A \subset \widetilde{A} \subset \widetilde{D}$ and $B \subset \widetilde{B} \subset \widetilde{D}$. By Lemma 5.7.3 (p. 210) there exists a smooth strongly pseudoconvex Cartan pair (D_0, D_1) in $\mathbb{C}^n$ such that $A \subset D_0 \subset \widetilde{A}$ and $B \subset D_1 \subset \widetilde{B}$. If $\delta > 0$ is small enough then Proposition 5.8.1, applied to $\widetilde{\gamma}$ on the Cartan pair (D_0, D_1), furnishes holomorphic maps $\widetilde{\alpha}\colon D_0 \times rW \to D_0 \times \mathbb{C}^N$, $\widetilde{\beta}\colon D_1 \times rW \to D_1 \times \mathbb{C}^N$ satisfying the condition

$$\widetilde{\gamma} \circ \widetilde{\alpha} = \widetilde{\beta} \quad \text{on } (D_0 \cap D_1) \times rW.$$

Take $U_A = D_0 \cap X$, $U_B = D_1 \cap X$ and denote by α (resp. β) the restriction of $\widetilde{\alpha}$ (resp. $\widetilde{\beta}$) to $U_A \times rW$ (resp. to $U_B \times rW$). $\square$

5.9 Gluing Local Holomorphic Sprays

In this section we show how Proposition 5.8.1 (the splitting lemma) can be used to glue holomorphic sprays over a Cartan pair in the base space.

Assume that X and Z are complex spaces and $\pi\colon Z \to X$ is a holomorphic submersion. For each point $z \in Z$ we denote by $\mathrm{VT}_z Z = \ker \mathrm{d}\pi_z$ the vertical tangent space of Z (the tangent space to the fiber $\pi^{-1}(\pi(z))$ at z).

Definition 5.9.1. *Let D be a domain in X and let P be an open set in $\mathbb{C}^N$ containing the origin. A* local holomorphic fiber-spray (or π-spray) over D with the parameter set P *is a holomorphic map $f\colon D \times P \to Z$ satisfying*

$$\pi(f(x,w)) = x, \qquad x \in D,\ w \in P. \tag{5.13}$$

Such f is dominating *on a subset $K \subset D$ if the partial differential*

$$\partial_w|_{w=0} f(x,w)\colon \mathrm{T}_0\mathbb{C}^N = \mathbb{C}^N \to \mathrm{VT}_{f(x,0)} Z$$

is surjective for all $x \in K$. We call $f_0 = f(\cdot\,, 0)$ the central (core) section *of f. If D is a relatively compact domain with boundary of class $\mathcal{C}^l$ contained in X_{reg} then a* π-spray of class $\mathcal{A}^l(D)$ with the parameter set P *is a $\mathcal{C}^l$ map $f\colon \bar{D} \times P \to Z$ that is holomorphic on $D \times P$ and satisfies (5.13).*

The following result is our main *Heftungslemma* for holomorphic sprays; it can be viewed as a solution of a nonlinear Cousin-I problem.

Proposition 5.9.2. [122, Proposition 2.4] *Assume that Z is a complex manifold, X is a Stein manifold, $\pi\colon Z \to X$ is a holomorphic submersion, and (D_0, D_1) is a strongly pseudoconvex Cartan pair of class $\mathcal{C}^\ell$ ($\ell \geq 2$) in X (Def. 5.7.1 on p. 209). Set $D = D_0 \cup D_1$, $D_{0,1} = D_0 \cap D_1$. Let $l \in \{0, 1, \ldots, \ell\}$. Given a π-spray $f\colon \bar{D}_0 \times P_0 \to X$ ($P_0 \subset \mathbb{C}^N$) of class $\mathcal{A}^l(D_0)$ which is dominating on $\bar{D}_{0,1}$, there is an open set $P \subset \mathbb{C}^N$ with $0 \in P \subset P_0$ satisfying the following properties:*

1. *For every π-spray $g\colon \bar{D}_1 \times P_0 \to X$ of class $\mathcal{A}^l(D_1)$ which is sufficiently close to f in $\mathcal{C}^l(\bar{D}_{0,1} \times P_0)$ there exists a π-spray $f'\colon \bar{D} \times P \to X$ of class $\mathcal{A}^l(D)$, close to f in $\mathcal{C}^l(\bar{D}_0 \times P)$ (depending on the $\mathcal{C}^l$-distance between f and g on $\bar{D}_{0,1} \times P_0$), whose core f'_0 is homotopic to f_0 on $\bar{D}_0$ and is homotopic to g_0 on $\bar{D}_1$.*
2. *If f and g agree to order $m \in \mathbb{Z}_+$ along $\bar{D}_{0,1} \times \{0\}$, then f' can be chosen to agree to order m with f along $\bar{D}_0 \times \{0\}$, and with g along $\bar{D}_1 \times \{0\}$.*
3. *If σ is the zero set of finitely many $\mathcal{A}^l(D_0)$ functions and $\sigma \cap \bar{D}_{0,1} = \emptyset$, then f' can be chosen such that f'_0 agrees with f_0 to a finite order on σ.*

In the proof we will need the following lemma which provides a transition map between the two sprays.

Lemma 5.9.3. (Hypotheses as in Proposition 5.9.2) *Let $\epsilon > 0$. If g approximates f sufficiently closely in the $\mathcal{C}^l$ topology on $\bar{D}_{0,1} \times P_0$, then there exist a domain P_1 in $\mathbb{C}^N$ with $0 \in P_1 \subset P_0$ and a map $\psi\colon \bar{D}_{0,1} \times P_1 \to \mathbb{C}^N$ of class $\mathcal{A}^l(D \times P_1)$ satisfying*

$$\psi(x, w) = w + c(x, w), \qquad ||c||_{\mathcal{C}^l(D_{0,1} \times P_1)} < \epsilon, \tag{5.14}$$

$$f(x, w) = g\big(x, \psi(x, w)\big), \quad (x, w) \in \bar{D}_{0,1} \times P_1. \tag{5.15}$$

If f and g agree to order m along $\bar{D} \times \{0\}$ then ψ can be chosen of the form $\psi(x, w) = w + \sum_{|J|=m} \widetilde{c}_J(x, w) w^J$ with $\widetilde{c}_J \in \mathcal{A}^l(D_{0,1} \times P_1)^N$.

Proof. Let E denote the $\mathcal{A}^l(D_{0,1})$-subbundle of $\bar{D}_{0,1} \times \mathbb{C}^N$ with fibers

$$E_x = \ker\left(\partial_w f(x, w)|_{w=0}\colon \mathbb{C}^N \to \mathrm{VT}_{f(x,0)} Z\right), \quad x \in \bar{D}_{0,1}. \tag{5.16}$$

By Theorem B for $\mathcal{A}^l$-bundles [262, 330], E is complemented in the sense that there is another $\mathcal{A}^l(D_{0,1})$-subbundle E' of $\bar{D}_{0,1} \times \mathbb{C}^N$ such that

$$\bar{D}_{0,1} \times \mathbb{C}^N = E \oplus E' \tag{5.17}$$

For each $x \in \bar{D}_{0,1}$ we write $\mathbb{C}^N \ni w = t_x \oplus t'_x \in E_x \oplus E'_x$. The map

$$\partial_w\big|_{w=0} f(\cdot\,, w)\colon E'|_{\bar{D}_{0,1}} \to \mathrm{VT}_{f_0(\bar{D}_{0,1})} Z$$

is an isomorphism. The implicit function theorem now gives an open neighborhood $P_1 \Subset P_0$ of $0 \in \mathbb{C}^N$ such that for each spray $g\colon \bar{D}_{0,1} \times P_0 \to Z$ which is sufficiently $\mathcal{C}^l$-close to f on $\bar{D}_{0,1} \times P_0$ there is a unique map

$$\widetilde{\psi}(x,w) = \widetilde{\psi}(x, t_x \oplus t'_x) = t_x \oplus \big(t'_x + \widetilde{c}(x,w)\big) \in E_x \oplus E'_x = \mathbb{C}^N$$

of class $\mathcal{A}^l(D_{0,1} \times P_1)$ solving $f(x, \widetilde{\psi}(x,w)) = g(x,w)$, and $||\widetilde{c}||_{\mathcal{A}^l(D_{0,1}\times P_1)}$ is controlled by the $\mathcal{C}^l$ distance between f and g on $\bar{D}_{0,1} \times P_0$. After shrinking P_1 the map $\widetilde{\psi}$ admits a fiberwise inverse $\psi(x,w) = t_x \oplus (t'_x + c'(x,w))$ which satisfies the conclusions of Lemma 5.9.3. □

Proof (of Proposition 5.9.2). By Lemma 5.9.3 there exist a domain $P_1 \Subset P_0$ containing the origin and a transition map $\gamma\colon \bar{D}_{0,1} \times P_1 \to \bar{D}_{0,1} \times \mathbb{C}^N$ of the form $\gamma(x,w) = (x, \psi(x,w))$ and of class $\mathcal{A}^l(D_{0,1} \times P_1)$, close to the identity map $\mathrm{Id}(x,w) = (x,w)$ in the $\mathcal{C}^l$ topology (the closeness depending on the $\mathcal{C}^l$ distance between f and g on $\bar{D}_{0,1} \times P_0$), satisfying

$$f = g \circ \gamma \qquad \text{on } \bar{D}_{0,1} \times P_1.$$

Let $P \Subset P_1$ be a domain containing the origin $0 \in \mathbb{C}^N$. If γ is sufficiently $\mathcal{C}^l$-close to the identity on $\bar{D}_{0,1} \times P_1$ then by Proposition 5.8.1 and Remark 5.8.3 (B) there exist maps

$$\alpha\colon \bar{D}_0 \times P \to \bar{D}_0 \times \mathbb{C}^N, \quad \beta\colon \bar{D}_1 \times P \to \bar{D}_1 \times \mathbb{C}^N,$$

of class $\mathcal{A}^l$ on their respective domains, satisfying

$$\gamma \circ \alpha = \beta \qquad \text{on } \bar{D}_{0,1} \times P.$$

From this and $f = g \circ \gamma$ it follows that

$$f \circ \alpha = g \circ \beta \quad \text{on } \bar{D}_{0,1} \times P.$$

Hence $f \circ \alpha$ and $g \circ \beta$ amalgamate into a π-spray $f'\colon \bar{D} \times P \to Z$ with the stated properties. □

Remark 5.9.4. (A) If the cores of the two sprays in Proposition 5.9.2 agree on $D_{0,1}$ (as in part 2.) then it suffices to assume that g is defined on $(\bar{D}_{0,1} \times P_0) \cup (\bar{D}_1 \times P')$ for some domain $P' \subset \mathbb{C}^N$ containing the origin, but possibly much smaller than P_0.

(B) Proposition 5.9.2 holds with the same additions as Proposition 5.8.1; see Remark 5.8.3 on p. 213. In particular, one can glue sprays defined on not necessarily trivial holomorphic vector bundles $E \to \bar{D}$.

(C) The same proof gives the analogue of Proposition 5.9.2 for gluing local holomorphic fiber-sprays defined over open neighborhoods of an arbitrary Cartan pair (A,B) in a possibly singular Stein space (these neighborhoods are allowed to shrink during the gluing process). To this end one uses Proposition 5.8.4 instead of Proposition 5.8.1. The proof of Lemma 5.9.3 applies in this setting without any changes and is even simpler. □

5.10 Noncritical Strongly Pseudoconvex Extensions

By applying the Heftungslemma from the previous section we now prove the following major step in the proof of Theorem 5.4.4.

Proposition 5.10.1. *Assume that $A \subset A'$ are compact strongly pseudoconvex domains in a Stein manifold X such that there exists a strongly plurisubharmonic function ρ in an open set $V \supset \overline{A'\backslash A}$ satisfying*

$$A \cap V = \{x \in V : \rho(x) \le c\}, \quad A' \cap V = \{x \in V : \rho(x) \le c'\},$$

and such that ρ has no critical values in $[c, c']$. (Such A' is said to be a noncritical strongly pseudoconvex extension *of A.) Assume that $\pi : Z \to X$ is a holomorphic submersion such that $Z|_V \to V$ is a holomorphic fiber bundle whose fiber enjoys* CAP. *Then every holomorphic section $f : U \to Z|_U$ over an open set $U \supset A$ can be approximated, uniformly on A, by sections f' which are holomorphic on an open neighborhood of A'.*

The main step in the proof is to extend a holomorphic section to a special convex bump that we now introduce. Proposition 5.10.1 follows immediately from Lemmas 5.10.3 and 5.10.5 below.

Definition 5.10.2. *Let X be a complex space. A pair of compact sets (A, B) in X is a* special Cartan pair, *and B is a* special convex bump on A, *if*

(i) *(A, B) is a Cartan pair in X (Def. 5.7.1 (I), p. 209), and*

(ii) *there are a compact set $\widetilde{Q} \subset X_{\mathrm{reg}}$ (a* window *for B), containing B in its interior, and a holomorphic coordinate map ϕ from an open neighborhood of $\widetilde{Q}$ to $\mathbb{C}^n$ such that $Q = \phi(\widetilde{Q})$ is a cube (5.5) (p. 192), and the sets*

$$K = \phi(A \cap \widetilde{Q}), \qquad K' = \phi((A \cup B) \cap \widetilde{Q})$$

are special convex sets in Q (Def. 5.4.2, p. 192).

The situation is illustrated in Figure 5.2.

Lemma 5.10.3. *If $A \subset A' \subset X$ are as in Proposition 5.10.1 then there is a finite sequence of compact strongly pseudoconvex domains $A = A_0 \subset A_1 \subset \cdots \subset A_m = A'$ such that for every $k = 0, 1, \ldots, m-1$ we have $A_{k+1} = A_k \cup B_k$, where B_k is a special bump on A_k (Def. 5.10, p. 218). In addition, given an open cover $\mathcal{U} = \{U_j\}$ of $\overline{A'\backslash A}$, we can choose the above sequence such that a window $\widetilde{Q}_k \supset B_k$ for each B_k is contained in some U_j.*

Proof. Given a pair of numbers a, b with $c \le a \le b \le c'$ we set

$$X_{a,b} = \{x \in V : a \le \rho(x) \le b\}.$$

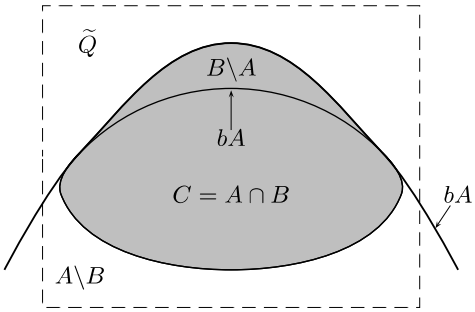

Fig. 5.2. A special convex bump B in a window $\widetilde{Q}$. (Modified from [180, p. 860, Fig. 2])

If $b-a>0$ is sufficiently small, Narasimhan's lemma (on local convexification of a strongly pseudoconvex hypersurface) shows that we can cover $X_{a,b}$ by the interiors of finitely many compact sets $\widetilde{Q}_1,\dots,\widetilde{Q}_N \subset V$ such that for every $j=1,\dots,N$ there is a holomorphic coordinate map $\phi_j\colon U_j \to \phi_j(U_j) \subset \mathbb{C}^n$ on a neighborhood U_j of $\widetilde{Q}_j$ satisfying

(a) $\phi_j(\widetilde{Q}_j) = Q \subset \mathbb{C}^n$ is a cube of the form (5.5),
(b) the function $\widetilde{\rho}_j = \rho \circ \phi_j^{-1}\colon Q \to \mathbb{R}$ is strongly convex,
(c) let $Q = Q_1 \times Q_2$, with $Q_1 \subset \mathbb{C}^{n-1} \times \mathbb{R}$ and $Q_2 \subset \mathrm{i}\mathbb{R}$; then

$$\phi_j(X_{a,b} \cap \widetilde{Q}_j) \cap (Q_1 \times bQ_2) = \emptyset.$$

Property (c) shows that for every $t \in [a,b]$ the set

$$\phi_j(\{\rho \le t\} \cap \widetilde{Q}_j) = \{z \in Q\colon \widetilde{\rho}_j(z) \le t\}$$

is a special convex set (5.4) in Q.

By compactness we can partition $[c,c']$ into finitely many subintervals $c = c_0 < c_1 < \cdots < c_m = c'$ such that the above conditions hold on each subinterval $[c_{l-1}, c_l]$. It suffices to explain the construction of the bumps on each of the subintervals, and hence we may assume that the initial interval $[a,b] = [c,c']$ satisfies the above properties.

Using the established notation we choose a smooth partition of unity $\{\chi_j\}$ in a neighborhood of $X_{a,b} = \overline{A'\backslash A}$, with $\operatorname{supp}\chi_j \subset \operatorname{Int}\widetilde{Q}_j$ for $j=1,\dots,N$. For a sufficiently small $\epsilon > 0$ each of the functions $\rho_k = \rho - \sum_{j=1}^k \epsilon\chi_j$ is strongly plurisubharmonic, and the functions $\widetilde{\rho}_{j,k} = \rho_k \circ \phi_j^{-1}\colon \widetilde{Q}_j \to \mathbb{R}$ are strongly convex. Furthermore, for $\epsilon > 0$ small, the sets $\{z \in P\colon \widetilde{\rho}_{j,k} \le c\}$ for $c \in [a,b]$ are special convex sets of the form (5.4) (p. 192). Decreasing $\epsilon > 0$ if necessary we insure that $b - a = \epsilon l$ for some $l \in \mathbb{N}$.

Let $A_k = A \cup \{x \in V : \rho_k(x) \le a\}$ for $k = 0, 1, \ldots, N$. Since $\rho_k = \rho_{k-1} - \epsilon\chi_k$ and $\operatorname{supp} \chi_k \subset \operatorname{Int} \widetilde{Q}_k$, we see that

$$A_k = A \cup \left\{x \in V : \rho_{k-1}(x) \le a + \epsilon\chi_k(x)\right\} = A_{k-1} \cup B_k$$

where B_k is a special convex bump in the window $\widetilde{Q}_k$. Since $\sum_{j=1}^N \chi_j = 1$ in $X_{a,b}$, we see that $A_N = A \cup \{\rho \le a + \epsilon\}$. We now repeat the same procedure of attaching bumps to A_N. Explicitly, we take

$$A_{N+1} = A \cup \left\{x \in V : \rho(x) \le a + \epsilon + \epsilon\chi_1(x)\right\}$$

etc. By attaching N bumps, each contained in one of the windows $\widetilde{Q}_j$, we reach the set $A_{2N} = A \cup \{\rho \le a + 2\epsilon\}$. In l steps of this kind we reach the set $A' = A \cup \{\rho \le b\}$. $\square$

Lemma 5.10.4. (Thickening a section.) *Assume that $\pi : Z \to X$ is a holomorphic submersion onto a complex space X, X' is a closed complex subvariety of X, and $A_0 \subset A$ are Stein compacts in X such that $A_0 \subset X \backslash X'$. Given open Stein sets $V \Subset V_0 \subset X$ containing A, a holomorphic section $f : V_0 \to Z$ and an integer $r \in \mathbb{N}$, there exist $N \in \mathbb{N}$, an open set $W \subset \mathbb{C}^N$ containing 0, and a holomorphic map $F : V \times W \to Z$ such that*

(i) *$F_w = F(\cdot, w)$ is a section of $Z|_V$ for every fixed $w \in W$, with $F_0 = f$,*
(ii) *F_w agrees with f to order r along $V \cap X'$ for every $w \in W$, and*
(iii) *$\partial_w F(x, w)|_{w=0} : \mathbb{C}^N \to \mathrm{VT}_{f(x)} Z$ is surjective for every $x \in A_0$.*

If A_0 is $\mathcal{O}(A)$-convex and it admits a contractible Stein neighborhood in $V_0 \backslash X'$ then the above conclusion holds with $N = \dim \pi^{-1}(x)$ $(x \in A_0)$.

Proof. The image $f(V_0)$ is a closed Stein subvariety of $Z|_{V_0}$, and hence it admits an open Stein neighborhood $\Omega \subset Z$. By Cartan's Theorem A there exist finitely many holomorphic vector fields $v_1, \ldots, v_N$ on Ω that are tangent to the fibers of π, that span $\mathrm{VT}Z$ at every point of $f(A_0)$, and that vanish to order r on the subvariety $\pi^{-1}(X') \cap \Omega$. Let θ_j^t denote the flow of v_j. The map

$$F(x, w_1, \ldots, w_N) = \theta_1^{w_1} \circ \cdots \circ \theta_N^{w_N} \circ f(x)$$

is defined and holomorphic for all $x \in V$ and for all w in an open set $W \subset \mathbb{C}^N$ containing the origin. Since $\partial_{w_j} F(x, w)|_{w=0} = v_j(f(x))$ and since the vector fields v_j span $\mathrm{VT}Z$ on $f(A_0)$, we see that the differential $\partial_w F(x, w)|_{w=0} : \mathbb{C}^N \to \mathrm{VT}_{f(x)} Z$ is surjective for every $x \in A_0$.

This proves Lemma 5.10.4, except for the last claim that one can take $N = \dim \pi^{-1}(x)$ with $x \in A_0$. To prove this, we need the following special case of Theorem 5.3.1 on p. 190: *Every holomorphic vector bundle on a contractible Stein space is holomorphically trivial.* This is of course well known from the classical Oka-Grauert theory, but it also follows from our proof since

the particular value of N is not needed in the remainder of the proof of Proposition 5.10.5 (and hence of the proof of Theorem 5.4.4.) Since a complex homogeneous space satisfies CAP (see Proposition 5.5.1 on p. 198), we thus obtain Theorem 5.3.1, and in particular the above statement.

We now complete the proof of Lemma 5.10.4. Since the set A_0 admits a contractible Stein neighborhood, $f^*\mathrm{VT}Z$ is a trivial holomorphic vector bundle of rank $N = \dim \pi^{-1}(x)$ $(x \in A_0)$ in a neighborhood of A_0. Therefore $\mathrm{VT}Z$ is generated over a neighborhood of $f(A_0)$ in Z by N holomorphic sections. As the set A_0 is $\mathcal{O}(A)$-convex, we can approximate them by sections (vertical holomorphic vector fields) in a neighborhood of $f(A)$ that vanish to order r on $\pi^{-1}(X')$. If the approximation is sufficiently close then these new sections still generate $\mathrm{VT}Z$ on $f(A_0)$. Hence Lemma 5.10.4 holds with $N = \dim \pi^{-1}(x)$. (As we shall see below, this will prove Theorem 5.4.4 when each fiber Y satisfies CAP_{n+N} where $n = \dim X$ and $N = \dim Y$.) $\square$

Lemma 5.10.5. (Extension across a special convex bump) *Assume that X is a complex space, $X' \subset X$ is closed complex subvariety containing X_{sing}, and (A, B) is a special Cartan pair in X with a window $B \subset \widetilde{Q} \subset X \backslash X'$. Assume that $\pi\colon Z \to X$ is a holomorphic submersion whose restriction $Z|_U$ to an open set $U \supset \widetilde{Q}$ is equivalent to a trivial bundle $U \times Y$ whose fiber Y enjoys* CAP*. Every holomorphic section $f\colon V_0 \to Z$ over an open set $V_0 \supset A$ can be approximated uniformly on A by sections f' that are holomorphic over an open neighborhood of $A \cup B$ and agree with f to a given order $m \in \mathbb{N}$ along the subvariety X'. If in addition the set $A \cap B$ is $\mathcal{O}(A)$-convex then the same conclusion holds if Y satisfies* CAP_{n+N}*, where $N = \dim Y$ and n is the dimension of the component of X_{reg} containing B.*

Proof. Step 1: Thickening. Lemma 5.10.4 furnishes an open set V in X with $A \subset V \subset V_0$, an open set $0 \in W \subset \mathbb{C}^N$, and a holomorphic spray of sections $F\colon V \times W \to Z$ with $f = F(\cdot\,, 0)$ on V. Using the trivialization $Z|_U \cong U \times Y$ we can write $F(x, w) = (x, F'(x, w)) \in X \times Y$ for $x \in U \cap V$.

Step 2: Approximation. Set $K = \phi(A \cap \widetilde{Q})$ and $Q = \phi(\widetilde{Q})$; then (K, Q) is a special convex pair in $\mathbb{C}^n$ by the definition of a special Cartan pair. Choose a compact cube $Q' \subset W$ containing $0 \in \mathbb{C}^N$ in its interior; then $(K \times Q', Q \times Q')$ is a special convex pair in $\mathbb{C}^n \times \mathbb{C}^N$. Since Y enjoys CAP, we can approximate F' uniformly on a neighborhood of $(A \cap \widetilde{Q}) \times Q'$ by a holomorphic map $G'\colon U' \times Q' \to Y$, where U' is an open set satisfying $\widetilde{Q} \subset U' \subset U$. (For this approximation we need to assume that Y enjoys CAP_{n+N}. This is the unique place in the proof where CAP of the fiber is invoked.)

Step 3: Gluing. Set $G(x, w) = (x, G'(x, w))$. If the approximation in Step 2 is sufficiently close then by Proposition 5.9.2 and Remark 5.9.4 (C) we can glue F and G into a holomorphic spray $F'\colon V' \times rQ' \to Z$ $(0 < r < 1)$ over an open neighborhood V' of $A \cup B$ in X. The section $f' = F'(\cdot\,, 0)\colon V' \to Z$ then satisfies the conclusion of Lemma 5.10.5.

For the last conclusion we use Lemma 5.10.4 with $N = \dim Y$. $\square$

5.11 Proof of the Main Theorem: The Basic Case

In this section we prove Theorem 5.4.4 in the case when $\pi: Z \to X$ is a holomorphic fiber bundle over a Stein manifold X, $X' = \emptyset$, P is a singleton and $P_0 = \emptyset$. We assume that the fiber Y enjoys CAP. Let $K \subset X$ be a compact $\mathcal{O}(X)$-convex subset, and let $f_0: X \to Z$ be a continuous section that is holomorphic in an open set $U \subset X$ containing K.

Choose a smooth strongly plurisubharmonic Morse exhaustion function $\rho: X \to \mathbb{R}$ with nice critical points (Def. 3.9.2, p. 89) such that $\rho < 0$ on K and $\rho > 0$ on $X \backslash U$. Let $p_1, p_2, p_3, \ldots$ be the critical points of ρ in $\{\rho > 0\}$, ordered so that $0 < \rho(p_1) < \rho(p_2) < \rho(p_3) < \cdots$. Choose a sequence of numbers

$$0 = c_0 < c_1 < c_2 < \cdots, \quad \lim_{j\to\infty} c_j = +\infty$$

such that $c_{2j-1} < \rho(p_j) < c_{2j}$ for every $j = 1, 2, \ldots$. If there are only finitely many p_j's, we choose the remainder of the sequence c_j arbitrarily so that $\lim_{j\to\infty} c_j = +\infty$. We subdivide the parameter interval $[0, 1]$ of the homotopy into subintervals $I_j = [t_j, t_{j+1}]$ with $t_j = 1 - 2^{-j}$ $(j = 0, 1, 2, \ldots)$.

Choose a distance function dist on Z induced by a complete Riemannian metric. Fix an $\epsilon > 0$. We shall construct a homotopy of sections $f_t: X \to Z$ $(0 \le t < 1)$ such that for every $j \in \mathbb{Z}_+$ and $t \in [t_j, t_{j+1}]$ the section f_t is holomorphic in a neighborhood of the set $K_j = \{x \in X: \rho(x) \le c_j\}$ and

$$\sup\bigl\{\mathrm{dist}(f_t(x), f_{t_j}(x)): x \in K_j,\ t \in [t_j, t_{j+1}]\bigr\} < 2^{-j-1}\epsilon.$$

The limit section $f_1 = \lim_{t\to 1} f_t: X \to Z$ is then holomorphic on X and satisfies $\sup\{\mathrm{dist}(f_1(x), f_0(x)): x \in K_0\} < \epsilon$.

Assuming inductively that a homotopy $\{f_t\}$ with the stated properties has been constructed for $t \in [0, t_j]$, we explain how to find it for $t \in [t_j, t_{j+1}]$.

The noncritical case: If j is even then ρ has no critical points in $K_{j+1} \backslash K_j$. By Proposition 5.10.1 (p. 218) we can approximate f_{t_j} as well as desired, uniformly on a neighborhood of K_j, by a section $f_{t_{j+1}}$ which is holomorphic in a neighborhood of K_{j+1}. We also get a homotopy of global continuous sections f_t $(t \in [t_j, t_{j+1}])$ which are holomorphic in a neighborhood of K_{t_j}.

The critical case: If j is odd then ρ has a unique critical point $p \in K_{j+1} \backslash K_j$. Denote by E the stable manifold of p for the gradient flow of ρ (Fig. 5.3). For a small $\delta > 0$ Proposition 3.10.4 on p. 96 (applied with $q = 1$) gives a smooth strongly plurisubharmonic function τ on $\{x \in X: \rho(x) < \rho(p) + 3\delta\}$ satisfying

(i) $\{\rho \le c_j\} \cup E \subset \{\tau \le 0\} \subset \{\rho \le \rho(p) - \delta\} \cup E$,

(ii) $\{\rho \le \rho(p) + \delta\} \subset \{\tau \le 2\delta\} \subset \{\rho < \rho(p) + 3\delta\}$, and

(iii) τ has no critical values in $(0, 3\delta)$.

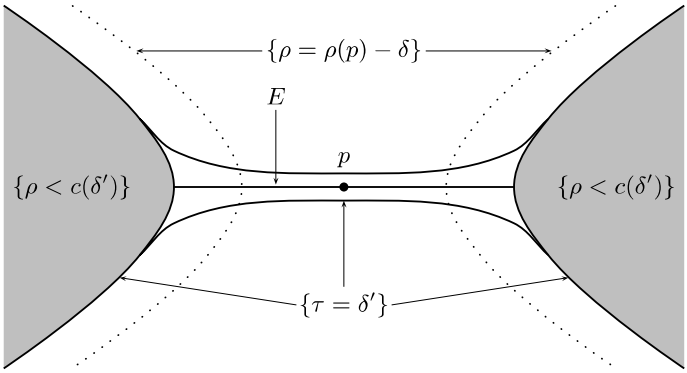

Fig. 5.3. The level sets of τ. (A modification of Fig. 3.5 on p. 94)

A typical level set $\{\tau = \delta'\}$ for small $\delta' > 0$ is shown on Fig. 5.3. Outside of a neighborhood of the critical point the level set $\{\tau = \delta'\}$ coincides with a certain level set $\{\rho = c(\delta')\}$ of ρ.

We proceed in four steps, hence dividing $[t_j, t_{j+1}]$ into four subintervals.

Step 1: By the noncritical case we deform f_{t_j} to a section that is holomorphic on a neighborhood of $\{\rho \leq \rho(p) - \delta\}$. The deformation consists of sections which are holomorphic and close to f_{t_j} near K_j.

Step 2: Theorem 3.7.2 on p. 81 allows us to deform the final section from Step 1 to a section that is holomorphic in a neighborhood of $\{\rho \leq \rho(p) - \delta\} \cup E$. The deformation consists of sections which are holomorphic and close to the initial section on a neighborhood of $\{x \in X : \rho(x) \leq \rho(p) - \delta\}$.

Step 3: By (i) the set $\{\tau \leq \delta'\}$ for a sufficiently small $\delta' > 0$ is contained in the region where the final section from Step 2 is holomorphic. Applying the noncritical case (Proposition 5.10.1 on p. 218) with the function τ we deform it to a holomorphic section on a neighborhood of $\{\tau \leq 2\delta\}$. (We disregard a part of the domain of the section from Step 2 on which $\tau > \delta'$.)

Step 4: By (ii), the section from Step 3 is holomorphic in a neighborhood of $\{\rho \leq \rho(p) + \delta\}$. Applying the noncritical case with ρ we deform it to a holomorphic section on a neighborhood of $K_{j+1} = \{\rho \leq c_{j+1}\}$. (When switching from τ back to ρ we again disregard a part of the domain.)

These four steps together give a homotopy $\{f_t : t \in [t_j, t_{j+1}]\}$, consisting of sections which are holomorphic and close to f_{t_j} near K_j and continuous elsewhere on X. The induction may proceed.

This completes the proof of the special case of Theorem 5.4.4.

5.12 Proof of the Main Theorem: Stratified Fiber Bundles

In this section we prove Theorem 5.4.4 in the nonparametric case $(P, P_0) = (\{p\}, \emptyset)$. The proof proceeds by a double induction: the outer one over a normal exhaustion of X, and the inner one over the strata in a suitable stratification. The following proposition provides the main step.

Proposition 5.12.1. *Assume that X is a Stein space, $M_1 \subset M_0$ are closed complex subvarieties of X such that $S = M_0 \backslash M_1$ is a complex manifold with $\bar{S} = M_0$, and $\pi\colon Z \to X$ is a holomorphic submersion such that $Z|_S \to S$ is a holomorphic fiber bundle whose fiber enjoys* CAP. *Given compact $\mathcal{O}(X)$-convex subsets $K \subset L$ of X and a continuous section $f\colon X \to Z$ that is holomorphic in an open neighborhood of $K_1 = K \cup (L \cap M_1)$, there exists for every $\epsilon > 0$ and $\ell \in \mathbb{N}$ a homotopy of continuous sections $f_t\colon X \to Z$ $(t \in [0,1])$ that are holomorphic in a neighborhood of K_1 satisfying the following:*

(i) *f_t agrees with $f_0 = f$ to order ℓ along $M_1 \cap L$ for each $t \in [0,1]$,*
(ii) *$\sup_{x \in K,\ t \in [0,1]} \operatorname{dist}\big(f_t(x), f_0(x)\big) < \epsilon$, and*
(iii) *f_1 is holomorphic in a neighborhood of $K_0 = K \cup (L \cap M_0)$ in X.*

Proof of Theorem 5.4.4. Assume Proposition 5.12.1 for the moment. Choose a sequence of compact $\mathcal{O}(X)$-convex sets

$$K = K_0 \subset K_1 \subset K_2 \subset \cdots \subset \bigcup_{k=0}^{\infty} K_k = X$$

Set $t_k = 1 - 2^{-k}$ and $I_k = [t_k, t_{k+1}]$ for $k = 0, 1, \ldots$; thus $\cup_{k=0}^{\infty} I_k = [0, 1)$.

Let $f = f_0\colon X \to Z$ be a continuous section that is holomorphic on a complex subvariety X' of X and in an open neighborhood of $K = K_0$ in X. Fix a number $\epsilon > 0$. By induction on $k \in \mathbb{Z}_+$ we shall construct homotopies of sections $f_t\colon X \to Z$, $t \in I_k$, that agree at the common endpoint t_{k+1} of the adjacent intervals I_k, I_{k+1} and such that the following hold:

- for every $k = 0, 1, \ldots$ and $t \in I_k$ the section f_t is holomorphic in an open neighborhood of the set K_k and it satisfies the condition

$$\sup_{x \in K_k} \operatorname{dist}\big(f_t(x), f_{t_k}(x)\big) < 2^{-k-1}\epsilon,$$

- the homotopy $\{f_t\}_{t \in [0,1)}$ is fixed on the subvariety X'.

These properties clearly imply that the limit section $f_1 = \lim_{t \to 1} f_t\colon X \to Z$ exists and is holomorphic on X, and it satisfies

$$\sup_{x \in K_0} \operatorname{dist}\big(f_1(x), f_0(x)\big) < \epsilon, \quad f_1|_{X'} = f_0|_{X'}.$$

Thus the homotopy $\{f_t\}_{t\in[0,1]}$ satisfies the conclusion of Theorem 5.4.4.

Since all inductive steps are of the same kind, we shall explain how to get the first homotopy for $t \in I_0 = [0, \frac{1}{2}]$. Set $K = K_0$ and $L = K_1$. By the assumption there exists an open set $U \subset X$ containing L such that $Z|_U$ is a stratified holomorphic fiber bundle whose strata satisfy CAP. Since L is assume to be $\mathcal{O}(X)$-convex, there is a relatively compact Stein domain Ω in X with $L \subset \Omega \subset U$. Choose a stratification

$$\Omega = X_0 \supset X_1 \supset \cdots \supset X_m = \emptyset$$

such that the restriction of $\pi\colon Z|_\Omega \to \Omega$ to each stratum $S \subset X_k \backslash X_{k+1}$ is a fiber bundle whose fiber enjoys CAP. Taking $X'_k = X_k \cup (X' \cap \Omega)$ we obtain a stratification

$$\Omega = X'_0 \supset X'_1 \supset \cdots \supset X'_m = \Omega \cap X'$$

with regular strata $X'_k \backslash X'_{k+1} = X_k \backslash (X_{k+1} \cup X')$, ending with $X'_m = \Omega \cap X'$.

By Theorem 3.4.1 (p. 68) we can assume that the initial section f_0 is holomorphic in an open neighborhood of $K \cup (L \cap X')$. Let $\{f_t\}_{t\in[0,1/2m]}$ be a homotopy furnished by Proposition 5.12.1 for the pair of subvarieties $M_1 = X'_m$ and $M_0 = X'_{m-1}$ of Ω, with ϵ replaced by $\frac{\epsilon}{2m}$. Then $f_{\frac{1}{2m}}$ is holomorphic in a neighborhood of $K \cup (L \cap X'_{m-1})$, the homotopy is fixed on X', and $\sup_{x\in K_0} \operatorname{dist}\big(f_t(x), f_0(x)\big) < \frac{\epsilon}{2m}$ for all $t \in [0, \frac{1}{2m}]$.

Next we apply Proposition 5.12.1 with the 'initial' section $f = f_{\frac{1}{2m}}$ and the pair of subvarieties $M_1 = X'_{m-1}$, $M_0 = X'_{m-2}$ to get a homotopy $\{f_t\}_{t\in[\frac{1}{2m},\frac{2}{2m}]}$ that is fixed on X' such that the section $f_{\frac{2}{2m}} = f_{\frac{1}{m}}$ is holomorphic in a neighborhood of $K \cup (L \cap X'_{m-2})$, and $\sup_{x\in K_0} \operatorname{dist}\big(f_t(x), f_{\frac{1}{2m}}(x)\big) < \frac{\epsilon}{2m}$ for all $t \in [\frac{1}{2m}, \frac{2}{2m}]$.

Continuing in this way we obtain after m steps a homotopy $\{f_t\}_{t\in[0,\frac{1}{2}]}$ with the required properties. In particular, the section $f_{\frac{1}{2}}$ is holomorphic in a neighborhood of the set $L = K_1$ and on the subvariety X' (where it agrees with f_0). We can extend this homotopy to all of X (without changing it near $L = K_1$) by using a cut-off function in the parameter. When the initial section $f = f_0$ is already holomorphic in a neighborhood of $K \cup X'$, we can use Theorem 3.4.5 (p. 70) and Remark 3.4.4 in order to keep the sections f_t in our homotopy holomorphic in a neighborhood of $K \cup X'$ and tangent to f_0 to a chosen order ℓ along X'.

This proves Theorem 5.4.4 in the nonparametric case, provided that Proposition 5.12.1 holds. □

Proof (of Proposition 5.12.1). We first consider the case when X is a Stein manifold and $M_0 = X$; the general case will be explained in the end.

We may assume that $L = \{x \in X\colon \rho(x) \le 0\}$ where $\rho\colon X \to \mathbb{R}$ is a smooth strongly plurisubharmonic exhaustion function on X such that $\rho|_K < 0$ and $\mathrm{d}\rho \neq 0$ on $bL = \{\rho = 0\}$. By the assumption f is holomorphic in an open

set $U \supset K \cup M_1$. Since the compact set $K' = K \cup (M_1 \cap \{\rho \le 1\}) \subset U$ is $\mathcal{O}(X)$-convex, there exists a smooth strongly plurisubharmonic exhaustion function $\tau\colon X \to \mathbb{R}$ such that $\tau < 0$ on K' and $\tau > 0$ on $X \backslash U$. By general position we may assume that 0 is a regular value of τ and the hypersurfaces $\{\rho = 0\} = bL$ and $\{\tau = 0\}$ intersect transversely along $\Sigma = \{\rho = 0\} \cap \{\tau = 0\}$. Hence $D_0 := \{\tau \le 0\} \subset U$ is a strongly pseudoconvex domain with smooth boundary. For each $s \in [0,1]$ let

$$\rho_s = \tau + s(\rho - \tau) = (1-s)\tau + s\rho, \quad D_s = \{\rho_s \le 0\} = \{\tau \le s(\tau - \rho)\}. \tag{5.18}$$

We have that $D_0 = \{\tau \le 0\}$ and $D_1 = \{\rho \le 0\} = L$.

Let $\Omega = \{\rho < 0,\ \tau > 0\} \subset D_1 \backslash D_0$ and $\Omega' = \{\rho > 0,\ \tau < 0\} \subset D_0 \backslash D_1$. As s increases from 0 to 1, $D_s \cap L$ increases to $D_1 = L$ while $D_s \backslash L \subset D_0$ decrease to $\emptyset$. All hypersurfaces $bD_s = \{\rho_s = 0\}$ intersect along Σ. Since $\mathrm{d}\rho_s = (1-s)\mathrm{d}\tau + s\mathrm{d}\rho$ and the differentials $\mathrm{d}\tau$, $\mathrm{d}\rho$ are linearly independent along Σ, bD_s is smooth near Σ. Finally, bD_s is strongly pseudoconvex at every smooth point, in particular at every point where $\mathrm{d}\rho_s \ne 0$. The situation is shown on Fig. 5.4.

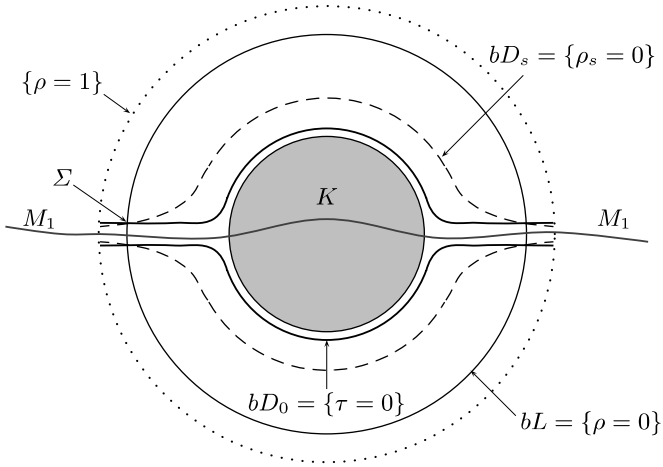

Fig. 5.4. The sets D_s. (Modified from [121, p. 245, Fig. 4] and also [180, p. 859, Fig. 1])

We investigate the singular points of $bD_s = \{\rho_s = 0\}$ inside Ω. (The remaining singular points will be irrelevant.) The defining equation of $D_s \cap \Omega$ can be written as $\tau \le s(\tau - \rho)$ and, after dividing by $\tau - \rho > 0$, as

$$D_s \cap \Omega = \left\{ x \in \Omega\colon h(x) = \frac{\tau(x)}{\tau(x) - \rho(x)} \le s \right\}.$$

The critical point equation $\mathrm{d}h = 0$ is equivalent to

$$(\tau - \rho)\mathrm{d}\tau - \tau(\mathrm{d}\tau - \mathrm{d}\rho) = \tau\mathrm{d}\rho - \rho\mathrm{d}\tau = 0.$$

A generic choice of ρ and τ insures that there are at most finitely many solutions $p_1, \ldots, p_m \in \Omega$ and no solution on $b\Omega$. A calculation shows that at each critical point the complex Hessians satisfy

$$(\tau - \rho)^2 H_h = \tau H_\rho - \rho H_\tau.$$

Since $\tau > 0$ and $-\rho > 0$ on Ω, we conclude that $H_h > 0$ at such points. By a small modification of h we can assume that all its critical points are nice (Def. 3.9.2, p. 89) and belong to different levels of h.

Let s_0, s_1 $(0 \le s_0 < s_1 \le 1)$ be regular values of h on Ω such that h has at most one critical point in $\Omega_{s_0,s_1} = \{x \in \Omega : s_0 < h(x) < s_1\}$. Suppose inductively that we have already found a homotopy $f_t : X \to Z$, $t \in [0, s_0]$, satisfying the conditions in Proposition 5.12.1 and such that f_{s_0} is holomorphic in a neighborhood of D_{s_0}. We wish to deform f_{s_0} to a section f_{s_1} which is holomorphic in a neighborhood of D_{s_1} by a homotopy which is fixed on M_1 and consist of sections that are holomorphic near $D_{s_0} \cap D_{s_1}$; the proof is then completed by a finite induction. We consider two cases as in §5.11.

The noncritical case: h has no critical values in Ω_{s_0,s_1}. By subdividing $[s_0, s_1]$ into finitely many small subintervals and replacing $[s_0, s_1]$ by one such subinterval we can assume that D_{s_1} is obtained from $D_{s_0} \cap D_{s_1}$ by attaching to the latter set finitely many special convex bumps contained in $X \backslash M_1$ (Def. 5.10.2 on p. 218; see Fig. 5.5). On each bump we apply Proposition 5.10.5 exactly as in §5.11. In finitely many steps we accomplish our task.

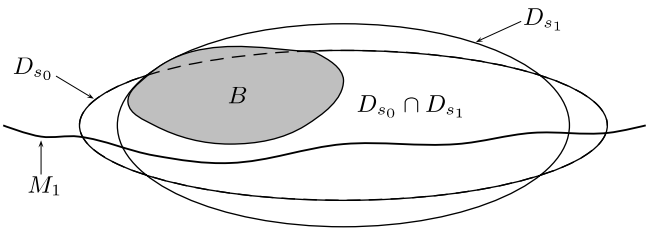

Fig. 5.5. A special bump B on $D_{s_0} \cap D_{s_1}$. (Modified from [180, p. 861, Fig. 3])

The critical case: h has a unique critical point $p \in \Omega_{c_0,c_1}$. We apply the critical case construction in §5.11, using the function h in place of ρ.

This proves the special case of Proposition 5.12.1.

The general case: Now X is a Stein space and $M_1 \subset M_0$ are closed complex subvarieties of X whose difference $S = M_0 \backslash M_1$ is a complex manifold with $\bar{S} = M_0$. By the assumption, f is holomorphic in an open set $U \subset X$ containing

$K_1 = K \cup (M_1 \cap L)$. As before we may assume that $L = \{x \in X : \rho(x) \le 0\}$, where $\rho : X \to \mathbb{R}$ is a smooth strongly plurisubharmonic exhaustion function whose restriction to $S = M_0 \backslash M_1$ has no critical points on the set $bL \cap S = \{\rho = 0\} \cap S$. Note that only finitely many connected components of S intersect the compact set L, and hence we can refine the stratification and assume that S is connected. We embed a relatively compact neighborhood of L in X holomorphically into $\mathbb{C}^N$. Since the set K_1 is $\mathcal{O}(X)$-convex, there is a smooth strongly plurisubharmonic function $\tau : X \to \mathbb{R}$ such that $\tau < 0$ on K_1 and $\tau > 0$ on $X \backslash U$. Set

$$D_0 = \{x \in M_0 : \tau(x) \le 0\} \subset U.$$

By general position we may assume that 0 is a regular value of $\tau|_S$ and that the hypersurfaces $\{\rho = 0\} \cap S = bL \cap S$ and $\{\tau = 0\} \cap S$ in S intersect transversely along the real codimension two submanifold $\Sigma = \{\rho = 0\} \cap \{\tau = 0\} \cap S$ of S. We define ρ_s as in (5.18) and set

$$D_s = \big\{x \in M_0 : \rho_s(x) \le 0\big\} = \big\{x \in M_0 : \tau(x) \le s\big(\tau(x) - \rho(x)\big)\big\}.$$

As s increases from 0 to 1, $D_s \cap L$ increases from $D_0 \cap L \subset M_0$ to $D_1 = L \cap M_0$. Like in the special case we successively attach to the set $A_0 = K \cup (D_0 \cap L)$ convex bumps and handles contained in the submanifold S, thereby reaching the set $K_0 = K \cup (L \cap M_0)$ in finitely many steps. Note that A_0 is $\mathcal{O}(X)$-convex and contains a collar around the set $K_1 = K \cup (L \cap M_1)$ in $L \cap M_0$.

Consider a typical step in the *noncritical case*. (For details see §5.11 above.) Assume that (A, B) is a Cartan pair in X such that A is obtained by attaching to A_0 finitely many bumps and handles contained in $S \cap L$ and $B \subset S \cap L$ is a convex bump attached to $A \cap S$. Then (A, B) is also a Cartan pair in $\mathbb{C}^N$.

Assume inductively that a section $f : X \to Z$ is holomorphic in a neighborhood of $K_1 = K \cup (L \cap M_1)$ in X and in a relative neighborhood of $A \cap S$ in S. Using Lemma 5.10.4 (p. 220) we embed f (on its domain of holomorphicity) into a family of holomorphic sections $F(x, w)$ of Z, depending holomorphically on a parameter w in an open neighborhood W of the origin in some $\mathbb{C}^p$, such that $f = F(\cdot\,, 0)$, $F(x, w) = f(x)$ for $x \in M_1$ and $w \in W$, and F is submersive in the w-variable for all x in a neighborhood of $A \cap B$ in S.

Since the fiber of the bundle $Z|_S \to S$ satisfies CAP, we can approximate F, uniformly in a neighborhood of $A \cap B$ in S, by a family G of holomorphic sections defined in a neighborhood of B in S and depending holomorphically on the parameter $w \in W$. If the approximation is sufficiently close, Lemma 5.9.3 (p. 216) furnishes a holomorphic transition map of the form

$$\gamma(x, w) = \big(x, \psi(x, w)\big),$$

close to the identity map and satisfying $F = G \circ \gamma$ on the intersection of their domains in $S \times \mathbb{C}^p$. Applying Proposition 5.8.4 on p. 214 we decompose

$$\gamma = \beta \circ \alpha^{-1}$$

over a pair of neighborhoods $(\widetilde{A}, \widetilde{B})$ of the Cartan pair (A, B) in X, where α and β are biholomorphic maps of the same type as γ and close to the identity on $\widetilde{A}$, resp. on $\widetilde{B}$. (The parameter set W again shrinks a little.) In addition, the map α can be chosen to agree with the identity map to order ℓ along the intersection of its domain with the subvariety M_1. From

$$F = G \circ \gamma = G \circ \beta \circ \alpha^{-1}$$

we see that $F \circ \alpha = G \circ \beta$ holds near $A \cap B$, and hence twe two sides define a family of holomorphic sections $\widetilde{F}$ of $Z \to X$ over a neighborhood of $A \cup B$. By the construction the section $\widetilde{F}(\cdot\,, 0)$ approximates f uniformly on K, it agrees with f to order ℓ along the subvariety M_1, and is homotopic to f by a homotopy satisfying the required properties. The induction may now proceed.

We deal with the critical points of the function $h = \frac{\tau}{\tau - \rho}$ in $S \cap L$ exactly as before by reducing to the noncritical case (see the critical case above).

In finitely many such steps we obtain a homotopy $\{f_t\}_{t \in [0,1]}$ with the required properties such that f_1 is holomorphic in a neighborhood of K_1 in X, and also in a relative neighborhood of $L \cap M_0$ in the subvariety M_0.

By Theorem 3.4.1 (p. 68) there is a holomorphic section $\widetilde{f}_1$ in a neighborhood of $K_0 = K \cup (L \cap M_0)$ in X such that $\widetilde{f}_1$ is as close as desired to f_1 on K, $\widetilde{f}_1 = f_1$ on $L \cap M_0$, and $\widetilde{f}_1$ agrees with f_1 to order ℓ along $L \cap M_1$. Replace f_1 by $\widetilde{f}_1$ and adjust the homotopy $\{f_t\}$ accordingly. By using a cut-off function in the parameter of the homotopy we can extend $\{f_t\}$ continuously to all of X without changing it near K_0 and on M_1. This complete the proof of Proposition 5.12.1. $\square$

5.13 Proof of the Main Theorem: The Parametric Case

In this section we complete the proof of Theorem 5.4.4 by establishing the parametric case. We begin by introducing a parametric version of the convex approximation property CAP.

Definition 5.13.1. *A complex manifold Y enjoys the* Parametric Convex Approximation Property (PCAP) *for a certain pair of compact Hausdorff spaces $P_0 \subset P$ if the following holds. Given a special convex pair (K, Q) in $\mathbb{C}^n$* (Def. 5.4.2, p. 192) *and a continuous map $f\colon P \times Q \to Y$ such that $f(p, \cdot)\colon Q \to Y$ is holomorphic for every $p \in P_0$, and is holomorphic on K for every $p \in P$, f can be approximated uniformly on $P \times K$ by continuous maps $\widetilde{f}\colon P \times Q \to Y$ such that $\widetilde{f}(p, \cdot)$ is holomorphic on Q for all $p \in P$, and $\widetilde{f} = f$ on $P_0 \times Q$.*

Theorem 5.13.2. *If $\pi\colon Z \to X$ is a stratified fiber bundle over a Stein space X such that all fibers $Z_x = \pi^{-1}(x)$ $(x \in X)$ enjoy* PCAP *for a certain pair of*

compact Hausdorff spaces $P_0 \subset P$, then sections $X \to Z$ satisfy the parametric Oka principle with approximation and jet interpolation for this pair $P_0 \subset P$.

Proof. We follow the proof of Theorem 5.4.4, but using the parametric versions of all main ingredients. Each of the basic steps is of the following two types:

- extension to a special bump (Proposition 5.10.5 on p. 221), or
- extension across a totally real disc (Theorem 3.7.2 on p. 81).

The first step consists of the following four substeps:

(i) embedding a holomorphic section into a spray of sections (Lemma 5.10.4 on p. 220),
(ii) approximation of the spray (by invoking CAP),
(iii) finding a transition map (Lemma 5.9.3 on p. 216), and
(iv) splitting the transition map (Proposition 5.8.1 on p. 211).

Each of these substeps can be performed simultaneously for all sections in the family, keeping continuous dependence on the parameter:

- In substep (i) we apply a continuous partition of unity on P to find a family of holomorphic sprays depending continuously on $p \in P$.
- Substep (ii) is a tautology by the definition of PCAP. The sections belonging to the values of the parameter $p \in P_0$ are left unchanged.
- For substep (iii) observe that the transition map furnished by (the proof of) Lemma 5.9.3 is uniquely determined by a choice of a complementary bundle E' (5.17) to the bundle E (5.16). In the parametric case E_p depends continuously on the parameter $p \in P$, and we need a continuous family of complementary subbundles E'_p. Clearly the same bundle E' can be used for an open set of points in the parameter space P, and we find the family E'_p by a partition of unity on P. (See also Lemma 6.5.2 on p. 261 below.)
- Substep (iv) is immediate since Proposition 5.8.1 includes parameters.

Similar arguments apply in the second step. □

The parametric case of Theorem 5.4.4 now follows from Theorem 5.13.2 and the following result from [179].

Theorem 5.13.3. *If a complex manifold enjoys* CAP*, then it also enjoys* PCAP *for each pair $P_0 \subset P$ of compact subsets of a Euclidean space $\mathbb{R}^m$.*

Proof. Assume that Y enjoys CAP. Then Y also enjoys the Basic Oka Property according to the nonparametric case of Theorem 5.4.4 proved above. Assume that $P_0 \subset P$ are compacts in $\mathbb{R}^m \subset \mathbb{C}^m$, (K, Q) is a special convex pair in $\mathbb{C}^n$, $U \supset K$ and $V \supset Q$ are open convex neighborhoods of K resp. of Q in $\mathbb{C}^n$, and $f\colon P \times V \to Y$ is a continuous map such that

(i) the map $f(p,\cdot)\colon V \to Y$ is holomorphic for every $p \in P_0$, and

(ii) $f(p,\cdot)$ is holomorphic on U for every $p \in P$.

We shall obtain the conclusion of Def. 5.13.1 for this f. After shrinking the set $V \supset Q$ we can assume by Proposition 5.6.3 (p. 208) that $f(p,\cdot)$ is holomorphic on V for all p in a neighborhood $P_0' \subset \mathbb{C}^m$ of P_0. We may assume that $0 \in \mathbb{C}^n$ belongs to $\mathrm{Int}K$. Choose a continuous function $\tau\colon P \to [0,1]$ such that $\tau = 0$ on P_0 and $\tau = 1$ on $P \backslash P_0'$. Set

$$f_t(p,z) = f\big(p,(1-(1-t)\tau(p))z\big) \in Y, \quad p \in P,\ z \in V,\ t \in [0,1].$$

Then f_t has the same properties as $f = f_1$, the homotopy f_t is fixed for $p \in P_0$, and the map $f_0(p,\cdot)$ is holomorphic on V for all $p \in P$.

Set $Z = \mathbb{C}^m \times \mathbb{C}^n \times Y$. For every $t \in [0,1]$ let

$$F_t(p,z) = (p,z,f_t(p,z))\,, \quad \Sigma_t = F_t(P \times K) \subset Z, \quad S_0 = F_0(P \times Q) \subset Z.$$

By Corollary 3.6.6 on p. 77 the sets S_0 and Σ_t ($t \in [0,1]$) are Stein compacta in Z. Hence there are numbers $0 = t_0 < t_1 < \cdots < t_N = 1$ and Stein domains $\Omega_0, \ldots, \Omega_{N-1} \subset Z$ such that

$$\Sigma_t \subset \Omega_j \text{ when } t_j \le t \le t_{j+1} \text{ and } j = 0,1,\ldots,N-1. \tag{5.19}$$

Let $E \to Z$ denote the vertical tangent bundle of the projection map $Z \to \mathbb{C}^m \times \mathbb{C}^n$. By Proposition 3.3.2 (p. 67) there exists for every Stein domain $\Omega \subset Z$ a Stein neighborhood $W \subset E|_\Omega$ of the zero section $\Omega \subset E|_\Omega$ and a holomorphic map $s\colon W \to Z$ that maps the fiber $W_{(\zeta,z,y)}$ over $(\zeta,z,y) \in Z$ biholomorphically onto a neighborhood of this point in $\{(\zeta,z)\} \times Y$ such that s preserves the zero section Ω of $E|_\Omega$. (Such an s is a local fiber dominating spray in the sense of Def. 5.9.1, p. 215.) We may assume that W is Runge in $E|_\Omega$ and its fibers are convex domains in the fibers of E.

Since $E|_\Omega$ is Stein and Y enjoys the Basic Oka Property, s can be approximated uniformly on compacts in W by a holomorphic fiber-dominating spray $\widetilde{s}\colon E|_\Omega \to Z$ that agrees with s to the second order along the zero section Ω. Thus $\widetilde{s}$ maps each fiber $E_{(\zeta,z,y)}$ into the corresponding fiber $\{(\zeta,z)\} \times Y$, and its differential in the fiber direction is an isomorphism along the zero section.

This shows that after refining our subdivision $\{t_j\}$ of $[0,1]$ and shrinking the set $U \supset K$ there are Stein domains $\Omega_0, \ldots, \Omega_{N-1}$ as in (5.19), sprays $s_j\colon E|_{\Omega_j} \to Z$, and homotopies of z-holomorphic sections ξ_t ($t \in [t_j, t_{j+1}]$) of the restricted bundle $E|_{F_{t_j}(P\times U)}$ such that ξ_{t_j} is the zero section, $\xi_t(p,\cdot)$ is independent of t when $p \in P_0$ (hence it is the zero section), and

$$s_j \circ \xi_t \circ F_{t_j} = F_t \quad \text{on } P \times U, \quad t \in [t_j, t_{j+1}]. \tag{5.20}$$

Furthermore, the existence of such liftings ξ_t is stable under sufficiently small perturbations of the homotopy F_t. (See Fig. 5.6.)

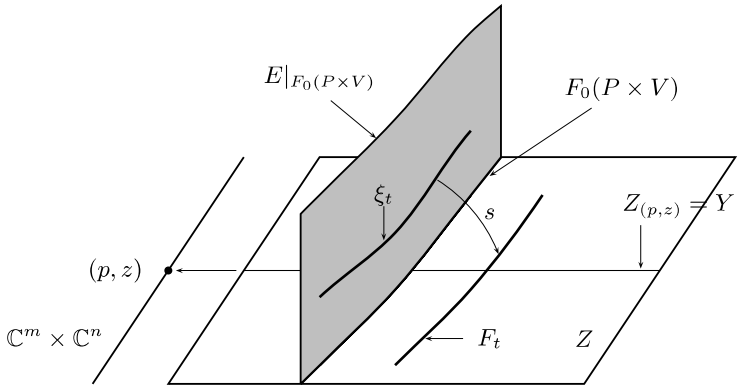

Fig. 5.6. Lifting sections F_t to the spray bundle $E|_{F_0(P\times V)}$. (Modified from [182, p. 148, Fig. 1])

Consider the homotopy of sections ξ_t of $E|_{F_0(P\times U)}$ for $t \in [0,t_1]$. By the Oka-Weil theorem we can approximate ξ_t uniformly on $P \times K$ by z-holomorphic sections $\widetilde{\xi}_t$ of $E|_{F_0(P\times V')}$ for an open convex set $V' \subset \mathbb{C}^n$ with $Q \subset V' \subset V$. (This parametric version of the Oka-Weil theorem is obtained by using a continuous partition of unity with respect to the parameter.) Further, we may choose $\widetilde{\xi}_t = \xi_t$ for $t = 0$ and on $P_0 \times V'$.

By Corollary 3.6.6 (p. 77) there is a Stein neighborhood $\Omega \subset Z$ of S_0 such that Σ_0 is $\mathcal{O}(\Omega)$-convex. Hence $E|_{\Sigma_0}$ is exhausted by $\mathcal{O}(E|_\Omega)$-convex compact sets. Since $E|_\Omega$ is a Stein manifold and Y enjoys the Basic Oka Property, we conclude that the fiber spray $s_0\colon E|_{\Omega_0} \to Z$ can be approximated on the range of the homotopy $\{\xi_t\colon t \in [0,t_1]\}$ by a fiber spray $\widetilde{s}_0\colon E|_\Omega \to Z$ that agrees with s_0 to the second order along the zero section. The maps

$$\widetilde{f}_t = \pi_Y \circ \widetilde{s}_0 \circ \widetilde{\xi}_t \circ F_0\colon P \times V' \to Y, \quad t \in [0,t_1]$$

are then z-holomorphic on $V' \supset Q$, and they approximate f_t uniformly on $P \times K$. If the approximation is sufficiently close, we obtain a new homotopy $\{f_t\colon t \in [0,1]\}$ that agrees with $\widetilde{f}_t$ for $t \in [0,t_1]$ (hence is z-holomorphic on Q for these values of t), and that agrees with the initial homotopy for $t \in [t_1',1]$ for some $t_1' > t_1$ close to t_1.

We now repeat the same argument with the parameter interval $[t_1,t_2]$, using f_{t_1} as the new reference map. Let F_{t_1} denote the section corresponding to f_{t_1}. We must ensure that the new homotopy f_t for $t \in [t_1,t_2]$ can be lifted by the spray s_1 to a family of sections ξ_t ($t \in [t_1,t_2]$) of $E|_{F_{t_1}(P\times U)}$ such that (5.20) holds. But this is so if the approximations in the first step were sufficiently close since the existence of liftings is stable under small perturbations of the homotopy F_t. This gives us a new homotopy that is z-holomorphic on

Q for $t \in [0, t_2]$. After finitely many steps of this kind we obtain a desired homotopy whose final map $\widetilde{f}$ at $t = 1$ satisfies Def. 5.13.1.

This proves Theorem 5.13.3 and hence completes the proof of the parametric case of Theorem 5.4.4. □

5.14 Existence Theorems for Holomorphic Sections

Theorem 5.4.4 gives a holomorphic section homotopic to a given continuous section under suitable assumptions on the fibers. We now add a connectivity condition on the fibers that insures the existence of sections.

Theorem 5.14.1. *Assume that X is a finite dimensional Stein space, $\pi\colon Z \to X$ is a holomorphic submersion, and $X = X_0 \supset X_1 \supset \cdots \supset X_m = \emptyset$ is a stratification of X such that for each connected component S of $X_k \backslash X_{k+1}$ the restriction $Z|_S \to S$ is a holomorphic fiber bundle whose fiber Y_S is an Oka manifold and $\pi_q(Y_S) = 0$ for all $q < \dim S$. Then there exists a holomorphic section $X \to Z$. Furthermore, given a closed complex subvariety $X' \subset X$, a compact $\mathcal{O}(X)$-convex subset $K \subset X$, an open set $U \supset K$ and a holomorphic section $f\colon U \cup X' \to Z$, there exists a holomorphic section $\widetilde{f}\colon X \to Z$ such that $\widetilde{f}|_{X'} = f|_{X'}$ and $\widetilde{f}$ approximates f as close as desired uniformly on K. If f is holomorphic in a neighborhood of $K \cup X'$ then $\widetilde{f}$ can be chosen to agree with f to any given finite order along the subvariety X'.*

Proof. The only place in the proof of Theorem 5.4.4 where a topological condition on the fiber is needed is to cross a critical point p of index $k \geq 1$ of a strongly plurisubharmonic Morse function ρ on a stratum S. (See the *critical case* in the proof of Proposition 5.12.1 on p. 224.) At such a point $p \in S$ we must be able to extend a given holomorphic section, defined on a sublevel set $\{\rho \leq c\}$ for some $c < \rho(p)$ close to the critical level $\rho(p)$, to a continuous section over a k-dimensional totally real disc $E \subset S$, attached with its boundary $(k-1)$-sphere bE to $\{\rho = c\}$, such that $\{\rho \leq c\} \cup E$ is a strong deformation retraction of a sublevel set $\{\rho \leq c'\}$ for some $c' > \rho(p)$.

Since $Z|_S \to S$ is a fiber bundle with fiber Y_S, sections coincide with maps into the fiber. We see that a required extension exists if and only if the map $f\colon bE \to Y_S$ is null-homotopic in Y_S, and this is certainly the case if the homotopy group $\pi_{k-1}(Y_S)$ vanishes. Since all Morse indexes of a strongly plurisubharmonic function on S are $\leq \dim S$, the condition that $\pi_q(Y_S) = 0$ for $q < \dim S$ therefore insures the existence of a continuous extension of a section at each critical point on every stratum. □

Corollary 5.14.2. *Assume that X is a Stein space of dimension n, K is a compact $\mathcal{O}(X)$-convex subset of X, X' is a closed complex subvariety of X, U is an open set in X containing K, and $f\colon U \cup X' \to Y$ is a holomorphic map.*

If Y is an Oka manifold and if $\pi_q(Y) = 0$ for $q = 1, \ldots, n-1$ then there exists a holomorphic map $\widetilde{f}: X \to Y$ such that $\widetilde{f}|_{X'} = f|_{X'}$ and $\widetilde{f}$ approximates f uniformly on K as close as desired. This holds in particular if $Y = \mathbb{C}^N \backslash A$, where A is a closed algebraic subvariety of codimension $q \geq \max\{2, \frac{n+1}{2}\}$.

Proof. The first conclusion follows from Theorem 5.14.1. For the second part note that $\mathbb{C}^N \backslash A$ is an Oka manifold if $q \geq 2$ (Proposition 5.5.8, p. 201), and $\pi_k(\mathbb{C}^N \backslash A) = 0$ for $k \leq 2q - 2$ by the transversality theorem. □

Essentially the same argument gives the following result.

Corollary 5.14.3. *Assume that $X = X_0 \supset X_1 \supset \cdots \supset X_m = X'$ is a stratification of a Stein space X and $\pi: Z \to X$ is a stratified holomorphic fiber bundle whose fiber Y_S over every stratum $S \subset X_k \backslash X_{k+1}$ is of the form $Y_S = \mathbb{C}^N \backslash A_S$, where $A_S \subset \mathbb{C}^N$ is an algebraic subvariety whose codimension $q_S = \operatorname{codim}_{\mathbb{C}} A_S$ satisfies $q_S \geq 2$ and $2q_S > \dim S$. Then every holomorphic section $f': X' \to Z|_{X'}$ extends to a holomorphic section $f: X \to Z$.*

Remark 5.14.4. The above results use only the most elementary obstruction theory. More refined results on the existence of continuous sections are obtained as follows (see e.g. [458]). Suppose that $E \to X$ is a topological fiber bundle with fiber Y over a CW complex X. If

$$H^{q+1}\big(X; \pi_q(Y)\big) = 0, \qquad q = 1, 2, \ldots, \tag{5.21}$$

then there exists a continuous section $X \to E$. The condition enables one to construct a section by induction over skeleta of increasing dimension in a CW decomposition of X. The same method applies to the problem of extending a section from a CW subcomplex. Suppose that $E \to X$ is a topological fibration, X' is a CW subcomplex of X and the restriction $E|_{X \backslash X'} \to X \backslash X'$ is a fiber bundle with fiber Y. If (5.21) holds then any continuous section $f: X' \to E|_{X'}$ extends to a continuous section $X \to E$. When coupled with the Oka principle, this gives a strong tool for constructing holomorphic sections of stratified holomorphic fiber bundles over Stein spaces. □

5.15 Equivalences Between Oka Properties

Recall from §5.4 that a complex manifold Y is said to enjoy a certain Oka property if the corresponding form of the Oka principle holds for maps from every Stein space X to Y (see Theorem 5.4.4 on p. 193). Every Oka property comes in two forms, the *basic* and the *parametric*. When speaking of the basic properties we take the parameter set P to be a singleton and $P_0 = \emptyset$. The parametric properties refer to any pair of compact subsets $P_0 \subset P$ of a Euclidean space.

Consider the following *Oka properties* of a complex manifold Y.

CAP – *Convex Approximation Property.* (See Def. 5.4.3 on p. 192.) Note that CAP is the same as BOPA applied in the model case when K is a special compact convex set in a Euclidean space $X = \mathbb{C}^n$.

BOPA – *Basic Oka Property with Approximation:* Every continuous map $f_0 \colon X \to Y$ from a Stein space X that is holomorphic on (a neighborhood of) a compact $\mathcal{O}(X)$-convex subset $K \subset X$ can be deformed to a holomorphic map $f_1 \colon X \to Y$ by a homotopy of maps that are holomorphic near K and arbitrary close to f_0 on K.

BOPI – *Basic Oka Property with Interpolation:* Given a diagram

$$\begin{tikzcd} X' \arrow[d, "\iota"'] \arrow[dr] & \\ X \arrow[r, "f"'] & Y \end{tikzcd}$$

where $\iota \colon X' \hookrightarrow X$ is the inclusion of a closed complex subvariety into a Stein space X and $f \colon X \to Y$ is a continuous map such that $f|_{X'} \colon X' \to Y$ is holomorphic, there is a homotopy $f_t \colon X \to Y$ $(t \in [0,1])$ that is fixed on X' from $f = f_0$ to a holomorphic map $f_1 \colon X \to Y$.

BOPJI – *Basic Oka Property with Jet Interpolation:* This is the same as BOPI, except that the given continuous map $f \colon X \to Y$ is assumed to be holomorphic in a neighborhood of a subvariety X', and we ask that there exist a homotopy $f_t \colon X \to Y$ $(t \in [0,1])$ as above that is fixed to a given finite order along X'.

We introduce a restricted version of BOPI as follows. Let D be a convex domain in $\mathbb{C}^m$ and $\phi \colon D \hookrightarrow \mathbb{C}^k$ a proper holomorphic embedding. Let

$$X' = \{(x, \phi(x)) \colon x \in D\} \subset \mathbb{C}^n = \mathbb{C}^m \times \mathbb{C}^k.$$

CIP – *Convex Interpolation Property:* Given a subvariety $X' \subset \mathbb{C}^n$ as above, every holomorphic map $f \colon X' \to Y$ extends to a holomorphic map $\mathbb{C}^n \to Y$. (The existence of a continuous extension is trivial in this case.) This property was introduced by Lárusson [323]. Note that CIP plays the same role versus BOPI as CAP does versus BOPA.

BOPAI – *Basic Oka Property with Approximation and Interpolation:* Combine BOPA and BOPI.

BOPAJI – *Basic Oka Property with Approximation and Jet Interpolation:* Combine BOPA and BOPJI. This means the full conclusion of Theorem 5.4.4 for the trivial parameter space $(P, P_0) = (\{p\}, \emptyset)$.

The corresponding parametric Oka properties are introduced in an obvious way (see Theorem 5.4.4):

POPA – *Parametric Oka Property with Approximation.* Note that PCAP (Def. 5.13.1, p. 229) is the restricted version of POPA.

POPI – *Parametric Oka Property with Interpolation.*

POPJI – *Parametric Oka Property with Jet Interpolation.*

PCIP – *Parametric Convex Interpolation Property.* This is the restricted version of POPI, analogous to CIP.

POPAI – *Parametric Oka Property with Approximation and Interpolation.*

POPAJI – *Parametric Oka Property with Approximation and Jet Interpolation.* This means the full conclusion of Theorem 5.4.4, and is equivalent to Gromov's Ell_∞*-property* [237, §3.1] when the parameter pairs (P, P_0) are finite polyhedra and their subpolyhedra.

We thus have thirteen ostensibly different Oka properties. It turns out that they are all equivalent, and hence they all characterize the class of Oka manifolds. The simplest properties to verify in concrete examples are CAP and CIP (which is not to say that the verification is simple!).

Proposition 5.15.1. [175, 322] *The following properties of a complex manifold are equivalent, and hence any of them characterizes the class of Oka manifolds:* CAP, CIP, BOPA, BOPI, BOPJI, BOPAI, BOPAJI, PCAP, PCIP, POPA, POPI, POPAI, POPJI, POPAJI.

Proof. By Theorem 5.4.4, CAP implies all the other properties. Conversely, all properties containing the letter A in their name trivially imply CAP. Each of the remaining properties (CIP, BOPI, BOPJI, PCIP, POPI, POPJI) trivially implies CIP. To complete the proof it remains to show that CIP implies CAP.

Let K be a special convex set in $\mathbb{C}^n$ and $f\colon U \to Y$ a holomorphic map from an open convex neighborhood $U \subset \mathbb{C}^n$ of K. Choose a proper holomorphic embedding $\phi\colon U \hookrightarrow \mathbb{C}^r$. Its graph $\Sigma = \{(x, \phi(x))\colon x \in U\} \subset \mathbb{C}^{n+r}$ is a closed complex subvariety of $\mathbb{C}^{n+r}$ of the type used in the definition of CIP. Consider the holomorphic map $g\colon \Sigma \to Y$ defined by $g(x, \phi(x)) = f(x)$. Assuming that Y enjoys CIP, there exists a holomorphic map $G\colon \mathbb{C}^{n+r} \to Y$ such that $G|_\Sigma = g$. Thus $G(x, \phi(x)) = f(x)$ for $x \in U$.

By the Oka-Weil theorem there is a sequence $\phi_k\colon \mathbb{C}^n \to \mathbb{C}^r$ $(k = 1, 2, \ldots)$ of holomorphic maps that converges to ϕ uniformly on K. Then

$$f_k(x) = G(x, \phi_k(x)), \quad x \in \mathbb{C}^n$$

is a sequence of holomorphic maps $\mathbb{C}^n \to Y$ such that $\lim_{k\to\infty} f_k = f$ on K. This shows that Y enjoys CAP. $\square$

By Corollary 5.5.12 (p. 204), a subelliptic manifold is an Oka manifold. The converse is not known in general, but it holds on Stein manifolds.

Proposition 5.15.2. [237, 3.2.A.] *A Stein Oka manifold is elliptic.*

Proof. Assume that Y is a Stein manifold; its tangent bundle $E = \mathrm{T}Y$ is then also a Stein manifold. We identify Y with the zero section of E. By Proposition 3.3.2 (p. 67) there is a neighborhood $\Omega \subset E$ of the zero section and a holomorphic spray $s\colon \Omega \to Y$; that is, s is the identity on Y and for every $y \in Y$ the differential $\mathrm{d}s_y\colon \mathrm{T}_yE \to \mathrm{T}_yY$ maps the vertical subspace E_y of T_yE onto T_yY. We may assume that s extends to a continuous map $E \to Y$ and that Ω is Runge in E. Since E is Stein and Y is Oka, there is a holomorphic map $\widetilde{s}\colon E \to Y$ that agrees with s to second order along Y. Clearly such $\widetilde{s}$ is a dominating spray on Y. $\square$

It follows from the definition that every Oka manifold Y of dimension n is *strongly dominable* by $\mathbb{C}^n$, in the sense that for every point $y \in Y$ there exists a holomorphic map $f\colon \mathbb{C}^n \to Y$ such that $f(0) = y$ and $\mathrm{d}f_0\colon \mathrm{T}_0\mathbb{C}^n \to \mathrm{T}_yY$ is surjective. Hence all infinitesimal Kobayashi-Eisenman metrics of Y [126, 299] vanish identically at every point $y \in Y$. A compact complex manifold Y which is dominable by $\mathbb{C}^n$ is not of Kodaira general type, i.e., its Kodaira dimension κ_Y is strictly smaller than $\dim Y$ [70, 301]. Hence we get

Corollary 5.15.3. *A compact Oka manifold is not of Kodaira general type.*

The following result summarizes the relations between the flexibility properties considered in Chaps. 5 and 6. In the algebraic category, CAP is interpreted as the validity of the conclusion of Corollary 7.10.2 (p. 325).

Corollary 5.15.4. *The following hold for any complex manifold:*

$$\begin{gathered}\text{homogeneous} \Longrightarrow \text{elliptic} \Longrightarrow \text{subelliptic} \Longrightarrow \text{CAP} \Longleftrightarrow \text{Oka} \\ \Longrightarrow \text{jet transversality theorem} \Longrightarrow \text{strongly dominable} \Longrightarrow \text{dominable.}\end{gathered}$$

In the algebraic category we have the implications

$$\begin{array}{ccccc} \text{elliptic} \Longrightarrow & \text{subelliptic} & \Longrightarrow & \text{CAP} \\ & \Downarrow & & \Downarrow \\ & \text{jet transversality} & \Longrightarrow & \text{dominable.} \end{array}$$

5.16 Open Problems

In the last section of this chapter we collect open problems concerning the convex approximation property (CAP) and Oka manifolds.

Clearly we have $\mathrm{CAP}_1 \Leftarrow \mathrm{CAP}_2 \Leftarrow \mathrm{CAP}_3 \cdots$. The following example shows that these implications are not reversible in the range of integers up to $\dim Y$.

Example 5.16.1. For every pair of integers $1 < k \le p$ there exists a p-dimensional complex manifold which satisfies CAP_{k-1} but not CAP_k. For $k = p$ we can take $Y = \mathbb{C}^p \backslash A$ where A is a rigid discrete subset of $\mathbb{C}^p$ (see Theorem 4.7.5 on p. 122). Thus CAP_p fails, but CAP_{p-1} holds since a generic holomorphic map $\mathbb{C}^{p-1} \to \mathbb{C}^p$ avoids A by dimension reasons. For $k < p$ we take a Brody k-hyperbolic manifold $Y = \mathbb{C}^p \backslash \phi(\mathbb{C}^{p-k})$ where $\phi\colon \mathbb{C}^{p-k} \hookrightarrow \mathbb{C}^p$ is a proper holomorphic embedding furnished by Theorem 4.18.8 on p. 170; then CAP_k fails but CAP_{k-1} holds by dimension reasons. Another example is $Y = (\mathbb{C}^k \backslash A) \times \mathbb{C}^{p-k}$, where A is a rigid discrete set in $\mathbb{C}^k$ (see §4.7). $\square$

Problem 5.16.2. Given integers $1 < k \le p$, is there a *compact* p-dimensional complex manifold which satisfies CAP_{k-1} but not CAP_k? $\square$

Problem 5.16.3. Do the CAP_n properties stabilize at some integer, i.e., is there a $p \in \mathbb{N}$ depending on Y, or perhaps only on $\dim Y$, such that

$$\mathrm{CAP}_p \Longrightarrow \mathrm{CAP}_n \text{ for all } n > p\,?$$

Does this hold for $p = \dim Y$?

Problem 5.16.4. Let K be a compact convex set in $\mathbb{C}^n$ for some $n > 1$. Is $\mathbb{C}^n \backslash K$ an Oka manifold? In particular, iss the complement of a closed ball (or a closed polydisc) an Oka manifold?

Note that the complement of any compact convex set in $\mathbb{C}^n$ for $n > 1$ is a union of Fatou-Bieberbach domains [420].

Problem 5.16.5. [237, p. 881, 3.4.(D)] Suppose that every holomorphic map from a ball $B \subset \mathbb{C}^n$ to Y (for any $n \in \mathbb{N}$) can be approximated by entire maps $\mathbb{C}^n \to Y$. Does it follow that Y is an Oka manifold?

Problem 5.16.6. Let $\pi\colon E \to B$ be a holomorphic fiber bundle with fiber Y. Assuming that E is an Oka manifold, does it follow that the base B and the fiber Y are also Oka manifolds?

Problem 5.16.7. Which compact complex surfaces are Oka? In particular, are any (or all) K3 surfaces Oka? $\square$

A particularly interesting and promising case among K3 surfaces seem to be Kummer surfaces. Every Kummer surface Y contains 16 pairwise disjoint embedded rational curves $E_1, \ldots, E_{16} \subset Y$ such that the complement $Y \backslash \cup_{j=1}^{16} E_j$ is an Oka manifold. (This complement is a nonramified holomorphic quotient of a complex torus with 16 points removed, and hence is an Oka manifold by Proposition 5.5.2 and Corollary 5.5.15.) Also, each $E_j \cong \mathbb{P}^1$ is an Oka manifold. The question whether it follows from this that the union Y is also Oka is a test case for the following 'union problem'.

Problem 5.16.8. (The union problem for Oka manifolds.) Let Y be a complex manifold and $Y' \subset Y$ a closed complex submanifold. If Y' and $Y \backslash Y'$ are Oka manifolds, does it follow that Y is also an Oka manifold?

Among the classes of complex manifolds that are believed to have no volume hyperbolicity (a necessary condition to be Oka) are also *Calabi-Yau manifolds* and *hyper-Kähler manifolds*. A Calabi-Yau manifold has $\pi_1(Y) = 0$, $c_1(Y) = 0$, $K_Y \cong Y \times \mathbb{C}$, and $H^0(Y, \Omega^p) = \mathbb{C}$ in dimensions $p = 0$ and $p = \dim Y$, and it equals zero otherwise. The definition of a hyper-Kähler manifold is similar as that of a Calabi-Yau manifold, except that it admits a holomorphic symplectic form $\alpha \in H^0(X, \Omega^2)$. Both these classes generalize the class of K3 surfaces.

Problem 5.16.9. Suppose that Y is an Oka manifold with $\dim Y > 1$. Is the complement $Y \backslash \{p\}$ of an arbitrary point $p \in Y$ still an Oka manifold? Is the blowup of Y at a single point an Oka manifold?

A positive answer to this problem is known in the class of complex algebraic manifolds that are Zariski locally Euclidean (see Def. 6.4.5 on p. 252). Perhaps one could get a counterexample by considering holomorphic quotients of $\mathbb{C}^n$ by nonlinear discrete subgroups $\Gamma \subset \operatorname{Aut} \mathbb{C}^n$ acting freely and properly discontinuously. If there exists such Γ such that the orbit Γp of a certain point $p \in \mathbb{C}^n$ is an unavoidable sequence in $\mathbb{C}^n$ (Def. 4.7.1 on p. 120), then the quotient $Y = \mathbb{C}^n / \Gamma$ is an Oka manifold by Proposition 5.5.2 (p. 199), but $Y \backslash \{p\}$ is not Oka since its covering space $\mathbb{C}^n \backslash \Gamma p$ is not Oka.

Problem 5.16.10. Are there any restrictions on the homotopy type of a compact Oka manifold? Are there any contractible Stein Oka manifolds other than the Euclidean spaces?

Problem 5.16.11. (Variation of the Oka property in holomorphic families.) Assume that Z is a complex manifold and $\pi \colon Z \to \mathbb{D} = \{t \in \mathbb{C} \colon |t| < 1\}$ is a proper holomorphic submersion.

(a) If the fiber $Z_t = \pi^{-1}(t)$ is an Oka manifold for every $t \in \mathbb{D} \backslash \{0\}$, does it follow that the central fiber Z_0 is also an Oka manifold?

(b) Is there an example when the central fiber Z_0 is Oka, but Z_t is not Oka for some $t \neq 0$ arbitrarily near 0?

Problem 5.16.12. Is there a geometric characterization of Oka manifolds in the spirit of Gromov's ellipticity (see Def. 5.5.11)? In particular, is every Oka manifold elliptic or subelliptic?

Let us call a complex manifold X *good* if it is the image of an Oka map $\pi \colon E \to X$ (see Def. 6.14.7) from a Stein manifold E, and *very good* if X carries

a holomorphic affine bundle $\pi\colon E \to X$ whose total space E is Stein. (These definitions are slightly different from those introduced in [322]. A holomorphic fiber bundle projection $\pi\colon E \to X$ is an Oka map if and only if its fiber is Oka.) In such case the manifold X is Oka if and only if E is Oka. (For fiber bundles this is Theorem 5.5.4.) If this holds then E, being Stein, is also elliptic by Proposition 5.15.2. Hence in the class of good manifolds we have the following geometric characterization of the Oka property.

Proposition 5.16.13. *Let X be a good complex manifold. Then X is an Oka manifold if and only if there is a surjective Oka map $\pi\colon E \to X$ such that E is an elliptic Stein manifold.*

How large is the class of good (or very good) manifolds? By definition these classes include all Stein manifolds. By Example 1.4.3 on p. 12 there is an affine holomorphic bundle $E \to \mathbb{P}^n$ with fiber $\mathbb{C}^n$ whose total space E is Stein, so $\mathbb{P}^n$ is very good. By restricting the bundle $E \to \mathbb{P}^n$ to a closed complex submanifold $X \subset \mathbb{P}^n$ we see that every complex projective manifold is very good. (Note that $E|_X$, being a closed complex submanifold of a Stein manifold E, is Stein.) More generally, all quasi-projective algebraic manifolds are very good [322]. Lárusson showed that the classes of good, resp. very good, complex manifolds are closed under taking products, covering spaces, finite branched covering spaces, submanifolds, and complements of analytic hypersurfaces, so they seem to be quite large.

Problem 5.16.14. To what extent is the class of Oka manifolds invariant under proper modifications of (projective) algebraic manifolds? In particular, is it invariant under blowing up and/or blowing down?

The answer to the first question is affirmative for manifolds of Class $\mathcal{A}$ (Def. 6.4.5, p. 252) as this class is closed under blowups (Proposition 6.4.7).

We have already mentioned in §5.5 that the complement of a projective algebraic hypersurface may fail to be Oka. However, the complement of a smooth cubic curve in $\mathbb{P}^2$ is dominable by $\mathbb{C}^2$ [69, Proposition 5.1], and hence is it reasonable to ask whether it is Oka.

Problem 5.16.15. Is the complement of every smooth cubic curve in $\mathbb{P}^2$ an Oka manifold?

The analysis in [69] shows that an affirmative answer to the following problem would imply an affirmative answer to Problem 5.16.15.

Problem 5.16.16. Let $\pi\colon X \to X'$ be a proper finite holomorphic map. If X is an Oka manifold, does it follow that X' is an Oka manifold?

The converse implication clearly fails as is shown by meromorphic functions on compact Riemann surfaces of genus $g > 1$; such a function defines a finite branched holomorphic map $X \to \mathbb{P}^1$ onto the Riemann sphere which is Oka, but X is not Oka since it is hyperbolic.

6

Elliptic Complex Geometry and Oka Principle

In geometric constructions one often encounters the problem of finding global holomorphic sections of holomorphic maps that are more general than (stratified) fiber bundles. In the present chapter we consider this problem for holomorphic submersions. This eliminates most of the difficulties that one encounters with maps to non smooth varieties.

The main results of this chapter are Gromov's Oka principle for sections of elliptic submersions $h\colon Z \to X$ over Stein base manifolds, and its extension to stratified subelliptic submersions over Stein spaces (Theorem 6.2.2 on p. 243). A holomorphic submersion $Z \to X$ is *elliptic* in the sense of Gromov [237] if there exists a fiber-dominating holomorphic spray on the restricted submersion $Z|_U \to U$ over small open sets $U \subset X$. *Subellipticity*, a presumably weaker condition introduced in [170], means the existence of a finite fiber-dominating family of holomorphic sprays on $Z|_U$ over small open sets $U \subset X$ (Def. 6.1.2 on p. 242). Each of these conditions implies a certain Runge type *Homotopy Approximation Property* (HAP) concerning homotopies of holomorphic sections of the restricted submersion $Z|_U$ (see §6.6). The crux of the matter is the following (see Theorem 6.6.6 on p. 266):

If X is a Stein space and if every point $x \in X$ admits an open neighborhood $U \subset X$ such that the restricted submersion $Z|_U \to U$ satisfies HAP*, then global sections $X \to Z$ satisfy all forms of the Oka principle. For stratified submersions it suffices to have* HAP *over small open sets in every stratum.*

The proof uses similar analytic techniques as in the locally trivial case considered in Chapter 5, but the underlying induction scheme is more involved. This extends the results of the previous chapter, unless the fibers are compact and biholomorphic to each other; in the latter case we have a holomorphic fiber bundle in view of the Fischer-Grauert theorem [141].

Some of the applications in Chapters 7 and 8 depend on the most general version of the Oka principle proved in this chapter.

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6.1 Holomorphic Fiber-Sprays and Elliptic Submersions

The notion of a dominating spray on a complex manifold was introduced in §5.5. We now define the notion of a fiber-dominating spray.

Definition 6.1.1. [237, §1.1.B] *Let X and Z be complex spaces and let $h\colon Z \to X$ be a holomorphic submersion.*

(a) *An h-*spray *(or a* fiber-spray*) on Z is a triple (E, π, s), where $\pi\colon E \to Z$ is a holomorphic vector bundle and $s\colon E \to Z$ is a holomorphic map such that for each $z \in Z$ we have*

$$s(0_z) = z, \quad s(E_z) \subset Z_{h(z)} = h^{-1}(h(z)).$$

(b) (E, π, s) *is* dominating *at $z \in Z$ if the derivative $\mathrm{d}s_{0_z}\colon \mathrm{T}_{0_z}E \to \mathrm{T}_z Z$ maps the subspace $E_z \subset \mathrm{T}_{0_z}E$ surjectively onto $\mathrm{VT}_z Z = \ker \mathrm{d}h_z$. The spray is* dominating *(on Z) if it is dominating at every point $z \in Z$.*

(c) *A family of h-sprays (E_j, π_j, s_j) $(j = 1, \ldots, m)$ on Z is dominating at the point $z \in Z$ if*

$$(\mathrm{d}s_1)_{0_z}(E_{1,z}) + (\mathrm{d}s_2)_{0_z}(E_{2,z}) \cdots + (\mathrm{d}s_m)_{0_z}(E_{m,z}) = \mathrm{VT}_z Z. \tag{6.1}$$

If (6.1) holds at every point $z \in Z$ then the family is dominating.

We denote the restriction of the differential $\mathrm{d}s_{0_z}\colon \mathrm{T}_{0_z}E \to \mathrm{VT}_z Z$ to the subspace $E_z \subset \mathrm{T}_{0_z}E$ by

$$\mathrm{V}\mathrm{d}s_z = \mathrm{d}s_{0_z}|_{E_z}\colon E_z \to \mathrm{VT}_z Z \tag{6.2}$$

and call it the *vertical derivative* of s at the point $z \in Z$.

Definition 6.1.2. *A holomorphic submersion $h\colon Z \to X$ is said to be* elliptic *(resp.* subelliptic*) if for each point $x_0 \in X$ there is an open neighborhood $U \subset X$ such that the restricted submersion $h\colon Z|_U \to U$ admits a dominating h-spray (resp. a finite dominating family of h-sprays).*

Comparing with Def. 5.5.11 (p. 203) we see that a spray on a manifold Y is a fiber-spray associated to the trivial submersion $Y \to$ *point*. A holomorphic fiber bundle map is elliptic (resp. subelliptic) if and only if its fiber is such.

Example 6.1.3. If U is an open set in X and $V_1, \ldots, V_n$ are $\mathbb{C}$-complete holomorphic vector fields on $Z|_U$ that are tangential to the fibers Z_x and that span $\mathrm{VT}_z Z$ at every point of $z \in Z|_U$, then the composition of their flows

$$s(z, t_1, \ldots, t_n) = \phi_1^{t_1} \circ \phi_2^{t_2} \circ \cdots \circ \phi_n^{t_n}(z) \tag{6.3}$$

is a fiber dominating spray on $Z|_U$. (Compare with Example 5.5.13 (B).) □

Further examples of elliptic and subelliptic submersions are given in §6.4.

6.2 Gromov's Oka Principle

The following main theorem of this chapter gives an advanced form of the Oka principle for sections of certain holomorphic submersions onto Stein spaces.

Definition 6.2.1. *Let X and Z be complex spaces. A holomorphic submersion $h\colon Z \to X$ is said to be* stratified elliptic (resp. stratified subelliptic) *if there exists a stratification of X by closed complex subvarieties*

$$X = X_0 \supset X_1 \supset \cdots \supset X_m = \emptyset \tag{6.4}$$

such that every difference $S_k = X_k \backslash X_{k+1}$ is nonsingular and the restricted submersion $h\colon Z|_{S_k} \to S_k$ is elliptic (resp. subelliptic) (Def. 6.1.2, p. 242).

Theorem 6.2.2. *If $h\colon Z \to X$ is a stratified subelliptic submersion onto a Stein space X then sections $X \to Z$ satisfy the parametric Oka property with approximation and interpolation (the conclusion of Theorem 5.13.2 holds). In particular, the inclusion $\Gamma_{\mathcal{O}}(X, Z) \hookrightarrow \Gamma(X, Z)$ of the space of holomorphic sections into the space of continuous sections is a weak homotopy equivalence.*

In the analogous result for stratified fiber bundles with Oka fibers (Theorem 5.13.2 on p. 229) the parameter sets $P_0 \subset P$ were assumed to be Euclidean compacts. The parametric Oka property in Theorem 6.2.2 refers to an arbitrary pair of compact Hausdorff parameter space $P_0 \subset P$. However, some of the arguments can be simplified by restricting to Euclidean compacts and using Stein neighborhoods constructed in §3.6.

Theorem 6.2.2 has a complex genesis. For elliptic submersions over Stein manifolds, and without interpolation, it is due to Gromov [237, 4.5]; a detailed exposition in [191] is one of the main sources for this chapter. Interpolation on subvarieties was added in [192]. The proof for stratified elliptic submersions was indicated without details in [192, §7]. The notion of a subelliptic submersion was introduced in [170]. The proof Theorem 6.2.2 for a Stein base space X, also in the stratified case, was given in [180] by following the argument for stratified fiber bundles with Oka fibers (Theorem 5.4.4, p. 193).

The following main induction step in the proof of Theorem 6.2.2 is sometimes used independently.

Theorem 6.2.3. *Let $h\colon Z \to X$ be a holomorphic submersion of a complex space Z onto a Stein space X. Assume that $X_1 \subset X_0$ are closed complex subvarieties of X and K is a compact $\mathcal{O}(X)$-convex subset of X. Let $f\colon X \to Z$ be a continuous section which is holomorphic in an open neighborhood of K and on X_1. If the submersion $Z \to X$ is subelliptic over $X_0 \backslash (X_1 \cup K)$, then there is a homotopy of section $f_t\colon X \to Z$ ($t \in [0,1]$), with $f_0 = f$, such that the homotopy is fixed on X_1, every f_t is holomorphic and close to f in a neighborhood of K, and the section f_1 is holomorphic on X_0. The analogous conclusion holds for a continuous family of sections with these properties.*

In some applications we assume that the initial section f_0 is actually holomorphic in an open set $U \subset X$ containing $X_1 \cup K$, and we obtain a homotopy $f_t \colon X \to Z$ as in Theorem 6.2.3 which remains holomorphic and tangential to f_0 to a prescribed order along the subvariety X_1.

The proof of these results occupies §6.9 – §6.12. We use an induction scheme proposed in [237] and elaborated in [191]. Our exposition mainly follows [191, 192], with improvements in many places. In §6.3 – §6.5 we collect the preparatory material on holomorphic sprays, and we give examples of (sub-) elliptic submersions. In §6.6 we prove the main analytic ingredient, an Oka-Weil type approximation theorem for homotopies of holomorphic sections of subelliptic submersions over Stein spaces. In §6.7 we explain how to glue homotopic families of holomorphic sections. In §6.8 we introduce the notion of a holomorphic complex, the main bookkeeping tool that is used in the proof.

Outline of proof of Theorem 6.2.2. For simplicity of exposition we focus on the nonparametric case, and without the interpolation condition. The main problem is the following:

Given a compact $\mathcal{O}(X)$-convex set K in X and a continuous section $a \colon X \to Z$ that is holomorphic in an open set $U_0 \supset K$, find a homotopy of sections $H^s \colon X \to Z$ ($s \in [0,1]$) such that $H^0 = a$, the section $f = H^1$ is holomorphic on X, and every section H^s is holomorphic in a neighborhood of K and it approximates a uniformly on K (also uniformly in the parameter).

We begin by constructing a locally finite cover $\mathcal{A} = \{A_0, A_1, A_2, \ldots\}$ of X by compact sets A_j, and a family of holomorphic sections $a_{(j)} \colon U_j \to Z$ over open sets $U_j \supset A_j$, satisfying the following (Proposition 6.10.1):

- $K \subset A_0 \subset U_0$;
- for each $n \geq 1$ the sequence $(A_0, A_1, \ldots, A_n)$ is a $\mathcal{C}$-*string* (Def. 6.9.1);
- for every $j = 1, 2, \ldots$ the restricted submersion $h \colon Z|_{U_j} \to U_j$ admits a finite dominating family of fiber-sprays;
- for each $j \in \mathbb{Z}_+$ the section $a_{(j)} \colon U_j \to Z$ is homotopic to $a|_{U_j}$;
- for each pair of indexes $i \neq j$ such that $A_i \cap A_j \neq \emptyset$ there is a holomorphic homotopy between $a_{(i)}$ and $a_{(j)}$ in $U_{(i,j)} = U_i \cap U_j$;
- more generally, for each multiindex $J = (j_0, j_1, \ldots, j_n)$ such that $A_J = A_{j_0} \cap \cdots \cap A_{j_n} \neq \emptyset$ there is an n-dimensional homotopy $a_J(t)$ of holomorphic sections on $U_J = U_{j_0} \cap \cdots \cap U_{j_n}$, with the parameter t belonging to the standard n-simplex $\triangle^n \subset \mathbb{R}^n$, such that for every t in a boundary face of $\triangle^n$, determined by a sub-multiindex $I \subset J$, we have $a_J(t) = a_I(t)|_{U_J}$.

The parameter space of our collection of holomorphic sections and of homotopies between them is the geometric realization of a simplicial complex called the *nerve* of the cover $\mathcal{A}$ (§6.8). The sets U_j will shrink around A_j, but the A_j's will remain fixed during the entire construction.

The family $\{a_{(j)}: j \in \mathbb{Z}_+\}$, together will all the connecting homotopies, is a puzzle whose pieces are systematically reassembled into a holomorphic section $f: X \to Z$, homotopic to the initial continuous section a. All modifications of the complex will consist of holomorphic homotopies.

Suppose inductively that the sections $a_{(0)}, \ldots, a_{(n-1)}$ have already been joined into a holomorphic section f^{n-1} over a neighborhood of $A^{n-1} = A_0 \cup A_1 \cup \cdots \cup A_{n-1}$. The inductive step amounts to gluing f^{n-1} with the next section $a_{(n)}$ in the initial collection. To accomplish this, we need a holomorphic homotopy between the two sections in a neighborhood of $A^{n-1} \cap A_n$. Our inductive construction is such that for each $j = 0, 1, \ldots, n-1$ we have a holomorphic homotopy between f^{n-1} and $a_{(n)}$ in a neighborhood of $A_j \cap A_n$, inherited from the initial homotopy between $a_{(j)}$ and $a_{(n)}$. We now patch these n partial homotopies into a homotopy defined in a neighborhood of $A^{n-1} \cap A_n = \cup_{j=0}^{n-1}(A_j \cap A_n)$. This can be done by induction on n, provided that the sequence $(A_0 \cap A_n, A_1 \cap A_n, \ldots, A_{n-1} \cap A_n)$ is also a $\mathcal{C}$-string. Once this is achieved, we use that (A^{n-1}, A_n) is a $\mathcal{C}$-pair to glue f^{n-1} and $a_{(n)}$ into a section f^n over A^n by appealing to Proposition 6.7.2, thereby completing the induction step. The main induction step is furnished by Proposition 6.11.1. The resulting sequence of sections f^n converges, uniformly on compacts in X, to a holomorphic section $f: X \to Z$ which solves the problem.

In the stratified case we apply the above argument within a stratum in X, and we proceed from one stratum to the next one just as was done in the case of a stratified fiber bundle in §5.12.

6.3 Composed and Iterated Sprays

In this section we explain some basic operations with sprays, in particular the notions of *composed* and *iterated* sprays, which will be used in the proof of the Oka-Weil approximation theorems for homotopies of holomorphic sections (§6.6). These constructions are due to Gromov [237].

Lemma 6.3.1. *If $s_j: E_j \to Y$ $(j = 1, 2, \ldots, k)$ is a dominating family of sprays on Y, defined on trivial bundles $E_j \cong Y \times \mathbb{C}^{m_j}$, then Y admits a dominating spray. The analogous result holds for fiber-sprays.*

Proof. Assume that s_j is defined on $E_j = Y \times \mathbb{C}^{m_j}$ for each j. We inductively define sprays $s^{(j)}: E_1 \oplus \cdots \oplus E_j = Y \times \mathbb{C}^{m_1+\cdots+m_j} \to Y$ by $s^{(1)} = s_1$ and

$$s^{(j)}(y, e_1, \ldots, e_j) = s_j\big(s^{(j-1)}(y, e_1, \ldots, e_{j-1}), e_j\big), \quad j = 2, \ldots, k.$$

Clearly we have

$$\mathrm{d}s^{(k)}_{0_y}(E_{1,y} \oplus \cdots \oplus E_{k,y}) = (\mathrm{d}s_1)_{0_y}(E_{1,y}) + \cdots + (\mathrm{d}s_k)_{0_y}(E_{k,y}).$$

Hence $(s_1, \ldots, s_k)$ is a dominating family of sprays on Y (Def. 5.5.11) if and only if $s^{(k)}$ is a dominating spray. $\square$

Lemma 6.3.2. *Assume that* (E_j, π_j, s_j) $(j = 1, 2, \ldots, k)$ *is a dominating family of sprays on* Y*. If every bundle* $E_j \to Y$ *is generated by finitely many holomorphic sections over* Y *then there exists a dominating spray on* Y*. The analogous result holds for fiber-sprays.*

Proof. Assume that E_j is generated by sections $f_{j,l}$ $(l = 1, \ldots, m_j)$. Let $\sigma_j \colon E'_j = Y \times \mathbb{C}^{m_j} \to E_j$ be the holomorphic vector bundle epimorphism which takes the l-th standard basis section of $Y \times \mathbb{C}^{m_j}$ to $f_{j,l}$. Then $s'_j = s_j \circ \sigma_j \colon E'_j \to Y$ is a holomorphic spray, and Lemma 6.3.1 applies to the family (E'_j, π'_j, s'_j) $(j = 1, 2, \ldots, k)$. □

Lemma 6.3.3. *A subelliptic Stein manifold is elliptic. More generally, if* Z *is a Stein space then every subelliptic submersion* $h \colon Z \to X$ *is elliptic.*

Proof. Since every holomorphic vector bundle over a Stein manifold is generate by finitely many holomorphic sections by Cartan's Theorem A, the first claim follows directly from Lemma 6.3.2.

Assume now that Z is a Stein space and $h \colon Z \to X$ is a subelliptic submersion. Fix a point $x_0 \in X$ and choose a neighborhood $x_0 \in U \subset X$ such that $Z|_U \to U$ admits a finite dominating family of fiber-sprays (E_j, p_j, s_j). Decreasing U if necessary we may assume that U is Stein. Since Z is Stein, it follows that $Z|_U = h^{-1}(U)$ is also Stein and hence Lemma 6.3.2 applies to the bundles E_j restricted to $Z|_U$. □

Lemma 6.3.4. *The Cartesian product of any finite family of elliptic (resp. subelliptic) manifolds is elliptic (resp. subelliptic).*

Proof. It suffices to prove the result for the Cartesian product of two manifolds. Let $Y = Y_1 \times Y_2$ and let $\pi_j \colon Y \to Y_j$ $(j = 1, 2)$ denote the projection $\pi_j(y_1, y_2) = y_j$. If (E_j, π_j, s_j) is a spray on Y_j for $j = 1, 2$, we get a spray $s_1 \times s_2$ on the bundle $E = \pi_1^* E_1 \oplus \pi_2^* E_2 \to Y$ by

$$(s_1 \times s_2)(y_1, y_2, e_1, e_2) = \big(s_1(y_1, e_1), s_2(y_2, e_2)\big).$$

If s_1 is dominating on Y_1 and s_2 is dominating on Y_2 then $s_1 \times s_2$ is dominating on $Y_1 \times Y_2$.

Similarly, if a family of sprays $\{s_j \colon j = 1, \ldots, j_0\}$ is dominating on Y_1 and a family of sprays $\{\sigma_k \colon k = 1, \ldots, k_0\}$ is dominating on Y_2, then the family $\{s_j \times \sigma_k \colon j = 1, \ldots, j_0,\ k = 1, \ldots, k_0\}$ is dominating on $Y_1 \times Y_2$. □

Definition 6.3.5. *Let* (E, π, s)*,* (E_1, π_1, s_1) *and* (E_2, π_2, s_2) *be fiber-sprays associated to a holomorphic submersion* $h \colon Z \to X$*.*

(a) *The* composed spray $(E_1 * E_2, \pi_1 * \pi_2, s_1 * s_2)$ *over* Z *is defined by*

$$E_1 * E_2 = \{(e_1, e_2) \in E_1 \times E_2 \colon \ s_1(e_1) = \pi_2(e_2)\},$$

$$\pi_1 * \pi_2(e_1, e_2) = \pi_1(e_1), \quad s_1 * s_2(e_1, e_2) = s_2(e_2).$$

(b) *For each integer* $k = 1, 2, \ldots$ *the* k*-th* iterated spray $(E^{(k)}, \pi^{(k)}, s^{(k)})$ *is*

$$\begin{aligned} E^{(k)} = \big\{ e = (e_1, e_2, \ldots, e_k) \colon e_j \in E \ \text{ for } j = 1, 2, \ldots, k, \\ s(e_j) = \pi(e_{j+1}) \text{ for } j = 1, 2, \ldots, k-1 \big\}, \\ \pi^{(k)}(e) = \pi(e_1), \quad s^{(k)}(e) = s(e_k). \end{aligned}$$

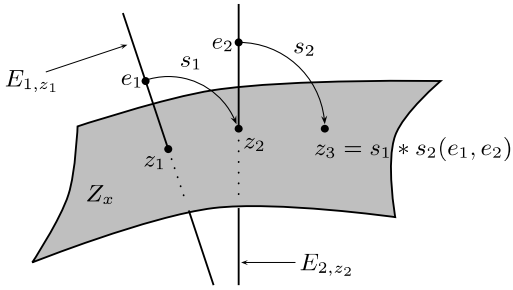

Fig. 6.1. A composed spray $s_1 * s_2$ on a fiber Z_x

Observe that $E_1 * E_2 = s_1^*(E_2)$ is the pull-back of the second vector bundle $\pi_2 \colon E_2 \to Z$ by the first spray map $s_1 \colon E_1 \to Z$. The following diagram shows all relevant maps in a composed spray; ι is the natural map over s_1, and $s_1 * s_2 = s_2 \circ \iota \colon E_1 * E_2 \to Z$ is the composed spray map.

$$\begin{tikzcd} s_1^* E_2 = E_1 * E_2 \arrow[r, "\iota"] \arrow[d] & E_2 \arrow[r, "s_2"] \arrow[d, "\pi_2"] & Z \\ E_1 \arrow[r, "s_1"] \arrow[d, "\pi_1"'] & Z & \\ Z & & \end{tikzcd}$$

Each of the two projections in $E_1 * E_2 \longrightarrow E_1 \xrightarrow{\pi_1} Z$ in the first column is a holomorphic vector bundle, but their composition $\pi_1 * \pi_2 \colon E_1 * E_2 \to Z$ does not have a natural holomorphic vector bundle structure. Nevertheless it has a well defined zero section which we identify with the base space Z.

The composition extends to any finite number of factors and is associative. In particular, the k-th iterate $(E^{(k)}, \pi^{(k)}, s^{(k)})$ is obtained by composing k copies of the spray (E, π, s).

Let $h \colon Z \to X$ be a holomorphic submersion and (E, π, s) a composed fiber-spray on Z. We denote by $\mathrm{VT}(E) = \ker \mathrm{d}\pi$ the vertical tangent bundle of E and by $\mathrm{VT}(E)|_Z$ its restriction to the zero section $Z \subset E$. If (E, π, s) is obtained by composing the h-sprays (E_j, π_j, s_j) $(j = 1, \ldots, m)$ then

$$\mathrm{VT}E|_Z \cong E_1 \oplus E_2 \oplus \cdots \oplus E_m.$$

Since $s\colon E \to Z$ preserves the zero section and maps the fiber $E_z = \pi^{-1}(z)$ to the fiber $Z_{h(z)}$ for every $z \in Z$, its differential $\mathrm{d}s$ at the zero section maps $\mathrm{VT}E|_Z$ to $\mathrm{VT}Z$. In analogy to (6.2) we denote this map by

$$\mathrm{Vd}s\colon \mathrm{VT}E|_Z \to \mathrm{VT}Z \tag{6.5}$$

and call it the *vertical derivative* of s. A composed spray (E, π, s) is said to be *fiber-dominating* if $\mathrm{Vd}s$ is surjective; in such case $\ker \mathrm{Vd}s \subset \mathrm{VT}E|_Z$ is a holomorphic vector subbundle of the vertical tangent bundle $\mathrm{VT}E|_Z$.

Lemma 6.3.6. *A family of h-sprays (E_j, π_j, s_j) $(j = 1, \ldots, m)$ is fiber-dominating at $z \in Z$ if and only if the composed h-spray $s = s_1 * \cdots * s_m\colon E = E_1 * \cdots * E_m \to Z$ is fiber-dominating at z.*

Proof. Observe that $s(z, 0, \ldots, e_j, \ldots, 0) = s_j(z, e_j)$, where $e_j \in E_{j,z}$ appears at the j-th spot. Hence the range (in $\mathrm{VT}_z Z$) of the vertical derivative of $s = s_1 * \cdots * s_k$ at the point $z \in E_1 * \cdots * E_k$ equals the vector sum of the vertical derivatives $\mathrm{Vd}s_j$ of the individual sprays s_j at the points $0_z \in E_{j,z}$. The lemma now follows from (6.1). □

Lemma 6.3.7. *Assume that $h\colon Z \to X$ is a holomorphic submersion. Given a family (E_j, π_j, s_j) $(j = 1, \ldots, m)$ of h-sprays on Z, let (E, π, s) be the associated composed spray. For any Stein subset $\Omega \subset Z$ (either an open Stein subset, or a Stein subvariety) there is a fiber preserving biholomorphic map*

$$\Theta\colon E|_\Omega = E_1 * \cdots * E_m|_\Omega \to E_1 \oplus E_2 \oplus \cdots \oplus E_m|_\Omega$$

which preserves the zero section and whose differential at the zero section equals the identity map.

Proof. It suffices to prove the result for $m = 2$ and apply induction. Recall that $E_1 * E_2 = s_1^*(E_2)$ is the pull-back of $\pi_2\colon E_2 \to Z$ by $s_1\colon E_1 \to Z$. Let $E_1' = E_1|_\Omega$, and let $\pi_1'\colon E_1' \to \Omega$ (resp. $s_1'\colon E_1' \to Z$) denote the restriction of π_1 (resp. of s_1) to E_1'. Consider the homotopy of holomorphic maps

$$g_t\colon E_1' \to Z, \quad g_t(z, e) = s_1(z, te), \quad t \in \mathbb{C}.$$

Note that $g_0 = \pi_1'$ and $g_1 = s_1'$. Since Ω is Stein, the total space E_1' is also a Stein space. By pulling the bundle $\pi_2\colon E_2 \to Z$ back to E_1' by $g_t\colon E_1' \to Z$ we obtain a family of holomorphically isomorphic vector bundles according to Corollary 7.3.9. In particular, we have an isomorphism between $g_1^* E_2 = (s_1')^* E_2 = E_1 * E_2|_\Omega$ and $g_0^* E_2 = (\pi_1')^*(E_2|_\Omega)$ (as holomorphic vector bundles over E_1'). The total space of $g_0^* E_2$ is clearly isomorphic to $E_1 \oplus E_2|_\Omega$ which is a holomorphic vector bundle over Ω. This furnishes the desired isomorphism $\Theta\colon E_1 * E_2|_\Omega \to E_1 \oplus E_2|_\Omega$. □

Pull-back of sprays. Given a holomorphic submersion $h\colon Z \to X$ and a holomorphic map $f\colon Y \to X$, consider the submersion $\widetilde{h}\colon \widetilde{Z} \to Y$ defined by

$$\widetilde{Z} = \big\{(y,z) \in Y \times Z\colon f(y) = \pi(z)\big\}, \quad \widetilde{h}(y,z) = y. \tag{6.6}$$

Let $\iota\colon \widetilde{Z} \to Z$ denote the map $\iota(y,z) = z$ for $(y,z) \in \widetilde{Z}$, covering the base map $f\colon Y \to X$. Given a fiber-spray (E,π,s) on $h\colon Z \to X$, let $\tilde{\pi}\colon \widetilde{E} \to \widetilde{Z}$ denote the pull-back of the vector bundle $\pi\colon E \to Z$ by the map ι, and let $\tilde{\iota}\colon \widetilde{E} \to E$ be the natural map over ι. There is a fiber-spray $\tilde{s}\colon \widetilde{E} \to \widetilde{Z}$ associated to the submersion $\widetilde{h}\colon \widetilde{Z} \to Y$ such that the following diagram commutes:

$$\begin{tikzcd}
\widetilde{E} \arrow[r, "\tilde{s}"] \arrow[d, "\tilde{\iota}"'] & \widetilde{Z} \arrow[r, "\tilde{h}"] \arrow[d, "\iota"'] & Y \arrow[d, "f"] \\
E \arrow[r, "s"'] & Z \arrow[r, "h"'] & X
\end{tikzcd}$$

The triple $(\widetilde{E},\tilde{\pi},\tilde{s})$ is the pull-back of the spray (E,p,s) by the map $f\colon Y \to X$. It is easily verified that a fiber-dominating spray (resp. a fiber dominating family of sprays) on $Z \to X$ pulls back to a fiber-dominating spray (resp. to a fiber dominating family of sprays) on $\widetilde{Z} \to Y$. Similarly, a fiber-spray on $Z|_U$, where U is an open subset of X, pulls back to a fiber-spray on $\widetilde{Z}|_{\widetilde{U}}$, where $\widetilde{U} = f^{-1}(U) \subset Y$. In conclusion, we have the following result.

Proposition 6.3.8. *If $h\colon Z \to X$ is an elliptic (resp. a subelliptic) submersion and $f\colon Y \to X$ is a holomorphic map, then the pull-back submersion $\widetilde{Z} = f^*Z \to Y$ (6.6) is also elliptic (resp. subelliptic).*

6.4 Examples of Subelliptic Manifolds and Submersions

Complements of Tame Subvarieties

Proposition 6.4.1. (a) *If A is an algebraic subvariety of $\mathbb{C}^n$ of dimension $\dim A \le n-2$ then $\mathbb{C}^n \backslash A$ is algebraically elliptic.*

(b) *If $A \subset \mathbb{C}^n$ is a tame analytic subvariety (Def. 4.48) and $\dim A \le n-2$ then $\mathbb{C}^n \backslash A$ is elliptic.*

(c) *If Y is a complex Grassmann manifold and $A \subset Y$ is a closed complex (=algebraic) subvariety of codimension at least two then $Y \backslash A$ is algebraically subelliptic. This holds in particular when $Y = \mathbb{P}^n$.*

Proof. Parts (a) and (b) are Proposition 5.5.14. (See also Example 5.5.13 (C).)

Assume now that $Y = \mathbb{P}^n$. Choose a hyperplane $\Lambda \subset \mathbb{P}^n$ and homogeneous coordinates $z = [z_0\colon z_1\colon \cdots\colon z_n]$ on $\mathbb{P}^n$ such that $\Lambda = \{z_0 = 0\}$. Set $U_j = \{z \in$

$\mathbb{P}^n\colon z_j \neq 0\} \cong \mathbb{C}^n$ for $j = 0, 1 \ldots, n$; hence $\mathbb{P}^n = U_0 \cup \Lambda$. Let $L \to \mathbb{P}^n$ denote the holomorphic line bundle $L = [\Lambda]^{-1}$, where $[\Lambda]$ is the line bundle determined by the divisor of Λ (see Example 1.5.7). The bundle L admits holomorphic vector bundle trivializations $\phi_j\colon L|_{U_j} \to U_j \times \mathbb{C}$ with the transition maps

$$\phi_{ik}(z,t) = \phi_i \circ \phi_k^{-1}(z,t) = (z, tz_i/z_k)\,, \quad z \in U_{ik} = U_i \cap U_k,\ t \in \mathbb{C}.$$

On $U_0 \cong \mathbb{C}^n$ we use the affine coordinates $\zeta = (\zeta_1, \ldots, \zeta_n)$ with $\zeta_j = \frac{z_j}{z_0}$. Choose $v = (v_1, \ldots, v_n) \in \mathbb{C}^n$ such that the projection $\pi\colon \mathbb{C}^n \to \mathbb{C}^{n-1}$ with kernel $\mathbb{C}v$ is proper when restricted to $A \cap U_0$. (Since $\dim A \leq n-2$, this holds for almost every $v \in \mathbb{C}^n$.) Then $A' = \pi(A \cap U_0) \subset \mathbb{C}^{n-1}$ is a proper algebraic subvariety of $\mathbb{C}^{n-1}$. If p is a holomorphic polynomial on $\mathbb{C}^{n-1}$ that vanishes on A', then the map $U_0 \times \mathbb{C} \to U_0$, given by

$$s(\zeta, t) = \zeta + tp(\pi\zeta)v = \zeta + tf(\zeta), \qquad \zeta \in \mathbb{C}^n,\ t \in \mathbb{C},$$

is an algebraic spray satisfying $\frac{\partial}{\partial t} s(\zeta, 0) = p(\pi\zeta)v = f(\zeta)$. Although s does not extend to a spray $\mathbb{P}^n \times \mathbb{C} \to \mathbb{P}^n$ on the trivial line bundle due to singularities on $\Lambda = \{z_0 = 0\}$, we shall see that s induces an algebraic spray $\tilde{s}\colon L^{\otimes m} \to \mathbb{P}^n$, where $m \in \mathbb{N}$ is the degree of the polynomial p.

The line bundle $E = L^{\otimes m}$ admits vector bundle charts $\theta_i\colon E|_{U_i} \to U_i \times \mathbb{C}$ $(i = 0, 1, \ldots, n)$ with the transition maps

$$\theta_{ik}(z,t) = \big(z, t(z_i/z_k)^m\big), \qquad z \in U_{ik},\ t \in \mathbb{C}.$$

Set $\tilde{s} = s \circ \theta_0\colon E|_{U_0} \to \mathbb{P}^n$. We claim that $\tilde{s}$ extends to an algebraic spray $E \to \mathbb{P}^n$. Writing $z = [z_0\colon z']$, we see that the spray s has the following expression in the homogeneous coordinates $z \in U_0 \subset \mathbb{P}^n$:

$$s(z,t) = \big[1\colon s(z'z_0^{-1}, t)\big] = \big[1\colon z'z_0^{-1} + tf(z'z_0^{-1})\big] = \big[z_0\colon z' + tz_0 f(z'z_0^{-1})\big]\,.$$

Hence we get for $k = 1, \ldots, n$ and $z \in U_0 \cap U_k$ that

$$\tilde{s} \circ \theta_k^{-1}(z,t) = s \circ \theta_{0k}(z,t) = s\left(z, t(z_0 z_k^{-1})^m\right) = \big[z_0\colon z' + t\, z_0^{m+1} z_k^{-m} f(z'z_0^{-1})\big]\,.$$

By the choice of m, $z_0^{m+1} z_k^{-m} f(z'z_0^{-1})$ vanishes on $\{z_0 = 0\} \cap U_k$, and hence $\tilde{s} \circ \theta_k^{-1}$ is holomorphic on U_k. Thus $\tilde{s}\colon E \to \mathbb{P}^n$ is a spray satisfying

- $\tilde{s}(E|_{U_0 \setminus A}) \subset U_0 \setminus A$, and
- $\tilde{s}(z,t) = z$ for all $z \in A \cup \Lambda$ and $t \in E_z$.

For each point $p \in U_0 \setminus A$ we can find finitely many sprays of this kind which together dominate at p, and hence they dominate over a Zariski open set containing the point p. Repeating this construction at other points, and for different choices of the hyperplane $\Lambda \subset \mathbb{P}^n$, we obtain a finite dominating family of algebraic sprays on $\mathbb{P}^n \setminus A$.

The same proof applies to a Grassmannian $Y = G_{k,n}$ since it is covered by finitely many Zariski open sets $U_j \cong \mathbb{C}^{k(n-k)}$ (Example 1.2.8). $\square$

Localization of Algebraic Subellipticity

Algebraic subellipticity can be localized as follows (see [237, 3.5.B., 3.5.C.] and [170, Proposition 1.3]). No comparable localization result is known for ellipticity.

Proposition 6.4.2. *If Y is a quasi-projective algebraic manifold such that each point $y \in Y$ has a Zariski open neighborhood $U \subset Y$ and algebraic sprays $s_j \colon E_j \to Y$ $(j = 1, 2, \ldots, k)$, defined on algebraic vector bundles $p_j \colon E_j \to U$ and satisfying*

$$(\mathrm{d}s_1)_{0_y}(E_{1,y}) + (\mathrm{d}s_2)_{0_y}(E_{2,y}) \cdots + (\mathrm{d}s_k)_{0_y}(E_{k,y}) = \mathrm{T}_y Y \tag{6.7}$$

for every $y \in U$, then Y is algebraically subelliptic.

Proof. We repeatedly use the observation that for every closed algebraic subvariety A in a quasi-projective manifold Y and for every point $y_0 \in Y \backslash A$ there exists an algebraic hypersurface $\Lambda \subset Y$ such that $A \subset \Lambda$ but $y_0 \notin \Lambda$.

Fix a point $y_0 \in Y$ and let $U \subset Y$ be a Zariski neighborhood of y_0 with finitely many algebraic sprays $s_j \colon E_j \to Y$ $(j = 1, \ldots, k)$ such that (6.7) holds at the point $y = y_0$. Replacing U by a smaller Zariski neighborhood of y_0 we may assume that $\Lambda = Y \backslash U$ is an algebraic hypersurface in Y and that the bundle $E_j|_U \to U$ is algebraically trivial for each j. Composing an algebraic trivialization of $E_j|_U$ with the spray s_j we may therefore assume that s_j is defined on the product bundle $U \times \mathbb{C}^{N_j}$ and has values in Y. To remove the singularities of s_j along the hypersurface Λ we replace the product bundle by $N_j[\Lambda]^{-m_j}$ for a sufficiently large integer $m_j \in \mathbb{N}$ (see the proof of Proposition 6.4.1). This gives finitely many algebraic sprays on Y which together dominate at y_0, and hence over a Zariski neighborhood of y_0. Finitely many such families of sprays then dominate on all of Y. $\square$

Proposition 6.4.3. *Let A be a closed algebraic subvariety of codimension at least two in a quasi-projective algebraic manifold Y. Suppose that each point $y_0 \in Y \backslash A$ has a Zariski open neighborhood $U \subset Y$ and an algebraic spray $s \colon E \to Y$, defined on a vector bundle $p \colon E \to U$, such that s is dominating at y_0 and $s^{-1}(A) \subset E$ contains no hypersurfaces. (This holds in particular if $s \colon E \to Y$ is a submersion of E onto Y.) Then $Y \backslash A$ is subelliptic.*

Proof. After removing an algebraic hypersurface which does not contain the point y_0 we may assume that $E|_U = U \times \mathbb{C}^N$ and that each fiber $\widetilde{A}_y = \{t \in \mathbb{C}^N \colon s(y,t) \in A\}$ of $\widetilde{A} := s^{-1}(A)$ has codimension at least two in $\mathbb{C}^N$. Note that $0 \notin \widetilde{A}_y$ for $y \in U \backslash A$. Let $t = (t_1, \ldots, t_N) \in \mathbb{C}^N$. For each $k = 0, 1, \ldots, N$ we set $\mathbb{C}_{(k)} = \mathbb{C}^k \times \{0\}^{N-k}$. Let $\pi_k \colon U \times \mathbb{C}_{(k)} \to U \times \mathbb{C}_{(k-1)}$ denote the projection $\pi_k(y, t_1, \ldots, t_k) = (y, t_1, \ldots, t_{k-1})$. After a linear change of coordinates on $\mathbb{C}^N$ and removing another algebraic hypersurface from U we may assume that

(i) for each $k = 1, \ldots, N$, $A_{(k)} = (U \times \mathbb{C}_{(k)}) \cap \widetilde{A}$ is a subvariety of $U \times \mathbb{C}_{(k)}$ with fibers of codimension at least two (in particular $A_{(1)} = \emptyset$), and

(ii) $\pi_k(A_{(k)}) \subset U \times \mathbb{C}_{(k-1)}$ is an algebraic subvariety of $U \times \mathbb{C}_{(k-1)}$ that does not contain the point $(y, 0, \ldots, 0)$ for any $y \in U$. (Note that $\pi_k(A_{(k)})$ contains $A_{(k-1)}$.)

Condition (ii) insures that for each $k = 2, \ldots, N$ there exists an algebraic function p_k on $U \times \mathbb{C}_{(k-1)}$ that vanishes on $\pi_k(A_{(k)})$ and satisfies $p_k(y_0, 0, \ldots, 0) \neq 0$. Consider the map $g: U \times \mathbb{C}^N \to U \times \mathbb{C}^N$ defined by

$$g(y,t) = \big(y, t_1, p_2(y,t_1)t_2, \ldots, p_N(y,t_1,\ldots,t_{N-1})t_N\big).$$

Clearly $g(y,0) = (y,0)$, the map $t \to g(y_0,t)$ is nondegenerate at $t = 0$, and the image of g avoids $\widetilde{A}$. Thus $\sigma = s \circ g: U \times \mathbb{C}^N \to Y$ is an algebraic spray that is dominating at y_0 and satisfies $\sigma((U \backslash A) \times \mathbb{C}^N) \subset Y \backslash A$. Since $y_0 \in Y \backslash A$ was arbitrary, $Y \backslash A$ is subelliptic by Proposition 6.4.2. $\square$

Problem 6.4.4. Let Y be an algebraically subelliptic manifold. Is the complement $Y \backslash A$ of every thin algebraic subvariety algebraically subelliptic?

Zariski Locally Affine Manifolds

We consider a class of algebraically subelliptic manifolds that contains all manifolds in Proposition 6.4.1.

Definition 6.4.5. *Let Y be a quasi-projective algebraic manifold.*

(a) *Y is of* Class $\mathcal{A}_0$ *if it is covered by finitely many Zariski open sets, each of them biregularly isomorphic to the affine space $\mathbb{C}^p$ with $p = \dim Y$.*

(b) *Y is of* Class $\mathcal{A}$ *if $Y = \widehat{Y} \backslash A$ where $\widehat{Y}$ is of class $\mathcal{A}_0$ and A is a thin (of codimension at least two) algebraic subvariety of $\widehat{Y}$.*

Class $\mathcal{A}_0$ includes all complex affine and projective spaces as well as Grassmannians. Another example is the total space W of a holomorphic fiber bundle $\pi: W \to Y$ where the base Y is a manifold of Class $\mathcal{A}_0$, the fiber is $F = \pi^{-1}(y) \in \{\mathbb{C}^m, \mathbb{P}^m\}$, and the structure group is $GL_m(\mathbb{C})$ respectively $PGL_m(\mathbb{C})$. Every such bundle is algebraic by the GAGA principle [441], and its restriction to any affine Zariski open set $\mathbb{C}^p \cong U \subset Y$ is algebraically trivial, $\pi^{-1}(U) \cong U \times F$. Hence W is covered by Zariski open sets biregularly isomorphic to $\mathbb{C}^{p+m}$, i.e., W is of Class $\mathcal{A}_0$. An example are the *Hirzebruch surfaces* H_l, $l = 0, 1, 2, \ldots$ [33, p. 191]; these are $\mathbb{P}^1$-bundles over $\mathbb{P}^1$, and each of them is birationally equivalent to $\mathbb{P}^2$. By Corollary 6.4.8 below every compact rational surfaces if of class $\mathcal{A}_0$.

Proposition 6.4.6. *Every manifold of Class $\mathcal{A}$ is algebraically subelliptic, and hence an Oka manifold.*

Proof. By Proposition 6.4.1 (a), or by Example 5.5.13 (C), the manifold $\mathbb{C}^p \backslash A$ is algebraically elliptic for each closed affine algebraic subvariety $A \subset \mathbb{C}^p$ of codimension at least two. The conclusion now follows from the localization principle (Proposition 6.4.2). □

Manifolds of Class $\mathcal{A}$ have been considered by Gromov who called them *Ell-regular*. This class is stable under blowing up points:

Proposition 6.4.7. [237, §3.5.D"] *If Y is a manifold of Class $\mathcal{A}$ (respectively of Class $\mathcal{A}_0$) then any manifold $\widetilde{Y}$, obtained from Y by blowing up finitely many points, is also of Class $\mathcal{A}$ (respectively of Class $\mathcal{A}_0$).*

Proof. By localization it suffices to show that the manifold L, obtained by blowing up $\mathbb{C}^q$ at the origin, is of Class $\mathcal{A}$. Note that L is the total space of a holomorphic line bundle $\pi\colon L \to \mathbb{P}^{q-1}$ (the universal bundle); L is trivial over the complement of each hyperplane $\mathbb{P}^{q-2} \subset \mathbb{P}^{q-1}$ (which equals $\mathbb{C}^{q-1}$), and hence every point in L has a Zariski neighborhood of the form $\pi^{-1}(\mathbb{P}^{q-1} \backslash \mathbb{P}^{q-2})$ that is biregularly isomorphic to $\mathbb{C}^q$. □

Corollary 6.4.8. *Every compact rational surface is of Class $\mathcal{A}_0$ and hence an Oka manifold.*

Proof. Every compact rational surface X is obtained by blowing up a certain Hirzebruch surface Y at finitely many points. Now Y is either a $\mathbb{P}^2$ or a holomorphic fiber bundle $Y \to \mathbb{P}^1$ with fiber $\mathbb{P}^1$ over $\mathbb{P}^1$. In either case Y is easily seen to be of class $\mathcal{A}_0$. Now apply Proposition 6.4.7. □

Covering Spaces

By Proposition 5.5.2, the Oka property passes up and down in unramified holomorphic coverings. We consider the same question for ellipticity.

Proposition 6.4.9. *Let $\pi\colon \widetilde{Y} \to Y$ be an unramified holomorphic covering map. If Y is elliptic (resp. subelliptic) then so is $\widetilde{Y}$.*

Proof. Let (E, p, s) be a spray on Y. Denote by $\widetilde{E} = \pi^*(E) \to \widetilde{Y}$ the pull-back of the bundle $p\colon E \to Y$ by the map $\pi\colon \widetilde{Y} \to Y$:

$$\widetilde{E} = \big\{(\tilde{y}, e)\colon \tilde{y} \in \tilde{Y},\ e \in E,\ \pi(\tilde{y}) = p(e)\big\}.$$

Let $\sigma\colon \widetilde{E} \to Y$ be defined by $\sigma(\tilde{y}, e) = s(y, e)$ where $y = \pi(\tilde{y}) \in Y$. Fix $\tilde{y} \in \widetilde{Y}$. Since the fiber $\widetilde{E}_{\tilde{y}}$ is simply connected and π is a holomorphic covering, the map $\sigma(\tilde{y}, \cdot)\colon \widetilde{E}_{\tilde{y}} \to Y$ has a unique holomorphic lifting $\tilde{s}(\tilde{y}, \cdot)\colon \widetilde{E}_{\tilde{y}} \to \widetilde{Y}$ (i.e., $\pi(\tilde{s}(\tilde{y}, e)) = \sigma(\tilde{y}, e)$) with $\tilde{s}(\tilde{y}, 0) = \tilde{y}$. Clearly $\tilde{s}\colon \widetilde{E} \to \widetilde{Y}$ is a spray on $\widetilde{Y}$, and $\tilde{s}$ is dominating when the original spray s is dominating. Hence $\widetilde{Y}$ is elliptic whenever Y is elliptic. A similar argument applies to families of sprays, showing that subellipticity of Y implies subellipticity of $\widetilde{Y}$. □

Problem 6.4.10. Let $\pi\colon \widetilde{Y} \to Y$ be an unramified holomorphic covering map. Does (sub-) ellipticity of $\widetilde{Y}$ imply (sub-) ellipticity Y ?

A good test case may be complex tori $X = \mathbb{C}^n/\Gamma$. Denote by $\pi\colon \mathbb{C}^n \to X$ the quotient map and let $p_0 = \pi(0) \in X$; then $\pi^{-1}(p_0) = \Gamma$. By Corollary 5.5.15, $\mathbb{C}^n \backslash \Gamma$ is elliptic. Is the punctured torus $X \backslash \{p_0\}$ also elliptic? (A torus with finitely many punctures is an Oka manifold by Corollary 5.5.15.)

Blowing Up a Tame Sequence in $\mathbb{C}^n$

We apply the construction from §6.4 to prove the following result.

Proposition 6.4.11. *Any manifold X, obtained from $\mathbb{C}^n$ by blowing up each point of a tame discrete set (Def. 4.6.1 on p. 118), is weakly subelliptic, and hence an Oka manifold.*

Proof. For simplicity we consider the case $n = 2$. Let (z, w) be complex coordinates on $\mathbb{C}^2$. It suffices to prove the result for the tame sequence $e_k = (k, 0)$, $k = 0, 1, 2, \ldots$. Let $X_0 = \widehat{\mathbb{C}^2}$ be the surface obtained by blowing up $\mathbb{C}^2$ at $e_0 = (0, 0)$. Denote by $\pi_0\colon X_0 \to \mathbb{C}^2$ the natural projection and let $S_0 = \pi_0^{-1}(e_0) \cong \mathbb{P}^1$ be the exceptional fiber with the homogeneous coordinates $[z\colon w]$. Let $\Lambda_0 \subset X_0$ be the proper transform of the line $\{w = 0\}$:

$$\Lambda_0 = \overline{\pi_0^{-1}\left(\{(z, 0)\colon z \neq 0\}\right)} \cong \mathbb{C}.$$

Note that $\Lambda_0 \cap S_0 = [1\colon 0] \in S_0$ and $S_0 \backslash \Lambda_0 = S_0 \backslash \{[1\colon 0]\} \cong \mathbb{C}$. The complement $X_0 \backslash \Lambda_0$ is a holomorphic line bundle over $S_0 \backslash \Lambda_0 \cong \mathbb{C}$, and hence is biholomorphic to $\mathbb{C}^2$. As in the proof of Proposition 6.4.1 we obtain an algebraic spray (E_0, p_0, s_0) on X_0 which is dominating over $X_0 \backslash \Lambda_0 \cong \mathbb{C}^2$ and has values in this set, and that vanishes over Λ_0.

Let X_1 be obtained by blowing up the point $e_1 = (1, 0) \in X_0$ (we are identifying $e_1 \in \mathbb{C}^2$ with $\pi_0^{-1}(e_1) \in X_0$). Let $\pi_1\colon X_1 \to X_0$ be the associated projection which is biholomorphic over $X_0 \backslash \{e_1\}$ and has the exceptional fiber $S_1 \cong \mathbb{P}^1$ over e_1. We may consider $X_0 \backslash \Lambda_0$ as an open subset of $X_1 \backslash \pi_1^{-1}(\Lambda_0)$. Since the spray (E_0, p_0, s_0) vanishes over Λ_0, it extends to a spray on X_1 that vanishes on the curve $\pi_1^{-1}(\Lambda_0)$.

Next let $\pi_2\colon X_2 \to X_1$ be obtained by blowing up X_1 at $e_2 = (2, 0)$; the same argument as before shows that $X_0 \backslash \Lambda_0 \cong \mathbb{C}^2$ is an open subset of X_2 and (E_0, p_0, s_0) extends to a spray on X_2 that vanishes on the curve $(\pi_2 \circ \pi_1)^{-1}(\Lambda_0)$.

Continuing this process we obtain a complex surface X ($\mathbb{C}^2$ blown up at each point e_k, $k = 0, 1, \ldots$), with a projection $\pi\colon X \to X_0$, and a complex curve $\Sigma_0 = \pi^{-1}(\Lambda_0)$ such that $U_0 = X \backslash \Sigma_0 \cong X_0 \backslash \Lambda_0 \cong \mathbb{C}^2$. Note that Σ_0 is a normal crossing divisor consisting of a copy of $\mathbb{C}$ that is intersected by a rational curve $\mathbb{P}^1$ at each of the points $e_1, e_2, \ldots$, and these rational curves

are pairwise disjoint. Furthermore, we get a spray (E_0, p_0, s_0) on X that is dominating over the open set $U_0 \cong \mathbb{C}^2$ in X, with values in this set and vanishing over $\Sigma_0 = X \backslash U_0$. Considering the projection $\pi_0 \circ \pi \colon X \to \mathbb{C}^2$, we see that Σ_0 is the total transform of the line $\{w = 0\}$ at each point $e_1, e_2, \ldots$, and is the exact transform of this line at the point e_0.

In the same way get for every $k \in \mathbb{Z}_+$ a complex curve $\Sigma_k \subset X$ by taking the exact transform of the line $\{w = 0\}$ at e_k and its total transform at every other point e_j $(j \neq k \in \mathbb{Z}_+)$. We also get a spray (E_k, p_k, s_k) on X that is dominating on $U_k = X \backslash \Sigma_k$ and that vanishes over Σ_k.

Let $\Sigma \subset X$ denote the exact transform of $\{w = 0\}$ at every point e_k $(k \in \mathbb{Z}_+)$. Then $X \backslash \Sigma = \cup_{k=0}^{\infty} U_k$. The collections of sprays (E_k, p_k, s_k) dominates over $X \backslash \Sigma$. We now add another family of sprays to get domination over all of X. For this purpose consider the shear $\phi \in \operatorname{Aut} \mathbb{C}^2$ defined by

$$\phi(z, w) = \big(z, w + \sin(\pi z)\big), \quad (z, w) \in \mathbb{C}^2.$$

We get an induced automorphism $\Phi \in \operatorname{Aut} X$ such that the following diagram commutes:

$$\begin{array}{ccc} X & \xrightarrow{\Phi} & X \\ \downarrow & & \downarrow \\ \mathbb{C}^2 & \xrightarrow{\phi} & \mathbb{C}^2 \end{array}$$

Since $\sin(\pi z)$ vanishes at the integer values of z, Φ maps the exceptional fiber $S_k \cong \mathbb{P}^1$ of $X \to \mathbb{C}^2$ over e_k to itself for every $k \in \mathbb{Z}_+$, and $\Phi \colon S_k \to S_k$ is the nontrivial automorphism of $S_k \cong \mathbb{P}^1$ sending $[1\colon 0]$ to $[1\colon (-1)^k \pi]$ and fixing the point $[0\colon 1]$. Hence $\Phi(\Sigma) \cap \Sigma = \emptyset$ in X. By adding to the family of sprays (E_k, p_k, s_k) $(k \in \mathbb{Z}_+)$ all their pull-backs by Φ (see §6.3) we thus obtain a collection of sprays that dominates at each point of X, and a finite subcollection dominates over a given compact in X. $\square$

Let us call a complex torus any quotient $\mathbb{C}^n / \Gamma$ of $\mathbb{C}^n$ by a lattice, not necessarily of maximal rank.

Corollary 6.4.12. *A complex torus of dimension > 1, blown up at finitely many points, is an Oka manifold.*

Proof. Let Γ be a lattice in $\mathbb{C}^n$ and $\pi \colon \mathbb{C}^n \to \mathbb{T} = \mathbb{C}^n / \Gamma$ the quotient projection. Fix points $p_1, \ldots, p_m \in \mathbb{T}$. Choose points $z_1, \ldots, z_m \in \mathbb{C}^n$ such that $\pi(z_k) = p_k$ for $k = 1, \ldots, m$. Then $\pi^{-1}(p_k) = z_k + \Gamma$, and the union

$$\Lambda = \pi^{-1}(\{p_1, \ldots, p_m\}) = \bigcup_{k=1}^{m} (z_k + \Gamma) \subset \mathbb{C}^n$$

is a tame discrete set in $\mathbb{C}^n$ ([69, Proposition 4.1], [63]). Let Y be obtained from $\mathbb{T}$ by blowing up each of the points $p_1, \ldots, p_m$. Let X be obtained from

$\mathbb{C}^n$ by blowing up each point of Λ. Every translations of $\mathbb{C}^n$ by an element $\gamma \in \Gamma$ induce a biholomorphism $\widetilde{\gamma}\colon X \to X$. In this way Γ becomes a discrete subgroup of $\operatorname{Aut} X$, and we have a covering map $\widetilde{\pi}\colon X \to X/\Gamma \cong Y$ making the following diagram commute.

$$\begin{tikzcd} X \arrow[r, "\widetilde{\pi}"] \arrow[d] & Y \arrow[d] & \cong X/\Gamma \\ \mathbb{C}^n \arrow[r, "\pi"] & \mathbb{T} & \cong \mathbb{C}^n/\Gamma \end{tikzcd}$$

The manifold X is weakly subelliptic by Proposition 6.4.11, and hence an Oka manifold by Corollary 5.5.12. Proposition 5.5.2 implies that Y is Oka. $\square$

Complements of Subvarieties in Fiber Bundles

Assume that X and Z are complex spaces, $\pi\colon Z \to X$ is a holomorphic fiber bundle with an elliptic fiber Y. (Our main examples will be bundles with fiber $\mathbb{C}^n$ or $\mathbb{P}^n$.) Let A be a closed complex subvariety of Z whose fiber $A_x = A \cap Z_x$ has codimension at least two in $Z_x \cong Y$ for every $x \in X$.

Question 6.4.13. Under which condition on the subvariety A is the restricted submersion $\pi\colon Z \backslash A \to X$ elliptic, or subelliptic?

Proposition 6.4.1 gives sufficient conditions for the trivial submersion $Z = Y \backslash A \to$ *point* when Y equals $\mathbb{C}^n$ or $\mathbb{P}^n$. We now extend it to fiber bundles.

Proposition 6.4.14. *Assume that $\pi\colon Z \to X$ is a holomorphic fiber bundle with fiber $\mathbb{C}^n$ $(n \geq 2)$ and $A \subset Z$ is a subvariety that is* locally uniformly tame subvariety *in the following sense: For every point $x_0 \in X$ there are a neighborhood $U \subset X$, a fiber bundle isomorphism $\Phi\colon Z|_U \to U \times \mathbb{C}^n$ and a number $C > 0$ such that for every $x \in U$ we have*

$$\Phi_x(A_x) \subset \{(z', z_n) \in \mathbb{C}^n \colon |z_n| \leq C(1 + |z'|)\}. \tag{6.8}$$

If every fiber $A_x = A \cap \pi^{-1}(x)$ has codimension at least two in Z_x then the restricted submersion $\pi\colon Z \backslash A \to X$ is elliptic. Therefore sections $X \to Z$ whose images avoid A satisfy all forms of the Oka principle.

Proof. Let $A' = \Phi(A \cap Z|_U) \subset U \times \mathbb{C}^n$. The condition (6.8) means that

$$\omega(A'_x) \subset \Gamma = \{[v', v_n] \in \mathbb{P}^{n-1} \colon |v_n| \leq C|v'|\}, \quad x \in U.$$

(Here $\omega(A'_x)$ is the limits set at infinity, defined by (4.47) on p. 139.) Since Γ is a proper closed subset of $\mathbb{P}^{n-1}$, the proof of Proposition 5.5.14 on p. 205 shows that there exist complete holomorphic vector fields $V_1, \ldots, V_m$ on $U \times \mathbb{C}^n$ that

are tangential to the fibers $\{x\} \times \mathbb{C}^n$, that vanish on A', and that span the tangent space to the fiber at every point in $(U \times \mathbb{C}^n)\backslash A'$. The composition of the flows of these vector fields (see 5.10 on p. 205) is a dominating fiber-spray on $(U \times \mathbb{C}^n)\backslash A'$. This proves that the submersion $Z\backslash A \to X$ is elliptic. $\square$

To a holomorphic vector bundle $\pi\colon E \to X$ of rank n we associate a holomorphic fiber bundle $\widehat{\pi}\colon \widehat{E} \to X$ whose fiber $\widehat{E}_x \cong \mathbb{P}^n$ is obtained by compactifying $E_x \cong \mathbb{C}^n$ with the hyperplane at infinity $\Lambda_x \cong \mathbb{P}^{n-1}$.

Proposition 6.4.15. *Let $\pi\colon E \to X$ be a holomorphic vector bundle of rank n over a Stein space X, and let $\widehat{\pi}\colon \widehat{E} \to X$ be the associated bundle with fiber $\mathbb{P}^n$. Assume that A is a closed analytic subset of E such that*

(a) *the fiber A_x has codimension ≥ 2 in $E_x \cong \mathbb{C}^n$ for every $x \in X$, and*

(b) *the closure of A in $\widehat{E}$ does not contain any hyperplane at infinity Λ_x.*

Then the restricted submersion $\pi\colon E\backslash A \to X$ is elliptic.

Proof. Fix a point $p \in X$ and choose a local trivialization for $\widehat{E}|_U \cong U \times \mathbb{P}^n$ in a neighborhood $U \subset X$ of p such that $E|_U = U \times \mathbb{C}^n$. By the hypothesis the set $\Lambda_p\backslash\bar{A}$ is nonempty. Since the set $\bar{A} \cap (\widehat{E}\backslash E)$ is closed, after shrinking U around p there is a closed set with nonempty interior $\sigma \subset \mathbb{P}^{n-1}$ such that $\bar{A} \cap (U \times \sigma) = \emptyset$. By a linear change coordinates we may assume that the point $[0\colon\cdots\colon 0\colon 1] \in \mathbb{P}^{n-1}$ belongs to the interior of σ. Let $\Gamma \subset \mathbb{C}^n$ be the open complex cone determined by the set $\mathbb{P}^{n-1}\backslash\sigma$. It is easily seen that $A \subset R\mathbb{B} \cup \Gamma$ for a large $R > 0$. This means that (6.8) holds for a suitably large constant $C > 0$ and hence Proposition 6.4.14 applies. $\square$

In a similar way one can generalize the fact that the complement of a thin subvariety in a complex Grassmannian (Proposition 6.4.1) to fiber bundles. We leave the proof of the following proposition to the reader.

Proposition 6.4.16. *Let $\pi\colon Z \to X$ be a holomorphic fiber bundle whose fiber is a projective space or a complex Grassmannian. If A is a closed complex subvariety of Z whose fiber $A_x = A \cap Z_x$ has complex codimension at least two in Z_x for every $x \in X$ then $\pi\colon Z\backslash A \to X$ is a stratified subelliptic submersion.*

The following result, which generalizes Corollary 5.14.3, follows by combining Theorem 6.2.2 and Proposition 6.4.15. The topological condition needed to extend the section is seen as in Corollary 5.14.3.

Corollary 6.4.17. *Let $\pi\colon E \to X$ be a holomorphic vector bundle of rank n over a Stein space X, and let A be a closed complex subvariety of E. Assume that $X = X_0 \supset X_1 \supset \cdots \supset X_m = X'$ is a stratification such that for every stratum $S \subset X_k\backslash X_{k+1}$ the intersection $A \cap E|_S$ is locally uniformly tame in $E|_S$ (Proposition 6.4.14), and for every point $x \in S$ we have*

$$n - \dim A_x = \operatorname{codim} A_x \geq 2, \quad 2 \operatorname{codim} A_x > \dim S.$$

Given a compact $\mathcal{O}(X)$-convex subset K in X, a neighborhood U of $K \cup X'$ and a holomorphic section $f\colon U \to E|_U$ such that $f(K \cup X') \cap A = \emptyset$, there exists a holomorphic section $\widetilde{f}\colon X \to E$ that approximates f uniformly on K, agrees with f to a given finite order along X', and satisfies $\widetilde{f}(X) \cap A = \emptyset$.

The following example of J.-P. Rosay (personal communication) shows that the Oka principle fails in general for sections avoiding complex hypersurfaces.

Proposition 6.4.18. *For every sufficiently large $k > 0$ the graph of any holomorphic function $f\colon \mathbb{D} = \{|z| < 1\} \to \mathbb{C}$ intersects the analytic set*

$$A_k = \{w = 0\} \cup \{w = 1\} \cup \{w = kz\} \cup \{zw = 1\} \subset \mathbb{C}^2_{(z,w)}.$$

Proof. Suppose that $f\colon \mathbb{D} \to \mathbb{C}\backslash\{0,1\}$ is a holomorphic function omitting 0 and 1. Denote by dist the complete Kobayashi distance function on $\mathbb{C}\backslash\{0,1\}$. Let l denote the length of the circle $C = \{|z| = \frac{1}{2}\}$ with respect to the Kobayashi (=Poincaré) metric on the disc $\mathbb{D}$. Set

$$k_0 = \sup\left\{|\zeta| \in \mathbb{C}\backslash\{0,1\}\colon \inf_\theta \operatorname{dist}(\zeta, 2\mathrm{e}^{\mathrm{i}\theta}) \leq l\right\} < +\infty.$$

Case 1: There exists a $\theta \in \mathbb{R}$ such that $|f(\mathrm{e}^{\mathrm{i}\theta}/2)| \leq 2$. Observe that the Kobayashi length of $f(C) \subset \mathbb{C}\backslash\{0,1\}$ is at most l. Then for any $\gamma \in \mathbb{R}$ we have $|f(\mathrm{e}^{\mathrm{i}\gamma}/2)| \leq k_0$ by the choice of k_0. Rouché's theorem shows that for every $k > 2k_0$ the equation $kz - f(z) = 0$ has a solution z_0 with $|z_0| < \frac{1}{2}$, and at this point the graph of f intersects the line $\{w = kz\} \subset A_k$.

Case 2: For every $\theta \in \mathbb{R}$ we have $|f(\mathrm{e}^{\mathrm{i}\theta}/2)| > 2$. Since f has values in $\mathbb{C}\backslash\{0,1\}$, so does $g = 1/f$, and the assumption on f gives $|g(\mathrm{e}^{\mathrm{i}\theta}/2)| < \frac{1}{2}$ for every θ. Rouché's theorem implies that $z - g(z) = z - \frac{1}{f(z)}$ has a zero on $\{|z| < \frac{1}{2}\}$. At this point the graph of f intersects the hyperbola $\{zw = 1\} \subset A_k$. $\square$

Open Problems

Problem 6.4.19. Is there a subelliptic manifold that is not elliptic? In particular, is there an algebraic subvariety $A \subset \mathbb{P}^n$ of codimension ≥ 2 such that $\mathbb{P}^n\backslash A$ fails to be elliptic?

Recall that a map $X \to Y$ is a Serre fibration if it enjoys the homotopy lifting property for cells (see [512, p. 8]). If X and Y are manifolds then a Serre fibration $X \to Y$ is also a Hurewicz fibration.

Problem 6.4.20. Let X and Y be complex manifolds and $h\colon X \to Y$ a surjective elliptic (resp. subelliptic) Serre fibration. Assuming that Y is elliptic (resp. subelliptic), does it follows that X is also elliptic (resp. subelliptic)?

Problem 6.4.21. Let $h\colon X \to Y$ be a holomorphic covering map. If X is elliptic (resp. subelliptic), does it follow that Y is elliptic (resp. subelliptic)?

Problem 6.4.22. Is every Oka manifold elliptic or subelliptic?

Problem 6.4.23. Kaliman and Kutzschebauch gave many examples of contractible algebraic hypersurfaces satisfying the algebraic density property [288]. Each such manifold is algebraically subelliptic, and hence Oka by Theorem 5.5.18. It would be interesting to know whether all these examples are biholomorphic to a Euclidean space.

6.5 Lifting Homotopies to Spray Bundles

The following result on lifting of a homotopy to a spray bundle will be used in the proof of the Runge approximation theorems in §6.6.

Proposition 6.5.1. *Assume that X is a Stein space, $h\colon Z \to X$ is a holomorphic submersion, and (E, π, s) is a dominating composed h-spray on Z (Def. 6.1.1). Given a homotopy of holomorphic sections $f_t\colon X \to Z$ $(t \in [0,1])$ and an open relatively compact subset $V \Subset X$, there are an integer $k > 0$ and a homotopy of holomorphic sections ξ_t $(t \in [0,1])$ of the iterated spray bundle $E^{(k)}$ over the set $f_0(V) \subset Z$ such that*

$$\xi_0(z) = z, \quad s^{(k)}(\xi_t(z)) = f_t(h(z)), \qquad z \in f_0(V),\ t \in [0,1].$$

More generally, given a compact Hausdorff space P and a family of holomorphic sections $f_{p,t}\colon X \to Z$ depending continuously on $(p,t) \in P \times [0,1]$, there are an integer $k \in \mathbb{N}$ and a continuous family of holomorphic sections $\xi_{p,t}$ of $E^{(k)}|_{f_{p,0}(V)}$ satisfying the following conditions:

$$\xi_{p,0}(z) = z, \quad s^{(k)}(\xi_{p,t}(z)) = f_{p,t}(h(z)), \quad z \in f_{p,0}(V),\ p \in P,\ t \in [0,1].$$

On a sufficiently short subinterval $[0, t_1] \subset I$ this holds for $k = 1$.

Here we assume the existence of a dominating composed h-spray on all of Z. In a subelliptic submersion $Z \to X$ this result will be applied small open sets $U \subset X$ such that $h^{-1}(U)$ admits a dominating spray.

Proof. Consider first the basic case. We shall find numbers $0 = t_0 < t_1 < \cdots < t_k = 1$, and for each $j = 0, 1, \ldots, k-1$ a homotopy $\{\xi_t^j\colon t \in [t_j, t_{j+1}]\}$ of holomorphic sections of the restricted bundle $E|_{f_{t_j}(V)}$ satisfying

$$s \circ \xi_t^j(f_{t_j}(x)) = f_t(x), \qquad x \in V,\ t \in [t_j, t_{j+1}]. \tag{6.9}$$

In particular, $s \circ \xi^j_{t_{j+1}}(f_{t_j}(x)) = f_{t_{j+1}}(x)$ for $j = 0, 1, \ldots, k-1$. Comparing these compatibility conditions with those defining the iterated spray bundle $E^{(k)}$ we see that these k homotopies join into a homotopy of holomorphic sections $\xi_t \colon f_0(V) \to E^{(k)}|_{f_0(V)}$ ($t \in [0,1]$) satisfying Proposition 6.5.1. Explicitly, using the notation in Def. 6.3.5 (b), we define for $x \in V$ and $t \in [t_j, t_{j+1}]$

$$\xi_t(f_0(x)) = \left(\xi^0_{t_1}(f_0(x)), \xi^1_{t_2}(f_{t_1}(x)), \ldots, \xi^{j-1}_{t_j}(f_{t_{j-1}}(x)), \xi^j_t(f_{t_j}(x)), \overbrace{0, \ldots, 0}^{k-j-1}\right).$$

Now to the construction of the homotopies ξ^j_t. We begin by the following consideration. Assume that Ω is a Stein open subset of Z. By Lemma 6.3.7 the restricted bundle $E|_\Omega$ admits a holomorphic vector bundle structure. Since $\mathrm{V}\mathrm{d}s \colon \mathrm{VT}E \cong E \to \mathrm{VT}Z$ is surjective, we have a holomorphic direct sum splitting $E|_\Omega = \ker \mathrm{V}\mathrm{d}s|_\Omega \oplus E'$ such that $\mathrm{V}\mathrm{d}s \colon E' \xrightarrow{\cong} \mathrm{VT}Z|_\Omega$ is an isomorphism. Given an open subset $U \subset X$ and a holomorphic section $f \colon U \to Z|_U$ such that $\overline{f(U)} \subset \Omega$, we get an isomorphism $\mathrm{V}\mathrm{d}s \colon E'|_{f(U)} \to \mathrm{VT}Z|_{f(U)}$. By the inverse function theorem s maps an open neighborhood of the zero section in $E'|_{f(U)}$ biholomorphically onto an open neighborhood of $f(U)$ in Z. Assuming that $V \Subset U$ and that $g \colon V \to Z$ is a holomorphic section which is sufficiently close to $f|_V$, there is a unique holomorphic lifting $\xi \colon f(V) \to E'|_{f(V)}$ satisfying $s \circ \xi(f(x)) = g(x)$ for $x \in V$ (Fig. 6.2). In particular, given a homotopy of holomorphic sections $f_t \colon X \to Z$ and an open set $U \Subset X$ such that $f_{t_0}(\overline{U}) \subset \Omega$ for some $t_0 \in [0,1]$, we can lift $f_t|_U$ for t near t_0 to a holomorphic homotopy $\xi_t \colon U \to E|_{f_{t_0}(U)}$.

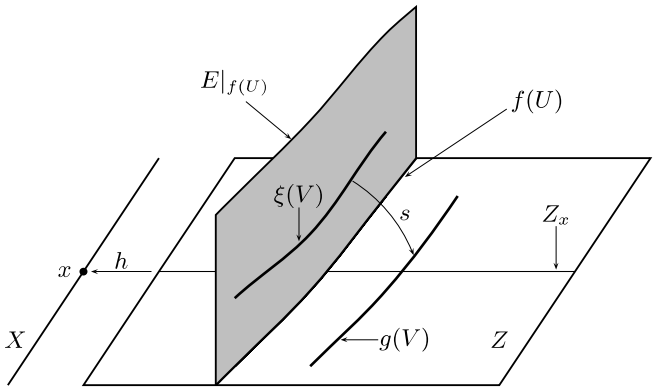

Fig. 6.2. Lifting a section g to the spray bundle $E|_{f(U)}$. (Modified from [182, p. 148, Fig. 1])

Assume now that $f_t \colon X \to Z$ ($t \in [0,1]$) is a homotopy of holomorphic sections and $V \Subset X$. Choose a compact $\mathcal{O}(X)$-convex set $K \subset X$ containing V in its interior. Then $f_t(K)$ is $\mathcal{O}(f_t(X))$-convex. Since $f_t(X)$ is a closed Stein subspace of Z, $f_t(K)$ admits an open Stein neighborhood in Z by Theorem

3.2.1. By compactness of the set $\{f_t(x): x \in K,\ t \in [0,1]\}$ there are numbers $0 < c_1 < c_2 < \cdots < c_m = 1$ and Stein open sets $\Omega_1, \ldots, \Omega_m \subset Z$ such that $f_t(K) \subset \Omega_j$ for all $t \in [c_{j-1}, c_j]$ and $j = 1, \ldots, m$. The above lifting argument now shows that by subdividing each segment $[c_{j-1}, c_j]$ into smaller segments we obtain a subdivision $0 = t_0 < t_1 < \cdots < t_k = 1$ and homotopies ξ_t^j satisfying (6.9). This proves the basic case.

The parametric case is not a straightforward extension since several steps in the proof depend on the choice of a Stein domain $\Omega \subset Z$ containing a part of our homotopy. We need a couple of lemmas.

Lemma 6.5.2. *Let $h: Z \to X$ and (E, π, s) be as in Proposition 6.5.1. Assume that P is a compact Hausdorff space and $f_p: X \to Z$ is a family of holomorphic sections depending continuously on $p \in P$. For any open relatively compact subset $U \Subset X$ there exist a family $G_p: \mathrm{VT}Z|_{f_p(U)} \to \mathrm{VT}E|_{f_p(U)}$ of holomorphic splittings of $\mathrm{V}ds: \mathrm{VT}E|_Z \to \mathrm{VT}Z$, and holomorphic direct sums*

$$\mathrm{VT}E|_{f_p(U)} = \ker \mathrm{V}ds|_{f_p(U)} \oplus E'_p, \tag{6.10}$$

both depending continuously on the parameter $p \in P$.

Proof. For each fixed $p \in P$, $f_p(X)$ is a closed Stein subspace of Z, and it is therefore contained in an open Stein domain $D_p \subset Z$ (Theorem 3.2.1). By compactness of P and $\overline{U}$ there exist an open cover $P_1, \ldots, P_{j_0}$ of P and a family of open Stein subsets $D_1, \ldots, D_{j_0}$ of Z such that $f_{p,0}(U) \subset D_j$ for all $p \in P_j$. Choose a continuous partition of unity $\{\chi_j\}$ on P subordinate to the cover $\{P_j\}$. Let $H_j: \mathrm{VT}Z|_{D_j} \to \mathrm{VT}E|_{D_j}$ be a holomorphic splitting of $\mathrm{V}ds$ over D_j (such H_j exists since D_j is Stein). Then the family

$$G_p = \sum_{j=1}^{j_0} \chi_j(p) H_j\colon\ \mathrm{VT}Z|_{f_p(U)} \to \mathrm{VT}E|_{f_p(U)}, \qquad p \in P$$

satisfies Lemma 6.5.2. G_p is well defined since $\chi_j(p) = 0$ if $f_{p,0}(U) \not\subset D_j$, and it is a splitting of $\mathrm{V}ds$ since it is a convex linear combination of splittings. Taking $E'_p = G_p(\mathrm{VT}Z|_{f_p(U)})$ we get (6.10). $\square$

In the following lemma we keep the assumptions from Lemma 6.5.2, identifying Z with the zero section of a composed spray bundle $\pi: E \to Z$. The image $f(X)$ of any section $f: X \to Z$ is correspondingly identified with the zero section of the restricted bundle $E|_{f(X)}$.

Lemma 6.5.3. *Let $E'_p \subset \mathrm{VT}E|_{f_p(X)}$ be a family of holomorphic vector subbundles of $\mathrm{VT}E$ depending continuously on $p \in P$. For each compact set K in X there exist an open set $\Omega \subset E$ containing $\cup_{p \in P} f_p(K)$ and a family of (not necessarily closed) complex subspaces Σ_p of $\Omega \cap E|_{f_p(X)}$, depending continuously on $p \in P$ and with regular fibers, such that $\Omega \cap f_p(X) \subset \Sigma_p$ and the vertical tangent bundle to Σ_p along $\Omega \cap f_p(X)$ equals E'_p for each $p \in P$.*

Proof. Since $f_p(X)$ admits an open Stein neighborhood in Z for each $p \in P$, we can find a cover $P_1, \ldots, P_m$ of P by compact sets and open Stein sets $D_1, \ldots, D_m \subset Z$ such that $f_p(K) \subset D_j$ for every $p \in P_j$ and $j = 1, \ldots, m$. By continuity there is an open set $U \subset X$ containing K and compact sets $P'_j \subset P$ such that $P_j \subset \mathrm{Int} P'_j$ and $f_p(U) \subset D_j$ for every $p \in P'_j$. By Lemma 6.3.7 there is for each $j = 1, \ldots, m$ a biholomorphic map $\Theta_j \colon E|_{D_j} \to \mathrm{VT}E|_{D_j}$ that is tangent to the identity map along the zero section of the two bundles. Replace E'_p by its restriction to $f_p(U)$ and consider it as a complex vector subbundle of $\mathrm{VT}E|_{f_p(U)}$. Then

$$\Sigma_{p,j} := \Theta_j^{-1}(E'_p) \subset E|_{f_p(U)}, \quad p \in P'_j$$

is a family of complex subspaces satisfying Lemma 6.5.3 for the values $p \in P'_j$. In particular, the vertical tangent bundle of $\Sigma_{p,j}$ along the zero section equals E'_p. Set $\Phi_{j,p} = \Theta_j^{-1}|_{E'_p} \colon E'_p \hookrightarrow E|_{f_p(U)}$.

The families of subspaces $\Sigma_{j,p}$ do not necessarily agree on the intersections $P_j \cap P_k$. We combine them by the *method of successive patching* which was first applied on p. 71; here are the details. Choose a compact set $\widetilde{P}_1 \subset P$ such that $P_1 \subset \mathrm{Int}\widetilde{P}_1$ and $\widetilde{P}_1 \subset \mathrm{Int} P'_1$. Also choose a continuous function $\chi \colon P \to [0,1]$ such that $\chi = 1$ on $\widetilde{P}_1$ and $\mathrm{supp}\, \chi \subset P'_1$. For $p \in \widetilde{P}_1 \cup P'_2$ we set

$$\Phi_p = \chi(p)\Phi_{1,p} + (1 - \chi(p))\Phi_{2,p}, \quad \Sigma_p = \Phi_p(E'_p) \subset E|_{f_p(U)}.$$

For $p \in \widetilde{P}_1$ we have $\chi(p) = 1$ and hence $\Phi_p = \Phi_{1,p}$. The convex linear combination therefore only occurs for $p \in P'_2 \backslash \widetilde{P}_1 \subset P'_2$ where we use the holomorphic vector bundle structure on $E|_{D_2}$ induced by Θ_2. Since the two maps are tangent at the zero section, the resulting family of subspaces Σ_p satisfies Lemma 6.5.3 for $p \in \widetilde{P}_1 \cup P'_2$, and the latter set contains $P_1 \cup P_2$ in its interior.

Next we patch $\{\Phi_{3,p} \colon p \in P'_3\}$ with the family $\{\Phi_p \colon p \in \widetilde{P}_1 \cup P'_2\}$, using the complex vector bundle structure on $E|_{D_3}$. This gives a new family satisfying the required properties for p in an open neighborhood of $P_1 \cup P_1 \cup P_3$. After m steps of this kind we obtain a solution for all $p \in P$. $\square$

We now conclude the proof Proposition 6.5.1. Choose an open set $U \Subset X$ with $\overline{V} \subset U$. Let $\Sigma_{p,t}$ for $(p,t) \in P \times [0,1]$ be a family of complex submanifolds of $E|_{f_{p,t}(U)}$ given by Lemma 6.5.3. By the construction, the vertical derivative $\mathrm{V}ds \colon \mathrm{VT}E|_Z \to \mathrm{VT}Z$ maps the vertical tangent bundle to $\Sigma_{p,t}$ (namely $E'_{p,t}$) isomorphically onto $\mathrm{VT}Z|_{f_{p,t}(U)}$. By the inverse function theorem, the restriction of s to $\Sigma_{p,t}$ maps a neighborhood of $f_{p,t}(U)$ in $\Sigma_{p,t}$ biholomorphically onto a neighborhood of $f_{p,t}(U)$ in Z. Denote its inverse by $u_{p,t}$. By the continuous dependence on $p \in P$ and the compactness of P, the domains of $u_{p,t}$ are uniform with respect to $p \in P$. Hence there are numbers $0 = t_0 < t_1 < \cdots < t_k = 1$ such that for all $t \in [t_j, t_{j+1}]$ and $p \in P$, $f_{p,t}(V)$ belongs to the domain of u_{p,t_j}. The sections

$$\xi_{p,t} = u_{p,t_j} \circ f_{p,t} \circ h \colon f_{p,t_j}(V) \to E$$

for $t \in [t_j, t_{j+1}]$ and $j = 0, 1, \ldots, k-1$ then satisfy (6.9) for every $p \in P$. $\square$

6.6 Runge's Theorem for Sections of Subelliptic Submersions

We prove that the existence of a finite dominating family of fiber-sprays on a holomorphic submersion $Z \to X$ over a Stein space implies a homotopy approximation property (HAP) for its sections; see Theorem 6.6.2. This generalizes the approximation results of Grauert and all [221, 222, 226, 407] for maps of Stein spaces to complex homogeneous manifolds. In §7.10 we prove analogous approximation results for algebraic maps from affine algebraic varieties to algebraically subelliptic manifolds.

We first give the basic case without parameters.

Theorem 6.6.1. ([237, 190, 170]) *Assume that X is a Stein space and $h \colon Z \to X$ is a holomorphic submersion with a finite dominating family of h-sprays (Def. 6.1.1). Let K be a compact $\mathcal{O}(X)$-convex set in X. Given an open set $U \Subset X$ containing K and a homotopy of holomorphic sections $f_t \colon U \to Z$ ($t \in [0,1]$) such that f_0 extends to a holomorphic section $f_0 \colon X \to Z$, there exists for every $\epsilon > 0$ a homotopy of holomorphic sections $\tilde{f}_t \colon X \to Z$ ($t \in [0,1]$) such that $\tilde{f}_0 = f_0$ and*

$$\sup \Big\{ \operatorname{dist}\big(\tilde{f}_t(x), f_t(x)\big) \colon x \in K,\ t \in [0,1] \Big\} < \epsilon.$$

Proof. By the assumption there exist h-sprays (E_j, π_j, s_j) ($j = 1, 2, \ldots, m$) on Z satisfying the domination condition (6.1) at each point $z \in Z$. Let (E, π, s), with $\pi \colon E = E_1 * E_2 \cdots * E_m \to Z$ and $s = s_1 * s_2 * \cdots * s_m \colon E \to Z$, denote the composed h-spray and $(E^{(k)}, \pi^{(k)}, s^{(k)})$ its k-th iterate (Def. 6.3.5). Choose an open set V in X with $K \subset V \Subset U$. By Proposition 6.5.1 the homotopy f_t ($t \in [0,1]$) lifts to a homotopy ξ_t of sections of the bundle $E^{(k)}|_{f_0(V)} \to f_0(V)$ for some $k \in \mathbb{N}$ such that ξ_0 is the zero section.

By Lemma 6.3.7 the restricted bundle $E^{(k)}|_{f_0(X)}$ admits the structure of a holomorphic vector bundle. By the Oka-Weil theorem we can approximate $\{\xi_t\}_{t \in [0,1]}$ uniformly on $f_0(K)$ by a homotopy of global holomorphic sections $\tilde{\xi}_t \colon f_0(X) \to E^{(k)}|_{f_0(X)}$ such that $\tilde{\xi}_0$ is the zero section. Then

$$\tilde{f}_t(x) := s^{(k)} \circ \tilde{\xi}_t \circ f_0(x) \in Z, \qquad x \in X,\ t \in [0,1]$$

is a homotopy of holomorphic sections $X \to Z$ satisfying Theorem 6.6.1. $\square$

The following is a parametric version of Theorem 6.6.1.

Theorem 6.6.2. ([237, 190, 170]) *Let $h\colon Z \to X$ and $K \subset U \subset X$ be as in Theorem 6.6.1, and let $P_0 \subset P$ be compact Hausdorff spaces. Set*

$$Q = P \times [0,1], \quad Q_0 = (P \times \{0\}) \cup (P_0 \times [0,1]).$$

Assume that $f\colon X \times Q \to Z$ is a continuous map such that

(i) *for every $q = (p,t) \in Q$, $f_q = f(\cdot\,,q)\colon X \to Z$ is a continuous section of $Z \to X$ that is holomorphic on U, and*

(ii) *for every $q \in Q_0$ the section f_q is holomorphic on X.*

Given $\epsilon > 0$, there is a continuous map $\tilde{f}\colon X \times Q \to Z$ such that for each $q \in Q$, $\tilde{f}_q = \tilde{f}(\cdot\,,q)\colon X \to Z$ is a holomorphic section, $\tilde{f}_q = f_q$ for $q \in Q_0$, and

$$\operatorname{dist}(\tilde{f}(x,q), f(x,q)) < \epsilon, \quad x \in K,\ q \in Q.$$

Furthermore, there is a homotopy $f^\tau\colon X \times Q \to Z$ ($\tau \in [0,1]$) that is fixed on Q_0 such that each f^τ satisfies the same conditions as $f^0 = f$ and $f^1 = \tilde{f}$.

It will be convenient to use the following terminology.

Definition 6.6.3. *Let P be a nonempty compact Hausdorff space and let P_0 be a closed subset of P (possibly empty). Let $h\colon Z \to X$ be a holomorphic map.*

(a) *A P-*section *of $h\colon Z \to X$ is a continuous map $f\colon X \times P \to Z$ such that $f_p = f(\cdot\,,p)\colon X \to Z$ is a section of h for each $p \in P$. Such f is* holomorphic *if f_p is holomorphic on X for each fixed $p \in P$. If K is a compact set in X and if X' is a closed complex subvariety of X, then f is* holomorphic on $K \cup X'$ *if there is an open set $U \subset X$ containing K such that the restrictions $f_p|_U$ and $f_p|_{X'}$ are holomorphic for every $p \in P$.*

(b) *A* homotopy of P-sections *is a $P \times [0,1]$-section, i.e., a continuous map $H\colon X \times P \times [0,1] \to Z$ such that $H_t = H(\cdot\,,\cdot\,,t)\colon X \times P \to Z$ is a P-section for each $t \in [0,1]$. Such homotopy H is holomorphic if $H_{p,t} = H(\cdot\,,p,t)\colon X \to Z$ is holomorphic for each fixed $(p,t) \in P \times [0,1]$.*

(c) *A* (P,P_0)-section *of h is a P-section $f\colon X \times P \to Z$ such that $f_p = f(\cdot\,,p)\colon X \to Z$ is holomorphic on X for each $p \in P_0$. A (P,P_0)-section is holomorphic on a subset $U \subset X$ if $f_p|_U$ is holomorphic for every $p \in P$.*

In the sense of Def. 6.6.3, a map f as in Theorem 6.6.2 is a (Q,Q_0)-section of $Z \to X$ which is holomorphic on U, $\tilde{f}$ is a Q-section which is holomorphic on X, and $\{f^\tau\}_{\tau\in[0,1]}$ is a homotopy of (Q,Q_0)-sections which are holomorphic on an open neighborhood of K.

Proof (of Theorem 6.6.2). Choose an open Stein set $V \Subset X$ that is Runge in X such that $K \subset U \Subset V$. It suffices to find a map $\tilde{f}\colon V \times Q \to Z$ satisfying the

conclusion of Theorem 6.6.2 on V and then apply induction over an increasing sequence of Runge Stein domains $V = V_0 \subset V_1 \subset \cdots \subset \cup_{j=0}^{\infty} V_j = X$.

By Proposition 5.6.3 we may assume that there is an open set P' in P, with $P_0 \subset P'$, such that $f_{p,t}$ is holomorphic on X for all (p,t) in the set

$$Q' = (P' \times [0,1]) \cup (P \times \{0\}).$$

By the hypothesis there exists a dominating composed h-spray (E, π, s) on Z. For each $k \in \mathbb{N}$ we denote by $(E^{(k)}, \pi^{(k)}, s^{(k)})$ the k-th iterated spray. After shrinking the open sets $P' \supset P_0$ (in P) and $U \supset K$ (in X), Proposition 6.5.1 gives an integer $k \in \mathbb{N}$ and a continuous family of sections $\xi_{p,t}$ of $E^{(k)}|_{f_{p,0}(V)}$ satisfying the following properties:

- $\xi_{p,0}$ is the zero section of $E^{(k)}|_{f_{p,0}(V)}$ for all $p \in P$,
- $\xi_{p,t}$ is holomorphic on $f_{p,0}(U)$ for all $(p,t) \in Q$,
- $\xi_{p,t}$ is holomorphic on $f_{p,0}(V)$ for all $(p,t) \in Q'$, and
- $s^{(k)} \circ \xi_{p,t}(f_{p,t}(x)) = f_{p,t}(x)$ for all $(x,p,t) \in (V \times Q') \cup (U \times Q)$.

As in the proof of Lemma 6.5.3 we find a cover of P by compact sets $P_1, \ldots, P_m$, and we find open Stein sets $D_2, \ldots, D_m \subset Z$ such that

- $P_0 \subset \mathrm{Int} P_1 \subset P'$,
- $P_0 \cap P_j = \emptyset$ for $j = 2, \ldots, m$, and
- $f_{p,0}(K) \subset D_j$ for all $p \in P_j$ and $j = 2, \ldots, m$.

Since $E^{(k)}|_{D_j}$ admits a holomorphic vector bundle structure for each $j = 2, \ldots, m$ (Lemma 6.3.7), we can apply the classical Oka-Weil theorem to approximate $\xi_{p,t}$ for $(p,t) \in P_j \times [0,1]$, uniformly on $f_{p,0}(K)$, by a continuous family of holomorphic sections $\xi^j_{p,t} \colon f_{p,0}(V) \to E^{(k)}|_{f_{p,0}(V)}$ such that $\xi^j_{p,0}$ is the zero section for each $p \in P_j$. For $j = 1$ we take $\xi^1_{p,t} = \xi_{p,t}$ since these sections are already holomorphic over $f_{p,0}(V)$.

Finally we glue the families $\{\xi^j_{p,t}\}$ $(j = 1, \ldots, m)$ into a single family $\tilde{\xi}_{p,t}$ by *successive patching* as on p. 262. The family of sections

$$\tilde{f}_{p,t} = s^{(k)} \circ \tilde{\xi}_{p,t} \circ f_{p,0} \colon V \to Z, \quad p \in P,\ t \in [0,1]$$

satisfies the conclusion of Theorem 6.6.2. The existence of a homotopy $\{f^\tau\}_{\tau \in [0,1]}$ from $f^0 = f$ to $f^1 = \tilde{f}$ follows from the proof since all steps were made by homotopies. $\square$

Remark 6.6.4. Theorems 6.6.1 and 6.6.2 can also be proved by the following argument that avoids composed sprays. By compactness of the set $\cup_{t \in [0,1]} f_t(K) \subset Z$ we can find an open set V in X, with $K \subset V \Subset U$, numbers $0 = t_0 < t_1 < \cdots < t_k = 1$, and Stein domains $\Omega_1, \ldots, \Omega_k \subset Z$ such that $f_t(V) \subset \Omega_j$ for every $t \in [t_{j-1}, t_j]$ and $j = 1, \ldots, k$. The restriction $E|_{\Omega_j}$ of

the composed spray bundle E to each of the Stein domains Ω_j admits a holomorphic vector bundle structure according to Lemma 6.3.7. The domination property of the composed spray $s\colon E \to Z$ implies that, possibly after refining our subdivision of the parameter interval $[0,1]$, for every $j = 0, 1, \ldots, k-1$ the homotopy f_t ($t \in [t_j, t_{j+1}]$) lifts to a homotopy of holomorphic sections ξ_t of $E|_{f_{t_j}(V)}$ satisfying

$$s \circ \xi_t \circ f_{t_j}(x) = f_t(x), \quad x \in V,\ t \in [t_j, t_{j+1}].$$

By the Oka-Weil theorem we can approximate $\{\xi_t\}_{t\in[0,t_1]}$ uniformly on a neighborhood of $f_0(K)$ in $f_0(X)$ by a homotopy of holomorphic sections $\tilde{\xi}_t\colon f_0(X) \to E|_{f_0(X)}$ ($t \in [0,t_1]$) such that $\tilde{\xi}_0$ is the zero section. The holomorphic sections $\tilde{f}_t = s \circ \tilde{\xi}_t \circ f_0$ ($t \in [0,t_1]$) are then close to f_t near K. In particular, we may assume that $\tilde{f}_{t_1}(K)$ is contained in a Stein tubular neighborhood of $f_{t_1}(V)$ in Z, and hence we can connect these two sections by a homotopy of nearby holomorphic sections over a neighborhood of K. We thus get a homotopy of sections $\tilde{f}_t\colon X \to Z$ ($t \in [0,1]$), close to the original one in a neighborhood of K, such that $\tilde{f}_0 = f_0$, $\tilde{f}_t$ is holomorphic on X for $t \in [0,t_1]$, and it is holomorphic in a neighborhood of K for $t \in (t_1, 1]$.

We now use $\tilde{f}_{t_1}$ as the new base section and repeat the same procedure for the next segment $t \in [t_1, t_2]$. Assuming that $\tilde{f}_{t_1}$ is sufficiently close to f_{t_1} near K, the choice of t_2 implies that we can lift all section $\tilde{f}_t$ for $t \in [t_1, t_2]$ to sections $\tilde{\xi}_t$ of $E|_{\tilde{f}_{t_1}(X)}$. Repeating the above procedure we obtain a new homotopy of sections which are holomorphic on X for $t \in [0, t_2]$, and on a neighborhood of K for $t \in (t_2, 1]$. After k steps the proof is completed.

The same proof works in the parametric case (Theorem 6.6.2), with P a compact set in a Euclidean space $\mathbb{R}^m$, by using Stein neighborhoods constructed in §3.6. (See the proof of Theorem 5.13.3.) □

Remark 6.6.5. The above proof only uses the domination property of sprays on the range of the given homotopy, which is a compact subset of Z. This shows that Theorems 6.6.1 and 6.6.2 hold under the following weaker hypothesis: *For every compact set L in Z there exists a composed fiber-spray (E, π, s) on Z that is dominating over L.* This holds for the trivial submersion $Z = X \times Y \to X$ if the fiber Y is weakly subelliptic (Def. 5.5.11). □

We say that a holomorphic submersion $h\colon Z \to X$ satisfies the *Homotopy Approximation Property* (HAP) if the conclusion of Theorem 6.6.2 holds for all data $K \subset X$, $P_0 \subset P$, and $f\colon X \times Q \to Z$ as in that theorem. Condition HAP is a natural replacement for CAP in the case of holomorphic submersions. Our proof of the main results in §6.2 actually gives the following.

Theorem 6.6.6. *Assume that X is a Stein space and $h\colon Z \to X$ is a stratified holomorphic submersion such that the restriction $Z|_S \to S$ to every stratum S satisfies* HAP *over small open sets in S. Then sections $X \to Z$ satisfy the parametric Oka principle.*

6.7 Gluing Holomorphic Sections on $\mathcal{C}$-Pairs

Proposition 6.7.2 below is the main *Heftungslemma* used in this chapter. It plays the analogous role as Proposition 5.9.2 in Chapter 5.

Definition 6.7.1. *An ordered pair of compact sets (A, B) in a complex space X is a $\mathcal{C}$-*pair *if*

(i) *each of the sets A, B, $D = A \cup B$ is a Stein compact,*
(ii) $\overline{A \backslash B} \cap \overline{B \backslash A} = \emptyset$ *(the separation condition), and*
(iii) *the set $C = A \cap B$ is $\mathcal{O}(B)$-convex. (C may be empty.)*

As a consequence of (i) the set $C = A \cap B$ is also a Stein compact. Comparing with Def. 5.7.1 (p. 209) we see that (A, B) is a $\mathcal{C}$-pair if and only if it is a Cartan pair and, in addition, A and B are Stein compacts and (iii) holds.

Proposition 6.7.2. *Assume that $h\colon Z \to X$ is a holomorphic submersion onto a Stein space X, X' is a closed complex subvariety of X, and (A, B) is a $\mathcal{C}$-pair in X (Def. 6.7.1) such that $C = A \cap B \subset X \backslash X'$. Suppose that $\widetilde{A} \supset A$ and $\widetilde{B} \supset B$ are open set such that the restriction $Z|_{\widetilde{B}} \to \widetilde{B}$ admits a finite dominating family of sprays (Def. 6.1.1 (iii), p. 242). Let $P_0 \subset P$ be compact Hausdorff spaces. Assume that $a\colon \widetilde{A} \times P \to Z|_{\widetilde{A}}$ and $b\colon \widetilde{B} \times P \to Z|_{\widetilde{B}}$ are holomorphic P-sections (Def. 6.6.3, p. 264) whose restrictions to $\widetilde{C} = \widetilde{A} \cap \widetilde{B}$ are homotopic by a homotopy of P-sections. Then there exist open sets A', B' in X, with $A \subset A' \subset \widetilde{A}$ and $B \subset B' \subset \widetilde{B}$, and for every $\epsilon > 0$ there exist homotopies of holomorphic P-sections $a^t\colon A' \times P \to Z$, $b^t\colon B' \times P \to Z$ $(t \in [0, 1])$, satisfying the following properties:*

(i) $a^0 = a|_{A' \times P}$, $b^0 = b|_{A' \times P}$,
(ii) $a^1 = b^1$ *on* $(A' \cap B') \times P$,
(iii) $\operatorname{dist}(a^t_p(x), a_p(x)) < \epsilon$ *for all* $x \in C' = A' \cap B'$, $p \in P$ *and* $t \in [0, 1]$,
(iv) *The homotopy a^t is fixed to a given finite order over $A' \cap X'$, and*
(v) *If the homotopy between a and b is fixed for all $p \in P_0$, then a^t_p and b^t_p can be chosen independent of $t \in [0, 1]$ for all $p \in P_0$.*

Proof. The homotopies a^t, b^t are found in two steps. In the first step we deform b to a new P-section $\tilde{b}$ which is close to a on a neighborhood of C; this is done by applying Theorem 6.6.2 (p. 264) to the given homotopy between a and b, with b serving as the initial section. In the second step we glue a and $\tilde{b}$ into a holomorphic P-section over a neighborhood of $D = A \cup B$. This is done by 'thickening' both P-sections and applying the gluing techniques from §5.8 – §5.9. All moves are made by homotopies of P-sections.

We may assume that $\widetilde{A}$, $\widetilde{B}$ and $\widetilde{C} = \widetilde{A} \cap \widetilde{B}$ are relatively compact Stein domains and $\widetilde{C} \cap X' = \emptyset$. We first consider the nonparametric case; thus

we are given holomorphic sections $a\colon \widetilde{A} \to Z$, $b\colon \widetilde{B} \to Z$, and a homotopy of holomorphic sections $b_t\colon \widetilde{C} \to Z$ ($t \in [0,1]$) such that $b_0 = a|_{\widetilde{C}}$ and $b_1 = b|_{\widetilde{C}}$.

We first thicken all the sections b_t in the given homotopy by using local flows of vector fields. As in Lemma 5.10.4 (p. 220) we find an open Stein neighborhood $\Omega \subset Z$ of $a(\widetilde{A})$ and finitely many holomorphic vector fields $v_1, \ldots, v_N$ on Ω, tangent to the fibers of h, which span $\mathrm{VT}Z$ at every point of $a(\widetilde{C})$ and vanish to a chosen order $m \in \mathbb{N}$ on the subvariety $h^{-1}(X') \cap \Omega$. Let $\theta_{j,y}$ denote the flow of v_j for time $y \in \mathbb{C}$, with $\theta_{j,0}$ being the identity map on Z. We extend our submersion $h\colon Z \to X$ by adding the variables $y = (y_1, \ldots, y_N) \in \mathbb{C}^N$; that is, we take $\widetilde{X} = X \times \mathbb{C}^N$, $\widetilde{Z} = Z \times \mathbb{C}^N$, and let $\tilde{h}\colon \widetilde{Z} \to \widetilde{X}$ be defined by $\tilde{h}(z,y) = (h(z), y)$ for $(z,y) \in \widetilde{Z}$. To simplify the notation we identify a point $z \in Z$ over $x = h(z) \in X$ with the point $(z,y) \in \widetilde{Z}$ over $(x,y) \in \widetilde{X}$ when there is no danger of confusion. In particular, a holomorphic section $a(x,y)$ of $\widetilde{Z} \to \widetilde{X}$ may be considered as a family of sections $a(\cdot\,, y)$ of $Z \to X$ depending holomorphically on the parameter y. A fiber-dominating family of sprays on $Z|_U$ trivially extends to a family with the same property on $\widetilde{Z}|_{\widetilde{U}}$, where $\widetilde{U} = U \times \mathbb{C}^N$.

By shrinking the set $\widetilde{A} \supset A$ we may assume that the closure of $a(\widetilde{A})$ is contained in Ω. Choose a number $0 < t_1 \le 1$ such that $b_t(\widetilde{C}) \subset \Omega$ for $t \in [0, t_1]$. Then choose a continuous function $\chi\colon [0,1] \to [0,1]$ which equals one near $t = 0$ and has support contained in $[0, t_1)$. Consider the following sections of $\widetilde{Z} \to \widetilde{X}$ for $y = (y_1, \ldots, y_N)$ in a small ball $r\mathbb{B} \subset \mathbb{C}^N$:

$$a'(x,y) = \theta_{1,y_1} \circ \cdots \circ \theta_{N,y_N} \circ a(x), \qquad x \in \widetilde{A}, \tag{6.11}$$

$$b'_t(x,y) = \theta_{1,\chi(t)y_1} \circ \cdots \circ \theta_{N,\chi(t)y_N} \circ b_t(x), \quad x \in \widetilde{C}. \tag{6.12}$$

Using the above mentioned identification we have

$$a'(x,0) = a(x), \quad b'_t(x,0) = b_t(x), \quad a'(x,y) = b'_0(x,y) \quad \text{for } x \in \widetilde{C}.$$

If $t \in [t_1, 1]$ then $\chi(t) = 0$, hence $\theta_{j,\chi(t)y} = \theta_{j,0} = \mathrm{Id}$ and we get

$$b'_t(x,y) = b_t(x) \quad \text{for } x \in \widetilde{C},\ y \in \mathbb{C}^N,\ t \in [t_1, 1].$$

Now apply the h-Runge approximation theorem (Theorem 6.6.1 on p. 263) to the homotopy b'_t ($t \in [0,1]$), with $b'_1(x,y) = b(x)$ ($x \in \widetilde{B},\ y \in r\mathbb{B}$) serving as the initial section. The result is a homotopy $\tilde{b}'_t$ of holomorphic sections over $\widetilde{B} \times r\mathbb{B} \subset \widetilde{X}$ such that $\tilde{b}'_1 = b'_1 = b$, and $\tilde{b}'_t$ approximates b'_t as close as desired on $\widetilde{C} \times r\mathbb{B}$. (We must shrink the domains a little.) At $t = 0$, the section $b' := \tilde{b}'_0$ approximates a' as close as desired on $\widetilde{C} \times r\mathbb{B}$.

Choose an open Stein set C', with $C \subset C' \Subset \widetilde{C}$, and open sets $A' = U_A$ and $B' = U_B$ as in Proposition 5.8.4 (p. 214) such that

$$A \subset A' \Subset \widetilde{A}, \quad B \subset B' \Subset \widetilde{B}, \quad C' = A' \cap B' \Subset \widetilde{C}.$$

Observe that the family of sections $a'(\cdot\,, y)$ is a local dominating spray around the core section $a = a'(\cdot\,, 0)$ as in Lemma 5.10.4 (p. 220); the analogous observation holds for b'. If b' is close enough to a' over $\widetilde{C} \times r\mathbb{B}$ then we can glue them into a family of holomorphic sections over the union $D' = A' \cup B'$ by following the proof of Proposition 5.9.2 (p. 216). We first appeal to Lemma 5.9.3 to find a transition map between them, and then to Proposition 5.8.4 to split the transition map. Since all steps are made by homotopies, we also get homotopies a^t, b^t as in Proposition 6.7.2. Interpolation over the subvariety X' is built into the construction in view of Proposition 5.8.4.

All tools that we have used are available with a continuous dependence on a parameter $p \in P$. We cover the image of the initial P-section a by a finite family of open Stein sets $D_j \subset Z$ and choose a finite cover $\{P_j\}$ of P such that $a_p(\widetilde{A}) \subset D_j$ for all $p \in P_j$. For each j choose holomorphic vector fields $v_1^j, \ldots, v_{N_j}^j$ on D_j as above and multiply them by a continuous partition of unity on P subordinate to $\{P_j\}$. This allows us to define families of sections $a'(x, y, p)$ (resp. $b'_t(x, y, p)$) by (6.11) (resp. by (6.12)), depending continuously on the parameter $p \in P$ and such that a' is submersive in the y-variable for all $(x, p) \in \widetilde{C} \times P$. The subsequent steps (pushing them close together and gluing them) can also be made continuously with respect to the parameter $p \in P$ as is clear from the results used in the process. $\square$

6.8 Complexes of Holomorphic Sections

A *holomorphic complex* is a collection of holomorphic sections of a certain submersion $Z \to X$, defined on subdomains of X, and of (multi parameter) homotopies between these sections on the intersections of their domains (Def. 6.8.3 on p. 271). This is a convenient bookkeeping device in the proof of Theorem 6.2.2. In the following sections we describe operations on complexes which will enable us to assemble the collection of local sections into a global section, thereby proving Theorem 6.2.2.

We begin by describing the nerve complex and its geometric realization.

Definition 6.8.1. *Let $\mathcal{A} = \{A_0, A_1, A_2, \ldots\}$ be a locally finite family of nonempty subsets of a set X. The* nerve *of $\mathcal{A}$ is the combinatorial simplicial complex, $\mathcal{K}(\mathcal{A})$, consisting of all multiindexes $J = (j_0, j_1, \ldots, j_k) \in \mathbb{Z}_+^{k+1}$ ($k \in \mathbb{Z}_+$), with $0 \le j_0 < j_1 < \cdots < j_k$, for which*

$$A_J = A_{j_0} \cap A_{j_1} \cap \cdots \cap A_{j_k} \neq \emptyset.$$

Recall [271] that the *geometric realization*, $K(\mathcal{A})$, of a simplicial complex $\mathcal{K}(\mathcal{A})$ is a topological space obtained as follows. Each multiindex $J = (j_0, j_1, \ldots, j_k) \in \mathcal{K}(\mathcal{A})$ of length $k+1$ determines a closed k-dimensional face $|J| \subset K(\mathcal{A})$, homeomorphic to the standard k-simplex $\triangle^k \subset \mathbb{R}^k$ (the closed

convex hull of the set $\{0, e_1, e_2, \ldots, e_k\} \subset \mathbb{R}^k$ containing the origin and the standard basis vectors), and any k-dimensional face of $K(\mathcal{A})$ is of this form. The face $|J|$ is called the *body* (or *carrier*) of J, and J is the *vertex scheme* of $|J|$. The vertexes of $K(\mathcal{A})$ correspond to the individual sets $A_j \in \mathcal{A}$, i.e., to singletons $\{j\} \in \mathcal{K}(\mathcal{A})$. Given $I, J \in \mathcal{K}(\mathcal{A})$ we have $|I| \cap |J| = |I \cap J|$. For any two (bodies of) simplexes in $K(\mathcal{A})$, either one is a subset of the other, or else their intersection is a simplex of lower dimension, possibly empty.

For each $n \in \mathbb{Z}_+$ we denote by

$$\mathcal{K}^n(\mathcal{A}) = \mathcal{K}(A_0, A_1, \ldots, A_n) \subset \mathcal{K}(\mathcal{A})$$

the nerve of the finite subfamily $\mathcal{A}_n = \{A_0, \ldots, A_n\}$, and by $K^n(\mathcal{A})$ its body. Clearly $\mathcal{K}^n(\mathcal{A}) \subset \mathcal{K}^{n+1}(\mathcal{A})$ for each n, and $\mathcal{K}(\mathcal{A}) = \bigcup_{n=0}^{\infty} \mathcal{K}^n$. More generally, for any multiindex $J = (j_0, j_1, \ldots, j_k) \in \mathbb{Z}_+^{k+1}$ (not necessarily belonging to $\mathcal{K}(\mathcal{A})$) we denote by

$$\mathcal{K}_J(\mathcal{A}) = \mathcal{K}(A_{j_0}, A_{j_1}, \ldots, A_{j_k})$$

the nerve of the indicated family of sets, and by $K_J(\mathcal{A})$ its body. Note that $\mathcal{K}_J(\mathcal{A})$ is a finite subcomplex of $\mathcal{K}(\mathcal{A})$ whose body is a k-dimensional simplex if and only if $J \in \mathcal{K}(\mathcal{A})$; otherwise it is a union of simplexes of lower dimension. Occasionally we shall write simply $\mathcal{K}$ instead of $\mathcal{K}(\mathcal{A})$ when it is clear from the context which family $\mathcal{A}$ is meant.

Example 6.8.2. Fig. 6.3 shows the body $|K(A_0, \ldots, A_9)|$ in which the nonempty intersections of two sets are shown by firm lines, and the nonempty triple intersections are $A_{3,4,9}$ and $A_{6,7,8}$.

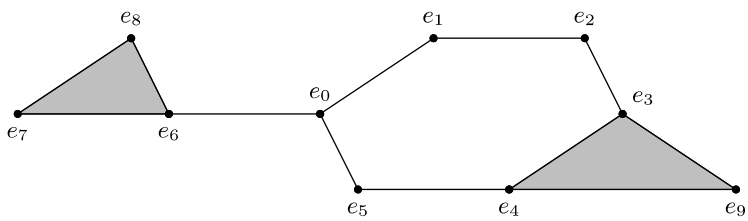

Fig. 6.3. The body $|K(A_0, \ldots, A_9)|$

Assume now that X is a locally compact Hausdorff space and $\mathcal{A} = \{A_0, A_1, A_2, \ldots\}$ is a (finite or countable) locally finite family of compact subsets of X. An open neighborhood of $\mathcal{A}$ is a family $\mathcal{U} = \{U_0, U_1, U_2, \ldots\}$ of open subsets of X such that $A_j \subset U_j$ for each index j. Such a neighborhood $\mathcal{U}$ is *faithful* if $\mathcal{K}(\mathcal{U}) = \mathcal{K}(\mathcal{A})$, that is, the two families have the same

nerve complex. It is easily seen that any locally finite family $\mathcal{A}$ of compact subsets admits an open faithful neighborhood. As before we write for each $J = (j_0, j_1, \dots, j_k)$

$$U^J = U_{j_0} \cup U_{j_1} \cup \dots \cup U_{j_k}, \quad U_J = U_{j_0} \cap U_{j_1} \cap \dots \cap U_{j_k}.$$

We now explain how the body of a simplicial complex is used to parametrize a collection of sections. Assume that Z and X are complex spaces and $h\colon Z \to X$ is a holomorphic submersion. Given an open set $U \subset X$, we denote by $\Gamma_{\mathcal{O}}(U, Z)$ (resp. $\Gamma(U, Z)$) the set of all holomorphic (resp. continuous) sections $f\colon U \to Z|_U$ of $h\colon Z \to X$ over U.

Definition 6.8.3. *Assume that $h\colon Z \to X$ is a holomorphic submersion and $\mathcal{A} = \{A_0, A_1, A_2, \dots\}$ is a locally finite family of compacts in X.*

1. *A* holomorphic $\mathcal{K}(\mathcal{A})$-complex *with values in Z is a continuous family of holomorphic sections*

$$f_* = \{f_J\colon |J| \to \Gamma_{\mathcal{O}}(U_J, Z), \quad J \in \mathcal{K}(\mathcal{A})\},$$

where $\mathcal{U} = \{U_0, U_1, U_2, \dots\}$ is a faithful neighborhood of $\mathcal{A}$, which satisfy the following compatibility conditions:

$$I, J \in \mathcal{K}(\mathcal{A}),\ I \subset J \Longrightarrow f_J(t) = f_I(t)|_{U_J},\ t \in |I|.$$

2. *A* continuous $\mathcal{K}(\mathcal{A})$-complex *with values in Z is a continuous family*

$$f_* = \{f_J\colon |J| \to \Gamma(U_J, Z), \quad J \in \mathcal{K}(\mathcal{A})\}$$

satisfying the same compatibility conditions as in (i).

3. *If f_* is a $\mathcal{K}(\mathcal{A})$-complex and $\mathcal{K}'$ is a subcomplex of $\mathcal{K}(\mathcal{A})$, we denote by $f_*|_{\mathcal{K}'}$ the restriction of f_* to $\mathcal{K}'$.*
4. *A $\mathcal{K}(\mathcal{A})$-complex f_*, defined on a faithful neighborhood $\mathcal{U} = \{U_j\}$ of $\mathcal{A}$, is* constant *if there is a section $g\colon \cup_j U_j \to Z$ such that $f_J(t)|_{U_J} = g|_{U_J}$ for each $J \in \mathcal{K}(\mathcal{A})$ and each $t \in |J|$.*

Thus a complex of sections of $Z \to X$ is a family of sections, depending continuously on the parameter $t \in K(\mathcal{A})$, such that the domain of the section corresponding to a point $t \in K(\mathcal{A})$ is the union of all sets U_J for which $t \in |J|$; equivalently, it equals U_J where $J \in \mathcal{K}(\mathcal{A})$ is the shortest multiindex for which $t \in |J|$. For each $J = (j_0, \dots, j_k) \in \mathcal{K}(\mathcal{A})$ we have a family of holomorphic sections $f_J(t)\colon U_J \to Z$, depending continuously on $t \in |J|$, which we consider as a homotopy of sections over the set $U_J = U_{j_0} \cap U_{j_1} \cap \dots \cap U_{j_k}$, with the parameter t belonging to the simplex $|J| \subset K(\mathcal{A})$. For each face I of J and for every $t \in |I| \subset |J|$ the section $f_J(t)$ agrees with $f_I(t)$, restricted from its original domain U_I to the subdomain U_J.

We consider $\mathcal{K}(\mathcal{A})$-complexes in the sense of their germs: two $\mathcal{K}(\mathcal{A})$-complexes f_* and g_* are considered *equivalent* if there is an open faithful neighborhood $\mathcal{U} = \{U_j\}$ of $\mathcal{A} = \{A_j\}$ such that for each $J \in \mathcal{K}(\mathcal{A})$ and $t \in |J|$, the sections $f_J(t)$ and $g_J(t)$ are defined and equal in U_J.

We also need the notion of a multiparameter homotopy of $\mathcal{K}(\mathcal{A})$-complexes. A suitable concept is the following.

Definition 6.8.4. *Let $h\colon Z \to X$ and $\mathcal{A}$ be as in Def. 6.8.3 and let $k \in \mathbb{Z}_+$.*

1. *A* holomorphic $(\mathcal{K}(\mathcal{A}), k)$-prism*, or a* k-prism over $\mathcal{K}(\mathcal{A})$*, with values in Z, is a family*

$$f_* = \{f_J\colon |J| \times [0,1]^k \to \Gamma_{\mathcal{O}}(U_J, Z), \quad J \in \mathcal{K}(\mathcal{A})\},$$

where $\mathcal{U}$ is a faithful open neighborhood of $\mathcal{A}$, such that for each fixed $y \in [0,1]^k$ the associated family

$$f_{*,y} = \{f_{J,y} = f_J(\cdot\,, y)\colon |J| \to \Gamma_{\mathcal{O}}(U_J, Z), \quad J \in \mathcal{K}(\mathcal{A})\}$$

is a holomorphic $\mathcal{K}(\mathcal{A})$-complex.

2. *A* continuous $(\mathcal{K}(\mathcal{A}), k)$-prism *with values in Z is a family*

$$f_* = \{f_J\colon |J| \times [0,1]^k \to \Gamma(U_J, Z), \quad J \in \mathcal{K}(\mathcal{A})\}$$

such that $f_{*,y}$ is a continuous $\mathcal{K}(\mathcal{A})$-complex for each $y \in [0,1]^k$.

3. *A prism f_* is* sectionally constant *if there is an open set $U \supset \cup_{j\geq 0} A_j$ such that the complex $f_{*,y}$ is represented by a section $f_y\colon U \to Z|_U$ for each fixed $y \in [0,1]^k$. If this holds for all y in a subset $Y \subset [0,1]^k$, we say that f_* is sectionally constant on Y.*

Thus a $(\mathcal{K}(\mathcal{A}), 0)$-prism is a $\mathcal{K}(\mathcal{A})$-complex, a $(\mathcal{K}(\mathcal{A}), 1)$-prism is the same thing as a homotopy of $\mathcal{K}(\mathcal{A})$-complexes, a 2-prism is a homotopy of 1-prisms, etc. A sectionally constant $(\mathcal{K}(\mathcal{A}), k)$-prism is a k-parameter homotopy of sections over an open neighborhood of the union $\cup_j A_j$ of all the sets in $\mathcal{A}$.

6.9 $\mathcal{C}$-Strings

In order to be able to glue pairs of sections (or their homotopies) in a holomorphic $\mathcal{K}(\mathcal{A})$-complex, the sets in the collection $\mathcal{A} = (A_0, A_1, \ldots)$ must satisfy certain analytic conditions, generalizing those of a $\mathcal{C}$-pair (Def. 6.7.1). The following notion was introduced by Gromov [237, 4.2.D'].

Definition 6.9.1. *An ordered sequence $(A_0, A_1, \ldots, A_n)$ of compact sets in a complex space X is a* $\mathcal{C}$-string of length $n+1$ *if*

(i) $(A_0 \cup \cdots \cup A_{n-1}, A_n)$ *is a* $\mathcal{C}$*-pair (Def. 6.7.1 on p. 267), and*
(ii) $(A_0, \ldots, A_{n-1})$ *and* $(A_0 \cap A_n, \ldots, A_{n-1} \cap A_n)$ *are* $\mathcal{C}$*-strings of length* n*.*

An ordered locally finite cover $\mathcal{A} = (A_0, A_1, \ldots)$ *of* X *by compact sets is a* $\mathcal{C}$-cover *if* $(A_0, A_1, \ldots, A_n)$ *is a* $\mathcal{C}$*-string for each* $n \in \mathbb{N}$*.*

The order of sets in a $\mathcal{C}$-string $(A_0, A_1, \ldots, A_n)$ is clearly important. Each of the sets A_k, and the union $A^k = \cup_{i=0}^k A_i$, is a Stein compact in X.

$\mathcal{C}$-strings enjoy the following hereditary property.

Proposition 6.9.2. *If* $(A_0, A_1, \ldots, A_n)$ *is a* $\mathcal{C}$*-string in* X *and if* B *is a Stein compact in* X*, then* $(A_0 \cap B, \ldots, A_n \cap B)$ *is also a* $\mathcal{C}$*-string in* X*.*

Proof. We proceed by induction on the length $n \in \mathbb{N}$. Note that, if A and B are Stein compacts then so is $C = A \cap B$. For $n = 1$ we have a $\mathcal{C}$-pair (A_0, A_1), and we claim that $(A_0 \cap B, A_1 \cap B)$ is also a $\mathcal{C}$-pair. Clearly it satisfies properties (i) and (ii) in Def. 6.7.1. Property (iii) follows from the following lemma.

Lemma 6.9.3. *Let* $D_0 \subset D_1$ *and* Ω *be open Stein domains in a complex space* X*. If* D_0 *is Runge in* D_1*, then* $D_0 \cap \Omega$ *is Runge in* $D_1 \cap \Omega$*.*

Proof. Choose a compact set $K \Subset D_0 \cap \Omega$. We denote by $\widehat{K}_D$ the holomorphically convex hull of K with respect to a domain $D \subset X$ containing K. Clearly $\widehat{K}_{D_1 \cap \Omega} \subset \widehat{K}_{D_1} \cap \widehat{K}_\Omega$. Since D_0 is Runge in D_1 and both domains are Stein, we have $\widehat{K}_{D_0} = \widehat{K}_{D_1}$ [267, Theorem 2.7.3]. Therefore

$$\widehat{K}_{D_1 \cap \Omega} \subset \widehat{K}_{D_1} \cap \widehat{K}_\Omega = \widehat{K}_{D_0} \cap \widehat{K}_\Omega \Subset D_0 \cap \Omega.$$

It follows that $D_0 \cap \Omega$ is Runge in $D_1 \cap \Omega$ [267, Theorem 2.7.3] . □

This settles the case $n = 1$. Assume inductively that the result holds for some n. Let $(A_0, A_1, \ldots, A_{n+1})$ be a $\mathcal{C}$-string of length $n + 2$. To see that $(A_0 \cap B, \ldots, A_{n+1} \cap B)$ is also a $\mathcal{C}$-string, we must verify the following:

- The pair of sets $(A_0 \cap B) \cup \cdots \cup (A_n \cap B) = (A_0 \cup \cdots \cup A_n) \cap B$ and $A_{n+1} \cap B$ is a $\mathcal{C}$-pair. Since $(A_0 \cup \cdots \cup A_n, A_{n+1})$ is a $\mathcal{C}$-pair, this follows from the case $n = 1$ proved above.
- $(A_0 \cap B, \ldots, A_n \cap B)$ and $(A_0 \cap A_{n+1} \cap B, \ldots, A_n \cap A_{n+1} \cap B)$ are $\mathcal{C}$-strings of length $n + 1$. This follows immediately from Definition 6.9.1 and from the inductive hypothesis. □

Corollary 6.9.4. *If* $\mathcal{A} = (A_0, A_1, A_2, \ldots)$ *is a sequence of compact sets in a complex space* X *such that for each* $n \in \mathbb{N}$ *the pair* $(A_0 \cup \cdots \cup A_{n-1}, A_n)$ *is a* $\mathcal{C}$*-pair, then* $(A_0, A_1, \ldots, A_n)$ *is a* $\mathcal{C}$*-string for each* $n \in \mathbb{N}$*.*

Proof. This follows from Proposition 6.9.2 by induction on n. □

Another hereditary property of $\mathcal{C}$-strings is the following; the straightforward proof is left to the reader.

Proposition 6.9.5. *If $(A_0, A_1, \ldots, A_n)$ is a $\mathcal{C}$-string in a complex space X then for each closed complex subspace X' of X, $(A_0 \cap X', \ldots, A_n \cap X')$ is a $\mathcal{C}$-string in X'.*

We now show that each Stein space admits arbitrarily fine $\mathcal{C}$-covers.

Proposition 6.9.6. *For each open cover $\mathcal{U} = \{U_j\}$ of a Stein space X there exists a $\mathcal{C}$-cover $\mathcal{A} = (A_0, A_1, \ldots)$ of X which is subordinate to $\mathcal{U}$, in the sense that each A_i is contained in some U_j. Moreover, if $K \subset X$ is a compact $\mathcal{O}(X)$-convex subset and $U_0 \subset X$ is an open set containing K, we can choose $\mathcal{A}$ such that $K \subset A_0 \subset U_0$ and $A_i \cap K = \emptyset$ for $i \geq 1$.*

Proof. We consider first the case when X is a Stein manifold. The conditions imply that there is a strongly plurisubharmonic Morse exhaustion function $\rho\colon X \to \mathbb{R}$ with nice critical points (Def. 3.9.2 on p. 89) such that $K \subset \{\rho < 0\} \Subset U_0$, and 0 is a regular value of ρ. Let $p_1, p_2, p_3, \ldots$ be the critical points of ρ in $\{\rho > 0\}$, ordered so that $0 < \rho(p_1) < \rho(p_2) < \cdots$. Choose a sequence of numbers $0 = c_0 < c_1 < c_2 < \cdots$, with $\lim_{j\to\infty} c_j = +\infty$, such that $c_{2j-1} < \rho(p_j) < c_{2j}$ for every $j = 1, 2, \ldots$. The numbers c_{2j-1} and c_{2j} can be chosen as close as desired to $\rho(p_j)$. If there are only finitely many critical points p_j, we choose the remainder of the sequence c_j arbitrarily.

Set $A_0 = \{\rho \leq 0\}$. Since the set $\{\rho \leq c_1\}$ is a noncritical strongly pseudoconvex extension of A_0, Lemma 5.10.3 provides a finite sequence of compact strongly pseudoconvex domains $A_0 \subset A'_1 \subset \cdots \subset A'_{m_1-1} = \{\rho \leq c_1\}$ such that for every $k = 0, 1, \ldots, m_1 - 2$ we have $A'_{k+1} = A'_k \cup A_k$, where A_k is a special bump on A'_k contained in one of the sets $U_j \in \mathcal{U}$. Hence (A'_k, A_k) is a $\mathcal{C}$-pair for each k, and $(A_0, \ldots, A_{m_1-1})$ is a $\mathcal{C}$-string by Corollary 6.9.4.

To pass the critical point p_1 we attach to the set $A'_{m_1-1} = \{\rho \leq c_1\}$ a strongly pseudoconvex handle (a 'bone') A_{m_1-1}, described in [259, §2], such that (A'_{m_1-1}, A_{m_1-1}) is a $\mathcal{C}$-pair, and the sublevel set $\{\rho \leq c_2\}$ is a noncritical strongly pseudoconvex extension of $A'_{m_1} = A'_{m_1-1} \cup A_{m_1-1}$ for a possibly different strongly plurisubharmonic function τ. (This means that the above sets are sublevel sets of τ, and τ has no critical points in between). If c_1 and c_2 were chosen sufficiently close to $\rho(p_1)$ then A_{m_1-1} can be chosen small enough to be contained in a set $U_j \in \mathcal{U}$. Using the same procedure as above with τ (instead of ρ) we continue attaching small bumps to reach the next level $\{\rho \leq c_2\}$; then we revert back to the original function ρ and continue attaching bumps to reach the level $\{\rho \leq c_3\}$. Next we attach a bone to pass the critical level of ρ at p_2, etc. Corollary 6.9.4 shows that the resulting string $\mathcal{A} = (A_0, A_1, A_2, \ldots)$, obtained in this way, is a $\mathcal{C}$-cover of X.

To do the same on a Stein space X we choose an injective proper holomorphic map $\Phi\colon X \hookrightarrow \mathbb{C}^N$ which is a topological embedding. Using the

above procedure we find a $\mathcal{C}$-cover $(\widetilde{A}_0, \widetilde{A}_1, \ldots)$ of $\mathbb{C}^N$ such that the $\mathcal{C}$-string $(A_0, A_1, A_2, \ldots)$, where $A_k = X \cap \widetilde{A}_k$, is a $\mathcal{C}$-cover of X with the stated properties (Corollary 6.9.5).

When proving Theorem 6.2.2 for stratified subelliptic submersions we use $\mathcal{C}$-strings $(A_0, A_1, \ldots)$ such that the sets $A_1, A_2, \ldots$ are contained in a certain regular stratum in a stratification of X. $\square$

6.10 Construction of the Initial Holomorphic Complex

The following proposition provides a homotopy of complexes $a_{*,s}$ (a 1-prism) connecting the initial continuous section $a\colon X \to Z$ (which is thought of as a constant continuous complex $a_{*,0}$) to a certain holomorphic complex $a_{*,1}$.

Proposition 6.10.1. (The initial holomorphic complex.) *Let X be a Stein space, $K \subset X$ a compact $\mathcal{O}(X)$-convex subset, and $h\colon Z \to X$ a holomorphic submersion. Given a continuous section $a\colon X \to Z$ which is holomorphic in an open set $U_0 \supset K$, there exist a $\mathcal{C}$-cover $\mathcal{A} = (A_0, A_1, \ldots)$ of X and a continuous $(\mathcal{K}(\mathcal{A}), 1)$-prism $a_* = \{a_{*,s}\colon s \in [0,1]\}$ with values in Z (Def. 6.8.4), such that*

(i) *$K \subset A_0 \subset U_0$, $K \cap A_i = \emptyset$ for $i \geq 1$, and $a_{(0),s} = a|_{U_0}$ for $s \in [0,1]$,*
(ii) *$a_{*,0}$ is the constant $\mathcal{K}(\mathcal{A})$-complex given by the section $a\colon X \to Z$, and*
(iii) *$a_{*,1}$ is a holomorphic $\mathcal{K}(\mathcal{A})$-complex.*

If the submersion $Z \to X$ is subelliptic over $X \backslash K$ (Def. 6.1.2 on p. 242) then $\mathcal{A}$ can be chosen such that, in addition to the above, there exists for each $j \geq 1$ an open set $U_j \supset A_j$ such that the restricted submersion $Z|_{U_j} \to U_j$ admits a finite dominating family of fiber-sprays.

Proof. For simplicity of exposition we consider the case when X is a Stein manifold; the extension to a Stein space is immediate, either directly or by using local holomorphic embeddings of X into $\mathbb{C}^n$.

Let $N = \dim Z = n + m$, where $n = \dim X$ and m is the dimension of the fibers $h^{-1}(x)$, $x \in X$. Denote by P^N the unit open polydisc in $\mathbb{C}^N$ with the complex coordinates $\zeta = (\zeta', \zeta'')$, where $\zeta' \in \mathbb{C}^n$ and $\zeta'' \in \mathbb{C}^m$. Let $\pi\colon P^N \to P^n$ be the projection $\pi(\zeta', \zeta'') = \zeta'$.

Since $h\colon Z \to X$ is a submersion, there exist for each point $z_0 \in Z$ open neighborhoods $V \subset Z$ of z_0, $U \subset X$ of $x_0 = h(z_0)$, and biholomorphic maps $\Phi\colon V \to P^N$, $\phi\colon U \to P^n$, such that $\pi \circ \Phi = \phi \circ h$ on V and $\Phi(z_0) = 0$. Such Φ induces a linear structure on the fibers of $h|_V$ which lets us add sections of $h\colon V \to U$ and take their convex linear combinations.

If z_0 belongs to the graph of a, we can choose the above neighborhoods and maps such that $a(U) \subset V$. In this case Φ maps $a(U)$ onto the graph of

a section $\tilde{a}(\zeta') = (\zeta', a''(\zeta'))$ $(\zeta' \in P^n)$ of the projection $\pi\colon P^N \to P^n$. The family $a_s\colon U \to V$, given by

$$a_s(x) = \Phi^{-1}\big(\phi(x), (1-s)a''(\phi(x))\big), \qquad x \in U,\ s \in [0,1],$$

is a homotopy of continuous sections of h over U such that $a_s(U) \subset V$ for each $s \in [0,1]$, $a_0 = a|_U$, and the section a_1 is holomorphic. By shrinking U around x_0 and replacing V by $V \cap h^{-1}(U)$ we may insure in addition that the graph of the homotopy $\{a_s\}$ stays in a prescribed open neighborhood of $\overline{a(U)}$.

Let $U_0 \subset X$ be an open set containing K such that a is holomorphic in U_0. Set $V_0 = h^{-1}(U_0) \subset Z$ and $a_{(0),s} = a|_{U_0}$ for $s \in [0,1]$. Using the above argument we can cover the graph of a outside of V_0 by open neighborhoods $V_j \subset Z$ $(j = 1, 2, \ldots)$, biholomorphic to P^N and with $U_j = h(V_j) \subset X$ biholomorphic to P^n, such that for each $j \in \mathbb{N}$ we have a homotopy of continuous sections $a_{(j),s}\colon U_j \to V_j$ $(s \in [0,1])$ of h, satisfying

- $a_{(j),0} = a|_{U_j}$,
- the section $a_{(j),1}$ is holomorphic in U_j,
- for each $j \geq 1$, $U_j \cap K = \emptyset$ and $Z|_{U_j} \to U_j$ admits a finite dominating family of fiber-sprays, and
- if $U_{(i,j)} = U_i \cap U_j \neq \emptyset$ then $a_{(i),s}(U_{(i,j)}) \subset V_j$ for each $s \in [0,1]$.

The last property insures that for each pair of indexes $i, j \in \mathbb{Z}_+$ such that $U_{(i,j)} \neq \emptyset$ there is a homotopy of sections $a_{(i,j),s}(t)\colon U_{(i,j)} \to Z$, depending continuously on $t, s \in [0,1]$, such that

- $a_{(i,j),s}(0) = a_{(i),s}|_{U_{(i,j)}}$,
- $a_{(i,j),s}(1) = a_{(j),s}|_{U_{(i,j)}}$,
- $a_{(i,j),0}(t) = a|_{U_{(i,j)}}$, and
- the section $a_{(i,j),1}(t)$ is holomorphic on $U_{(i,j)}$ for each $t \in [0,1]$.

We get $a_{(i,j),s}(t)$ by taking the convex linear combinations (with respect to $t \in [0,1]$) of the sections $a_{(i),s}$ and $a_{(j),s}$, restricted to $U_{(i,j)}$. At least one of the indexes i, j is positive, say $i > 0$, and in this case the convex combinations are taken by using the linear structure on the fibers of $h|_{V_i}$ induced by Φ_i.

Likewise, if $U_{(i,j,k)} \neq \emptyset$ for some $J = (i,j,k)$, we can use the linear structure on the fibers of one of the sets V_i, V_j, V_k to get a homotopy of sections $a_{J,s}(t)\colon U_J \to Z$, with t belonging to the standard 2-simplex $\triangle^2 \subset \mathbb{R}^2$, whose restriction to the sides of the simplex equals the respective homotopy obtained in the previous step. Continuing in this way we build a 1-prism a_* on the cover $\mathcal{U} = \{U_0, U_1, U_2, \ldots\}$ of X. By Proposition 6.9.6 there is a $\mathcal{C}$-cover $\mathcal{A} = (A_0, A_1, \ldots)$ of X, subordinate to $\mathcal{U}$ and such that $K \subset A_0 \subset U_0$. Then a_* defines a $(\mathcal{K}(\mathcal{A}), 1)$-prism with the required properties. $\square$

6.11 The Main Modification Lemma

The following result is the key step in the proof of Theorem 6.2.2.

Proposition 6.11.1. *Let $h\colon Z \to X$ be a holomorphic submersion of a complex space Z onto a complex space X, and let X' be a closed complex subvariety of X. Given a $\mathcal{C}$-string $(A_0, \dots, A_n)$ in X, with $A_1, \dots, A_n \subset X \backslash X'$, we denote by $\mathcal{K}^n = \mathcal{K}(A_0, \dots, A_n)$ its nerve. Assume that for each $i = 1, \dots n$ there is an open set $U_i \supset A_i$ in $X \backslash X'$ such that $h\colon Z|_{U_i} \to U_i$ admits a finite dominating family of fiber-sprays. If f_* is a holomorphic $(\mathcal{K}^n, k)$-prism with values in Z which is sectionally constant (Def. 6.8.4) on a compact set $Y \subset [0,1]^k$, then there is a homotopy f_*^u $(u \in [0,1])$ of holomorphic $(\mathcal{K}^n, k)$-prisms such that the following properties hold:*

(i) *$f_*^0 = f_*$ is the given prism,*
(ii) *the prism f_*^1 is sectionally constant,*
(iii) *for each $y \in [0,1]^k$ and $u \in [0,1]$, the section $f_{(0),y}^u$ approximates $f_{(0),y}$ on A_0 as close as desired,*
(iv) *$f_{*,y}^u = f_{*,y}$ for all $y \in Y$ and $u \in [0,1]$, and*
(v) *the homotopy $f_{*,y}^u$ is fixed to a given finite order along X'.*

If the restriction of f_* to the subcomplex $\mathcal{K}^{n-1} = \mathcal{K}(A_0, \dots, A_{n-1})$ is sectionally constant then f_*^u can be chosen such that, in addition to the above, the prism $f_*^u|_{\mathcal{K}^{n-1}}$ is sectionally constant for each $u \in [0,1]$, and the corresponding sections $f_{*,y}^u|_{\mathcal{K}^{n-1}}$ $(y \in [0,1]^k)$, which are holomorphic in a neighborhood of A^{n-1}, approximate $f_{*,y}|_{\mathcal{K}^{n-1}}$ uniformly on A^{n-1}.

Proof. Replacing X by a suitable Stein neighborhood of $A^n = \cup_{j=0}^n A_j$ we may assume that X is Stein. The proof is by induction on $n \geq 0$, and for $n = 0$ there is nothing to prove.

The case $n = 1$. Our data consists of a $\mathcal{C}$-pair (A_0, A_1) in X, with $A_1 \subset X \backslash X'$, and a holomorphic $(\mathcal{K}(A_0, A_1), k)$-prism f_* which is sectionally constant on a compact set $Y \subset [0,1]^k$. Such f_* is determined by the following data:

- a pair of open sets $U_0 \supset A_0$ and $U_1 \supset A_1$, with $U_1 \Subset X \backslash X'$,
- families of holomorphic sections

$$a_y = f_{(0),y}\colon U_0 \to Z|_{U_0}, \quad b_y = f_{(1),y}\colon U_1 \to Z|_{U_1},$$

  depending continuously on $y \in [0,1]^k$, and
- a family of holomorphic sections

$$c_{y,t} = f_{(0,1),y}(t)\colon U_{(0,1)} = U_0 \cap U_1 \to Z|_{U_0 \cap U_1}$$

  depending continuously on $t \in [0,1]$ and $y \in [0,1]^k$,

such that $a_y|_{U_{(0,1)}} = c_{y,0}$, $b_y|_{U_{(0,1)}} = c_{y,1}$, and for each $y \in Y$ the section $c_{y,t}$ is independent of $t \in [0,1]$. Hence for $y \in Y$ the family $\{c_{y,t}\colon t \in [0,1]\}$ determines a holomorphic section $c_y\colon U_0 \cup U_1 \to Z$ such that $c_y|_{U_0} = a_y$ and $c_y|_{U_1} = b_y$. Write $f_* = (a_*, b_*, c_*)$, where $*$ indicates the missing parameters.

Our goal is to construct a homotopy $f_*^u = (a_*^u, b_*^u, c_*^u)$ $(u \in [0,1])$ of holomorphic $(\mathcal{K}(A_0, A_1), k)$-prisms over smaller open sets $U_0 \supset A_0$ and $U_1 \supset A_1$ such that $f_*^0 = f_*$ and f_*^1 is a constant prism, that is, a family of holomorphic sections $f_y^1\colon U_0 \cup U_1 \to Z$. Furthermore, the homotopy must be fixed for $y \in Y$, it must approximate the sections a_y over A_0 for $y \in [0,1]^k$, and the homotopy a_*^u must be fixed to a finite order along the subvariety X'. We denote the data in the homotopy f_*^u by the same letters as above, adding the upper index u.

The homotopy f_*^u will be found in two steps. For convenience we use the parameter interval $u \in [0,2]$ and later rescale it to $[0,1]$. In the first step we apply the h-Runge approximation theorem (Theorem 6.6.2 on p. 264) to obtain a homotopy $\{f_*^u\}_{u\in[0,1]}$ from $f_*^0 = f_*$ to another prism f_*^1 such that we do not move the section a_y (i.e., $a_y^u = a_y\colon U_0 \to Z$ for all u and y), and such that the section $b_y^1\colon U_1 \to Z$ approximates a_y in a neighborhood of $A_0 \cap A_1$ for each $y \in [0,1]^k$. In the second step we apply the Heftungslemma, Proposition 6.7.2 (p. 267), to obtain homotopies of sections

$$a_y^u\colon U_0 \to Z, \quad b_y^u\colon U_1 \to Z, \quad c_{y,t}^u\colon U_{(0,1)} \to Z, \qquad u \in [1,2]$$

such that at $u = 2$, $a_y^2 = b_y^2$ over $U_{(0,1)}$ for each $y \in [0,1]^k$; hence these two sections amalgamate to a holomorphic section $f_y^2\colon U_0 \cup U_1 \to Z$.

Consider the first step. Since the set $A_{(0,1)} = A_0 \cap A_1$ is $\mathcal{O}(A_1)$-convex and the submersion $h\colon Z \to X$ admits a finite dominating family of fiber-sprays over a neighborhood of A_1, the h-Runge approximation theorem (Theorem 6.6.2 on p. 264) gives (after shrinking the sets $U_0 \supset A_0$ and $U_1 \supset A_1$) a homotopy of holomorphic sections $g_{y,t}^s\colon U_{0,1} \to Z$ $(s \in [0,1])$, depending continuously on t, s, y and satisfying the following properties:

- $g_{y,t}^0 = c_{y,t}$ for each y and t,
- $g_{y,1}^s = c_{y,1} = b_y|_{U_0 \cap U_1}$ for each s and y,
- $g_{y,t}^1$ extends to a holomorphic section over U_1 for each y and t,
- the homotopy is fixed on Y, i.e., for $y \in Y$ we have $g_{y,t}^s = c_y|_{U_0 \cap U_1}$ for each s and t, and
- $g_{y,t}^s$ approximates $c_{y,t}$ in a neighborhood of $A_{(0,1)}$ as close as desired, uniformly with respect to all parameters.

We define $f_*^u = (a_*^u, b_*^u, c_*^u)$ for $u \in [0,1]$ by

$$a_y^u = a_y, \quad b_y^u = g_{y,1-u}^1,$$

$$c_{y,t}^u = \begin{cases} c_{y,2t(1-u)} & \text{if } 0 \le t \le \frac{1}{2}; \\ g_{y,1-u}^{2t-1} & \text{if } \frac{1}{2} \le t \le 1. \end{cases}$$

This satisfies all required properties. In particular, at $u = 1$, $b^1_y = g^1_{y,0}$ approximates $c_{y,0} = a_y$ in a neighborhood of $A_{(0,1)}$.

We now apply Proposition 6.7.2 (p. 267) to get homotopies of sections $a^u_y\colon U_0 \to Z$ and $b^u_y\colon U_1 \to Z$ for $u \in [1,2]$ such that a^u_y approximates $a^1_y = a_y$ on A_0 for each $u \in [1,2]$, the homotopy is fixed along X', and $a^2_y = b^2_y$ in $U_{(0,1)}$. Moreover, over a neighborhood of the set A_0, the graphs of all sections a^u_y, b^u_y ($u \in [1,2]$) and $c^1_{y,t}$ ($t \in [0,1]$) lie in small tubular neighborhood of the image of the section a_y in Z. Using a vector bundle structure on such a neighborhood, we see that the triangle of homotopies, formed by these families, is contractible and can be filled by a 2-parameter homotopy $c^u_{y,t}$ ($t \in [0,1],\ u \in [1,2]$) over a neighborhood of $A_{(0,1)}$. This proves Proposition 6.11.1 for $n = 1$.

The induction step $n \Rightarrow n+1$. Suppose that Proposition 6.11.1 holds for all $\mathcal{C}$-strings of length $n+1$ for some $n \geq 1$ and for all $k \geq 0$. Let $\mathcal{A} = (A_0, \dots, A_{n+1})$ be a $\mathcal{C}$-string of length $n+2$ with the nerve $\mathcal{K}^{n+1} = \mathcal{K}(\mathcal{A})$, and let f_* be a holomorphic $(\mathcal{K}^{n+1}, k)$-prism with values in Z that is sectionally constant on a compact subset $Y \subset [0,1]^k$. Let $\mathcal{K}^n = \mathcal{K}(A_0, \dots, A_n) \subset \mathcal{K}^{n+1}$. The proof consists of the following three steps, each of which is accomplished by constructing a suitable homotopy of prisms.

Step 1: Reduction to the case when $f_*|_{\mathcal{K}^n}$ is a sectionally constant prism.

Step 2: Reduction to the case when f_* represents a $(k+1)$-prism over the $\mathcal{C}$-pair (A^n, A_{n+1}), where $A^n = A_0 \cup A_1 \cup \dots \cup A_n$.

Step 3: Applying the case $n = 1$ to the prism in Step 2 to get a sectionally constant $(\mathcal{K}^{n+1}, k)$-prism.

We begin by some general considerations. We denote the coordinates on $\mathbb{R}^{n+1}$ by $t = (t', t_{n+1})$, where $t' = (t_1, \dots, t_n) \in \mathbb{R}^n$, and identify $\mathbb{R}^n$ with the coordinate hyperplane $\mathbb{R}^n \times \{0\} \subset \mathbb{R}^{n+1}$. The body K^{n+1} of the nerve $\mathcal{K}^{n+1}$ can be represented as the union of certain faces of the standard simplex $\triangle^{n+1} \subset \mathbb{R}^{n+1}$. (In fact, $K^{n+1} = \triangle^{n+1}$ if and only if $A_0 \cap A_1 \cap \dots \cap A_{n+1} \neq \emptyset$.) The body $K^n \subset \mathbb{R}^n$ of the subcomplex $\mathcal{K}^n = \mathcal{K}(A_0, \dots, A_n) \subset \mathcal{K}^{n+1}$ is precisely the base $K^{n+1} \cap \{t_{n+1} = 0\}$ of K^{n+1}. We shall also need the complex

$$\mathcal{K}^n_1 = \mathcal{K}(A_0 \cap A_{n+1}, \dots, A_n \cap A_{n+1}) \subset \mathcal{K}^n. \tag{6.13}$$

Note that $\mathcal{K}^n_1 = \{J \in \mathcal{K}^n\colon (J, n+1) \in \mathcal{K}^{n+1}\}$. Its body K^n_1 is a subset of K^n which equals $\overline{(K^{n+1} \setminus K^n)} \cap K^n$. Moreover, for each $s \in [0,1]$, the section $K^{n+1} \cap \{t_{n+1} = s\}$ is homeomorphic to K^n_1. The map

$$r\colon \mathbb{R}^{n+1} \to \mathbb{R}^{n+1}, \quad r(t,s) = (t(1-s), s), \qquad t \in \mathbb{R}^n,\ s \in \mathbb{R}$$

maps the prism $\triangle^n \times [0,1]$ onto the standard simplex $\triangle^{n+1}$ (it is homeomorphic outside the level $s = 1$), and it maps $K^n_1 \times \{s\}$ homeomorphically onto $K^{n+1} \cap \{t_{n+1} = s\}$ for each $s \in (0,1)$. (See Fig. 6.4 on p. 282.)

Proof of Step 1. Since $\tilde{f}^0_* = f_*|_{\mathcal{K}^n} = \{f_J\colon J \in \mathcal{K}^n\}$ is a k-prism over a $\mathcal{C}$-string of length $n+1$, the induction hypothesis provides a holomorphic homotopy

$\tilde{f}_* = \{\tilde{f}_*^u : u \in [-1,0]\}$ such that each $\tilde{f}_*^u$ is a $(\mathcal{K}^n, k)$-prism, the homotopy is fixed for all $y \in Y$, and the prism $\tilde{f}_*^{-1}$ is sectionally constant. The parameter space of the prism f_* is $K^{n+1} \times [0,1]^k$ while the parameter space of $\tilde{f}_*$ is $K' \times [0,1]^k$, where

$$K' = \{(t', u) \in \mathbb{R}^n \times \mathbb{R} \colon\ t' \in K^n,\ -1 \le u \le 0\}.$$

Note that f_* and $\tilde{f}_*$ agree on the intersections of their domains $K^n \times [0,1]^k$ and hence define a family of sections, parametrized by the set $L \times [0,1]^k$, where $L = K^{n+1} \cup K' \subset \mathbb{R}^{n+1}$. We denote this family by $\{g_y(t) \colon t \in L,\ y \in [0,1]^k\}$.

For each $s \in [0,1]$ we denote by $L_s \subset \mathbb{R}^{n+1}$ the set

$$L_s = (K^{n+1} \backslash K^n) \cup \{(t', t_{n+1}) \colon t' \in K_1^n,\ -s \le t_{n+1} \le 0\} \cup \{(t', -s) \colon t' \in K^n\}.$$

Intuitively speaking, L_s is obtained by pushing the base K^n of K^{n+1} for s units in the negative t_{n+1} direction and then adding to this the vertical sides $K_1^n \times [-s, 0]$. Clearly $L_0 = K^{n+1}$, and L_s is homeomorphic to K^{n+1} for each $s \in [0,1]$. In fact, there is a continuous family of homeomorphisms $\Theta_s \colon K^{n+1} \to L_s$ $(0 \le s \le 1)$ such that Θ_0 is the identity, each Θ_s preserves the top vertex $(0, \ldots, 0, 1) \in K^{n+1}$ and the cellular structure of the two sets, and Θ_s maps K^n (the base of K^{n+1}) onto $K^n \times \{-s\}$ (the base of L_s) by a downward shift for s units. By 'respecting the cellular structure' we mean the following. Each face $J \in \mathcal{K}_1^n$ determines a face $\widetilde{J} = (J, n+1) \in \mathcal{K}^{n+1}$, and Θ_s maps its body $|\widetilde{J}| \subset K^{n+1}$ onto the set

$$|\widetilde{J}| \cup \{(t', t_{n+1}) \colon t' \in |J|,\ -s \le t_{n+1} \le 0\} \subset L_s.$$

We define a homotopy H_*^u $(0 \le u \le 1)$ of $(\mathcal{K}^{n+1}, k)$-prisms

$$H_J^u(t) = g_y(\Theta_u(t)) \quad (J \in \mathcal{K}^{n+1},\ t \in |J| \subset K^{n+1}).$$

Clearly $H_*^0 = f_*$ and $H_*^1|_{\mathcal{K}^n} = \tilde{f}_*^{-1}|_{\mathcal{K}^n}$ is sectionally constant.

Proof of Step 2: By Step 1 we may assume that the prism f_* is such that $f_*|_{\mathcal{K}^n}$ is sectionally constant. The next step is to modify f_* by a homotopy of holomorphic prisms into another prism which is sectionally constant also in the direction of the last variable t_{n+1}. Let $\mathcal{K}_1^n$ be the complex (5.1). We associate to f_* a holomorphic $(\mathcal{K}_1^n, k+1)$-prism

$$F_* = \{F_{J,(y,s)} \colon |J| \to \mathcal{O}_h(U_{(J,n+1)}, Z), \quad J \in \mathcal{K}_1^n,\ y \in [0,1]^k,\ s \in [0,1]\}$$

where

$$F_{J,(y,s)}(t) = f_{(J,n+1),y}(r(t,s)) \quad (t \in |J|,\ y \in [0,1]^k,\ s \in [0,1]).$$

The set

$$Y_1 = (Y \times [0,1]) \cup ([0,1]^k \times \{0,1\}) \subset [0,1]^{k+1} \tag{6.14}$$

is a compact subset of $[0,1]^{k+1}$. Since $f_*|_{\mathcal{K}^n}$ is sectionally constant, the prism F_* (which is associated to the complex $\mathcal{K}_1^n$ of length $n+1$) satisfies the induction hypothesis with respect to the set Y_1 (6.14). Hence there is a homotopy F_*^u ($u \in [0,1]$) of holomorphic $(\mathcal{K}_1^n, k+1)$-prisms, beginning with $F_*^0 = F_*$, such that the homotopy is fixed for $(y,s) \in Y_1$ and such that F_*^1 is sectionally constant. This means that for each fixed $(y,s) \in [0,1]^{k+1}$ the $\mathcal{K}_1^n$-complex $F_{*,(y,s)}^1$ is constant, i.e., it represents a holomorphic section $F_y^1(s)\colon V \to Z|_V$ over an open set $V \supset A^n \cap A_{n+1}$. Since the homotopy is fixed for $s=0$ and $s=1$, $F_y^1(0)$ coincides with the section represented by the constant complex $f_{*,y}|_{\mathcal{K}^n}$, and $F_y^1(1)$ coincides with the section $f_{(0,\dots,0,1),y}$ associated to A_{n+1}.

Proof of Step 3: We consider the family of sections obtained in Step 2,

$$F_*^1 = \{F_y^1(s)\colon V \to Z|_V \colon s \in [0,1],\ y \in [0,1]^k\},$$

as a holomorphic k-prism over the complex $\mathcal{K}' = \mathcal{K}(A^n, A_{n+1})$ determined by the $\mathcal{C}$-pair (A^n, A_{n+1}). The parameter s is the variable in the body $|\mathcal{K}'| = [0,1]$. For each $y \in [0,1]^k$ the section $F_y^1(0)$ extends holomorphically to a neighborhood of A^n, and $F_y^1(1)$ extends to a neighborhood of A_{n+1}.

The case $n=1$ of Proposition 6.11.1 gives a homotopy G_*^u ($u \in [1,2]$) of holomorphic $(\mathcal{K}', k)$-prisms such that $G_*^1 = F_*^1$, G_y^2 is a constant $\mathcal{K}'$-complex for each $y \in [0,1]^k$ (i.e., a holomorphic section over an open neighborhood of $A^{n+1} = A_0 \cup \dots \cup A_{n+1}$), the homotopy is fixed for $y \in Y$ (where $G_y^1 = F_y^1$ is already a section over A^{n+1}), it is fixed to a finite order along X', and the section $G_y^u(0)$ (which is holomorphic over a neighborhood of A^n) approximates the section $F_y^1(0) = f_{*,y}|_{\mathcal{K}^n}$ on the set A^n for each $u \in [1,2]$.

The collection $\{f_*^u \colon u \in [0,1]\}$, defined by

$$f_*^u = \begin{cases} F_*^{2u} & \text{if } u \in [0, \frac{1}{2}]; \\ G_*^{2u} & \text{if } u \in [\frac{1}{2}, 1], \end{cases}$$

is a homotopy of holomorphic $(\mathcal{K}^{n+1}, k)$-prisms, beginning at $u=0$ with f_* and ending at $u=1$ with the sectionally constant prism G_*^2.

If we assume in addition that the restriction $f_*|_{\mathcal{K}^n}$ is sectionally constant on $[0,1]^k$ (so $f_{*,y}|_{\mathcal{K}^n}$ is a holomorphic section in a neighborhood of A^n for each $y \in [0,1]^k$), we can skip the initial step in the proof of the inductive step. By the construction, the restriction $F_*^u|_{\mathcal{K}^n}$ is independent of $u \in [0,1]$ since the homotopy F_*^u is fixed on the set Y_1 (6.14), and the homotopy G_*^u is such that the complex $G_{*,y}^u|_{\mathcal{K}^n}$ is represented by a holomorphic section in a neighborhood of A^n which approximates the section $F_{*,y}^1|_{\mathcal{K}^n} = f_{*,y}|_{\mathcal{K}^n}$ on A^n, uniformly with respect to $u \in [0,1]$ and $y \in [0,1]^k$. Hence the section $f_{*,y}^u|_{\mathcal{K}^n}$ approximates $f_{*,y}|_{\mathcal{K}^n}$ on A^n, the approximation being uniform with respect to $u \in [0,1]$ and $y \in [0,1]^k$. $\square$

The next proposition shows that a holomorphic 1-prism can be extended from a finite subcomplex to the entire complex such that the 0-level of the

prism matches a given complex. This does not require any analytic tools and hence the result applies to any locally finite family $\mathcal{A}$.

Proposition 6.11.2. *Assume that $h\colon Z \to X$ is a holomorphic submersion. Let $\mathcal{A} = \{A_0, A_1, \ldots\}$ be a locally finite family of compact sets in X. Denote its nerve by $\mathcal{K}(\mathcal{A})$, and let $\mathcal{K}^n = \mathcal{K}(A_0, \ldots, A_n) \subset \mathcal{K}(\mathcal{A})$ for each $n \in \mathbb{N}$. If f_* is a holomorphic $\mathcal{K}(\mathcal{A})$-complex with values in Z, and if g_* is a holomorphic $(\mathcal{K}^n, 1)$-prism for some $n \in \mathbb{N}$ such that $g_{*,0} = f_*|_{\mathcal{K}^n}$, then there exists a holomorphic $(\mathcal{K}(\mathcal{A}), 1)$-prism G_* such that $G_{*,0} = f_*$ and $G_*|_{\mathcal{K}^n} = g_*$. Similarly, if f_* is a holomorphic $(\mathcal{K}(\mathcal{A}), k)$-prism and g_* is a holomorphic $(\mathcal{K}^n, k+1)$-prism with the base $f_*|_{\mathcal{K}^n}$, then g_* extends to a holomorphic $(\mathcal{K}(\mathcal{A}), k+1)$-prism G_* with $G_{*,0} = f_*$.*

Proof. We choose representatives of f_* and g_* defined on a faithful open neighborhood $\mathcal{U}$ of $\mathcal{A}$ (Def. 6.8.3). Write $A^n = A_0 \cup \cdots \cup A_n$ as before. Let $m \geq n$ be the smallest integer such that $A_k \cap A^n = \emptyset$ for all $k \geq m$ (such m exists since the family $\{A_j\}$ is locally finite). We represent the body $K^m = K(A_0, \ldots, A_m)$ of the subcomplex $\mathcal{K}^m \subset \mathcal{K}(\mathcal{A})$ as a subset of $\mathbb{R}^m$. Denote the coordinates on $\mathbb{R}^{m+1}$ by (t, s), with $t \in \mathbb{R}^m$ and $s \in \mathbb{R}$, and identify $\mathbb{R}^m$ with $\mathbb{R}^m \times \{0\} = \{s = 0\} \subset \mathbb{R}^{m+1}$. Similarly we identify a set $K \subset \mathbb{R}^m$ with $K \times \{0\} \subset \mathbb{R}^{m+1}$ and write $K \times [0,1] = \{(t,s)\colon t \in K,\ s \in [0,1]\}$. For each face $J \in \mathcal{K}^m$ we denote by $b|J| \subset K^m$ the boundary of its body $|J|$.

Lemma 6.11.3. *There exists a retraction*

$$r\colon K^m \times [0,1] \to K^m \cup (K^n \times [0,1]) \subset \mathbb{R}^{m+1}$$

such that for each face $J \in \mathcal{K}^m \backslash \mathcal{K}^n$ we have

(i) $r(|J| \times [0,1]) \subset |J| \cup (b|J| \times [0,1])$,
(ii) *if $|J| \cap K^n = \emptyset$ then $r(t,s) = t$ for each $t \in |J|$ and $s \in [0,1]$.*

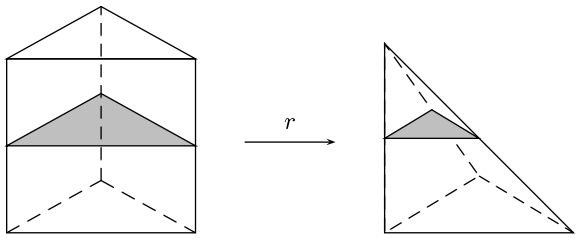

Fig. 6.4. Retracting a prism onto a simplex

Proof. The retraction r is shown on Fig. 6.4. We first define r over those faces $J \in \mathcal{K}^m$ for which either $|J| \subset K^n$ (in such case we let r be the identity on $|J| \times [0,1]$), or $|J| \cap K^n = \emptyset$ (we let $r(t,s) = t$ for $t \in |J|$). We also define r to be the identity map on the bottom side $K^m = K^m \times \{0\}$. On the remaining faces $|J|$, $J \in \mathcal{K}^m$, we define r inductively with respect to the dimension of J. Suppose that r has already been defined on all faces of dimension $< k$ and let $J = (j_0, \ldots, j_k) \in \mathcal{K}^m$. Then r is already defined on $|J| \cup (b|J| \times [0,1])$ and it satisfies (i); it also satisfies (ii) on those sides of $b|J|$ which are disjoint from K^n. Moreover, r is the identity on $|J| = |J| \times \{0\}$. It is now clear that r extends from $|J| \cup (b|J| \times [0,1])$ to $|J| \times [0,1]$ so that (i) holds. $\square$

Let r be as in Lemma 6.11.3. Write $r(t,s) = (r_0(t,s), u(t,s))$, where $r_0(t,s) \in K^m$ and $u(t,s) \in [0,1]$. We define a holomorphic $(\mathcal{K}^m, 1)$-prism G_* by setting for each $J \in \mathcal{K}^m$, $t \in |J|$ and $s \in [0,1]$

$$G_{J,s}(t) = \begin{cases} f_J(r_0(t,s)) & \text{if } u(t,s) = 0; \\ g_{J,u(t,s)}(r_0(t,s)) & \text{if } u(t,s) > 0. \end{cases}$$

Property (i) in Lemma 6.11.3 implies that the section $G_{J,s}(t)$ for $t \in |J|$ is defined (and holomorphic) in the set U_J. (It may even be holomorphic in a larger set if $r_0(t,s) \in b|J|$, but in such case we restrict it to U_J.) The family $G_* = \{G_{J,s} \colon J \in \mathcal{K}^m,\ s \in [0,1]\}$ is then a holomorphic $(\mathcal{K}^m, 1)$-prism which extends g_* and satisfies $G_{*,0} = f_*$. Property (ii) of the retraction r lets us extend G_* to a prism over the entire complex $\mathcal{K}(\mathcal{A})$ by observing that for those faces $J \in \mathcal{K}(\mathcal{A})$ which do not belong to $\mathcal{K}^m$ we have $|J| \cap K^n = \emptyset$ (by the definition of m), and therefore $r(t,s) = t$ for $t \in |J| \cap K^m$. Thus we can simply take $G_{J,s}(t) = f_J(t)$ for $t \in |J|$ and $s \in [0,1]$. $\square$

6.12 Proof of Gromov's Oka Principle

In this section we prove Theorems 6.2.2 and 6.6.6. We begin with the basic case when $Z \to X$ is subelliptic submersion onto a Stein space X and $X' = \emptyset$ (i.e., without the interpolation condition). For simplicity of exposition we concentrate on a single section; the proof in the parametric case is an immediate extension since all tools have been prepared in the required generality. Given a compact $\mathcal{O}(X)$-convex subset K of X and a continuous section $a \colon X \to Z$ which is holomorphic in an open set $U_0 \supset K$, our goal is to find a homotopy $H_s \colon X \to Z$ ($s \in [0,1]$) of continuous sections such that $H_0 = a$, the section H_1 is holomorphic on X, and for each $s \in [0,1]$ the section H_s is holomorphic near K and it approximates a on K.

Let $\mathcal{A} = \{A_0, A_1, \ldots\}$ be a $\mathcal{C}$-cover of X such that $K \subset A_0 \subset U_0$ and $K \cap A_i = \emptyset$ for all $i = 1, 2, \ldots$ (see Proposition 6.9.6 on p. 274). Let $a_* = \{a_{*,s}\}_{s \in [0,1]}$ be a continuous $(\mathcal{K}(\mathcal{A}), 1)$-prism furnished by Proposition 6.10.1

(p. 275). The 0-level complex $a_{*,0}$ agrees with the initial section a, the final complex $a_{*,1}$ is a holomorphic, and $a_{(0),s} = a|_{U_0}$ for each $s \in [0,1]$.

Let dist be a complete distance function on Z. Fix an $\epsilon > 0$. We construct a sequence of holomorphic $\mathcal{K}(\mathcal{A})$-complexes f_*^n, and a sequence of holomorphic $(\mathcal{K}(\mathcal{A}),1)$-prisms $G_*^n = \{G_{*,s}^n\}_{s\in[0,1]}$ $(n = 0,1,\ldots)$, satisfying the following:

(a) $f_*^0 = a_{*,1}$ is the initial holomorphic complex,

(b) $G_{*,0}^n = f_*^n$ and $G_{*,1}^n = f_*^{n+1}$ for each $n \in \mathbb{Z}_+$ (hence G_*^n is a homotopy of complexes connecting f_*^n and f_*^{n+1}),

(c) for each $n, k \in \mathbb{Z}_+$, $n \geq k$, and each $s \in [0,1]$ the complexes $f_*^n|_{\mathcal{K}^k}$ and $G_{*,s}^n|_{\mathcal{K}^k}$ are constant, i.e., they are given by holomorphic sections f^n, resp. G_s^n, in an open neighborhood of $A^k = A_0 \cup \cdots \cup A_k$, and

(d) for each $n \in \mathbb{Z}_+$ and $s \in [0,1]$ we have

$$\operatorname{dist}\big(G_s^n(x), f^n(x)\big) < \epsilon 2^{-n-1}, \quad x \in A^n.$$

In particular, $\operatorname{dist}\big(f^{n+1}(x), f^n(x)\big) < \epsilon 2^{-n-1}$ for $x \in A^n$. (Here we are using the notation for sections established in property (c) above.)

The property (d) implies that the sequence of sections $f^n\colon A^n \to Z$ converges, uniformly on compacts in X, to a holomorphic section $f^\infty = \lim_{n\to\infty} f^n\colon X \to Z$ satisfying

$$\operatorname{dist}\big(f^\infty(x), a(x)\big) = \operatorname{dist}\big(f^\infty(x), f^0(x)\big) < \epsilon, \quad x \in A_0.$$

To find a homotopy $H_s\colon X \to Z$ $(s \in [0,1])$, with $H_0 = a$ and $H_1 = f^\infty$, we first construct a continuous $(\mathcal{K}(\mathcal{A}),1)$-prism h_* with $h_{*,0} = a$ and $h_{*,1} = f^\infty$. It suffices to collect all individual 1-prisms a_* and G_*^n $(n \in \mathbb{Z}_+)$ into a single 1-prism as follows. For each $n \in \mathbb{Z}_+$ set $I_n = [1-2^{-n}, 1-2^{-n-1}]$, and let $\lambda_n\colon I_n \to [0,1]$ be the linear bijection $\lambda_n(s) = 2^{n+1}(s-1+2^{-n})$. Then $\cup_{n=0}^\infty I_n = [0,1)$. For $s \in [0,1]$ we define

$$h_{*,s} = \begin{cases} a_{*,2s} & \text{if } s \in I_0 = [0,\frac{1}{2}]; \\ G_{*,\lambda_n(s)}^{n-1} & \text{if } s \in I_n,\ n \geq 1; \\ f^\infty & \text{if } s = 1. \end{cases}$$

The two definitions of $h_{*,s}$ at the values $s = 1-2^{-n}$ are compatible by (b).

Properties (c) and (d) imply that $\lim_{s\to 1} h_{*,s} = f^\infty$, uniformly on compacts in X. Indeed, each compact set $L \Subset X$ is contained in some A^m, and for $n \geq m$ the complex $G_{*,s}^n$ is constant on A^m, i.e., it is represented there by a holomorphic section. Hence for $1-2^{-n-1} \leq s < 1$ the complex $h_{*,s}|_{\mathcal{K}^n}$ is a holomorphic section in a neighborhood of A^n. As $s \to 1$, these sections converge uniformly on compacts to f^∞. This proves that $h_* = \{h_{*,s}\}_{s\in[0,1]}$ is indeed a continuous $(\mathcal{K}(\mathcal{A}),1)$-prism. Notice also that the restriction of h_* to the trivial subcomplex $\mathcal{K}(A_0)$ is a homotopy of holomorphic sections h_s $(s \in [0,1])$ in a neighborhood of A_0, connecting $h_0 = a$ to $h_1 = f^\infty$, such that

$$\operatorname{dist}(h_s(x), a(x)) < \epsilon, \qquad x \in A_0,\ s \in [0,1].$$

To complete the proof we apply Proposition 6.11.1 (p. 277) to modify the 1-prism h_* by a homotopy of 1-prisms (keeping the ends $s = 0$ and $s = 1$ fixed) into a 1-prism H_* which is *sectionally constant*, i.e., such that H_* represents a homotopy of continuous sections $H_s\colon X \to Z$ ($s \in [0,1]$). Moreover, we can achieve that H_s and h_s agree in a neighborhood of A_0. This concludes the proof of Theorem 6.2.2 in the basic case.

The parametric case is proved by the same tools by introducing the parameter space P into the definition of holomorphic (and continuous) complexes and prisms and repeating the same arguments in this setting. All required technical tools used in the process have been proved in this generality.

It remains to consider the stratified case; this will also yields Theorem 6.2.3 as an induction step. We follow the scheme of proof of Theorem 5.4.4. Given a stratification (6.4), we replace each X_k by $X_k \cup X'$ and get another stratification

$$X = X_0 \supset X_1 \supset \cdots \supset X_m = X',$$

with regular strata $S_j = X_j \backslash X_{j+1}$, such that $Z|_{S_j}$ is a subelliptic submersion for $j = 0, \ldots, m-1$.

In a typical induction step we have compact $\mathcal{O}(X)$-convex sets $K \subset L$ in X and a section $f\colon X \to Z$ which is holomorphic in an open neighborhood of K and whose restriction to the subvariety X' is holomorphic. (Alternatively, for the jet interpolation case, f is holomorphic in a neighborhood of $K \cup X'$.) By a homotopic deformation which is fixed on X' we must find a new section that is holomorphic on a neighborhood of the larger set L.

As in §5.12 we inductively extend the section across the strata $S_k = X_k \backslash X_{k+1}$. (We replace every stratum by its intersection with the compact subset L, so there are only finitely many connected components to consider.) By the inductive hypothesis we have a section $f\colon X \to Z$ that is holomorphic in an open set $U \subset X$ containing $K \cup (L \cap X_{k+1})$. Choose a $\mathcal{C}$-string $\mathcal{A} = (A_0, A_1, \ldots, A_n)$ in X such that

- $K \cup (L \cap X_{k+1}) \subset A_0 \subset U$,
- the sets $A_1, \ldots, A_n$ are contained in the stratum S_k, and
- $K \cup (L \cap X_k) \subset K \cup (\cup_{j=1}^n A_j)$.

The set A_0 is chosen as the union of $K \cup (L \cap X_{k+1})$ and a small collar of this set inside the stratum $S_k = X_k \backslash X_{k+1}$. Each of the sets $A_1, \ldots, A_n$ is a convex bump or a handle contained in S_k and attached to the previous set so that they form a $\mathcal{C}$-pair (see the *general case* in the proof of Proposition 5.12.1). Then $\mathcal{A}$ is a $\mathcal{C}$-string by Corollary 6.9.4 on p. 273.

Proposition 6.10.1 (p. 275) furnishes a holomorphic $\mathcal{K}(\mathcal{A})$-complex in which f is the holomorphic section associated to the initial set A_0.

Following the proof of Theorem 6.2.2 in §6.12 and using Proposition 6.11.1 we obtain a new section $\widetilde{f}: X \to Z$, homotopic to f relative to the subvariety X_{k+1}, which is holomorphic over a neighborhood of A_0 in X, and also over a relative neighborhood of $\cup_{j=1}^n A_j$ in the stratum S_k. This is accomplished in finitely many moves. Finally, by Theorem 3.4.1 on p. 68 we can deform $\widetilde{f}$ to a section that is holomorphic in an open neighborhood of the set $\cup_{j=0}^n A_j$ in X, thereby completing the induction step. The remaining details of the proof are exactly as in §5.12.

6.13 Relative Oka Principle on 1-Convex Spaces

The following the relative Oka principle over 1-convex complex spaces is due to J. Prezelj [402].

Theorem 6.13.1. [402, Theorem 1.1] *Let X be a 1-convex complex space, S its exceptional variety, Z a complex space, and $h: Z \to X$ a holomorphic submersion whose restriction $h: Z \backslash h^{-1}(S) \to X \backslash S$ is a stratified subelliptic submersion (Def. 6.2.1), or a stratified fiber bundle whose fibers are Oka manifolds (Def. 1.4.1). Let K be a compact $\mathcal{O}(X)$-convex subset of X containing S. Given a continuous section $f: X \to Z$ that is holomorphic in a neighborhood of K, there exists a homotopy of sections $f_t: X \to Z$ ($t \in [0,1]$) such that the following properties hold:*

(i) *$f_t(x) = f_0(x)$ for $x \in S$,*

(ii) *for every $t \in [0,1]$ the sections f_t is holomorphic near K and it approximates f_0 uniformly on K as well as desired, and*

(iii) *the section f_1 is holomorphic on X.*

The analogous result holds for families of sections depending continuously on a parameter in a compact Hausdorff space (compare with Theorem 6.2.2).

In the classical case, with $h: Z \to X$ a principal holomorphic fiber bundle with a Lie group fiber, this Oka principle is due to Henkin and Leiterer [259] and Colţoiu [90] (when X is nonsingular), and to Leiterer and Vâjâitu [332] for 1-convex spaces with singularities. Theorem 6.13.1 implies the relative Oka-Grauert principle concerning the classification of holomorphic vector bundles over 1-convex spaces (see Corollary 7.3.10).

Proof (sketch). We follow the proof of Theorem 6.2.2. A nontrivial addition comes in the construction of a local spray of sections. Recall from §6.7 that, for the purpose of gluing a pair of holomorphic sections over a Cartain pair in X, one of the sections $f: U \subset X \to Z$ must first be embedded into a local fiber-dominating spray of sections. If X is Stein then such a spray is furnished by

Lemma 5.10.4 on p. 220. This result crucially depends on Siu's theorem on the existence of a Stein neighborhood of the section in the total space Z (Theorem 3.1.1). Granted such a Stein neighborhood, we find vertical holomorphic vector fields generating the vertical tangent bundle $\mathrm{VT}Z$ near the section and then compose their local flows to get a local fiber-dominating spray. If X is only 1-convex then $f(U)$ does not have any Stein neighborhood in Z. Instead we choose a complex hypersurface $X' \subset X$ containing the exceptional subvariety S and find a conical Stein neighborhood $\Omega \subset Z$ of $f(U \backslash X')$ as in Theorem 3.2.3 (p. 64). On such Ω we use the same tools as before to find a local spray of holomorphic sections that is fixed along X' and is dominating over $U \backslash X'$. Finitely many such sprays together are fiber-dominating over $U \backslash S$. This enables the gluing of the given spray with another local spray, provided that the intersection of their domains in X does not meet the exceptional set S. The proof now proceeds as in Theorem 6.2.2. $\square$

A similar but simpler proof gives the following relative Oka principle for sections $X \to Z$ that are holomorphic near the set of critical values of a holomorphic map $h\colon Z \to X$.

Theorem 6.13.2. [172, Theorem 2.1] *Let X be a Stein space and $h\colon Z \to X$ a holomorphic map of a complex space Z onto X. Assume that X' is a closed complex subvariety of X such that h is a subelliptic submersion over $X \backslash X'$, and K is a compact $\mathcal{O}(X)$-convex set in X. (The map h need not be a submersion over X'.) Given a continuous section $f\colon X \to Z$ that is holomorphic in an open set containing $K \cup X'$, there is for any $k \in \mathbb{N}$ a homotopy $f_t\colon X \to Z$ $(t \in [0,1])$ of continuous sections such that $f_0 = f$, each f_t is holomorphic near $K \cup X'$, is uniformly close to f_0 on K and tangent to f to order k along X', and the section f_1 is holomorphic on X.*

As before, the nontrivial addition is the construction of a local spray that is dominating over the complement of X' (Proposition 2.2 in [172]). In the same paper we discuss Oka principle for multivalued sections of a ramified map $Z \to X$ onto a Stein space; the graph of such a section is a complex subvariety of Z with a proper finite-to-one projection onto X.

6.14 Oka Maps

So far we considered Oka properties of maps $X \to Y$ from Stein spaces X, and of sections of submersions $h\colon Z \to X$ onto a Stein base X. A natural generalization arises by considering maps in the following diagram:

$$\begin{tikzcd}
P_0 \times X \arrow[r, "F"] \arrow[d, "incl"'] & E \arrow[d, "\pi"] \\
P \times X \arrow[r, "f"] \arrow[ur, "F"] & B
\end{tikzcd}$$

Here $\pi\colon E \to B$ is a holomorphic submersion of a complex space E onto a complex space B, X is a Stein space, $P_0 \subset P$ are compact Hausdorff spaces (the parameter spaces), and $f\colon P \times X \to B$ is an X-holomorphic map, meaning that $f(p,\cdot)\colon X \to B$ is holomorphic on X for every fixed $p \in P$. A map $F\colon P \times X \to E$ such that $\pi \circ F = f$ is said to be a *lifting* of f; such F is X–holomorphic on P_0 if $F(p,\cdot)$ is holomorphic for every $p \in P_0$.

Definition 6.14.1. *A holomorphic map $\pi\colon E \to B$ between reduced complex spaces enjoys the* Parametric Oka Property (POP) *if for any collection $(X, X', K, P, P_0, f, F_0)$ where X is a reduced Stein space, X' is a closed complex subvariety of X, $P_0 \subset P$ are compact Hausdorff spaces, $f\colon P \times X \to B$ is an X-holomorphic map, and $F_0\colon P \times X \to E$ is a continuous map such that $\pi \circ F = f$, the map $F_0(p,\cdot)$ is holomorphic on X for all $p \in P_0$ and is holomorphic on $K \cup X'$ for all $p \in P$, there exists a homotopy $F_t\colon P \times X \to E$ such that the following hold for all $t \in [0,1]$:*

(i) $\pi \circ F_t = f$,

(ii) $F_t = F_0$ *on* $(P_0 \times X) \cup (P \times X')$,

(iii) F_t *is X-holomorphic on K and uniformly close to F_0 on $P \times K$, and*

(iv) *the map $F_1\colon P \times X \to E$ is X-holomorphic.*

The map $\pi\colon E \to B$ enjoys the Basic Oka Property (BOP) *if the above holds for the case when P is a singleton and $P_0 = \emptyset$.*

Neglecting approximation and interpolation, POP of a map $\pi\colon E \to B$ can be illustrated by the following commuting diagram:

$$\begin{tikzcd}
P_0 \arrow[r] \arrow[d, "incl"'] & \mathcal{O}(X,E) \arrow[r] \arrow[d, "F_0"] & \mathcal{C}(X,E) \arrow[d, "\pi"] \\
P \arrow[r, "f"'] \arrow[ur, "F_1"] \arrow[urr] & \mathcal{O}(X,B) \arrow[r] & \mathcal{C}(X,B)
\end{tikzcd}$$

By comparing definitions we see that a complex manifold E enjoys a certain Oka property if and only if the trivial map $\pi\colon E \to$ *point* does. For a complex manifold Y all Oka properties (for parameter spaces $P_0 \subset P \subset \mathbb{R}^m$) are equivalent (Proposition 5.15.1), and such Y is called an Oka manifold.

Observe that the problem of lifting a single holomorphic map $f\colon X \to B$ to a holomorphic map $F\colon X \to E$ immediately reduces to the problem of finding a suitable section. As an example we prove the following result.

Corollary 6.14.2. (i) *Every stratified subelliptic submersion enjoys* BOP.

(ii) *Every stratified holomorphic fiber bundle with Oka fibers enjoys* BOP.

Proof. We proceed as in Corollary 5.4.11. Let $\widetilde{\pi}\colon f^*E \to X$ denote the pull-back of a holomorphic submersion $\pi\colon E \to B$ by a holomorphic map $f\colon X \to B$. If $\pi\colon E \to B$ is a stratified subelliptic submersion (case (i)) then $\widetilde{\pi}\colon f^*E \to X$ is also a stratified subelliptic submersion. (Stratify X such that each stratum is mapped by f into a stratum of B over which the submersion π is subelliptic.) Then sections of $\widetilde{\pi}\colon f^*E \to X$ are in bijective correspondence with liftings $X \to E$ of f. Since sections of $f^*E \to X$ satisfy BOP by Theorem 6.2.2 (p. 243), π also satisfies BOP. The same proof applies in case (ii). □

The above argument no longer holds in the parametric case as there is no holomorphic dependence of the pull-back $f^*E \to X$ on the map $f\colon X \to B$ in the given family. Nevertheless, BOP$\Longrightarrow$POP still holds.

Theorem 6.14.3. [182] *For every holomorphic submersion $\pi\colon E \to B$ of reduced complex spaces we have the implication* BOP $\Longrightarrow$ POP*, where* POP *is restricted to parameter spaces $P_0 \subset P$ that are Euclidean compacts.*

The proof is similar to that of Theorem 5.13.3. We first show that BOP implies a certain parametric homotopy approximation property (PHAP) for liftings, similar to the one expressed (for sections) by Theorem 6.6.2. In [181, Theorem 4.2] it is shown that, by using PHAP, we can assemble complexes of local holomorphic liftings over $\mathcal{C}$-strings into holomorphic liftings exactly as was done for sections in the proof of Theorem 6.2.2 above.

The following consequence of Theorem 6.14.3 and Corollary 6.14.2 gives the most general presently known Oka principle.

Corollary 6.14.4. (i) *Every stratified subelliptic submersion enjoys* POP.
(ii) *Every stratified holomorphic fiber bundle with Oka fibers enjoys* POP.

Example 6.14.5. We say that f is a *stratified holomorphic function* on a complex space X if there is a stratification $X = X_0 \supset X_1 \supset \cdots \supset X_m = \emptyset$ such that f is defined over the union of some of the strata $S_j = X_j \backslash X_{j+1}$, it is holomorphic over every stratum S_j, and its graph $\Gamma_f \subset X \times \mathbb{C}$ is closed. Then $Z = (X \times \mathbb{C}) \backslash \Gamma_f \to X$ is a submersion, and the restriction $Z|_S \to S$ to any stratum S in the above stratification is a holomorphic fiber bundle with fiber $\mathbb{C}$ or $\mathbb{C}^*$. Hence $Z \to X$ is a stratified elliptic submersion and consequently it enjoys POP. In particular, if X is Stein then sections $X \to Z$ satisfy the Oka property. (These are holomorphic functions $g \in \mathcal{O}(X)$ such that $g(x) \neq f(x)$ for each x.) A particular example of this type is a meromorphic function f on X with the polar locus $X_1 \subset X$ and without the indeterminacy set; $f|_{X_1}$ can again be a function of the same type, etc. □

Another interesting point is that POP is a local property. The proof of the following result [181, Theorem 4.7] uses the fact that POP follows from PHAP applied over small open subsets of the base space.

Theorem 6.14.6. (Localization principle for POP) *A holomorphic submersion $\pi\colon E \to B$ of a complex space E onto a complex space B satisfies* POP *if and only if every point $x \in B$ admits an open neighborhood $U_x \subset B$ such that the restricted submersion $\pi\colon E|_{U_x} \to U_x$ satisfies* POP*.*

In analogy to the class of Oka manifolds (Def. 5.4.1) we introduce the class of *Oka maps*. This class of maps was first defined in [182].

Definition 6.14.7. *A holomorphic map $\pi\colon E \to B$ of complex spaces is said to be an* Oka map *if it is a Serre fibration (see [512]) and it enjoys* POP*.*

A holomorphic map is an Oka map precisely when it is an *intermediate fibrations* in Lárusson's model category [321, 322]. Corollary 6.14.4 implies the following result.

Corollary 6.14.8. *A holomorphic fiber bundle projection with Oka fiber is an Oka map. A stratified subelliptic submersion, or a stratified holomorphic fiber bundle with Oka fibers, is an Oka map if and only if it is a Serre fibration.*

If a holomorphic submersion $\pi\colon E \to B$ enjoys the Oka property, then by considering liftings of constant maps $X \to b \in B$ we see that every fiber $E_b = \pi^{-1}(b)$ is an Oka manifold. The converse fails in general, even in the simplest case when the base is the unit disc $\mathbb{D} \subset \mathbb{C}$ and the fiber over each point $z \in \mathbb{D}$ is $E_z = \mathbb{C}\backslash\{g(z)\}$, where g is a complex valued function on the disc. This failure is nicely illustrated by the following result of independent interest, due to Eremenko, which shows that the missing value of an entire function $f\colon \mathbb{C} \to \mathbb{C}$ depends holomorphically on f.

Theorem 6.14.9. [136, Theorem 2] *Assume that D is a domain in $\mathbb{C}$ and $f\colon D \times \mathbb{C} \to \mathbb{C}$ a holomorphic function such that $\mathbb{C} \ni w \mapsto f(z,w)$ is non-constant for all $z \in D$. Assume that for some function $a\colon D \to \mathbb{C}$ we have $f(z,w) \neq a(z)$ for all $w \in \mathbb{C}$. Then there exists a discrete set $\Lambda = \{z_j\} \subset D$ such that a is holomorphic in $D\backslash\Lambda$ and $a(z) \to \infty$ as $z \to z_j$ for every $z_j \in \Lambda$.*

The function a in Eremenko's theorem can take arbitrary values at the points $z_j \in \Lambda$, and hence it is stratified holomorphic (Example 6.14.5).

Corollary 6.14.10. *Let $g\colon D \to \mathbb{C}$ be a continuous function on a domain $D \subset \mathbb{C}$. The holomorphic submersion $\pi\colon E_g = D \times \mathbb{C}\backslash\Gamma_g \to D$ is an Oka map if and only if g is holomorphic.*

Proof. If g is holomorphic then $E_g \cong D \times \mathbb{C}^*$ via $(z,w) \mapsto (z, w - a(z))$, and hence $\pi\colon E_g \to D$ is an Oka map by Corollary 6.14.8 (i). Conversely, assume that the projection $\pi\colon E_g \to D$ is an Oka map. Then the continuous lifting

$$F_0\colon D \times \mathbb{C}^* \to E_g, \quad (z,w) \mapsto (z, w + g(z))$$

of the projection map $D \times \mathbb{C}^* \to D$ can be deformed to a holomorphic lifting $F_1(z,w) = (z, f(z,w))$. Note that f satisfies the hypothesis of Theorem 6.14.9, and hence a is holomorphic. $\square$

7

Applications

In this and the following chapter we apply the results and methods from the previous two chapters to a variety of problems. We begin with applications of the classical Oka-Grauert principle (§7.1 – §7.5). Gromov's Oka principle and its generalizations from Chap. 6 are used mainly in §7.6 (elimination of intersections of holomorphic maps with complex subvarieties), §7.7 (the holomorphic Vaserstein problem), and in the construction of proper holomorphic embeddings and immersions of Stein spaces to Euclidean spaces of minimal dimension (§8.2–§8.4). In §7.8 and §7.9 we discuss transversality theorems for holomorphic and algebraic maps. In §7.10 we extend the homotopy version of the Runge approximation theorem (§6.6) to algebraic maps from affine algebraic varieties to a certain class of algebraic manifolds; this is used in Chap. 8. In §7.11 we mention a few existing quantitative results of Oka theory. Further applications of the method of splitting and gluing of holomorphic sprays, developed in §5.8 and §5.9, can be found in the papers [121, 122, 123, 178].

7.1 Principal Fiber Bundles

In this section we recall some basic notions on principal G-bundles in preparation for Grauert's classification theorem in the following section.

Definition 7.1.1. *Let G be a (real or complex) Lie group. A* topological principal G-bundle *is fiber bundle $\pi\colon E \to X$ with fiber $E_x \cong G$, endowed with a faithful right action of G on E whose orbits are the fibers $E_x = \pi^{-1}(x)$ $(x \in X)$ and which is locally trivial in the following sense: Every point $x_0 \in X$ admits an open neighborhood $U \subset X$ and a homeomorphism $\theta\colon E|_U \to U \times G$, $\theta(e) = (\pi(e), \vartheta(e))$, satisfying*

$$\theta(eg) = \big(\pi(e), \vartheta(e)\, g\big), \quad e \in E,\ g \in G. \tag{7.1}$$

*(Such θ is said to be G-*equivariant*.) If in addition X and E are complex spaces, the map $\pi\colon E \to X$ is a holomorphic fiber bundle projection whose*

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fiber G is a complex Lie group acting holomorphically on E, and there exist G-equivariant local holomorphic trivializations θ as above, then we have a holomorphic principal G-bundle structure *on $E \to X$.*

The same definition applies in the smooth and the real analytic category. It follows that every G-bundle $\pi\colon E \to X$ admits a G-atlas $\Theta = \{(U_j, \theta_j)\}_{j\in J}$ where $\mathcal{U} = \{U_j\}_{j\in J}$ is an open cover of X, every map

$$\theta_j\colon E|_{U_j} = \pi^{-1}(U_j) \to U_j \times G, \quad j \in J$$

is a homeomorphism (or a biholomorphism) as in (7.1), and the transition maps $\theta_{ij} = \theta_i \circ \theta_j^{-1}\colon U_{ij} \times G \to U_{ij} \times G$ (where $U_{ij} = U_i \cap U_j$) are of the form

$$\theta_{ij}(x, v) = \big(x, g_{ij}(x)\, v\big), \quad x \in U_{ij},\ v \in G, \tag{7.2}$$

where $g_{ij}\colon U_{ij} \to G$ are continuous (resp. holomorphic) maps to G. The collection $g = \{g_{ij}\}_{i,j\in J}$ clearly satisfies the 1-cocycle condition

$$g_{ii} = 1, \quad g_{ij}g_{ji} = 1, \quad g_{ij}g_{jk}g_{ki} = 1, \quad i, j, k \in J; \tag{7.3}$$

such g is a *1-cocycle* with values in the sheaf $\mathcal{C}^G = \mathcal{C}^G_X$ (resp. $\mathcal{O}^G = \mathcal{O}^G_X$) of germs of continuous (resp. holomorphic) maps $X \to G$. (The fibers E_x do not admit a natural Lie group structure since the transition maps are not Lie group automorphisms.) Conversely, every 1-cocycle (7.3) is the transition cocycle of a G-atlas on a principal G-bundle $E \to X$: Define E as the quotient of the disjoint union of the products $U_j \times G$ by the equivalence relation which identifies $(x, v) \in U_j \times G$ with $(x', v') \in U_i \times G$ if and only if $x = x' \in U_{ij}$ and $v' = g_{ij}(x)v$. The right multiplication of G on itself induces a right action of G on E, and the local trivializations are just the identifications of $U_j \times G$ with the corresponding subsets in E.

Two G-atlases $\Theta = \{(U_i, \theta_i)\}$ and $\Theta' = \{(U'_j, \theta'_j)\}$ on E are *equivalent* if their union $\Theta \cup \Theta'$ is again a G-atlas; this holds if and only if the transition maps $\theta'_j \circ \theta_i^{-1}$ are of the form (7.2). A *principal G-bundle structure* is determined by an equivalence class of G-atlases or, equivalently, by a maximal G-atlas obtained as the union of all atlases in an equivalence class.

If $\mathcal{V} = \{V_j\}_{j\in J'}$ is an open cover of X which is a refinement of $\mathcal{U} = \{U_j\}_{j\in J}$, with a refinement map $\alpha\colon J' \to J$ such that $V_j \subset U_{\alpha(j)}$ for every $j \in J'$, then any G-atlas on $\mathcal{U}$ induces a G-atlas on $\mathcal{V}$ by restriction. This allows us to consider any pair of G-atlases over a common refinement.

Suppose that $\Theta = \{(U_j, \theta_j)\}$ and $\Theta' = \{(U_j, \theta'_j)\}$ are equivalent G-atlases on an open cover $\mathcal{U} = \{U_j\}_{j\in J}$. For each index $j \in J$ the transition map $\theta'_j \circ \theta_j^{-1}\colon U_j \times G \to U_j \times G$ is given by $(x, v) \to (x, c_j(x)v)$ for some continuous map $c_j\colon U_j \to G$. If g_{ij} (resp. g'_{ij}) is the transition 1-cocycle of Θ (resp. of Θ') then $g'_{ij}c_j = c_i g_{ij}$ on U_{ij}; equivalently,

$$g'_{ij} = c_i g_{ij} c_j^{-1} \quad \text{on } U_{ij}. \tag{7.4}$$

Conversely, for every 1-cocycle $g = \{g_{ij}\}$ and every collection $c = \{c_j\}_{j\in J}$ of maps $c_j\colon U_j \to G$ (such c is called a 0-*cochain with values in the sheaf* $\mathcal{C}^G$) we can define another 1-cocycle $g' = \{g'_{ij}\}$ by (7.4). This operation is called *twisting of the 1-cocycle g by the 0-cochain c* and is denoted by

$$g' = c \,\square\, g.$$

If $\Theta = \{(U_j, \theta_j)\}_{j\in J}$ is a G-bundle atlas on $E \to X$ with the transition cocycle g, and if we define $\theta'_j\colon E|_{U_j} \to U_j \times G$ by $\theta'_j = C_j \circ \theta_j$, where $C_j(x, v) = (x, c_j(x)v)$ for $x \in U_j$ and $v \in G$, then $\Theta' = \{(U_j, \theta'_j)\}$ is a G-atlas on E with the transition cocycle $g' = c \,\square\, g$ which is equivalent to Θ.

Let $\pi\colon E \to X$ and $\pi'\colon E' \to X$ be principal G-bundles. A map $\Phi\colon E \to E'$ is a *G-bundle map* if $\pi' \circ \Phi = \pi$ and Φ is G-equivariant:

$$\Phi(eg) = \Phi(e)g, \quad e \in E,\ g \in G.$$

In any pair of G-bundle charts such Φ is given as left multiplication by a map from the base into the group G. More precisely, given G-atlases $\Theta = \{(U_j, \theta_j)\}$ for E and $\Theta' = \{(U_j, \theta'_j)\}$ for E' over an open cover $\mathcal{U} = \{U_j\}_{j\in J}$ of X, each map $\Phi_j = \theta'_j \circ (\Phi|_{E|_{U_j}}) \circ \theta_j^{-1}\colon U_j \times G \to U_j \times G$ is of the form

$$\Phi_j(x, v) = \big(x, \phi_j(x)v\big), \quad x \in U_j,\ v \in G$$

for some continuous (resp. holomorphic) map $\phi_j\colon U_j \to G$. The 0-cochain $\phi = \{\phi_j\}_{j\in J}$ with values in the sheaf $\mathcal{C}^G$ (resp. $\mathcal{O}^G$) then represent Φ with respect to the given pair of G-atlases. Denoting the transition 1-cocycle of Θ, Θ' by (g_{ij}), (g'_{ij}), respectively, we have

$$g'_{ij}\phi_j = \phi_i\, g_{ij} \quad \text{on } U_{ij}. \tag{7.5}$$

Conversely, any 0-cochain $\phi = \{\phi_j\}$ satisfying (7.5) defines a G-bundle map $\Phi\colon E \to E'$. Such Φ is automatically invertible and hence a G-bundle isomorphism; its inverse given by the 0-cochain $\{\phi_j^{-1}\}$. Note that (7.5) means $g' = \phi \,\square\, g$ exactly as in (7.4). This is explained as follows: By Φ we pull the G-atlas Θ' back to the G-atlas $\widetilde{\Theta} = \Phi^*\Theta'$ (on E) with the bundle charts

$$\widetilde{\theta}_j = \theta'_j \circ \Phi|_{E_{U_j}}\colon E_j \to U_j \times G.$$

The transition cocycle of $\widetilde{\Theta}$ is g' (the same as for Θ'), and ϕ is now seen as the transition 0-cochain between Θ and $\widetilde{\Theta}$. Similarly we can use Φ to push forward a G-atlas Θ on E to a G-atlas $\Phi_*\Theta$ on E'.

Denote by $Z^1(\mathcal{U}, \mathcal{C}^G)$ the set of all 1-cocycles, and by $C^0(\mathcal{U}, \mathcal{C}^G)$ the set of all 0-cochains on a cover $\mathcal{U}$ with values in the sheaf $\mathcal{C}^G$. We declare 1-cocycles $g, g' \in Z^1(\mathcal{U}, \mathcal{C}^G)$ equivalent if and only if $g' = c \,\square\, g$ for some $c \in C^0(\mathcal{U}, \mathcal{C}^G)$. Denote by $H^1(\mathcal{U}; \mathcal{C}^G)$ the set of equivalence classes. Then

$$H^1(X;\mathcal{C}^G) = \varinjlim H^1(\mathcal{U};\mathcal{C}^G)$$

(the direct limit over open covers, called *the first Čech cohomology group with the coefficients in* $\mathcal{C}^G$) is precisely the set of equivalence classes of topological principal G-bundles on X. In the holomorphic category,

$$H^1(X;\mathcal{O}^G) = \varinjlim H^1(\mathcal{U};\mathcal{O}^G)$$

is the set of all holomorphic equivalence classes of holomorphic principal G-bundles on X. The natural map $H^1(X;\mathcal{O}^G) \to H^1(X;\mathcal{C}^G)$ induced by the inclusion $\mathcal{O}^G \hookrightarrow \mathcal{C}^G$ is in general neither injective nor surjective as we have already seen in §5.3 in the case when $G = \mathbb{C}^*$.

7.2 The Oka-Grauert Principle for G-Bundles

In this section we prove the following theorem of Grauert.

Theorem 7.2.1. [222, 223] *Let X be a Stein space. For every complex Lie group G the holomorphic and the topological classes of principal G-bundles over X are in one to one correspondence, i.e., the map $H^1(X;\mathcal{O}^G) \to H^1(X;\mathcal{C}^G)$ is bijective. More precisely:*

(i) *For every topological G-bundle isomorphism $\Phi_0\colon E \to E'$ of holomorphic principal G-bundles over X there is a homotopy $\Phi_t\colon E \to E'$ ($t \in [0,1]$) of topological G-bundle isomorphisms such that Φ_1 is a holomorphic G-bundle isomorphism. If in addition Φ_0 is holomorphic on a closed complex subvariety $M \subset X$ and also in an open neighborhood of a compact $\mathcal{O}(X)$-convex subset K of X, then the homotopy $\{\Phi_t\}$ can be chosen fixed on $E|_M$ and uniformly close to Φ_0 on $E|_K$.*

(ii) *Every principal G-bundle $E \to X$ admits a holomorphic G-bundle structure. If K and M are as in (i), then a holomorphic G-bundle structure on $E|_{K\cup M}$ extends to a holomorphic G-bundle structure on E.*

Parts (i) and (ii) in Theorem 7.2.1 (without the approximation and the interpolation condition) are equivalent to the following:

(i') Given 1-cocycles $g, g' \in Z^1(\mathcal{U},\mathcal{O}^G)$ and a 0-cochain $\phi^0 \in C^0(\mathcal{U},\mathcal{C}^G)$ such that $\phi^0 \,\square\, g = g'$, there exists a homotopy of 0-cochains $\phi^t \in C^0(\mathcal{U},\mathcal{C}^G)$ ($t \in [0,1]$) such that $\phi^t \,\square\, g = g'$ for all $t \in [0,1]$ and $\phi^1 \in C^0(\mathcal{U},\mathcal{O}^G)$.

(ii') Let $\mathcal{U}$ be a Stein open cover of X. For every 1-cocycle $g \in Z^1(\mathcal{U},\mathcal{C}^G)$ there exists a 0-cochain $c \in C^0(\mathcal{U},\mathcal{C}^G)$ such that $c \,\square\, g \in Z^1(\mathcal{U},\mathcal{O}^G)$ is a holomorphic 1-cocycle.

Hence the natural map of Čech cohomology groups $H^1(\mathcal{U};\mathcal{O}^G) \to H^1(\mathcal{U};\mathcal{C}^G)$ is injective for every open cover by (i'), and is surjective for every Stein open cover by (ii'). The equivalence (i) $\Leftrightarrow$ (i') follows from the discussion in §7.1. For (ii) $\Leftrightarrow$ (ii') we need to observe that, by (i), a holomorphic G-bundle $E \to X$ which admits a topological atlas over a Stein open cover $\mathcal{U}$ of X also admits a holomorphic atlas over the same cover.

Proof (of Theorem 7.2.1, part (i)). Choose an open cover $\mathcal{U} = \{U_j\}_{j\in J}$ of X and holomorphic G-atlases $\Theta = \{(U_j, \theta_j)\}$ on E and $\Theta' = \{(U_j, \theta'_j)\}$ on E'. Let $g = \{g_{ij}\}$, $g' = \{g'_{ij}\}$ denote the holomorphic transition cocycles of Θ, Θ', respectively. The isomorphism $\Phi_0\colon E \to E'$ is then given by a 0-cochain $\phi^0 = \{\phi^0_j\}_{j\in J} \in C^0(\mathcal{U},\mathcal{C}^G)$ satisfying $\phi^0 \,\square\, g = g'$ (7.5); if Φ_0 is holomorphic over $M \cup V$ then ϕ^0_j is holomorphic on $U_j \cap (M \cup V)$ for every $j \in J$. We can view ϕ^0 as a section of the holomorphic fiber bundle $H \to X$ with fiber G which is defined over the cover $\mathcal{U}$ by the transition maps

$$\psi_{ij}(x, v) = \big(x, g'_{ij}(x)\, v\, g_{ij}^{-1}(x)\big), \quad x \in U_{ij},\ v \in G.$$

By Corollary 5.4.6 (p. 194) there exists a homotopy $\phi^t\colon X \to H$ ($t \in [0,1]$) of continuous sections such that ϕ^1 is a holomorphic section, the homotopy is fixed on M, and every ϕ^t is holomorphic on an open set $V' \supset K$ and is uniformly close to ϕ^0 on K (also uniformly in $t \in [0,1]$). Viewing ϕ^t again as a 1-cocycle on the cover $\mathcal{U}$ we have $\phi^t \,\square\, g = g'$, and hence ϕ^t determines a topological G-bundle isomorphism $\Phi_t\colon E \to E'$. At $t = 1$ we get a holomorphic G-bundle isomorphism $\Phi_1\colon E \to E'$. This proves part (i).

Applying the above argument with $E = E'$ and Φ_0 the identity map on E, considered as a topological equivalence between a pair of holomorphic atlases on E, we obtain the following corollary.

Corollary 7.2.2. *For any pair of topologically equivalent holomorphic G-atlases Θ_0 and Θ_1 on a principal G-bundle $E \to X$ there exists a homotopy of topological G-bundle automorphisms $\Phi_t\colon E \to E$ ($t \in [0,1]$) such that Φ_0 is the identity and $(\Phi_1)_*\Theta_0 = \Theta_1$. Furthermore, if K, V and M are as in Theorem 7.2.1 and if Θ_0 and Θ_1 define the same holomorphic structure on $E|_{M\cup V}$, then the homotopy Φ_t can be chosen fixed on M, holomorphic on an open set $V' \supset K$, and uniformly close to the identity on K.*

If $\{\Phi_t\}_{t\in[0,1]}$ is as in Corollary 7.2.2 then $\Theta_t = (\Phi_t)_*\Theta_0$ is a homotopy of holomorphic G-atlases on E connecting Θ_0 and Θ_1, and the holomorphic structure on $E|_{M\cup V'}$ defined by Θ_t does not depend on $t \in [0,1]$. $\square$

Proof (of Theorem 7.2.1, part (ii)). We first consider the special case when X is a Stein manifold and $M = \emptyset$. Then $X = \bigcup_{k=0}^{\infty} A_k$ where each A_k is a compact strongly pseudoconvex domain and $A_{k+1} = A_k \cup B_k$, where B_k is a small compact strongly pseudoconvex domain such that E is trivial over

an open neighborhood of B_k. If we are given a holomorphic structure on $E|_V$ over an open neighborhood V of a compact $\mathcal{O}(X)$-convex subset $K \subset X$, then A_0 may be chosen such that $K \subset A_0 \subset V$; otherwise take $A_0 = \emptyset$.

Assume inductively that we have an open set $V \supset A_k$ and a holomorphic G-bundle structure on $E|_V$. (This holds for $k = 0$.) By shrinking V around A_k we may assume that V is Stein. By the assumption there are an open Stein neighborhood $W \supset B_k$ and a trivialization $\theta\colon E|_W \to W \times G$; in particular, E is trivial over the Stein domain $U = V \cap W$. By part (i) there exists a trivialization $\theta'\colon E|_U \to U \times G$ which is holomorphic with respect to the holomorphic principal bundle structure on $E|_V$. Then

$$\Phi(x,v) := \theta' \circ \theta^{-1}(x,v) = \big(x, g(x)v\big), \quad x \in U,\ v \in G$$

for some continuous map $g\colon U \to G$. Since U is Stein, Corollary 5.4.6 on p. 194 provides a homotopy $g_t\colon U \to G$ ($t \in [0,1]$) such that $g_0 = g$ and g_1 is holomorphic. Set

$$\Phi_t(x,v) = \big(x, g_t(x)v\big), \quad x \in U,\ v \in G,\ t \in [0,1];$$

thus $\Phi_0 = \Phi$. Choose a smooth function $\chi\colon X \to [0,1]$ such that $\operatorname{supp}\chi \subset V$ and $\chi = 1$ in an open set $V' \supset A_k$. Define $\Psi_t\colon W \times G \to W \times G$ by

$$\Psi_t(x,v) = \big(x, g_{t\chi(x)}^{-1} g_0(x)\, v\big), \quad x \in W,\ v \in G,\ t \in [0,1].$$

Although g_t is only defined on $U = V \cap W$, Ψ_t extends to $W \times G$ as the identity on $(W \backslash V) \times G$ since $\chi = 0$ on $W \backslash V$. Then $\{\Psi_t\}_{t\in[0,1]}$ is a homotopy of continuous automorphisms of the trivial G-bundle over W, with Ψ_0 the identity map. Over $U' = V' \cap W$ we have $\Phi_t \circ \Psi_t = \Phi_0$ since $\chi = 1$ there. Now define

$$\theta_t = \Psi_t \circ \theta\colon E|_W \to W \times G, \quad t \in [0,1];$$

this is a homotopy of G-bundle charts on $E|_W$. On $E|_{U'}$ we have

$$\theta' = \Phi_0 \circ \theta = \Phi_t \circ \Psi_t \circ \theta = \Phi_t \circ \theta_t, \quad t \in [0,1].$$

Since Φ_1 is holomorphic, the holomorphic structure on $E|_{V'}$ agrees with the holomorphic structure on $E|_W$ induced by the chart $\theta_1\colon E|_W \to W \times G$. Our change of the trivialization of $E|_W$ did not affect the G-bundle structure of E, and we obtained a holomorphic G-bundle structure on $E|_{V' \cup W}$ extending the structure on $E|_{V'}$. Since $V' \cup W \supset A_{k+1}$, the induction may proceed. This completes the proof of (ii) in the special case.

This produces a holomorphic G-bundle structure by a stepwise extension that does not require any convergence arguments. Together with Corollary 7.2.2 we get the following lemma.

Lemma 7.2.3. (Gluing holomorphic G-bundle structures.) *Let $E \to X$ be a topological principal G-bundle over a complex manifold X, and let M be a*

closed complex subvariety of X. Let $V, W \subset X$ be open sets such that $E|_V$ and $E|_W$ admit holomorphic G-bundle structures that coincide over $M \cap V \cap W$, and that are topologically compatible with the structure on E. If $V \cap W$ is Stein then for every open set $V' \subset V$ with $\overline{V}' \cap W \subset V$ there is a holomorphic G-bundle structure on $E|_{V' \cup W}$ that agrees with the initial structures on $E|_{V'}$ (inherited from $E|_V$) and on $E|_{M \cap (V' \cup W)}$.

Proof. Since $U = V \cap W$ is Stein, Corollary 7.2.2 furnishes a homotopy of topological G-bundle isomorphisms $\Phi_t \colon E|_U \to E|_U$ $(t \in [0,1])$ such that Φ_0 is the identity, the homotopy is fixed on $M \cap U$, and Φ_1 pushes forward to holomorphic structure on $E|_W$ to the holomorphic structure on $E|_V$ over U. The assumption $\overline{V}' \cap W \subset V$ implies that there is a smooth function $\chi \colon W \to [0,1]$ which equals one on $\overline{V}' \cap W$ and equals zero on $W \backslash V$. Then $\widetilde{\Phi}_t(x) = \Phi_{t\chi(x)}(x) \colon E_x \to E_x$ $(x \in W,\ t \in [0,1])$ is a homotopy of topological G-bundle automorphisms of $E|_W$ such that $\widetilde{\Phi}_0$ is the identity, the homotopy is fixed on $M \cap W$, and $\widetilde{\Phi}_t = \Phi_t$ on $E|_{V' \cap W}$. The push-forward by $\widetilde{\Phi}_1$ of the holomorphic structure on $E|_W$ is a holomorphic structure on $E|_W$ that agrees with the structure on $E|_V$ over the set $V' \cap W$. $\square$

The following lemma provides the inductive step for extending a holomorphic structure over a Stein stratum in a stratified complex space.

Lemma 7.2.4. *Assume that $\pi \colon E \to X$ and $M, K \subset X$ are as in Theorem 7.2.1 (i). Let $M_1 \subset M_0$ be closed complex subvarieties of M such that $S = M_0 \backslash M_1$ is a Stein manifold. Given an open set $U_1 \supset M_1 \cup K$ and a holomorphic G-bundle structure on $E|_{M \cup U_1}$, there exist an open set $U_0 \supset M_0 \cup K$ and a holomorphic G-bundle structure on $E|_{M \cup U_0}$ which agrees with the given structure over $M \cup U_1'$ for an open neighborhood $U_1' \subset U_1$ of $K \cup M_1$.*

Proof. The Stein manifold $S = M_0 \backslash M_1$ admits an open Stein neighborhood W in X and a holomorphic retraction $\tau \colon W \to S$ (Theorem 3.3.3 on p. 67; here we may consider a neighborhood of S in X as embedded in some $\mathbb{C}^N$). Using τ we extend the given holomorphic structure on $E|_{M_0 \cap W}$ to a holomorphic structure on $E|_W$. By Lemma 7.2.3 we can assume that this extension agrees with the given structure on $E|_M$ over the set $M \cap W$. (It suffices to push it forward by a G-bundle automorphism of $E|_W$ which is the identity on $E|_{M_0 \cap W}$.) By shrinking $U_1 \supset K \cup M_1$ we may assume that U_1 is Stein. Lemma 7.2.3 then shows that we can correct the structure on $E|_W$ (without changing it on $E|_{M \cap W}$) to make it agree with the structure on $E|_{U_1}$ over $W \cap U_1'$, where U_1' is an open set with $K \cup M_1 \subset U_1' \subset U_1$. Setting $U_0 = U_1' \cup W \supset K \cup M_0$ we get a holomorphic G-bundle structure on $E|_{M \cup U_0}$.

Proof of Theorem 7.2.1 (ii): the general case. The first task is to find an open set $U_0 \supset M \cup K$ in X and a holomorphic structure on $E|_{U_0}$ which extends the given holomorphic structure on $E|_{K \cup M}$. Choose a stratification

$M = M_0 \supset M_1 \supset \cdots \supset M_k = \emptyset$ with smooth Stein strata $S_j = M_j \backslash M_{j+1}$. (To find such stratification, choose $f \in \mathcal{O}(X)$ that vanishes on M_{sing} but does not vanish identically on any irreducible component of maximal dimension in M, take $M_1 = M \cap \{f = 0\}$ and $S_1 = M \backslash M_1$, and continue inductively.) Assume inductively that for some index $j \in \{1, \ldots, k\}$ we have an open set $U_j \subset X$ containing $K \cup M_j$ and a holomorphic extension of the given structure to $E|_{M \cup U_j}$. (By the hypothesis this holds for $j = k$ since $M_k = \emptyset$.) Since $S_{j-1} = M_{j-1} \backslash M_j$ is a Stein manifold, Lemma 7.2.4 provides an extension to a holomorphic structure on $E|_{M \cup U_{j-1}}$, where U_{j-1} is a neighborhood of $K \cup M_{j-1}$. At $j = 0$ we obtain a holomorphic structure on $E|_{U_0}$ over an open set $U_0 \supset M \cup K$ which extends the original structure.

It remains to extend this structure to all of E. Choose a stratification $X = X_0 \supset X_1 \supset \cdots \supset X_m = M$ whose strata $X_j \backslash X_{j+1}$ are Stein manifolds and proceed again by a downward induction on j, starting with $j = m$. In the j-step we have a holomorphic structure over an open set $W_j \supset X_j \cup K$. (For $j = m$ take $W_m = U_0$.) By (the proof of) Proposition 5.12.1 (p. 224), W_j can be chosen such that the subvariety X_{j-1} is obtained by attaching strongly pseudoconvex bumps to $W_j \cap X_{j-1}$. We extend the holomorphic structure across each bump exactly as in the special case when X is a Stein manifold. In this way we extend the holomorphic structure first to $E|_{X_{j-1}}$, and then to an open neighborhood $W_{j-1} \supset X_{j-1} \cup K$ by Lemma 7.2.4. The induction may proceed. At $j = 0$ we obtain a holomorphic G-bundle structure on E that extends the original structure over $M \cup K$. $\square$

Remark 7.2.5. (A) In the case of principal $GL_r(\mathbb{C})$-bundles, Theorem 7.2.1 is equivalent to Theorem 5.3.1 (p. 294) since we can associate to every holomorphic vector bundle $E \to X$ with fiber $\mathbb{C}^r$ the principal $GL_r(\mathbb{C})$-bundle whose transition maps are precisely those of E, and vice versa.

(B) In our construction of a holomorphic G-structure, each step is accomplished by adapting the structure on the new set to the already given one. This sheme avoids any convergence process and is well adapted to the problem of extending the structure from a subvariety. For the classical approach with induction on analytic cubes see the papers [223, 78, 331].

(C) Another construction of a holomorphic vector bundle structure on a complex vector bundle $E \to X$ over a Stein space was given by Cornalba and Griffiths [92, Chapter D]. A holomorphic structure is determined by its sheaf of holomorphic sections. The latter is chosen as the sheaf of sections which are annihilated by a $(0,1)$-connection operator $\nabla = \nabla^{0,1}$ satisfying $\nabla^2 = 0$. Using such flat $(0,1)$-connection they define a complex structure operator on $\mathrm{T}E$ whose associated $\overline{\partial}$-operator on section $X \to E$ is just ∇. The condition $\nabla^2 = 0$ implies that this structure is involutive and hence integrable by the Newlander-Nirenberg theorem. This procedure works well over the regular part of X, and the induction argument uses a suitable stratification of X in essentially the same way as above. $\square$

7.3 Homomorphisms and Generators of Vector Bundles

Let $\pi\colon E \to X$, $\pi'\colon E' \to X$ be holomorphic vector bundles over a complex space X. A complex vector bundle map $\Phi\colon E \to E'$ is said to be of *maximal rank* if for every $x \in X$, $\Phi_x\colon E_x \to E'_x$ is a $\mathbb{C}$-linear map of maximal rank (injective if $\operatorname{rank} E \le \operatorname{rank} E'$, resp. surjective if $\operatorname{rank} E \ge \operatorname{rank} E'$). The following is an Oka principle for vector bundle maps of maximal rank.

Theorem 7.3.1. *Let $\pi\colon E \to X$ and $\pi'\colon E' \to X$ be holomorphic vector bundles of rank r, resp. r', over an n-dimensional Stein space X.*

(a) *A topological complex vector bundle map $\Phi\colon E \to E'$ of maximal rank is homotopic through complex vector bundle maps of maximal rank to a holomorphic vector bundle map of maximal rank.*

(b) *If Φ in (a) is holomorphic over a closed complex subvariety X' of X and in a neighborhood of a compact $\mathcal{O}(X)$-convex subset $K \subset X$ then the homotopy in (a) can be chosen to be fixed on X', holomorphic on a neighborhood of K, and uniformly close to Φ on K.*

(c) *If $|r-r'| \ge \left[\frac{n}{2}\right]$ then there exists a holomorphic vector bundle map $E \to E'$ of maximal rank. Furthermore, given a closed complex subvariety Y of X, any holomorphic vector bundle map $E|_Y \to E'|_Y$ of maximal rank extends to a holomorphic vector bundle map $E \to E'$ of maximal rank.*

Proof. We identify complex vector bundle maps $\Phi\colon E \to E'$ with sections of the holomorphic vector bundle $\operatorname{Hom}(E, E') = E^* \otimes_{\mathbb{C}} E'$ over the base X. The fiber of the latter bundle is $M_{r',r} = \mathbb{C}^{r'r}$, the set of all complex $r' \times r$ matrices. The set $\Sigma \subset M_{r',r}$ consisting of all matrices of less than maximal rank is an algebraic subvariety which is given locally at every point by at least $|r'-r|+1$ independent equations (this is the number of independent minors of maximal size); hence Σ has complex codimension $q = |r'-r|+1$ in $M_{r',r}$. (See Example 1.2.7 on p. 6 or Lemma 7.9.2 on p. 324 below.) The complement $V_{r',r} = M_{r',r} \backslash \Sigma$ (the Stiefel manifold) is the homogeneous space of the Lie group $GL_N(\mathbb{C})$, $N = \max\{r, r'\}$, acting by matrix multiplication (on the left if $r \ge r'$, resp. on the right if $r \le r'$).

Let A be the closed complex subvariety of (the total space of) $\operatorname{Hom}(E, E')$ whose fiber $A_x \subset \operatorname{Hom}(E_x, E'_x) = E_x^* \otimes E'_x$ at a point $x \in X$ corresponds to the set $\Sigma \subset M_{r',r}$ under an identification $M_{r',r} \cong E_x^* \otimes E'_x$. (We can use any pair of holomorphic vector bundle charts to make this identification; the set A_x will be independent of the choices.) Then a complex vector bundle map $\Phi\colon E \to E'$ is of maximal rank if and only $\Phi_x \in \operatorname{Hom}(E_x, E'_x) \backslash A_x$ for all $x \in X$. The conclusion (a) now follows from Corollary 5.4.6 (p. 194) which is a special case of Theorem 5.4.4 (p. 193). The latter result also contains the additions concerning the approximation and interpolation to get part (b).

Part (c) follows from Corollary 5.14.3 on p. 234; the condition $2 \operatorname{codim} A_x > \dim X$, which insures the existence of a continuous section of $\operatorname{Hom}(E, E') \backslash A$, reads $2(|r'-r|+1) > n$ which is equivalent to $|r-r'| \ge \left[\frac{n}{2}\right]$. $\square$

Applying Theorem 7.3.1 (c) with E' being a trivial bundle we obtain

Corollary 7.3.2. *Let $E \to X$ be a holomorphic vector bundle of rank r over an n-dimensional Stein space X.*

1. *There is a holomorphic vector bundle embedding $E \hookrightarrow X \times \mathbb{C}^{r+\left[\frac{n}{2}\right]}$.*
2. *If $r = \left[\frac{n}{2}\right] + d$ for some $d > 0$ then there exists a holomorphic vector bundle embedding $X \times \mathbb{C}^d \hookrightarrow E$. Hence E admits d pointwise linearly independent holomorphic sections, and it splits into a holomorphic direct sum $E = E' \oplus \mathbb{T}^d$ of a holomorphic vector bundle E' of rank $\left[\frac{n}{2}\right]$ and a trivial vector bundle of rank d.*
3. *There is a surjective holomorphic vector bundle map $X \times \mathbb{C}^{r+\left[\frac{n}{2}\right]} \to E$; hence E admits $r + \left[\frac{n}{2}\right]$ holomorphic sections which span each fiber E_x.*

All statements in Corollary 7.3.2 also hold in the relative form: A holomorphic vector bundle map of the given type extends from a restricted bundle $E|_Y$ over a subvariety $Y \subset X$ to the whole bundle. The last statement in part 3 is a special case of Theorem 7.4.3 below, due to Forster and Ramspott [157], which gives an upper bound for the number of generators of a coherent analytic sheaf on a Stein space.

Example 7.3.3. In [158, Satz 3] the authors show that for every pair of integers $n, r \geq 1$ there exists a holomorphic vector bundle $E \to X = (\mathbb{C}^*)^n$ of rank r which is not generated by fewer than $r + \left[\frac{n}{2}\right]$ sections. □

In [158] it is also shown how dual Chern classes of a complex vector bundle can be used to estimate the number of generators. (Recall that the total dual Chern class $\tilde{c}(E)$ of a complex vector bundle E is defined by $c(E) \cdot \tilde{c}(E) = 1$, where $c(E) = 1 + c_1(E) + c_2(E) + \cdots$ is the total Chern class of E.) For example, we have the following result that we quote without proof.

Theorem 7.3.4. [158, Satz 2] *Let X be an n-dimensional Stein space and $E \to X$ a holomorphic vector bundle of rank r. Given an integer $q \geq \frac{n}{2}$, the module $H^0(X; E)$ can be generated by $r + q - 1$ holomorphic sections if and only if the q-th dual Chern class $\tilde{c}_q(E) = 0$ vanishes.*

We state the following special case of Theorem 7.3.1 for future applications.

Corollary 7.3.5. *Let E, E' be holomorphic vector bundles over a Stein space X. Every topological complex vector bundle isomorphism $\Phi\colon E \to E'$ is homotopic (in the space of topological complex vector bundle isomorphisms) to a holomorphic one. If Φ is holomorphic over a closed complex subvariety X' of X and over a (neighborhood of a) $\mathcal{O}(X)$-convex subset K of X, then the homotopy can be chosen fixed on X' and holomorphic and close to Φ on K.*

The Oka principle in Corollary 7.3.5 has the following converse due to M. Putinar [404]. For references to earlier converse results see the MathSciNet review MR1237106 of the paper [404]. We do not prove this result here.

Theorem 7.3.6. [404] *Let X be a Stein space of finite Zariski dimension d and let $r = \max\{\dim_{\mathbb{C}} X, d-1\}$. Then an open subset Y of X is Stein if and only if every analytic, topologically trivial vector bundle of rank r on Y is analytically trivial.*

We denote by $\mathrm{Vect}^k_{\mathbb{R}}(X)$ (resp. $\mathrm{Vect}^k_{\mathbb{C}}(X)$) the topological isomorphism classes of vector bundles of rank k over a CW-complex X. By $\mathbb{T}^k_{\mathbb{R}}$ (resp. $\mathbb{T}^k_{\mathbb{C}}$) we denote the trivial vector bundle of rank k over a given base. Recall the following standard result (see e.g. [270]).

Theorem 7.3.7. *Let X be an n-dimensional CW complex.*

(a) *The map $\mathrm{Vect}^k_{\mathbb{R}}(X) \to \mathrm{Vect}^{k+r}_{\mathbb{R}}(X)$, $E \mapsto E \oplus \mathbb{T}^r_{\mathbb{R}}$ $(k, r \geq 1)$ is surjective if $k \geq n$, and is bijective if $k \geq n+1$.*

(b) *The map $\mathrm{Vect}^k_{\mathbb{C}}(X) \to \mathrm{Vect}^{k+r}_{\mathbb{C}}(X)$, $E \mapsto E \oplus \mathbb{T}^r_{\mathbb{C}}$ $(k, r \geq 1)$ is surjective when $k \geq \left[\frac{n}{2}\right]$, and is bijective when $k \geq \left[\frac{n+1}{2}\right]$. In particular, if $E \to X$ is a nontrivial complex vector bundle of rank $k \geq \left[\frac{n+1}{2}\right]$ then the bundle $E \oplus \mathbb{T}^r_{\mathbb{C}}$ is also nontrivial for each $r \in \mathbb{N}$.*

This result applies in particular when X is an n-dimensional Stein space; in that case Corollary 7.3.5 gives $\mathrm{Vect}^r_{\mathrm{holo}}(X) = \mathrm{Vect}^r_{\mathbb{C}}(X)$. Thus part (b) of Theorem 7.3.7 implies the following.

Corollary 7.3.8. ([157], [431, Satz 1.4]) *If X is an n-dimensional Stein space then the map*

$$\mathrm{Vect}^k_{\mathrm{holo}}(X) \to \mathrm{Vect}^{k+r}_{\mathrm{holo}}(X), \quad E \mapsto E \oplus \mathbb{T}^r_{\mathbb{C}}, \qquad k, r \geq 1$$

is surjective for $k \geq \left[\frac{n}{2}\right]$, and is bijective for $k \geq \left[\frac{n+1}{2}\right]$. In particular, if E, E' are holomorphic vector bundles of rank $k \geq \left[\frac{n+1}{2}\right]$ over X such that $E \oplus \mathbb{T}_{\mathbb{C}} \cong E' \oplus \mathbb{T}_{\mathbb{C}}$, then also $E \cong E'$.

Corollary 7.3.9. *Let X be a Stein space, $D \subset \mathbb{C}^n$ a contractible open set containing the origin, and $\rho\colon X \times D \to X \times \{0\}$ the map $\rho(x, \zeta) = (x, 0)$. A holomorphic vector bundle $E \to X \times D$ is holomorphically equivalent to the bundle $\widetilde{E} = \rho^*(E|_{X \times \{0\}}) \to X \times D$.*

Proof. There is a topological isomorphism $\Phi\colon E \to \widetilde{E}$ of complex vector bundles which is the identity over $X \times \{0\}$ (where the two bundles coincide). Thus Φ may be considered as a continuous section, holomorphic over $X \times \{0\}$, of an associated fiber bundle $Z \to X \times D$ (see the proof of Theorem 5.3.1 on p. 190). By Corollary 5.4.6 (p. 194) we can deform Φ to a holomorphic section $\widetilde{\Phi}$ by a homotopy which is fixed on $X \times \{0\}$. Then $\widetilde{\Phi}$ defines a holomorphic isomorphism $E \to \widetilde{E}$ which equals the identity over $X \times \{0\}$. $\square$

From Theorem 6.13.1 we obtain the following classification of vector bundles over 1-convex complex spaces.

Corollary 7.3.10. *Let X be a 1-convex space with the exceptional set S.*

(a) *If $E \to X$ is a topological complex vector bundle which is holomorphic over a neighborhood of S, then there exists a holomorphic vector bundle structure on E that is holomorphically compatible with the existing structure over a small neighborhood of S.*

(b) *Assume that $E \to X$ and $E' \to X$ are holomorphic vector bundles. If $\Phi_0 \colon E \to E'$ is a topological complex vector bundle isomorphism that is holomorphic over a neighborhood of S in X, then Φ_0 can be deformed to a holomorphic vector bundle isomorphism $\Phi_1 \colon E \to E'$ by a homotopy of complex vector bundle isomorphisms $\Phi_t \colon E \to E'$ ($t \in [0,1]$) that is holomorphic over a neighborhood of S and is fixed over S.*

We now give a few examples illustrating the failure of the Oka-Grauert principle for complex line bundles over non-Stein manifolds.

Example 7.3.11. (A) Let $X = \mathbb{C}^2 \backslash \{0\}$. Then $H^1(X;\mathcal{O})$ is infinite dimensional. (In fact, for a domain $X \subset \mathbb{C}^n$, the group $H^p(X;\mathcal{O})$ is either zero or infinite dimensional for any $p \geq 1$ by Laufer [327].) Clearly X is homotopic to the 3-sphere and hence $H^1(X;\mathbb{Z}) = H^2(X;\mathbb{Z}) = 0$. The top row in the exact sequence (5.2) (p. 189) shows that the exponential map $\mathcal{O} \to \mathcal{O}^*$ induces an isomorphism $H^1(X;\mathcal{O}) = H^1(X;\mathcal{O}^*)$. Thus the family of holomorphic line bundle structures on $X \times \mathbb{C}$ is infinite dimensional, and all these holomorphic line bundles are topologically trivial.

(B) Let X be a compact Riemann surface of genus g. Then $H^1(X;\mathcal{O}) = \mathbb{C}^g$, $H^1(X;\mathbb{Z}) = \mathbb{Z}^{2g}$, and the homomorphism $H^1(X;\mathbb{Z}) \to H^1(X;\mathcal{O})$ induced by the inclusion $\mathbb{Z} \hookrightarrow \mathcal{O}$ is an injective homomorphism $\mathbb{Z}^{2g} \hookrightarrow \mathbb{C}^g$ whose image is a lattice $\Gamma \subset \mathbb{C}^g$. Furthermore, $H^2(X;\mathbb{Z}) = \mathbb{Z}$ is generated by the fundamental class of X, and $H^2(X;\mathcal{O}) = 0$ by the Dolbeault isomorphism theorem. (See e.g. [154] for these facts.) From the exponential sheaf sequence (see (5.2) on p. 189) we obtain an exact sequence

$$0 \to \mathbb{T}^g = \mathbb{C}^g/\Gamma \hookrightarrow H^1(X;\mathcal{O}^*) \xrightarrow{c_1} \mathbb{Z} \to 0.$$

The kernel $\mathbb{T}^g \subset H^1(X;\mathcal{O}^*)$ of the 1-st Chern class map c_1 is the *Jacobi torus* of X which parametrizes the distinct classes of holomorphic line bundle structures on the topologically trivial bundle $X \times \mathbb{C}$.

(C) Example (A) above shows that the map $H^1(X;\mathcal{O}^*) \xrightarrow{c_1} H^2(X;\mathbb{Z})$ is non-injective on certain simple domains in $\mathbb{C}^2$. It fails to be surjective on any compact Kähler manifold X of dimension > 1 with $H^{(2,0)}(X) \neq 0$. This follows from the Hodge decomposition $H^2(X;\mathbb{C}) = H^{(2,0)}(X) \oplus H^{(1,1)}(X) \oplus H^{(0,2)}(X)$ since the class $c_1(E) \in H^2(X,\mathbb{Z})$ of any complex line bundle $E \to X$ is

represented by the de Rham cohomology class $\left[\frac{\mathrm{i}}{2\pi}\Theta\right] \in H^{(1,1)}(X)$, where Θ is the Chern curvature form of any Hermitian metric on E.

(D) There also exist domains in $\mathbb{C}^n$ on which the first Chern class map c_1 fails to be surjective (J. Winkelmann, private communication). Let $S \subset \mathbb{P}^n$ be a compact projective surface with a nontrivial holomorphic 2-form, so $H^2(S;\mathbb{Z}) \to H^2(S;\mathcal{O})$ is not the zero map. (There are plenty of examples: any complex 2-torus, any K3 surface, any hypersurface in $\mathbb{P}^3$ of degree at least four, etc.) Consider the diagram

$$\begin{array}{ccc} Z & \hookrightarrow & \mathbb{C}_*^{n+1} \\ {\scriptstyle\pi}\downarrow & & \downarrow{\scriptstyle\pi} \\ S & \hookrightarrow & \mathbb{P}^n \end{array}$$

where $\mathbb{C}_*^{n+1} = \mathbb{C}^{n+1}\backslash\{0\}$, $\pi\colon \mathbb{C}_*^{n+1} \to \mathbb{P}^n$ is the universal $\mathbb{C}^*$-bundle over $\mathbb{P}^n$, and $Z = \pi^{-1}(S)$. Choose a domain $X \subset \mathbb{C}_*^{n+1}$ which admits a strong deformation retraction onto Z. Then $c_1\colon H^1(X;\mathcal{O}^*) \to H^2(X;\mathbb{Z})$ is not surjective. To see this, consider the commutative diagram

$$\begin{array}{ccc} H^2(X;\mathbb{Z}) & \longrightarrow & H^2(X;\mathcal{O}) \\ \| & & \downarrow \\ H^2(Z;\mathbb{Z}) & \longrightarrow & H^2(Z;\mathcal{O}) \\ \uparrow & & \uparrow \\ H^2(S;\mathbb{Z}) & \longrightarrow & H^2(S;\mathcal{O}) \end{array}$$

By the choice of S the map in the bottom row is not the zero map. Using that $Z \to S$ is a principal $\mathbb{C}^*$ bundle it can be shown that the bottom right vertical map $H^2(S;\mathcal{O}) \to H^2(Z;\mathcal{O})$ is injective; we omit the details. Hence the composed map $H^2(S;\mathcal{O}) \to H^2(Z;\mathcal{O}) \to H^2(Z;\mathcal{O})$ is nonzero. It follows by diagram chasing that the map $H^2(X;\mathbb{Z}) \to H^2(X;\mathcal{O})$ in the top row is nonzero as well. From exactness of (5.2) we see that $c_1\colon H^1(X;\mathcal{O}^*) \to H^2(X;\mathbb{Z})$ is surjective if and only if the next map $H^2(X;\mathbb{Z}) \to H^2(X;\mathcal{O})$ in the long exact cohomology sequence is the zero map. It follows that the map $H^1(X;\mathcal{O}^*) \to H^2(X;\mathbb{Z})$ is not surjective.

(E) K. Stein [460] gave an example of a nontrivial complex line bundle over a domain of holomorphy X in $\mathbb{C}^n$ whose restriction to each relatively compact subset of X is trivial. □

7.4 Generators of Coherent Analytic Sheaves

In this section we look for a minimal number of generators of a coherent analytic sheaf $\mathcal{F}$ over a finite dimensional Stein space X. For a locally trivial sheaf (i.e., the sheaf of sections of a holomorphic vector bundle) the answer is given by Corollary 7.3.2. For a general coherent analytic sheaf this problem

was treated by Forster and Ramspott [157] using the Oka principle for *Oka pairs of sheaves* from [156]. As we shall see, the problem of decreasing the number of generators reduces to the problem of finding a holomorphic section of a certain stratified holomorphic fiber bundle with homogeneous fibers, so their results follow from Theorems 5.4.4 and 5.14.1 in Chapter 5.

Let $(X, \mathcal{O})$ be a complex space and $\mathcal{F}$ a coherent analytic sheaf over X. Let $\mathfrak{m}_x$ denote the maximal ideal in the ring $\mathcal{O}_x$. Then $\mathfrak{m}_x\mathcal{F}_x$ is a submodule of $\mathcal{F}_x$ and the quotient $L_x(\mathcal{F}) = \mathcal{F}_x/\mathfrak{m}_x\mathcal{F}_x$ is a finite dimensional vector space over $\mathcal{O}_x/\mathfrak{m}_x\mathcal{O}_x = \mathbb{C}$. Let us denote by

$$\lambda_x \colon \mathcal{F}_x \to L_x(\mathcal{F}) = \mathcal{F}_x/\mathfrak{m}_x\mathcal{F}_x \cong \mathbb{C}^{k_x}$$

the quotient projection. A collection $f_1, \ldots, f_r \in \mathcal{F}_x$ generates the stalk $\mathcal{F}_x$ as an $\mathcal{O}_x$-module if and only if their images $\lambda_x(f_1), \ldots, \lambda_x(f_r) \in L_x(\mathcal{F})$ span the vector space $L_x(\mathcal{F})$ over $\mathbb{C}$. By choosing a basis for the vector space $L_x(\mathcal{F})$ we may represent $\lambda_x(f_j)$ by a row vector with k_x components. We assemble the germs $f_j \in \mathcal{F}_x$ into a vector $f = (f_1, \ldots, f_r)^t \in \mathcal{F}_x^r$ and consider $\lambda_x(f) \in M_{r,k_x}$ as a complex $r \times k_x$ matrix.

There exists a stratification $X = X_0 \supset X_1 \supset X_2 \supset \cdots$ (possibly infinite) by closed complex subvarieties

$$X_x = X_k(\mathcal{F}) = \{x \in X \colon \dim L_x(\mathcal{F}) \geq k\}. \tag{7.6}$$

If $\mathcal{F}$ is finitely generated, in the sense that there exist finitely many global sections $f_1, \ldots, f_r \in \Gamma(X, \mathcal{F}) = \mathcal{F}(X)$ whose germs at any point $x \in X$ generate the stalk $\mathcal{F}_x$ as an $\mathcal{O}_x$-module, then clearly $r \geq \dim L_x(\mathcal{F})$ for every $x \in X$ and hence

$$k(\mathcal{F}) := \sup_{x \in X} \dim L_x(\mathcal{F}) < +\infty. \tag{7.7}$$

Conversely, if X is a Stein space then $\mathcal{F}$ is finitely generated if and only if $k(\mathcal{F}) < +\infty$ [150, Corollary 4.4]. The proof uses Cartan's Theorem A and proceeds by induction on the dimension of the exceptional subvariety of X over which a given collection of sections of $\mathcal{F}$ fails to generate.

Assume now that X is a Stein space and the sheaf $\mathcal{F}$ is finitely generated, say by the sections $f_1, \ldots, f_r \in \mathcal{F}(X)$. We have the short exact sequence

$$0 \longrightarrow \mathcal{R} \longrightarrow \mathcal{O}^r \xrightarrow{\sigma} \mathcal{F} \longrightarrow 0$$

where σ is the sheaf epimorpism $\sigma(\alpha_1, \ldots, \alpha_r) = \sum_{j=1}^r \alpha_j f_j$ and $\mathcal{R} = \ker \sigma$ is the sheaf of relations (2.4). We have $\operatorname{rank} \lambda_x(f) = k_x = \dim L_x(\mathcal{F})$ for every $x \in X$, and the above stratification of X ends with the term $X_{r+1} = \emptyset$. We wish to know whether $\mathcal{F}$ is generated by a smaller number, say $s < r$, of sections $g_1, \ldots, g_s \in \mathcal{F}(X)$. (A clear necessary condition is that $s \geq k(\mathcal{F})$.) By Cartan's Theorem B any s-tuple $g = (g_1, \ldots, g_s)^t \in \mathcal{F}(X)^s$ can be expressed as $g = Af$ for some holomorphic matrix function $A \colon X \to M_{s,r}$. We can easily see that

$$\lambda_x(g) = \lambda_x(Af) = A(x)\lambda_x(f), \quad x \in X.$$

Hence the components of $g = Af \in \mathcal{F}(X)^s$ generate $\mathcal{F}_x$ if and only if

$$\operatorname{rank}(A(x)\lambda_x(f)) = \dim L_x(\mathcal{F}).$$

Thus A must be a holomorphic section of the projection $\pi\colon E \to X$ where

$$E = E(\mathcal{F}, f, s) = \big\{(x, T) \in X \times M_{s,r}\colon \operatorname{rank}\,(T\lambda_x(f)) = \dim L_x(\mathcal{F})\big\}. \quad (7.8)$$

In [157] this is called the *Endromisbündel*.

Recall that $V_{s,k} \subset M_{s,k}$ is the Stiefel manifold of all $s \times k$ matrices of full rank k (assuming that $s \geq k$). Since $GL_s(\mathbb{C})$ acts transitively on $V_{s,k}$ by left multiplication, $V_{s,k}$ is an Oka manifold.

Proposition 7.4.1. [157, Satz 1] *Let $\mathcal{F}$ be a coherent analytic sheaf with r generators over a complex space X, and let $X_k \subset X$ be the subvariety defined by (7.6). The set E (7.8) is an open subset of $X \times M_{s,r}$, and the restriction of the submersion $E \to X$ to the subvariety $Y_k = X_k \backslash X_{k+1}$ is a holomorphic fiber bundle with the fiber $F_{s,r,k} = V_{s,k} \times M_{s,r-k} \cong V_{s,k} \times \mathbb{C}^{s(r-k)}$.*

The proof uses elementary sheaf theory and linear algebra; we refer the reader to the original paper. This means that $E \to X$ is a stratified fiber bundle whose fibers $F_{s,r,k}$ are Oka manifolds, and hence the Oka principle applies to sections $X \to E$ in view of Theorem 5.4.4. In particular:

Corollary 7.4.2. [157, Satz 6] *Let X be a Stein space and $\mathcal{F}$ a coherent analytic sheaf over X with generators $f = (f_1, \ldots, f_r)$. Then the $\mathcal{O}(X)$-module $\mathcal{F}(X) = H^0(X; \mathcal{F})$ admits s generators for some integer $s \in \mathbb{N}$ if and only if the bundle $E(\mathcal{F}, f, s)$ (7.8) admits a continuous section.*

By using elementary obstruction theory we now find a condition for the existence of a continuous section $X \to E = E(\mathcal{F}, f, s)$ (compare with Corollary 5.14.3 on p. 234). Observe that each variety $Y_k = X_k \backslash X_{k+1}$ can be substratified such that all strata are Stein manifolds. We need that for each integer $k = 1, \ldots, k(\mathcal{F})$ the homotopy groups $\pi_q(F_{s,r,k}) = \pi_q(V_{s,k})$ of the fiber over Y_k vanish for all $q = 0, 1, \ldots, \dim Y_k - 1$. Since $\pi_q(V_{s,k}) = 0$ for $q \leq 2(s-k)$ (Example 1.2.7 on p. 6), we see that E admits a continuous section when

$$\dim Y_k - 1 \leq 2s - 2k, \quad \forall k = 1, \ldots, k(\mathcal{F}).$$

This gives the following result of Forster and Ramspott [157], [159, Satz 6].

Theorem 7.4.3. *Let $\mathcal{F}$ be a coherent analytic sheaf with finitely many generators over a finite dimensional Stein space X. Set*

$$s = s(\mathcal{F}) = \max_{k \in \mathbb{N}} \left\{ k + \left[\frac{1}{2} \dim Y_k \right] \right\} \quad (7.9)$$

where $Y_k = \{x \in X\colon \dim L_x(\mathcal{F}) = k\}$. Then $\mathcal{F}$ admits s generators.

If $\mathcal{F} = \mathcal{O}^k$ is a locally free sheaf or rank k then $X = Y_k$ and we get $s = k + \left[\frac{n}{2}\right]$ generators; this coincides with part 2 in Corollary 7.3.2. This number can not be decreased in general.

In this connection we mention another interesting problem that was first considered by Henri Cartan in his 1940 paper on holomorphic matrices [73].

Suppose that $\mathcal{F}$ is a coherent analytic sheaf on a Stein space X and $f = (f_1, \dots, f_r)$, $g = (g_1, \dots, g_r)$ are two sets of generators for the algebra of global sections $\mathcal{F}(X)$. It follows from Cartan's Theorem B that there exist holomorphic $r \times r$ matrix functions $A, B\colon X \to M_{r,r}(\mathbb{C}) \cong \mathbb{C}^{r^2}$ such that $Af = g$ and $Bg = f$. However, Cartan showed by a simple example over the polydisc in $\mathbb{C}^2$ that in general there does not exist an *invertible* holomorphic matrix $A\colon X \to GL_r(\mathbb{C})$ satisfying $Af = g$. The obstruction in his example is purely topological, and he conjectured that this problem has an analytic solution if it has a topological solution. The Oka principle for this problem was established in 1968 by Forster and Ramspott [159]. Define

$$E = \big\{(x, A)\colon x \in X,\ A \in GL_r(\mathbb{C}),\ A\lambda_x(f) = \lambda_x(g)\big\} \subset X \times GL_r(\mathbb{C})$$

and let $\pi\colon E \to X$ be the projection $\pi(x, A) = x$.

Theorem 7.4.4. [159, Satz 7] *If there exists a continuous section $X \to E$, then there also exists a holomorphic section $T = (\mathrm{Id}_X, A)\colon X \to E$. In such case the holomorphic matrix function $A\colon X \to GL_r(\mathbb{C})$ satisfies $Af = g$.*

With X_k defined by (7.6) and $Y_k = X_k \backslash X_{k+1}$, the restriction $E|_{Y_k}$ is a holomorphic fiber bundle with the fiber $F_{r,k}$ consisting of all matrices of the form $\begin{pmatrix} I & Q \\ 0 & R \end{pmatrix}$ with $R \in GL_{r-k}(\mathbb{C})$ and $Q \in M_{k,r-k}(\mathbb{C})$ [159, Hilfssatz 6]. Even though the projection $\pi\colon E \to X$ is not a submersion, and hence not a stratified fiber bundle in our sense, it is still possible to extend a holomorphic section $X_{k+1} \to E$ to an open neighborhood of X_{k+1} in X and then proceed in the usual way to extend it to a holomorphic section over X_k.

If the matrix function A in Theorem 7.4.4 is null-homotopic (this is always the case if X is contractible) then the homotopy $A_t\colon X \to GL_r(\mathbb{C})$ $(t \in [0, 1])$ from $A = A_0$ to $A_1 = I$ provides an isotopy $A_t f$ of generators of $H^0(X; \mathcal{F})$ connecting f to g. This problem is treated in more details in [159].

7.5 The Number of Equations Defining a Subvariety

Every affine algebraic subvariety Y of the affine complex space $\mathbb{A}^n$ is the common zero set of finitely many polynomials in n complex variables. In the holomorphic category, Cartan's Theorem A implies that every closed complex subvariety Y in a Stein manifold X (or in a finite dimensional Stein space) is the common zero set of finitely many global holomorphic functions on X. We now look at the following classical problem.

Question 7.5.1. How many holomorphic functions are needed to define a closed complex subvariety Y in a Stein space X? What is the answer when $X = \mathbb{C}^n$?

We show how the Oka principle sheds light on this subject in the holomorphic category. The reader is referred to the papers [29, 50, 153, 431, 432] for additional results on complete intersections in the holomorphic, affine algebraic and differentiable categories.

There are two natural interpretations of 'defining a subvariety'. In the set theoretic sense we are looking for functions $f_1, \ldots, f_k \in \mathcal{O}(X)$ (or polynomials in $\mathbb{C}[z_1, \ldots, z_n]$ in the affine case) such that

$$Y = \big\{x \in X \colon f_1(x) = 0, \ldots, f_k(x) = 0\big\}. \tag{7.10}$$

If such Y is of pure codimension k in X then Y is said to be a *set theoretic complete intersection*. Replacing $\mathbb{C}^n$ by $\mathbb{P}^n$ and functions by projective hypersurfaces gives the corresponding notion in the projective category.

A classical theorem of Kronecker [310] asserts that any affine algebraic subvariety Y of an affine algebraic variety $X \subset \mathbb{C}^n$ is the common zero set of $n+1$ polynomials on $\mathbb{C}^n$ restricted to X; if $X = \mathbb{C}^n$ then n polynomials suffice [468, 125]. Kronecker's problem whether every affine algebraic curve is an algebraic set theoretic complete intersection is apparently still open. On the analytic side it was proved by Grauert [220] that each complex subvariety in an n-dimensional Stein space is the common zero set of $n+1$ functions; this was improved to n by Forster and Ramspott [155] (Corollary 7.5.5 below).

A stronger notion of defining a subvariety is the ideal theoretic one. Let $\mathcal{J} = \mathcal{J}_Y \subset \mathcal{O}_X$ denote the ideal sheaf of Y in X. In addition to (7.10) we now ask that $f_1, \ldots, f_k$ generate the ideal $\mathcal{J}_x \subset \mathcal{O}_x$ of Y at any point $x \in Y$. This is equivalent to asking that the functions $f_1, \ldots, f_k$ generate the algebra of global sections $\mathcal{J}(Y)$ of the sheaf $\mathcal{J}$. (Note that $\mathcal{J}(Y)$ is a closed ideal in the Fréchet algebra $\mathcal{O}(X)$. A collection of functions generating the ideal $\mathcal{J}(Y)$ also define Y in the sense of (7.10), but the converse need not hold.) A subvariety Y in an n-dimensional Stein space X can be defined by finitely many functions in the ideal theoretic sense if and only if its ideal sheaf $\mathcal{J}$ satisfies the condition $k(\mathcal{J}) < +\infty$ (7.7); in such case Theorem 7.4.3 gives a bound on the number of generators.

Definition 7.5.2. *Let Y be a complex subvariety of pure codimension k in a complex space X.*

(a) *Y is an* ideal theoretic complete intersection *if its ideal sheaf $\mathcal{J} = \mathcal{J}_Y$ is generated by k sections $f_1, \ldots, f_k \in \mathcal{J}(X)$.*

(b) *Y is an (ideal theoretic)* local complete intersection *if the ideal $\mathcal{J}_x$ at any point $x \in Y$ is generated by k elements.*

Note that every complex submanifold of a complex manifold is a local complete intersection.

Proposition 7.5.3. *Any complete intersection Stein submanifold Y in a parallelizable complex manifold X is itself parallelizable.*

Proof. From $\mathrm{T}Y \oplus \mathrm{N}_{Y/X} = \mathrm{T}X|_Y \cong Y \times \mathbb{C}^n$ and $\mathrm{N}_{Y/X}$ trivial we infer by Corollary 7.3.8 that $\mathrm{T}Y$ is trivial. □

Assume now that Y is a local complete intersection of pure codimension k in an an n-dimensional Stein space X. The sheaf $\mathcal{J}/\mathcal{J}^2$ is then locally free of rank k over Y, and hence it is the sheaf of sections of a holomorphic vector bundle over Y. The dual bundle of $\mathcal{J}/\mathcal{J}^2$ is called the *normal bundle of Y in X*, denoted $\mathrm{N}_{Y/X}$. (In the nonsingular case this is the usual normal bundle of Y.) The sheaf $\mathcal{J}/\mathcal{J}^2$ is clearly trivial over $X\backslash Y$. Recall from §7.4 that $L_x(\mathcal{J}) = \mathcal{J}_x/\mathfrak{m}_x\mathcal{J}_x$; in the present case we have $\dim L_x(\mathcal{J}) = k$ if $x \in Y$ and $\dim L_x(\mathcal{J}) = 1$ if $x \in X\backslash Y$. Stratifying X by $X \supset Y \supset \emptyset$ we see that the number $s = s(\mathcal{J})$ (7.9) equals

$$s = \max\left\{k + \left[\frac{n-k}{2}\right], 1 + \left[\frac{n}{2}\right]\right\}.$$

Hence Theorem 7.4.3 implies the following result [157, Satz 9].

Theorem 7.5.4. *Let Y be a local complete intersection subvariety of pure codimension k in an n-dimensional Stein space X. Then the ideal sheaf $\mathcal{J}_Y$ is generated by $\left[\frac{n+k}{2}\right]$ functions if $k \geq 2$, and by $1 + \left[\frac{n}{2}\right]$ functions if $k = 1$.*

Corollary 7.5.5. (a) *Every nonsingular analytic curve in a connected Stein manifold of dimension $n \geq 3$ is a complete intersection.*

(b) *Every discrete set in a Stein manifold X is defined by $n = \dim X$ functions.*

Assume now that Y is a local complete intersection in X. When is Y also a global complete intersection? According to Boratyński, a local complete intersection in $\mathbb{C}^n$ is a set theoretic complete intersection; see [153]. However, if we ask the question in the ideal theoretic sense then a clear necessary condition is that the normal bundle $\mathrm{N}_{Y/X}$ be trivial. Conversely, triviality of $\mathrm{N}_{Y/X}$ guarantees that Y is a complete intersection in an open neighborhood $U \subset X$ of Y, defined by a holomorphic map $f = (f_1, \ldots, f_k): U \to \mathbb{C}^k$ with $Y = f^{-1}(0)$. If f can be chosen such that it extends continuously to all of X and it maps $X\backslash Y$ to $\mathbb{C}^k\backslash\{0\}$, then we can deform f to a holomorphic map $\widetilde{f}: X \to \mathbb{C}^k$ which defines Y as a complete intersection in X.

Theorem 7.5.6. (Oka principle for complete intersections.) *Let X be a Stein manifold and $f = (f_1, \ldots, f_k): X \to \mathbb{C}^k$ a continuous map that is holomorphic in an open neighborhood $U \subset X$ of the set $Y = f^{-1}(0)$. Given a coherent analytic sheaf of ideals $\mathcal{S} \subset \mathcal{O}_X$ supported on Y, there exists a holomorphic map $\widetilde{f}: X \to \mathbb{C}^k$ which is homotopic to f relative to Y such that $\widetilde{f}^{-1}(0) = X$ and $\widetilde{f} - f$ is a section of $\mathcal{S}$. If f defines Y as an ideal theoretic complete intersection in U, then $\widetilde{f}$ can be chosen to define Y as an ideal theoretic complete intersection in X.*

Proof. Since the manifold $\mathbb{C}^k\backslash\{0\}$ is complex homogeneous and hence an Oka manifold, this follows from Theorem 7.6.1 in the following section. $\square$

The problem of extending f continuously to X as a zero-free map on $X\backslash Y$ can be treated by the obstruction theory as in the following result.

Theorem 7.5.7. [157, Satz 10], [432, Theorem 2.5] *Assume that X is a Stein manifold and $Y \subset X$ is a pure k-codimensional local complete intersection with trivial normal bundle $\mathrm{N}_{Y/X}$. Then the ideal sheaf $\mathcal{J}_Y$ is generated by $k+1$ functions. If in addition $H^q(X, Y; \pi_q(S^{2k-1})) = 0$ for all $q \geq 2k-1$ then Y is an ideal theoretic complete intersection in X.*

Proof. The conditions imply that Y is a complete intersection in an open neighborhood $U \subset X$, defined by k functions $f_1, \ldots, f_k \in \mathcal{O}(U)$. By Cartan's Theorem B we can find functions $g_j \in \mathcal{O}(X)$ such that $g_j - f_j$ is a section of the coherent sheaf of ideals $\mathcal{J}_Y^2$ in a neighborhood of Y for every $j = 1, \ldots, k$. This implies that the map $g = (g_1, \ldots, g_k)\colon X \to \mathbb{C}^k$ defines Y as a complete intersection in an open neighborhood of Y. We have $g^{-1}(0) = Y \cup Y'$ where the (possibly empty) subvariety Y' is disjoint from Y. Choose a function $g_{k+1} \in \mathcal{O}(X)$ that vanishes on Y and equals one on Y'; then the functions $g_1, \ldots, g_{k+1}$ generate the ideal sheaf $\mathcal{J}_Y$.

Since $\mathbb{C}^k\backslash\{0\}$ is homotopy equivalent to the sphere S^{2k-1}, the cohomological condition implies that $f = (f_1, \ldots, f_k)$ extends to a continuous map $\widetilde{f}\colon X \to \mathbb{C}^k$ which agrees with f in a neighborhood $U_1 \subset U$ of Y and satisfies $Y = \widetilde{f}^{-1}(0)$ (see Remark 5.14.4 on p. 234), so Theorem 7.5.6 applies. $\square$

Corollary 7.5.8. [157, Satz 11, Satz 12] *Let Y be a closed complex submanifold of a Stein manifold X. If $\dim Y < \frac{1}{2}\dim X$, then Y is an ideal theoretic complete intersection in X if and only if the normal bundle $\mathrm{N}_{Y/X}$ is trivial. If $X = \mathbb{C}^n$ then the same conclusion holds when $\dim Y \leq \frac{2}{3}(n-1)$.*

Proof. Let $n = \dim X$ and $k = n - \dim Y$. Then $2k > n$ and $H^q(Y; A) = H^{q+1}(X; A) = 0$ for $q \geq n$ and for every Abelian group A (see §3.11). From the exact cohomology sequence

$$\cdots \longrightarrow H^q(Y; A) \longrightarrow H^{q+1}(X, Y; A) \longrightarrow H^{q+1}(X; A) \longrightarrow \cdots$$

we infer that $H^{q+1}(X, Y; A) = 0$ for all $q \geq 2k-1$. Taking $A = \pi_q(S^{2k-1})$ we see that the condition in Theorem 7.5.7 holds.

In the case $X = \mathbb{C}^n$ we have $H^{q+1}(\mathbb{C}^n; A) = 0$ for all $q \geq 0$. The above exact cohomology sequence gives $H^q(Y; A) = H^{q+1}(\mathbb{C}^n, Y; A)$ for $q \geq 1$, and hence $H^{q+1}(\mathbb{C}^n, Y; A) = 0$ for $q > d = \dim Y$. If $d \leq \frac{2}{3}(n-1)$ then $d < 2k-1$ and the conclusion follows from Theorem 7.5.7. $\square$

Example 7.5.9. In [169] an example is given of a closed complex submanifold $X^3 \subset \mathbb{C}^5$ which is a differentiable complete intersection, but is not a holomorphic complete intersection. In fact, for any compact orientable 2-surface M of genus $g \geq 2$ there exists a three dimensional Stein manifold X, homotopy equivalent to M, whose tangent bundle $\mathrm{T}X$ is trivial as a real vector bundle, but is nontrivial as a complex vector bundle. The normal bundle of any proper holomorphic embedding of such X in $\mathbb{C}^5$ or $\mathbb{C}^7$ is trivial as a real vector bundle, but nontrivial as a complex vector bundle. $\square$

7.6 Elimination of Intersections

The Oka principle for complete intersections (Theorem 7.5.6) is a special case of Theorem 7.6.1 below. The main problem is the following.

Assume that X is a Stein space, $h\colon Z \to X$ is a holomorphic submersion onto X, and $\Sigma \subset Z$ is a closed complex subvariety of Z whose fiber $\Sigma_x = \Sigma \cap Z_x$ is a proper subvariety of $Z_x = h^{-1}(x)$ for every $x \in X$. Suppose that $f\colon X \to Z$ is a holomorphic section and that

$$f^{-1}(\Sigma) = Y \cup Y', \quad Y \cap Y' = \emptyset,$$

where Y and Y' are unions of connected components of $f^{-1}(\Sigma)$. The problem is to find a homotopic deformation of f that is fixed on Y and that pushes Y' out of the intersection set with Σ.

Theorem 7.6.1. (The Oka principle for elimination of intersections.) *Under the above assumptions, suppose that there exists a homotopy of continuous sections $g_t\colon X \to Z$ $(t \in [0,1])$ which is fixed near Y such that $g_0 = f$ and $g_1^{-1}(\Sigma) = Y$. Let $\mathcal{S} \subset \mathcal{O}_X$ be a coherent sheaf of ideals supported on Y.*

(a) *If the submersion $h\colon Z\backslash\Sigma \to X$ is stratified subelliptic over $X\backslash Y$ (Def. 6.2.1 on p. 243), then there exists a holomorphic section $\widetilde{f}\colon X \to Z$ such that $(\widetilde{f})^{-1}(\Sigma) = Y$ and $\widetilde{f}$ is $\mathcal{S}$-tangent to f along Y (Def. 1.3.2 on p. 9).*

(b) *If in addition the submersion $h\colon Z \to X$ is stratified subelliptic over $X\backslash Y$, then there is a homotopy of holomorphic sections $f_t\colon X \to Z$ such that $f_0 = f$, $f_1 = \widetilde{f}$, and f_t is $\mathcal{S}$-tangent to f along Y for every $t \in [0,1]$.*

The same conclusions hold if the submersions $Z \to X$ and $Z\backslash\Sigma \to X$ are stratified fiber bundles with Oka fibers.

Proof. If $Y = \emptyset$, this is the Oka principle for sections $X \to Z\backslash\Sigma$ avoiding Σ (a special case of Theorem 6.2.2). The proof in the general case requires minor adjustments that we now explain. Let $\mathcal{J}_\Sigma$ denote the sheaf of ideals of the subvariety $\Sigma \subset Z$. Let $\mathcal{R} \subset \mathcal{O}_X$ be the coherent sheaf of ideals whose stalk $\mathcal{R}_x$ at each point $x \in Y$ is the pull-back of $\mathcal{J}_\Sigma$ by f, while for $x \in X\backslash Y$ we

take $\mathcal{S}_x = \mathcal{O}_{X,x}$. (More precisely, if $x \in Y$ and if $\mathcal{J}_{\Sigma,f(x)}$ is generated by the germs $h_1, \ldots, h_m$, we take the functions $h_j \circ f$ $(j = 1, \ldots, m)$ as the generators of $\mathcal{S}_x$.) It suffices to prove the theorem with $\mathcal{S}$ replaced by the product sheaf $\mathcal{SR}\mathcal{J}_Y$, where $\mathcal{J}_Y$ is the ideal sheaf of Y in X. It is easily seen that for every section $f' \colon U \to Z|_U$ in an open set $U \supset Y$ such that f' is $\mathcal{S}$-tangent to f along Y there is a smaller open set $V \supset Y$ such that $(f')^{-1}(\Sigma) = Y$. (See the proof of Theorem 1.3 in [169] for the details when $\Sigma \subset \mathbb{C}^n$.)

Assume now that $g_1 \colon X \to Z$ is a continuous section which agrees with f near Y and satisfies $g_1^{-1}(\Sigma) = Y$. We apply to g_1 the same procedure as in the proof of Theorem 6.2.2 to deform it to a holomorphic section $\widetilde{f} \colon X \to Z$ that is $\mathcal{S}$-tangential to f along Y and satisfies $\widetilde{f}^{-1}(\Sigma) = Y$. The condition that the sections in the homotopy are holomorphic near Y and $\mathcal{S}$-tangential to f implies that no new intersections with Σ appear in a sufficiently small open neighborhood V of Y, while on $X \backslash V$ we can insure this by the construction.

In a typical inductive step we choose a relatively compact Runge domain $W \Subset X$ and stratify it by a descending chain of closed complex subvarieties $W = W_0 \supset W_1 \supset \cdots \supset W_m = Y \cap W$ such that the submersion $Z \backslash \Sigma \to X$ is subelliptic over each (smooth) stratum $W_j \backslash W_{j+1}$. This allows the construction of a holomorphic section with the stated properties by following the proof of Theorem 6.2.2, resp. of Theorem 5.4.4 in the case of a stratified fiber bundle with Oka fibers. (See §5.12 and §6.12.) The proof is concluded by another induction over a normal exhaustion of X by Runge domains.

By the construction we have a continuous homotopy of sections $X \to Z$ connecting f to $\widetilde{f}$ that are holomorphic near Y and $\mathcal{S}$-tangential to f along Y. If $Z \to X$ is a stratified subelliptic submersion over $X \backslash Y$, Theorem 6.2.2 gives a deformation (with fixed ends at $t = 0, 1$) to a homotopy of holomorphic sections that are $\mathcal{S}$-tangential to f along Y. $\square$

Corollary 7.6.2. *Assume that $h \colon Z \to X$ is a holomorphic fiber bundle with fiber $\mathbb{C}^n$ and $\Sigma \subset Z$ is a closed complex subvariety that is locally uniformly tame (see Proposition 6.4.14 on p. 256) and satisfies* $\dim \Sigma_x \le n-2$ *for every point $x \in X$. Then the conclusion of Theorem 7.6.1 holds. The same is true if the fiber is $\mathbb{P}^n$ and* $\dim \Sigma_x \le n-2$ *for every $x \in X$.*

Proof. In the first case with fiber $\mathbb{C}^n$, the submersion $Z \backslash \Sigma \to X$ is stratified elliptic by Proposition 6.4.14, and obviously $Z \to X$ is elliptic. In the second case with fiber $\mathbb{P}^n$ the subvariety Σ has algebraic fibers, and $E \backslash \Sigma \to X$ is a stratified subelliptic submersion by Proposition 6.4.15. $\square$

The following is a special case of Corollary 7.6.2; complete intersections correspond to $\Sigma = \{0\} \subset \mathbb{C}^n$. It is easily seen that this fails if Σ is a non-avoidable discrete set in $\mathbb{C}^n$ (see Theorem 4.7.2 on p. 120).

Corollary 7.6.3. *If Σ is a closed complex analytic subvariety of $\mathbb{C}^n$ such that $\mathbb{C}^n \backslash \Sigma$ is an Oka manifold then the conclusion of Theorem 7.6.1 holds for maps from any Stein space X to $\mathbb{C}^n$.*

7.7 The Holomorphic Vaserstein Problem

In this section we outline the solution to *Gromov's Vaserstein problem* obtained by Ivarsson and Kutzschebauch [272, 273]. For this application one needs the Oka principle for sections of stratified elliptic submersions.

Let I_m denote the identity matrix of size m. It is well known that $SL_m(\mathbb{C})$ is generated by elementary matrices $I_m + \alpha e_{ij}$ $i \neq j$, i.e., matrices with 1's on the diagonal and all entries outside the diagonal except one equal to zero. Equivalently, every $A \in SL_m(\mathbb{C})$ can be written as a finite product of upper and lower diagonal unipotent matrices (in interchanging order). The same question for matrices in $SL_m(R)$, where R is a commutative ring instead of the field $\mathbb{C}$, can be much more delicate. For example, if R is the ring of complex valued functions (continuous, smooth, algebraic or holomorphic) from a space X, the problem amounts to finding for a given map $f\colon X \to SL_m(\mathbb{C})$ a factorization as a product of upper and lower triangular unipotent matrices

$$f(x) = \begin{pmatrix} 1 & 0 \\ G_1(x) & 1 \end{pmatrix} \begin{pmatrix} 1 & G_2(x) \\ 0 & 1 \end{pmatrix} \cdots \begin{pmatrix} 1 & G_N(x) \\ 0 & 1 \end{pmatrix},$$

where the G_i are maps $G_i\colon X \to \mathbb{C}^{m(m-1)/2}$. Since any product of such matrices is homotopic to a constant map (multiplying each entry outside the diagonals by $t \in [0,1]$ we get a homotopy to the identity matrix), one has to assume that the given map $f\colon X \to SL_m(\mathbb{C})$ is homotopic to a constant map.

This problem has been studied in the case of polynomials of n variables. For $n = 1$, i.e., $f\colon X \to SL_m(\mathbb{C})$ a polynomial map (the ring R equals $\mathbb{C}[z]$), it is an easy consequence of the fact that $\mathbb{C}[z]$ is an Euclidean ring that such f factors through a product of upper and lower diagonal unipotent matrices. For $m = 2$ it was shown by Cohn [87] that the matrix

$$\begin{pmatrix} 1 - z_1 z_2 & z_1^2 \\ -z_2^2 & 1 + z_1 z_2 \end{pmatrix} \in SL_2(\mathbb{C}[z_1, z_2])$$

does not decompose as a finite product of unipotent matrices.

For $m \geq 3$ and any n it is a result of Suslin [481] that any matrix in $SL_m(\mathbb{C}^{[n]})$ decomposes as a finite product of unipotent matrices. Further results in the algebraic setting can be found in [481] and [239]. For a connection to the Jacobian conjecture see [528].

In the case of continuous complex valued functions on a topological space X the problem was partially solved by Thurston and Vaserstein [492], and completely solved by Vaserstein [502, Theorem 4].

It is natural to consider the Vaserstein problem for rings of holomorphic functions on Stein spaces, in particular on $\mathbb{C}^n$. This problem was posed by Gromov [237, sec 3.5.G] and was solved by Ivarsson and Kutzschebauch.

Theorem 7.7.1. [272, 273] *Let X be a finite dimensional reduced Stein space and let $f\colon X \to SL_m(\mathbb{C})$ be a null-homotopic holomorphic mapping. Then there exist $k \in \mathbb{N}$ and holomorphic mappings $G_1, \ldots, G_k\colon X \to \mathbb{C}^{m(m-1)/2}$ such that*

$$f(x) = \begin{pmatrix} 1 & 0 \\ G_1(x) & 1 \end{pmatrix} \begin{pmatrix} 1 & G_2(x) \\ 0 & 1 \end{pmatrix} \cdots \begin{pmatrix} 1 & G_k(x) \\ 0 & 1 \end{pmatrix}$$

is a product of upper and lower diagonal unipotent matrices. In particular, if X is contractible then this holds for every holomorphic map $f\colon X \to SL_m(\mathbb{C})$.

An effective version of Theorem 7.7.1 with a bound on the number of factors was obtained in [274] for the group $SL_2(\mathbb{C})$.

By the definition of the Whitehead K_1-group [421, p. 61] the above result has the following consequence.

Corollary 7.7.2. *Let $(X, \mathcal{O}_X)$ be a finite dimensional reduced Stein space that is topologically contractible. Then $SK_1(\mathcal{O}(X))$ is trivial, and the determinant induces an isomorphism* $\det\colon K_1(\mathcal{O}(X)) \to \mathcal{O}(X)^\star$.

Proof (of Theorem 7.7.1 for maps $X \to SL_2(\mathbb{C})$). Define the map $\Psi_k\colon \mathbb{C}^k \to SL_2(\mathbb{C})$ by

$$\Psi_k(z_1, \ldots, z_k) = \begin{pmatrix} 1 & 0 \\ z_1 & 1 \end{pmatrix} \begin{pmatrix} 1 & z_2 \\ 0 & 1 \end{pmatrix} \cdots \begin{pmatrix} 1 & z_k \\ 0 & 1 \end{pmatrix}$$

(matrix product). We wish to find a holomorphic map $G = (G_1, \ldots, G_k)\colon X \to \mathbb{C}^k$ such that the following diagram commutes:

$$\begin{tikzcd} & \mathbb{C}^k \arrow[d, "\Psi_k"] \\ X \arrow[ur, "G"] \arrow[r, "f"'] & SL_2(\mathbb{C}) \end{tikzcd}$$

Vaserstein's result gives a continuous lifting of f. We shall deform it to a holomorphic lifting by applying the Oka principle to the diagram

$$\begin{tikzcd} & \mathbb{C}^k \arrow[d, "\pi_2 \circ \Psi_k = \Phi_k"] \\ X \arrow[ur, "F"] \arrow[r, "\pi_2 \circ f"'] & \mathbb{C}^2 \backslash \{0\} \end{tikzcd}$$

where $\pi_2\colon SL_2(\mathbb{C}) \to \mathbb{C}^2\backslash\{0\}$ is the projection of a matrix to its second row.

The following lemma is easily verified.

Lemma 7.7.3. *The mapping $\Phi_k = \pi_2 \circ \Psi_k\colon \mathbb{C}^k \to \mathbb{C}^2\backslash\{0\}$ is a holomorphic submersion exactly at points $\mathbb{C}^k\backslash S_k$, where for $k \geq 2$,*

$$S_k = \{(z_1, \ldots, z_k) \in \mathbb{C}^k : z_1 = \cdots = z_{k-1} = 0\}.$$

The submersion $\Phi_k = \pi_2 \circ \Psi_k\colon \mathbb{C}^k\backslash S_k \to \mathbb{C}^2\backslash\{0\}$ is surjective when $k \geq 3$.

The following lemma enables us to apply the Oka principle.

Lemma 7.7.4. *If $k \geq 3$ then the map*

$$\Phi_k \colon \mathbb{C}^k \backslash S_k \to \mathbb{C}^2 \backslash \{0\} \tag{7.11}$$

is a stratified elliptic submersion.

Proof. Write $\Phi_k(z_1, \dots, z_k) = (P_k(z_1, \dots, z_k), Q_k(z_1, \dots, z_k))$. Note that

$$\begin{aligned} P_k(z_1, \dots, z_k) &= P_{k-1}(z_1, \dots, z_{k-1}), \\ Q_k(z_1, \dots, z_k) &= Q_{k-1}(z_1, \dots, z_{k-1}) + z_k P_{k-1}(z_1, \dots, z_{k-1}) \end{aligned}$$

when k is even, and $P_k = z_k P_{k-1} + Q_{k-1}$, $Q_k = Q_{k-1}$ when k is odd. We concentrate on the case when k is even; the odd case is handled similarly. Let $(a, b) \in \mathbb{C}^2 \backslash \{0\}$ and study the fiber $P_k = a$, $Q_k = b$. When $a \neq 0$, we have $z_k = \frac{1}{a}(b - Q_{k-1})$, so the fiber is a graph in $\mathbb{C}^{k-1} \times \mathbb{C}_{z_k}$ over the fiber $\{P_{k-1} = a\} \subset \mathbb{C}^{k-1}$. When $a = 0$, the fiber is $\Phi_{k-1}^{-1}(0, b) \times \mathbb{C}_{z_k}$ and, since in this case $b \neq 0$, $\Phi_{k-1}^{-1}(0, b)$ is a graph in $\mathbb{C}^{k-2} \times \mathbb{C}_{z_{k-1}}$ over $\{Q_{k-2} = b\} \subset \mathbb{C}^{k-2}$. So in both cases the fiber is a graph over a hypersurface given by a single polynomial equation.

We now find globally integrable tangential holomorphic vector fields spanning the tangent space at each point of the fiber; a dominating spray is then obtained by composing their flows. We stratify $\mathbb{C}^2 \backslash \{0\}$ by

$$\mathbb{C}^2 \backslash \{0\} \supset \big\{(a, b) \in \mathbb{C}^2 \backslash \{0\} \colon a = 0\big\} \supset \emptyset.$$

We construct a spray over the highest stratum $\{(a, b) \in \mathbb{C}^2 \backslash \{0\} \colon a \neq 0\}$; the stratum $\{(a, b) \in \mathbb{C}^2 \backslash \{0\} \colon a = 0\}$ is handled similarly. We need to find complete holomorphic vector fields that are tangential to $\{P_{k-1} = a \neq 0\}$ and span the tangent space at each point. We claim that the vector fields

$$V_{ij} = \frac{\partial P_{k-1}}{\partial z_i} \frac{\partial}{\partial z_j} - \frac{\partial P_{k-1}}{\partial z_j} \frac{\partial}{\partial z_i}, \quad 1 \leq i < j \leq k$$

have the required properties. They obviously are tangential, and one easily checks the spanning property. (It helps to observe that the nonsmooth points of $\{P_{k-1} = a\}$ are contained in S_k.) To see that they are globally integrable, note that P_{k-1} is a polynomial that is no more than linear in each variable separably, so $\partial P_{k-1}/\partial z_i$ is independent of z_i and $\partial P_{k-1}/\partial z_j$ is independent of z_j. Hence the flow of V_{ij} is the solution of a system of two independent differential equations that are both globally integrable. $\square$

We can now conclude the proof. By [502, Theorem 4] we have a continuous map $F \colon X \to \mathbb{C}^{k'}$ for some $k' \in \mathbb{N}$ such that $f(x) = \Psi_{k'}(F(x))$. By Lemma 7.7.3 we see that $F = (F_1, \dots, F_{k'}, 1, 0, -1)$ gives a map from X into $\mathbb{C}^{k'+3} \backslash S_{k'+3}$.

Putting $k = k'+3$ we have $f(x) = \Psi_k(F(x))$. It follows that $\Psi_k(F(x))f(x)^{-1} = I_2$. Since (7.11) is a stratified elliptic submersion, Corollary 6.14.4 (p. 289) gives a holomorphic map $G\colon X \to \mathbb{C}^k \backslash S_k$ such that

$$\Phi_k(F(x)) = \pi_2(f(x)) = \Phi_k(G(x));$$

that is, the matrices $\Psi_k(F(x))$ and $\Psi_k(G(x))$ have the same last row. Hence

$$\Psi_k(G(x))f(x)^{-1} = \begin{pmatrix} a(x) & b(x) \\ 0 & 1 \end{pmatrix}$$

with a, b holomorphic. Moreover, $a = 1$ since the matrix is in $SL_2(\mathbb{C})$. Thus

$$f(x) = \Psi_k(G(x)) \begin{pmatrix} 1 & -b(x) \\ 0 & 1 \end{pmatrix}$$

which solves the problem. $\square$

The proof for $m > 2$ is by induction on the size of the matrices. The difficult part is to show Lemma 7.7.4. One can reduce stratawise the m polynomial equations which define the fibers of Φ_k to a single equation. The special form of the occurring polynomials allows one to find finitely many complete holomorphic vector fields which span the tangent space of the fibers at each point. The details can be found in [273].

Since a stratified elliptic submersion also enjoys the parametric Oka property (POP) according to Corollary 6.14.4 (p. 289), the same proof yields the following parametric version of Theorem 7.7.1.

Theorem 7.7.5. *Assume that X is a finite dimensional reduced Stein space, P is a compact subset of $\mathbb{R}^m$, and $f\colon P \times X \to SL_n(\mathbb{C})$ is a null-homotopic X-holomorphic mapping. Then there exist a natural number N and X-holomorphic mappings $G_1, \ldots, G_N\colon P \times X \to \mathbb{C}^{n(n-1)/2}$ such that*

$$f(p,x) = \begin{pmatrix} 1 & 0 \\ G_1(p,x) & 1 \end{pmatrix} \begin{pmatrix} 1 & G_2(p,x) \\ 0 & 1 \end{pmatrix} \cdots \begin{pmatrix} 1 & G_N(p,x) \\ 0 & 1 \end{pmatrix}.$$

7.8 Transversality Theorems for Holomorphic Maps

If X and Y are smooth manifolds and $k \in \{0, 1, 2, \ldots\}$ then $\mathrm{J}^k(X, Y)$ is the manifold of k-jets of smooth maps $X \to Y$ (see §1.10). If Z is a smooth closed submanifold of $\mathrm{J}^k(X, Y)$ then for a generically chosen smooth map $f\colon X \to Y$ the k-jet extension map $\mathrm{j}^k f\colon X \to \mathrm{J}^k(X, Y)$ is transverse to Z ([488, 489]; for extensions see [4, 219, 364, 495, 514]).

The analogous result only rarely holds for holomorphic maps between complex manifolds. Indeed, the transversality theorem for 1-jets of holomorphic

maps $\mathbb{C}^n \to Y$ implies that all Kobayashi-Eisenman metrics of Y vanish identically, and that Y is *dominable* by $\mathbb{C}^n$ if $n = \dim Y$. If such Y is compact and connected, its Kodaira dimension $\kappa = \kappa_Y$ [33, p. 29] satisfies $\kappa < \dim Y$, i.e., *Y is not Kodaira general type* [70, 301].

In the positive direction, the basic transversality theorem holds for holomorphic maps to any complex manifold with a submersive family of holomorphic self-maps [4]; a classical example is *Bertini's theorem* that almost all projective hyperplanes in $\mathbb{P}^n$ intersect a given complex submanifold $Z \subset \mathbb{P}^n$ transversally [219, p. 150]. The jet transversality theorem holds for holomorphic maps of Stein manifolds to Euclidean spaces [152]. Kaliman and Zaidenberg [292] proved the jet transversality theorem for holomorphic maps from Stein manifolds to any complex manifold provided that one shrinks the domain of the map (Theorem 7.8.12 on p. 321 below).

We shall use the following notion of a *Whitney stratification.*

Theorem 7.8.1. [514, Theorem 8.5] *Every complex analytic subvariety A in a complex space X admits a stratification $\{A_\alpha\}$ satisfying the following Whitney's condition (a):*

(a) *Suppose that $A_\alpha \subset \overline{A_\beta}$, $x_i \in A_\beta$ $(i = 1, 2, \ldots)$ is a sequence converging to a point $x \in A_\alpha$, and in some local coordinates in a neighborhood of $x \in X$ the tangent spaces $\mathrm{T}_{x_i} A_\beta$ converge to a plane $\tau \subset \mathrm{T}_x X$. Then $\mathrm{T}_x A_\alpha \subset \tau$.*

We recall how Whitney's condition (a) is used in transversality arguments. Given stratified subvarieties $A \subset X$, $B \subset Y$, let $\mathrm{NT}_{A,B} \subset \mathrm{J}^1(X, Y)$ consist of all one-jets $(x, y, \lambda) \in \mathrm{J}^1(X, Y)$ such that, if x belongs to a stratum A_α of A and y belongs to a stratum B_β of B, then

$$\lambda(\mathrm{T}_x A_\alpha) + \mathrm{T}_y B_\beta \neq \mathrm{T}_y Y.$$

If $x \notin A$ or $y \notin B$ then $(x, y, \lambda) \notin \mathrm{NT}_{A,B}$.

The following elementary lemma is proved in [219, p. 38].

Lemma 7.8.2. *If $A \subset X$ and $B \subset Y$ are Whitney stratified complex subvarieties then $\mathrm{NT}_{A,B}$ is closed in $\mathrm{J}^1(X, Y)$. $\mathrm{NT}_{A,B}$ is also closed in $\mathrm{J}^1(X, Y)$ if B is a closed smooth submanifold of Y.*

Let $A \subset X$ and $B \subset Y$ be stratified complex subvarieties. Given a holomorphic map $f \in \mathcal{O}(X, Y)$, we say that $f|_A$ *is transverse to* B if the range of the first jet extension map $\mathrm{j}^1 f \colon X \to \mathrm{J}^1(X, Y)$ does not intersect the set $\mathrm{NT}_{A,B}$. Equivalently,

$$(x \in A_\alpha,\ f(x) \in B_\beta) \implies \mathrm{d}f_x(\mathrm{T}_x A_\alpha) + \mathrm{T}_{f(x)}(B_\beta) = \mathrm{T}_{f(x)} Y.$$

The forthcoming discussion is based on the following condition introduced by Gromov [236, pp. 71–73]. (See also [177].)

Definition 7.8.3. [236, pp. 71–73] *Let X and Y be complex (resp. algebraic) manifolds. Holomorphic (resp. algebraic) maps $X \to Y$ satisfy* Condition Ell_1 *if for every holomorphic (resp. algebraic) map $f\colon X \to Y$ there is a holomorphic (resp. algebraic) map $F\colon X \times \mathbb{C}^N \to Y$ for some $N \geq \dim Y$ such that $F(\cdot\,,0) = f$ and $F(x,\cdot\,)\colon \mathbb{C}^N \to Y$ has rank $\dim Y$ at $0 \in \mathbb{C}^N$ for every $x \in X$.*

Such F is called a *dominating spray over f*. For validity of Ell_1 see Proposition 7.8.10 on p. 321 below. Here are a couple of elementary examples.

Example 7.8.4. If $Y = \mathbb{C}^n$ then $F(x,t) = f(x) + t$ $(x \in X,\ t \in \mathbb{C}^n)$ is a dominating spray over a map $f\colon X \to \mathbb{C}^n$. More generally, assume that a complex Lie group G acts holomorphically and transitively on Y. Let $\exp\colon \mathfrak{g} \to G$ denote the exponential map. Then $F(x,v) = \exp v \cdot f(x)$ $(x \in X,\ v \in \mathfrak{g})$ is a dominating spray over the map $f\colon X \to Y$. $\square$

Theorem 7.8.5. *Let X be a Stein manifold. If holomorphic maps $X \to Y$ to a complex manifold Y satisfy Condition* Ell_1 *then for every pair of closed, Whitney stratified complex analytic subvarieties $A \subset X$, $B \subset \mathrm{J}^k(X,Y)$ the set*

$$\big\{f \in \mathcal{O}(X,Y)\colon \mathrm{j}^k f|_A \text{ is transverse to } B\big\}$$

is residual in $\mathcal{O}(X,Y)$. The same holds if B is a smooth closed submanifold of $\mathrm{J}^k(X,Y)$. For $k = 0$ the conclusion holds even if X is not Stein.

In the algebraic category we have the analogous result on compact sets:

Theorem 7.8.6. *Let X and Y be algebraic manifolds, with X affine algebraic. If algebraic maps $X \to Y$ satisfy Condition* Ell_1 *then for every compact set $K \subset X$ and every pair of Whitney stratified complex subvarieties $A \subset X$, $B \subset \mathrm{J}^k(X,Y)$, the set*

$$\big\{f \in \mathcal{O}(X,Y)\colon \mathrm{j}^k f|_A \text{ is transverse to } B \text{ on } A \cap K\big\}$$

is open and dense in $\mathcal{O}(X,Y)$. The same holds if B is a smooth closed submanifold of $\mathrm{J}^k(X,Y)$. For $k = 0$ this holds without assuming that X be affine.

In particular, for $Y = \mathbb{C}^n$ we obtain the theorem of Forster [152]:

Corollary 7.8.7. *Holomorphic maps of any Stein manifold X to $\mathbb{C}^n$ satisfy the jet transversality theorem with respect to any pair of closed, Whitney stratified complex analytic subvarieties $A \subset X$, $B \subset \mathrm{J}^k(X,\mathbb{C}^n)$. If X is affine algebraic then the same conclusion holds for algebraic maps $X \to \mathbb{C}^n$.*

Proof. The proofs of Theorems 7.8.5 and 7.8.6 are parallel up to the point where the Baire property of the space $\mathcal{O}(X,Y)$ is invoked; in the algebraic case this leaves us with the weaker statement. We shall follow Abraham's reduction [4] to Sard's theorem.

Lemma 7.8.8. *Let X and Y be complex manifolds and let $A \subset X$, $B \subset \mathrm{J}^k(X,Y)$ be closed, Whitney stratified complex analytic subvarieties. For every compact subset K of X the set*

$$\mathcal{T}_{A,B,K} = \big\{ f \in \mathcal{O}(X,Y) \colon \mathrm{j}^k f|_A \ \text{ is transverse to } B \text{ on } A \cap K \big\}$$

is open in $\mathcal{O}(X,Y)$. The same holds if B is a smooth closed submanifold.

Proof. Consider the basic case with $B \subset Y$. Given a map $f \colon X \to Y$ and a compact set $K \subset X$, the restriction $f|_A \colon A \to B$ is transverse to B at each point of $A \cap K$ if and only if $(\mathrm{j}^1 f)(K) \cap \mathrm{NT}_{A,B} = \emptyset$. Assuming this to be the case, and taking into account that $\mathrm{NT}_{A,B}$ is closed in $\mathrm{J}^k(X,Y)$ by Whitney's condition, there is a compact set $L \subset X$, with $K \subset \operatorname{Int} L$, such that $(\mathrm{j}^1 f)(L) \cap \mathrm{NT}_{A,B} = \emptyset$. If $g \in \mathcal{O}(X,Y)$ is sufficiently uniformly close to f on L, then $\mathrm{j}^1 g$ is close to $\mathrm{j}^1 f$ on K, and hence $(\mathrm{j}^1 g)(K) \cap \mathrm{NT}_{A,B} = \emptyset$. In the general case one applies the same argument with f replaced by the k-jet extension $\mathrm{j}^k f \colon X \to \mathrm{J}^k(X,Y)$. $\square$

To prove Theorem 7.8.5 it suffices to show that for every compact K in X the set $\mathcal{T}_{A,B,K} \subset \mathcal{O}(X,Y)$, which is open by Lemma 7.8.8, is everywhere dense in $\mathcal{O}(X,Y)$. Since $\mathcal{O}(X,Y)$ is a Baire space, the conclusion then follows by taking the intersection of such sets over a countable family of compacts exhausting X. In the algebraic case we must omit the last step.

Consider first the basic case with $A = X$ and $B \subset Y$. Let $f \colon X \to Y$ be a holomorphic (resp. an algebraic) map. Choose a map $F \colon X \times \mathbb{C}^N \to Y$ as in Def. 7.8.3. Let $\pi \colon X \times \mathbb{C}^N \to \mathbb{C}^N$ denote the projection $\pi(x,t) = t$. Fix a compact set K in X. Since $\partial_t F(x,0) \colon \mathrm{T}_0\mathbb{C}^N \to \mathrm{T}_{f(x)}Y$ is surjective for every $x \in X$, there are a ball $D \subset \mathbb{C}^N$ around the origin and an open set $U \subset X$ containing K such that F is a submersion of $V = U \times D$ to Y. Hence $B' = F^{-1}(B) \cap V$ is a closed, Whitney stratified, complex analytic subvariety of V. (The strata B_β of B pull back by $F|_V$ to the strata B'_β of B').

Set $f_t = F(\cdot\,,t) \colon X \to Y$ for $t \in \mathbb{C}^N$. If $(x,t) \in B'_\beta$ then $y = f_t(x) \in B_\beta$, and by inspecting the definitions we see that the following are equivalent (compare with [219, p. 40]):

(a) $\mathrm{d}(f_t)_x(\mathrm{T}_x X) + \mathrm{T}_y B_\beta = \mathrm{T}_y Y$;

(b) (x,t) is a regular point of the restricted projection $\pi \colon B'_\beta \to D$.

By Sard's theorem, applied inductively to the components of a projection π in (b), we see that the set of regular values of all projections in (b) is residual in D. Choosing t in this set and close to 0 we get a map $f_t \colon X \to Y$ which is transverse to B on U and approximates $f = f_0$ uniformly on K. The same argument applies if B is a smooth closed submanifold of Y.

If A is a Whitney stratified complex subvariety of X, one applies the above argument with U replaced by $U \cap A_\alpha$ for a fixed stratum A_α of A (f and F

are still defined globally on X). This gives a residual set of points $t \in D \subset \mathbb{C}^N$ for which $f_t|_{A_\alpha \cap U}$ is transverse to a stratum B_β of B. Since A and B have at most countably many strata and $\mathbb{C}^N$ is a Baire space, we find $t \in \mathbb{C}^N$ arbitrarily close to 0 such that $f_t|_{A\cap U}$ is transverse to B.

This proves the basic transversality theorem for holomorphic maps $X \to Y$. All steps hold for algebraic maps as well, even if the subvarieties A and B are non-algebraic, and we did not need any special properties of X and Y other than Condition Ell_1 for algebraic maps $X \to Y$.

Consider now the case $k > 0$. Fix a map $f\colon X \to Y$ and a compact set $K \subset X$. The goal is to prove that f can be approximated uniformly on K by holomorphic (resp. algebraic) maps $X \to Y$ whose k-jet extension $\mathrm{j}^k f$ is such that $\mathrm{j}^k f|_A\colon A \to \mathrm{J}^k(X,Y)$ is transverse to the subvariety $B \subset \mathrm{J}^k(X,Y)$ at each point of $A \cap K$.

Let $F\colon X \times \mathbb{C}^N \to Y$ be as in Def. 7.8.3, with $F(\cdot\,,0) = f$. Recall that X is Stein (in Theorem 7.8.5), resp. affine algebraic (in Theorem 7.8.6). Thus we may assume that X is a closed holomorphic (resp. algebraic) submanifold of a Euclidean space $\mathbb{C}^n$.

Let $\mathcal{W}$ denote the complex vector space of all holomorphic polynomial maps $P\colon \mathbb{C}^n \to \mathbb{C}^N$ of degree at most k. Consider the holomorphic (resp. algebraic) map $G\colon X \times \mathcal{W} \to Y$ defined by

$$G(x,P) = F\big(x, P(x)\big), \qquad x \in X,\ P \in \mathcal{W}.$$

For each $P \in \mathcal{W}$ set $G_P = G(\cdot\,,P)\colon X \to Y$; then $G_0(x) = F(x,0) = f(x)$.

Lemma 7.8.9. *The map $\Phi\colon X \times \mathcal{W} \to J^k(X,Y)$, defined by $\Phi(x,P) = \mathrm{j}^k_x G_P$, is a submersion in an open neighborhood of $X \times \{0\}$ in $X \times \mathcal{W}$.*

Proof. The argument is local and hence we may assume that $X = \mathbb{C}^n$. Write $P = (P_1, \ldots, P_N) \in \mathcal{W}$, and let $t = (t_1, \ldots, t_N)$ be the coordinates on $\mathbb{C}^N$. For every multiindex $I = (i_1, \ldots, i_n)$ we have

$$\partial_x^I(G_P) = \sum_{j=1}^N \frac{\partial}{\partial t_j} F(x,P(x))\, \partial_x^I P_j(x) + H_I(x),$$

where $H_I(x)$ contains only terms $\partial_x^J P$, with $|J| < |I|$, multiplied by various partial derivatives of F. Hence the k-jet extension map $\mathrm{j}^k_x G_P$ is triangular with respect to the components of $\mathrm{j}^k_x P$, and the diagonal terms are nondegenerate at $P = 0$ (since $G_0(x) = F(x,0)$ and $\partial_t F(x,0)$ is nondenegenerate). $\square$

Sard's theorem, applied to the map Φ in Lemma 7.8.9, shows that for most $P \in \mathcal{W}$ the map $\mathrm{j}^k G_P|_A$ is transverse to the subvariety $B \subset \mathrm{J}^k(X,Y)$ at every point of $A \cap K$. This proves Theorems 7.8.5 and 7.8.6. $\square$

Combining Theorems 7.8.5 and 7.8.6 with the following proposition gives several transversality theorems. Compare with [236, p. 72].

Proposition 7.8.10. *Let X and Y be complex manifolds.*

(a) *If Y admits a dominating spray $s\colon Y \times \mathbb{C}^N \to Y$ defined on a trivial bundle over Y then holomorphic maps $X \to Y$ satisfy Condition* Ell_1. *This holds in particular if Y is a complex homogeneous space.*

(b) *Let X be Stein. If Y is subelliptic (Def. 5.5.11 on p. 203), or if Y is an Oka manifold (see §5.4), then holomorphic maps $X \to Y$ satisfy* Ell_1.

(c) *If X and Y are algebraic manifolds and Y admits a dominating algebraic spray $s\colon Y \times \mathbb{C}^N \to Y$ then algebraic maps $X \to Y$ satisfy* Ell_1.

(d) *If X is affine algebraic and Y is algebraically subelliptic then algebraic maps $X \to Y$ satisfy* Ell_1.

Proof. Fix a holomorphic map $f\colon X \to Y$. If Y admits a dominating spray (E, p, s) then $f^*E \to X$ is a holomorphic vector bundle, and there is a fiberwise bijective holomorphic map $\iota\colon f^*E \to E$ covering f. The map $F = s \circ \iota\colon f^*E \to Y$ satisfies Ell_1 for f, except that f^*E need not be a trivial bundle over X. In case (a) the bundle $E \to Y$ is assumed to be trivial, hence f^*E is also trivial and (a) follows. The analogous argument proves (c).

If X is a Stein manifold then by Cartan's Theorem A the holomorphic vector bundle $f^*E \to X$ is generated by finitely many (say N) holomorphic sections. Hence there is a surjective holomorphic vector bundle map $\tau\colon X \times \mathbb{C}^N \to f^*E$. The map $F = s \circ \iota \circ \tau\colon X \times \mathbb{C}^N \to Y$ satisfies Condition Ell_1 with respect to $f = F(\cdot\,, 0)$. The analogous argument holds in the algebraic case by appealing to Serre's Theorem A. This proves (b) (resp. (d)) for an elliptic (resp. algebraically elliptic) target manifold Y.

Assume now that Y is (algebraically) subelliptic, and let X be an affine manifold (Stein, resp. affine algebraic). Let (E_j, p_j, s_j) for $j = 1, \ldots, k$ be a finite dominating family of holomorphic (resp. algebraic) sprays on Y (Def. 5.5.11, p. 203). Let $E_1' = f^*E_1 \to X$ be the pull-back of $\pi_1\colon E_1 \to Y$ by the map $f\colon X \to Y$, and define $\sigma_1\colon E_1' \to Y$ by $\sigma_1(x, e) = s_1(f(x), e)$. As before, there is a surjective complex vector bundle map $X \times \mathbb{C}^{n_1} \to E_1'$ for some $n_1 \in \mathbb{N}$. By composing it with σ_1 we obtain a map $f_1\colon X_1 = X \times \mathbb{C}^{n_1} \to Y$ satisfying $f_1(x, 0) = f(x) = y \in Y$ and

$$\partial_t f_1(x,t)\big|_{t=0}(\mathrm{T}_0\mathbb{C}^{n_1}) = (\mathrm{d}s_1)_y(E_{1,y}) \subset \mathrm{T}_yY.$$

Repeating the construction with $f_1\colon X_1 \to Y$ and with the second spray $s_2\colon I_2 \to Y$, we find an integer $n_2 \in \mathbb{N}$ and a holomorphic map $f_2\colon X_2 = X_1 \times \mathbb{C}^{n_2} = X \times \mathbb{C}^{n_1} \times \mathbb{C}^{n_2} \to Y$ satisfying $f_2(x, t, 0) = f_1(x, t)$ (hence $f_2(x, 0, 0) = f(x) = y$) and

$$\partial_u f_2(x,0,u)|_{u=0}(\mathrm{T}_0\mathbb{C}^{n_2}) = (\mathrm{d}s_2)_y(E_{2,y}) \subset \mathrm{T}_yY.$$

After k steps we obtain a map $F\colon X \times \mathbb{C}^N \to Y$ ($N = n_1 + \cdots + n_k$) satisfying the following for every $x \in X$ and $y = f(x) \in Y$:

$$F(x,0) = f(x), \quad \partial_t F(x,t)\big|_{t=0}(\mathrm{T}_0\mathbb{C}^N) = \sum_{j=1}^{k} (\mathrm{d}s_j)_y(E_{j,y}) = \mathrm{T}_y Y.$$

This completes the proof of (b) for a subelliptic Y; the same proof applies in the algebraic case (d) by appealing to the Theorem A of Serre when passing at each step to a trivial bundle.

It remains to prove (b) when X is a Stein manifold and Y is an Oka manifold. Let $f\colon X \to Y$ be a holomorphic map. Consider the associated embedding $x \in X \to (x, f(x)) \in X \times Y$ with the normal bundle $E = f^*\mathrm{T}Y \to X$. By the Docquier–Grauert theorem (Theorem 3.3.3 on p. 67) there are an open neighborhood $V \subset E$ of the zero section $X \subset E$ and a biholomorphic map $G\colon V \to G(V) \subset X \times Y$ of the form

$$G(x,\xi) = (x, g(x,\xi)), \quad x \in X,\ \xi \in E_x$$

satisfying $g(x, 0_x) = f(x)$. We can extend g to a continuous map $E \to Y$ without changing its values on a smaller neighborhood of the zero section $X \subset E$. Since E is a Stein manifold and Y is an Oka manifold, there exists a holomorphic map $\widetilde{g}\colon E \to Y$ that agrees with g to the second order along the zero section $X \subset E$ (see Theorem 5.4.4 on p. 193). For some $N \in \mathbb{N}$ there exists a surjective holomorphic vector bundle map $\iota\colon X \times \mathbb{C}^N \to E$. The composition $F = \widetilde{g} \circ \iota\colon X \times \mathbb{C}^N \to Y$ is then a dominating spray over f. $\square$

Corollary 7.8.11. *Holomorphic maps from any complex manifold X to a complex homogeneous manifold Y satisfy the basic transversality theorem (for zero-jets). The same holds if $Y = \mathbb{C}^n \backslash A$, where A is a thin algebraic subvariety; for such A the basic transversality theorem also holds for algebraic maps from any algebraic manifold to $\mathbb{C}^n \backslash A$.*

In the case $Y = \mathbb{P}^n$ we recover Bertini's theorem [219, p. 150].

Kaliman and Zaidenberg proved the following theorem without any restriction on the target manifold, but the domain of the map shrinks.

Theorem 7.8.12. [292] *Assume that X is a Stein manifold, Y is a complex manifold, and $A \subset X$, $B \subset \mathrm{J}^k(X,Y)$ are closed, Whitney stratified, complex analytic subvarieties. For every holomorphic map $f \in \mathcal{O}(X,Y)$ and for every compact set $K \subset X$ there is a holomorphic map $g\colon U \to Y$, defined in an open neighborhood of K, such that $\mathrm{j}^k g|_{A\cap U}$ is transverse to B, and g approximates f as close as desired uniformly on K.*

Proof. By enlarging K we may assume that K is $\mathcal{O}(X)$-convex, and hence the set $\{(x, f(x)) \colon x \in K\} \subset X \times Y$ has an open Stein neighborhood $\Omega \subset X \times Y$ by Corollary 3.1.1 (p. 57). Hence there exist holomorphic vector fields $V^1, \ldots, V^N$ in Ω tangent to the fibers of the projection $X \times Y \to X$ and generating the tangent space of the fiber at each point. Let θ_t^j denote the flow of V^j. For

x in an open set $U \supset K$ and for small $t_1, \ldots, t_N \in \mathbb{C}$, the map defined by $F(x, t_1, \ldots, t_N) = \pi_Y \circ \theta^1_{t_1} \circ \cdots \circ \theta^N_{t_N}(x, f(x))$ satisfies Condition Ell_1 along $U \times \{0\}$. Theorem 7.8.12 now follows as in the proof of Theorem 7.8.5. $\square$

Remark 7.8.13. Theorem 7.8.12 gives the following alternative proof of the jet transversality theorem for holomorphic maps from a Stein manifold X to an Oka manifold Y. Let $f\colon X \to Y$ be a holomorphic map. Choose compact $\mathcal{O}(X)$-convex subsets $K, L \subset X$ with $K \subset \operatorname{Int} L$. By Theorem 7.8.12 we can approximate f uniformly on L by a holomorphic map $g\colon U \to Y$ on an open set $U \supset L$ such that $\mathrm{j}^k g|_{U \cap A}$ is transverse to a given Whitney stratified subvariety B of $\mathrm{J}^k(X, Y)$. If the approximation is sufficiently close, there is a smooth map $\widetilde{g}\colon X \to Y$ which agrees with g in a neighborhood of L and with f on $X \backslash U$. Since Y is an Oka manifold, the map $\widetilde{g}$ can be approximated uniformly on L by holomorphic maps $\widetilde{f}\colon X \to Y$. If the approximation is sufficiently close then $\widetilde{f}$ still satisfies the given transversality condition on K by Lemma 7.8.8. This shows the density of transverse maps on compacts in X. $\square$

We also have an interpolation version of Theorem 7.8.5. Given a closed complex subvariety X_0 of X, $f_0 \in \mathcal{O}(X, Y)$ and $r \in \{0, 1, \ldots\}$, the set

$$\mathcal{O}(X, Y; X_0, f_0, r) = \left\{ f \in \mathcal{O}(X, Y) \colon \mathrm{j}^r f|_{X_0} = \mathrm{j}^r f_0|_{X_0} \right\}$$

is a closed metric subspace of $\mathcal{O}(X, Y)$, hence a Baire space.

Theorem 7.8.14. *Assume that X is a Stein manifold and that Y is an Oka manifold. Let $A \subset X$ and $B \subset \mathrm{J}^k(X, Y)$ be closed, Whitney stratified subvarieties. If $f_0 \in \mathcal{O}(X, Y)$ and $\mathrm{j}^k f_0|_A$ is transverse to B at all points of $A \cap X_0$, then for every integer $r \geq k$ there is a residual set of $f \in \mathcal{O}(X, Y; X_0, f_0, r)$ for which $\mathrm{j}^k f|_A$ is transverse to B.*

Proof. Since $r \geq k$, the set of all $f \in \mathcal{O}(X, Y; X_0, f_0, r)$ for which $\mathrm{j}^k f|_A$ is transverse to B at all points of $A \cap X_0$ is open in $\mathcal{O}(X, Y; X_0, f_0, r)$. It thus suffices to show that we can approximate the initial map f_0, uniformly on any compact $\mathcal{O}(X)$-convex subset $K \subset X$, by $f \in \mathcal{O}(X, Y; X_0, f_0, r)$ such that $\mathrm{j}^k f|_A$ is transverse to B at every point of $A \cap K$.

Since Y is an Oka manifold, holomorphic maps $X \to Y$ satisfy Condition Ell_1. Let $F\colon X \times \mathbb{C}^N \to Y$ be as in Def. 7.8.3, with $F(\cdot, 0) = f_0$. Consider the basic case $k = 0$, $B \subset Y$. There exist functions $g_1, \ldots, g_l \in \mathcal{O}(X)$ which vanish to order $r+1$ on the subvariety $X_0 = \{x \in X \colon g_j(x) = 0,\ j = 1, \ldots, l\}$. For every $x \in X$ let $\sigma_x\colon (\mathbb{C}^N)^l \to \mathbb{C}^N$ be defined by

$$\sigma_x(t_1, \ldots, t_l) = \sum_{j=1}^{l} t_j g_j(x), \qquad t_j \in \mathbb{C}^N,\ j = 1, 2, \ldots, l.$$

Clearly σ_x is surjective if $x \in X \backslash X_0$ and is the zero map if $x \in X_0$. The map $\widetilde{F}\colon X \times \mathbb{C}^{Nl} \to Y$, defined by

$$\widetilde{F}(x,t) = \widetilde{F}(x,t_1,\dots,t_l) = F(x,\sigma_x(t_1,\dots,t_l)),$$

is a submersion with respect to t (at $t=0$) if $x \in X\backslash X_0$, and is degenerate (constant) if $x \in X_0$. Hence the proof of Theorem 7.8.5 applies over $X\backslash X_0$.

Let $f_t = \widetilde{F}(\cdot\,,t)\colon X \to Y$ for $t \in \mathbb{C}^{Nl}$. By construction $\mathrm{j}^r f_t|_{X_0} = \mathrm{j}^r f_0|_{X_0}$ for every t. Choose a compact set $K \subset X$. By the assumption $f_0|_A$ is transverse to B on $A \cap X_0$. Hence there is an open neighborhood $U \subset X$ of $A \cap X_0 \cap K$ such that $f_t|_{A\cap U}$ is transverse to B for every t sufficiently close to 0 (Lemma 7.8.8). The set $K' = K\backslash U \subset X\backslash X_0$ is compact, and hence for most values of t the map $f_t|_A$ is transverse to B on $A \cap K'$. Thus $f_t|_A$ is transverse to B on $K \cap A$ for most t close to 0 which concludes the proof for $k=0$.

Similarly one obtains the proof for $k>0$ by following the arguments in the proof of Theorem 7.8.5. We also get a semiglobal version of Theorem 7.8.14, analogous to Theorem 7.8.12, without any restriction on the manifold Y. □

In the algebraic category the global transversality theorem holds under the following stronger assumption on Y.

Proposition 7.8.15. [177, Proposition 4.10] *If Y is an algebraic manifold with a submersive algebraic spray $s\colon E \to Y$ (i.e., such that $s\colon E_y \to Y$ is a submersion for every $y \in Y$), then algebraic maps $X \to Y$ from any affine algebraic manifold X to Y satisfy the jet transversality theorem with respect to closed complex analytic subvarieties.*

Proof. Let $f_0\colon X \to Y$ be an algebraic map. By pulling back the submersive algebraic spray $s\colon E \to Y$ by the map f_0 we obtain an algebraic submersion $F\colon X \times \mathbb{C}^N \to Y$ satisfying $f_0 = F(\cdot\,,0)$ (compare with the proof of Proposition 7.8.10). Given closed complex subvarieties $A \subset X$ and $B \subset Y$, Sard's theorem shows that for a generic choice of $t \in \mathbb{C}^N$ the algebraic map $f_t|_A = F(\cdot\,,t)|_A$ is transverse to B. (See the proof of Theorem 7.8.5.) Similarly we obtain the jet transversality theorem by considering maps $x \to F(x,P(x))$ for polynomials $P\colon \mathbb{C}^n \to \mathbb{C}^N$, where X is an affine subvariety in $\mathbb{C}^n$. □

7.9 Singularities of Holomorphic Maps

Let X, Y be complex manifolds of dimension n, m, respectively. We consider the following *singularity sets* associated to a holomorphic map $f\colon X \to Y$:

$$\Delta_f^2 = \big\{(x_1,x_2) \in X \times X\colon\ x_1 \neq x_2,\ f(x_1) = f(x_2)\big\}, \tag{7.12}$$

$$\Sigma_f^i = \big\{x \in X\colon \dim\ker \mathrm{d}f_x = i\big\}, \qquad i = 1,2,\dots \tag{7.13}$$

$$\Lambda_f^i = \cup_{j\geq i}\Sigma_f^j, \qquad j = 1,2,\dots. \tag{7.14}$$

Thus $\Delta_f^2 \subset X \times X$ is the variety of double points (self-intersections) of f, and $\Delta_f^2 = \emptyset$ if and only if f is injective. For every $i \in \mathbb{N}$ the set Λ_f^i is a complex

subvariety of X consisting of all points $x \in X$ at which the differential $\mathrm{d}f_x$ has at least i-dimensional kernel.

If $n \le m$, we have $\Lambda_f^1 = \emptyset$ if and only if f is an immersion. Thus f is an injective immersion if and only if $\Lambda_f^1 = \emptyset$ and $\Delta_f^2 = \emptyset$.

If $n > m$ then $\Sigma_f^i = \emptyset$ for $i = 1, \dots, n-m-1$, and f is a submersion if and only if $X = \Sigma_f^{n-m}$ (equivalently, $\Lambda_f^i = \emptyset$ for $i > n-m$).

Theorem 7.9.1. *Assume that X is an n-dimensional Stein manifold and that Y is an m-dimensional Oka manifold (Def. 5.4.1, p. 192). Then there is a residual set in $\mathcal{O}(X,Y)$ consisting of maps $f\colon X \to Y$ satisfying*

$$\dim \Delta_f^2 \le 2n - m, \quad \operatorname{codim} \Sigma_f^i = i(m-n+i),\ i = 1, 2, \dots \tag{7.15}$$

This holds in particular for holomorphic maps $X \to \mathbb{C}^m$. The same estimates hold for a generic algebraic map $\mathbb{C}^n \to \mathbb{C}^m$.

Proof. By the results in §7.8, holomorphic maps from a Stein manifold X to an Oka manifold Y satisfy the jet transversality theorem. In particular, the following conditions hold for a generic holomorphic map $f\colon X \to Y$.

- The map $f \times f\colon X \times X \to Y \times Y$, $(x_1, x_2) \to (f(x_1), f(x_2))$, is transverse to the diagonal $\Delta_Y = \{(y,y)\colon y \in Y\}$ of $Y \times Y$.
- For every $i = 1, 2, \dots$ the 1-jet extension $\mathrm{j}^1 f\colon X \to \mathrm{J}^1(X,Y)$ is transverse to the subvariety $\Lambda^i \subset \mathrm{J}^1(X,Y)$ consisting of all 1-jets $(x, y, \lambda) \in \mathrm{J}^1(X,Y)$ such that $\dim \ker \lambda \ge i$.

Note that $\Delta_f^2 = (f \times f)^{-1}(\Delta_Y) \backslash \Delta_X$. Since codimensions are preserved under transverse maps, we have $\operatorname{codim} \Delta_f^2 = \operatorname{codim} \Delta_Y = m$ and hence $\dim \Delta_f^2 = 2n - m$ at every point of Δ_f^2. This gives the first estimate in (7.15).

To prove the second estimate we need the following lemma.

Lemma 7.9.2. [214, Proposition 5.3, p. 60] *Let V and W be vector spaces of dimension n and m, respectively, over a field $F \in \{\mathbb{R}, \mathbb{C}\}$. The set $L^i(V,W) = \{S \in \operatorname{Hom}_F(V,W)\colon \dim_F \ker S = i\}$ is a (real resp. complex) submanifold of $\operatorname{Hom}_F(V,W)$ satisfying $\operatorname{codim} L^i(V,W) = i(m-n+i)$.*

The proof amounts to finding the number of independent minors of an $m \times n$ matrix whose vanishing implies that it has i-dimensional kernel.

Let $\Sigma^i \subset \mathrm{J}^1(X,Y)$ be the set of all complex 1-jets with i-dimensional kernel. Then $\Sigma_f^i = (\mathrm{j}^1 f)^{-1}(\Sigma^i)$. Since the fiber $\Sigma_{(x,y)}^i$ of Σ^i over any point $(x,y) \in X \times Y$ is isomorphic to $L^i(\mathrm{T}_x X, \mathrm{T}_y Y)$, the second estimate in (7.15) follows from Lemma 7.9.2 and the jet transversality theorem.

The same arguments apply to algebraic (polynomial) maps $\mathbb{C}^n \to \mathbb{C}^m$. $\square$

Corollary 7.9.3. *Let X^n and Y^m be as in Theorem 7.9.1.*

(a) *If $2n \le m$ then a generic holomorphic map $X \to Y$ is an immersion.*

(b) *If $2n+1 \le m$ then a generic holomorphic map $X \to Y$ is an injective immersion.*

(c) *If $n > m$ then a generic holomorphic map $f\colon X \to Y$ satisfies the condition*

$$\operatorname{codim}\left\{x \in X\colon \operatorname{rank}(\mathrm{d}f_x) < m\right\} \ge n - m + 1.$$

Parts (a) and (b) follow from Theorem 7.9.1 applied with $i = 1$, while (c) follows by taking $i = n - m + 1$.

7.10 Approximation by Algebraic Maps

The problem of approximating holomorphic maps by algebraic maps is of central importance in analytic geometry. Algebraic approximations in general do not exist even for maps between very simple affine algebraic manifolds (there are no nontrivial algebraic morphisms $\mathbb{C} \to \mathbb{C}^* = \mathbb{C}\backslash\{0\}$). However, we establish an analogue of Theorem 6.6.1 (p. 263) for algebraic maps from affine algebraic varieties to algebraically subelliptic manifolds. This will be used in Chap. 8 in the construction of holomorphic submersions from Stein manifolds. All algebraic maps are assumed to be morphisms (without singularities).

Theorem 7.10.1. [177, Theorem 3.1] *Assume that X is an affine algebraic variety and Y is an algebraically subelliptic manifold (Def. 5.5.11 (e)). Given an algebraic map $f_0\colon X \to Y$, a compact $\mathcal{O}(X)$-convex subset K of X, an open set $U \subset X$ containing K, a homotopy $f_t\colon U \to Y$ of holomorphic maps ($t \in [0,1]$) and an $\epsilon > 0$, there exists an algebraic map $F\colon X \times \mathbb{C} \to Y$ such that $F(\cdot\,,0) = f_0$ and $\sup_{x\in K,\, t\in[0,1]} \operatorname{dist}(F(x,t), f_t(x)) < \epsilon$. In particular, a holomorphic map $X \to Y$ that is homotopic to an algebraic map through a family of holomorphic maps can be approximated by algebraic maps.*

Letting K be a convex compact set in $X = \mathbb{C}^n$ we obtain

Corollary 7.10.2. (Algebraic CAP) *If Y is an algebraically subelliptic manifold, then every holomorphic map $K \to Y$ from a compact convex subset $K \subset \mathbb{C}^n$ can be approximated, uniformly on K, by algebraic maps $\mathbb{C}^n \to Y$.*

Proof. We may assume that $0 \in K$ and that f is defined on a convex open set $U \supset K$. Let $f_t(z) = f(tz)$ ($z \in U$, $t \in [0,1]$). The map $f_0(z) = f(0)$ is constant and therefore algebraic, so Theorem 7.10.1 applies. □

Theorem 7.10.1 is a special case of the following result.

Theorem 7.10.3. *Let $h\colon Z \to X$ be an algebraic submersion from an algebraic variety Z onto an affine algebraic variety X. Assume that h is* algebraically subelliptic*, in the sense that Z admits a family of algebraic fiber-sprays (E_j, π_j, s_j) $(j = 1, \ldots, k)$ satisfying the fiber-domination property (6.1) (p. 242) at every point $z \in Z$. Let $K \subset X$ be a compact $\mathcal{O}(X)$-convex set and $f_t\colon K \to Z$ $(t \in [0,1])$ a homotopy of holomorphic sections such that f_0 extends to an algebraic section $X \to Z$. For every $\epsilon > 0$ there is an algebraic map $F\colon X \times \mathbb{C} \to Z$ such that*

- *$h(F(x,t)) = x$ for all $x \in X$ and $t \in \mathbb{C}$,*
- *$F(\cdot\,, 0) = f_0$, and*
- *$\mathrm{dist}(F(x,t), f_t(x)) < \epsilon$ for every $x \in K$ and $t \in [0,1]$.*

Proof. Let (E, p, s) be the composed algebraic fiber-spray on Z obtained from the fiber-sprays (E_j, p_j, s_j) $(j = 1, \ldots, k)$ (Def. 6.3.5, p. 246). Choose open Stein neighborhoods $V \Subset U$ of K in X such that the homotopy f_t $(t \in [0,1])$ in Theorem 7.10.3 is defined in U. Set $V_t = f_t(V) \subset Z$ for $t \in [0,1]$.

Lemma 7.10.4. *There are numbers $l \in \mathbb{N}$ and $0 = t_0 < t_1 < \cdots < t_l = 1$ such that for every $j = 0, 1, \ldots, l-1$ there exists a homotopy of holomorphic sections ξ_t of the restricted bundle $E|_{V_{t_j}} \to V_{t_j}$ $(t \in I_j = [t_j, t_{j+1}])$, such that ξ_{t_j} is the zero section and $s(\xi_t(z)) = f_t(h(z))$ for all $t \in I_j$ and $z \in V_{t_j}$.*

Proof. Assume first that there exists a Stein open set $\Omega \subset Z$ containing $\cup_{t\in[0,1]} \overline{V}_t$. By Lemma 6.3.7 (p. 248), the restriction $E|_\Omega \to \Omega$ admits the structure of a holomorphic vector bundle. Choose a holomorphic direct sum splitting $E|_\Omega = H \oplus H'$, where H' is the kernel of $\mathrm{d}s$ at the zero section of E. The domination condition

$$(\mathrm{d}s)_{0_z}(\mathrm{T}_{0_z} E_z) = \mathrm{VT}_z Z, \qquad z \in Z$$

implies that for every $z \in \Omega$ the restriction $s\colon H_z \to Z_{h(z)} = h^{-1}(h(z))$ maps a neighborhood of 0_z in H_z biholomorphically onto a relative neighborhood of z in the fiber $Z_{h(z)}$. The size of this neighborhood, and of its image in the corresponding fiber of Z, can be chosen uniform for points in the compact set $\cup_{t\in[0,1]} \overline{V}_t \subset \Omega$. Hence there is a $\delta > 0$ such that for every $t \in [0,1]$ the local inverse of $s\colon H|_{V_t} \to Z$ at the zero section gives a homotopy of sections ξ_τ of $H|_{V_t}$ $(\tau \in J_t = [t, t+\delta] \cap [0,1])$, with ξ_t being the zero section, such that

$$s(\xi_\tau(z)) = f_\tau(h(z)), \qquad \tau \in J_t,\ z \in V_t.$$

This proves Lemma 7.10.4 in the special case. For the general case observe that $f_t(U)$, being a closed Stein submanifold of $Z|_U$, admits an open Stein neighborhood in Z by Corollary 3.1.1 (p. 57). By compactness there are Stein open sets $\Omega_j \subset Z$ $(j = 1, 2, \ldots, m)$, and a partitition $[0,1] = \cup_{j=1}^m I_j$ into adjacent closed subintervals I_j, such that $\cup_{t\in I_j} \overline{V}_t \subset \Omega_j$. It remains to apply the above argument separately for each I_j. $\square$

Let $(E^{(l)}, p^{(l)}, s^{(l)})$ be the l-th iterated bundle of (E, p, s) (Def. 6.3.5 (b), p. 246). Let (E', p', s') denote the pull-back of $(E^{(l)}, p^{(l)}, s^{(l)})$ to X by the algebraic map $f_0 \colon X \to Z$; this is an algebraic composed spray bundle over X.

Lemma 7.10.5. *There is a homotopy $\eta_t \colon V \to E'|_V$ $(t \in [0,1])$, consisting of holomorphic sections of the restricted bundle $E'|_V \to V$, such that η_0 is the zero section and $s'(\eta_t(x)) = f_t(x)$ for every $x \in V$ and $t \in [0,1]$.*

Proof. It suffices to assemble the individual homotopies $\{\xi_t \colon t \in [t_j, t_{j+1}]\}$ $(j = 0, \ldots, l-1)$, furnished by Lemma 7.10.4, into a homotopy of sections $\widetilde{\xi}_t \colon V_0 \to E^{(l)}|_{V_0}$ $(t \in [0,1])$ of the iterated bundle $E^{(l)}$ over the open subset $V_0 = f_0(V)$ in the algebraic submanifold $f_0(X)$ of Z. Clearly $\widetilde{\xi}_t$ corresponds to a holomorphic sections $\eta_t \colon V \to E' = f_0^*(E^{(l)})$ of the pull-back bundle, with η_0 being the zero section. (For details see the proof of Prop. 6.5.1.) $\square$

Lemma 7.10.6. *Let $\{\eta_t\}_{t \in [0,1]}$ be as in Lemma 7.10.5. For every $\delta > 0$ there is an algebraic map $\eta' \colon X \times \mathbb{C} \to E'$ satisfying*

(i) $\eta'(x, 0) = 0_x \in E'_x$ $(x \in X)$, *and*
(ii) $\operatorname{dist}(\eta'(x,t), \eta_t(x)) < \delta$ *for all* $x \in K$ *and* $t \in [0,1]$.

Proof. By construction of the composed bundle $E' \to X$ there is a sequence

$$E' = E^{m,0} \longrightarrow E^{m-1,0} \longrightarrow \cdots \longrightarrow E^{1,0} \longrightarrow X, \tag{7.16}$$

with $m = kl$ and $E^{(1,0)} = f_0^* E_1 \to X$, in which every map $E^{j,0} \to E^{j-1,0}$ is an algebraic vector bundle projection. Here k is the number of the initial sprays in Theorem 7.10.3, and l is the number in Lemma 7.10.4.

Since X is an affine algebraic manifold, the algebraic vector bundle $E^{1,0} \to X$ is generated by finitely many (say n_1) algebraic sections according to Serre's Theorem A. This gives a surjective algebraic map $\pi_1 \colon E^{1,1} = X \times \mathbb{C}^{n_1} \to E^{1,0}$ from the trivial rank n_1 bundle onto $E^{1,0}$. Pulling back the sequence (7.16) to the new total space $E^{1,1}$ we obtain a commutative diagram

$$\begin{array}{ccccccccccc}
E^{m,1} & \longrightarrow & E^{m-1,1} & \longrightarrow & \cdots & \longrightarrow & E^{2,1} & \longrightarrow & E^{1,1} & \longrightarrow & X \\
\Big\downarrow \pi_m & & \Big\downarrow \pi_{m-1} & & & & \Big\downarrow \pi_2 & & \Big\downarrow \pi_1 & & \Big\| \\
E^{m,0} & \longrightarrow & E^{m-1,0} & \longrightarrow & \cdots & \longrightarrow & E^{2,0} & \longrightarrow & E^{1,0} & \longrightarrow & X
\end{array}$$

in which all horizontal maps are algebraic vector bundle projections and the vertical maps π_j for $j \geq 2$ are the induced natural maps which are bijective on fibers. More precisely, let $E^{2,1} \to E^{1,1}$ be the pull-back of the vector bundle $E^{2,0} \to E^{1,0}$ (in the bottom row) by the vertical morphism $\pi_1 \colon E^{1,1} \to E^{1,0}$, and denote by $\pi_2 \colon E^{2,1} \to E^{2,0}$ the associated natural map which makes the respective diagram commute. Moving one step to the left, $E^{3,1} \to E^{2,1}$ is the pull-back of the bundle $E^{3,0} \to E^{2,0}$ in the bottom row by the vertical

morphism $\pi_2\colon E^{2,1} \to E^{2,0}$, and $\pi_3\colon E^{3,1} \to E^{3,0}$ is the associated natural map; etc. There is an algebraic spray map $s^1\colon E^{m,1} \to Z$ which is the composition of $\pi_m\colon E^{m,1} \to E^{m,0}$ with the initial spray $s\colon E^{m,0} = E \to Z$.

We claim that the homotopy η_t of holomorphic sections of $E^{m,0}|_V = E'|_V \to V$, furnished by Lemma 7.10.5, lifts to a homotopy η_t^1 of holomorphic sections of $E^{m,1}|_V \to V$ such that $s^1(\eta_t^1) = f_t$ for all $t \in [0,1]$, and η_0^1 is the zero section. It suffices to see that the $E^{1,0}$-component of η_t (i.e., the projection of η_t under the composed projection $E^{m,0} \to E^{1,0}$) lifts to $E^{1,1}$; the rest of the lifting is then obtained by applying the inverses of the fiberwise isomorphic vertical maps. But this follows from the fact that the surjective vector bundle map $\pi_1\colon E^{1,1} \to E^{1,0}$ admits a holomorphic splitting $\sigma_1\colon E^{1,0} \to E^{1,1}$ over X, with $\pi_1 \circ \sigma_1$ the identity on $E^{1,0}$.

Repeating the same argument with the bundle $E^{2,1} \to E^{1,1} = X_1$ over the affine manifold $X_1 = X \times \mathbb{C}^{n_1}$ we obtain a surjective algebraic vector bundle map $E^{2,2} = X_1 \times \mathbb{C}^{n_2} = X \times \mathbb{C}^{n_1+n_2} \to E^{2,1}$. As before we lift the top line in the above diagram to a new level

$$E^{m,2} \longrightarrow E^{m-1,2} \longrightarrow \cdots \longrightarrow E^{2,2} \longrightarrow E^{1,1} = X_1 = X \times \mathbb{C}^{n_1}.$$

Note that $E^{2,2} = X_1 \times \mathbb{C}^{n_2} = X \times \mathbb{C}^{n_1+n_2}$ (algebraic equivalence). The homotopy η_t^1 lifts to a homotopy $\eta_t^2\colon V \to E^{m,2}|_V$, with η_0^2 the zero section, and we have a new spray map $s^2\colon E^{m,2} \to Z$ satisfying $s^2(\eta_t^2) = f_t$ for all $t \in [0,1]$.

Continuing inductively we obtain after m steps a lifting of the homotopy η_t to a homotopy $\eta_t^m\colon V \to E^{m,m}|_V = V \times \mathbb{C}^N$ ($t \in [0,1]$) of holomorphic sections of $E^{m,m} = X \times \mathbb{C}^N$ ($N = n_1 + n_2 + \cdots + n_m$) over the open subset $V \subset X$, with η_0^m being the zero section. By construction there is an algebraic spray $s^m\colon E^{m,m} \to Z$ such that $s^m(\eta_t^m) = f_t\colon V \to Z|_V$ for all $t \in [0,1]$.

Recall that X is a closed algebraic submanifold of an affine space $\mathbb{C}^n$. The $\mathcal{O}(X)$-convex set $K \subset X$ is then polynomially convex in $\mathbb{C}^n$, and $K \times [0,1]$ is polynomially convex in $\mathbb{C}^{n+1}$. (We have identified $[0,1] \subset \mathbb{R}$ with its image in $\mathbb{C}$.) By the Oka-Weil theorem we can approximate the homotopy $\{\eta_t^m\}_{t\in[0,1]}$ (which is continuous in $(x,t) \in V \times [0,1]$ and holomorphic with respect to $x \in V$ for every fixed $t \in [0,1]$), uniformly on the set $K \times [0,1]$, by a holomorphic polynomial map $\mathbb{C}^n \times \mathbb{C} \to \mathbb{C}^n \times \mathbb{C}^N$ of the form $\widetilde{g}(x,t) = (x, g(x,t))$, with $g(x,0) = 0$ for $x \in \mathbb{C}^n$. By projecting the point $\widetilde{g}(x,t) \in E^{m,m} = X \times \mathbb{C}^N$ ($x \in X,\ t \in \mathbb{C}$) back to the bundle $E' = E^{m,0}$ we obtain an algebraic map $\eta'(x,t)$ satisfying Lemma 7.10.6. $\square$

If s', η' and δ are as in Lemmas 7.10.5 and 7.10.6, with $\delta > 0$ chosen sufficiently small, then the algebraic map

$$F(x,t) = s'(\eta'(x,t)) \in Z, \quad (x,t) \in X \times \mathbb{C}$$

satisfies the conclusion of Theorem 7.10.3. $\square$

In connection with Theorem 7.10.1 it is natural to ask whether algebraic maps $X \to Y$ from an affine algebraic variety X to an algebraically subelliptic manifold Y satisfy the Oka principle, i.e., is every continuous map homotopic to an algebraic map? The following example of Loday [345] shows that the answer is negative in general. (I wish to thank Burt Totaro for pointing this out in a private communication.)

Example 7.10.7. Let Σ^n denote the complex n-sphere, i.e., the affine variety

$$\Sigma^n = \big\{(z_0, \dots, z_n) \in \mathbb{C}^{n+1} \colon z_0^2 + \cdots + z_n^2 = 1\big\}. \tag{7.17}$$

Then Σ^n retracts onto the real n-sphere S^n. Also, Σ^n is algebraically subelliptic for $n \geq 2$ because it is homogeneous for the complex Lie group $SO(n{+}1, \mathbb{C})$, and $\mathrm{Hom}(SO(n+1, \mathbb{C}), \mathbb{C}^*) = 1$ for $n \geq 2$. Loday [345] showed that every algebraic map $\Sigma^p \times \Sigma^q \to \Sigma^{p+q}$ is null-homotopic when p and q are odd. By contrast, there is always a homotopically nontrivial continuous map $\Sigma^p \times \Sigma^q \to \Sigma^{p+q}$, in fact, an isomorphism of the homology groups H_{p+q}. So the algebraic Oka principle fails when p and q are odd. $\square$

Probably the best explanation for Loday's result comes from Deligne's *mixed Hodge theory* [98]. For every complex algebraic variety X, Deligne defined an increasing filtration on the rational cohomology groups of X, the *weight filtration*. For X smooth, the weight filtration has the form

$$\{0\} \subset W_i H^i(X; \mathbb{Q}) \subset W_{i+1} H^i(X; \mathbb{Q}) \subset \cdots \subset W_{2i} H^i(X; \mathbb{Q}) = H^i(X; \mathbb{Q}).$$

For the affine variety Σ^n (7.17), $H^n(\Sigma^n; \mathbb{Q})$ is in weight n if n is even (meaning that $W_{n-1} = 0$ and W_n is the whole group), while $H^n(\Sigma^n; \mathbb{Q})$ is in weight $n{+}1$ if n is odd. Loday's theorem follows from Deligne's powerful theorem that every algebraic map $X \to Y$ is strictly compatible with the weight filtration. That is, the pullback map sends $W_j H^i(Y; \mathbb{Q})$ into $W_j H^i(X; \mathbb{Q})$ for all i and j, and moreover (this is the 'strict' part) any element of $W_j H^i(X; \mathbb{Q})$ that is in the image of $H^i(Y; \mathbb{Q})$ is in fact in the image of $W_j H^i(Y; Q)$.

So we could ask if continuous maps from an affine variety X to an algebraically subelliptic manifold are homotopic to algebraic maps provided that Deligne's weight filtration gives no obstruction. However, there is another obstruction coming from the Hodge theory.

Example 7.10.8. (B. Totaro, private communication.) Let X be an affine algebraic surface of the form $X = M - D$, where M is a smooth quartic surface in $\mathbb{P}^3$ (a K3 surface) and D is a smooth hyperplane section in M. Let $Y = \mathbb{P}^n$ where we choose $n \geq 3$. Then to ask whether continuous maps $X \to Y$ are homotopic to algebraic maps amounts to asking whether every continuous line bundle on X is isomorphic to an algebraic line bundle. Equivalently, we are asking whether all of $H^2(X; \mathbb{Z})$ is 'algebraic', that is, spanned by the Chern classes of algebraic curves in X.

The weight filtration does not gives any obstruction: The restriction map $H^2(M,\mathbb{Z}) \to H^2(X,\mathbb{Z})$ is surjective, and therefore $H^2(X,\mathbb{Z})$ is all of weight two by the properties of the weight filtration. (In general, classes of algebraic divisors in H^2 of a smooth affine variety are always in weight 2.)

However, there is another obstruction coming from the Hodge filtration. Since the K3 surface M has $H^2(M;\mathbb{C}) \neq 0$, not all of $H^2(M;\mathbb{Q})$ is algebraic. (The algebraic classes all map into the $(1,1)$ part of the Hodge decomposition of $H^2(M;\mathbb{C})$.) We have $H^2(X;\mathbb{Z}) = H^2(M;\mathbb{Z})/(D)$, and every algebraic curve on X has closure which is an algebraic curve in M. Therefore, not all of $H^2(X;\mathbb{Q})$ is algebraic (otherwise $H^2(M;\mathbb{Q})$ would be all algebraic). $\square$

7.11 Towards Quantitative Oka Theory

The results on Oka theory presented so far are purely qualitative, without any consideration of the growth of solutions. Here we mention without proof a few existing results on what can be called *quantitative Oka theory* where one is looking for analytic solutions to Oka type problems with growth estimates. This field offers interesting and natural problems.

The first quantitative results seem to be those of E. L. Stout [469, 471] and Andreotti and Stoll [21]. They considered the second Cousin problem with bounded data, thereby obtaining bounded holomorphic functions defining certain complex hypersurfaces. Further developments on this subject were summarized by A. Nagel [365] who proved the vanishing of the positive dimensional cohomology groups of certain domains in $\mathbb{C}^n$ with coefficients in sheaves of germs of holomorphic functions satisfying certain boundary conditions that are axiomatically formulated. His result includes as special cases L^p, differentiable, Hölder continuous and bounded.

In the mid-1970's G. M. Henkin [254] and H. Skoda [451] independently obtained interesting results on defining analytic hypersurfaces by functions of the Nevanlinna class. If $D = \{z \in \mathbb{C}^n \colon \rho(z) < 0\}$, let $D_\varepsilon = \{z \in D \colon \rho(z) < -\varepsilon\}$ for $\varepsilon > 0$. The Nevanlinna class $N(D)$ is defined as follows:

$$N(D) = \left\{ f \in \mathcal{O}(D) \colon \limsup_{\varepsilon \to 0^+} \int_{\partial D_\varepsilon} \max\{0, \log|f|\} < \infty \right\}.$$

It is well known that the zero set $X = Z(f) = \{f = 0\}$ of $f \in N(D)$, $f \not\equiv 0$, is a complex hypersurface in D which satisfies the Blaschke condition $\int_X \delta(z)\,\mathrm{d}\sigma(z) < \infty$, where σ is the area measure of X and $\delta(z)$ is the boundary distance. In the case $n = 1$ and $D = \mathbb{D} = \{z \in \mathbb{C} \colon |z| < 1\}$ the Blaschke condition for X implies that $X = Z(f)$ for some $f \in H^\infty(\mathbb{D})$, namely the Blaschke product corresponding to X. The analogue of this result fails in higher dimensions. Nevertheless we have the following remarkable theorem.

Theorem 7.11.1. [254, 451] *Let $D \subset \mathbb{C}^n$ be a strictly pseudoconvex domain with $H^1(D, \mathbb{R}) = 0$. Suppose that X is a complex hypersurface in D determining the trivial class in $H^2(D, \mathbb{Z})$. If X satisfies the Blaschke condition, then X is the zero set of some function in the Nevanlinna class $N(D)$.*

It follows in particular that every algebraic hypersurface is the zero set of a Nevanlinna function.

The next result is due to Forster and Ohsawa [160] and it concerns complete intersections in $\mathbb{C}^n$. Examples show that Corollary 7.5.6 (p. 308) fails in the algebraic category, i.e., there exist algebraic subvarieties $Y \subset \mathbb{C}^n$ that are holomorphic but not algebraic complete intersections. It is therefore natural to ask whether algebraic complete intersections in $\mathbb{C}^n$ can be defined by entire functions satisfying certain growth conditions.

A holomorphic function $f\colon \mathbb{C}^n \to \mathbb{C}$ is *of finite order* if $|f(z)| \le \mathrm{e}^{P(|z|)}$ for some real polynomial P. Answering a problem posed by M. Cornalba and P. Griffiths in their fundamental paper [92], Forster and Ohsawa [160] proved the following results concerning affine algebraic submanifolds of $\mathbb{C}^n$ which are complete intersections in the finite order category.

Theorem 7.11.2. [160] *An algebraic submanifold $X \subset \mathbb{C}^n$ of codimension 2 with topologically trivial canonical bundle is an ideal theoretic complete intersection in the finite order category, i.e., the ideal of X is generated by two entire functions of finite order on $\mathbb{C}^n$. The ideal of any smooth affine algebraic curve $C \subset \mathbb{C}^n$ for $n \ge 3$ is defined by $n-1$ entire functions of finite order.*

Since there exist smooth algebraic curves in $\mathbb{C}^3$ that fail to be algebraic complete intersections, Theorem 7.11.2 can not be improved.

Finally we mention a more recent result of Berndtsson and Rosay [44]. We have seen in §7.3 that a topological isomorphism between a pair of complex vector bundles E, E' over a Stein space X is homotopic to a holomorphic isomorphism. Suppose now that h, h' are Hermitian metrics on E, E', respectively. Given a complex vector bundle isomorphism $\Phi\colon E \to E'$, we can pull back the metric h' by Φ to another Hermitian metric $\Phi^* h'$ on E. Hence there exists a positive continuous function $c\colon X \to [1, +\infty)$ such that

$$\frac{1}{c(x)}\, h_x(v \otimes v) \le (\Phi^* h')_x(v \otimes v) \le c(x)\, h_x(v \otimes v)$$

holds for all $x \in X$ and $v \in E_x$.

Question 7.11.3. Is there an isomorphism Φ such that the function c in the above estimates grows at a controlled rate?

The Hermitian bundles (E, h), (E', h') are *isometric* if this holds with $c \equiv 1$, and *quasi-isometric* if it holds for some constant $c \ge 1$.

Berndtsson and Rosay [44] studied this problem for Hermitian holomorphic vector bundles over the unit disc $\mathbb{D} \subset \mathbb{C}$. Since every such bundle is holomorphically trivial by Corollary 7.3.5, the problem is to decide when is a Hermitian holomorphic vector bundle (E, h) over $\mathbb{D}$ quasi-isometric to the trivial bundle with the standard metric. Such bundle carries a unique Chern connection, i.e., connection compatible with the metric such that the $(0,1)$-part of its covariant derivative is the $\overline{\partial}$-operator. In the standard holomorphic coordinate of the disc, the curvature operator Θ_h is an h-Hermitian endomorphism; it is said to be *semi-negative* if $h(\Theta_E u, u) \leq 0$ for any $u \in E$.

Theorem 7.11.4. [44] *Let (E, h) be a Hermitian holomorphic vector bundle over the unit disc $\mathbb{D}$. If there exists a bounded subharmonic function ψ on $\mathbb{D}$ such that the curvature of the metric $\mathrm{e}^{\psi} h$ is seminegative, then (E, h) is quasi-isometric to the trivial bundle over the disc.*

The comparability constant is given explicitly in terms of the sup-norm of ψ. This allows an effective generalization of Cartan's factorization lemma over the disk, extending the result of E. L. Stout for line bundles [469, 471].

Here are a couple problems in this area that seem worth considering. The first one is to suitably quantify the Runge approximation property in the definition of the class of Oka manifolds (Def. 5.4.1). Suppose that $K \subset U \Subset V$ are bounded convex sets in $\mathbb{C}^n$, where K is compact and U, V are open. Let Y be an Oka manifold. Fix a complete Hermitian metric h on Y. Given a (large) open set $W \subset Y$, a holomorphic map $f\colon U \to W$ and a number $\epsilon > 0$, we look for a holomorphic map $F\colon V \to Y$ that is ϵ-close to f on K and such that F^*h has the smallest sup-norm (or L^2-norm) on V. Taking the infimum of these norms over all admissible maps f gives a function of the data (K, U, V, W, ϵ). How does this function behave when deforming the complex structure on Y? In particular, when do we have a uniform upper bound in a smooth family of manifolds Y_t, $t \in \mathbb{D}$? This might be useful when studying Problem 5.16.11 on the variation of Oka property in a smooth family of complex manifolds.

Another intriguing problem is to generalize the Ohsawa-Takegoshi extension theorems [381, 380] to maps into an Oka manifold. Let us consider a very special case of this extension problem. Suppose that D is a bounded strongly pseudoconvex domain in $\mathbb{C}^n$ and $H \subset \mathbb{C}^n$ is an affine hyperplane intersecting bD transversely such that $D \cap H$ is a retract of D. This implies that every continuous map $D \cap H \to Y$ extends to a continuous map $D \to Y$; if Y is an Oka manifold then also every holomorphic map $f\colon D \cap H \to Y$ extends to a holomorphic map $F\colon D \to Y$. In the case $Y = \mathbb{C}$, Ohsawa and Takegoshi proved that for every plurisubharmonic function ρ on D we can choose the extension such that $\int_D |F|^2 \mathrm{e}^{-\rho} \leq \mathrm{const} \int_{D \cap H} |f|^2 \mathrm{e}^{-\rho}$. Is there an analogous results for maps to Oka manifolds, and in particular to complex Lie groups?

8

Embeddings, Immersions and Submersions

In this chapter we apply Oka theory to study *regular* holomorphic maps – immersion, embeddings, and submersions – of Stein manifolds and Stein spaces to complex Euclidean spaces and certain other complex manifolds.

We begin in §8.1 with a classical result of Bishop on the existence of almost proper maps $X \to \mathbb{C}^n$ to Euclidean spaces of dimension $n = \dim X$, with dimension estimates of the degeneration subvarieties of a generic map. In §8.2 – §8.4 we present results on proper holomorphic embeddings and immersions of Stein spaces of dimension > 1 to Euclidean spaces of minimal dimension, due to Eliashberg and Gromov and to Schürmann. The main ingredient is the Relative Embedding Theorem (Theorem 8.2.5) whose proof in §8.3 uses the Oka principle for sections of holomorphic vector bundles that avoid certain complex subvarieties of the total space. In §8.5 we outline the proof of Gromov's homotopy principle for holomorphic immersions of Stein manifolds to Euclidean spaces, analogous to the Hirsch-Smale theorem for smooth immersions. In §8.6 we look at the Oka principle for proper holomorphic maps of strongly pseudoconvex Stein domains to q-convex manifolds.

In §8.7 we prove a theorem on splitting biholomorphic maps close to the identity on a Cartan pair. This result is essential in the construction of proper holomorphic embeddings of certain open Riemann surfaces into $\mathbb{C}^2$, due to Wold and the author (§8.8 – §8.10), and in the construction of noncritical holomorphic functions on Stein manifolds (§8.12 – §8.14).

In §8.11 – §8.14 we prove the homotopy principle for holomorphic submersions of Stein manifolds X to Euclidean spaces $\mathbb{C}^q$ for $q < \dim X$, and to certain other complex manifolds. In particular, we construct noncritical holomorphic functions on any Stein manifold, holomorphic functions with prescribed discrete critical locus, and closed holomorphic one-forms without zeros in all cohomology classes in $H^1(X;\mathbb{C})$. In §8.15 we give applications to holomorphic foliations on Stein manifolds.

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8.1 Generic Almost Proper Mappings

Let X and Y be locally compact topological spaces. Recall that a continuous mapping $f\colon X \to Y$ is *proper* if for every compact set K in Y the inverse image $f^{-1}(K)$ is compact. The map f is *almost proper* if for every compact $K \subset Y$ each connected component of $f^{-1}(K)$ is compact. A proper map with finite fibers is called a *finite map*. If X is Stein then any proper holomorphic map $f\colon X \to Y$ is finite, and any almost proper map has discrete fibers.

A subset of a Fréchet space is *residual* if it is the intersection of at most countably many dense open subsets; such set is everywhere dense.

Theorem 8.1.1. ([48], [241, Theorem 2, p. 220]) *Let X be a Stein manifold of dimension n. There is a residual set $\mathcal{A}$ in the Fréchet space $\mathcal{O}(X)^n$ such that for every $f \in \mathcal{A}$ the map $f = (f_1, \ldots, f_n)\colon X \to \mathbb{C}^n$ is almost proper and satisfies the estimates*

$$\dim\big\{x \in X\colon \dim\ker \mathrm{d}f_x \geq i\big\} \leq n - i^2, \quad i = 1, 2, \ldots \tag{8.1}$$

Recall from §7.9 that the set in (8.1) is a complex subvariety of X.

Proof (sketch). An *analytic polyhedron* in a complex manifold X is a compact set of the form $P = \{x \in U\colon |f_k(x)| \leq 1,\ k = 1, \ldots, r\}$ where f_k are holomorphic functions in an open set U containing K. Since $f = (f_1, \ldots, f_r)$ maps the interior of P properly onto the polydisc $\mathbb{D}^r \subset \mathbb{C}^r$, the number r is $\geq n = \dim X$. An analytic polyhedron which can be defined by $\dim X$ functions is said to be *special*. Bishop proved that every analytic polyhedron can be approximated from the outside by special analytic polyhedra (see [48, p. 691] or [241, Corollary 4, p. 218]). His proof uses a reduction of the number of functions defining the polyhedron to $n = \dim X$. It follows that every Stein manifold X can be exhausted by an increasing sequence $P_1 \subset P_2 \subset \cdots \subset \cup_{j=1}^{\infty} P_j = X$ of special analytic polyhedra

$$P_j = \big\{x \in U_j\colon |f_{j,k}(x)| \leq 1,\ k = 1, \ldots, n\big\} \Subset X$$

where $f_{j,k} \in \mathcal{O}(X)$ for all $j \in \mathbb{N}$ and $k = 1, \ldots, m$.

Let $h \in \mathcal{O}(X)^n$ and let K be a compact set in X. Choose $j_0 \in \mathbb{N}$ large enough such that K is contained in the interior of P_{j_0}, and choose real numbers $\alpha_1 < \alpha_2 < \cdots$ with $\lim_{j\to\infty} \alpha_j = +\infty$. Given $\epsilon > 0$, we can find a holomorphic map $f = (f_1, \ldots, f_n)\colon X \to \mathbb{C}^n$, with the components of the form

$$f_k(x) = \sum_{j \geq j_0} a_j (f_{j,k}(x))^{t_j} + h_k(x), \quad k = 1, \ldots, n,$$

such that $|f - h| < \epsilon$ on K and $|f| > \alpha_j$ on bP_j for all $j \geq j_0$. These conditions are easily achieved by an inductive choice of the constants $a_j > 0$ and of the exponents $t_j \in \mathbb{N}$. (See [241, Theorem 2, p. 220] for further details.)

For every $j \in \mathbb{N}$ let $\mathcal{H}(j)$ denote the set of all $f \in \mathcal{O}(X)^n$ such that there exist j different integers $k_1, \ldots, k_j \in \mathbb{N}$ with the property that $|f(x)| > k_i$ for all $x \in bP_{k_i}$ and $i = 1, \ldots, j$. Clearly $\mathcal{H}(j)$ is open in $\mathcal{O}(X)^n$, and by the first part of the proof it is also dense. Hence $\mathcal{H} = \cap_{j=1}^{\infty} \mathcal{H}(j)$ is residual in $\mathcal{O}(X)^n$. Every $f \in \mathcal{H}$ determines an almost proper map $f\colon X \to \mathbb{C}^n$, since we get an infinite sequence of integers $k \in \mathbb{N}$ such that $|f(x)| > k$ holds for all $x \in bP_k$.

By Theorem 7.9.1 on p. 324 (which is an application of the jet transversality theorem) there is a residual set $\mathcal{H}' \subset \mathcal{O}(X)^n$ such that every $f \in \mathcal{H}'$ satisfies the estimates (8.1). The set $\mathcal{H} \cap \mathcal{H}'$ is still residual in $\mathcal{O}(X)^n$, and every $f \in \mathcal{H} \cap \mathcal{H}'$ is an almost proper map $X \to \mathbb{C}^n$ satisfying (8.1). $\square$

Assume that X is a Stein n-manifold and that $f = (f_1, \ldots, f_n)\colon X \to \mathbb{C}^n$ is an almost proper holomorphic map. Then there is a dense set of functions $f_{n+1} \in \mathcal{O}(X)$ such that $(f_1, \ldots, f_{n+1})\colon X \to \mathbb{C}^{n+1}$ is a proper holomorphic map [241, Theorem 4, p. 221]. (It suffices to choose f_{n+1} to be large enough on the compact connected components of each sublevel set $\{|f| \le k\}$ for every $k \in \mathbb{N}$.) An argument similar to that in the proof of Theorem 8.1.1 gives the following result.

Theorem 8.1.2. *For every n-dimensional Stein manifold X there is a residual set $\mathcal{P} \subset \mathcal{O}(X)^{n+1}$ such that every $f \in \mathcal{P}$ is a proper holomorphic map $f\colon X \to \mathbb{C}^{n+1}$ satisfying the estimates*

$$\dim\big\{x \in X\colon \dim\ker \mathrm{d}f_x \ge i\big\} \le n - i(i+1), \quad i = 1, 2 \ldots \tag{8.2}$$

Variants of Theorems 8.1.1 and 8.1.2, with interpolation on a discrete set in X, were proved in [400, §2].

Assume now that $(X, \mathcal{O}_X)$ is a (possibly nonreduced) Stein space of dimension n and of finite embedding dimension

$$n_0 = \mathrm{embdim} X = \sup_{x \in X} \mathrm{embdim}_x X < \infty. \tag{8.3}$$

According to Theorem 2.2.8 (c) (p. 49) such X is biholomorphic to a closed (nonreduced) complex subspace of $\mathbb{C}^m$ with $m = \max\{n + n_0, 2n + 1\}$. We consider the more general situation that X is a closed complex subspace of an m-dimensional Stein manifold $\widetilde{X}$. Choose a stratification

$$X = X_0 \supset X_1 \supset X_2 \supset \cdots \supset X_l = \emptyset$$

by closed complex subvarieties of X such that $\mathrm{embdim}_x X$ is constant on every (smooth) stratum $X_r^* = X_r \backslash X_{r+1}$. After refining the stratification may assume that it is Whitney regular (see Theorem 7.8.1, p. 316), and hence the jet transversality theorem applies (Theorem 7.8.5, p. 317). For every $k = 0, 1, \ldots, n_0 = \mathrm{embdim} X$ we let

$$n(k) = \dim\big\{x \in X\colon \mathrm{embdim}_x X = k\big\} \le k. \tag{8.4}$$

Note that $n(k)$ is the maximal complex dimension of the strata Z in the above stratification of X such that $\operatorname{embdim}_x X = k$ for all $x \in Z$. We also set

$$b_X = \max\left\{k + \left[\frac{n(k)}{2}\right] : k = 0, 1, \ldots, n_0\right\}, \tag{8.5}$$

$$N(n) = n + q(n) = \max\left\{\left[\frac{3n}{2}\right] + 1, b_X, 3\right\}. \tag{8.6}$$

Note that the invariant b_X (8.5) is denoted $b'(X)$ in [436].

Let $\mathcal{J}_X$ denote the sheaf of ideals of the subvariety X in $\widetilde{X}$. Then $\mathcal{J}(X) = \Gamma(\widetilde{X}, \mathcal{J}_X)$ is a closed subspace of $\mathcal{O}(X)$, and the set

$$\mathcal{H} = \mathcal{O}(\widetilde{X})^n \times \mathcal{J}(X)^{m-n} \subset \mathcal{O}(\widetilde{X})^m$$

is a Fréchet space.

For a holomorphic map between complex spaces, $\mathrm{d}f_x \colon \mathrm{T}_x X \to \mathrm{T}_{f(x)} Y$ denotes its differential on the Zariski tangent spaces (see §1.3).

Theorem 8.1.3. [436, Theorem 1.1] *If X is a closed complex subspace of an m-dimensional Stein manifold $\widetilde{X}$ then the set all $H = (h, h') \in \mathcal{H}$ satisfying the following two conditions is residual in $\mathcal{H}$:*

(i) *$H \colon \widetilde{X} \to \mathbb{C}^m$ is almost proper (and hence so is $h|_X \colon X \to \mathbb{C}^n$), and*

(ii) $\dim\{x \in X \colon \dim\ker(\mathrm{d}h|_X)_x \geq i\} < 2(q(n) - i + 1)$ *for $i = 1, \ldots, n_0$, where $q(n)$ is defined by (8.6) above.*

Proof. The proof of (i) is similar to the first part of proof of Theorem 8.1.1, the main addition being that we can exhaust the manifold $\widetilde{X}$ by special analytic polyhedra defined by maps from $\mathcal{H}$. (For the details see [48] or [241].) We now turn to (ii). As in §7.9 we consider the varieties

$$\Sigma_h^i = \left\{x \in X \colon \dim\ker(\mathrm{d}h|_X)_x = i\right\}, \quad \Lambda_h^i = \cup_{j \geq i} \Sigma_h^j.$$

Assume that $Z \subset X_r^* = X_r \backslash X_{r+1}$ is a stratum with $\dim \mathrm{T}_x X = k$ for all $x \in Z$; then $\dim Z \leq n(k) \leq k$ by (8.4). By Theorem 7.9.1 (p. 324) there is a residual set of holomorphic maps $h \colon \widetilde{X} \to \mathbb{C}^n$ that satisfy

$$\dim \Lambda_h^i \cap Z \leq n(k) - i(n + i - k) \tag{8.7}$$

for $k = 0, \ldots, n_0$ and $\max\{1, k - n\} \leq i \leq k$. (This estimate is trivial for the other values of i.) From the definition of $q(n)$ (8.6) it follows by an elementary computation (see Appendix A in [436]) that

$$n(k) - i(n + i - k) < 2\big(q(n) - i + 1\big)$$

holds for the indicated range of i and k. Since a generic map $h \colon \widetilde{X} \to \mathbb{C}^n$ satisfies (8.7) on every stratum Z in a given stratification of X, there is a residual set of maps $h \in \mathcal{O}(\widetilde{X})^n$ that satisfy Theorem 8.1.3 (ii). The product of this set with $\mathcal{J}(X)^{m-n}$ is clearly residual in $\mathcal{H}$. $\square$

8.2 Embedding Stein Manifolds into Euclidean Spaces of Minimal Dimension

In this and the following two sections we present results on the existence of proper holomorphic immersions and embeddings of Stein manifolds of dimension $n > 1$ to Euclidean spaces of minimal dimension. The constructions essentially depend on the Oka-Gromov principle from Chapter 6.

The following precise embedding and immersion theorem was proved by Eliashberg and Gromov; an improvement of the embedding dimension by one for odd values of $n \in \mathbb{N}$ is due to Schürmann.

Theorem 8.2.1. [133, 436] *Every Stein manifold of dimension $n > 1$ embeds properly holomorphically into $\mathbb{C}^N$ for $N = \left[\frac{3n}{2}\right] + 1$, and it immerses properly holomorphically into $\mathbb{C}^M$ for $M = \left[\frac{3n+1}{2}\right]$.*

This was conjectured by O. Forster [152, p. 184]. He gave the following example showing that these dimensions are minimal for every $n > 1$.

Example 8.2.2. [152, Proposition 3, p. 183] Let Y be the Stein surface

$$Y = \{[x\colon y\colon z] \in \mathbb{P}^2\colon\ x^2 + y^2 + z^2 \neq 0\}.$$

Given an integer $n \geq 2$, let X denote the n-dimensional Stein manifold

$$X = \begin{cases} Y^m & \text{if } n = 2m, \\ Y^m \times \mathbb{C} & \text{if } n = 2m+1. \end{cases}$$

It is easily seen that the Stiefel-Whitney class $w_{2m}(\mathrm{T}X)$ of the tangent bundle of X is the nonzero element of the group $H^{2m}(X;\mathbb{Z}_2) = H^{2m}((\mathbb{RP}^2)^m;\mathbb{Z}_2) = \mathbb{Z}_2$, and consequently the Chern class $c_m(\mathrm{T}X)$ is the nonzero element of $H^{2m}(X;\mathbb{Z}) = \mathbb{Z}_2$ [152, Proposition 3]. It follows by purely topological reasons [270, p. 263] that X does not embed properly holomorphically in $\mathbb{C}^{[3n/2]}$, and it does not immerse in $\mathbb{C}^{[3n/2]-1}$. $\square$

In 1971 Eliashberg and Gromov announced in [132] that every Stein n-manifold for $n > 1$ embeds properly holomorphically in $\mathbb{C}^N$ with $N = \left[\frac{3n}{2}\right] + 2$. Twenty years later they proved the embedding theorem with $N = \left[\frac{3n+1}{2}\right] + 1$ [133]; this is the optimal number in Theorem 8.2.1 if n even, and it is larger by one if n is odd. The optimal result for odd dimensional manifolds was obtained by Schürmann [436].

In dimension $n = 1$, Theorem 8.2.1 does not give any improvement over Remmert's theorem [411] that every open Riemann surface X admits a proper holomorphic embedding in $\mathbb{C}^3$. The problem of embedding open Riemann surfaces in $\mathbb{C}^2$ is discussed in §8.9 – §8.10 below.

Recall that every real analytic manifold M of dimension n admits a complexification X, a complex manifold of complex dimension n containing M as

a total real submanifold. Furthermore, M admits a basis of Stein neighborhoods in X. Theorem 8.2.1 implies the following embedding theorem for real analytic manifolds, improving Grauert's result in [224].

Corollary 8.2.3. *Every real analytic manifold M of dimension n admits a proper real analytic embedding onto a totally real submanifold of $\mathbb{C}^{[\frac{3n}{2}]+1}$.*

Schürmann also proved the following embedding theorem for Stein spaces of finite embedding dimension; this includes Theorem 8.2.1 since $b_X = \left[\frac{3n}{2}\right]$ when X is nonsingular and $\dim X = n$.

Theorem 8.2.4. [436, Theorem 0.5] *An n-dimensional Stein space X with $n_0 = \operatorname{embdim} X < +\infty$ (8.3) embeds properly holomorphically into $\mathbb{C}^N$ with $N = \max\left\{\left[\frac{3n}{2}\right]+1, b_X, 3\right\}$, where $b_X = \max\{k + [\frac{n(k)}{2}]: k = 1, \dots, n_0\}$ and $n(k) = \dim\{x \in X: \operatorname{embdim}_x X \geq k\}$. Such X immerses properly holomorphically to $\mathbb{C}^M$ with $M = \max\left\{\left[\frac{3n+1}{2}\right], b_X\right\}$.*

If X does not have bounded embedding dimension then Theorem 2.2.8 (b) still seems the best existing results. However, for Stein spaces with isolated singular points see also Theorem 8.4.3 below.

Let $h: X \to Y$ be a holomorphic map. We say that a holomorphic map $f: X \to \mathbb{C}^q$ is a *relative embedding* (resp. a *relative immersion*) over h if $(h, f): X \to Y \times \mathbb{C}^q$ is an embedding (resp. an immersion). Let

$$\begin{aligned} \Delta_h^2 &= \{(x_1, x_2) \in X \times X:\ x_1 \neq x_2,\ h(x_1) = h(x_2)\}, \\ \Sigma_h^i &= \{x \in X: \dim \ker \mathrm{d}h_x = i\}, \qquad i = 1, 2, \dots. \end{aligned}$$

The main ingredient in the proof of Theorem 8.2.1 is the following *Relative Embedding Theorem* [133, 2.A., p. 126] proved in the next section.

Theorem 8.2.5. *Let X be a Stein space with $\dim X = n$ and $\operatorname{embdim} X = n_0 < \infty$. Assume that Y is a Stein manifold and that $h: X \to Y$ is a proper holomorphic map satisfying the following conditions for some integer $q \geq 2$:*

(i) $\dim \Delta_h^2 < 2q$, *and*
(ii) $\dim \Sigma_h^i < 2(q - i + 1)$ *for* $i = 1, 2, \dots, n_0$.

Let K be a compact $\mathcal{O}(Y)$-convex set in Y, and let L be a union of connected components of $h^{-1}(K)$. Given a neighborhood $U \subset X$ of L and a relative embedding $f_0: U \to \mathbb{C}^q$ over h, there exists for every $\epsilon > 0$ a relative embedding $f: X \to \mathbb{C}^q$ over h such that $\operatorname{dist}(f(x), f_0(x)) < \epsilon$ for all $x \in L$. If $q = 1$ then the above conclusion holds if the set Δ_h^3 of triple points of h is 0-dimensional.

If h satisfies (ii) then there exists a relative immersion $f: X \to \mathbb{C}^q$ over h.

Assuming Theorem 8.2.5 for the time being, we now give

Proof (of Theorem 8.2.4). We assume that $(X, \mathcal{O}_X)$ is reduced; in the general case we can work on an ambient Stein manifold $\widetilde{X}$ as in Theorem 8.1.3.

Write $N = n + q$. Theorem 8.1.1 (p. 334) furnishes an almost proper holomorphic map $h\colon X \to \mathbb{C}^n$ satisfying Theorem 8.2.5 (ii). Since $q \geq \left[\frac{n}{2}\right] + 1$, we have $n < 2q$ and hence condition (i) holds as well. Choose an exhaustion $K = K_1 \subset K_2 \subset \cdots \subset \cup_{j=1}^{\infty} K_j = \mathbb{C}^n$ by compact $\mathcal{O}(Y)$-convex sets such that K_j is contained in the interior of K_{j+1} for every $j \in \mathbb{N}$. Let L_j be a union of finitely many (compact) connected components of the set $h^{-1}(K_j) \subset X$ such that $L_1 = L$ is as in the theorem, $L_j \subset \mathrm{Int} L_{j+1}$ for every j, and $\cup_{j=1}^{\infty} L_j = X$. Note that $h\colon \mathrm{Int} L_j \to \mathrm{Int} K_j$ is a proper holomorphic map for every $j \in \mathbb{N}$. The set $Q_j = h^{-1}(K_j) \cap L_{j+1} \backslash L_j$ is compact and $\mathcal{O}(X)$-convex. Set $K_0 = \emptyset$, $L_0 = Q_0 = \emptyset$. We construct a sequence of holomorphic maps $f_j\colon U_j \to \mathbb{C}^q$ on open neighborhoods $U_j \supset L_j$ and a sequence of positive real numbers $1 = \epsilon_0 > \epsilon_1 > \cdots$ such that the following properties hold for $j = 1, 2, \ldots$:

(a) the map $(h, f_j)\colon U_j \to \mathbb{C}^n \times \mathbb{C}^q = \mathbb{C}^N$ is an embedding on L_j,
(b) $|f_j(x) - f_{j-1}(x)| < 2^{-j}\epsilon_{j-1}$ for all $x \in L_{j-1}$,
(c) $|f_j| > j$ on Q_{j-1}, and
(d) for every holomorphic map $\widetilde{f}\colon U_j \to \mathbb{C}^q$ satisfying $|\widetilde{f} - f_j|_{L_j} < \epsilon_j$ the map $(h, \widetilde{f})$ is an embedding on L_{j-1}.

Choose a number $\epsilon = \epsilon_1 \in (0, 1)$ and let $f_1 = f$; hence (a) holds on L_1 while properties (b)–(d) are vacuous for $j = 1$. Assume inductively that we have constructed maps $f_1, \ldots, f_j$ and numbers $\epsilon_1, \ldots, \epsilon_j$ such that (a)–(d) hold up to index j. Since Q_j is $\mathcal{O}(X)$-convex, Theorem 8.2.5 furnishes a holomorphic map $g_j\colon V_j \to \mathbb{C}^q$ on a neighborhood $V_j \supset Q_j$ such that (h, g_j) is an embedding on Q_j. By adding a large constant to g_1 we insure that $|g_1| > j + 2$ on Q_j. Another application of Theorem 8.2.5 gives a holomorphic map $f_{j+1}\colon U_{j+1} \to \mathbb{C}^q$ on a neighborhood U_{j+1} of L_{j+1} such that (h, f_{j+1}) is an embedding on L_{j+1} and $|f_{j+1} - f_j| < 2^{-j-1}\epsilon_j$ on $L_j \cup Q_j$. It follows that conditions (a)–(c) hold for the index $j+1$. Since embeddings are stable under perturbations, we can choose a number $\epsilon_{j+1} \in (0, \epsilon_j)$ such that (d) holds for $j + 1$. This completes the induction step, and the induction may proceed.

By the construction the limit $f = \lim_{j\to\infty} f_j\colon X \to \mathbb{C}^q$ exists and is holomorphic on X. Indeed, on L_j we have $f = f_j + \sum_{l=j}^{\infty}(f_{l+1} - f_l)$ and hence

$$|f - f_j| \leq \sum_{l=j}^{\infty} 2^{-l-1}\epsilon_l < 2^{-j}\epsilon_j \quad \text{on } L_j \text{ for } j = 1, 2, \ldots.$$

Property (d) implies that (h, f) is an embedding on L_{j-1}. Since this holds for every j, f is a relative embedding over h. Finally, by (c) we have $|f| > j$ on Q_j for all $j \in \mathbb{N}$. It follows that $(h, f)^{-1}(K_j \times j\overline{\mathbb{B}}) \subset L_j$ for every j, and hence this set is compact. (Here $\mathbb{B}$ is the unit ball in $\mathbb{C}^q$.) Since the sets $K_j \times j\overline{\mathbb{B}}$ exhaust $Y \times \mathbb{C}^q$, the map $(h, f)\colon X \to Y \times \mathbb{C}^q$ is proper.

If we replace N by $M = \max\left\{\left[\frac{3n+1}{2}\right], b_X\right\} = n+q$, then condition (ii) in Theorem 8.2.5 still holds (since if follows from $n+q \le b_X$) and we get a proper immersion $(h,f)\colon X \to \mathbb{C}^M$. (See also Remark 8.3.5 on p. 346.) $\square$

Remark 8.2.6. Eliashberg and Gromov [133] begin instead with a generic *proper* background map $h\colon X \to \mathbb{C}^{n+1}$; this gives the same result for even values of n, but is off by one for odd values of n. Almost proper maps were introduced to this proof in [436]. $\square$

8.3 Proof of the Relative Embedding Theorem

In this section we prove Theorem 8.2.5 (p. 338). We begin with a few reductions. We may assume that the proper holomorphic map $h\colon X \to Y$ is onto Y, and hence $\dim Y = n$. (In the general case we replace Y by the closed complex subvariety $f(X) \subset Y$; singularities do not present any difficulties in the proof.) Next we observe that it suffices to prove the approximation result in the special case when $L = h^{-1}(K)$. Indeed, let Q be the union of those connected components of $h^{-1}(K)$ that do not meet L; then Q is compact and $\mathcal{O}(X)$-convex. Since h maps suitably chosen small open neighborhoods of Q in X properly onto small neighborhoods of K in Y, the absolute version of the theorem (without approximation), applied to this restricted map, gives a relative embedding f_0' on a neighborhood of Q. By taking f_0 near L and f_0' near Q we get a relative embedding defined near $h^{-1}(K)$, and it suffices to prove the approximation statement for this new map.

We stratify X and Y so that the basic topological characteristics of the map $h\colon X \to Y$ are constant along each stratum in the following sense.

Definition 8.3.1. *Let $S \subset Y$ be a locally closed connected complex submanifold of Y and set $\widetilde{S} = h^{-1}(S) \subset X$. We say that S is* equisingular *with respect to a proper holomorphic map $h\colon X \to Y$ if the following hold:*

(i) *the map $h|_{\widetilde{S}}\colon \widetilde{S} \to S$ is a (proper) immersion,*

(ii) $\dim\ker \mathrm{d}h_x$ *is constant on every connected component of $\widetilde{S}$, and*

(iii) $\ker \mathrm{d}h_x$ *is transverse to $\widetilde{S}$ at every point $x \in \widetilde{S}$.*

It follows from (i) that $h\colon \widetilde{S} \to S$ is a finitely sheeted holomorphic covering projection. Property (ii) means that each connected component of $\widetilde{S}$ is contained in one of the sets $\Sigma_h^i \setminus \Sigma_h^{i+1}$, where $\Sigma_h^i = \{x \in X\colon \dim\ker \mathrm{d}h_x = i\}$. Note that (iii) is a consequence of (ii).

The proof of the following lemma is straightforward.

Lemma 8.3.2. ([133, Proposition 3.A$_2$], [496, Proposition 2.1]) *Every subvariety $Y_0 \subset Y$ admits an equisingular stratification, that is, a descending*

sequence of closed complex subvarieties $Y_0 \supset Y_1 \supset \cdots \supset Y_m$, where the strata (the connected components of differences $S_i = Y_i \backslash Y_{i+1}$) are smooth locally closed subvarieties of Y that are equisingular with respect to h.

The proof of Theorem 8.2.5 consists of a repeated application of two basic operations: (A) separations of points over a stratum, and (B) elimination of the kernel of the differential over a stratum.

(A) *Separation of points over a stratum.* Let $Y_1 \subset Y_0$ be closed complex subvarieties of Y such that $S = Y_0 \backslash Y_1$ is a connected complex submanifold that is equisingular with respect to h. (In a typical situation, Y_0 and Y_1 are two subvarieties from an equisingular stratification of Y.) Let $X_j = h^{-1}(Y_j)$ for $j = 0, 1$; these are closed complex subvarieties of X such that $\widetilde{S} = X_0 \backslash X_1$ is smooth and $h\colon \widetilde{S} \to S$ is a finite holomorphic covering map.

Lemma 8.3.3. [133, Lemma 3.B$_1$] *Let $K \subset Y$ and $L \subset X$ be as in Theorem 8.2.5. Assume that $f\colon X \to \mathbb{C}^q$ is a holomorphic map such that $(h, f)\colon X \to Y \times \mathbb{C}^q$ is an embedding on a neighborhood of $Y_1 \cup L$. If $q \geq 2$ and $2q > \dim Y_0$ then for a suitably small neighborhood L' of L and for every $\epsilon > 0$ there exists a holomorphic map $f'\colon X \to \mathbb{C}^q$ such that*

(i) *$(h, f')\colon X \to Y \times \mathbb{C}^q$ is injective on $\widetilde{S}$,*
(ii) *$f - f'$ vanishes to the second order along the subvariety X_1, and*
(iii) *$|f(x) - f'(x)| < \epsilon$ for $x \in L'$.*

If $q = 1$ and $\dim Y_0 = 1$ then such f' exists provided that h has no triple points over $S = Y_0 \backslash Y_1$ (i.e., every fiber $h^{-1}(y)$ for $y \in S$ has at most two points).

Proof. The assumptions imply that $f(x_1) \neq f(x_2)$ for any pair of distinct points $x_1, x_2 \in X$ such that $h(x_1) = h(x_2) \in Y_1 \cup K$, and $\mathrm{d}f_x \cdot v \neq 0$ for every vector $0 \neq v \in \ker \mathrm{d}h_x$, $x \in X_1 \cup L$. The same is then true if we replace K and L by a pair of slightly larger $\mathcal{O}(Y)$-convex sets $K_1 \subset Y$, resp. $L_1 \subset X$.

We seek a map f' with desired properties in the form

$$f'(x) = f_\alpha(x) = f(x) + \sum_{j=1}^{N} \alpha_j(h(x))\psi_j(x), \quad x \in X, \tag{8.8}$$

where $\alpha_j \in \mathcal{O}(Y)$ are holomorphic functions on Y (to be determined), and $\psi_1, \ldots, \psi_N\colon X \to \mathbb{C}^q$ (with $N \geq dq$) are holomorphic maps chosen such that

(a) each ψ_j vanishes to the second order along the subvariety X_1, and
(b) for every point $y \in S$ let $h^{-1}(y) = \{x_1, \ldots, x_d\} \in \widetilde{S}$ (distinct points); then the following $dq \times N$ matrix has maximal rank dq:

$$\begin{pmatrix} \psi_1(x_1) & \psi_2(x_1) & \ldots & \psi_N(x_d) \\ \psi_1(x_1) & \psi_2(x_1) & \ldots & \psi_N(x_d) \\ \cdots & \cdots & \cdots & \cdots \\ \psi_1(x_1) & \psi_2(x_1) & \ldots & \psi_N(x_d) \end{pmatrix}$$

The existence of such maps $\psi_1, \dots, \psi_N$ follows from Cartan's Theorem A. It is trivial to achieve this rank condition at one point $y \in S$, and hence the same holds outside a proper subvariety; by adding more functions we complete the proof by induction on the dimension of the exceptional set.

Property (b) of the collection $\{\psi_j\}$ implies that for every fixed $y \in S$ and $h^{-1}(y) = \{x_1, \dots, x_d\} \in \widetilde{S}$ the linear map $\Phi_y \colon \mathbb{C}^N \to \mathbb{C}^{dq}$ given by

$$\alpha = (\alpha_1, \dots, \alpha_N) \overset{\Phi_y}{\longmapsto} \left(f(x_1) + \sum_{j=1}^N \alpha_j \psi_j(x_1), \dots, f(x_d) + \sum_{j=1}^N \alpha_j \psi_j(x_d) \right)$$

is surjective. Consider the trivial bundle $\pi \colon E = Y \times \mathbb{C}^N \to Y$. Let $\Sigma \subset E$ be the subset of the total space whose fiber Σ_y over any point $y \in S$ consists of all $\alpha = (\alpha_1, \dots, \alpha_N) \in \mathbb{C}^N$ such that

$$f(x_i) + \sum_{j=1}^N \alpha_j \psi_j(x_i) = f(x_k) + \sum_{j=1}^N \alpha_j \psi_j(x_k)$$

holds for at least one pair of indexes $1 \le i \ne k \le d$. (For $y \in Y \backslash S$ set $\Sigma_y = \emptyset$.) By introducing the codimension q linear subspaces

$$\Lambda_{i,k} = \left\{ (b_1, \dots, b_d) \in (\mathbb{C}^q)^d \colon b_i = b_k \right\} \subset \mathbb{C}^{dq}$$

we see that

$$\Sigma_y = \bigcup_{1 \le i \ne k \le d} \Phi_y^{-1}(\Lambda_{ik}) \subset \mathbb{C}^N.$$

Since the map $\Phi_y \colon \mathbb{C}^N \to \mathbb{C}^{dq}$ is affine linear and surjective, Σ_y is a union of $\binom{d}{2}$ affine linear subspaces of codimension q in $\mathbb{C}^N$. Furthermore, the points $x_i(y)$ in the fiber $h^{-1}(y)$ depend locally holomorphically on $y \in S$, and hence Σ_y are the fibers of a complex subvariety $\Sigma \subset E|_S$.

To complete the proof, it suffices to find a holomorphic section $A(y) = (y, \alpha(y))$ of the trivial vector bundle $Y \times \mathbb{C}^n \to Y$ whose image avoids Σ. Indeed, for such α the map $f' = f_\alpha \colon X \to \mathbb{C}^q$ (8.8) satisfies the separation property over the stratum S, and the remaining properties can also be arranged. We are thus looking for a holomorphic map $\alpha \colon Y \to \mathbb{C}^N$ such that

$$\alpha(y) \notin \Sigma_y \quad \text{for all } y \in Y. \tag{8.9}$$

For points y in a neighborhood of $Y_1 \cup K$ this holds for the constant map $\alpha = 0$ since f is a relative embedding there. The assumption $2q > \dim Y_0$ insures the existence of a continuous extension α_0 to $Y_0 \cup K$ satisfying (8.9) and such that $\alpha_0 = 0$ on $Y_1 \cup K$. (We can then extend it an arbitrary way to Y, although this will not be needed.)

Clearly the subvariety $\Sigma \subset E|_S$ is locally uniformly tame in the sense of Proposition 6.4.14 on p. 256. If $q \ge 2$ then the cited proposition implies that

the restricted submersion $\pi\colon E|_S\backslash\Sigma \to S$ is elliptic. If $q = 1$ and $d = 2$ then Σ_y consists of a single affine hyperplane in $\mathbb{C}^N$ depending holomorphically on $y \in S$ (as $\binom{d}{2} = 1$ in this case); hence $\mathbb{C}^N\backslash\Sigma_y \cong \mathbb{C}^{N-1} \times \mathbb{C}^*$ is complex homogeneous, and thus elliptic. In each of these two case, Theorem 6.2.2 (the Oka principle for sections of stratified elliptic submersions) shows that α_0 is homotopic to a holomorphic section α avoiding Σ such that $\alpha|_{Y_1} = 0$ and α approximates $\alpha_0 = 0$ uniformly on a neighborhood of K. (See also Theorem 6.2.3 on p. 243 that pertains to this situation.) The corresponding map $f' = f_\alpha\colon X \to \mathbb{C}^q$ (8.8) then satisfies Lemma 8.3.3. Since $f' - f$ vanishes to the second order on X_1, f' is still a relative embedding on X_1, and on L this holds if the approximation is close enough over a neighborhood of L. □

(B) *Elimination of the kernel.* Let $Y_1 \subset Y_0 \subset Y$, $S = Y_0\backslash Y_1$, $X_1 \subset X_0 \subset X$ and $\widetilde{S} = h^{-1}(S)$ be as in (A) above, and let $f'\colon X \to \mathbb{C}^q$ be furnished by Lemma 8.3.3. Thus (h, f') is an embedding over a neighborhood of $X_1 \cup L$ and it is injective on $\widetilde{S} = h^{-1}(S)$, but it need not be an embedding on X_0 since its differential may have nontrivial kernel transverse to $\mathrm{T}\widetilde{S}$. We now explain how to change f' to a immersion over X_0.

Lemma 8.3.4. [133, Lemma 3.C$_1$] *If* $\dim \Sigma_h^i < 2(q - i + 1)$ *holds for some integer* $q \in \mathbb{N}$ *and for all* $i = 1, 2, \ldots, n_0$*, then for a suitably small neighborhood* $L' \subset X$ *of* L *and for every* $\epsilon > 0$ *there exists a holomorphic map* $f''\colon X \to \mathbb{C}^q$ *satisfying the following properties:*

(i) $f'' = f'$ *on* X_0*,*
(ii) $f'' - f'$ *vanishes to second order along* X_1*,*
(iii) $|f''(x) - f'(x)| < \epsilon$ *for all* $x \in L'$*, and*
(iv) $\ker \mathrm{d}h_x \cap \ker \mathrm{d}f''_x = \{0\}$ *for every* $x \in X_0$*.*

Property (i) shows that (h, f'') is injective on X_0 (as it equals (h, f') there), (ii) shows that (h, f'') is an embedding on X_1 (since it agrees along X_1 to the second order with (h, f') which is an embedding on X_1), (iii) shows that (h, f'') remains an embedding on a neighborhood of L if the approximation is close enough, and (iv) insures that (h, f'') is and immersion, and hence an embedding, also on the larger subvariety X_0.

Proof (of Lemma 8.3.4). The submanifold $\widetilde{S} = h^{-1}(S) = X_0\backslash X_1$ of X is mapped by h properly onto the connected manifold $S = Y_0\backslash Y_1$, and hence $\widetilde{S}$ is the union of at most finitely many connected manifold $\widetilde{S}_1, \ldots, \widetilde{S}_m$. Since S is equisingular with respect to h, every component $\widetilde{S}_k$ is contained in $\Sigma_h^i\backslash\Sigma_h^{i+1}$ for some $i = i(k)$, that is, $\dim \ker \mathrm{d}h_x = i$ for every point $x \in \widetilde{S}_k$. The kernels $\ker \mathrm{d}h_x$ form a holomorphic vector subbundle $E^k \subset \mathrm{T}X|_{\widetilde{S}_k}$ of rank i.

By Cartan's Theorem A there exist finitely many holomorphic maps $\psi_1, \ldots, \psi_N\colon X \to \mathbb{C}^q$ satisfying the following properties:

- the ψ_j vanish on the subvariety X_0,
- the ψ_j vanish to the second order on the subvariety X_1, and
- $\big(\cap_{j=1}^N \ker(\mathrm{d}\psi_j)_x\big) \cap \ker \mathrm{d}h_x = \{0\}$ for every $x \in \widetilde{S}$.

The last property means that, setting $\psi = (\psi_1, \dots, \psi_N)\colon X \to (\mathbb{C}^q)^N$, the differential $\mathrm{d}\psi_x$ is injective on $E_x^k = \ker \mathrm{d}h_x$ for every $x \in \widetilde{S}_k$ and $k = 1, \dots, m$. This is satisfied for a generically chosen ψ provided that N is large enough. We now seek a solution $f''\colon X \to \mathbb{C}^q$ in the form

$$f''(x) = f'(x) + \sum_{j=1}^N \alpha_j(x)\psi_j(x), \quad x \in X \tag{8.10}$$

where $\alpha_j \in \mathcal{O}(X)$ must be chosen. Since ψ vanishes on X_0, we have

$$\mathrm{d}f''_x = \mathrm{d}f'_x + \sum_{j=1}^N \alpha_j(x)\,(\mathrm{d}\psi_j)_x, \quad x \in X_0.$$

Hence the injectivity of $\mathrm{d}f''_x$ on $\ker \mathrm{d}h_x$ for $x \in \widetilde{S}$ leads to the algebraic condition that the graph of the map $\alpha = (\alpha_1, \dots, \alpha_N)\colon X \to \mathbb{C}^N$ avoids a certain subvariety $\varSigma \subset X \times \mathbb{C}^N$ which projects to $X_0 \backslash X_1$. Indeed, given a point $x \in \widetilde{S}_k \subset \widetilde{S}$, the 'bad' set $\varSigma_x \subset \mathbb{C}^N$ consists of those N-tuples $\alpha = (\alpha_1, \dots, \alpha_N) \in \mathbb{C}^N$ for which the $\mathbb{C}$-linear map

$$\theta_x(\alpha) = \mathrm{d}f'_x + \sum_{j=1}^N \alpha_j\,(\mathrm{d}\psi_j)_x\colon \mathrm{T}_x X \to \mathrm{T}_{f'(x)}\mathbb{C}^q$$

fails to be injective on the i-dimensional subspace $E_x^k \subset \mathrm{T}_x X$ (where $i = i(k)$).

Choose a local holomorphic trivialization near the point x of the holomorphic vector bundle $\mathrm{T}X|_{\widetilde{S}_k}$ and of its rank i holomorphic vector subbundle $E^k = \ker \mathrm{d}h \subset \mathrm{T}X|_{\widetilde{S}_k}$. For every $\alpha \in \mathbb{C}^N$ let

$$\varTheta_x(\alpha) \in M_{q,i} \cong \mathbb{C}^q \times \mathbb{C}^i$$

denote the $q \times i$ matrix representing $\theta_x(\alpha)|_{E_x^k} \in \mathrm{Hom}(E_x^k, \mathrm{T}_{f'(x)}\mathbb{C}^q)$. Our choice of the maps $\psi_j\colon X \to \mathbb{C}^q$ insures that the affine map

$$\mathbb{C}^N \ni \alpha \mapsto \theta_x(\alpha)|_{E_x^k} \in \mathrm{Hom}(E_x^k, \mathrm{T}_{f'(x)}\mathbb{C}^q) \cong \mathrm{Hom}(\mathbb{C}^i, \mathbb{C}^q)$$

is surjective for every $x \in \widetilde{S}$ and it depends holomorphically on x. Hence

$$\varSigma_x = \varTheta_x^{-1}(\varLambda) \subset \mathbb{C}^N, \quad \varLambda = \big\{A \in M_{q,i}\colon \mathrm{rank}\, A < \min\{i, q\}\big\}.$$

By Lemma 7.9.2 (p. 324), $\varLambda$ is a subvariety of codimension $|q-i|+1 \geq 1$ whose complement $M_{q,i} \backslash \varLambda$ is $\mathbb{C}$-homogeneous. Hence $\varSigma_x$ is an algebraic subvariety

of $\mathbb{C}^N$ with homogeneous complement, and $\widetilde{S} \times \mathbb{C}^N \backslash \Sigma \to \widetilde{S}$ is a holomorphic fiber bundle with homogeneous fibers. Thus the Oka principle (Theorem 6.2.3 on p. 243) applies to sections $\alpha: X \to X \times \mathbb{C}^N$ whose graphs avoid Σ.

The properties of f' insure that the zero section $\alpha_0 = 0$ avoids Σ for all points in a neighborhood of $X_1 \cup L$. By the assumption, the estimate

$$\dim \widetilde{S}_k \leq \dim \Sigma_h^i < 2(q - i + 1)$$

holds for every connected component $\widetilde{S}_k$ of $\widetilde{S}$. If $q < i$ then $\widetilde{S}_k = \emptyset$ and hence this case need not be considered. If $q \geq i$ then the above condition reads

$$\dim \widetilde{S}_k < 2 \operatorname{codim} \Sigma_x, \quad x \in \widetilde{S}.$$

Since $\widetilde{S} = X_0 \backslash X_1$ and X_0 is Stein, this conditions suffices to extend the constant map $\alpha_0 = 0$ from a neighborhood of $X_1 \cup L$ to a continuous map $\alpha_0: X \to \mathbb{C}^N$ such that $\alpha_0(x) \notin \Sigma_x$ for all $x \in X_0$. (See Theorem 5.14.1 and Corollary 5.14.2 on p. 233 for the details of this argument.) By the Oka principle (Theorem 6.2.3, p. 243) the map α_0 is homotopic to a holomorphic map $\alpha: X \to \mathbb{C}^q$ such that $\alpha = 0$ on X_1, α is arbitrary close to 0 on a neighborhood of L, and $\alpha(x) \notin \Sigma_x$ for all $x \in X_0$. (Theorem 6.2.3 shows that we can make α holomorphic over X_0, and we can then extend it to X in an arbitrary way since the subvariety Σ projects to X_0.) The corresponding holomorphic map f'' given by (8.10) then satisfies Lemma 8.3.4, and hence $(h, f''): X \to Y \times \mathbb{C}^q$ is an embedding over a neighborhood of $X_0 \cup L$. $\square$

Proof (of Theorem 8.2.5). Let K be a compact $\mathcal{O}(Y)$-convex set in Y and $L = f^{-1}(K)$. Assume that $f: U \to \mathbb{C}^q$ is a holomorphic map on an open set $U \supset L$ such that f is a relative embedding over h. Choose another compact $\mathcal{O}(Y)$-convex set $K_1 \subset Y$ containing K and set $L_1 = f^{-1}(K_1) \subset X$. It suffices to show that f can be approximated uniformly on L by a holomorphic map $f_1: U_1 \to \mathbb{C}^q$ on a neighborhood $U_1 \supset L_1$ such that (h, f_1) is an embedding over L_1; the proof is then completed by induction over a suitable exhaustion of X by compact holomorphically convex sets.

We stratify Y as in Lemma 8.3.2 above and let $X_j = h^{-1}(Y_j)$ be the corresponding stratification of X. Although every difference $Y_k \backslash Y_{k+1}$ may have infinitely many connected components (strata), only finitely many of them intersect the compact set K_1. Replacing Y and X by a neighborhood of K_1 and L_1, respectively, and by refining the stratifications we may assume that every difference $Y_k \backslash Y_{k+1}$ is connected and there are only finitely many strata. Hence the induction described below ends in finitely many steps.

We begin by extending f to a map $f: X \to \mathbb{C}^q$ that is a relative embedding over $L \cup X_m$ where $X_m = h^{-1}(Y_m)$ is zero-dimensional (discrete). We then proceed to change f in finitely many steps so as to make it a relative embedding over the successive strata in X. In a typical step we have a pair of closed complex subvarieties $Y_{k+1} \subset Y_k$ of Y whose difference $S = Y_k \backslash Y_{k+1}$ is

equisingular with respect to h, and a holomorphic map $f\colon X \to \mathbb{C}^q$ such that (h, f) is an embedding over a neighborhood of the set $X_{k+1} \cup L$ in X. We now change f so that we get an embedding over $X_k \cup L$. The situation is exactly as described above for the pair $Y_1 \subset Y_0$: First use Lemma 8.3.3 to separate points over $X_k = h^{-1}(Y_k)$, and subsequently apply Lemma 8.3.4 to eliminate the kernel of the differential over X_k. The dimension estimates in Theorem 8.2.5 insure that each of these steps can be carried out. □

Remark 8.3.5. In the proof of Theorem 8.2.4, with $M = \max\left\{\left[\frac{3n+1}{2}\right], b_X\right\} = n + q$, condition (i) in Theorem 8.2.5 holds for all strata S of dimension $< n$ since $2q \geq 2\left[\frac{n+1}{2}\right] \geq n$. Hence we obtain a holomorphic map $f\colon X \to \mathbb{C}^q$ such that $(h, f)\colon X \to \mathbb{C}^M$ is a proper immersion that is injective on all strata of dimension $< n$ in a given stratification of X. However, if $2q = n$, we might not be able to separate points over the n-dimensional stratum.

If we take $q = n + 1$ then the separation of points and the elimination of the kernel (in the proof of Theorem 8.2.5) can be achieved by transversality theorem. This gives a proper holomorphic embedding $X^n \hookrightarrow \mathbb{C}^{2n+1}$ and a proper holomorphic immersion $X^n \looparrowright \mathbb{C}^{2n}$ (Theorem 2.2.8 on p. 49). □

8.4 Weakly Regular Embeddings and Interpolation

In this section we present further results on embedding Stein spaces to complex Euclidean spaces. We begin with a theorem of J. Prezelj [400] on *weakly regular embeddings* of Stein spaces with isolated singular points. Before stating her result, we must recall the following notions.

For every point x in a complex space X let $C_4(X, x) \subset C_5(X, x)$ denote the Whitney cones of X at the point x [515], [85, p. 91]. Embedding a neighborhood of x in X as a complex subvariety of some $\mathbb{C}^m$, we have

$$\begin{aligned} C_4(X,x) = \{v \in \mathbb{C}^m\colon &\text{there exist a sequence } x_j \in X_{\mathrm{reg}} \text{ such that } x_j \to x, \\ &\text{and a sequence } v_j \in \mathrm{T}_{x_j}X \text{ such that } v_j \to v\}; \\ C_5(X,x) = \{v \in \mathbb{C}^m\colon &\text{there exist sequences } x_j, y_j \in X_{\mathrm{reg}} \text{ with } x_j, y_j \to x, \\ &\text{and a sequence } \lambda_j \in \mathbb{C} \text{ with } \lambda_j(x_j - y_j) \to v\}. \end{aligned}$$

The cones $C_4(X, x) \subset C_5(X, x)$ are closed algebraic sets in $\mathbb{C}^m$ contained in the Zariski tangent space $\mathrm{T}_x X$, and we have

$$n \leq \dim C_4(X,x) \leq \dim C_5(X,x) \leq 2n, \quad n = \dim_x X.$$

If $x \in X_{\mathrm{reg}}$ then clearly $\dim C_4(X, x) = \dim C_5(X, x) = n$; conversely, $\dim C_5(X, x) = n$ implies that $x \in X_{\mathrm{reg}}$. For these and other properties of Whitney cones see [515] and [85].

Definition 8.4.1. [400, Def. 2.11] *A holomorphic map* $f\colon X \to \mathbb{C}^N$ *is* weakly regular *at a point* $x \in X$ *if* $\ker \mathrm{d}f_x \cap C_5(X,x) = \{0\}$*;* f *is weakly regular on* X *is this holds at every point* $x \in X$*.*

The notions of regular and weakly regular coincide at a regular point $x \in X_{\mathrm{reg}}$. The following lemma explains the significance of this condition.

Lemma 8.4.2. [400, Corollary 2.10] *If a holomorphic map* $f\colon X \to \mathbb{C}^N$ *is weakly regular at a point* $x \in X$ *then there is a neighborhood* $U \subset X$ *of* x *such that* $f|_U\colon U \to \mathbb{C}^N$ *is an injective map that is an immersion on* $U \cap X_{\mathrm{reg}}$*.*

Proof. We first show that f is regular on X_{reg} near the point x. If not, there exist a sequences $x_j \in X_{\mathrm{reg}}$ converging to x and tangent vectors $v_j \in \mathrm{T}_{x_j}X$ of length $|v_j| = 1$ such that $\mathrm{d}f_{x_j} \cdot v_j = 0$ for all $j = 1, 2, \ldots$. Passing to a subsequence we have $\lim_{j\to\infty} v_j = v \in \mathrm{T}_x X$. Then $0 \neq v \in C_4(X,x)$ and $\mathrm{d}f_x \cdot v = 0$, contradicting the assumption that f is weakly regular at x.

If f fails to be injective in every neighborhood of x, there exist sequences $x_j \neq y_j$ in X converging to x such that $f(x_j) - f(y_j) = 0$ for $j = 1, 2, \ldots$. Let $v_j = \frac{x_j - y_j}{|x_j - y_j|}$. From $0 = f(x_j) - f(y_j) = \mathrm{d}f_{x_j} \cdot (x_j - y_j) + o(|x_j - y_j|)$ we see that $\mathrm{d}f_{x_j} \cdot v_j \to 0$ as $j \to \infty$. By passing to a subsequence we may assume that $v_j \to v$, and hence $\mathrm{d}f_x \cdot v = 0$. Since $v \in C_5(X,x)$ and $|v| = 1$, this again contradicts the assumption that f is weakly regular at x. $\square$

Theorem 8.4.3. (Prezelj [400, Theorem 1.1]) *Assume that* X *is a Stein space of dimension* n *with isolated singular points. Let*

$$N(n) = \max\left\{\left[\frac{3n}{2}\right] + 1, \max_{x\in X} \dim C_5(X,x), 3\right\}. \tag{8.11}$$

There exists a proper, injective holomorphic map $f\colon X \hookrightarrow \mathbb{C}^{N(n)}$ *which is regular (an immersion) on the regular locus* X_{reg}*.*

Since $\dim C_5(X,x) \le 2\dim_x X$, we have $N(n) \le 2n$ for $n > 1$, and hence this result improves Theorem 2.2.8 (b). Observe that $f\colon X \to f(X) \subset \mathbb{C}^{N(n)}$ is a homeomorphism of X onto its image.

Proof. Without loss of generality we may assume that the space X is reduced. By the assumption the singular locus $X_{\mathrm{sing}} = \{x_1, x_2, \ldots\}$ is discrete. Let $N = N(n) = n + q$. The condition $N \ge \dim C_5(X,x)$ insures the existence of a holomorphic map $g_0 = (h_0, f_0)\colon X \to \mathbb{C}^{n+q} = \mathbb{C}^N$ such that

$$\ker(\mathrm{d}g_0)_{x_j} \cap C_5(X, x_j) = \{0\}, \quad j = 1, 2, \ldots.$$

In view of Lemma 8.4.2 it follows that, locally near every singular point x_j, g_0 is regular on X_{reg} and injective. By a minor improvement of Theorem 8.1.3 (see [400, Proposition 3.1]) we find an almost proper map $h\colon X \to \mathbb{C}^n$ such

that $h - h_0$ vanishes to a given finite order $m_j \in \mathbb{N}$ at each of the singular points x_j, and such that $\ker \mathrm{d}h$ satisfies the estimates in Theorem 8.1.3 (ii) on the regular locus X_{reg}. We then find a holomorphic map $f\colon X \to \mathbb{C}^q$ such that $f - f_0$ vanishes to a high order at each point $x_j \in X_{\mathrm{sing}}$, and such that the map $(h, f)\colon X \to \mathbb{C}^N$ is proper, injective, and regular on X_{reg}. This is done as in the proof of Theorem 8.2.4. If $g = (h, f)$ agrees with $g_0 = (h_0, f_0)$ to a sufficiently high order at each point $x_j \in X_{\mathrm{sing}}$ (which can be insured by the construction), then we have

$$\ker \mathrm{d}g_{x_j} \cap C_5(X, x_j) = \{0\}, \quad x_j \in X_{\mathrm{sing}}.$$

Hence g is injective and regular on X_{reg} near x_j. Elsewhere we insure injectivity, and regularity on X_{reg}, just as in the proof of Theorem 8.2.4. $\square$

Remark 8.4.4. As pointed out by Prezelj in [400], Theorem 8.4.3 gives embedding results for smooth, compact, strongly pseudoconvex, integrable CR manifolds of dimension $2n-1 \geq 5$ and CR dimension $n-1 \geq 2$. Any such M bounds a relatively compact domain D in a pure n-dimensional Stein space X with finitely many isolated singularities; hence the problem of finding a CR embedding $M \hookrightarrow \mathbb{C}^N$ with minimal N reduces to the problem of finding a weakly regular embedding $X \hookrightarrow \mathbb{C}^N$. $\square$

We now consider the extension problem for proper regular maps and embeddings. Given a Stein space X, a closed complex subspace X' of X, and a proper holomorphic embedding $\varphi\colon X' \hookrightarrow \mathbb{C}^N$, the problem is to find a holomorphic map $f\colon X \to \mathbb{C}^N$ that extends φ and satisfies some other regularity properties (such as being proper, injective, regular on X_{reg}, or an embedding). The following result of Acquistapace, Broglia and Tognoli generalizes the theorems of Bishop and Narasimhan (Theorem 2.2.8 on p. 49).

Theorem 8.4.5. [6, Theorem 1] *Assume that X is a reduced Stein space of dimension n, X' is a closed complex subspace of X, and $\varphi\colon X' \hookrightarrow \mathbb{C}^N$ is a proper holomorphic embedding for some $N \geq 2n+1$. Then the set of all holomorphic maps $f\colon X \to \mathbb{C}^N$ that extend φ and are proper, injective and regular on X_{reg}, is dense in the space of all holomorphic maps $X \to \mathbb{C}^N$ extending φ. In particular, if X is nonsingular (a Stein manifold) then the above holds for proper holomorphic embeddings $X \hookrightarrow \mathbb{C}^N$ extending φ.*

The same authors proved the following relative version of the theorem of Wiegmann [517].

Theorem 8.4.6. [6, Theorem 2] *Let $(X, \mathcal{O}_X)$ be a possibly nonreduced Stein space of dimension n and of embedding dimension m. Let X' be a closed subspace of X and let $\varphi\colon X' \hookrightarrow \mathbb{C}^N$ be an embedding with $N \geq m+n$. Then the set of all embeddings $f\colon X \hookrightarrow \mathbb{C}^N$ which extend φ is dense in the set of all holomorphic maps $X \to \mathbb{C}^N$ extending φ.*

The analogue of both results is true for real analytic spaces if X and X' carry a coherent structure. We do not prove these results here.

The following example shows that Theorem 8.4.5 is close to optimal – interpolation fails in general for proper holomorphic embeddings $X^n \hookrightarrow \mathbb{C}^N$ when $N < 2n$. We identify $\mathbb{C}^{n-1}$ with the hyperplane $\mathbb{C}^{n-1} \times \{0\} \subset \mathbb{C}^n$.

Proposition 8.4.7. *For every integer $n > 1$ there exists a proper holomorphic embedding $\varphi\colon \mathbb{C}^{n-1} \hookrightarrow \mathbb{C}^{2n-1}$ that can not be extended to an injective holomorphic map $f\colon \mathbb{C}^n \hookrightarrow \mathbb{C}^{2n-1}$.*

Proof. Theorem 4.18.10 on p. 171 furnishes a proper holomorphic embedding $\varphi\colon \mathbb{C}^{n-1} \hookrightarrow \mathbb{C}^{2n-1}$ such that the complement of its image $\Omega = \mathbb{C}^{2n-1} \backslash \varphi(\mathbb{C}^{n-1})$ is Eisenman n-hyperbolic. Write $z = (z', z_n) \in \mathbb{C}^n$. Assume that $f\colon \mathbb{C}^n \to \mathbb{C}^{2n-1}$ is an injective holomorphic map extending φ; then f has rank n at a generic point. Note that $f(z', 0) = \varphi(z')$ and $f(z', z_n) \notin \varphi(\mathbb{C}^{n-1})$ if $z_n \neq 0$. The entire map $F(z', z_n) = f(z', \mathrm{e}^{z_n})$ maps $\mathbb{C}^n$ to Ω and has rank n at a generic point. This contradicts to the choice of φ. $\square$

A similar construction gives for every pair of integers $1 \leq k < n$ a proper holomorphic embedding $\varphi\colon \mathbb{C}^k \hookrightarrow \mathbb{C}^{n+k}$ that can not be extended from $\mathbb{C}^k = \mathbb{C}^k \times \{0\}^{n-k} \subset \mathbb{C}^n$ to an injective holomorphic map $\mathbb{C}^n \to \mathbb{C}^{n+k}$.

In light of Theorem 8.2.1 and Example 8.4.7, the best possible result concerning the interpolation of holomorphic embeddings would the following.

Problem 8.4.8. Let X be an n-dimensional Stein manifold and $X' \subset X$ a k-dimensional complex submanifold. Let

$$N(n,k) = \max\left\{\left[\frac{3n}{2}\right] + 1, n + k + 1, 3\right\}.$$

Does every proper holomorphic embedding $X' \hookrightarrow \mathbb{C}^N$ for $N \geq N(n,k)$ extend to a proper holomorphic embedding $X \hookrightarrow \mathbb{C}^N$? $\square$

Not much seems known beyond Theorems 8.4.5 and 8.4.6, except if X' is discrete; in the latter case we have the following analogue of Theorem 8.2.1. Let b_X denote the number (8.5) on p. 336.

Theorem 8.4.9. [185, 400] *If X is an n-dimensional Stein space with finite embedding dimension and if $N \geq \max\left\{\left[\frac{3n}{2}\right] + 1, b_X, 3\right\}$, then for every pair of discrete sequences $\{a_j\} \subset X$, $\{b_j\} \subset \mathbb{C}^N$ without repetition there is a proper holomorphic embedding $f\colon X \hookrightarrow \mathbb{C}^N$ satisfying $f(a_j) = b_j$ for $j = 1, 2, \ldots$. If X is a Stein manifold of dimension $n > 1$ then this holds for $N \geq \left[\frac{3n}{2}\right] + 1$.*

Proof. Write $N = n + q$. The proof of Theorem 8.2.1 furnishes a proper holomorphic embedding $G = (h, g)\colon X \hookrightarrow \mathbb{C}^{n+q} = \mathbb{C}^N$ (where $h\colon X \to \mathbb{C}^n$ is

an almost proper map) such that the discrete sequence $\{G(a_j)\}_{j\in\mathbb{N}}$ is tame in $\mathbb{C}^N$ (Definition 4.6.1, p. 118). In fact, it suffices to insure in the construction of $g\colon X \to \mathbb{C}^q$ that the sequence $|g(a_j)|$ tends to infinity fast enough so that the projection of the set $\{G(a_j)\}_{j\in\mathbb{N}} \subset \mathbb{C}^{n+q}$ onto $\{0\}^n \times \mathbb{C}^q$ is proper; tameness then follows from Theorem 4.6.2 (a) on p. 118. By Theorem 4.17.1 (p. 164) there exist a Fatou-Bieberbach domain $\Omega \subset \mathbb{C}^N$, containing the complex subvariety $G(X)$ of $\mathbb{C}^N$, and a biholomorphic map $\Phi\colon \Omega \to \mathbb{C}^N$ onto $\mathbb{C}^N$, such that $\Phi(G(a_j)) = b_j$ for $j = 1, 2, \ldots$. The composed map $f = \Phi \circ G\colon X \hookrightarrow \mathbb{C}^N$ then satisfies the conclusion of the theorem. $\square$

Theorem 8.4.9 was first proved by Prezelj in [400, Theorem 1.1] under the assumption that X is a Stein n-manifold and $N \geq \left[\frac{3n+1}{2}\right] + 1$; this number is the same as in [133]. She obtained similar interpolation results for immersions, proper maps, and almost proper maps.

8.5 The H-Principle for Holomorphic Immersions

In the proof of Lemma 8.3.4 we have not used the hypothesis that the manifold Y was Stein or that the background map $h\colon X \to Y$ was proper. Essentially the same proof gives the following *relative Oka principle for immersions*, due to Eliashberg and Gromov [133], for more general background maps.

Given a holomorphic map $h\colon X \to Y$, we define

$$\dim_h X = \sup_{y\in Y} \dim h^{-1}(y).$$

Thus $\dim_h X = 0$ precisely when the fiber $h^{-1}(y)$ over any point $y \in Y$ is 0-dimensional, and hence a discrete subset of X.

Theorem 8.5.1. [133, 2.E$_1$, p. 128] *Let X be a Stein space and $h\colon X \to Y$ a holomorphic map. If $\dim_h X = 0$ then relative immersions $f\colon X \to \mathbb{C}^q$ over h satisfy the h-principle in the sense of Gromov: There exists a holomorphic map $f\colon X \to \mathbb{C}^q$ such that $(h, f)\colon X \to Y \times \mathbb{C}^q$ is an immersion if and only if there exists a continuous complex vector bundle map $\theta\colon \mathrm{T}X \to \mathrm{T}\mathbb{C}^q$ such that $\mathrm{d}h \oplus \theta\colon \mathrm{T}X \hookrightarrow \mathrm{T}(Y \times \mathbb{C}^q)$ is injective.*

Proof (sketch). We stratify X such that for every stratum S, $h|_S\colon S \to Y$ is an immersion and $\dim \ker \mathrm{d}h_x$ is independent of the point $x \in S$. (It follows that $\ker \mathrm{d}h|_S$ is transverse to S and is a holomorphic vector subbundle of $\mathrm{T}X|_S$.) We then construct a relative immersion $f\colon X \to \mathbb{C}^q$ by passing from one stratum to the next one as in the proof of Theorem 8.2.5 (see the end of the previous section), using Lemma 8.3.4 at every step. The necessary topological condition to apply the Oka principle is the existence of a map θ as in the statement of the theorem. $\square$

Theorem 8.5.1 also holds with approximation on a compact $\mathcal{O}(X)$-convex subset K in X: If $f_0\colon X \to \mathbb{C}^q$ is a continuous map that is holomorphic on a neighborhood of K, and if there exists a vector bundle map $\theta\colon \mathrm{T}X \to \mathrm{T}\mathbb{C}^q$ as in the theorem such that $\theta = \mathrm{d}f_0$ on a neighborhood of K, then we can deform f_0 to a holomorphic map $f\colon X \to \mathbb{C}^q$ such that (h, f) is an immersion on X, and such that f approximates f_0 as close as desired on K.

The proof of Lemma 8.3.4 shows that the dimension estimates on $\Sigma_h^i = \{x \in X\colon \dim\ker \mathrm{d}h_x = i\}$ in the following corollary imply the existence of a map $\theta\colon \mathrm{T}X \to \mathrm{T}\mathbb{C}^q$ in Theorem 8.5.1.

Corollary 8.5.2. [133, 2.E$_2$.] *Let X be a Stein space of finite embedding dimension and let $h\colon X \to Y$ be a holomorphic map with $\dim_h X = 0$. If*

$$\dim \Sigma_h^i \le 2(q - i + 1), \quad i = 1, 2, \ldots, \mathrm{embdim} X,$$

for some $q \in \mathbb{N}$ then there exists a holomorphic immersion $(h, f)\colon X \to Y \times \mathbb{C}^q$.

It is not known whether *proper* holomorphic immersions or embeddings of Stein manifolds satisfy the h-principle. However, non-proper holomorphic immersions of Stein manifolds satisfy the following h-principle (Eliashberg and Gromov [132]; see also [236, §2.1.5]).

Theorem 8.5.3. *If X is a Stein manifold whose complex cotangent bundle $\mathrm{T}^*X \cong \mathrm{T}^{*1,0}X$ is generated by q differential $(1,0)$-forms $\theta_1, \ldots, \theta_q$ for some $q > \dim X$, then there exists a holomorphic immersion $X \to \mathbb{C}^q$. More precisely, every such q-tuple $\theta = (\theta_1, \ldots, \theta_q)$ can be changed by a homotopy (through q-tuples generating T^*X) to a q-tuple $\mathrm{d}f = (\mathrm{d}f_1, \ldots, \mathrm{d}f_q)$ of differentials, where $f = (f_1, \ldots, f_q)\colon X \to \mathbb{C}^q$ is a holomorphic immersion. Every n-dimensional Stein manifold admits a (not necessarily proper) holomorphic immersion into $\mathbb{C}^{[\frac{3n}{2}]}$.*

The idea of the proof of Theorem 8.5.3 is the following. By the Oka-Grauert principle (§7.2) we may assume that θ_j are holomorphic 1-forms. In the first step θ_q is replaced by the differential $\mathrm{d}f_q$ of a holomorphic function on X such that $\theta_1, \ldots, \theta_{q-1}, \mathrm{d}f_q$ still generate T^*X. Since $q > \dim X$, we may assume that the forms $\theta_1, \ldots, \theta_{q-1}$ generate T^*X outside a proper complex subvariety $\Sigma \subset X$, and f_q must satisfy an algebraic condition on its jet along Σ provided that the forms $\theta_1, \ldots, \theta_{q-1}$ are chosen generically (this requires a small perturbation). Once f_q has been chosen, one proceeds in the same way with the forms $\theta_1, \ldots, \theta_{q-2}, \mathrm{d}f_q$; after a small perturbation of these forms we can replace θ_{q-1} with an exact differential $\mathrm{d}f_{q-1}$ such that $\theta_1, \ldots, \theta_{q-2}, \mathrm{d}f_{q-1}, \mathrm{d}f_q$ span T^*X. In q steps all forms are replaced with holomorphic differentials.

For a complete exposition and some generalizations (including the approximation on compact $\mathcal{O}(X)$-convex subsets, and the 1-parametric h-principle) see the paper [308].

8.6 The Oka Principle for Proper Holomorphic Maps

A classical subject in complex analysis concerns the existence and boundary regularity of proper holomorphic mappings $f\colon D \to D'$ between domains $D \Subset \mathbb{C}^n$ and $D' \Subset \mathbb{C}^N$ for $1 \le n \le N$ (or, more generally, between domains in Stein manifolds). This is closely related to questions concerning CR maps. Indeed, if a proper holomorphic map $f\colon D \to D'$ extends continuously (or smoothly) to $\bar{D}$, then its restriction to the boundary $f|_{bD}\colon bD \to bD'$ is a CR map of bD into bD'. Conversely, if the domains are bounded, bD is connected and $n = \dim > 1$, then every CR map $bD \to bD'$ extends to a proper holomorphic map $D \to D'$ by Bochner's extension theorem.

Here are some typical questions that have been considered in the literature:

(A) *Existence of maps:* Which domains $D \subset \mathbb{C}^n$ can be mapped properly holomorphically to a given domain $D' \subset \mathbb{C}^N$? What is the answer when D' is a model domain (the ball $\mathbb{B}^N$, the polydisc $\mathbb{D}^N$, a bounded symmetric domain, etc.)? And what is the answer if the source D is one of these models?

(B) *Boundary regularity, equidimensional domains:* Is every proper holomorphic mapping $f\colon D \to D'$ between smoothly bounded domains $D, D' \Subset \mathbb{C}^n$ for $n > 1$ smooth up to the boundary? If the boundaries are real analytic, does every such map extend holomorphically across the boundary?

(C) *Boundary regularity, positive codimension:* When $n < N$, there exist proper holomorphic maps $\mathbb{B}^n \to \mathbb{B}^N$ that do not extend continuously up to the boundary of $\mathbb{B}^n$, and maps that are continuous at the boundary need not be smooth. The main question is the following: Assuming that $D \Subset \mathbb{C}^n$ and $D' \Subset \mathbb{C}^N$ are bounded domains with smooth (or real analytic) boundaries, is there an integer $k = k(D, D') \in \mathbb{Z}_+$ such that every $\mathcal{C}^k$ map $f\colon \bar{D} \to \bar{D}'$ that is holomorphic in D and maps bD to bD' is automatically $\mathcal{C}^\infty$ on $\bar{D}$ (extends holomorphically to a neighborhood of $\bar{D}$ in the $\mathcal{C}^\omega$ case)? If D and D' are strongly pseudoconvex, does this hold for $k = 1$?

(D) *Regularity of CR mappings* (the local version of questions (B) and (C)): Which geometric conditions on a pair of smooth (real analytic, real algebraic) CR manifolds $M \subset \mathbb{C}^n$, $M' \subset \mathbb{C}^{n'}$, insure that a CR map $f\colon M \to M'$ with some initial regularity is actually smooth (analytic, algebraic)?

A proper discussion of this subject would require another volume, and it is not our intention to do it here. We refer the reader to the monograph [30] (especially for the regularity theory), and to the survey articles [35, 43, 165] for results up to 1992.

However, since question (A) concerning the *existence* of proper maps is compatible with the principal theme of this book – Oka theory, we describe some of the main results in this direction. Until recently this problem has been considered mainly for maps between domains in Euclidean spaces, using the Euclidean structure and solution to the $\overline{\partial}$-equation. (The case when the

target manifold is Stein allows a simple reduction to the Euclidean case by the standard embedding and retraction theorems.) The highlights include, but are not limited to, the construction of:

- a proper holomorphic embedding of every strongly pseudoconvex domain in a Stein manifold to some strongly convex domain in a high dimensional Euclidean space [142, 255];
- proper holomorphic maps and embeddings of strongly pseudoconvex domains to polydiscs and balls [346, 161, 209];
- nonrational proper holomorphic maps of balls $\mathbb{B}^n \to \mathbb{B}^N$ for any $2 \le n < N$ [242, 462];
- proper holomorphic maps of strongly pseudoconvex domains to smoothly bounded pseudoconvex domains, and to q-convex domains for suitable values of q [117, 118, 119];
- proper holomorphic maps from the disc $\mathbb{D}$; see the introduction to the paper [121] and the references therein.

The main idea in the proof of the convex embedding theorem [142, 255] is the following. Let D be a bounded strongly pseudoconvex domain in a Stein manifold X of dimension n. For every boundary point $p \in bD$, Narasimhan's lemma gives a local biholomorphic map $f = (f_1, \ldots, f_n)$ from a neighborhood $U \subset X$ of p onto a domain in $\mathbb{C}^n$ such that the image $f(D \cap U) \subset \mathbb{C}^n$ is strongly convex along the hypersurface $f(bD \cap U)$. The main point now is that there exists an f with this property which is holomorphic in a neighborhood of all of $\bar{D}$. (One finds such f by solving a suitable Cousin problem.) A finite collection of such maps, convexifying different boundary points of bD, embeds D properly into a product of convex domains in some high dimensional space $\mathbb{C}^N$. By smoothing the corners one can then show that the image of $\bar{D}$ is actually contained in a smoothly bounded strongly convex domain Ω in $\mathbb{C}^N$. If bD is real analytic then we can also choose $\Omega \Subset \mathbb{C}^N$ to have real analytic boundary [161, Theorem 1.2].

The constructions of proper holomorphic maps of a (strongly) pseudoconvex domain $D \Subset \mathbb{C}^n$ (or in a Stein manifold) into a prescribed domain $\Omega \subset \mathbb{C}^m$ for $m > n$, or to a more general complex manifold, is substantially more difficult. All known constructions exploit the Levi geometry of a well chosen exhaustion function $\rho\colon \Omega \to \mathbb{R}$ on the target domain.

We explain the main idea in the simplest case when D is the unit disc $\mathbb{D} \subset \mathbb{C}$. We wish to map the disc $\mathbb{D}$ properly into some complex manifold X of dimension $n > 1$. Assume that X admits a Morse exhaustion function $\rho\colon X \to \mathbb{R}$ whose Levi form $\mathcal{L}_\rho$ has at least two positive eigenvalues at every point of X. (This means that X is 2-complete, see Def. 1.8.9.) The construction proceeds by induction. Assume that $f\colon \overline{\mathbb{D}} \to X$ is a map of class $\mathcal{A}(\mathbb{D})$ (continuous on $\overline{\mathbb{D}}$ and holomorphic in $\mathbb{D}$) such that the boundary curve $f(b\mathbb{D}) \subset X$ does not contain any critical point of ρ. At every step of the induction we change

f so that the value of $\rho \circ f$ increases by a controlled amount (depending on the geometry of ρ) on a boundary arc $C \subset b\mathbb{D}$, while at the same time insuring that the new map $f_1 \colon \overline{\mathbb{D}} \to X$ approximates f sufficiently well on a smaller disc $r\mathbb{D}$ and that $\rho \circ f$ drops very little everywhere on $\overline{\mathbb{D}}$. This is achieved by considering the geometry of ρ in local holomorphic coordinates $z = (z_1, \ldots, z_n)$ near a point $p_0 = f(\zeta_0) \in f(b\mathbb{D}) \subset X$, where $\zeta_0 \in b\mathbb{D}$. For simplicity assume that $z(p_0) = 0$. The Taylor expansion of ρ at 0 (see §1.8) is

$$\rho(z) = \rho(0) + \Re\left(2\sum_{j=1}^{n} \frac{\partial \rho}{\partial z_j}(0) z_j + \sum_{j,k=1}^{n} \frac{\partial^2 \rho}{\partial z_j \partial z_k}(0) z_j z_k\right) + \frac{1}{2}\mathcal{L}_{\rho,0}(z) + o(|z|^2).$$

Let Σ be the smooth local complex hypersurface, defined by the vanishing of the quadratic holomorphic polynomial under the parentheses in the first term on the right hand side. For points $z \in \Sigma$ we thus have

$$\rho(z) = \rho(0) + \frac{1}{2}\mathcal{L}_{\rho,0}(z) + o(|z|^2).$$

Since $\mathcal{L}_{\rho,0}$ is assumed to have at least two positive eigenvalues, there is a holomorphic disc $z = g(\tau) \in \Sigma$ ($\tau \in \overline{\mathbb{D}}$), $g(0) = 0$, such that $\rho(g(\tau)) \geq c|\tau|^2$ for some $c > 0$. In fact, this happens if we choose g such that $\mathcal{L}_{\rho,0}(g'(0)) > 0$. This construction can be done uniformly for all points in some arc $C \subset b\mathbb{D}$ around any point $\zeta_0 \in b\mathbb{D}$. This gives a family of analytic discs $g_\zeta \colon \overline{\mathbb{D}} \to X$, depending continuously on the point $\zeta \in C \subset b\mathbb{D}$, such that

$$g_\zeta(0) = f(\zeta), \quad \rho(g(\tau)) \geq \rho(f(\zeta)) + c|\tau|^2$$

holds for all $\zeta \in C$ and $\tau \in \overline{\mathbb{D}}$. The constant $c > 0$ in the above estimate can be chosen the same for all discs in the family, and it only depends on the geometry of ρ near the point $p_0 = f(\zeta_0) \in X$.

Choose a slightly smaller arc $C' \subset C$ around ζ_0. The main point now is to find a new holomorphic disc $f_1 \colon \overline{\mathbb{D}} \to X$ such that for every $\zeta \in C'$ the point $f_1(\zeta)$ lies very close to the curve $\Gamma_\zeta = f(\zeta) + g_\zeta(b\mathbb{D}) \subset X$, while at the same time $f_1(\zeta)$ is close to $f(\zeta)$ for all $\zeta \in \overline{\mathbb{D}}$ outside of a small neighborhood of the arc C'. Moreover, for every $\zeta = re^{i\theta} \in \overline{\mathbb{D}}$ the point $f_1(\zeta)$ lies close to the set $f(\mathbb{D}) \cup g_{e^{i\theta}}(\overline{\mathbb{D}}) \subset X$. We first finds a solution h over a small disc $\Delta \subset \mathbb{D}$ such that $C \subset b\Delta \cap b\mathbb{D}$; here we can work in local holomorphic coordinates on X and take an approximate solution of a Riemann-Hilbert type problem. We then glue h with f by the method of sprays to get an analytic disc f_1 satisfying the desired properties. This technique is explained in §5.8 and §5.9.

With this method we can lift the boundary of an analytic disc in X in finitely many steps from a level $\rho = c_1$ to a higher level $\rho = c_2$, as long as ρ has no critical values on the interval $[c_1, c_2]$. (This is the so called noncritical case.) It remains to consider the situation when the boundary curve $f(b\mathbb{D})$ comes close to some critical point $p \in X$ of ρ. By a general position argument

we can insure that $f(b\mathbb{D})$ avoids the stable manifold of p. Lemma 3.10.3 (p. 95) then allows us to push the boundary curve over the critical level $\rho = \rho(p)$ by using the noncritical procedure with another 2-convex function. (For a Stein manifold X this was first explained in [211].)

This method generalizes to the case when $\mathbb{D}$ is replaced by any smoothly bounded strongly pseudoconvex domain D in a Stein manifold S, and X is a complex manifold with a q-convex exhaustion function for a suitable value of $q = q(n)$. We have the following result.

Theorem 8.6.1. [123, Theorem 1.1] *Assume that X is an n-dimensional complex manifold, Ω is an open subset of X, $\rho\colon \Omega \to (0, +\infty)$ is a smooth Morse function whose Levi form has at least r positive eigenvalues at every point of Ω for some $r \le n$, and for any pair of numbers $0 < c_1 < c_2$ the set $\{x \in \Omega\colon c_1 \le \rho(x) \le c_2\}$ is compact. Let $D \Subset S$ be a smoothly bounded strongly pseudoconvex domain in a Stein manifold S, and let $f_0\colon \bar{D} \to X$ be a continuous map that is holomorphic in D and satisfies $f_0(bD) \subset \Omega$. Suppose that at least one of the following conditions hold:*

(a) *$r \ge 2d$, where $d = \dim_{\mathbb{C}} D$;*

(b) *$r \ge d + 1$ and ρ has no critical points of index $> 2(n - d)$ in Ω.*

Then f_0 can be approximated uniformly on compacts in D by holomorphic maps $f\colon D \to X$ such that $f(z) \in \Omega$ for every $z \in D$ sufficiently close to bD, and $\lim_{z \to bD} \rho(f(z)) = +\infty$. Moreover, given an integer $k \in \mathbb{Z}_+$, f can be chosen to agree with f_0 to order k at each point in a given finite set $\sigma \subset D$.

When X is a domain in $\mathbb{C}^n$, Theorem 8.6.1 is due to A. Dor [119]. In the special case when D is a bordered Riemann surface (i.e., $\dim D = 1$), this result coincides with [121, Theorem 1.1]. For a survey of this subject see [121, 123] and the references therein.

Recall that a smooth function ρ on an n-dimensional complex manifold X whose Levi form $\mathcal{L}_\rho$ has at least r positive eigenvalues at every point in an open set $\Omega \subset X$ is said to be $(n - r + 1)$-convex on Ω (Def. 3.1.2, p. 58). All Morse indexes of such function are $\le r + 2(n - r) = 2n - r$. (See Lemma 3.9.4 on p. 91 for the quadratic normal form of such function at a critical point.) The Morse condition can always be achieved by a small perturbation of ρ in the fine $\mathcal{C}^2$-topology. Hence condition (a) in Theorem 8.6.1 implies that all Morse indexes of ρ in Ω are $\le 2(n - d)$, and therefore condition (b) holds as well. When $d = 1$ (i.e., D is a bordered Riemann surface), conditions (a) and (b) are both equivalent to $r \ge 2$. When $d > 1$, (b) is weaker than (a).

Example 8.6.2. [123, Example 1.3] We show that condition (b) in Theorem 8.6.1 can not be weakened for any pair of dimensions $1 \le d < n$. Hence the Oka principle may fail for *proper* holomorphic maps, even when it holds for ordinary holomorphic maps.

Given integers $1 \le d < n$, set $m = n - d + 1 \in \{2, \ldots, n\}$. Let $\mathbb{T}^m = \mathbb{C}^m/\Gamma$ be a complex torus that does not contain any closed complex curves. (Most tori of dimension > 1 are such; an explicit example with $m = 2$ can be found in [508, p. 222].) Set

$$X = (\mathbb{T}^m \backslash \{p\}) \times \mathbb{C}^{n-m} = (\mathbb{T}^m \backslash \{p\}) \times \mathbb{C}^{d-1}. \tag{8.12}$$

Note that X is an Oka manifold. Choose a function $\tau \colon \mathbb{T}^m \backslash \{p\} \to \mathbb{R}$ that equals $|y - y(p)|^{-2}$ in some local holomorphic coordinates y on $\mathbb{T}^m$ near p. The exhaustion function $\rho(y, w) = \tau(y) + |w|^2$ on X has no critical points in a deleted neighborhood of p, and its Levi form has $1 + n - m = d$ positive eigenvalues near $\{p\} \times \mathbb{C}^{d-1}$. Thus X satisfies condition (b) in Theorem 8.6.1 for domains D of dimension $< d$, but not for domains of dimension $\ge d$. It is easily seen that no d-dimensional Stein manifold D admits a proper holomorphic map to the manifold X. Indeed, suppose that $f \colon D \to X$ is such a map. Let $\pi \colon X \to \mathbb{C}^{d-1}$ denote the projection $\pi(y, w) = w$, and consider the map $\pi \circ f \colon D \to \mathbb{C}^{d-1}$. By dimension reasons there exists a point $w \in \mathbb{C}^{d-1}$ for which $\Sigma = \{z \in D \colon \pi(f(z)) = w\}$ is a subvariety of positive dimension in D. Since D is Stein, Σ contains a one dimensional subvariety C, and $f(C)$ is then a closed complex curve in $\mathbb{T}^m \backslash \{p\} \times \{w\}$. Since a point is a removable singularity for positive dimensional analytic subvarieties, $\overline{f(C)}$ is a nontrivial closed complex curve in $\mathbb{T}^m \times \{w\}$, a contradiction to the choice of $\mathbb{T}^m$. $\square$

Let us consider two examples of particular interest. In the first we assume that $\rho \colon X \to \mathbb{R}$ is a smooth Morse exhaustion function, and we apply Theorem 8.6.1 with $\Omega = \{\rho > c\}$ for some $c \in \mathbb{R}$.

Corollary 8.6.3. *Let X be an n-dimensional complex manifold, and let D be a d-dimensional strongly pseudoconvex domain as in Theorem 8.6.1. Assume that $2d \le n$ and $q \in \{1, \ldots, n - 2d + 1\}$. Then the following hold:*

(a) *If X is q-convex then there exists a proper holomorphic map $D \to X$.*

(b) (Oka principle) *If X is q-complete then every continuous map $\bar{D} \to X$ that is holomorphic in D can be approximated, uniformly on compacts in D, by proper holomorphic immersions $D \to X$ (embeddings if $2d + 1 \le n$).*

Taking $q = 1$, Corollary 8.6.3 gives the following Oka principle for proper immersions and embeddings into Stein manifolds.

Corollary 8.6.4. [123, Corollary 1.5] *Assume that D is a smoothly bounded, relatively compact, strongly pseudoconvex Stein domain, X is a Stein manifold, and $f_0 \colon \bar{D} \to X$ is a continuous map that is holomorphic in D.*

(i) *If $\dim X \ge 2 \dim D$, then f_0 can be approximated uniformly on compacts in D by proper holomorphic immersions $D \looparrowright X$.*

(ii) *If* $\dim X \geq 2\dim D + 1$, *then* f_0 *can be approximated uniformly on compacts in* D *by proper holomorphic embeddings* $D \hookrightarrow X$.

If X is a complex manifold and $A \subset X$ is a compact complex submanifold of codimension q with Griffiths positive normal bundle, then $X \backslash A$ admits a function $\rho: X \backslash A \to \mathbb{R}$ which tends to $+\infty$ along A and is q-convex and noncritical in a deleted neighborhood of A [429]. This holds in particular for every projective submanifold $A \subset \mathbb{P}^n$ [32]; the manifold $\mathbb{P}^n \backslash \mathbb{P}^{n-q}$ is even q-complete. Thus condition (b) in Theorem 8.6.1 holds when $r = n - q + 1 > \dim D$ or, equivalently, $\dim D \leq \dim A$. This gives the following corollary [123, Corollary 1.4] generalizing Bishop's theorem on the existence of proper holomorphic maps $D \to \mathbb{C}^n = \mathbb{P}^n \backslash \mathbb{P}^{n-1}$ for $n > \dim D$.

Corollary 8.6.5. *If* A *is a compact complex submanifold of* $\mathbb{P}^n$ *then every smoothly bounded, relatively compact, strongly pseudoconvex Stein domain* D *of dimension* $\dim D \leq \dim A$ *admits a proper holomorphic map* $D \to \mathbb{P}^n \backslash A$. *In particular, if* $\dim D < n$ *then* D *admits a proper holomorphic map into* $\mathbb{P}^n \backslash A$ *for any nonsingular complex hypersurface* A *in* $\mathbb{P}^n$. *The analogous conclusion holds for maps* $D \to X \backslash A$, *where* A *is a compact complex submanifold with Griffiths positive normal bundle in a complex manifold* X.

Outline of proof of Theorem 8.6.1. The proof combines three techniques. The first ingredient is a method of lifting small pieces of the boundary of D (considered as a subset of X) to higher levels of the function ρ. As was explained before the statement of Theorem 8.6.1, this technique is fairly elementary when $\dim D = 1$, but it becomes rather delicate in higher dimension. The main lifting lemma [123, Lemma 5.3] employs special holomorphic peak functions that reach their maximum along certain complex tangential submanifolds of maximal real dimension in the boundary of D. Its proof relies on the work of A. Dor [117, 118, 119], building upon earlier results of many authors [347, 161, 209, 242, 462]. (The idea of using such peak functions goes back to the construction of inner functions by Hakim and Sibony [243] and Løw [346].)

Each local modification is patched with the global map $\bar{D} \to X$ by the method of gluing holomorphic sprays as was explained (in the special case when D is the disc) before the statement of Theorem 8.6.1. To pass a critical level of the exhaustion function ρ on X one applies Lemma 3.10.1 and the general position argument. For further details see [121, 123].

Recently Coupet, Sukhov and Tumanov constructed proper pseudoholomorphic discs in bounded strongly pseudoconvex domains in almost complex surfaces [93]. They pull back the almost complex structure from the target surface to an almost complex structure on the bidisc in $\mathbb{C}^2$ and then solve an appropriate Riemann-Hilbert problem for this structure.

8.7 A Splitting Lemma for Biholomorphic Maps

In this section we prove a result from [173] on compositional splitting of a biholomorphic map close to the identity on a Cartan pair. This is used in several subsequent results; in particular, in the construction of noncritical holomorphic functions on Stein manifolds, and in the construction of proper holomorphic embeddings of certain open Riemann surfaces into $\mathbb{C}^2$.

Let X be a complex manifold of dimension n. Suppose that $\mathcal{F}$ is a nonsingular holomorphic foliation of X of dimension p and codimension $q = n - p$. Every $x \in X$ is contained in a *distinguished chart* (U, ϕ), where $U \subset X$ is an open subset containing x and $\phi: U \to \mathbb{D}^n \subset \mathbb{C}^n$ is a biholomorphic map onto the open unit polydisc in $\mathbb{C}^n$ such that, in the coordinates (z, w) on $\mathbb{D}^n = \mathbb{D}^p \times \mathbb{D}^q$, $\phi(\mathcal{F}|_U)$ is the horizontal foliation $\{w = c\}$, $c \in \mathbb{D}^q$.

Fix a number $0 < r < 1$. For any distinguished chart (U, ϕ) on X let $U' \subset U$ be defined by $\phi(U') = (r\mathbb{D}^p) \times \mathbb{D}^q$. Given $V \Subset X$, there exists a finite collection of distinguished charts $\mathcal{U} = \{(U_j, \phi_j): 1 \le j \le N\}$ such that $\overline{V} \subset \cup_{j=1}^N U'_j$ and $\mathcal{U}$ is *$\mathcal{F}$-regular* in the sense of Godbillon [213, Definition 1.5]. Such $\mathcal{U}$ will be called a regular $\mathcal{F}$-atlas on V.

Definition 8.7.1. (Notation as above) *An injective holomorphic map $\gamma: V \to X$ is said to be an $\mathcal{F}$-*map *if there exists a a regular $\mathcal{F}$-atlas $\mathcal{U}$ on V such that for every $(U_j, \phi_j) \in \mathcal{U}$ the restriction of γ to $V \cap U'_j$ has range contained in U_j and is of the form $(z, w) \to (c_j(z, w), w)$ in the distinguished holomorphic coordinates (z, w) on U_j.*

Thus an $\mathcal{F}$-map preserves the leaves of $\mathcal{F}$ and does not permute the connected components of the intersection of a global leaf with any of the distinguished sets U_j. The definition is good since the transition map between a pair of distinguished charts preserves this form of the map. Any γ preserving the leaves of $\mathcal{F}$ (in the sense that x and $\gamma(x)$ belong to the same leaf) and which is sufficiently close to the identity map in the *fine topology* on X, defined by $\mathcal{F}$, is of this form. (The restriction of the fine topology to any distinguished local chart $U \cong \mathbb{D}^p \times \mathbb{D}^q$ is the product of the usual topology on $\mathbb{D}^p$ and the discrete topology on $\mathbb{D}^q$. For further details see [213, pp. 2–3, pp. 71–75].)

Let dist be a smooth distance function on X. Given an open set $V \subset X$ and a map $\gamma: V \to X$, we set

$$\mathrm{dist}_V(\gamma, \mathrm{Id}) = \sup_{x \in V} \mathrm{dist}(\gamma(x), x).$$

The following result is a key ingredient in several subsequent constructions. We shall use the notion of a *Cartan pair*, and of a *strongly pseudoconvex Cartan pair of class $\mathcal{C}^\ell$* (see Def. 5.7.1 on p. 209).

Theorem 8.7.2. [173, Theorem 4.1] *Let (A, B) be a Cartan pair in a complex manifold X. Given an open set $\widetilde{C} \subset X$ containing $C = A \cap B$, there exist open sets $A' \supset A$, $B' \supset B$, $C' \supset C$, with $C' \subset A' \cap B' \subset \widetilde{C}$, satisfying the following. For every $\eta > 0$ there is $\epsilon_\eta > 0$ such that for each injective holomorphic map $\gamma\colon \widetilde{C} \to X$ with $\mathrm{dist}_{\widetilde{C}}(\gamma, \mathrm{Id}) < \epsilon_\eta$ there exist injective holomorphic maps $\alpha\colon A' \to X$, $\beta\colon B' \to X$, depending continuously on γ and satisfying*

$$\gamma = \beta \circ \alpha^{-1} \text{ on } C', \quad \mathrm{dist}_{A'}(\alpha, \mathrm{Id}) < \eta, \quad \mathrm{dist}_{B'}(\beta, \mathrm{Id}) < \eta.$$

If in addition $\mathcal{F}$ is a holomorphic foliation of X and γ is an $\mathcal{F}$-map on $\widetilde{C}$, then we can choose α and β to be $\mathcal{F}$-maps on A', resp. on B'. Further, given a closed complex subvariety X_0 of X such that $X_0 \cap C = \emptyset$, we can choose α and β tangent to the identity map to any finite order along X_0.

Proof. The proof is based on Nash-Moser type rapidly convergent iteration in which the domain of the map shrinks in a controlled way at every step.

By Proposition 5.7.3 (p. 210) we may assume that X is a Stein manifold and (A, B) is a smooth strongly pseudoconvex Cartan pair in X. We assume this to be the case for the rest of the proof.

By Cartan's Theorem A, the tangent bundle $\mathrm{T}\mathcal{F}$ (a holomorphic subbundle of $\mathrm{T}X$) is spanned by finitely many holomorphic vector fields $L_1, L_2, \ldots, L_m$ on X. Denote by $\theta_t^j(x)$ the flow of L_j for time $t \in \mathbb{C}$, solving $\frac{\partial}{\partial t}\theta_t^j(x) = L_j(\theta_t^j(x))$ and $\theta_0^j(x) = x$. The map θ^j is defined and holomorphic for (x, t) in an open neighborhood of $X \times \{0\}$ in $X \times \mathbb{C}$. Their composition

$$\theta(x, t) = \theta(x, t_1, \ldots, t_m) = \theta_{t_m}^m \circ \cdots \circ \theta_{t_2}^2 \circ \theta_{t_1}^1(x) \in X$$

is a holomorphic map in an open neighborhood $U \subset X \times \mathbb{C}^m$ of the zero section $X \times \{0\}^m$, and it satisfies $\theta(x, t) \in \mathcal{F}_x$ for all $(x, t) \in U$ and

$$\theta(x, 0) = x, \qquad \frac{\partial}{\partial t_j}\,\theta(x, t)|_{t=0} = L_j(x) \quad (x \in X,\ j = 1, \ldots, m).$$

Hence $\Theta = \partial_t \theta|_{t=0}$ maps the trivial bundle $X \times \mathbb{C}^m$ surjectively onto the tangent bundle $\mathrm{T}\mathcal{F}$ of the foliation $\mathcal{F}$. Splitting $X \times \mathbb{C}^m = E \oplus \ker \Theta$ into a holomorphic direct sum, we infer that $\Theta\colon E \to T\mathcal{F}$ is an isomorphism of holomorphic vector bundles. In any holomorphic vector bundle chart on E we have a Taylor expansion

$$\theta(x, t_1, \ldots, t_m) = x + \sum_{j=1}^m t_j\, L_j(x) + O(|t|^2) \tag{8.13}$$

where the remainder $O(|t|^2)$ is uniform on any compact subset of the base set.

Choose a Hermitian metric $|\cdot|_E$ on E. Given an open set $V \subset X$ and a section $c\colon V \to E|_V$ we shall write $||c||_V = \sup_{x \in V} |c(x)|_E$. By the construction of θ and E, $x \to \theta(x, c(x))$ is an $\mathcal{F}$-map provided $||c||_V$ is sufficiently small.

Lemma 8.7.3. *For every open relatively compact set $V \Subset X$ there exist constants $M_1 \geq 1$ and $\epsilon_0 > 0$ satisfying the following property. For every $\mathcal{F}$-map $\gamma\colon V \to X$ with $\mathrm{dist}_V(\gamma, \mathrm{Id}) < \epsilon_0$ there is a unique holomorphic section $c\colon V \to E$ of $E|_V \to V$ such that for every $x \in V$ we have $\theta(x, c(x)) = \gamma(x)$ and the estimates*

$$M_1^{-1}|c(x)| \leq \mathrm{dist}(\gamma(x), x) \leq M_1|c(x)|.$$

Proof. This is an immediate application of the implicit function theorem. If $\mathcal{F}$ is the trivial foliation with X as the only leaf, Lemma 8.7.3 asserts that every biholomorphic map $\gamma\colon V \to X$ sufficiently close to the identity map has the form $\gamma(x) = \theta(x, c(x))$ for some holomorphic section $c\colon V \to \mathrm{T}V$. □

From now on all our sets in X will be assumed contained in a fixed relatively compact set for which Lemma 8.7.3 holds with a constant M_1.

Lemma 8.7.4. *Let $V \Subset X$. There are constants $\delta_0 > 0$ (small) and $M_2 > 0$ (large) with the following property. Let $0 < \delta < \delta_0$ and $0 < 4\epsilon < \delta$. Assume that $\alpha, \beta, \gamma\colon V(\delta) \to X$ are $\mathcal{F}$-maps which are ϵ-close to the identity on $V(\delta)$. Then $\widetilde{\gamma} := \beta^{-1} \circ \gamma \circ \alpha\colon V \to X$ is a well defined $\mathcal{F}$-map on V. Write*

$$\begin{aligned} \alpha(x) &= \theta(x, a(x)), \qquad & \beta(x) &= \theta(x, b(x)), \\ \gamma(x) &= \theta(x, c(x)), \qquad & \widetilde{\gamma}(x) &= \theta(x, \widetilde{c}(x)), \end{aligned}$$

where a, b, c are sections of $E|_{V(\delta)} \to V(\delta)$ and $\widetilde{c}$ is a section of $E|_V \to V$ given by Lemma 8.7.3. Then

$$||\widetilde{c} - (c + a - b)||_V \leq M_2\delta^{-1}\epsilon^2. \tag{8.14}$$

If $c = b - a$ on $V(\delta)$, then

$$||\widetilde{c}||_V \leq M_2\delta^{-1}\epsilon^2, \quad \mathrm{dist}_V(\widetilde{\gamma}, \mathrm{Id}) \leq M_1M_2\delta^{-1}\epsilon^2.$$

Proof. The conditions imply that $\gamma\circ\alpha$ maps V biholomorphically onto a subset of $V(2\epsilon)$. Since β is ϵ-close to the identity map on $V(\delta)$, its range contains $V(\delta - \epsilon)$. Hence β^{-1} is defined on $V(\delta - \epsilon)$ and is ϵ-close to the identity on this set. Since $4\epsilon < \delta$, it follows that $\widetilde{\gamma} = \beta^{-1} \circ \gamma \circ \alpha$ is defined on V and maps V biholomorphically onto a subset of $V(3\epsilon) \Subset V(\delta)$.

To prove the estimate (8.14), we choose a holomorphic vector bundle chart on $\pi\colon E \to X$ over an open set $\widetilde{U} \subset X$ and let $U \Subset \widetilde{U}$. We shall use the expansion (8.13) for θ on $\pi^{-1}(U) \subset E$; this suffices since $\overline{V(\delta)}$ can be covered by finitely many such sets U. We replace the fiber variable t in (8.13) by one of the functions $a(x), b(x)$, or $c(x)$. These are bounded on $V(\delta)$ by $M_1\epsilon$, where M_1 is the constant from Lemma 8.7.3. This gives for $x \in U \cap V(\delta)$:

$$\alpha(x) = x + \sum_{j=1}^{m} a_j(x)L_j(x) + O(\epsilon^2),$$
$$\beta(x) = x + \sum_{j=1}^{m} b_j(x)L_j(x) + O(\epsilon^2),$$
$$\gamma(x) = x + \sum_{j=1}^{m} c_j(x)L_j(x) + O(\epsilon^2).$$

The remainder $O(\epsilon^2)$ is uniform with respect to $x \in U \cap V(\delta)$. For $x \in U \cap V$ this gives

$$\begin{aligned}\gamma(\alpha(x)) &= \alpha(x) + \sum_{j=1}^{m} c_j(\alpha(x))L_j(\alpha(x)) + O(\epsilon^2)\\ &= x + \sum_{j=1}^{m} \left(a_j(x) + c_j(x)\right) L_j(x)\\ &\qquad + \sum_{j=1}^{m} \big(c_j(\alpha(x))L_j(\alpha(x)) - c_j(x)L_j(x)\big) + O(\epsilon^2).\end{aligned}$$

To estimate the last sum we fix j and write $g(x) = c_j(x)L_j(x)$ for $x \in U \cap V(\delta)$. Since $||c_j||_{V(\delta)} < M_1\epsilon$ and $4\epsilon < \delta$, the Cauchy estimates imply $||\mathrm{d}c_j||_{U\cap V(\epsilon)} = O(\epsilon\delta^{-1})$. Since L_j is holomorphic in a neighborhood of $\overline{V(\delta)}$, we may assume that its expression in the local coordinates on U is uniformly bounded and has uniformly bounded differential. This gives $||\mathrm{d}g||_{U\cap V(\epsilon)} = O(\epsilon\delta^{-1})$. Since $\mathrm{dist}(x, \alpha(x)) < \epsilon$, there is a smooth arc $\lambda\colon [0,1] \to U$, of length comparable to ϵ, such that $\lambda(0) = x$ and $\lambda(1) = \alpha(x)$. Then

$$|g(\alpha(x)) - g(x)| \le \int_0^1 |\mathrm{d}g(\lambda(\tau)|\cdot |\lambda'(\tau)|\,\mathrm{d}\tau = O(\epsilon^2\delta^{-1}).$$

(The additional ϵ is contributed by the length of λ). This gives for $x \in U \cap V$

$$\gamma(\alpha(x)) = x + \sum_{j=1}^{m} \left(a_j(x) + c_j(x)\right) L_j(x) + O(\epsilon^2\delta^{-1}).$$

The same argument holds for the composition of several maps provided that ϵ is small in comparison to δ; the error term remains of order $O(\epsilon^2\delta^{-1})$.

It remains to find the expansion of β^{-1} on the set $U \cap V(2\epsilon)$, where U is a local chart as above. Set $\widetilde{\beta}(x) = x - \sum_{j=1}^{m} b_j(x)L_j(x)$ for $x \in U \cap V(\delta)$. Assuming that $\beta(x) \in U \cap V(2\epsilon)$ we obtain

$$\widetilde{\beta}(\beta(x)) = \beta(x) - \sum_{j=1}^{m} b_j(\beta(x)L_j(\beta(x))$$

$$= x + \sum_{j=1}^{m} \big(b_j(x)L_j(x) - b_j(\beta(x))L_j(\beta(x))\big) + O(\epsilon^2)$$
$$= x + O(\epsilon^2\delta^{-1}).$$

The terms in the parentheses on the middle line were estimated by $O(\epsilon^2\delta^{-1})$ exactly as above. Writing $\beta(x) = y \in U \cap V(2\epsilon)$, $x = \beta^{-1}(y)$, the above gives $\widetilde{\beta}(y) = \beta^{-1}(y) + O(\epsilon^2\delta^{-1})$ and therefore

$$\beta^{-1}(y) = y - \sum_{j=1}^{m} b_j(y)L_j(y) + O(\epsilon^2\delta^{-1}).$$

The same argument as before gives

$$\widetilde{\gamma}(x) = (\beta^{-1}\gamma\alpha)(x) = x + \sum_{j=1}^{m} \big(c_j(x) + a_j(x) - b_j(x)\big)L_j(x) + O(\epsilon^2\delta^{-1})$$

for $x \in U \cap V$. This proves the estimate (8.14). □

Remark 8.7.5. The proof of Lemmas 8.7.3 and 8.7.4 shows that for each fixed open set $V_0 \Subset X$ the constants M_1, M_2, δ_0 may be chosen independent of V for any open set $V \subset V_0$. In this case $O(\epsilon^2\delta^{-1})$ means an expression satisfying $\leq C\epsilon^2\delta^{-1}$ for some C independent of ϵ, δ and V. □

Lemma 8.7.6. *Let $U, V \subset X$ be open sets such that $\overline{U \backslash V} \cap \overline{V \backslash U} = \emptyset$ and $D = U \cup V$ is a relatively compact strongly pseudoconvex domain with $\mathcal{C}^2$ boundary in X. (This holds in particular if (U, V) is a strongly pseudoconvex Cartan pair of class $\mathcal{C}^2$.) Set $W = U \cap V$. There is a constant $M_3 \geq 1$ such that for every bounded holomorphic section $c\colon W \to E|_W$ there exist bounded holomorphic sections $a\colon U \to E|_U$, $b\colon V \to E|_V$ satisfying*

$$c = b|_W - a|_W, \quad ||a||_U < M_3||c||_W, \quad ||b||_V < M_3||c||_W.$$

Such a and b are given by bounded linear operators between the spaces of bounded holomorphic sections of E on the respective sets. The constant M_3 can be chosen uniform for all such pairs (U, V) in X close to an initial pair (U_0, V_0), provided that $D = U \cup V$ is sufficiently $\mathcal{C}^2$-close to $D_0 = U_0 \cup V_0$. If X_0 is a closed complex subvariety of X such that $X_0 \cap \overline{W} = \emptyset$ then for every $s \in \mathbb{N}$ we can insure in addition that a and b vanish to order s on X_0.

Proof. If E is the trivial bundle, this follows by applying Lemma 5.8.2 (p. 212) with $W = \{0\}$ (i.e., without the parameter w). The general case reduces to this one by considering E as a complemented holomorphic vector subbundle of a trivial bundle. □

Lemma 8.7.7. *There are constants $r_0 > 0, \delta_0 > 0$ (small) and $M_4, M_5 > 1$ (large) satisfying the following. Let $0 < r \le r_0$, $0 < \delta \le \delta_0$ and $s \in \mathbb{N}$. For every $\mathcal{F}$-map $\gamma\colon C(r+\delta) \to X$ satisfying $4M_4 \mathrm{dist}_{C(r+\delta)}(\gamma, \mathrm{Id}) < \delta$ there exist $\mathcal{F}$-maps $\alpha\colon A(r+\delta) \to X$ and $\beta\colon B(r+\delta) \to X$, tangent to the identity map to order s along X_0, such that $\widetilde{\gamma} = \beta^{-1} \circ \gamma \circ \alpha$ is an $\mathcal{F}$-map on $C(r)$ satisfying*

$$\mathrm{dist}_{C(r)}(\widetilde{\gamma}, \mathrm{Id}) < M_5 \delta^{-1} \mathrm{dist}_{C(r+\delta)}(\gamma, \mathrm{Id})^2. \tag{8.15}$$

Proof. If r_0 and δ_0 are chosen sufficiently small then $D(t)$ is a small $\mathcal{C}^2$-perturbation of the strongly pseudoconvex domain $D = A \cup B$ for every $t \in [0, r_0 + \delta_0]$. Hence we can use the same constant as a bound on the sup-norm of an operator solving the $\overline{\partial}$-problem on $D(t)$.

Let $\epsilon = \mathrm{dist}_{C(r+\delta)}(\gamma, \mathrm{Id})$. By Lemma 8.7.3 there is a holomorphic section $c\colon C(r+\mathrm{d}) \to E$, with $||c||_{C(r+\delta)} \le M_1 \epsilon$, such that $\gamma(x) = \theta(x, c(x))$. (Here we can use the constant M_1 for the set $D(r_0 + \delta_0)$.) By Lemma 8.7.6 we have $c = b - a$, where a is a holomorphic section of E over $A(r+\delta)$ and b is a holomorphic section of E over $B(r+\delta)$. The sup-norms of a and b on their respective domains are bounded by $M_1 M_3 \epsilon$, where the constant M_3 from Lemma 8.7.6 can be chosen independent of r and δ. Set

$$\begin{aligned} \alpha(x) &= \theta(x, a(x)), \qquad x \in A(r+\delta), \\ \beta(x) &= \theta(x, b(x)), \qquad x \in B(r+\delta). \end{aligned}$$

By Lemma 8.7.3 we have

$$\mathrm{dist}_{A(r+\delta)}(\alpha, \mathrm{Id}) < M_1^2 M_3 \epsilon, \quad \mathrm{dist}_{B(r+\delta)}(\beta, \mathrm{Id}) < M_1^2 M_3 \epsilon.$$

Set $M_4 = M_1^2 M_3$. If $0 < 4M_4 \epsilon < \delta$ then by Lemma 8.7.4 the composition $\widetilde{\gamma} = \beta^{-1} \circ \gamma \circ \alpha$ is an $\mathcal{F}$-map on $C(r)$ satisfying the estimate (8.15) with $M_5 = M_2 M_4^2 = M_1^4 M_2 M_3^2$. This completes the proof. □

We now complete the proof of Theorem 8.7.2. Choose a sufficiently small number $0 < r_0 < 1$ such that the initial $\mathcal{F}$-map γ is defined on the set $C_0 = C(r_0)$ and such that Lemma 8.7.7 holds for all $\delta, r > 0$ with $\delta + r \le r_0$. For each $k = 0, 1, 2, \ldots$ we set

$$r_k = r_0 \prod_{j=1}^{k} (1 - 2^{-j}), \quad \delta_k = r_k - r_{k+1} = r_k 2^{-k-1}.$$

The sequence $r_k > 0$ is decreasing, $r^* = \lim_{k\to\infty} r_k > 0$, $\delta_k > r^* 2^{-k-1}$ for all k, and $\sum_{k=0}^{\infty} \delta_k = r_0 - r^*$. Set $A_k = A(r_k)$, $B_k = B(r_k)$, $C_k = C(r_k)$. Choose $r_0 > 0$ small enough such that $C_k = A_k \cap B_k$ for all k (Lemma 5.7.4).

Let $\epsilon_0 = \mathrm{dist}_{C_0}(\gamma, \mathrm{Id})$. Assuming that $4M_4 \epsilon_0 < \delta_0 = r_0/2$, Lemma 8.7.7 gives $\mathcal{F}$-maps $\alpha_0\colon A_0 \to X$ and $\beta_0\colon B_0 \to X$ such that $\gamma_1 = \beta_0^{-1} \circ \gamma \circ \alpha_0\colon C_1 \to X$ is an $\mathcal{F}$-map defined on C_1 and satisfying

$$\mathrm{dist}_{C_1}(\gamma_1, \mathrm{Id}) < M_5\epsilon_0^2\delta_0^{-1} < 2M\epsilon_0^2, \quad M = \frac{M_5}{r^*}.$$

Define $\epsilon_1 = \mathrm{dist}_{C_1}(\gamma_1, \mathrm{Id})$, so $\epsilon_1 < 2M\epsilon_0^2$. Assuming for a moment that $4M_4\epsilon_1 < \delta_1$, we apply Lemma 8.7.7 to obtain a pair of $\mathcal{F}$-maps $\alpha_1\colon A_1 \to X$, $\beta_1\colon B_1 \to X$ such that $\gamma_2 = \beta_1^{-1} \circ \gamma_1 \circ \alpha_1\colon C_2 \to X$ is an $\mathcal{F}$-map satisfying

$$\mathrm{dist}_{C_2}(\gamma_2, \mathrm{Id}) < M_5\epsilon_1^2\delta_1^{-1} < 2^2M\epsilon_1^2.$$

Define $\epsilon_2 = \mathrm{dist}_{C_2}(\gamma_2, \mathrm{Id})$. Continuing inductively we get sequences of $\mathcal{F}$-maps

$$\alpha_k\colon A_k \to X, \quad \beta_k\colon B_k \to X, \quad \gamma_k\colon C_k \to X$$

such that $\gamma_{k+1} = \beta_k^{-1} \circ \gamma_k \circ \alpha_k\colon C_{k+1} \to X$ is an $\mathcal{F}$-map satisfying

$$\epsilon_{k+1} = \mathrm{dist}_{C_{k+1}}(\gamma_{k+1}, \mathrm{Id}) < M_5\epsilon_k^2\delta_k^{-1} < 2^{k+1}M\epsilon_k^2. \tag{8.16}$$

The necessary condition for the induction step is that $4M_4\epsilon_k < \delta_k$ holds for each k. Since $\delta_k > r^*2^{-k-1}$, it suffices to have

$$4M_4\epsilon_k < r^*2^{-k-1}, \qquad k = 0, 1, 2, \ldots. \tag{8.17}$$

In order to obtain convergence of this process we need the following elementary lemma whose proof is found in [173, Lemma 4.8].

Lemma 8.7.8. *Let $M, M_4 \geq 1$. Let the sequence $\rho_k > 0$ be defined recursively by $\rho_0 = \epsilon_0 > 0$ and $\rho_{k+1} = 2^{k+1}M\rho_k^2$ for $k = 0, 1, \ldots$. If $\epsilon_0 < r^*/32MM_4$ then $\rho_k < (4M\epsilon_0)^{2^k} < (1/8)^{2^k}$ and $4M_4\rho_k < r^*2^{-k-1}$ for all $k = 0, 1, 2, \ldots$.*

From (8.16) we see that $\epsilon_k \leq \rho_k$, where ρ_k is the sequence from Lemma 8.7.8. From the assumption $\epsilon_0 < r^*/32MM_4$ we obtain

$$q := 4M\epsilon_0 < r^*/8M_4 < 1/8$$

(since $0 < r^* < 1$ and $M_4 \geq 1$). Hence the sequence

$$\epsilon_k = \mathrm{dist}_{C_k}(\gamma_k, \mathrm{Id}) < q^{2^k} < (1/8)^{2^k}$$

converges to zero very rapidly as $k \to \infty$. The second estimate on ρ_k in Lemma 8.7.8 insures that (8.17) holds, so the induction in the proof works. Set

$$\widetilde{\alpha}_k = \alpha_0 \circ \alpha_1 \circ \cdots \circ \alpha_k\colon A_k \to X, \quad \widetilde{\beta}_k = \beta_0 \circ \beta_1 \circ \cdots \circ \beta_k\colon B_k \to X.$$

Then $\gamma_{k+1} = \widetilde{\beta}_k^{-1} \circ \gamma \circ \widetilde{\alpha}_k$ on C_{k+1} for $k = 0, 1, 2, \ldots$. As $k \to \infty$, the sequences $\widetilde{\alpha}_k$ resp. $\widetilde{\beta}_k$ converge, uniformly on $A(r^*)$ resp. on $B(r^*)$, to $\mathcal{F}$-maps $\alpha\colon A(r^*) \to X$ resp. $\beta\colon B(r^*) \to X$. Furthermore, the sequence γ_k converges uniformly on $C(r^*)$ to the identity map according to (8.16) and Lemma 8.7.8. In the limit we obtain $\beta^{-1} \circ \gamma \circ \alpha = \mathrm{Id}$ on $C(r^*)$, and hence $\gamma = \beta \circ \alpha^{-1}$ on $\alpha(C(r^*))$. If $\epsilon_0 > 0$ is chosen sufficiently small then the latter set contains a neighborhood of C. $\square$

8.8 Conformal Diffeomorphisms of Bordered Riemann Surfaces

In this section we construct certain conformal diffeomorphisms of bordered Riemann surfaces that will be used in the construction of proper holomorphic embeddings of their interior into $\mathbb{C}^2$ (Theorem 8.9.3 on p. 370 below). The main result of this section is the following.

Theorem 8.8.1. [198, Theorem 2.3] *Assume that D is a relatively compact smoothly bounded domain in a Riemann surface R. Choose finitely many pairwise distinct points $a_1, \ldots, a_k \in bD$, $b_1, \ldots, b_k \in R \backslash D$, and $c_1, \ldots, c_l \in D \backslash \{a_1, \ldots, a_k\}$ such that for each $j = 1, \ldots, k$ the points a_j and b_j are connected by a Jordan arc $\gamma_j \subset R \backslash D$ with $\gamma_j \cap \bar{D} = \{a_j\}$. Suppose that for every $j = 1, \ldots, k$ we are given neighborhoods $U_j' \Subset U_j$ of the point a_j and a neighborhood $V_j \subset R$ of γ_j. Then for every integer $N \in \mathbb{N}$ there exists a smooth diffeomorphism $\phi \colon \bar{D} \to \bar{D}'$ onto a smoothly bounded domain $D' \subset R$ such that the following hold:*

1. *$\phi \colon D \to D'$ is biholomorphic,*
2. *ϕ is as close as desired to the identity map in $\mathcal{C}^\infty(\bar{D} \backslash \cup_{j=1}^k U_j')$,*
3. *ϕ is tangent to the identity map to order N at each point c_j, and*
4. *$\phi(a_j) = b_j$ and $\phi(\bar{D} \cap U_j') \subset U_j \cup V_j$ for $j = 1, \ldots, k$.*

We begin with a lemma on conformal mappings. Denote by $\mathbb{D}$ the open unit disc in $\mathbb{C}$, and by $r\mathbb{D}$ the disc of radius $r > 0$.

Lemma 8.8.2. *Assume that R is a connected open Riemann surface, $G \Subset R$ is an open simply connected domain with smooth boundary, $V' \Subset V'' \subset R$ are small neighborhoods of a boundary point $a \in bG$, b is a point in $R \backslash \overline{G}$, γ is a smooth Jordan arc with endpoints a and b such that $\gamma \cap \overline{G} = \{a\}$ and the tangent lines to γ and bG at the point a are transverse, and V is a neighborhood of γ. Then there exists a sequence of smooth diffeomorphisms $\psi_n \colon \overline{G} \to \psi_n(\overline{G}) \subset R$ that are holomorphic on G and satisfy the following:*

(i) *$\psi_n \to \mathrm{id}$ locally uniformly on G as $n \to \infty$,*
(ii) *$\psi_n(a) = b$ for $n = 1, 2 \ldots$, and*
(iii) *$\psi_n(\overline{V}' \cap \overline{G}) \subset V'' \cup V$ for $n = 1, 2 \ldots$.*

Proof. Since $\overline{G} \cup \gamma$ admits a simply connected neighborhood in R and since we are going to construct maps with images near $\overline{G} \cup \gamma$, we may assume that we are working in the complex plane, that a is the origin, and that the strictly positive real axis lies in the complement of $\overline{G}$ near the origin.

For each $n \in \mathbb{N}$ let l_n denote the line segment between 0 and $\frac{1}{n}$ in $\mathbb{R} \subset \mathbb{C}$. Let $\widetilde{V}$ be a neighborhood of the origin with $V' \Subset \widetilde{V} \Subset V''$. By approximation (see Theorem 3.7.2) there are neighborhoods U_n of $\overline{G} \cup l_n$ and holomorphic injections $f_n \colon U_n \to \mathbb{C}$ such that the following hold for all $n \in \mathbb{N}$:

(1) $f_n \to \mathrm{id}$ uniformly on $\overline{G}$ as $n \to \infty$,

(2) $f_n(l_n)$ approximates γ, with $f_n(\frac{1}{n}) = b$ and $f_n(l_n) \subset V$, and

(3) $f_n(\overline{G} \cap \widetilde{V}) \subset V''$. (This is a consequence of (1) for large enough n.)

For small $\epsilon > 0$ let Ω_ϵ denote a domain obtained by adding an ϵ-strip around l_n to G, containing the point $\frac{1}{n}$ in the boundary $b\,\Omega_\epsilon$, and smoothing the corners. Let $R_\epsilon = \Omega_\epsilon \backslash G$. Choose a sequence $\epsilon_n \to 0$ such that $\overline{\Omega}_{\epsilon_n} \subset U_n$ for each $n \in \mathbb{N}$. Write $\Omega_n = \Omega_{\epsilon_n}$ and $R_n = R_{\epsilon_n}$. By choosing the ϵ_n's small enough we get that

(4) $f_n(R_n) \subset V$ for each $n \in \mathbb{N}$.

Next we choose a point $p \in G$ and a sequence of conformal maps $g_n \colon G \to \Omega_n$ such that $g_n(p) = p$ and $g_n'(p) > 0$ for $n = 1, 2, \ldots$. Since our domains are smoothly bounded, the map g_n extends to a smooth diffeomorphism of $\overline{G}$ onto $\overline{\Omega}_n$. Furthermore, since the domains $\overline{\Omega}_n$ converge to $\overline{G}$ as $n \to \infty$, we conclude by Rado's theorem (see e.g. [399, Corollary 2.4, p. 22]) that

(5) $g_n \to \mathrm{id}$ uniformly on $\overline{G}$ as $n \to \infty$.

Hence for n large enough we have $g_n(\overline{V'} \cap \overline{G}) \subset (\widetilde{V} \cap \overline{G}) \cup R_n$ (keeping in mind that g_n is injective and close to the identity). Combining this with (3) and (4) we see that $(f_n \circ g_n)(\overline{V'} \cap \overline{G}) \subset V'' \cup V$ if n is large enough. Hence, by defining $\psi_n = f_n \circ g_n$ we get property (iii) for all large n, and we clearly also get property (i).

To see that property (ii) holds, let $a_n \in b\,G$ denote the point that g_n sends to $\frac{1}{n} \in b\Omega_n$. By (5) the sequence a_n has to converge to the origin, and so there is a sequence of conformal automorphisms φ_n of G fixing the point p, sending the origin to a_n, with $\varphi_n \to \mathrm{id}$ uniformly on $\overline{G}$. Replacing the maps g_n by $g_n \circ \varphi_n$ in the above argument also gives (ii). $\square$

The next lemma provides the main step in the proof of Theorem 8.8.1.

Lemma 8.8.3. (Assumptions as in Theorem 8.8.1.) *There is a smooth diffeomorphism $\phi \colon \bar{D} \to \bar{D}' \subset R$ such that $\phi \colon D \to D'$ is biholomorphic, $\phi(a_1) = b_1$, ϕ is tangent to the identity map to order N at each of the points $a_2, \ldots, a_k$ and $c_1, \ldots, c_l$, ϕ is as close as desired to the identity map in the smooth topology on $\bar{D} \backslash U_1'$, and $\phi(\bar{D} \cap U_1') \subset U_1 \cup V_1$.*

Proof. We may assume that $N > 2$. Write $\gamma = \gamma_1$. By approximation we may assume that γ intersects bD transversely at a_1. Then γ has an open, connected and simply connected neighborhood $W \subset R$ which is conformally equivalent to the disc. Let z denote the corresponding holomorphic coordinate on W, chosen such that $z(a_1) = 0$. By shrinking the neighborhood $U' = U_1'$ of the point a_1 we may assume that $\overline{U}' \subset W$, that $\overline{U}'$ does not contain

any of the points $a_2, \ldots, a_k, c_1, \ldots, c_l$, and that $z(U') = r'\mathbb{D} \subset \mathbb{C}$ for some $r' > 0$. Choose a number $r'' \in (0, r')$ and let $U'' \subset U'$ be chosen such that $z(U'') = r''\mathbb{D}$. Choose a connected and simply connected domain $G \subset W$ with smooth boundary, with a defining function ρ such that $G = \{\rho < 0\}$ and $d\rho \neq 0$ on bG, satisfying the following properties (see Fig. 8.1):

(i) $\bar{D} \cap U' \subset G \cup \{a_1\}$,
(ii) $-\rho(z) \geq \text{const} \cdot \text{dist}(z, a_1)^2$ for points $z \in bD$ close to a_1, and
(iii) $\gamma \cap \overline{G} = \{a_1\}$.

Property (iii) can be achieved since the arc γ is transverse to bD at a_1.

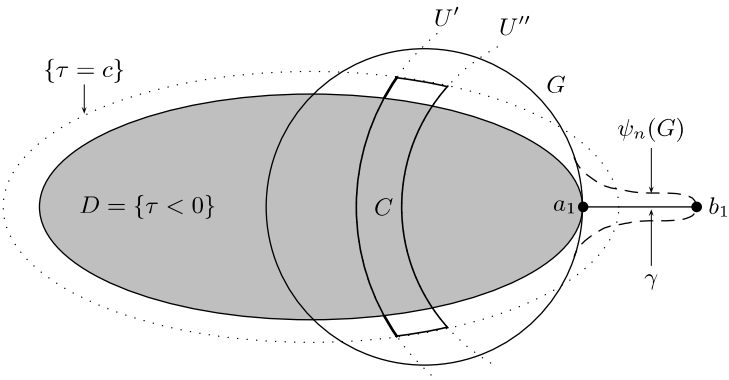

Fig. 8.1. The domains D and G. ([197, p. 104, Fig. 1])

Choose a smooth function τ on R such that $D = \{\tau < 0\}$ and $d\tau \neq 0$ on $bD = \{\tau = 0\}$. Choose a small number $c > 0$ and let

$$A = \{\tau \leq c\} \backslash U'', \quad B = \{\tau \leq c\} \cap \overline{U}', \quad C = \{\tau \leq c\} \cap (\overline{U}' \backslash U'').$$

By choosing $c > 0$ small enough we insure that C is a compact set contained in G (see Fig. 8.1), and we have

$$A \cup B = \{\tau \leq c\}, \quad A \cap B = C.$$

On Fig. 8.1, the set C is bounded by the two circular arcs (left and right) and by the two arcs in the larger dotted ellipse representing the level set $\{\tau = c\}$. The set A is the part of the dotted ellipse lying on the left hand side of the right boundary arc of C, and B is the part of the dotted ellipse on the right hand side of the left boundary arc of C.

Choose open neighborhoods $V' \Subset V''$ of the point a_1 such that $\overline{V}''$ is contained in the interior of the set $B \backslash A$. By shrinking the neighborhood $V =$

V_1 around the arc γ we may assume that $\overline{V} \cap \overline{A \backslash B} = \emptyset$. Let $\psi_n \colon \overline{G} \to \psi_n(\overline{G})$ be a sequence of conformal maps satisfying Lemma 8.8.2 with respect to the sets V, V', V''. Choose an open set $C' \Subset G$ containing the compact set C. As $n \to +\infty$, ψ_n converges to the identity uniformly on C', and hence also in the smooth topology (by the Cauchy estimates). For every sufficiently large integer $n \in \mathbb{N}$, Theorem 8.7.2 on p. 358 furnishes a splitting

$$\alpha_n = \psi_n \circ \beta_n \quad \text{near } C \tag{8.18}$$

where α_n is a small holomorphic perturbation of the identity map on a fixed neighborhood of A (independent of n) that is tangent to the identity to order N at each of the points $a_2, \ldots, a_k, c_1, \ldots, c_l$, and β_n is a small holomorphic perturbation of the identity map on a neighborhood of B that is tangent to the identity to order N at the point a_1. The closeness of α_n (resp. of β_n) to the identity in any $\mathcal{C}^r$ norm on A (resp. on B) can be estimated by the closeness of ψ_n to the identity on C'. If the approximations are sufficiently close (which holds for n large enough) then the two sides in (8.18), restricted to $A \cap \bar{D}$ resp. to $B \cap \bar{D}$, define a diffeomorphism $\phi_n \colon \bar{D} \to \phi_n(\bar{D}) \subset R$ that is holomorphic in D and such that $\phi_n(a_1) = b_1$, ϕ_n is tangent to the identity map to order N at each of the points $a_2, \ldots, a_k$, $c_1, \ldots, c_l$, ϕ_n converges to the identity map in the smooth topology on $\bar{D} \backslash U'$ as $n \to +\infty$, while $\phi_n(\bar{D} \cap U'') \subset U' \cup V$ for all large n. Indeed, both sides α_n and $\psi_n \circ \beta_n$ satisfy the stated properties on their respective domain. For α_n this is clear from the construction. For β_n we need a more precise argument to see that it maps $B \cap \bar{D}$ into $G \cup \{a_1\}$ for sufficiently large $n \in \mathbb{N}$. By the construction, its Taylor expansion in a local holomorphic coordinate z near a_1, with $z(a_1) = 0$, equals

$$\beta_n(z) = z + M_n z^N + O(z^{N+1}).$$

The size of the constant M_n, and of the remainder term, can be estimated (using the Cauchy estimates) by $\operatorname{dist}(\beta_n, \mathrm{Id})$ on B, and hence by $\operatorname{dist}(\psi_n, \mathrm{Id})$ on the set C'. Since G osculates D from the outside to the second order at the point a_1 by property (ii) above, it follows that for a sufficiently small neighborhood U' of the point a_1 and for all large enough $n \in \mathbb{N}$ we have

$$\beta_n(\bar{D} \cap U'') \subset (G \cup \{a_1\}) \cap V'. \tag{8.19}$$

On the complement $(B \cap \bar{D}) \backslash U''$, β_n is close to the identity for large n, and hence it maps this set into a fixed compact set in G. Thus the composition $\psi_n \circ \beta_n$ is well defined on $B \cap \bar{D}$ and it satisfies the stated properties.

It is also easily seen that ϕ_n is injective if n is large enough. Indeed, each of the two expressions defining ϕ_n on $A \cap \bar{D}$, resp. on $B \cap \bar{D}$, is injective by the construction. Hence it suffices to verify that no point from $(A \backslash B) \cap \bar{D}$ is identified with a point from $(B \backslash A) \cap \bar{D}$ under ϕ_n for large n. By the construction, the points from the first set remain nearby since α_n is close to the identity. Consider now points $x \in (B \backslash A) \cap \bar{D}$. If $x \in U''$ then $\beta_n(x) \in$

$(G \cup \{a_1\}) \cap V'$ by (8.19), and hence $\psi_n \circ \beta_n(x) \in V'' \cup V$ by property (iii) in Lemma 8.8.2. Since the set $V'' \cup V$ is at a positive distance from $(A \backslash B) \cap \bar{D}$, we see that $\psi_n \circ \beta_n(x) \neq \alpha_n(x')$ for any point $x' \in (A \backslash B) \cap \bar{D}$ provided that n is large enough. The remaining set $(B \backslash A) \cap \bar{D} \backslash U''$ is compactly contained in $B \cap \bar{D} \cap G$ where $\psi_n \circ \beta_n$ is close to the identity for large n, and hence no point from this set can get identified with a point from $(A \backslash B) \cap \bar{D}$. $\square$

Proof (of Theorem 8.8.1). We may assume that $N \geq 2$ and the neighborhoods $V_j \subset R$ of the arcs γ_j (with the endpoints a_j and b_j) for $j = 1, \ldots, k$ have pairwise disjoint closures. We seek ϕ of the form

$$\phi = \phi_k \circ \phi_{k-1} \circ \cdots \circ \phi_2 \circ \phi_1 \colon \bar{D} \to \bar{D}'.$$

Lemma 8.8.3 furnishes a diffeomorphism $\phi_1 \colon \bar{D} \to \bar{D}_1 = \phi_1(\bar{D}) \subset R$ such that $\phi_1(a_1) = b_1$, ϕ_1 is tangent to the identity to order N at each of the points $a_2, \ldots, a_k, c_1, \ldots, c_l$, ϕ_1 is close to the identity map on $\bar{D} \backslash U_1'$, and $\phi_1(\bar{D} \cap U_1') \subset U_1 \cup V_1$. Hence the points $b_1 = \phi_1(a_1)$, $a_2, \ldots, a_k$ lie on bD_1, while $c_1, \ldots, c_l \in D_1$. In the second step we apply Lemma 8.8.3, with $\bar{D}$ replaced by $\bar{D}_1 = \phi_1(\bar{D})$, to find a diffeomorphism $\phi_2 \colon \bar{D}_1 \to \phi_2(\bar{D}_1) = \bar{D}_2$ that is holomorphic in the interior and such that $\phi_2(a_2) = b_2$, ϕ_2 is tangent to the identity to order N at the points $b_1, a_3, \ldots, a_k, c_1, \ldots, c_l$, ϕ_2 is close to the identity map on $\bar{D}_1 \backslash U_2'$, and $\phi_2(\bar{D}_1 \cap U_2') \subset U_2 \cup V_2$. Continuing inductively we obtain after k steps a map ϕ satisfying the conclusion of Theorem 8.8.1 with $D' = D_k$. The final domain $D' = D_k = \phi(D)$ contains the points $b_1, \ldots, b_k$ in the boundary, it is close to D away from a small neighborhood of each point a_j, and at a_j it includes a spike reaching out to b_j. $\square$

8.9 Embedding Bordered Riemann Surfaces in $\mathbb{C}^2$

One of the oldest open problems in the classical function theory is whether every open Riemann surface, D, is biholomorphically equivalent to a topologically closed smooth complex curve in $\mathbb{C}^2$. Equivalently,

Problem 8.9.1. Does every open Riemann surface embed properly holomorphically in $\mathbb{C}^2$?

See the survey by Bell and Narasimhan [43, Conjecture 3.7, p. 20]. Such D embeds properly holomorphically in $\mathbb{C}^3$ by the Remmert-Bishop-Narasimhan theorem (Theorem 2.2.8 on p. 49). The general method for embedding higher dimensional Stein manifolds (§8.2 – §8.3 above) breaks down in this lowest dimensional case due to a hyperbolicity obstruction.

The first positive answers date back to 1970's for the case when D is a unit disc (Kasahara and Nishino [459]), an annulus [326], or the punctured disc $\mathbb{D} \backslash \{0\}$ [14]. These were essentially the only known results at the time of

the survey by Bell and Narasimhan [43]. The first positive result on Problem 8.9.1 for a large class of domains was obtained in 1995 by J. Globevnik and B. Stensønes who proved the following result.

Theorem 8.9.2. [212] *Every finitely connected domain in $\mathbb{C}$ without isolated boundary points embeds properly holomorphically into $\mathbb{C}^2$.*

They initially embed the closure $\bar{D}$ of the given domain properly into the cylinder $\overline{\mathbb{D}} \times \mathbb{C}$, choosing the first coordinate to be an inner function that is continuous on $\bar{D}$. Then the boundary of D is pushed to infinity by the method from §4.4, using compositions of shears in coordinate directions. For technical reasons they make small modifications of the conformal structure on D, due to cutting away small pieces of D near the boundary; hence an additional argument using the uniformization theory is needed to see that every planar domain of the indicated type can be embedded in this way.

Improvements and (minor) extensions of Theorem 8.9.2, using the solution of a Riemann-Hilbert boundary value problem, can be found in [95, 96]. By [95, Theorem 1.1] every bordered surface has a complex structure in which it admits a proper holomorphic embedding into $\mathbb{C}^2$.

A considerably more general method was developed by E. F. Wold [523, 524, 525]. In [523] he extended the Globevnik–Stensønes result to all finitely connected and some infinitely connected plane domains. In subsequent papers he developed a method that, together with another key addition from [197], culminated in the proof of Theorem 8.9.3 below.

A *bordered Riemann surface* is a compact one dimensional complex manifold, $\bar{D}$, not necessarily connected, with smooth boundary bD consisting of finitely many closed Jordan curves, and without any isolated boundary points. The problem of finding a proper holomorphic embedding of its interior, D, into $\mathbb{C}^2$ naturally decouples in the following two problems:

(A) find a holomorphic embedding $f\colon \bar{D} \hookrightarrow \mathbb{C}^2$ onto a smooth compact complex curve with boundary $\overline{\Sigma} = f(\bar{D}) \subset \mathbb{C}^2$;

(B) push the boundary $b\Sigma = f(bD)$ to infinity without introducing any interior double points.

The following theorem gives a complete answer to problem (B). Clearly it includes Theorem 8.9.2.

Theorem 8.9.3. [197, Corollary 1.2] *Assume that $\bar{D}$ is a bordered Riemann surface with $\mathcal{C}^r$ boundary for some $r > 1$ and that $f\colon \bar{D} \hookrightarrow \mathbb{C}^2$ is a $\mathcal{C}^1$ embedding that is holomorphic in D. Then f can be approximated, uniformly on compacts in D, by proper holomorphic embeddings $D \hookrightarrow \mathbb{C}^2$.*

Not much is known about problem (A). In view of the above result it would suffice to solve (A) whenever D is the complement of a small disc in a compact Riemann surface.

Theorem 8.9.3 applies to any bordered Riemann surface D whose boundary bD consists of finitely many Jordan curves; such D is conformally equivalent to a domain D' with smooth real analytic boundary in an open Riemann surface R. Furthermore, if bD is of class $\mathcal{C}^r$ and $r > 1$ is a noninteger then any biholomorphic map $D \xrightarrow{\cong} D'$ extends to a $\mathcal{C}^r$ diffeomorphism $\bar{D} \to \bar{D}'$. (See the discussion in [197, §6].) Hence we assume from now on that D is a relatively compact smoothly bounded domain in a Riemann surface R.

Theorem 8.9.3, together with the main result of [318], implies the following result on embeddings with interpolation on a discrete set. (For the case when D is the disc $\mathbb{D}$ see also [210].)

Corollary 8.9.4. *Let D be as in Theorem 8.9.3. Given discrete sequences of points $\{a_j\} \subset D$ and $\{b_j\} \subset \mathbb{C}^2$ without repetitions, there is a proper holomorphic embedding $\varphi\colon D \hookrightarrow \mathbb{C}^2$ such that $\varphi(a_j) = b_j$ for $j = 1, 2, \ldots$.*

Example 8.9.5. ([525], [197, Ex. 1.4]) Let R be a smooth closed algebraic curve in the projective plane $\mathbb{P}^2$. If $U_1, ..., U_k$ are pairwise disjoint smoothly bounded discs in R whose union contains the intersection of R with a projective line $\mathbb{P}^1 \subset \mathbb{P}^2$, then the bordered Riemann surface $D = R \backslash \cup_{i=1}^k \overline{U}_i \subset \mathbb{P}^2 \backslash \mathbb{P}^1 = \mathbb{C}^2$ embeds properly holomorphically into $\mathbb{C}^2$ according to Theorem 8.9.3. In particular, since every one dimensional complex torus embeds as a smooth cubic curve in $\mathbb{P}^2$, we obtain

Corollary 8.9.6. *Every finitely connected subset without isolated boundary points in a torus embeds properly holomorphically into $\mathbb{C}^2$.*

Example 8.9.7. ([95], [197, Example 1.5]) A compact Riemann surface R is called *hyperelliptic* if it admits a meromorphic function of degree two, i.e., a two-sheeted branched holomorphic covering $R \to \mathbb{P}^1$ (see [233, p. 247]). Such R is the normalization of a complex curve in $\mathbb{P}^2$ given by $w^2 = \Pi_{j=1}^k (z - z_j)$ for some choice of points $z_1, \ldots, z_k \in \mathbb{C}$. A bordered Riemann surface D is said to be hyperelliptic if its double is hyperelliptic. (The double of D is obtained by gluing two copies of $\bar{D}$, the second one with the conjugate conformal structure, along their boundaries.) A hyperelliptic bordered Riemann surface $\bar{D}$ admits a holomorphic embedding into the closed bidisc $\overline{\mathbb{D}}^2 \subset \mathbb{C}^2$ by a pair of inner functions mapping bD to the torus $(b\mathbb{D})^2$ [422]. Hence Theorem 8.9.3 implies

Corollary 8.9.8. *Every smoothly bounded domain in a hyperelliptic bordered Riemann surface admits a proper holomorphic embedding in $\mathbb{C}^2$.*

In the proof of Theorem 8.9.3 we follow [197]. (Another proof, using the uniformization theory of bordered Riemann surfaces, is outlined in [197, §6].) The following notion from [524] plays the key role.

Definition 8.9.9. *Let $\pi\colon \mathbb{C}^2 \to \mathbb{C}$ denote the projection $\pi(z_1, z_2) = z_1$. Assume that $\Sigma \subset \mathbb{C}^2$ is a locally closed complex curve, possibly with boundary. A point $p = (p_1, p_2) \in \Sigma$ is* exposed *(with respect to π) if the complex line*

$$\Lambda_p = \pi^{-1}(\pi(p)) = \{(p_1, \zeta)\colon \zeta \in \mathbb{C}\}$$

intersects Σ precisely at p, and the intersection is transverse. If $\Sigma = f(R)$, where R is a Riemann surface and $f\colon R \to \mathbb{C}^2$ is a holomorphic map, then a point $a \in R$ is said to be f-exposed if the point $p = f(a) \in \Sigma$ is exposed.

Theorem 8.9.3 now follows immediately from the following two results.

Theorem 8.9.10. [197, Theorem 4.2] *An embedding $f\colon \bar{D} \hookrightarrow \mathbb{C}^2$ as in Theorem 8.9.3 can be approximated by holomorphic embeddings $F\colon \bar{D} \hookrightarrow \mathbb{C}^2$ such that every boundary component of $F(\bar{D})$ contains an exposed point. The approximation is in the $\mathcal{C}^1$ topology on $\bar{D}\backslash \cup_j U_j$ where U_j is a small neighborhood of a point a_j in each connected component C_j of bD. We can insure that F agrees with f to a finite order at finitely many interior points of D.*

Theorem 8.9.11. [524, Theorem 1.1] *Let $\overline{\Sigma} = f(\bar{D}) \subset \mathbb{C}^2$ be as in Theorem 8.9.3. If every boundary component of Σ contains an exposed point (Def. 8.9.9) then the conclusion of Theorem 8.9.3 holds.*

Proof (of Theorem 8.9.10). We realize $\bar{D}$ as a domain with $\mathcal{C}^\infty$ smooth boundary in an open Riemann surface R; the corresponding biholomorphic map is of class $\mathcal{C}^1$ up to the boundary. By Mergelyan's theorem we can approximate f in the $\mathcal{C}^1(\bar{D})$ topology by a holomorphic map $g\colon U \to \mathbb{C}^2$ on an open neighborhood $U \subset R$ of $\bar{D}$. Replacing f by g and R by a sufficiently small open neighborhood of $\bar{D}$ in R, we may therefore assume that $f\colon R \hookrightarrow \mathbb{C}^2$ is a (nonproper) holomorphic embedding.

We have $bD = \cup_{j=1}^m C_j$ where each C_j is a closed curve. For every j we choose a point $a_j \in C_j$ and a smooth embedded arc $\gamma_j \subset R$ that is transversely attached to $\bar{D}$ at its endpoint $a_j \in bD$. The rest of the arc, $\gamma_j \backslash \{a_j\}$, is contained in $R\backslash\bar{D}$. Let b_j denote the other endpoint of γ_j.

Choose an open set $U \Subset R\backslash\{b_1, \ldots, b_m\}$ that contains $\bar{D}$ and such that $\gamma_j \cap U = \widetilde{\gamma}_j$ is an arc with an endpoint a_j. In $\mathbb{C}^2$ we choose for every $j = 1, \ldots, m$ a smooth embedded arc λ_j that agrees with the arc $f(\widetilde{\gamma}_j)$ near the endpoint $q_j = f(a_j)$, while $\lambda_j \backslash f(\widetilde{\gamma}_j)$ does not intersect $f(U)$. We also insure that the arcs $\lambda_1, \ldots, \lambda_m$ are pairwise disjoint and that the other endpoint p_j of λ_j is an exposed point for the set $f(\bar{D}) \cup (\cup_{j=1}^m \lambda_j) \subset \mathbb{C}^2$ (Fig. 8.2).

Consider the compact set $K = \bar{D} \cup (\cup_{j=1}^m \gamma_j)$ in the Riemann surface R. Let $f'\colon U \cup (\cup_{j=1}^m \gamma_j) \to \mathbb{C}^2$ be a smooth map that agrees with f on U and that maps each arc $\gamma_j \subset R$ diffeomorphically onto the corresponding arc $\lambda_j \subset \mathbb{C}^2$. In particular, the endpoint b_j of γ_j is mapped by f' to the exposed endpoint

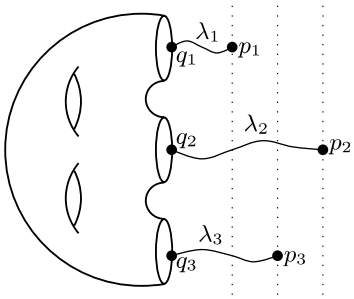

Fig. 8.2. A Riemann surface with exposed tails. ([197, p. 109, Fig. 2])

p_j of λ_j. By Mergelyan's theorem we can approximate f', uniformly on a neighborhood of $\bar{D}$ in R and in the $\mathcal{C}^1$ topology on each of the arcs γ_j, by a holomorphic map $\widetilde{f}\colon V \to \mathbb{C}^2$ from an open neighborhood of K in R. At the same time we insure that $\widetilde{f}$ agrees with f' to a high order at each of the points $a_1, \ldots, a_m, b_1, \ldots, b_m$. If the approximation is close enough, the neighborhood $V \supset K$ is chosen small enough, and the interpolation at the indicated points is to a sufficiently high order, then $\widetilde{f}\colon V \to \mathbb{C}^2$ is a (non-proper) holomorphic embedding, the complex curve $\widetilde{\Sigma} = \widetilde{f}(D) \subset \mathbb{C}^2$ is biholomorphic to the curve $\Sigma = f(D)$, and the points $p_j = \widetilde{f}(b_j)$ are exposed in $\widetilde{f}(V)$.

Theorem 8.8.1 on p. 365 furnishes a diffeomorphism $\phi\colon \bar{D} \to \phi(\bar{D}) \subset V$ that is holomorphic in D, that sends the point $a_j \in bD$ to the point b_j for every $j = 1, \ldots, m$, and that is close to the identity map outside a small neighborhood of $\{a_1, \ldots, a_m\}$ in $\bar{D}$. The composition $F = \widetilde{f} \circ \phi\colon \bar{D} \to \mathbb{C}^2$ maps $\bar{D}$ onto the domain $F(\bar{D})$ in the complex curve $\widetilde{f}(V) \subset \mathbb{C}^2$ such that each point $p_j = F(a_j)$ for $j = 1, \ldots, m$ is an exposed boundary point of $F(\bar{D})$. In addition, by the properties of ϕ given by Theorem 8.8.1, we can insure that $F(\bar{D})$ lies in an arbitrary given open neighborhood of the set $f(\bar{D}) \cup (\cup_{j=1}^k \lambda_j)$. Finally we can approximate F in the smooth topology on $\bar{D}$ by an embedding that is holomorphic in a neighborhood of $\bar{D}$. $\square$

Proof (of Theorem 8.9.11). Let $bD = \cup_{j=1}^m C_j$, and assume that $a_j \in C_j$ is an f-exposed point for each $j = 1, \ldots, m$ with respect to the first coordinate projection $\pi(z, w) = z$. Let $\pi_2\colon \mathbb{C}^2 \to \mathbb{C}$ be the second coordinate projection: $\pi_2(z, w) = w$. Define a rational shear

$$g(z, w) = \left(z, w + \sum_{j=1}^{m} \frac{\alpha_j}{z - \pi(f(a_j))}\right). \tag{8.20}$$

The numbers $\alpha_j \in \mathbb{C}\backslash\{0\}$ can be chosen such that π_2 maps the unbounded curves $\lambda_j = (g \circ f)(C_j \backslash \{a_j\}) \subset \mathbb{C}^2$ properly to pairwise disjoint unbounded curves $\gamma_j = \pi_2(\lambda_j) \subset \mathbb{C}$, and $\pi_2\colon \lambda_j \to \gamma_j$ is a diffeomorphism near infinity

for every j (see [524, Theorem 1.1] for the details). Thus $\cup_{j=1}^m \lambda_j$ has the nice projection property with respect to π_2 (Def. 4.14.2, p. 149). At the same time, the absolute values $|\alpha_j|$ can be chosen as small as desired, thereby insuring that the embedding $g \circ f \colon \bar{D} \backslash \{a_j\}_{j=1}^m \hookrightarrow \mathbb{C}^2$ is close to f on a given compact set in D. The embedded complex curve $X = (g \circ f)(D) \subset \mathbb{C}^2$ with boundary $bX = \cup_{j=1}^m \lambda_j$ is then biholomorphic to $\Sigma = f(D)$. Since bX has the nice projection property, the conclusion follows from Theorem 4.14.6. $\square$

The following result generalizes Theorem 8.9.3 to bordered complex curves in $\mathbb{C}^2$ with interior singularities.

Theorem 8.9.12. [197, Theorem 1.1] *If $\overline{\Sigma}$ is a (possibly reducible) compact complex curve in $\mathbb{C}^2$ with boundary $b\Sigma$ of class $\mathcal{C}^r$ for some $r > 1$, then the inclusion map $\iota \colon \Sigma = \overline{\Sigma} \backslash b\Sigma \hookrightarrow \mathbb{C}^2$ can be approximated, uniformly on compacts in Σ, by proper holomorphic embeddings $\Sigma \hookrightarrow \mathbb{C}^2$. In particular, a smoothly bounded relatively compact domain Σ in an affine complex curve $A \subset \mathbb{C}^2$ admits a proper holomorphic embedding in $\mathbb{C}^2$.*

Proof. The precise assumption on $\overline{\Sigma}$ is that locally near each boundary point $p \in b\Sigma$ it is a one dimensional complex manifold with boundary of class $\mathcal{C}^r$, while the interior, Σ, is a pure one dimensional analytic subvariety. To prove Theorem 8.9.12, we first find a bordered Riemann surface $\bar{D}$ with $\mathcal{C}^r$ boundary and a $\mathcal{C}^r$ map $f \colon \bar{D} \to \mathbb{C}^2$ such that the following hold:

- $f(\bar{D}) = \overline{\Sigma}$ and $f(bD) = b\Sigma$,
- f is a diffeomorphism near bD, and
- $f \colon D \to \Sigma$ is a holomorphic normalization of Σ. In particular, f is biholomorphic over the regular locus of Σ.

This is entirely elementary [197, Lemma 3.1]. We then apply the same proof as in Theorem 8.9.3, exposing a boundary point in each connected component of $\Sigma \subset \mathbb{C}^2$, sending these exposed points to infinity by a rational shear, and finally pushing the boundary curves of the new complex curve to infinity. The only addition is to choose the new map at every inductive step to be tangent to the existing map to a sufficiently high order at every point of the finite set $f^{-1}(\Sigma_{\text{sing}}) \subset D$. This insures that the proper holomorphic map $\widetilde{f} \colon D \to \mathbb{C}^2$ obtained by this process has the image $\widetilde{f}(D) \subset \mathbb{C}^2$ biholomorphic to $\Sigma = f(D)$. (See [197, Lemma 3.3] for further details.) $\square$

We also have the following result for Riemann surfaces with punctures.

Theorem 8.9.13. [197, Theorem 5.2] *Assume that $f \colon \bar{D} \to \mathbb{C}^2$ satisfies the hypotheses of Theorem 8.9.3. Let $\pi \colon \mathbb{C}^2 \to \mathbb{C}$ be a linear projection, $b_1, ..., b_k \in \mathbb{C}$, and $\{c_1, \ldots, c_l\} = (\pi \circ f)^{-1}(\{b_1, \ldots, b_k\}) \subset D$. Then the punctured Riemann surface $D \backslash \{c_1, \ldots, c_l\}$ embeds properly holomorphically in $\mathbb{C}^2$.*

Proof. We may assume that π is the first coordinate projection. Theorem 8.9.10 furnishes an embedding $F\colon \bar{D} \hookrightarrow \mathbb{C}^2$ with an exposed point $a_j \in bD$ in each boundary component and satisfying $F(c_j) = f(c_j)$ for $j = 1, \ldots, l$. We can avoid creating any new intersections of $F(\bar{D})$ with the complex lines $\pi^{-1}(b_j)$ for $j = 1, \ldots, k$, so that we have $(\pi \circ F)^{-1}(\{b_1, \ldots, b_k\}) = \{c_1, \ldots, c_l\} \subset D$. Let g be a shear (8.20) with simple poles at all points $(\pi \circ F)(a_j)$ $(j = 1, \ldots, m)$ and $b_1, \ldots, b_k$. Then $g \circ F$ embeds $D \backslash \{c_1, \ldots, c_l\}$ onto a complex curve $X \subset \mathbb{C}^2$, and we push bX to infinity by applying Theorem 4.14.6 (p. 152). $\square$

The techniques in this section have been extended by T. Ritter [416] to the construction of proper holomorphic embeddings of finitely connected planar domains without isolated boundary components into $\mathbb{C} \times \mathbb{C}^*$. Every such domain is conformally equivalent to a circular domain.

Theorem 8.9.14. [416, Main Theorem] *If D is a circular domain in $\mathbb{C}$, then every continuous map $D \to \mathbb{C} \times \mathbb{C}^*$ is homotopic to a proper holomorphic embedding $D \hookrightarrow \mathbb{C} \times \mathbb{C}^*$.*

8.10 Some Infinitely Connected Riemann Surfaces in $\mathbb{C}^2$

Theorems 8.9.3 and 8.9.12 were extended to certain infinitely connected Riemann surfaces by I. Majcen. The following is her main result.

Theorem 8.10.1. [350, Theorem 1] *Assume that $\bar{R}$ is a bordered Riemann surface and $f\colon \bar{R} \hookrightarrow \mathbb{C}^2$ is an embedding which is holomorphic in the interior R. Let $C_1, \ldots, C_m$ be the boundary components of R. Let D be an open set in R obtained by removing countably many closed disjoint discs $\overline{\Delta}_i$. Assume that for every $j = 1, \ldots, m$ there is a point $a_j \in C_j$ such that any point in $(bD \backslash bR) \cup \{a_1, \ldots, a_m\}$ has a neighborhood in $\bar{R}$ meeting only finitely many of the discs $\overline{\Delta}_i$. Given a compact set K in D and an $\epsilon > 0$, there exists a proper holomorphic embedding $F\colon D \hookrightarrow \mathbb{C}^2$ such that $||F - f||_K < \epsilon$.*

The discs $\overline{\Delta}_i$ in this theorem are clustering only on the boundary curves $C_1, \ldots, C_m$ of R, and they stay away from the points $a_j \in C_j$ $(j = 1, \ldots, m)$. It is not clear whether the latter condition is really necessary, but the induction procedure used in the proof can not be controlled without it.

Proof. We use the same techniques as in the proof of Theorem 8.9.3, but the induction scheme and the technical details are substantially more involved. Choose an exhaustion $K \Subset K_1 \Subset K_2 \Subset \cdots \Subset \cup_{j=1}^{\infty} K_j = D$ by compact $\mathcal{O}(D)$-convex sets with smooth boundaries. Let $z = (z_1, z_2)$ be coordinates on $\mathbb{C}^2$ with the corresponding projections $\pi_i(z_1, z_2) = z_i$ for $i = 1, 2$. At the n-th step of the inductive construction we shall obtain the following:

- integers $m_n, k_n \in \mathbb{N}$,
- a number $\epsilon_n > 0$ such that $2\epsilon_n \le \epsilon_{n-1}$,
- curves $\Gamma_j = b\Delta_j$ $(j = 1, \dots, k_n)$ which are the intersection of $bD \backslash bR$ with the $\mathcal{O}(R)$-convex hull of the set K_{m_n},
- the domain $D_n = R \backslash \cup_{j=1}^{k_n} \overline{\Delta}_j$ with boundary $bD_n = bR \cup \left(\cup_{j=1}^{k_n} \Gamma_j\right)$,
- points $c_j \in \Gamma_j$ for $j = 1, \dots, k_n$,
- a holomorphic embedding $f_n \colon \bar{D}_n \hookrightarrow \mathbb{C}^2$ that exposes the points $a_1, \dots, a_m$ and $c_1, \dots, c_{k_n}$ with respect to the projection π_1 (see Theorem 8.9.10),
- a rational shear

$$g_n(z_1, z_2) = \left(z_1, z_2 + \sum_{j=1}^{m} \frac{\alpha_j}{z_1 - \pi_1(f_n(a_j))} + \sum_{j=1}^{k_n} \frac{\beta_j}{z_1 - \pi_1(f_n(c_j))}\right)$$

  with poles at the exposed points $f(a_j)$ and $f(c_j)$ of $f_n(\bar{D}_n)$, and
- an automorphism ϕ_n of $\mathbb{C}^2$,

such that, setting

$$\Phi_n = \phi_n \circ \Phi_{n-1} = \phi_n \circ \phi_{n-1} \cdots \circ \phi_1, \qquad F_{n-1} = \Phi_{n-1} \circ g_n \circ f_n,$$

the following conditions hold:

$$|g_n \circ f_n(x) - g_{n-1} \circ f_{n-1}(x)| < \epsilon_n, \quad x \in K_{m_n}, \tag{8.21}$$

$$\overline{\mathbb{B}}_{j-1} \cap F_{n-1}(D) \subset F_{n-1}(\mathring{K}_{m_j}), \quad j = 1, \dots, n, \tag{8.22}$$

$$|\phi_n(z) - z| < \frac{\epsilon}{2^{n+1}}, \quad z \in \overline{\mathbb{B}}_{n-1} \cup F_{n-1}(K_{m_n}), \tag{8.23}$$

$$|\phi_n(z)| > n, \quad z \in F_{n-1}(bD). \tag{8.24}$$

Note that $R \supset D_1 \supset D_2 \supset \cdots \supset \cap_{j=1}^{\infty} D_j = D$.

We begin the induction with $n = 0$, $m_0 = k_0 = 0$, $\epsilon_0 = \epsilon$, $K_0 = \emptyset$, $f_0 = f$, $g_0 = \mathrm{Id}$, and $\Phi_0 = \mathrm{Id}$. In this case (8.21) – (8.23) are vacuous, and (8.24) holds if $(0,0) \notin f(bR)$ which can be arranged by a small translation.

We now explain the inductive step $n \to n+1$. By (8.24) there exists an integer $m_{n+1} > m_n$ such that

$$\overline{\mathbb{B}}_n \cap \big(\Phi_n \circ g_n \circ f_n(D)\big) \subset \Phi_n \circ g_n \circ f_n(\mathring{K}_{m_{n+1}}).$$

The $\mathcal{O}(D)$-convex hull of $K_{m_{n+1}}$ contains the boundary curves $\Gamma_1, \dots, \Gamma_{k_n}$ of D and at most finitely many other boundary curves which we label $\Gamma_{k_n+1}, \dots, \Gamma_{k_{n+1}}$. Let $D_{n+1} = R \backslash \cup_{j=1}^{k_{n+1}} \overline{\Delta}_j$. Choose a point $c_j \in \Gamma_j$ for each $j = k_n + 1, \dots, k_{n+1}$. Also choose a number $\epsilon_{n+1} \in (0, \epsilon_n/2)$ such that any holomorphic map $h \colon D \to \mathbb{C}^2$ satisfying $||h - g_n \circ f_n||_{K_{m_{n+1}}} < 2\epsilon_{n+1}$ is an embedding on the smaller set K_{m_n}.

Theorem 8.9.10 furnishes a holomorphic embedding $f_{n+1}\colon \bar{D}_{n+1} \hookrightarrow \mathbb{C}^2$ that agrees with f_n at the points $a_1, \ldots, a_m$, $c_1, \ldots, c_{k_n}$, it exposes the boundary points $a_1, \ldots, a_m$, $c_1, \ldots, c_{k_{n+1}}$, and it approximates f_n as close as desired outside small neighborhoods of these points. The image $f_{n+1}(\bar{D}_{n+1})$ stays as close as desired to the union of $f_n(\bar{D}_{n+1}) \subset \mathbb{C}^2$ with the family of arcs that were attached to this domain in order to expose the points $c_{k_n+1}, \ldots, c_{k_{n+1}}$. In particular, we insure that none of the complex lines $z_1 = \pi_1(f_{n+1}(c_j))$ for $j = 1, \ldots, k_{n+1}$ intersects the set $\varPhi_n^{-1}(\overline{\mathbb{B}}_n)$. The rational shear

$$g_{n+1}(z_1, z_2) = g_n(z_1, z_2) + \left(0, \sum_{j=k_n+1}^{k_{n+1}} \frac{\beta_j}{z_1 - \pi_1(f_{n+1}(c_j))}\right)$$

sends the exposed points $f_{n+1}(a_1), \ldots, f_{n+1}(a_m)$, $f_{n+1}(c_1), \ldots, f_{n+1}(c_{k_{n+1}})$ to infinity. A suitable choice of the arguments of the numbers $\beta_j \in \mathbb{C}^*$ for $j = k_n + 1, \ldots, k_{n+1}$ insures that the unbounded curve $g_{n+1} \circ f_{n+1}(bD_{n+1})$ has the nice projection property with respect to π_2; at the same time we can choose their absolute values $|\beta_j|$ arbitrarily small. Set $F_n = \varPhi_n \circ g_{n+1} \circ f_{n+1}$. If the approximations of f_n, g_n by f_{n+1}, g_{n+1}, respectively, were close enough then the conditions (8.21) and (8.22) hold with n replaced by $n+1$.

Observe that $X_n = F_n(D_{n+1})$ is a complex curve in $\mathbb{C}^2$ satisfying the hypotheses of Theorem 4.14.6 (p. 152) (controlling unbounded curves by automorphisms of $\mathbb{C}^n$). In particular, bX_n consists of unbounded arcs diffeomorphic to $\mathbb{R}$, and $\varPhi_n^{-1}(bX_n) = g_{n+1} \circ f_{n+1}(bD_{n+1})$ has the nice projection property with respect to the projection π_2. Consider the discs $\overline{\varDelta}_j$ that are contained in D_{n+1} (this holds for all but finitely many of them), and let $\varLambda_n \subset X_n$ denote the union of their images with respect to the map F_n. Set

$$L_n = \overline{\mathbb{B}}_n \cup F_n(K_{m_{n+1}}). \tag{8.25}$$

By (the proof of) Theorem 4.14.6 there is an automorphism ϕ_{n+1} satisfying

$$|\phi_{n+1}(z) - z| < \frac{\epsilon}{2^{n+2}}, \quad z \in L_n, \tag{8.26}$$

$$|\phi_{n+1}(z)| > n+1, \quad z \in bX_n \cup \varLambda_n. \tag{8.27}$$

Such ϕ_{n+1} is found by a small modification in the proof of Lemma 4.14.4 (p. 150). By the argument that we gave there, it suffices to explain the construction when bX_n has the nice projection property with respect to π_2. (The general case is obtained by conjugation with the automorphism $\alpha = \varPhi_n^{-1}$.) We look for a composition

$$\phi_{n+1} = \varphi_2 \circ \varphi_1 \circ \psi$$

of three automorphisms. In the first step we choose a long (but finite) arc λ_j on each of the boundary curves of X_n and apply Corollary 4.13.5 (p. 148) to find an automorphism $\varphi_1 \in \operatorname{Aut} \mathbb{C}^2$ that it is close to the identity map near L_n

and that sends each of the arcs λ_j out of the ball $\overline{\mathbb{B}}_{n+1}$. We may assume that the first m arcs $\lambda_1, \ldots, \lambda_m$ belong to those boundary curves in bX_n that are F_n-images of the boundary curves C_j of R. Choosing the arcs λ_j sufficiently long, the condition in the theorem implies that the discs in Λ_n cluster only on $\lambda = \cup_{j=1}^m \lambda_j$.

We now explain the construction of φ_2 and ψ. Choose a smoothly bounded, simply connected closed domain $E_n \subset \overline{X}_n$ that is a relative neighborhood of λ in $\overline{X}_n$, is disjoint from the set L_n (8.25), and such that every disc $\overline{\Delta}_i \subset \Lambda_n$ is either entirely contained in E_n or else is disjoint from E_n. (Such E_n is found by entirely elementary geometric considerations.) Let Λ'_n be the union of those finitely many discs in Λ_n that remained out of E_n. Since L_n is polynomially convex and $E_n \cup \Lambda'_n$ is a union of finitely many pairwise disjoint, smoothly bounded holomorphic discs contained in the complex curve $\overline{X}_n$, the sets $E_n \cup L_n$ and $E_n \cup L_n \cup \Lambda'_n$ are also polynomially convex by [467]. (See the argument in the proof of Theorem 4.14.6, property 1.)

It is easily seen that the set Λ'_n can be moved out of the ball $\overline{\mathbb{B}}_{n+1}$ by an isotopy of biholomorphic maps that are close to identity near the compact polynomially convex set $E_n \cup L_n$ and such that the trace of this isotopy is polynomially convex in the extended phase space. (We start by contracting the holomorphic discs belonging to Λ'_n to make them very small, and then we expel them out of $\overline{\mathbb{B}}_{n+1}$ without doing anything near $E_n \cup L_n$.) By Theorem 4.12.1 on p. 143 (which uses the Andersén-Lempert theory) we can therefore find $\varphi_2 \in \operatorname{Aut}\mathbb{C}^2$ that is close to the identity near $E_n \cup L_n$ and that maps Λ'_n out of $\overline{\mathbb{B}}_{n+1}$. It follows that

$$\{z \in bX_n \colon |\varphi_2 \circ \varphi_1(z)| \le n+1\} \subset \bigcup_{j=1}^m F_n(C_j \backslash \{a_j\}) \backslash \lambda.$$

If the arcs $\lambda_j \subset \lambda$ have been chosen long enough, then this remaining intersection of $\varphi_2 \circ \varphi_1(bX_n)$ with the ball $\overline{\mathbb{B}}_{n+1}$ can be removed by precomposing $\varphi_2 \circ \varphi_1$ by a shear $\psi(z_1, z_2) = \big(z_1 + h(z_2), z_2\big)$. (See (4.50) in the proof of Lemma 4.14.4 for the construction of such ψ.)

This completes the construction of $\phi_{n+1} = \varphi_2 \circ \varphi_1 \circ \psi$ satisfying (8.26) and (8.27). Hence the map $\Phi_{n+1} = \phi_{n+1} \circ \Phi_n$ satisfies

$$|\Phi_{n+1}(z)| > n+1, \quad z \in g_{n+1} \circ f_{n+1}(bD).$$

Thus condition (8.24) holds for $n+1$ which completes the induction step.

By (8.21) and the choice of ϵ_n we see that $G = \lim_{n\to\infty} g_n \circ f_n \colon D \hookrightarrow \mathbb{C}^2$ is a holomorphic embedding. Condition (8.23) implies that the sequence of automorphisms $\Phi_n \in \operatorname{Aut}\mathbb{C}^n$ converges on the set $\Omega = \cup_{n=2}^\infty \Phi_n^{-1}(\overline{\mathbb{B}}_{n-1})$ to a Fatou-Bieberbach map $\Phi = \lim_{n\to\infty} \Phi_n \colon \Omega \to \mathbb{C}^2$ in view of Corollary 4.4.2. From (8.22) and (8.24) we see that G embeds D properly into Ω. Hence the map $F = \Phi \circ G \colon D \hookrightarrow \mathbb{C}^2$ is a proper holomorphic embedding of D into $\mathbb{C}^2$ satisfying $||F - f||_K < \epsilon$. $\square$

For planar domains, Wold proved the following stronger result whose proof we omit. (For domains without punctures this is due to Globevnik and Stensøness [212].)

Theorem 8.10.2. [523] *Any finitely connected domain D in $\mathbb{C}$ can be embedded properly holomorphically into $\mathbb{C}^2$. Moreover, let $\{p_j\} \subset D$ be a sequence converging to a point p in the boundary bD (we allow $p = \infty$), and assume there exists a continuous curve $\gamma\colon [0,1] \to \bar{D}$ such that $\gamma([0,1)) \subset D$, $\gamma(1) = p$, and $\{p_j\} \subset \gamma$. Then $D \backslash \{p_j\}_{j=1}^{\infty}$ embeds properly holomorphically in $\mathbb{C}^2$.*

Proper holomorphic embeddings of bordered Riemann surfaces can be almost prescribed on certain smooth curves approaching the boundary. A result of this type is Theorem 4.14.1 on p. 149. In the following result, due to I. Majcen, $\mathbb{C}$ is replaced by a finitely connected domain $D \subset \mathbb{C}$ and $\mathbb{R}$ is replaced by a union of arcs which may terminate at boundary points of D (Fig. 8.3).

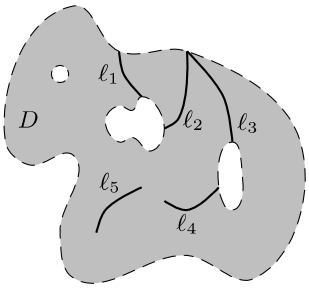

Fig. 8.3. Planar domain D and curves ℓ_i in Theorem 8.10.3

Theorem 8.10.3. [351] *Let $n \geq 2$ and $r \geq 1$ be integers. Let D be a finitely connected domain in $\mathbb{C}$. Let $\ell_i = \{\gamma_i(t)\colon t \in [0,1]\}$ $(i = 1, 2, \ldots, k)$ be smooth embedded curves in $\mathbb{C}$ such that $\gamma_i((0,1)) \subset D$ and $\ell_i \cap \ell_j \cap D = \emptyset$ for all $1 \leq i < j \leq k$. Set $\ell = \left(\cup_{i=1}^{k} \ell_i\right) \backslash bD$. Given a proper $\mathcal{C}^r$ embedding $\lambda\colon \ell \hookrightarrow \mathbb{C}^n$ and a positive continuous function $\eta\colon \ell \to (0,\infty)$ there exists a proper holomorphic embedding $f\colon D \hookrightarrow \mathbb{C}^n$ such that $|f^{(s)}(x) - \lambda^{(s)}(x)| < \eta(x)$ holds for all $x \in \ell$ and $s = 0, 1, \ldots, r$.*

Observe that some arcs $\ell_i = \gamma_i([0,1])$ may be entirely contained in D, and in such case every embedding $\ell_i \hookrightarrow \mathbb{C}^2$ is proper. On the other hand, an endpoint of ℓ_i that happens to belong to bD is sent by a proper embedding of $\ell_i \backslash bD$ to infinity. Theorem 8.10.3 is obtained by following the proof of Theorem 8.9.3 but using Lemma 4.14.4 on p. 150 (in place of Corollary 4.14.5) when dealing with the curves in ℓ.

We conclude this section with an interesting example of a rather different type due to S. Yu. Orevkov.

Theorem 8.10.4. [389] *There exists a Cantor set K in $\mathbb{P}^1$ such that the open Riemann surface $\mathbb{P}^1\backslash K$ admits a proper holomorhic embedding in $\mathbb{C}^2$.*

Proof. We construct such a curve in $\mathbb{C}^2$ as a limit of algebraic curves A_n obtained from each other by a birational transformation $F_n\colon \mathbb{C}^2 \to \mathbb{C}^2$. For some exhaustion of $\mathbb{C}^2$ by nested bidiscs $B_1 \subset B_2 \subset \cdots$ the topological type of $A_n \cap B_n$ will not change under further transformations.

Let (x, y) be complex coordinates on $\mathbb{C}^2$. Choose a sequence of complex numbers $a_1, a_2, \ldots$ whose absolute values are strictly increasing and tend to infinity. We inductively define a sequence of birational mappings $F_n\colon \mathbb{C}^2 \to \mathbb{C}^2$ by setting F_0 to be the identity mapping and by setting $F_n = f_n \circ F_{n-1}$, where

$$f_n(x, y) = \begin{cases} \big(x, y + g_n(x)\big), & \text{if } n \text{ is odd,} \\ \big(x + g_n(y), y\big), & \text{if } n \text{ is even,} \end{cases}$$

where $g_n(t) = \dfrac{\epsilon_n}{t - a_n}$ and the sequence $0 < \epsilon_n \ll \epsilon_{n-1}$ tend to zero sufficiently fast (to be specified below). Let $\overline{\mathbb{C}}^2 = \mathbb{C}^2 \cup \{\infty\}$ be the one point compactification of $\mathbb{C}^2$. Let $\gamma_n\colon \mathbb{P}^1 \to \overline{\mathbb{C}}^2$ be defined by $\gamma_n(z) = F_n(z, 0)$. Then, for a suitable choice of ϵ_n, the limit $\gamma = \lim_{n\to\infty} \gamma_n$ is a continuous mapping $\gamma\colon \mathbb{P}^1 \to \overline{\mathbb{C}}^2$ such that $K = \gamma^{-1}(\infty)$ is a Cantor set, and the restriction of γ to $\mathbb{P}^1\backslash K$ is a proper embedding of the open Riemann surface $\mathbb{P}^1\backslash K$ into $\mathbb{C}^2$.

Let us describe the choice of the parameters ϵ_n and explain why γ satisfies the stated properties. Choose numbers $R_n > 0$ such that $|a_n| < R_n < |a_{n+1}|$ for every $n \in \mathbb{N}$. Let

$$A_n = F_n(\mathbb{P}^1), \qquad D_n = \{z \in \mathbb{C}\colon |z| < R_n\}.$$

Denote the projection $(z_1, z_2) \to z_i$ by π_i $(i = 1, 2)$. Set

$$C_n^{(i)} = \pi_i^{-1}(D_n), \quad B_n = C_n^{(1)} \cap C_n^{(2)} = D_n \times D_n, \quad C_n = C_n^{(1)} \cup C_n^{(2)}.$$

Then $B_1 \subset B_2 \subset \cdots$ and $\cup_{n=1}^{\infty} B_n = \mathbb{C}^2$. We define the sequence $\epsilon_n > 0$ inductively so that they satisfy the following properties for every $n \in \mathbb{N}$:

1. $A_n \subset C_n$,
2. $A_n \cap (C_n^{(i)}\backslash B_n)$ $(i = 1, 2)$ has a finite number of connected components each being mapped by π_i biholomorphically onto $\mathbb{C}\backslash D_n$,
3. all curves $A_p \cap B_n$ for $p \geq n$ are isotopic to each other in B_n and they converge in the $\mathcal{C}^\infty$ topology to an analytic curve which is also isotopic to all of them, and
4. $\lim_{n\to\infty} d_n = 0$, where d_n is the maximum of diameters (with respect to some fixed metric on $\overline{C}$) of the connected components of $F_n^{-1}(A_n\backslash B_n)$.

Let us call a boundary component of $A_n \cap B_n$ horizontal if it is contained in $bD_n \times D_n$, and vertical if it is contained in $D_n \times bD_n$. (Condition 1 shows that there are no other components.) If the numbers ϵ_n are chosen as described then, up to a small perturbation, $A_{2n+1} \cap B_{2n+1}$ is obtained from $A_{2n} \cap B_{2n}$ by attaching an annulus to each vertical boundary component, and by attaching a pair of pants (an annulus with a hole) to each horizontal boundary component. So each vertical component at the $2n$-th step provides a single vertical component at the next step, but each horizontal component provides one horizontal and one vertical component at the next step. When passing from $A_{2n+1} \cap B_{2n+1}$ to $A_{2n+2} \cap B_{2n+2}$, the roles of vertical and horizontal boundary components are exchanged. $\square$

Problem 8.10.5. Does the complement $\mathbb{P}^1 \backslash K$ of every Cantor set $K \subset \mathbb{P}^1$ admit a proper holomorphic embedding into $\mathbb{C}^2$? $\square$

8.11 Approximation of Holomorphic Submersions

In this section we prove results on approximation of holomorphic submersions that will be used in the following section. We begin with the simplest case of approximating noncritical holomorphic functions on polynomially convex subsets of $\mathbb{C}^n$ by entire noncritical functions.

Theorem 8.11.1. [173, Theorem 3.1] *Let K be a compact polynomially convex subset of $\mathbb{C}^n$, and let f be a holomorphic function in an open set $U \supset K$ satisfying $\mathrm{d}f_z \neq 0$ for every $z \in U$. Given $\epsilon > 0$, there exists a function $g \in \mathcal{O}(\mathbb{C}^n)$ satisfying $\mathrm{d}g \neq 0$ on $\mathbb{C}^n$ and $\sup_K |f - g| < \epsilon$.*

Proof. Choose a point $z_0 \in K$ and a compact polynomially convex set $L \subset U$ with smooth boundary and containing K in the interior. Consider first the case $n = 1$. Then L is the union of finitely many compact, connected and simply connected sets (discs). Let P be a holomorphic polynomial on $\mathbb{C}$ such that $\sup_L |f - P| < \frac{\epsilon}{2}$ and $P' \neq 0$ on L. The critical set $\Sigma = \{P' = 0\}$ consists of finitely many points in $\mathbb{C} \backslash L$. Hence there is a connected and simply connected domain $V \subset \mathbb{C} \backslash \Sigma$ with $L \subset V$. Choose a smoothly bounded, connected and simply connected compact set $M \subset V$ containing L. Since $P' \neq 0$ on V, we have $P'|_V = \mathrm{e}^h$ for some $h \in \mathcal{O}(V)$. We approximate h uniformly on M by a holomorphic polynomial η and set

$$g(z) = g(z_0) + \int_{z_0}^{z} \mathrm{e}^{\eta(\zeta)}\, \mathrm{d}\zeta, \quad z \in \mathbb{C}.$$

The integral is clearly independent of the choice of a path from z_0 to z. Then $g' = \mathrm{e}^\eta \neq 0$ on $\mathbb{C}$, and $|g' - P'| = |\mathrm{e}^\eta - \mathrm{e}^h|$ is small on M. For every point $z \in M$ the integration path can be chosen in M, with length bounded by

$C|z - z_0|$ for some constant $C \geq 1$ independent of the point $z \in M$. If the approximation of h by η is close enough on M then $|g - P| < \frac{\epsilon}{2}$ on M, and hence $|g - f| < \epsilon$ holds on K.

Assume now that $n \geq 2$. Since L is polynomially convex, there exists a holomorphic polynomial h on $\mathbb{C}^n$ satisfying $\sup_L |f - h| < \frac{\epsilon}{2}$. If $\epsilon > 0$ is chosen sufficiently small, then $\mathrm{d}h \neq 0$ on K. For a generic choice of h, its critical set $\Sigma = \{z \in \mathbb{C}^n : \mathrm{d}h_z = 0\} \subset \mathbb{C}^n \backslash K$ is finite (since it is given by n polynomial equations $\frac{\partial h}{\partial z_1} = 0, \ldots, \frac{\partial h}{\partial z_n} = 0$). Let $c = \sup_{z \in L} |\mathrm{d}h_z|$. Choose a number δ such that

$$0 < \delta < \min\left\{\mathrm{dist}(K, \mathbb{C}^n \backslash L), \frac{\epsilon}{2c}\right\}.$$

By Corollary 4.12.7 (p. 146) there exists a biholomorphic map ϕ of $\mathbb{C}^n$ onto a subset $\Omega \subset \mathbb{C}^n \backslash \Sigma$ such that $\sup_{z \in K} |\phi(z) - z| < \delta$. The entire function $g = h \circ \phi$ then satisfies $\mathrm{d}g_z = \mathrm{d}h_{\phi(z)} \cdot \mathrm{d}\phi_z \neq 0$ for every $z \in \mathbb{C}^n$ (since $\phi(z) \in \Omega \subset \mathbb{C}^n \backslash \Sigma$ and $\mathrm{d}h \neq 0$ on $\mathbb{C}^n \backslash \Sigma$). Our choice of δ insures that for every point $z \in K$ the line segment connecting z and $\phi(z)$ is entirely contained in L. Hence

$$|g(z) - h(z)| = |h(\phi(z)) - h(z)| \leq c|\phi(z) - z| < c\delta < \frac{\epsilon}{2},$$

and therefore $|g(z) - f(z)| < \epsilon$ for every $z \in K$. $\square$

By replacing the use of Corollary 4.12.7 by Corollary 4.12.2 we obtain the following approximation result for submersions $\mathbb{C}^n \to \mathbb{C}^q$, $1 \leq q < n$.

Theorem 8.11.2. *Let K be a compact convex set in $\mathbb{C}^n$. Given a holomorphic submersion $f : K \to \mathbb{C}^q$ $(1 \leq q < n)$ and an $\epsilon > 0$, there exists a holomorphic submersion $g : \mathbb{C}^n \to \mathbb{C}^q$ such that $\sup_K |f - g| < \epsilon$.*

Proof. By the assumption, $f : U \to \mathbb{C}^q$ is a holomorphic submersion on an open set $U \supset K$. Choose a compact convex set $L \subset U$ containing K in the interior. By the Oka-Weil theorem there is a polynomial map $h : \mathbb{C}^n \to \mathbb{C}^q$ satisfying $\sup_L |f - h| < \frac{\epsilon}{2}$. A generic choice of h insures that the set

$$\Sigma = \{z \in \mathbb{C}^n : \mathrm{rank}\, \mathrm{d}h_x < q\}$$

is an algebraic subvariety of dimension $q - 1 \leq n - 2$ that does not intersect L (Theorem 7.9.1, p. 324). By Corollary 4.12.2 (p. 144) there exists for every $\delta > 0$ a biholomorphic map ϕ of $\mathbb{C}^n$ onto a subset $\Omega \subset \mathbb{C}^n \backslash \Sigma$ such that $\sup_{z \in K} |\phi(z) - z| < \delta$. The entire map $g = h \circ \phi : \mathbb{C}^n \to \mathbb{C}^q$ is then a submersion at each point, and if $\delta > 0$ is chosen small enough then $\sup_K |g - h| < \frac{\epsilon}{2}$. It follows that $\sup_K |g - f| < \epsilon$. $\square$

Observe that the proof of Theorem 8.11.2 breaks down when $q = n$: In this case the subvariety $\Sigma = \{z \in \mathbb{C}^n : \mathrm{rank}\, \mathrm{d}h_z < n\}$ (the zero locus of the Jacobian determinant $\mathrm{J}(h) = \det(\frac{\partial h_j}{\partial z_k})$) is a hypersurface in $\mathbb{C}^n$, and in general there do not exist any nondegenerate holomorphic maps $\mathbb{C}^n \to \mathbb{C}^n \backslash \Sigma$.

Problem 8.11.3. Let $n > 1$. Is it possible to approximate every locally biholomorphic map $\mathbb{C}^n \supset U \to \mathbb{C}^n$ on a neighborhood of a compact convex set $K \subset \mathbb{C}^n$, uniformly on K, by entire locally biholomorphic maps $\mathbb{C}^n \to \mathbb{C}^n$?

Theorem 8.11.2 has been extended to holomorphic maps $f\colon \mathbb{C}^n \to Y$ to manifolds Y of Class $\mathcal{A}$ (Def. 6.4.5, p. 252) such that the rank of f satisfies a pointwise estimate from below. The following result generalizes [177, Proposition 5.4] and the main result of [306].

Theorem 8.11.4. *Let n, q, r be integers satisfying $(n-r+1)(q-r+1) \geq 2$, and let Y be a manifold of Class $\mathcal{A}$ and of dimension q. If K is a compact convex set in $\mathbb{C}^n$ and if $f\colon K \to Y$ is a holomorphic map such that* $\operatorname{rank} \mathrm{d}f_z \geq r$ *for every point $z \in K$, then f can be approximated uniformly on K by entire maps $g\colon \mathbb{C}^n \to Y$ such that* $\operatorname{rank} \mathrm{d}g_z \geq r$ *for every point $z \in \mathbb{C}^n$.*

If $Y = \mathbb{C}^q$, then by taking $n > q = r$ we get Theorem 8.11.2, while for $n = r < q$ we get the following result for immersions $\mathbb{C}^n \looparrowright \mathbb{C}^q$.

Corollary 8.11.5. *Let K be a compact convex set in $\mathbb{C}^n$. Every holomorphic immersion $K \to \mathbb{C}^q$ for $q > n$ can be approximated, uniformly on K, by entire immersions $\mathbb{C}^n \looparrowright \mathbb{C}^q$.*

Proof (of Theorem 8.11.4). The proof for $Y = \mathbb{C}^q$ is a straightforward generalization of the proof of Theorem 8.11.2. The condition $(n-r+1)(q-r+1) \geq 2$ insures that for a generic polynomial map $h\colon \mathbb{C}^n \to \mathbb{C}^q$, the algebraic subvariety $\Sigma = \{z \in \mathbb{C}^n \colon \operatorname{rank} \mathrm{d}h_z < r\}$ has codimension at least two in $\mathbb{C}^n$ (see Theorem 7.9.1 on p. 324). The rest of the proof is exactly as before.

If the target Y is a manifold of Class $\mathcal{A}$, we need a more careful argument since we do not have a global jet transversality theorem for algebraic maps $\mathbb{C}^n \to Y$. We give the details for the case of a submersion, i.e., $n > q = r$.

We may assume that Y is connected. Let $f\colon K \to Y$ be a holomorphic submersion from an open neighborhood of compact convex set $K \subset \mathbb{C}^n$. By Proposition 6.4.6 (p. 252) every manifold Y of Class $\mathcal{A}$ is algebraically subelliptic. By Corollary 7.10.2 (p. 325) it follows that f can be approximated uniformly on K by algebraic maps (morphisms) $\mathbb{C}^n \to Y$. Thus we may assume that $f\colon \mathbb{C}^n \to Y$ is an algebraic map which is a submersion on K.

The set $\Sigma = \{z \in \mathbb{C}^n \colon \operatorname{rank} \mathrm{d}f_z < q = \dim Y\}$ is an algebraic subvariety of $\mathbb{C}^n$ which does not intersect K. We will show that, by a small perturbation of f, we get an algebraic map for which $\dim \Sigma \leq n-2$. If this holds, the proof is concluded exactly as in Theorem 8.11.2.

Assume to the contrary that $\dim \Sigma = n-1$. We shall inductively remove all $(n-1)$-dimensional irreducible components from Σ, changing the map only a little at every step. Choose an irreducible component $\Sigma' \subset \Sigma$ of dimension $n-1$ and a point $z_0 \in \Sigma'$ that does not belong to any other irreducible

component of Σ. By the definition of Class $\mathcal{A}$ (Def. 6.4.5) we have $Y = \widehat{Y}\backslash A$, where $\widehat{Y}$ is a connected manifold of Class $\mathcal{A}_0$ and A is a thin (of codimension at least two) algebraic subvariety of $\widehat{Y}$. Let $U \subset \widehat{Y}$ be a Zariski open set isomorphic to $\mathbb{C}^q$ and containing the point $y_0 = f(z_0)$. Let $s_0\colon U \times \mathbb{C}^q \to U \cong \mathbb{C}^q$ denote the spray $s_0(y,t) = y+t$. Choose a closed algebraic subvariety Y_0 of $\widehat{Y}$ of pure dimension $q-1$ (a hypersurface) such that $\widehat{Y} = Y_0 \cup U$ and $y_0 \notin Y_0$. Let $[Y_0] \to \widehat{Y}$ denote the holomorphic line bundle defined by the divisor of Y_0, and let $L = [Y_0]^{-1}$. Set $\mathbb{T}^q = \widehat{Y} \times \mathbb{C}^q$, the trivial bundle of rank q over $\widehat{Y}$.

By Proposition 6.4.2 (p. 251) there are an integer $m \in \mathbb{N}$ and an algebraic spray $s\colon E = \mathbb{T}^q \otimes L^{\otimes m} \to \widehat{Y}$ such that $s(y,t) = y$ for all $y \in Y_0$ and $t \in E_y$, and such that s equals s_0 over the open set $\widehat{Y}\backslash Y_0 \subset U$ (using an obvious identification of $E|_U$ with $\mathbb{T}^q|_U$). Note that L is trivial over $\widehat{Y}\backslash Y_0$.

By Serre's Theorem A the algebraic vector bundle $f^*(E)$ over $\mathbb{C}^n$ is generated by finitely many (say p) algebraic sections, and hence there is a surjective algebraic vector bundle map $\rho\colon \mathbb{C}^n \times \mathbb{C}^p \to f^*E$. Let $\iota\colon f^*E \to E$ be the natural vector bundle map covering the base map $f\colon \mathbb{C}^n \to Y$. Set $Z = f^{-1}(Y_0) \subset \mathbb{C}^n$. The algebraic map

$$F = s \circ \iota \circ \rho\colon \mathbb{C}^n \times \mathbb{C}^p \to \widehat{Y}$$

then satisfies the following properties:

(a) $F(z,0) = f(z)$ for $z \in \mathbb{C}^n$,
(b) $F(z,t) = f(z)$ for $z \in Z$ and $t \in \mathbb{C}^p$, and
(c) $F(z,\cdot)\colon \mathbb{C}^p \to \widehat{Y}$ is a submersion for every point $z \in \mathbb{C}^n\backslash Z$.

The proof of Theorem 7.8.5 (see especially Lemma 7.8.9 and Proposition 7.8.15 on p. 323) gives a polynomial map $P\colon \mathbb{C}^n \to \mathbb{C}^p$ such that the algebraic map

$$f_1\colon \mathbb{C}^n \to \widehat{Y}, \quad f_1(z) = F(z, P(z)) \quad (z \in \mathbb{C}^n)$$

satisfies the following properties:

(i) f_1 approximates f as close as desired on a neighborhood of K,
(ii) $\mathrm{j}^1_z f = \mathrm{j}^1_z f_1$ for every $z \in Z = f^{-1}(Y_0)$,
(iii) $f_1|_{\mathbb{C}^n\backslash Z}$ is transverse to the subvariety $A \subset \widehat{Y}$, and
(iv) the ramification locus of $f_1|_{\mathbb{C}^n\backslash Z}$ has dimension $\leq n-2$.

To obtain (iv), choose P such that $\mathrm{j}^1 f_1\colon X \to J^1(\mathbb{C}^n, Y)$ is transverse to the subvariety of $J^1(\mathbb{C}^n, Y)$ of all one-jets of rank $< q = \dim Y$. By Lemma 7.9.2 (p. 324) this subvariety has codimension $n-q+1 \geq 2$ which implies (iv). Let $C \subset \mathbb{C}^n$ denote the ramification locus of the map $f_1\colon \mathbb{C}^n \to \widehat{Y}$; thus $\dim C\backslash Z \leq n-2$ by (iv). The set

$$\Sigma_1 = (\Sigma \cap Z) \cup f_1^{-1}(A) \cup C$$

is an algebraic subvariety of $\mathbb{C}^n$ which does not intersect K, provided that the approximation of f by f_1 is sufficiently close near K. The restriction of f_1 to $\mathbb{C}^n \backslash \Sigma_1$ maps the latter set submersively to $Y = \widehat{Y} \backslash A$. We claim that Σ_1 has fewer $(n-1)$-dimensional irreducible components than Σ. Observe that $\dim(\Sigma_1 \backslash Z) \leq n-2$ by properties (iii) and (iv) of f_1. Next we show that

$$\dim \left[(Z \backslash \Sigma) \cap (f_1^{-1}(A) \cup C) \right] \leq n-2.$$

If $z \in Z \backslash \Sigma$, then f is unramified at z by the definition of Σ. Furthermore, $\mathrm{j}^1_z f_1 = \mathrm{j}^1_z f$ by property (ii), and hence f_1 is also unramified at such point z, thereby showing that $(Z \backslash \Sigma) \cap C = \emptyset$. This also implies that $f_1|_{Z \backslash \Sigma}$ is transverse to A and hence $\dim f_1^{-1}(A) \cap (Z \backslash \Sigma) \leq n-2$. It follows that the $(n-1)$-dimensional irreducible components of Z_1 are the same as those of $\Sigma \cap Z$. Since $z_0 \in \Sigma' \backslash Z$, the component Σ' of Σ is not among them, which proves the claim.

Repeating this argument with the pair (f_1, Σ_1) gives an algebraic map $f_2 \colon \mathbb{C}^n \to \widehat{Y}$, and an algebraic subvariety $\Sigma_2 \subset \mathbb{C}^n$ with fewer $(n-1)$-dimensional components than Σ_1, such that $f_2 \colon \mathbb{C}^n \backslash \Sigma_2 \to Y$ is a submersion which approximates f_1 (and hence f) uniformly on K. In finitely many steps of this kind we obtain an algebraic map $\widetilde{f} \colon \mathbb{C}^n \to Y$ that approximates f on K and whose ramification locus $\widetilde{\Sigma}$ satisfies $\dim \widetilde{\Sigma} \leq n-2$. $\square$

In the next section we shall need the following result on the existence of a holomorphic transition map, close to the identity map, between a pair of nearby holomorphic submersions in a neighborhood of a Stein compact.

Lemma 8.11.6. [173, Lemma 5.1] *Let K be a Stein compact in a complex manifold X. Assume that $U \subset X$ is an open set containing K and $f \colon U \to \mathbb{C}^q$ is a holomorphic submersion for some $q \leq \dim X$. Then there exist constants $\epsilon_0 > 0$, $M > 0$, and an open set $V \subset X$ with $K \subset V \Subset U$, satisfying the following property. Given a number $\epsilon \in (0, \epsilon_0)$ and a holomorphic submersion $g \colon U \to \mathbb{C}^q$ with $\sup_{x \in U} |f(x) - g(x)| < \epsilon$, there is an injective holomorphic map $\gamma \colon V \to U$ satisfying $f = g \circ \gamma$ on V and $\sup_{x \in V} \mathrm{dist}(\gamma(x), x)) < M\epsilon$.*

Proof. This is similar to the proof of Lemma 5.9.3 (p. 216). We may assume that U is Stein. Hence $\mathrm{T}X|_U = \ker \mathrm{d}f \oplus E$ for some trivial rank q holomorphic subbundle $E \subset \mathrm{T}X|_U$. Thus E is spanned by q pointwise independent holomorphic vector fields on U. Denote by $\phi(x, t_1, \ldots, t_q) = \phi^1_{t_1} \circ \cdots \circ \phi^q_{t_q}(x)$ the composition of their local flows. Then ϕ is defined in an open set $\Omega \subset U \times \mathbb{C}^q$ containing $U \times \{0\}^q$. For $x \in U$ write $\Omega_x = \{t \in \mathbb{C}^q \colon (x,t) \in \Omega\}$. After shrinking Ω we may assume that for each $x \in U$ the fiber Ω_x is connected and that $F_x = \{\phi(x,t) \colon t \in \Omega_x\} \subset X$ is a local complex submanifold of X which intersects the level set $\{f = f(x)\}$ transversely at x (since $\mathrm{T}_x F_x = E_x$ is complementary to the kernel of $\mathrm{d}f_x$). The implicit function theorem shows that, after shrinking $\Omega \supset U \times \{0\}^q$, the map $\Omega_x \ni t \to f(\phi(x,t)) \in \mathbb{C}^q$

maps Ω_x biholomorphically onto a neighborhood of the point $f(x)$ in $\mathbb{C}^q$. If $g\colon U \to \mathbb{C}^q$ is sufficiently uniformly close to f, then the same holds for the map $\Omega_x \ni t \to g(\phi(x,t))$, provided that we restrict x to a compact subset of U. It follows that, if $K \subset V \Subset U$ and if g is sufficiently close to f on U, there exists for every $x \in V$ a unique point $c(x) \in \Omega_x$ such that $g(\phi(x, c(x))) = f(x)$. Clearly $c\colon V \to \mathbb{C}^q$ is holomorphic, and the map $\gamma(x) = \phi(x, c(x)) \in X$ $(x \in V)$ satisfies Lemma 8.11.6. $\square$

8.12 Noncritical Holomorphic Functions

Let X be a Stein manifold. A holomorphic map $f = (f_1, \ldots, f_q)\colon X \to \mathbb{C}^q$ is a submersion if its differential $\mathrm{d}f_x\colon \mathrm{T}_x X \to \mathrm{T}_{f(x)}\mathbb{C}^q \cong \mathbb{C}^q$ is surjective at every point $x \in X$. Equivalently, $\mathrm{d}f_1 \wedge \mathrm{d}f_2 \wedge \cdots \wedge \mathrm{d}f_q \neq 0$. Thus the differential of a holomorphic submersion $X \to \mathbb{C}^q$ induces a surjective complex vector bundle map $\mathrm{T}X \to X \times \mathbb{C}^q$. We prove in this section that this necessary condition for the existence of a holomorphic submersion $X \to \mathbb{C}^q$ is also sufficient, except perhaps in the equidimensional case $\dim X = q > 1$. The following result was proved for Riemann surfaces by Gunning and Narasimhan [240]; for $\dim X > q \geq 1$ this is [173, Theorem II].

Theorem 8.12.1. (H-principle for holomorphic submersions to Euclidean spaces.) *If X is a Stein manifold of dimension n and if $1 \leq q < n$, or if $q = n = 1$, then every surjective complex vector bundle map $\mathrm{T}X \to X \times \mathbb{C}^q$ is homotopic (through surjective complex vector bundle maps $\mathrm{T}X \to X \times \mathbb{C}^q$) to the tangent map of a holomorphic submersion $X \to \mathbb{C}^q$. In particular, every open Riemann surface can be represented as a Riemann domain over $\mathbb{C}$.*

Theorem 7.3.1 (c) on p. 299 implies that the tangent bundle $\mathrm{T}X$ of an n-dimensional Stein manifold admits a surjective complex vector bundle map onto the trivial bundle $X \times \mathbb{C}^{[(n+1)/2]}$. (This result is due to Ramspott [406] who showed the equivalent dual statement: The cotangent bundle of an n-dimensional Stein manifold admits $\left[\frac{n+1}{2}\right]$ pointwise linearly independent sections.) Hence Theorem 8.12.1 gives the following result on the existence of noncritical holomorphic functions on Stein manifolds.

Corollary 8.12.2. [173] *Every Stein manifold admits a holomorphic function without critical points. More precisely, an n-dimensional Stein manifold admits $\left[\frac{n+1}{2}\right]$ holomorphic functions with pointwise independent differentials, and this number is maximal for every $n \in \mathbb{N}$.*

Theorem 8.12.1 is the holomorphic analogue of the h-principle for submersions of smooth open manifolds [234, 394]. It is not known whether it remains valid for $q = n > 1$. In this equidimensional case, submersions and immersions coincide with locally biholomorphic maps.

Problem 8.12.3. [236, p. 70] Assume that X is a Stein manifold of dimension $n > 1$ with trivial holomorphic tangent bundle $\mathrm{T}X$. Does X admit a holomorphic immersion $f\colon X \to \mathbb{C}^n$? Equivalently, is every such X a Riemann domain over $\mathbb{C}^n$? Is every complex vector bundle trivialization $\mathrm{T}X \xrightarrow{\cong} X \times \mathbb{C}^n$ homotopic to the differential of a holomorphic immersion $X \to \mathbb{C}^n$?

Our proof Theorem 8.12.1 implies the following result.

Theorem 8.12.4. *If Problem 8.11.3 (p. 383) has a positive answer, then for every n-dimensional Stein manifold X with trivial complex tangent bundle $\mathrm{T}X$, each complex vector bundle trivialization $\mathrm{T}X \xrightarrow{\cong} X \times \mathbb{C}^n$ is homotopic to the differential of a locally biholomorphic map $X \to \mathbb{C}^n$.*

Example 8.12.5. (Parallelizable Stein manifolds) A complex manifold X whose complex tangent bundle $\mathrm{T}X$ is holomorphically trivial is said to be (holomorphically) parallelizable. Examples are holomorphic quotients of $\mathbb{C}^n$ and its subdomains. If X is Stein then, by the Oka-Grauert principle (Corollary 7.3.5), $\mathrm{T}X$ is topologically trivial if and only if it is holomorphically trivial.

Every closed smooth complex hypersurface $X \subset \mathbb{C}^{n+1}$ is parallelizable [151]. To see this, note that the normal bundle N of X in $\mathbb{C}^{n+1}$ is trivial by the solution of the Cousin II problem; since $\mathrm{T}X \oplus N = \mathrm{T}\mathbb{C}^{n+1}|X \cong X \times \mathbb{C}^{n+1}$ is trivial, Corollary 7.3.8 (p. 301) implies that $\mathrm{T}X$ is itself trivial. The same argument shows that a closed complex (Stein) submanifold $X^n \subset \mathbb{C}^m$, $n \geq \left[\frac{m}{2}\right]$, with trivial normal bundle is parallelizable. Only a few of these examples are known to immerse holomorphically to $\mathbb{C}^n$. J. J. Loeb found explicit holomorphic immersions $X \to \mathbb{C}^n$ of algebraic hypersurfaces

$$X = \left\{(z_0, z_1, \ldots, z_k)\colon z_0^d + P_1(z_1) + \cdots + P_k(z_k) = 1\right\} \subset \mathbb{C}^{n+1},$$

where $z_0 \in \mathbb{C}$, $z_j \in \mathbb{C}^{n_j}$, P_j is a homogeneous polynomial of some degree d_j on $\mathbb{C}^{n_j}$ for every $j = 1, \ldots, k$, and $n_1 + \cdots + n_k = n$ (see [43, p. 19]). These manifolds are even algebraically parallelizable, but they do not admit any algebraic immersions to $\mathbb{C}^n$. An example of this type is the complex n-sphere $\Sigma^n = \{z \in \mathbb{C}^{n+1}\colon \sum z_j^2 = 1\}$. In another direction, Y. Nishimura found explicit holomorphic immersions $\mathbb{P}^2 \backslash C \to \mathbb{C}^2$, where C is an irreducible cuspidal cubic curve in $\mathbb{P}^2$ [376]. Further examples and remarks on parallelizable Stein manifolds can be found in [151]. $\square$

Example 8.12.6. There exist Stein Riemann domains $X \to \mathbb{C}^n$ such that X is not biholomorphic (not even homeomorphic) to a domain in $\mathbb{C}^n$. For example, if X is a complex torus $\mathbb{C}/\Gamma$ with one point removed, then X contains a pair of embedded closed curves intersecting transversely at one point. Such X is a Riemann domain over $\mathbb{C}$ by the Gunning-Narasimhan theorem [240], but is not homeomorphic to a domain in $\mathbb{C}$ since a pair of closed curves in $\mathbb{C}$ intersecting transversely meet at an even number of points. (This is standard in the smooth intersection theory; for the relevant topological intersection theory see [114, Corollary 4.9, p. 199].) $\square$

Example 8.12.7. Stout [473] constructed a domain in $\mathbb{C}^n$ for every $n > 1$ whose envelope of holomorphy is not homeomorphic to a domain in $\mathbb{C}^n$. (Much earlier, Stout and Zame [477] used a similar idea to find a domain in $\mathbb{C}^n$ whose universal covering space is not homeomorphic to a domain in $\mathbb{C}^n$.) Choose a pair of smooth embedded totally real submanifolds $M_1, M_2 \subset \mathbb{C}^n$ of real dimension n which intersect transversely at precisely two points p, q. Let V be a thin tube of variable radius around $M_1 \cup M_2$. Pick a ball $B \subset \mathbb{C}^n$ around p such that $V \cap B = V_1 \cup V_2$ is the union of tubes $V_j \supset M_j$ of radii $\delta_j > 0$ for $j = 1, 2$. We choose V such that the number δ_1 is much smaller than δ_2. Now let V_2' be obtained by removing from V_2 the tube of radius $2\delta_1$ around $M_1 \cap B$, and let $\Omega = (V \backslash B) \cup V_1 \cup V_2'$. For suitable choices of the numbers $\delta_1 >> \delta_2 > 0$ the envelope of holomorphy of V_2 contains the set $M_2 \cap B$. (This follows from a result on removable singularities of CR functions and the fact that the totally real sphere $M_1 \cap bB \subset bB$ is polynomially convex.) Hence the envelope of holomorphy $\widetilde{\Omega} \to \mathbb{C}^n$ of Ω contains totally real submanifolds $\widetilde{M_1}$, $\widetilde{M_2}$ (preimages of M_1, M_2, respectively), that intersect only over q, but not over p; hence their intersection index equals ± 1. This is impossible in $\mathbb{C}^n$. $\square$

Example 8.12.8. We show that the Stein manifold X^n from Forster's Example 8.2.2 (p. 337) does not admit a holomorphic submersion to $\mathbb{C}^{[(n+1)/2]+1}$.

Let $m = \left[\frac{n}{2}\right]$. We have seen in Example 8.2.2 that the Chern class $c_m(\mathrm{T}X)$ is the nonzero element of $H^{2m}(X;\mathbb{Z}) = \mathbb{Z}_2$. Hence $c_m(\mathrm{T}^*X) = (-1)^m c_m(\mathrm{T}X) \neq 0$ [359, p. 168]. This implies that the cotangent bundle T^*X does not contain a trivial complex subbundle of rank $n - m + 1 = \left[\frac{n+1}{2}\right] + 1$. (Indeed, if $\mathrm{T}^*X = E \oplus E'$ for some trivial subbundle E', then $0 \neq c_m(T^*X) = c_m(E)$ [359, Lemma 14.3]. This means that $\operatorname{rank} E \geq m$, and consequently $\operatorname{rank} E' \leq n - m = \left[\frac{n+1}{2}\right]$.) Hence there does not exist a surjective complex vector bundle map $\mathrm{T}X \to X \times \mathbb{C}^{[(n+1)/2]+1}$.

The same conclusion holds for every n-dimensional Stein manifold X such that $c_{[n/2]}(\mathrm{T}X) \neq 0$: Such X admits a submersion $X \to \mathbb{C}^{[(n+1)/2]}$, but every map $X \to \mathbb{C}^q$ for $q > \left[\frac{n+1}{2}\right]$ has a nontrivial ramification locus. $\square$

We now turn to the proof of Theorem 8.12.1, beginning with preliminaries.

Every complex vector bundle map $\Theta\colon \mathrm{T}X \to X \times \mathbb{C}^q$ is of the form $\Theta(x, v) = (x, \theta v)$, where $\theta = (\theta_1, \ldots, \theta_q)$ is a q-tuple of differential $(1,0)$-forms on X; that is, a q-tuple of sections of the holomorphic cotangent bundle $\mathrm{T}^{*1,0}X$. Furthermore, Θ is surjective if and only if the $(1,0)$-forms $\theta_1, \ldots, \theta_q$ are $\mathbb{C}$-linearly independent at every point $x \in X$:

$$\theta_{1,x} \wedge \cdots \wedge \theta_{q,x} \neq 0, \quad x \in X. \tag{8.28}$$

A q-tuple satisfying (8.28) will be called a *q-coframe* on X. Such a q-coframe spans a trivial rank q complex subbundle E^* of the complex cotangent bundle $\mathrm{T}^{*1,0}X \cong \mathrm{T}^*X$. Conversely, every q-coframe induces a surjective complex

vector bundle map $\Theta\colon \mathrm{T}X \to X \times \mathbb{C}^q$. Clearly we may speak of holomorphic q-coframes, homotopies of q-coframes, etc. If every component θ_j of a q-coframe θ is exact holomorphic, $\theta_j = \mathrm{d}f_j$ for some $f_j \in \mathcal{O}(X)$, then we write $\theta = \mathrm{d}f$ and say that θ is an *exact holomorphic q-coframe.*

We may view q-coframes as sections of the holomorphic fiber bundle $V^q(\mathrm{T}^*X) \to X$ whose fiber V_x^q is the Stiefel manifold of all q-tuples of $\mathbb{C}$-independent elements in T_x^*X. (Ex. 1.2.7). Since the fibers are $GL_n(\mathbb{C})$-homogeneous ($n = \dim X$), the Oka-Grauert principle applies to sections of $V^q(\mathrm{T}^*X)$; hence every q-coframe is homotopic to a holomorphic one.

Theorem 8.12.1 is an immediate consequence of the following result that includes the approximation and interpolation conditions.

Theorem 8.12.9. [173, Theorem 2.5] *Assume that X is a Stein manifold of dimension n, X' is a closed complex subvariety of X, and K is a compact $\mathcal{O}(X)$-convex subset of X. Assume that $q \in \{1, \ldots, n-1\}$ or $q = n = 1$. Let $\epsilon > 0$ and $r \in \mathbb{N}$. Given a q-coframe θ^0 on X that $\theta^0 = \mathrm{d}f^0$ is exact holomorphic in an open set $U \supset X' \cup K$, there exists a homotopy of q-coframes θ^t ($t \in [0,1]$) such that $\theta^1 = \mathrm{d}f$ is the differential of a holomorphic submersion $f = (f_1, \ldots, f_q)\colon X \to \mathbb{C}^q$ and the following hold for every $t \in [0,1]$:*

(i) *$\theta^t = \mathrm{d}f^t$ is exact holomorphic in a neighborhood of $X' \cup K$,*

(ii) *$\sup_{x \in K} |f^t(x) - f^0(x)| < \epsilon$, and*

(iii) *$f^t - f^0$ vanishes to order r on the subvariety X'.*

Proof. We follow the scheme of proof of Theorem 5.4.4 (see §5.11). Consider first the case when $X' = \emptyset$. Choose a smooth strongly plurisubharmonic Morse exhaustion function $\rho\colon X \to \mathbb{R}$ with nice critical points (Def. 3.9.2, p. 89) such that $\rho < 0$ on K and $\rho > 0$ on $X \backslash U$. We shall inductively modify the given q-coframe θ such that, at every step, we make it exact holomorphic on a somewhat larger set, taking care to satisfy conditions (i)–(iii). There are two main cases to consider.

The noncritical case: This amounts to passing from $\{\rho \le c\}$ to $\{\rho \le c'\}$ when ρ has no critical values on the interval $[c, c']$. By Lemma 5.10.3 (p. 218) this is accomplished in finitely many basic steps, each consisting of attaching a special convex bump B to a compact strongly pseudoconvex domain A; i.e., such that (A, B) is a *special Cartan pair* in X (Def. 5.10.2, p. 218).

We explain the procedure for such pair (A, B). By the assumption we have a q-coframe θ on X that is exact holomorphic in an open neighborhood U_A of A: $\theta = \mathrm{d}f$, where $f\colon U_A \to \mathbb{C}^q$ is a holomorphic submersion. The goal is to approximate f uniformly on A by a holomorphic submersion $f'\colon U'_D \to \mathbb{C}^q$ over a neighborhood U'_D of $D = A \cup B$. Then $\theta' = \mathrm{d}f'$ is exact holomorphic near D and is homotopic to θ there. Outside of a larger neighborhood of D we patch θ' with θ by using a cut-off function in the parameter of the homotopy (see the arguments in §5.11).

In suitable local holomorphic coordinates on X in a neighborhood of B the sets B and $C = A \cap B$ are compact convex subsets of $\mathbb{C}^n$ (see Fig. 5.2 on p. 219). By Theorem 8.11.2 on p. 382 (if $q < n$), resp. by Theorem 8.11.1 (if $q = 1 \leq n$), we can approximate f uniformly on a compact neighborhood C' of C by a holomorphic submersion $g\colon U_B \to \mathbb{C}^n$ on neighborhood of B. If g is sufficiently close to f then Lemma 8.11.6 furnishes an injective holomorphic map $\gamma\colon U_C \to X$ in some neighborhood of C such that γ is uniformly close to the identity map (depending on $\mathrm{dist}_{C'}(f,g)$), and it satisfies $f = g \circ \gamma$ near C. If g is close enough to f on C', then γ is so close to the identity map on U_C that Theorem 8.7.2 (p. 358) applies and gives $\gamma = \beta \circ \alpha^{-1}$, where α, β are injective holomorphic maps close to the identity map near A, B, respectively. It follows that $f \circ \alpha = g \circ \beta$ holds near C. These two maps clearly define a holomorphic submersion f' from a neighborhood of $D = A \cup B$ to $\mathbb{C}^q$ which approximates f uniformly on A. This concludes the noncritical case.

The critical case: We use the local normal form of ρ at a nice critical point p (see §3.9 – §3.10). We may assume that $\rho(p) = 0$. Let $k \in \{0, 1, \ldots, n\}$ denote the Morse index of ρ at p. If $k = 0$, a new connected component of the sublevel set appears at p, and we can take an arbitrary submersion in this component. Assume now that $k \geq 1$. In suitable local holomorphic coordinates $z = (z_1, \ldots, z_n) = (z', z'') \in \mathbb{C}^k \times \mathbb{C}^{n-k}$ around the critical point p, $\rho(z)$ is a normal form given by Lemma 3.9.1. In particular, the stable manifold of p,

$$E = \big\{(x' + iy', z'')\colon y' = 0,\ z'' = 0,\ |x'|^2 \leq |c|\big\},$$

is a k-dimensional totally real disc, attached from the outside to the sublevel set $\{\rho \leq c\}$ along the $(k-1)$-sphere $bE \subset \{\rho = c\}$. Choose numbers $c < r < 0$ close enough to 0 such that $\theta = \mathrm{d}f$ is exact holomorphic on $\{\rho < r\}$ and such that ρ has no critical values on $[c, 0)$. The extension across the critical level is accomplished in three steps (see the *critical case* in §5.11):

1. Smooth extension of f to a neighborhood of E such that $\overline{\partial} f|_E = 0$ and $\mathrm{d}f|_E = \partial f|_E$ has maximal rank q along E.
2. Approximation of f from Step 1 by a holomorphic submersion g defined in a neighborhood of the handlebody $\{\rho \leq c\} \cup E$.
3. Applying the noncritical case with another strongly plurisubharmonic function τ to extend g across the critical level $\{\rho = 0\}$.

Let us explain Step 1. The components θ_j of the q-coframe θ are expressed in the z-coordinates by $\theta_j(z) = \sum_{l=1}^n \theta_{j,l}(z)\,\mathrm{d}z_l$, where $\theta_{j,l}$ are continuous functions and the $q \times n$ matrix $(\,\theta_{j,l}\,)$ has complex rank q at each point. For $z \in E$ near bE we have $\theta = \mathrm{d}f$, and hence $\theta_{j,l} = \frac{\partial f_j}{\partial z_l} = \frac{\partial f_j}{\partial x_l}$.

Let $M_{q,n} \cong \mathbb{C}^{q \times n}$ denote the set of all complex $q \times n$ matrices, and let $M^*_{q,n}$ consist of all matrices in $M_{q,n}$ of maximal rank q.

Lemma 8.12.10. *The map f extends smoothly to a neighborhood of the set $\{\rho \leq c\} \cup E$ such that the following hold:*

(i) *the extension satisfies* $\overline{\partial} f = 0$ *at every point of* E,

(ii) *the Jacobian matrix* $f' = \left(\frac{\partial f_j}{\partial z_l}\right)$ *has complex rank* q *along* E, *and*

(iii) f' *is homotopic to* $(\theta_{j,l})$ *by a homotopy of maps* $E \to M^*_{q,n}$ *that is fixed near* $bE = \{\rho \le c\} \cap E$.

Proof. We identify $x \in \mathbb{R}^n$ with $x + \mathrm{i}0 \in \mathbb{C}^n$. Choose a small $\delta > 0$. Let $A \subset D \subset \mathbb{R}^n$ be subsets of X, given in the local coordinates by

$$D = \big\{(x', x''): |x'|^2 \le |c|,\ |x''| \le \delta\big\},\ \ A = \big\{(x', x''): |r| \le |x'|^2 \le |c|,\ |x''| \le \delta\big\}.$$

Choosing $\delta > 0$ small enough, f is a holomorphic submersion to $\mathbb{C}^q$ in a neighborhood of the annular set $A \subset \mathbb{R}^n$, and hence its Jacobian matrix defines a map $f': A \to M^*_{q,n}$. By the assumption we have $f' = \theta$ on A.

Lemma 8.12.11. *There exists a smooth map* $h = (h_1, \ldots, h_q): D \to \mathbb{C}^q$ *such that* $h = f$ *on* A, $h' = \left(\frac{\partial h_j}{\partial x_l}\right): D \to M^*_{q,n}$, *and the map* $\theta: D \to M^*_{q,n}$ *is homotopic to* h' *(relative to* A*) in the space of maps* $D \to M^*_{q,n}$.

Proof. We have $M^*_{q,n} = M_{q,n} \backslash \Sigma$, where Σ consists of all matrices of rank less than q. By Lemma 7.9.2 (p. 324), applied with $i = n - q + 1$, Σ is an algebraic subvariety of complex codimension $n - q + 1$ in $M_{q,n} \cong \mathbb{C}^{q \times n}$.

If $k < 2(n-q+1)$ (which is equivalent to $q \le n - \left[\frac{k}{2}\right]$), the jet transversality theorem shows that a generically chosen extension of f from A to D satisfies $f'(x) \notin \Sigma$ for every point x in the k-dimensional disc $D_k = \{(x', 0): |x'| \le |c|\}$, and hence also in an open neighborhood $V \subset \mathbb{R}^n$ of $A \cup D_k$. Clearly there exists a diffeomorphism $\psi: D \to \psi(D) \subset V$ which equals the identity on A. Then $h = f \circ \psi$ has the desired properties. The Jacobian h' is homotopic to θ on D (relative to A) by dimension reasons.

If $k \ge 2(n-q+1)$ then the maximal rank condition is no longer generic, and we use instead Gromov's h-principle for ample differential relations (Theorem 1.10.5 on p. 42). Consider $M_{q,n}$ as the space of all one-jets of smooth maps $D \to \mathbb{C}^q$ at any point $x \in D$ (that is, the space of all first order partial derivatives at x, ignoring the image point). The open set $\Omega = M^*_{q,n} \subset M_{q,n}$ defines a differential relation of order one. We show that this relation is ample in the coordinate directions (Def. 1.10.4 on p. 42). Choose $l \in \{1, \ldots, n\}$ and fix in an arbitrary way the entries of a $q \times n$ matrix which do not belong to the column l. Let $\Omega' \subset \mathbb{C}^q$ consist of all vectors whose insertion in the l-th column gives a matrix of maximal rank q. Note that Ω' is either empty, the complement of a complex hyperplane in $\mathbb{C}^q$, or all of $\mathbb{C}^q$, depending on the rank of the initial $q \times (n-1)$ matrix; in the latter two cases Ω' is connected and its convex hull equals $\mathbb{C}^q$. This shows that Ω is ample in the coordinate directions. The existence of h now follows from Gromov's Lemma 1.10.6. □

Since D is contained in the real subspace $\mathbb{R}^n$ of $\mathbb{C}^n$, we can apply Lemma 3.5.5 (p. 73) to extend the map h from Lemma 8.12.11 smoothly to a neighborhood of D in $\mathbb{C}^n$ (considered as a subset of X) such that $\overline{\partial} h = 0$ on D and

the extension agrees with f near A. By setting $f = h$ near D we thus obtain an extension of f satisfying Lemma 8.12.10. $\square$

Clearly the homotopy of the q-coframe θ on D can be extended to a homotopy on X such that $\theta = \mathrm{d}f$ holds on $\{\rho \le c\} \cup D$. This completes Step 1 in the critical case. Now apply Theorem 3.7.2 (p. 81) to approximate f in the $\mathcal{C}^1$ topology on $\{\rho \le c\} \cup E$ by a holomorphic map g from a neighborhood of this set to $\mathbb{C}^q$. If the approximation is sufficiently close then g is a submersion, and this completes Step 2. Step 3 is accomplished by using Lemma 3.10.1 (p. 92) just as in §5.11.

This proves Theorem 8.12.9 when $X' = \emptyset$. The general case with interpolation is obtained exactly as in the proof of Theorem 5.4.4 (see §5.12). $\square$

The above construction allows improvements in several directions. First, we can add the interpolation condition for finite order jets at a discrete set of points, obtaining holomorphic functions on a Stein manifold with a prescribed critical set. Furthermore, we can add interpolation along a subvariety.

Theorem 8.12.12. [173, Theorem 2.1] *Let X be a Stein manifold, X' a closed complex subvariety of X, and K a compact $\mathcal{O}(X)$-convex subset of X. Let $U \subset X$ be an open set containing $X' \cup K$ and let $f \in \mathcal{O}(U)$ be a holomorphic function with discrete critical set $P = \{p_1, p_2, \ldots\} \subset X' \cup K$. Given numbers $\epsilon > 0$ and $r, n_1, n_2, \ldots \in \mathbb{N}$, there exists $g \in \mathcal{O}(X)$ with the critical set P and satisfying the following additional properties:*

(i) $|f(x) - g(x)| < \epsilon$ *for all* $x \in K$,
(ii) $\mathrm{j}^r_x f = \mathrm{j}^r_x g$ *for all* $x \in X'$, *and*
(iii) $\mathrm{j}^{n_k}_{p_k} f = \mathrm{j}^{n_k}_{p_k} g = 0$ *for* $k = 1, 2, \ldots$.

In particular, for any discrete set of points $P = \{p_j\} \subset X$ there exists a holomorphic function $f \in \mathcal{O}(X)$ whose critical locus equals P.

In the proof Theorem 8.12.9, the topological assumption concerning the existence of a q-coframe θ on X was used only in the Critical case – Step 1 (extension of a submersion across the stable manifold of a critical point p of ρ). If the Morse index k of p satisfies $k < 2(n-q+1)$ (equivalently, $q \le n - \left[\frac{k}{2}\right]$), then this maximal rank condition holds for a generic map on the handle (see the proof of Lemma 8.12.11). Hence we obtain the following corollary.

Corollary 8.12.13. *Let X be a Stein manifold of dimension n. Assume that $\rho\colon X \to \mathbb{R}$ is a strongly plurisubharmonic Morse exhaustion function, c is a regular value of ρ, and every critical point of ρ in $\{x \in X\colon \rho(x) > c\}$ has Morse index $\le k$. If $q \le q(k,n) = \min\left\{n - \left[\frac{k}{2}\right], n-1\right\}$ then every holomorphic submersion $f\colon \{x \in X\colon \rho(x) < c\} \to \mathbb{C}^q$ can be approximated uniformly on compacts by holomorphic submersions $X \to \mathbb{C}^q$. In particular, a Stein manifold X^n of geometric dimension k admits a holomorphic submersion $X \to \mathbb{C}^{q(k,n)}$.*

8.13 The H-Principle for Holomorphic Submersions

The h-principle in Theorem 8.12.1 can be extended to holomorphic submersions of Stein manifolds to a certain class of complex manifolds that satisfy the following approximation property analogous to CAP.

Definition 8.13.1. *A complex manifold Y satisfies* Property $\mathrm{S_n}$ *for some integer $n \geq \dim Y$ if every holomorphic submersion $K \to Y$ from a compact convex set $K \subset \mathbb{C}^n$ can be approximated by holomorphic submersions $\mathbb{C}^n \to Y$.*

Note that a manifold of Class $\mathcal{A}$ (Def. 6.4.5) satisfies Property $\mathrm{S_n}$ for any $n > \dim Y$ according to Theorem 8.11.4 (p. 383). We have the following h-principle which is extends the results for smooth submersions, due to Phillips [394] and Gromov [234, 236].

Theorem 8.13.2. [174, Theorem 1.1] *Assume that Y is a complex manifold of dimension q satisfying Property $\mathrm{S_n}$ for some $n \geq q$. Let X be a Stein manifold of dimension n. A continuous map $f\colon X \to Y$ is homotopic to a holomorphic submersion $X \to Y$ if and only if there exists a surjective complex vector bundle map $\iota\colon \mathrm{T}X \to f^*\mathrm{T}Y$.*

If Y satisfies a 1-parametric version of Property $\mathrm{S_n}$ [174, Def. 2.1 (b)] then we also have the following

One-parametric h-principle for holomorphic submersions: *Regular homotopy classes of holomorphic submersion $X \to Y$ from n-dimensional Stein manifolds X to Y are in one-to-one correspondence with homotopy classes of fiberwise surjective complex vector bundle maps $\mathrm{T}X \to \mathrm{T}Y$.*

The proof of Theorem 8.13.2 is similar to that of Theorem 8.12.9. The only analytic property of the target manifold $\mathbb{C}^q$ that was used in the proof is the approximation property for holomorphic submersions $\mathbb{C}^n \to \mathbb{C}^q$ on compact convex subsets of $\mathbb{C}^n$ (Theorem 8.11.2, p. 382); this is precisely Property $\mathrm{S_n}$ of $\mathbb{C}^q$ for $n > q$. The last statement follows from Theorem 8.11.4.

Property $\mathrm{S_n}$ obviously passes up and down in an unramified holomorphic covering $\widetilde{Y} \to Y$. For further functorial properties see [174]. While its relationship with CAP (characterizing the class of Oka manifolds) is unclear, $\mathrm{S_n}$ has been verified in most cases when CAP holds and $n > \dim Y$.

Problem 8.13.3. Does every Oka manifold Y enjoy Property $\mathrm{S_n}$ for each integer $n > \dim Y$? Does $\mathbb{C}^n$ for $n > 1$ enjoy Property $\mathrm{S_n}$? (Note that this is a rephrasing of Problem 8.11.3.)

Problem 8.13.4. Assume that X is a Stein manifold of dimension $n > 1$ and $f_1, \ldots, f_q$ are holomorphic functions on X satisfying $\mathrm{d}f_1 \wedge \cdots \wedge \mathrm{d}f_q \neq 0$. Assume that θ is a $(1,0)$-form on X such that

$$\mathrm{d}f_1 \wedge \cdots \wedge \mathrm{d}f_q \wedge \theta \neq 0. \tag{8.29}$$

Is θ homotopic to an exact holomorphic differential $\theta_1 = \mathrm{d}g$ in the space of $(1,0)$-forms satisfying the independence condition (8.29)?

That is, we are looking for a function $g \in \mathcal{O}(X)$ whose restriction to every fiber $f^{-1}(z)$ ($z \in \mathbb{C}^q$) of the holomorphic submersion $f = (f_1, \ldots, f_q)\colon X \to \mathbb{C}^q$ is noncritical on that fiber. At the moment this is open even for $q = 1$, i.e., when the fibers are hypersurfaces. A positive solution for $q = n-1$ (when the fibers are complex curves) would clearly imply that every Stein manifold X with trivial tangent bundle admits a locally biholomorphic map to $\mathbb{C}^n$ with $n = \dim X$ (see Problem 8.12.3 on p. 387). Another way of solving Problem 8.12.3 would be to give an affirmative answer to the following question which is open even for polynomial vector fields on $\mathbb{C}^n$ (c.f. [236, p. 70]).

Problem 8.13.5. Let L be a nowhere vanishing holomorphic vector field on a Stein manifold X. Does there exist a holomorphic function $f \in \mathcal{O}(X)$ such that $Lf \neq 0$ at every point of X? (Simple examples show that in general one can not solve the equation $Lf = g$ for a given $g \in \mathcal{O}(X)$.)

If Problem 8.13.5 is solvable, we get a solution of Problem 8.12.3 as follows. Assume that X is a Stein manifold of dimension $n > 1$ with trivial complex tangent bundle $\mathrm{T}X$. By Theorem 8.12.1 there exists a holomorphic submersion $f = (f_1, \ldots, f_{n-1})\colon X \to \mathbb{C}^{n-1}$ such that the line bundle $E = \ker \mathrm{d}f \subset \mathrm{T}X$ is trivial. Let L be a nowhere vanishing section of E, that is, a holomorphic vector field without zeros that is tangential to the fibers of f. If $g \in \mathcal{O}(X)$ satisfies $Lg \neq 0$, then $(f, g)\colon X \to \mathbb{C}^n$ is a locally biholomorphic map.

8.14 Closed Holomorphic One-Forms Without Zeros

By a theorem of J.-P. Serre [439, Theorem 1] each element of the de Rham cohomology group $H^1(X;\mathbb{C})$ on a Stein manifold X is represented by a closed holomorphic 1-form. By Corollary 8.12.2 on p. 386, the zero class in $H^1(X;\mathbb{C})$ is also represented by a closed holomorphic 1-form *without zeros*, namely the differential $\mathrm{d}f$ a noncritical holomorphic function $f\colon X \to \mathbb{C}$. The following result of I. Majcen shows that *every* cohomology class in $H^1(X;\mathbb{C})$ is represented by a closed holomorphic 1-form without zeros.

Theorem 8.14.1. [349] *Let X be a Stein manifold of dimension n whose holomorphic cotangent bundle T^*X admits a trivial complex subbundle of rank q for some $1 \le q < n$. Given closed 1-forms $\theta_1, \ldots, \theta_q$ on X, there exist closed holomorphic 1-forms $\omega_1, \ldots, \omega_q$ on X satisfying*

$$[\omega_j] = [\theta_j] \in H^1(X;\mathbb{C}) \ (j = 1, \ldots, q), \quad \omega_1 \wedge \omega_2 \wedge \cdots \wedge \omega_q \neq 0.$$

In particular, every cohomology class in $H^1(X;\mathbb{C})$ is represented by a closed holomorphic 1-form without zeros.

On open Riemann surfaces this was first shown by Kusunoki and Sainouchi [314]. Theorem 8.14.1 fails on general complex manifolds. For example, if X is a compact Riemann surface of genus g, the Riemann-Roch theorem shows that each nonconstant holomorphic 1-form has $2g-2$ zeros.

Proof. We explain the main idea in the case of a single closed 1-form θ. By Serre's theorem cited above we may assume that θ is a closed holomorphic 1-form. If ω is another such form, de Rham's theorem implies that $[\theta]=[\omega]\in H^1(X;\mathbb{C})$ if and only if $\int_C \omega = \int_C \theta$ for every closed path C in X. Thus we need to find a closed holomorphic 1-form without zeros, and with prescribed periods on a set of curves forming a basis of the free part of $H_1(X,\mathbb{Z})$. We follow the proof of Theorem 8.12.9 and explain the necessary modifications.

Choose a strongly plurisubharmonic exhaustion function $\rho\colon X\to\mathbb{R}$ with nice critical points (Def. 3.9.2). Assume that ω is a 1-form with the desired properties on a neighborhood of a certain sublevel set $\{\rho\le c\}$; we wish to approximate it by another such form $\widetilde{\omega}$ in a neighborhood of $\{\rho\le c'\}$ for some $c'>c$. We may assume that c and c' are regular values of ρ.

Consider first the case when ρ has no critical values on $[c,c']$. The topology of the sublevel sets does not change from c to c', and no new closed curves appear. It suffices to explain how to extend ω across a convex bump B attached to a certain set A such that (A,B) is a special convex pair (Def. 5.4.2, p. 192). In a neighborhood of $A\cap B$ (which is a compact convex set in $\mathbb{C}^n$ in some local coordinates) the form ω is exact holomorphic, $0\neq\omega=\mathrm{d}f$. Now approximate f by a noncritical holomorphic function g in a neighborhood of B, find a holomorphic transition map γ close to the identity near $A\cap B$ such that $f=g\circ\gamma$, and decompose $\gamma=\beta\circ\alpha^{-1}$ where α and β are small perturbations of the identity map near A and B, respectively. (See the proof of Theorem 8.12.9, p. 389). Thus $f\circ\alpha=g\circ\beta$ holds near $A\cap B$. Define a holomorphic 1-form ω' in a neighborhood of $A\cup B$ by taking $\omega'=\alpha^*\omega$ near A and $\omega'=\beta^*\mathrm{d}g$ near B. Near the intersection $A\cap B$ these two expressions agree:

$$\beta^*\mathrm{d}g=\mathrm{d}(g\circ\beta)=\mathrm{d}(f\circ\alpha)=\alpha^*\mathrm{d}f=\alpha^*\omega.$$

In finitely many such steps we accomplish the job. The closed holomorphic 1-form $\widetilde{\omega}$ without zeros on $\{\rho\le c'\}$, obtained in this way, is conjugate to ω near $\{\rho\le c\}$ by a biholomorphism close to the identity map, and hence the periods over closed curves in $\{\rho\le c\}$ remain the same.

It remain to explain how to extend ω across a critical level of ρ. Suppose that p is a critical point of ρ and let k be its Morse index. Let E be the local stable manifold of p (a totally real disc of dimension k).

If $k=0$, a new connected component of the sublevel set appears at p, and in this component we can take $\omega=\mathrm{d}f\neq 0$ for any local noncritical holomorphic function f near p.

If $k\in\{2,\dots,n\}$ then no nontrivial new curves appear when passing the critical level at p. Furthermore, we claim that ω is exact holomorphic in a

neighborhood of the $(k-1)$-sphere bE. This is obvious if $k > 2$, while for $k = 2$ (when $bE \cong S^1$) we have $\int_{bE} \omega = \int_{bE} \theta = \int_E \mathrm{d}\theta = 0$ by Stokes' theorem; the claim follows. Hence $\omega = \mathrm{d}f$ in a neighborhood of bE for some noncritical holomorphic function f. As before we find a noncritical function g near E that approximates f well enough near bE; hence $\mathrm{d}g \approx \omega$ near bE. As in the noncritical case explained above we glue ω and $\mathrm{d}g$ into a closed holomorphic 1-form ω' without zeros in a neighborhood of $\{\rho \le c\} \cup E$. We complete this step by applying the noncritical case with a different strongly plurisubharmonic function (proof of Theorem 8.12.9, critical case, Step 3).

It remains to consider the case $k = 1$. There are two possibilities:

- The segment E joins two distinct connected components of the sublevel set. In this case no new curve appears and we proceed as before.
- E is attached with both endpoints to the same component of $\{\rho \le c\}$.

In the latter case we add to E another segment $E' \subset \{\rho \le c\}$ so that $C = E \cup E'$ is a closed loop in X. This loop is a nontrivial new element of the first homology group when passing the critical level at p. We proceed as before, but taking care to insure that $\int_C \omega = \int_C \theta$. This can be arranged by a suitable choice of the noncritical holomorphic function f along E such that $\mathrm{d}f \approx \omega$ near bE. Since the value of the integral changes a little when performing approximation and patching, we choose a submersive family of functions and use the implicit function theorem to obtain an extension with the correct period within this family. For further details see [349]. □

8.15 Holomorphic Foliations on Stein Manifolds

The results on holomorphic submersions in the previous two sections give interesting consequences concerning the existence and approximation of nonsingular holomorphic foliations on Stein manifolds. The results in this section are taken from [173]. We begin with two immediate consequences of the existence of noncritical functions.

Corollary 8.15.1. *Every Stein manifold X admits a nonsingular holomorphic foliation by closed complex hypersurfaces; in addition, such a foliation may be chosen to be transverse to a given closed complex submanifold of X.*

Proof. A closed complex submanifold V of a Stein manifold X is itself a Stein manifold, and hence it admits a noncritical function $f \in \mathcal{O}(V)$ by Corollary 8.12.2. By Cartan's theorem f extends to a holomorphic function on X. Since the extension remains noncritical on V, Theorem 8.12.9 (p. 389), applied with $q = 1$ and $X' = V$, gives a noncritical function $g \in \mathcal{O}(X)$ such that $g|_V = f$. The levels sets $\{g = c\}$ $(c \in \mathbb{C})$ form a foliation of X by closed smooth complex hypersurfaces that are transverse to V. □

Corollary 8.15.2. *If V is a smooth closed complex hypersurface with trivial normal bundle in a Stein manifold X, then V is a union of leaves in a nonsingular holomorphic foliation of X by closed complex hypersurfaces. This holds in particular if $H^2(V;\mathbb{Z}) = 0$, or if $X = \mathbb{C}^n$. A smooth connected complex curve in a Stein surface is a leaf in a nonsingular holomorphic foliation.*

Proof. Choose a holomorphic trivialization $N = \mathrm{T}X|_V/\mathrm{T}V \xrightarrow{\cong} V \times \mathbb{C}$ of the normal bundle of V in X. The projection $h\colon N \to \mathbb{C}$ onto the second factor is a noncritical holomorphic function on N. We identify the zero section $\{h = 0\}$ of N with V. By Theorem 3.3.3 (the Docquier-Grauert theorem) there is an open neighborhood Ω of V in X and an injective holomorphic map $\phi\colon \Omega \to N$ which equals the identity on V. Then $f = h \circ \phi$ is a noncritical holomorphic function on Ω such that $\{f = 0\} = V$. Applying Theorem 8.12.9 (p. 389) with $q = 1$ and $X' = V$ we obtain a noncritical function $g \in \mathcal{O}(X)$ which vanishes on V; the foliation $\{g = c\}$ $(c \in C)$ clearly satisfies Corollary 8.15.2.

The second statement follows from $\mathrm{Pic}(V) = H^1(V;\mathcal{O}^*) \cong H^2(V;\mathbb{Z})$ (see Theorem 5.2.1 on p. 189); the latter group vanishes if V is an open Riemann surface. Since every divisor on $\mathbb{C}^n$ is a principal divisor, the normal bundle of any closed complex hypersurface $V \subset \mathbb{C}^n$ is trivial. $\square$

The h-principle for holomorphic submersions $X \to \mathbb{C}^q$ (Theorem 8.12.1 and Corollary 8.12.13, p. 392) implies the following corollary; the last statement follows from the Oka-Grauert principle in §7.2.

Corollary 8.15.3. *Let X be a Stein manifold. Every trivial complex subbundle $\Theta \subset \mathrm{T}^*X$ of rank $q < \dim X$ is homotopic to a subbundle generated by independent holomorphic differentials $\mathrm{d}f_1, \ldots, \mathrm{d}f_q$. If Θ is holomorphic then the homotopy can be chosen through holomorphic subbundles of T^*X.*

Corollary 8.15.3 admits the following dual formulation in terms of subbundles of the tangent bundle $\mathrm{T}X$. See also Theorem 8.15.7 below.

Corollary 8.15.4. *Let X be a Stein manifold of dimension n. Every complex subbundle $E \subset \mathrm{T}X$ of rank $k \geq 1$ with trivial quotient bundle $\mathrm{T}X/E$ is homotopic to an integrable holomorphic subbundle of the form $\ker \mathrm{d}f \subset \mathrm{T}X$, where $f\colon X \to \mathbb{C}^{n-k}$ is a holomorphic submersion.*

Proof. The complex subbundle $\Theta = E^\perp \subset \mathrm{T}^*X$ with fibers

$$\Theta_x = \left\{\lambda \in \mathrm{T}^*_x X\colon \lambda(v) = 0 \text{ for all } v \in E_x\right\}$$

(the complex conormal bundle of E) satisfies $\Theta \simeq (\mathrm{T}X/E)^*$, and hence is trivial. Corollary 8.15.3 gives a homotopy of complex subbundles $\Theta^t \subset \mathrm{T}^*X$ $(t \in [0,1])$ from $\Theta^0 = \Theta$ to a subbundle $\Theta^1 \subset \mathrm{T}^*X$ that is spanned by $n-k$ independent holomorphic differentials $\mathrm{d}f_1, \ldots, \mathrm{d}f_{n-k}$. The homotopy of subbundles $E^t = (\Theta^t)^\perp \subset \mathrm{T}X$ then satisfies Corollary 8.15.4. $\square$

We now mention a couple of results on the existence of submersion foliations that either contain a given submanifold as a union of leaves, or else are transverse to it. These generalize Corollaries 8.15.1 and 8.15.2 to foliations of lower dimension. The proof is similar but slightly more involved, depending on Theorem 8.12.9 (see [173, p. 181]).

Corollary 8.15.5. *Let X be an n-dimensional Stein manifold and $V \subset X$ a closed complex submanifold. If the tangent bundle $\mathrm{T}X$ admits a trivial complex subbundle N satisfying $\mathrm{T}X|_V = \mathrm{T}V \oplus N|_V$, then there is a holomorphic submersion $f\colon X \to \mathbb{C}^q$ $(q = n - \dim V)$ such that V is a union of connected components of the fiber $f^{-1}(0)$. If $\dim V \geq \left[\frac{n}{2}\right]$ then the above conclusion holds if V has a trivial normal bundle in X.*

Corollary 8.15.6. *Let X be a Stein manifold of dimension n, $\iota\colon V \hookrightarrow X$ a closed complex submanifold, and $f = (f_1, \ldots, f_q)\colon V \to \mathbb{C}^q$ a holomorphic submersion. If there is a q-coframe $\theta = (\theta_1, \ldots, \theta_q)$ on X satisfying $\iota^*\theta_j = \mathrm{d}f_j$ for $j = 1, \ldots, q$, then there exists a holomorphic submersion $F\colon X \to \mathbb{C}^q$ with $F|_V = f$. Such F always exists if $q \leq \left[\frac{n+1}{2}\right]$.*

All results mentioned so far give submersion foliations. Corollary 8.15.4 generalizes to subbundles $E \subset \mathrm{T}X$ with flat quotients $N = \mathrm{T}X/E$, furnishing foliations that are not necessarily given by submersions. Recall that a complex vector bundle $N \to X$ is flat if it admits a vector bundle atlas with locally constant transition functions. (See Example 1.4.8 on p. 13).

Theorem 8.15.7. [173, Theorem 7.1] *Let X be a Stein manifold. If E is a complex subbundle of the tangent bundle $\mathrm{T}X$ such that the quotient bundle $N = \mathrm{T}X/E$ is flat, then E is homotopic (through complex subbundles of $\mathrm{T}X$) to the tangent bundle of a nonsingular holomorphic foliation of X.*

Analogous results concerning smooth foliations on smooth open manifolds were proved by Gromov [234] and Phillips [395, 396, 397], and on closed manifolds by Thurston [490, 491]. See also [213, pp. 65–66].

Proof. We may assume that X is connected. Assume that $N = \mathrm{T}X/E$ has rank q. Let $\pi\colon N \to X$ denote the vector bundle projection. By flatness of N there is an open cover $\mathcal{U} = \{U_i\}_{i\in\mathbb{N}}$ of X and vector bundle charts $\phi_i\colon N|_{U_i} = \pi^{-1}(U_i) \to U_i \times \mathbb{C}^q$ with transition maps of the form

$$\phi_{ij}(x, z) = \phi_i \circ \phi_j^{-1}(x, z) = \big(x, h_{ij}(z)\big), \quad x \in U_i \cap U_j,\ z \in \mathbb{C}^q,$$

where $h_{ij} \in GL_q(\mathbb{C})$ is independent of the base point $x \in U_i \cap U_j$. The latter property insures that the product foliations of $N|_{U_i} \cong U_i \times \mathbb{C}^q$ over the sets $U_j \in \mathcal{U}$ amalgamate into a global holomorphic foliation $\mathcal{H}$ of N, with the zero section of N as a leaf. Precisely, if $U_i \cap U_j \neq \emptyset$ and $z \in \mathbb{C}^q$, then $\phi_i^{-1}(U_i \times \{h_{ij}(z)\})$ and $\phi_j^{-1}(U_j \times \{z\})$ belong to the same leaf of $\mathcal{H}$.

The tangent bundle of N decomposes as $\mathrm{T}N = H \oplus V$, where the *horizontal component* $H = \mathrm{T}\mathcal{H}$ is the tangent bundle of the foliation $\mathcal{H}$, and the *vertical component* V is the tangent bundle of the vertical foliation $N_x = \pi^{-1}(x)$ ($x \in X$). Denote by $\tau\colon \mathrm{T}N \to V$ the projection onto V with $\ker \tau = H$. Observe that $V = \pi^* N$ is the pull-back of the vector bundle $N \to X$ to the total space N by the projection map $\pi\colon N \to X$, and for every section $f\colon X \to N$ of π we have $f^*V = N$. If $f\colon X \to N$ is a holomorphic section that is transverse to the horizontal foliation $\mathcal{H}$, then the intersection of its graph $f(X) \subset N$ with the horizontal foliation $\mathcal{H}$ on N defines a holomorphic foliation $\mathcal{H}_f$ on X, of dimension $k = n - q$, whose tangent bundle $\mathrm{T}\mathcal{H}_f$ (a holomorphic subbundle of $\mathrm{T}X$) has fibers

$$\mathrm{T}_x\mathcal{H}_f = \big\{\xi \in \mathrm{T}_x X\colon \mathrm{d}f_x(\xi) \in H_{f(x)}\big\} = \big\{\xi \in \mathrm{T}_x X\colon\ \tau \circ \mathrm{d}f_x(\xi) = 0\big\}.$$

Transversality of f to the foliation $\mathcal{H}$ means that the vector bundle map

$$f' = f^* \circ \tau \circ \mathrm{d}f\colon \mathrm{T}X \to f^*V = N \tag{8.30}$$

is surjective, and hence it induces an isomorphism $\mathrm{T}X/\mathrm{T}\mathcal{H}_f \xrightarrow{\cong} N$. In particular, N is the normal bundle of any foliation $\mathcal{H}_f$ obtained in this way.

To prove Theorem 8.15.7 we construct a holomorphic section $f\colon X \to N$ transverse to $\mathcal{H}$ and a continuous complex vector bundle injection $\iota\colon N \to \mathrm{T}X$ such that the subbundle $\mathrm{T}\mathcal{H}_f \subset \mathrm{T}X$ is homotopic to E, and such that $f' \circ \iota\colon N \to N$ is a complex vector bundle automorphism homotopic to the identity through complex vector bundle automorphisms of N.

On every sufficiently small open set $U \subset X$ we have $N|_U \cong U \times \mathbb{C}^q$, and the restriction of $\mathcal{H}$ to N_U has leaves $U \times \{z\}$ ($z \in \mathbb{C}^q$). A section of N over such U is of the form $f(x) = (x, \widetilde{f}(x))$ where $\widetilde{f}\colon U \to \mathbb{C}^q$; such f is transverse to $\mathcal{H}$ if and only if $\widetilde{f}$ is a submersion to the fiber $\mathbb{C}^q$. This reduces every local problem in the construction of a transverse section to the corresponding problem for submersions.

From this point on the construction follows the proof of Theorem 8.12.9. Choose a strongly plurisubharmonic Morse exhaustion function $\rho\colon X \to \mathbb{R}$ and a holomorphic vector bundle embedding $\tau\colon N \hookrightarrow \mathrm{T}X$ such that $\mathrm{T}X = E \oplus \iota(N)$. Suppose f is a transverse holomorphic section, defined on a sublevel set $\{\rho \le c\}$, such that $\ker(\tau \circ \mathrm{d}f)$ is complementary to $\iota(N)$ and $f' \circ \iota$ is homotopic to the identity over the domain of f. (Here f' is defined by (8.30).)

We now inductively enlarge the domain of f as in the proof of Theorem 8.12.9. Whenever we change f, the injection ι is changed accordingly (by a homotopy of vector bundle injections $N \to \mathrm{T}X$) so that the composition $f' \circ \iota$ remains homotopic to the identity on N. The noncritical case is essentially the same as before. In the critical case we extend f across a totally real handle by using Gromov's h-principle for differential relations that are ample in the coordinate directions (see Theorem 1.10.5 on p. 42). The necessary topological condition for the existence of such extension is precisely that $f' \circ \iota\colon N \to N$

is a complex vector bundle automorphism homotopic to the identity. The remaining steps are just as before. See [173, pp. 183–184] for the details. □

Problem 8.15.8. Assume that $\mathcal{F}$ is a nonsingular holomorphic foliation in a neighborhood of a compact $\mathcal{O}(X)$-convex set K in a Stein manifold X.

1. Is it possible to approximate $\mathcal{F}$ on K by global holomorphic foliations $\mathcal{F}'$ of X (possibly with singularities on $X \backslash K$)?
2. Assume that the tangent bundle $\mathrm{T}\mathcal{F}$ of the foliation $\mathcal{F}$ extends to a complex subbundle E of $\mathrm{T}X$. Can we deform E to an integrable holomorphic subbundle E' of $\mathrm{T}X$ (the tangent bundle of a nonsingular foliation $\mathcal{F}'$) by a deformation that is holomorphic and close to the identity near K?
3. Does the above hold on $X = \mathbb{C}^n$? Can every holomorphic foliation on a convex (or Runge) open set $\Omega \subset \mathbb{C}^n$ be approximated by a holomorphic or an algebraic foliation of $\mathbb{C}^n$?
4. Let M be a compact, totally real, polynomially submanifold of $\mathbb{C}^n$ with trivial complex normal. Does there exist an algebraic foliation $\mathcal{F}$ of $\mathbb{P}^n$ that is nonsingular near M and is transverse to M?
5. Let $\mathcal{F}$ be a nonsingular holomorphic foliation of a Stein manifold X. Is every leaf $F \in \mathcal{F}$ a Stein manifold?
6. Let $\mathcal{F}$ be as above, and assume that K is a compact set contained in a leaf F. Consider its holomorphically convex hull (or its envelope of holomorphy) in F, and in X. Are the two always the same?

There are two possible notions of closeness of a pair of holomorphic foliations $\mathcal{F}$, $\mathcal{F}'$ on a compact set $K \subset X$: (i) their tangent bundles $\mathrm{T}\mathcal{F}, \mathrm{T}\mathcal{F}'$ are close (as subbundles of $\mathrm{T}X|_K$), or (ii) $\mathcal{F}'$ is conjugate to $\mathcal{F}$ in a neighborhood of K by a biholomorphism close to the identity. The latter stronger notion of closeness implies that the two foliations have the same leaf space in a neighborhood of K. Results of this type are furnished by Theorem 8.12.9 and Theorem 8.14.1.

Problem 4 is related to the following question attributed to Bogomolov.

Problem 8.15.9. Given a complex manifold M of dimension m, do there exist a totally real embedding $M \hookrightarrow \mathbb{P}^n$ for some $n \geq 2m$ and an algebraic foliation on $\mathbb{P}^n$ with leaves of dimension $n - m$ that is transverse to M and determines the given complex structure on M? □

9

Topological Methods in Stein Geometry

We begin by exploring implications of Gromov's h-principle for totally real immersions and embeddings (§9.1). A large part of the chapter is devoted to the study of closed real surfaces S embedded or immersed in a complex surface X. A generic embedding has isolated complex points; the algebraic number $I(S)$ of such points, or the numbers $I_\pm(S)$ if S is oriented, are isotopy invariants called *Lai indexes* (see §9.2 – §9.4). Every surface is isotopic to one with minimal number of complex points (see §9.5). The *Seiberg-Witten theory* shows that every closed oriented real surface S, smoothly embedded in a compact Kähler surface X with $b^+(X) > 1$ and such that S is not a homologically trivial sphere, satisfies the *adjunction inequality* $I_\pm(S) \leq 0$; similar arguments yield the *generalized Thom conjecture* (see §9.7 – §9.8). On the other hand, the inequalities $I_\pm(S) \leq 0$ imply that S is isotopic to a surface with a basis of open tubular Stein neighborhoods in X, and this gives interesting results on the existence of Stein domains of a given diffeomorphism type in compact Kähler surfaces (a form of the Oka principle).

Another interesting direction in Stein geometry comes from the Eliashberg-Gompf construction of integrable Stein structures on almost complex manifolds with a suitable handlebody decomposition (see §9.10 – §9.12). This is a highly nontrivial converse to the fact that an n-dimensional Stein manifold has the homotopy type of a CW complex of dimension at most n (see §3.11). On 4-manifolds, these Stein structures are typically *exotic* due to framing obstructions coming from symplectic geometry and the adjunction inequality in Stein surfaces. One of the main result proved in this chapter is the following version of Oka principle (see §9.10):

The Soft Oka principle: Any continuous map $X \to Y$ from a Stein manifold (X, J) to a complex manifold Y is homotopic to a holomorphic map in some Stein structure $\widetilde{J}$ on X that is homotopic to J. If $\dim_{\mathbb{R}} X = 4$ then this holds after a change of the smooth structure on X.

There is a number of interesting open problems in this interface between Stein geometry, symplectic geometry and 4-dimensional topology.

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9.1 The H-Principle for Totally Real Immersions

Let M be a smooth m-dimensional manifold and (X, J) a complex manifold. Recall that a $\mathcal{C}^1$ immersion $f\colon M \looparrowright X$ is *totally real* if

$$\mathrm{d}f_p(\mathrm{T}_pM) \cap J(\mathrm{d}f_p(\mathrm{T}_pM)) = \{0\}, \quad p \in M.$$

Assume that $E \to M$ is a real vector bundle and $F \to X$ is a complex vector bundle. A vector bundle map $\Phi\colon E \to F$ (over a base map $f\colon M \to X$) is *totally real* if for every $p \in M$ the fiber map $\Phi_p\colon E_p \to F_{f(p)}$ is injective and the image $\Phi_p(E_p) \subset F_{f(p)}$ is a totally real subspace of the complex vector space $F_{f(p)}$. Equivalently, the complexified map $\Phi^{\mathbb{C}}\colon E \otimes \mathbb{C} \to F$ is a complex vector bundle injection. We denote by $\mathrm{Hom}_{\mathrm{tr}}(E, F)$ the set of all totally real vector bundle maps $E \to F$. Thus a $\mathcal{C}^1$ map $f\colon M \to X$ is a totally real immersion if and only if its tangent map $\mathrm{T}f\colon \mathrm{T}M \to \mathrm{T}X$ is a totally real vector bundle map of the real tangent bundle $\mathrm{T}M$ to the complex tangent bundle $\mathrm{T}X \cong \mathrm{T}^{(1,0)}X$.

The following h-principle for totally real immersions is due to Gromov [235, p. 332], [236, p. 192].

Theorem 9.1.1. *Let M be a smooth real manifold and X a complex manifold.*

(a) *For every $\Phi \in \mathrm{Hom}_{\mathrm{tr}}(\mathrm{T}M, \mathrm{T}X)$ there exists a totally real immersion $f\colon M \looparrowright X$ whose tangent map $\mathrm{T}f$ is homotopic to Φ in $\mathrm{Hom}_{\mathrm{tr}}(\mathrm{T}M, \mathrm{T}X)$.*

(b) *If $f_0, f_1\colon M \looparrowright X$ are totally real immersions whose tangent maps are homotopic in $\mathrm{Hom}_{\mathrm{tr}}(\mathrm{T}M, \mathrm{T}X)$, then f_0 and f_1 are regularly homotopic through totally real immersions.*

Proof. Consider the open subset of the jet manifold $\mathrm{J}^1(M, X)$ (see §1.10),

$$\Omega = \big\{(p, x, \lambda)\colon p \in M,\ x \in X,\ \lambda \in \mathrm{Hom}_{\mathrm{tr}}(\mathrm{T}_pM, \mathrm{T}_xX)\big\}. \tag{9.1}$$

Note that a $\mathcal{C}^1$ map $f\colon M \to X$ is a totally real immersion if and only if its 1-jet $\mathrm{j}^1 f$ is a section of Ω. Choosing a real basis of T_pM and a complex basis of T_xX, we can identify $\mathrm{Hom}_{\mathrm{tr}}(\mathrm{T}_pM, \mathrm{T}_xX)$ with the Stiefel manifold $V_{m,n}$ of complex m-frames in $\mathbb{C}^n$, i.e., with the set of complex $m \times n$ matrices of rank m (see Example 1.2.7 on p. 6). Theorem 9.1.1 follows immediately from Theorem 1.10.5 on p. 42 (the h-principle for differential relations that are ample in the coordinate directions) in view of the following lemma.

Lemma 9.1.2. *The differential relation Ω (9.1) controlling total reality is ample in the coordinate directions (Def. 1.10.4 on p. 42).*

Proof. In any pair of local smooth coordinates $u = (u_1, \ldots, u_m)$ on M and local holomorphic coordinates $z = (z_1, \ldots, z_n)$ on X the 1-jet of a map $f\colon M \to X$ with $f(u) = z$ is represented by $z^{(1)} = (u, z; v_1, \ldots, v_m) \in \mathrm{J}^1(M, X)$, where

$v_k = \frac{\partial f}{\partial u_k}(u) \in \mathbb{C}^n$ for $k = 1, \ldots, m$. Clearly $z^{(1)}$ belongs to Ω if and only if the vectors $v_1, \ldots, v_m \in \mathbb{C}^n$ are $\mathbb{C}$-linearly independent. To verify ampleness, consider a restricted 1-jet $z_k^{(1)} = (u, z; v_1, \ldots, \hat{v}_k, \ldots, v_m)$. The subset

$$\big\{v \in \mathbb{C}^n \colon (u, z; v_1, \ldots, v_{k-1}, v, v_{k+1}, \ldots, v_m) \in \Omega\big\}$$

is either empty or else the complement of a complex subspace of $\mathbb{C}^n$. In the latter case the convex hull of each of its connected components equals $\mathbb{C}^n$. $\square$

Theorem 9.1.1 can be coupled with the h-principle for ordinary immersions. Denote by $\mathrm{Hom}_{\mathrm{inj}}(E, F)$ the set of all fiberwise injective vector bundle maps $E \to F$. For maps of smooth manifolds $M \to X$ we consider the differential relation

$$\Omega_0 = \big\{(p, x, \lambda) \colon p \in M,\ x \in X,\ \lambda \in \mathrm{Hom}_{\mathrm{inj}}(\mathrm{T}_p M, \mathrm{T}_x X)\big\} \subset \mathrm{J}^1(M, X).$$

Clearly a $\mathcal{C}^1$ map $f \colon M \to X$ is an immersion if and only if $\mathrm{j}^1 f \colon M \to \mathrm{J}^1(M, X)$ is a section of Ω_0. If $\dim_{\mathbb{R}} M = m$ and $\dim_{\mathbb{R}} X = N$ then the fiber of the projection $\Omega_0 \to M \times X$ is the Stiefel manifold $V_{m,N}^{\mathbb{R}}$ of real m-frames in $\mathbb{R}^N$ (Example 1.2.7). It is easily seen that Ω_0 is ample in the coordinate directions if and only if $m < N$. Indeed, the set of vectors $v \in \mathbb{R}^N$ that complete a restricted 1-jet $z_k^{(1)}$ to a 1-jet $z^{(1)} \in \Omega_0$ is either empty or the complement of a real subspace of codimension $N - m - 1$ in $\mathbb{R}^N$. Theorem 1.10.5 applied to Ω_0 yields the Hirsch-Smale h-principle for immersions when $\dim M < \dim X$ [455, 264]. If $\dim M = \dim X = n$ and M is open and connected, we triangulate M and retract it by a diffeotopy into a neighborhood of its $(n-1)$-skeleton, so the h-principle for immersions $M \looparrowright X$ still holds.

Corollary 9.1.3. *Let M be a smooth manifold and X a complex manifold.*

(a) *An immersion $f \colon M \looparrowright X$ is regularly homotopic to a totally real immersion if and only if its tangent map $\mathrm{T}f \colon \mathrm{T}M \to \mathrm{T}X$ is homotopic in $\mathrm{Hom}_{\mathrm{inj}}(\mathrm{T}M, \mathrm{T}X)$ to a totally real vector bundle map $\mathrm{T}M \hookrightarrow \mathrm{T}X$.*

(b) *A regulary homotopy of immersions $f_t \colon M \looparrowright X$ ($t \in I = [0,1]$) such that f_0 and f_1 are totally real can be deformed (rel. $bI = \{0,1\}$) to a regular homotopy of totally real immersions if and only if the path $\mathrm{T}f_t \in \mathrm{Hom}_{\mathrm{inj}}(\mathrm{T}M, \mathrm{T}X)$ can be deformed (rel. bI) to a path in $\mathrm{Hom}_{\mathrm{tr}}(\mathrm{T}M, \mathrm{T}X)$.*

Proof. The condition in (a) is clearly necessary. Conversely, assume that $\tau_t \in \mathrm{Hom}_{\mathrm{inj}}(\mathrm{T}M, \mathrm{T}X)$ ($t \in [0,1]$) is a homotopy with $\tau_0 = \mathrm{T}f$ and $\tau_1 \in \mathrm{Hom}_{\mathrm{tr}}(\mathrm{T}M, \mathrm{T}X)$. By Theorem 9.1.1 (i) there is a totally real immersion $f_1 \colon M \looparrowright X$ such that $\mathrm{T}f_1$ is homotopic in $\mathrm{Hom}_{\mathrm{tr}}(\mathrm{T}M, \mathrm{T}X)$ to τ_1. Coupling this homotopy from $\mathrm{T}f_1$ to τ_1 with the homotopy $\{\tau_t\}_{t \in [0,1]}$ gives a path in $\mathrm{Hom}_{\mathrm{inj}}(\mathrm{T}M, \mathrm{T}X)$ connecting $\mathrm{T}f_0$ to $\mathrm{T}f_1$. By the h-principle for the usual immersions $M \looparrowright X$ we infer that f_0 and f_1 are regularly homotopic through immersions. A similar argument proves (b). $\square$

A totally real vector bundle map $\Phi\colon \mathrm{T}M \to \mathrm{T}\mathbb{C}^n$ over $f\colon M \to \mathbb{C}^n$ induces a complex vector bundle injection $\Phi^{\mathbb{C}}\colon \mathbb{C}\mathrm{T}M \hookrightarrow f^*\mathrm{T}X \cong M \times \mathbb{C}^n$. If $\dim_{\mathbb{R}} M = n$ then this map is a complex vector bundle isomorphism $\mathbb{C}\mathrm{T}M \xrightarrow{\cong} M \times \mathbb{C}^n$. Hence for $X = \mathbb{C}^n$ we have the following statement.

Proposition 9.1.4. *Let M be a smooth real manifold.*

(a) *Regular homotopy classes of totally real immersions $M \looparrowright \mathbb{C}^n$ are in one-to-one correspondence with the homotopy classes of injective complex vector bundle maps $\mathbb{C}\mathrm{T}M \hookrightarrow M \times \mathbb{C}^n$.*

(b) *In particular, M admits a totally real immersion into $\mathbb{C}^n$ with $n = \dim_{\mathbb{R}} M$ if and only if the complexified tangent bundle $\mathbb{C}\mathrm{T}M$ is trivial. In this case the regular homotopy classes of totally real immersions $M \looparrowright X$ are in bijective correspondence with the homotopy classes of maps $M \to GL_n(\mathbb{C})$ (that is, with the elements of the complex K-group $K^1(M)$).*

(c) *A compact smooth orientable manifold M^n which admits a totally real embedding in $\mathbb{C}^n$ has Euler number $\chi(M) = 0$. In particular, if $n = 2$ then M is the torus.*

Proof. Part (a) follows from Corollary 9.1.3, and (b) is a special case of (a). The last statement in (b) follows by observing that any two complex vector bundle trivializations $\mathbb{C}\mathrm{T}M \xrightarrow{\cong} M \times \mathbb{C}^n$ differ by a map $M \to GL_n(\mathbb{C})$.

Part (c), due to Wells [507], is seen as follows. For any totally real immersion $f\colon M^n \looparrowright X^n$ the complex structure operator J on X induces an isomorphism between the tangent bundle $\mathrm{T}M$ and the normal bundle N_f of the immersion, so we have $\mathbb{C}\mathrm{T}M \cong \mathrm{T}M \oplus \mathrm{T}M \cong \mathrm{T}M \oplus \mathrm{N}_f \cong f^*\mathrm{T}X$. Hence the Euler number $\chi(M)$ equals the normal Euler number $\chi_n(f) = \chi(\mathrm{N}_f)$ of the immersion. If f is an embedding then the normal Euler number is the self-intersection number of $f(M)$ in X. When $X = \mathbb{C}^n$, this number is zero since we can remove the image $f(M)$ off itself; hence $\chi(M) = 0$. $\square$

Totally real immersions are closely related to *Lagrangian immersions*. Let ω be a *symplectic form* on a smooth manifold X^{2n}, that is, a closed 2-form with $\omega^n \neq 0$. (A Kähler form (p. 28) on a complex manifold is also a symplectic form.) An immersion $f\colon M \looparrowright X$ of a real manifold M into X is *ω-isotropic* if $f^*\omega = 0$; this requires $\dim_{\mathbb{R}} M \le n$. If $\dim M = n$ then an isotropic immersion $M \looparrowright X$ is called an *ω-Lagrangian immersion*. The standard symplectic (Kähler) form on $\mathbb{R}^{2n} = \mathbb{C}^n$ is

$$\omega = \sum_{j=1}^n \mathrm{d}x_j \wedge \mathrm{d}y_j = \mathrm{d}\alpha, \quad \alpha = \sum_{j=1}^n x_j\,\mathrm{d}y_j.$$

An immersion $M^n \looparrowright \mathbb{C}^n$ is Lagrangian if $f^*\omega = 0$. This implies $\mathrm{d}f^*\alpha = f^*\mathrm{d}\alpha = f^*\omega = 0$ and hence $f^*\alpha$ is a closed 1-form on M. The immersion f is *exact Lagrangian* if $f^*\alpha$ is an *exact* 1-form, $f^*\alpha = \mathrm{d}\beta$ for some $\beta\colon M \to \mathbb{R}$. We have the following result (see Lees [328] and Gromov [236, pp. 60–61]).

Theorem 9.1.5. *Let M be a smooth compact manifold of real dimension n and let (X, ω) be a Kähler manifold of complex dimension n.*

(a) *A totally real immersion $f\colon M \looparrowright \mathbb{C}^n$ with $[f^*\omega] = 0 \in H^2(M;\mathbb{R})$ is regularly homotopic through totally real immersions to a Lagrangian immersion.*

(b) *If M is stably parallelizable (i.e., $\mathrm{T}M \oplus \mathbb{R} \cong M \times \mathbb{R}^{n+1}$ is a trivial bundle) then M admits an exact $\mathcal{C}^\infty$ Lagrangian immersion in $\mathbb{C}^n$.*

(c) *The n-sphere S^n admits an exact smooth Lagrangian immersion $S^n \looparrowright \mathbb{C}^n$ for each $n \in \mathbb{N}$.*

Explicit examples of immersed Lagrangian spheres $S^n \looparrowright \mathbb{C}^n$ in part (c) were given by Weinstein [505, p. 26].

By a more refined version of Gromov's h-principle one also obtains the following h-principle for totally real embeddings. (By the transversality theorem and the fact that total reality is an open condition, this is a consequence of Theorem 9.1.1 except when $\dim_{\mathbb{R}} M = \dim_{\mathbb{C}} X$.) For real surfaces in complex surfaces see also Corollary 9.6.1 below.

Theorem 9.1.6. [236, p. 192] *Let $f_0\colon M \hookrightarrow X$ be a smooth embedding into a complex manifold X. If the tangent map $\mathrm{T}f_0\colon \mathrm{T}M \to \mathrm{T}X$ is homotopic in the space $\mathrm{Hom}_{\mathrm{inj}}(\mathrm{T}M, \mathrm{T}X)$ to a totally real map $\tau_1 \in \mathrm{Hom}_{\mathrm{tr}}(\mathrm{T}M, \mathrm{T}X)$, then f_0 is isotopic to a totally real embedding $f_1\colon M \hookrightarrow X$.*

For results on classification of totally real immersions and embeddings we refer to [236] and the papers [27, 162, 478, 454, 106]. The last mentioned paper contains a more complete set of references. Here we mention without proofs some of the main results on totally real immersions and embeddings of compact n-manifolds in $\mathbb{C}^n$.

Let us begin by considering the spheres S^n. By part (c) in Theorem 9.1.5, S^n admits a totally real immersion in $\mathbb{C}^n$ for each $n \in \mathbb{N}$. The question arises for which values of n does there exist a totally real embedding $S^n \hookrightarrow \mathbb{C}^n$. According to Kervaire [296] every embedding of the n-sphere in $\mathbb{C}^n$ has trivial normal bundle. If the embedding is totally real then the normal bundle is isomorphic to the tangent bundle $\mathrm{T}S^n$ and hence the latter must be trivial. This holds precisely when $n \in \{1, 3, 7\}$. (See e.g. [53] for these facts.) For $n = 1$ the standard inclusion $S^1 \hookrightarrow \mathbb{C}$ is totally real. Explicit totally real embeddings $S^3 \hookrightarrow \mathbb{C}^3$ were found by Ahern and Rudin [11]. The 7-sphere does not admit a totally real embedding into $\mathbb{C}^7$ [236, p. 193], [478]. Summarizing we have the following result.

Theorem 9.1.7. *The n-sphere S^n admits a totally real embedding in $\mathbb{C}^n$ if and only if $n = 1$ or $n = 3$.*

In dimension three we have the following more general result.

Theorem 9.1.8. [162, Theorem 1.4] *Every immersion of a compact smooth orientable 3-manifold M into $\mathbb{C}^3$ is regularly homotopic to a totally real immersion. In particular, such M admits a totally real embedding in $\mathbb{C}^3$.*

Corollary 9.1.9. *For every smooth compact orientable 3-manifold M there exists a domain of holomorphy in $\mathbb{C}^3$ which is diffeomorphic to $M \times \mathbb{R}^3$.*

Stout and Zame proved the following more precise result concerning S^7.

Theorem 9.1.10. [478] *The 7-dimensional Stein manifold*

$$\Sigma^7 = \left\{ (z_1, \ldots, z_8) \in \mathbb{C}^8 \colon \sum_{j=1}^{8} z_j^2 = 1 \right\}$$

(a complexification of the real 7-sphere S^7) is real analytically equivalent to a domain in $\mathbb{C}^7$, but is not holomorphically equivalent to a domain in $\mathbb{C}^7$. In particular, S^7 does not admit a totally real embedding in $\mathbb{C}^7$.

We shall see below (Example 9.8.3 and Corollary 9.12.3) that there also exist Stein surfaces which are homeomorphic to a domain in $\mathbb{C}^2$, but not diffeomorphic to a domain in $\mathbb{C}^2$. In particular, there exists an exotic Stein structure on the topological manifold $S^2 \times \mathbb{R}^2$ with this property.

Further results concerning totally real embeddings $S^k \hookrightarrow \mathbb{C}^n$ for $k < n$ were obtained by X. Gong [218]. We also mention the following example.

Theorem 9.1.11. (Fiedler [138]) *The 2-torus $T = S^1 \times S^1$ admits a pair of totally real embeddings in $\mathbb{C}^2$ that are isotopic, but are not isotopic within the class of totally real tori.*

The h-principle also applies to generic immersions $M \looparrowright \mathbb{C}^n$ of smooth manifolds of dimension $\dim_{\mathbb{R}} M > n$ to $\mathbb{C}^n$ (see [275]).

9.2 Real Surfaces in Complex Surfaces

In this section S denotes a real two dimensional surface of class $\mathcal{C}^2$ embedded in a complex surface (X, J). Given a point $p \in S$, the maximal complex tangent space $\Lambda_p = \mathrm{T}_p S \cap J(\mathrm{T}_p S)$ is a complex subspace of the tangent space $\mathrm{T}_p X$ that is contained in $\mathrm{T}_p S$. Hence we have precisely two possibilities:

- $\Lambda_p = \{0\}$, in which case p is a totally real point of S, or
- $\Lambda_p = \mathrm{T}_p S$ is a complex line in $\mathrm{T}_p X$. Such p is a *complex point* of S.

Locally near any point $p \in S$ the surface S can be represented in suitably chosen local holomorphic coordinates (z, w) as the graph $\{w = f(z)\} \subset \mathbb{C}^2$ of a $\mathcal{C}^2$ function over a domain in $\mathbb{C}$. It is easily seen that $p = (z, f(z))$ is a complex point of S if and only if the Cauchy-Riemann equation holds at z:

$$2\frac{\partial f}{\partial \bar{z}}(z) = \frac{\partial f}{\partial x}(z) + \mathrm{i}\frac{\partial f}{\partial y}(z) = 0.$$

We may further assume that p corresponds to the origin $(0,0) \in \mathbb{C}^2$ and that $\mathrm{T}_0 S = \{w = 0\}$ (equivalently, $\mathrm{d}f_0 = 0$). The complex point is said to be *nondegenerate* if the second order Taylor polynomial of f at 0 does not vanish identically; assume that this holds. If $\frac{\partial^2 f}{\partial z \partial \bar{z}}(0) \neq 0$ then by Bishop [49] there exist local holomorphic coordinates at $(0,0)$ in which S is given by

$$w = |z|^2 + \lambda(z^2 + \bar{z}^2) + o(|z|^2) = (1 + 2\lambda)x^2 + (1 - 2\lambda)y^2 + o(|z|^2) \tag{9.2}$$

for some real number $\lambda \geq 0$ that is a biholomorphic invariant of the complex point. Note that $1 \pm 2\lambda$ are the eigenvalues of the real Hessian of f at 0. In particular, up to an error term $o(|z|^2)$, S is locally contained in the real hyperplane $\{\Im w = 0\} \cong \mathbb{C} \times \mathbb{R}$. If $\frac{\partial^2 f}{\partial z \partial \bar{z}}(0) = 0$ but the second order Taylor polynomial of f at 0 is nonzero then we can simplify S to the form

$$w = z^2 + \bar{z}^2 + o(|z|^2). \tag{9.3}$$

This case corresponds to $\lambda = +\infty$ in (9.2).

A complex point $p \in S$ is said to be *quadratic* if, in some local holomorphic coordinates near p, S is of the form (9.2) or (9.3) without the remainder term. The point p is *flat* if S is locally near p of the form (9.2) or (9.3) with $\Im o(|z|^2) = 0$; that is, if S is locally contained in $\mathbb{C} \times \mathbb{R}$.

By using the invariant $\lambda = \lambda(p) \geq 0$ we classify nondegenerate complex points as follows:

- *elliptic* if $0 \leq \lambda < \frac{1}{2}$ (the real Hessian of f at 0 is positive definite),
- *parabolic* if $\lambda = \frac{1}{2}$ (the Hessian of f is degenerate), or
- *hyperbolic* if $\lambda > \frac{1}{2}$ (the eigenvalues of the Hessian have opposite signs); this includes the case $\lambda = \infty$ when S is given by (9.3).

Elliptic and hyperbolic complex points are isolated and stable under small $\mathcal{C}^2$ deformations of S in X. Parabolic and degenerate complex points need not be isolated. For example, taking $\lambda = \frac{1}{2}$ and $o(|z|^2) = 0$ in (9.2) gives the surface $w = 2x^2$ with the line of parabolic complex points $\{x = 0,\ w = 0\}$. A generic compact real surface in a complex surface has at most finitely many complex points, each of them either elliptic or hyperbolic. Parabolic complex points may occur in a generic homotopy of real surfaces in a complex surface.

A more precise normalization near an elliptic complex point was obtained by Moser and Webster [363] when S is real analytic: There exist local holomorphic coordinates $z = x + \mathrm{i}y$, $w = u + \mathrm{i}v$ on X at p in which S equals

$$u = |z|^2 + \Gamma(u)(z^2 + \bar{z}^2), \quad v = 0,$$

where $\Gamma(u) = \lambda + \delta u^s$, $\lambda \in (0, \frac{1}{2})$ is as above, $\delta = \pm 1$, and $s \in \mathbb{Z}_+$, or $\Gamma = \lambda$ ($s = \infty$). In particular, an elliptic point in a real analytic surface is flat. The triple (λ, δ, s) completely determines the germ of S at p up to a local biholomorphic equivalence.

Complex analytic properties of elliptic points are very different from those of hyperbolic points. At an elliptic point $p \in S$ there exists a one parameter family of closed, embedded, pairwise disjoint holomorphic discs $\{\Delta_t\}_{t\in(0,c]}$ in X, with boundaries $b\Delta_t \subset S$ covering a deleted neighborhood of p in S, such that Δ_t shrinks to the point p as $t \to 0$. The union $\Sigma = \cup_{t\in(0,c]}\Delta_t$ of these discs is a Levi flat hypersurface with boundary $b\Sigma \subset S$ that is contained in the local rational hull of S. (This is so since every circle $b\Delta_t$ also bounds a disc $D_t \subset S$ around p, and hence any complex curve intersecting Δ_t must also intersect D_t since the total intersection number with $\Delta_t \cup D_t$ equals zero.) Furthermore, $\Sigma \cup \{p\}$ is the local envelope of holomorphy of S at p. In particular, S does not admit a basis of Stein neighborhoods at an elliptic complex point. This is a special case of the results obtained by Bishop in his seminal paper [49]; for related works see König and Webster [309] and Hill and Taiani [263]. Bishop's work had a profound impact on the study of envelopes of holomorphy of real surfaces in complex surfaces and on related questions concerning extension and approximation of holomorphic functions near real surfaces. Bishop's theorem was extended to real surfaces in almost complex surfaces by Sukhov and Tumanov [479].

In contrast to the above, a real surface is locally holomorphically convex at a hyperbolic point (Forstnerič and Stout [196]). This is seen by considering the preimage of the surface S (9.2) by the proper two-to-one quadratic map $\Phi\colon \mathbb{C}^2 \to \mathbb{C}^2$ given by

$$\Phi(z, w) = \left(z, zw + \lambda(z^2 + w^2)\right).$$

A calculation shows that $\Phi^{-1}(S) = S_1 \cup S_2$, where S_1 and S_2 are small discs in totally real surfaces that are tangential at the origin to the totally real 2-planes

$$\Sigma_1 = \{(\zeta, \bar{\zeta})\colon \zeta \in \mathbb{C}\}, \quad \Sigma_2 = \left\{\left(\zeta, -\frac{1}{\lambda}\zeta - \bar{\zeta}\right)\colon \zeta \in \mathbb{C}\right\}.$$

Assuming that $\lambda > \frac{1}{2}$, one can see that for sufficiently small $\epsilon > 0$ (and taking S small around the origin) the polynomial $\psi(z, w) = \frac{1}{4}(z^2 - w^2) + \epsilon zw$ maps S_1 and S_2 to cones in $\mathbb{C}$ whose only common point is their vertex at the origin. By Kallin's lemma ([294], [474, p. 62]) it follows that $S_1 \cup S_2$ is polynomially convex. Hence $S = \Phi(S_1 \cup S_2)$ is also polynomially convex.

If S is smooth and a hyperbolic complex point $p \in S$ is flat, then locally near p the surface S admits a basis of tubular strongly pseudoconvex Stein neighborhoods in X [453]. This is easily seen in the following special case.

Example 9.2.1. A complex point $p \in S$ is *special hyperbolic* if there exist local holomorphic coordinates (z, w) on X near p in which S is given by $w = z^2 + \bar{z}^2$. The nonnegative function

$$\rho(z, w) = \left|w - z^2 - \bar{z}^2\right|^2 = \left|w - z^2\right|^2 + |z|^4 - 2\Re\left((w - z^2)z^2\right)$$

is clearly plurisubharmonic and it vanishes quadratically on S. The sublevel sets $\{\rho < t\}$ form a basis of tubular Stein neighborhoods of S. $\square$

This simple idea does not work directly with any other type of hyperbolic points, not even for the quadric $w = |z|^2 + \lambda(z^2 + \bar{z}^2)$ with $\frac{1}{2} < \lambda < \infty$, since the corresponding function ρ fails to be plurisubharmonic in a full neighborhood of the origin. The construction of Stein neighborhoods in [453] uses certain nonhomogeneous squared distance functions and is rather delicate. From the existence of local Stein neighborhoods at hyperbolic points one easily obtains global Stein neighborhoods of S as in the following result.

Theorem 9.2.2. (Slapar [453, Theorem 2]) *Let $S \hookrightarrow X$ be a compact real surface smoothly embedded in a complex surface X and having only flat hyperbolic points $\{p_1, \ldots, p_k\}$. Then there exists a $\mathcal{C}^\infty$ function ρ in a neighborhood $U \subset X$ of S such that $S = \{\rho = 0\} = \{\mathrm{d}\rho = 0\}$ and ρ is strictly plurisubharmonic on $U \backslash \{p_1, \ldots, p_k\}$. The sublevel sets $\{\rho < t\}$ for small $t > 0$ define a basis of tubular strongly pseudoconvex Stein neighborhoods of S in X.*

It is remains to consider degenerate complex point. Before proceeding, we define the *index* of an isolated complex point $p \in S$. Assume as before that in some local holomorphic coordinates the point p corresponds to the origin in $\mathbb{C}^2$ and $S = \{w = f(z)\}$. Then $z = 0$ is an isolated zero of the function $\frac{\partial f}{\partial \bar{z}}$.

Definition 9.2.3. (Notation as above.) *Let S be an embedded real surface of class $\mathcal{C}^1$ in a complex surface X, and let p be an isolated complex point of S. The index $I(p) = I(p; S)$ is defined as the winding number of the function $\frac{\partial f}{\partial \bar{z}}$ along any small enough positively oriented circle $|z| = \epsilon$, $\epsilon > 0$. The index of a totally real point is defined to be* 0.

It is easily seen that the index is a local holomorphic invariant of S at p (see e.g. [164]). If S is given by (9.2) (p. 407) then

$$\frac{\partial f}{\partial \bar{z}} = z + 2\lambda\bar{z} + o(|z|).$$

By looking at the winding number along small circles $|z| = \epsilon$ we conclude that

$$I(p) = +1 \text{ if } p \text{ is an elliptic point}, \quad I(p) = -1 \text{ if } p \text{ is a hyperbolic point.}$$

Another characterization of the index was given by Webster [504]. Let $\tau\colon \mathrm{T}X|_S \to \mathrm{N}_{S/X}$ be the projection onto the normal bundle of S in X with

$\ker \tau = \mathrm{T}S$. Choose a local orientation on $\mathrm{T}S$ and coorient $\mathrm{N}_{S/X}$ so that the two orientations add up to the standard orientation of the complex bundle $\mathrm{T}X|_S = \mathrm{T}S \oplus \mathrm{N}_{S/X}$. Let J denote the complex structure operator on X. Assume that p is an isolated complex point of S. Choose a tangent vector field V on S such that $V_p \neq 0$. Then $\widetilde{V} = \tau(JV)$ is a normal vector field (a section of $\mathrm{N}_{S/X}$) with an isolated zero at p.

Lemma 9.2.4. (Notation as above.) *The index $I(p;S)$ of an isolated complex point $p \in S$ equals the index of the normal vector field $\widetilde{V}$ at p (i.e., the winding number of the fiber coordinate of the map $S \ni x \to \widetilde{V}_x$ around p).*

Proof. Choose local coordinates (z,w) near $p=0$ in which $S = \{w = f(z)\}$. We can take $\mathrm{N}_{S/X} = S \times (\{0\} \oplus \mathbb{C})$. The vector fields

$$V = \frac{\partial}{\partial z} + \frac{\partial f}{\partial x}(z)\frac{\partial}{\partial w}, \qquad W = \mathrm{i}\frac{\partial}{\partial z} + \frac{\partial f}{\partial y}(z)\frac{\partial}{\partial w} \tag{9.4}$$

form a real basis of the tangent space $\mathrm{T}_{(z,f(z))}S$ for all points z near $0 \in \mathbb{C}$. Then $JV = \mathrm{i}\frac{\partial}{\partial z} + \mathrm{i}\frac{\partial f}{\partial x}(z)\frac{\partial}{\partial w}$, and by subtracting W we get a vector field whose first component equals zero (a section of the normal bundle $\mathrm{N}_{S/X}$). Hence the latter vector field is precisely $\widetilde{V} = \tau(JV)$:

$$\widetilde{V} = JV - W = \left(\mathrm{i}\frac{\partial f}{\partial x}(z) - \frac{\partial f}{\partial y}\right)\frac{\partial}{\partial w} = 2\mathrm{i}\,\frac{\partial f}{\partial \bar z}(z)\frac{\partial}{\partial w}.$$

The index of $\widetilde{V}$ at 0 equals the winding number of $\frac{\partial f}{\partial \bar z}$ around 0, and this coincides with our definition of the index $I(p)$. We leave it to the reader to verify that the index of $\widetilde{V}$ at p does not depend on the choice of V. $\square$

Example 9.2.5. For every $k \in \mathbb{N}$ the surface $w = z^k\bar z$ has an isolated complex point of index k at $z=0$, while the surface $w = \bar z^{k+1}$ has an isolated complex point of index $-k$ at $z=0$. These surfaces are denerate if $k>1$. $\square$

Remark 9.2.6. By Chern and Spanier [84] one can interpret $I(p;S)$ as a local intersection index in the following way. The Grassmann manifold $G_2(X)$ of all oriented real 2-planes in the complex tangent bundle $(\mathrm{T}X, J)$ has two disjoint oriented submanifolds $\Sigma_\pm$ consisting of all complex lines with positive and negative orientation, respectively. Let $\tau\colon S \to G_2(X)$ denote the Gauss map of an oriented embedded surface $S \hookrightarrow X$, sending a point $p \in S$ to its oriented tangent plane $\mathrm{T}_pS \in G_2(X)$. Then $p \in S$ is a complex point if and only if $\tau(p) \in \Sigma = \Sigma_+ \cup \Sigma_-$. The index $I(p;S)$ of an isolated complex point equals the local intersection index of $\tau(S) \subset G_2(X)$ with Σ at p. $\square$

The last two characterizations of the index $I(p;S)$ also apply to real surfaces in almost complex (not necessarily integrable) surfaces.

The analysis near parabolic and degenerate complex points was first studied by Wiegerinck [516]. A thorough investigation of the local hull at an isolated parabolic point $p \in S$ ($\lambda = \frac{1}{2}$) was made by Jöricke [282]. She proved that the index $I(p)$ can equal $+1$, -1 or 0, the latter case being generic. If $I(p) = -1$ then S is locally polynomially convex near p. To analyse the other cases we assume that S is contained in the boundary $b\Omega$ of a bounded strongly pseudoconvex domain $\Omega \subset \mathbb{C}^2$. Given a compact set K in $b\Omega$, we let $\widehat{K}_{\text{ess}} = \widehat{K} \backslash \overline{K}$ denote the *essential part* of the $\mathcal{O}(\overline{\Omega})$-convex hull of K, and we call $K_{\text{tr}} = \overline{\widehat{K}_{\text{ess}}} \cap K$ the *trace* of $\widehat{K}_{\text{ess}}$ on K. Jöricke proved that if $S \subset b\Omega$ is a $\mathcal{C}^2$ surface and $p \in S$ is an isolated parabolic point of index 0 then either S is locally polynomially convex near p (the generic case) or, for sufficiently small closed discs K in S about p, K_{tr} is a single *onion*, a notion defined in her paper. The latter case can actually occur. Finally, if $p \in S$ is an isolated parabolic point of index $+1$ and if $K \subset S$ is a sufficiently small closed disc containing p, then K_{tr} is nonempty; in this case a more precise description of K_{tr} (4 cases, at least 3 of which occur) is given. In her analysis Jöricke uses properties of the characteristic foliation of S near p (with singularity at p) and tools from the theory of dynamical systems.

Further results concerning local polynomial convexity of a real surface in $\mathbb{C}^2$ were obtained by Bharali [45, 46].

All results mentioned so far, with the exception of Theorem 9.2.2, concern local behavior of a surface near a complex tangent. There also exist a number of results concerning global holomorphic, polynomial and rational convexity of real surfaces (see [474], especially Sec. 5.3 and Chapter 6). One of the most interesting global results in this subject is due to Bedford and Klingenberg [36] and Kruzhilin [313]. They proved that the envelope of holomorphy of a generic smooth 2-sphere S, contained in the boundary of a strongly pseudoconvex domain $\Omega \Subset \mathbb{C}^2$, is a real three dimensional Levi flat hypersurface contained in $\overline{\Omega}$ and foliated by analytic discs. This fails in general for spheres that are not contained in a strongly pseudoconvex boundary. The corresponding result for continuous graphs over spheres in $\mathbb{C} \times \mathbb{R}$ was obtained by Shcherbina [443]. The Bedford-Klingenberg theorem had a profound impact on this field. For topological applications of the 'filling by discs' methods see [130].

9.3 Invariants of Smooth 4-Manifolds

The geometry of a smooth oriented 4-manifold X is to a large extent determined by the smooth real surfaces that it contains; in particular, by the intersection form on the second homology group $H_2(X; \mathbb{Z})$. In this section we briefly recall the relevant topological invariants of compact smooth 4-manifolds that are used in the sequel; for details see any of the standard sources such as [233, 246, 205, 206, 207]. The section can be skipped altogether by a knowledgeable reader.

Let X be an oriented smooth 4-manifold. An embedded oriented closed surface S in X determines an integral homology class $[S] \in H_2(X;\mathbb{Z})$; if X is closed (compact and without boundary) then every element of $H_2(X;\mathbb{Z})$ is of this form. To a pair of compact smooth oriented surfaces $S, S' \subset X$ we associate their intersection number $S\cdot S' \in \mathbb{Z}$. Assuming that S and S' are intersecting transversely, $S\cdot S'$ is simply the number of intersection points counted with orientation signs. The intersection number $S\cdot S'$ only depends on the respective homology classes $\alpha = [S], \alpha' = [S'] \in H_2(X;\mathbb{Z})$, and we write $\alpha\cdot\alpha'$ accordingly. We denote by $S\cdot S = [S]^2$ the self-intersection number of S in X, obtained by deforming one copy of S into a generic position S' and taking $S\cdot S'$. (We avoid writing S^2 to avoid confusion with the 2-sphere.)

Assume now that X is a closed oriented 4-manifold. Then the intersection pairing $H_2(X;\mathbb{Z})\times H_2(X;\mathbb{Z}) \to \mathbb{Z}$ is a nondegenerate bilinear form on the free part $\widetilde{H}_2(X;\mathbb{Z})$ of $H_2(X;\mathbb{Z})$ (the quotient of $H_2(X;\mathbb{Z})$ modulo torsion) and it vanishes identically on the torsion part. Considering $\widetilde{H}_2(X;\mathbb{Z})$ as a lattice in the real homology group $H_2(X;\mathbb{R}) = H_2(X;\mathbb{Z})\otimes\mathbb{R}$, the intersection form extends to a nondegenerate real-valued bilinear form on the finite dimensional vector space $H_2(X;\mathbb{R})$. Denote by $b_2^+(X)$ and $b_2^-(X)$ the number of positive and negative eigenvalues, respectively. Then

$$b_2^+(X) + b_2^-(X) = b_2(X) = \dim H_2(X;\mathbb{R}), \quad b_2^+(X) - b_2^-(X) = \sigma(X)$$

are the *second Betti number* and the *signature* of X, respectively.

A nondegenerate intersection pairing on the second cohomology group $H^2(X;\mathbb{R})$ is induced by the cup product evaluated on the fundamental class of X. Consider $H^2(X;\mathbb{R})$ as the second de Rham group of X. Given a pair of classes $[\alpha], [\beta] \in H^2(X;\mathbb{R})$, their intersection product is defined by

$$[\alpha]\cdot[\beta] = \int_X \alpha\wedge\beta \in \mathbb{R}. \tag{9.5}$$

Considering $H^2(X;\mathbb{R})$ as the dual of $H_2(X;\mathbb{R})$, this pairing gives rise to the isomorphism of vector spaces $PD\colon H_2(X;\mathbb{R}) \xrightarrow{\cong} H^2(X;\mathbb{R})$ called *Poincaré duality*. Explicitly, given a smooth oriented closed real surface $S \subset X$, its Poincaré dual $PD([S]) = [\beta_S] \in H^2(X;\mathbb{R})$ is represented by a closed 2-form β_S on X such that for every $[\alpha] \in H^2(X;\mathbb{R})$ we have

$$\langle[\alpha],[S]\rangle = \int_S \alpha = \int_X \alpha\wedge\beta_S. \tag{9.6}$$

We often write $PD(S)$ instead of $PD([S])$. It is customary to denote the inverse of PD by the same sign, thereby writing $PD([\alpha]) = PD(\alpha) \in H_2(X;\mathbb{R})$ for a class $[\alpha] \in H^2(X;\mathbb{R})$.

Poincaré duality maps integral elements of $H_2(X;\mathbb{R})$ (those coming from $H_2(X;\mathbb{Z})$) to integral elements of $H^2(X;\mathbb{R})$. The intersection pairing (9.5) on

$H^2(X;\mathbb{R})$ is then Poincaré dual to the intersection pairing of homology classes in $H_2(X;\mathbb{Z})$ in the sense that $[S]\cdot[S'] = PD(S)\cdot PD(S')$.

Assume now that X is a complex surface. The first Chern class of its complex tangent bundle $\mathrm{T}X$ will be denoted $c_1(X) \in H^2(X;\mathbb{Z})$. Recall that $c_1(X) = c_1(\Lambda^2\mathrm{T}X)$ where $\Lambda^2\mathrm{T}X$ is the determinant bundle of $\mathrm{T}X$ (a holomorphic line bundle over X). The standard definition of the Chern class $c_1(L) \in H^2(X;\mathbb{Z})$ of a complex line bundle $L \to X$ is via the exact exponential sheaf sequence

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathcal{C} \xrightarrow{\mathrm{e}^{2\pi\mathrm{i}\cdot}} \mathcal{C}^* \longrightarrow 0$$

which induces an isomorphism $c_1\colon H^1(X;\mathcal{C}^*) \xrightarrow{\cong} H^2(X;\mathbb{Z})$. (See (5.1) on p. 189).) The dual bundle $(\Lambda^2\mathrm{T}X)^* = \Lambda^2\mathrm{T}^*X = K_X$ is the *canonical bundle* of X, and its Poincaré dual $PD(K_X) = [K_X] \in H_2(X;\mathbb{Z})$ is called the *canonical class* of X. In subsequent formulas we follow the custom of writing K_X both for the line bundle and for the corresponding homology class $[K_X] \in H_2(X;\mathbb{Z})$.

Given a smooth embedded surface $S \subset X$, the integer $c_1(X)\cdot S = -c_1(K_X)\cdot S$ is the degree of the restricted bundle $\mathrm{T}X|_S$, which is just the oriented self-intersection number of the zero section in $\mathrm{T}X|_S$.

It is customary to identify a second homology or cohomology class with a particular representative, or with its Poincaré dual. In the same vein, a complex line bundle L on X is often identified with its Chern class $c_1(L) \in H^2(X;\mathbb{Z})$, and also with its Poincaré dual $PD(c_1(L)) \in H_2(X;\mathbb{Z})$. If X is a complex surface and L is a line bundle determined by the divisor of a complex curve $C \subset X$, then $PD(c_1(L)) = [C] \in H_2(X;\mathbb{Z})$ is the homology class determined by C. If L, L' are line bundles over X determined by the divisors of C, C', respectively, then $L\cdot L' = c_1(L)\cdot c_1(L') = C\cdot C'$. Further, if S is a smooth closed real surface in X then $L\cdot S$ stands for $c_1(L)\cdot[S] = C\cdot S$. In the same spirit we identify the canonical bundle K_X with its Poincaré dual $[K_X] \in H_2(X;\mathbb{Z})$ and write $c_1(X)\cdot S = -[K_X]\cdot[S] = -K_X\cdot S$.

9.4 Lai Indexes and Index Formulas

Recall that an orientable compact real surface of genus $g \geq 0$ is the two-sphere if $g = 0$, and is a connected sum of g tori if $g > 0$. The Euler number of such surface equals $\chi(S) = 2 - 2g$. A nonorientable surface of genus $g > 0$ is a connected sum of g copies of the real projective plane; its Euler number equals $\chi(S) = 2 - g$. Assume now that S is smoothly embedded or immersed into a complex surface X. After a generic small perturbation of S in X we may assume that it has only isolated complex points. If S is compact then it has only finitely many complex points $p_1, \ldots, p_k \in S$, and we define the *index* of the embedding $S \hookrightarrow X$ as the integer

$$I(S) = \sum_{p \in S} I(p; S) = \sum_{j=1}^{k} I(p_j; S), \tag{9.7}$$

where $I(p_j; S)$ is the index of the complex point p_j in S (Def. 9.2.3). Similarly we define the index $I(\pi)$ of an immersion $\pi\colon S \looparrowright X$.

If S is oriented, a complex point $p \in S$ is either *positive* or *negative*, depending on whether the orientation on $\mathrm{T}_p S$, induced by the complex structure, agrees or disagrees with the orientation on S. Let $I_\pm(S) \in \mathbb{Z}$ denote the sum of indexes over positive (resp. negative) complex points of S; hence $I(S) = I_+(S) + I_-(S)$. If S has only elliptic and hyperbolic points then, denoting by $e_\pm(S)$, $h_\pm(S)$ their respective number, we have

$$I_+(S) = e_+(S) - h_+(S), \quad I_-(S) = e_-(S) - h_-(S).$$

These numbers are called *Lai indexes* after H. F. Lai [319], although they were already present in the work of Chern and Spanier [84] and Bishop [49] who studied surface in $\mathbb{C}^2$. In view of Remark 9.2.6, the Lai index $I_\pm(S)$ equals the intersection index of the Gauss map $\tau\colon S \to G_2(X, J)$ with the submanifold $\Sigma_\pm$ of $G_2(X, J)$ consisting of all complex lines with positive or negative orientation, respectively. Lai [319] studied the relationship between these numbers and the topological invariants of the embedding or immersion, also for immersions of compact real k-dimensional manifolds to complex n-dimensional manifolds for $k \geq n$.

Suppose now that $\pi\colon S \looparrowright X$ is an immersion all with only simple (transverse) double points such that both tangent planes at any double point are totally real. At each double point π has self-intersection index ± 1 which is independent of the choice of the orientation on S. Double points with index $+1$ are called positive, and those with index -1 are negative. If π has δ_+ positive and δ_- negative double points, then $\delta(\pi) = \delta_+ - \delta_-$ is the (geometric) *self-intersection index* of π which only depends on its regular homotopy class.

The following result summarizes most index theorems concerning complex points of immersed real surfaces in complex surfaces (c.f. [319, 134, 164, 171, 217, 372, 504]). The notation is explained in the previous section. In particular, $[\pi(S)]^2 \in \mathbb{Z}$ is the self-intersection number of the homology class $[\pi(S)] \in H_2(X; \mathbb{Z})$, and $c_1(X) \cdot \pi(S)$ is the value of the first Chern class $c_1(X) = c_1(\mathrm{T}X) \in H^2(X; \mathbb{Z})$ on $[\pi(S)]$.

Theorem 9.4.1. *Let S be a closed connected real surface with genus $g(S)$ and Euler number $\chi(S)$. For every immersion $\pi\colon S \looparrowright X$ into a complex surface X we have*

$$I(\pi) = \chi(S) + \chi_n(\pi), \tag{9.8}$$

where $\chi_n(\pi)$ is the normal Euler number of π. If S is oriented then

$$I(\pi) = 2 - 2g(S) - 2\delta(\pi) + [\pi(S)]^2, \tag{9.9}$$

$$I_\pm(\pi) = 1 - g(S) - \delta(\pi) + \frac{1}{2}\left([\pi(S)]^2 \pm c_1(X) \cdot \pi(S)\right). \tag{9.10}$$

Proof. Let J denote the almost complex structure operator on X. Choose a tangent vector field V to S with isolated zeros that do not coincide with any of the complex points of S. Let $\tau\colon \mathrm{T}X|_S \to \mathrm{N}_{S/X} = \mathrm{T}X|_S/\mathrm{T}S$ denote the projection onto the normal bundle of S in X. The vector field $W = \tau(JV)$ is then a section of $\mathrm{N}_{S/X}$ whose zeros are the zeros of V and the complex points of S. A choice of local orientation of S coorients the normal bundle $\mathrm{N}_{S/X}$ by the complex orientation of $\mathrm{T}X|_S = \mathrm{T}S \oplus \mathrm{N}_{S/X}$. The map $\tau \circ J\colon \mathrm{T}S \to \mathrm{N}_{S/X}$ is orientation reversing over the totally real part of S. It follows that at every zero of V the vector field W has zero of index opposite to that of V. At a complex point p of S the index of W equals $I(p;S)$ by Lemma 9.2.4. Since the sum of indexes of V is the Euler number $\chi(S)$ while the sum of indexes of W is the normal Euler number of S in X, (9.8) follows. The same argument applies to immersions if V is chosen to be nonzero at any double point.

Suppose now that $\pi\colon S \looparrowright X$ is an immersed oriented surface with simple double point. Let $\mathrm{N}_\pi \to S$ denote the normal bundle of the immersion; this is an oriented 2-plane bundle whose fiber over a point $p \in S$ equals $\mathrm{T}_{\pi(p)}X/\mathrm{d}\pi_p(\mathrm{T}_pS)$. Note that π extends to an immersion (still denoted π) of a neighborhood of the zero section of N_π (which we identify with S) onto a neighborhood of $\pi(S)$ in X; this map is an embedding if $\pi\colon S \hookrightarrow X$ is an embedding. The normal Euler number $\chi_n(\pi)$ is the oriented self-intersection number of the zero section S in N_π. Choosing a small generic deformation $S' \subset \mathrm{N}_\pi$ of S that does not intersect S at any of the double points of π, we have $\chi_n(\pi) = S\cdot S'$. Every intersection point of S and S' in N_π gives an intersection point of the image surfaces $\pi(S), \pi(S')$ in X with the same intersection index. In addition, each simple double point of π creates two nearby intersection point of $\pi(S)$ and $\pi(S')$ of the same sign in X. (Look at the cross $(\mathbb{R}^2 \times \{0\}^2) \cup (\{0\}^2 \times \mathbb{R}^2) \subset \mathbb{R}^4$ and translate it off itself.) This shows that

$$\chi_n(\pi) + 2\delta(\pi) = [\pi(S)]^2. \tag{9.11}$$

Substituting in (9.8) yields (9.9). To prove (9.10) it remains to show that

$$I_+(\pi) - I_-(\pi) = c_1(X)\cdot \pi(S). \tag{9.12}$$

Assume first that S is embedded. Choose a nonvanishing 2-form ω on S (such exists if S is oriented). For a local frame (V,W) of $\mathrm{T}S$ the expression $\omega(V,W)^{-1}V\wedge W$ is independent of the choice of the frame (V,W), and hence it gives a well defined global section ξ of the determinant bundle $\Lambda^2_{\mathbb{C}}(\mathrm{T}X)|_S$. The zeros of ξ are precisely the complex points of S. Locally near a complex point we represent S as the graph $\{w = f(z)\} \subset \mathbb{C}^2$ and use the local frame (V,W) for $\mathrm{T}S$ given by (9.4) on p. 410. Then

$$V\wedge W = \left(\frac{\partial f}{\partial y}(z) - \mathrm{i}\frac{\partial f}{\partial x}(z)\right)\frac{\partial}{\partial z}\wedge\frac{\partial}{\partial w} = -2\mathrm{i}\,\frac{\partial f}{\partial \bar z}(z)\cdot\frac{\partial}{\partial z}\wedge\frac{\partial}{\partial w}.$$

This shows that the index of ξ at the complex point at $0 \in \mathbb{C}^2$ equals $I(0;S)$ if the point is positive, and it equals $-I(0;S)$ if the point is negative. Hence the

algebraic number of zeros of ξ is $I_+(S) - I_-(S)$. But this number also equals $c_1(X)\cdot S$ since ξ is a section of $\Lambda^2_{\mathbb{C}}(\mathrm{T}X)|_S$. This proves (9.12) and hence (9.10). The same argument applies in the immersed case. □

Remark 9.4.2. We give another derivation of the index formula (9.10) for an immersed surface $\pi(S) \subset X$ from the corresponding formula for embedded surfaces by using surgery that replaces every double point by a handle. This argument, which is also used in the proof of the adjunction inequality for immersed surfaces in §9.7, clearly exposes the different roles played by the positive and the negative double points.

Assume that $\pi(S)$ has only simple double points, say δ_+ positive ones and δ_- negative ones, so $\delta_+ - \delta_- = \delta(\pi)$. After a regular homotopy we may assume that in certain local holomorphic coordinates $(z, w) = (x+\mathrm{i}y, u+\mathrm{i}v)$ on X near a double point $p \in \pi(S)$, with p corresponding to the origin $(0,0) \in \mathbb{C}^2$, the immersed surface $\pi(S)$ intersects a small ball $\mathbb{B} \subset \mathbb{C}^2$ in the union $D_1 \cup D_2$ of discs contained in the Lagrangian planes $\Lambda_1 = \{y = 0,\ v = 0\} \subset \mathbb{C}^2$, $\Lambda_2 = \{x = 0,\ u = 0\} \subset \mathbb{C}^2$, respectively. We orient Λ_1 by $\partial_x \wedge \partial_u$ and Λ_2 by $\kappa(p)\,\partial_v \wedge \partial_y$, where $\kappa(p) = \pm 1$ is the self-intersection index of $\pi(S)$ at p. We now remove from $\pi(S)$ the union of intersecting discs $D_1 \cup D_2$ and glue in a smooth annular handle Σ diffeomorphic to $S^1 \times [0,1]$. In topological language, the pair of oriented circles $bD_1 \cup bD_2$ is a *Hopf link* in the 3-sphere $b\mathbb{B} \subset \mathbb{R}^4$, and Σ is a *Seifert surface* for this link. This surgery removes the double point at p and increases the genus of the surface by one.

If $\kappa(p) = +1$ then an appropriate handle matching the orientation on $bD_1 \cup bD_2$ is $\Sigma_+ = \{(x+\mathrm{i}u)(y-\mathrm{i}v) = \epsilon\} \cap \mathbb{B}$ for a small $\epsilon \neq 0$. (This is a totally real analogue of the modification $zw = \epsilon$ of the complex double point of $zw = 0$ at $(0,0)$.) Outside of $\mathbb{B}$ we can smoothly patch Σ_+ with $(\Lambda_1 \backslash D_1) \cup (\Lambda_2 \backslash D_2)$ without introducing any new complex points provided that $\epsilon > 0$ is small enough. A calculation shows that Σ_+ is totally real for every $\epsilon \neq 0$, so this surgery does not change the indexes $I_\pm$.

If $\kappa(p) = -1$, an appropriate handle is $\Sigma_- = \{(x+\mathrm{i}u)(y+\mathrm{i}v) = \epsilon\} \cap \mathbb{B}$ for small $\epsilon \neq 0$. It is easily seen that Σ_- has four complex points, located at $x = y = \pm\sqrt{\frac{\epsilon}{2}}$, $v = -u = \pm\sqrt{\frac{\epsilon}{2}}$, two positive ones and two negative ones, all of them hyperbolic. Hence $I_\pm(\Sigma_-) = -2$, so this surgery decreases each of the indexes $I_\pm$ by two.

After replacing all double points of $\pi(S)$ by handles we obtain a smooth embedded surface $S' \subset X$ of genus $g(S') = g(S) + \delta_+ + \delta_-$, homology class $[S'] = [\pi(S)] \in H_2(X;\mathbb{Z})$, and Lai indexes $I_\pm(S') = I_\pm(S) - 2\delta_-$. Inserting these data in $I_\pm(S') = 1 - g(S') + \frac{1}{2}\,(S'\cdot S' \pm c_1(X)\cdot S')$ yields (9.10). □

Corollary 9.4.3. *Let $\pi\colon S \looparrowright X$ is a smooth immersion of a compact oriented real surface S in a complex surface X. If $[\pi(S)] = 0 \in H_2(X;\mathbb{Z})$ then*

$$I_\pm(\pi) = 1 - g(S) - \delta(\pi). \tag{9.13}$$

This holds in particular if $X = \mathbb{C}^2$.

Proof. This follows from (9.9) since $[\pi(S)] = 0$. For $X = \mathbb{C}^2$ this is due to Chern and Spanier [84] (for the 2-sphere) and to Bishop [49]. □

If C is a complex curve in X with its natural orientation induced by the complex structure, then a generic small perturbation of C in X yields a surface with only positive complex points, so in this case we have

$$I_-(C) = 0, \quad I_+(C) = c_1(X)\cdot C = -K_X\cdot C.$$

The same holds if C is symplectic curve for a symplectic form ω on X that is tamed by J, in the sense that $\langle \omega, v \wedge Jv\rangle > 0$ for every $0 \neq v \in \mathrm{T}X$. From the expression (9.10) for $I_-(C)$ we get the following:

Corollary 9.4.4. (Genus formula) *If C is a smoothly embedded complex or symplectic curve in a complex (resp. symplectic) surface X then*

$$2g(C) - 2 = [C]^2 - c_1(X)\cdot C = [C]^2 + K_X\cdot C. \tag{9.14}$$

The standard proof of (9.14) uses the normal bundle exact sequence

$$0 \longrightarrow \mathcal{T}_C \longrightarrow \mathcal{T}|_C \longrightarrow \mathcal{N}_{C/X} \longrightarrow 0$$

which gives $\Lambda^2\mathrm{T}X|_C = \mathrm{T}C \otimes \mathrm{N}_{C/X}$. Here $\mathcal{T}$ and $\mathcal{N}$ denote the tangent and the normal sheaf, respectively. Since $\Lambda^2\mathrm{T}X|_C = (K_X|_C)^{-1}$, $\mathrm{T}C = (K_C)^{-1}$ and $\mathcal{N}_{C/X} = \mathcal{O}_X(C)|_C$, where $\mathcal{O}_X(C)$ is the line bundle on X determined by the divisor of C, we obtain $K_C = K_X|_C \otimes \mathcal{O}_X(C)$ as line bundles over C. Taking the degrees and noting that $\deg K_C = 2g(C) - 2$ gives (9.14).

Example 9.4.5. The group $H_2(\mathbb{P}^2;\mathbb{Z}) = \mathbb{Z}$ is generated by the homology class $[H]$ of the projective line $H = \mathbb{P}^1 \hookrightarrow \mathbb{P}^2$. If $C \hookrightarrow \mathbb{P}^2$ is a compact complex or symplectic curve of degree $d > 0$ then $[C] = d[H]$ and hence $[C]^2 = d^2$. Furthermore, we have $c_1(\mathbb{P}^2)\cdot H = 3$ and hence $c_1(\mathbb{P}^2)\cdot C = c_1(\mathbb{P}^2)\cdot 3H = 3d$. Thus for $X = \mathbb{P}^2$ the genus formula (9.14) reads

$$g(C) = \frac{1}{2}(d-1)(d-1). \tag{9.15}$$

9.5 Cancelling Pairs of Complex Points

In this section we prove a theorem of Eliashberg and Kharlamov that it is possible to cancel a pair of an elliptic and a hyperbolic point of the same sign by a $\mathcal{C}^0$-small deformation of the given real surface in a complex surface. This shows that Lai indexes $I_\pm$ (p. 413) are the only topological invariants of an embedded or immersed real surface with respect to the complex structure of the ambient complex surface (a form of the h-principle).

Assume that $S \hookrightarrow X$ is a real surface smoothly embedded in a complex surface X. Let $\gamma \subset S$ be an arc connecting a pair of complex points $p, q \in S$. We say that γ is orientation preserving if the orientation of the tangent plane $\mathrm{T}_q S$, obtained by transporting along γ the standard orientation of $\mathrm{T}_p S$ induced by the complex structure, agrees with the standard orientation on $\mathrm{T}_q S$; in the opposite case γ is orientation reversing. (This does not require S to be oriented.) The same definition applies to immersions $S \looparrowright X$. If S is oriented then any arc in S connecting a pair of complex points is orientation preserving if its endpoints are of the same sign, and is orientation reversing if they are of opposite signs. If however S is connected and nonorientable then any pair of complex points can be connected by an orientation preserving arc.

Theorem 9.5.1. [134] *Let $p, q \in S$ be isolated complex points of an immersion $\pi_0 \colon S \looparrowright X$ of a real surface S to a complex surface X, and let $\gamma \subset S$ be an orientation preserving simple smooth arc connecting p to q. Assume that there is a neighborhood $U \subset S$ of γ such that $\overline{U}$ does not contain any other complex point or double point of π_0. Given $\epsilon > 0$, there is a regular homotopy of immersions $\pi_t \colon S \looparrowright X$ $(t \in [0,1])$ satisfying the following properties:*

(a) $\pi_t = \pi_0$ *on* $S \backslash U$ *for all* $t \in [0,1]$,

(b) $\sup_{x \in S} \mathrm{dist}\big(\pi_t(x), \pi_0(x)\big) < \epsilon$ *for all* $t \in [0,1]$,

(c) *every* π_t *has the same number of double points as* π_0, *and*

(d) $\pi_1|_U$ *has a single complex point of index* $I(s;\pi_1) = I(p;\pi_0) + I(q;\pi_0)$.

In particular, if $I(p;\pi_0) + I(q;\pi_0) = 0$ then π_1 can be chosen totally real on U. If π_0 is an embedding then $\{\pi_t\}_{t \in [0,1]}$ can be chosen an isotopy of embeddings.

This formulation of the cancellation theorem is taken from [164, Theorem 1.1]. The same result holds if (X, J) is an almost complex surface.

Corollary 9.5.2. *Let S be an immersed connected real surface with isolated complex points in a complex surface X. If S is oriented then every pair of an elliptic and a hyperbolic point of the same orientation class can be cancelled by a $\mathcal{C}^0$-small smooth deformation of S in X which does not create any new complex points or double points. If S is nonorientable then any pair of an elliptic and a hyperbolic complex point can be cancelled in this way.*

The key step in the proof of Theorem 9.5.1 is the following $\overline{\partial}$-modification lemma on the disc $\overline{\mathbb{D}} = \{z \in \mathbb{C} \colon |z| \le 1\} \subset \mathbb{C}$.

Lemma 9.5.3. *Let f be a smooth complex function on the closed unit disc $\overline{\mathbb{D}} \subset \mathbb{C}$ such that $\overline{\partial} f \neq 0$ on the circle $b\mathbb{D}$ and the winding number of $\overline{\partial} f$ along $b\mathbb{D}$ equals zero. Given an $\epsilon > 0$ there exists a smooth function g on $\overline{\mathbb{D}}$ that equals f in a neighborhood of $b\mathbb{D}$ such that $\overline{\partial} g \neq 0$ on $\overline{\mathbb{D}}$ and $|g - f| < \epsilon$ on $\overline{\mathbb{D}}$.*

Proof. We give a simple complex analytic proof due to Nemirovski [372, p. 735]. A similar result holds for a nonintegrable almost complex structure on $\mathbb{C}^2$ that is close to the standard structure along the disc $\mathbb{D} \times \{0\}$; the proof of [164, Lemma 4.1] can be adapted to this situation.

Recall that if ψ is a smooth function supported in the disc $\mathbb{D}$ then its Cauchy-Greene transform

$$u(z) = \frac{1}{2\pi \mathrm{i}} \int_{\mathbb{C}} \frac{\psi(\zeta + z)}{\zeta} \, \mathrm{d}\zeta \wedge \mathrm{d}\bar{\zeta} \tag{9.16}$$

solves the equation $\overline{\partial} u = \psi$ with a uniform estimate

$$||u||_{\mathcal{C}^0(\mathbb{D})} \leq \mathrm{const}\, ||\psi||_{L^p(\mathbb{D})}, \qquad p > 2.$$

(Here $\overline{\partial} u = \frac{\partial u}{\partial \bar{z}}$.) This well known fact follows from the Hölder inequality since the Cauchy kernel $1/\zeta$ belongs to L^q_{loc} for all $q < 2$. Fix a number $p > 2$.

By perturbing f away from $b\mathbb{D}$ we insure that $\overline{\partial} f$ has only finitely many zeros $z_1, \ldots, z_k$ in $\mathbb{D}$ and no zeros on $b\mathbb{D}$. Choose a simple smooth arc γ in $\mathbb{D}$ connecting these points. Pick a number $r \in (0,1)$ such that $\gamma \subset r\mathbb{D}$. Let $m = \inf\{|\overline{\partial} f(z)| : r \leq |z| \leq 1\} > 0$. Choose a smooth function $\chi \geq 0$ with support in $\mathbb{D}$ that equals 1 on $r\mathbb{D}$. Decrease the number $\epsilon > 0$ if necessary so that $m - \epsilon|\overline{\partial}\chi| > 0$. Note that $\overline{\partial} f$ has winding number zero around the arc γ. Hence there is a smooth function ψ, supported in any given neighborhood $V \subset r\mathbb{D}$ of γ, such that $\overline{\partial} f + \psi \neq 0$ on $\overline{\mathbb{D}}$ and $||\psi||_{\mathcal{C}^0(\mathbb{D})} \leq 2\|\overline{\partial} f\|_{\mathcal{C}^0(\mathbb{D})}$. By taking the area of V small enough we can insure that $||\psi||_{L^p}$ is so small that the corresponding function u (9.16) satisfies $||u||_{\mathcal{C}^0(\mathbb{D})} < \epsilon$. Set $g = f + \chi u$. Then $g = f$ near $b\mathbb{D}$ and $||f - g||_{\mathcal{C}^0(\mathbb{D})} = ||\chi u||_{\mathcal{C}^0(\mathbb{D})} \leq ||u||_{\mathcal{C}^0(\mathbb{D})} < \epsilon$. We claim that $\overline{\partial} g \neq 0$ on $\overline{\mathbb{D}}$. This is clear on $r\mathbb{D}$ where $\overline{\partial} g = \overline{\partial} f + \overline{\partial} u = \overline{\partial} f + \psi$. Outside of $r\mathbb{D}$ we have $\overline{\partial} u = \psi = 0$ and hence $|\overline{\partial} g| = |\overline{\partial} f + u\overline{\partial}\chi| \geq |\overline{\partial} f| - \epsilon|\overline{\partial}\chi| \geq m - \epsilon|\overline{\partial}\chi| > 0$. This concludes the proof. $\square$

Proof (of Theorem 9.5.1). We begin by showing that a neighborhood of γ in S can be holomorphically spread as a graph of a complex function over a simply connected domain in $\mathbb{C}$. Choose a pair of smooth $\mathbb{R}$-independent $(1,0)$ vector fields V and W in a neighborhood of γ in X that are tangent to S, with V tangent to the arc γ. The assumptions imply that V and W are $\mathbb{C}$-independent near γ, except at the two endpoints p and q. Pick another vector field V' near γ that is $\mathbb{C}$-linearly independent of V. Then $W = aV + bV'$ for some smooth complex valued functions a and b, and b vanishes precisely at the endpoints p, q of γ. For any function c the vector field $Z = cV + V'$ is $\mathbb{C}$-linearly independent of V, and we have $W = (a - bc)V + bZ$. Write $a - bc = \alpha + \mathrm{i}\beta$ with α, β real. We wish to choose c such that $\beta(x) \neq 0$ for all $x \in \gamma$. Since γ is orientation preserving, $\Im a$ is of the same sign at both endpoints of γ, say positive. Taking $c(x) = -\mathrm{i}C\overline{b(x)}$ for a sufficiently large $C > 0$ gives $\beta(x) = \Im a(x) + C|b(x)|^2 > 0$ for $x \in \gamma$, and hence the requirement is fulfilled.

Let (z,w) be complex coordinates on $\mathbb{C}^2$. There is a neighborhood Ω of γ in X, with $\Omega \cap S \subset U$, and a smooth diffeomorphism $\Phi\colon \Omega \to \Omega' \subset \mathbb{C}^2$ satisfying the following properties:

(i) $\Phi(\gamma) = [0,T] \times \{0\}$ for some $T > 0$,

(ii) the differential $\mathrm{d}\Phi = \Phi_*$ is $\mathbb{C}$-linear at each point of γ, and

(iii) $\Phi_* V = \frac{\partial}{\partial z}$ and $\Phi_* Z = \frac{\partial}{\partial w}$ at every point of γ.

It follows that along γ we have

$$\Phi_* W_x = (\alpha(x) + \mathrm{i}\beta(x))\frac{\partial}{\partial z} + b(x)\frac{\partial}{\partial w}, \qquad x \in \gamma.$$

Note that the tangent space $\Phi_*(\mathrm{T}_x S) \subset \mathrm{T}_{\Phi(x)}\mathbb{C}^2 \cong \mathbb{C}^2$ to $\Phi(S \cap \Omega)$ is spanned over $\mathbb{R}$ by the vectors $\Phi_* V_x$ and $\Phi_* W_x$. Since $\beta > 0$ on γ, we see that $\Phi_*(\mathrm{T}_x S)$ projects isomorphically onto $\mathbb{C} \times \{0\}$ for each point $x \in \gamma$. The implicit function theorem now shows that for a small enough neighborhood $\Omega \subset X$ of γ the set $\Phi(S \cap \Omega)$ is a smooth graph over a domain in $\mathbb{C}$.

The smooth arc γ has a basis of Stein neighborhoods in X by Corollary 3.5.2. Since the differential of Φ is $\mathbb{C}$-linear along γ, we can approximate Φ in the $\mathcal{C}^1$ topology on γ by a holomorphic mapping $\Psi = (\psi_1, \psi_2)$ on a neighborhood of γ in X (Theorem 3.5.4 on p. 73). Then Ψ is biholomorphic on a neighborhood Ω_1 of γ in X and the image $\Psi(S \cap \Omega_1)$ is a graph

$$S' = \Psi(S \cap \Omega_1) = \big\{(z, f(z))\colon z \in D\big\} \subset \mathbb{C}^2$$

of a smooth function f on a domain $D \subset \mathbb{C}$. By construction, the only complex points of S' are the endpoints $p' = \Psi(p)$ and $q' = \Psi(q)$ of the curve $\gamma' = \Psi(\gamma)$. Since the index of a complex point is preserved under a biholomorphism, we have $I(p;S) = I(p';S')$ and $I(q;S) = I(q';S')$. Write $p' = (z_0, f(z_0))$ and $q' = (z_1, f(z_1))$. The first coordinate projection $\psi_1(\gamma)$ of the arc $\Psi(\gamma)$ is a simple smooth arc in D with the endpoints z_0 and z_1.

The conclusion is now immediate if $k = 0$ (when the pair of complex point p, q should be cancelled). Choose a simply connected domain $D_0 \Subset D$ containing the arc $\psi_1(\gamma)$. The winding number $\mathcal{W}(\overline{\partial} f; bD_0)$ equals the sum of winding numbers of $\overline{\partial} f$ along small circles around the points z_0 and z_1, and the latter equals $I(p';S') + I(q',S') = 0$. By Lemma 9.5.3 we can approximate f by a smooth function F on D_0 such that $F = f$ near bD_0 and $\overline{\partial} F \neq 0$ on D_0. The graph of $f_t = (1-t)f + tF$ over D_0 for $t \in [0,1]$ then provides an isotopy $S'_t \subset \mathbb{C}^2$, fixed near the boundary, from the initial surface $S'_0 = S'$ to a totally real surface S'_1. The isotopy $\Psi^{-1}(S'_t) \subset X$ extends to the rest of S as the identity and makes the surface totally real near the arc γ.

In the general case we let $s \in \gamma$ be the point at which we wish to produce a complex point of index $k = I(p;S) + I(q;S)$. Let $\Psi(s) = s' = (z_2, f(z_2)) \in \gamma'$. Pick a number $\eta > 0$. Choose a pair of smoothly bounded simply connected domains D_0, D_1 in $\mathbb{C}$ such that $\psi_1(\gamma) \subset D_1 \Subset D_0 \Subset D$. Choose a smooth

function g on $\mathbb{C}$ whose graph has an isolated complex point of index k at z_2 and $g(z_2) = f(z_2)$. Let $D_2 \Subset D_1$ be a small disc centered at z_2 such that $|f - g| < \eta$ on $\bar{D}_2$. Choose a thin strip G connecting the disc D_2 with the annular region $D_0 \backslash D_1$ and satisfying $G \cap \psi_1(\gamma) = \emptyset$. We can find a smooth function F on the closed annular region $\bar{A}_* = (\bar{D}_0 \backslash D_1) \cup \bar{G} \cup \bar{D}_2$ such that $F = f$ on $\bar{D}_0 \backslash D_1$, $F = g$ on $\bar{D}_2$, $|F - f| < \eta$ on $\bar{A}_*$, and $\overline{\partial} F \neq 0$ on $\bar{A}_* \backslash \{z_2\}$. The complement $D_* = D_0 \backslash \bar{A}_* = D_1 \backslash (\overline{G} \cup \overline{D}_2) \Subset D_0$ is a disc whose boundary curve τ_* is homologous in $\bar{A}_* \backslash \{z_2\}$ to the cycle $bD_1 - bD_2$. We then have

$$\begin{aligned} \mathcal{W}(\overline{\partial} F; \tau_*) &= \mathcal{W}(\overline{\partial} F; bD_1) - \mathcal{W}(\overline{\partial} F; bD_2) = \mathcal{W}(\overline{\partial} f; bD_1) - \mathcal{W}(\overline{\partial} g; bD_2) \\ &= I(p'; S') + I(q'; S') - k \end{aligned}$$

which equals zero by the choice of k. By Lemma 9.5.3 we can smoothly extend F from $\bar{A}_*$ to $\bar{D}_0$ such that $\overline{\partial} F \neq 0$ on $\bar{D}_*$ and $|F - f| < \eta$ on $\bar{D}_0$. Thus F is a smooth function on $\bar{D}_0$ such that $F = f$ near bD_0, $F = g$ near the point z_2, $\overline{\partial} F \neq 0$ on $\bar{D}_0 \backslash \{z_2\}$, and $\sup_{D_0} |F - f| < \eta$. We now conclude the proof exactly as before by letting $S'_t \subset D_0 \times \mathbb{C}$ be the graph of $F_t = (1-t)f + tF$ and considering the isotopy $\Psi^{-1}(S'_t) \subset X$. $\square$

9.6 Applications of the Cancellation Theorem

The following is an immediate corollary to Theorem 9.5.1.

Corollary 9.6.1. *Let S be a compact connected real surface smoothly embedded in a complex surface X.*

(a) *If S is oriented and $I_+(S) = I_-(S) = 0$, or if S is nonorientable and $I(S) = 0$, then S is isotopic to an embedded totally real surface $S' \subset X$ by an isotopy that is $\mathcal{C}^0$-close to S.*

(b) *If S is oriented and $I_\pm(S) \leq 0$, or if S is nonorientable and $I(S) \leq 0$, then S is isotopic to an embedded surface with only hyperbolic complex points.*

The analogous conclusions hold for immersed surfaces.

Observe that the pair of inequalities $I_\pm(S) \leq 0$ for an embedded oriented surface $S \subset X$ is equivalent to

$$2g(S) - 2 \geq [S]^2 + |c_1(X) \cdot S| = [S]^2 + |K_X \cdot S|. \tag{9.17}$$

Since the right hand side only depends on the homology class $[S] \in H_2(X; \mathbb{Z})$, this *adjunction inequality* imposes a lower bound on the genus of an embedded surface $S \subset X$ representing a given homology class in $H_2(X; \mathbb{Z})$. Seiberg-Witten theory shows that the inequality (9.17) holds for most real surfaces in compact Kähler surfaces (Theorem 9.7.1, p. 427).

Part (b) of Corollary 9.6.1 together with Theorem 9.2.2 (p. 409) implies the following.

Corollary 9.6.2. *Let S be a closed oriented real surface smoothly embedded in a complex surface X. If (9.17) holds then S can be $\mathcal{C}^0$ approximated by an isotopic surface $S' \subset X$ with a basis of tubular Stein neighborhoods.*

A very useful operation to find explicit examples is the *connected sum*; let us recall this briefly. Assume that X_1 and X_2 are smooth n-manifolds. Choose embedded n-discs $D_j \subset X_j$ for $j = 1, 2$ and let $\phi: bD_1 \to bD_2$ be a smooth diffeomorphism of their boundaries. The identification space

$$X_1 \# X_2 = (X_1 \backslash \mathrm{int} D_1) \cup_\phi (X_2 \backslash \mathrm{int} D_2)$$

has the structure of a smooth manifold called the connected sum of X_1 and X_2. The smoothing process can be visualized by connecting the two pieces $X_j \backslash \mathrm{int} D_j$ by a cylinder $\Sigma = [-1, 1] \times S^{n-1}$ glued along its boundary spheres $(\{-1\} \times S^{n-1}) \cup (\{+1\} \times S^{n-1})$ to the spheres bD_1, bD_2, respectively. If both X_1 and X_2 are oriented then $X_1 \# X_2$ is also oriented provided that the gluing diffeomorphism $\phi: bD_1 \to bD_2$ is orientation reversing. If $X_1 = X_2 = X$ and ϕ is the identity map on $bD \subset X$ then we must reverse the orientation on one of the copies of X; in this case we write $X \# \overline{X}$. For example, if X is a complex n-manifold then the connected sum $X \# \overline{\mathbb{P}^n}$ of X and the complex projective space $\mathbb{P}^n$ with reversed orientation is the complex manifold obtained by blowing up a point in X. The connected sum operation extends to several terms; one writes $X_1 \# k X_2$ for the connected sum of X_1 with k copies of X_2.

Suppose now that $S_1, S_2 \subset X$ are embedded or immersed real surfaces in general position in a complex surface X. We can realize their *interior connected sum* $S_1 \# S_2$ as an immersed surface of genus $g(S_1 \# S_2) = g(S_1) + g(S_2)$ in X whose double points are the double points of S_1, S_2 and the intersection points $S_1 \cap S_2$. We obtain $S_1 \# S_2$ by removing a pair of totally real discs $D_j \subset S_j$ $(j = 1, 2)$ and gluing in a tube $\Sigma \cong [-1, 1] \times S^1$, identifying its boundary circles with bD_1 and bD_2. After an isotopy we may assume that we are working in a local chart in $\mathbb{C}^2$, with D_1 and D_2 being discs of center $(0, 0)$ and radius $\epsilon > 0$ in the totally real planes $\Lambda^\pm = \{(\pm 1 + \mathrm{i}y, u) : (u, y) \in \mathbb{R}^2\} \subset \mathbb{C}^2$. A standard tube used for gluing is

$$\Sigma = \{(x + iy, u + \mathrm{i}0) \in \mathbb{C}^2 : -1 \le x \le 1,\ y^2 + u^2 = h(x)^2\}$$

where $h: [-1, 1] \to (0, \epsilon]$ is an even continuous function which is smooth and strictly convex on $(-1, +1)$, strictly increasing on $(0, 1]$ and satisfies $h(1) = \epsilon$ and $\lim_{t \to +1} h'(t) = +\infty$. These conditions imply that Σ has a pair of hyperbolic points of different orientation class at $(0, \pm h(0))$ and no elliptic points, so $I_\pm(\Sigma) = 1$. Assuming that $S_1 \cap S_2 = \emptyset$ we thus have

$$I(S_1 \# S_2) = I(S_1) + I(S_2) - 2. \tag{9.18}$$

If both surfaces S_1 and S_2 are oriented and the gluing was done by an orientation reversing diffeomorphism of the circles $bD_1 \to bD_2$ then we also have $[S_1 \# S_2] = [S_1] + [S_2] \in H_2(X; \mathbb{Z})$ and

$$I_+(S_1\#S_2) = I_+(S_1) + I_+(S_2) - 1, \quad I_-(S_1\#S_2) = I_-(S_1) + I_-(S_2) - 1.$$

By taking the connected sum of any real surface $S \subset X$ with a homologically trivial torus $T \subset X$ we increase the genus of the surface by one and decrease the index by two. (In the oriented case each of the indexes $I_\pm$ decreases by one.) After attaching k such trivial handles, each of the indexes $I_\pm$ decreases by k. From Theorem 9.5.1 (p. 418) we get the following.

Corollary 9.6.3. *Let $S \subset X$ be a closed, connected, oriented real surface smoothly embedded in a complex surface X. If $k = \max\{I_+(S), I_-(S), 0\}$ then S is homologous to an embedded surface $S' \subset X$ of genus $g(S') = g(S) + k$ with a basis of tubular Stein neighborhods.*

We now consider surfaces in $\mathbb{C}^2$. The index formulas simplify to

$$I(\pi) = \chi(S) + \chi_n(\pi); \quad I_\pm(\pi) = 1 - g(S) - \delta(\pi) \ \text{ if } S \text{ is orientable.} \tag{9.19}$$

For an oriented embedded surface $S \hookrightarrow \mathbb{C}^2$ we thus have $I_\pm(S) = 1 - g(S)$ which is ≤ 0 unless S is the 2-sphere. The standard 2-sphere

$$S = \{(x + \mathrm{i}y, u + \mathrm{i}v) \in \mathbb{C}^2\colon\ x^2 + y^2 + u^2 = 1,\ v = 0\} \tag{9.20}$$

has two complex points $(0, \pm 1)$ which are both elliptic.

Assume now that S is an nonorientable surface of genus $g(S)$ and Euler number $h = \chi(S) = 2 - g(S) \leq 1$. By Whitney and Massey [352] the set of normal Euler numbers of embeddings $S \hookrightarrow \mathbb{C}^2$ is $\{2h-4, 2h, 2h+4, \ldots, 4-2h\}$, and hence the set of indexes of embeddings $S \hookrightarrow \mathbb{C}^2$ is $\{3h-4, 3h, \ldots, 4-h\}$. This set always contains a nonpositive number. In summary:

Corollary 9.6.4. [164, Theorem 1.8] *Every closed real surface S other than the 2-sphere admits a smooth embedding in $\mathbb{C}^2$ with only hyperbolic complex points and with a basis of tubular strongly pseudoconvex Stein neighborhoods. Moreover, if S is orientable and $g(S) > 0$ then every embedding $S \hookrightarrow \mathbb{C}^2$ is isotopic to an embedding with this property.*

An explicit construction involves the connected sum of a suitable number of copies of the torus and the real projective plane [164]. Similarly we obtain the following result concerning totally real embeddings into $\mathbb{C}^2$.

Corollary 9.6.5. *A closed real surface S of genus $g \geq 0$ admits a totally real embedding into $\mathbb{C}^2$ if and only if S is orientable and $g = 1$ (S is the 2-torus), or S is nonorientable and $g = 2 \pmod 4$ (S is the connected sum of an odd number $\frac{g}{2}$ of Klein bottles).*

Proof. If S is orientable then (9.9) gives $I_\pm(S) = 1 - g$ which equal zero precisely when $g = 1$. The standard embedding of the torus $S^1 \times S^1 \hookrightarrow \mathbb{C}^2$ as the product of unit circles in both copies of $\mathbb{C}$ is Langrangian.

If S is a nonorientable surface with Euler number $h = 2 - g$ then the set of Lai indexes of embeddings $S \hookrightarrow \mathbb{C}^2$ is $\{3h-4, 3h, \ldots, 4-h\}$ (see above). Since this set includes the number 0 if and only if $2 - g = h = 0 \pmod 4$, S admits a totally real embedding in $\mathbb{C}^2$ if and only if it is the connected sum of an odd number $\frac{g}{2}$ of Klein bottles. An explicit totally real embedding of the Klein bottle into $\mathbb{C}^2$ was given by Rudin [424] (see Example 9.6.6 below). Givental [208] found a Lagrangian embedding of any nonorientable surface with Euler number $h \le -4$, $h = 0 \pmod 4$. The Klein bottle itself does not admit a Langrangian embedding into $\mathbb{C}^2$ [445]. $\square$

Example 9.6.6. We recall Rudin's construction [424] of a totally real Klein bottle in $\mathbb{C}^2$. Let (x, y) be coordinates on $\mathbb{R}^2$. Pick numbers $a > b > 0$, put

$$g(x,y) = (a + b\cos y)\mathrm{e}^{\mathrm{i}x}, \quad h(y) = \sin y + \mathrm{i}\sin 2y$$

and define the map $F(x, y) = (z, w)$ by

$$z = g(x,y)^2, \quad w = g(x,y)h(y).$$

Then $F(x, y+2\pi) = F(x, y) = F(x+\pi, -y)$, so F identifies the opposite sides of the rectangle $Q = \{(x, y) \in \mathbb{R}^2 \colon 0 \le x \le \pi,\ -\pi \le y \le \pi\}$ consistently with the Klein bottle construction. It is easily verified that no other identifications in Q are made. Its Jacobian equals

$$z_x w_y - z_y w_x = 2g^2 g_x h' = 2\mathrm{i}g^3 h' \neq 0.$$

Hence $F(\mathbb{R}^2) = K$ is a totally real embedded Klein bottle in $\mathbb{C}^2$. $\square$

It is not surprising that there are fewer restrictions on immersed surfaces in $\mathbb{C}^2$. We give the following result in the oriented case.

Corollary 9.6.7. *Let S be a closed orientable surface of genus g. Every totally real immersion $\pi \colon S \looparrowright \mathbb{C}^2$ satisfies $\delta(\pi) = 1 - g$. There exists a totally real immersion of S into $\mathbb{C}^2$ with precisely $|1-g|$ simple double points.*

Proof. The first statement follows from (9.31) since $[\pi(S)] = 0$. To find a totally real immersion with $|1-g|$ normal crossings we begin with the case $g = 0$. Consider the immersion $\pi \colon S \looparrowright \mathbb{C}^2$ given by

$$\pi(x,y,u) = \big(x(1+2\mathrm{i}u), y(1+2\mathrm{i}u)\big), \quad x^2 + y^2 + u^2 = 1. \tag{9.21}$$

Its image $S^+ = \pi(S) \subset \mathbb{C}^2$ is a Lagrangian immersed sphere with one double point; this double point is positive and is located at the origin $(0,0) \in \mathbb{C}^2$. We have $\delta(S^+) = +1$ and $\chi_n(S^+) = -2$. (A 'figure eight' immersion $S^n \looparrowright \mathbb{C}^n$ with these properties exists for every n; see [505, p. 26].)

Let S^- be the image of S^+ under the map $(z, w) \mapsto (z, \overline{w})$. This is an immersed sphere with a negative double point at the origin and $I(S^-) =$

$2-2\delta(S^-)=4$. Let $T=S^1\times S^1=\{e^{i\theta},e^{i\phi}):\theta,\phi\in\mathbb{R}\}\subset\mathbb{C}^2$ be the standard Lagrangian torus. The connected sum $T_1=T\#S^-$ is an immersed torus with one negative double point and $I(T_1)=I(T)+I(S^-)-2=2$ in view of (9.18). For any integer $g>0$ the orientable immersed surface of genus g given by

$$M_g=T\#\overbrace{T_1\#\cdots\#T_1}^{(g-1)\text{ times}}\subset\mathbb{C}^2$$

has $g-1$ (negative) double points. We see from (9.18) that $I(M_g)=0$ and hence it can be deformed into a totally real immersion. □

Example 9.6.8. (Real surfaces in $\mathbb{P}^2$.) The group $H_2(\mathbb{P}^2;\mathbb{Z})=\mathbb{Z}$ is generated by the homology class $h=[H]$ of the projective line $H=\mathbb{P}^1\hookrightarrow\mathbb{P}^2$. We have $c_1(\mathbb{P}^2)=3PD(h)$, where $PD(h)\in H^2(\mathbb{P}^2;\mathbb{Z})$ is the Poincaré dual of h. Let $S\subset\mathbb{P}^2$ be an oriented embedded surface of genus $g(S)$ and of degree $d\neq 0$, meaning that $[S]=d[H]\in H_2(\mathbb{P}^2;\mathbb{Z})$. After reversing the orientation of S if needed it suffices to consider the case $d>0$. We have $[S]^2=[dH]\cdot[dH]=d^2$ and $c_1(\mathbb{P}^2)\cdot S=[3H]\cdot[S]=3d$ which gives

$$I_\pm(S)=1-g(S)+\frac{d^2\pm 3d}{2}.$$

In particular, the condition $I_\pm(S)\le 0$ is equivalent to

$$g(S)\ge\frac{(d+1)(d+2)}{2}. \tag{9.22}$$

Comparing with the genus formula $g(C)=\frac{1}{2}(d-1)(d-1)$ that holds for a smooth complex curve $C\subset\mathbb{P}^2$ of degree d (see (9.15) on p. 417) we see that one must attach at least $3d$ homologically trivial torus handles to C to obtain a surface $S\subset\mathbb{P}^2$ of degree d satisfing $I_\pm(S)\le 0$.

From $I_+(S)-I_-(S)=c_1(\mathbb{P}^2)\cdot S=3d$ we also infer the following.

Corollary 9.6.9. *A homologically nontrivial embedded oriented surface in $\mathbb{P}^2$ is never totally real.*

On the other hand, $\mathbb{P}^2$ contains many totally real embedded nonorientable surfaces with nontrivial homology class in $H_2(\mathbb{P}^2;\mathbb{Z}_2)=\mathbb{Z}_2$. An obvious example is the real projective plane $\mathbb{RP}^2=\{[x:y:z]\in\mathbb{P}^2:x,y,z\in\mathbb{R}\}$. □

It is a natural question whether there exist complex surfaces into which every closed connected real surface embeds as a totally real surface. This question was studied by M. Slapar in [454]; here is a summary of his results. (For a discussion of elliptic and K3 surfaces see [33].)

- Every closed, connected real surface S admits a totally real embedding into a $K3$ surface blown up at a single point. If S is orientable then the blowup is not necessary.

- Every closed, connected real surface S admits a totally real embedding into any elliptic surface of type $E(3)$.
- Let $D(n,g)$ be an open disk bundle with Euler number n over a closed, connected, orientable real surface of genus g. If $n \leq 2g-2$ then every elliptic surface of type $E(-n+2g)$ contains a Stein domain diffeomorphic to $D(n,g)$.
- Let $\widetilde{D}(n,\chi)$ be the disk bundle with Euler number n over a closed, connected, nonorientable surface with Euler number χ. If $n+\chi \leq 0$ then there is a Stein domain diffeomorphic to $\widetilde{D}(n,\chi)$ inside a complex surface obtained by blowing up $\mathbb{P}^2$ at finitely many points.

Derdzinski and Januszkiewicz [106] studied totally real immersions and embeddings of real surfaces into almost complex 4-dimensional manifolds. By using the h-principle they established a one-to-one correspondence between the set of regular homotopy classes of such immersions $S \looparrowright X$ and the set of homotopy classes of maps $S \to E(X)$ subject to a simple cohomological condition, where $E(X)$ is an $\mathbb{RP}^1$-bundle over X naturally associated with the almost complex structure. For the complex surfaces $\mathbb{C}^2$, $\mathbb{P}^1 \times \mathbb{P}^1$, $\mathbb{P}^2$ and $\mathbb{P}^2 \# m\overline{\mathbb{P}^2}$, $1 \leq m \leq 7$, they provided explicit examples of immersions and embeddings which exhaust all possible equivalence classes.

9.7 The Adjunction Inequality in Kähler Surfaces

In this and the following section we explore implications of Seiberg-Witten theory to the geometry of complex surfaces. We also explain the connection between the adjunction inequality and the *generalized Thom Conjecture*.

Let S be a compact oriented surface smoothly embedded in a complex surface X. Recall that $I_\pm(S)$ is the number of complex points of S in the given orientation class (see §9.2). By Theorem 9.4.1 (p. 414) we have

$$2I_\pm(S) = 2 - 2g(S) + [S]^2 \pm c_1(X)\cdot S = 2 - 2g(S) + [S]^2 \mp K_X \cdot S,$$

where $K_X = -c_1(X)$ denotes the canonical class of X. If S has no negative complex points then $I_-(S) = 0$ which is equivalent to the genus formula

$$2g(S) - 2 = [S]^2 - c_1(X)\cdot S = [S]^2 + K_X \cdot S. \tag{9.23}$$

This holds in particular if S is a smooth complex curve in a complex surface X, or a symplectic surface embedded in a symplectic 4-manifold (X,ω).

The next theorem summarizes some of the results of Fintushel and Stern [139], Kronheimer and Mrowka [311], [312, Theorem 1.7], Morgan, Szabó and Taubes [362], and Ozsváth and Szabó [391]. We restrict our attention to Kähler surfaces; a similar result holds in many other 4-manifolds [217, Theorem 2.4.8,

p. 53]. We denote by $b_2^+(X)$ the dimension of a maximal linear subspace $H^+(X;\mathbb{R})$ of $H^2(X;\mathbb{R})$ on which the intersection form is positive definite (see §9.3).

Theorem 9.7.1. *Assume that X is a compact Kähler surface with $b_2^+(X) > 1$ and $S \hookrightarrow X$ is a smoothly embedded closed oriented real surface of genus $g(S)$ which is not a 2-sphere with $[S] = 0 \in H_2(X;\mathbb{R})$. If $[S]^2 \geq 0$, or if $[S]^2 < 0$ and $g(S) > 0$, then S satisfies the adjunction inequality*

$$[S]^2 + |K_X \cdot S| \leq 2g(S) - 2. \tag{9.24}$$

If the canonical bundle K_X is ample then (9.24) also holds if S is an embedded sphere with $[S]^2 < 0$. If $\pi\colon S \looparrowright X$ is an immersion with $\delta_+(\pi)$ positive (simple) double points and if $g(S) + \delta_+(\pi) > 0$ then

$$[\pi(S)]^2 + |K_X \cdot \pi(S)| \leq 2g(S) + 2\delta_+(\pi) - 2. \tag{9.25}$$

A sketch of proof is given at the end of the section.

The main interest of the adjunction inequality (9.24) to Stein geometry is that by (9.12) it is equivalent to $I_\pm(S) \leq 0$, and by Corollary 9.6.1 (p. 421) such S can be $\mathcal{C}^0$ approximated by an isotopic embedding with tubular Stein neighborhoods. Conversely, Theorem 9.8.2 below shows that the adjunction inequality also holds for embedded surfaces in a Stein surface thanks to a Kähler embedding theorem (Theorem 9.8.1, p. 434). This connection, which was observed by Lisca and Matić [343] and Nemirovski [372], has interesting applications and gives nontrivial obstructions to the existence of Stein structures on smooth 4-manifolds. For immersed surfaces the situation is less clear since $I_\pm(\pi) \leq 0$ is equivalent to the stronger inequality

$$[\pi(S)]^2 + |K_X \cdot \pi(S)| + 2\delta_-(\pi) \leq 2g(S) + 2\delta_+(\pi) - 2 \tag{9.26}$$

obtained by adding the contribution of negative double points. This is false in general since the number $\delta_-(\pi)$ can be increased at will by attaching to $\pi(S)$ sufficiently many homologically trivial immersed spheres with a negative double point (see the proof of Corollary 9.6.7, p. 424). However, see Theorem 9.8.6 below for immersed surfaces in Stein surfaces.

Remark 9.7.2. (A) If the homology class of S is a torsion element of $H_2(X;\mathbb{Z})$ then the right hand side of (9.24) clearly vanishes; in this case the adjunction inequality trivially holds when $g(S) > 0$ and it fails for the 2-sphere. Thus it suffices to consider surfaces which are *essential* in the sense that $[S]$ is an element of infinite order in $H_2(X;\mathbb{Z})$; equivalently, $[S] \neq 0 \in H_2(X;\mathbb{R})$.

(B) Comparing (9.24) with the genus formula (9.23) (p. 426) we see that a smooth complex curve C in a complex surface X satisfies the adjunction inequality (9.24) if and only if $K_X \cdot C \geq 0$. If this holds for all complex curves

in X then K_X is said to be *numerically effective* (nef). If K_X is ample then $[K_X]^2 > 0$ and $K_X \cdot C > 0$ for every complex curve $C \subset X$; a surface with ample canonical bundle is minimal and of general type.

If on the other hand we have $K_X \cdot C < 0$ for a complex curve $C \subset X$ then (9.24) clearly fails for C, and possibly also for some surfaces of higher genus representing the homology class $[C]$. This happens for example in the projective plane $X = \mathbb{P}^2$ as was seen in the previous section, so the hypothesis $b_2^+(X) > 1$ can not be removed in general. However, by Proposition 9.7.6 below this can not happen if X is a compact Kähler surface with $b_2^+(X) > 1$ and $g(C) > 0$. Here is an alternative argument. On a Kähler surface X, $b_2^+(X) > 1$ is equivalent to $h^{2,0}(X) > 0$ (the Morse index theorem says that there can be only one positive eigenvalue of the intersection form on $H^{1,1}(X)$). This implies that X can not be uniruled. Hence it has a minimal model Y whose canonical class K_Y is nef, and $K_X = \mu^* K_Y + \sum PD(E_i)$ where $\mu\colon X \to Y$ is the projection and $E_i \subset X$ are the exceptional curves. The only curves in X with $K_X \cdot C < 0$ are the exceptional rational curves E_i. All these things can be found in [33]. $\square$

The adjunction inequality (9.24) is closely related to the *Thom conjecture* stating that a smooth embedded complex curve C in a complex surface X has the smallest genus among all oriented smoothly embedded surfaces $S \hookrightarrow X$ with $[C] = [S] \in H_2(X;\mathbb{Z})$. The following summarizes results of several authors (Fintushel and Stern [139], Kronheimer and Mrowka [311, 312], Morgan, Szabo and Taubes [362], Ozsváth and Szabó [391]).

Theorem 9.7.3. (Generalized Thom Conjecture) *A smooth embedded complex curve in a compact Kähler surface is genus minimizing in its homology class. More generally, an embedded symplectic surface in a closed symplectic 4-manifold is genus minimizing in its homology class.*

To see the connection with the adjunction inequality (9.24), assume that C is a smooth complex or symplectic curve in X and $S \subset X$ is an embedded surface with $[S] = [C] \in H_2(X;\mathbb{Z})$. We have

$$[S]^2 + |c_1(X) \cdot S| = C \cdot C + |c_1(X) \cdot C| \geq 2g(C) - 2,$$

where the last inequality follows from the genus formula (9.23). If (9.24) is valid for any such S, it follows that $g(S) \geq g(C)$ and hence the Thom conjecture holds for the class $[C] \in H_2(X;\mathbb{Z})$. Conversely, if C satisfies the Thom conjecture and if $K_X \cdot C \geq 0$, then $K_X \cdot C = |K_X \cdot C| = |K_X \cdot S|$; thus

$$g(S) \geq g(C) = 1 + \frac{1}{2}\left([C]^2 + K_X \cdot C\right) = 1 + \frac{1}{2}\left([S]^2 + |K_X \cdot S|\right)$$

and hence S satisfies (9.24). Although Theorem 9.7.3 does not formally imply Theorem 9.7.1 (since not all homology classes in $H_2(X;\mathbb{Z})$ are represented by complex curves), the proofs of both results follow the same pattern.

Before proceeding we recall the basic notions and results of Seiberg-Witten theory. We refer to Morgan's monograph [361] for a more complete exposition. An excellent introductory survey is Appendix 2.4 in [217].

If X is a closed, oriented, smooth 4-manifold with a Riemannian metric g then the *Seiberg-Witten invariant* of X is an integer valued function

$$SW_{X,g}\colon Spin^c(X) \to \mathbb{Z} \tag{9.27}$$

defined on the set of all spin^c structures on X. We explain what is a spin^c structure. Recall that the special orthogonal group $SO(4)$ has the fundamental group equal to $\mathbb{Z}_2 = \mathbb{Z}/2\mathbb{Z}$. The universal (double) covering group of $SO(4)$ is $Spin(4) = SO(4) \times \{\pm 1\}/\{\pm(I,1)\}$. The Lie group $Spin^c(4)$ is defined by

$$Spin^c(4) = Spin(4) \times S^1/\{\pm(I,1)\},$$

where $S^1 = U(1)$ is the unit circle in $\mathbb{C}$. By projecting $Spin^c(4)$ onto the first factor $Spin(4)$ and noting that, by doing so, we actually pass to the quotient $Spin(4)/\{\pm I\} = SO(4)$, we obtain an exact sequence (an S^1-fibration)

$$1 \longrightarrow S^1 \longrightarrow Spin^c(4) \xrightarrow{\tau} SO(4) \longrightarrow 1.$$

Let $P \to X$ denote the principal $SO(4)$-bundle consisting of oriented g-orthonormal frames of the tangent bundle $\mathrm{T}X$. A spin^c *structure* on X is a lifting of P to a principal $Spin^c(4)$-bundle $\widetilde{P} \to X$, together with a bundle map $\tau\colon \widetilde{P} \to P$ over X which equals the map $\tau\colon Spin^c(4) \to SO(4)$ fiberwise, so it gives an identification $\widetilde{P} \times_\tau SO(4) = P$. More precisely, there is an open cover $\{U_j\}$ of X and a 1-cocyle (g_{ij}) with values in the sheaf $\mathcal{C}_X^{Spin^c(4)}$ of smooth maps $X \to Spin^c(4)$ defining the principal $Spin^c(4)$-bundle $\widetilde{P} \to X$ such that the corresponding 1-cocycle $(\tau \circ g_{ij})$ with values in the sheaf $\mathcal{C}_X^{SO(4)}$ defines the bundle P. Such lifting of the principal $SO(4)$-bundle $P \to X$ to a principal $Spin^c(4)$-bundle $\widetilde{P} \to X$ always exists.

The group $Spin^c(4)$ also admits the presentation

$$Spin^c(4) = \{(A,B) \in U(2) \times U(2)\colon \det A = \det B\}. \tag{9.28}$$

This gives projections $\mu^\pm\colon Spin^c(4) \to U(2)$ onto the first and the second factor. Using $\mu^\pm$ we associate to a spin^c structure $(\widetilde{P},\tau) \in Spin^c(X)$ a pair of unitary vector bundles with fiber $\mathbb{C}^2$,

$$W^+ = \widetilde{P} \times_{\mu^+} \mathbb{C}^2 \to X, \quad W^- = \widetilde{P} \times_{\mu^-} \mathbb{C}^2 \to X,$$

called the bundles of *positive and negative spinors*, respectively. This gives a presentation of a spin^c structure on X as a triple $\mathfrak{s} = (\rho, W^+, W^-)$, where W^+, W^- are smooth $U(2)$-bundles over X with isomorphic $U(1)$ determinant bundles $\det W^+ = \det W^-$ and

$$\rho\colon \mathrm{T}^*X \otimes \Gamma(X; W^+) \to \Gamma(X; W^-) \tag{9.29}$$

is the *Clifford multiplication*, a purely algebraic operation (see e.g. [361]). The bundles $W^\pm$ and the Cliffor multiplication ρ can also be explained by the following representations of $Spin^c(4)$ on the field of quaternions $\mathbb{H}$ (see [217, p. 57]). Let us use the presentation

$$Spin^c(4) = S^1 \times SU(2) \times SU(2)/\{\pm(1, I, I)\}.$$

Identifying $SU(2)$ with the group of unit quaternions in $\mathbb{H}$, an element of $a = (\lambda, q_1, q_2) \in Spin^c(4)$ is given by a unit complex number $|\lambda| = 1$ and two unit quaternions q_1, q_2. The representations $\rho_0(\lambda, q_1, q_2)(h) = q_1 h \bar{q}_2$ ($h \in \mathbb{H}$, quaternionic multiplication) results in $\widetilde{P} \times_{\rho_0} \mathbb{H} = \mathrm{T}X$. The representations $\rho_+(\lambda, q_1, q_2)(h) = q_1 h \lambda$ and $\rho_-(\lambda, q_1, q_2)(h) = q_2 h \lambda$ result in $\widetilde{P} \times_{\rho_\pm} \mathbb{H} = W^\pm$. It follows easily that $\mathrm{T}X \otimes \mathbb{C} \cong \mathrm{Hom}_{\mathbb{C}}(W^+, W^-)$, and this isomorphism gives the Clifford multiplication $\Gamma(X; \mathrm{T}X \otimes W^+) \to \Gamma(X; W^-)$. By using the Riemannian metric g on X we identify $\mathrm{T}X$ with T^*X, and this gives the Clifford product (9.29).

The Seiberg-Witten monopole equation is an elliptic system of two nonlinear partial differential equations that involve a smooth section $\psi\colon X \to W^+$ of the positive spinor bundle and a smooth $U(1)$-connection A on the unitary line bundle $L = \det W^+ = \det W^-$ over X. The connection A on L, together with the Riemannian connection on $\mathrm{T}X$ determined by the metric g, determine a covariant derivative ∇_A on any spin^{c} structure on X. This derivative maps smooth sections of W^+ to smooth sections of $\mathrm{T}^*X \otimes W^+$. By composing ∇_A with the Clifford product (9.29) we get the *Dirac operator*

$$\partial\!\!\!/_A = \rho \circ \nabla_A\colon \Gamma(X, W^+) \to \Gamma(X, W^-).$$

Given a pair (ψ, A), where A is a unitary connection on $L = \det W^+$ and ψ is a smooth section of W^+, the *unperturbed Seiberg-Witten equations* are

$$\partial\!\!\!/_A \psi = 0, \quad F_A^+ = \mathrm{i}q(\psi) = \mathrm{i}\left(\psi \otimes \psi^* - \frac{|\psi|^2}{2}\mathrm{Id}\right)$$

where F_A^+ is the self-dual part (with respect to the Hodge operator $*_g$ associated to the metric g) of the curvature form F_A of the connection A, and $q(\psi)$ is naturally identified with a self-dual 2-form. (See [361, §4.1].) For technical reasons one replaces the second equation by a perturbed equation

$$F_A^+ = \mathrm{i}q(\psi) + \mathrm{i}\eta$$

for some self-dual two-form η on X. The gauge transformations $X \to U(1) = S^1$ act on the space of solutions (A, ψ) of this system. For a generic choice of the metric g on X and of the perturbation η, the moduli space $\mathcal{M}_X(\mathfrak{s})$ of solutions (A, ψ) of the perturbed Seiberg-Witten equations $\partial\!\!\!/_A \psi = 0$, $F_A^+ = \mathrm{i}q(\psi) + \mathrm{i}\eta$ is a compact smooth oriented manifold of dimension

$$d(\mathfrak{s}) = \dim_{\mathbb{R}} \mathcal{M}_X(\mathfrak{s}) = \frac{c_1(\mathfrak{s})^2 - 2\chi(X) - 3\sigma(X)}{4}. \tag{9.30}$$

Here $c_1(\mathfrak{s})^2 \in \mathbb{Z}$ is the square of the Chern class $c_1(\mathfrak{s}) \in H^2(X;\mathbb{Z})$ (an element of $H^4(X;\mathbb{Z})$) evaluated on the fundamental class $[X] \in H_4(X;\mathbb{Z})$, $\chi(X)$ is the Euler number of X, and $\sigma(X) = b_2^+(X) - b_2^-(X)$ is the signature of the intersection form on $H_2(X;\mathbb{Z})$. In the most interesting case when $d(\mathfrak{s}) = 0$, $SW_{X,g}(\mathfrak{s})$ is simply the number of solutions of the Seiberg-Witten equation (up to gauge transformations $X \to U(1)$ and counted with orientation signs). The definition of $SW_{X,g}$ is somewhat more involved when $d(\mathfrak{s}) > 0$.

Remark 9.7.4. The case of interest to us is when X is a compact Kähler surface. Recall that a cohomology class $c \in H^2(X;\mathbb{Z})$ is a *characteristic class* if c is an integral lift of the second Stiefel-Whitney class $w_2(X) \in H^2(X;\mathbb{Z}_2)$, that is, if $c \equiv w_2(X) \pmod 2$. We denote the set of all characteristic classes by $\mathcal{C}_X$. By a theorem of Hirzebruch and Hopf [265], a smooth, compact, oriented 4-manifold X admits an almost complex structure J if and only if there is a characteristic class $c \in \mathcal{C}_X \subset H^2(X;\mathbb{Z})$ satisfying

$$\langle c^2, [X]\rangle - 2\chi(X) - 3\sigma(X) = 0;$$

in this case there exists an almost complex structure J on X with $c_1(X,J) = c$. Thus $d(\mathfrak{s}) = 0$ for any spin^c structure $\mathfrak{s}$ with $c_1(\mathfrak{s}) = c_1(X,J) = -K_X$. $\square$

The hypothesis $b_2^+(X) > 1$ insures that the Seiberg-Witten invariant (9.27) is independent of the generic choice of the metric g on X, so it is an invariant SW_X of the $\mathcal{C}^\infty$ structure on X. Furthermore, for any smooth map $f\colon X \to Y$ and spin^c structure $\mathfrak{s} \in Spin^c(Y)$ we have $SW_X(f^*\mathfrak{s}) = \pm SW_Y(\mathfrak{s})$. Assume this to be the case from now on.

There is a finite-to-one map $\iota\colon Spin^c(X) \to \mathcal{C}_X \subset H^2(X;\mathbb{Z})$ of $Spin^c(X)$ onto the set $\mathcal{C}_X$ of characteristic classes of X. The map ι associates to any spin^c structure $\mathfrak{s} = (\rho, W^+, W^-)$ the first Chern class $c_1(W^+)$ of the positive spinor bundle W^+. We write $\iota(\mathfrak{s}) = c_1(\mathfrak{s}) = c_1(W^+)$. One also defines a function $SW_X\colon \mathcal{C}_X \to \mathbb{Z}$ by $SW_X(c) = \sum_{\iota(\mathfrak{s})=c} SW_X(\mathfrak{s})$. If the group $H_2(X;\mathbb{Z})$ has no 2-torsion (which holds in particular if X is simply connected) then ι is bijective and hence the above distinction becomes unnecessary.

A spin^c structure $\mathfrak{s}$ on X is said to be a *Seiberg-Witten structure* if $SW_X(\mathfrak{s}) \neq 0$; a characteristic class $c \in \mathcal{C}_X$ is a *Seiberg-Witten class* if $SW_X(\mathfrak{s}) \neq 0$ for some $\mathfrak{s} \in Spin^c(X)$ with $c_1(\mathfrak{s}) = c$. At most finitely many characteristic classes (and the associated spin^c structures) are Seiberg-Witten classes. A smooth 4-manifold X is said to be of *Seiberg-Witten simple type* if for any $\mathfrak{s} \in Spin^c(X)$ with $SW_X(\mathfrak{s}) \neq 0$ we have $d(\mathfrak{s}) = 0$; that is, for every $\mathfrak{s} \in Spin^c(X)$ with $SW_X(\mathfrak{s}) \neq 0$ the associated Seiberg-Witten equations have finitely many solutions up to gauge equivalence. So far there seem to be no known examples of 4-manifolds that are not of Seiberg-Witten simple type.

We quote the following result of Witten and Taubes.

Theorem 9.7.5. [521, 485, 486] *Every compact Kähler surface X satisfying $b_2^+(X) > 1$ is of Seiberg-Witten simple type and satisfies $SW_X(\pm K_X) \neq 0$, where K_X is the canonical class. If such X is also minimal and of general type then $\pm K_X$ are the only Seiberg-Witten classes of X.*

Proof (of Theorem 9.7.1.). Assume first that S is embedded in X.

Case 1: $[S]^2 \geq 0$. In this case S can not be a sphere. Indeed, the existence of an embedded homologically essential sphere S with $[S]^2 \geq 0$ in a compact 4-manifold X implies that the Seiberg-Witten invariant $SW_X\colon Spin^c(X) \to \mathbb{Z}$ is the zero function [139, Lemma 5.1, p. 154]. (The proof amounts to showing that the existence of a Seiberg-Witten class would produce infinitely many Seiberg-Witten classes in X, a contradiction.) In our case this is not so according to Theorem 9.7.5. Assume now that $g(S) > 0$. Then Theorem 9.7.1 is a consequence of Proposition 4.2 in [362, p. 714] (the key step in the proof of Thom Conjecture). We state the special case of interest to us.

Proposition 9.7.6. [362, Proposition 4.2] *Assume that X is a compact Kähler surface with $b_2^+(X) > 1$ and $S \subset X$ is a smooth embedded real surface with $[S]^2 \geq 0$ and $g(S) > 1$. If for some Seiberg-Witten class $K \in \mathcal{C}_X$ we have $K \cdot S + [S]^2 = 2g(S) - 2$, then S has the smallest genus among all smooth embedded real surface $S' \subset X$ with $[S'] = [S] \in H_2(X;\mathbb{Z})$.*

For the canonical class $K = K_X$, the hypothesis in Proposition 9.7.6 is precisely that S satisfies the genus formula for complex curves. The proof reduces to the case $[S]^2 = 0$ as follows. Let $\widetilde{X} = X \# k\overline{\mathbb{P}^2} \xrightarrow{\mu} X$ be obtained by blowing up X at $k = [S]^2 \geq 0$ distinct points of S. Let $E_1, \ldots, E_k \subset \widetilde{X}$ be the exceptional fibers of the blown-up points; so $E_j \cong \mathbb{P}^1$ and $[E_j]^2 = -1$. Let $S' \subset \widetilde{X}$ be the connected sum of S with $E_1, \ldots, E_k$. Note that $g(S') = g(S)$ and that S' is homologous to the total transform $\mu^{-1}(S) \subset \widetilde{X}$ of S. Since K is a Seiberg-Witten class in X, $K' = \mu^* K + \sum_{j=1}^k PD(E_j) \in H^2(\widetilde{X};\mathbb{Z})$ is a Seiberg-Witten class in $\widetilde{X}$ [139]. It is easily seen that $S' \cdot S' = [S]^2 - k = 0$ and $K' \cdot S' = K \cdot S + k$. It now suffices to prove Proposition 9.7.6 for the pair (K', S') in $\widetilde{X}$. This is done by using a product formula for Seiberg-Witten invariants [362, Theorem 3.1].

To prove Theorem 9.7.1 in the case $[S]^2 \geq 0$, assume that the adjunction inequality (9.24) fails for a smooth embedded surface $S \subset X$ of genus $g(S) > 0$. Then $[S]^2 + |K_X \cdot S| = 2g(S) - 2 + k$ for a positive integer $k > 0$. Choose $\epsilon = \pm 1$ so that $\epsilon K_X \cdot S = |K_X \cdot S|$. Let $C \subset X$ be a surface obtained by adding k homologically trivial handles to S. Then $[C] = [S] \in H_2(X;\mathbb{Z})$ and $g(C) = g(S) + k > 1$, so $C \cdot C + \epsilon K_X \cdot C = g(C) - 2$. Since $SW_X(\epsilon K_X) = \pm 1$, Proposition 9.7.6 implies that C has the smallest genus in its homology class, a contradiction to $g(S) < g(C)$.

Case 2: $[S]^2 < 0$. If $g(S) > 0$, one gets the adjunction inequality by applying the following result with the canonical class $K = K_X$.

Theorem 9.7.7. (Ozsváth and Szabó [391, Corollary 1.7]) *Let X be a compact oriented 4-manifold of Seiberg-Witten simple type with $b_2^+(X) > 1$, and let $S \subset X$ be a smoothly embedded, oriented closed real surface of genus $g(S) > 0$ and with $[S]^2 < 0$. Then for all Seiberg-Witten classes $K \in \mathcal{C}_X$*

$$[S]^2 + |K \cdot S| \leq 2g(S) - 2. \tag{9.31}$$

This is an immediate consequence of the main result in [391], Theorem 1.3 on p. 96. The latter asserts that if the adjunction inequality (9.31) fails for a certain Seiberg-Witten class $K \in \mathcal{C}_X$ then $K' = K + 2\epsilon PD(S) \in \mathcal{C}_X$ is also a Seiberg-Witten class; here $PD(S)$ is the Poincaré dual of $[S]$ and $\epsilon = \pm 1$ is the sign of $K \cdot S$. (Remark 1.6 in [391, p. 97] explains the presence of the coefficient 2, as opposed to [391, Theorem 1.3] which pertains to a spin^c structure $\mathfrak{s}$ with $c_1(\mathfrak{s}) = K$.) From (9.30) we get

$$d(K') = \frac{1}{4}\big[(K + 2\epsilon PD(S))^2 - 2\chi(X) - 3\sigma(X)\big] = d(K) + \epsilon K \cdot S + PD(S)^2.$$

If (9.31) fails then $\epsilon K \cdot S + PD(S)^2 = |K \cdot S| + [S]^2 \geq 2g \geq 2$. This implies $d(K') \geq 2$, contradicting the assumption that X is of simple type.

The last conclusion fails if S is the sphere ($g = 0$). However, under the stated conditions on X, its only Seiberg-Witten classes are $\pm K_X$ by Theorem 9.7.5. Applying the above argument with $K = K_X$ we see as before that $K' = K_X + 2\epsilon PD(S)$ is a Seiberg-Witten class. Hence either $K' = K_X$ which gives $PD(S) = 0$, a contradiction since S is essential, or $K' = -K_X$ which gives $PD(S) = \pm K_X$. In the latter case we have $[S]^2 = PD(S)^2 = K_X^2 > 0$ since K_X is assumed to be ample. This contradicts the assumption $[S]^2 < 0$ and thus proves the adjunction inequality (9.31) also for spheres.

It remains to consider immersed surfaces. Assume that $\pi\colon S \looparrowright X$ is an immersion with simple double points, say δ_+ positive ones and δ_- negative ones. We replace each of the positive double points by an embedded totally real handle in X as described in Remark 9.4.2 on p. 416. This replaces $\Sigma = \pi(S) \subset X$ by a homologically equivalent immersed surface $\Sigma' = \pi'(S') \subset X$ of genus $g(S') = g(S) + \delta_+$. Let Y be the complex surface obtained by blowing up X at each of the remaining (negative) double points p_j of Σ'. Let $\iota\colon Y \to X$ denote the projection and $E_j = \iota^{-1}(p_j) \cong \mathbb{P}^1 \subset Y$ the exceptional fibers. Then Y is a compact Kähler surface with $b_2^+(Y) > 1$ containing an embedded copy of S', namely the proper transform A of $\Sigma' \subset X$. We claim that $[A] = [\iota^{-1}(\Sigma')] \in H_2(\widetilde{X}; \mathbb{Z})$. To see this, observe that a local model for Σ' at any negative double point p_j is $L_1 \cup \bar{L}_2 \subset \mathbb{C}^2$, where $L_1 \neq L_2$ is a pair of complex lines through $p = 0 \in \mathbb{C}^2$ and the bar on L_2 means the reversed orientation. The class of the total transform $[\iota^{-1}(\Sigma')]$ is obtained from the proper transform by adding a copy of $[E_j]$ (the contribution of L_1) and also a copy of $-[E_j]$ (the contribution of $\bar{L}_2$), so $[E_j]$ cancels out. This proves the claim. It follows that $[A]^2 = [\Sigma']^2$ and $K_Y \cdot A = K_X \cdot \Sigma'$. The first property is obvious, while the second follows from $K_Y = \iota^*(K_X) + \sum_j PD(E_j)$ and

$PD(E_j)\cdot[A] = [E_j]\cdot[A] = [E_j]\cdot[\iota^{-1}(\Sigma')] = 0$. Hence the left hand side of the inequality (9.25) remains unchanged when passing from $\Sigma' \subset X$ to $A \subset Y$, and the right hand side equals $2g(S') - 2$. Since $g(S') = g(S) + \delta_+ > 0$, the adjunction inequality (9.24) for embedded surfaces gives (9.25).

This leaves the case when $\Sigma \subset X$ is an immersed sphere with only negative double points. We still get the same conclusion if either $[\Sigma]^2 \geq 0$, or if $[\Sigma]^2 < 0$ and K_X is ample (so $\pm K_X$ are the only Seiberg-Witten classes of X). To see this, let $Y \to X$ and $A \subset Y$ be as above, so A is an embedded homologically nontrivial sphere. The case $[A]^2 \geq 0$ can be ruled out as before. Assume now that $[A]^2 < 0$ and that K_X is ample. Then the Seiberg-Witten classes of Y are $\pm\iota^*(K_X) + \sum_j \pm PD(E_j)$ (see [217, p. 54]). If (9.25) fails, we conclude as in the embedded case that $K_Y + 2\epsilon PD(A) \in H^2(Y;\mathbb{Z})$ is a Seiberg-Witten class for at least one of the choices of $\epsilon = \pm 1$. Hence $PD(A) = \pm K_Y$ modulo the exceptional classes $PD(E_j)$. Dualizing gives $[A] = \pm[K_Y]$ in $H^2(Y;\mathbb{Z})$ modulo the exceptional curves $[E_j]$. Pushing down to X by the map $\iota\colon Y \to X$ we get $[\Sigma] = \pm[K_X]$ and hence $[A]^2 = [\Sigma]^2 = [K_X]^2 > 0$, a contradiction to the initial assumption that $[A]^2 < 0$. $\square$

9.8 The Adjunction Inequality in Stein Surfaces

It was observed by Nemirovski [372] and Lisca and Matić [343] that Theorem 9.7.1 (the adjunction inequality in compact Kähler surfaces) also implies the adjunction inequality for real surfaces in Stein surfaces. The main point of the reduction is the following embedding theorem (see Theorem 3.2 and Corollary 1.4 in [343]). Similar results without the Kähler condition (which is actually not used here) were obtained earlier by Stout [472] and by Demailly, Lempert and Schiffman [105, 336].

Theorem 9.8.1. *Let X be a Stein manifold and $\rho\colon X \to \mathbb{R}$ a smooth strongly plurisubharmonic function. Let $r \in \mathbb{R}$ be a regular value of ρ. Then there exists a biholomorphic map $\phi\colon X_r = \{x \in X\colon \rho(x) < r\} \to \Omega$ onto a domain in a smooth minimal projective variety Y with ample canonical bundle K_Y, having a Kähler form ω whose pull-back to X_r equals the Kähler form ω_ρ determined by the function ρ. If $\dim_{\mathbb{C}} X = 2$ then Y may be chosen such that $b_2^+(Y) > 1$. Furthermore, if $S \subset X_r$ is an embedded real surface with $[S] \neq 0 \in H_2(X;\mathbb{Z})$ then we can arrange that $\phi(S)$ is a homologicaly essential surface in Y.*

Corollary 9.8.2. (Lisca and Matić [343], Nemirovski [372]) *If S is a closed, connected, oriented real surface smoothly embedded in a Stein surface X and S is not a homologically trivial two-sphere, then*

$$[S]^2 + |K_X\cdot S| \leq 2g(S) - 2.$$

In particular, a homologically nontrivial embedded sphere in a Stein surface satisfies $[S]^2 \leq -2$. If $\pi\colon S \looparrowright X$ is an immersion and $g(S) + \delta_+(\pi) > 0$ then

$$\pi(S)^2 + |K_X \cdot \pi(S)| \leq 2g(S) + 2\delta_+(S) - 2.$$

Proof. A Stein surface has no homology above level 2 and the group $H_2(X;\mathbb{Z})$ is without torsion (see §3.11). Hence $[S] \neq 0 \in H_2(X;\mathbb{Z})$ implies that S is essential in X. By Theorem 9.8.1 there exists a domain $\Omega \subset X$ containing S and a biholomorphic map $\phi\colon \Omega \to \Omega'$ onto a domain in a minimal algebraic (hence Kähler) surface Y of general type, with ample canonical bundle K_Y and satisfying the condition $b_2^+(Y) > 1$. The construction in [343] also insures that $\phi(S)$ is essential in Y. The right hand side of the adjunction inequality (9.24) does not change if we replace $S \subset X$ with its image $\phi(S) \subset Y$ since the numbers $[S]^2$ and $K_X \cdot S$ only depend on the neighborhood of S in the ambient surface. Therefore Corollary 9.8.2 follows from Theorem 9.7.1 applied to $\phi(S) \subset Y$. The same argument holds in the immersed case. □

Example 9.8.3. The smooth 4-manifold $S^2 \times \mathbb{R}^2$ does not admit any Stein structure in view Corollary 9.8.2 since it contains the homologically essential embedded sphere $S^2 \times \{0\}$ with self-intersection number zero. □

Example 9.8.4. If X is a Stein surface then a surface Y obtained by blowing up finitely many points in X does not admit any Stein structure. In fact, the exceptional fiber of a blown up point is an embedded rational curve with self-intersection number -1, so the claim follows from Corollary 9.8.2. □

We have seen in §9.6 that the adjunction inequality holds for an embedded oriented surface S of degree $d > 0$ in $\mathbb{P}^2$ if and only if $2g(S) \geq \mathrm{d}^2 + 3d + 2 \geq 6$. This fails if S is obtained by adding at most $3d - 1$ homologically trivial handles to a smooth complex curve of degree $d > 0$; hence such a surface does not admit any Stein neighborhoods. Going a step further, S. Nemirovski proved the following result that in particular answers a question of Vitushkin.

Theorem 9.8.5. [372, Theorem 10] *Let $S \subset \mathbb{P}^2$ be a smoothly embedded oriented surface of genus g and of degree $d > 0$. If there exists a nonconstant holomorphic function in a neighborhood of S then $2g(S) \geq d^2 + 3d + 2 \geq 6$. In particular, if S is a homologically nontrivial embedded sphere in $\mathbb{P}^2$ then every holomorphic function in a neighborhood of S is constant.*

Proof. Assume that there is a nonconstant holomorphic function in a neighborhood $U \subset \mathbb{P}^2$ of S. Then the envelope of holomorphy of U is a Riemann domain $\pi\colon \Omega \to \mathbb{P}^2$ whose total space Ω is a Stein surface. (The envelope of holomorphy exists for every domain $U \subset \mathbb{P}^n$; see the references in [372, §1.4]. If U does not admit any nonconstant holomorphic functions then its envelope equals $\mathbb{P}^n$; otherwise it is a Stein manifold.) Note that Ω contains a domain $\widetilde{U}$ that projects by π bijectively onto U. The surface $S' = (\pi|_{\widetilde{U}})^{-1}(S) \subset \Omega$ is homologically nontrivial in Ω (otherwise $S = \pi(S')$ would be homologically trivial in $\mathbb{P}^2$), and it clearly satisfies

$$[S']^2 + |K_\Omega \cdot S'| = [S]^2 + |K_{\mathbb{P}^2} \cdot S| = d^2 + 3d.$$

By Corollary 9.8.2 the surface $S' \subset \Omega$ satisfies the adjunction inequality:

$$2g(S') - 2 \geq [S']^2 + |K_\Omega \cdot S'|.$$

Since $g(S) = g(S')$, the result follows. □

We have seen that the stronger inequality (9.26) for an immersed surface $\pi: S \looparrowright X$ is false in general. However, Nemirovski proved in [374] that the *essential negative double points* still contribute to (9.26). Let us explain this notion. A double point $x \in \pi(S) \subset X$ is the image of two different points $p \neq q \in S$. Let us connect p and q by an arc in S. Its image in X is a loop at x. The double point x is called *essential* if the homotopy class of any such 'double point loop' at x does not belong to the image $\pi_*(\pi_1(S))$ of the fundamental group of S in the fundamental group $\pi_1(X; x)$.

Theorem 9.8.6. [374, Theorem 4.1] *Let S be a closed oriented surface of genus g and $\pi: S \looparrowright X$ an immersion with simple double points into a Stein surface X. Assume that π has δ_+ positive double points and δ_-^{ess} essential negative double points. Then either $\Sigma = \pi(S)$ is a homotopically trivial 2-sphere in X, or else we have*

$$[\Sigma]^2 + |K_X \cdot \Sigma| + 2\delta_-^{\mathrm{ess}} \leq 2g + 2\delta_+ - 2.$$

In particular, a smoothly embedded homotopically nontrivial 2-sphere $S \hookrightarrow X$ satisfies $[S]^2 + |K_X \cdot S| \leq -2$.

If $S \subset X$ is homotopically trivial then it is also homologically trivial, but the converse need not hold, so Theorem 9.8.6 is stronger than Corollary 9.8.2.

Proof. We first consider embedded spheres. Let $h: \widetilde{X} \to X$ be the universal covering of X. Then $\widetilde{X}$ is a simply connected Stein surface, and h induces an isomorphism $\pi_2(\widetilde{X}) \xrightarrow{\cong} \pi_2(X)$. (By a theorem of K. Stein [461] any covering space of a Stein manifold is Stein.) Assume that $\pi: S \hookrightarrow X$ is a smooth embedded 2-sphere in X whose image $\pi(S)$ is homotopically nontrivial in X. Since S is simply connected, π lifts to an embedding $\tilde{\pi}: S \hookrightarrow \widetilde{X}$, and $\widetilde{S} = \tilde{\pi}(S) \subset \widetilde{X}$ is homotopically nontrivial in $\widetilde{X}$. By Hurewicz theorem we have $\pi_2(\widetilde{X}) = H_2(\widetilde{X}; \mathbb{Z})$, and hence $\widetilde{S}$ is also homologically nontrivial in $\widetilde{X}$. It is easily seen that $[S]^2 = [\widetilde{S}]^2$ and $K_X \cdot S = K_{\widetilde{X}} \cdot \widetilde{S}$. Thus

$$[S]^2 + |K_X \cdot S| = [\widetilde{S}]^2 + K_{\widetilde{X}} \cdot \widetilde{S} \leq -2,$$

where the last inequality holds by Corollary 9.8.2.

Assume now that $\pi: S \looparrowright X$ is an immersed surface with δ_+ positive double points and with δ_-^{ess} essential negative double points. We replace each positive

double point by a handle as before; this does not change the number $g + \delta_+$ or the homology class of the image $\Sigma = \pi(S)$ in X. Let $h\colon \widetilde{X} \to X$ be a covering space determined by the condition that $h_*(\pi_1(\widetilde{X})) = \pi_*(\pi_1(S))$. This means that every loop in X which does not come (up to homotopy) from a nontrivial loop in S opens up in $\widetilde{X}$. Then the immersion $\pi\colon S \looparrowright X$ lifts to an immersion $\widetilde{\pi}\colon S \looparrowright \widetilde{X}$ which removes all essential negative double points (since the preimages $p \neq q \in S$ of any such point are mapped by $\widetilde{\pi}$ to different levels in $\widetilde{X}$). One can easily see that $\widetilde{\Sigma} = \widetilde{\pi}(S) \subset \widetilde{X}$ satisfies $[\widetilde{\Sigma}]^2 = [\Sigma]^2 + 2\delta_-^{\text{ess}}$. Applying Corollary 9.8.2 to $\widetilde{\Sigma} \subset \widetilde{X}$ gives the conclusion. □

Theorem 9.8.6 has several interesting applications. For example, if Y is a smooth open 3-manifold then $X = Y \times \mathbb{R}$ admits a handlebody decomposition without handles of index > 2, and hence X is *homeomorphic* to a Stein surface according to Theorem 9.10.1 (b) on p. 446 below. However, such X is in general not *diffeomorphic* to a Stein surface as is shown by the following result.

Corollary 9.8.7. [374, Corollary 2.1] *If Y is a smooth open 3-manifold such that $Y \times \mathbb{R}$ is diffeomorphic to a Stein surface, then every smooth embedded 2-sphere in Y bounds a 3-ball.*

Proof. Assume that $Y \times \mathbb{R}$ admits the structure of a Stein surface. Let $S \subset Y$ be an embedded 2-sphere. Then $S \times \{0\} \subset Y \times \{0\} \subset Y \times \mathbb{R}$ is an embedded 2-sphere with self-interesection number zero; hence its homotopy class in $Y \times \mathbb{R}$ is trivial by Theorem 9.8.6. By composing with the projection $Y \times \mathbb{R} \to Y$ we see that S is homotopically trivial in Y. A null-homotopic embedded 2-sphere S in a 3-manifold Y bounds a homotopy 3-ball B in Y [249, Proposition 3.10]. Gluing to B a 3-ball along $bB = S$ we get a homotopy 3-sphere Σ. By Perelman's solution of the Poincaré Conjecture [392] (*a simply connected closed 3-manifold is homeomorphic to the 3-sphere*), Σ is homeomorphic to the 3-sphere. Since S is smoothly embedded in Y, S is a locally flat 2-sphere in the 3-sphere Σ, and hence each of the two sides determined by S are homeomorphic to a 3-ball. □

Another application is the following result which is related to the Bedford-Gaveau-Klingenberg-Kruzhilin theorem concerning the envelope of holomorphy of certain embedded 2-spheres in $\mathbb{C}^2$ [36, 313].

Corollary 9.8.8. (Nemirovski [374]) *A smoothly embedded 2-sphere S in a domain $U \subset \mathbb{C}^2$ is homotopically trivial in the envelope of holomorphy of U.*

Proof. The envelope of holomorphy of $U \subset \mathbb{C}^2$ is a Riemann domain $\widetilde{U} \to \mathbb{C}^2$ containing a biholomorphic copy of U; this embeds S in $\widetilde{U}$. Note that $[S]^2 = 0$ and hence the adjunction inequality fails (in U and also in $\widetilde{U}$). Since $\widetilde{U}$ is a Stein surface, Theorem 9.8.6 implies that S is homotopically trivial in $\widetilde{U}$. □

Remark 9.8.9. Corollary 9.8.8 is particular to dimension two. For example, the 3-sphere admits a totally real embedding $S \subset \mathbb{C}^3$ [11]; a thin tubular neighborhood of S is a strongly pseudoconvex domain in $\mathbb{C}^3$ in which S is homotopically nontrivial. Also, for any $n \geq 3$ the n-sphere admits a homologically nontrivial embedding in the (Stein) complement $X = \mathbb{C}^n \backslash \Sigma$ of a generic algebraic hypersurface Σ of degree ≥ 3 in $\mathbb{C}^n$ (Nemirovski [373]). See also Example 9.12.4 on p. 460. $\square$

9.9 Well Attached Handles

In this section we prove a key lemma that will be used in the Eliashberg-Gompf construction of integrable Stein manifold structures on a smooth almost complex manifold with a correct handlebody decomposition (see §9.10 below). We mainly follow the exposition in the papers [194, 195].

Let J_{st} denote the standard complex structure on $\mathbb{C}^n$. For a fixed $k \in \{1, \ldots, n\}$ let $z = (z_1, \ldots, z_n) = (x' + \mathrm{i}y', x'' + \mathrm{i}y'')$, with $z_j = x_j + \mathrm{i}y_j$, denote the coordinates on $\mathbb{C}^n$ corresponding to the decomposition

$$\mathbb{C}^n = \mathbb{C}^k \oplus \mathbb{C}^{n-k} = \mathbb{R}^k \oplus \mathrm{i}\mathbb{R}^k \oplus \mathbb{R}^{n-k} \oplus \mathrm{i}\mathbb{R}^{n-k}.$$

Let $D = D^k \subset \mathbb{R}^k$ be the closed unit ball in $\mathbb{R}^k$ and $S = S^{k-1} = bD$ its boundary $(k-1)$-sphere. We shall often identify D^k with its image in $\mathbb{R}^k \oplus \{0\}^{2n-k} \subset \mathbb{C}^n$. We introduce the following (trivial) bundles over D^k:

$$\begin{aligned}
\nu' &= \mathrm{Span}\Big\{\frac{\partial}{\partial y_1}, \ldots, \frac{\partial}{\partial y_k}\Big\}\Big|_D = D \times \big(\{0\}^k \oplus \mathbb{R}^k \oplus \{0\}^{2n-2k}\big),\\
\nu'' &= \mathrm{Span}\Big\{\frac{\partial}{\partial x_j}, \frac{\partial}{\partial y_j} \colon j = k+1, \ldots, n\Big\}\Big|_D = D \times \big(\{0\}^{2k} \oplus \mathbb{R}^{2n-2k}\big),\\
\nu &= \nu' \oplus \nu'' = D \times \big(\{0\}^k \oplus \mathbb{R}^{2n-k}\big).
\end{aligned}$$

Clearly ν is the real normal bundle to D^k in $\mathbb{C}^n$ and we have

$$\nu' = J_{st}(\mathrm{T}D), \quad \mathrm{T}^{\mathbb{C}}D = \mathrm{T}D \oplus \nu', \quad \mathrm{T}\mathbb{C}^n|_D = \mathrm{T}D \oplus \nu = \mathrm{T}^{\mathbb{C}}D \oplus \nu''.$$

Let $v \to S = bD$ denote the (trivial) real line bundle over S spanned by the radial vector field $\sum_{j=1}^k x_j \frac{\partial}{\partial x_j}$. Over S we then have further decompositions

$$\mathrm{T}D|_S = v \oplus \mathrm{T}S, \quad \nu'|_S = J_{st}(v) \oplus J_{st}(\mathrm{T}S), \quad \mathrm{T}D|_S \oplus \nu'|_S \simeq v^{\mathbb{C}} \oplus \mathrm{T}^{\mathbb{C}}S.$$

Note that $\mathrm{T}^{\mathbb{C}}S$ is a trivial complex vector bundle over S.

Let X be a smooth $2n$-dimensional manifold. Given a smooth embedding (or immersion) $G \colon D \to X$ of the disc $D = D^k \subset \mathbb{C}^n$, a *normal framing* over G is a homomorphism $\beta \colon \nu \to \mathrm{T}X|_{G(D)}$ such that

$$\mathrm{d}G_x \oplus \beta_x \colon \mathrm{T}_x D \oplus \nu_x = \mathrm{T}_x \mathbb{C}^n \to \mathrm{T}_{G(x)} X$$

is a linear isomorphism for every $x \in D$.

Assume now that (X, J) is an almost complex manifold of real dimension $2n$. Let W be an open, relatively compact domain with smooth strongly J-pseudoconvex boundary $\Sigma = bW$ in X. Choose a real line subbundle $w \subset \mathrm{T}X|_\Sigma$ such that $Jw \subset \mathrm{T}\Sigma$; this implies

$$\mathrm{T}X|_\Sigma = w \oplus \mathrm{T}\Sigma = w \oplus Jw \oplus \xi, \quad \xi = \mathrm{T}\Sigma \cap J(\mathrm{T}\Sigma).$$

Let $D = D^k$, $S = S^{k-1} = bD$ and v be as above. An *embedding of a pair* $G \colon (D, S) \hookrightarrow (X \backslash W, \Sigma)$ is a smooth embedding $G \colon D \hookrightarrow X \backslash W$ such that $G(S) = G(D) \cap \Sigma$ and G is transverse to Σ along $G(S)$. Such G is said to be *normal to* Σ if $\mathrm{d}G_x(v_x) = w_{G(x)}$ for every $x \in S$, where w is as above. The analogous definition applies to immersions.

The following lemma of Eliashberg [129] is a key result in Stein geometry. The present formulation is taken from [194, Lemma 3.1].

Lemma 9.9.1. (Existence of well attached handles) *Let W be an open relatively compact domain with smooth strongly J-pseudoconvex boundary $\Sigma = bW$ in an almost complex manifold (X, J). Let $1 \le k \le n = \dim_{\mathbb{C}} X$, $D = D^k \subset \mathbb{R}^k$ the closed k-disc and $S = bD \cong S^{k-1}$. Given a smooth embedding $G_0 \colon (D, S) \hookrightarrow (X \backslash W, \Sigma)$, there exists a regular homotopy of immersions $G_t \colon (D, S) \looparrowright (X \backslash W, \Sigma)$ $(t \in [0, 1])$ that is $\mathcal{C}^0$ close to G_0 such that the immersion $G_1 \colon D \looparrowright X \backslash W$ is J-totally real and normal to Σ, and $g_1 = G_1|_S \colon S \hookrightarrow \Sigma$ is a Legendrian (complex tangential) embedding. If $k < n$, or if $k = n \neq 2$, there exists an isotopy of embeddings G_t with these properties. If J is integrable in a neighborhood of $\overline{W}$ and $\Sigma = bW$ is real-analytic, then G_1 can be chosen real analytic.*

Lemma 9.9.1 shows in particular that one can attach an embedded totally real disc of any dimension $\le n = \dim X \neq 2$ to a strongly pseudoconvex domain $\overline{W}$ in a Stein manifold X along a complex tangential sphere in bW. When coupled with Theorem 3.8.2, this lemma is a strong tool for constructing Stein structures (see §9.10 below). The main point is that after a small perturbation of the set $\overline{W} \cup G_1(D)$ and straightening the disc $G_1(D)$ we can obtain the standard situation described in Lemma 3.8.3; hence the Stein structure on a neighborhood of $\overline{W}$ extends to a Stein structure on a strongly pseudoconvex handlebody around $\overline{W} \cup M$.

As pointed out by Eliashberg in [129], the framing obstruction in dimension $n = 2$ is essential. In particular, it is impossible to attach an embedded totally real two-disc $D \subset \mathbb{C}^2 \backslash \mathbb{B}$ to the ball $\mathbb{B} \subset \mathbb{C}^2$ along a complex tangential curve $C \subset b\mathbb{B}$. If this were possible, we would obtain a strongly pseudoconvex domain in $\mathbb{C}^2$ containing an embedded homologically essential sphere with self-intersection number zero. (To get such a sphere we glue a 2-disc $D' \subset \mathbb{B}$ to D along $bD' = C = bD$.) This contradicts Corollary 9.8.2 (p. 434).

Proof. We follow the construction of a *special handle attaching triple* in [129, §2], with an essential additional argument in the critical case $k = n \neq 2$. (The details are taken from [194, Lemma 3.1].) First we find a regular homotopy from the initial disc $G_0 \colon D \hookrightarrow X \backslash W$ to an immersed disc $G_1 \colon D \looparrowright X \backslash W$ which is attached with a correct normal framing to W along an embedded Legendrian sphere in bW. Next we apply the h-principle for totally real immersions to deform G_1 by a regular homotopy that is fixed near the boundary to a totally real immersed disc G_2 (Fig. 9.1). Finally we show that, unless $k = n = 2$, the construction can be done by isotopies of embeddings.

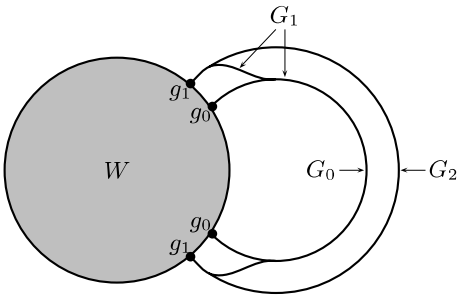

Fig. 9.1. Deformations of an attached disc. ([194, p. 622, Fig. 2])

Set $g_0 = G_0|_S \colon S \hookrightarrow bW$. By a correction of G_0 along S (keeping g_0 fixed) we may assume that it is normal to Σ, i.e., such that $\ell_0 := \mathrm{d}G_0|_v$ maps v to $w|_{g_0(S)}$. Choose a complex vector bundle isomorphism

$$\phi_0 \colon \mathrm{T}\mathbb{C}^n|_D = D \times \mathbb{C}^n \to \mathrm{T}X|_{G_0(D)}, \quad \phi_0 \circ J_{st} = J \circ \phi_0$$

covering G_0. We use the coordinates on $\mathbb{C}^n$ introduced at the beginning of the section. The vector field $\tau = \sum_{j=1}^k x_j \frac{\partial}{\partial x_j}$ is the outer radial field to the sphere $S = bD$ in $\mathbb{R}^k \times \{0\}^{2n-k}$. Let $\widetilde{\tau}$ be the unique nonvanishing vector field on $\mathbb{C}^n$ over S satisfying $\phi_0(\widetilde{\tau}_x) = \ell_0(\tau_x)$ for every $x \in S$. By dimension reasons there exists a map $A \colon D \to GL_n(\mathbb{C})$ satisfying $A_x \tau_x = \widetilde{\tau}_x$ for all $x \in S$. Replacing ϕ_0 by $\phi_0 \circ A$ we may assume from now on that $\phi_0|_v = \ell_0$. A further homotopic correction of ϕ_0 insures that

$$\phi_0(\mathrm{T}^{\mathbb{C}} S \oplus \nu''|_S) = \xi|_{g_0(S)},$$

thereby providing a trivialization of the latter bundle.

Write $\phi_0 = \phi_0' \oplus \phi_0''$, where $\phi_0' = \phi_0|_{\mathrm{T}^{\mathbb{C}} D}$ and $\phi_0'' = \phi_0|_{\nu''}$. (We are using the notation introduced before Lemma 3.8.3.) Setting $\psi_0 := \phi_0|_{\mathrm{T}^{\mathbb{C}} S}$ we thus have the following:

$$\phi_0'|_{\mathrm{T}^{\mathbb{C}} D|_S} = \ell_0^{\mathbb{C}} \oplus \psi_0 \colon v^{\mathbb{C}} \oplus \mathrm{T}^{\mathbb{C}} S \to \mathrm{T}X|_{g_0(S)} = w^{\mathbb{C}} \oplus \xi|_{g_0(S)}.$$

Note that $\psi_0 \oplus \phi_0'' \colon \mathrm{T}^{\mathbb{C}} S \oplus \nu''|_S \to \xi|_{g_0(S)}$ is a complex vector bundle isomorphism. Furthermore, there is a homotopy of real vector bundle monomorphisms $\iota_s \colon \mathrm{T}D \hookrightarrow \mathrm{T}X|_{G_0(D)}$ $(s \in [0,1])$ satisfying

$$\iota_0 = \mathrm{d}G_0, \quad \iota_1 = \phi_0|_{\mathrm{T}D}, \quad \iota_s|_v = \ell_0 \colon v \to w|_{g_0(S)} \quad (s \in [0,1]).$$

Consider the pair (g_0, ψ_0) consisting of the embedding $g_0 \colon S \hookrightarrow \Sigma$ and the $\mathbb{C}$-linear embedding $\psi_0 \colon \mathrm{T}^{\mathbb{C}} S \hookrightarrow \xi|_{g_0(S)}$ over g_0 of the complexified tangent bundle of S (a trivial complex vector bundle of rank $k-1$) into the contact subbundle $\xi \subset \mathrm{T}\Sigma$. By Gromov's Legendrization theorem [236, p. 339, (B')] there exists a Legendrian embedding $g_1 \colon S \hookrightarrow \Sigma$ whose complexified differential $\psi_1 := \mathrm{d}^{\mathbb{C}} g_1$ is homotopic to ψ_0 by a homotopy of $\mathbb{C}$-linear vector bundle embeddings $\psi_t \colon \mathrm{T}^{\mathbb{C}} S \hookrightarrow \xi$ $(t \in [0,1])$.

Let $\mathrm{Hom}_{\mathrm{inj}}(\mathrm{T}S, \mathrm{T}\Sigma)$ denote the space of fiberwise injective real vector bundle maps $\mathrm{T}S \to \mathrm{T}\Sigma$. Consider the path in $\mathrm{Hom}_{\mathrm{inj}}(\mathrm{T}S, \mathrm{T}\Sigma)$ from $\mathrm{d}g_0$ to $\mathrm{d}g_1$, consisting of the homotopy $\iota_s|_{\mathrm{T}S}$ $(s \in [0,1])$ followed by the homotopy $\psi_t|_{\mathrm{T}S}$ $(t \in [0,1])$ (left and top side of the square in Fig. 9.2). By Hirsch's one-parametric h-principle for immersions [236, 264] this path can be deformed in the space $\mathrm{Hom}_{\mathrm{inj}}(\mathrm{T}S, \mathrm{T}\Sigma)$ (with fixed ends) to a path of differentials $\mathrm{d}g_t \colon \mathrm{T}S \hookrightarrow \mathrm{T}\Sigma|_{g_t(S)}$, where $g_t \colon S \looparrowright \Sigma$ $(t \in [0,1])$ is a regular homotopy of immersions from g_0 to g_1. We can insure that ψ_t covers the base map g_t for all $t \in [0,1]$. This gives a homotopy $\theta_{t,s} \in \mathrm{Hom}_{\mathrm{inj}}(\mathrm{T}S, \mathrm{T}\Sigma)$ for $(t,s) \in [0,1]^2$ satisfying the following conditions (Fig. 9.2):

(i) $\theta_{t,0} = \mathrm{d}g_t$ (bottom side),
(ii) $\theta_{t,1} = \psi_t|_{\mathrm{T}S}$ (top side),
(iii) $\theta_{0,s} = \iota_s|_{\mathrm{T}S}$ (left side; hence $\theta_{0,0} = \mathrm{d}g_0$ and $\theta_{0,1} = \psi_0|_{\mathrm{T}S}$),
(iv) $\theta_{1,s} = \mathrm{d}g_1$ (right side), and
(v) $\theta_{t,s}$ covers g_t for every $t, s \in [0,1]$.

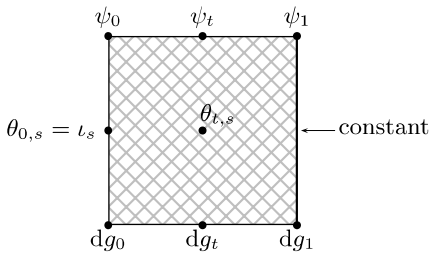

Fig. 9.2. The homotopy $\theta_{t,s}$. ([194, p. 623, Fig. 3])

We can extend g_t to a regular homotopy $G_t \colon (D, S) \looparrowright (X \backslash W, \Sigma)$ $(t \in [0,1])$ consisting of immersions normal to Σ, beginning at $t = 0$ with the map G_0.

Let $\ell_t := \mathrm{d}G_t|_v : v \to w|_{g_t(S)}$. By the homotopy lifting theorem there exists a homotopy of $\mathbb{C}$-linear complex vector bundle isomorphisms ϕ_t covering G_t,

$$\phi_t = \phi_t' \oplus \phi_t'' : \mathrm{T}\mathbb{C}^n|_D = \mathrm{T}^{\mathbb{C}} D \oplus \nu'' \to \mathrm{T}X|_{G_t(D)}, \quad t \in [0,1],$$

beginning at $t = 0$ with the map ϕ_0, such that over $S = bD$ we have

$$\phi_t' = \ell_t^{\mathbb{C}} \oplus \psi_t, \quad t \in [0,1],$$

and $\mathrm{d}G_1 = \phi_1$ on $\mathrm{T}D|_S$. Set $\widetilde{\theta}_{t,s} = \ell_t \oplus \theta_{t,s} : \mathrm{T}D|_S \hookrightarrow \mathrm{T}X|_{g_t(S)}$ for $t, s \in [0,1]$; this is a real vector bundle monomorphism over the map g_t. From the above properties (i)–(v) of $\theta_{t,s}$ we obtain the following:

(i') $\widetilde{\theta}_{t,0} = \ell_t \oplus \mathrm{d}g_t = \mathrm{d}G_t|_{\mathrm{T}D|_S}$ (bottom side),

(ii') $\widetilde{\theta}_{t,1} = \ell_t \oplus \psi_t|_{\mathrm{T}S} = \phi_t|_{\mathrm{T}D|_S}$ (top side),

(iii') $\widetilde{\theta}_{0,s} = \iota_s|_{\mathrm{T}D|_S}$ (left side),

(iv') $\widetilde{\theta}_{1,s} = \ell_1 \oplus \mathrm{d}g_1 = \mathrm{d}G_1|_{\mathrm{T}D|_S}$ (right side), and

(v') $\widetilde{\theta}_{t,s}$ covers g_t for every $t, s \in [0,1]$.

We wish to extend the monomorphisms $\widetilde{\theta}_{t,s} : \mathrm{T}D|_S \hookrightarrow \mathrm{T}X|_{g_t(S)}$ to real vector bundle monomorphisms $\Theta_{t,s} : \mathrm{T}D \to \mathrm{T}X$ ($t, s \in [0,1]$) covering the immersions $G_t : D \hookrightarrow X$. Such extension already exists for (t,s) in the bottom, top and left face of the parameter square $[0,1]^2$ where we take $\mathrm{d}G_t$, $\phi_t|_{\mathrm{T}D}$ and ι_s, respectively (see the properties (i'), (ii') and (iii')). The homotopy lifting property provides an extension $\Theta_{t,s}$ for all $(t,s) \in [0,1]^2$ with the given boundary values on the bottom, top and left side of $[0,1]^2$. (See Fig. 9.3. The front and the back face belong to the homotopy $\widetilde{\theta}_{t,s}$ over $S = bD$; compare also with Fig. 9.2.) Over the right face $\{t = 1\}$ we thus obtain a homotopy $\Theta_{1,s} \in \mathrm{Hom}_{\mathrm{inj}}(\mathrm{T}D, \mathrm{T}X|_{G_1(D)})$ ($s \in [0,1]$) satisfying

$$\Theta_{1,0} = \mathrm{d}G_1 : \mathrm{T}D \to \mathrm{T}X|_{G_1(D)}, \quad \Theta_{1,1} = \phi_1|_{\mathrm{T}D} : \mathrm{T}D \to \mathrm{T}X|_{G_1(D)}.$$

The homotopy $\Theta_{1,s}$ is fixed over S where it coincides with $\widetilde{\theta}_{1,s} = \mathrm{d}G_1|_{\mathrm{T}D|_S}$ by property (iv'). (On Fig. 9.3 the map $\Theta_{1,s}$ appears on the right face of the cube, with bold vertical sides indicating that it is constant on $\mathrm{T}D|_S$ where it equals $\ell_1 \oplus \mathrm{d}g_1$.)

Since $\phi_1 : \mathrm{T}\mathbb{C}^n|_D \to \mathrm{T}X|_{G_1(D)}$ is a $\mathbb{C}$-linear vector bundle isomorphism, the h-principle for totally real immersions (see §9.1) provides a regular homotopy of immersions $G_t : D \looparrowright X \backslash W$ ($t \in [1,2]$) that is fixed near S such that G_2 is J-totally real and its complexified differential $\mathrm{d}^{\mathbb{C}} G_2$ is homotopic to ϕ_1 in the space of $\mathbb{C}$-linear maps $\mathrm{T}\mathbb{C}^n|_D \to \mathrm{T}X$ of maximal rank. If in addition G_1 is an embedding, then we can deform it to a totally real embedding G_2 by an isotopy that is fixed near S. This follows from the fact that totally real embeddings also satisfy the h-principle. (See §9.1 above.) For $k = n = 2$ this

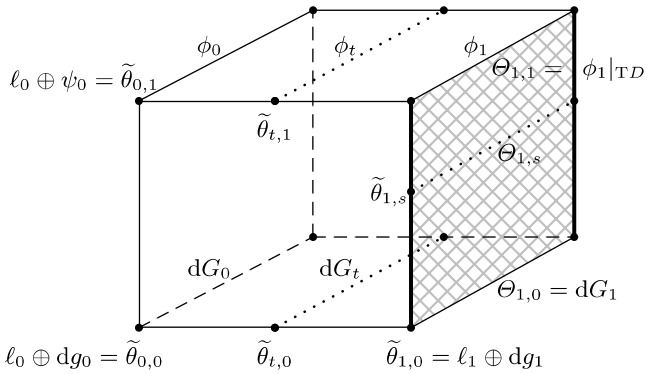

Fig. 9.3. The homotopy $\Theta_{t,s}$. ([194, p. 624, Fig. 4])

follows from Theorem 9.5.1 (p. 418) on cancellation of complex points of real surfaces in complex surfaces.

Finally we reparametrize the family $\{G_t : t \in [0,2]\}$ back to the parameter interval $[0,1]$ and obtain a regular homotopy with the required properties.

It remains to show that, unless $k = n = 2$, there also exists an *isotopy of embeddings* $\{G_t\}$ with these properties. If $k < n$, a small perturbation of $\{g_t\}$ with fixed ends at $t = 0, 1$ gives an isotopy which can be realized by an ambient diffeotopy, and we get an isotopy of embedded discs $G_t : D \hookrightarrow X \backslash W$ with $G_t|_S = g_t$. For $k = n = 1$ the conclusion of Lemma 9.9.1 obviously holds for any attached 1-disc (segment).

In the sequel we consider the critical case $k = n > 2$. A generic choice of the isotopy $g_t : S \cong S^{n-1} \looparrowright \Sigma$ insures that g_t is an embedding for all but finitely many values $t \in [0,1]$, and it has a simple double point at each of the exceptional parameter values. We wish to change the Legendrian embedding g_1 by a regular homotopy of Legendrian immersions $g_t : S \hookrightarrow \Sigma$ $(t \in [1,2])$ to another Legendrian embedding g_2 so that the resulting regular homotopy $\{g_t : t \in [0,2]\}$ will have self-intersection index zero. More precisely, the map

$$\widetilde{g} : \widetilde{S} = S \times [0,2] \to \widetilde{\Sigma} = \Sigma \times [0,2], \quad \widetilde{g}(x,t) = (g_t(x), t)$$

is an immersion of the n-dimensional oriented manifold $\widetilde{S}$ into the $2n$-dimensional oriented manifold $\widetilde{\Sigma}$ such that the double points of $\widetilde{g}$ correspond to the double points of the regular homotopy $\{g_t\}$. We define the *index* $i(\{g_t\})$ as the number of double points of $\widetilde{g}$ counted with the orientation signs.

If this index equals zero then a foliated version of the Whitney trick [513] allows us to deform $\{g_t\}_{t \in [0,2]}$ with fixed ends to an isotopy of embeddings. This is done by connecting a chosen pair of double points $q_0, q_1 \in \widetilde{g}(\widetilde{S})$ of the

opposite sign, lying over two different values $t_0 < t_1$ of the parameter, by a pair of curves

$$\lambda_j(t) = \widetilde{g}(c_j(t), t), \qquad t \in [t_0, t_1],\ j = 1, 2$$

which together bound an embedded Whitney disc $D^2 \subset \widetilde{\Sigma}$ such that $D^2 \cap (\Sigma_t \times \{t\})$ is an arc connecting $\lambda_1(t)$ to $\lambda_2(t)$ for every $t \in [t_0, t_1]$, and it degenerates to q_0 resp. q_1 over the endpoints t_0 resp. t_1. The rest of the procedure, removing this pair of double points by pulling $\widetilde{g}(\widetilde{S})$ across D^2, is standard [513]. Performing this operation finitely many times one can remove all double points and change $\{g_t\}$ to an isotopy of embeddings. The rest of the proof can be completed exactly as before: We extend g_t to an isotopy of embedded discs $G_t\colon D \hookrightarrow X \backslash W$, with $G_t|_S = g_t$, covered by a homotopy of $\mathbb{C}$-linear isomorhisms $\phi_t\colon \mathrm{T}\mathbb{C}^n|_D \to \mathrm{T}X|_{G_t(D)}$. Observe that $\{\mathrm{d}g_t\}$ still has the correct homotopy property so that the final embedding G_2 can be deformed (with fixed boundary) to a totally real embedding.

It remains to see that the index can be changed to zero by a Legendrian isotopy in Σ which introduces the correct number of double points. (A similar deformation is used in [129, §2.4]. It is here that the hypothesis $n \neq 2$ is used.) The set $L = g_1(S)$ is an embedded Legendrian sphere in Σ. Choose a point $a \in L$. In suitable local coordinates $(z, q, p) \in \mathbb{R}^{2n-1}$ on Σ, with a corresponding to $0 \in \mathbb{R}^{2n-1}$, the contact form is $\eta = \mathrm{d}z - \sum_{j=1}^{n-1} p_j \mathrm{d}q_j$ and

$$L = \left\{ (z, q, p) \in \mathbb{R}^{2n-1} \colon z^2 = q_1^3,\ p_1^2 = \frac{9}{4} q_1,\ p_2 = \cdots = p_{n-1} = 0 \right\}$$

(see [129, §2.4]). Let $\pi\colon \mathbb{R}^{2n-1} \to \mathbb{R}^{n-1}$ denote the projection $\pi(z, q, p) = q$. Choose a small number $q_1^0 > 0$ and let $\Delta \subset \mathbb{R}^{n-1}$ denote the closed ball of radius $q_1^0/2$ centered at $(q_1^0, 0, \ldots, 0)$. Let $\phi\colon \Delta \to \mathbb{R}$ be a smooth function that equals 0 near $b\Delta$. Set

$$h_t(q) = q_1^{3/2}\big(1 + (t-1)\phi(q)\big), \quad t \in [1, 2].$$

Let L_t equal L outside of $\pi^{-1}(\Delta)$ and equal

$$\left\{ (z, q, p) \colon z = h_t(q),\ p = \frac{\partial h_t}{\partial q}(q) \right\} \cup \left\{ (z, q, p) \colon z = -h_t(q),\ p = -\frac{\partial h_t}{\partial q}(q) \right\}$$

over Δ. We choose ϕ with sufficiently small derivative to insure that we remain in the given coordinate patch; this can be done if $q_1^0 > 0$ is chosen small enough. Let $g_t\colon S \looparrowright \Sigma$ ($t \in [1, 2]$) be the regular homotopy such that $g_t(S) = L_t$. The deformation is illustrated by Fig. 9.4. The top diagrams show the projection onto the (z, q)-plane at three typical stages, with the cusp at $(z, q) = (0, 0)$ and with a self-intersection shown in the middle figure.

The index of $\{g_t\}_{t \in [1,2]}$ equals the intersection number between the manifolds $M_\pm$ defined by

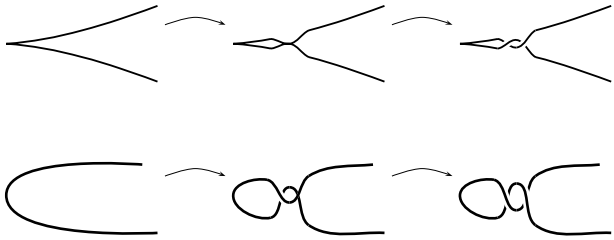

Fig. 9.4. Changing the index of a regular homotopy by +1. ([194, p. 626, Fig. 5])

$$M_\pm = \left\{ (z,q,p,t) \in \mathbb{R}^{2n} \colon z = \pm h_t(q),\ p = \pm \frac{\partial h_t}{\partial q}(q),\ q \in \Delta,\ t \in [1,2] \right\}.$$

The intersection points of M_+ and M_- are solutions of the equations

$$1 + (t-1)\phi = 0, \quad \frac{\partial \phi}{\partial q} = 0, \qquad t \in [1,2].$$

This is precisely the set of all critical points of ϕ with the critical values belonging to $(-\infty, -1]$. By a generic choice of ϕ we can insure that -1 is not a critical value. A computation shows that each point (q,t) satisfying the above equations adds ± 1 to the index $i(\{g_t\})$, depending on the sign of the determinant of the Hessian of ϕ at q; hence we get $+1$ at a critical point of even Morse index and -1 at a critical point of odd Morse index. Similarly, as we increase $c \in \mathbb{R}$, the Euler characteristic of the sublevel set $\{\phi \le c\}$ increases by one at every critical point of ϕ of even Morse index, and it decreases by one at every critical point of odd Morse index.

We conclude that $i = i(\{g_t\}_{t\in[1,2]})$ equals the Euler number of the set $\{q \in \Delta \colon \phi(q) \le -1\}$. If $n > 2$, this can be arranged to equal any preassigned integer by a suitable choice of ϕ, and hence we can arrange the index i to equal zero. If $n = 2$ then the index i can be arranged to be any nonnegative number (since $\{\phi \le -1\}$ is a union of segments), but it can not be negative. This completes the proof of Lemma 9.9.1 in the smooth case.

Assume now that J is integrable in a neighborhood of $\overline{W}$ in X and the hypersurface $\Sigma = bW$ is real analytic near the attaching sphere $G_0(S) \subset \Sigma$. We wish to find a real analytic disc satisfying Lemma 9.9.1.

Since the embedded disc G_1 constructed above can be chosen arbitrarily $\mathcal{C}^0$-close to G_0, we may assume that the same conditions on J and Σ also hold near $G_1(D)$. By Gray's theorem on real analytic approximation of Legendrian embeddings (see [129, Lemma 2.5.1.]) it is possible to approximate G_1 in the $\mathcal{C}^1$ topology by an embedded disc $G_1' \colon (D,S) \hookrightarrow (X \backslash W, \Sigma)$ which is real analytic near S such that $G_1'|_S \colon S \hookrightarrow \Sigma$ is Legendrian and G_1' is normal to Σ along S. It remains to perturb G_1' to a nearby real analytic embedding

$G_1'': D \hookrightarrow X$ that agrees with G_1' to the second order along S and to replace G_1 by G_1''. $\square$

9.10 Stein Structures and the Soft Oka Principle

Recall from §3.11 that every n-dimensional Stein manifold admits a handlebody decomposition without handles of index $> n$, induced by a strongly plurisubharmonic Morse exhaustion function. The following seminal result in the opposite direction is due to Eliashberg and Gompf.

Theorem 9.10.1. *Assume that (X, J) is a smooth almost complex manifold of real dimension $2n$ admitting a Morse exhaustion function $\rho: X \to \mathbb{R}$ without critical points of index $> n$.*

(a) [129] *If $n \neq 2$ then J is homotopic to an integrable Stein structure.*

(b) [215] *If $n = 2$ then there exists a Stein surface $(\widetilde{X}, \widetilde{J})$ and an orientation preserving homeomorphism $h: X \to \widetilde{X}$ such that $h^*\widetilde{J}$ is homotopic to J (i.e., (a) holds for a possibly exotic Stein structure on X).*

More precisely, (a) asserts that there exists a homotopy $\{J_t\}_{t\in[0,1]}$ of smooth almost complex structure on X connecting the given structure $J_0 = J$ to an integrable Stein structure $J_1 = \widetilde{J}$. The hypothesis concerning ρ is equivalent to asking that X is a CW complex of dimension at most $n = \dim_{\mathbb{C}} X$. Eliashberg also showed that $\widetilde{J}$ can be chosen such that the sublevel sets $\{x \in X: \rho(x) < c\}$ are strongly $\widetilde{J}$-pseudoconvex. Regarding part (b), the pullback $h^*\widetilde{J}$ of an almost complex structure by a homeomorphism $h: X \to \widetilde{X}$ gives a well-defined homotopy class of almost complex structures on X [215, p. 645]. A change of the smooth structure on X is necessary in general as shown by Examples 9.8.3 and 9.8.4 on p. 435.

Theorem 9.10.1 is proved in §9.11 for $n \neq 2$, or if $n = 2$ and X is a handlebody without handles of index > 1. The case $\dim_{\mathbb{R}} X = 4$ is discussed in §9.12. For $n \neq 2$ the result follows easily from Lemma 9.9.1 (providing well attached totally real handles) and Theorem 3.8.2 (p. 83) which enables one to extend the Stein structure across the handle.

When $\dim_{\mathbb{C}} X = 2$, the framing obstruction in Lemma 9.9.1 can not be avoided in general, and one must deal with immersed (also called *kinky*) 2-handles. The superfluous double point loops are cancelled by again attaching kinky handles, and so on. The resulting procedure converges to an attached *Casson handle* which is homeomorphic, but not diffeomorphic, to the standard 4-handle $D^2 \times \mathrm{Int} D^2$, thereby changing the underlying $\mathcal{C}^\infty$ structure.

We now describe a certain soft version of the Oka principle which fits in the context of Theorem 9.10.1. We have seen in Chapter 5 that it is only rarely possible to find a holomorphic representative in a given homotopy class

of maps $X \to Y$ from a Stein manifold X to a complex manifold Y. The main problem is holomorphic rigidity of Y; the manifold X may be holomorphically 'too large' to fit into Y, at least in some homotopy classes of maps. To some extent this obstruction can be explained by the distance decreasing property of holomorphic maps in standard biholomorphically invariant metrics (such as the Kobayashi metric). We can expect the Oka principle to hold in two cases: Either Y has no holomorphic rigidity (this is the class of Oka manifolds, see Def. 5.4.1), or else X is holomorphically small enough in a certain sense. We now show that the the source Stein manifold X can always be made holomorphically 'smaller' so as to accomodate a holomorphic map $X \to Y$ in a given homotopy class, while still retaining its essential topological (and smooth in dimension $\neq 2$) characteristics. We give two different formulations; the first one fits in the context of Theorem 9.10.1. There is no restriction whatsoever on the target manifold Y in Theorems 9.10.2 and 9.10.3.

Theorem 9.10.2. (The Soft Oka Principle; [194, Theorem 1.1]) *Let (X, J) satisfy the hypotheses in Theorem 9.10.1, and let $f\colon X \to Y$ be a continuous map to a complex manifold (Y, J_Y).*

(i) *If $\dim_{\mathbb{C}} X \neq 2$, there exist an integrable Stein structure $\widetilde{J}$ on X, homotopic to J, and a $(\widetilde{J}, J_Y)$-holomorphic map $\widetilde{f}\colon X \to Y$ homotopic to f.*

(ii) *If $\dim_{\mathbb{C}} X = 2$ then there exist an orientation preserving homeomorphism $h\colon X \to X'$ onto a Stein surface X' and a holomorphic map $f'\colon X' \to Y$ such that the map $\widetilde{f} = f' \circ h\colon X \to Y$ is homotopic to f.*

A family of maps $f_p\colon X \to Y$ depending continuously on the parameter p in a compact Hausdorff space can be deformed to a family of holomorphic maps with respect to some Stein structure $\widetilde{J}$ on X that is homotopic to J.

If (X, J) is an integrable complex manifold with a suitable handlebody structure, we can obtain a holomorphic map in a given homotopy class on a Stein domain $\Omega \subset X$ that is diffeotopic (homeotopic if $\dim_{\mathbb{R}} X = 4$) to X:

Theorem 9.10.3. [194, Theorem 1.2] *Let X be a complex manifold with a Morse exhaustion function without critical points of index $> n = \dim_{\mathbb{C}} X$. Let P be a compact Hausdorff space and $f\colon X \times P \to Y$ a continuous map to a complex manifold Y.*

(i) *If $n \neq 2$, or if $n = 2$ and ρ has no critical points of index > 1, then there exist an open Stein domain Ω in X, a diffeomorphism $h\colon X \to \Omega$ that is diffeotopic to Id_X, and a map $f'\colon \Omega \times P \to Y$ such that $f'_p = f'(\cdot\,, p)\colon \Omega \to Y$ is holomorphic for every $p \in P$ and the map $f' \circ h\colon X \times P \to Y$ is homotopic to f.*

(ii) *If $n = 2$ then the conclusion in (i) holds for a homeomorphism $h\colon X \to \Omega$ that is homeotopic to the identity map on X.*

Theorems 9.10.1 and 9.10.3 together imply Theorem 9.10.2 as follows. First we change J to an integrable Stein structure on X by applying Theorem 9.10.1. Let $h_t\colon X \to h_t(X) \subset X$ ($t \in [0,1]$) be a diffeotopy from $h_0 = \mathrm{Id}_X$ to $h_1 = h\colon X \to \Omega$, furnished by Theorem 9.10.3. Then $J_t = h_t^*(J)$ is a homotopy of integrable complex structures on X connecting $J = J_0$ to the Stein structure $J_1 = h^*(J|_{\mathrm{T}\Omega})$, and $f' \circ h\colon X \to Y$ is a (J_1, J_Y)-holomorphic map homotopic to f. Although there is no formal implication in the opposite direction, the proofs of all results stated above follow the same pattern.

The following simple example illustrates the failure of Theorems 9.10.2 and 9.10.3 for noncompact families of maps.

Example 9.10.4. (Mappings of annuli.) Let $X = A_r = \{z \in \mathbb{C}\colon \frac{1}{r} < |z| < r\}$ for some $r > 1$, and let $Y = A_R$ for another $R > 1$. We have $[X, Y] = \pi_1(S^1) = \mathbb{Z}$, and the homotopy class of a map $f\colon X \to Y$ is determined by its winding number around the unit circle. By the classical theory, a homotopy class represented by $k \in \mathbb{Z}$ admits a holomorphic representative if and only if $r^{|k|} \le R$, and in this case a representative is $z \to z^k$. Since every complex structure on an annulus is biholomorphic to A_r for some $r > 1$, we see that at most finitely many homotopy classes of maps between a pair of annuli contain a holomorphic map. The conclusion of Theorem 9.10.2 is obtained in this example by a radial dilation, decreasing the value of $r > 1$ to another value satisfying $r^k \le R$; this amounts to a homotopic deformation of the complex structure on X. This allows us to simultaneously deform any compact family of maps $X \to Y$ to a family of holomorphic maps, but it is impossible to do the same for a sequence of maps belonging to different homotopy classes.

The same phenomenon occurs whenever $\pi_1(Y)$ contains an element $[\alpha]$ of infinite order such that the minimal Kobayashi length l_N of loops in Y representing the class $N[\alpha] \in \pi_1(Y)$ tends to $+\infty$ as $N \to +\infty$. $\square$

Problem 9.10.5. Assume that (X, J) is a Stein manifold of dimension $\neq 2$. Given a continuous map $f\colon X \to Y$, is there a homotopy $\{J_t\}_{t\in[0,1]}$ consisting of integrable *Stein* structures J_t on X, with $J_0 = J$, such that there exists a (J_1, J_Y)-holomorphic map $f_1\colon X \to Y$ homotopic to f? The almost complex structures $\{J_t\}_{t\in[0,1]}$ obtained in Theorem 9.10.3 are all integrable, but they need not be Stein for the intermediate values $t \in (0,1)$. $\square$

Problem 9.10.6. If (X, J) is an integrable Stein surface and $f\colon X \to Y$ is a continuous map, does the conclusion of Theorem 9.10.3 (b) hold for some *diffeomorphism* h? That is, can we change the given Stein structure J to another nonexotic Stein surface structure $\widetilde{J}$ on X in which the map f admits a holomorphic representative? Can this be done through a homotopy of Stein surface structures on X? $\square$

For a Stein source manifold, Theorem 9.10.3 holds with approximation on compact holomorphically convex subsets and interpolation on closed complex subvarieties, in analogy to the standard Oka principle in §5.

Theorem 9.10.7. (Theorems 1.2 and 4.1 in [195]) *Assume that X is a Stein manifold, A is a closed complex subvariety of X, and $K \subset X$ is a compact $\mathcal{O}(X)$-convex subset. Let Y be a complex manifold and $f\colon X \to Y$ a continuous map that is holomorphic in a neighborhood of $A \cup K$. If $\dim_{\mathbb{C}} X \neq 2$ then there exist a Stein domain $\Omega \subset X$ containing $A \cup K$, a holomorphic map $\widetilde{f}\colon \Omega \to Y$, and a diffeomorphism $h\colon X \to \Omega$ which is diffeotopic to Id_X by a diffeotopy that is fixed on a neighborhood $A \cup K$ such that $\widetilde{f}|_A = f|_A$, $\widetilde{f}$ approximates f uniformly on K as close as desired, and the map $\widetilde{f} \circ h\colon X \to Y$ is homotopic to f relative to $A \cup K$. If $\dim_{\mathbb{C}} X = 2$ then the same conclusion holds for a homeomorphism h that is homeotopic to Id_X.*

Our construction also gives holomorphic maps of maximal rank (immersions or submersions) in the absence of topological obstructions. The following is a simplified version of Theorem 6.3 in [194].

Theorem 9.10.8. (Soft Oka principle for maps of pointwise maximal rank.) *Let X be a Stein manifold with $\dim_{\mathbb{C}} X \neq 2$ and $f\colon X \to Y$ a continuous map to a complex manifold Y. If there is a complex vector bundle map $\iota\colon \mathrm{T}X \to f^*(\mathrm{T}Y)$ of fiberwise maximal rank, then there exist a Stein structure $\widetilde{J}$ on X, homotopic to J, and a $\widetilde{J}$-holomorphic map $\widetilde{f}\colon X \to Y$ of pointwise maximal rank which is homotopic to f. The same holds if $\dim X = 2$ and X admits a Morse exhaustion function without critical points of index > 1.*

This shows that Theorem 8.12.1 on p. 386 (the h-principle for holomorphic submersions) holds without any assumptions on the target manifold Y, at the cost of allowing a homotopic change of the Stein structure on the source manifold X. In particular, taking $Y = \mathbb{C}^n$ with $n = \dim_{\mathbb{C}} X$ we obtain the following corollary which shows that every parallelizable Stein manifold becomes a Riemann domain over $\mathbb{C}^n$ after a homotopic change of its Stein structure. (Compare with Problem 8.12.3 on p. 387.)

Corollary 9.10.9. *If (X, J) is a Stein manifold of dimension $n \neq 2$ whose holomorphic tangent bundle $\mathrm{T}X$ is trivial, then there are a Stein structure $\widetilde{J}$ on X, homotopic to J, and a $\widetilde{J}$-holomorphic immersion $\pi\colon X \to \mathbb{C}^n$.*

Theorem 9.10.7 was extended by Prezelj and Slapar [403] to 1-convex source manifolds X, provided that the initial map $f\colon X \to Y$ is holomorphic in a neighborhood of the maximal compact subvariety of X.

The results stated in this section are proved in the following two sections. Theorem 3.7.2 is used to obtain a holomorphic extension of the map across the handle at every inductive step. The essential difference from the standard Oka principle in Chapters 5 and 6 is that one can not enlarge the domain of existence of a holomorphic map in the absence of a Runge approximation property for maps $\mathbb{C}^n \to Y$, so we only get a holomorphic map on a thin Stein handlebody $\Omega \subset X$ that is diffeotopic (resp. homeotopic) to X.

9.11 The Case $\dim_{\mathbb{R}} X \neq 4$

In this section we prove the results stated in the previous section when $\dim_{\mathbb{R}} X \neq 4$, or when X is a 4-manifold without handles of index > 1.

Let P denote a compact Hausdorff space. A *P-map* $X \to Y$ is a continuous map $f\colon X \times P \to Y$. If X and Y are complex manifolds then f is a *holomorphic P-map* if $f_p = f(\cdot\,, p)\colon X \to Y$ is holomorphic for every $p \in P$. (Compare with Def. 6.6.3.) The following result includes Theorem 9.10.1 (a) and Theorem 9.10.2 (i) as special cases. Except for the fact that we also find a holomorphic map, the construction is essentially the one of Eliashberg [129].

Theorem 9.11.1. [194, Theorem 6.1] *Let (X, J) be a smooth almost complex manifold of real dimension $2n$ endowed with a Morse exhaustion function $\rho\colon X \to \mathbb{R}$ without critical points of index $> n$. Assume that for some $c \in \mathbb{R}$ the structure J is integrable in $X_c = \{x \in X\colon \rho(x) < c\}$ and ρ is strongly J-plurisubharmonic in X_c. Let Y be a complex manifold with a complete distance function* dist_Y *and $f\colon X \times P \to Y$ a P-map which is J-holomorphic in X_c. If $n \neq 2$, or if $n = 2$ and ρ has no critical points of index > 1 in the set $\{x \in X\colon \rho(x) \geq c\}$, then for every compact set $K \subset X_c$ and for every $\epsilon > 0$ there exist a Stein structure $\widetilde{J}$ on X and a homotopy of P-maps $f^t\colon X \times P \to Y$ $(t \in [0,1])$ such that $f^0 = f$ and the following hold:*

(a) *the P-map $\widetilde{f} := f^1$ is $\widetilde{J}$-holomorphic on X;*
(b) *there is a homotopy J_t of almost complex structures on X which is fixed in a neighborhood of K such that $J_0 = J$ and $J_1 = \widetilde{J}$. If J is integrable on X then J_t can be chosen integrable on X for all $t \in [0,1]$;*
(c) *for every $t \in [0,1]$ the P-map f^t is J-holomorphic in a neighborhood of K and satisfies* $\sup\{\mathrm{dist}_Y(f^t(x,p), f(x,p))\colon x \in K,\ p \in P\} < \epsilon$.

Proof. We give the proof for the nonparametric case; the parameteric case is essentially the same. Pick a compact set $K \subset X_c$ and a regular value $c_0 \in \mathbb{R}$ of ρ such that $K \subset X_{c_0} \Subset X_c$. Let $p_1, p_2, \ldots$ be the critical points of ρ in the set $\{x \in X\colon \rho(x) > c_0\}$, ordered so that $\rho(p_j) < \rho(p_{j+1})$ for $j = 1, 2, \ldots$. Choose numbers c_j satisfying $c_{-1} = -\infty < c_0 < \rho(p_1) < c_1 < \rho(p_2) < c_2 < \cdots$. Let k_j denote the Morse index of ρ at p_j. For each $j = 0, 1, \ldots$ we set

$$X_j = \{x \in X\colon \rho(x) < c_j\}, \quad \Sigma_j = bX_j = \{x \in X\colon \rho(x) = c_j\}.$$

Set $J_0 = J$. We construct a sequence of almost complex structures J_j on X and maps $f_j\colon X \to Y$ satisfying the following properties for $j = 0, 1, 2, \ldots$:

(i) J_j is integrable in a neighborhood of $\overline{X}_j$ and (X_j, J_j) is a Stein manifold with strongly pseudoconvex boundary,
(ii) $J_j = J_{j-1}$ in a neighborhood of $\overline{X}_{j-1}$,
(iii) the set $\overline{X}_{j-1}$ is $\mathcal{O}(X_j, J_j)$-convex,

(iv) the map f_j is J_j-holomorphic in a neighborhood of $\overline{X}_j$,

(v) $\sup_{x \in X_{j-1}} \operatorname{dist}_Y(f_j(x), f_{j-1}(x)) < \epsilon\, 2^{-j-1}$, and

(vi) there is a homotopy from f_{j-1} to f_j which is J_j-holomorphic and uniformly close to f_{j-1} in a neighborhood of $\overline{X}_{j-1}$ (satisfying (v)).

These conditions hold for $j = 0$; in this case (ii), (iii), (v) and (vi) are vacuous. Assume inductively that we have found a pair (J_l, f_l) for all indexes up to $j - 1 \geq 0$. If $k_j = 0$ then a new connected component of the sublevel set $\{\rho < t\}$ appears at the critical point p_j when t passes the value $\rho(p_j)$, and it is trivial to find the next pair (J_j, f_j). Assume now that $k_j > 0$. By Morse theory X_j is diffeotopic to a handlebody obtained by smoothly thickening $\overline{X}_{j-1} \cup M_j$ for an embedded disc $M_j \subset X_j \backslash X_{j-1}$ of dimension k_j (the stable manifold for the gradient flow of ρ at p_j). The manifold X admits a real analytic structure that is equivalent to the underlying smooth structure and that is induced near $\overline{X}_{j-1}$ by the complex structure J_{j-1}. By perturbing $\Sigma_{j-1} = bX_{j-1}$ we may assume that it is real analytic. Applying Lemma 3.8.3 (p. 84) we can isotopically deform M_j to an embedded real analytic disc that is well attached to $\overline{X}_{j-1}$ along $bM_j \subset \Sigma_{j-1}$. Theorem 3.8.2 (p. 83) now furnishes a strongly pseudoconvex Stein handlebody $(\widetilde{W}_j, \widetilde{J}_j)$ in X such that

(a) $\overline{X}_{j-1} \subset \widetilde{W}_j \subset X_j$,

(b) X_j is diffeotopic to $\widetilde{W}_j$ inside $X_j \backslash X_{j-1}$,

(b) $\widetilde{J}_j$ coincides with J_{j-1} in a neighborhood of $\overline{X}_{j-1}$, and

(c) the sets $\overline{X}_{j-1}$ and $\overline{X}_{j-1} \cup M_j$ are $\mathcal{O}(\widetilde{W}_j, \widetilde{J}_j)$-convex.

By choosing $\widetilde{W}_j$ thin enough around $\overline{X}_{j-1} \cup M_j$, Theorem 3.7.2 (p. 81) furnishes a map $\widetilde{f}_j \colon X \to Y$ that is homotopic to f_{j-1}, it is $\widetilde{J}_j$-holomorphic on $\widetilde{W}_j$, and it approximates f_{j-1} uniformly on X_{j-1} so that Condition (v) holds.

There is a smooth diffeotopy $h_t \colon X \to X$ $(t \in [0,1])$ that is fixed in a neighborhood of $\overline{X}_{j-1}$ such that h_0 is the identity map on X and $h_1(X_j) = \widetilde{W}_j$. Taking $J_j = h_1^*(\widetilde{J}_j)$ and $f_j = \widetilde{f} \circ h_1$ completes the inductive step. A homotopy from f_{j-1} to f_j is obtained by composing the homotopy from f_{j-1} to $\widetilde{f}$ by the map h_1. The induction may proceed.

By properties (i) and (ii) there is a unique integrable complex structure $\widetilde{J}$ on X which agrees with J_j on X_j for every j. Note that $\widetilde{J}$ is homotopic to $J = J_0$ since the structure J_j at the j-th stage of the construction was chosen homotopic to J_{j-1} by a homotopy that is fixed near $\overline{X}_{j-1}$. The complex manifold $(X, \widetilde{J})$ is exhausted by the increasing sequence of Stein domain X_j, and the Runge property (iii) implies that $(X, \widetilde{J})$ is Stein. Properties (iv) and (v) insure that the sequence $f_j \colon X \to Y$ converges uniformly on compacts in X to the $\widetilde{J}$-holomorphic map $\widetilde{f} = \lim_{j\to\infty} f_j \colon X \to Y$ satisfying $\sup_{x \in X_0} \operatorname{dist}_Y(\widetilde{f}(x), f_0(x)) < \epsilon$. Property (vi) insures that the homotopies

from f_{j-1} to f_j also converge uniformly on compacts in X to a homotopy $f_t\colon X \to Y$ from the initial map f_0 to $f_1 = \widetilde{f}$. If the initial structure J on X is integrable then clearly all intermediate structures are also integrable. $\square$

Proof (of Theorem 9.10.3). This requires a few minor modifications in the proof of Theorem 9.11.1. We do not change the complex structure J on X during the construction at the cost of remaining on subsets of X which are only diffeomorphic to sublevel sets of ρ. We get approximation at no additional cost, but for the interpolation condition in Theorem 9.10.7 one must employ the exhaustion method that has been used in §5.12.

We use the same notation as above. Thus $W_0 = X_0$ is a sublevel set of a Morse exhaustion function $\rho\colon X \to \mathbb{R}$ without critical points of index $> n$ in $X \backslash W_0$, ρ is strongly plurisubharmonic in a neighborhood of $\overline{W}_0$, and $f_0 = f\colon X \to Y$ is holomorphic in a neighborbood of $\overline{W}_0$. Let $X_j = \{\rho < c_j\}$ where the constants c_j are chosen so that ρ has a unique critical point p_j in $X_j \backslash X_{j-1}$. Pick $\epsilon > 0$. Assume that $n \neq 2$. We inductively construct a sequence of relatively compact, strongly pseudoconvex domains $W_1 \subset W_2 \subset \cdots \subset X$, maps $f_j\colon X \to Y$, and diffeomorphisms $h_j\colon X \to X$ such that the following hold for all $j = 1, 2, \ldots$:

(i) $\overline{W}_{j-1}$ is $\mathcal{O}(W_j)$-convex,

(ii) f_j is holomorphic in a neighborhood of $\overline{W}_j$ and is homotopic to f_{j-1} by a homotopy $f_{j,t}\colon X \to Y$ ($t \in [0,1]$) such that each $f_{j,t}$ is holomorphic near $\overline{W}_{j-1}$ and satisfies $\sup_{x \in W_{j-1}} \mathrm{dist}_Y\big(f_{j,t}(x), f_{j-1}(x)\big) < \epsilon 2^{-j}$,

(iii) $h_j(X_j) = W_j$, and

(iv) $h_j = g_j \circ h_{j-1}$, where $g_j\colon X \to X$ is a diffeomorphism of X which is diffeotopic to Id_X by a diffeotopy that is fixed in a neighborhood of $\overline{W}_{j-1}$. (In particular, h_j agrees with h_{j-1} near $\overline{W}_{j-1}$.)

It is easily seen that the map $\widetilde{f} = \lim_{j\to\infty} f_j\colon \Omega = \cup_{j=1}^{\infty} W_j \to Y$ and the diffeomorphism $h = \lim_{j\to\infty} h_j\colon X \to \Omega$ satisfy Theorem 9.10.3.

To prove the inductive step we begin by attaching to $W_{j-1} = h_{j-1}(X_{j-1})$ the disc $M_j = h_{j-1}(D_j)$, where $D_j \subset X_j \backslash X_{j-1}$ (with $bD_j \subset bX_{j-1}$) is the stable manifold for the gradient flow of ρ at the unique critical point $p_j \in X_j \backslash X_{j-1}$. By Lemma 3.8.3 (p. 84) we can isotope M_j to a real analytic totally real disc in $X \backslash W_{j-1}$ that is well attached to the domain $\overline{W}_{j-1}$ along a Legendrian sphere $bM_j \subset bW_{j-1}$. Applying Theorem 3.7.2 (p. 81) we find the next map $f_j\colon X \to Y$ that is holomorphic in a thin strongly pseudoconvex handlebody $W_j \supset \overline{W}_{j-1} \cup M_j$. The next diffeomorphism h_j is furnished by Morse theory.

With a bit more care one can insure that $b\Omega$ is smoothly bounded and strongly pseudoconvex, but in general we can not choose Ω to be relatively compact in X, unless X admits an exhaustion function with at most finitely many critical points. $\square$

9.12 Exotic Stein Structures on Smooth 4-Manifolds

In this section we discuss the construction of Stein surface structures on a smooth oriented 4-manifold X with a handlebody decomposition without handles of index > 2. In particular, we prove Theorems 9.10.1 and 9.10.2 (p. 446) in the case $\dim_{\mathbb{R}} X = 4$. We first indicate Gompf's original construction [215], but for the actual proof we use the modification from [194] which is better suited to the task of finding a holomorphic map in the given homotopy class. We also give a construction inside a given integrable complex surface as in [216], thereby proving Theorem 9.10.3 for $\dim_{\mathbb{R}} X = 4$.

We begin with the main point of the construction – extending a Stein structure across a handle. Assume that X is a smooth oriented 4-manifold, $W \Subset X$ is a smoothly bounded domain in X, and J is an integrable Stein structure on a neighborhood of $\overline{W}$ such that the boundary $\Sigma = bW$ is strongly J-pseudoconvex. The problem is to extend J to a Stein structure over a suitably chosen handle attached to W. Handles of index 0 or 1 do not present any problem (see the previous section), so we focus on handles of index 2.

Let $D = D^2 \subset \mathbb{R}^2 \times \{\mathrm{i}0\}^2 \subset \mathbb{C}^2$ be the standard 2-disc and $S = bD$ its boundary circle. Let $G\colon (D, S) \hookrightarrow (X \backslash W, \Sigma)$ be an embedded disc with boundary $G(S) \subset \Sigma$. After an isotopy given by Lemma 3.8.3 (p. 84) we can assume that $C = G(S)$ is a Legendrian knot in Σ (tangential to the contact subbundle $\xi = \mathrm{T}\Sigma \cap J(\mathrm{T}\Sigma)$ of $\mathrm{T}\Sigma$), and the embedded disc $M = G(D) \subset X \backslash W$ is J-normal to Σ along C, in the sense that $J\mathrm{T}_x M \in \mathrm{T}_x \Sigma$ for all $x \in C$. This implies that the embedding G is J-real near S. Let $\nu = D \times \mathbb{R}^2$ denote the normal bundle of D in $\mathbb{C}^2$. The normal bundle ν_M of M in X is a trivial 2-plane bundle; a trivialization $\beta\colon \nu \to \nu_M$ covering G is called a *normal framing* of M. If β can be chosen such that

$$\beta \circ J_{st} = J \circ \mathrm{d}G \quad \text{on } \mathrm{T}_x D, \quad x \in S, \tag{9.32}$$

then (after perturbing M slightly) we can extend J to a Stein structure in a neighborhood of $\overline{W} \cup M$ exactly as in the proof of Theorem 9.10.1 (a).

The key problem is that, when $n = 2$, the framing condition (9.32) can not be achieved in general by an isotopy of the disc G. Let us analyze the underlying geometry more carefully. Assume that C is an oriented Legendrian knot in the contact 3-manifold (Σ, ξ). (For the moment C need not bound a disc.) We may assume that the global orientation of X agrees with the orientation induced by J near $\overline{W}$. We coorient the boundary $\Sigma = bW$ by the outer normal vector field w, chosen such that $Jw \in \mathrm{T}\Sigma$. These choices coorient the normal bundle ν_C of C in Σ so that the orientations on $\mathrm{T}\Sigma|_C = \mathrm{T}C \oplus \nu_C$ add up correctly. Let τ be a nonvanishing vector field tangent to C in the chosen orientation class. Then $J\tau$ is tangent to ξ since C is Legendrian, and $(\tau, J\tau)$ is a framing of 2-plane bundle $\xi|_C$. The pair of vector fields $(J\tau, Jw)$ spans the normal bundle $\nu_C \subset \mathrm{T}\Sigma|_C$ of the knot C in Σ. This is a *canonical framing*, also called a *Thurston-Bennequin (TB) framing* of the normal bundle ν_C of

the Legendrian knot C in (Σ, ξ). Any pair of TB framings are homotopic to each other (as framings of ν_C). Observe that the triple $(\tau, J\tau, Jw)$ is a positive framing of $\mathrm{T}\Sigma|_C$. Indeed, w is the outer normal to Σ and $(w, \tau, J\tau, Jw)$ defines the same orientation of $\mathrm{T}X$ as $(\tau, J\tau, w, Jw)$, hence positive. Thus any TB framing $(J\tau, Jw)$ defines a positive orientation of the normal bundle ν_C.

Assume now that C bounds an embedded 2-disc $M = G(D) \subset X \backslash W$ that is transverse to Σ along C. By an isotopic correction, keeping the boundary curve $C = bM$ fixed, we insure that M is J-orthogonal to Σ along C. Let (x_1, x_2) be coordinates on $\mathbb{R}^2$. The vector fields

$$w_0 = -x_1\partial_{x_1} - x_2\partial_{x_2}, \quad \tau_0 = -x_2\partial_{x_1} + x_1\partial_{x_2}$$

are the inner normal and the tangential field to $S = \{x_1^2 + x_2^2 = 1\} = bD$, respectively. Their images $w = G_*w_0$, $\tau = G_*\tau_0$ are vector field on X along $C = G(S)$ spanning the tangent bundle $\mathrm{T}M|_C$. Since w is outer normal to $\Sigma = bW$ along C, the pair $(J\tau, Jw)$ is a TB framing of ν_C. We coorient ν_M so that the orientations on $\mathrm{T}X|_M = \mathrm{T}M \oplus \nu_M$ add up correctly. Choose a normal framing $\beta\colon \nu_M \to \mathrm{T}X|_M$ over M such that $\beta(\nu|_S) = \nu_C$ and the orientations match up. We thus have two framings of ν_C, namely β and the TB framing. Since ν_C is a trivial oriented 2-plane bundle over the circle C, any two framings in the same orientation class differ up to homotopy by a map $C \cong S^1 \to SO(2) = S^1$, hence by an integer. We can thus write

$$[\beta] = TB + k(\beta), \quad k(\beta) \in \mathbb{Z}.$$

The integer $k = k(\beta)$ is called the *framing index* of β.

Note that $\partial_{y_1} = J_{st}\partial_{x_1}$, $\partial_{y_2} = J_{st}\partial_{x_2}$ is a negative framing of the normal bundle ν of D. The framing condition (9.32) that we wish to achieve is

$$\beta\left(\partial_{y_1}\right) = J\left(G_*\partial_{x_1}\right), \quad \beta\left(\partial_{y_2}\right) = J\left(G_*\partial_{x_2}\right)$$

along the circle S. In terms of the vector fields $w = G_*w_0$ and $\tau = G_*\tau_0$ introduced above this is equivalent to

$$\beta\left(-x_2\partial_{y_1} + x_1\partial_{y_2}\right) = J\tau, \quad \beta\left(-x_1\partial_{y_1} - x_2\partial_{y_2}\right) = Jw.$$

Since the framing $(-x_2\partial_{y_1} + x_1\partial_{y_2}, -x_1\partial_{y_1} - x_2\partial_{y_2})$ makes one right twist against the (positive) framing $(\partial_{y_2}, \partial_{y_1})$ of ν as the point (x_1, x_2) traces the circle S, we see that β is obtained by one left (negative) twist from the TB framing $(J\tau, Jw)$. Hence the framing condition (9.32) is equivalent to

$$k(\beta) = -1 \Longleftrightarrow [\beta] = TB - 1. \tag{9.33}$$

When the normal framing β of M satisfies (9.33) then J extends to an integrable complex structure in a neighborhood of $\overline{W} \cup M$ in X such that M is J-real. If this ideal situation occurs for all 2-handles in $X \backslash W$ then the construction of a Stein structure on X, and of a holomorphic map $X \to Y$ in a given homotopy class, can be completed exactly as in the previous section.

Suppose now that $k = [\beta] - TB \neq -1$ for some 2-handle. A basic fact [24, 128] is that for any Legendrian knot C in a contact 3-manifold there is a $\mathcal{C}^0$-small isotopy preserving the knot type, but changing its Legendrian knot type, which adds a desired number of left (negative) twists to the TB framing. (One adds small spirals to the knot C.) Since the homotopy class $[\beta]$ of the framing is preserved under an isotopy of C in Σ, we see that $k = [\beta] - TB$ can be increased by any number of units. If $k < -1$, it is therefore possible to obtain an isotopic embedding $(D, S) \hookrightarrow (X \backslash W, \Sigma)$ satisfying (9.33), thereby reducing the problem to the previous case.

The problem becomes nontrivial when $k \geq 0$ since it is in general impossible to add right twists to the TB framing (equivalently, to decrease the framing index k). This is only possible in a contact structure which is *overtwisted*, in the sense that it contains a topologically unknotted Legendrian knot K with the Thurston-Bennequin index $tb(K) = 0$; adding such knot to a Legendrian knot adds a positive twist to the TB framing, making it possible to decrease $k = [\beta] - TB$ and hence reach $k = -1$. However, Eliashberg proved in [128] that contact structures arising as boundaries of strongly pseudoconvex Stein manifolds are never overtwisted. (Contact structures that are not overtwisted are called *tight*.) A 2-handle for which we can not find an isotopy of the boundary circle to a Legendrian knot so that (9.33) holds will be called in the sequel *a wrongly attached handle*.

Gompf [215] circumvented the problem by replacing a wrongly attached 2-handle by a handle with the correct framing index -1 and then introducing sufficiently many positive self-plumbings on the core disc. These self-plumbings restore the second homology group of the manifold to the correct group of the initial manifold, at the cost of introducing new superfluous generators to its fundamental group. We end up with an immersed correctly attached 2-handle to which the Stein structure can be extended. In order to get the topologically correct manifold, one must then trivialize all double point loops coming from the self-plumbings (double points) on the core disc. This is again done by correctly attaching an immersed 2-disc along each double point loop, etc. This procedure does not terminate in a finite number of steps, and it converges to an attached Casson handle that is homeomorphic, but not necessarily diffeomorphic, to the standard 4-handle. For a summary of Gompf's construction see [194, p. 639].

We now recall from [194] a slightly different constrution whose advantage is that we remain inside the given manifold X at all steps.

As said before, the TB invariant of a Legendrian knot can be increased by an arbitrary integer. Since the homotopy class $[\beta]$ of the normal framing does not change by such isotopy, we can assume that the framing coefficient is odd, $k(\beta) = [\beta] - TB = -1 + 2k$ for some $k \in \mathbb{Z}$. If $k < 0$, we add left twists to the TB framing to get (9.33) and we are done. If $k > 0$, we glue onto the core disc M k copies of homologically trivial Weinstein immersed sphere $K = F(S^2)$ (9.21), given as the image of the map

$$F(x,y,z) = \big(x(1+2\mathrm{i}z), y(1+2\mathrm{i}z)\big) \in \mathbb{C}^2 \tag{9.34}$$

where $(x,y,z) \in \mathbb{R}^3$, $x^2+y^2+z^2 = 1$. (Precisely, we take the connected sum of M with k copies of K.) Recall (p. 424) that the normal bundle ν_K of $K \subset \mathbb{C}^2$ has Euler number $\chi(\nu_K) = -2$; hence a copy K glued to M reduces the framing coefficient of M by two units. Adding k copies of K will therefore reduce the framing coefficient to -1.

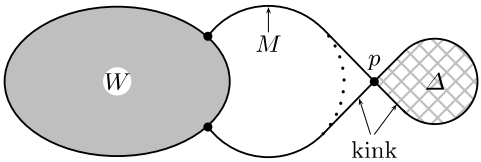

Fig. 9.5. A kinky disc M with a trivializing 2-cell Δ. ([194, p. 640, Fig. 8] and also [195, p. 356, Fig. 4])

With F as in (9.34), let $\Delta = \{F(0,y,z) \colon y \geq 0,\ y^2+z^2 \leq 1\} \subset \mathbb{C}^2$. This 2-disc is embedded except along the side $\{y=0\}$ which gets pinched to $0 \in \mathbb{C}^2$. Note that $b\Delta \subset F(S^2)$, and the union $F(S^2) \cup \Delta$ has a tubular neighborhood diffeomorphic to $S^2 \times \mathbb{R}^2$.

In order to make a self-intersection at a point p in the core disc M of a handle h in our 4-manifold X, we replace a small disc in M around p by a copy of the standard kink K. (See Fig. 9.5; we removed the small dotted disc and smoothly attached along its boundary a kinky disc.) This surgery reduces the relative Euler number over the immersed disc M by 2 for each kink. Adding k kinks on M inside X and then taking a tubular neighborhood has the same effect as first removing the handle h from X, reattaching it with a framing of the boundary reduced by $2k$, and then performing k self-plumbings on h (as was done in [215]). This shows that the manifold constructed in Gompf's proof can be seen as a submanifold of the original manifold X, changed only by a surgery in a small coordinate neighborhood of each of the kinked points on the core disc of the handle h. We can also explicitly see the trivializing 2-cell Δ that needs to be added to each of the kinks in order to reconstruct the desired manifold. In the next stage of the construction every such disc Δ will also have to receive a kink in order to correct its framing coefficient. This begins the *Casson tower* procedure which will converge to a Casson handle in place of the original removed disc in M.

We are now ready to prove the following more precise version of Theorem 9.10.2 when $\dim_{\mathbb{R}} X = 4$. Unlike in the case $n > 2$ we do not need to assume that the almost complex structure J is defined on all of X since the obstruction to extending J only appears for handles of index > 2. However, if J is already given on all of X then one can choose (X', J') in the theorem such that the almost complex structure $h^*(J')$ on X is homotopic to J; this notion makes sense for orientation preserving homeomorphisms [215, p. 645].

Theorem 9.12.1. *Let X be a smooth oriented 4-manifold exhausted by a Morse function $\rho\colon X \to \mathbb{R}$ without critical points of index > 2. Assume that for some $c \in \mathbb{R}$ there is a Stein structure J on $X_c = \{x \in X\colon \rho(x) < c\}$ such that $\rho|_{X_c}$ is strongly J-plurisubharmonic. Let Y be a complex manifold, P a compact Hausdorff space, and $f\colon X \times P \to Y$ a P-map which is J-holomorphic in X_c. Given a compact set $K \subset X_c$ and a number $\epsilon > 0$, there exist a Stein surface (X', J'), an orientation preserving homeomorphism $h\colon X \to X'$ which is biholomorphic in a neighborhood of K, and a holomorphic P-map $f'\colon X' \times P \to Y$ such that the P-map $\widetilde{f}\colon X \times P \to Y$, defined by $\widetilde{f}(x,p) = f'(h(x),p)$, is homotopic to f and satisfies*

$$\sup_{x\in K,\ p\in P} \operatorname{dist}_Y\big(f(x,p), \widetilde{f}(x,p)\big) < \epsilon.$$

Proof. Assume that our smooth 4-manifold X is constructed by successively attaching handles $h_1, h_2, h_3, \ldots$ of index ≤ 2, beginning with the compact domain $\overline{W} \subset X$ with smooth boundary $\Sigma = bW$. By the assumption we also have an integrable complex structure J in a neighborhood of $\overline{W}$ such that W is Stein and its boundary Σ is strongly pseudoconvex.

Let $M_1, M_2, \ldots$ be the cores of the handles $h_1, h_2, \ldots$, chosen such that their union is a smoothly embedded CW complex inside X. Since we have not assumed that our handlebody is finite, we can not ask for the ordering of the handles with regard to their indexes. However, due to local compactness we can ask that when a handle h_j with core M_j is being attached, all handles whose core discs intersect the boundary bM_j have already been attached in earlier steps. We can also assume that bM_j consists only of the core discs of handles of lower indexes. We can now proceed with the induction as in the proof of Theorem 9.11.1, but with the following modifications:

(1) When a 2-handle is attached with a wrong framing, we insert the right number of kinks to its core disc in order to change the framing coefficient to -1. This insures that we can extend J to a Stein structure in a tubular strongly pseudoconvex neighborhood of the immersed disc. (The disc is totally real in this structure and has a special double point at each kink that corresponds in local coordinates to the union of totally real coordinate planes $\mathbb{R}^2 \times \{\mathrm{i}0\}^2 \cup \{0\}^2 \times \mathrm{i}\mathbb{R}^2$ in $\mathbb{C}^2$.)

(2) Each time before proceeding to the next handle h_{j+1} we perform one more step on each of the kinked discs appearing in the sequence before. More precisely, we add a new kinked disc which cancels the double point loop at the self-intersection point introduced in the previous step.

The first condition is essential since we need to build a manifold that is Stein. The second condition insures that each handle is properly worked upon, thereby producing a Casson tower at every place where a kink was made in the initial 2-disc. At every step we also approximate the given map,

which has already been made holomorphic in a tubular strongly pseudoconvex neighborhood of our partial (finite) subcomplex, by a map holomorphic in a tubular neighborhood of the previous domain with all core discs that have been added at the given step.

Note that whenever a handle is wrongly attached, the above process is never finite. The reason is that in the standard kink K, the disc Δ needed to be added to reconstruct the original manifold requires exactly one positive kink in order to be able to extend the Stein structure to its neighborhood.

The proof can now be concluded as in Theorem 9.11.1. We construct an increasing sequence of Stein domains $X_1 \subset X_2 \subset \cdots$ inside the smooth 4-manifold X such that X_j is Runge in X_{j+1} for every j, and a sequence of maps $f_j \colon X \to Y$ $(j = 1, 2, \ldots)$ such that f_j is holomorphic on X_j, it approximates f_{j-1} uniformly on X_{j-1}, and is homotopic to f_{j-1} by a homotopy which is holomorphic and uniformly close to f_{j-1} on X_{j-1}. Note that the complex structure on X_j does not change from the j-th step of the construction on, so we get a limit complex structure J' on the union $X' = \cup_j X_j \subset X$. The Runge property of each pair $X_j \subset X_{j+1}$ insures that (X', J') is a Stein manifold. By the construction X' is homeomorphic to X. (It is also diffeomorphic to X provided that no Casson handles were used in the construction.) A small ambient topological deformation moves the initial CW complex into X'; see [216] for the details. By the construction the limit map $f' = \lim_{j\to\infty} f_j \colon X' \to Y$ is holomorphic, and the map $f' \circ h \colon X \to Y$ is homotopic to f.

The same proof applies to any smoothly embedded 2-complex M inside X: After a small ambient topological deformation we find a new embedding $M' \hookrightarrow X$ with a Stein thickening $M' \subset X' \subset X$ such that a given continuous map $M \to Y$ admits a holomorphic representative $X' \to Y$. $\square$

Proof of Theorem 9.10.3 for $n = 2$. The proof, taken from [194], is similar to Gompf's construction in [216]. Assume that (X, J) is a complex surface, not necessarily Stein, such that the underlying smooth manifold X has a handlebody decomposition without handles of index > 2. As in the case $n > 2$ we construct an increasing sequence of open Stein domains $X_1 \subset X_2 \subset \cdots$ in X and J-holomorphic maps $f_j \colon X_j \to Y$ such that each X_j is Runge in X_{j+1} and the other properties are as before, the difference being that their union $\Omega = \cup_{j=1}^{\infty} X_j$ is only homeomotopic (and not diffeotopic) to X.

The main inductive step requires certain 'kinky handles'. To this end one could use totally real immersed discs furnished by Lemma 3.8.3 (p. 84). We give another construction which depends on the cancellation theorem for complex points of real surface in complex surfaces (Theorem 9.5.1, p. 418). Let K be the immersed Lagrangian sphere in $\mathbb{C}^2$ (9.34) with one positive double point. From the index formula (9.10) (p. 414) we see that the connected sum of an oriented real surface M in a complex surface X with a homologically trivial copy of K decreases each of the Lai indexes $I_\pm(M)$ by one. (Lai indexes were introduced in §9.4. They are well defined for surfaces that are totally real

along the boundary curves, and are invariant under isotopies preserving this condition.) As before we say that we added a positive kink to the surface. Adding a sufficient number of positive kinks to M we thus get a homologically equivalent immersed surface M' satisfying $I_\pm(M') \le 0$ that agrees with M near bM. By Theorem 9.5.1 (p. 418) we can deform M' to a surface with only (special) hyperbolic complex points, keeping the boundary fixed. Hence the new surface has a basis of tubular Stein neighborhoods by Theorem 9.2.2 (p. 409). Theorem 3.7.2 (concerning holomorphic approximation on handlebodies) easily adapts to immersed totally real surfaces with isolated special double points (see [194, Theorem 4.1] for the details).

We now complete the proof of Theorem 9.10.3 in the case $\dim_{\mathbb{R}} X = 4$. Assume that X is obtained from a strongly pseudoconvex domain $W \subset X$ by successively attaching handles $h_1, h_2, \ldots$ with core discs $M_1, M_2, \ldots$, where the ordering of these handles satisfies the same condition as in the proof of Theorem 9.12.1 above. We use the same notation as in the proof of Theorem 9.10.3 for $\dim_{\mathbb{R}} X \neq 4$, beginning with $W_0 = W$ and $f_0 = f$. In the inductive step we have a smoothly bounded, strongly pseudoconvex domain $W_j \subset X$ and a map $f_j \colon X \to Y$ that is holomorphic in a neighborhod of $\overline{W}_j$. The set W_j is a tubular neighborhood of the union of $\overline{W}$ with the cores of handles attached in the earlier steps. Since these cores may have received kinks, W_j does not have the correct homeomorphic type, but this will be corrected in the limit by the Casson handles resulting from the construction.

We now attach to $\overline{W}_j$ the next handle in the sequence. As before, 0- and 1-handles do not pose any problem. For a 2-handle h we first apply Gromov's Legendrization theorem [236, p. 339, (B')] to make sure that the boundary of its core disc M is a Legendrian curve in bW_j and the attachment is J-orthogonal. We then add enough positive kinks to the core disc M in order to cancel all elliptic points as explained above. The new immersed disc M' obtained in this way is made real analytic by a small deformation that does not destroy any of the above properties. It follows from Theorem 3.8.2 (p. 83) that $\overline{W}_j \cup M'$ has a basis of tubular, strongly pseudoconvex Stein neighborhoods in X. In the same way we add to $\overline{W}_j$ a new kinky disc at each of the kinks from the earlier stages of the construction (trivializing the double point loop), making sure that the conditions (1) and (2) in the proof of Theorem 9.12.1 are satisfied. These additional kinky discs $\Delta_1, \ldots, \Delta_k$ can be chosen such that the set $L_j = \overline{W}_j \cup M' \cup (\cup_{l=1}^k \Delta_l)$ admits a basis of tubular, strongly pseudoconvex, Stein neighborhoods in X. We then we approximate the map f_j from the previous step (which is holomorphic near $\overline{W}_j$) uniformly on the set L_j by a map $f_{j+1} \colon X \to Y$ that is holomorphic in a neighborhood U_j of L_j. By Theorem 3.8.2 (p. 83) there is a strongly pseudoconvex tubular neighborhood W_{j+1} of L_j, contained in U_j, such that W_j is Runge in W_{j+1}. This completes the induction step. In the limit we obtain a holomorphic map $f' \colon \Omega \to Y$ on the Stein domain $\Omega = \cup_j W_j \subset X$ with the stated properties. $\square$

The following corollary is obtained by combining Theorem 9.12.1 (p. 457) with Corollary 3.2 and Theorem 3.3 in [215, p. 648].

Corollary 9.12.2. *Let X be a smooth, closed, oriented 4-manifold. There exists a smooth, finite wedge of circles $\Gamma \subset X$ such that for every continuous map $f\colon X\backslash\Gamma \to Y$ to a complex manifold Y there is a (possibly exotic) Stein structure on $X\backslash\Gamma$ and a holomorphic map $\widetilde{f}\colon X\backslash\Gamma \to Y$ homotopic to f. In $X = \mathbb{P}^2$ this holds after removing a single point, and in this case any Stein structure on $\mathbb{P}^2\backslash\{p\}$ is exotic. The analogous result holds for every open oriented smooth 4-manifold after removing a suitably chosen smooth 1-complex.*

The point is that there is a wedge of circles Γ in X such that $X\backslash\Gamma$ admits a handle decomposition without any 3-handles or 4-handles. The complex projective plane $\mathbb{P}^2$ has a single 4-cell (and no 3-cells) in its handlebody decomposition, hence removing a point leaves only cells of index ≤ 2.

Here is another result obtained by combining Gompf's methods from [216] with the proof of Theorem 9.10.2.

Corollary 9.12.3. *Let M be a tame, topologically embedded CW n-complex in a complex manifold X^n, and let U be an open neighborhood of M in X. For every continuous map $f\colon M \to Y$ to a complex manifold Y there exist a topological isotopy $h_t\colon X \to X$ satisfying $h_0 = \mathrm{Id}_X$ and $h_t(M) \subset U$ for all $t \in [0,1]$, a Stein thickening $\Omega \subset U$ of the CW complex $h_1(M)$, and a holomorphic map $\widetilde{f}\colon \Omega \to Y$ such that $\widetilde{f} \circ h_1\colon M \to Y$ is homotopic to f.*

Gompf showed that the necessary adjustment of the initial complex $M \subset X$ is quite mild from the topological point of view. When $n = 2$ then Stein domains Ω obtained in this way will typically have nonsmooth boundaries in X and may be chosen to realize uncountably many distinct diffeomorphism types. If M is a smooth CW complex and if all 2-cells satisfy the correct framing conditions then it is possible to find Stein thickenings of a $\mathcal{C}^0$-small smooth perturbation of M in X which even have the *diffeomorphism type* of a smooth handlebody with core M. (This is always possible in dimension $n \neq 2$ by the methods explained in §9.11.) This gives interesting consequences concerning envelopes of holomorphy of real submanifolds of dimension $\leq n$ in $\mathbb{C}^n$ and in other Stein manifolds. Here is a simple example.

Example 9.12.4. Assume that $M \hookrightarrow \mathbb{C}^n$ is an embedded n-dimensional submanifold of $\mathbb{C}^n$. Let $U \subset \mathbb{C}^n$ be an open set containing M. If $n \neq 2$ then there exists an isotopic deformation $M' \subset U$ of M and a Stein domain $\Omega \subset U$ containing M'. In particular, the (possibly multisheeted) envelope of holomorphy can remain very close to M, something which definitely fails for the polynomial hull $\widehat{M}$ [15]. $\square$

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The darkness has come in from the Mediterranean... Time to go! Time to go!

M. A. Bulgakov, *The Master and Margarita*

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